

Class 11 Physics

Master Reference

Complete Chapter Compendium

Fourteen chapters of the NCERT Class 11 Physics syllabus, prepared for CBSE Board examinations and for JEE Main and JEE Advanced. Every subtopic carries concept intuition, key formulas with validity conditions, board derivations, worked examples laddered from CBSE to JEE Advanced, practice exercises, MCQ distractor analysis, common pitfalls, and a cheat sheet. Each chapter is followed by a supplement and a Round 2 addendum, cross-checked against NCERT and a full JEE reference library, closing the gaps between the Board syllabus and what JEE actually tests. Corrections to the printed text are collected in the errata at the back.

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Chapter 1: Units and Measurements

Sub-topic 01: Systems of Units and SI Units

Exam Relevance: This subtopic shows up as 1 direct question in JEE Main most years (often unit-conversion or "which is a base unit" type) and underpins every dimensional-analysis problem in JEE Advanced. NEET asks it rarely and only at the definition level. CBSE Boards favour 1–2 mark short answers on base units, supplementary units, or the SI definitions.

Section 1: Concept Intuition

Imagine you call up a friend in another city and tell him, "Yaar, I just bought a table that's *five* long." Five *what*? Five feet? Five metres? Five hand-spans? The number "five" by itself is useless — it tells him nothing until you attach a *standard of comparison* to it. That standard is what we call a **unit**. Every measurement in the universe is really a sentence with two words: a **number** and a **unit**. Drop either one and the sentence becomes nonsense.

Here's the deeper idea. When you measure something, you're secretly asking: "How many times does my chosen standard fit into this thing?" If your standard is the metre and the table swallows it three-and-a-half times, you say the table is 3.5 m. So a measurement is just a *ratio* — the size of the thing divided by the size of your reference. This immediately tells you something neat: if you switch to a *smaller* unit, the number gets *bigger* (more of the small unit fit in), and vice versa. A wall that is 3 m is also 300 cm — same wall, smaller unit, bigger number.

Now, we don't need a separate fundamental standard for *everything*. That would be chaos. Instead, scientists picked a small handful of quantities to act as the "founding members" — **base (fundamental) quantities** — and built everything else out of them. Length, mass, and time are the obvious three. Speed? That's just length per time. Force? Mass times length per time-squared. These built-from-base quantities are called **derived quantities**, and their units are **derived units**. It's like a language: a few root words (base units) combine to make an entire dictionary (derived units).

Over history, different groups of people chose different "founding members" and different sizes for them. The British used the **FPS system** (foot, pound, second). Scientists liked the **CGS system** (centimetre, gram, second). Engineers leaned on the **MKS system** (metre, kilogram, second). The problem? A pharma company in Pune, a NASA engineer, and a French physicist all measuring the same thing got three different numbers. So in 1971 the world finally agreed on one universal standard — the **Système International d'Unités**, or **SI** — an upgraded, "rationalised" MKS system with **seven base units** and **two supplementary units**.

Think about it this way... picture the seven SI base units as the seven primary notes of Indian classical music — *Sa, Re, Ga, Ma, Pa, Dha, Ni*. Just seven notes, yet every raga, every Bollywood song, every symphony is built by combining them. Similarly, length, mass, time, electric current, temperature, amount of substance, and luminous intensity are the seven "notes" of physics, and *every* physical quantity — from the energy of a falling mango to the magnetic field of a galaxy — is a "song" composed from these seven.

One more practical wrinkle worth meeting now: even with a single agreed system, the *same* base quantity often needs differently-sized "rulers" for different jobs. You'd measure the distance between two cities in kilometres, the thickness of a hair in micrometres, the gap between atoms in angstroms, and the span of the solar system in astronomical units — yet all of these are *length*. SI handles this gracefully through **prefixes** (kilo-, milli-, nano-,

and so on) and a handful of accepted **practical units**, so the scientist never has to write a number trailing twenty zeros. The base unit stays the metre; the prefixes are just convenient costumes it wears.

Limiting Conditions & Assumptions. A few quiet rules govern this whole system:

- The choice of base quantities is a matter of *convention*, not nature. There's nothing sacred about choosing length and time as fundamental — you *could* build a system where speed and time are fundamental and length becomes derived. SI's seven are simply the agreed-upon, practical choice.
- A good unit must be **well-defined, invariant (unchanging with time/place/temperature), reproducible anywhere, and imperishable**. This is exactly why, in 2019, the world abandoned the physical platinum cylinder that defined the kilogram — a lump of metal can get scratched or gain dust — and switched to definitions based on **fixed values of fundamental constants of nature**, which never change.
- **Supplementary units** (radian, steradian) are special: they are genuinely **dimensionless**, even though they describe angles. Don't treat them like ordinary base units.

Section 2: Key Formulas & Definitions

The seven SI base quantities, their units, symbols, and dimensional formulae:

Base Quantity	Unit	Symbol	Dimensional Formula
Length	metre	m	$[L]$
Mass	kilogram	kg	$[M]$
Time	second	s	$[T]$
Electric current	ampere	A	$[A]$
Thermodynamic temperature	kelvin	K	$[K]$
Amount of substance	mole	mol	$[\text{mol}]$
Luminous intensity	candela	cd	$[\text{cd}]$

The two SI supplementary units:

$$\text{Plane angle:} \quad = \frac{ds}{r} \quad (\text{unit: radian, rad})$$

where ds is the arc length and r the radius. Dimensionless: $[M^0 L^0 T^0]$.

$$\text{Solid angle:} \quad d\Omega = \frac{dA}{r^2} \quad (\text{unit: steradian, sr})$$

where dA is the area of the patch on a sphere and r the radius. Dimensionless: $[M^0 L^0 T^0]$.

The seven defining constants of the SI (the basis of the 2019 redefinition). Every base unit is now fixed by assigning an *exact* numerical value to a constant of nature:

- Speed of light in vacuum: $c = 299792458 \text{ m s}^{-1} \rightarrow$ defines the **metre**
- Caesium hyperfine frequency: $\Delta\nu_{\text{Cs}} = 9192631770 \text{ Hz} \rightarrow$ defines the **second**
- Planck constant: $h = 6.62607015 \times 10^{-34} \text{ J s} \rightarrow$ defines the **kilogram**
- Elementary charge: $e = 1.602176634 \times 10^{-19} \text{ C} \rightarrow$ defines the **ampere**
- Boltzmann constant: $k_B = 1.380649 \times 10^{-23} \text{ J K}^{-1} \rightarrow$ defines the **kelvin**

- Avogadro constant: $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1} \rightarrow$ defines the **mole**
- Luminous efficacy: $K_{cd} = 683 \text{ lm W}^{-1} \rightarrow$ defines the **candela**

Common derived units (built from the base seven). A derived unit that is just a product/quotient of base units with no extra numerical factor is called **coherent** — and SI is deliberately built to be coherent, which is why conversions inside SI never need awkward constants. A few you must know cold:

- Force: newton, $\text{N} = \text{kg m s}^{-2}$, dimension $[MLT^{-2}]$
- Energy/work: joule, $\text{J} = \text{kg m}^2 \text{s}^{-2}$, dimension $[ML^2T^{-2}]$
- Power: watt, $\text{W} = \text{kg m}^2 \text{s}^{-3}$, dimension $[ML^2T^{-3}]$
- Pressure: pascal, $\text{Pa} = \text{kg m}^{-1} \text{s}^{-2}$, dimension $[ML^{-1}T^{-2}]$
- Charge: coulomb, $\text{C} = \text{A s}$, dimension $[AT]$

SI prefixes. Physical quantities span an enormous range — from the $\sim 10^{-15} \text{ m}$ size of a nucleus to the $\sim 10^{26} \text{ m}$ distance of a far quasar. Rather than drowning in zeros, SI uses standard decimal prefixes:

Factor	Prefix	Symbol		Factor	Prefix	Symbol
10^{12}	tera	T		10^{-2}	centi	c
10^9	giga	G		10^{-3}	milli	m
10^6	mega	M		10^{-6}	micro	μ
10^3	kilo	k		10^{-9}	nano	n
10^2	hecto	h		10^{-12}	pico	p
10^1	deca	da		10^{-15}	femto	f

Practical (non-SI but accepted) units. For convenience, physics still uses some handy units alongside SI:

- *Length:* 1 angstrom = $1 \text{ \AA} = 10^{-10} \text{ m}$ (atomic sizes); 1 fermi = $1 \text{ fm} = 10^{-15} \text{ m}$ (nuclear sizes); 1 astronomical unit = $1 \text{ AU} \approx 1.496 \times 10^{11} \text{ m}$ (Earth–Sun distance); 1 light year $\approx 9.46 \times 10^{15} \text{ m}$; 1 parsec $\approx 3.086 \times 10^{16} \text{ m}$.
- *Mass:* 1 atomic mass unit = $1 \text{ u} \approx 1.66 \times 10^{-27} \text{ kg}$; 1 tonne = 10^3 kg ; 1 quintal = 10^2 kg .
- *Time:* derived directly from the second (minute, hour, day) — but the second remains the SI base.

The unit-conversion master relation. The physical size of a quantity is fixed, so the product of its numerical value n and unit size u is a constant across systems:

$$n_1 u_1 = n_2 u_2$$

For a quantity with dimensional formula $[M^a L^b T^c]$:

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c$$

Section 3: Key Procedures & Logical Framework

Since "Systems of Units and SI Units" has no formal physics derivation, here are the three logical procedures every student must own.

Procedure A — Expressing a derived unit in terms of base units.

1. Start from the defining relation of the quantity. Example: pressure is force per area, $P = F/A$.
2. Write each piece in base units. Force (from $F = ma$) is $\text{kg} \times \text{m s}^{-2} = \text{kg m s}^{-2}$, and area is m^2 . Why? Because every derived unit must ultimately trace back to the seven base units — that's the whole point of a coherent system.
3. Combine. $\frac{\text{kg m s}^{-2}}{\text{m}^2} = \text{kg m}^{-1} \text{s}^{-2}$. This is the pascal (Pa) in base units. The reasoning, not the memorised result, is what protects you when the quantity is unfamiliar — in an exam you will sometimes meet a quantity you have never seen, and this base-unit chain is the only safe way to handle it.

Procedure B — Converting a value between two systems.

1. Find the dimensional formula of the quantity, $[M^a L^b T^c]$. (This is the step students skip — and then get conversions wrong.)
2. Apply the master relation $n_2 = n_1 \left(\frac{M_1}{M_2}\right)^a \left(\frac{L_1}{L_2}\right)^b \left(\frac{T_1}{T_2}\right)^c$. Why raise to the powers? Because if length appears cubed in the quantity (like in volume or density), then the length-unit ratio must be cubed too — the dimension counts how many times that unit "participates."
3. Sanity check the direction. Converting to a *smaller* unit should give a *larger* number. If your answer breaks this, you've inverted a ratio.

Procedure C — Building a "new system of units" (the trick behind redefinition).

1. Identify which quantities are declared fundamental. In the real 2019 SI, fundamental *constants* (like c , h) are declared fixed.
2. Express the desired unit as a product of powers of the fundamentals, with unknown exponents.
3. Equate dimensions on both sides and solve for the exponents. This is dimensional bookkeeping — each of M , L , T gives one equation. The philosophy: if you anchor a few unchanging constants, every unit follows automatically and can be reproduced in any lab in the cosmos.

Figure 1.1 A central box labelled "7 SI Base Units (m , kg , s , A , K , mol , cd)" with seven arrows fanning outward to derived-unit boxes — e.g., "speed = m/s ", "force = $kg \cdot m/s^2$ (N)", "pressure = $kg/(m \cdot s^2)$ (Pa)", "energy = $kg \cdot m^2/s^2$ (J)". Below, a separate small box "Supplementary: radian, steradian (dimensionless)" sits to one side, deliberately not connected, to emphasise it is not a base unit. (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Problem. The universal gas constant R has the SI unit $\text{J mol}^{-1} \text{K}^{-1}$. Express this unit entirely in terms of the SI base units.

Solution. Given: unit of R is $\text{J mol}^{-1} \text{K}^{-1}$.

Step 1 — Break down the non-base unit (joule). Energy = work = force $\times$ distance, so

$$\text{J} = \text{N} \times \text{m} = (\text{kg m s}^{-2}) \times \text{m} = \text{kg m}^2 \text{s}^{-2}.$$

Step 2 — Substitute back.

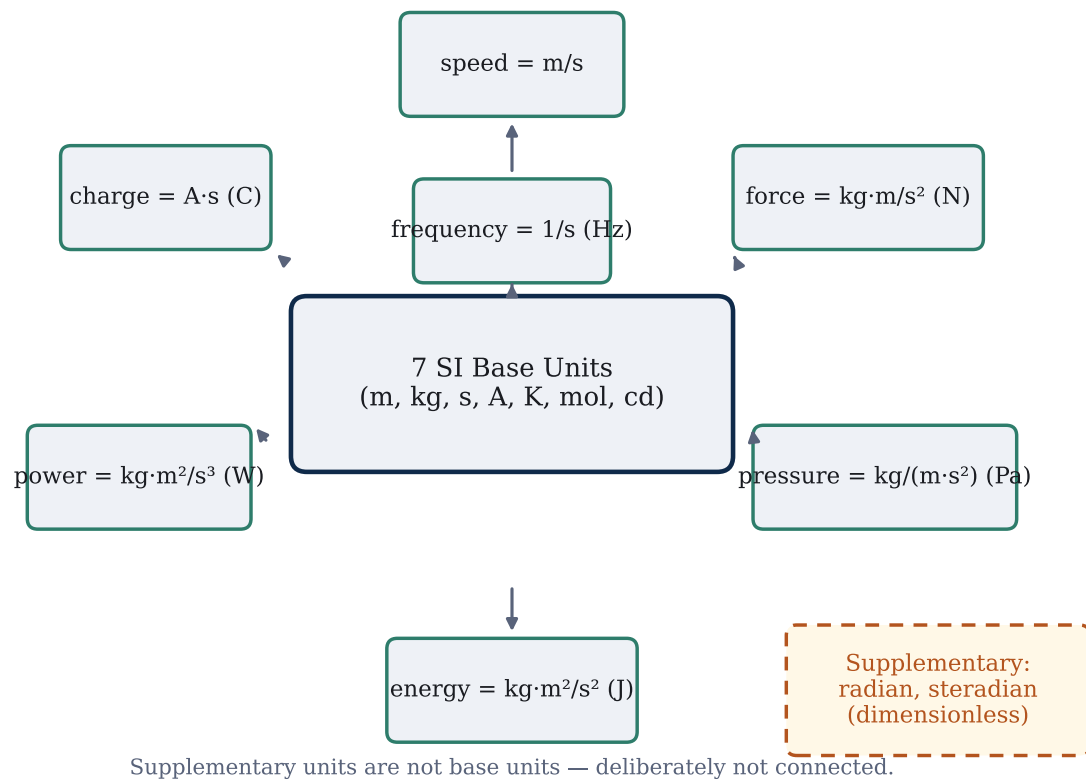


Figure 1.1 A central box labelled "7 SI Base Units (m, kg, s, A, K, mol, cd)" with seven arrows fanning outward to derived-unit boxes — e.g., "speed = m/s", "force = kg·m/s² (N)", "pressure = kg/(m·s²) (Pa)", "energy = kg·m²/s² (J)". Below, a separate small box "Supplementary: radian, steradian (dimensionless)" sits to one side, deliberately not connected, to emphasise it is not a base unit.

$$\text{unit of } R = \frac{\text{kg m}^2 \text{s}^{-2}}{\text{mol} \cdot \text{K}} = \text{kg m}^2 \text{s}^{-2} \text{mol}^{-1} \text{K}^{-1}.$$

Result: The SI base-unit form of the gas constant is $\boxed{\text{kg m}^2 \text{s}^{-2} \text{mol}^{-1} \text{K}^{-1}}$. Notice how mole and kelvin survive as base units — they cannot be broken down further.

Example 2 — [NEET Speed Trap]

Problem. Which one of the following is a **base (fundamental) quantity** in the SI system? (A) Electric charge (B) Force (C) Electric current (D) Energy

The trap. Electric charge *feels* like it should be fundamental — it's such a basic idea! Most students tick (A) on reflex. **Wrong.** In SI, the chosen electrical base quantity is **electric current**, not charge. Charge is *derived*: since current is charge per unit time, charge = current $\times$ time, giving the coulomb as $C = A \text{ s}$.

Fast elimination. Force (kg m s^{-2}) and energy ($\text{kg m}^2 \text{s}^{-2}$) are visibly built from other units $\rightarrow$ derived. That leaves charge vs current. Recall the rule: *SI picked current as base; everything electrical is derived from the ampere.*

Answer: (C) Electric current.

Pitfall to remember: "Fundamental in feeling" $\neq$ "fundamental in SI." The seven base quantities are a fixed, memorised list — don't reason from intuition here.

Example 3 — [JEE Main Level]

Problem. A student measures the surface tension of a soap film as 36 dyne cm^{-1} in the CGS system. Convert this value into SI units. (*This combines two ideas: identifying the dimensional formula of surface tension, and applying the system-conversion relation.*)

Solution. Step 1 — Dimensional formula of surface tension. Surface tension = force per unit length:

$$[\text{Surface tension}] = \frac{[MLT^{-2}]}{[L]} = [MT^{-2}], \quad \text{i.e., } a = 1, b = 0, c = -2.$$

Step 2 — Apply the conversion relation. CGS: $M_1 = \text{g}$, $L_1 = \text{cm}$, $T_1 = \text{s}$. SI: $M_2 = \text{kg}$, $L_2 = \text{m}$, $T_2 = \text{s}$.

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^1 \left(\frac{L_1}{L_2} \right)^0 \left(\frac{T_1}{T_2} \right)^{-2} = 36 \times \left(\frac{\text{g}}{\text{kg}} \right) \times 1 \times 1$$

Step 3 — Plug the ratio. Since $1 \text{ g} = 10^{-3} \text{ kg}$:

$$n_2 = 36 \times 10^{-3} = 0.036.$$

Result: Surface tension = $\boxed{0.036 \text{ N m}^{-1}}$. *Sanity check:* the kilogram is a much bigger mass unit than the gram, so the number should shrink — and it did ($36 \rightarrow 0.036$).

Example 4 — [JEE Advanced Level]

Problem. Suppose we build a "natural" system of units in which the **speed of light** c , the **gravitational constant** G , and **Planck's constant** h are each taken as fundamental and set numerically equal to 1. Derive, using dimensional analysis, the expression for the **unit of length** in this system in terms of c , G , and h , and estimate its order of magnitude.

Solution. This mirrors exactly how the modern SI anchors units to constants of nature — here we anchor length to three constants.

Step 1 — Write length as a product of powers.

$$L = c^x G^y h^z$$

Step 2 — Insert dimensional formulae.

$$[c] = [LT^{-1}], \quad [G] = [M^{-1}L^3T^{-2}], \quad [h] = [ML^2T^{-1}]$$

So:

$$[L^1] = [LT^{-1}]^x [M^{-1}L^3T^{-2}]^y [ML^2T^{-1}]^z$$

$$[L] = [M^{-y+z} L^{x+3y+2z} T^{-x-2y-z}]$$

Step 3 — Match exponents of M, L, T (this is the rigorous core).

$$M : -y + z = 0 \Rightarrow z = y$$

$$T : -x - 2y - z = 0 \Rightarrow x = -2y - z = -3y$$

$$L : x + 3y + 2z = 1 \Rightarrow -3y + 3y + 2y = 1 \Rightarrow y = \frac{1}{2}$$

Hence $z = \frac{1}{2}$ and $x = -\frac{3}{2}$.

Step 4 — Assemble the result.

$$L = c^{-3/2} G^{1/2} h^{1/2} = \sqrt{\frac{Gh}{c^3}}$$

Step 5 — Order of magnitude. Using $G \approx 6.67 \times 10^{-11}$, $h \approx 6.63 \times 10^{-34}$, $c \approx 3 \times 10^8$ (all SI):

$$L = \sqrt{\frac{(6.67 \times 10^{-11})(6.63 \times 10^{-34})}{(3 \times 10^8)^3}} \approx \sqrt{1.64 \times 10^{-69}} \approx 4 \times 10^{-35} \text{ m.}$$

Result: the natural unit of length is $\boxed{\sqrt{Gh/c^3} \approx 4 \times 10^{-35} \text{ m}}$ — essentially the Planck length, the scale at which gravity and quantum mechanics collide. This is the same logic the SI now uses: fix the constants, and every unit emerges automatically.

Section 5: Practice Exercises

1. **[CBSE]** Express the SI unit of power (the watt) entirely in terms of base units.
2. **[NEET]** State the total number of base quantities and the number of supplementary units in the SI system.
3. **[JEE Main]** The density of a metal is 5 g cm^{-3} . Convert this value into SI units (kg m^{-3}).
4. **[JEE Main]** In a new system of units, the unit of force is 1000 N, the unit of length is 100 m, and the unit of time is 10 s. Find the unit of mass in this system (in kg).
5. **[JEE Advanced]** If energy (E), velocity (v), and time (T) are chosen as the fundamental quantities, find the dimensions of **mass** and of **length** in terms of E, v , and T .

Final Answers:

1. $\text{kg m}^2 \text{s}^{-3}$
2. 7 base quantities; 2 supplementary units (radian, steradian)

3. 5000 kg m^{-3}
 4. 1000 kg (since $\text{mass unit} = F \cdot T^2 / L = 1000 \times 10^2 / 100$)
 5. $\text{Mass} = [E v^{-2}]$; $\text{Length} = [v T]$
-

Section 6: Crack the MCQ

Q1. Which of the following is **NOT** an SI base unit? (A) mole (B) kelvin (C) coulomb (D) candela

Answer: (C) coulomb. The coulomb is a *derived* unit, $C = A \text{ s}$, since the ampere (current) — not charge — is the electrical base unit. (A) mole and (D) candela trap students who forget that "amount of substance" and "luminous intensity" made it onto the base list — both are legitimate base units. (B) kelvin is a base unit; students sometimes wrongly demote it because "temperature feels derived from energy," but SI defines it as base.

Q2. The 2019 redefinition of the SI base units was based primarily on: (A) carefully preserved physical artifacts (B) fixed numerical values of fundamental constants of nature (C) properties of water at standard conditions (D) fractions of the Earth's dimensions

Answer: (B). Every base unit is now fixed by assigning an exact value to a constant (c , h , e , etc.), making the units permanent and reproducible anywhere. (A) describes the *old* kilogram (the platinum–iridium cylinder), which was abandoned precisely because artifacts degrade — a classic distractor. (C) recalls the old Celsius/temperature and litre conventions tied to water — outdated. (D) recalls the *original* 1790s metre (one ten-millionth of the Earth's quadrant) — historically true but not the 2019 basis.

Q3. Plane angle and solid angle in the SI system are best described as: (A) base quantities with their own dimensions (B) derived quantities with dimension $[L]$ (C) dimensionless supplementary units (D) fundamental units with dimension $[L^2]$

Answer: (C). Radian (ds/r) and steradian (dA/r^2) are ratios of like quantities, so all dimensions cancel — they are dimensionless **supplementary** units. (A) wrongly promotes them to base quantities. (B) and (D) fall for the illusion that "angle involves length/area, so it must carry length dimensions" — but the division by r or r^2 kills every dimension.

Q4. When the same physical quantity is expressed in two different systems of units, the product (numerical value $\times$ size of the unit) is: (A) doubled (B) always constant (C) dependent on the system chosen (D) equal to zero

Answer: (B). The physical magnitude is fixed by nature, so $n_1 u_1 = n_2 u_2$ always holds — this single relation powers every conversion. (A) and (D) are nonsensical distractors for students guessing. (C) is the subtle trap: the *number* and the *unit* each change with the system, which fools students into thinking the *product* changes too — but they change in exactly compensating ways.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Calling charge, force, or energy "fundamental."** Only **electric current** is the electrical base quantity. Memorise the seven base quantities as a fixed list and never reason them out from intuition.
2. **Sloppy unit symbols.** It's "kg" not "Kg" or "kgs"; "K" for kelvin but "k" for the prefix kilo; "N" (capital symbol) but "newton" (lowercase name); "s" not "sec". Unit symbols never take a plural "s" and never a full stop (except at a sentence's end). Boards deduct marks for this.

3. **Forgetting the dimensional power in conversions.** Converting density ($[ML^{-3}]$) means the length ratio must be **cubed**. Skipping the power is the #1 conversion error.
4. **Giving radian/steradian dimensions.** They are dimensionless. Treating them as length/area units corrupts later dimensional analysis.

Pro-Tip (speed + accuracy): For *any* conversion, write the dimensional formula *first*, then mechanically plug into $n_2 = n_1 \left(\frac{M_1}{M_2}\right)^a \left(\frac{L_1}{L_2}\right)^b \left(\frac{T_1}{T_2}\right)^c$. Finish with the direction check: smaller unit $\Rightarrow$ bigger number. This one habit eliminates almost every conversion mistake in seconds.

Section 8: Cheat Sheet Box

SYSTEMS OF UNITS & SI UNITS — QUICK REVISION

- **SI has 7 base units:** metre (m), kilogram (kg), second (s), ampere (A), kelvin (K), mole (mol), candela (cd).
- **2 supplementary units:** radian (rad) for plane angle, steradian (sr) for solid angle — both dimensionless.
- **Plane angle** $d\theta = \frac{ds}{r}$; **Solid angle** $d\Omega = \frac{dA}{r^2}$.
- **2019 SI:** every base unit is fixed by a constant of nature ($c, \Delta\nu_{Cs}, h, e, k_B, N_A, K_{cd}$) — permanent, reproducible, artifact-free.
- **Master conversion relation:** $n_1 u_1 = n_2 u_2 \Rightarrow n_2 = n_1 \left(\frac{M_1}{M_2}\right)^a \left(\frac{L_1}{L_2}\right)^b \left(\frac{T_1}{T_2}\right)^c$.
- **Older systems:** FPS (foot-pound-second), CGS (cm-g-s), MKS (m-kg-s); SI is rationalised MKS.

Memory aids:

- **Seven base quantities** $\rightarrow$ "*Little Miss Tiger Caught A Mango Curry*" = Length, Mass, Time, Current, (temperature in) K... use units instead: *metre, kilogram, second, ampere, kelvin, mole, candela*.
- **Charge is NOT base** $\rightarrow$ "*Current is king; charge is its child* ($C = A \cdot s$)."
- **Conversion direction** $\rightarrow$ "*Smaller unit, bigger number.*"
- **Supplementary units** $\rightarrow$ "*Angles are ratios, so they're dimension-less.*"

Chapter 1: Units and Measurements

Sub-topic 02: Dimensional Analysis and Formulas

Exam Relevance: This is one of the highest-yield subtopics in the whole chapter. JEE Main almost always carries 1 question (checking a relation, deriving a formula, or unit conversion), and JEE Advanced loves it as a tool buried inside harder problems. NEET asks it occasionally as "match the dimensions." CBSE Boards typically set a 2–3 mark question on checking correctness or finding a dimensional formula.

Section 1: Concept Intuition

Every physical quantity is, at heart, a blend of just a few "ingredients" — how much length is in it, how much mass, how much time, and so on. Think of it like a recipe. A samosa isn't a fundamental thing; it's *potato + flour + spices* in certain proportions. In exactly the same way, **speed** isn't fundamental — it's *one part length, one part inverse-time* (length $\div$ time). **Force** is *one part mass, one part length, two parts inverse-time*. The list of "how many parts of each base ingredient" a quantity contains is called its **dimensions**, and writing it out compactly is its **dimensional formula**.

We use square brackets and the symbols M (mass), L (length), T (time), A (current), K (temperature) to record the recipe. Speed has dimensional formula $[LT^{-1}]$. Force has $[MLT^{-2}]$. The beauty is that this recipe is completely *independent of units*: whether you measure speed in m/s, km/h, or furlongs per fortnight, its dimensional formula is always $[LT^{-1}]$. Dimensions capture the *nature* of a quantity, stripped of the accident of which ruler you happened to use.

Why does that unit-independence matter so much? Because it means dimensions describe something *real and permanent* about a quantity, while units are just our changeable labels. Two students in two countries using two unit systems will write different *numbers* for the same force, but they will write the *same dimensional formula*. This is why dimensions can act as a universal cross-check: they sit underneath all the bookkeeping of units, watching whether the physics makes sense. A handy consequence: any quantity that is a *ratio of two similar things* — strain (length/length), refractive index (speed/speed), relative density (density/density) — comes out **dimensionless**, with formula $[M^0 L^0 T^0]$, because the ingredients cancel perfectly.

Now here comes the single most powerful idea in this entire subtopic. **You can only add or compare things of the same nature.** You'd never say "3 kilograms plus 5 seconds" — it's gibberish. The same rule governs every equation in physics: **every term that is added, subtracted, or equated must have identical dimensions.** This is the **principle of homogeneity of dimensions**, and it's the engine behind everything we do here. It lets you (1) *check* whether an equation could be right, (2) *derive* unknown relationships, and (3) *convert* a quantity between unit systems — all without doing the underlying physics in detail.

Think about it this way... imagine you're proofreading a friend's physics answer but you don't have time to redo the whole problem. You can still catch a blunder instantly: just check that the "recipe" on the left matches the "recipe" on the right. If your friend writes that pressure equals force times area, you check — pressure is $[ML^{-1}T^{-2}]$ but force times area is $[MLT^{-2}][L^2] = [ML^3T^{-2}]$. Mismatch! You've caught the error in five seconds without knowing a thing about the actual scenario. Dimensional analysis is the physicist's spell-checker.

What makes this genuinely powerful — not just a classroom trick — is that the *same* check runs in reverse to *predict* physics you haven't been taught. If you know a quantity depends on a few others, demanding that the recipes balance often forces the relationship almost entirely into shape, leaving only an unknown number out front. Physicists have used exactly this reasoning to estimate everything from the energy released by an explosion to the speed of ripples on water, long before doing the full theory. So treat dimensions not as a chore but as a fast, surprisingly deep first weapon you reach for whenever a relation is unknown or a formula looks suspicious.

Limiting Conditions & Assumptions. This power comes with sharp boundaries you must respect, because examiners test them relentlessly:

- A dimensionally **correct** equation is **not guaranteed to be physically correct**. The method is blind to pure numbers — it cannot see the $\frac{1}{2}$ in $\frac{1}{2}mv^2$ or the 2π in a pendulum's period. So $s = ut + at^2$ (missing the $\frac{1}{2}$) is dimensionally fine but physically wrong.
- It **cannot determine dimensionless constants** ($\frac{1}{2}$, 2π , coefficients).
- It **fails when a quantity depends on more than three** base-dimension variables, because we only have three equations (M , L , T) to solve.
- It **cannot handle equations with sums of terms** when deriving (e.g., it can't produce $ut + \frac{1}{2}at^2$ as a single dimensional expression).
- The **arguments of trigonometric, logarithmic, and exponential functions must be dimensionless** — $\sin(\theta)$, e^x , $\ln(y)$ only make sense when θ , x , y carry no dimensions.
- It **cannot distinguish two quantities with the same dimensions** — work and torque are both $[ML^2T^{-2}]$, yet they are physically very different.

Section 2: Key Formulas & Definitions

Dimensional formula: an expression showing which base quantities, and to what powers, make up a physical quantity. Written as $[M^a L^b T^c \dots]$, where a, b, c are the **dimensions** of the quantity in mass, length, and time.

Dimensional equation: the statement equating a physical quantity to its dimensional formula, e.g., $[\text{Force}] = [MLT^{-2}]$.

Essential dimensional formulae to know by heart:

Quantity	Relation	Dimensional Formula	SI Unit
Velocity	displacement / time	$[M^0 L T^{-1}]$	m s^{-1}
Acceleration	velocity / time	$[M^0 L T^{-2}]$	m s^{-2}
Force	mass $\times$ acceleration	$[MLT^{-2}]$	N
Work / Energy / Torque	force $\times$ distance	$[ML^2 T^{-2}]$	J
Power	work / time	$[ML^2 T^{-3}]$	W
Pressure / Stress / Young's modulus	force / area	$[ML^{-1} T^{-2}]$	Pa
Momentum / Impulse	mass $\times$ velocity	$[MLT^{-1}]$	kg m s^{-1}
Density	mass / volume	$[ML^{-3}]$	kg m^{-3}
Surface tension	force / length	$[ML^0 T^{-2}]$	N m^{-1}

Quantity	Relation	Dimensional Formula	SI Unit
Coefficient of viscosity	—	$[ML^{-1}T^{-1}]$	Pa s
Planck's constant	energy / frequency	$[ML^2T^{-1}]$	J s
Gravitational constant G	—	$[M^{-1}L^3T^{-2}]$	N m ² kg ⁻²

Principle of homogeneity: In any physically valid equation,

$$[\text{Term}_1] = [\text{Term}_2] = \dots = [\text{Both sides}]$$

i.e., every additive term shares the same dimensional formula.

Dimensionless quantities: Pure numbers (π , 2), ratios of like quantities (strain, refractive index, relative density), and angles (radian, steradian) all have dimensional formula $[M^0 L^0 T^0]$.

Section 3: CBSE Board Derivations / Key Procedures

The logical foundation — why homogeneity must hold. Consider any physical equation, say $A = B + C$. Each symbol represents a physical quantity with some dimensional formula. Suppose $[B] \neq [C]$. Then we would be adding, for instance, a length to a time — which is meaningless, exactly as "2 metres + 3 seconds" is meaningless. Therefore, for the equation to have any physical sense, we *must* have $[B] = [C]$, and since A equals their sum, $[A] = [B] = [C]$. This is the principle of homogeneity, and notice it is *forced* on us by the simple impossibility of adding unlike quantities — not an arbitrary rule.

Procedure 1 — Checking the correctness of an equation.

1. Write the dimensional formula of the **left-hand side (LHS)**.
2. Write the dimensional formula of **each term** on the **right-hand side (RHS)** separately.
3. If LHS matches every RHS term, the equation is **dimensionally consistent** (and *may* be correct). If even one term differs, the equation is **definitely wrong**. Why "*may*"? Because dimensions can't see numerical factors — consistency is necessary but not sufficient.

Procedure 2 — Deriving a relationship between quantities.

1. Assume the unknown quantity is a product of powers of the quantities it depends on, times a dimensionless constant C : e.g., $Q = C x^a y^b z^c$.
2. Replace every symbol with its dimensional formula.
3. Apply homogeneity: equate the powers of M , L , and T on both sides — this gives you simultaneous equations.
4. Solve for a, b, c and substitute back. *Why this works:* if the relation is a single power-law product, its exponents are fully fixed by requiring dimensional balance. *Remember:* C stays unknown — the method is blind to it.

Procedure 3 — Converting a quantity between unit systems.

1. Write the dimensional formula of the quantity, $[M^a L^b T^c]$.
2. Use the fact that the physical magnitude is fixed, so $n_1 u_1 = n_2 u_2$, giving

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c.$$

3. Substitute the unit ratios and compute. *Why the powers appear:* each base unit "participates" as many times as its exponent in the dimensional formula, so its conversion ratio must be raised to that exponent.

Quick worked conversion — express 1 newton in dynes. Force has dimensions $[MLT^{-2}]$, so $a = 1, b = 1, c = -2$. Going from SI to CGS: $M_1/M_2 = \text{kg/g} = 10^3$, $L_1/L_2 = \text{m/cm} = 10^2$, $T_1/T_2 = \text{s/s} = 1$:

$$n_2 = 1 \times (10^3)^1 (10^2)^1 (1)^{-2} = 10^5.$$

So $1 \text{ N} = 10^5 \text{ dyne}$ — the same trick as in Subtopic 01, now seen as one of the three pillars of dimensional analysis.

Worked derivation (pendulum period). Let us derive how the time period t of a simple pendulum depends on its length ℓ , the bob's mass m , and gravity g . Assume:

$$t = C \ell^a m^b g^c$$

Insert dimensions ($[t] = [T]$, $[\ell] = [L]$, $[m] = [M]$, $[g] = [LT^{-2}]$):

$$[T] = [L]^a [M]^b [LT^{-2}]^c = [M^b L^{a+c} T^{-2c}]$$

Equate powers:

$$M : b = 0, \quad T : -2c = 1 \Rightarrow c = -\frac{1}{2}, \quad L : a + c = 0 \Rightarrow a = \frac{1}{2}$$

Therefore:

$$t = C \ell^{1/2} g^{-1/2} = C \sqrt{\frac{\ell}{g}}$$

Two deep lessons fall out instantly: the period is **independent of the bob's mass** ($b = 0$), and it grows as $\sqrt{\ell}$. The actual constant $C = 2\pi$ — which dimensional analysis *cannot* give us. That gap is the method's signature limitation.

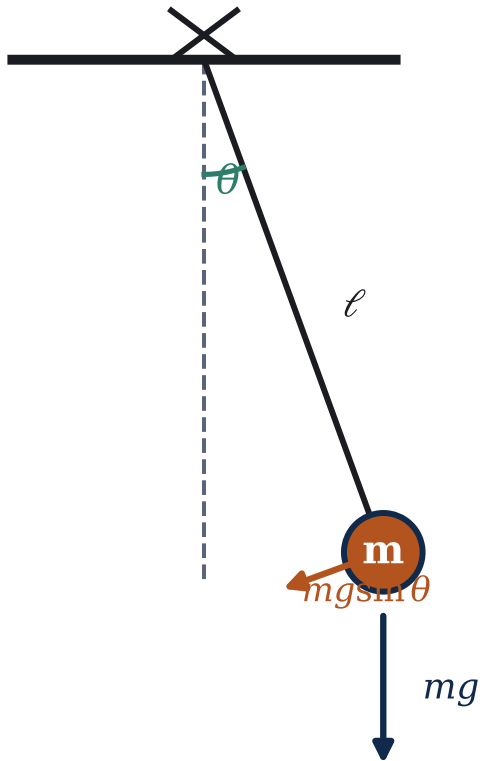
A word on when the method breaks down. Procedure 2 works cleanly only when the unknown depends on **three or fewer** quantities, because we get just three balancing equations (one each from M, L, T). Suppose the drag force on a sphere moving through air were assumed to depend on its radius, its speed, the air's density, *and* the air's viscosity — that is four unknowns but still only three equations. The system is then under-determined, and dimensional analysis alone cannot pin down a unique answer; we would have to bring in extra physical insight or experiment. Recognising *when not to trust the tool* is as important as knowing how to use it, and questions testing exactly this judgment appear in JEE Advanced.

Figure 1.2 A simple pendulum of length l hanging from a fixed support, bob of mass m at the bottom, displaced by a small angle θ from the vertical dashed line. Arrows show the restoring component of gravity. A caption box reads " $t = C\sqrt{l/g}$; mass m cancels out." (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Problem. A student writes the displacement of a particle as $s = ut + \frac{1}{2}at^3$, where u is initial velocity, a is acceleration, and t is time. Check whether this equation is dimensionally correct.



$$t = C\sqrt{\ell/g}; \text{ mass } m \text{ cancels out}$$

Figure 1.2 *A simple pendulum of length ℓ hanging from a fixed support, bob of mass m at the bottom, displaced by a small angle θ from the vertical dashed line. Arrows show the restoring component of gravity. A caption box reads " $t = C\sqrt{\ell/g}$; mass m cancels out."*

Solution. Step 1 — LHS. Displacement: $[s] = [L]$.

Step 2 — RHS term by term.

- First term: $[ut] = [LT^{-1}][T] = [L]$. ✓ (matches LHS)
- Second term: $[\frac{1}{2}at^3] = [LT^{-2}][T^3] = [LT]$. ✗ (does **not** match $[L]$)

Step 3 — Conclusion. Since the second term has dimensions $[LT]$ while the LHS is $[L]$, the principle of homogeneity is violated.

Result: The equation is **dimensionally incorrect**, so it cannot be physically correct. (The right term should be $\frac{1}{2}at^2$.) This shows how a single dimensional check exposes the error without solving any motion.

Example 2 — [NEET Speed Trap]

Problem. Which of the following pairs does **NOT** have the same dimensional formula? (A) Work and Torque (B) Impulse and Momentum (C) Pressure and Stress (D) Force and Surface Tension

The trap. Surface tension *sounds* like a force — students see "tension" and instantly pair it with force, ticking some other option. **Wrong.** Surface tension is force *per unit length*: $[\text{force}]/[\text{length}] = [MLT^{-2}]/[L] = [MT^{-2}]$, whereas force is $[MLT^{-2}]$. They differ by one power of length.

Fast check of the others: Work and torque are both $[ML^2T^{-2}]$ ✓; impulse and momentum are both $[MLT^{-1}]$ ✓; pressure and stress are both $[ML^{-1}T^{-2}]$ ✓.

Answer: (D) Force and Surface Tension.

Pitfall to remember: Never judge dimensions by a quantity's *name*. "Tension" in "surface tension" does not make it a force — always reduce to base dimensions.

Example 3 — [JEE Main Level]

Problem. The speed v of a transverse wave on a stretched string depends on the tension T in the string and the mass per unit length μ of the string. Using dimensional analysis, derive the formula for v . (*This blends wave mechanics with dimensional reasoning.*)

Solution. Step 1 — Assume a power-law form.

$$v = CT^a\mu^b$$

Step 2 — Insert dimensions. $[v] = [LT^{-1}]$, $[T] = [MLT^{-2}]$ (tension is a force), $[\mu] = [ML^{-1}]$:

$$[LT^{-1}] = [MLT^{-2}]^a [ML^{-1}]^b = [M^{a+b} L^{a-b} T^{-2a}]$$

Step 3 — Equate powers (homogeneity).

$$M : a + b = 0, \quad T : -2a = -1 \Rightarrow a = \frac{1}{2}, \quad L : a - b = 1$$

From $a = \frac{1}{2}$ and $a + b = 0$, we get $b = -\frac{1}{2}$ (and this satisfies $a - b = 1$ ✓).

Step 4 — Assemble.

$$v = CT^{1/2}\mu^{-1/2} = C\sqrt{\frac{T}{\mu}}$$

Result: $v = C\sqrt{T/\mu}$. The full physics gives $C = 1$, but dimensional analysis has already captured the entire *structure* — that wave speed rises with tension and falls with string heaviness — which is the examiner's real target.

Example 4 — [JEE Advanced Level]

Problem. A self-gravitating spherical cloud of gas pulsates. Its natural frequency of oscillation f is expected to depend on its radius R , its mean density ρ , and the gravitational constant G . Use dimensional analysis to find how f depends on these quantities, and explain the striking result that emerges.

Solution. Step 1 — Assume the power-law.

$$f = C R^a \rho^b G^c$$

Step 2 — Dimensions. $[f] = [T^{-1}]$, $[R] = [L]$, $[\rho] = [ML^{-3}]$, $[G] = [M^{-1}L^3T^{-2}]$:

$$[M^0 L^0 T^{-1}] = [L]^a [ML^{-3}]^b [M^{-1}L^3T^{-2}]^c = [M^{b-c} L^{a-3b+3c} T^{-2c}]$$

Step 3 — Equate powers.

$$M : b - c = 0 \Rightarrow b = c$$

$$T : -2c = -1 \Rightarrow c = \frac{1}{2}, \quad \text{so } b = \frac{1}{2}$$

$$L : a - 3b + 3c = 0 \Rightarrow a - 3b + 3b = 0 \Rightarrow a = 0$$

Step 4 — Assemble.

$$f = C R^0 \rho^{1/2} G^{1/2} = C \sqrt{G\rho}$$

Result: $f = C\sqrt{G\rho}$ — and remarkably, **the radius R drops out entirely** ($a = 0$).

Why this is "Advanced": the insight is physical, not just algebraic. It tells us that the pulsation frequency of a self-gravitating body depends *only* on its density, not its size — a big star and a small star of the same density oscillate at the same rate. This $\sqrt{G\rho}$ scaling is genuinely used by astrophysicists to estimate stellar pulsation timescales, and it falls out of pure dimensional reasoning.

Section 5: Practice Exercises

1. [CBSE] Find the dimensional formula of (a) coefficient of viscosity and (b) Planck's constant.
2. [NEET] Name a physical quantity whose dimensional formula is $[ML^2T^{-2}]$.
3. [JEE Main] The escape velocity v_e of a body from a planet depends on the acceleration due to gravity g at its surface and the planet's radius R . Derive the dependence of v_e on g and R using dimensional analysis.
4. [JEE Main] Using dimensional analysis, convert 1 joule into ergs (the CGS unit of energy).
5. [JEE Advanced] The time period t of small oscillations of a liquid drop depends on its radius r , density ρ , and surface tension S . Use dimensional analysis to derive the form of t .

Final Answers:

1. (a) $[ML^{-1}T^{-1}]$ (b) $[ML^2T^{-1}]$
2. Work / Energy / Torque (any one)
3. $v_e = C\sqrt{gR}$

$$4. 1 \text{ J} = 10^7 \text{ erg}$$

$$5. t = C \sqrt{\frac{\rho r^3}{S}}$$

Section 6: Crack the MCQ

Q1. The dimensional formula $[ML^{-1}T^{-2}]$ corresponds to which of the following? (A) Force (B) Pressure (C) Energy (D) Power

Answer: (B) Pressure. Pressure (also stress, Young's modulus, energy density) is force per area:

$[MLT^{-2}]/[L^2] = [ML^{-1}T^{-2}]$. (A) Force is $[MLT^{-2}]$ — students who forget to divide by area pick this. (C) Energy is $[ML^2T^{-2}]$ — confused by the similar look of the symbols. (D) Power is $[ML^2T^{-3}]$ — chosen by those who muddle the time exponents.

Q2. A given physical equation is found to be dimensionally correct. We can conclude that the equation: (A) is definitely physically correct (B) may or may not be physically correct (C) is definitely wrong (D) must contain a dimensionless constant

Answer: (B). Dimensional correctness is *necessary but not sufficient*: a missing $\frac{1}{2}$ or an extra dimensionally-identical term passes the check yet makes the equation wrong. (A) is the single most common conceptual error in the chapter — consistency does not prove correctness. (C) is the opposite mistake. (D) is a tempting-sounding statement but false; many correct equations (like $v = u + at$) have no separate dimensionless constant at all.

Q3. The principle of dimensional analysis **cannot** be used to: (A) check the correctness of an equation (B) determine the value of a dimensionless constant (C) convert units between systems (D) find the dimensional formula of a derived quantity

Answer: (B). The method is fundamentally blind to pure numbers, so it can never deliver factors like $\frac{1}{2}$ or 2π — this is its defining limitation. (A), (C), and (D) are precisely the three jobs dimensional analysis *does* perform, so a student who knows the applications eliminates them at once.

Q4. Which of the following pairs has the **same** dimensional formula? (A) Work and Power (B) Impulse and Momentum (C) Force and Pressure (D) Energy and Force

Answer: (B). Impulse (= force $\times$ time = $[MLT^{-1}]$) and momentum (= mass $\times$ velocity = $[MLT^{-1}]$) match exactly — this equality is itself the impulse–momentum theorem in disguise. (A) work $[ML^2T^{-2}]$ vs power $[ML^2T^{-3}]$ differ by a time. (C) force vs pressure differ by an area. (D) energy vs force differ by a length. Each wrong option is a near-miss differing by one base quantity, the classic distractor design.

Section 7: Common Student Pitfalls & Pro-Tips

- 1. Believing "dimensionally correct" means "correct."** It only means the equation *survives* a basic check. The real formula may differ by a numerical factor or contain extra same-dimension terms.
- 2. Forgetting dimensionless constants exist.** When you derive a relation, always write the unknown constant C and state explicitly that dimensional analysis cannot evaluate it. Boards expect this caveat.
- 3. Assuming equal dimensions $\implies$ same quantity.** Work and torque, or energy and torque, share $[ML^2T^{-2}]$ but are physically distinct. Dimensions cannot tell them apart.
- 4. Putting dimensional quantities inside sin, ln, or e .** The argument of any trig/log/exponential function *must* be dimensionless. Spotting $\sin(\omega t)$ tells you instantly that ωt has no dimensions, so $[\omega] = [T^{-1}]$.

Pro-Tip (speed): To find the dimensions of an unknown buried in a formula, use homogeneity directly: the unknown must make its term match the others (for added terms) or make a function's argument dimensionless. Example: in $y = A \sin(kx - \omega t)$, instantly $[k] = [L^{-1}]$ and $[\omega] = [T^{-1}]$ because kx and ωt must be pure numbers. This turns a scary expression into a two-second read.

Section 8: Cheat Sheet Box

DIMENSIONAL ANALYSIS — QUICK REVISION

- **Dimensional formula** = the "recipe" $[M^a L^b T^c]$ of a quantity; unit-independent.
- **Principle of homogeneity:** every added/equated term must have identical dimensions.
- **Three uses:** (1) check correctness, (2) derive power-law relations, (3) convert units via $n_1 u_1 = n_2 u_2$.
- **Cannot do:** find dimensionless constants ($\frac{1}{2}$, 2π); handle > 3 variables; produce sums of terms; distinguish same-dimension quantities (work vs torque).
- **Golden rule:** dimensionally correct $\nRightarrow$ physically correct; but dimensionally wrong $\Rightarrow$ definitely wrong.
- **Function rule:** arguments of $\sin$, $\cos$, $\ln$, e^x are always dimensionless.
- **Must-know:** Energy/Work/Torque $[ML^2 T^{-2}]$; Power $[ML^2 T^{-3}]$; Pressure/Stress $[ML^{-1} T^{-2}]$; G is $[M^{-1} L^3 T^{-2}]$; h is $[ML^2 T^{-1}]$.

Memory aids:

- "Same recipe, different dish" $\rightarrow$ equal dimensions don't mean equal quantities (work vs torque).
- "Dimensions catch the liar, not the sloppy" $\rightarrow$ they expose a wrong equation but miss a missing $\frac{1}{2}$.
- "Inside sin, no dimension within."

Chapter 1: Units and Measurements

Sub-topic 03: Errors in Measurement

Exam Relevance: This is a perennial scorer. JEE Main reliably carries 1–2 questions, almost always on combination/propagation of errors (finding the maximum percentage error in a derived quantity). JEE Advanced embeds error analysis inside experiment-based questions. NEET asks 1 question, usually conceptual (accuracy vs precision, or a quick percentage-error calculation). CBSE Boards set a 2–3 mark question on types of errors or computing mean/absolute/percentage error.

Section 1: Concept Intuition

Picture yourself at a railway station, watching the big platform clock and timing how long a train takes to pull in. You press an imaginary stopwatch when the engine crosses the pole and stop it when the last coach clears. Did you press *exactly* at the right instants? Of course not — your thumb was a fraction early once, a fraction late the next. No matter how hard you try, your measurement carries a tiny, unavoidable "wrongness." That built-in wrongness is what we call **error**.

The very first thing to get straight — and students mix this up constantly — is that **error is not the same as a mistake**. Writing $7 \times 8 = 54$ is a *mistake*; you could have avoided it by being careful. But measuring a desk as 92.3 cm when its true length is 92.34 cm is *not* carelessness — it's the **limit of your instrument and senses**. Mistakes can be eliminated; errors can only be *estimated and reduced*. Accepting this is the whole spirit of experimental physics.

So where do errors sneak in? They come in three flavours that behave very differently:

- **Systematic errors** are the *consistent, one-directional* ones. Imagine a weighing scale that reads 20 g even with nothing on it. Every reading is now 20 g too high — same size, same sign, every single time. Zero errors in a screw gauge, a wrongly calibrated thermometer, a stopwatch that runs slow: all systematic. The good news is that because they're predictable, they can be *identified and corrected*.
- **Random errors** are the *jittery, unpredictable* ones — your thumb on the stopwatch, tiny fluctuations in temperature or voltage, slight changes in how you line up your eye. These scatter on *both* sides of the true value with no fixed pattern. You can't fix a single random error, but you *can* beat them down by taking many readings and averaging.
- **Gross errors** are plain human blunders — misreading 23 as 32, copying a wrong value, forgetting a unit. We don't model these mathematically; we simply stay alert.

There's also a special case worth naming: the **least-count error**, the smallest amount your instrument can resolve. A metre scale graduated in millimetres simply cannot tell you anything finer than a millimetre, so a single reading carries a least-count uncertainty.

How do we fight back against all this? Each error type has its own remedy, and matching the cure to the disease is half the skill. *Systematic* errors are defeated by better technique — checking and removing zero errors, calibrating against a trusted standard, correcting for known biases like temperature drift. *Random* errors are defeated by repetition — take many readings and average, because the scatter cancels. And *least-count* error is defeated by choosing a finer instrument: a screw gauge (least count 0.01 mm) beats a vernier (typically 0.1 mm), which beats

a plain scale (1 mm). A thoughtful experimenter therefore doesn't just measure — they first ask, "Which kind of error dominates here, and what's the right weapon against it?"

Think about it this way... an archer firing at a target makes the difference between two prized words crystal clear. If all her arrows land in a tight little cluster but off in the top corner, she is **precise** (arrows agree with each other) but **inaccurate** (they miss the bull's-eye) — that's *systematic* error. If her arrows are scattered all around the centre, some left, some right, she is **imprecise**, but their *average* may sit right on the bull's-eye — that's *random* error, which averaging can tame. Burn this picture in: **precision = agreement among readings; accuracy = closeness to the truth**. A measurement can have one without the other, and examiners adore probing this gap.

Limiting Conditions & Assumptions. A few quiet assumptions hold this topic together:

- We assume a **true value** exists even though we can never know it exactly; in practice we treat the **arithmetic mean** of many readings as our best stand-in. This works *only* for random errors. If a systematic error is present, the mean is shifted too — and averaging will **not** rescue you.
- The "half the least count" rule applies to a *single* reading where no repeated data exists.
- The combination-of-error formulas we will derive give the **maximum possible error** — a deliberately pessimistic, worst-case bound where all the individual errors conspire in the same unlucky direction. The *typical* error is usually smaller.

Section 2: Key Formulas & Definitions

Let $a_1, a_2, \dots, a_n$ be n readings of the same quantity.

1. Mean (best estimate of true value):

$$a_{\text{mean}} = \frac{1}{n} \sum_{i=1}^n a_i$$

2. Absolute error of the i -th reading (same units as the quantity):

$$\Delta a_i = |a_{\text{mean}} - a_i|$$

3. Mean absolute error:

$$\Delta a_{\text{mean}} = \frac{1}{n} \sum_{i=1}^n \Delta a_i$$

Result reported as $a = a_{\text{mean}} \pm \Delta a_{\text{mean}}$.

4. Relative (fractional) error — dimensionless, $[M^0 L^0 T^0]$:

$$\text{Relative error} = \frac{\Delta a_{\text{mean}}}{a_{\text{mean}}}$$

5. Percentage error:

$$\delta a = \frac{\Delta a_{\text{mean}}}{a_{\text{mean}}} \times 100\%$$

6. Least-count error (single reading): absolute error = $\frac{1}{2} \times (\text{least count})$, unless otherwise stated.

Combination of errors (the heart of exam questions):

Operation	Result	Rule for error
Sum $x = a + b$	—	Absolute errors add: $\Delta x = \Delta a + \Delta b$
Difference $x = a - b$	—	Absolute errors still add : $\Delta x = \Delta a + \Delta b$
Product $x = ab$	—	Relative errors add: $\frac{\Delta x}{x} = \frac{\Delta a}{a} + \frac{\Delta b}{b}$
Quotient $x = \frac{a}{b}$	—	Relative errors add: $\frac{\Delta x}{x} = \frac{\Delta a}{a} + \frac{\Delta b}{b}$
Power $x = a^p b^q c^{-r}$	—	$\frac{\Delta x}{x} = p \frac{\Delta a}{a} + q \frac{\Delta b}{b} + r \frac{\Delta c}{c}$

Note the two "surprises" students forget: in a **difference**, the absolute errors *add* (not subtract), and in a **power**, the exponent *multiplies* the relative error (a square root contributes *half*).

Section 3: CBSE Board Derivations

These three derivations are frequently asked for full marks. The single guiding idea is the **worst case**: we let every individual error push the result in the *same* direction so that the total error is as large as it can possibly be.

Derivation A — Error in a sum (and difference). Let $x = a + b$, where a and b carry absolute errors $\pm \Delta a$ and $\pm \Delta b$. The measured extremes are $(a \pm \Delta a)$ and $(b \pm \Delta b)$, so

$$x \pm \Delta x = (a \pm \Delta a) + (b \pm \Delta b) = (a + b) \pm (\Delta a + \Delta b).$$

The largest the error can be occurs when both errors add with the same sign, giving

$$\Delta x = \Delta a + \Delta b.$$

Now the surprising part — the **difference** $x = a - b$:

$$x \pm \Delta x = (a \pm \Delta a) - (b \pm \Delta b) = (a - b) \pm \Delta a \mp \Delta b.$$

Why do the errors still add? Because we must take the **worst case**: the maximum error happens when Δa is positive while $-(\mp \Delta b)$ also acts positively, i.e. the errors reinforce. So even for subtraction,

$$\Delta x = \Delta a + \Delta b.$$

This has a sharp practical consequence: subtracting two nearly-equal numbers keeps the *absolute* error the same but shrinks the *result*, so the **percentage error explodes** — a trap we'll see in Example/Practice.

Derivation B — Error in a product (and quotient). Let $x = ab$. Then $x \pm \Delta x = (a \pm \Delta a)(b \pm \Delta b)$.

Expanding:

$$x \pm \Delta x = ab \pm a \Delta b \pm b \Delta a \pm \Delta a \Delta b.$$

Why drop the last term? Because $\Delta a \Delta b$ is a product of two tiny quantities — negligibly small compared to the others. Dividing through by $x = ab$:

$$1 \pm \frac{\Delta x}{x} = 1 \pm \frac{\Delta b}{b} \pm \frac{\Delta a}{a}.$$

Taking the worst case (errors reinforcing):

$$\frac{\Delta x}{x} = \frac{\Delta a}{a} + \frac{\Delta b}{b}.$$

A near-identical argument for $x = a/b$ gives the *same* result — **relative errors add for both multiplication and division**.

Derivation C — Error in a power. Let $x = a^p$. Write it as a repeated product $x = a \cdot a \cdots a$ (p times). By the product rule, the relative errors of all p identical factors add:

$$\frac{\Delta x}{x} = \underbrace{\frac{\Delta a}{a} + \frac{\Delta a}{a} + \cdots}_{p \text{ terms}} = p \frac{\Delta a}{a}.$$

So for a general expression $x = \frac{a^p b^q}{c^r}$:

$$\boxed{\frac{\Delta x}{x} = p \frac{\Delta a}{a} + q \frac{\Delta b}{b} + r \frac{\Delta c}{c}.}$$

The exponent multiplies the relative error — and a quantity in the denominator contributes with a **plus** sign (worst case again), never a minus.

Putting the rules together — a quick illustration. Suppose a rectangular metal plate has length $\ell = (16.30 \pm 0.05)$ cm and breadth $b = (13.80 \pm 0.05)$ cm, and we want its area $A = \ell b$ with uncertainty. Since area is a product, we add *relative* errors:

$$\frac{\Delta A}{A} = \frac{\Delta \ell}{\ell} + \frac{\Delta b}{b} = \frac{0.05}{16.30} + \frac{0.05}{13.80} = 0.00307 + 0.00362 = 0.00669.$$

The area is $A = 16.30 \times 13.80 = 224.94 \text{ cm}^2$, so the absolute uncertainty is $\Delta A = 0.00669 \times 224.94 \approx 1.5 \text{ cm}^2$, giving $A = (225 \pm 2) \text{ cm}^2$ after sensible rounding. Notice how we moved fluidly between relative and absolute error — that fluency is exactly what the rules buy you.

Figure 1.3 A flow-style chart with two columns. Left column "Absolute errors ADD" lists $x = a \pm b \rightarrow \Delta x = \Delta a + \Delta b$. Right column "Relative (%) errors ADD" lists $x = a^p b^q / c^r \rightarrow \Delta x/x = p(\Delta a/a) + q(\Delta b/b) + r(\Delta c/c)$. A caption strip across the bottom reads "ALWAYS worst case: every error pushes the same way." (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Problem. The time period of a simple pendulum is measured five times, giving:

2.56 s, 2.54 s, 2.59 s, 2.55 s, 2.51 s. Calculate the mean period, the mean absolute error, the relative error, and the percentage error, and express the result properly.

Solution. Given: five readings, $n = 5$.

Step 1 — Mean period.

$$T_{\text{mean}} = \frac{2.56 + 2.54 + 2.59 + 2.55 + 2.51}{5} = \frac{12.75}{5} = 2.55 \text{ s}.$$

Step 2 — Absolute errors $\Delta T_i = |T_{\text{mean}} - T_i|$:

$$0.01, \quad 0.01, \quad 0.04, \quad 0.00, \quad 0.04 \quad (\text{all in s}).$$

Step 3 — Mean absolute error.

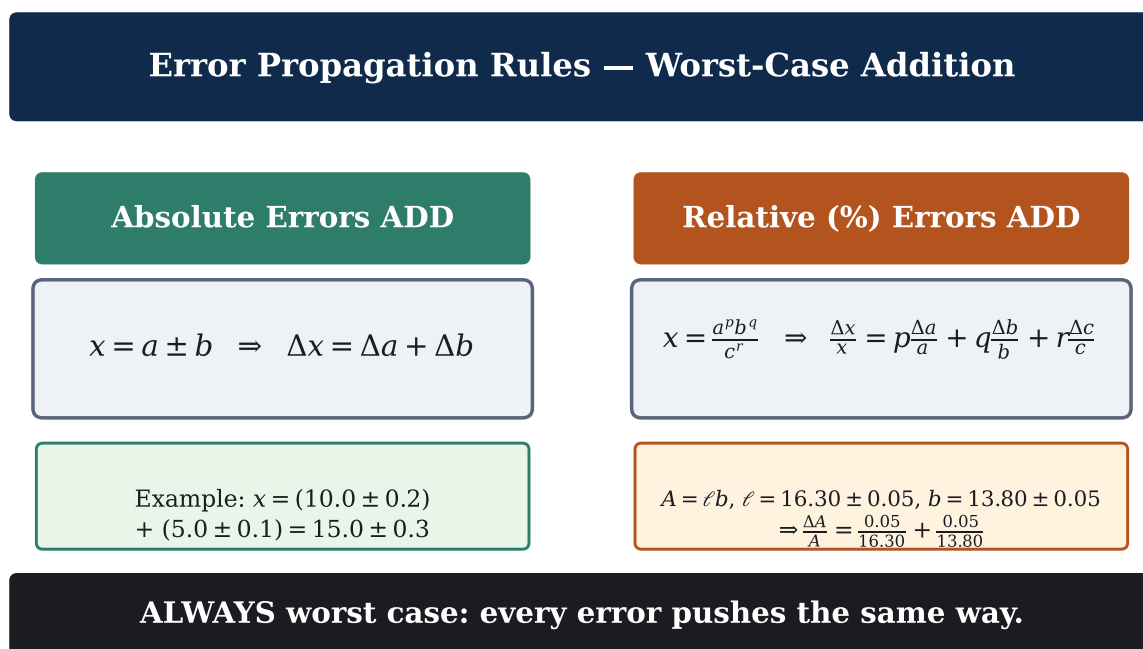


Figure 1.3 A flow-style chart with two columns. Left column "Absolute errors ADD" lists $x = a \pm b \rightarrow \Delta x = \Delta a + \Delta b$. Right column "Relative (%) errors ADD" lists $x = a^p b^q / c^r \rightarrow \Delta x / x = p(\Delta a / a) + q(\Delta b / b) + r(\Delta c / c)$. A caption strip across the bottom reads "ALWAYS worst case: every error pushes the same way."

$$\Delta T_{\text{mean}} = \frac{0.01 + 0.01 + 0.04 + 0.00 + 0.04}{5} = \frac{0.10}{5} = 0.02 \text{ s.}$$

Step 4 — Relative and percentage error.

$$\frac{\Delta T_{\text{mean}}}{T_{\text{mean}}} = \frac{0.02}{2.55} = 0.0078, \quad \delta T = 0.0078 \times 100\% \approx 0.78\%.$$

Result: $T = (2.55 \pm 0.02) \text{ s}$, with a percentage error of about 0.78%.

Example 2 — [NEET Speed Trap]

Problem. A quantity is computed as $Z = \frac{A^2 B}{\sqrt{C}}$. The percentage errors in A , B , and C are 2%, 1%, and 4% respectively. Find the maximum percentage error in Z .

The trap. The lazy approach just adds $2 + 1 + 4 = 7\%$ — **wrong** on two counts: it ignores the *square* on A and the *square root* on C .

Fast solution (apply the power rule mentally).

$$\delta Z = 2(\delta A) + 1(\delta B) + \frac{1}{2}(\delta C) = 2(2\%) + 1\% + \frac{1}{2}(4\%) = 4\% + 1\% + 2\% = 7\%.$$

The number happens to be 7% here too — but only because the factors worked out that way. The *method* is what matters: **multiply each percentage error by the power of its quantity** ($\sqrt{C} = C^{1/2}$ contributes $\frac{1}{2}$).

Pitfall to remember: Always attach the exponent. A square doubles a quantity's error contribution; a square root halves it. Forgetting this is the single most common NEET/JEE error-analysis slip.

Example 3 — [JEE Main Level]

Problem. The density of a small steel sphere is found from its mass and diameter. The mass is measured as $m = (50.0 \pm 0.1) \text{ g}$ and the diameter as $d = (2.00 \pm 0.01) \text{ cm}$. Find the maximum percentage error in the calculated density. (*This combines the geometry of a sphere's volume with error propagation.*)

Solution. Step 1 — Set up the formula. Density $\rho = \frac{m}{V}$, and for a sphere $V = \frac{\pi}{6}d^3$, so

$$\rho = \frac{6m}{\pi d^3}.$$

Step 2 — Identify the powers. $\rho \propto m^1 d^{-3}$ (the constant $\frac{6}{\pi}$ is exact and contributes no error).

Step 3 — Apply the power rule.

$$\frac{\Delta \rho}{\rho} = 1 \cdot \frac{\Delta m}{m} + 3 \cdot \frac{\Delta d}{d}.$$

Step 4 — Substitute percentage errors.

$$\begin{aligned} \frac{\Delta m}{m} &= \frac{0.1}{50.0} = 0.2\%, & \frac{\Delta d}{d} &= \frac{0.01}{2.00} = 0.5\%. \\ \delta \rho &= 0.2\% + 3(0.5\%) = 0.2\% + 1.5\% = 1.7\%. \end{aligned}$$

Result: The maximum percentage error in the density is 1.7%. Notice how the *diameter* dominates the error budget — because it enters as a *cube*, its modest 0.5% becomes 1.5%. A clear lesson: in an experiment, measure the highest-power quantity most carefully.

Example 4 — [JEE Advanced Level]

Problem. Two resistors, $R_1 = (100 \pm 2) \Omega$ and $R_2 = (200 \pm 4) \Omega$, are connected in **parallel**. Find the equivalent resistance and its absolute error. (*The simple product/quotient rule does not apply directly here — you must use calculus.*)

Solution. Step 1 — Equivalent resistance. For parallel resistors,

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow R = \frac{R_1 R_2}{R_1 + R_2} = \frac{(100)(200)}{300} = 66.7 \Omega.$$

Step 2 — Why not the quotient rule? Because R_1 and R_2 appear in *both* numerator and denominator in a coupled way; blindly adding relative errors would double-count. The clean route is to **differentiate the reciprocal relation**, which separates the variables.

Step 3 — Differentiate $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$. Treating each term:

$$-\frac{\Delta R}{R^2} \rightarrow \Delta R = R^2 \left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2} \right),$$

taking magnitudes (worst case, all errors reinforcing).

Step 4 — Substitute.

$$\Delta R = (66.7)^2 \left(\frac{2}{(100)^2} + \frac{4}{(200)^2} \right) = 4448.9 (0.0002 + 0.0001) = 4448.9 \times 0.0003 \approx 1.3 \Omega.$$

Result: $R = (66.7 \pm 1.3) \Omega$.

Why this is "Advanced": it forces you to recognise *when the standard rules fail* and to fall back on differentials. The relation $\Delta R = R^2 \sum \Delta R_i / R_i^2$ is exactly how error propagates through any "reciprocals add" situation — parallel resistors, thin-lens combinations, capacitors in series — so this single technique unlocks a whole family of experiment questions.

Section 5: Practice Exercises

1. [CBSE] The mass of an object is measured five times: 5.12 g, 5.16 g, 5.10 g, 5.14 g, 5.13 g. Find the mean mass, the mean absolute error, and the percentage error.
2. [NEET] A quantity is given by $Z = \frac{A^3}{B^2}$. If the percentage errors in A and B are 1% and 2%, find the maximum percentage error in Z .
3. [JEE Main] In an experiment, resistance is found from $R = \frac{V}{I}$, with $V = (8.0 \pm 0.2) \text{ V}$ and $I = (2.0 \pm 0.1) \text{ A}$. Find R and its maximum percentage error.
4. [JEE Main] A pendulum's period is found by timing 20 oscillations: total time $t = (40.0 \pm 0.1) \text{ s}$, length $L = (80.0 \pm 0.1) \text{ cm}$. Using $g = \frac{4\pi^2 L}{T^2}$, find the maximum percentage error in g .
5. [JEE Advanced] Two lengths are measured as $a = (5.00 \pm 0.02) \text{ cm}$ and $b = (4.00 \pm 0.02) \text{ cm}$. Find the percentage error in the difference $x = a - b$, and comment on the result.

Final Answers:

1. Mean = 5.13 g, mean absolute error = 0.016 g, $\delta \approx 0.31\%$
2. 7% (from $3(1\%) + 2(2\%)$)

3. $R = 4.0 \, \Omega$, $\delta R = 7.5\%$

4. $\delta g \approx 0.63\%$ (from $\delta L + 2 \delta T = 0.125\% + 2(0.25\%)$)

5. $x = (1.00 \pm 0.04) \, \text{cm}$, $\delta x = 4\%$ — the percentage error is large because subtracting nearly-equal values shrinks the result while the absolute error stays fixed. **Avoid such subtractions in experiment design.**

Section 6: Crack the MCQ

Q1. Random errors in a measurement can be reduced by: (A) correcting the instrument's zero error (B) taking a large number of readings and averaging (C) recalibrating the instrument against a standard (D) using a clean formula

Answer: (B). Random errors scatter both ways, so their average tends toward zero as readings pile up — averaging is the cure. (A) zero-error correction and (C) recalibration tackle *systematic* errors, not random ones — the classic confusion this question targets. (D) is irrelevant; a formula doesn't reduce measurement scatter.

Q2. For a quantity $R = \frac{AB}{C}$, the maximum relative error in R equals: (A) the sum of the absolute errors in A , B , C (B) the sum of the relative errors in A , B , C (C) the difference of the relative errors (D) the product of the relative errors

Answer: (B). For products and quotients, relative errors add (worst case), with the denominator C contributing a **plus** sign too. (A) wrongly mixes absolute with relative — a dimensional impossibility. (C) is the seductive trap: students think the denominator should "subtract," but worst-case analysis forbids that. (D) confuses adding errors with multiplying quantities.

Q3. A measurement is described as having *high precision but low accuracy*. This most likely indicates: (A) large random error, small systematic error (B) small random error, large systematic error (C) both errors large (D) both errors negligible

Answer: (B). High precision means readings agree tightly (small random error); low accuracy means they sit far from the truth (large systematic error) — the clustered-but-off archer. (A) describes the opposite (imprecise but possibly accurate on average). (C) and (D) describe neither the "precise" nor the "inaccurate" half correctly.

Q4. If a physical quantity is given by $Z = A^n$, the relative error in Z is: (A) equal to the relative error in A (B) n times the relative error in A (C) $\frac{1}{n}$ times the relative error in A (D) n times the absolute error in A

Answer: (B). The power rule: an exponent multiplies the relative error, so $\Delta Z/Z = n (\Delta A/A)$. (A) forgets the exponent entirely. (C) inverts it (a common slip for roots — but even a root, $n = \frac{1}{2}$, follows rule B). (D) illegally mixes relative on one side with absolute on the other.

Section 7: Common Student Pitfalls & Pro-Tips

- Subtracting absolute errors in a difference.** For both $a + b$ and $a - b$, the absolute errors **add**. The minus sign in the quantity never becomes a minus sign in the error — worst case always reinforces.
- Dropping the power factor.** A squared term contributes *twice* its relative error; a square root contributes *half*. This is the most frequent JEE/NEET mistake in this topic.
- Comparing absolute errors of different quantities.** "0.5 cm is worse than 0.1 s" is meaningless — different units. Convert to **percentage error** before judging precision.

4. **Believing averaging fixes everything.** Averaging crushes *random* error but is completely blind to *systematic* error. A precise but uncalibrated instrument gives a very repeatable *wrong* answer.

Pro-Tip (speed): For any "find the maximum percentage error" problem, never compute absolute errors first. Write the expression as a product of powers, then add percentage errors directly with their exponents: for $x = a^p b^q / c^r$, just say $\delta x = p \delta a + q \delta b + r \delta c$ and plug the numbers. A three-minute calculation collapses to ten seconds. Reserve absolute errors only for sums/differences and for "reciprocals add" cases needing differentials. One more sanity habit: after computing a maximum percentage error, glance at which term contributed most — it is almost always the quantity with the highest power, and that tells you instantly where a real experiment should spend its care.

Section 8: Cheat Sheet Box

ERRORS IN MEASUREMENT — QUICK REVISION

- **Error $\neq$ mistake.** Errors are unavoidable; estimate and reduce, never eliminate.
- **Types:** *Systematic* (fixed sign, correctable, kills accuracy), *Random* (random sign, averaged away, kills precision), *Gross* (blunder), *Least-count* ($\frac{1}{2} \times \text{LC}$ for a single reading).
- **Precision** = readings agree with each other; **Accuracy** = readings close to the true value.
- **Best estimate** = mean: $a_{\text{mean}} = \frac{1}{n} \sum a_i$; **mean absolute error** $\Delta a_{\text{mean}} = \frac{1}{n} \sum |\Delta a_i|$.
- **Relative error** = $\frac{\Delta a_{\text{mean}}}{a_{\text{mean}}}$ (dimensionless); **percentage** = $\times 100\%$.
- **Sum/Difference:** absolute errors **add** ($\Delta x = \Delta a + \Delta b$ for both).
- **Product/Quotient:** relative errors **add**.
- **Power** $x = a^p b^q / c^r$: $\frac{\Delta x}{x} = p \frac{\Delta a}{a} + q \frac{\Delta b}{b} + r \frac{\Delta c}{c}$.
- **All formulas give the worst-case maximum error.**

Memory aids:

- "Plus or minus, errors always add" → even subtraction adds absolute errors.
- "Powers come down to multiply" → exponent $\times$ relative error.
- "Average kills the random, calibration kills the systematic."
- "Subtract close numbers → percentage error blows up."

Chapter 1: Units and Measurements

Sub-topic 04: Significant Figures

Exam Relevance: A high-frequency, easy-to-score subtopic. JEE Main carries roughly 1 question most years (counting significant figures, or reporting a calculated result correctly). NEET asks a quick counting/rounding question fairly often. CBSE Boards almost always include a 1–2 mark question on counting significant figures or expressing an answer to the correct number of figures. The rules are few — the marks are easy if you are precise.

Section 1: Concept Intuition

Imagine you measure the length of your pencil with an ordinary plastic ruler marked in millimetres. You can confidently read it as 14.2 cm — the 1 and the 4 are certain, and that last 2 is your best honest estimate of the bit between two millimetre marks. Now suppose you wrote down 14.2000 cm. That would be *lying* — it claims you knew the length to the nearest ten-thousandth of a centimetre, which your humble ruler simply cannot deliver. The digits you are *genuinely entitled to write down* — the certain ones plus the first uncertain one — are called the **significant figures** of your measurement.

Here's the heart of the idea. **Every digit in a measured value carries a promise about how precise your instrument was.** The number 14.2 cm quietly says "I measured to about a millimetre." The number 14.20 cm says "I measured ten times more finely, to a tenth of a millimetre." These are *different claims about the real world*, even though mathematically $14.2 = 14.20$. In pure mathematics trailing zeros are decoration; in physics they are *information about precision*. This is the single mental shift that makes significant figures click.

Think of it like a *kirana* shopkeeper weighing dal. A cheap spring balance might read "2 kg" — two significant figures, trustworthy only to the nearest hundred grams or so. A good digital scale reads "2.000 kg" — four significant figures, trustworthy to the gram. Both describe the same dal, but the second instrument *earned* those extra digits. You can never manufacture precision you didn't measure; significant figures are simply the rule of honesty that stops you from doing so.

Why does this honesty matter beyond exams? Because measurements travel. The length you record gets multiplied, divided, and combined by other people into final results, and if you over-state your precision, that false confidence silently corrupts everything downstream — a bridge tolerance, a drug dosage, a satellite trajectory. Significant figures are the universal shorthand scientists and engineers use to pass along not just a *number* but an honest statement of *how much that number can be trusted*. When a result is reported as 9.8 m s^{-2} versus 9.81 m s^{-2} , the writer is telling you something real about the experiment behind it. Learning to read and write that signal correctly is a genuine professional skill, not just a marking rule.

Think about it this way... picture significant figures as the "load-bearing" digits in a number — the ones actually holding up the weight of your measurement. Some zeros are load-bearing (they report real precision), and some are just scaffolding (they only position the decimal point). The whole skill of this topic is learning to tell the *structural* digits from the *scaffolding*. In 0.00450 kg, the three zeros on the left are pure scaffolding — they just push the decimal around — while the 4, 5, and the final 0 are load-bearing. So that number has **three** significant figures, not six.

Limiting Conditions & Assumptions.

- Significant figures apply to **measured (and calculated) quantities**, not to **exact** ones. The number of students in a class (say 42), pure counts, defined constants, and pure numbers like 2 in $2\pi r$ have **infinite** significant figures and never limit your answer.
- Trailing zeros in a whole number without a decimal point are ambiguous.** Does 4500 have two, three, or four significant figures? You genuinely can't tell. The only honest fix is **scientific notation** — 4.5×10^3 (two), 4.50×10^3 (three), 4.500×10^3 (four).
- The number of significant figures is **independent of the unit chosen**. 5.6 cm, 0.056 m, and 56 mm all carry two significant figures — changing units only moves the decimal point and adds scaffolding zeros, never structural ones.
- Significant figures are a *convention for reporting precision*, not a replacement for a proper error analysis. They are a quick, standardised shorthand for "this is how far I trust this number."

Section 2: Key Formulas & Definitions

Significant figures is a topic of **rules rather than formulas**, but the key definitions and quantitative conventions are:

Significant figures: all the digits in a measured value that are known reliably, plus the first uncertain (estimated) digit.

Implied uncertainty: a measured value is taken to be uncertain by about ± 1 in its last significant digit (some conventions use $\pm \frac{1}{2}$). Thus:

$$3.5 \text{ cm} \Rightarrow 3.5 \pm 0.1 \text{ cm}, \quad 3.50 \text{ cm} \Rightarrow 3.50 \pm 0.01 \text{ cm}.$$

Adding one significant figure tightens the implied precision roughly **tenfold**.

Scientific notation: any measured value written as $m \times 10^n$, where $1 \leq m < 10$ is the **mantissa** and n is an integer power of ten. The number of significant figures is read **entirely from the mantissa m** ; the power of ten contributes none.

$$6.30 \times 10^4 \Rightarrow 3 \text{ significant figures (from 6.30)}.$$

The two arithmetic precision rules (quantitative):

- Multiplication/Division:** the result keeps as many *significant figures* as the factor with the **fewest** significant figures.
- Addition/Subtraction:** the result keeps as many *decimal places* as the term with the **fewest** decimal places.

These two rules are different and must never be swapped — that distinction is the most heavily tested point in the whole subtopic.

Order of magnitude (a close cousin): Closely tied to scientific notation is the *order of magnitude* of a quantity — the power of ten nearest to it, found by writing the value as $m \times 10^n$ with $\frac{1}{2} \leq m < \text{about } 5$ (i.e. rounding the mantissa to the nearer power of ten) and reading off n . For example, the diameter of a hydrogen atom, about $1.06 \times 10^{-10} \text{ m}$, has order of magnitude 10^{-10} m , while the mass of an electron, $9.1 \times 10^{-31} \text{ kg}$, rounds up to order 10^{-30} kg (since 9.1 is closer to 10 than to 1). Orders of magnitude let physicists compare wildly different scales at a glance and make rapid "back-of-the-envelope" estimates without precise data.

Section 3: Key Procedures & Logical Framework

There is no derivation here; instead, master three clean procedures, each backed by a reason.

Procedure 1 — Counting significant figures. Apply these rules in order:

1. **All non-zero digits are significant.** (42.3 has 3.) *Why?* Every non-zero digit reports a measured magnitude.
2. **Zeros between non-zero digits ("sandwiched zeros") are significant.** (5.004 has 4.) *Why?* They sit inside the measured span, so they carry genuine information.
3. **Leading zeros (left of the first non-zero digit) are never significant.** (0.0067 has 2.) *Why?* They are pure scaffolding — they only locate the decimal point.
4. **Trailing zeros after a decimal point are significant.** (4.330 has 4; 0.05000 has 4.) *Why?* You would only write them if you had actually measured that far.
5. **Trailing zeros in a whole number with no decimal point are ambiguous** and conventionally *not* counted. (4500 → treat as 2.) *Why?* There's no way to know if they're measured or placeholders — hence use scientific notation.
6. **In scientific notation, only the mantissa counts;** the power of ten is never significant.
7. **Significant figures don't depend on the system of units.** (164 mm = 0.164 m, both 3.)

A quick counting walkthrough. Let's apply these to four tricky numbers at once. In 0.003080, the leading 0.00 is scaffolding, while 3, the sandwiched 0, 8, and the trailing 0 are all significant → **4**. In 5060, the sandwiched 0 is significant but the trailing 0 (no decimal) is ambiguous → conventionally **3**. In 5060. (note the deliberate decimal point), that trailing zero now is significant → **4**. And in 7.00×10^8 , only the mantissa 7.00 counts → **3**, regardless of the large power of ten. Notice how a single decimal point or a switch to scientific notation completely changes the count — that is why careful writers reach for scientific notation whenever zeros could be misread.

Procedure 2 — Rounding off. To round a number to the desired number of significant figures, look at the digit immediately to the right of the cut-off:

1. If it is **less than 5**, drop it and leave the last kept digit unchanged. (6.24 → 6.2.)
2. If it is **greater than 5** (or 5 followed by any non-zero digit), increase the last kept digit by 1. (6.27 → 6.3; 14.252 → 14.3.)
3. If it is **exactly 5** (or 5 followed only by zeros), use the **round-half-to-even** rule: leave the last kept digit unchanged if it is *even*, increase it if it is *odd*. (6.250 → 6.2, since 2 is even; 6.350 → 6.4, since 3 is odd.)
Why this odd-looking rule? Because always rounding 5 up would bias a long column of data slightly upward; rounding to the nearest even digit cancels that bias over many values.

Procedure 3 — Carrying significant figures through a calculation.

1. For $\times$ or $\div$, count *significant figures* in each input; the answer gets the smallest count.
2. For $+$ or $-$, count *decimal places* in each input; the answer gets the smallest count. *Why the difference?* Addition is limited by the *position* of the last trustworthy digit (decimal place), while multiplication is limited by the *total* relative precision (significant figures).
3. **Round only once, at the very end.** Keep one or two extra "guard" digits through intermediate steps; rounding early injects fresh error. *Why?* Repeated rounding compounds, dragging the final answer away from the truth.

A quick addition illustration. Add 124.5 g + 0.36 g + 7.02 g. The raw sum is 131.88 g. Now look at *decimal places*, not significant figures: 124.5 has one decimal place, 0.36 has two, 7.02 has two — the fewest is **one**. So the answer is rounded to one decimal place: 131.9 g. Contrast this with multiplication, where we would have counted *significant figures* instead. Running the same three numbers through the wrong rule (treating it as

significant figures) would give a different, incorrect answer — which is exactly why the two rules must be kept firmly apart.

Figure 1.4 A two-column decision card. Left column titled " $\times$ or $\div \rightarrow$ count SIGNIFICANT FIGURES, keep the least" with example $2.5 \times 3.42 = 8.55 \rightarrow 8.6$ (2 sf). Right column titled " $+$ or $- \rightarrow$ count DECIMAL PLACES, keep the least" with example $12.5 + 0.0021 = 12.5021 \rightarrow 12.5$ (1 dp). A banner across the top reads "Round ONCE, at the end." (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Problem. A rectangular glass sheet has a measured length of 2.165 m and breadth of 1.2 m. Calculate its area, expressed to the correct number of significant figures.

Solution. Given: length = 2.165 m (4 significant figures), breadth = 1.2 m (2 significant figures).

Step 1 — Multiply the raw values.

$$\text{Area} = 2.165 \times 1.2 = 2.598 \text{ m}^2.$$

Step 2 — Apply the multiplication rule. The result can carry only as many significant figures as the *least precise* factor. Here breadth has just 2 significant figures.

Step 3 — Round to 2 significant figures. The digit after the cut-off is 9 (> 5), so we round up:

$$\text{Area} = 2.6 \text{ m}^2.$$

Result: The area is $\boxed{2.6 \text{ m}^2}$. Even though the multiplication produced four digits, honesty forces us back to two — the breadth measurement was the weak link.

Example 2 — [NEET Speed Trap]

Problem. How many significant figures are there in the measured value 0.002030 kg?

The trap. Students rush and either (a) count the three leading zeros as significant, shouting "6!", or (b) drop the *final* zero, shouting "3!". Both are wrong.

Fast solution. Scan left to right:

- 0.00 $\rightarrow$ leading zeros, pure scaffolding $\rightarrow$ **not significant**.
- 2 $\rightarrow$ non-zero $\rightarrow$ significant.
- 0 $\rightarrow$ sandwiched between 2 and 3 $\rightarrow$ significant.
- 3 $\rightarrow$ non-zero $\rightarrow$ significant.
- final 0 $\rightarrow$ trailing zero *after* a decimal point $\rightarrow$ significant.

Count the load-bearing digits: 2, 0, 3, 0 $\rightarrow$ **4 significant figures**.

Answer: 4.

Pitfall to remember: Leading zeros never count; trailing zeros *after a decimal* always count. Rewriting as 2.030×10^{-3} kg makes the four figures unmistakable.

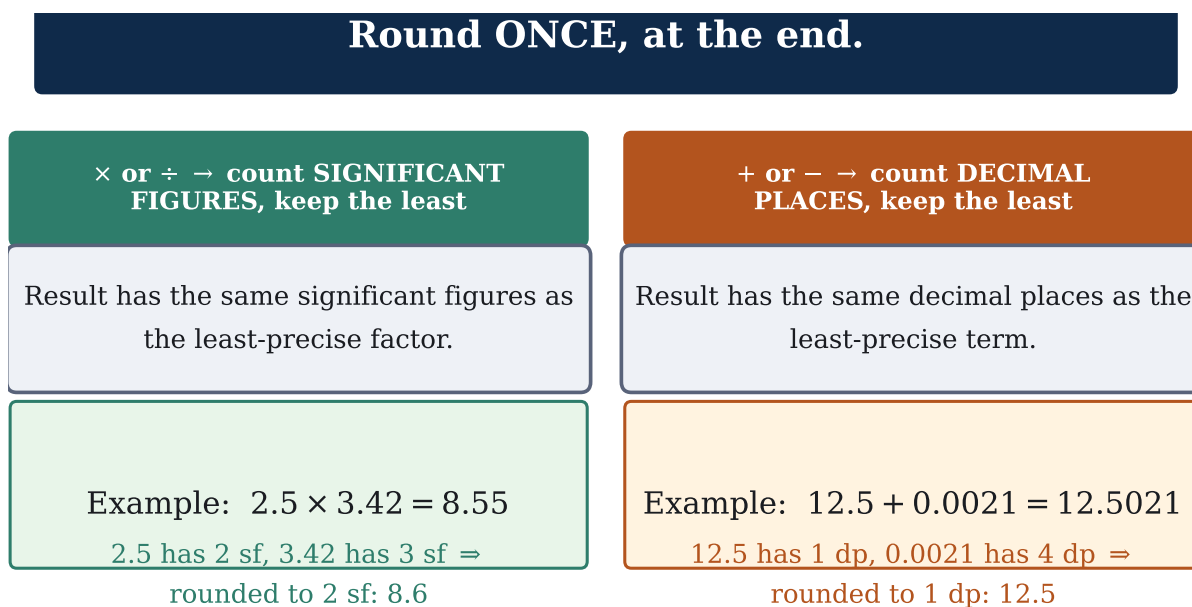


Figure 1.4 A two-column decision card. Left column titled "× or ÷ → count SIGNIFICANT FIGURES, keep the least" with example $2.5 \times 3.42 = 8.55 \rightarrow 8.6$ (2 sf). Right column titled "+ or − → count DECIMAL PLACES, keep the least" with example $12.5 + 0.0021 = 12.5021 \rightarrow 12.5$ (1 dp). A banner across the top reads "Round ONCE, at the end."

Example 3 — [JEE Main Level]

Problem. A thin cylindrical wire has radius 1.5 mm, length 30.0 cm, and mass 8.0 g. Taking $\pi = 3.14$ (exact), find the density of the wire in g cm^{-3} to the correct number of significant figures. (This blends the geometry of a cylinder with the significant-figure rules.)

Solution. Step 1 — Convert and note significant figures.

- radius $r = 1.5 \text{ mm} = 0.15 \text{ cm}$ (2 sig figs)
- length $\ell = 30.0 \text{ cm}$ (3 sig figs)
- mass $m = 8.0 \text{ g}$ (2 sig figs)
- $\pi = 3.14$ is exact $\rightarrow$ infinite sig figs, so it never limits the answer.

Step 2 — Compute the volume $V = \pi r^2 \ell$ (keep guard digits):

$$V = 3.14 \times (0.15)^2 \times 30.0 = 3.14 \times 0.0225 \times 30.0 = 2.1195 \text{ cm}^3.$$

Step 3 — Compute density $\rho = \frac{m}{V}$:

$$\rho = \frac{8.0}{2.1195} = 3.774 \text{ g cm}^{-3}.$$

Step 4 — Round to the least significant figures among the measured inputs. The fewest is 2 (from r and m):

$$\rho = 3.8 \text{ g cm}^{-3}.$$

Result: The density is $\boxed{3.8 \text{ g cm}^{-3}}$. The exact π is a free pass — only the *measured* quantities decide the final precision.

Example 4 — [JEE Advanced Level]

Problem. A student records the side of a small metal cube as 2.5 cm. (a) Using the implied-uncertainty convention, find the percentage uncertainty in the side. (b) Hence find the volume and its percentage uncertainty, and state the volume to the correct significant figures. (c) Explain quantitatively how the situation would improve if the side were instead recorded as 2.50 cm.

Solution. (a) *Implied uncertainty of the side.* A value 2.5 cm is uncertain by ± 1 in its last digit, i.e. $\pm 0.1 \text{ cm}$:

$$\delta(\text{side}) = \frac{0.1}{2.5} \times 100\% = 4\%.$$

(b) *Volume and its uncertainty.* Volume $V = a^3$, so by the power rule the percentage uncertainty triples:

$$\delta V = 3 \times \delta(\text{side}) = 3 \times 4\% = 12\%.$$

Numerically $V = (2.5)^3 = 15.625 \text{ cm}^3$. Since the side has only 2 significant figures, we report

$$V = 16 \text{ cm}^3.$$

And indeed 12% of 16 is about 2, consistent with $V = (16 \pm 2) \text{ cm}^3$.

(c) *The effect of an extra significant figure.* Writing 2.50 cm implies uncertainty $\pm 0.01 \text{ cm}$:

$$\delta(\text{side}) = \frac{0.01}{2.50} \times 100\% = 0.4\%, \quad \delta V = 3 \times 0.4\% = 1.2\%.$$

Result: A single extra significant figure cuts the side's percentage uncertainty from 4% to 0.4% — a **tenfold** improvement — and the volume's from 12% to 1.2%.

Why this is "Advanced": it fuses significant figures with error propagation and reveals the *deep meaning* of a trailing zero — it is not cosmetic but a tenfold tightening of your claim about reality. That conceptual link is exactly what separates rote rule-counting from real understanding.

Section 5: Practice Exercises

1. [CBSE] State the number of significant figures in each: (a) 0.00708 (b) 5.040×10^4 (c) 6500 (d) 100.00.
2. [NEET] How many significant figures are there in the measured value 2.0040 m?
3. [JEE Main] Evaluate $24.36 + 0.0245 + 105.2$ and express the answer to the correct number of significant figures.
4. [JEE Main] Evaluate 5.42×0.6 and express the answer to the correct number of significant figures.
5. [JEE Advanced] The edge of a cube is measured as 2.5 cm. Find its volume to the correct significant figures and state the implied percentage uncertainty in both the edge and the volume.

Final Answers:

1. (a) 3 (b) 4 (c) 2 (ambiguous; taken as 2) (d) 5
 2. 5
 3. 129.6 (addition rule → least decimal places is 1, from 105.2)
 4. 3 (multiplication rule → least significant figures is 1, from 0.6)
 5. $V \approx 16 \text{ cm}^3$; edge $\approx 4\%$, volume $\approx 12\%$
-

Section 6: Crack the MCQ

Q1. The number of significant figures in 0.0500 is: (A) 1 (B) 2 (C) 3 (D) 4

Answer: (C) 3. The two leading zeros are scaffolding (not significant); the 5 and the two trailing zeros *after the decimal* are all significant → 5, 0, 0 = 3. (A) and (B) come from wrongly dropping the trailing zeros, the most common error. (D) comes from wrongly counting a leading zero. Rewriting as 5.00×10^{-2} makes the three figures obvious.

Q2. When two measured quantities are multiplied, the number of significant figures in the result equals: (A) the largest among the factors (B) the smallest among the factors (C) the sum of the factors' significant figures (D) the smallest number of decimal places among the factors

Answer: (B). A product can be no more precise than its least precise factor, so it keeps the *fewest* significant figures. (A) over-claims precision the data doesn't support. (C) is nonsensical — multiplying figures together. (D) is the killer trap: decimal places govern *addition*, not multiplication. Mixing these two rules is the most-tested mistake in the chapter.

Q3. Which of the following measured values has exactly **four** significant figures? (A) 0.0024 (B) 2400 (C) 2.400 (D) 24000

Answer: (C) 2.400. The 2, 4, and two trailing zeros after the decimal are all significant → 4. (A) has only 2 (leading zeros don't count). (B) and (D) have ambiguous trailing zeros with no decimal point, conventionally

counted as 2 — students who assume "more digits = more significant figures" fall for these.

Q4. Rounded to **three** significant figures using the standard round-half-to-even rule, 4.7350 becomes: (A) 4.73 (B) 4.74 (C) 4.735 (D) 4.70

Answer: (B) 4.74. The digit being dropped is exactly 5 (followed only by a zero), so we apply round-half-to-even: the preceding digit is 3, which is **odd**, so we round it *up* to 4 → 4.74. (A) is what you'd wrongly get by leaving the odd digit unchanged. (C) ignores the instruction to round to three figures. (D) drops a significant digit incorrectly. This is the exact scenario where round-half-to-even is tested.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Counting leading zeros as significant.** Zeros to the *left* of the first non-zero digit are pure scaffolding. 0.0042 has two significant figures, not four.
2. **Mishandling trailing zeros.** After a decimal point they *are* significant (3.50 has 3); in a bare whole number they are ambiguous (350). Always resolve the ambiguity with scientific notation.
3. **Swapping the two arithmetic rules.** Use **significant figures** for $\times/\div$ and **decimal places** for $+/-$. They are not interchangeable, and exams deliberately bait this confusion.
4. **Rounding too early.** Rounding at intermediate steps compounds error. Carry one or two guard digits and round only the final answer — and remember exact numbers (counts, defined constants, π taken as exact) have infinite significant figures and never limit the result. A handy self-check: your reported answer should never look *more* precise than your least precise measurement allowed.

Pro-Tip (accuracy): When a number's zeros make you unsure, mentally rewrite it in scientific notation — the mantissa instantly reveals the count, with zero ambiguity. And in any multi-step calculation, do the *whole* computation first keeping extra digits, decide the governing rule (least sig figs for $\times/\div$, least decimal places for $+/-$), and round exactly once at the end. This habit alone prevents the majority of significant-figure mark losses. A final guard: physical constants supplied in a question (like $g = 9.8 \text{ m s}^{-2}$ or $\pi = 3.14$) carry their own significant figures, so check whether the question intends them as exact or as measured — when they are the least-precise value in your calculation, *they* set the precision of your answer.

Section 8: Cheat Sheet Box

SIGNIFICANT FIGURES — QUICK REVISION

- **Significant figures** = all reliably known digits + the first uncertain one. They report *precision*.
- **Counting:** non-zero digits ✓; sandwiched zeros ✓; leading zeros ✗; trailing zeros after a decimal ✓; trailing zeros in a bare whole number = ambiguous (use scientific notation); powers of ten ✗.
- **Unit-independent:** $5.6 \text{ cm} = 0.056 \text{ m}$, both 2 sig figs.
- **Exact numbers** (counts, defined constants, π) → infinite sig figs; never limit the answer.
- $\times, \div$ → keep the **least number of significant figures**.
- $+, -$ → keep the **least number of decimal places**.
- **Rounding:** < 5 drop; > 5 round up; exactly 5 → round-half-to-even.

- **Round only once, at the end;** keep guard digits in between.
- **Implied uncertainty** $\approx \pm 1$ in the last digit; one extra sig fig $\approx$ tenfold tighter precision.

 Memory aids:

- "Leading zeros lead nowhere; trailing decimals tell the truth."
- "Times-divide $\rightarrow$ figures; plus-minus $\rightarrow$ places."
- "When zeros confuse, scientific notation defuses."
- "Round half to even keeps the data honest."
- "Exact numbers are infinitely precise — they never set the limit."

Chapter 1: Units and Measurements

Sub-topic 05: Measuring Instruments (Vernier Callipers & Screw Gauge)

Exam Relevance: A favourite of examiners because it blends a formula, a reading procedure, and a sign convention. JEE Main and JEE Advanced both ask least-count and zero-error questions almost every year, often as "modified instrument" twists. NEET sets a quick least-count or reading question. CBSE Boards include it as a 2–3 mark numerical or in the practical/viva, especially the least-count calculation and zero-error correction.

Section 1: Concept Intuition

An ordinary plastic ruler can measure to about a millimetre — but what if you need the diameter of a wire thinner than that, or the thickness of a coin to a hundredth of a millimetre? You'd be stuck squinting between two marks, guessing. The genius of the **vernier callipers** and the **screw gauge** is that they let you read *beyond* the smallest mark on the main scale, turning a guess into a precise number. They are the instruments that take you from "about 2 mm" to "2.34 mm, confidently."

The whole magic rests on one clever idea: **least count** — the smallest length an instrument can reliably measure. A plain scale has a least count of 1 mm. A good vernier callipers gets down to 0.1 mm, and a screw gauge to 0.01 mm — ten times finer still. The smaller the least count, the more decimal places you're entitled to write, and the more *precise* your measurement. So this subtopic is really the practical, hands-on face of everything you learned about significant figures and errors: the least count *is* the uncertainty in a single reading.

How does a vernier read between the millimetre marks? Through a beautiful trick of *mismatched* scales. The sliding **vernier scale** has divisions that are very slightly *smaller* than the main scale's divisions — for example, 10 vernier divisions span only 9 millimetres. Because of this tiny, deliberate mismatch, exactly *one* vernier line will line up perfectly with a main-scale line, and *which* line it is tells you the fractional part. It's like two slightly different combs sliding past each other: only one pair of teeth ever aligns at a time, and that alignment pinpoints the gap.

The **screw gauge** uses a different trick — the magnifying power of a screw. When you turn a screw by one full rotation, it advances by a fixed small distance called the **pitch**. By wrapping a finely divided circular scale around that rotation, a movement far too small to see on a straight ruler gets "stretched out" around the circle, where you can read it easily. Turning rotational motion into tiny, readable linear motion is the screw's superpower.

Think about it this way... imagine trying to measure a tiny gap with your fingers — impossible directly. But if you had a ramp so gentle that walking a long way along it raised you only a centimetre, you could measure that centimetre by measuring your *long walk* instead. The screw's threads are exactly that gentle ramp: they convert a hard-to-see linear distance into an easy-to-read rotation. The vernier and the screw gauge are both "precision amplifiers" — they take a measurement too fine for the eye and blow it up into something the eye can read.

This also explains *which* instrument a good experimenter reaches for. You don't always want the finest tool — you want the one matched to the job. To measure the length of a table you'd use a metre scale; the vernier's extra precision would be wasted effort. To measure the internal diameter of a pipe or the length of a small cylinder, the

vernier earns its keep. And for something genuinely tiny — the diameter of a wire, the thickness of a glass slide — only the screw gauge will do. There is a quiet trade-off here: finer instruments measure smaller ranges. A metre scale spans a metre but resolves a millimetre; a screw gauge resolves a hundredth of a millimetre but typically spans only a couple of centimetres. Choosing the right instrument is itself part of good measurement.

Limiting Conditions & Assumptions.

- The least count sets the **best-case** precision; the *actual* uncertainty can be worse if the instrument has a **zero error** (a built-in offset when the jaws are closed) that you forget to correct.
- A single reading carries an uncertainty equal to the least count (or, by some conventions, half of it) — these instruments do not magically remove all error.
- The vernier formula assumes the vernier divisions are *uniformly* slightly smaller than the main-scale divisions; a damaged or worn scale breaks this.
- **Backlash error** (play in the screw threads when you reverse direction) and over-tightening can spoil a screw-gauge reading, which is why a *ratchet* is used to apply consistent, gentle pressure.

A useful link back to error analysis: a **zero error is a textbook systematic error** — it has a fixed size and sign and shifts *every* reading the same way, so no amount of averaging will remove it. The only cure is to *measure* it (close the jaws and read the offset) and *correct* for it, exactly as we said systematic errors must be handled. By contrast, the irreducible least-count uncertainty behaves like the random/limiting error in a single reading. So these instruments are where the abstract ideas of the previous subtopic — systematic vs random error, precision vs accuracy — become something you can physically see and fix with your own hands.

Section 2: Key Formulas & Definitions

Least count (LC): the smallest length an instrument can measure — equivalently, the uncertainty in one reading.

Vernier callipers:

$$LC = 1 \text{ MSD} - 1 \text{ VSD}$$

where MSD = main scale division and VSD = vernier scale division. If n vernier divisions coincide with $(n - 1)$ main scale divisions, this simplifies to the working formula:

$$LC = \frac{\text{Value of 1 main scale division}}{\text{Number of vernier scale divisions } (n)}$$

Reading of a vernier:

$$\text{Reading} = \text{MSR} + (\text{coinciding vernier division} \times LC) - \text{zero error}$$

where MSR is the main scale reading just before the vernier zero.

Screw gauge (micrometer):

$$\text{Pitch} = \frac{\text{distance moved on linear scale}}{\text{number of complete rotations}}$$

$$LC = \frac{\text{Pitch}}{\text{Number of divisions on the circular scale}}$$

Reading of a screw gauge:

$$\text{Reading} = \text{MSR} + (\text{coinciding circular division} \times LC) - \text{zero error}$$

Zero error and correction:

$$\text{Corrected reading} = \text{Observed reading} - \text{zero error}$$

- **Positive zero error:** when the jaws are closed, the moving zero lies *ahead* of the main zero, giving an excess reading. The zero error is **positive**, so we **subtract** it.
- **Negative zero error:** the moving zero lies *behind* the main zero. The zero error is **negative**; subtracting a negative value means we effectively **add** its magnitude.

All of these are *length* quantities, with dimensional formula $[L]$ and SI unit metre.

Section 3: Key Procedures & Logical Framework

There is no grand derivation here, but the least-count formula *can* be reasoned out, and the reading procedures must be airtight.

Reasoning out the vernier least count. Suppose n vernier divisions exactly span $(n - 1)$ main scale divisions. Then:

$$n (\text{VSD}) = (n - 1) (\text{MSD}) \Rightarrow 1 \text{ VSD} = \frac{n - 1}{n} \text{ MSD}.$$

The least count is the *gap* between one main division and one vernier division:

$$\text{LC} = 1 \text{ MSD} - 1 \text{ VSD} = \text{MSD} \left(1 - \frac{n - 1}{n} \right) = \frac{\text{MSD}}{n}.$$

Why this is the smallest measurable length: the vernier zero advances by exactly this gap before the *next* vernier line takes its turn to coincide — so this gap is the finest distinction the instrument can register. This is *why* more vernier divisions (n larger) means a finer instrument.

Procedure A — Reading a vernier callipers.

1. Note the **main scale reading (MSR)** — the last main-scale mark *just before* the vernier zero.
2. Find which **vernier division coincides** exactly with any main-scale line; call it the vernier scale reading (VSR).
3. Compute: $\text{Reading} = \text{MSR} + (\text{VSR} \times \text{LC})$.
4. **Apply zero-error correction:** subtract the zero error (with its sign).

Procedure B — Reading a screw gauge.

1. Determine the **pitch** = (linear distance moved) $\div$ (number of full rotations). *Don't assume it's 1 mm* — always check.
2. Compute $\text{LC} = \text{pitch} \div (\text{circular scale divisions})$.
3. Note the **MSR** on the linear scale, and the **circular division** in line with the reference line.
4. Compute: $\text{Reading} = \text{MSR} + (\text{circular division} \times \text{LC})$, then **correct for zero error**.

Procedure C — Handling zero error (the part students fail).

1. Close the jaws fully (no object). If the moving zero does *not* line up with the main zero, there is a zero error.
2. If the moving zero sits *past* the main zero (excess), the error is **positive** $\rightarrow$ corrected reading = observed — (positive value).

3. If the moving zero sits *short of* the main zero, the error is **negative** → corrected reading = observed – (negative value) = observed + magnitude.
4. *Why subtract in both cases?* Because "corrected = observed – zero error" is one universal rule; the sign of the zero error automatically does the right thing. Memorise the *formula*, not two separate cases.

A quick end-to-end illustration. Suppose a vernier callipers has 1 MSD = 1 mm and 10 vernier divisions over 9 mm, so $LC = 0.1 \text{ mm} = 0.01 \text{ cm}$. When the jaws are closed, the 2nd vernier line coincides with a main mark *past* the zero, giving a positive zero error of $2 \times 0.01 = 0.02 \text{ cm}$. Now an object is measured: main scale reads 4.5 cm and the 8th vernier division coincides. The observed reading is $4.5 + (8 \times 0.01) = 4.58 \text{ cm}$. Applying the correction, the true length is $4.58 - 0.02 = 4.56 \text{ cm}$. Every measurement you ever take with these instruments follows this same three-beat rhythm: **find the LC, take the reading, correct the zero.**

Figure 1.5 Three side-by-side vernier zero-position sketches when jaws are closed. (i) "No zero error": vernier 0 aligns exactly with main 0. (ii) "Positive zero error": vernier 0 sits to the right of main 0, with the 4th vernier line coinciding, labelled "+0.04 cm → subtract". (iii) "Negative zero error": vernier 0 sits to the left of main 0, with the 6th-from-top line coinciding, labelled "-0.04 cm → add". A caption reads "Corrected = Observed – Zero error (sign included)." (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Problem. A vernier callipers has a main scale graduated in millimetres (1 MSD = 1 mm), and 10 vernier divisions coincide with 9 main scale divisions. While measuring a rod, the main scale reads 1.2 cm and the 6th vernier division coincides with a main scale mark. There is no zero error. Find the length of the rod.

Solution. *Step 1 — Least count.* With $n = 10$ vernier divisions and 1 MSD = 1 mm:

$$LC = \frac{1 \text{ MSD}}{n} = \frac{1 \text{ mm}}{10} = 0.1 \text{ mm} = 0.01 \text{ cm}.$$

Step 2 — Apply the reading formula.

$$\text{Length} = \text{MSR} + (\text{VSR} \times \text{LC}) = 1.2 \text{ cm} + (6 \times 0.01 \text{ cm}) = 1.2 + 0.06 = 1.26 \text{ cm}.$$

Result: The length of the rod is 1.26 cm. Note how the vernier added a confident second decimal place that the main scale alone could never provide.

Example 2 — [NEET Speed Trap]

Problem. A screw gauge shows a reading of +0.05 mm when its jaws are completely closed. While measuring a wire, the instrument reads 2.34 mm. What is the correct diameter of the wire?

The trap. Seeing the wire reads 2.34 mm, the hasty student either reports it as is, or — worse — *adds* the zero error to get 2.39 mm. Both are wrong. A **positive** zero error means the instrument reads *too high*, so it must be **subtracted**.

Fast solution.

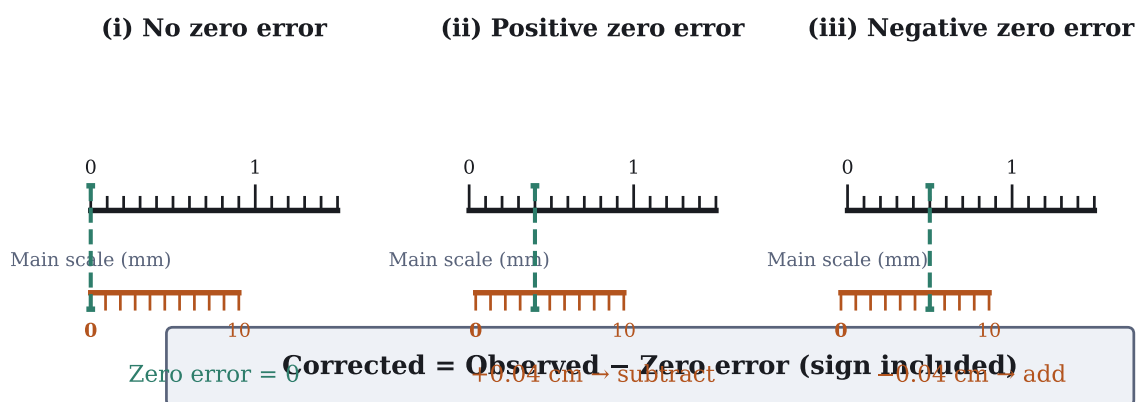


Figure 1.5 Three side-by-side vernier zero-position sketches when jaws are closed. (i) "No zero error": vernier 0 aligns exactly with main 0. (ii) "Positive zero error": vernier 0 sits to the right of main 0, with the 4th vernier line coinciding, labelled "+0.04 cm → subtract". (iii) "Negative zero error": vernier 0 sits to the left of main 0, with the 6th-from-top line coinciding, labelled "-0.04 cm → add". A caption reads "Corrected = Observed - Zero error (sign included)."

$$\text{Corrected} = \text{Observed} - \text{zero error} = 2.34 - (+0.05) = 2.29 \text{ mm.}$$

Answer: 2.29 mm.

Pitfall to remember: Positive zero error $\rightarrow$ *subtract*. The single safe rule is **corrected = observed – zero error**, letting the sign handle the rest. Adding when you should subtract is the most common instrument mistake in NEET.

Example 3 — [JEE Main Level]

Problem. A screw gauge has a pitch of 1 mm and 100 divisions on its circular scale. While measuring the diameter of a wire, the main scale reads 2 mm and the 45th circular division coincides with the reference line. The instrument has a **negative zero error** of 5 circular divisions. Find (a) the corrected diameter, and (b) the maximum percentage error in the cross-sectional area of the wire, taking the least count as the uncertainty. (*This blends instrument reading, zero-error correction, and error propagation through area.*)

Solution. Step 1 — *Least count.*

$$\text{LC} = \frac{\text{pitch}}{\text{circular divisions}} = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm.}$$

Step 2 — *Observed reading.*

$$\text{Observed} = \text{MSR} + (45 \times \text{LC}) = 2 + (45 \times 0.01) = 2.45 \text{ mm.}$$

Step 3 — *Zero error.* A negative zero error of 5 divisions means

$$\text{zero error} = -5 \times 0.01 = -0.05 \text{ mm.}$$

Corrected diameter:

$$d = \text{Observed} - \text{zero error} = 2.45 - (-0.05) = 2.50 \text{ mm.}$$

Step 4 — *Percentage error in area.* The cross-sectional area is $A = \frac{\pi d^2}{4}$, so $A \propto d^2$. By the power rule:

$$\frac{\Delta A}{A} = 2 \frac{\Delta d}{d} = 2 \times \frac{0.01}{2.50} = 2 \times 0.004 = 0.008 = 0.8\%.$$

Result: (a) the corrected diameter is 2.50 mm; (b) the maximum percentage error in the area is 0.8%. Watch how the negative zero error *increased* the reading, and how the diameter's error doubles on its way into the area.

Example 4 — [JEE Advanced Level]

Problem. A precision lab has two instruments. In a **vernier callipers**, 1 cm on the main scale is divided into 20 equal parts, and 25 vernier divisions coincide with 24 main scale divisions. In a **screw gauge**, the linear (main) scale is graduated in millimetres, one complete rotation of the circular scale advances the spindle by 3 main-scale divisions, and the circular scale has 200 divisions. (a) Find the least count of the vernier. (b) Find the least count of the screw gauge. (c) State which instrument is more precise.

Solution. (a) *Vernier least count.* First, the value of one main scale division:

$$1 \text{ MSD} = \frac{1 \text{ cm}}{20} = 0.05 \text{ cm} = 0.5 \text{ mm.}$$

With 25 VSD = 24 MSD, we have $n = 25$, so

$$\text{LC}_{\text{vernier}} = \frac{1 \text{ MSD}}{n} = \frac{0.5 \text{ mm}}{25} = 0.02 \text{ mm.}$$

(b) *Screw gauge least count.* The linear scale is in millimetres, so 1 main-scale division = 1 mm, and one rotation advances 3 such divisions:

$$\text{pitch} = 3 \times 1 \text{ mm} = 3 \text{ mm}.$$

$$\text{LC}_{\text{screw gauge}} = \frac{\text{pitch}}{200} = \frac{3 \text{ mm}}{200} = 0.015 \text{ mm}.$$

(c) *Comparison.* The smaller the least count, the finer the instrument:

$$0.015 \text{ mm (screw gauge)} < 0.02 \text{ mm (vernier)}.$$

Result: The screw gauge is more precise (LC = 0.015 mm versus 0.02 mm).

Why this is "Advanced": every number is non-standard — you cannot assume a 1 mm main division, a 1 mm pitch, or the usual $(n - 1)$ relationship. You must rebuild each least count from its definition and then compare. Modified-instrument problems like this are exactly how JEE Advanced separates students who *memorised* a formula from those who *understand* where it comes from.

Section 5: Practice Exercises

1. [CBSE] A vernier callipers has 1 MSD = 1 mm and 20 vernier divisions coinciding with 19 main scale divisions. Find its least count.
2. [NEET] A screw gauge has a pitch of 0.5 mm and 100 divisions on the circular scale. Find its least count.
3. [JEE Main] A screw gauge (least count 0.01 mm) reads a main scale value of 4 mm with the 32nd circular division coinciding. The instrument has a positive zero error of 0.03 mm. Find the corrected diameter.
4. [JEE Main] A vernier callipers (least count 0.01 cm) reads 3.2 cm on the main scale with the 7th vernier division coinciding. It has a negative zero error of 4 divisions. Find the corrected length.
5. [JEE Advanced] In a vernier callipers, 50 vernier divisions coincide with 49 main scale divisions, and 1 MSD = 0.5 mm. Find the least count, and how many such least counts make up exactly 1 cm.

Final Answers:

1. 0.05 mm
2. 0.005 mm
3. 4.29 mm (*observed 4.32 – zero error 0.03*)
4. 3.31 cm (*observed 3.27 – (–0.04)*)
5. LC = 0.01 mm; and 1000 least counts make 1 cm

Section 6: Crack the MCQ

Q1. The least count of a vernier callipers is given by: (A) 1 MSD + 1 VSD (B) 1 MSD – 1 VSD (C) 1 VSD – 1 MSD (D) 1 MSD × 1 VSD

Answer: (B). The least count is the small *gap* by which a vernier division falls short of a main division, i.e. 1 MSD – 1 VSD. (C) reverses the subtraction and would give a negative number — impossible for a length. (A) and (D) have no physical meaning; they come from blindly combining the symbols without understanding the "mismatched scales" idea.

Q2. The least count of a screw gauge equals: (A) $\text{pitch} \times \text{number of circular divisions}$ (B) $\text{pitch} \div \text{number of circular divisions}$ (C) $\text{number of circular divisions} \div \text{pitch}$ (D) $\text{pitch} + \text{number of circular divisions}$

Answer: (B). One full rotation (the pitch) is shared out among all the circular divisions, so each division corresponds to $\text{pitch} \div (\text{number of divisions})$. (A) would make the least count *larger* than the pitch — absurd. (C) inverts the ratio, giving wrong units. (D) is dimensionally meaningless.

Q3. A measuring instrument has a positive zero error. To obtain the correct reading, the zero error should be: (A) added to the observed reading (B) subtracted from the observed reading (C) ignored, as it is constant (D) multiplied with the least count

Answer: (B). A positive zero error means the instrument reads systematically *high*, so the excess must be subtracted: $\text{corrected} = \text{observed} - \text{zero error}$. (A) doubles the error — the classic sign mistake. (C) is wrong because a constant offset is precisely a *systematic* error that must be removed. (D) confuses the correction with the least-count calculation.

Q4. Among the following, the instrument capable of the *most precise* length measurement is the one with the: (A) largest least count (B) smallest least count (C) longest main scale (D) most main scale divisions

Answer: (B). Precision improves as the least count *shrinks*, since the least count is the uncertainty in a single reading. (A) is exactly backwards. (C) a longer scale measures a bigger *range*, not more finely. (D) the number of *main* scale divisions doesn't set the least count — it's the *vernier/circular* divisions that refine it.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Getting the zero-error sign wrong.** Positive zero error → subtract; negative zero error → effectively add its magnitude. Always use the one rule **corrected = observed – zero error** and let the sign work for you.
2. **Assuming the screw gauge pitch is 1 mm.** The pitch must be computed from how far the spindle moves per rotation. In modified problems it is often *not* 1 mm.
3. **Confusing MSD with VSD.** The least count is $1 \text{ MSD} - 1 \text{ VSD}$ (main minus vernier). Swapping them gives a meaningless negative value.
4. **Using the coinciding division number directly.** The vernier/circular division that coincides must be *multiplied by the least count* before adding — it is not a length on its own. For instance, "the 45th division" is not 45 mm; it is $45 \times \text{LC}$, which for a screw gauge of $\text{LC} = 0.01 \text{ mm}$ is only 0.45 mm.

Pro-Tip (speed + accuracy): Compute the least count *first, every time* — for a vernier, $\text{LC} = (\text{smallest main division}) \div (\text{number of vernier divisions})$; for a screw gauge, find the pitch, then divide by the circular divisions. With the LC in hand, every reading is just $\text{MSR} + (\text{coinciding division} \times \text{LC})$, followed by the universal correction $\text{corrected} = \text{observed} - \text{zero error}$. Nail this fixed sequence and these questions become near-automatic. One more time-saver for the exam hall: the answer's last meaningful decimal place should match the least count — if your "corrected diameter" carries more precision than the LC allows, you've slipped somewhere, so the LC doubles as a built-in sanity check on your final figure.

Section 8: Cheat Sheet Box

MEASURING INSTRUMENTS — QUICK REVISION

- **Least count (LC)** = smallest measurable length = uncertainty in one reading. Smaller LC $\Rightarrow$ more precise.
- **Vernier LC** = $1 \text{ MSD} - 1 \text{ VSD} = \frac{\text{value of 1 main division}}{\text{number of vernier divisions}}$.
- **Screw gauge:** $\text{pitch} = \frac{\text{linear distance moved}}{\text{number of rotations}}$; $\text{LC} = \frac{\text{pitch}}{\text{circular divisions}}$.
- **Reading** = MSR + (coinciding division $\times$ LC), then correct for zero error.
- **Zero-error rule (universal):** corrected = observed – zero error (sign included). Positive $\rightarrow$ subtract; negative $\rightarrow$ add magnitude.
- **Typical least counts:** metre scale 1 mm; vernier 0.1 mm; screw gauge 0.01 mm.

Memory aids:

- "Main minus vernier" $\rightarrow$ LC of callipers.
- "Pitch over circle" $\rightarrow$ LC of screw gauge.
- "Corrected = observed minus zero error" $\rightarrow$ one rule, every sign.
- "Smaller least count, sharper the count."
- "Zero error is systematic — measure it, then correct it; averaging won't help."

Sub-topic 06: Measurement of Length, Mass and Time

Exam Relevance: This subtopic is the “indirect measurement” toolkit of the chapter. CBSE Boards favour 1–2 mark questions on the parallax method, angular size, or the size-of-molecule estimate, and love order-of-magnitude reframing questions. JEE Main occasionally sets a parallax/angular-diameter or distance-by-echo numerical, and quietly tests parsec vs light-year. NEET asks the occasional definition or order-of-magnitude question (range of lengths/masses/times). JEE Advanced rarely asks it directly but uses the *estimation mindset* it builds. The marks here are easy and conceptual — provided you keep parsec, light year and AU straight, and never confuse half-angle parallax with full-baseline parallax.

Section 1: Concept Intuition

Stretch out a measuring tape and you can read off the length of a room, the height of a friend, the width of a notebook. But now try to measure three things with that same tape: the distance to the Moon, the diameter of an oil molecule, and the distance to a star. The tape is useless for all three — two are absurdly large, one is absurdly small. This is the central problem of this subtopic: **direct measurement only works in the narrow “human-sized” band**, roughly a millimetre to a few hundred metres. Outside that band, we must be clever and measure things *indirectly*.

Here is the first clever trick, and you already use it without knowing. Hold your thumb up at arm’s length and look at it with your left eye only, then your right eye only. Your thumb appears to jump against the far wall. That apparent jump is **parallax** — the shift in a nearby object’s position when you view it from two different places. The closer the object, the bigger the jump. Turn this around and you have a distance-measuring machine: if you know how far apart your two eyes (or two telescopes) are, and you measure how big the jump is, you can compute the distance. Astronomers measure the distance to the Moon and to nearby stars in exactly this way — the two “eyes” become two observatories on Earth, or the same telescope six months apart at opposite ends of Earth’s orbit.

The second trick is **timing an echo**. Shout into a canyon and the time for the echo to return tells you the canyon wall’s distance, because sound travels at a known speed. Replace sound with a radio pulse (RADAR), or a laser pulse (LIDAR), or underwater sound (SONAR), and the same idea measures the distance to an aircraft, the Moon, or the seabed. The rule is always the same: the pulse goes out and comes back, so the distance is *half* the round-trip — and forgetting that factor of two is the classic blunder.

For the impossibly small — the size of a single molecule — we use a third trick: **spread it thin and divide**. If you take a tiny, known volume of oil and let it spread on water until it is just *one molecule thick*, then the thickness equals the molecule’s size. And thickness is simply volume divided by area, both of which you can measure. A measurement too small to see directly becomes an easy volume and an easy area.

Think about it this way... measuring extreme sizes is like weighing a single grain of rice. You can’t — your kitchen scale won’t budge for one grain. So instead you weigh a thousand grains and divide by a thousand. You never measured one grain directly; you measured something big and easy, then scaled down. Every method in this subtopic is a version of that same move: convert the un-measurable into something measurable, then do arithmetic.

Mass and time hide the same story. A chemical balance handles grams to kilograms, but the mass of an electron ($\sim 10^{-30}$ kg) or of a galaxy ($\sim 10^{42}$ kg) needs indirect methods — a **mass spectrograph** bends charged particles in a magnetic field to weigh atoms, while planetary orbits and gravity “weigh” stars and galaxies. Time stretches from the 10^{-22} s life of an unstable particle to the $\sim 10^{17}$ s age of the universe, and the most precise clocks today are **atomic clocks**, which count the unimaginably steady vibrations of caesium atoms. Stepping back, the quantities of physics span an astonishing range — about 10^{41} in length, 10^{85} in mass, 10^{39} in time — and learning the *order of magnitude* of things across that range is itself a skill examiners test.

Limiting Conditions & Assumptions. A few quiet rules hold this topic together:

- The parallax relation $D = b/\theta$ requires the parallactic angle θ to be **in radians** and assumes θ is small

(so the arc equals the chord). It works only when the object is far compared to the basis b .

- The angular-size relation $\alpha = d/D$ likewise needs α in radians and a small angle, and treats the object's diameter as an arc of a circle of radius D .
- The molecular-size estimate assumes the oil film is **exactly one molecule thick** (a monolayer) and that molecules pack without gaps — so it gives an *estimate* of molecular diameter, good to an order of magnitude, not a precise value.
- Echo methods assume the pulse speed is constant along the whole path; in air or water this can vary with temperature, introducing error.

Section 2: Key Formulas & Definitions

Parallax method (distance of a far object). Observe the object from two points A and B separated by a basis b . Let the object subtend the parallax angle θ (in radians) between the two lines of sight. Then

$$\theta = \frac{b}{D} \implies D = \frac{b}{\theta}.$$

For a planet, b is the distance between two observatories on Earth. For a star, the basis is the diameter of the Earth's orbit, $b = 2 \text{ AU}$.

Angular size (actual size of a distant object). If an object of true diameter d at distance D subtends an angular diameter α (in radians), then

$$\alpha = \frac{d}{D} \implies d = \alpha D.$$

Echo / RADAR / SONAR / LIDAR (distance from round-trip time). A pulse travels to the target and back in total time t at speed v :

$$D = \frac{vt}{2}.$$

Here $v = c = 3 \times 10^8 \text{ m s}^{-1}$ for radio/laser pulses, and $v \approx 1500 \text{ m s}^{-1}$ for sound in water.

Size of a molecule (oleic-acid monolayer). A drop of known oleic-acid volume V spreads into a one-molecule-thick film of area A . The molecular diameter is the film thickness:

$$t = \frac{V}{A}.$$

Standard length units (practical, non-SI but accepted):

- 1 fermi = 1 fm = 10^{-15} m (nuclear sizes)
- 1 angstrom = 1 Å = 10^{-10} m (atomic sizes)
- 1 astronomical unit = 1 AU $\approx 1.496 \times 10^{11} \text{ m}$ (mean Earth-Sun distance)
- 1 light year = 1 ly $\approx 9.46 \times 10^{15} \text{ m}$ (distance light travels in one year)
- 1 parsec = 1 pc $\approx 3.08 \times 10^{16} \text{ m} \approx 3.26 \text{ ly}$

Definition of the parsec: the distance at which 1 AU subtends an angle of 1 arc-second ($1'' = 1/3600$ degree). Equivalently, a star whose parallax (the angle 1 AU subtends at it) is p arc-seconds lies at a distance D (in parsec) = $1/p$.

Mass units and methods:

- 1 atomic mass unit = 1 u = $\frac{1}{12}$ (mass of a ^{12}C atom) $\approx 1.66 \times 10^{-27} \text{ kg}$.
- 1 tonne = 10^3 kg ; 1 quintal = 10^2 kg .

- Ordinary masses: common/beam balance. Atomic/sub-atomic masses: **mass spectrograph**, where a charged particle of charge q and speed v in a magnetic field B moves on a circle of radius $r = mv/(qB)$, so $m = qBr/v$. Astronomical masses: from gravitation (a body's orbit reveals the central mass via Newton's law of gravitation).
- **Gravitational mass** (measured by the gravitational force a body feels) and **inertial mass** (measured by $F = ma$) are found by experiment to be equal — the basis of Einstein's equivalence principle.

Time and atomic clocks: The SI second is defined by 9 192 631 770 oscillations of the radiation from the caesium-133 atom. **Atomic (caesium) clocks** are accurate to about 1 part in 10^{13} — a drift of roughly 1 s in $\sim 10^5$ years.

Order of magnitude: the power of ten nearest a quantity, found by writing it as $m \times 10^n$ with $1 \leq m < 10$ and rounding the mantissa (mantissa $< \sqrt{10} \approx 3.16$ rounds down, otherwise up). Approximate ranges worth knowing:

- **Lengths:** proton/nucleus $\sim 10^{-15}$ m $\rightarrow$ atom $\sim 10^{-10}$ m $\rightarrow$ human $\sim 10^0$ m $\rightarrow$ Earth's radius $\sim 10^7$ m $\rightarrow$ Earth-Sun $\sim 10^{11}$ m $\rightarrow$ observable universe $\sim 10^{26}$ m. (Span $\sim 10^{41}$.)
- **Masses:** electron $\sim 10^{-30}$ kg $\rightarrow$ proton $\sim 10^{-27}$ kg $\rightarrow$ human $\sim 10^2$ kg $\rightarrow$ Earth $\sim 10^{25}$ kg $\rightarrow$ Sun $\sim 10^{30}$ kg $\rightarrow$ observable universe $\sim 10^{55}$ kg.
- **Time intervals:** light to cross a nucleus $\sim 10^{-24}$ s $\rightarrow$ period of light wave $\sim 10^{-15}$ s $\rightarrow$ human heartbeat $\sim 10^0$ s $\rightarrow$ one year $\sim 10^7$ s $\rightarrow$ age of universe $\sim 10^{17}$ s.

Section 3: Key Procedures & Logical Framework

There is no formal derivation here, but four procedures must be airtight, each backed by a reason.

Procedure A — Distance by the parallax method.

1. Pick a basis b : two points whose separation you know (two observatories, or the two ends of Earth's orbit, $b = 2$ AU, viewed six months apart).
2. Sight the distant object from both ends and measure the parallactic angle θ between the two lines of sight. Convert θ to radians.
3. Apply $D = b/\theta$. *Why this works:* the object, the basis and the lines of sight form a thin isosceles triangle; for a small angle the basis is the arc of a circle of radius D , so arc = radius $\times$ angle gives $b = D\theta$.
4. Sanity check: a *larger* parallax means a *closer* object. If your answer says otherwise, you have inverted the ratio.

Procedure B — Actual size from angular size.

1. Measure the angular diameter α of the object (radians) and know its distance D (often found first by parallax).
2. Apply $d = \alpha D$. *Why:* the diameter d is the arc subtended by angle α at radius D .

Procedure C — Distance by echo (RADAR/SONAR/LIDAR).

1. Send a pulse; measure the total time t for it to return.
2. Halve it — the one-way travel time is $t/2$ — and multiply by the pulse speed: $D = vt/2$. *Why halve?* The pulse covers the distance twice. Skipping the $\frac{1}{2}$ doubles your answer; this is the single most common error.

Procedure D — Size of a molecule (oleic-acid monolayer).

1. Dilute oleic acid in alcohol to a known, very small concentration (e.g. 1 cm³ of acid in 20 cm³ alcohol, then 1 cm³ of *that* in 20 cm³ — giving a concentration $\frac{1}{20} \times \frac{1}{20} = \frac{1}{400}$).
2. Sprinkle lycopodium powder on clean water and place **one drop** of the solution on it. The alcohol dissolves away and the oleic acid spreads into a thin circular film, its edge marked by the powder.
3. Measure the film's area A (from its diameter). Compute the volume of *pure oleic acid* in the drop: (drop volume) $\times$ (concentration).

4. The film is assumed one molecule thick, so the molecular diameter is $t = V/A$. Why this is only an estimate: it assumes a perfect, gap-free monolayer — true to about an order of magnitude.

Figure 1.6 A thin isosceles triangle for the parallax method: the distant object S at the apex; the base is the basis b joining two observation points A and B on Earth; the equal long sides are the lines of sight of length D . The small apex angle θ is the parallactic angle, with the relation " $b = D \times \theta$, so $D = b / \theta$ " written beside it. A small inset shows a thumb at arm's length jumping against a far wall when viewed by each eye in turn, captioned " $\text{parallax} = \text{the apparent shift.}$ " (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Problem. The Moon subtends an angular diameter of about 0.5° at an observer on Earth. Taking the Earth-Moon distance as 3.84×10^8 m, estimate the diameter of the Moon.

Solution. Step 1 — Convert the angle to radians.

$$\alpha = 0.5^\circ = 0.5 \times \frac{\pi}{180} \text{ rad} = 8.73 \times 10^{-3} \text{ rad}.$$

Step 2 — Apply the angular-size relation $d = \alpha D$.

$$d = (8.73 \times 10^{-3}) \times (3.84 \times 10^8 \text{ m}) = 3.35 \times 10^6 \text{ m}.$$

Result: The Moon's diameter is about $3.35 \times 10^6 \text{ m} \approx 3350 \text{ km}$, close to the accepted value 3475 km. Notice we never went to the Moon — an angle and a distance were enough.

Example 2 — [NEET Speed Trap]

Problem. A nearby star lies at a distance of 4.0 light years from Earth. Its distance in parsec is closest to: (A) 13.0 pc (B) 1.2 pc (C) 4.0 pc (D) 0.31 pc

The trap. Students who half-remember " $1 \text{ pc} \approx 3.26 \text{ ly}$ " often *multiply* ($4.0 \times 3.26 \approx 13$, option A) instead of dividing. A parsec is the *bigger* unit, so the number of parsecs must be *smaller* than the number of light years — the same "bigger unit, smaller number" logic from unit conversion.

Fast solution.

$$D = \frac{4.0 \text{ ly}}{3.26 \text{ ly/pc}} = 1.2 \text{ pc}.$$

Answer: (B) 1.2 pc. Option (A) is the multiply-instead-of-divide trap; (C) wrongly assumes $1 \text{ pc} = 1 \text{ ly}$; (D) inverts and also mis-scales.

Pitfall to remember: $1 \text{ parsec} \approx 3.26 \text{ light years} \approx 3.08 \times 10^{16} \text{ m}$ — parsec is the largest of the three astronomical length units, then light year, then AU.

Example 3 — [JEE Main Level]

Problem. In the oleic-acid experiment, 1 cm^3 of oleic acid is dissolved in alcohol to make 20 cm^3 of solution; then 1 cm^3 of this solution is further diluted to 20 cm^3 . One drop of the final solution (volume 0.05 cm^3) is placed on water and spreads into a circular film of radius 10 cm. Estimate the diameter of an oleic-acid molecule.

Solution. Step 1 — Concentration of oleic acid.

$$\text{conc.} = \frac{1}{20} \times \frac{1}{20} = \frac{1}{400}.$$

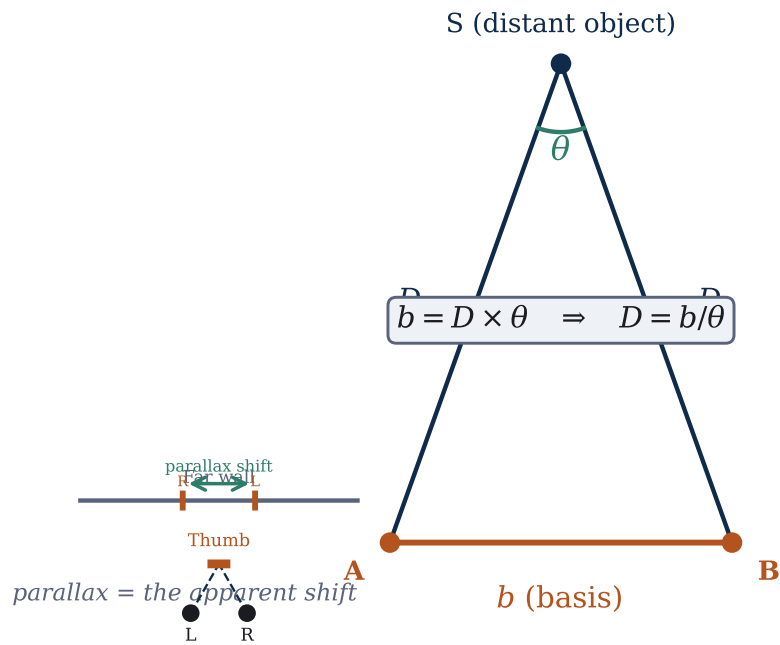


Figure 1.6 A thin isosceles triangle for the parallax method: the distant object S at the apex; the base is the basis b joining two observation points A and B on Earth; the equal long sides are the lines of sight of length D . The small apex angle θ is the parallactic angle, with the relation " $b = D \times \theta$, so $D = b / \theta$ " written beside it. A small inset shows a thumb at arm's length jumping against a far wall when viewed by each eye in turn, captioned "parallax = the apparent shift."

Step 2 — Volume of pure oleic acid in one drop.

$$V = (0.05 \text{ cm}^3) \times \frac{1}{400} = 1.25 \times 10^{-4} \text{ cm}^3.$$

Step 3 — Area of the circular film.

$$A = \pi r^2 = 3.14 \times (10 \text{ cm})^2 = 314 \text{ cm}^2.$$

Step 4 — Thickness = molecular diameter.

$$t = \frac{V}{A} = \frac{1.25 \times 10^{-4} \text{ cm}^3}{314 \text{ cm}^2} = 3.98 \times 10^{-7} \text{ cm} \approx 4.0 \times 10^{-9} \text{ m}.$$

Result: The molecular diameter is about $4 \times 10^{-9} \text{ m}$, i.e. a few nanometres — the right order of magnitude for a long-chain organic molecule. The whole estimate rests on the monolayer assumption: one molecule thick means thickness *equals* molecular size.

Example 4 — [JEE Advanced Level]

Problem. The parallax of a star, measured using the diameter of the Earth's orbit ($\approx 3.0 \times 10^{11} \text{ m}$) as the basis, is $4.0 \times 10^{-6} \text{ rad}$. (a) Find the star's distance, in metres, light years and parsec. (b) The star's angular diameter is measured as $4.0 \times 10^{-8} \text{ rad}$. Find its actual diameter and express it as a multiple of the Sun's diameter ($1.39 \times 10^9 \text{ m}$).

Solution. (a) Distance by parallax, $D = b/\theta$:

$$D = \frac{3.0 \times 10^{11} \text{ m}}{4.0 \times 10^{-6}} = 7.5 \times 10^{16} \text{ m}.$$

Convert:

$$D = \frac{7.5 \times 10^{16}}{9.46 \times 10^{15}} \text{ ly} \approx 7.9 \text{ ly}, \quad D = \frac{7.5 \times 10^{16}}{3.08 \times 10^{16}} \text{ pc} \approx 2.4 \text{ pc}.$$

(b) Actual diameter from angular diameter, $d = \alpha D$:

$$d = (4.0 \times 10^{-8}) \times (7.5 \times 10^{16} \text{ m}) = 3.0 \times 10^9 \text{ m}.$$

As a multiple of the Sun:

$$\frac{d}{d_{\odot}} = \frac{3.0 \times 10^9}{1.39 \times 10^9} \approx 2.2.$$

Result: The star is at $7.5 \times 10^{16} \text{ m} \approx 7.9 \text{ ly} \approx 2.4 \text{ pc}$, with a diameter about 2.2 times the Sun's.

Why this is "Advanced": it chains two indirect methods — parallax first gives the distance, and only with that distance in hand does the angular diameter yield a real size — exactly the two-step reasoning real astronomers use, with three unit systems to juggle cleanly.

Section 5: Practice Exercises

- [CBSE] The Sun's angular diameter seen from Earth is $1920''$ (arc-seconds). Taking the Earth-Sun distance as $1.496 \times 10^{11} \text{ m}$, estimate the Sun's diameter. ($1'' = 4.85 \times 10^{-6} \text{ rad}$.)
- [NEET] Arrange in increasing order of length: 1 AU, 1 light year, 1 parsec, 1 angstrom.
- [JEE Main] A RADAR pulse sent towards a planet returns after 7.0 minutes. Taking $c = 3.0 \times 10^8 \text{ m s}^{-1}$, find the distance of the planet from Earth.

- [JEE Main] In a mass spectrograph, a singly-ionised atom ($q = 1.6 \times 10^{-19}$ C) moving at $v = 2.0 \times 10^5$ m s⁻¹ describes a circular arc of radius $r = 0.10$ m in a magnetic field $B = 0.50$ T. Find the atom's mass (use $m = qBr/v$) and express it in atomic mass units.
- [JEE Advanced] A star has a parallax of $0.76''$ measured with 1 AU as the basis (the standard stellar parallax). Find its distance in parsec, light years and metres.

Final Answers:

- $d = \alpha D = (1920 \times 4.85 \times 10^{-6}) \times (1.496 \times 10^{11}) \approx 1.39 \times 10^9$ m.
- $1 \text{ \AA} < 1 \text{ AU} < 1 \text{ ly} < 1 \text{ pc}$ (i.e. $10^{-10} < 1.5 \times 10^{11} < 9.46 \times 10^{15} < 3.08 \times 10^{16}$ m).
- $D = ct/2 = (3.0 \times 10^8) \times (420)/2 = 6.3 \times 10^{10}$ m.
- $m = qBr/v = (1.6 \times 10^{-19})(0.50)(0.10)/(2.0 \times 10^5) = 4.0 \times 10^{-26}$ kg ≈ 24 u.
- $D = 1/0.76 = 1.32 \text{ pc} \approx 4.3 \text{ ly} \approx 4.06 \times 10^{16}$ m.

Section 6: Crack the MCQ

Q1. The parallax method is best suited to measure: (A) the thickness of a sheet of paper (B) the distance of a nearby star (C) the size of a molecule (D) the mass of an atom

Answer: (B). Parallax exploits the apparent shift of a relatively nearby object against a far background, ideal for the Moon, planets and nearby stars. (A) and (C) are small-length problems (screw gauge; monolayer method), and (D) is a mass measurement (spectrograph) — none involve a line-of-sight shift.

Q2. A radar signal takes 2.6 s to return after reflecting off the Moon. The Earth-Moon distance is closest to: (A) 7.8×10^8 m (B) 3.9×10^8 m (C) 1.95×10^8 m (D) 3.9×10^5 m

Answer: (B). $D = ct/2 = (3 \times 10^8)(2.6)/2 = 3.9 \times 10^8$ m. (A) is the trap — forgetting to halve the round trip, the single most common echo-method error. (C) wrongly divides by an extra factor; (D) is a units slip.

Q3. Which of the following correctly orders the units from smallest to largest? (A) fermi, angstrom, AU, light year, parsec (B) angstrom, fermi, AU, parsec, light year (C) fermi, angstrom, AU, parsec, light year (D) angstrom, fermi, light year, AU, parsec

Answer: (A). $10^{-15} < 10^{-10} < 1.5 \times 10^{11} < 9.46 \times 10^{15} < 3.08 \times 10^{16}$ m. The two classic traps are placing the fermi after the angstrom (it is smaller) and swapping light year with parsec ($1 \text{ pc} \approx 3.26 \text{ ly}$, so parsec is larger).

Q4. The equality of inertial mass and gravitational mass is the experimental basis of: (A) Newton's first law (B) the principle of homogeneity (C) the equivalence principle (D) the conservation of momentum

Answer: (C). That a body's resistance to acceleration ($F = ma$) and its response to gravity are measured to be identical underpins Einstein's equivalence principle. (A), (B) and (D) are unrelated principles offered as distractors.

Section 7: Common Student Pitfalls & Pro-Tips

- Forgetting the factor of two in echo methods.** The pulse travels to the target *and back*, so $D = vt/2$, never vt . Halve the round-trip time first.
- Mixing up parsec, light year and AU.** From smallest to largest: $\text{AU} < \text{ly} < \text{pc}$, with $1 \text{ pc} \approx 3.26 \text{ ly} \approx 3.08 \times 10^{16}$ m. Since parsec is the biggest, a distance has the *fewest* parsecs — divide, don't multiply, when converting ly to pc.
- Using degrees instead of radians.** Both $D = b/\theta$ and $d = \alpha D$ demand the angle in radians. Convert first: $1^\circ = \pi/180 \text{ rad}$, $1'' = 4.85 \times 10^{-6} \text{ rad}$.
- Treating the molecular-size estimate as exact.** The monolayer method gives only an order-of-magnitude diameter ($\sim 10^{-9}$ m); it assumes a perfect one-molecule-thick film. Report it as an estimate, not a precise figure.

Pro-Tip (speed + accuracy): For any “indirect length” question, first name the method from the clue word — “subtends an angle” $\rightarrow$ parallax or angular size ($D = b/\theta$, $d = \alpha D$); “signal/echo returns in time t ” $\rightarrow D = vt/2$; “film/drop/monolayer” $\rightarrow t = V/A$. Then convert every angle to radians *before*

substituting. One more habit: after any astronomical answer, glance at the order of magnitude against the known ranges (Moon $\sim 10^8$ m, nearby star $\sim 10^{16}$ m, molecule $\sim 10^{-9}$ m) — if you are off by powers of ten, you have dropped a factor or mishandled a unit.

Section 8: Cheat Sheet Box

MEASUREMENT OF LENGTH, MASS & TIME — QUICK REVISION

- **Parallax:** $D = \frac{b}{\theta}$ (θ in radians); basis $b = 2$ AU for stars. Bigger parallax $\rightarrow$ closer object.
- **Angular size:** $d = \alpha D$ (α in radians) — find D first (often by parallax), then the size.
- **Echo / RADAR / SONAR / LIDAR:** $D = \frac{vt}{2}$ — always halve the round-trip.
- **Molecule size (oleic-acid monolayer):** $t = \frac{V}{A}$, with $V = (\text{drop volume}) \times (\text{concentration})$; gives $\sim 10^{-9}$ m.
- **Astronomical lengths:** 1 AU $\approx 1.496 \times 10^{11}$ m; 1 ly $\approx 9.46 \times 10^{15}$ m; 1 pc $\approx 3.08 \times 10^{16}$ m ≈ 3.26 ly.
- **Small lengths:** 1 Å = 10^{-10} m (atoms); 1 fermi = 10^{-15} m (nuclei).
- **Mass:** 1 u $\approx 1.66 \times 10^{-27}$ kg; spectrograph $m = qBr/v$; inertial mass = gravitational mass (equivalence principle).
- **Time:** SI second = 9 192 631 770 caesium oscillations; atomic clocks accurate to ~ 1 part in 10^{13} .
- **Ranges (order of magnitude):** length 10^{-15} to 10^{26} m; mass 10^{-30} to 10^{55} kg; time 10^{-22} to 10^{17} s.

Memory aids:

- “Two eyes, one distance” $\rightarrow$ parallax needs a basis b and an angle θ ; $D = b/\theta$.
- “There and back, so halve the track” $\rightarrow$ echo distance is $vt/2$.
- “AU, then ly, then pc” $\rightarrow$ smallest to largest astronomical unit; parsec is biggest, so fewest parsecs.
- “One molecule thick” $\rightarrow$ film thickness is the molecular size, $t = V/A$.
- “Radians always” $\rightarrow$ convert every angle before using $D = b/\theta$ or $d = \alpha D$.

Chapter 1: Units and Measurements — Round 2

Addendum

The chapter gave you the seven base units, the conversion relation, the three jobs of dimensional analysis, the worst-case error rules, the significant-figure discipline, the vernier and the screw gauge, and the indirect-length toolkit. Four examinable patterns are still standing outside the door, and each one is a marriage of two things the chapter already teaches separately. Addendum A pushes homogeneity through derivatives and integrals, which is how JEE extracts the dimensions of every coefficient hiding in a differential equation. Addendum B finishes the error-propagation toolkit: the chapter derived the sum rule but never once worked a percentage-error-in-a-sum problem, and never systematised the “differentiate the reciprocal” route beyond one resistor example. Addendum C builds the dimensional formulae of the constants the chapter’s tables skip — ε_0 , μ_0 , B , R , capacitance — straight from the laws that define them, and weaponises the dimensionless-argument rule. Addendum D faces the chapter’s admitted limitation head-on: what dimensional analysis can still tell you when there are four variables, plus the estimation habits behind every Fermi-style numerical.

Nothing here is new theory. Every method below is homogeneity or worst-case error, applied one layer deeper.

Addendum A — Dimensions Through Calculus: Derivatives and Integrals Carry Units

Sub-topic 02 checks static formulas like $s = ut + \frac{1}{2}at^2$. But JEE’s favourite hunting ground is the *dynamic* equation — $m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$ — where the quantities carrying unknown dimensions sit next to derivative symbols. Nothing in the chapter tells you what $\frac{d^2x}{dt^2}$ does to dimensions. One rule fixes everything, and it follows directly from the definition of the derivative.

Method

The rule, derived. Let y have dimensional formula $[Y]$ and x have $[X]$. A change Δy is a difference of two values of y , so it carries the dimension $[Y]$; likewise Δx carries $[X]$. The quotient $\frac{\Delta y}{\Delta x}$ therefore carries $[Y][X]^{-1}$, and taking the limit $\Delta x \rightarrow 0$ changes no units — shrinking a denominator does not strip dimensions. Hence

$$\left[\frac{dy}{dx}\right] = [Y][X]^{-1},$$

and repeating the argument,

$$\left[\frac{d^n y}{dx^n} \right] = [Y] [X]^{-n}.$$

An integral runs the arrow backwards: $\int y dx$ is the limit of a sum of terms $y \Delta x$, each carrying $[Y][X]$, so

$$\left[\int y dx \right] = [Y] [X].$$

Operation	Dimensional rule
$\frac{dy}{dx}$	$[Y][X]^{-1}$
$\frac{d^n y}{dx^n}$	$[Y][X]^{-n}$
$\int y dx$	$[Y][X]$
mixed derivative $\frac{\partial^2 y}{\partial x \partial t}$	$[Y][X]^{-1}[T]^{-1}$

Here $[Y]$, $[X]$ denote the dimensions of y and x respectively, and n is a positive integer.

The engine. Any differential equation must satisfy homogeneity *term by term*, exactly as in Procedure 1 of Sub-topic 02 — the derivative symbols contribute their powers, and every term must land on the same dimensional formula. That requirement converts each unknown coefficient into a one-step division.

Example A.1 — [CBSE Board Level]

Find the dimensions of b and k in the damped-oscillator equation $m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0$.

Step 1 — Dimensions of the derivatives. Position x carries $[L]$ and time carries $[T]$, so

$$\left[\frac{dx}{dt} \right] = L T^{-1}, \quad \left[\frac{d^2 x}{dt^2} \right] = L T^{-2}.$$

Step 2 — Anchor term. The first term is $m \cdot L T^{-2}$, giving $M L T^{-2}$ — a newton, as it must be, since Newton's second law guarantees this.

Step 3 — Extract b . Homogeneity forces the second term onto $M L T^{-2}$ as well:

$$[b] (L T^{-1}) = M L T^{-2} \implies [b] = \frac{M L T^{-2}}{L T^{-1}} = M T^{-1}.$$

Step 4 — Extract k . Same demand on the third term:

$$[k] (L) = M L T^{-2} \implies [k] = M T^{-2}.$$

Result: $[b] = M T^{-1}$ (SI unit kg s^{-1}) and $[k] = M T^{-2}$ (SI unit kg s^{-2} , equivalently N m^{-1} — exactly the spring constant you will meet in oscillations).

Check. Both divisions can be read backwards: $\text{kg s}^{-1} \times \text{m s}^{-1} = \text{kg m s}^{-2}$ and $\text{kg s}^{-2} \times \text{m} = \text{kg m s}^{-2}$. All three terms land on the newton. Homogeneity holds.

Example A.2 — [JEE Main Level]

Show that the wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ forces c to be a speed, whatever the wave.

Left side: y is a displacement, so

$$\left[\frac{\partial^2 y}{\partial t^2} \right] = L (T^{-2}) = L T^{-2}.$$

Right side: $\frac{\partial^2 y}{\partial x^2}$ carries $L(L)^{-2} = L^{-1}$, so the right side carries $[c^2] L^{-1}$. Homogeneity demands

$$[c^2] L^{-1} = L T^{-2} \implies [c^2] = L^2 T^{-2} \implies [c] = L T^{-1}.$$

Result: c has the dimensions of a speed (m s^{-1} in SI). The equation itself announces that waves propagate — no wave theory required.

Check. Substitute $[c] = L T^{-1}$ into the right side: $(L^2 T^{-2})(L^{-1}) = L T^{-2}$, matching the left side exactly.

Example A.3 — [JEE Advanced Level]

Air drag on a slow projectile is modelled by $F = -\gamma v - \lambda v^2$. Find $[\gamma]$ and $[\lambda]$, and use them to locate the crossover speed at which the two drag terms are equal.

Both terms must carry the dimensions of force, $M L T^{-2}$, and v carries $L T^{-1}$.

Linear term:

$$[\gamma] (L T^{-1}) = M L T^{-2} \implies [\gamma] = M T^{-1}.$$

Quadratic term: v^2 carries $L^2 T^{-2}$, so

$$[\lambda] (L^2 T^{-2}) = M L T^{-2} \implies [\lambda] = M L^{-1}.$$

Crossover speed. Set the magnitudes of the two terms equal, $\gamma v = \lambda v^2$; the nonzero solution is $v_* = \gamma/\lambda$. Check its dimensions before trusting it:

$$\left[\frac{\gamma}{\lambda} \right] = \frac{M T^{-1}}{M L^{-1}} = L T^{-1}.$$

Result: $[\gamma] = M T^{-1}$ (kg s^{-1}), $[\lambda] = M L^{-1}$ (kg m^{-1}), and the crossover speed $v_* = \gamma/\lambda$ comes out with dimensions of speed automatically — a consistency proof that costs nothing.

Why it is Advanced-level. The examiner is not asking for two divisions; the mark-winning move is the third act — forming a new physical quantity from the two coefficients and *demanding* that it come out dimensionally sensible. If your extracted dimensions were wrong, v_* would fail the speed test instantly. The dimensions audit each other.

Practice — Addendum A

1. [CBSE] In $v = a + bt$, where v is speed and t is time, find $[a]$ and $[b]$.
2. [NEET] A population grows according to $\frac{dy}{dt} = ky$. Find $[k]$ and state its SI unit.
3. [JEE Main] Viscous force obeys $F = -\eta A \frac{dv}{dx}$. Find $[\eta]$ and compare with the chapter's table value for coefficient of viscosity.
4. [JEE Main] Heat conduction obeys $\frac{dQ}{dt} = -kA \frac{dT}{dx}$, where Q is heat energy, T temperature, A area and x distance. Find $[k]$.
5. [JEE Advanced] An LR circuit obeys $L \frac{di}{dt} + Ri = E$, where i is current and E is emf with $[E] = \text{ML}^2 \text{T}^{-3} \text{A}^{-1}$. Find $[L]$, $[R]$, and show that the ratio L/R has the dimension of time.

Answers: 1. $[a] = \text{LT}^{-1}$, $[b] = \text{LT}^{-2}$ — each term must be a speed. 2. $[k] = \text{T}^{-1}$; SI unit s^{-1} , since $[y][T]^{-1} = [k][y]$. 3. $\frac{dv}{dx}$ carries T^{-1} , so $[\eta] = \text{MLT}^{-2}/(\text{L}^2 \text{T}^{-1}) = \text{ML}^{-1} \text{T}^{-1}$ — identical to the chapter's table entry, now derived rather than memorised. 4. $\frac{dQ}{dt}$ carries $\text{ML}^2 \text{T}^{-3}$ and $\frac{dT}{dx}$ carries KL^{-1} , so $[k] = \text{MLT}^{-3} \text{K}^{-1}$ (thermal conductivity; SI unit $\text{W m}^{-1} \text{K}^{-1}$). 5. $\frac{di}{dt}$ carries AT^{-1} , so $[L] = \text{ML}^2 \text{T}^{-3} \text{A}^{-1}/(\text{AT}^{-1}) = \text{ML}^2 \text{T}^{-2} \text{A}^{-2}$ (henry); $[R] = [E]/[A] = \text{ML}^2 \text{T}^{-3} \text{A}^{-2}$ (ohm); $[L]/[R] = \text{T}$ — the circuit's time constant, dimensionally guaranteed.

Addendum B — Percentage Errors in Sums, Differences and Reciprocals

Sub-topic 03 derives $\Delta(a+b) = \Delta a + \Delta b$ and then spends every worked example on products, quotients and powers. Two consequences follow that the chapter never demonstrates, and JEE Main sets both almost every year. First: when the input errors arrive as *percentages* and the quantity is a *sum*, you must detour through absolute errors — percentages of different bases never add. Second: for “reciprocals-add” relations (thin lenses, spherical mirrors, parallel anything), the clean route is to differentiate the reciprocal relation, and the recipe deserves to be written down once, properly.

Method

B1 — Sums with percentage inputs. For $x = a + b$ with percentage errors δa and δb :

1. Convert each percentage to an absolute error: $\Delta a = \frac{\delta a}{100} a$, $\Delta b = \frac{\delta b}{100} b$ — keeping the units of a and b .
2. Add the *absolute* errors: $\Delta x = \Delta a + \Delta b$ (Derivation A of Sub-topic 03).
3. Convert back: $\delta x = \frac{\Delta x}{x} \times 100\%$.

Why the detour is compulsory: δa is a fraction *of* a and δb is a fraction *of* b — different denominators. Adding them produces a number attached to neither. Mini-counterexample: 100 ± 1 plus 1 ± 0.5 gives 101 ± 1.5 , i.e. 1.49% — not 1% + 50%.

B2 — Reciprocals-add relations by differentials. For any smooth relation, small changes propagate as

$$\Delta x \approx \left| \frac{\partial f}{\partial a} \right| \Delta a + \left| \frac{\partial f}{\partial b} \right| \Delta b + \dots,$$

with every contribution taken positive (worst case, as always). Apply this to the reciprocal relation $\frac{1}{x} = \frac{1}{a} + \frac{1}{b}$. Differentiating both sides,

$$-\frac{\Delta x}{x^2} = -\frac{\Delta a}{a^2} - \frac{\Delta b}{b^2} \Rightarrow \Delta x = x^2 \left(\frac{\Delta a}{a^2} + \frac{\Delta b}{b^2} \right),$$

where the minus signs have been absorbed by taking magnitudes. Note the structure: the *absolute* error in x picks up each input's absolute error weighted by $(x/a)^2$ — not the relative-error rule of Derivation B, because a and b are entangled on both sides of the defining relation.

Sanity habit. After any sum-or-reciprocal computation, compare the percentage errors of the inputs and the output. A sum's percentage error is dragged toward the *smaller* input's percentage; a reciprocal-add output usually lands *worse* than either input.

Example B.1 — [JEE Main Level]

Two masses are weighed as $m_1 = (10.0 \pm 0.1)$ g and $m_2 = (20.0 \pm 0.1)$ g. Find the maximum percentage error in the total mass $M = m_1 + m_2$, and explain why adding the individual percentage errors fails.

Step 1 — Individual percentage errors. $\delta m_1 = \frac{0.1}{10.0} \times 100\% = 1\%$ and $\delta m_2 = \frac{0.1}{20.0} \times 100\% = 0.5\%$.

Step 2 — Convert to absolute errors (both in grams):

$$\Delta m_1 = \frac{1}{100}(10.0 \text{ g}) = 0.1 \text{ g}, \quad \Delta m_2 = 0.1 \text{ g}.$$

Step 3 — Add absolutes, convert back:

$$\Delta M = 0.1 + 0.1 = 0.2 \text{ g}, \quad M = 30.0 \text{ g},$$

$$\delta M = \frac{0.2}{30.0} \times 100\% = 0.67\%.$$

Result: $M = (30.0 \pm 0.2)$ g, maximum percentage error $\approx 0.67\%$.

Why the shortcut fails. Adding percentages gives $1\% + 0.5\% = 1.5\%$ — more than double the truth. Each percentage divides by its own mass; the sum divides by the larger total. Notice also the structural lesson: the sum inherits the percentage error of the *dominant* mass (0.67% sits between 0.5% and 1%), whereas for a *product* the percentage errors genuinely add. Same data, opposite rules — the distinction the chapter flags but never exercises.

Example B.2 — [JEE Advanced Level]

In a thin-lens experiment (magnitudes; ray-optics signs suppressed), the object and image distances are measured as $u = (30.0 \pm 0.1)$ cm and $v = (20.0 \pm 0.1)$ cm, with $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$. Find f and its maximum absolute and percentage error.

Step 1 — Central value. $\frac{1}{20.0} - \frac{1}{30.0} = \frac{3-2}{60} = \frac{1}{60} \text{ cm}^{-1}$, so

$$f = 60.0 \text{ cm}.$$

Step 2 — Differentiate the reciprocal relation. From $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$,

$$-\frac{\Delta f}{f^2} = -\frac{\Delta v}{v^2} - \frac{\Delta u}{u^2} \Rightarrow \Delta f = f^2 \left(\frac{\Delta v}{v^2} + \frac{\Delta u}{u^2} \right),$$

all magnitudes, worst case.

Step 3 — Substitute, keeping units throughout. $\Delta u = \Delta v = 0.1 \text{ cm}$, so each fraction carries units $\text{cm}/\text{cm}^2 = \text{cm}^{-1}$:

$$\frac{\Delta v}{v^2} = \frac{0.1}{(20.0)^2} = 2.5 \times 10^{-4} \text{ cm}^{-1}, \quad \frac{\Delta u}{u^2} = \frac{0.1}{(30.0)^2} = 1.11 \times 10^{-4} \text{ cm}^{-1}.$$

Then

$$\Delta f = (60.0)^2 \times (2.5 + 1.11) \times 10^{-4} \text{ cm} = 3600 \times 3.61 \times 10^{-4} \text{ cm} \approx 1.3 \text{ cm}.$$

Result: $f = (60.0 \pm 1.3) \text{ cm}$, a maximum percentage error of $\frac{1.3}{60.0} \times 100\% \approx 2.2\%$.

Check. The inputs carry errors of 0.33% (u) and 0.5% (v), yet the output lands at 2.2% — worse than either, exactly as the method warns for entangled reciprocals. A naive “add the relative errors” attempt would give 0.83%, wrong by a factor of nearly three; the entanglement punishes the shortcut.

Why it is Advanced-level. The relation cannot be reduced to a product of powers, so the chapter’s flagship rule does not apply. Recognising that fact — and switching to differentials of the reciprocal — is the entire assessment objective. The same template handles mirrors, capacitors in series, and spring combinations.

Example B.3 — [JEE Main Level]

In a pendulum experiment, $L = (100.0 \pm 0.1) \text{ cm}$ and $T = (2.00 \pm 0.01) \text{ s}$ over one oscillation. Using $g = 4\pi^2 L/T^2$, find g with its maximum percentage and absolute error, and identify which measurement limits the precision.

Step 1 — Central value. $L = 1.000 \text{ m}$, so

$$g = \frac{4\pi^2 \times 1.000}{(2.00)^2} \text{ m s}^{-2} = \frac{39.48}{4.00} \text{ m s}^{-2} = 9.87 \text{ m s}^{-2}.$$

Step 2 — Percentage errors of the inputs. $\delta L = \frac{0.1}{100.0} \times 100\% = 0.1\%$; $\delta T = \frac{0.01}{2.00} \times 100\% = 0.5\%$. Since g is a product of powers, the chapter’s rule applies directly:

$$\delta g = \delta L + 2\delta T = 0.1\% + 2(0.5\%) = 1.1\%.$$

Step 3 — Absolute error:

$$\Delta g = \frac{1.1}{100} \times 9.87 \text{ m s}^{-2} = 0.11 \text{ m s}^{-2}.$$

Result: $g = (9.87 \pm 0.11) \text{ m s}^{-2}$, maximum percentage error 1.1%.

Where the error lives. Timing contributes $2 \times 0.5\% = 1.0\%$ of the 1.1%; the length contributes only 0.1%. The fix is procedural, not instrumental: time N oscillations together so that δT falls roughly N -fold — timing 50 oscillations here would cut the timing contribution to about 0.02% and make the length the limiter instead.

Practice — Addendum B

1. [CBSE] $m_1 = (15.0 \pm 0.1) \text{ g}$ and $m_2 = (5.0 \pm 0.1) \text{ g}$. Find the maximum percentage error in (a) the sum and (b) the difference.
2. [NEET] $Z = A + B$ with $A = (50 \pm 1)$ units and $B = (20 \pm 1)$ units. Write Z with its error and give the percentage error.
3. [JEE Main] A concave mirror forms a real image with $u = (20.0 \pm 0.1) \text{ cm}$ and $v = (20.0 \pm 0.1) \text{ cm}$, using $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ (magnitudes). Find f with its maximum absolute error.
4. [JEE Main] In a Searle's-apparatus determination, $Y \propto m L (r^2 e)^{-1}$. The percentage errors are $\delta m = 1\%$, $\delta L = 0.2\%$, $\delta r = 0.5\%$, $\delta e = 1.5\%$. Find δY .
5. [JEE Advanced] A pendulum gives $L = (99.8 \pm 0.1) \text{ cm}$ and $T = (2.00 \pm 0.02) \text{ s}$. Find the maximum percentage error in $g = 4\pi^2 L/T^2$ and state which input dominates it.

Answers: 1. (a) Sum = 20.0 g, $\Delta = 0.2 \text{ g}$, 1.0%. (b) Difference = 10.0 g, $\Delta = 0.2 \text{ g}$, 2.0% — halving the result doubles the percentage while the absolute error stays put. 2. $Z = (70 \pm 2)$ units; $\delta Z = \frac{2}{70} \times 100\% \approx 2.9\%$. 3. $\frac{1}{f} = \frac{2}{20.0}$ gives $f = 10.0 \text{ cm}$; $\Delta f = f^2 \left(\frac{0.1}{400} + \frac{0.1}{400} \right) = 100 \times 5 \times 10^{-4} \text{ cm} = 0.05 \text{ cm}$, so $f = (10.00 \pm 0.05) \text{ cm}$, 0.5%. 4. Exponents 1, 1, -2, -1: $\delta Y = 1 + 0.2 + 2(0.5) + 1.5 = 3.7\%$ — the radius, squared, quietly out-bids the extension. 5. $\delta g = 0.1\% + 2(1\%) = 2.1\%$, dominated entirely by the timing (2% of the 2.1%); time many oscillations per reading.

Addendum C — Dimensional Formulae from Physical Laws: The Constants the Chapter's Tables Skip

The chapter's dimensional table stops at viscosity, G and h . JEE Main happily asks for the dimensional formula of ϵ_0 , B , resistance or capacitance — quantities defined not by a schoolbook formula but by a *law of physics*. The rescue is a principle you already own: every physical law is itself dimensionally homogeneous, so isolating the target constant in its defining law and substituting dimensional formulae must succeed. Alongside this sits the dimensionless-argument rule from the pitfalls section, promoted here to a full method for dissecting constants buried inside sin, exp and ln.

Method

C1 — Isolate and substitute. For a constant C appearing in a law:

1. Rearrange the law algebraically until C stands alone.
2. Replace every other symbol with its dimensional formula (use the chapter's master table; recall $[q] = \text{AT}$ from charge = current $\times$ time).
3. Simplify the powers of M, L, T, A, K.
4. Name the SI unit if you know it — a built-in cross-check, since the named unit must reduce to the same base-unit string.

C2 — Dimensionless arguments. The quantities $\sin(\cdot)$, $\cos(\cdot)$, $e^{(\cdot)}$, $\ln(\cdot)$ take pure numbers only. Therefore, in any valid equation, every argument is dimensionless. Reading off “which products of symbols multiply to nothing” hands you the dimensions of the constants inside the argument, one equation per symbol.

Standard results (each derived below or left to practice):

Constant	Defining law	Dimensional formula	Named unit
ε_0	Coulomb's law	$\text{M}^{-1} \text{L}^{-3} \text{T}^4 \text{A}^2$	farad per metre
μ_0	force per length, parallel wires	$\text{M} \text{L} \text{T}^{-2} \text{A}^{-2}$	henry per metre
B	$F = qvB$	$\text{M} \text{T}^{-2} \text{A}^{-1}$	tesla
R (electrical)	$V = IR$	$\text{M} \text{L}^2 \text{T}^{-3} \text{A}^{-2}$	ohm
C (capacitance)	$Q = CV$	$\text{M}^{-1} \text{L}^{-2} \text{T}^4 \text{A}^2$	farad

Example C.1 — [CBSE / NEET Level]

Find the dimensional formula of ε_0 from Coulomb's law $F = \frac{q_1 q_2}{4\pi \varepsilon_0 r^2}$.

Isolate: $\varepsilon_0 = \frac{q_1 q_2}{4\pi F r^2}$, with 4π a pure number contributing nothing.

Charge: $[q] = \text{AT}$, so the numerator carries $(\text{AT})^2 = \text{A}^2 \text{T}^2$. Denominator: $[F][r^2] = (\text{MLT}^{-2})(\text{L}^2) = \text{ML}^3 \text{T}^{-2}$. Dividing,

$$[\varepsilon_0] = \frac{\text{A}^2 \text{T}^2}{\text{ML}^3 \text{T}^{-2}} = \text{M}^{-1} \text{L}^{-3} \text{T}^4 \text{A}^2.$$

Check. The SI unit of ε_0 is F m^{-1} ; the farad reduces to $\text{A}^2 \text{s}^4 \text{kg}^{-1} \text{m}^{-2}$, and dividing by metre gives $\text{A}^2 \text{s}^4 \text{kg}^{-1} \text{m}^{-3}$ — the same string as above with $\text{s} \leftrightarrow \text{T}$. Agreement.

Example C.2 — [JEE Main Level]

(a) Find $[B]$ from the Lorentz force $F = qvB$. (b) Find $[\mu_0]$ from the force per unit length between parallel wires, $F/\ell = \mu_0 I_1 I_2 / (2\pi d)$. Confirm the two are mutually consistent.

(a) Isolate $B = F/(qv)$. Numerator: MLT^{-2} . Denominator: $(\text{AT})(\text{LT}^{-1}) = \text{AL}$. Hence

$$[B] = \frac{\text{M L T}^{-2}}{\text{A L}} = \text{M T}^{-2} \text{A}^{-1}.$$

This is the tesla — and the cross-check against the wire law $F = BI\ell$ gives $[B] = \text{M L T}^{-2}/(\text{A L})$, identical.

(b) Rearrange: $\mu_0 = \frac{(F/\ell) 2\pi d}{I_1 I_2}$, with 2π inert. Force per length carries $\text{M L T}^{-2}/\text{L} = \text{M T}^{-2}$; multiplying by d (dimension L) restores M L T^{-2} ; dividing by $I^2 = \text{A}^2$:

$$[\mu_0] = \text{M L T}^{-2} \text{A}^{-2}.$$

Consistency. The electromagnetic wave-speed combination $1/\sqrt{\varepsilon_0\mu_0}$ must be a speed. Multiply the two results: $[\varepsilon_0\mu_0] = \text{T}^2 \text{L}^{-2}$, so $[1/\sqrt{\varepsilon_0\mu_0}] = \text{L T}^{-1}$. Two independently derived dimensional formulae click together into Maxwell's prediction that light is an electromagnetic wave — dimensional analysis alone gets you to the doorstep.

Example C.3 — [JEE Main Level]

The van der Waals equation reads $(P + a/V^2)(V - b) = RT$ for one mole. Find $[a]$ and $[b]$.

Homogeneity forbids adding unlike quantities, so inside the first bracket,

$$\left[\frac{a}{V^2}\right] = [P] \implies [a] = [P][V]^2 = (\text{M L}^{-1} \text{T}^{-2})(\text{L}^3)^2 = \text{M L}^5 \text{T}^{-2}.$$

Inside the second bracket, $[b] = [V] = \text{L}^3$.

Check. The right side, RT , carries $\text{M L}^2 \text{T}^{-2} \text{mol}^{-1}$ per the chapter's Example 1 — energy per mole. The left side must match: $[P][V] = (\text{M L}^{-1} \text{T}^{-2})(\text{L}^3) = \text{M L}^2 \text{T}^{-2}$, and the per-mole character rides along because the equation is stated for one mole. Consistent.

Example C.4 — [JEE Advanced Level]

A travelling wave is written $y = A \sin(kx - \omega t)$. Deduce $[k]$, $[\omega]$, $[A]$, and then extract a prediction from the ratio ω/k .

The sine's argument must be dimensionless. The argument splits into two pieces added together, so each piece is individually dimensionless:

$$[k][x] = 1 \implies [k] = \text{L}^{-1}, \quad [\omega][t] = 1 \implies [\omega] = \text{T}^{-1}.$$

Since y is a displacement and the sine is a pure number, $[A] = [y] = \text{L}$.

The prediction. Form the ratio:

$$\left[\frac{\omega}{k}\right] = \frac{\text{T}^{-1}}{\text{L}^{-1}} = \text{L T}^{-1}.$$

A ratio of the two constants *built into the wave's shape* has the dimensions of speed. Dimensional analysis thus predicts that the wave profile moves at ω/k — the phase velocity — before any wave mechanics is done.

Why it is Advanced-level. Step one is routine argument-hunting. The assessed step is the last one: noticing that dimensional analysis lets you manufacture a speed from two constants whose individual dimensions you have just certified, and recognising that a dimensionally forced speed in a wave equation must be the wave's speed. The chapter's pro-tip planted the seed; this is the harvest.

Practice — Addendum C

1. [CBSE] Given $[V] = \text{ML}^2 \text{T}^{-3} \text{A}^{-1}$, find $[R]$ from Ohm's law $V = IR$.
2. [NEET] Find the dimensional formula of capacitance from $Q = CV$.
3. [JEE Main] In $y = a \sin(bx + ct)$, find $[a]$, $[b]$, $[c]$, and the dimensions of bc .
4. [JEE Main] Kinetic theory gives $\frac{3}{2}k_B T = \frac{1}{2}mv^2$ for a gas molecule. Find $[k_B]$.
5. [JEE Advanced] A charging capacitor obeys $q = q_0 e^{-t/RC}$. Using your results for $[R]$ and $[C]$, prove that $[RC] = \text{T}$, confirming the exponential's argument is dimensionless.

Answers: 1. $[R] = \text{ML}^2 \text{T}^{-3} \text{A}^{-1} / \text{A} = \text{ML}^2 \text{T}^{-3} \text{A}^{-2}$ (ohm). 2. $[C] = [Q]/[V] = \text{AT}/(\text{ML}^2 \text{T}^{-3} \text{A}^{-1}) = \text{M}^{-1} \text{L}^{-2} \text{T}^4 \text{A}^2$ (farad). 3. $[a] = \text{L}$, $[b] = \text{L}^{-1}$, $[c] = \text{T}^{-1}$; $[bc] = \text{LT}^{-1}$ — again a speed, the same phase-velocity logic as Example C.4. 4. $[k_B] = [mv^2]/[T] = \text{ML}^2 \text{T}^{-2} \text{K}^{-1}$ (joule per kelvin). 5. $[RC] = (\text{ML}^2 \text{T}^{-3} \text{A}^{-2})(\text{M}^{-1} \text{L}^{-2} \text{T}^4 \text{A}^2) = \text{T}$: mass, length and current cancel pairwise, leaving exactly one power of time — the argument t/RC is a pure number, as the exponential demands.

Addendum D — When Three Equations Are Not Enough: Dimensionless Groups and Estimation

Sub-topic 02 confesses the limitation honestly: with four or more governing quantities and only three balancing equations, the power-law method stalls. What it does not say is what JEE Advanced actually expects you to *do* at that point. The answer has two halves. First, the stall leaves exactly one degree of freedom — and that freedom can always be repackaged as an unknown *function of a dimensionless group*, which is itself a publishable, examinable result. Second, when even that is beyond reach, physics falls back on order-of-magnitude estimation, the skill quietly tested whenever a question says “estimate.”

Method

D1 — The leftover-exponent manoeuvre. Suppose Q depends on four quantities q_1, q_2, q_3, q_4 . Write $Q = C q_1^a q_2^b q_3^c q_4^d$ and apply homogeneity: three equations, four unknowns. Everything is determined except one free exponent — say d . Solve a, b, c in terms of d , regroup the result as

$Q = (\text{a fully determined power-law product}) \times \Phi(\text{a dimensionless combination}),$

and verify by direct substitution that the combination inside Φ truly carries $M^0 L^0 T^0$. The method's honest verdict: it fixes everything *except* the dimensionless function Φ .

D2 — Estimation protocol. For a Fermi-style quantity:

1. Model the object with the simplest formula containing measurable or tabulated quantities.
2. Insert orders of magnitude, keeping one guard digit.
3. Audit the result against the chapter's range tables (lengths 10^{-15} to 10^{26} m; masses 10^{-30} to 10^{55} kg). A landing inside the expected decade is the estimate's certificate.

Example D.1 — [JEE Advanced Level]

Drag on a sphere moving slowly through a fluid is expected to depend on its radius r , speed v , the fluid's density ρ , and the fluid's viscosity η . Show that dimensional analysis forces F into the form $\rho r^2 v^2 \Phi(\rho v r / \eta)$, and verify the grouping is dimensionless.

Assume $F = C r^a v^b \rho^c \eta^d$ and insert the dimensional formulae $[F] = M L T^{-2}$, $[r] = L$, $[v] = L T^{-1}$, $[\rho] = M L^{-3}$, $[\eta] = M L^{-1} T^{-1}$:

$$M L T^{-2} = L^a (L T^{-1})^b (M L^{-3})^c (M L^{-1} T^{-1})^d = M^{c+d} L^{a+b-3c-d} T^{-b-d}.$$

Three equations, four unknowns — exactly the predicted stall. Mass:

$$c + d = 1 \implies c = 1 - d.$$

Time:

$$-b - d = -2 \implies b = 2 - d.$$

Length:

$$a + b - 3c - d = 1 \implies a = 1 - b + 3c + d = 1 - (2 - d) + 3(1 - d) + d = 2 - d.$$

Regroup, pulling the free exponent d into one lump:

$$F = C r^{2-d} v^{2-d} \rho^{1-d} \eta^d = C \rho r^2 v^2 \left(\frac{\eta}{\rho v r} \right)^d.$$

Since d is unknowable, absorb it into a function of the reciprocal grouping:

$$F = \rho r^2 v^2 \Phi \left(\frac{\rho v r}{\eta} \right).$$

Verification of the grouping:

$$\left[\frac{\rho v r}{\eta} \right] = \frac{(\text{M L}^{-3})(\text{L T}^{-1})(\text{L})}{\text{M L}^{-1} \text{T}^{-1}} = \frac{\text{M L}^{-1} \text{T}^{-1}}{\text{M L}^{-1} \text{T}^{-1}} = \text{M}^0 \text{L}^0 \text{T}^0. \checkmark$$

Result: $F = \rho r^2 v^2 \Phi(\rho v r / \eta)$ — every dimensional content pinned, the residue quarantined inside a dimensionless function (physicists call its argument the Reynolds number). The full fluid dynamics lives entirely inside Φ ; dimensional analysis correctly refuses to guess it.

Check. Set Φ to a constant and the surviving product carries $(\text{M L}^{-3})(\text{L}^2)(\text{L}^2 \text{T}^{-2}) = \text{M L T}^{-2}$ — a force, as required.

Example D.2 — [CBSE Board / JEE Main Level]

Estimate the mass of the Earth, using $g = 9.8 \text{ m s}^{-2}$, $R = 6.37 \times 10^6 \text{ m}$, and $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$.

Model: surface gravity obeys $g = GM/R^2$, so

$$M = \frac{gR^2}{G}.$$

Compute R^2 : $(6.37)^2 = 40.58$, so $R^2 = 4.058 \times 10^{13} \text{ m}^2$. Multiply:

$$gR^2 = 9.8 \times 4.058 \times 10^{13} \text{ m}^3 \text{ s}^{-2} = 39.77 \times 10^{13} = 3.98 \times 10^{14} \text{ m}^3 \text{ s}^{-2}.$$

Divide:

$$M = \frac{3.98 \times 10^{14}}{6.67 \times 10^{-11}} \text{ kg} = 5.96 \times 10^{24} \text{ kg}.$$

Audit. The chapter's mass-range table puts the Earth at $\sim 10^{25} \text{ kg}$; the estimate lands squarely in that decade. No laboratory ever weighed the Earth — a falling apple, a radius, and a constant did.

Example D.3 — [JEE Main Level]

Estimate the radius of a copper atom. Density of copper $\rho = 9.0 \times 10^3 \text{ kg m}^{-3}$; molar mass $M_{\text{mol}} = 63.5 \times 10^{-3} \text{ kg mol}^{-1}$; Avogadro constant $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$.

Model: one mole occupies M_{mol}/ρ cubic metres and contains N_A atoms packed edge to edge, so the volume per atom is

$$V_{\text{atom}} = \frac{M_{\text{mol}}}{\rho N_A} = \frac{63.5 \times 10^{-3}}{(9.0 \times 10^3)(6.02 \times 10^{23})} \text{ m}^3.$$

Denominator first: $9.0 \times 6.02 = 54.2$, so $\rho N_A = 5.42 \times 10^{27} \text{ kg m}^{-3} \text{ mol}^{-1}$. Then

$$V_{atom} = \frac{6.35 \times 10^{-2}}{5.42 \times 10^{27}} \text{ m}^3 = 1.17 \times 10^{-29} \text{ m}^3.$$

Treating the atom as a sphere, $V_{atom} = \frac{4}{3}\pi r^3$, so

$$r = \left(\frac{3V_{atom}}{4\pi} \right)^{1/3} = \left(\frac{3 \times 1.17 \times 10^{-29}}{12.57} \right)^{1/3} \text{ m} = (2.79 \times 10^{-30})^{1/3} \text{ m} \approx 1.4 \times 10^{-10} \text{ m}.$$

Audit. The result is $1.4 \text{ \AA}$, sitting exactly on the atomic rung (10^{-10} m) of the chapter's length ladder. One density and one molar mass buy you the size of matter's brick.

Practice — Addendum D

1. [JEE Advanced] Gas kinetic theory suggests pressure p depends on number density n (molecules per volume), molecular mass m , and typical molecular speed v . Derive the form of p by homogeneity.
2. [JEE Main] A torsional pendulum's period depends on moment of inertia I and torsional constant C as $T = 2\pi\sqrt{I/C}$. Find $[C]$ by homogeneity alone.
3. [CBSE] Estimate the number of seconds in a 70-year human lifetime (take $1 \text{ year} \approx 3.16 \times 10^7 \text{ s}$).
4. [JEE Main] A room measures $6 \text{ m} \times 5 \text{ m} \times 4 \text{ m}$. Estimate the number of air molecules in it at $P = 1.0 \times 10^5 \text{ Pa}$, $T = 300 \text{ K}$, using $N = PV/(k_B T)$ with $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$.
5. [JEE Advanced] Capillary rise h depends on surface tension S , liquid density ρ , tube radius r , and g . Show that $h = r\Phi(\rho g r^2/S)$, and identify the choice of the free exponent that yields the familiar form $h = CS/(\rho g r)$.
6. [CBSE] Estimate the number of heartbeats in that same 70-year lifetime, taking roughly 1 beat per second.

Answers: 1. Write $p = C n^a m^b v^c$ with $[n] = \text{L}^{-3}$: mass gives $b = 1$; time gives $c = 2$; length gives $-3a + 2 = -1$, so $a = 1$. Hence $p = C n m v^2$ — kinetic theory's headline result, dimensionally forced. 2. $[C] = [I]/[T]^2 = (\text{M L}^2)(\text{T}^{-2}) = \text{M L}^2 \text{T}^{-2}$ (joule per radian squared, radians being dimensionless). 3. $70 \times 3.16 \times 10^7 \text{ s} \approx 2.2 \times 10^9 \text{ s}$. 4. $PV = 10^5 \times 120 = 1.2 \times 10^7 \text{ J}$; $k_B T = 1.38 \times 10^{-23} \times 300 = 4.14 \times 10^{-21} \text{ J}$; $N = 1.2 \times 10^7 / 4.14 \times 10^{-21} \approx 2.9 \times 10^{27}$ molecules. 5. With $h = C S^a \rho^b r^c g^d$: mass gives $b = -a$; time gives $a = -d$; length gives $c = 1 + 2d$, so $h = C r (\rho g r^2/S)^d = r\Phi(\rho g r^2/S)$, the grouping being dimensionless by substitution. Choosing the free exponent $d = -1$ gives $h = CS/(\rho g r)$ — check: $(\text{M T}^{-2})/[(\text{M L}^{-3})(\text{L T}^{-2})(\text{L})] = \text{L}$. 6. $2.2 \times 10^9 \text{ s} \times 1 \text{ s}^{-1} \approx 2 \times 10^9$ beats — a heart is a counter with a billion-count warranty.

Motion in a Straight Line

Chapter 2 — Complete Study Material

Physics · JEE / NEET / CBSE

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Sub-topic 01: Kinematic Quantities

Exam Relevance: This is the bedrock of all mechanics. CBSE Boards ask 1–2 marks on the distance-vs-displacement distinction or a clean average-speed calculation. JEE Main reliably carries 1 question — typically an “average speed” trap or a calculus-based position/velocity numerical. NEET loves the equal-distance vs equal-time average-speed trap as a fast MCQ. JEE Advanced rarely tests these definitions alone but builds every harder kinematics and dynamics problem on top of them, so a shaky foundation here costs marks everywhere downstream.

Section 1: Concept Intuition

Picture a local train pulling out of Dadar station, looping along the Mumbai coast, and stopping at Bandra. If you asked the motorman, “How far did we travel?” he’d point to the odometer — every metre of track the train rolled over. But if you stood at Dadar with a measuring tape and stretched it straight to Bandra, you’d get a much shorter, straight-line gap. These are two genuinely different questions, and physics gives them two different names.

The odometer reading — the **total length of the actual path** covered — is called **distance**. It only ever grows; it cannot shrink, and it doesn’t care which way you went. The straight tape from start to finish — the **shortest gap from the initial position to the final position, together with the direction** — is called **displacement**. Distance is a plain number (a *scalar*); displacement carries a direction (a *vector*). This single split is the seed from which the whole topic grows.

From this pair, two more pairs sprout. **Speed** is how fast distance piles up: rate of change of *distance*. **Velocity** is how fast displacement changes, *with direction*: rate of change of

displacement. So speed is the scalar twin and velocity is the vector twin — same units (m/s), same dimensions, but velocity also tells you *which way*. A car’s speedometer shows speed (always positive); it never shows velocity, because it doesn’t know your direction.

Think about it this way... Imagine a cricketer taking a quick single. He sprints from one crease to the other (that’s his displacement — a fixed 17.68 m down the pitch) but if he has to scramble back because the call was cancelled, his *distance* keeps growing while his displacement shrinks back toward zero. Run there and back, and his displacement is **zero** even though he’s gasping for breath from all the distance covered. This is the famous result: **a round trip has zero displacement but plenty of distance**. That gap is the heart of kinematics.

Now zoom in on time. Over a whole journey we talk about **average** values — total distance over total time (average speed), or net displacement over total time (average velocity). But at a single instant — the exact moment the speed gun clocks a bowler’s delivery — we need the **instantaneous** value. Instantaneous velocity is what you get when you shrink the time interval down to nearly nothing: mathematically, the derivative of position with respect to time. And the magnitude of instantaneous velocity is the instantaneous speed — at a single instant, there’s no path left to bend, so the scalar and vector twins finally agree exactly.

Finally, **acceleration** is the rate of change of *velocity*. Crucially, velocity can change in **magnitude** (speeding up / slowing down) *or* in **direction**, so acceleration is not just “speeding up.” When acceleration points opposite to velocity, the body slows — this is **retardation** (deceleration).

Limiting conditions & assumptions.

- All of this sub-topic, as taught here, lives in **one dimension** — motion along a straight line — so a direction reduces to a simple **+ or – sign** once you fix a positive direction. Choosing that positive direction *first* is non-negotiable.
- **Distance** $\geq$ **|displacement|** always; they are equal *only* when the motion is along a straight line **without reversing direction**.
- Consequently **average speed** $\geq$ **|average velocity|** always, with equality under the same no-reversal condition.
- “Zero velocity” does **not** mean “zero acceleration” — at the top of a thrown ball’s flight, velocity is momentarily zero but acceleration is still *g* downward.

Section 2: Key Formulas & Definitions

Quantity	Definition	Symbol & SI unit	Dimensional formula
Distance	total path length (scalar, ≥ 0)	s , m	$[L]$
Displacement	change in position (vector)	$\Delta x = x_f - x_i$, m	$[L]$
Average speed	$\frac{\text{total distance}}{\text{total time}}$	m/s	$[L T^{-1}]$
Average velocity	$\frac{\text{displacement}}{\text{time interval}}$	m/s	$[L T^{-1}]$
Instantaneous velocity	$\frac{dx}{dt}$	m/s	$[L T^{-1}]$

Quantity	Definition	Symbol & SI unit	Dimensional formula
Instantaneous speed	$\left \frac{dx}{dt} \right $	m/s	$[L T^{-1}]$
Acceleration	$\frac{dv}{dt} = \frac{d^2x}{dt^2}$	m/s ²	$[L T^{-2}]$

Core relations:

$$v_{\text{avg}} = \frac{x_2 - x_1}{t_2 - t_1}, \quad v_{\text{inst}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

$$a_{\text{avg}} = \frac{v_2 - v_1}{t_2 - t_1}, \quad a_{\text{inst}} = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

Two special-case averaging results (memorise the distinction — it is the single most-tested trap):

$$\text{Equal **distances** at } v_1, v_2 : \quad v_{\text{avg}} = \frac{2v_1v_2}{v_1 + v_2} \text{ (harmonic mean)}$$

$$\text{Equal **times** at } v_1, v_2 : \quad v_{\text{avg}} = \frac{v_1 + v_2}{2} \text{ (arithmetic mean)}$$

Key inequalities: distance $\geq$ |displacement| and average speed $\geq$ |average velocity|.

Section 3: Key Procedures & Logical Framework

There is no single grand “derivation” for these definitions, but one piece of genuine reasoning matters — *why* instantaneous velocity is a derivative — plus two procedures you must perform without hesitation.

The logic of instantaneous velocity (from average to instantaneous). Start with

average velocity over an interval Δt : $v_{\text{avg}} = \frac{\Delta x}{\Delta t}$.

1. On a position-time (x - t) graph, this ratio is the **slope of the secant line** (chord) joining the two points. Reasoning: rise (Δx) over run (Δt) is slope.
2. Now squeeze the interval: let t_2 slide toward t_1 , so $\Delta t \rightarrow 0$. The chord pivots and rotates until it just kisses the curve — it becomes the **tangent** at that point.
3. The slope of that tangent is the instantaneous velocity:

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}.$$

This is *why* we use calculus: velocity at an instant is the limiting slope as the measuring window collapses to a point.

Figure 2.1 An x - t curve rising and levelling off. Mark two points A (t_1, x_1) and B (t_2, x_2); draw the secant chord AB labelled “slope = average velocity”. At point A draw the tangent line labelled “slope = instantaneous velocity”. Show Δt and Δx 3 (see next page)

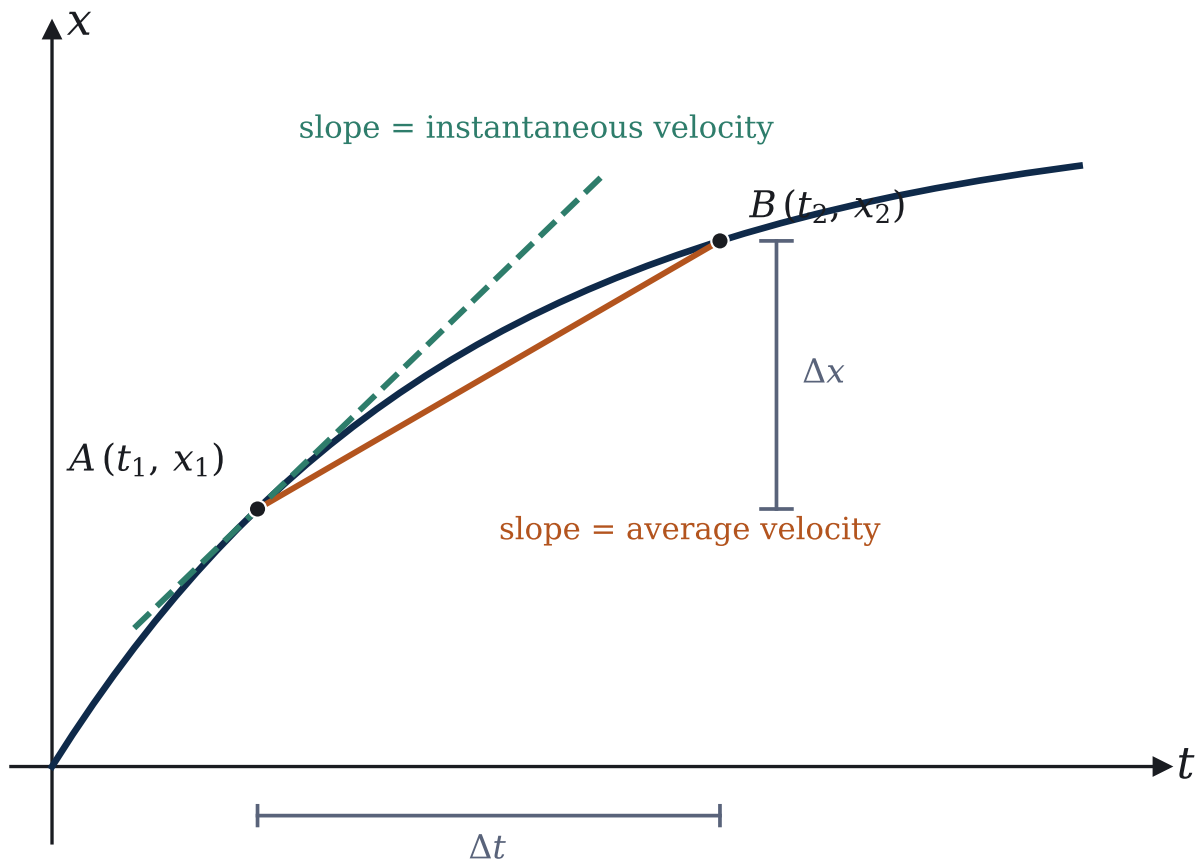


Figure 2.1 An x - t curve rising and levelling off. Mark two points $A(t_1, x_1)$ and $B(t_2, x_2)$; draw the secant chord AB labelled “slope = average velocity”. At point A draw the tangent line labelled “slope = instantaneous velocity”. Show Δt and Δx

as the run and rise of the chord, with an arrow indicating B sliding toward A as $\Delta t \rightarrow 0$.

Procedure A — Distance vs displacement for a 1-D trip with reversals. 1. Fix a positive direction (say, rightward = +). 2. Split the journey into legs where the direction of motion does not change. 3. **Displacement** = algebraic sum of the legs (respect + / - signs); equivalently $x_{\text{final}} - x_{\text{initial}}$. 4. **Distance** = sum of the *magnitudes* of all legs (drop the signs). 5. Sanity check: distance must be $\geq |\text{displacement}|$; if not, you slipped a sign.

Procedure B — Average speed when the data is “by distance” vs “by time”. 1. Identify whether the equal pieces are equal **distances** or equal **times** — this decides everything. 2. Always return to first principles: $v_{\text{avg}} = \frac{\text{total distance}}{\text{total time}}$. 3. For equal distances, total time is $\sum d/v_i$ (small speeds dominate the time, dragging the average *down* $\rightarrow$ harmonic mean). For equal times, total distance is $\sum v_i t$ ($\rightarrow$ arithmetic mean). 4. Quick reasonableness check: the average must lie **between** the smallest and largest speed, and for equal distances it sits **closer to the slower** speed.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A jogger runs 300 m due east in 100 s, then turns and runs 100 m due west in 50 s. Find (a) total distance, (b) displacement, (c) average speed, (d) average velocity.

Given: Leg 1: 300 m east, 100 s. Leg 2: 100 m west, 50 s. Take east as positive.

Solution: (a) Total distance = $300 + 100 = 400$ m (magnitudes added). (b) Displacement = $(+300) + (-100) = +200$ m, i.e. 200 m **east**. (c) Total time = $100 + 50 = 150$ s.

$$\text{Average speed} = \frac{400 \text{ m}}{150 \text{ s}} \approx 2.67 \text{ m/s.}$$

(d)

$$\text{Average velocity} = \frac{200 \text{ m}}{150 \text{ s}} \approx 1.33 \text{ m/s, east.}$$

Notice average speed $> |\text{average velocity}|$ — exactly as the inequality promises, because the jogger reversed direction.

Example 2 — [NEET Speed Trap] A bus covers the first half of a route at 40 km/h and the second half (equal distance) at 60 km/h. Its average speed for the whole route is:

(A) 50 km/h (B) 48 km/h (C) 52 km/h (D) 24 km/h

The trap: the eye jumps to $\frac{40+60}{2} = 50$ — option (A) — but that arithmetic mean is **wrong for equal distances**.

Fast solution: Equal distances $\Rightarrow$ harmonic mean:

$$v_{\text{avg}} = \frac{2v_1v_2}{v_1 + v_2} = \frac{2(40)(60)}{40 + 60} = \frac{4800}{100} = 48 \text{ km/h.}$$

(B) 48 km/h

Speed cue: the harmonic mean is **always less than** the arithmetic mean, so the instant you see “equal distances, two speeds,” eliminate the simple average and pick the slightly smaller value. You barely need to compute.

Example 3 — [JEE Main Level] A particle moving along the x -axis has position $x(t) = t^3 - 6t^2 + 9t + 5$ (metres, t in seconds). (a) Find the instants when it is momentarily at rest. (b) Find its velocity at the instant its acceleration is zero. (c) Find the distance and displacement between $t = 0$ and $t = 3$ s.

Why it’s multi-concept: it fuses differentiation with the distance-vs-displacement distinction.

Solution:

$$v = \frac{dx}{dt} = 3t^2 - 12t + 9 = 3(t-1)(t-3), \quad a = \frac{dv}{dt} = 6t - 12.$$

(a) At rest $\Rightarrow v = 0 \Rightarrow t = 1$ s and $t = 3$ s.

(b) $a = 0 \Rightarrow 6t - 12 = 0 \Rightarrow t = 2$ s. Then

$$v(2) = 3(2-1)(2-3) = 3(1)(-1) = -3 \text{ m/s}.$$

The negative sign means it is moving in the $-x$ direction at that instant.

(c) Positions: $x(0) = 5$, $x(1) = 9$, $x(3) = 5$.

- From $t = 0$ to $t = 1$: velocity positive, moves $+4$ m.
- From $t = 1$ to $t = 3$: velocity negative, moves back -4 m.

$$\text{Displacement} = x(3) - x(0) = 5 - 5 = \boxed{0}, \quad \text{Distance} = |+4| + |-4| = \boxed{8 \text{ m}}.$$

The particle ends where it started (zero displacement) yet covered 8 m of path — the reversal at $t = 1$ s is exactly what makes the two disagree.

Example 4 — [JEE Advanced Level] A particle moves along a straight line with velocity $v(t) = t^2 - 6t + 8$ (m/s) over the interval $0 \leq t \leq 5$ s. Find (a) the instants it reverses direction, (b) the total displacement, (c) the total distance, and (d) the average speed and average velocity, verifying $\bar{v}_{\text{speed}} \geq |\bar{v}_{\text{vel}}|$.

Reasoning: Direction reverses wherever v changes sign. Distance needs $\int |v| dt$ piecewise, while displacement is the plain $\int v dt$.

Solution: Factor: $v = (t-2)(t-4)$, so $v = 0$ at $t = 2$ s and $t = 4$ s. Sign of v : positive on $[0, 2]$, negative on $[2, 4]$, positive on $[4, 5]$ → **two reversals**, at $t = 2$ and $t = 4$ s.

Let the antiderivative be $F(t) = \frac{t^3}{3} - 3t^2 + 8t$. Then:

$$F(0) = 0, \quad F(2) = \frac{20}{3}, \quad F(4) = \frac{16}{3}, \quad F(5) = \frac{20}{3}.$$

(b) **Displacement** $= F(5) - F(0) = \frac{20}{3} \approx 6.67$ m.

(c) **Distance** $= |F(2) - F(0)| + |F(4) - F(2)| + |F(5) - F(4)|$
 $= \frac{20}{3} + \left| \frac{16}{3} - \frac{20}{3} \right| + \left| \frac{20}{3} - \frac{16}{3} \right| = \frac{20}{3} + \frac{4}{3} + \frac{4}{3} = \frac{28}{3} \approx 9.33$ m.

(d) Over total time 5 s:

$$\bar{v}_{\text{vel}} = \frac{20/3}{5} = \frac{4}{3} \approx 1.33 \text{ m/s}, \quad \bar{v}_{\text{speed}} = \frac{28/3}{5} = \frac{28}{15} \approx 1.87 \text{ m/s}.$$

Indeed $1.87 \geq 1.33$ — the inequality holds, and the gap is precisely the “wasted” backtracking between $t = 2$ and $t = 4$ s.

Section 5: Practice Exercises

1. **[CBSE]** A hiker walks 4 km north, then 3 km east, taking 30 min in all. Find the distance, the magnitude of displacement, the average speed, and the magnitude of average velocity.
2. **[NEET]** A drone covers two equal-distance stretches at 30 km/h and 20 km/h. What is its average speed?
3. **[JEE Main]** A particle moves with $x(t) = 5 + 3t + 2t^2$ (m). Find its initial velocity, its acceleration, and its velocity at $t = 2$ s.
4. **[JEE Main]** Position is $x(t) = 4t - t^2$ (m). Find the velocity at $t = 1$ s and the instant at which the particle is momentarily at rest.
5. **[JEE Advanced]** A particle has velocity $v(t) = 3t^2 - 12t + 9$ (m/s) for $0 \leq t \leq 4$ s. Find the total distance travelled.

Answers: 1. distance 7 km, displacement 5 km, avg speed 14 km/h, avg velocity 10 km/h | 2. 24 km/h | 3. $u = 3$ m/s, $a = 4$ m/s², $v(2) = 11$ m/s | 4. $v(1) = 2$ m/s, at rest at $t = 2$ s | 5. 12 m (displacement is only 4 m).

Section 6: Crack the MCQ

Q1. For *any* moving body, which relation is always correct? (A) displacement $\geq$ distance (B) distance $\geq$ |displacement| (C) distance = |displacement| (D) distance $<$ |displacement|

Answer: (B). The path can only be longer than (or equal to) the straight-line gap, never shorter. (A) reverses the true inequality. (C) holds *only* for straight-line motion with no reversal — true sometimes, not always, so it fails “always.” (D) is impossible; if a student picks it they’ve confused which quantity is the shortcut route.

Q2. A runner completes exactly one full lap of a circular track of radius R . Over this lap, the distance and the magnitude of displacement are respectively: (A) $2\pi R$, $2\pi R$ (B) $2\pi R$, 0 (C) 0, $2\pi R$

Answer: (B). Distance is the full circumference $2\pi R$; but the runner ends exactly where they began, so displacement is 0. (A) wrongly equates the two — the classic round-trip error. (C) swaps them. (D) is the half-lap answer (distance πR , displacement = diameter $2R$); picked by students who misread “full” as “half.”

Q3. The magnitude of a particle’s average velocity equals its average speed **only** when the particle: (A) moves on a circular path (B) moves along a straight line without reversing direction (C) has constant acceleration (D) returns to its starting point

Answer: (B). Equality of distance and |displacement| requires an unbent, one-way path — that’s exactly straight-line motion with no reversal. (A) maximises the gap between them, the opposite case. (C) constant acceleration still allows a turnaround (e.g. a ball thrown up), so they can differ. (D) gives **zero** average velocity but non-zero average speed — they’re maximally unequal, the trap for hasty readers.

Q4. A particle’s position is $x(t) = 2t^2$ (m). Its instantaneous velocity at $t = 3$ s is: (A) 6 m/s (B) 12 m/s (C) 18 m/s (D) 4 m/s

Answer: (B). $v = \frac{dx}{dt} = 4t$, so $v(3) = 12$ m/s. (A) comes from differentiating as if $x = 2t$ (forgetting the power rule’s factor of 2). (C) is the *position* $x(3) = 2(9) = 18$ m — answering the wrong question. (D) reports just the coefficient 4 without substituting $t = 3$.

Section 7: Common Student Pitfalls & Pro-Tips

- **Treating distance and displacement as interchangeable.** They coincide only for straight, one-way motion. The moment direction reverses (or the path curves), distance pulls ahead. Always ask: did the body turn around?
- **Arithmetic-averaging speeds over equal distances.** Two equal stretches at 40 and 60 km/h is **not** 50 — it’s the harmonic mean 48. Arithmetic mean is correct only for equal *times*.
- **Assuming** $v = 0 \Rightarrow a = 0$. At a turning point the body is instantaneously at rest yet still accelerating (think of the topmost point of a vertical throw). Velocity and acceleration are independent at an instant.
- **Forgetting to fix a positive direction in 1-D.** Without a chosen sign convention, “−3 m/s” is meaningless and sign errors cascade through the whole solution.

Pro-Tip: Lock in one phrase — “*same distance → harmonic, same time → arithmetic.*” And for average velocity, never integrate the wiggly path: it is simply $\frac{x_{\text{final}} - x_{\text{initial}}}{\text{total time}}$, endpoints only. When a velocity function changes sign, split the time axis at the roots **before** integrating for distance, or you’ll silently cancel the back-tracking and report distance as displacement.

Section 8: Cheat Sheet Box

- **Distance** = total path length (scalar, never decreases); **Displacement** = $x_f - x_i$ (vector, sign matters).
- **Speed** = |velocity|; same units (m/s, $[L T^{-1}]$) but speed is scalar, velocity is vector.
- $v_{\text{avg (speed)}} = \frac{\text{total distance}}{\text{total time}}$, $v_{\text{avg (velocity)}} = \frac{\text{displacement}}{\text{time}}$.
- $v_{\text{inst}} = \frac{dx}{dt}$ = slope of the x - t tangent; $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$ (m/s², $[L T^{-2}]$).
- **Always:** distance $\geq$ |displacement| and average speed $\geq$ |average velocity|; equal only for unidirectional straight-line motion.
- **Equal distances → harmonic mean** $\frac{2v_1v_2}{v_1 + v_2}$; **equal times → arithmetic mean**

$$\frac{v_1 + v_2}{2}.$$

- *Memory aid:* "Distance is how far you walked; displacement is how far you ended up."

Sub-topic 02: Graphical Analysis

Exam Relevance: Motion graphs are a perennial favourite. CBSE Boards often ask a 2–3 mark “find displacement from this v - t graph” or “what does the slope represent.” JEE Main almost guarantees 1 question, usually reading acceleration (slope) and displacement (area) off a piecewise graph. NEET leans on conceptual graph-interpretation MCQs (slope = velocity, area = displacement) where one wrong reading sinks the answer. JEE Advanced uses graphs as a reasoning tool inside multi-part problems and loves the a - $t \rightarrow v$ - $t \rightarrow x$ - t conversion chain.

Section 1: Concept Intuition

A motion graph is a story told without words. Instead of describing where an object is and how fast it moves in sentences, we draw a picture and let its **shape** do the talking. The trick — and the entire skill of this sub-topic — is learning to read two features of any motion graph: its **slope** (how steeply it rises or falls) and the **area** trapped between the curve and the time axis. Master those two readings and almost every graph question collapses into something easy.

Start with the **position-time (x - t) graph**, where the vertical axis is position and the horizontal axis is time. Its slope answers “how fast is the position changing?” — which is exactly *velocity*. A steep line means high velocity; a gentle line means slow; a flat horizontal line means the position isn’t changing at all, i.e. the body is **at rest**. If the graph is a straight slanted line, the slope is constant, so the velocity is constant (uniform motion). If it *curves*, the slope is changing, so the velocity is changing — that’s **acceleration**.

Think about it this way... Imagine the live train-tracker app for a Rajdhani Express, plotting the train’s position against time. While the train sits at a platform, the line is flat. When it races between stations, the line shoots up steeply. When it slows for a junction, the line bends and flattens. You never see the actual tracks — yet from the *shape of the line alone* you can tell exactly when the train was stopped, sprinting, or braking. That is graphical analysis: the trajectory may be a dead-straight railway line, but the graph’s curve encodes the *changing speed*, not a curving path. A common beginner error is to think a curved x - t graph means the train turned a corner — it doesn’t; the motion is still in a straight line.

Next, the **velocity-time (v - t) graph**. Here the slope answers “how fast is velocity changing?” — which is *acceleration*. But the v - t graph has a second, even more useful reading: the **area** between the curve and the time axis equals the **displacement**. Why? Over a tiny sliver of time Δt , the strip of area is (velocity $\times \Delta t$) = a little bit of displacement; add up all the slivers and you get the total. Crucially, area *below* the time axis (where velocity is negative) counts as **negative** displacement. So the *net signed area* gives displacement, while the *total area ignoring sign* gives distance.

Finally, the **acceleration-time (a - t) graph**: the area under it gives the **change in velocity** Δv , by the same slicing logic.

So the chain runs: **slope takes you one step “down”** ($x \rightarrow v \rightarrow a$), and **area takes you one step “up”** ($a \rightarrow v \rightarrow x$).

Limiting conditions & assumptions.

- An x - t graph can **never be vertical** — that would mean the body is in two places at

once (infinite velocity). For the same reason, the graph can't fold back over a single time value.

- “Slope” means the slope of the **tangent** for an instantaneous reading; the slope of a **chord** gives the *average* over that interval.
- Reading area as displacement assumes you respect signs. Forgetting this is the No. 1 graph mistake.
- These tools are for one-dimensional (rectilinear) motion; sign (+/–) stands in for direction.

Section 2: Key Formulas & Definitions

The “formulas” here are really *reading rules*. Commit this table to memory:

Graph	What the slope gives	What the area gives
Position-time ($x-t$)	velocity $v = \frac{dx}{dt}$	— (not standard)
Velocity-time ($v-t$)	acceleration $a = \frac{dv}{dt}$	displacement (signed) / distance (unsigned)
Acceleration-time ($a-t$)	jerk (rarely needed)	change in velocity Δv

Slope (instantaneous vs average):

$$v_{\text{inst}} = \left(\frac{dx}{dt} \right)_{\text{tangent}}, \quad v_{\text{avg}} = \left(\frac{\Delta x}{\Delta t} \right)_{\text{chord}}$$

Area as integral:

$$\text{Displacement} = \int_{t_1}^{t_2} v \, dt = (\text{net signed area under } v-t), \quad \Delta v = \int_{t_1}^{t_2} a \, dt$$

Shape dictionary for the $x-t$ graph:

- Horizontal line $\rightarrow$ at rest ($v = 0$).
- Straight slanted line $\rightarrow$ uniform velocity ($a = 0$).
- Concave-up curve $\rightarrow$ velocity increasing $\rightarrow$ **positive** acceleration.
- Concave-down curve $\rightarrow$ velocity decreasing $\rightarrow$ **negative** acceleration.

Shape dictionary for the $v-t$ graph:

- Horizontal line $\rightarrow$ uniform velocity ($a = 0$).
- Straight slanted line $\rightarrow$ uniform acceleration.
- Line crossing the time axis $\rightarrow$ particle **reverses direction** at the crossing.

Figure 2.2 A 2×3 panel reference chart. Top row, three $x-t$ graphs: (1) horizontal line labelled “at rest”, (2) straight slanted line labelled “uniform velocity”, (3) upward-curving (concave up) line labelled “positive acceleration”. Bottom row, three $v-t$ graphs: (1) horizontal line labelled “constant velocity, $a = 0$ ”, (2) straight slanted line from origin labelled “uniform acceleration”, (3) a line starting positive and crossing the t -axis into negative, with the region above the axis 10 (see next page)

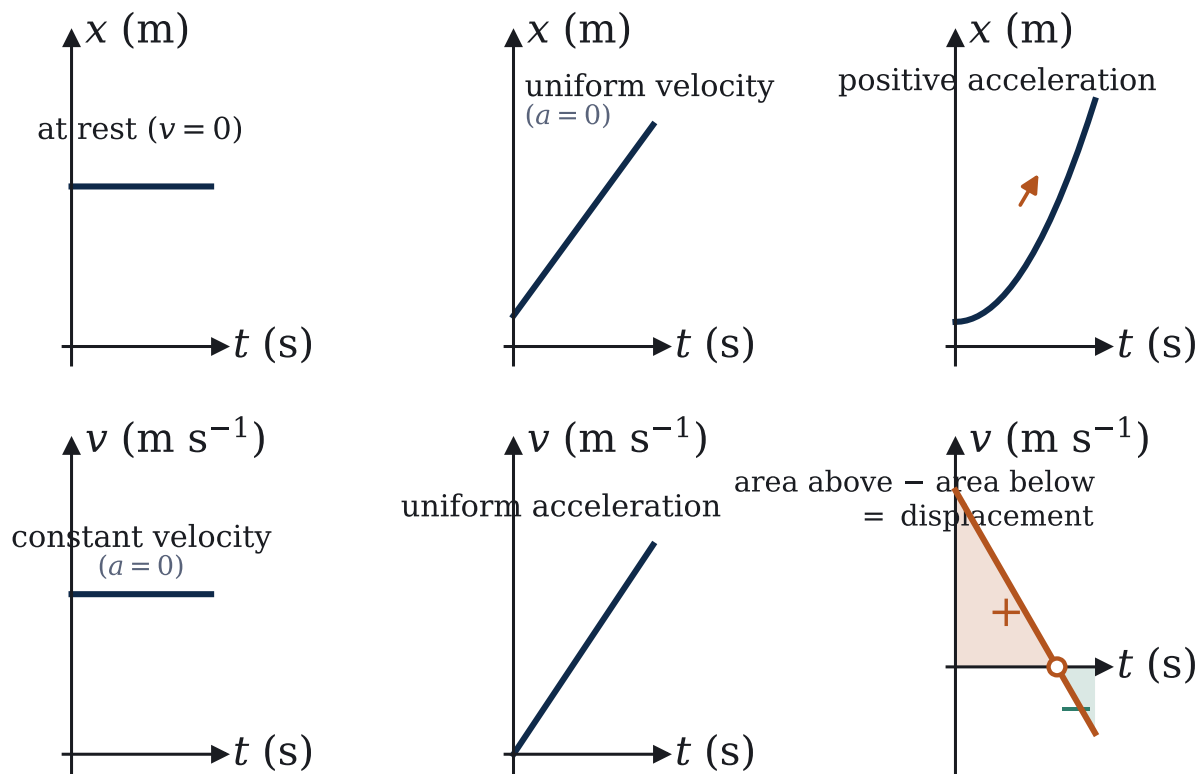


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shaded '+' and below shaded '-', labelled "area above – area below = displacement".

Section 3: Key Procedures & Logical Framework

No formal derivation here — the skill is *procedural reading*. Two procedures cover almost everything.

Procedure A — Extracting motion data from a single graph. 1. **Identify the axes first.** Is the vertical axis x , v , or a ? Every rule depends on this. Most wrong answers come from applying v - t rules to an x - t graph. 2. **Need a rate (velocity or acceleration)? Take the slope.** For an instant, draw the tangent and compute rise/run; for an average, use the straight chord between the two endpoints. 3. **Need displacement or Δv ? Take the area.** Break the shape into triangles, rectangles, and trapezia. Compute each area. 4. **Apply signs.** Area *below* the time axis is negative. For **displacement**, add areas with their signs; for **distance**, add the magnitudes. 5. **Sanity check.** Distance $\geq$ |displacement|; a reasonable acceleration; velocity continuous (no instantaneous jumps unless idealised).

Procedure B — Converting between graph types (the reasoning chain). The graphs are linked like a ladder: $x \xrightarrow{\text{slope}} v \xrightarrow{\text{slope}} a$, and back up by area $a \xrightarrow{\text{area}} v \xrightarrow{\text{area}} x$.

1. **Going down (differentiate / take slope):** to get v - t from x - t , plot the slope of x - t at each instant. A straight x - t line (constant slope) becomes a horizontal v - t line. A parabola becomes a sloped line.
2. **Going up (integrate / accumulate area):** to get v - t from a - t , start from the known initial velocity and add the running area under a - t . To then get position, add the running area under v - t .
3. **The reasoning behind it:** slope is the derivative (instantaneous rate of change), and area is the integral (accumulated total). Differentiation and integration are inverse operations, which is precisely why the ladder works both ways.
4. **Anchor the constants.** Going *up*, you always need the starting value (v_0 or x_0); area gives the *change*, not the absolute value.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A scooter starts from rest and speeds up uniformly to 20 m/s in 4 s, cruises at 20 m/s for the next 6 s, then brakes uniformly to rest in 5 s. Using its v - t graph, find (a) the acceleration in each phase and (b) the total displacement.

Figure 2.3 A v - t graph shaped like a trapezium. It rises as a straight line from (0, 0) to (4, 20), runs flat from (4, 20) to (10, 20), then falls as a straight line from (10, 20) to (15, 0). The three regions are shaded and labelled "triangle", "rectangle", "triangle". (see next page)

Solution: (a) Acceleration = slope of each segment.

- Phase 1: $a_1 = \frac{20 - 0}{4} = 5 \text{ m/s}^2$.
- Phase 2: $a_2 = 0$ (flat line).
- Phase 3: $a_3 = \frac{0 - 20}{5} = -4 \text{ m/s}^2$ (deceleration).

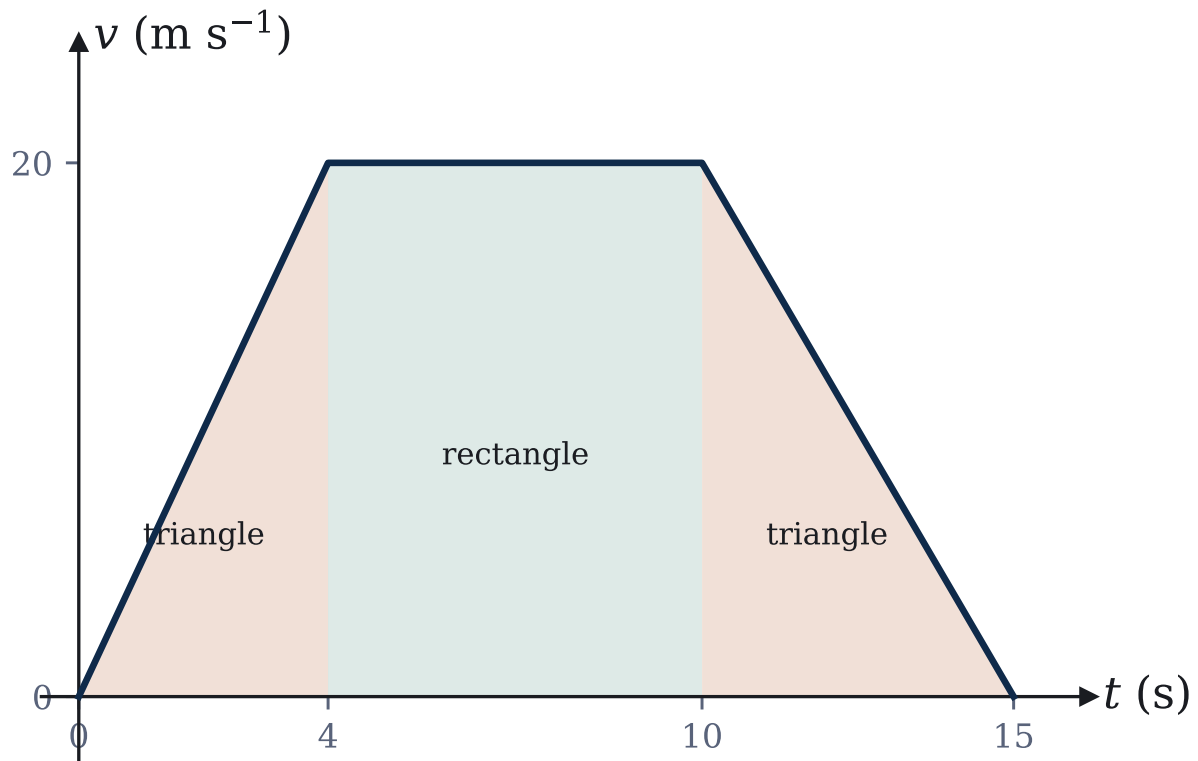


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(b) Displacement = area under the graph:

- Triangle: $\frac{1}{2}(4)(20) = 40$ m.
- Rectangle: $(6)(20) = 120$ m.
- Triangle: $\frac{1}{2}(5)(20) = 50$ m.

$$\boxed{\text{Total displacement} = 40 + 120 + 50 = 210 \text{ m}}$$

All area is above the axis, so distance equals displacement here.

Example 2 — [NEET Speed Trap] *The position-time graph of a body is a straight line inclined at a fixed angle to the time axis. The body is:*

(A) speeding up (B) moving with uniform velocity (C) uniformly accelerating (D) at rest

The trap: “the line goes up, so the body must be speeding up.” That instinct treats every rising graph alike — but on an $x-t$ graph, *rising* only means *moving forward*, not *speeding up*.

Fast solution: On an $x-t$ graph, slope = velocity. A straight line has **constant slope**, so velocity is constant $\rightarrow$ uniform velocity, zero acceleration.

$$\boxed{(B) \text{ moving with uniform velocity}}$$

Speed cue: memorise the one-line rule — “straight $x-t$ line = steady speed; only a **curved** $x-t$ line means acceleration.” (A) and (C) describe a curved graph; (D) would be a flat horizontal line. Eliminate in two seconds.

Example 3 — [JEE Main Level] *A particle’s $v-t$ graph is piecewise linear: it rises from 0 to 12 m/s during $0 \leq t \leq 3$ s, stays at 12 m/s until $t = 5$ s, then falls steadily to -4 m/s at $t = 9$ s. Find (a) the instant the particle reverses direction, (b) the total displacement, and (c) the total distance, over 0 to 9 s.*

Figure 2.4 *A $v-t$ graph: straight rise from (0,0) to (3,12); flat from (3,12) to (5,12); straight fall from (5,12) through the t -axis at $t=8$ to (9,-4). Region above the t -axis shaded ‘+’, the small triangle below (from $t=8$ to 9) shaded ‘-’. (see next page)*

Why it’s multi-concept: it combines slope (acceleration), signed-area (displacement), and the distance-vs-displacement idea from Sub-topic 1.

Solution: The falling segment goes from 12 m/s to -4 m/s over 4 s, slope = -4 m/s^2 . It crosses zero when $12 - 4(t - 5) = 0 \Rightarrow t = 8$ s.

(a) The velocity changes sign at $t = 8$ s $\rightarrow$ **reverses direction there.**

Now the areas:

- 0–3 s (triangle): $\frac{1}{2}(3)(12) = 18$ m
- 3–5 s (rectangle): $(2)(12) = 24$ m
- 5–8 s (triangle, above axis): $\frac{1}{2}(3)(12) = 18$ m
- 8–9 s (triangle, below axis): $\frac{1}{2}(1)(4) = 2$ m $\rightarrow$ contributes -2 m

(b) Displacement = $18 + 24 + 18 - 2 = \boxed{58 \text{ m}}$

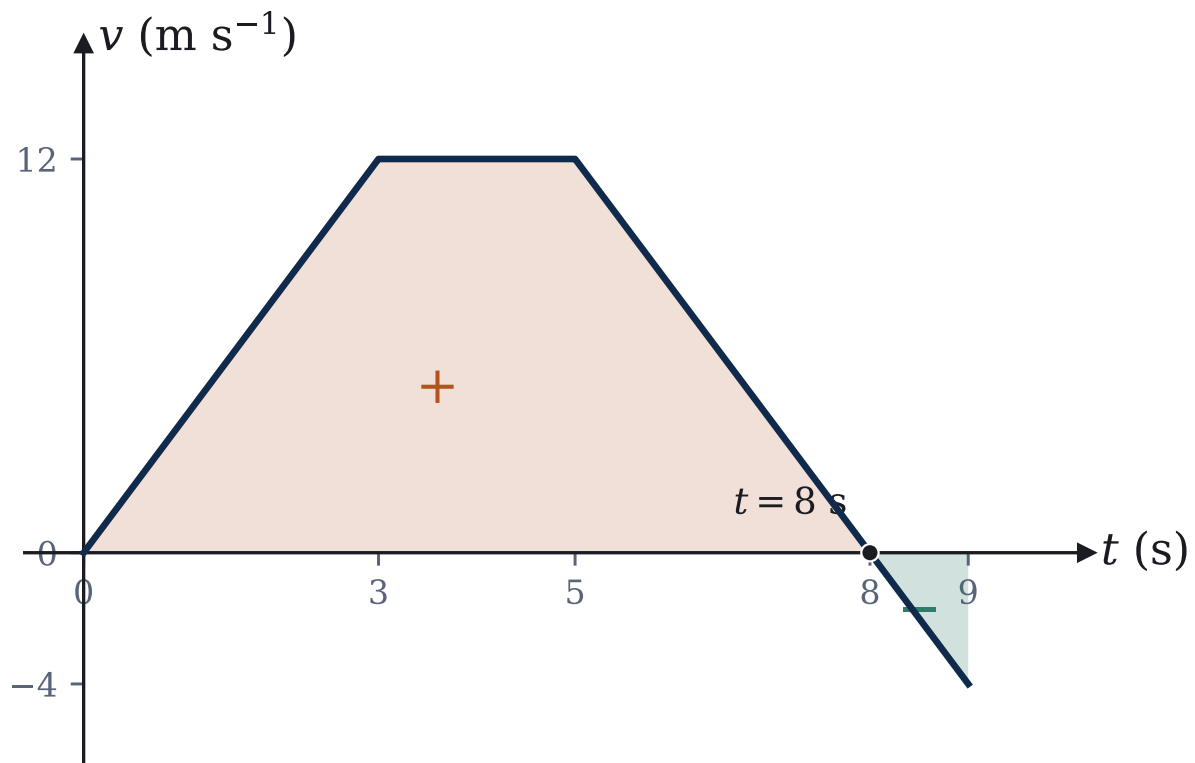


Figure 2.4 A v - t graph: straight rise from (0,0) to (3,12); flat from (3,12) to (5,12); straight fall from (5,12) through the t -axis at $t=8$ to (9,-4). Region above the t -axis shaded '+', the small triangle below (from $t=8$ to 9) shaded '-'.

(c) Distance = $18 + 24 + 18 + 2 = \boxed{62 \text{ m}}$

The 4 m gap between distance and displacement is exactly twice the back-tracked area — the body crept 2 m backward after $t = 8$ s.

Example 4 — [JEE Advanced Level] A particle starts from rest at the origin. Its acceleration-time graph is: $a = +6 \text{ m/s}^2$ (constant) for $0 \leq t \leq 2 \text{ s}$, then $a = -3 \text{ m/s}^2$ (constant) for $2 \leq t \leq 8 \text{ s}$. Find (a) the maximum velocity and when it occurs, (b) the instant the particle reverses direction, (c) the total displacement and (d) the total distance over 0 to 8 s.

Reasoning: Climb the ladder. Area under a - t gives Δv , building the v - t graph; area under that v - t graph then gives displacement.

Figure 2.5 Two stacked graphs sharing a time axis. Top: a - t graph, a step at $+6$ for t in $[0,2]$ then a step at -3 for t in $[2,8]$. Bottom: the derived v - t graph, rising straight from $(0,0)$ to $(2,12)$, then falling straight from $(2,12)$ through zero at $t=6$ down to $(8,-6)$; region $[2,6]$ above axis and $[6,8]$ below axis shaded.] (see next page)

Solution — build the v - t graph (start from $v_0 = 0$):

- Phase 1: $\Delta v = (6)(2) = +12 \text{ m/s}$, so $v(2) = 12 \text{ m/s}$, rising linearly.
- Phase 2: slope -3 , so $v(t) = 12 - 3(t - 2)$. Then $v(6) = 0$ and $v(8) = -6 \text{ m/s}$.

(a) Velocity peaks where acceleration switches from $+$ to $-$: $v_{\text{max}} = 12 \text{ m/s}$ at $t = 2 \text{ s}$.

(b) $v = 0$ at $t = 6 \text{ s} \rightarrow$ **reverses direction** at $t = 6 \text{ s}$.

(c)/(d) Areas of the v - t graph:

- $0-2 \text{ s}$ (triangle): $\frac{1}{2}(2)(12) = 12 \text{ m}$
- $2-6 \text{ s}$ (triangle, above): $\frac{1}{2}(4)(12) = 24 \text{ m}$
- $6-8 \text{ s}$ (triangle, below): $\frac{1}{2}(2)(6) = 6 \text{ m} \rightarrow -6 \text{ m}$

$$\text{Displacement} = 12 + 24 - 6 = \boxed{30 \text{ m}}, \quad \text{Distance} = 12 + 24 + 6 = \boxed{42 \text{ m}}$$

The subtlety that separates strong students: even while the *acceleration* turns negative at $t = 2 \text{ s}$, the particle keeps moving **forward** until $t = 6 \text{ s}$ — negative acceleration first slows it, and only *then* drives it backward. Reading “negative acceleration” as “moving backward” is the classic Advanced-level trap.

Section 5: Practice Exercises

1. **[CBSE]** A car moves at a constant 8 m/s for 5 s , then decelerates uniformly to rest in 4 s . From the v - t graph, find the total displacement.
2. **[NEET]** A body’s position-time graph is a horizontal straight line parallel to the time axis. State what kind of motion this represents.
3. **[JEE Main]** A v - t graph is a straight line rising from $(0, 4 \text{ m/s})$ to $(5 \text{ s}, 14 \text{ m/s})$. Find the acceleration and the displacement in this interval.
4. **[JEE Main]** At a certain instant, the slope of a particle’s x - t graph is negative. What does this tell you about its motion at that instant?

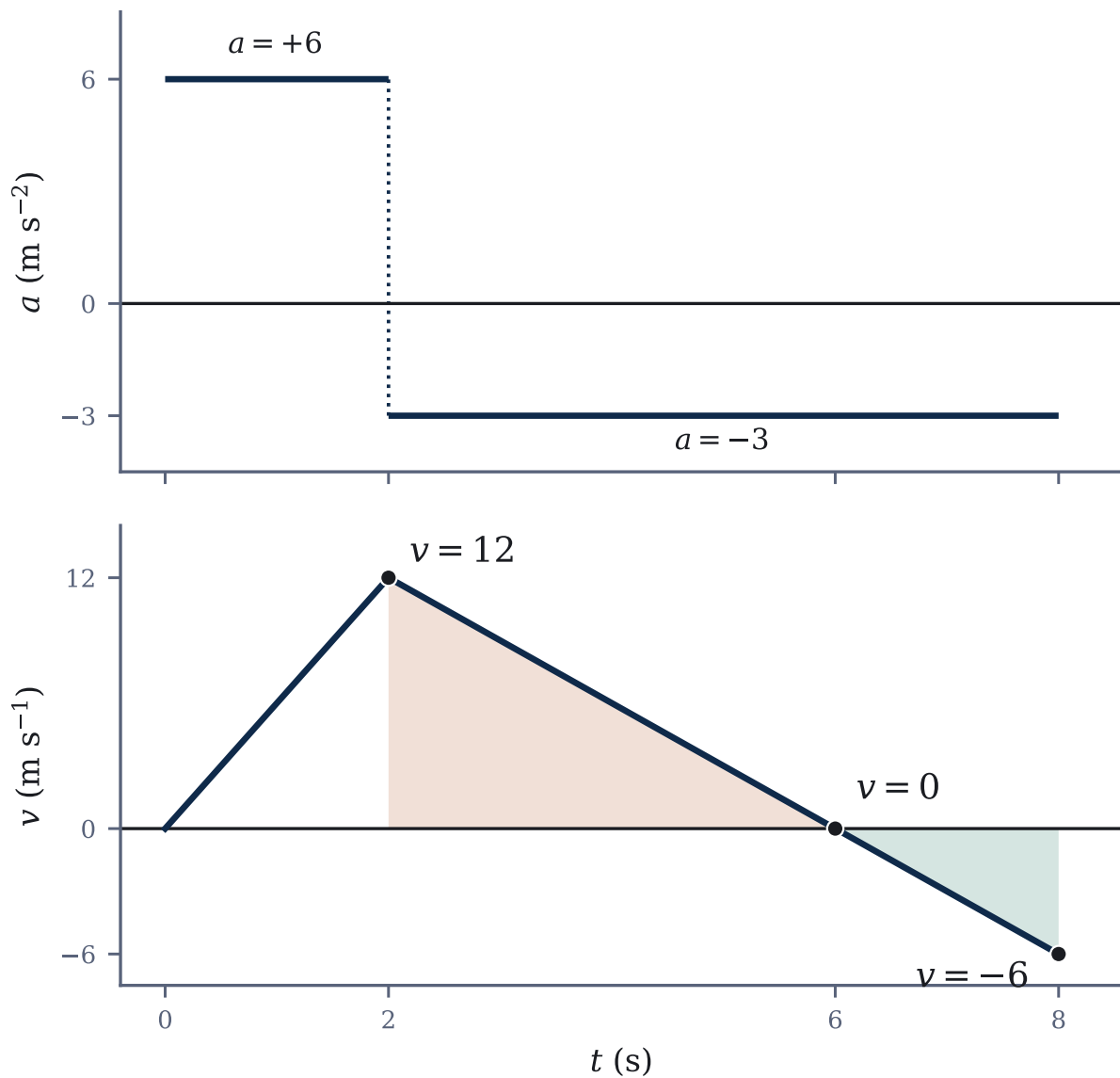


Figure 2.5 Two stacked graphs sharing a time axis. Top: a - t graph, a step at $+6$ for t in $[0, 2]$ then a step at -3 for t in $[2, 8]$. Bottom: the derived v - t graph, rising straight from $(0, 0)$ to $(2, 12)$, then falling straight from $(2, 12)$ through zero at $t = 6$ down to $(8, -6)$; region $[2, 6]$ above axis and $[6, 8]$ below axis shaded.]

5. **[JEE Advanced]** A particle starts from rest with an acceleration that varies as $a = 2t \text{ m/s}^2$ for $0 \leq t \leq 3 \text{ s}$. Find its velocity and displacement at $t = 3 \text{ s}$.

Answers: 1. 56 m | 2. at rest (zero velocity) | 3. $a = 2 \text{ m/s}^2$, displacement = 45 m | 4. moving in the negative direction (velocity is negative) at that instant | 5. $v(3) = 9 \text{ m/s}$, $x(3) = 9 \text{ m}$.

Section 6: Crack the MCQ

Q1. The area enclosed between a velocity-time graph and the time axis represents the: (A) acceleration (B) displacement (C) distance, always (D) velocity

Answer: (B). Net signed area under v - t is displacement. (A) is the *slope* of the v - t graph, not its area. (C) is a tempting half-truth: total area *ignoring sign* gives distance, but with the sign convention the area gives displacement — “always distance” is wrong whenever the curve dips below the axis. (D) confuses the curve’s height with the region beneath it.

Q2. The slope of the tangent to a position-time graph at a point gives the: (A) acceleration (B) instantaneous velocity (C) total distance (D) average velocity

Answer: (B). Slope of x - t is velocity, and the *tangent* (single point) makes it the instantaneous value. (A) would be the slope of a v - t graph, one rung down the ladder. (C) is unrelated to slope. (D) comes from the slope of a **chord**, not a tangent — the trap for students who don’t distinguish the two.

Q3. A particle’s position-time graph is a parabola that is concave up (opens upward). The motion has: (A) zero acceleration (B) constant positive acceleration (C) negative acceleration (D) constant velocity

Answer: (B). A concave-up parabola in x - t means the slope (velocity) steadily increases $\rightarrow$ constant positive acceleration. (A) and (D) describe a *straight* line, not a curve. (C) describes a concave-**down** curve, where velocity decreases — picked by students who don’t link concavity to the sign of acceleration.

Q4. A particle’s v - t graph is a horizontal line at $v = -3 \text{ m/s}$. Over 4 s, its displacement and distance are respectively: (A) -12 m , 12 m (B) 12 m , 12 m (C) -12 m , -12 m (D) 0 , 12 m

Answer: (A). Displacement = signed area = $(-3)(4) = -12 \text{ m}$; distance = magnitude = 12 m . (B) drops the negative sign on displacement — the body genuinely moved backward. (C) reports a *negative distance*, which is impossible by definition. (D) wrongly assumes constant negative velocity gives zero net displacement.

Section 7: Common Student Pitfalls & Pro-Tips

- **Applying the wrong graph’s rule.** “Slope = velocity” is true for x - t , but on a v - t graph slope is *acceleration*. Always read the vertical axis label **before** reaching for a rule.

- **Reading “rising graph” as “speeding up.”** A rising $x-t$ line just means moving in the $+$ direction at steady speed; a rising $v-t$ line means accelerating. Same shape, different meaning.
- **Ignoring signed area.** Area below the time axis is negative for displacement. Students who add all areas as positive get distance and mislabel it displacement.
- **Confusing negative acceleration with backward motion.** A body can have negative acceleration while still moving forward (it’s merely slowing) — it only goes backward once velocity actually crosses zero.
- **Thinking a curved $x-t$ graph means a curved path.** The motion is along a straight line; the curve in the graph encodes changing velocity, not a bend in the road.

Pro-Tip: Carry one mental ladder — **slope steps down** ($x \rightarrow v \rightarrow a$), **area climbs up** ($a \rightarrow v \rightarrow x$). Before summing any $v-t$ area, shade the pieces with $+$ and $-$ signs first; then “all magnitudes” gives distance and “with signs” gives displacement, in one pass. For graph MCQs, the fastest filter is almost always: *straight line or curve?* — it instantly separates the constant-rate options from the changing-rate ones.

Section 8: Cheat Sheet Box

- **$x-t$ graph:** slope = velocity. Flat $\rightarrow$ at rest; straight slope $\rightarrow$ uniform velocity; concave up $\rightarrow +a$; concave down $\rightarrow -a$.
- **$v-t$ graph:** slope = acceleration; **area = displacement** (with signs), $|\text{area}| = \text{distance}$; line crossing the t -axis $\rightarrow$ direction reversal.
- **$a-t$ graph:** area = change in velocity Δv .
- **The ladder:** $x \xrightarrow{\text{slope}} v \xrightarrow{\text{slope}} a$; $a \xrightarrow{\text{area}} v \xrightarrow{\text{area}} x$ (going up needs the initial value).
- **Forbidden:** a vertical $x-t$ segment (infinite velocity); a graph that loops back over one time value.
- **Mnemonic:** “**Slope steps Down, Area climbs Up.**”

Sub-topic 03: Equations of Motion and Free Fall

Exam Relevance: This is the most heavily examined slice of the whole chapter. CBSE Boards regularly ask for a derivation of one of the three equations (2-3 marks) plus a free-fall numerical. JEE Main carries 1-2 questions every year — free fall, stopping distance, or the “distance in the n th second” formula. NEET asks fast plug-in numericals and the Galileo odd-number ratio. JEE Advanced rarely tests the bare equations but threads them through multi-body and last-second free-fall problems where sign convention and setup are everything.

Section 1: Concept Intuition

Everything so far has described motion. Now we get the **tools to predict it**. If a body moves with *uniform acceleration* — acceleration that stays constant in size and direction — then three compact equations let you connect its initial velocity, final velocity, acceleration, time, and displacement. Give any three of these, and the equations hand you the rest. That predictive power is why they’re called the **equations of motion**.

The catch — and it is the single most important sentence in this sub-topic — is the phrase *uniform acceleration*. These equations are **not** universal laws of motion; they are the special-case solution that appears only when acceleration doesn’t change. The moment acceleration varies with time or position, you must return to calculus. Treat the three equations as a trusted shortcut with a clearly stamped expiry condition: “constant a only.”

Think about it this way... Picture an auto-rickshaw pulling away from a red light. If the driver presses the accelerator by exactly the same amount the whole time, the speed climbs in a perfectly steady, predictable way — gain the same chunk of speed every second. *That* steadiness is uniform acceleration, and it’s what makes the future calculable. But the instant he eases off, then floors it again, the neat pattern breaks and the three equations no longer apply to the whole trip. They describe the disciplined driver, not the erratic one.

The most important real-world case of uniform acceleration is **free fall** — motion under gravity alone, with air resistance ignored. Near Earth’s surface, *every* object, regardless of mass, accelerates downward at the same rate $g \approx 9.8 \text{ m/s}^2$ (often rounded to 10 for quick work). A feather and a cricket ball fall identically in a vacuum — Galileo’s famous insight. Because g is constant, the three equations apply directly to anything thrown, dropped, or launched vertically, *provided* you fix a sign convention and keep it.

A clean consequence worth feeling in your bones: a ball thrown straight up rises, slows, stops for an instant at the top, and falls back — and by symmetry it returns to the launch point with the **same speed** it left with, taking **equal time** up and down. At the very top, its velocity is momentarily zero, but its acceleration is still a full g downward (gravity never switches off).

Limiting conditions & assumptions.

- All three equations require **constant acceleration**. Variable $a \rightarrow$ use $v = \frac{dx}{dt}$, $a = \frac{dv}{dt}$ and integrate instead.
- Free fall assumes **no air resistance**, g effectively constant (valid near the surface), and the body treated as a point mass.
- You must **fix a positive direction first**; a sign slip on u , a , or s is the most common

source of wrong answers here.

- g is an acceleration ($\approx 9.8 \text{ m/s}^2$), not a force; the gravitational *force* on a mass m is mg .

Section 2: Key Formulas & Definitions

The three equations of motion (uniform acceleration a):

$$v = u + at \quad (1)$$

$$s = ut + \frac{1}{2}at^2 \quad (2)$$

$$v^2 = u^2 + 2as \quad (3)$$

where u = initial velocity, v = final velocity, a = (constant) acceleration, t = time elapsed, s = displacement. All SI: m/s, m/s^2 , s, m.

Useful companions:

$$s = \left(\frac{u + v}{2} \right) t \quad (\text{average-velocity form})$$

$$s_{n^{\text{th}}} = u + \frac{a}{2}(2n - 1) \quad (\text{displacement during the } n^{\text{th}} \text{ second})$$

Free fall (substitute $a = \pm g$, with a chosen sign convention):

- Dropped from rest (taking down as positive): $v = gt$, $h = \frac{1}{2}gt^2$, $v^2 = 2gh$.
- Thrown up with speed u (taking up as positive, $a = -g$): max height $H = \frac{u^2}{2g}$, time to

$$\text{top} = \frac{u}{g}, \text{ total flight time (back to start)} = \frac{2u}{g}.$$

Galileo's law of odd numbers: for a body starting from rest under constant acceleration, the displacements in successive equal time intervals are in the ratio $1 : 3 : 5 : 7 : \dots$

Stopping distance: with final velocity 0 and deceleration of magnitude a , equation (3)

gives $d = \frac{v_0^2}{2a}$, so stopping distance $\propto v_0^2$.

Section 3: CBSE Board Derivations

Take a body moving with constant acceleration a , initial velocity u , and let its velocity be v after time t , having covered displacement s .

Derivation 1 — $v = u + at$ (from the definition of acceleration). Constant acceleration means $a = \frac{dv}{dt}$ is fixed. Rearranging and integrating from the start ($v = u$ at $t = 0$) to time t :

$$\int_u^v dv = \int_0^t a dt \Rightarrow v - u = at \Rightarrow \boxed{v = u + at}.$$

Why this step: acceleration is the rate of change of velocity; if that rate is constant, velocity grows linearly with time, gaining exactly a each second.

Derivation 2 — $s = ut + \frac{1}{2}at^2$ (from the area under the v - t graph). Since velocity rises linearly from u to v , the v - t graph is a straight line, and displacement is the area beneath it — a rectangle of height u sitting under a triangle.

Figure 2.6 *A v - t graph: a straight line starting at height u on the v -axis (at $t=0$) and rising to height v at time t . The area beneath is split into a lower rectangle of width t and height u (labelled “ ut ”) and an upper right-triangle of base t and height $(v-u)$ (labelled “ $\frac{1}{2} \times t \times at$ ”). Caption: area = displacement s . $s = (u)(t) + \frac{1}{2}(t)(v-u)$ rectangle + triangle. Substitute $v-u = at$ from equation (1): $s = ut + \frac{1}{2}(t)(at) = s = ut + \frac{1}{2}at^2$. Why this step: the rectangle is the distance the body would cover at its starting speed; the triangle is the extra distance the speeding-up contributes. (see next page)*

Derivation 3 — $v^2 = u^2 + 2as$ (eliminating time). This time-independent form is what you reach for when t is neither given nor wanted. Use $v dv = a ds$ (obtained from $a = \frac{dv}{dt} = \frac{dv}{ds} \cdot \frac{ds}{dt} = v \frac{dv}{ds}$) and integrate:

$$\int_u^v v dv = \int_0^s a ds \Rightarrow \frac{v^2 - u^2}{2} = as \Rightarrow \boxed{v^2 = u^2 + 2as}.$$

Why this step: it links velocities directly to displacement, sidestepping time entirely — perfect for stopping-distance problems.

Bonus — Galileo’s odd-number law. For a body starting from rest ($u = 0$), the displacement in the n^{th} second is $s_{n^{\text{th}}} = \frac{a}{2}(2n-1)$, which is proportional to $(2n-1)$. So successive seconds give $1 : 3 : 5 : 7 : \dots$ — exactly the pattern Galileo measured for falling bodies four centuries ago.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A ball is dropped from rest from the top of a 45 m tall building. Taking $g = 10 \text{ m/s}^2$ and ignoring air resistance, find (a) the time it takes to reach the ground and (b) its velocity just before impact.

Given: $u = 0$, $h = 45 \text{ m}$, $g = 10 \text{ m/s}^2$ (take downward as positive).

Solution: (a) Using $h = ut + \frac{1}{2}gt^2$ with $u = 0$:

$$45 = \frac{1}{2}(10)t^2 = 5t^2 \Rightarrow t^2 = 9 \Rightarrow t = 3 \text{ s}.$$

(b) Using $v = u + gt$:

$$v = 0 + (10)(3) = 30 \text{ m/s (downward)}.$$

$$\boxed{t = 3 \text{ s}, \quad v = 30 \text{ m/s}}$$

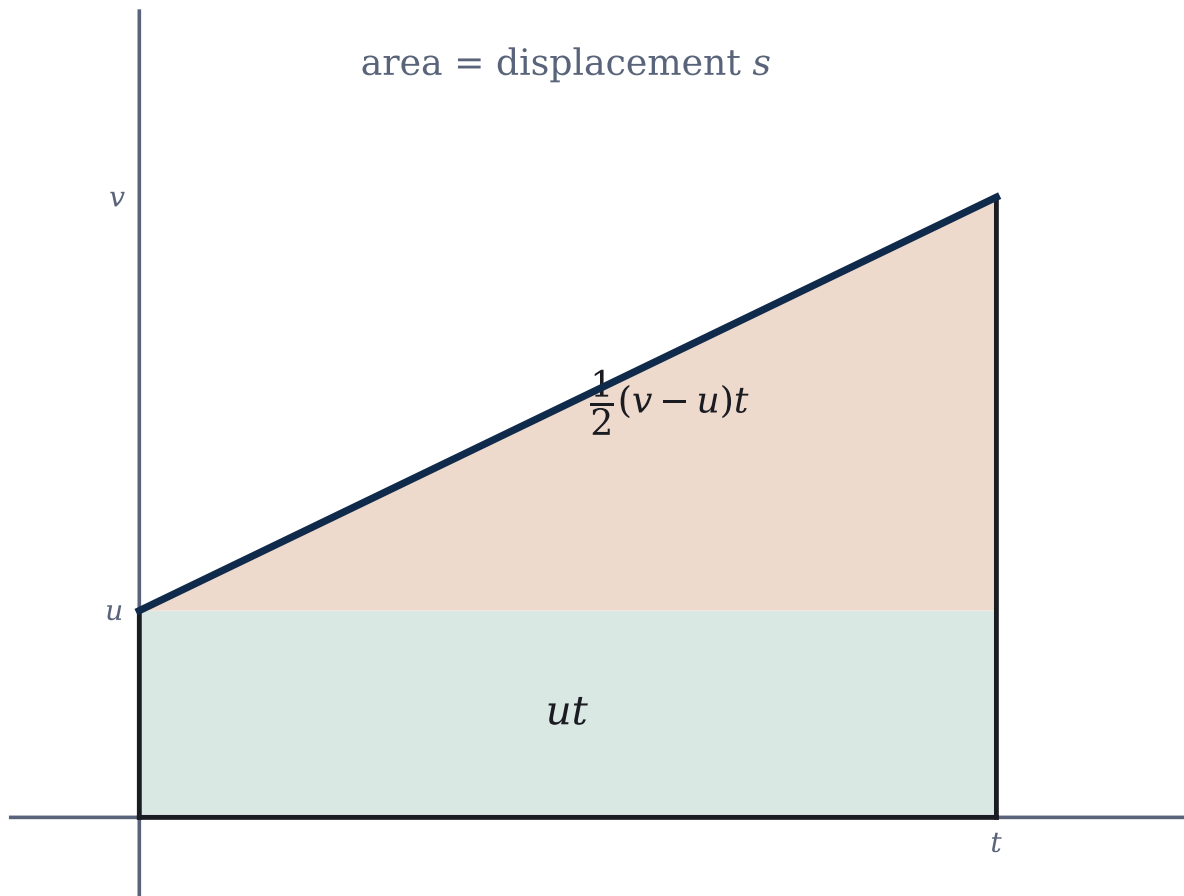


Figure 2.6 A v - t graph: a straight line starting at height u on the v -axis (at $t=0$) and rising to height v at time t . The area beneath is split into a lower rectangle of width t and height u (labelled " ut ") and an upper right-triangle of base t and height $(v-u)$ (labelled " $\frac{1}{2} \times t \times (v-u)$ "). Caption: area = displacement s . $s = (u)(t) + \frac{1}{2}(t)(v-u)$ triangle. Substitute $v-u = at$ from equation (1): $s = ut + \frac{1}{2}(t)(at) \Rightarrow s = ut + \frac{1}{2}at^2$. Why this step: the rectangle is the distance the body would cover at its starting speed; the triangle is the extra distance the speeding-up contributes.

Example 2 — [NEET Speed Trap] A car braking at a fixed deceleration stops in a distance d from a speed v . If it instead approaches at speed $2v$ (same deceleration), its stopping distance is:

(A) d (B) $2d$ (C) $3d$ (D) $4d$

The trap: “double the speed $\rightarrow$ double the distance,” so $2d$. That linear instinct is wrong because braking distance grows with the **square** of speed.

Fast solution: From $v^2 = u^2 + 2as$ with final velocity 0, stopping distance $d = \frac{v^2}{2a} \propto v^2$.

Doubling v multiplies d by $2^2 = 4$.

(D) $4d$

Speed cue: whenever you see “speed doubled/tripled, same braking,” jump straight to the ratio $d \propto v^2$ — no numbers needed. This is also why highway speed limits matter so much: a small speed rise inflates stopping distance disproportionately.

Example 3 — [JEE Main Level] From the top of a 25 m tower, a stone is thrown vertically upward with a speed of 20 m/s. Taking $g = 10 \text{ m/s}^2$, find (a) the maximum height the stone reaches above the ground, (b) the total time before it hits the ground, and (c) its speed on impact.

Why it’s multi-concept: the stone rises above the launch point, then falls past it to the ground — you must handle the sign convention and a quadratic carefully.

Solution: Take **up as positive**, with the origin at the launch point (top of tower). Then $u = +20$, $a = -10$, and the ground is at $s = -25 \text{ m}$.

(a) Maximum height above the launch point: at the top $v = 0$, so by $v^2 = u^2 + 2as$:

$$0 = 20^2 - 2(10)H \Rightarrow H = \frac{400}{20} = 20 \text{ m above the tower top.}$$

Height above the ground = $25 + 20 =$

45 m

.

(b) Position: $s = ut + \frac{1}{2}at^2 = 20t - 5t^2$. Set $s = -25$ (ground):

$$20t - 5t^2 = -25 \Rightarrow t^2 - 4t - 5 = 0 \Rightarrow (t - 5)(t + 1) = 0 \Rightarrow t = 5 \text{ s.}$$

(c) $v^2 = u^2 + 2as = 20^2 - 2(10)(-25) = 400 + 500 = 900 \Rightarrow v = 30 \text{ m/s (downward).}$

$H_{\text{ground}} = 45 \text{ m, } t = 5 \text{ s, } v = 30 \text{ m/s}$

The negative root $t = -1 \text{ s}$ is discarded as unphysical (it’s the “time before launch” the math politely offers).

Example 4 — [JEE Advanced Level] A ball is dropped from rest from a height H . During the **last second** of its fall it covers $\frac{9}{25}$ of the total height H . Taking $g = 10 \text{ m/s}^2$, find the total time of fall and the height H .

Reasoning: “Last second” means the interval from $(T - 1)$ to T , where T is the total fall time. Express that interval’s distance as a fraction of the whole.

Solution: From rest, distance fallen in time t is $\frac{1}{2}gt^2$. So

$$\frac{\text{distance in last second}}{H} = \frac{\frac{1}{2}gT^2 - \frac{1}{2}g(T-1)^2}{\frac{1}{2}gT^2} = \frac{T^2 - (T-1)^2}{T^2} = \frac{2T-1}{T^2}.$$

Set this equal to $\frac{9}{25}$:

$$25(2T-1) = 9T^2 \Rightarrow 9T^2 - 50T + 25 = 0.$$

$$T = \frac{50 \pm \sqrt{2500 - 900}}{18} = \frac{50 \pm 40}{18} \Rightarrow T = 5 \text{ s (or } \frac{5}{9} \text{ s, rejected as } < 1).$$

Then $H = \frac{1}{2}gT^2 = \frac{1}{2}(10)(25) = 125 \text{ m}.$

$T = 5 \text{ s, } H = 125 \text{ m}$

The rejected root is smaller than 1 s, so “the last second” wouldn’t even exist — a neat reminder to always test roots against the physics, not just the algebra.

Section 5: Practice Exercises

1. **[CBSE]** A car speeds up uniformly from 10 m/s to 25 m/s in 5 s. Find its acceleration and the distance covered.
2. **[NEET]** A ball thrown vertically upward takes 2 s to reach its highest point ($g = 10 \text{ m/s}^2$). Find its initial speed and the maximum height.
3. **[JEE Main]** A body starts from rest with constant acceleration. Find the ratio of the distance it covers in the 3rd second to that in the 5th second.
4. **[JEE Main]** A stone dropped from a cliff reaches the ground in 4 s ($g = 10 \text{ m/s}^2$). Find the height of the cliff and the stone’s speed on impact.
5. **[JEE Advanced]** A ball is thrown vertically upward from the ground with 30 m/s ($g = 10 \text{ m/s}^2$). For how long is the ball at a height of 40 m or more above the ground?

Answers: 1. $a = 3 \text{ m/s}^2$, $s = 87.5 \text{ m}$ | 2. $u = 20 \text{ m/s}$, $H = 20 \text{ m}$ | 3. $5 : 9$ | 4. height 80 m, impact speed 40 m/s | 5. 2 s (from $t = 2 \text{ s}$ to $t = 4 \text{ s}$).

Section 6: Crack the MCQ

Q1. The equation $v^2 = u^2 + 2as$ can be applied only when: (A) the acceleration varies with time (B) the acceleration is uniform (C) the initial velocity is zero (D) the motion is vertical

Answer: (B). All three equations of motion are derived assuming constant acceleration. (A) is the exact case where they fail. (C) is unnecessary — u can be any value. (D) confuses the general equation with the special free-fall application; horizontal uniform acceleration works just as well.

Q2. A body falling freely from rest covers distances in the 1st, 2nd, and 3rd seconds in the ratio: (A) 1 : 2 : 3 (B) 1 : 3 : 5 (C) 1 : 4 : 9 (D) 1 : 2 : 4

Answer: (B). By Galileo's law, distances in successive seconds go as $(2n - 1)$: 1, 3, 5. (C) is the seductive trap — that's the ratio of *total* distances after 1, 2, 3 seconds ($\propto n^2$), not the distance *in* each second. (A) is the ratio of the speeds (or times). (D) fits no kinematic pattern. The (B)-vs-(C) confusion (distance *in* the n th second vs *up to* n seconds) is the most common error here.

Q3. A ball is thrown vertically upward with speed u and returns to the thrower's hand (ignore air resistance). Which statement is correct? (A) it returns with speed greater than u (B) it returns with speed equal to u (C) the time going up exceeds the time coming down (D) its acceleration is zero at the highest point

Answer: (B). By symmetry (and energy conservation), it returns with the same speed u . (A) violates energy conservation. (C) is false — the up and down times are equal. (D) is the classic trap: at the top the *velocity* is zero, but the *acceleration* is still g downward — gravity never turns off.

Q4. A car travelling at speed v stops in a distance d when the brakes are applied. If it travels at $3v$ with the same braking deceleration, the stopping distance becomes: (A) $3d$ (B) $6d$ (C) $9d$ (D) $d/3$

Answer: (C). Stopping distance $\propto v^2$, so tripling the speed gives $3^2 = 9$ times the distance. (A) assumes linear scaling — the standard mistake. (B) has no basis. (D) inverts the relationship entirely.

Section 7: Common Student Pitfalls & Pro-Tips

- **Using the equations when acceleration isn't constant.** This is the cardinal error. If a depends on t or position, the three equations are simply invalid — go back to calculus.
- **Sign-convention chaos in free fall.** Mixing the signs of u , a , and s wrecks vertical-motion problems. Pick one positive direction and apply it to *every* quantity in the problem.
- **Confusing “distance in the n^{th} second” with “distance in n seconds.”** The first is $s_{n^{\text{th}}} = u + \frac{a}{2}(2n - 1)$; the second is $s = un + \frac{1}{2}an^2$. They are not the same number.
- **Believing acceleration is zero at the top of a throw.** Velocity is zero there; acceleration is still g . The ball doesn't hang weightless — it's about to fall.
- **Mixing $g = 9.8$ and $g = 10$ within one problem.** Read which value the question intends and use it consistently.

Pro-Tip: Lock in “up = positive, $a = -g$ ” for *every* vertical problem and never re-decide mid-solution. For “speed doubled/tripled” braking questions, reach for the proportionality $d \propto v^2$ instead of recomputing from scratch. And when you need to find *when* a body is at a given height, set up the quadratic in t — the **two roots** are usually the moment it passes that height going up and again coming down, which is often the whole point of the question.

Section 8: Cheat Sheet Box

- **Three equations (constant a only):** $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $s = \left(\frac{u+v}{2}\right)t$.
- **Distance in the n^{th} second:** $s_{n^{\text{th}}} = u + \frac{a}{2}(2n-1)$.
- **Free fall from rest:** $v = gt$, $h = \frac{1}{2}gt^2$, $v^2 = 2gh$.
- **Thrown up with u :** $H_{\text{max}} = \frac{u^2}{2g}$, $t_{\text{up}} = \frac{u}{g}$, $t_{\text{flight}} = \frac{2u}{g}$; returns with the same speed u .
- **Galileo's odd numbers** (from rest): successive-second distances $1 : 3 : 5 : 7 : \dots$
- **Stopping distance:** $d = \frac{v_0^2}{2a} \propto v_0^2$.
- *Key reminders:* $g \approx 9.8 \text{ m/s}^2$ (use 10 for speed); at the top of a throw $v = 0$ but $a = g$; equations need **constant acceleration**.

Sub-topic 04: Relative Velocity

Exam Relevance: A compact but reliable scorer. CBSE Boards ask a 1–2 mark conceptual or short numerical (relative velocity of two trains). JEE Main carries roughly 1 question — usually a pursuit/overtaking problem or two bodies meeting. NEET tests the same-direction-subtract / opposite-direction-add idea as a quick MCQ. JEE Advanced uses relative velocity as a *time-saving lens* for two-body problems, especially the elegant free-fall result where relative acceleration vanishes. (The 2-D river-boat and rain problems build on this same idea but are taught in *Motion in a Plane*.)

Section 1: Concept Intuition

Sit in a moving train and watch a second train on the next track. If both crawl forward at nearly the same speed, the other train seems almost frozen beside you — even though a person on the platform sees both racing past. If the other train moves the opposite way, it appears to *whoosh* by at a frightening rate. Nothing about either train’s actual motion changed; what changed is **who is watching**. That is the whole idea of **relative velocity**: velocity is never absolute — it is always measured with respect to some observer, some **frame of reference**.

When we quote a car’s speed as “60 km/h,” we silently mean *relative to the ground*. The ground is just our default frame. But we are free to ask how fast one object moves **as seen from another object**, and that is the velocity of one relative to the other. The velocity of A with respect to B — written $\vec{v}_{AB}$ — is what an observer riding on B would measure for A:

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B.$$

In words: to find how A moves *in B’s eyes*, subtract B’s velocity from A’s. Because we are in this chapter restricting ourselves to a single straight line, this vector subtraction collapses to ordinary subtraction of signed numbers once a positive direction is fixed.

Think about it this way... Imagine two motorcyclists on a long, straight highway. If both ride north at 80 and 80 km/h, each sees the other parked alongside — relative velocity zero. If one rides north at 80 and the other south at 80, each sees the other approaching at 160 km/h. Same individual speeds; wildly different *relative* speeds. The rule falls out naturally: **same direction → subtract** (the speeds partly cancel), **opposite directions → add** (they pile up).

Two more ideas complete the picture. First, relative velocity comes in two flavours by *what it does to the gap* between the bodies: a **relative velocity of approach** shrinks their separation (they’re closing in), while a **relative velocity of separation** grows it. Second, just as positions and velocities are relative, so is acceleration: the **relative acceleration** of A with respect to B is $\vec{a}_{AB} = \vec{a}_A - \vec{a}_B$. This yields one of the most beautiful shortcuts in kinematics — if two bodies have the *same* acceleration (for instance, both in free fall, each accelerating at g), their relative acceleration is **zero**, so one sees the other moving at a *constant* relative velocity, no matter how fast both are actually accelerating.

Limiting conditions & assumptions.

- This treatment is **one-dimensional** — motion along a common straight line — so direction reduces to $+$ / $-$. The full vector version (river-boat, rain-umbrella) lives in *Motion in a Plane*.

- $\vec{v}_{AB} = -\vec{v}_{BA}$ always; A's view of B is the exact opposite of B's view of A.
 - Velocities add the simple Galilean way only at everyday speeds ($v \ll c$). Near the speed of light this breaks down (relativistic velocity addition) — not in the JEE/NEET syllabus, but worth knowing the rule has a domain.
-

Section 2: Key Formulas & Definitions

- **Relative velocity of A with respect to B:** $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$. (1-D: $v_{AB} = v_A - v_B$, with signs.)
- **Antisymmetry:** $\vec{v}_{AB} = -\vec{v}_{BA}$.
- **Same direction,** speeds v_A, v_B : relative speed = $|v_A - v_B|$.
- **Opposite directions,** speeds v_A, v_B : relative speed = $v_A + v_B$.
- **Relative acceleration:** $\vec{a}_{AB} = \vec{a}_A - \vec{a}_B$.

Equations of motion in relative form (valid when relative acceleration is constant):

$$v_{\text{rel}} = u_{\text{rel}} + a_{\text{rel}} t, \quad s_{\text{rel}} = u_{\text{rel}} t + \frac{1}{2} a_{\text{rel}} t^2, \quad v_{\text{rel}}^2 = u_{\text{rel}}^2 + 2 a_{\text{rel}} s_{\text{rel}}$$

where $u_{\text{rel}}, v_{\text{rel}}, a_{\text{rel}}, s_{\text{rel}}$ are the initial relative velocity, final relative velocity, relative acceleration, and relative displacement (separation).

Time to meet (two bodies closing along a line):

$$t_{\text{meet}} = \frac{\text{initial separation}}{\text{relative speed of approach}}$$

Crossing / overtaking (objects of finite length): the relevant relative displacement is the **sum of the lengths** (two trains crossing) or one length (a train passing a point), covered at the relative speed.

All quantities SI: velocity m/s, acceleration m/s², separation m, time s. Dimensions are those of ordinary velocity/acceleration: $[L T^{-1}]$, $[L T^{-2}]$.

Section 3: Key Procedures & Logical Framework

The logical core — where the relations come from. Let A and B have positions x_A and x_B measured in the ground frame. The position of A *as seen from B* is the separation $x_{AB} = x_A - x_B$. Differentiate once with respect to time:

$$v_{AB} = \frac{d}{dt}(x_A - x_B) = v_A - v_B,$$

and once more:

$$a_{AB} = \frac{d}{dt}(v_A - v_B) = a_A - a_B.$$

Why this matters: the relative quantities obey the **same** equations of motion as ordinary ones, because they're just differences of quantities that each obey those equations. So you may treat B as a stationary "lab" and analyse A's motion using $u_{\text{rel}}, a_{\text{rel}}, s_{\text{rel}}$ exactly as you'd treat a single particle.

Procedure A — Solving a 1-D relative-velocity problem. 1. **Fix a positive direction** (say, rightward = +) and write every velocity/acceleration with its correct sign in the ground frame. 2. **Choose the frame** you'll work in — usually the ground, or one of the two bodies. Stick to it. 3. If working in B's frame, compute $u_{\text{rel}} = v_A - v_B$ and $a_{\text{rel}} = a_A - a_B$. 4. **Translate the question into the relative quantity.** "When do they meet?" → relative displacement equals the initial separation. "When are they d apart?" → relative displacement equals $\pm d$. 5. Apply the relative-form equation, solve, and **check signs/roots** against the physics (reject negative times, etc.).

Procedure B — Approach vs separation (sign-free quick method). 1. Same direction: subtract speeds; opposite directions: add speeds — this gives the magnitude of relative velocity. 2. Decide approach or separation by asking whether the gap is currently shrinking or growing. 3. For "time to meet," divide the initial gap by the relative *speed of approach*. For overtaking a finite-length object, use the total length that must pass. 4. Sanity check: a faster follower always eventually catches a slower leader; two bodies moving apart never meet.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Two trains run on parallel straight tracks. Train A moves at 72 km/h and train B at 54 km/h. Find the velocity of A relative to B when they move (a) in the same direction and (b) in opposite directions.

Given: $v_A = 72 \text{ km/h} = 20 \text{ m/s}$, $v_B = 54 \text{ km/h} = 15 \text{ m/s}$. Take A's direction of travel as positive.

Solution: (a) Same direction: $v_B = +15 \text{ m/s}$.

$$v_{AB} = v_A - v_B = 20 - 15 = +5 \text{ m/s (in A's direction).}$$

(b) Opposite directions: now $v_B = -15 \text{ m/s}$.

$$v_{AB} = 20 - (-15) = +35 \text{ m/s (in A's direction).}$$

(a) 5 m/s,	(b) 35 m/s
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The opposite-direction case is much larger because the two motions reinforce rather than cancel.

Example 2 — [NEET Speed Trap] Two cars travel toward each other along a straight road, one at 54 km/h and the other at 36 km/h. The speed of one car relative to the other is:

$$(A) 18 \text{ km/h} \quad (B) 45 \text{ km/h} \quad (C) 90 \text{ km/h} \quad (D) 72 \text{ km/h}$$

The trap: the eye sees two speeds and subtracts — $54 - 36 = 18$ — but subtraction is the **same-direction** rule. These cars move *toward each other*.

Fast solution: Opposite directions → **add**:

$$v_{\text{rel}} = 54 + 36 = 90 \text{ km/h.}$$

$$(C) \ 90 \text{ km/h}$$

Speed cue: burn in the one-liner — “**same way, subtract; opposite, add.**” The instant a problem says *toward each other* or *opposite directions*, add the speeds without hesitation. Option (A) is there precisely to catch the reflexive subtractor.

Example 3 — [JEE Main Level] On a straight road, car A moves at a constant 20 m/s. At the same instant, car B — which is 75 m behind A on the same road — starts at 10 m/s and accelerates uniformly at 2 m/s² in the same direction. After how long does B catch A, and how far has B then travelled?

Why it’s multi-concept: it combines relative velocity, relative acceleration, and the equations of motion in one setup.

Solution: Work in **A’s frame** (A is unaccelerated, so it’s a valid inertial lab). Take the common direction of travel as positive.

$$u_{\text{rel}} = v_B - v_A = 10 - 20 = -10 \text{ m/s}, \quad a_{\text{rel}} = a_B - a_A = 2 - 0 = 2 \text{ m/s}^2.$$

B must close the 75 m gap, i.e. achieve a relative displacement of +75 m:

$$s_{\text{rel}} = u_{\text{rel}}t + \frac{1}{2}a_{\text{rel}}t^2 \Rightarrow 75 = -10t + t^2 \Rightarrow t^2 - 10t - 75 = 0.$$

$$(t - 15)(t + 5) = 0 \Rightarrow t = 15 \text{ s}.$$

Distance B travels in the ground frame:

$$s_B = (10)(15) + \frac{1}{2}(2)(15)^2 = 150 + 225 = 375 \text{ m}.$$

$$t = 15 \text{ s}, \quad s_B = 375 \text{ m}$$

(Check: A is $75 + 20 \times 15 = 375$ m from B’s start — they coincide. ✓) Notice that early on, u_{rel} being negative means the gap actually *widens* at first; B only overhauls A once its acceleration has overcome that initial deficit.

Example 4 — [JEE Advanced Level] A ball is thrown vertically upward from the ground with 25 m/s. At the very same instant, a second ball is released from rest from a height of 100 m, directly above the first, along the same vertical line. Taking $g = 10 \text{ m/s}^2$, find when and at what height above the ground the two balls meet.

Reasoning — the elegant part: both balls are in free fall, so each accelerates at g downward. Their **relative acceleration is zero**, which means their relative velocity is *constant* and the gap closes uniformly — no quadratic needed.

Figure 2.7 A vertical line. At the bottom, the ground, with ball 1 launched upward labelled “ $u = 25 \text{ m/s } \uparrow$ ”. At the top, height 100 m, ball 2 labelled “released from rest”. A bracket spans the 100 m initial separation, annotated “closes at constant 25 m/s”. A dot at 20 m marks the meeting point, labelled “meet at $t = 4 \text{ s}$, height 20 m”. (see next page)

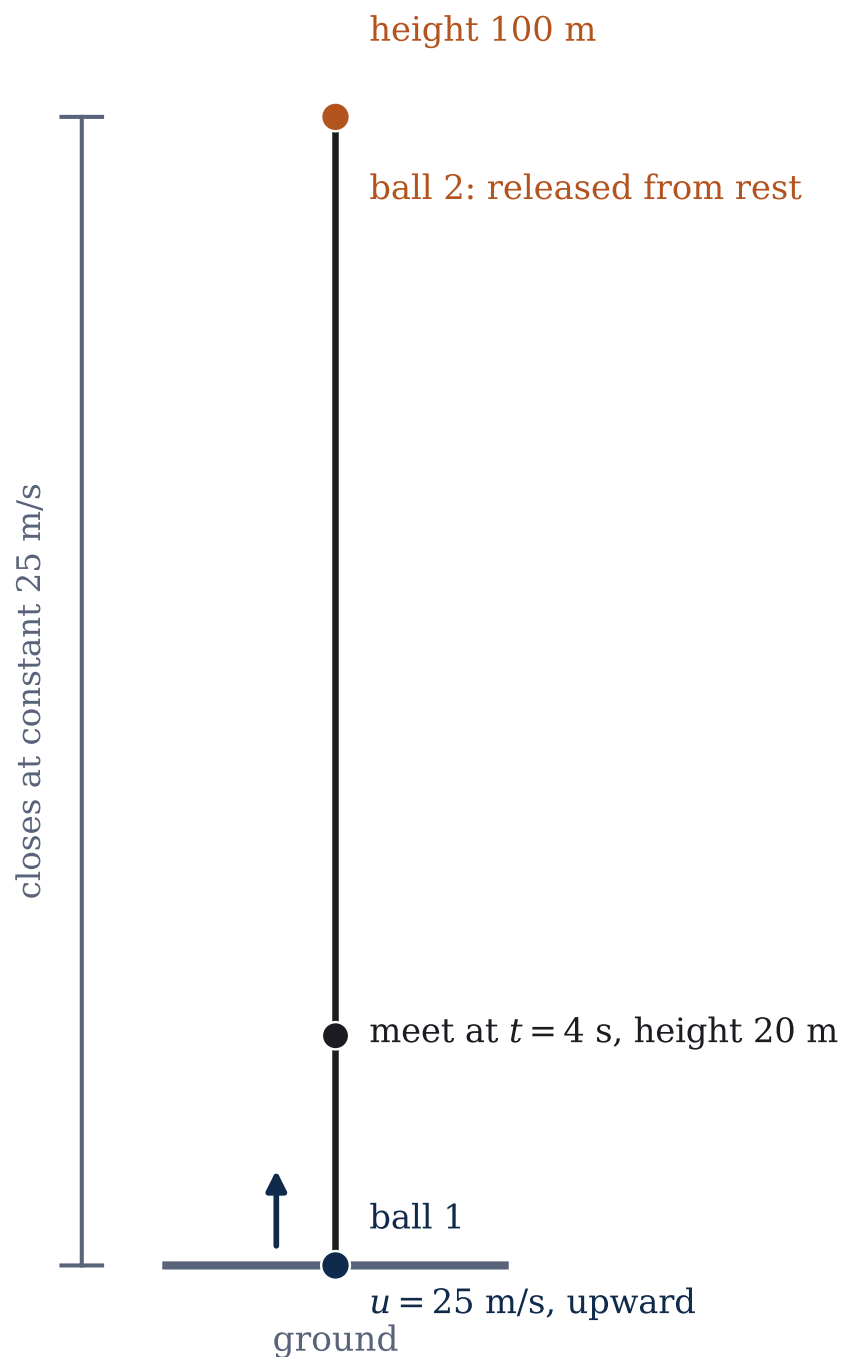


Figure 2.7 A vertical line. At the bottom, the ground, with ball 1 launched upward labelled " $u = 25 \text{ m/s}$ ". At the top, height 100 m, ball 2 labelled "released from rest". A bracket spans the 100 m initial separation, annotated "closes at constant 25 m/s". A dot at 20 m marks the meeting point, labelled "meet at $t = 4 \text{ s}$, height 20 m".

Solution: Take **up as positive**. In the ground frame, $u_1 = +25$, $u_2 = 0$, and both have $a = -g$.

$$u_{\text{rel}} = u_1 - u_2 = 25 - 0 = 25 \text{ m/s}, \quad a_{\text{rel}} = (-g) - (-g) = 0.$$

With zero relative acceleration, the separation of 100 m closes at a steady 25 m/s:

$$t_{\text{meet}} = \frac{100}{25} = 4 \text{ s}.$$

Height of ball 1 above the ground at $t = 4$ s:

$$y_1 = u_1 t - \frac{1}{2} g t^2 = 25(4) - \frac{1}{2}(10)(16) = 100 - 80 = 20 \text{ m}.$$

(Check with ball 2: $y_2 = 100 - \frac{1}{2}(10)(16) = 20 \text{ m}$. ✓)

$t = 4 \text{ s}, \quad \text{at } 20 \text{ m above the ground}$

Both balls are still airborne (ball 1's flight lasts 5 s), so the meeting is genuine. The lesson: spotting $a_{\text{rel}} = 0$ turned a messy two-body free-fall problem into a one-line division.

Section 5: Practice Exercises

1. **[CBSE]** Two cyclists ride directly toward each other on a straight road at 5 m/s and 7 m/s. Find their relative speed of approach.
2. **[NEET]** A boy runs at 3 m/s in the same direction as a bus moving at 12 m/s. Find the boy's velocity relative to the bus.
3. **[JEE Main]** Two cars, 120 m apart on a straight road, move toward each other at 8 m/s and 12 m/s. How long before they meet?
4. **[JEE Main]** A train 200 m long, moving at 20 m/s, overtakes a man walking at 2 m/s in the same direction along the track. How long does the train take to completely pass him?
5. **[JEE Advanced]** From the top of a 60 m tower a stone is dropped from rest. At the same instant another stone is thrown vertically upward from the ground at 20 m/s along the same line ($g = 10 \text{ m/s}^2$). Find when and at what height they meet.

Answers: 1. 12 m/s | 2. 9 m/s opposite to the bus's motion (i.e. -9 m/s) | 3. 6 s | 4. $\approx 11.1 \text{ s}$ ($200/18$) | 5. $t = 3 \text{ s}$, at 15 m above the ground.

Section 6: Crack the MCQ

Q1. The velocity of a body A relative to body B is correctly written as: (A) $\vec{v}_A + \vec{v}_B$ (B) $\vec{v}_A - \vec{v}_B$ (C) $\vec{v}_B - \vec{v}_A$ (D) $\frac{1}{2}(\vec{v}_A + \vec{v}_B)$

Answer: (B). By definition $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$. (C) is the velocity of B relative to A ($\vec{v}_{BA}$) — the exact negative, a sign trap. (A) gives only the *magnitude* in the special opposite-direction case, not the general definition. (D) is an average, which has no meaning here.

Q2. Two particles move along the same straight line in opposite directions with speeds v_1 and v_2 . The magnitude of their relative velocity is: (A) $|v_1 - v_2|$ (B) $v_1 + v_2$ (C) $\frac{1}{2}(v_1 + v_2)$ (D) zero

Answer: (B). Opposite directions add: relative speed $= v_1 + v_2$. (A) is the **same-direction** result — picked by students who memorise “subtract” without checking direction. (C) and (D) have no physical basis here; (D) would require equal velocities in the *same* direction.

Q3. Two stones are dropped from rest from the top of a tall tower, the second released a short moment after the first. As both fall, the distance between them: (A) remains constant (B) increases (C) decreases (D) first increases, then decreases

Answer: (B). Both fall with the same acceleration g , so $a_{\text{rel}} = 0$; but the first stone is always faster (it had a head start), giving a constant non-zero relative velocity. A constant relative velocity makes the separation grow **steadily**. (A) is the seductive trap: “same acceleration must mean a fixed gap.” Same acceleration fixes the *relative velocity*, not the *separation* — the gap widens at a constant rate. (C) and (D) contradict the steadily increasing relative speed of the leading stone.

Q4. Two trains, each 100 m long, travel toward each other on parallel tracks at 15 m/s and 25 m/s. The time taken for them to completely cross each other is: (A) 5 s (B) 10 s (C) 2.5 s (D) 8 s

Answer: (A). Relative speed (opposite directions) $= 15 + 25 = 40$ m/s. To fully cross, the trains’ fronts-to-tails must sweep the **sum of the lengths**, $100 + 100 = 200$ m. Time $= 200/40 = 5$ s. (B) uses only one train’s length ($100/40$ is even smaller — this comes from a different slip), (C) halves incorrectly, (D) mixes up the numbers. The key insight the distractors miss is that crossing requires the *total* length to pass, not a single point.

Section 7: Common Student Pitfalls & Pro-Tips

- **Adding when you should subtract (and vice-versa).** Same direction $\rightarrow$ subtract; opposite $\rightarrow$ add. The fix is to assign $+/-$ signs in the ground frame *before* combining, so the rule takes care of itself.
- **Confusing $\vec{v}_{AB}$ with $\vec{v}_{BA}$.** They’re equal in size but opposite in sign. State clearly which body is the “observer.”
- **Treating a crossing as a point event.** Overtaking or crossing a finite-length object requires the *whole length* (or sum of lengths) to pass at the relative speed, not just a single meeting point.
- **“Same acceleration $\Rightarrow$ constant gap.”** False. Equal accelerations make the *relative velocity* constant; the separation then changes *linearly*, not stays fixed.
- **Mixing frames mid-solution.** Switching between the ground frame and a moving body’s frame partway through scrambles the signs.

Pro-Tip: Commit to **one frame** for the whole problem — usually the ground, or a non-accelerating body treated as a stationary lab. The single most powerful shortcut: for two bodies in free fall, $a_{\text{rel}} = 0$, so they drift apart or together at a *constant* relative velocity — time to meet is just (initial separation) $\div$ (relative speed of approach), no quadratic required. Reach for that whenever two projectiles share gravity as their only acceleration.

Section 8: Cheat Sheet Box

- **Definition:** $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$ (velocity of A as seen from B); and $\vec{v}_{AB} = -\vec{v}_{BA}$.
- **Same direction $\rightarrow$ subtract speeds; opposite direction $\rightarrow$ add speeds.**
- **Relative acceleration:** $\vec{a}_{AB} = \vec{a}_A - \vec{a}_B$. Two free-falling bodies $\Rightarrow a_{\text{rel}} = 0 \Rightarrow$ constant relative velocity.
- **Relative-form equations:** $v_{\text{rel}} = u_{\text{rel}} + a_{\text{rel}}t$, $s_{\text{rel}} = u_{\text{rel}}t + \frac{1}{2}a_{\text{rel}}t^2$.
- **Time to meet** = $\frac{\text{initial separation}}{\text{relative speed of approach}}$.
- **Crossing two trains:** distance = sum of lengths, at the relative speed.
- *Mnemonic: "Same way, subtract; opposite, add."*

Sub-topic 05: Motion with Variable Acceleration (Calculus Methods)

Exam Relevance: This is where kinematics grows up. CBSE Boards occasionally ask a 2–3 mark numerical with acceleration given as a function of time (integrate to get velocity and position). JEE Main reliably carries 1 question built on a variable acceleration — usually a as a function of velocity or position, where the three constant- a equations silently fail. NEET sets the occasional “ v given as a function of x , find a ” trap. JEE Advanced loves this slice: resistive forces ($a \propto -v$ or $-v^2$), position-dependent acceleration, and problems that hinge on choosing the *right* calculus relation. The single skill that unlocks all of it is knowing which form of acceleration to use — $\frac{dv}{dt}$ or $v\frac{dv}{dx}$ — for the data you are given.

Section 1: Concept Intuition

The three equations of motion you met in Sub-topic 03 are a beautiful shortcut — but read the fine print stamped on them: *constant acceleration only*. The instant acceleration changes with time, with speed, or with position, those equations stop being laws and become traps. Real motion is full of changing acceleration: a parachutist meets more air drag the faster she falls, a charged particle feels a force that grows as it nears a plate, a rocket’s acceleration climbs as it burns fuel and lightens. To handle any of these you must return to the two definitions that are *always* true — $v = \frac{dx}{dt}$ and $a = \frac{dv}{dt}$ — and use the calculus they contain.

Recall the ladder from the graphs sub-topic: differentiation steps you *down* the chain ($x \rightarrow v \rightarrow a$), and integration carries you back *up* ($a \rightarrow v \rightarrow x$). With constant acceleration, climbing that ladder gave the three tidy equations. With variable acceleration, you climb the very same ladder — you just have to do the integral honestly instead of reading off a formula. Nothing new in spirit; only the arithmetic is heavier.

Here is the one genuinely new idea, and it is a small masterpiece. Sometimes acceleration is handed to you as a function of *position*, not time — say a grows with x . You cannot integrate $a dt$ because you do not know a as a function of t . The escape is a chain-rule trick: write $a = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} = v \frac{dv}{dx}$.

That single rewrite, $a = v \frac{dv}{dx}$, converts a position-dependent acceleration into something you can integrate against x . It is the master key of this whole sub-topic.

Think about it this way... Remember the disciplined auto-rickshaw driver from Sub-topic 03, pressing the accelerator by exactly the same amount the whole way? The three equations described *him*. This sub-topic is about the erratic driver — easing off, flooring it, braking, speeding up again. His motion is still perfectly governed by physics; it is just that the rule changes moment to moment, so we must track it with calculus rather than a one-line formula. The good news: once you can read off whether the changing acceleration depends on t , v , or x , the right tool snaps into place almost automatically.

A velocity-dependent acceleration also explains one of the most familiar facts of everyday physics: **terminal velocity**. A raindrop or a skydiver falls under gravity but meets air drag that grows with speed, so the net acceleration is something like $a = g - kv$. As speed rises, the drag term climbs until it exactly cancels gravity; at that point $a = 0$ and the speed stops increasing. Setting $a = 0$ gives the terminal speed $v_{\text{term}} = g/k$ — the steady value the body coasts at. You did not need to solve the full motion to read off this limiting speed; you just asked “when does the velocity-dependent acceleration vanish?” That is the kind of physical insight a calculus treatment of $a = f(v)$ unlocks.

So the entire strategy reduces to a single question you ask at the start of every problem: **what does the acceleration depend on?** If it depends on time, integrate with respect to time. If it depends on velocity, you may integrate against time *or* position depending on what is asked. If it depends on position, reach for $a = v \frac{dv}{dx}$. Diagnose first; integrate second.

Limiting conditions & assumptions.

- The three equations of motion ($v = u + at$, etc.) are **forbidden** here whenever a is not constant. Using them is the cardinal sin of this sub-topic.
- The definitions $v = \frac{dx}{dt}$, $a = \frac{dv}{dt} = v \frac{dv}{dx}$ hold for *all* one-dimensional motion, uniform or not — they are the safe ground.
- Every integration produces a constant; you must fix it from an initial condition (v_0 , x_0 , or a stated value of v at some x). Forgetting the constant is the second most common error.
- This is still one-dimensional motion; a sign (+/−) carries the direction, so choose a positive direction first.

Section 2: Key Formulas & Definitions

The always-true definitions.

$$v = \frac{dx}{dt}, \quad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}.$$

The master chain-rule relation (use when a depends on x or v):

$$a = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} = v \frac{dv}{dx}.$$

The three integration templates — chosen by what the acceleration depends on:

Case 1 — acceleration depends on time, $a = f(t)$. Integrate directly with respect to time:

$$v = v_0 + \int_0^t f(t) dt, \quad x = x_0 + \int_0^t v dt.$$

Case 2 — acceleration depends on velocity, $a = f(v)$. Pick the pairing by what you want:

$$\text{want } v(t) : \int_{v_0}^v \frac{dv}{f(v)} = \int_0^t dt; \quad \text{want } v(x) : \int_{v_0}^v \frac{v dv}{f(v)} = \int_{x_0}^x dx.$$

Case 3 — acceleration depends on position, $a = f(x)$. Use $a = v \frac{dv}{dx}$ and separate:

$$\int_{v_0}^v v dv = \int_{x_0}^x f(x) dx \implies \frac{v^2 - v_0^2}{2} = \int_{x_0}^x f(x) dx.$$

Consistency check. Putting $f(x) = a = \text{const}$ in Case 3 gives $\frac{v^2 - v_0^2}{2} = a(x - x_0)$, i.e. $v^2 = u^2 + 2as$ — the familiar third equation of motion is just the constant-acceleration special case of $v dv = a dx$.

Going the other way (differentiation). If position is given as $x(t)$, then $v(t) = \frac{dx}{dt}$ and $a(t) = \frac{dv}{dt}$ — no integration, just differentiate. If v is given as a function of x , the acceleration is **not** $\frac{dv}{dx}$ but $a = v \frac{dv}{dx}$.

All quantities are SI: x in m, v in m s⁻¹, a in m s⁻², with dimensions $[L T^{-1}]$ and $[L T^{-2}]$.

Section 3: Key Procedures & Logical Framework

The logic behind $a = v \frac{dv}{dx}$. Acceleration is by definition $\frac{dv}{dt}$. But velocity may be known as a function of position rather than time. By the chain rule, $\frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt}$, and since $\frac{dx}{dt} = v$, this becomes $a = v \frac{dv}{dx}$. *Why it matters:* it lets you relate velocity and position **without ever solving for time** — exactly what stopping-distance and energy-like problems need.

Procedure A — Going down the ladder (given $x(t)$).

1. Differentiate once: $v(t) = \frac{dx}{dt}$.
2. Differentiate again: $a(t) = \frac{dv}{dt}$.
3. Answer the question — “at rest” means $v = 0$; “acceleration zero” means $a = 0$; substitute the relevant instant.

Procedure B — Acceleration depends on time, $a = f(t)$.

1. Integrate a with respect to t and add v_0 : $v(t) = v_0 + \int_0^t f(t) dt$.
2. Integrate v with respect to t and add x_0 : $x(t) = x_0 + \int_0^t v dt$.
3. Anchor both constants from the initial conditions before substituting.

Procedure C — Acceleration depends on velocity, $a = f(v)$.

1. Decide what the question wants. Time involved? Use $\frac{dv}{dt} = f(v)$, so $\int \frac{dv}{f(v)} = \int dt$.
2. Position/distance involved (no time)? Use $v \frac{dv}{dx} = f(v)$, so $\int \frac{v dv}{f(v)} = \int dx$.
3. Apply limits and solve. *Why the choice matters:* picking the wrong pairing forces an integral you cannot evaluate.

Procedure D — Acceleration depends on position, $a = f(x)$.

1. Write $v dv = f(x) dx$ (the chain-rule relation rearranged).
2. Integrate both sides with limits: $\frac{v^2 - v_0^2}{2} = \int_{x_0}^x f(x) dx$.
3. Solve for v as a function of x ; if you then need time, integrate $\frac{dx}{v(x)} = dt$.

Figure 2.8 A decision tree titled “What does a depend on?” with three branches. Branch 1, labelled “ $a = f(t)$ ”, leads to a box “integrate $a dt$ to get v , then $v dt$ to get x ”. Branch 2, labelled “ $a = f(v)$ ”, splits into two leaves: “want $v(t)$: use $dv/f(v) = dt$ ” and “want $v(x)$: use $v dv/f(v) = dx$ ”.

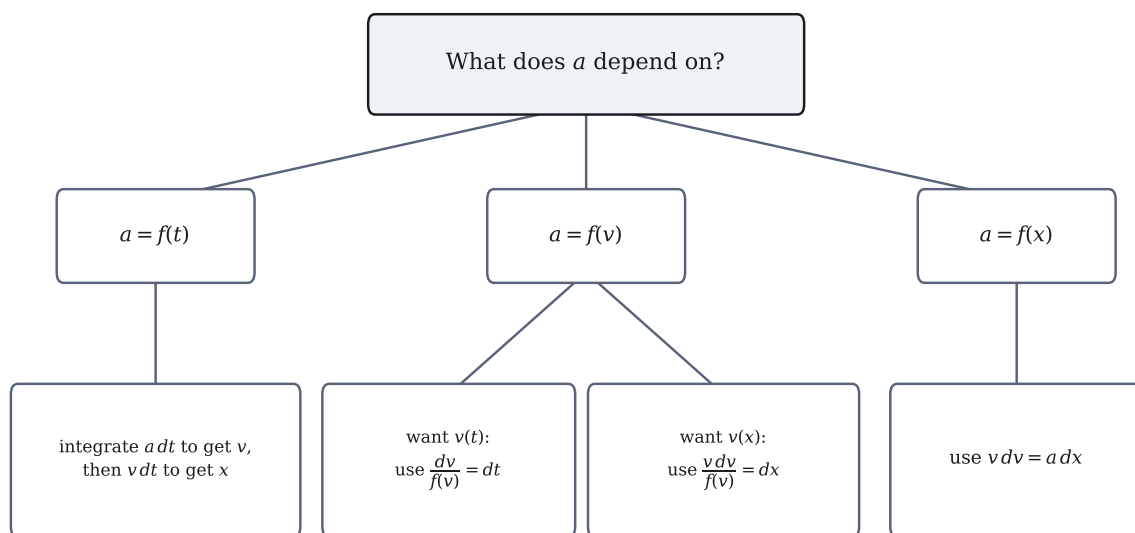


Figure 2.8 A decision tree titled “What does a depend on?” with three branches. Branch 1, labelled “ $a = f(t)$ ”, leads to a box “integrate $a dt$ to get v , then $v dt$ to get x ”. Branch 2, labelled “ $a = f(v)$ ”, splits into two leaves: “want $v(t)$: use $\frac{dv}{f(v)} = dt$ ” and “want $v(x)$: use $v dv/f(v) = dx$ ”. Branch 3, labelled

" $a = f(x)$ ", leads to a box "use $v dv = f(x) dx$ ". A side note reads "If v is given as $v(x)$, acceleration is $a = v dv/dx$, NOT dv/dx ."

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A particle starts from rest at the origin. Its acceleration varies with time as $a = 4t \text{ m s}^{-2}$. Find its velocity and displacement at $t = 3 \text{ s}$.

Given: $v_0 = 0$, $x_0 = 0$, $a = 4t$ (acceleration depends on time $\Rightarrow$ Case 1).

Solution: Integrate the acceleration for velocity:

$$v = v_0 + \int_0^t 4t dt = 0 + 2t^2 \Rightarrow v(3) = 2(3)^2 = 18 \text{ m s}^{-1}.$$

Integrate the velocity for displacement:

$$x = x_0 + \int_0^t 2t^2 dt = \frac{2}{3}t^3 \Rightarrow x(3) = \frac{2}{3}(27) = 18 \text{ m}.$$

$v(3) = 18 \text{ m s}^{-1}, \quad x(3) = 18 \text{ m}$

Note we could **not** have used $v = u + at$, because a is not constant — integration is the only honest route.

Example 2 — [NEET Speed Trap] A particle moves along the x -axis so that its velocity depends on position as $v = 4\sqrt{x}$ (SI units). Its acceleration is:

(A) not constant, growing with x (B) 8 m s^{-2} , constant (C) 4 m s^{-2} (D) 2 m s^{-2}

The trap: seeing $v = 4\sqrt{x}$, the hasty student writes $a = \frac{dv}{dx} = \frac{2}{\sqrt{x}}$ and concludes the acceleration depends on x . But $\frac{dv}{dx}$ is **not** acceleration — the correct relation is $a = v \frac{dv}{dx}$.

Fast solution:

$$a = v \frac{dv}{dx} = (4\sqrt{x}) \cdot \left(\frac{4}{2\sqrt{x}} \right) = (4\sqrt{x}) \cdot \frac{2}{\sqrt{x}} = 8 \text{ m s}^{-2}.$$

$(B) 8 \text{ m s}^{-2}, \text{ constant}$

Speed cue: whenever v^2 is linear in x (here $v^2 = 16x$), the acceleration is constant — because $v \frac{dv}{dx} = \frac{1}{2} \frac{d(v^2)}{dx}$, half the slope of the v^2 -vs- x line. Option (A) is the trap for anyone who forgets the factor of v .

Example 3 — [JEE Main Level] A particle starts from rest at $x = 1 \text{ m}$ and moves along the x -axis with a position-dependent acceleration $a = 4x \text{ m s}^{-2}$. Find its speed as it passes $x = 3 \text{ m}$.

Why it's multi-concept: acceleration depends on position, so the constant- a equations are useless; you must deploy $v dv = a dx$.

Solution: With $v_0 = 0$ at $x_0 = 1$, apply Case 3:

$$\int_0^v v dv = \int_1^3 4x dx \Rightarrow \frac{v^2}{2} = [2x^2]_1^3 = 2(9) - 2(1) = 16.$$

$$v^2 = 32 \Rightarrow v = \sqrt{32} \approx 5.66 \text{ m s}^{-1}.$$

$v \approx 5.66 \text{ m s}^{-1}$

The whole problem turned on recognising “ a depends on x ” and reaching for $v dv = a dx$ instead of a forbidden formula.

Example 4 — [JEE Advanced Level] A particle is projected along the x -axis with initial velocity u . It experiences a resistive acceleration $a = -kv^2$ (with $k > 0$ constant). Find its velocity (a) as a function of distance x travelled, and (b) as a function of time t .

Reasoning — the elegant part: acceleration depends on velocity, so this is Case 2 — and the two parts demand the *two different pairings*. Distance wants $v \frac{dv}{dx}$; time wants $\frac{dv}{dt}$.

Solution: (a) Velocity vs distance. Use $v \frac{dv}{dx} = -kv^2$:

$$\frac{dv}{v} = -k dx \Rightarrow \int_u^v \frac{dv}{v} = -k \int_0^x dx \Rightarrow \ln \frac{v}{u} = -kx.$$

$v = u e^{-kx}$

(b) Velocity vs time. Use $\frac{dv}{dt} = -kv^2$:

$$-\frac{dv}{v^2} = k dt \Rightarrow \int_u^v -\frac{dv}{v^2} = k \int_0^t dt \Rightarrow \frac{1}{v} - \frac{1}{u} = kt.$$

$v = \frac{u}{1 + ukt}$

Why this is “Advanced”: the same physical law gives two genuinely different-looking answers, and the only way to get each is to pair the acceleration with the right variable. Velocity falls *exponentially with distance* but only *reciprocally (hyperbolically) with time* — a distinction that surprises students and is a favourite Advanced discriminator.

Section 5: Practice Exercises

- [CBSE] A particle starts with velocity 2 m s^{-1} at the origin and has acceleration $a = 6t \text{ m s}^{-2}$. Find its velocity and position at $t = 2 \text{ s}$.

- [NEET] A particle moves so that $v = 3x$ (SI units). Find its acceleration at $x = 2$ m.
- [JEE Main] A particle starts from rest at $x = 2$ m with acceleration $a = 2x \text{ m s}^{-2}$. Find its speed at $x = 4$ m.
- [JEE Main] A body moving at 12 m s^{-1} decelerates with $a = -3v$ (SI units). Find the total distance it travels before (effectively) stopping.
- [JEE Advanced] A particle of initial velocity u has acceleration $a = -kv$ (with $k > 0$). Find (a) its velocity as a function of time and (b) the total distance it covers before stopping.

Answers: 1. $v = 2 + 3t^2$, so $v(2) = 14 \text{ m s}^{-1}$; $x = 2t + t^3$, so $x(2) = 12 \text{ m}$. | 2. $a = v dv/dx = 3x \cdot 3 = 9x = 18 \text{ m s}^{-2}$. | 3. $\frac{v^2}{2} = \int_2^4 2x dx = 12 \Rightarrow v = \sqrt{24} \approx 4.90 \text{ m s}^{-1}$. | 4. $v dv/dx = -3v \Rightarrow dv = -3 dx \Rightarrow v = 12 - 3x$; $v = 0$ at $x = 4 \text{ m}$. | 5. (a) $v = u e^{-kt}$; (b) $v dv/dx = -kv \Rightarrow dv = -k dx \Rightarrow \text{distance} = u/k$.

Section 6: Crack the MCQ

Q1. When the acceleration of a particle is given as a function of its position, $a = f(x)$, the most efficient relation to find velocity is: (A) $a = \frac{dv}{dt}$ (B) $v = \frac{dx}{dt}$ (C) $a = v \frac{dv}{dx}$ (D) $v = u + at$

Answer: (C). With a known as a function of x , $v dv = f(x) dx$ integrates directly in the x -variable. (A) is true but useless here — you cannot integrate $f(x) dt$ without knowing $x(t)$. (B) is just the definition of velocity. (D) is forbidden: a is not constant.

Q2. A particle's velocity varies with position as $v = c\sqrt{x}$ for a constant c . Its acceleration is: (A) zero (B) constant (C) proportional to x (D) proportional to $\sqrt{x}$

Answer: (B). $a = v \frac{dv}{dx} = c\sqrt{x} \cdot \frac{c}{2\sqrt{x}} = \frac{c^2}{2}$, a constant. (C) and (D) come from mistaking $\frac{dv}{dx}$ for acceleration. (A) ignores the motion entirely. This $v \propto \sqrt{x}$ pattern is exactly uniformly accelerated motion from rest.

Q3. For motion with a time-dependent acceleration $a = f(t)$, the change in velocity over $[t_1, t_2]$ equals: (A) $f(t_2) - f(t_1)$ (B) $\int_{t_1}^{t_2} f(t) dt$ (C) $f(t)(t_2 - t_1)$ (D) the slope of the a - t graph

Answer: (B). Change in velocity is the area under the a - t graph, i.e. the integral of a over time. (A) wrongly subtracts acceleration values. (C) assumes constant acceleration. (D) confuses area (the integral) with slope (the derivative).

Q4. A particle moves with $x(t) = t^3 - 6t^2 + 9t$ (SI). The magnitude of its acceleration at the instant it is momentarily at rest for the second time is: (A) 6 m s^{-2} (B) 12 m s^{-2} (C) 0 (D) 3 m s^{-2}

Answer: (A). $v = 3t^2 - 12t + 9 = 3(t-1)(t-3)$, zero at $t = 1$ and $t = 3$ s; the second is $t = 3$ s. Then $a = 6t - 12$, so $a(3) = 6 \text{ m s}^{-2}$. (B) is $|a|$ at $t = 0$ wrongly read; (C) assumes “at rest” forces $a = 0$ — false; (D) is the bare coefficient.

Section 7: Common Student Pitfalls & Pro-Tips

- Using the three equations of motion when a is not constant.** This is the defining error of the sub-topic. The moment a depends on t , v , or x , those formulas are invalid — integrate from the definitions instead.
- Mistaking $\frac{dv}{dx}$ for acceleration.** When v is given as a function of x , the acceleration is $a = v \frac{dv}{dx}$, never just $\frac{dv}{dx}$. Forgetting the factor of v is the most common single slip here.

- **Choosing the wrong pairing for $a = f(v)$.** If the question mentions time, pair with $\frac{dv}{dt}$; if it mentions distance or position, pair with $v\frac{dv}{dx}$. Pick wrong and you face an unsolvable integral.
- **Dropping the constant of integration.** Every indefinite integral needs an initial condition (v_0 , x_0 , or a stated v -at- x) to pin it down. Use definite integrals with proper limits to sidestep this entirely.

Pro-Tip: Open every problem with one diagnosis — “**does a depend on t , v , or x ?**” — and the correct relation follows immediately: $t \Rightarrow$ integrate $a dt$; v with time $\Rightarrow \int dv/f(v) = \int dt$; v or x without time $\Rightarrow v dv = a dx$. A second habit pays off constantly: $v\frac{dv}{dx} = \frac{1}{2} \frac{d(v^2)}{dx}$, so whenever you see acceleration in terms of x , think in v^2 — it linearises stopping-distance and “speed-at-a-point” problems in one stroke.

Section 8: Cheat Sheet Box

- **Always true:** $v = \frac{dx}{dt}$, $a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = v\frac{dv}{dx}$.
- **Master key:** $a = v\frac{dv}{dx}$ — use it whenever a or v is a function of position (relates v and x with no time).
- $a = f(t)$: $v = v_0 + \int a dt$, then $x = x_0 + \int v dt$.
- $a = f(v)$: want $v(t) \rightarrow \int \frac{dv}{f(v)} = \int dt$; want $v(x) \rightarrow \int \frac{v dv}{f(v)} = \int dx$.
- $a = f(x)$: $v dv = f(x) dx$, so $\frac{v^2 - v_0^2}{2} = \int f(x) dx$.
- **Consistency:** constant a in $v dv = a dx$ gives back $v^2 = u^2 + 2as$.
- **v given as $v(x)$:** acceleration is $a = v\frac{dv}{dx}$, **not** $\frac{dv}{dx}$.
- **Constants:** every integration needs an initial condition; prefer definite integrals with limits.
- **Mnemonic:** “Diagnose the dependence (t , v , or x), then integrate. Position-dependent? $v dv = a dx$.”

Chapter 2: Motion in a Straight Line — Round 2

Addendum

The chapter taught you the definitions, the graph ladder, the three equations, relative velocity, and the calculus of variable acceleration — but five examinable patterns sit just outside its fence. Addendum A generalises the average-speed machinery past the two-speed trap the chapter drills: n legs, uneven splits, and halts, where the harmonic-versus-arithmetic choice decides the mark. Addendum B separates the two *different* quantities both casually called “average velocity” — the time-average and the space-average — and derives the constant-acceleration formulas for each, a standing JEE discriminator. Addendum C derives the vertical-flight consequences the chapter quotes without proof: full-flight bookkeeping, time spent above a level, the root-product trick, delayed drops, and the meet-feasibility condition. Addendum D pushes relative velocity into pursuit and crossing problems where a catch condition, not just a time, is the answer. Addendum E finishes the drag story the chapter starts: terminal speed, the exponential approach, and the contrast between linear and quadratic drag.

Numbering starts at Addendum A. Nothing here introduces new theory; every result below is the chapter’s own machinery — first-principles averaging, the relative-frame equations, and the chain-rule relation — pointed at cases the chapter skipped.

Addendum A — Averaging Speeds: n Legs, Halts, and Which Mean Applies

The chapter proves the two-speed results: equal distances give the harmonic mean, equal times the arithmetic mean. JEE Main extends both in two directions — more than two legs, and journeys with halts — and the extension defeats anyone relying on memorised formulas instead of the defining ratio.

Method

There is exactly one law: average speed is total distance over total time. Every named mean is a shortcut whose validity must be checked against this ratio, never assumed.

R1 — n equal distances. Each leg has length d , speeds $v_1, v_2, \dots, v_n$. Total distance nd , total time $\sum d/v_i$, so

$$\bar{v} = \frac{nd}{d \left(\frac{1}{v_1} + \dots + \frac{1}{v_n} \right)} = \frac{n}{\frac{1}{v_1} + \dots + \frac{1}{v_n}}.$$

This is the harmonic mean of the n speeds. Units: km/h throughout if the speeds are in

km/h.

R2 — unequal distances. Return to the ratio:

$$\bar{v} = \frac{d_1 + d_2 + \cdots + d_n}{\frac{d_1}{v_1} + \frac{d_2}{v_2} + \cdots + \frac{d_n}{v_n}}.$$

No standard mean applies; compute the ratio directly. Notice slow legs contribute large times, dragging the average down.

R3 — equal or unequal times. For legs lasting $t_1, t_2, \dots$ at speeds $v_1, v_2, \dots$,

$$\bar{v} = \frac{v_1 t_1 + v_2 t_2 + \cdots}{t_1 + t_2 + \cdots},$$

which is a weighted arithmetic mean. Equal times collapse it to $(v_1 + v_2 + \cdots + v_n)/n$.

R4 — halts count. A halt contributes zero distance but nonzero time. Total time must include every stop. Forgetting this is the single most common error in board-level journey problems.

R5 — the ordering. For two speeds,

$$\frac{v_1 + v_2}{2} - \frac{2v_1 v_2}{v_1 + v_2} = \frac{(v_1 - v_2)^2}{2(v_1 + v_2)} \geq 0,$$

so harmonic $\leq$ arithmetic, with equality only when $v_1 = v_2$. Same data split by distance always scores lower than the same data split by time.

R6 — round trips. Any out-and-back along the *same route* has equal distances automatically, so $\bar{v} = 2v_1 v_2 / (v_1 + v_2)$ regardless of where the turn happens.

Example A.1 — [JEE Main Level]

A car covers three equal distances at 20, 30, and 60 km/h. Find its average speed.

By R1:

$$\bar{v} = \frac{3}{\frac{1}{20} + \frac{1}{30} + \frac{1}{60}} = \frac{3}{\frac{3+2+1}{60}} = \frac{3}{6} = 30 \text{ km/h.}$$

Cross-check. Take each leg 60 km. Times: $60/20 = 3$ h, $60/30 = 2$ h, $60/60 = 1$ h. Total: 180 km in 6 h gives 30 km/h.

Note the answer sits far closer to 20 than to 60: the slow leg eats 3 of the 6 hours.

Example A.2 — [CBSE Board Level]

A car travels 60 km at 60 km/h, halts 30 min, then covers 60 km at 40 km/h. Find the average speed.

Times: $60/60 = 1$ h, halt 0.5 h, $60/40 = 1.5$ h. Total time 3 h, total distance 120 km:

$$\bar{v} = \frac{120 \text{ km}}{3 \text{ h}} = 40 \text{ km/h.}$$

Without the halt the same computation gives $120/2.5 = 48$ km/h — the halt costs 8 km/h off the average. The trap option in an MCQ is almost always the halt-free value.

Example A.3 — [JEE Main Level]

Two drivers cover the same 120 km stretch containing two 60 km halves. Driver P drives each half at a steady speed: 30 km/h then 60 km/h. Driver Q drives the first half of the *time* at 30 km/h and the second half of the time at 60 km/h. Whose average speed is larger?

Driver P (equal distances, harmonic):

$$\bar{v}_P = \frac{2(30)(60)}{30 + 60} = \frac{3600}{90} = 40 \text{ km/h.}$$

Driver Q (equal times, arithmetic):

$$\bar{v}_Q = \frac{30 + 60}{2} = 45 \text{ km/h.}$$

Driver Q wins, 45 km/h against 40 km/h. Identical speeds, identical road — only the split variable differs. By R5 this outcome was forced before any arithmetic.

Practice — Addendum A

1. A cyclist covers four equal stretches at 10, 20, 30, and 60 km/h. Find the average speed.
2. A car goes 24 km out at 48 km/h and returns along the same road at 24 km/h. Find the average speed for the round trip.
3. A train covers 100 km at 80 km/h, halts 15 min, then covers 90 km at 90 km/h. Find the average speed.
4. Can the average speed over several legs equal *both* the harmonic and the arithmetic mean of the leg speeds? State the condition.
5. A man walks 2 km at 5 km/h and a further 2 km at 4 km/h. Find his average speed, to two decimal places.
6. A bus covers the first one-third of the *distance* at 30 km/h and the remaining two-thirds at 60 km/h. Find the average speed.

Answers: 1. $4/(1/10 + 1/20 + 1/30 + 1/60) = 4/(12/60) = 20$ km/h. 2. $2(48)(24)/(48 + 24) = 2304/72 = 32$ km/h. 3. Times $1.25 + 0.25 + 1 = 2.5$ h; $190 \text{ km}/2.5 \text{ h} = 76$ km/h. 4. Only when every leg speed is equal — R5 forces harmonic < arithmetic otherwise. 5. Harmonic: $2(5)(4)/9 = 40/9 \approx 4.44$ km/h. 6. Weighted harmonic: $3/(1/30 + 2/60) = 3/(4/60) = 45$ km/h — the slow third consumes half the total time.

Addendum B — Two Averages of Velocity: Over Time and Over Distance

“The average velocity” is always $\Delta x/T$. But JEE also deploys a second, differently built average — one that weights each *piece of the path* equally rather than each moment — and the two disagree whenever the speed changes. Confusing them costs the mark. This addendum defines both, derives their constant-acceleration forms, and collects the equal-splitting ratios for motion from rest.

Method

Definition 1 (time-average — the physical one).

$$\langle v \rangle_T = \frac{1}{T} \int_0^T v \, dt = \frac{x(T) - x(0)}{T}.$$

This is the chapter’s average velocity: endpoints only, units m/s.

Definition 2 (space-average — the weighting device).

$$\langle v \rangle_S = \frac{1}{S} \int_{\text{path}} v \, ds,$$

where the integral runs along the displacement S . Fast portions of the trip cover more distance, so they receive more weight here. This quantity appears in kinetic theory and in “averaged over the path” phrasing; it is *not* the average velocity of a journey.

Constant acceleration, from speed u to speed v . Since $v(t)$ is linear in t :

$$\langle v \rangle_T = \frac{u + v}{2}.$$

For the space average, use the chain-rule relation $v \, dv = a \, ds$, so $ds = v \, dv/a$:

$$\int_u^v v \cdot \frac{v \, dv}{a} = \frac{v^3 - u^3}{3a}, \quad S = \frac{v^2 - u^2}{2a}.$$

Dividing:

$$\langle v \rangle_S = \frac{(v^3 - u^3)/3a}{(v^2 - u^2)/2a} = \frac{2(v^2 + uv + u^2)}{3(u + v)}.$$

Comparison. Subtract the time-average from the space-average:

$$\langle v \rangle_S - \langle v \rangle_T = \frac{(v - u)^2}{6(u + v)} \geq 0,$$

with equality only when $u = v$. The space-average is always the larger — and it is symmetric in u and v , so it cannot tell speeding-up from slowing-down.

Splitting a from-rest run. With $s = \frac{1}{2}at^2$, the time to cover distance s is $t = \sqrt{2s/a}$, so:

- equal **times**: successive distances go as $1 : 3 : 5 : 7 : \dots$ (Galileo, Sub-topic 03);
- equal **distances**: successive times go as $1 : (\sqrt{2} - 1) : (\sqrt{3} - \sqrt{2}) : \dots$

The second list is the mirror image the chapter never writes down, and JEE asks for it.

Example B.1 — [JEE Main Level]

A particle accelerates uniformly from rest at $a = 2 \text{ m/s}^2$ for 6 s. Find its average velocity over the interval and its space-average over the distance covered.

Velocity after 6 s: $v = at = 12 \text{ m/s}$. Time-average: $(0 + 12)/2 = 6 \text{ m/s}$. Displacement: $S = \frac{1}{2}(2)(6)^2 = 36 \text{ m}$.

Space-average by the derived formula:

$$\langle v \rangle_S = \frac{2(12^2 + 0 + 0)}{3(12)} = \frac{288}{36} = 8 \text{ m/s}.$$

Direct verification. Parametrise: $v = 2t \text{ m/s}$, $s = t^2 \text{ m}$, so $ds = 2t dt$:

$$\int v ds = \int_0^6 (2t)(2t dt) = \frac{4t^3}{3} \Big|_0^6 = 288,$$

and $288/36 = 8 \text{ m/s}$. Both routes agree; the space-average exceeds the time-average by $(12)^2/(6 \cdot 12) = 2 \text{ m/s}$, exactly as the comparison formula demands.

Example B.2 — [JEE Advanced Level]

A body starts from rest with constant acceleration. Find the ratio of the time taken to cover the first half of the total distance S to the time taken for the second half.

From rest, $t(s) = \sqrt{2s/a}$:

$$t_1 = \sqrt{\frac{2(S/2)}{a}} = \sqrt{\frac{S}{a}}, \quad t_2 = \sqrt{\frac{2S}{a}} - \sqrt{\frac{S}{a}} = (\sqrt{2} - 1)\sqrt{\frac{S}{a}}.$$

So $t_1 : t_2 = 1 : (\sqrt{2} - 1) \approx 1 : 0.414$.

Numeric check. With $a = 2 \text{ m/s}^2$ and $S = 36 \text{ m}$: $t(18) = \sqrt{18} \approx 4.24 \text{ s}$, $t(36) = \sqrt{36} = 6 \text{ s}$, so the second half takes 1.76 s , and $4.24/1.76 \approx 2.41 = 1/0.414$.

Contrast with the time-split: the first and second halves of the *time* cover 9 m and 27 m ($1 : 3$). Same run, opposite asymmetries — the distance-split front-loads the *time*, the time-split back-loads the *distance*.

Example B.3 — [Conceptual, both exams]

Along a straight 100 m path, a cart moves at a steady 10 m/s for the first 50 m and a steady 20 m/s for the second 50 m. Find its average velocity and its space-average.

Average velocity: total time $= 50/10 + 50/20 = 5 + 2.5 = 7.5 \text{ s}$, so $\Delta x/T = 100/7.5 \approx 13.33 \text{ m/s}$.

Space-average: on each segment v is constant, so

$$\langle v \rangle_S = \frac{10 \times 50 + 20 \times 50}{100} = \frac{1500}{100} = 15 \text{ m/s.}$$

They differ — 13.33 against 15 m/s — and only the first is “the” average velocity. If an option list contains both, the question is testing whether you know which definition the stem asked for.

Practice — Addendum B

1. A particle accelerates uniformly from 4 m/s to 12 m/s. Find its time-average and space-average velocities.
2. From rest with $a = 8 \text{ m/s}^2$, find the times to cover each of the first two 8 m stretches, and their ratio.
3. A particle moves with $v = 3t^2$ (SI units). Find its average velocity over $0 \leq t \leq 2 \text{ s}$ and its space-average over the same trip.
4. For constant-acceleration motion, under what condition do the two averages coincide?
5. A body starting from rest covers distances in successive seconds in ratio 1 : 3 : 5. In what ratio does it cover the times for successive equal distances, over the first three such stretches?

Answers: 1. $\langle v \rangle_T = 8 \text{ m/s}$; $\langle v \rangle_S = 2(144 + 48 + 16)/(3 \cdot 16) = 416/48 = 26/3 \approx 8.67 \text{ m/s}$. 2. $t_1 = \sqrt{2} \approx 1.414 \text{ s}$, $t_2 = \sqrt{4} - \sqrt{2} \approx 0.586 \text{ s}$; ratio 1 : $(\sqrt{2} - 1)$. 3. Displacement $= \int_0^2 3t^2 dt = 8 \text{ m}$, so $\langle v \rangle_T = 4 \text{ m/s}$; with $s = t^3$, $\int v ds = \int_0^2 9t^4 dt = 57.6$, so $\langle v \rangle_S = 57.6/8 = 7.2 \text{ m/s}$. 4. Only when the speed never changes ($u = v$, hence $a = 0$). 5. 1 : $(\sqrt{2} - 1)$: $(\sqrt{3} - \sqrt{2})$.

Addendum C — Vertical Flight: The Results the Chapter Quotes but Never Derives

Sub-topic 03 hands you $H = u^2/2g$ and $t_{\text{flight}} = 2u/g$ as facts, and its practice sets rely on consequences — time above a height, delayed drops, whether two balls meet — that appear nowhere in the worked text. Every one follows in two or three lines from the same two equations, and each is a recurring MCQ.

Method

Take up as positive, launch from the origin at $t = 0$ with speed u , g positive constant downward, air resistance ignored.

R1 — full-flight bookkeeping. Flight time $T = 2u/g$; greatest height $H = u^2/2g$. Total distance $= 2H = u^2/g$ (up H , down H); net displacement $= 0$. Therefore

$$\bar{v}_{\text{vel}} = 0, \quad \bar{v}_{\text{speed}} = \frac{u^2/g}{2u/g} = \frac{u}{2}.$$

A full vertical flight is the extreme case of the chapter's inequality: maximum possible gap between the two averages.

R2 — time above a level h (with $h \leq H$). Solve $ut - \frac{1}{2}gt^2 = h$:

$$\frac{1}{2}gt^2 - ut + h = 0 \implies t = \frac{u \pm \sqrt{u^2 - 2gh}}{g}.$$

Duration above h : $t_2 - t_1 = 2\sqrt{u^2 - 2gh}/g$. By Vieta on the same quadratic:

$$t_1 t_2 = \frac{h}{g/2} = \frac{2h}{g}, \quad t_1 + t_2 = \frac{2u}{g} = T.$$

The sum being T is the symmetry of the flight made visible; the product $2h/g$ is *independent* of u — the favourite MCQ payoff.

R3 — delayed drops. Ball A released at $t = 0$, ball B at $t = \tau$, same point. For $t \geq \tau$:

$$s(t) = \frac{1}{2}gt^2 - \frac{1}{2}g(t - \tau)^2 = g\tau t - \frac{1}{2}g\tau^2 = g\tau \left(t - \frac{\tau}{2}\right).$$

Separation grows *linearly* at rate $g\tau$ — consistent with $a_{\text{rel}} = 0$ from Sub-topic 04, the constant relative velocity here being $g\tau$.

R4 — meet-or-not for a dropped-and-thrown pair. Ball A dropped from the top of a tower of height H ; simultaneously ball B thrown up from the foot with speed u . Both accelerate at g , so $a_{\text{rel}} = 0$ and the 100 m-style argument of the chapter applies at any height: the gap closes at the constant rate u , so

$$t_{\text{meet}} = \frac{H}{u}, \quad y_{\text{meet}} = H - \frac{1}{2}g \left(\frac{H}{u}\right)^2.$$

Meeting in mid-air needs $t_{\text{meet}} \leq T_B = 2u/g$ (B still airborne) *and* $t_{\text{meet}} \leq \sqrt{2H/g}$ (A not yet landed). Both reduce to the same condition:

$$u^2 \geq \frac{gH}{2}.$$

At equality all three times coincide: the balls meet exactly at the ground as both arrive.

Example C.1 — [JEE Main Level]

A ball is thrown up at $u = 30$ m/s ($g = 10$ m/s²). How long is it above 40 m?

Quadratic: $5t^2 - 30t + 40 = 0$, i.e. $t^2 - 6t + 8 = 0$, roots $t_1 = 2$ s, $t_2 = 4$ s.

Duration: 2 s. Checks: product $8 = 2(40)/10$; sum $6 = 2u/g = 6$; $H = 45$ m ≥ 40 m so the level is reached.

Example C.2 — [JEE Advanced Level]

Ball A is dropped from rest from a tower of height $H = 80$ m. Simultaneously, ball B is thrown upward from the base at $u = 25$ m/s. Do they meet in the air? When, and at what height?

Feasibility: $u^2 = 625$ against $gH/2 = 400$; the condition holds comfortably, so they meet airborne.

Since $a_{\text{rel}} = 0$, the 80 m gap closes at a steady 25 m/s:

$$t_{\text{meet}} = \frac{80 \text{ m}}{25 \text{ m/s}} = 3.2 \text{ s.}$$

Height: A has fallen $\frac{1}{2}(10)(3.2)^2 = 51.2$ m, so

$$y_{\text{meet}} = 80 - 51.2 = 28.8 \text{ m.}$$

Ground-frame cross-check with B: $y_B = 25(3.2) - 5(3.2)^2 = 80 - 51.2 = 28.8$ m. Both balls are airborne at 3.2 s (A lands at $\sqrt{16} = 4$ s, B at $2u/g = 5$ s), so the meeting is genuine.

Practice — Addendum C

1. A ball is thrown up at 20 m/s and caught at the launch point. Find its average velocity and average speed over the flight.
2. With $u = 40$ m/s, $g = 10$ m/s², find how long the ball is above 60 m, and verify the root-product relation.
3. Two balls are dropped from the same window, 1.5 s apart. Find their separation 4 s after the *first* release.
4. For $H = 125$ m, find the minimum launch speed u from the ground for a simultaneous dropped-and-thrown pair to meet before either lands.
5. Show that the time a ball thrown at speed u spends above *half* its maximum height equals $T/\sqrt{2}$, where $T = 2u/g$.

Answers: 1. Average velocity 0; average speed $u/2 = 10$ m/s. 2. Roots of $t^2 - 8t + 12$: $t = 2, 6$ s, duration 4 s; product $12 = 2(60)/10$, sum $8 = 2u/g$. 3. $g\tau(t - \tau/2) = 10(1.5)(4 - 0.75) = 48.75$ m — check directly: $80 - \frac{1}{2}(10)(2.5)^2 = 80 - 31.25 = 48.75$ m. 4. $u^2 = gH/2 = 625$, so $u_{\text{min}} = 25$ m/s. 5. Put $h = u^2/4g$ in R2: duration $= 2\sqrt{u^2 - 2(u^2/4g)g}/g = 2(u/\sqrt{2})/g = \sqrt{2}u/g = T/\sqrt{2}$.

Addendum D — Pursuit, Crossing Under Acceleration, and the Catch Condition

The chapter solves relative-velocity problems whose answers are times. JEE Advanced frequently wants a *verdict* — will the chaser ever catch up, will the braking cars collide — and that verdict comes from analysing the relative quadratic, not from computing one root. This addendum packages that analysis.

Method

Work in the frame of one body, exactly as Procedure A of Sub-topic 04 prescribes, with relative initial velocity u_{rel} , relative acceleration a_{rel} , and required relative displacement s_0 (the gap, or the sum of lengths for a crossing). The governing equation is

$$s_0 = u_{\text{rel}} t + \frac{1}{2} a_{\text{rel}} t^2.$$

R1 — verdict rules. - $a_{\text{rel}} > 0$: the right side is an upward-opening parabola in t , so it eventually exceeds any $s_0 > 0$. The catch *always* happens, however negative u_{rel} — the gap may widen first, but not forever. - $a_{\text{rel}} = 0$ and $u_{\text{rel}} \leq 0$: never catches. - $a_{\text{rel}} = 0$ and $u_{\text{rel}} > 0$: catches at $t = s_0/u_{\text{rel}}$ (the chapter's one-line division).

R2 — braking head-on collision test. Two bodies approach each other with speeds v_1, v_2 , gap d , each able to brake at magnitude a . Each needs $v_i^2/2a$ to stop. They collide if and only if

$$\frac{v_1^2 + v_2^2}{2a} > d.$$

If the sum of the stopping distances exceeds the gap, at least one body is still moving when the gap closes.

R3 — crossings with finite lengths. Replace s_0 by $L_1 + L_2$ (two trains) or L (train past a point) and apply R1; if $a_{\text{rel}} \neq 0$, the crossing time comes from the quadratic's *smaller* root — the larger root describes the fictitious continuation after overlap.

Example D.1 — [JEE Main Level]

Two cars approach each other on a straight road with a 60 m gap, at 20 m/s and 15 m/s. On seeing each other both brake at 5 m/s². Do they collide? If so, when and where?

Test R2: stopping distances $20^2/10 = 40$ m and $15^2/10 = 22.5$ m; sum 62.5 m $>$ 60 m — they collide.

Find when. Take car 1's initial position as origin, car 1 along +:

$$x_1 = 20t - 2.5t^2, \quad x_2 = 60 - 15t + 2.5t^2.$$

Setting $x_1 = x_2$:

$$20t - 2.5t^2 = 60 - 15t + 2.5t^2 \implies t^2 - 7t + 12 = 0 \implies t = 3 \text{ s (taking the smaller root).}$$

Place: $x_1 = 20(3) - 2.5(9) = 37.5$ m from car 1's start. Check with car 2: $60 - 15(3) + 2.5(9) = 60 - 45 + 22.5 = 37.5$ m. The root $t = 4$ s is discarded: the cars have already met and the model ceases to describe them.

Example D.2 — [JEE Advanced Level]

A thief cycles past a police post at a steady 10 m/s. The policeman starts from rest 5 s later and accelerates uniformly at 2 m/s². After how long, and at what distance from the post, is the thief caught?

Measure time t from the thief's passage. Equate distances from the post:

$$\frac{1}{2}(2)(t - 5)^2 = 10t \implies (t - 5)^2 = 10t \implies t^2 - 20t + 25 = 0.$$

$$t = \frac{20 \pm \sqrt{300}}{2} = 10 \pm 5\sqrt{3}.$$

Taking $t = 10 + 5\sqrt{3} \approx 18.66$ s. Distance: $10 \times (10 + 5\sqrt{3}) = 100 + 50\sqrt{3} \approx 186.6$ m from the post.

Verification. $(t - 5)^2 = (5 + 5\sqrt{3})^2 = 25(4 + 2\sqrt{3}) = 100 + 50\sqrt{3}$, matching $10t = 100 + 50\sqrt{3}$. The rejected root $t \approx 1.34$ s precedes the policeman's start — the algebra's ghost, discarded on physical grounds. Note the structure: $u_{\text{rel}} = -10$ m/s, $a_{\text{rel}} = +2$ m/s², so by R1 the catch was guaranteed from the outset despite the 5 s gift to the thief.

Practice — Addendum D

1. A car leads by 100 m at a constant 15 m/s; a chaser 100 m behind follows at 10 m/s accelerating at 2 m/s². Find the catch time.
2. Two trains 400 m apart approach each other at 20 m/s and 30 m/s, both braking at 2 m/s². Do they collide? If not, what gap remains when both have stopped?
3. A thief runs at 8 m/s; police start from rest 10 s later and accelerate at 3 m/s². Find the catch time and distance.
4. Chaser and leader move at 12 m/s and 10 m/s with the *same* constant acceleration 1 m/s², leader ahead by 50 m. Verdict?
5. Two trains of lengths 150 m each approach at 54 km/h and 36 km/h. Find the crossing time.

Answers: 1. $t^2 - 5t - 100 = 0$ gives $t = (5 + 5\sqrt{17})/2 \approx 12.8$ s (leader then ≈ 192 m from own start). 2. Stopping distances $100 + 225 = 325$ m $<$ 400 m: no collision; 75 m remain. 3. $3t^2 - 76t + 300 = 0$ gives $t = (76 + 8\sqrt{34})/6 \approx 20.4$ s; distance $8t \approx 164$ m. 4. Never catches: $a_{\text{rel}} = 0$ with $u_{\text{rel}} = -2$ m/s (rule R1, middle case). 5. Relative speed $15 + 10 = 25$ m/s over 300 m: 12 s.

Addendum E — Velocity-Dependent Drag: Terminal Speed and the Exponential Approach

Sub-topic 05 names terminal velocity in passing and solves $a = -kv^2$ in full, but the most examinable drag law — resistance proportional to v itself, giving the parachutist's smooth

approach to a terminal speed — is left with only a practice answer. This addendum derives that solution completely, extracts the limiting-speed shortcut, and tabulates how the two drag laws differ at long times.

Method

R1 — the terminal-speed shortcut. For $a = f(v)$ with f falling through zero, the body settles at the speed where f vanishes: setting $a = 0$ gives v_{term} instantly, no integration. For a fall with linear drag,

$$a = g - kv = 0 \implies v_{\text{term}} = \frac{g}{k},$$

with k in s^{-1} and v_{term} in m/s. Once at v_{term} the acceleration is zero, so the body stays there — a stable equilibrium.

R2 — the full solution of $a = g - kv$. Shift to the deficit $w = v - g/k$:

$$\frac{dw}{dt} = -kw \implies w = \left(u - \frac{g}{k}\right)e^{-kt},$$

so

$$v(t) = \frac{g}{k} + \left(u - \frac{g}{k}\right)e^{-kt}.$$

Integrating for position (launched at the origin):

$$x(t) = \frac{g}{k}t + \frac{u - \frac{g}{k}}{k}(1 - e^{-kt}).$$

Consistency check. As $k \rightarrow 0$, expand $e^{-kt} \approx 1 - kt + \frac{1}{2}k^2t^2$. The terms $\frac{g}{k}t$ cancel against the $-\frac{g}{k}t$ hidden in the second piece, leaving

$$x \rightarrow ut + \frac{1}{2}gt^2,$$

the drag-free result. The solution degrades gracefully to free fall.

R3 — the half-time. From rest, $v = v_{\text{term}}(1 - e^{-kt})$ reaches *half* the terminal speed at

$$t_{1/2} = \frac{\ln 2}{k}.$$

Each further half-time halves the *remaining gap*: $v/v_{\text{term}} = 1/2, 3/4, 7/8, \dots$

R4 — the $\frac{1}{2}d(v^2)/dx$ shortcut. Since $v dv/dx = \frac{1}{2}d(v^2)/dx$, any data giving v^2 as a function of x yields the acceleration immediately, and v^2 linear in x signals constant acceleration — the converse of $v^2 = u^2 + 2as$.

Example E.1 — [JEE Advanced Level]

A parachutist falls from rest with $a = g - kv$, where $g = 10 \text{ m/s}^2$ and $k = 0.5 \text{ s}^{-1}$. Find (a) the terminal speed, (b) her speed at $t = 2 \ln 2 \text{ s}$, and (c) her distance fallen by then.

(a) $v_{\text{term}} = g/k = 10/0.5 = 20 \text{ m/s}$.

(b) With $u = 0$: $v = 20(1 - e^{-0.5 \times 2 \ln 2}) = 20(1 - \frac{1}{2}) = 10 \text{ m/s}$ — exactly half of terminal, since $2 \ln 2$ is two half-times.

(c)

$$x = \frac{g}{k}t - \frac{g/k}{k}(1 - e^{-kt}) = 20(2 \ln 2) - 40(\frac{1}{2}) = 40 \ln 2 - 20 \approx 7.73 \text{ m}.$$

Sanity check. Uniform free fall would give $\frac{1}{2}(10)(1.386)^2 \approx 9.61 \text{ m}$; drag has cost her about 1.9 m — plausible, since her speed never exceeded 10 m/s .

Example E.2 — [JEE Main Level]

A particle is projected along the x -axis and obeys $v^2 = 100 - 4x$ (SI units). Find its acceleration, its stopping point, and the time to stop.

Acceleration by R4:

$$a = \frac{1}{2} \frac{d(v^2)}{dx} = \frac{1}{2}(-4) = -2 \text{ m/s}^2,$$

constant — as promised, v^2 linear in x means uniform deceleration. Initial speed $\sqrt{100} = 10 \text{ m/s}$. Stop when $v^2 = 0$: $x = 25 \text{ m}$. Cross-check with $v^2 = u^2 + 2as$: $0 = 100 + 2(-2)s$ gives $s = 25 \text{ m}$. Time: $t = u/|a| = 10/2 = 5 \text{ s}$.

The two drag laws compared

Both laws were solved in the chapter; here is what distinguishes them at long times, with the distance integrals shown.

Linear drag $a = -kv$ (projected at u): $v = ue^{-kt}$, so

$$x(t) = \int_0^t ue^{-kt'} dt' = \frac{u}{k}(1 - e^{-kt}) \rightarrow \frac{u}{k} \text{ as } t \rightarrow \infty.$$

Finite total distance: the body creeps asymptotically to rest.

Quadratic drag $a = -kv^2$: from the chapter, $v = u/(1 + ukt)$, so with $1/v = kt + 1/u$:

$$x(t) = \int_0^t \frac{u dt'}{1 + ukt'} = \frac{1}{k} \ln(1 + ukt) \rightarrow \infty.$$

Unbounded distance, growing only logarithmically: the body never truly stops. If an MCQ asks “under which resistive law does the body travel a finite total distance?”, the answer is

the linear law.

Practice — Addendum E

1. A diver falls from rest with $a = 10 - 0.25v$ (SI units). Find the terminal speed and the speed after $4 \ln 2$ s.
2. With $a = -kv$, $u = 20$ m/s, $k = 0.1 \text{ s}^{-1}$: find $v(t)$, the total distance travelled, and the distance covered in the first 10 s.
3. For $v^2 = 64 - 2x$ (SI), find a , the stopping point, and the stopping time.
4. Which drag law permits infinite total travel, and how does the distance grow at late times?
5. Starting from rest under linear drag, after how many half-times is a body within 5% of terminal speed?

Answers: 1. $v_{\text{term}} = 40$ m/s; after $4 \ln 2$ s (four half-times) $v = 40(1 - 2^{-4}) = 37.5$ m/s. 2. $v = 20e^{-0.1t}$ m/s; total distance $u/k = 200$ m; distance in 10 s $= 200(1 - e^{-1}) \approx 126.4$ m. 3. $a = -1$ m/s²; stops at $x = 32$ m; $t = 8$ s (initial speed 8 m/s). 4. Quadratic drag; $x = (1/k) \ln(1 + ukt)$ grows logarithmically. 5. Need $e^{-kt} \leq 0.05$, i.e. $t \geq (\ln 20)/k \approx 4.32$ half-times — between the fourth and fifth.

Motion in a Plane

Chapter 3 — Complete Study Material

Physics · JEE / NEET / CBSE

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Sub-topic 01: Vector Algebra and Resolution

Exam Relevance: This is the mathematical toolkit for *all* of two-dimensional physics — projectiles, forces, fields, everything downstream depends on it. CBSE Boards ask the parallelogram-law derivation (2–3 marks) and component resolution. JEE Main reliably carries 1–2 questions on resultants, dot/cross products, or unit vectors. NEET tests quick conceptual MCQs (scalar vs vector, resultant range, perpendicularity). JEE Advanced folds vector algebra silently into mechanics and electromagnetism, so fluency here pays off across the whole syllabus.

Section 1: Concept Intuition

Some quantities are fully described by a single number with a unit — the temperature in Jaipur today, the mass of a cricket ball, the time for a train journey. Tell me “35 °C” and there’s nothing left to ask. These are **scalars**: magnitude only. But other quantities are useless without a *direction*. If a friend says “the metro station is 500 m away,” your next question is *which way*? Quantities that need both a magnitude **and** a direction — and that combine by a special geometric rule — are **vectors**: displacement, velocity, acceleration, force, momentum.

That “special rule” is the whole reason vectors deserve their own algebra. Two scalars just add: $2\text{ kg} + 3\text{ kg} = 5\text{ kg}$, always. But two vectors do **not** simply add their magnitudes unless they point the same way. Walk 3 km east then 4 km north and you are not 7 km from home

— you are 5 km away, along a slanted line. Direction changes the arithmetic. Vectors add **head-to-tail** (the *triangle law*) or as the diagonal of a parallelogram (the *parallelogram law*), and that geometry is what we’re really learning.

Think about it this way... Picture two people pushing a stalled auto-rickshaw. If both push from behind in the same direction, their efforts add fully — a 300 N and 200 N push gives 500 N. If one pushes from behind and the other from the side, the rickshaw lurches off at an angle, and the *effective* combined push is less than 500 N and points somewhere between the two. The cart instinctively “computes” the parallelogram law for them. Every vector sum is just this: how do two arrows, pointing different ways, combine into one?

The reverse operation is just as important: **resolution**. Any single vector can be split into pieces along chosen directions — almost always the perpendicular x and y axes. A force pulling a sledge at an angle has one part dragging it forward (the x -component) and one part lifting it (the y -component). Resolving turns an awkward slanted vector into two clean, independent, perpendicular vectors that we can handle with ordinary arithmetic — and then recombine. Resolution is the single most powerful problem-solving move in this entire chapter.

Finally, vectors can be *multiplied* in two distinct ways, each capturing a different physical idea. The **dot (scalar) product** measures “how much of one vector lies along another” — it produces a scalar and underlies work and power. The **cross (vector) product** measures “how much two vectors are perpendicular” — it produces a new vector and underlies torque and angular momentum.

Limiting conditions & assumptions.

- Vector addition is **not** ordinary addition of magnitudes unless the vectors are **collinear** (same or opposite direction). The default is the parallelogram/component method.
- The angle θ in every formula is the angle between the two vectors when drawn **tail-to-tail** ($0 \leq \theta \leq 180^\circ$).
- A vector’s **components depend on the chosen axes**, but the vector itself does not — choose axes to make the problem easy.
- A **unit vector is dimensionless** (pure direction); a null vector has zero magnitude and undefined direction.
- The dot product yields a **scalar**; the cross product yields a **vector**. Never write one as the other.

Section 2: Key Formulas & Definitions

- **Unit vector:** $\hat{A} = \frac{\vec{A}}{|\vec{A}|}$, magnitude 1, dimensionless. Axis unit vectors: $\hat{i}, \hat{j}, \hat{k}$.
- **Component (Cartesian) form:** $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$.

Resolution of a vector $\vec{A}$ making angle θ with the x -axis:

$$A_x = A \cos \theta, \quad A_y = A \sin \theta, \quad A = \sqrt{A_x^2 + A_y^2}, \quad \tan \theta = \frac{A_y}{A_x}.$$

Addition (parallelogram law) — resultant of $\vec{A}$ and $\vec{B}$ with angle θ between them:

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}, \quad \tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

where α is the angle $\vec{R}$ makes with $\vec{A}$. **Range of resultant:** $|A - B| \leq R \leq A + B$.

Component addition: $\vec{R} = \vec{A} + \vec{B} \Rightarrow R_x = A_x + B_x, R_y = A_y + B_y$.

Scalar (dot) product:

$$\vec{A} \cdot \vec{B} = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z \quad (\text{a scalar}); \quad \cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}.$$

Properties: commutative ($\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$); $\hat{i} \cdot \hat{i} = 1, \hat{i} \cdot \hat{j} = 0$; $\vec{A} \cdot \vec{A} = A^2$; perpendicular $\Rightarrow \vec{A} \cdot \vec{B} = 0$.

Vector (cross) product:

$$\vec{A} \times \vec{B} = (AB \sin \theta) \hat{n} \quad (\text{a vector}), \quad |\vec{A} \times \vec{B}| = AB \sin \theta.$$

Direction $\hat{n}$ is perpendicular to the plane of $\vec{A}, \vec{B}$ by the right-hand rule. Anticommutative: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$; $\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$; $\vec{A} \times \vec{A} = \vec{0}$. Its magnitude equals the **area of the parallelogram** spanned by $\vec{A}$ and $\vec{B}$.

Section 3: CBSE Board Derivation — Parallelogram Law of Vector Addition

Statement: If two vectors are represented in magnitude and direction by the two adjacent sides of a parallelogram drawn from a point, their resultant is represented by the diagonal of the parallelogram drawn from that same point.

Derivation of the magnitude and direction. Let $\vec{A}$ and $\vec{B}$ be the two vectors with angle θ between them, drawn as adjacent sides OP and PQ (head-to-tail), so the resultant is $OQ = \vec{R}$.

Figure 3.1 A triangle/parallelogram construction. From O , vector A points right to P (length A). From P , vector B points up-and-right to Q at angle θ above the line OP (length B). The resultant $R = OQ$ runs from O to Q . Drop a perpendicular from Q to the extension of OP , meeting it at S . Mark $PS = B \cos \theta$ along the horizontal extension, and $QS = B \sin \theta$ vertical. Mark angle α at O between OP and OQ . Drop a perpendicular from Q to the line OP extended, meeting it at S . In the right triangle PQS : $PS = B \cos \theta, QS = B \sin \theta$. Now in the right triangle OQS , with $OS = OP + PS = A + B \cos \theta$: $OQ^2 = OS^2 + QS^2 = (A + B \cos \theta)^2 + (B \sin \theta)^2$. Expand: $R^2 = A^2 + 2AB \cos \theta + B^2 \cos^2 \theta + B^2 \sin^2 \theta = A^2 + B^2 + 2AB \cos \theta$, 3 (see next page)

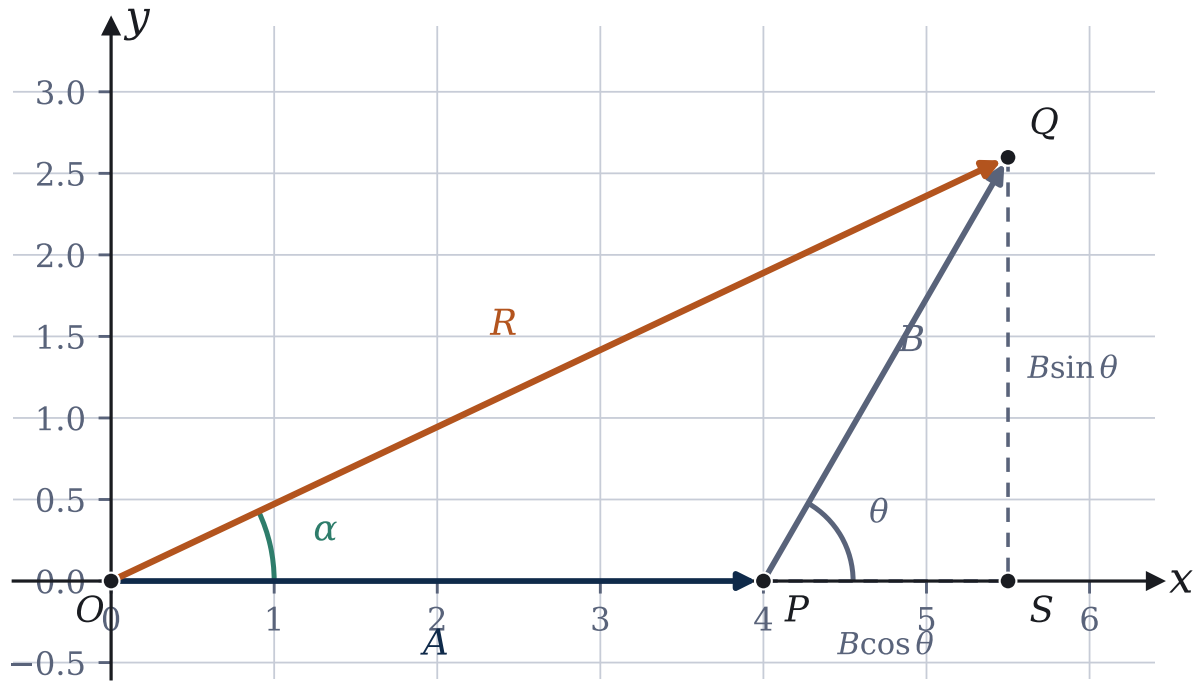


Figure 3.1 A triangle/parallelogram construction. From O , vector A points right to P (length A). From P , vector B points up-and-right to Q at angle θ above the line OP (length B). The resultant $R = OQ$ runs from O to Q . Drop a perpendicular from Q to the extension of OP , meeting it at S . Mark $PS = B \cos \theta$ along the horizontal extension, and $QS = B \sin \theta$ vertical. Mark angle α at O between OP and OQ . Drop a perpendicular from Q to the line OP extended, meeting it at S . In the right triangle PQS : $PS = B \cos \theta$, $QS = B \sin \theta$. Now in the right triangle OQS , with $OS = OP + PS = A + B \cos \theta$: $OQ^2 = OS^2 + QS^2 = (A + B \cos \theta)^2 + (B \sin \theta)^2$. Expand: $R^2 = A^2 + 2AB \cos \theta + B^2 \cos^2 \theta + B^2 \sin^2 \theta = A^2 + B^2 + 2AB \cos \theta$, 3

using $\cos^2 \theta + \sin^2 \theta = 1$. Therefore

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}.$$

Why each step: the perpendicular splits $\vec{B}$ into a part along $\vec{A}$ ($B \cos \theta$, which lengthens the base) and a part across it ($B \sin \theta$, the height); Pythagoras on the resulting right triangle then assembles the diagonal.

Direction. The resultant makes angle α with $\vec{A}$, where

$$\tan \alpha = \frac{QS}{OS} = \frac{B \sin \theta}{A + B \cos \theta}.$$

Useful special cases (read straight off the formula):

- $\theta = 0^\circ$: $R = A + B$ (maximum) — vectors aligned.
- $\theta = 180^\circ$: $R = |A - B|$ (minimum) — vectors opposed.
- $\theta = 90^\circ$: $R = \sqrt{A^2 + B^2}$ — the Pythagorean case.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A child pulls a toy cart with a string, applying a force of 20 N directed at 60° above the horizontal. Find the horizontal and vertical components of this force.

Given: $F = 20$ N, $\theta = 60^\circ$ to the horizontal.

Solution: Resolve along the horizontal (x) and vertical (y) axes:

$$F_x = F \cos \theta = 20 \cos 60^\circ = 20 \times 0.5 = 10 \text{ N},$$

$$F_y = F \sin \theta = 20 \sin 60^\circ = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3} \approx 17.3 \text{ N}.$$

$$F_x = 10 \text{ N (horizontal)}, \quad F_y \approx 17.3 \text{ N (vertical)}$$

The horizontal part drags the cart forward; the vertical part tends to lift it. A quick check: $\sqrt{F_x^2 + F_y^2} = \sqrt{100 + 300} = \sqrt{400} = 20$ N, recovering the original force. ✓

Example 2 — [NEET Speed Trap] Two forces of magnitudes 3 N and 4 N act at a point. Which of the following can **not** be the magnitude of their resultant?

(A) 1 N (B) 5 N (C) 7 N (D) 8 N

The trap: students compute one specific case (often the perpendicular $\sqrt{3^2 + 4^2} = 5$) and stop, never checking the *allowed range*.

Fast solution: The resultant of two vectors always lies in the band

$$|A - B| \leq R \leq A + B \Rightarrow |4 - 3| \leq R \leq 4 + 3 \Rightarrow 1 \leq R \leq 7.$$

So 1, 5 (90° case), and 7 (aligned) are all achievable, but $8 > 7$ is impossible.

$$\boxed{(D) \ 8 \text{ N}}$$

Speed cue: for any “which value of the resultant is possible/impossible” question, immediately write the range $[|A - B|, A + B]$ and test against it — usually no real calculation is needed.

Example 3 — [JEE Main Level] A person walks 6 m due east, then 8 m in a direction 60° north of east. Find the magnitude and direction of the net displacement, and verify it by two methods.

Why it’s multi-concept: it links component resolution, the Pythagorean magnitude, and the parallelogram-law formula as an independent cross-check.

Method 1 — components. Take east as $+x$, north as $+y$.

$$\vec{A} = (6, 0), \quad \vec{B} = (8 \cos 60^\circ, 8 \sin 60^\circ) = (4, 4\sqrt{3}).$$

$$\vec{R} = (6 + 4, 0 + 4\sqrt{3}) = (10, 4\sqrt{3}).$$

$$R = \sqrt{10^2 + (4\sqrt{3})^2} = \sqrt{100 + 48} = \sqrt{148} \approx 12.2 \text{ m.}$$

$$\tan \alpha = \frac{4\sqrt{3}}{10} = \frac{6.93}{10} = 0.693 \Rightarrow \alpha \approx 34.7^\circ \text{ north of east.}$$

Method 2 — parallelogram law (angle between the two legs is 60°):

$$R = \sqrt{6^2 + 8^2 + 2(6)(8) \cos 60^\circ} = \sqrt{36 + 64 + 96(0.5)} = \sqrt{148} \approx 12.2 \text{ m. } \checkmark$$

$$\boxed{R \approx 12.2 \text{ m at } 34.7^\circ \text{ north of east}}$$

Both routes agree — the component method is usually faster for three or more vectors, while the parallelogram formula is handy for exactly two.

Example 4 — [JEE Advanced Level] Given $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{B} = \hat{i} - \hat{j} + 2\hat{k}$, find (a) $\vec{A} \cdot \vec{B}$, (b) the angle between them, (c) $\vec{A} \times \vec{B}$, (d) a unit vector perpendicular to both, and (e) the area of the parallelogram they span.

Solution: (a) $\vec{A} \cdot \vec{B} = (2)(1) + (3)(-1) + (-1)(2) = 2 - 3 - 2 = -3$.

(b) $|\vec{A}| = \sqrt{4 + 9 + 1} = \sqrt{14}$, $|\vec{B}| = \sqrt{1 + 1 + 4} = \sqrt{6}$.

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|} = \frac{-3}{\sqrt{84}} = \frac{-3}{2\sqrt{21}} \approx -0.327 \Rightarrow \theta \approx 109^\circ.$$

(The negative dot product already told us the angle is obtuse.)

(c) Using the determinant:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 1 & -1 & 2 \end{vmatrix} = \hat{i}(6 - 1) - \hat{j}(4 + 1) + \hat{k}(-2 - 3) = 5\hat{i} - 5\hat{j} - 5\hat{k}.$$

(d) $|\vec{A} \times \vec{B}| = \sqrt{25 + 25 + 25} = 5\sqrt{3}$, so a unit normal is

$$\hat{n} = \frac{5\hat{i} - 5\hat{j} - 5\hat{k}}{5\sqrt{3}} = \frac{\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}.$$

(e) Area of the parallelogram $= |\vec{A} \times \vec{B}| = 5\sqrt{3} \approx 8.66$ square units.

$\vec{A} \cdot \vec{B} = -3, \quad \theta \approx 109^\circ, \quad \vec{A} \times \vec{B} = 5\hat{i} - 5\hat{j} - 5\hat{k}, \quad \hat{n} = \frac{\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}, \quad \text{Area} = 5\sqrt{3}$
--

Sanity check via $|\vec{A} \times \vec{B}| = AB \sin \theta = \sqrt{14}\sqrt{6} \sin 109^\circ = \sqrt{84}(0.946) \approx 8.67. \checkmark$

Section 5: Practice Exercises

1. **[CBSE]** A velocity of 100 m/s is directed 30° above the horizontal. Find its horizontal and vertical components.
2. **[NEET]** Two forces of 5 N and 12 N act at a point. Find the maximum and minimum possible magnitudes of their resultant.
3. **[JEE Main]** For $\vec{A} = 3\hat{i} + 4\hat{j}$, find $|\vec{A}|$ and the unit vector along $\vec{A}$.
4. **[JEE Main]** Find the angle that $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ makes with the x -axis.
5. **[JEE Advanced]** Given $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{B} = 3\hat{i} + 2\hat{j} + \hat{k}$, find $\vec{A} \times \vec{B}$ and the area of the parallelogram they span.

Answers: 1. $v_x = 50\sqrt{3} \approx 86.6$ m/s, $v_y = 50$ m/s | 2. max 17 N, min 7 N | 3. $|\vec{A}| = 5$, $\hat{A} = 0.6\hat{i} + 0.8\hat{j}$ | 4. $\cos \theta = 1/\sqrt{3}$, $\theta \approx 54.7^\circ$ | 5. $\vec{A} \times \vec{B} = -4\hat{i} + 8\hat{j} - 4\hat{k}$, area $= 4\sqrt{6} \approx 9.8$ square units.

Section 6: Crack the MCQ

Q1. Which of the following is a vector quantity? (A) work (B) speed (C) displacement (D) energy

Answer: (C). Displacement has both magnitude and direction and obeys the parallelogram law. (A) work and (D) energy are scalars (work is itself a dot product, which yields a scalar). (B) speed is the magnitude of velocity — the *direction* is exactly what speed throws away, making it scalar. The trap is associating “physics-y” quantities with vectors by default.

Q2. Two vectors have magnitudes 3 and 4 units. The magnitude of their resultant **cannot** be: (A) 1 (B) 5 (C) 7 (D) 8

Answer: (D). The resultant is bounded by $|4-3| = 1$ and $4+3 = 7$. 8 lies outside this range. (A) is the minimum (antiparallel), (C) the maximum (parallel), and (B) the perpendicular case $\sqrt{3^2 + 4^2} = 5$. Students who only check the perpendicular value miss that 8 exceeds the maximum.

Q3. The scalar product of two non-zero vectors is zero. The vectors must be: (A) parallel (B) antiparallel (C) perpendicular (D) identical

Answer: (C). $\vec{A} \cdot \vec{B} = AB \cos \theta = 0$ with $A, B \neq 0$ forces $\cos \theta = 0$, i.e. $\theta = 90^\circ$. (A) parallel gives $\cos 0^\circ = 1$ (maximum dot product), (B) antiparallel gives $\cos 180^\circ = -1$, and (D) identical vectors give $\vec{A} \cdot \vec{A} = A^2 \neq 0$. Only perpendicularity zeroes the dot product.

Q4. For two vectors, the magnitude of $\vec{A} \times \vec{B}$ equals $\vec{A} \cdot \vec{B}$ when the angle between them is: (A) 0° (B) 45° (C) 90° (D) 180°

Answer: (B). Setting $AB \sin \theta = AB \cos \theta$ gives $\tan \theta = 1$, so $\theta = 45^\circ$. (A) 0° : cross product is zero, dot is maximum. (C) 90° : dot is zero, cross is maximum — they're opposite extremes. (D) 180° : cross is zero again. Only at 45° do the sin and cos contributions match.

Section 7: Common Student Pitfalls & Pro-Tips

- **Adding magnitudes arithmetically.** $|\vec{A} + \vec{B}| \neq A + B$ in general — that equality holds *only* when the vectors are parallel. For everything else, use components or the parallelogram law.
- **Swapping sin and cos in resolution.** When θ is measured from the x -axis, the x -component uses cos and the y -component uses sin. If the angle is measured from the y -axis instead, they switch — always note *which axis* the angle is from.
- **Mixing up the two products.** The dot product is a **scalar** (write a number); the cross product is a **vector** (write components). Writing $\vec{A} \cdot \vec{B}$ as a vector, or $\vec{A} \times \vec{B}$ as a number, loses marks instantly.
- **Forgetting the cross product is anticommutative.** $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ — order matters, and the sign carries physical meaning (e.g. torque direction).
- **Giving a unit vector units.** A unit vector is pure direction, magnitude 1, dimensionless.

Pro-Tip: When in doubt, **resolve into components** — it converts every messy geometry problem into clean axis-by-axis arithmetic, and it scales effortlessly to three or more vectors where the parallelogram law gets clumsy. Keep two instant sanity checks handy: the resultant must satisfy $|A - B| \leq R \leq A + B$, and after resolving, $\sqrt{A_x^2 + A_y^2}$ must return the original magnitude. And remember the physical meaning to pick the right product: **dot** = “how much along” (cos); **cross** = “how much across” (sin).

Section 8: Cheat Sheet Box

- **Scalar:** magnitude only. **Vector:** magnitude + direction; adds by the parallelogram/triangle law.
- **Unit vector:** $\hat{A} = \vec{A}/|\vec{A}|$ (dimensionless); axis vectors $\hat{i}, \hat{j}, \hat{k}$.
- **Resolution:** $A_x = A \cos \theta$, $A_y = A \sin \theta$; $A = \sqrt{A_x^2 + A_y^2}$, $\tan \theta = A_y/A_x$.
- **Resultant:** $R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$; $\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$; range $|A - B| \leq R \leq A + B$.
- **Dot product:** $\vec{A} \cdot \vec{B} = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z$ (scalar); $\perp \Rightarrow 0$.
- **Cross product:** $|\vec{A} \times \vec{B}| = AB \sin \theta$, $\perp$ to both (right-hand rule); $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$; equals parallelogram area.
- *Mnemonic: "cos for components and dot; sin for cross."*

Sub-topic 02: Vector Products (Dot and Cross)

Exam Relevance: Vector multiplication is a workhorse that resurfaces across mechanics, electromagnetism, and modern physics. CBSE Boards ask the definitions, properties, and a work-done (dot product) numerical. JEE Main reliably carries a question on angle-between-vectors, projection, area, or the determinant form of the cross product. NEET tests quick conceptual MCQs (which quantity is a dot/cross product, perpendicularity). JEE Advanced exploits the scalar and vector triple products and uses these operations inside torque, angular momentum, and Biot-Savart problems.

Section 1: Concept Intuition

We already know how to *add* vectors. But multiplication is where vectors get genuinely interesting — because there are **two** completely different ways to multiply them, each answering a different physical question.

The first asks: “*how much do these two vectors agree — how much does one point along the other?*” This is the **dot product** (also called the scalar product). It’s largest when the vectors are parallel, shrinks as they turn apart, and is exactly **zero when they’re perpendicular**. The answer it returns is a plain number — a scalar — because “how much do they agree” is a quantity without direction. The physical archetype is **work**: only the part of a force that lies *along* the displacement does any work. Push a loaded thela (handcart) horizontally and the horizontal effort moves it; the vertical part of your push does nothing useful. Work is force “dotted” with displacement.

The second asks the opposite: “*how much do these two vectors disagree — how much does one point across the other?*” This is the **cross product** (vector product). It’s **zero when the vectors are parallel** and largest when they’re perpendicular — the mirror image of the dot product. And its answer is itself a *vector*, pointing perpendicular to both originals. The physical archetype is **torque**: a force applied along a spanner does nothing; a force applied *across* it, at right angles to the handle, turns the bolt most effectively. The “turning” lives in a direction perpendicular to both the lever arm and the force — which is exactly where the cross product points.

Think about it this way... Two cricketers pulling a heavy roller with ropes. The **dot product** is about *cooperation along the motion* — how much of each pull actually drags the roller forward (use $\cos\theta$: full when aligned, nothing when sideways). The **cross product** is about *twisting* — if their pulls aren’t aligned, there’s a turning tendency, strongest when the pulls are at right angles (use $\sin\theta$: nothing when aligned, full when perpendicular). One product measures “togetherness,” the other measures “crossness.” That single contrast — cos versus sin, scalar versus vector — is the heart of this sub-topic.

A neat consequence worth carrying: for the same two vectors, $\vec{A} \cdot \vec{B}$ and $|\vec{A} \times \vec{B}|$ are like a seesaw. When one is at its maximum (parallel $\rightarrow$ dot maximal, cross zero), the other is at its minimum, and vice versa (perpendicular $\rightarrow$ dot zero, cross maximal). Together they fully capture the geometric relationship between two vectors.

Limiting conditions & assumptions.

- The angle θ is always the angle between the vectors drawn **tail-to-tail** ($0 \leq \theta \leq 180^\circ$).

- The dot product is a **scalar** (write a number); the cross product is a **vector** (write components/direction). They are never interchangeable.
 - The cross product is meaningfully defined in **three dimensions** (its result needs a perpendicular direction); in 2-D it's treated as pointing out of/into the plane.
 - $\vec{A} \cdot \vec{B} = 0$ (with $A, B \neq 0$) means **perpendicular**; $\vec{A} \times \vec{B} = \vec{0}$ (with $A, B \neq 0$) means **parallel or antiparallel** — opposite conditions, easy to confuse.
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Section 2: Key Formulas & Definitions

Scalar (dot) product:

$$\vec{A} \cdot \vec{B} = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z \quad (\text{scalar}).$$

- Commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$; distributive: $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$.
- $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$; $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$; $\vec{A} \cdot \vec{A} = A^2$.
- **Angle:** $\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}$. **Projection (scalar component) of $\vec{A}$ along $\vec{B}$:** $\frac{\vec{A} \cdot \vec{B}}{B} = A \cos \theta$.

Vector (cross) product:

$$\vec{A} \times \vec{B} = (AB \sin \theta) \hat{n}, \quad |\vec{A} \times \vec{B}| = AB \sin \theta \quad (\text{vector}),$$

with $\hat{n}$ perpendicular to the plane of $\vec{A}, \vec{B}$ by the right-hand rule.

- Anticommutative: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$; distributive over addition; $\vec{A} \times \vec{A} = \vec{0}$.
- $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$ (cyclic); reverse order flips the sign.
- **Determinant form:**

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}.$$

- **Geometric meaning:** $|\vec{A} \times \vec{B}|$ equals the area of the parallelogram with sides $\vec{A}, \vec{B}$; the triangle's area is $\frac{1}{2}|\vec{A} \times \vec{B}|$.

Scalar triple product: $\vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = \text{volume of the parallelepiped};$
 $= 0 \Leftrightarrow$ **the three vectors are coplanar.**

Vector triple product (BAC-CAB rule): $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B}).$

Physical applications: dot — work $W = \vec{F} \cdot \vec{d}$, power $P = \vec{F} \cdot \vec{v}$, flux $\Phi = \vec{E} \cdot \vec{A}$; cross — torque $\vec{\tau} = \vec{r} \times \vec{F}$, angular momentum $\vec{L} = \vec{r} \times \vec{p}$, magnetic force $\vec{F} = q \vec{v} \times \vec{B}$.

Section 3: CBSE Board Derivation — Component Forms of the Products

Both component formulas come from one idea: **multiply out using the distributive law, then use the known products of the unit vectors.**

Dot product in component form. Write $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$. Distribute:

$$\vec{A} \cdot \vec{B} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) \cdot (B_x\hat{i} + B_y\hat{j} + B_z\hat{k}).$$

Expanding gives nine terms. Now apply $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ (parallel unit vectors) and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$ (perpendicular ones). Every “mixed” term dies, and only the three “like” terms survive:

$$\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y + A_zB_z.$$

Why it works: perpendicular directions contribute nothing to “agreement,” so only matching components multiply through.

Cross product in component form. Distribute the same way, but now use $\hat{i} \times \hat{i} = \vec{0}$ (parallel) and the cyclic rules $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$ (with a sign flip when reversed). Collecting terms:

$$\vec{A} \times \vec{B} = (A_yB_z - A_zB_y)\hat{i} + (A_zB_x - A_xB_z)\hat{j} + (A_xB_y - A_yB_x)\hat{k},$$

which is exactly the expansion of the determinant

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}.$$

Why it works: the “self” products vanish (a vector has no perpendicular component along itself), and the cyclic unit-vector products bookkeep exactly the alternating-sign pattern of a determinant.

Figure 3.2 Two side-by-side panels. Left (dot product): vectors A and B from a common origin with angle θ between; drop a perpendicular from the tip of B onto the line of A , marking the projection “ $B \cos\theta$ ” along A , captioned “dot = (length of A) $\times$ (projection of B on A)”. Right (cross product): vectors A and B in a horizontal plane with angle θ ; a vertical arrow $C = A \times B$ rising perpendicular out of the plane; curved arrow showing right-hand fingers curling from A to B , captioned “ $|A \times B| = AB \sin\theta = \text{area of the shaded parallelogram}$ ”. (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A constant force $\vec{F} = (3\hat{i} + 4\hat{j})\text{ N}$ acts on a body that undergoes a displacement $\vec{d} = (2\hat{i} + 3\hat{j})\text{ m}$. Find the work done.

Given: Work is the dot product $W = \vec{F} \cdot \vec{d}$.

Solution:

$$W = \vec{F} \cdot \vec{d} = (3)(2) + (4)(3) = 6 + 12 = 18\text{ J}.$$

$$W = 18\text{ J}$$

Work is a scalar (joules), and the dot product is what automatically picks out the part of the force that acts along the displacement — no need to find the angle separately.

dot = (length of A) $\times$ (projection of B on A)

$|A \times B| = AB \sin \theta = \text{area of shaded parallelogram}$

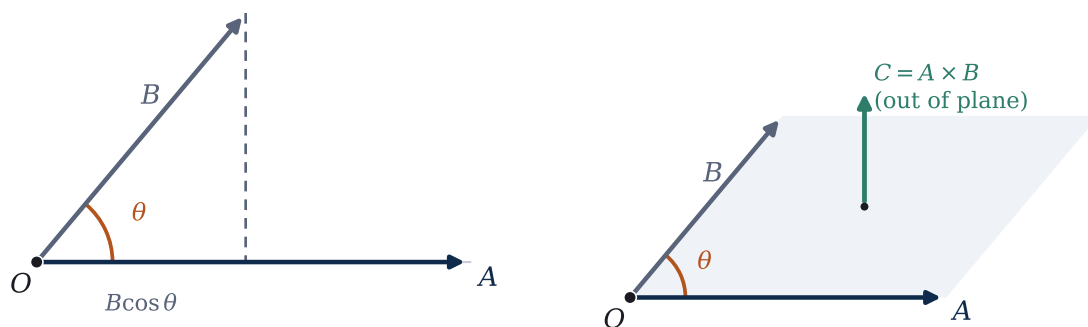


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Example 2 — [NEET Speed Trap] A force of 5 N is applied at the end of a spanner of length 0.2 m, at 30° to the handle. The magnitude of the torque about the bolt is:

(A) 0.5 N·m (B) 0.87 N·m (C) 1.0 N·m (D) 0 N·m

The trap: torque is a **cross** product, so it uses $\sin \theta$ — but under pressure students reach for $\cos \theta$ (the work/dot habit), giving the wrong 0.87.

Fast solution: $\tau = rF \sin \theta = (0.2)(5) \sin 30^\circ = (0.2)(5)(0.5) = 0.5 \text{ N}\cdot\text{m}$.

(A) 0.5 N·m

Speed cue: turning effects (torque, area, angular momentum, magnetic force) are *cross* products → use $\sin$; “along” effects (work, power, flux) are *dot* products → use $\cos$. Option (B) is the $\cos 30^\circ$ trap; only the $\sin$ form is physical for torque.

Example 3 — [JEE Main Level] For $\vec{A} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{B} = 6\hat{i} - 3\hat{j} + 2\hat{k}$, find (a) the angle between them, (b) the projection of $\vec{A}$ on $\vec{B}$, and (c) the magnitude of the component of $\vec{A}$ perpendicular to $\vec{B}$.

Why it’s multi-concept: it uses the dot product (angle, projection) and the cross product (perpendicular component) on the same pair of vectors.

Solution:

$$\vec{A} \cdot \vec{B} = (2)(6) + (2)(-3) + (1)(2) = 12 - 6 + 2 = 8.$$

$$|\vec{A}| = \sqrt{4 + 4 + 1} = 3, \quad |\vec{B}| = \sqrt{36 + 9 + 4} = 7.$$

$$(a) \cos \theta = \frac{8}{(3)(7)} = \frac{8}{21} \approx 0.381 \Rightarrow \theta \approx 67.6^\circ.$$

$$(b) \text{Projection of } \vec{A} \text{ on } \vec{B} = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} = \frac{8}{7} \approx 1.14 \text{ units.}$$

$$(c) \text{Perpendicular component magnitude} = A \sin \theta = \frac{|\vec{A} \times \vec{B}|}{|\vec{B}|}. \text{ Compute the cross product:}$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 6 & -3 & 2 \end{vmatrix} = \hat{i}(4 + 3) - \hat{j}(4 - 6) + \hat{k}(-6 - 12) = 7\hat{i} + 2\hat{j} - 18\hat{k}.$$

$$|\vec{A} \times \vec{B}| = \sqrt{49 + 4 + 324} = \sqrt{377} \approx 19.4, \quad A \sin \theta = \frac{19.4}{7} \approx 2.77 \text{ units.}$$

$\theta \approx 67.6^\circ, \quad \text{proj} \approx 1.14, \quad \text{perp comp} \approx 2.77$

Cross-check: $\sqrt{(\text{proj})^2 + (\text{perp})^2} = \sqrt{1.14^2 + 2.77^2} = \sqrt{1.30 + 7.67} = \sqrt{8.97} \approx 3 = |\vec{A}|$.
✓ — the along and across parts reassemble the original magnitude.

Example 4 — [JEE Advanced Level] Given $\vec{A} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{B} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{C} = 3\hat{i} + \hat{j} + 2\hat{k}$, find (a) the scalar triple product $\vec{A} \cdot (\vec{B} \times \vec{C})$ and state the volume of the parallelepiped and whether the three are coplanar, and (b) the vector triple product $\vec{A} \times (\vec{B} \times \vec{C})$ using the BAC-CAB rule.

Solution: (a) The scalar triple product is the determinant of the components:

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} 1 & 2 & -1 \\ 2 & -1 & 1 \\ 3 & 1 & 2 \end{vmatrix}.$$

Expand along the first row:

$$\begin{aligned} &= 1[(-1)(2) - (1)(1)] - 2[(2)(2) - (1)(3)] + (-1)[(2)(1) - (-1)(3)] \\ &= 1(-2 - 1) - 2(4 - 3) + (-1)(2 + 3) = -3 - 2 - 5 = -10. \end{aligned}$$

The volume of the parallelepiped is $|-10| = 10$ cubic units. Since the scalar triple product is **non-zero**, the three vectors are **not coplanar**.

(b) Apply $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$.

$$\vec{A} \cdot \vec{C} = (1)(3) + (2)(1) + (-1)(2) = 3 + 2 - 2 = 3,$$

$$\vec{A} \cdot \vec{B} = (1)(2) + (2)(-1) + (-1)(1) = 2 - 2 - 1 = -1.$$

$$\begin{aligned} \vec{A} \times (\vec{B} \times \vec{C}) &= 3\vec{B} - (-1)\vec{C} = 3\vec{B} + \vec{C} = 3(2\hat{i} - \hat{j} + \hat{k}) + (3\hat{i} + \hat{j} + 2\hat{k}) \\ &= (6\hat{i} - 3\hat{j} + 3\hat{k}) + (3\hat{i} + \hat{j} + 2\hat{k}) = 9\hat{i} - 2\hat{j} + 5\hat{k}. \end{aligned}$$

$$\boxed{\vec{A} \cdot (\vec{B} \times \vec{C}) = -10 \text{ (volume 10, not coplanar), } \vec{A} \times (\vec{B} \times \vec{C}) = 9\hat{i} - 2\hat{j} + 5\hat{k}}$$

The BAC-CAB rule sidesteps computing $\vec{B} \times \vec{C}$ first — recognising that the triple product collapses to a combination of $\vec{B}$ and $\vec{C}$ is the time-saving insight.

Section 5: Practice Exercises

- [CBSE]** Find the work done by a force $\vec{F} = (5\hat{i} + 2\hat{j})$ N in producing a displacement $\vec{d} = (3\hat{i} + 4\hat{j})$ m.
- [NEET]** Two vectors have magnitudes 4 and 5 units with 30° between them. Find the magnitude of their cross product.
- [JEE Main]** Find the angle between $\vec{A} = \hat{i} + 2\hat{j}$ and $\vec{B} = 2\hat{i} + \hat{j}$.
- [JEE Main]** Find the projection of $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$ on $\vec{B} = \hat{i} + \hat{j} + \hat{k}$.
- [JEE Advanced]** Determine whether $\vec{A} = \hat{i} + \hat{j} + \hat{k}$, $\vec{B} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{C} = \hat{i} + 4\hat{j} + 9\hat{k}$ are coplanar, using the scalar triple product.

Answers: 1. 23J | 2. 10 units | 3. $\cos \theta = 4/5$, $\theta \approx 36.9^\circ$ | 4. $6/\sqrt{3} = 2\sqrt{3} \approx 3.46$ | 5. scalar triple product = 2 $\neq$ 0, so **not** coplanar (volume 2).

Section 6: Crack the MCQ

Q1. The scalar product $\vec{A} \cdot \vec{B}$ is maximum when the angle between the vectors is: (A) 0° (B) 90° (C) 180° (D) 45°

Answer: (A). $\vec{A} \cdot \vec{B} = AB \cos \theta$ is largest when $\cos \theta = 1$, i.e. $\theta = 0^\circ$ (parallel). (B) 90° makes the dot product zero. (C) 180° gives the most negative value ($-AB$), a minimum, not maximum. (D) 45° gives an intermediate value. Students sometimes confuse “maximum magnitude” with the antiparallel case (C) — but that’s the minimum (most negative).

Q2. Which of the following physical quantities is defined as a cross product? (A) work (B) torque (C) power (D) kinetic energy

Answer: (B). Torque $\vec{\tau} = \vec{r} \times \vec{F}$ is a vector (cross) product. (A) work $\vec{F} \cdot \vec{d}$ and (C) power $\vec{F} \cdot \vec{v}$ are dot products (scalars). (D) kinetic energy is a scalar not formed from any vector product. The cross-product giveaway is that the result has a *direction* (torque points along the axis of rotation).

Q3. For two non-zero vectors, $\vec{A} \times \vec{B} = \vec{0}$. The vectors must be: (A) perpendicular (B) parallel or antiparallel (C) at 45° (D) equal in magnitude

Answer: (B). $|\vec{A} \times \vec{B}| = AB \sin \theta = 0$ with $A, B \neq 0$ forces $\sin \theta = 0$, i.e. $\theta = 0^\circ$ or 180° . (A) perpendicular is the condition for the *dot* product to vanish — the exact opposite case, and the most common mix-up. (C) and (D) place no such constraint on the cross product.

Q4. If the scalar triple product $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$ for three non-zero vectors, then the vectors are: (A) mutually perpendicular (B) coplanar (C) identical (D) all parallel to one axis

Answer: (B). The scalar triple product is the volume of the parallelepiped they span; zero volume means they lie in a single plane — they are coplanar. (A) mutually perpendicular vectors give a *non-zero* volume (a rectangular box). (C) identical and (D) all-parallel are special sub-cases that happen to be coplanar, but they are not what “ $= 0$ ” implies in general; the precise, complete condition is coplanarity.

Section 7: Common Student Pitfalls & Pro-Tips

- **Using cos where sin belongs (and vice versa).** Dot products take $\cos \theta$; cross products take $\sin \theta$. Reaching for the wrong one is the single most common slip — especially writing torque with $\cos$.
- **Treating the dot as a vector or the cross as a scalar.** $\vec{A} \cdot \vec{B}$ is a number; $\vec{A} \times \vec{B}$ is a vector with components and a direction. Mislabelling loses marks even when the arithmetic is right.
- **Forgetting anticommutativity of the cross product.** $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ — order matters, and the sign carries the physical direction (e.g. which way a torque turns).
- **Sign error in the 3×3 determinant.** The middle ($\hat{j}$) term carries a **minus** sign in the cofactor expansion; dropping it is a frequent error.
- **Confusing the two “zero” conditions.** $\vec{A} \cdot \vec{B} = 0 \Rightarrow$ perpendicular; $\vec{A} \times \vec{B} = \vec{0} \Rightarrow$

parallel. They are opposites.

Pro-Tip: Decide which product you need from the *physics*, not the algebra: “how much **along**” (work, power, flux, projection) → **dot, use cos**; “how much **across** / turning / area” (torque, angular momentum, magnetic force) → **cross, use sin**.

Two instant checks: the **sign** of $\vec{A} \cdot \vec{B}$ tells you whether the angle is acute (+) or obtuse (−); and for three vectors, a zero scalar triple product is the fastest possible **coplanarity test** — no need to draw anything.

Section 8: Cheat Sheet Box

- **Dot:** $\vec{A} \cdot \vec{B} = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z$ (scalar); proj of $\vec{A}$ on $\vec{B} = \vec{A} \cdot \vec{B} / B$; $\perp \Rightarrow 0$; max at $\theta = 0^\circ$.
- **Cross:** $|\vec{A} \times \vec{B}| = AB \sin \theta$ (vector, right-hand rule); determinant form; $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$; $\parallel \Rightarrow \vec{0}$; = parallelogram area ($\frac{1}{2}$ for a triangle).
- **Unit vectors:** $\hat{i} \cdot \hat{i} = 1$, $\hat{i} \cdot \hat{j} = 0$; $\hat{i} \times \hat{j} = \hat{k}$ (cyclic), $\hat{i} \times \hat{i} = \vec{0}$.
- **Scalar triple:** $\vec{A} \cdot (\vec{B} \times \vec{C})$ = parallelepiped volume; $= 0 \Rightarrow$ coplanar.
- **Vector triple:** $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$ (**BAC-CAB**).
- **Applications:** $W = \vec{F} \cdot \vec{d}$, $P = \vec{F} \cdot \vec{v}$, $\Phi = \vec{E} \cdot \vec{A}$ (dot); $\vec{\tau} = \vec{r} \times \vec{F}$, $\vec{L} = \vec{r} \times \vec{p}$, $\vec{F} = q\vec{v} \times \vec{B}$ (cross).
- **Mnemonic:** “**Dot agrees (cos); cross turns (sin).**”

Sub-topic 03: Motion in a Plane — Two-Dimensional Kinematics

Exam Relevance: This is the general framework that projectile and circular motion are special cases of. CBSE Boards ask the vector form of the equations of motion and position/velocity in component form (2-3 marks). JEE Main carries questions on 2-D constant-acceleration motion, velocity/position vectors, and trajectory shape. NEET tests the independence-of-motions concept and quick component-magnitude MCQs. JEE Advanced uses it for radius-of-curvature and time-of-closest-direction problems.

Section 1: Concept Intuition

In a straight line, a single number with a sign told us everything: position x , velocity v , acceleration a . But the moment a body can move *sideways as well as forward* — a footballer curling across the pitch, a boat angling across a river, a charged particle drifting in a field — one number is no longer enough. We need **vectors**: a position vector $\vec{r}$ that points from the origin to the particle, a velocity vector $\vec{v}$ tangent to its path, and an acceleration vector $\vec{a}$ that can point in any direction at all.

Here is the single most powerful idea of the whole chapter, and it's deceptively simple: **motion in a plane is just two independent one-dimensional motions happening at once**. The x -part of the motion evolves entirely on its own, governed only by the x -components of velocity and acceleration; the y -part does likewise. They never interfere. They share exactly one thing — the clock, t . This is the **principle of independence of perpendicular motions**, and it's what lets us take every tool from straight-line kinematics and apply it twice, once per axis.

Think about it this way... Imagine a carrom striker sliding across the board while you *separately* nudge it sideways with a steady puff of air. Its forward glide and its sideways drift don't argue with each other — the forward distance after two seconds is whatever the forward motion alone would give, and the sideways distance is whatever the sideways push alone would give. The actual path you see is just these two simple motions *superposed*. Resolve any 2-D motion into its x and y stories, solve each as an ordinary 1-D problem, then recombine the answers with vector addition. That recipe never fails.

Because vectors carry direction, the familiar equations of motion get a vector upgrade. For **constant acceleration in a plane**,

$$\vec{v} = \vec{v}_0 + \vec{a}t, \quad \vec{r} = \vec{r}_0 + \vec{v}_0t + \frac{1}{2}\vec{a}t^2.$$

These look identical to the 1-D versions — because they *are*, just promoted to vectors. Split each into components and you get two independent sets of scalar equations. Projectile motion is simply the special case where $\vec{a} = -g\hat{j}$ (gravity, vertical only); uniform circular motion is the case where $\vec{a}$ always points to a centre. Master the general framework and the special cases become effortless.

One subtlety worth flagging early: in two dimensions, **speed and the magnitude of velocity are equal, but average speed and the magnitude of average velocity are not** — because average velocity uses the straight-line displacement while average speed uses the actual (possibly curved) path length, exactly as in Chapter 2.

Limiting conditions & assumptions.

- The independence principle and the vector equations of motion above require **constant acceleration**. If $\vec{a}$ varies, return to $\vec{v} = d\vec{r}/dt$ and $\vec{a} = d\vec{v}/dt$ and integrate.
 - Components are taken along a chosen set of perpendicular axes; the *vector* is independent of that choice, but pick axes that simplify the problem (often along and across the acceleration).
 - “Independence” means the x and y motions don’t influence each other’s *components* — they are still tied together through the common time t .
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Section 2: Key Formulas & Definitions

Position, displacement, velocity, acceleration in the plane:

$$\begin{aligned}\vec{r} &= x\hat{i} + y\hat{j}, & \Delta\vec{r} &= \Delta x\hat{i} + \Delta y\hat{j}, \\ \vec{v} &= \frac{d\vec{r}}{dt} = v_x\hat{i} + v_y\hat{j}, & \vec{a} &= \frac{d\vec{v}}{dt} = a_x\hat{i} + a_y\hat{j}.\end{aligned}$$

Magnitudes and directions:

$$|\vec{v}| = \sqrt{v_x^2 + v_y^2}, \quad \tan \theta_v = \frac{v_y}{v_x}; \quad |\vec{r}| = \sqrt{x^2 + y^2}, \quad \tan \theta_r = \frac{y}{x}.$$

Average vs instantaneous:

$$\vec{v}_{\text{avg}} = \frac{\Delta\vec{r}}{\Delta t}, \quad \vec{a}_{\text{avg}} = \frac{\Delta\vec{v}}{\Delta t}.$$

Equations of motion (constant $\vec{a}$) in vector and component form:

$$\begin{aligned}\vec{v} &= \vec{v}_0 + \vec{a}t \Rightarrow v_x = v_{0x} + a_x t, \quad v_y = v_{0y} + a_y t, \\ \vec{r} &= \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2 \Rightarrow x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2, \quad y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2.\end{aligned}$$

Special cases: projectile ($a_x = 0$, $a_y = -g$); uniform circular motion ($\vec{a}$ of constant magnitude v^2/r , always toward the centre). All quantities SI: m, m/s, m/s².

Section 3: CBSE Board Derivation — Equations of Motion in a Plane

Start from the definition of acceleration as a vector, assumed **constant**: $\vec{a} = \frac{d\vec{v}}{dt}$. Integrating from the initial velocity $\vec{v}_0$ at $t = 0$ to $\vec{v}$ at time t :

$$\int_{\vec{v}_0}^{\vec{v}} d\vec{v} = \int_0^t \vec{a} dt \Rightarrow \vec{v} - \vec{v}_0 = \vec{a}t \Rightarrow \boxed{\vec{v} = \vec{v}_0 + \vec{a}t}.$$

Next use $\vec{v} = \frac{d\vec{r}}{dt}$ and substitute the expression just found:

$$\int_{\vec{r}_0}^{\vec{r}} d\vec{r} = \int_0^t \vec{v} dt = \int_0^t (\vec{v}_0 + \vec{a} t) dt = \vec{v}_0 t + \frac{1}{2} \vec{a} t^2,$$

$$\boxed{\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a} t^2}.$$

Why this proves independence. A vector equation is really two scalar equations, one per axis. Reading off the $\hat{i}$ and $\hat{j}$ parts:

$$v_x = v_{0x} + a_x t, \quad x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2,$$

$$v_y = v_{0y} + a_y t, \quad y = y_0 + v_{0y} t + \frac{1}{2} a_y t^2.$$

Notice the x -equations contain **only** x -quantities and the y -equations contain **only** y -quantities. Neither set mentions the other. That is the mathematical statement of the **independence of perpendicular motions**: each axis runs its own 1-D show, coupled to the other solely through the shared time t .

Figure 3.3 A curved 2-D trajectory in the x - y plane from point A (position vector $\vec{r}_0$ drawn from origin O) to point B (position vector $\vec{r}$). The displacement vector $\Delta \vec{r} = \vec{r} - \vec{r}_0$ connects A to B. At A, the velocity vector $\vec{v}_0$ is drawn tangent to the path; at B, the velocity $\vec{v}$ is tangent. Dashed lines drop from A and B to the x -axis showing the independent x -progress, and across to the y -axis showing the independent y -progress. (see next page)

Eliminating time on each axis also lets you reuse $v_x^2 = v_{0x}^2 + 2a_x \Delta x$ and $v_y^2 = v_{0y}^2 + 2a_y \Delta y$ when time isn't wanted — again, one independent relation per direction.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A particle starts from the origin with velocity $\vec{v}_0 = 4\hat{i}$ m/s and moves with constant acceleration $\vec{a} = 3\hat{j}$ m/s². Find its velocity and position after 2 s, and the magnitude of each.

Given: $\vec{r}_0 = 0$, $\vec{v}_0 = 4\hat{i}$, $\vec{a} = 3\hat{j}$, $t = 2$ s.

Solution:

$$\vec{v} = \vec{v}_0 + \vec{a} t = 4\hat{i} + (3\hat{j})(2) = 4\hat{i} + 6\hat{j} \text{ m/s}, \quad |\vec{v}| = \sqrt{4^2 + 6^2} = \sqrt{52} \approx 7.2 \text{ m/s}.$$

$$\vec{r} = \vec{v}_0 t + \frac{1}{2} \vec{a} t^2 = (4)(2)\hat{i} + \frac{1}{2}(3)(2)^2\hat{j} = 8\hat{i} + 6\hat{j} \text{ m}, \quad |\vec{r}| = \sqrt{8^2 + 6^2} = 10 \text{ m}.$$

$$\boxed{\vec{v} = 4\hat{i} + 6\hat{j} \text{ m/s } (\approx 7.2 \text{ m/s}), \quad \vec{r} = 8\hat{i} + 6\hat{j} \text{ m } (10 \text{ m})}$$

The x -motion (uniform, 4 m/s) and the y -motion (accelerating from rest) were solved separately and then combined.

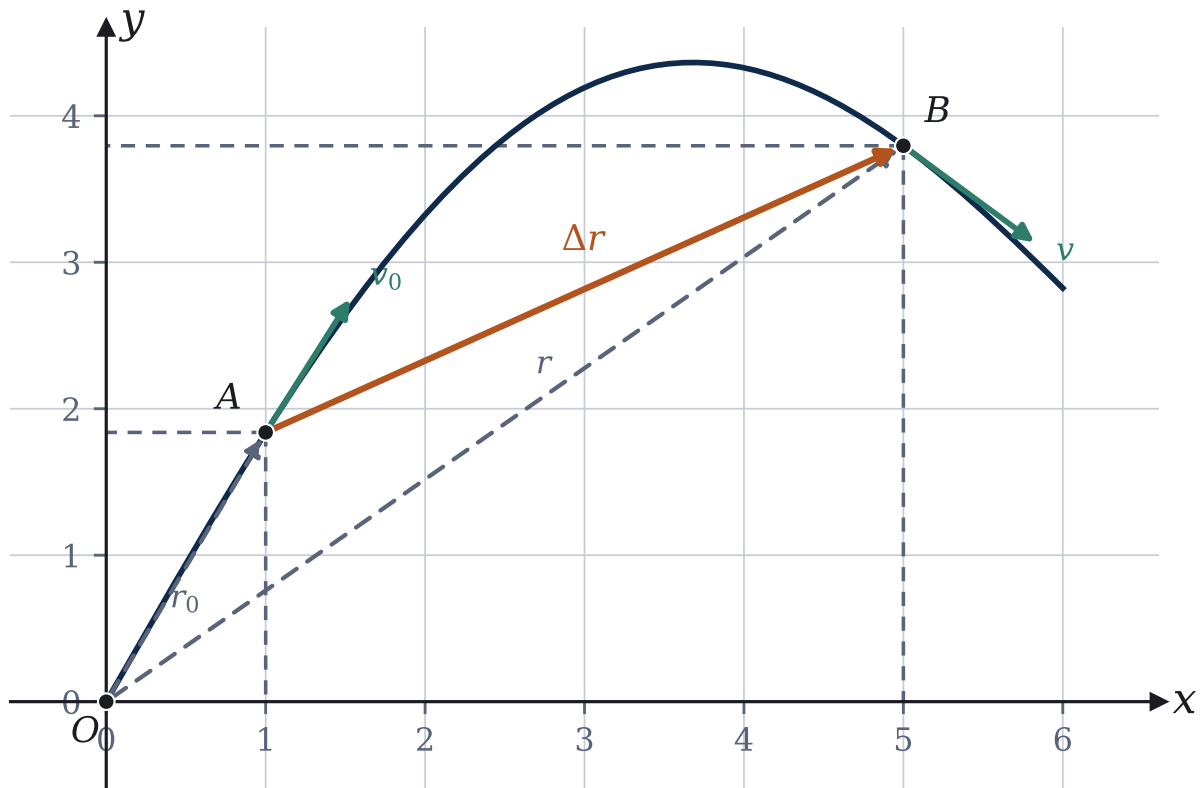


Figure 3.3 A curved 2-D trajectory in the x - y plane from point A (position vector r_0 drawn from origin O) to point B (position vector r). The displacement vector $\Delta r = r - r_0$ connects A to B. At A, the velocity vector v_0 is drawn tangent to the path; at B, the velocity v is tangent. Dashed lines drop from A and B to the x -axis showing the independent x -progress, and across to the y -axis showing the independent y -progress.

Example 2 — [NEET Speed Trap] At an instant a particle's velocity is $\vec{v} = 6\hat{i} + 8\hat{j}$ m/s. Its speed is:

- (A) 10 m/s (B) 14 m/s (C) 2 m/s (D) 48 m/s

The trap: adding the components ($6 + 8 = 14$) — but perpendicular components combine by Pythagoras, not by simple addition.

Fast solution: $|\vec{v}| = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ m/s.

(A) 10 m/s

Speed cue: spot the 6-8-10 (and 3-4-5, 5-12-13) right-triangle families instantly. Option (B) is the add-the-components trap; (C) subtracts them; (D) multiplies — all signs of forgetting that the components are perpendicular.

Example 3 — [JEE Main Level] A particle is launched with velocity $\vec{v}_0 = 2\hat{i} + 4\hat{j}$ m/s and experiences a constant acceleration $\vec{a} = -2\hat{j}$ m/s². Find (a) the time at which its velocity is directed purely along the x -axis, and (b) its speed at that instant.

Why it's multi-concept: it requires tracking both velocity components and recognising the geometric condition “velocity along x ” as $v_y = 0$.

Solution:

$$v_x = 2 \text{ m/s (constant),} \quad v_y = 4 + (-2)t = 4 - 2t.$$

(a) Velocity is along the x -axis when $v_y = 0$: $4 - 2t = 0 \Rightarrow t = 2$ s.

(b) At $t = 2$ s: $v_x = 2$, $v_y = 0$, so the speed is $|\vec{v}| = 2$ m/s.

$t = 2$ s, speed = 2 m/s

This is also the instant of **minimum speed**, since the speed $\sqrt{2^2 + (4 - 2t)^2}$ is smallest exactly when the y -component vanishes — the same logic as a projectile being slowest (horizontal) at the top of its arc.

Example 4 — [JEE Advanced Level] A particle is projected with velocity $\vec{v}_0 = 10\hat{i}$ m/s and moves under constant acceleration $\vec{a} = -10\hat{j}$ m/s² (a horizontal launch under gravity). Find the radius of curvature of its path at $t = 1$ s ($g = 10$ m/s²).

Reasoning: Radius of curvature relates to the component of acceleration **perpendicular** to the velocity: $r = \frac{v^2}{a_\perp}$. So we need the speed and the perpendicular slice of $\vec{a}$ at that instant.

Solution: At $t = 1$ s:

$$v_x = 10 \text{ m/s,} \quad v_y = 0 + (-10)(1) = -10 \text{ m/s,} \quad |\vec{v}| = \sqrt{10^2 + 10^2} = 10\sqrt{2} \text{ m/s.}$$

The velocity points at 45° below the horizontal. The acceleration $\vec{a} = -10\hat{j}$ (straight down) makes 45° with this velocity, so its perpendicular component is

$$a_\perp = a \cos 45^\circ = 10 \times \frac{1}{\sqrt{2}} = \frac{10}{\sqrt{2}} = 5\sqrt{2} \text{ m/s}^2.$$

Therefore

$$r = \frac{v^2}{a_{\perp}} = \frac{(10\sqrt{2})^2}{5\sqrt{2}} = \frac{200}{5\sqrt{2}} = \frac{40}{\sqrt{2}} = 20\sqrt{2} \approx 28.3 \text{ m.}$$

$$\boxed{r = 20\sqrt{2} \approx 28.3 \text{ m}}$$

The trick is splitting $\vec{a}$ into a part *along* the velocity (which changes speed) and a part *across* it (which curves the path); only the across-part sets the radius of curvature.

Section 5: Practice Exercises

1. **[CBSE]** A particle starts from the origin with $\vec{v}_0 = 5\hat{i}$ m/s and constant acceleration $\vec{a} = 2\hat{j}$ m/s². Find its position vector after 3 s.
2. **[NEET]** A particle's velocity is $\vec{v} = 9\hat{i} + 12\hat{j}$ m/s. Find its speed.
3. **[JEE Main]** A body has initial velocity $\vec{v}_0 = 3\hat{i} + 4\hat{j}$ m/s and acceleration $\vec{a} = 0.4\hat{i} + 0.3\hat{j}$ m/s². Find its velocity after 10 s and its magnitude.
4. **[JEE Main]** A particle moves so that $x = 3t$ and $y = 4t - 5t^2$ (SI units). Find its velocity vector at $t = 1$ s and the acceleration.
5. **[JEE Advanced]** A particle is projected horizontally at 20 m/s under gravity ($g = 10$ m/s²). Find the radius of curvature of its path at the instant of projection ($t = 0$).

Answers: 1. $\vec{r} = 15\hat{i} + 9\hat{j}$ m | 2. 15 m/s | 3. $\vec{v} = 7\hat{i} + 7\hat{j}$ m/s, $|\vec{v}| = 7\sqrt{2} \approx 9.9$ m/s | 4. $\vec{v} = 3\hat{i} - 6\hat{j}$ m/s, $\vec{a} = -10\hat{j}$ m/s² | 5. at $t = 0$ velocity is horizontal (20 m/s) and g is fully perpendicular, so $r = v^2/g = 400/10 = 40$ m.

Section 6: Crack the MCQ

Q1. For a particle moving in a plane with constant acceleration, the horizontal and vertical components of its motion are: (A) always equal (B) independent of each other (C) always opposite (D) coupled through acceleration

Answer: (B). The component equations contain only their own axis's quantities, so each axis evolves independently — linked only by the shared time. (A) and (C) impose relationships that don't generally hold. (D) is the common misconception: the two motions are *not* coupled through acceleration; the x and y accelerations act separately on their own components.

Q2. A particle has velocity components $v_x = 3$ m/s and $v_y = 4$ m/s. Its speed is: (A) 7 m/s (B) 1 m/s (C) 5 m/s (D) 12 m/s

Answer: (C). $|\vec{v}| = \sqrt{3^2 + 4^2} = 5$ m/s. (A) adds the components, (B) subtracts them, (D) multiplies — all ignore that perpendicular components combine by Pythagoras, not arithmetic.

Q3. The position of a particle is $\vec{r} = (3t)\hat{i} + (2t^2)\hat{j}$ (SI units). Its acceleration is: (A) $3\hat{i}$ m/s² (B) $4\hat{j}$ m/s² (C) $2\hat{j}$ m/s² (D) 0

Answer: (B). $\vec{v} = d\vec{r}/dt = 3\hat{i} + 4t\hat{j}$, so $\vec{a} = d\vec{v}/dt = 4\hat{j} \text{ m/s}^2$. (A) is the (constant) velocity's x -part, not acceleration. (C) forgets the factor of 2 from differentiating $2t^2$ twice. (D) wrongly assumes constant velocity — the y -motion is accelerated.

Q4. A particle moving in a plane with constant acceleration that is **not** parallel to its initial velocity follows a path that is: (A) a straight line (B) a circle (C) a parabola (D) a spiral

Answer: (C). Constant acceleration with a velocity component across it produces a parabolic path (projectile motion is the everyday example). (A) a straight line results only when $\vec{a}$ is parallel (or antiparallel) to $\vec{v}_0$. (B) a circle needs an acceleration that constantly *changes direction* toward a centre, not a constant one. (D) spirals require changing speed *and* changing radius — not constant acceleration.

Section 7: Common Student Pitfalls & Pro-Tips

- **Adding velocity components arithmetically.** v_x and v_y are perpendicular; the speed is $\sqrt{v_x^2 + v_y^2}$, never $v_x + v_y$.
- **Coupling the two axes.** A change in the x -motion does not alter the y -motion. Solve each axis as a separate 1-D problem; the only shared variable is time.
- **Forgetting the equations of motion are now vectors.** $\vec{v} = \vec{v}_0 + \vec{a}t$ must be applied component-by-component when $\vec{v}_0$ and $\vec{a}$ point in different directions.
- **Confusing average speed with the magnitude of average velocity.** In 2-D these differ whenever the path is curved — average velocity uses straight-line displacement, average speed uses path length.
- **Differentiation slips on position functions.** When given $x(t)$ and $y(t)$, differentiate *each* once for velocity and twice for acceleration; don't forget the chain-rule factors.

Pro-Tip: Treat every plane-motion problem as **two 1-D problems sharing a clock**. Resolve along convenient perpendicular axes (often along and across the acceleration), write the four scalar equations, solve each axis with ordinary kinematics, then recombine with vector addition at the end. Recognising a problem as “constant acceleration in a plane” instantly unlocks $\vec{v} = \vec{v}_0 + \vec{a}t$ and $\vec{r} = \vec{r}_0 + \vec{v}_0t + \frac{1}{2}\vec{a}t^2$ — the same equations you already trust from straight-line motion.

Section 8: Cheat Sheet Box

- **Vectors:** $\vec{r} = x\hat{i} + y\hat{j}$, $\vec{v} = d\vec{r}/dt = v_x\hat{i} + v_y\hat{j}$, $\vec{a} = d\vec{v}/dt = a_x\hat{i} + a_y\hat{j}$.
- **Magnitude/direction:** $|\vec{v}| = \sqrt{v_x^2 + v_y^2}$, $\tan \theta = v_y/v_x$.
- **Constant- $\vec{a}$ equations:** $\vec{v} = \vec{v}_0 + \vec{a}t$; $\vec{r} = \vec{r}_0 + \vec{v}_0t + \frac{1}{2}\vec{a}t^2$ (apply per component).
- **Independence:** x -motion uses only x -quantities, y -motion only y -quantities; shared variable = time.
- **Special cases:** projectile $a_x = 0$, $a_y = -g$; uniform circular motion $\vec{a}$ toward centre, magnitude v^2/r .

- **Radius of curvature:** $r = v^2/a_{\perp}$ (perpendicular component of acceleration).
- *Mnemonic:* **“Two motions, one clock — resolve, solve, recombine.”**

Sub-topic 04: Projectile Motion

Exam Relevance: One of the single most-tested topics in all of Class 11 mechanics. CBSE Boards reliably ask a derivation (time of flight, range, or maximum height) plus a numerical. JEE Main carries 1-2 questions every year — trajectory, range/height relations, velocity at an instant, or horizontal projectiles from a height. NEET loves the conceptual traps (velocity and acceleration at the top, complementary angles). JEE Advanced layers it with inclined planes, radius of curvature, and target-hitting constraints.

Section 1: Concept Intuition

Throw a cricket ball across the maidan and it traces a graceful arch before landing. That arch — the **trajectory** — looks complicated, but the secret of projectile motion is that it is really *two simple, independent motions happening at the same time*, stitched together only by the clock.

Once the ball leaves your hand, the only force on it (ignoring air) is gravity, and gravity pulls **straight down**. It has no sideways component at all. So think of the motion in two separate channels:

- **Horizontally:** there is *no* force, so by Newton's first law the horizontal velocity never changes. The ball drifts sideways at a perfectly constant speed — uniform motion.
- **Vertically:** gravity acts fully, so the vertical motion is exactly **free fall** — the ball rises, slows, stops momentarily, and comes back down, all governed by g .

The crucial idea is **independence**: what happens horizontally has *zero* effect on what happens vertically, and vice-versa. They share only one thing — the elapsed **time** t . This is why you resolve the launch velocity u into $u \cos \theta$ (horizontal) and $u \sin \theta$ (vertical) and then treat each channel with the tools you already know: constant-velocity motion sideways, the free-fall equations up-and-down.

Think about it this way... Fire one bullet horizontally from a gun and, at the very same instant, simply drop a second bullet from the same height. Which lands first? They land *together*. The fired bullet's huge horizontal speed does nothing to delay its fall, because horizontal motion and vertical motion don't talk to each other. Gravity pulls both downward identically. The fired one merely travels far sideways *while* falling the same way the dropped one does. Internalise this and projectile motion stops being scary — it's just "free fall with a sideways drift."

When you combine uniform horizontal drift with accelerated vertical fall, the path that emerges is a **parabola** — that's the shape of every (air-free) fountain jet, hose stream, and basketball shot. The familiar quantities follow naturally: how long it stays up (**time of flight**), how high it climbs (**maximum height**), and how far it travels (**horizontal range**), with the celebrated result that range is greatest at a launch angle of 45° .

Limiting conditions & assumptions.

- **Air resistance is neglected.** In reality drag reduces both range and height and breaks the up-down symmetry; the ideal parabola assumes a vacuum-like situation.
- g is taken **constant** (uniform near the Earth's surface) and the Earth flat over the flight.
- The standard symmetric formulas (T , H , R) assume **launch and landing at the**

same height. If the projectile lands higher or lower, you must go back to the component equations.

- Horizontal velocity is constant **only** because there's no horizontal force — introduce wind or thrust and this breaks.

Section 2: Key Formulas & Definitions

For a projectile launched from ground level with speed u at angle θ above the horizontal (take up as positive, $a_x = 0$, $a_y = -g$):

Velocity components (initial): $u_x = u \cos \theta$ (constant throughout), $u_y = u \sin \theta$.

Velocity at time t : $v_x = u \cos \theta$, $v_y = u \sin \theta - gt$, $|\vec{v}| = \sqrt{v_x^2 + v_y^2}$.

Position at time t : $x = (u \cos \theta) t$, $y = (u \sin \theta) t - \frac{1}{2}gt^2$.

Equation of trajectory (a parabola):

$$y = x \tan \theta - \frac{g x^2}{2u^2 \cos^2 \theta}.$$

Time of flight: $T = \frac{2u \sin \theta}{g}$.

Maximum height: $H = \frac{u^2 \sin^2 \theta}{2g}$.

Horizontal range: $R = \frac{u^2 \sin 2\theta}{g}$; $R_{\max} = \frac{u^2}{g}$ at $\theta = 45^\circ$.

Handy relations:

- Complementary angles θ and $(90^\circ - \theta)$ give the **same range**.
- For complementary angles, $H_1 + H_2 = \frac{u^2}{2g}$ (independent of θ).
- $R = 4H \cot \theta$; at 45° , $R_{\max} = 4H$.

Horizontal projectile from a height h (launched horizontally with speed u , so $u_y = 0$):

$$T = \sqrt{\frac{2h}{g}}, \quad R = u\sqrt{\frac{2h}{g}}, \quad y = \frac{g x^2}{2u^2}.$$

All quantities SI: u in m/s, g in m/s², H, R, h in m, T in s.

Section 3: CBSE Board Derivations

Set up axes with the origin at the launch point, x horizontal, y vertical (up positive). The launch velocity resolves into $u_x = u \cos \theta$ and $u_y = u \sin \theta$, with $a_x = 0$ and $a_y = -g$. Horizontal and vertical motions are then handled separately, sharing only the time t .

Figure 3.4 A parabolic trajectory from origin O rising to apex then landing at point B on the ground. At O , the launch velocity u is drawn at angle θ to the horizontal, with dashed components $u \cos \theta$ (horizontal) and $u \sin \theta$ (vertical). The apex is marked at height H , with a small horizontal velocity arrow ($u \cos \theta$) and a note " $v_y = 0$ ". The ground distance O -to- B is marked R . Velocity arrows shown at launch (up-right), apex (horizontal), and landing (down-right, symmetric to launch). (see next page)

Time of flight T . The projectile returns to launch level when $y = 0$ again:

$$y = (u \sin \theta) t - \frac{1}{2} g t^2 = 0 \Rightarrow t (u \sin \theta - \frac{1}{2} g t) = 0.$$

The root $t = 0$ is the launch instant; the other gives the flight time:

$$T = \frac{2u \sin \theta}{g}.$$

Why: the vertical motion is symmetric free fall — time up equals time down, so the total is twice the time to the top.

Maximum height H . At the apex the vertical velocity is momentarily zero, $v_y = u \sin \theta - g t_1 = 0$, giving $t_1 = \frac{u \sin \theta}{g}$ (exactly $T/2$). Substituting into the vertical displacement:

$$H = (u \sin \theta) t_1 - \frac{1}{2} g t_1^2 = \frac{u^2 \sin^2 \theta}{g} - \frac{u^2 \sin^2 \theta}{2g} = \frac{u^2 \sin^2 \theta}{2g}.$$

Why: only the vertical channel determines how high it climbs; the horizontal drift is irrelevant to height.

Horizontal range R . Horizontal motion is uniform, so distance = (horizontal speed) $\times$ (time of flight):

$$R = (u \cos \theta) T = (u \cos \theta) \cdot \frac{2u \sin \theta}{g} = \frac{2u^2 \sin \theta \cos \theta}{g} = \frac{u^2 \sin 2\theta}{g}.$$

Since $\sin 2\theta$ is largest ($= 1$) at $2\theta = 90^\circ$, the range is maximum at $\theta = 45^\circ$, where $R_{\max} = u^2/g$.

Equation of trajectory. Eliminate t between the two position equations. From $x = (u \cos \theta) t$, we get $t = \frac{x}{u \cos \theta}$. Substitute into $y = (u \sin \theta) t - \frac{1}{2} g t^2$:

$$y = (u \sin \theta) \frac{x}{u \cos \theta} - \frac{g}{2} \left(\frac{x}{u \cos \theta} \right)^2 = x \tan \theta - \frac{g x^2}{2u^2 \cos^2 \theta}.$$

This has the form $y = ax - bx^2$ — the equation of a **parabola**, confirming the arched path.

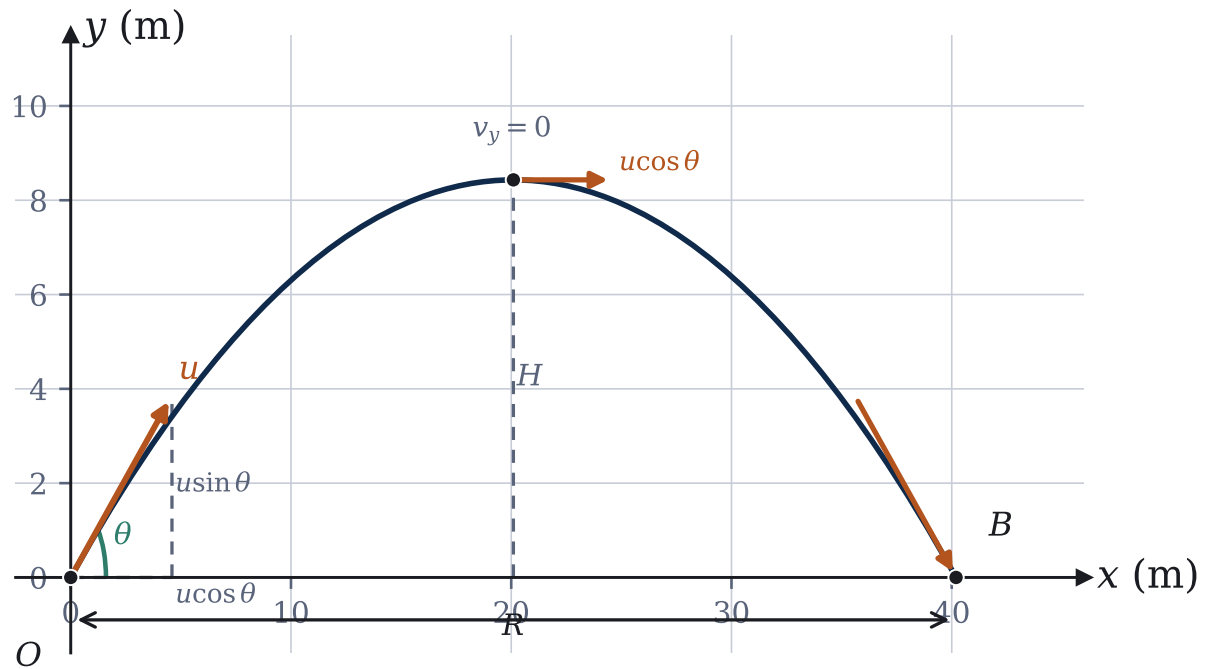


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Section 4: Worked Examples

Example 1 — [CBSE Board Level] A ball is projected from the ground with a speed of 20 m/s at 30° above the horizontal. Taking $g = 10 \text{ m/s}^2$, find (a) the time of flight, (b) the maximum height, and (c) the horizontal range.

Given: $u = 20 \text{ m/s}$, $\theta = 30^\circ$, $g = 10 \text{ m/s}^2$. ($\sin 30^\circ = 0.5$, $\sin 60^\circ \approx 0.866$.)

Solution: (a) $T = \frac{2u \sin \theta}{g} = \frac{2(20)(0.5)}{10} = \frac{20}{10} = 2 \text{ s}$.

$$(b) H = \frac{u^2 \sin^2 \theta}{2g} = \frac{(400)(0.25)}{20} = \frac{100}{20} = 5 \text{ m}.$$

$$(c) R = \frac{u^2 \sin 2\theta}{g} = \frac{(400) \sin 60^\circ}{10} = \frac{400 \times 0.866}{10} \approx 34.6 \text{ m}.$$

$$\boxed{T = 2 \text{ s}, \quad H = 5 \text{ m}, \quad R \approx 34.6 \text{ m}}$$

Example 2 — [NEET Speed Trap] A ball is thrown at 20 m/s at 30° to the horizontal. Its speed at the highest point of the trajectory is:

$$(A) 0 \quad (B) 10 \text{ m/s} \quad (C) 10\sqrt{3} \text{ m/s} \quad (D) 20 \text{ m/s}$$

The trap: students recall that “velocity is zero at the top” — but that’s only true for a ball thrown *straight up*. For an angled projectile, only the **vertical** part vanishes at the apex; the **horizontal** part sails on unchanged.

Fast solution: At the highest point $v_y = 0$, but $v_x = u \cos \theta = 20 \cos 30^\circ = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3} \approx 17.3 \text{ m/s}$.

$$\boxed{(C) 10\sqrt{3} \text{ m/s}}$$

Speed cue: at the top of any angled projectile, speed = $u \cos \theta$ (never zero unless launched vertically). Option (A) is the trap for those who forget the horizontal motion continues; (D) forgets that the vertical part is lost.

Example 3 — [JEE Main Level] A ball is launched from the ground at 20 m/s at 53° above the horizontal ($\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$, $g = 10 \text{ m/s}^2$). Find (a) the magnitude and direction of its velocity 1 s after launch, and (b) its height at that instant.

Why it’s multi-concept: it combines velocity-component tracking with the vertical kinematic equation, rather than just plugging into the standard $T/H/R$ formulas.

Solution: Resolve: $u_x = 20(0.6) = 12 \text{ m/s}$, $u_y = 20(0.8) = 16 \text{ m/s}$.

(a) At $t = 1 \text{ s}$: $v_x = 12 \text{ m/s}$ (unchanged), $v_y = u_y - gt = 16 - 10(1) = 6 \text{ m/s}$.

$$|\vec{v}| = \sqrt{12^2 + 6^2} = \sqrt{144 + 36} = \sqrt{180} \approx 13.4 \text{ m/s}.$$

$$\tan \phi = \frac{v_y}{v_x} = \frac{6}{12} = 0.5 \Rightarrow \phi \approx 26.6^\circ \text{ above the horizontal}.$$

$$(b) \ y = u_y t - \frac{1}{2}gt^2 = 16(1) - \frac{1}{2}(10)(1)^2 = 16 - 5 = 11 \text{ m.}$$

$$|\vec{v}| \approx 13.4 \text{ m/s at } 26.6^\circ \text{ above horizontal, } y = 11 \text{ m}$$

Note the velocity now points at a *smaller* angle than the launch — the projectile is climbing more gently as gravity erodes its upward speed while the horizontal speed stays fixed.

Example 4 — [JEE Advanced Level] Two balls are projected from the same point with the same speed 20 m/s ($g = 10 \text{ m/s}^2$), one at 30° and the other at 60° to the horizontal. (a) Show that their horizontal ranges are equal. (b) Find the maximum height of each, and verify that the sum of the two heights equals $u^2/2g$, independent of angle.

Reasoning: 30° and 60° are **complementary** ($30^\circ + 60^\circ = 90^\circ$), which is exactly the condition for equal range — a fact worth proving cleanly.

Solution: (a) $R = \frac{u^2 \sin 2\theta}{g}$. For $\theta = 30^\circ$: $\sin 60^\circ$. For $\theta = 60^\circ$: $\sin 120^\circ = \sin 60^\circ$. Since $\sin 2\theta$ is identical, the ranges match:

$$R_{30} = R_{60} = \frac{400 \times 0.866}{10} \approx 34.6 \text{ m. } \checkmark$$

(More generally, $\sin(2(90^\circ - \theta)) = \sin(180^\circ - 2\theta) = \sin 2\theta$, so complementary angles always share a range.)

$$(b) \ H = \frac{u^2 \sin^2 \theta}{2g}:$$

$$H_{30} = \frac{400(0.25)}{20} = 5 \text{ m,} \quad H_{60} = \frac{400(0.75)}{20} = 15 \text{ m.}$$

$$\text{Sum: } H_{30} + H_{60} = 20 \text{ m. Compare with } \frac{u^2}{2g} = \frac{400}{20} = 20 \text{ m. } \checkmark$$

General proof: for complementary angles θ and $90^\circ - \theta$,

$$H_1 + H_2 = \frac{u^2 \sin^2 \theta}{2g} + \frac{u^2 \sin^2(90^\circ - \theta)}{2g} = \frac{u^2(\sin^2 \theta + \cos^2 \theta)}{2g} = \frac{u^2}{2g},$$

which is independent of θ .

$$R_{30} = R_{60} \approx 34.6 \text{ m; } H_{30} + H_{60} = 20 \text{ m} = \frac{u^2}{2g}$$

Section 5: Practice Exercises

1. **[CBSE]** A projectile is fired from the ground at 40 m/s at 30° ($g = 10 \text{ m/s}^2$). Find its maximum height and horizontal range.
2. **[NEET]** A projectile launched at 45° has a horizontal range of 80 m. Find its maximum height.

3. **[JEE Main]** A stone is thrown horizontally at 20 m/s from the top of a 45 m tower ($g = 10 \text{ m/s}^2$). Find the time to reach the ground and the horizontal distance from the base.
4. **[JEE Main]** A projectile's horizontal range is 3 times its maximum height. Find the angle of projection.
5. **[JEE Advanced]** A ball is projected at 30 m/s at 37° ($\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$, $g = 10 \text{ m/s}^2$). Find the radius of curvature of its path at the highest point.

Answers: 1. $H = 20 \text{ m}$, $R \approx 138.6 \text{ m}$ | 2. $H = 20 \text{ m}$ (since $R = 4H$ at 45°) | 3. $T = 3 \text{ s}$, $R = 60 \text{ m}$ | 4. $\tan \theta = 4/3$, $\theta \approx 53^\circ$ | 5. at the top, speed $= u \cos \theta = 24 \text{ m/s}$, $a = g$ (centripetal), so $r = v^2/g = 576/10 = 57.6 \text{ m}$.

Section 6: Crack the MCQ

Q1. At the highest point of its trajectory, the velocity of a projectile (launched at an angle) is: (A) zero (B) horizontal (C) vertical (D) directed at 45°

Answer: (B). At the apex the vertical component is zero, leaving only the constant horizontal component $u \cos \theta$ — so the velocity is purely horizontal. (A) zero is true only for a vertically thrown body, not an angled projectile — the most common trap. (C) vertical describes the launch-and-fall extremes, not the top. (D) has no general basis.

Q2. The horizontal range of a projectile is maximum when the angle of projection is: (A) 30° (B) 45° (C) 60° (D) 90°

Answer: (B). $R = \frac{u^2 \sin 2\theta}{g}$ is greatest when $\sin 2\theta = 1$, i.e. $2\theta = 90^\circ \Rightarrow \theta = 45^\circ$. (A) and (C) are complementary and give equal — but sub-maximal — ranges. (D) 90° is a vertical throw: range zero. Students sometimes pick 90° thinking “steeper = farther,” which is exactly backwards.

Q3. Two projectiles thrown with the same speed cover equal horizontal ranges. If one is launched at 25° , the other is launched at: (A) 25° (B) 50° (C) 65° (D) 75°

Answer: (C). Equal ranges (same speed) require complementary angles: $90^\circ - 25^\circ = 65^\circ$. (A) is the same projectile, not a distinct one. (B) comes from doubling the angle — a misremembered rule. (D) doesn't sum to 90° with 25° . The key is $\theta + \theta' = 90^\circ$.

Q4. Neglecting air resistance, the acceleration of a projectile during its flight is: (A) zero at the highest point (B) g directed downward throughout (C) reversed in direction at the top (D) greatest at the moment of launch

Answer: (B). Gravity is the only force, so the acceleration is constant: g downward at every instant, including the apex. (A) confuses zero *velocity component* with zero acceleration — gravity never switches off. (C) gravity always points down; it never reverses. (D) acceleration is constant, not “greatest at launch.” This is the same misconception family as “ $v = 0 \Rightarrow a = 0$ ” from free fall.

Section 7: Common Student Pitfalls & Pro-Tips

- **“Velocity is zero at the top.”** Only the vertical component vanishes; the horizontal component $u \cos \theta$ continues. The projectile is moving fastest-horizontally precisely when it’s slowest-vertically.
- **“Acceleration is zero at the top.”** It’s always g downward. Velocity and acceleration are independent — one can be (partly) zero while the other isn’t.
- **Coupling the two motions.** Horizontal and vertical motions are independent; they share only the time t . Never let a horizontal quantity leak into a vertical equation or vice-versa.
- **Mixing the H and R formulas** ($\sin^2 \theta$ vs $\sin 2\theta$). Keep them distinct: height uses $\sin^2 \theta$, range uses $\sin 2\theta$.
- **Forgetting $u_y = 0$ for a horizontal launch.** A body fired horizontally from a height starts with zero vertical velocity — its fall is identical to a dropped object.

Pro-Tip: Solve every projectile by **splitting into two channels**: horizontal (constant velocity, $x = u \cos \theta \cdot t$) and vertical (free fall, $y = u \sin \theta \cdot t - \frac{1}{2}gt^2$), linked only by t . Find the time of flight from the vertical channel, then feed it into the horizontal one for range. Keep the shortcut shortcuts handy — complementary angles share a range, $R = 4H \cot \theta$, and $R_{\max} = 4H$ at 45° — they turn many problems into one-liners.

Section 8: Cheat Sheet Box

- **Components:** $u_x = u \cos \theta$ (constant); $u_y = u \sin \theta$; acceleration g downward always.
- **Time of flight:** $T = \frac{2u \sin \theta}{g}$; **max height:** $H = \frac{u^2 \sin^2 \theta}{2g}$; **range:** $R = \frac{u^2 \sin 2\theta}{g}$.
- **Max range** $= \frac{u^2}{g}$ at $\theta = 45^\circ$ (and then $R_{\max} = 4H$).
- **Trajectory:** $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$ (a parabola).
- **Complementary angles θ , $90^\circ - \theta$:** same range; $H_1 + H_2 = \frac{u^2}{2g}$; $R = 4H \cot \theta$.
- **Horizontal launch from height h :** $T = \sqrt{\frac{2h}{g}}$, $R = u\sqrt{\frac{2h}{g}}$.
- **At the apex:** $v_y = 0$, speed $= u \cos \theta$, acceleration still g .
- **Mnemonic:** “Horizontal coasts, vertical falls — time ties them together.”

Sub-topic 05: Relative Velocity in Two Dimensions

Exam Relevance: A classic, heavily tested application of vectors. CBSE Boards ask the river-boat crossing (time and drift) and the rain-umbrella problem (2-4 marks). JEE Main reliably carries one question — shortest path across a river, minimum time, or relative velocity of two moving bodies. NEET tests the rain-man angle and basic crossing. JEE Advanced layers it with closest-approach problems and currents stronger than the boat's own speed.

Section 1: Concept Intuition

We met relative velocity in one dimension in Chapter 2 — same line, just add or subtract. In a plane it becomes far richer, because now the velocities point in *different directions* and must be combined as **vectors**. The governing rule, though, is exactly the same:

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B,$$

the velocity of A *as seen from* B. The only new skill is doing that subtraction as a vector operation — by components or by the triangle law — rather than with signs alone.

Two beautiful real-world problems capture everything in this topic. The first is the **river-boat problem**. A boat's engine drives it at some velocity *relative to the water*, but the water itself is flowing relative to the ground. What an observer on the bank sees is the **vector sum** of these two: $\vec{v}_{\text{boat,ground}} = \vec{v}_{\text{boat,water}} + \vec{v}_{\text{water,ground}}$. This single equation answers every river question — how long to cross, where you land, which way to aim.

Think about it this way... Picture rowing straight across the Ganga toward a temple on the far bank. You point the boat dead ahead, perpendicular to the flow — yet you drift downstream and land well past the temple. Why? Because while your oars carry you *across*, the current simultaneously carries the *whole boat* downstream, and these two motions are independent (just like the x and y motions of the last sub-topic). To actually reach the temple, you must aim *upstream* at an angle, so that your across-velocity still carries you over while your upstream-component exactly cancels the current. There's a genuine trade-off: aim straight for the **least time**, or aim upstream for the **straightest path** — you can't always have both.

The second classic is the **rain-man problem**. Rain falling vertically looks slanted to someone walking, which is why you tilt your umbrella *forward* into your direction of motion. The rain's velocity relative to *you* is $\vec{v}_{\text{rain}} - \vec{v}_{\text{you}}$, and the umbrella must point along that relative velocity. Walk faster and the rain appears to come more horizontally, so you tilt the umbrella further forward. Both problems are the same physics — a vector subtraction — dressed in different clothes.

Finally, relative velocity lets us answer “**how close do two moving objects get?**” Two ships, two cars, an aircraft and a target — by sitting in one object's frame, the other appears to move in a straight line at the *relative* velocity, and the **closest approach** is just the perpendicular distance from your position to that straight line.

Limiting conditions & assumptions.

- The combining rule is **Galilean** vector addition, valid at everyday speeds ($v \ll c$).
- To reach a point **directly across** a river, the boat's own speed must exceed the current

($v_{bw} > v_w$); if the current is faster, you cannot avoid some downstream drift, and the goal shifts to *minimising* it.

- “Minimum time” and “shortest path (zero drift)” are generally **different** strategies requiring different headings — don’t conflate them.
- All velocities must be expressed relative to a **common reference** (usually the ground) before combining.

Section 2: Key Formulas & Definitions

General rule: $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$ (velocity of A relative to B); $\vec{v}_{AB} = -\vec{v}_{BA}$.

Chained (frame) relation: $\vec{v}_{Ag} = \vec{v}_{Ab} + \vec{v}_{bg}$ (A relative to ground = A relative to b + b relative to ground). For the boat: $\vec{v}_{\text{boat,ground}} = \vec{v}_{\text{boat,water}} + \vec{v}_{\text{water,ground}}$.

River-boat — river width d , boat speed in still water v_b , current speed v_r :

Minimum-time crossing (boat aimed straight across):

$$t_{\min} = \frac{d}{v_b}, \quad \text{drift downstream} = v_r t_{\min} = \frac{v_r d}{v_b}, \quad v_{\text{ground}} = \sqrt{v_b^2 + v_r^2}.$$

Shortest-path crossing (land directly opposite, zero drift; needs $v_b > v_r$): aim upstream at angle θ from the straight-across direction with

$$\sin \theta = \frac{v_r}{v_b}, \quad v_{\text{across}} = \sqrt{v_b^2 - v_r^2}, \quad t = \frac{d}{\sqrt{v_b^2 - v_r^2}}.$$

Rain-man — rain velocity $\vec{v}_r$ (relative to ground), walking velocity $\vec{v}_m$: the rain relative to the man is $\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$; the umbrella points along $\vec{v}_{rm}$. For vertical rain (speed v_r) and horizontal walking (speed v_m), the umbrella tilts forward from the vertical by

$$\tan \alpha = \frac{v_m}{v_r}.$$

Closest approach of two bodies: in B’s frame, A moves in a straight line at $\vec{v}_{AB}$; the closest distance is the perpendicular distance from B to that line (or minimise $|\vec{r}_{AB}(t)|$).

All SI: velocities m/s, distances m, time s.

Section 3: Key Procedures & Logical Framework

There’s no single grand derivation here; the skill is setting up the vectors correctly. Two procedures cover almost every problem.

Procedure A — The river-boat problem. 1. **Set axes:** let x be along the current (downstream +) and y straight across the river. 2. **Write the boat’s ground velocity** as the vector sum $\vec{v}_{bg} = \vec{v}_{bw} + \vec{v}_w$, resolving the boat’s heading into components. 3. **Crossing time depends only on the across (y) component**, because only y -motion covers the width: $t = \frac{d}{(v_{bg})_y}$. 4. **Drift depends on the net downstream (x) velocity** times the crossing time. 5. **Choose the strategy:**

- **Minimum time** → maximise the across-component → aim **straight across** ($v_{\text{across}} = v_b$). Drift is then unavoidable.
- **Zero drift / land opposite** → make the net x -velocity zero → aim **upstream** so $v_b \sin \theta = v_r$, giving $\sin \theta = v_r/v_b$ (possible only if $v_b > v_r$).

Figure 3.5 A river flowing left-to-right (current v_r arrows). Two boat scenarios from the same start point on the near bank. Scenario 1 (minimum time): boat heading drawn straight across (v_b up), resultant v_{ground} tilted downstream, landing point shifted right by the drift. Scenario 2 (shortest path): boat heading drawn aimed upstream at angle θ , resultant pointing straight across to the directly-opposite point. River width d marked vertically. (see next page)

Procedure B — The rain-man problem. 1. Write the rain's velocity relative to the ground, $\vec{v}_r$ (often straight down, sometimes slanted). 2. Write the man's velocity $\vec{v}_m$ (horizontal). 3. Compute the rain's velocity **relative to the man**: $\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$ (subtract the man's velocity — equivalently, add the *reverse* of his velocity to the rain). 4. The umbrella must point **along** $\vec{v}_{rm}$ (into the apparent rain). Its angle from the vertical satisfies $\tan \alpha = \frac{v_m}{v_r}$ for vertical rain. 5. *Reasoning*: in the man's own frame he is at rest, so the rain appears to approach along $\vec{v}_{rm}$ — that apparent direction is exactly what the umbrella must face.

Procedure C — Closest approach. Move into one body's frame so it is at rest; the other travels in a straight line at the relative velocity $\vec{v}_{AB}$. Either drop a perpendicular from the stationary body to that line, or write the separation $|\vec{r}_{AB}(t)|^2$ as a function of time and minimise it.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A river is 100 m wide and flows at 3 m/s. A boat that can move at 5 m/s in still water is steered straight across (perpendicular to the bank). Find (a) the time to cross, (b) how far downstream it drifts, and (c) its speed relative to the ground.

Given: $d = 100$ m, $v_r = 3$ m/s, $v_b = 5$ m/s, boat aimed straight across.

Solution: (a) Only the across-velocity (the full 5 m/s) covers the width:

$$t = \frac{d}{v_b} = \frac{100}{5} = 20 \text{ s.}$$

(b) During this time the current carries the boat downstream:

$$\text{drift} = v_r t = (3)(20) = 60 \text{ m.}$$

(c) The ground velocity is the vector sum of the perpendicular parts:

$$v_{\text{ground}} = \sqrt{v_b^2 + v_r^2} = \sqrt{5^2 + 3^2} = \sqrt{34} \approx 5.83 \text{ m/s.}$$

$t = 20 \text{ s,} \quad \text{drift} = 60 \text{ m,} \quad v_{\text{ground}} \approx 5.83 \text{ m/s}$

Aiming straight across gives the **shortest crossing time**, but the price is a 60 m downstream drift.

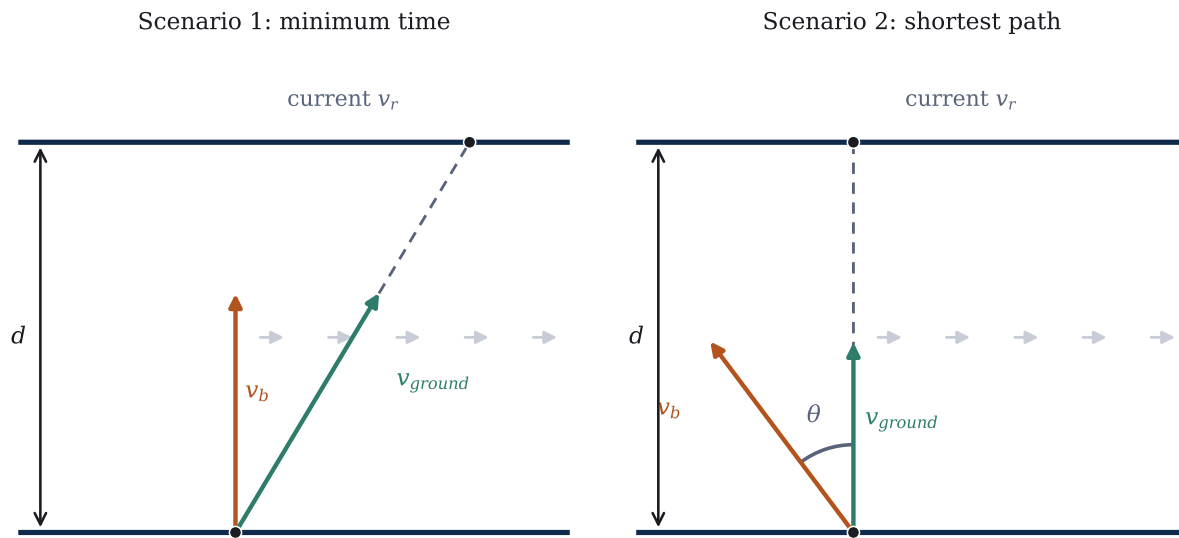


Figure 3.5 A river flowing left-to-right (current v_r arrows). Two boat scenarios from the same start point on the near bank. Scenario 1 (minimum time): boat heading drawn straight across (v_b up), resultant v_{ground} tilted downstream, landing point shifted right by the drift. Scenario 2 (shortest path): boat heading drawn aimed upstream at angle θ , resultant pointing straight across to the directly-opposite point. River width d marked vertically.

Example 2 — [NEET Speed Trap] Rain is falling vertically at 4 m/s. A man walks horizontally at 3 m/s. At what angle to the vertical should he tilt his umbrella?

- (A) 37° forward (B) 53° forward (C) 37° backward (D) 90°

The trap: students either tilt the umbrella *backward* (against motion) or invert the ratio to $\tan \alpha = v_r/v_m$.

Fast solution: The umbrella points along the rain's velocity relative to the man, tilted **forward** (into the walking direction) by

$$\tan \alpha = \frac{v_m}{v_r} = \frac{3}{4} = 0.75 \Rightarrow \alpha = 37^\circ.$$

(A) 37° forward

Speed cue: always **forward**, into your motion (the faster you walk, the more the rain seems to come at your face). The ratio is “walking speed over rain speed.” Option (C) gets the magnitude right but the direction wrong — a classic half-credit trap.

Example 3 — [JEE Main Level] A river is 80 m wide and flows at 6 m/s. A boat moves at 10 m/s in still water. The boatman wants to reach the point directly opposite his start. (a) In which direction must he steer, and (b) how long does the crossing take?

Why it's multi-concept: it combines vector resolution with the relative-velocity sum and the zero-drift condition.

Solution: To land directly opposite, the net downstream velocity must be zero, so the boat's upstream component must cancel the current. Aiming upstream at angle θ from the straight-across direction:

$$v_b \sin \theta = v_r \Rightarrow \sin \theta = \frac{v_r}{v_b} = \frac{6}{10} = 0.6 \Rightarrow \theta = 37^\circ \text{ upstream of straight-across.}$$

(b) The effective across-speed is the remaining component:

$$v_{\text{across}} = v_b \cos \theta = \sqrt{v_b^2 - v_r^2} = \sqrt{100 - 36} = \sqrt{64} = 8 \text{ m/s.}$$

$$t = \frac{d}{v_{\text{across}}} = \frac{80}{8} = 10 \text{ s.}$$

Steer 37° upstream of straight-across; $t = 10 \text{ s}$

Note this crossing (10 s) takes longer than the minimum-time route would ($80/10 = 8 \text{ s}$) — the cost of arriving with zero drift.

Example 4 — [JEE Advanced Level] Ship A sails due north at 10 km/h. At a certain instant, ship B is 20 km due east of A and sails due west at 10 km/h. Find (a) the time of closest approach and (b) the minimum distance between the ships.

Reasoning: Work in A's frame: B moves in a straight line at the relative velocity, and the closest approach is the minimum of the separation over time.

Solution: Take east as $+x$, north as $+y$, with A at the origin at $t = 0$.

$$\vec{v}_A = (0, 10), \quad \vec{v}_B = (-10, 0), \quad \vec{v}_{BA} = \vec{v}_B - \vec{v}_A = (-10, -10) \text{ km/h.}$$

Initial position of B relative to A: $\vec{r}_{BA}(0) = (20, 0) \text{ km}$. At time t (hours):

$$\vec{r}_{BA}(t) = (20 - 10t, -10t).$$

The separation squared:

$$s^2(t) = (20 - 10t)^2 + (10t)^2 = 400 - 400t + 100t^2 + 100t^2 = 200t^2 - 400t + 400.$$

Minimise: $\frac{d(s^2)}{dt} = 400t - 400 = 0 \Rightarrow t = 1 \text{ h}$.

At $t = 1 \text{ h}$: $\vec{r}_{BA} = (10, -10)$, so

$$s_{\min} = \sqrt{10^2 + 10^2} = \sqrt{200} = 10\sqrt{2} \approx 14.1 \text{ km.}$$

$t = 1 \text{ h}, \quad s_{\min} = 10\sqrt{2} \approx 14.1 \text{ km}$

In A's frame, B slides along a straight line heading south-west; the ships never collide, closing only to about 14.1 km before drawing apart again.

Section 5: Practice Exercises

1. **[CBSE]** A 120 m wide river flows at 4 m/s. A swimmer who manages 3 m/s in still water heads straight across. Find the time to cross and the downstream drift.
2. **[NEET]** Rain falls vertically at 10 m/s. A cyclist rides at 10 m/s. Find the angle from the vertical at which she should hold her umbrella.
3. **[JEE Main]** A boat moving at 13 m/s in still water must cross a 60 m river flowing at 5 m/s to land directly opposite. Find the steering angle (from straight-across) and the crossing time.
4. **[JEE Main]** To a man walking east at 5 m/s, the wind appears to blow from the north at 5 m/s. Find the actual velocity of the wind (magnitude and direction).
5. **[JEE Advanced]** Car A moves east at 20 m/s; car B, initially 100 m north of A, moves south at 20 m/s. Find their closest approach distance and the time at which it occurs.

v across = 12 m/s, $t = 5 \text{ s}$ | 4. wind blows from the north-west at $5\sqrt{2} \approx 7.07 \text{ m/s}$ (i.e. velocity $5\hat{i} - 5\hat{j}$, toward south-east) 5

$-5\hat{i} - 5\hat{j}$, toward south-west) | 5. relative velocity $(-20, -20)$... $s_{\min} = 50 \sqrt{2} \approx 70.7 \text{ m}$ at $t = 2.5 \text{ s}$.

Section 6: Crack the MCQ

Q1. A boat is steered straight across a flowing river. Compared with still water, its time to cross is: (A) longer (B) shorter (C) the same (D) zero

Answer: (C). The current acts purely downstream and does nothing to the across-component, so the crossing time d/v_b is unchanged — the current only causes drift, not delay. (A) and (B) wrongly assume the current affects the crossing time. (D) is nonsensical. The independence of perpendicular motions is the key: downstream flow can't speed up or slow down the across-progress.

Q2. A man walking forward tilts his umbrella to keep dry. The umbrella should be tilted: (A) backward, against his motion (B) forward, into his motion (C) vertically (D) horizontally

Answer: (B). In the man's frame the rain appears to come from the front, so the umbrella tilts forward into the apparent rain. (A) backward would expose him completely — a frequent wrong answer. (C) vertical only works when standing still. (D) horizontal would need an impossibly fast walk. The faster he walks, the more forward the tilt.

Q3. To cross a river and land at the point directly opposite the start, the boat's speed in still water must be: (A) less than the current speed (B) equal to the current speed (C) greater than the current speed (D) exactly zero

Answer: (C). Cancelling the downstream drift requires an upstream component $v_b \sin \theta = v_r$, which has a solution only if $v_b > v_r$. (A) if the boat is slower than the current, it cannot fully oppose the drift and *must* land downstream. (B) equal speeds would require $\theta = 90^\circ$ (heading straight upstream), giving zero across-progress — it never crosses. (D) zero speed means no motion at all.

Q4. Two objects move with the same constant velocity in the same direction. The velocity of one relative to the other is: (A) double their speed (B) zero (C) equal to their speed (D) perpendicular to their motion

Answer: (B). $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = \vec{0}$ when the velocities are identical — each sees the other as stationary. (A) double would apply to *opposite* velocities. (C) and (D) have no basis. This is why two vehicles cruising side by side at the same speed appear motionless to each other.

Section 7: Common Student Pitfalls & Pro-Tips

- **Confusing “minimum time” with “shortest path.”** Aiming straight across minimises *time* (but drifts); aiming upstream gives *zero drift* (but takes longer). They need different headings — decide which the question asks.
- **Tilting the umbrella the wrong way.** It always tilts *forward*, into the direction of motion, along the rain's velocity relative to you.
- **Inverting the rain-angle ratio.** For vertical rain and horizontal walking, $\tan \alpha = v_m/v_r$ (walking over rain), measured from the vertical — not the reverse.
- **Thinking the current changes the crossing time.** For a straight-across heading

it doesn't; the downstream flow only produces drift.

- **Mixing reference frames.** Express all velocities relative to a common frame (usually the ground) before adding or subtracting; the chain $\vec{v}_{Ag} = \vec{v}_{Ab} + \vec{v}_{bg}$ keeps the bookkeeping straight.

Pro-Tip: Set up clean perpendicular axes — **along the current and across the river** — and the river problem splits into two independent 1-D problems: the across-component sets the crossing time, the downstream-component sets the drift. For closest-approach questions, jump into one body's frame so the other moves in a straight line at the relative velocity, then it's pure geometry (perpendicular distance). And for rain: always tilt forward, ratio = walking speed over rain speed.

Section 8: Cheat Sheet Box

- **General:** $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$; chain $\vec{v}_{Ag} = \vec{v}_{Ab} + \vec{v}_{bg}$.
- **River, straight across (min time):** $t = \frac{d}{v_b}$, drift = $\frac{v_r d}{v_b}$, ground speed = $\sqrt{v_b^2 + v_r^2}$.
- **River, land opposite (zero drift, needs $v_b > v_r$):** $\sin \theta = \frac{v_r}{v_b}$, $v_{\text{across}} = \sqrt{v_b^2 - v_r^2}$,

$$t = \frac{d}{\sqrt{v_b^2 - v_r^2}}.$$
- **Rain-man:** umbrella along $\vec{v}_r - \vec{v}_m$; for vertical rain, $\tan \alpha = \frac{v_m}{v_r}$ (forward from vertical).
- **Closest approach:** in one body's frame, the other moves straight at $\vec{v}_{AB}$; minimise $|\vec{r}_{AB}(t)|$.
- **Mnemonic:** "Straight across = least time; angle upstream = no drift; umbrella tilts forward."

Sub-topic 06: Circular Motion Dynamics

Exam Relevance: A rich, high-yield topic that also seeds Rotational Motion and Gravitation later. CBSE Boards ask the centripetal-acceleration derivation and definitions of angular quantities (2–3 marks). JEE Main reliably carries a question on centripetal force, angular kinematics, or banking. NEET tests the conceptual core hard — direction of acceleration, “constant speed yet accelerating,” the centrifugal-force misconception. JEE Advanced layers it with non-uniform circular motion, vertical circles, and banking-with-friction.

Section 1: Concept Intuition

Swing a stone tied to a string in a horizontal circle at a steady rate. Its *speed* is constant — yet, surprisingly, the stone is **accelerating** the entire time. How can that be? Because velocity is a **vector**, and acceleration responds to *any* change in that vector — including a change in **direction**, not just in magnitude. The stone’s direction of motion is changing every instant, so its velocity is changing every instant, so it must be accelerating, even with the speedometer reading frozen.

Which way does this acceleration point? Toward the **centre** of the circle. We call it **centripetal acceleration** — Latin for “centre-seeking.” And by Newton’s second law, an acceleration toward the centre demands a **force** toward the centre: the **centripetal force**. This is the single most important idea of the topic, and it’s worth stating bluntly: *circular motion is impossible without a real, inward force*. The string’s tension supplies it for the whirling stone; friction supplies it for a car rounding a bend; gravity supplies it for a satellite orbiting Earth; the road’s normal force helps on a banked turn. Remove that inward force and the object obeys Newton’s first law — it flies off in a **straight line along the tangent**, not outward along the radius.

Think about it this way... Picture a *gilli* (or a ball) on a string being whirled overhead. People wrongly imagine that if you let go, it shoots straight *outward*, away from your hand. It doesn’t. The instant you release it, the inward pull vanishes and the stone simply continues in the straight line it was already travelling — the **tangent** — like mud flinging off a spinning bicycle tyre, always sideways, never radially out. The famous “outward” feeling you sense while spinning is the **centrifugal force**, and it is *not a real force at all* — it’s an illusion of being in a rotating frame. In the honest ground-frame view, there is only one force here, and it points **inward**.

To describe circular motion precisely we use **angular quantities**, mirrors of the linear ones: **angular displacement** θ (the angle swept, in radians), **angular velocity** $\omega = d\theta/dt$ (how fast the angle changes), and **angular acceleration** $\alpha = d\omega/dt$. They connect to the linear quantities through the radius: $v = \omega r$ and (for changing speed) $a_t = \alpha r$. When the speed is constant we have **uniform circular motion (UCM)**: ω constant, $\alpha = 0$, and only the centripetal acceleration acts. When the speed also changes, the motion is **non-uniform**, and there’s an additional **tangential** acceleration along the direction of travel.

Limiting conditions & assumptions.

- In **uniform** circular motion the *speed* is constant but the *velocity* is not, and the centripetal acceleration’s *magnitude* is constant while its *direction* continuously rotates.
- Centripetal force is **not a new kind of force** — it’s the *net* of real forces (tension,

friction, gravity, normal) that happens to point toward the centre. Always identify the physical source.

- **Centrifugal force** is a pseudo-force, valid only inside a rotating (non-inertial) frame; it has no place in a ground-frame analysis.
 - The relations $v = \omega r$, $a_t = \alpha r$ require θ and ω in **radians** — never degrees.
-

Section 2: Key Formulas & Definitions

Angular quantities and linear links:

$$s = r\theta, \quad v = \omega r, \quad a_t = \alpha r, \quad \omega = \frac{d\theta}{dt}, \quad \alpha = \frac{d\omega}{dt}.$$

Units: θ in radian (dimensionless), ω in rad/s, α in rad/s².

Uniform circular motion (period T , frequency f):

$$\omega = \frac{2\pi}{T} = 2\pi f, \quad v = \frac{2\pi r}{T} = \omega r.$$

Centripetal acceleration (directed toward the centre):

$$a_c = \frac{v^2}{r} = \omega^2 r = v\omega.$$

Centripetal force:

$$F_c = \frac{mv^2}{r} = m\omega^2 r.$$

Non-uniform circular motion — two perpendicular acceleration components:

$$a_t = \alpha r \text{ (tangential, changes speed)}, \quad a_c = \frac{v^2}{r} \text{ (radial, changes direction)}, \quad a = \sqrt{a_t^2 + a_c^2}.$$

Angular kinematics (constant α) — exact analogues of the linear equations:

$$\omega = \omega_0 + \alpha t, \quad \theta = \omega_0 t + \frac{1}{2}\alpha t^2, \quad \omega^2 = \omega_0^2 + 2\alpha\theta.$$

Applications:

- **Banking of a road (frictionless):** $\tan \theta = \frac{v^2}{rg}$.
 - **Maximum safe speed on an unbanked road** (friction μ): $v_{\max} = \sqrt{\mu rg}$.
-

Section 3: CBSE Board Derivation — Centripetal Acceleration

First, the linear-angular link $v = \omega r$. The arc length swept is $s = r\theta$. Differentiating with respect to time (with r constant):

$$v = \frac{ds}{dt} = r \frac{d\theta}{dt} = r\omega \Rightarrow \boxed{v = \omega r}.$$

Now the centripetal acceleration $a_c = v^2/r$. Consider a particle in uniform circular motion. In a small time Δt it moves from P to Q , turning through a small angle $\Delta\theta$. Its speed is unchanged, but the velocity vector rotates from $\vec{v}_1$ to $\vec{v}_2$ — also through the angle $\Delta\theta$, since velocity is always tangent to the circle (and the tangent turns by the same angle as the radius).

Figure 3.6 Two panels. Left: a circle with centre O ; a particle at P with velocity v_1 tangent there, and a moment later at Q with velocity v_2 tangent there; the radii OP and OQ subtend a small angle $\Delta\theta$ at O ; both speeds equal v . Right: the velocity-difference triangle — vectors v_1 and v_2 drawn tail-to-tail (equal length v) with the small angle $\Delta\theta$ between them, and the third side $\Delta v = v_2 - v_1$; an arrow shows Δv points roughly toward the centre O . Place Δv_1 and Δv_2 tail-to-tail. They have equal length v and a small angle $\Delta\theta$ between them, forming an isosceles triangle whose third side is the change in velocity Δv . For a small angle, the length of this third side is $|\Delta v| = v\Delta\theta$. The magnitude of the acceleration is therefore $a_c = \lim_{\Delta t \rightarrow 0} |\Delta v| / \Delta t = \lim_{\Delta t \rightarrow 0} v\Delta\theta / \Delta t = v\omega$. Using $v = \omega r$ to rewrite this: $a_c = v\omega = v^2/r = \omega^2 r$. (see next page)

Direction. As $\Delta t \rightarrow 0$, the vector $\Delta \vec{v}$ becomes perpendicular to the velocity and points **toward the centre** of the circle — hence “centripetal.”

Why each step matters: the speed staying constant is exactly why the velocity triangle is isosceles; the velocity vector turning by the *same* $\Delta\theta$ as the radius is what links the geometry of the path to the geometry of the velocity change; and taking the limit converts the average rate into the instantaneous centripetal acceleration.

Multiplying by mass gives the **centripetal force** $F_c = ma_c = \frac{mv^2}{r} = m\omega^2 r$, directed toward the centre.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A particle of mass 0.5 kg moves in a horizontal circle of radius 2 m with a constant angular speed of 3 rad/s . Find (a) its linear speed, (b) its centripetal acceleration, and (c) the centripetal force acting on it.

Given: $m = 0.5 \text{ kg}$, $r = 2 \text{ m}$, $\omega = 3 \text{ rad/s}$.

Solution: (a) $v = \omega r = (3)(2) = 6 \text{ m/s}$.

(b) $a_c = \omega^2 r = (3)^2(2) = 18 \text{ m/s}^2$ (toward the centre). [Check: $v^2/r = 36/2 = 18$. ✓]

(c) $F_c = ma_c = (0.5)(18) = 9 \text{ N}$ (toward the centre).

$v = 6 \text{ m/s}, \quad a_c = 18 \text{ m/s}^2, \quad F_c = 9 \text{ N}$
--

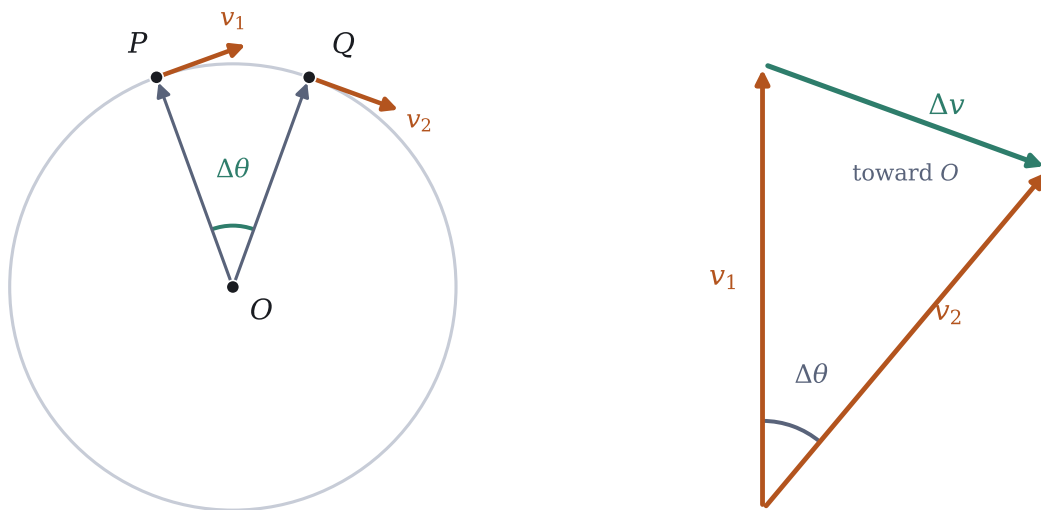


Figure 3.6 Two panels. Left: a circle with centre O ; a particle at P with velocity v_1 tangent there, and a moment later at Q with velocity v_2 tangent there; the radii OP and OQ subtend a small angle $\Delta\theta$ at O ; both speeds equal v . Right: the velocity-difference triangle — vectors v_1 and v_2 drawn tail-to-tail (equal length v) with the small angle $\Delta\theta$ between them, and the third side $\Delta v = v_2 - v_1$; an arrow shows Δv points roughly toward the centre O . Place $\square v_1$ and $\square v_2$ tail-to-tail. They have equal length v and a small angle $\Delta\theta$ between them, forming an isosceles triangle whose third side is the change in velocity $\Delta \square v$. For a small angle, the length of this third side is $|\Delta \square v| = v\Delta\theta$. The magnitude of the acceleration is therefore $a_c = \lim_{\Delta t \rightarrow 0} |\Delta \square v| / \Delta t = \lim_{\Delta t \rightarrow 0} v\Delta\theta / \Delta t = v\omega$. Using $v = \omega r$ to rewrite this: $a_c = v\omega = v^2 / r = \omega^2 r$.

Example 2 — [NEET Speed Trap] A car rounds a circular curve of fixed radius. If its speed is doubled, the centripetal force required to keep it on the curve becomes:

- (A) half (B) double (C) four times (D) unchanged

The trap: “double the speed $\rightarrow$ double the force” feels right, but the force depends on the **square** of the speed.

Fast solution: $F_c = \frac{mv^2}{r} \propto v^2$ (radius fixed). Doubling v multiplies F_c by $2^2 = 4$.

(C) four times

Speed cue: this is the same $\propto v^2$ scaling as stopping distance — whenever a quantity goes as v^2 , doubling the speed quadruples it. It’s also *why* taking a bend too fast is so dangerous: the friction needed quadruples while the friction available stays the same.

Example 3 — [JEE Main Level] A particle moves in a circle of radius 0.5 m, starting from rest, with its speed increasing uniformly at 4 m/s^2 . At $t = 1 \text{ s}$, find (a) its tangential acceleration, (b) its centripetal acceleration, and (c) the magnitude of its net (total) acceleration.

Why it’s multi-concept: non-uniform circular motion has two perpendicular acceleration components that must be found separately and then combined.

Solution: The “speed increasing at 4 m/s^2 ” is the tangential acceleration: $a_t = 4 \text{ m/s}^2$ (constant).

At $t = 1 \text{ s}$, the speed is $v = a_t t = (4)(1) = 4 \text{ m/s}$.

(a) $a_t = 4 \text{ m/s}^2$ (along the direction of motion).

(b) $a_c = \frac{v^2}{r} = \frac{16}{0.5} = 32 \text{ m/s}^2$ (toward the centre).

(c) Since $a_t \perp a_c$:

$$a = \sqrt{a_t^2 + a_c^2} = \sqrt{4^2 + 32^2} = \sqrt{16 + 1024} = \sqrt{1040} \approx 32.2 \text{ m/s}^2.$$

$a_t = 4 \text{ m/s}^2, \quad a_c = 32 \text{ m/s}^2, \quad a \approx 32.2 \text{ m/s}^2$

Notice the centripetal part dominates once the particle is moving quickly — the net acceleration points almost straight at the centre, tilted only slightly forward by the small tangential part.

Example 4 — [JEE Advanced Level] A car of mass 1000 kg negotiates a circular turn of radius 50 m (take $g = 10 \text{ m/s}^2$). (a) On a flat (unbanked) road with coefficient of friction $\mu = 0.5$, find the maximum speed for a safe turn. (b) The road is instead banked (frictionless) for a design speed of 15 m/s. Find the required banking angle.

Solution: (a) On a flat road, friction alone supplies the centripetal force. At the limit of safety, friction is maximal:

$$\mu mg = \frac{mv_{\max}^2}{r} \Rightarrow v_{\max} = \sqrt{\mu rg} = \sqrt{(0.5)(50)(10)} = \sqrt{250} \approx 15.8 \text{ m/s.}$$

(The mass cancels — the safe speed is independent of how heavy the car is.)

(b) On a frictionless banked road, the normal force N tilts inward. Resolving:

$$N \sin \theta = \frac{mv^2}{r} \quad (\text{horizontal, toward centre}), \quad N \cos \theta = mg \quad (\text{vertical}).$$

Dividing the two equations eliminates N and m :

$$\tan \theta = \frac{v^2}{rg} = \frac{(15)^2}{(50)(10)} = \frac{225}{500} = 0.45 \Rightarrow \theta = \tan^{-1}(0.45) \approx 24.2^\circ.$$

$v_{\max} \approx 15.8 \text{ m/s; banking angle } \theta \approx 24.2^\circ$

Figure 3.7 A car on a road banked at angle θ to the horizontal (cross-sectional view). Show the weight mg straight down, the normal force N perpendicular to the road surface, and N resolved into $N \cos \theta$ (vertical, balancing mg) and $N \sin \theta$ (horizontal, pointing toward the centre of the turn, providing the centripetal force). The centre of the circular turn is to the left at distance r . The banked design lets the road itself supply the inward force, so a car can corner safely even with no friction at all — the principle behind every highway interchange and velo-drome. (see next page)

Section 5: Practice Exercises

1. **[CBSE]** A body moves in a circle of radius 4 m at a constant speed of 8 m/s. Find its centripetal acceleration.
2. **[NEET]** A particle in uniform circular motion completes 300 revolutions per minute in a circle of radius 0.2 m. Find its angular velocity and linear speed.
3. **[JEE Main]** A wheel starts from rest with a constant angular acceleration of 2 rad/s^2 . Find its angular velocity and the total angle turned after 5 s.
4. **[JEE Main]** A 2 kg stone is whirled in a vertical circle of radius 1 m. At the topmost point its speed is 5 m/s ($g = 10 \text{ m/s}^2$). Find the tension in the string at the top.
5. **[JEE Advanced]** A car rounds an unbanked curve of radius 80 m with coefficient of friction $\mu = 0.4$ ($g = 10 \text{ m/s}^2$). Find the maximum safe speed.

Answers: 1. $a_c = 16 \text{ m/s}^2$ | 2. $f = 5 \text{ rev/s}$, $\omega = 10\pi \approx 31.4 \text{ rad/s}$, $v = 2\pi \approx 6.28 \text{ m/s}$ | 3. $\omega = 10 \text{ rad/s}$, $\theta = 25 \text{ rad}$ | 4. $T = \frac{mv^2}{r} - mg = 50 - 20 = 30 \text{ N}$ | 5. $v_{\max} = \sqrt{\mu rg} = \sqrt{320} \approx 17.9 \text{ m/s}$.

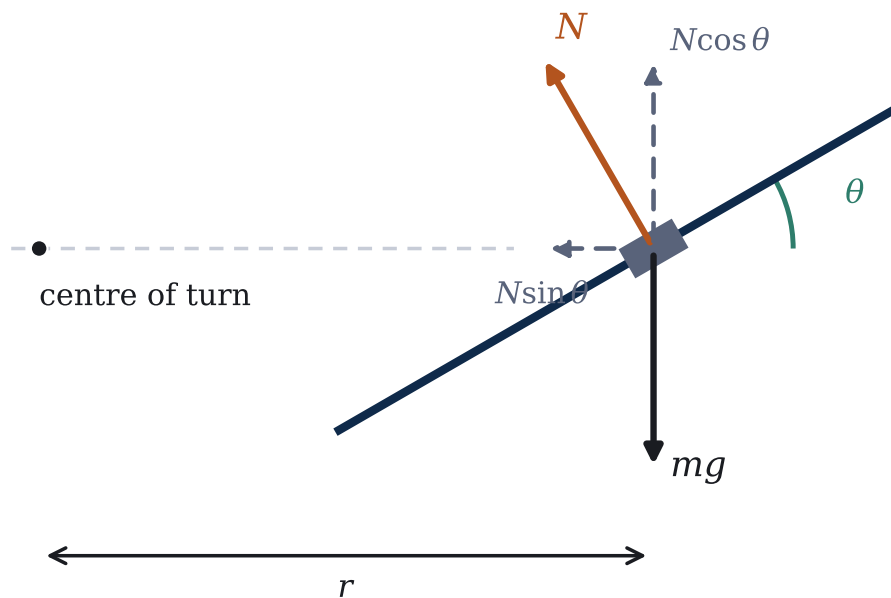


Figure 3.7 A car on a road banked at angle θ to the horizontal (cross-sectional view). Show the weight mg straight down, the normal force N perpendicular to the road surface, and N resolved into $N \cos \theta$ (vertical, balancing mg) and $N \sin \theta$ (horizontal, pointing toward the centre of the turn, providing the centripetal force). The centre of the circular turn is to the left at distance r . The banked design lets the road itself supply the inward force, so a car can corner safely even with no friction at all — the principle behind every highway interchange and velo- drome.

Section 6: Crack the MCQ

Q1. In uniform circular motion, which of the following remains constant? (A) velocity (B) acceleration (C) speed (D) the direction of the centripetal force

Answer: (C). The speed (magnitude of velocity) is constant by definition of *uniform* circular motion. (A) velocity changes because its direction continuously changes. (B) the acceleration has constant *magnitude* but its direction (always toward the centre) keeps rotating, so the acceleration *vector* is not constant. (D) the centripetal force points toward the centre, a direction that rotates as the particle moves. Only the scalar speed is truly fixed.

Q2. The centripetal acceleration of a particle in uniform circular motion is directed: (A) along the velocity (B) opposite to the velocity (C) toward the centre (D) away from the centre

Answer: (C). Centripetal means “centre-seeking”; the acceleration points radially inward, perpendicular to the (tangential) velocity. (A) and (B) describe tangential directions — those would change the *speed*, but in UCM the speed is constant. (D) “away from the centre” is the centrifugal misconception; no real acceleration points outward here.

Q3. A stone is whirled in a horizontal circle at the end of a string. If the string suddenly breaks, the stone flies off: (A) radially outward (B) along the tangent to the circle (C) toward the centre (D) spiralling outward

Answer: (B). With the inward force gone, the stone obeys Newton’s first law and continues in a straight line along the **tangent** — the direction it was moving at the instant of release. (A) radially outward is the single most common wrong answer, born of the centrifugal illusion. (C) toward the centre would need a force that no longer exists. (D) spiralling has no physical basis once all force is removed.

Q4. A particle in uniform circular motion has its speed doubled while the radius is kept the same. Its centripetal acceleration becomes: (A) double (B) half (C) four times (D) unchanged

Answer: (C). $a_c = \frac{v^2}{r} \propto v^2$ at fixed r , so doubling v gives $2^2 = 4$ times the acceleration. (A) assumes linear scaling, ignoring the square. (B) and (D) contradict the direct dependence on v^2 . The same v^2 law governs the centripetal force, which is why cornering speed is so safety-critical.

Section 7: Common Student Pitfalls & Pro-Tips

- **“Constant speed means no acceleration.”** In UCM the *direction* of velocity changes constantly, so there is always a centripetal acceleration — even though the speed never changes.
- **Treating centrifugal force as real.** In the ground (inertial) frame there is only the inward centripetal force. “Centrifugal force” is a pseudo-force that appears solely in the rotating frame — don’t put it in a free-body diagram drawn from the ground.
- **Thinking a released object flies outward.** It leaves along the **tangent**, not along

the radius. The inward force simply stops acting.

- **Forgetting the tangential component in non-uniform motion.** When the speed changes, there are *two* accelerations — tangential ($a_t = \alpha r$, changes speed) and centripetal ($a_c = v^2/r$, changes direction) — and the net is their perpendicular combination.
- **Using degrees for ω or θ .** The relations $v = \omega r$ and $a_t = \alpha r$ hold only with angles in **radians**.

Pro-Tip: For *every* circular-motion problem, first ask: *what real force is supplying the centripetal requirement?* — tension, friction, gravity, the normal force, or a combination. Then set that inward force equal to $\frac{mv^2}{r}$ and solve. Keep two facts at your fingertips: $a_c \propto v^2$ (so speed changes hit hard), and in non-uniform motion the tangential and centripetal accelerations are perpendicular, combining as $\sqrt{a_t^2 + a_c^2}$. Always work angles in radians.

Section 8: Cheat Sheet Box

- **Links:** $s = r\theta$, $v = \omega r$, $a_t = \alpha r$; $\omega = d\theta/dt$, $\alpha = d\omega/dt$ (radians!).
- **UCM:** $\omega = \frac{2\pi}{T} = 2\pi f$; $v = \frac{2\pi r}{T}$.
- **Centripetal:** $a_c = \frac{v^2}{r} = \omega^2 r = v\omega$ (toward centre); $F_c = \frac{mv^2}{r} = m\omega^2 r$.
- **Non-uniform:** tangential $a_t = \alpha r$ and radial $a_c = v^2/r$; net $a = \sqrt{a_t^2 + a_c^2}$.
- **Angular kinematics (const α):** $\omega = \omega_0 + \alpha t$; $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$; $\omega^2 = \omega_0^2 + 2\alpha\theta$.
- **Banking (frictionless):** $\tan \theta = \frac{v^2}{rg}$; **max speed (unbanked):** $v_{\max} = \sqrt{\mu rg}$.
- **Centrifugal force** = pseudo-force (rotating frame only); the real force is centripetal (inward).
- **Mnemonic:** “Speed steady, direction never — that’s why it accelerates inward.”

Chapter 3: Motion in a Plane — Round 2 Addendum

The chapter handed you the full toolkit — resolution, independence of perpendicular motions, the dot and cross products, level-ground projectiles, the river-boat dichotomy, and centripetal dynamics. Five examinable patterns still stand outside its door, and every one of them is a marriage of two things the chapter already teaches separately. Addendum A fires projectiles at targets that are *not* at launch level — a cliff below, an incline ahead — which is the JEE Main/Advanced staple the chapter’s own limiting-conditions note waves toward but never solves. Addendum B finishes the vertical circle, whose only appearance in the chapter is a practice answer asserting $T = mv^2/r - mg$ with no derivation. Addendum C merges the chapter’s two separate results — frictionless banking and flat-road friction — into the single safe-speed band JEE actually asks for. Addendum D settles the river the boat cannot beat ($v_b \leq v_r$) and replaces the calculus of closest approach with pure geometry.

Nothing here is new theory. Every method below is component resolution, the radial form of Newton’s second law, or energy conservation, applied one configuration over.

Addendum A — Projectiles That Do Not Land Where They Started

Sub-topic 04 derives T , H and R under one silent assumption: launch and landing at the same height. It then warns that “if the projectile lands higher or lower, you must go back to the component equations” — but never shows the return trip. Boards and JEE both test that return trip constantly: the tower shot (CBSE 2-3 marks), and the projectile on an inclined plane (a guaranteed JEE Main pattern, a recurring Advanced one).

Method

Case 1 — landing a depth h below the launch point. Keep the chapter’s axes: origin at launch, y up, so the landing condition is $y = -h$ instead of $y = 0$.

The vertical position at time t is

$$y(t) = (u \sin \theta) t - \frac{1}{2}gt^2.$$

Setting $y = -h$ and multiplying through by -2 :

$$gt^2 - 2u \sin \theta t - 2h = 0.$$

The quadratic formula gives two roots; only the positive one is physical:

$$t = \frac{u \sin \theta + \sqrt{u^2 \sin^2 \theta + 2gh}}{g}.$$

- u — launch speed in m/s; θ — launch angle above the horizontal; h — depth of the landing point below launch, in m; g in m/s², so t comes out in s.
- The discriminant $u^2 \sin^2 \theta + 2gh$ is exactly the square of the vertical impact speed, because energy gives $v_y^2 = u_y^2 + 2gh$. One square root, two readings.
- The horizontal range is then $R = (u \cos \theta) t$ — the horizontal channel never changes.
- Impact speed from energy alone: $v^2 = u^2 + 2gh$, valid whatever the launch angle.

Case 2 — the inclined plane. Rotate the axes to sit on the plane. Let the plane rise at angle β above the horizontal and the projectile be launched at angle θ above the horizontal, so the launch angle *relative to the incline* is $\theta - \beta$. Along the incline (x' , uphill positive) and perpendicular to it (y' , out of the plane positive), gravity resolves into two pieces:

$$a_{x'} = -g \sin \beta, \quad a_{y'} = -g \cos \beta,$$

and the launch velocity resolves into

$$u_{x'} = u \cos(\theta - \beta), \quad u_{y'} = u \sin(\theta - \beta).$$

This is the chapter's independence principle with rotated axes: two 1-D problems sharing the clock t .

Flight time. The projectile lands when y' returns to zero:

$$0 = u \sin(\theta - \beta) t - \frac{1}{2}(g \cos \beta) t^2,$$

so, discarding the root $t = 0$,

$$T = \frac{2u \sin(\theta - \beta)}{g \cos \beta}.$$

Range along the incline. The x' -displacement during time T is

$$R = u \cos(\theta - \beta) T - \frac{1}{2}(g \sin \beta) T^2.$$

Substituting T and factoring out $\frac{2u^2}{g \cos^2 \beta}$:

$$R = \frac{2u^2}{g \cos^2 \beta} \sin(\theta - \beta) \left[\cos(\theta - \beta) \cos \beta - \sin(\theta - \beta) \sin \beta \right].$$

The bracket is $\cos((\theta - \beta) + \beta) = \cos \theta$ by the cosine addition rule, so

$$R = \frac{2u^2 \sin(\theta - \beta) \cos \theta}{g \cos^2 \beta}.$$

Maximum range. Using $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$ with $A = \theta - \beta$, $B = \theta$:

$$R = \frac{u^2 [\sin(2\theta - \beta) - \sin \beta]}{g \cos^2 \beta}.$$

This peaks when $\sin(2\theta - \beta) = 1$, i.e. at

$$\theta = 45^\circ + \frac{\beta}{2}, \quad R_{\max} = \frac{u^2(1 - \sin \beta)}{g \cos^2 \beta} = \frac{u^2}{g(1 + \sin \beta)},$$

where the last step cancels $1 - \sin \beta$ against $\cos^2 \beta = (1 - \sin \beta)(1 + \sin \beta)$.

- If the angle α is given *from the incline* instead of from the horizontal, put $\theta = \alpha + \beta$ throughout.
- Sanity anchors: $\beta = 0$ collapses everything to the chapter's level-ground results, and $\theta = 90^\circ$ gives $R = 0$ (fired straight up, lands at your feet on the incline).

Example A.1 — [CBSE Board Level]

From the top of a tower of height 75 m, a ball is thrown at 20 m/s at 30° above the horizontal ($g = 10 \text{ m/s}^2$). Find (a) the time of flight, (b) the horizontal distance from the base of the tower where it lands, and (c) the impact speed.

Given: $u = 20 \text{ m/s}$, $\theta = 30^\circ$, $h = 75 \text{ m}$, so $u \sin \theta = 10 \text{ m/s}$ and $u \cos \theta = 10\sqrt{3} \text{ m/s} \approx 17.32 \text{ m/s}$.

(a) Apply the depth formula. The discriminant is

$$u^2 \sin^2 \theta + 2gh = (10)^2 + 2(10)(75) = 100 + 1500 = 1600 \text{ m}^2/\text{s}^2,$$

so $\sqrt{1600} = 40 \text{ m/s}$ and

$$t = \frac{10 + 40}{10} = 5 \text{ s}.$$

(b) The horizontal channel runs at constant speed:

$$R = (10\sqrt{3})(5) = 50\sqrt{3} \approx 86.6 \text{ m}.$$

(c) By energy, $v^2 = u^2 + 2gh = 400 + 1500 = 1900 \text{ m}^2/\text{s}^2$, so

$$v = \sqrt{1900} = 10\sqrt{19} \approx 43.6 \text{ m/s}.$$

Check. Component route: $v_y = u \sin \theta - gt = 10 - 50 = -40 \text{ m/s}$, and $\sqrt{(10\sqrt{3})^2 + (-40)^2} = \sqrt{300 + 1600} = \sqrt{1900}$. Same number — the discriminant of the flight-time quadratic really was the vertical impact speed. Note the flight lasts longer than the level-ground value $2u \sin \theta / g = 2 \text{ s}$: falling an extra 75 m buys extra time, hence extra range.

Example A.2 — [JEE Main Level]

A projectile is fired at 20 m/s at 60° above the horizontal, up a plane inclined at 30° to the horizontal ($g = 10 \text{ m/s}^2$). Find the time of flight and the range along the incline, and verify the range against the maximum-range formula.

Here $\theta - \beta = 60^\circ - 30^\circ = 30^\circ$, and $\cos \beta = \frac{\sqrt{3}}{2}$.

(a) Flight time:

$$T = \frac{2u \sin(\theta - \beta)}{g \cos \beta} = \frac{2(20)(0.5)}{10 \times \frac{\sqrt{3}}{2}} = \frac{20}{5\sqrt{3}} = \frac{4}{\sqrt{3}} \approx 2.31 \text{ s.}$$

(b) Range along the incline:

$$R = \frac{2u^2 \sin(\theta - \beta) \cos \theta}{g \cos^2 \beta} = \frac{2(400)(0.5)(0.5)}{10 \times \frac{3}{4}} = \frac{200}{7.5} \approx 26.7 \text{ m.}$$

(c) Verification. This angle happens to be the optimum, since $45^\circ + \beta/2 = 45^\circ + 15^\circ = 60^\circ$. The maximum-range formula must therefore agree:

$$R_{\max} = \frac{u^2}{g(1 + \sin \beta)} = \frac{400}{10 \times 1.5} = \frac{400}{15} \approx 26.7 \text{ m.} \checkmark$$

Two independent formulas, one number — and the agreement confirms both the rotated-axis setup and the trigonometric reduction in one stroke.

Practice — Addendum A

1. A ball rolls off a 45 m cliff horizontally at 10 m/s ($g = 10 \text{ m/s}^2$). Find its time of fall and where it lands.
2. A ball is thrown horizontally from a roof and lands 40 m from the base 4 s later ($g = 10 \text{ m/s}^2$). Find the roof's height and the launch speed.
3. Show that on a 30° incline, the range of a projectile of launch speed u is maximised at $\theta = 60^\circ$ from the horizontal, and evaluate that maximum range for $u = 10 \text{ m/s}$, $g = 10 \text{ m/s}^2$.
4. Derive the range formula for a projectile fired *down* an incline of angle β , launched at angle θ above the horizontal with the same speed u .
5. From a tower of height 20 m a ball is thrown at 15 m/s at 30° above the horizontal ($g = 10 \text{ m/s}^2$). Find the horizontal distance travelled and the impact speed.

Answers: 1. $T = \sqrt{2h/g} = \sqrt{9} = 3 \text{ s}$; $R = (10)(3) = 30 \text{ m}$ from the base. 2. $h = \frac{1}{2}gt^2 = 80 \text{ m}$; $u = 40/4 = 10 \text{ m/s}$. 3. The optimum is $\theta = 45^\circ + \beta/2 = 60^\circ$; $R_{\max} = u^2/[g(1 + \sin 30^\circ)] = 100/15 \approx 6.67 \text{ m}$. 4. Reverse the sign of the along-incline gravity term and of the geometry: $R = u^2[\sin(2\theta + \beta) + \sin \beta]/(g \cos^2 \beta)$, obtained by replacing $\beta \rightarrow -\beta$ in the uphill formula. 5. $t = (7.5 + \sqrt{56.25 + 400})/10 = (7.5 + 21.36)/10 \approx 2.89 \text{ s}$; $R \approx (12.99)(2.89) \approx 37.5 \text{ m}$; $v = \sqrt{u^2 + 2gh} = \sqrt{625} = 25 \text{ m/s}$ exactly — the energy check is the fast route to the impact speed.

Addendum B — The Vertical Circle, Done Properly

Sub-topic 06 builds all the machinery — centripetal force, real-force identification — but its only vertical-circle item is a practice answer that quotes $T = m\frac{v^2}{r} - mg$ at the top without ever deriving it. JEE Main asks for the minimum launch speed to complete a loop nearly every other year; JEE Advanced asks where the string goes slack. Both come from one radial equation plus one energy statement.

Method

A particle of mass m on a string of length l , speed u at the lowest point, g downward. Measure position by the angle θ from the bottom.

Step 1 — the radial equation at any angle. Toward the centre act the tension T (always) and the component of gravity along the radius, which is $-mg \cos \theta$ — inward when the particle is above the level of the centre ($\theta > 90^\circ$), outward when below. Newton's second law along the radius:

$$T - mg \cos \theta = \frac{mv^2}{l} \quad \Rightarrow \quad T = m \left(\frac{v^2}{l} + g \cos \theta \right).$$

Anchor it at the endpoints: $\theta = 0$ gives $T - mg = mv^2/l$ (bottom, correct); $\theta = 180^\circ$ gives $T + mg = mv^2/l$ (top, correct).

Step 2 — energy ties v to θ . The height above the bottom is $l(1 - \cos \theta)$, so with speed u at the bottom:

$$\frac{1}{2}mv^2 = \frac{1}{2}mu^2 - mgl(1 - \cos \theta) \quad \Rightarrow \quad v^2 = u^2 - 2gl(1 - \cos \theta).$$

Step 3 — three headline consequences.

Minimum speed at the top. A string can pull, never push, so staying on the circle requires $T \geq 0$. At the top ($\theta = 180^\circ$):

$$T = m \left(\frac{v_{\text{top}}^2}{l} - g \right) \geq 0 \quad \Rightarrow \quad v_{\text{top}} \geq \sqrt{gl}.$$

Energy between bottom and top (height change $2l$) gives $u^2 = v_{\text{top}}^2 + 4gl$, hence

$$u_{\text{min}} = \sqrt{5gl}.$$

Tension difference. At the bottom $T_1 = m\frac{u^2}{l} + mg$; at the top $T_2 = m\frac{v_{\text{top}}^2}{l} - mg$. Subtracting, and using $u^2 - v_{\text{top}}^2 = 4gl$:

$$T_1 - T_2 = \frac{m(u^2 - v_{\text{top}}^2)}{l} + 2mg = 4mg + 2mg = 6mg.$$

Independent of l , independent of u — provided the loop is completed. This $6mg$ is the single most quoted result of the topic.

The side point. At $\theta = 90^\circ$ gravity has no radial component, so $T_3 = m\frac{v_3^2}{l}$ with $v_3^2 = u^2 - 2gl$. For the critical launch $u^2 = 5gl$: $v_3^2 = 3gl$ and $T_3 = 3mg$.

Slack condition, generally. Setting $T = 0$ in Step 1 and substituting Step 2:

$$u^2 - 2gl(1 - \cos \theta) = -gl \cos \theta \implies u^2 = 2gl - 3gl \cos \theta.$$

A real solution with $0 \leq \theta \leq 180^\circ$ exists precisely when $2gl < u^2 < 5gl$: below $\sqrt{2gl}$ the particle merely oscillates; above $\sqrt{5gl}$ it completes the loop; between, it leaves the circle and becomes a projectile.

Rod versus string. A rigid rod can push. The loop then needs only $v_{\text{top}} \geq 0$, i.e. $u \geq 2\sqrt{gl}$ — and the tension difference $6mg$ no longer applies, since the rod's force at the top may be outward.

Example B.1 — [JEE Main Level]

A 2 kg stone on a 0.5 m string is whirled in a vertical circle with speed 6 m/s at the lowest point ($g = 10 \text{ m/s}^2$). Does it complete the loop? Find the tension at the top and at the bottom.

First the admission test: $u_{\text{min}} = \sqrt{5gl} = \sqrt{5 \times 10 \times 0.5} = \sqrt{25} = 5 \text{ m/s}$. Since $6 > 5$, the loop is completed.

Top: energy gives $v_{\text{top}}^2 = u^2 - 4gl = 36 - 20 = 16 \text{ m}^2/\text{s}^2$. Then

$$T_{\text{top}} = m \left(\frac{v_{\text{top}}^2}{l} - g \right) = 2 \left(\frac{16}{0.5} - 10 \right) = 2(32 - 10) = 44 \text{ N}.$$

Bottom:

$$T_{\text{bottom}} = m \left(\frac{u^2}{l} + g \right) = 2 \left(\frac{36}{0.5} + 10 \right) = 2(72 + 10) = 164 \text{ N}.$$

Check. $T_{\text{bottom}} - T_{\text{top}} = 120 \text{ N}$, and indeed $6mg = 6(2)(10) = 120 \text{ N}$.

Example B.2 — [JEE Advanced Level]

The stone of Example B.1's setup ($l = 0.5 \text{ m}$, $g = 10 \text{ m/s}^2$) is instead launched from the bottom with $u = \sqrt{4gl} = \sqrt{20} \approx 4.47 \text{ m/s}$. Show it neither oscillates nor completes the loop, and find the angle (from the bottom) at which the string goes slack, together with its speed there.

Since $\sqrt{2gl} < \sqrt{4gl} < \sqrt{5gl}$, the stone climbs past the horizontal but leaves the circle before the top — the intermediate regime.

Apply the slack condition with $u^2 = 4gl$:

$$4gl = 2gl - 3gl \cos \theta \implies \cos \theta = -\frac{2}{3} \implies \theta \approx 132^\circ.$$

Speed at release: from $T = 0$, $v^2 = -gl \cos \theta = \frac{2gl}{3} = \frac{2(10)(0.5)}{3} = \frac{10}{3} \approx 3.33 \text{ m}^2/\text{s}^2$, so $v \approx 1.83 \text{ m/s}$.

Sanity checks. The height at release is $l(1 - \cos \theta) = \frac{5l}{3} \approx 0.83 \text{ m}$, less than the diameter $2l = 1 \text{ m}$ — consistent with failing to reach the top. And the energy ledger closes: $\frac{1}{2}u^2 = 2gl = 10 \text{ J}$ per kilogram, of which $gh = 10 \times 0.833 \approx 8.33 \text{ J/kg}$ is potential at release, leaving $\approx 1.67 = \frac{1}{2}(1.83)^2 \text{ J/kg}$ kinetic. From the release instant onward the stone is an ordinary projectile under gravity.

Practice — Addendum B

- Find the minimum speed at the bottom needed to complete a vertical loop of radius 0.9 m ($g = 10 \text{ m/s}^2$).
- A bead completes a loop with $u^2 = 7gl$ at the bottom. Express the tension at the top and at the bottom in terms of mg .
- For the critical launch $u = \sqrt{5gl}$, find the speed and the tension when the string is horizontal.
- A bead threaded on a rigid circular wire is given $u = \sqrt{3gl}$ at the bottom. Does it complete the loop?
- Show that $T_{\text{bottom}} - T_{\text{top}} = 6mg$ contains neither l nor u , and explain physically why.

Answers: 1. $\sqrt{5gl} = \sqrt{45} \approx 6.71 \text{ m/s}$. 2. Top: $v_{\text{top}}^2 = 3gl$, so $T_{\text{top}} = 2mg$; bottom: $T_{\text{bottom}} = 8mg$; difference $6mg$ as always. 3. $v^2 = 3gl$, $v = \sqrt{3gl}$; $T = 3mg$ (gravity contributes nothing radially there). 4. Yes — a rod/wire can push, so the requirement is only $u \geq 2\sqrt{gl}$, and $\sqrt{3} > 2$ is false... check carefully: $\sqrt{3gl}$ compared with $2\sqrt{gl}$: squaring, $3gl$ versus $4gl$, so $u < 2\sqrt{gl}$ and it does **not** complete the loop; it rises past the horizontal and slides back (no slack issue exists on a track, so it oscillates about the bottom). 5. The difference reduces to $m(u^2 - v_{\text{top}}^2)/l + 2mg$, and energy fixes $u^2 - v_{\text{top}}^2 = 4gl$ regardless of the actual values — the extra tension at the bottom over the top must supply exactly the centripetal bookkeeping of climbing $2l$, which depends only on g .

Addendum C — Banking With Friction: The Complete Safe-Speed Band

Sub-topic 06 derives two separate results: $v_{\text{max}} = \sqrt{\mu rg}$ on a flat road, and $\tan \theta = \frac{v^2}{rg}$ on a frictionless bank. Real JEE questions put both ingredients on the same road — a banked curve *with* friction — and ask for the band of legal speeds. Neither chapter result survives alone.

Method

Road banked at θ , coefficient of static friction μ , curve radius r . The normal force N tilts inward from the vertical by θ ; friction f acts along the road surface. There are two regimes.

Upper limit — car tending to slide up and out. Friction opposes the tendency and acts *down* the slope, contributing an inward radial piece $f \cos \theta$ and a downward piece $f \sin \theta$. At the limit $f = \mu N$:

$$N \sin \theta + \mu N \cos \theta = \frac{mv^2}{r}, \quad N \cos \theta - \mu N \sin \theta = mg.$$

Divide the first by the second (N and m both cancel):

$$v_{\max} = \sqrt{rg \frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta}} = \sqrt{rg \frac{\mu + \tan \theta}{1 - \mu \tan \theta}}.$$

Lower limit — car tending to slide down the bank. Friction acts *up* the slope, outward radially. Repeat with signs flipped:

$$v_{\min} = \sqrt{rg \frac{\sin \theta - \mu \cos \theta}{\cos \theta + \mu \sin \theta}} = \sqrt{rg \frac{\tan \theta - \mu}{1 + \mu \tan \theta}}.$$

Three consistency checks you should run every time.

- $\mu = 0$: both collapse to $\sqrt{rg \tan \theta}$ — the chapter's frictionless design speed v_0 .
- $\theta = 0$: $v_{\max} = \sqrt{\mu rg}$ (the chapter's flat-road result), $v_{\min} = 0$.
- If $\tan \theta \leq \mu$, the numerator of $v_{\min}$ is non-positive: the bank is steep-frictioned enough to hold a *stationary* car, so the legal band is $0 \leq v \leq v_{\max}$.

Friction at the design speed. At $v = v_0 = \sqrt{rg \tan \theta}$, the pair $N \sin \theta = \frac{mv_0^2}{r}$ and $N \cos \theta = mg$ divides to exactly $\tan \theta = \frac{v_0^2}{rg}$ with $f = 0$ — the equations are satisfied with no friction at all, so the required friction there is zero however slippery the road.

Example C.1 — [JEE Main Level]

A curve of radius 20 m is banked at 37° (take $\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$). The coefficient of friction is 0.5 and $g = 10 \text{ m/s}^2$. Find $v_{\max}$, $v_{\min}$ and the friction-free design speed.

With $rg = 200 \text{ m}^2/\text{s}^2$:

$$v_{\max}^2 = 200 \times \frac{0.6 + (0.5)(0.8)}{0.8 - (0.5)(0.6)} = 200 \times \frac{1.0}{0.5} = 400,$$

so $v_{\max} = 20 \text{ m/s}$.

$$v_{\min}^2 = 200 \times \frac{0.6 - (0.5)(0.8)}{0.8 + (0.5)(0.6)} = 200 \times \frac{0.2}{1.1} \approx 36.4,$$

so $v_{\min} \approx 6.03$ m/s. Design speed: $v_0^2 = rg \tan 37^\circ = 200 \times 0.75 = 150$, so $v_0 \approx 12.2$ m/s.

Checks. Strip the friction ($\mu = 0$) and both limits become $\sqrt{150} \approx 12.2$ m/s — the band pinches to a point at the design speed, exactly as it must. Flatten the road ($\theta = 0$) and $v_{\max}$ becomes $\sqrt{\mu rg} = \sqrt{100} = 10$ m/s. The legal window here is roughly 6.0 m/s to 20 m/s: banking widens the band asymmetrically upward, which is the entire engineering point.

Example C.2 — [JEE Advanced Level]

On the curve of Example C.1, an 800 kg car travels at 10 m/s — below the design speed. Find the friction force on the car: magnitude, direction, and whether the available friction suffices.

Below v_0 the car tends to slide *down* the bank, so assume friction acts up the slope (its radial component then points outward). Newton's law radially and vertically:

$$N \sin \theta - f \cos \theta = \frac{mv^2}{r}, \quad N \cos \theta + f \sin \theta = mg.$$

To isolate f , multiply the first by $\cos \theta$ and the second by $\sin \theta$, then subtract the first from the second:

$$f = mg \sin \theta - \frac{mv^2 \cos \theta}{r} = (800)(10)(0.6) - \frac{(800)(100)(0.8)}{20} = 4800 - 3200 = 1600 \text{ N}.$$

Positive, so the assumed direction is right: friction acts **up the slope** with magnitude 1600 N. For feasibility we need N : multiply the first equation by $\sin \theta$ and the second by $\cos \theta$, then add:

$$N = \frac{mv^2 \sin \theta}{r} + mg \cos \theta = \frac{(800)(100)(0.6)}{20} + (800)(10)(0.8) = 2400 + 6400 = 8800 \text{ N}.$$

Required coefficient: $f/N = 1600/8800 \approx 0.18$, comfortably below the available 0.5.

Physical reading. Slow cars need friction holding them *up* the bank; fast cars need friction holding them *down*. Only at $v_0 \approx 12.2$ m/s does the road's geometry do all the work alone — which is why highway curves are banked for the typical traffic speed, not the maximum one.

Practice — Addendum C

1. A frictionless curve of radius 40 m is to be taken safely at 20 m/s ($g = 10 \text{ m/s}^2$). Find the banking angle.
2. For a curve with $r = 50$ m, $\theta = 30^\circ$ ($\tan 30^\circ \approx 0.577$, $\sin 30^\circ = 0.5$, $\cos 30^\circ \approx 0.866$), $\mu = 0.5$, $g = 10 \text{ m/s}^2$: find $v_{\max}$.
3. Prove that $v_{\min} = 0$ if and only if $\tan \theta \leq \mu$.
4. A car takes a banked curve at exactly v_0 . What friction force acts, and what supplies the centripetal force?
5. On a banked curve with $\tan \theta = 0.75$ and $\mu = 0.8$, what is the lower end of the legal speed band?

Answers: 1. $\tan \theta = v^2/rg = 400/400 = 1$, so $\theta = 45^\circ$. 2. $v_{\max}^2 = 500(0.5 + 0.433)/(0.866 - 0.289) = 500 \times 0.933/0.577 \approx 808$, so $v_{\max} \approx 28.4$ m/s. 3. $v_{\min}^2 \propto (\tan \theta - \mu)$; the expression under the root is non-negative precisely when $\tan \theta \geq \mu$, and otherwise the static-friction balance holds even for $v = 0$ — a parked car does not slide down a sufficiently rough bank. 4. Zero friction; the inward component of the normal force alone supplies mv_0^2/r . 5. Since $\tan \theta = 0.75 \leq \mu = 0.8$, $v_{\min} = 0$ — the car may stop on the bank.

Addendum D — When the Current Wins, and Closest Approach by Geometry

Sub-topic 05 handles the river cleanly for $v_b > v_r$, and its limiting-conditions box admits that when $v_b \leq v_r$ “the goal shifts to minimising” the drift — without doing it. Meanwhile its closest-approach work minimises $|\vec{r}|^2$ by calculus; the geometric shortcut is faster, less error-prone, and worth owning as a second method.

Method D.1 — Minimum drift when $v_b \leq v_r$

Axes: x along the current, y straight across. The boat heads upstream of straight-across by angle θ . Its velocity components relative to the ground:

$$v_y = v_b \cos \theta, \quad v_x = v_r - v_b \sin \theta.$$

Both are positive (the current cannot be fully cancelled when $v_r > v_b$), so the drift is

$$x(\theta) = \frac{v_r - v_b \sin \theta}{v_b \cos \theta} d = d \left[\frac{v_r}{v_b} \sec \theta - \tan \theta \right],$$

where d is the river width. Minimise over θ :

$$\frac{dx}{d\theta} = d \left[\frac{v_r}{v_b} \sec \theta \tan \theta - \sec^2 \theta \right] = 0 \quad \Rightarrow \quad \frac{v_r}{v_b} \sin \theta = 1 \quad \Rightarrow \quad \boxed{\sin \theta = \frac{v_b}{v_r}}.$$

With $\cos \theta = \frac{\sqrt{v_r^2 - v_b^2}}{v_r}$, substitution gives

$$t_{\text{cross}} = \frac{d v_r}{v_b \sqrt{v_r^2 - v_b^2}}, \quad x_{\min} = \frac{d \sqrt{v_r^2 - v_b^2}}{v_b}.$$

Limit check. As $v_r \rightarrow v_b$: $\theta \rightarrow 90^\circ$ (head straight upstream), $x_{\min} \rightarrow 0$, but $t_{\text{cross}} \rightarrow \infty$. You can hold station against an equal current forever and cross it never — the formula’s two limits disagree beautifully, and both are right. Below the critical heading, more upstream angle buys less drift at a growing price in time.

Method D.2 — Closest approach without calculus

Body B sits at relative position $\vec{r}_0$ from A at $t = 0$; their relative velocity is $\vec{v}$ (constant). Separation squared:

$$s^2(t) = |\vec{r}_0 + \vec{v}t|^2 = |\vec{r}_0|^2 + 2(\vec{r}_0 \cdot \vec{v})t + |\vec{v}|^2t^2.$$

This is a quadratic in t with positive leading coefficient, minimised at

$$t_* = -\frac{\vec{r}_0 \cdot \vec{v}}{|\vec{v}|^2}.$$

Substituting back and collecting terms as a perfect square:

$$s_{\min}^2 = |\vec{r}_0|^2 - \frac{(\vec{r}_0 \cdot \vec{v})^2}{|\vec{v}|^2} = \frac{|\vec{r}_0 \times \vec{v}|^2}{|\vec{v}|^2},$$

using $|\vec{r}_0|^2|\vec{v}|^2 = (\vec{r}_0 \cdot \vec{v})^2 + |\vec{r}_0 \times \vec{v}|^2$. Hence

$$s_{\min} = \frac{|\vec{r}_0 \times \vec{v}|}{|\vec{v}|}.$$

That is the perpendicular-distance formula: in A's frame, B moves along the straight line $\vec{r}_0 + \vec{v}t$, and the closest approach is the foot of the perpendicular from the origin onto it. Sign discipline: $\vec{r}_0 \cdot \vec{v} > 0$ makes $t_* < 0$ — the bodies are separating from the outset and the closest approach is at $t = 0$ itself.

Example D.1 — [JEE Main Level]

A river is 96 m wide and flows at 5 m/s. A boat manages 4 m/s in still water. Find the heading that minimises the downstream drift, the minimum drift, and the crossing time.

Since $v_b = 4 < v_r = 5$, zero drift is impossible. Aim upstream of straight-across with

$$\sin \theta = \frac{v_b}{v_r} = \frac{4}{5} = 0.8 \quad \Rightarrow \quad \theta = 53^\circ, \quad \cos \theta = 0.6.$$

Across-speed: $v_b \cos \theta = 2.4$ m/s. Crossing time:

$$t = \frac{96}{2.4} = 40 \text{ s}.$$

Minimum drift, by formula or directly as $v_x t = (5 - 4 \sin 53^\circ)(40)$:

$$x_{\min} = \frac{d\sqrt{v_r^2 - v_b^2}}{v_b} = \frac{96\sqrt{25 - 16}}{4} = \frac{96 \times 3}{4} = 72 \text{ m}.$$

Direct check: $v_x = 5 - (4)(0.8) = 1.8$ m/s, and $(1.8)(40) = 72$ m. The best possible outcome is still 72 m adrift — the current wins, and the art is losing by as little as possible.

Example D.2 — [JEE Advanced Level]

Re-derive the answer to Sub-topic 05, Example 4, by the geometric method: ship A at the origin sails due north at 10 km/h; ship B is initially at (20, 0) km and sails due west at 10 km/h. Find the time and distance of closest approach.

Relative velocity of B with respect to A, taking east as $\hat{i}$ and north as $\hat{j}$:

$$\vec{v} = \vec{v}_B - \vec{v}_A = (-10\hat{i} - 10\hat{j}) \text{ km/h}, \quad |\vec{v}|^2 = 200 \text{ km}^2/\text{h}^2.$$

Initial relative position: $\vec{r}_0 = 20\hat{i}$ km.

Dot product: $\vec{r}_0 \cdot \vec{v} = (20)(-10) + (0)(-10) = -200 \text{ km}^2/\text{h}$, so

$$t_* = -\frac{-200}{200} = 1 \text{ h}.$$

Cross product (only the $\hat{k}$ component survives): $\vec{r}_0 \times \vec{v} = (20)(-10)\hat{k} = -200\hat{k} \text{ km}^2/\text{h}$, so

$$s_{\min} = \frac{|-200|}{\sqrt{200}} = \frac{200}{10\sqrt{2}} = 10\sqrt{2} \approx 14.1 \text{ km}.$$

Both values match the chapter's calculus route exactly. The geometric route took four lines and no derivatives — and in an examination, the cross product doubles as a built-in error check, because a wrong relative velocity almost never produces a clean cancellation.

Practice — Addendum D

1. A river is 96 m wide, flows at 5 m/s, and a boat manages 4 m/s — but now the boatman simply aims *straight across*. Find the crossing time and drift, and compare with Example D.1.
2. Rain falls at 10 m/s tilted at 30° to the vertical, drifting horizontally in the same direction a man walks. At what walking speed does the rain appear to fall perfectly vertically on him?
3. Ship P sails north at 8 km/h. Ship Q is initially 30 km due east of P and sails with velocity $(-6\hat{i} - 8\hat{j})$ km/h. Find the time and distance of closest approach.
4. Derive $t_* = -(\vec{r}_0 \cdot \vec{v})/|\vec{v}|^2$ by differentiating $s^2(t)$, and state what a negative value of t_* means physically.
5. A swimmer can manage 2 m/s in still water; the river flows at 2 m/s. Describe the best possible crossing.

Answers: 1. $t = 96/4 = 24$ s, drift = $5 \times 24 = 120$ m — six seconds faster than the minimum-drift heading, but 48 m further adrift; the time-versus-drift trade-off made quantitative. 2. The rain's horizontal component is $10 \sin 30^\circ = 5$ m/s; walking at exactly 5 m/s cancels it, leaving a purely vertical relative velocity — the apparent rain is vertical. 3. $\vec{v}_{\text{rel}} = (-6, -16)$ km/h,

$|\vec{v}_{\text{rel}}|^2 = 292$; $\vec{r}_0 = (30, 0)$; $\vec{r}_0 \cdot \vec{v}_{\text{rel}} = -180$, so $t_* = 180/292 \approx 0.616 \text{ h} \approx 37 \text{ min}$; $s_{\text{min}} = |30 \times (-16)|/\sqrt{292} = 480/17.09 \approx 28.1 \text{ km}$. 4. Differentiating $s^2(t) = |\vec{r}_0|^2 + 2(\vec{r}_0 \cdot \vec{v})t + |\vec{v}|^2 t^2$ gives $2(\vec{r}_0 \cdot \vec{v}) + 2|\vec{v}|^2 t = 0$ at the minimum; a negative t_* means $\vec{r}_0 \cdot \vec{v} > 0$, i.e. the bodies are separating from the start and the true closest approach occurred at or before $t = 0$. 5. With $v_b = v_r$, the minimum-drift heading is straight upstream ($\theta = 90^\circ$), which crosses in infinite time; any finite crossing requires accepting some drift — formally, $x_{\text{min}} = 0$ is attainable only in the limit of never arriving.

Laws of Motion

Chapter 4 | Physics Content Bank — JEE / NEET / CBSE

June 2026

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Subtopic 1: Newton's Laws and Impulse

Exam Relevance: This is the spine of all of mechanics. JEE Main reliably pulls 1–2 questions a year from impulse and conservation of momentum, often disguised inside collision or variable-mass problems. NEET leans on the conceptual side — inertia, action–reaction, and the impulse “area under the graph” idea. JEE Advanced rarely asks it standalone but weaves it into multi-body and variable-mass setups. CBSE Boards almost always carries a 2–3 mark numerical on the impulse–momentum theorem.

Section 1: Concept Intuition

Imagine you're standing in a crowded Mumbai local train, no handrail in reach, when the train suddenly jerks forward to leave the platform. Your feet move with the floor, but your upper body — for just a heartbeat — stays exactly where it was. You lurch backward relative to the train. Nobody pushed you. So what threw you?

Nothing did. That backward lurch *is* the whole of Newton's First Law. Your body was at rest, and a body at rest genuinely wants to *stay* at rest. It has no built-in desire to speed up just because the floor underneath it did. We give this stubbornness a name: **inertia** — the reluctance of matter to change whatever state of motion it's already in. The train pulling away didn't push you backward; it simply ran out from under your lazy upper body.

Newton's First Law says: *a body continues in its state of rest or of uniform motion in a straight line unless compelled by a net external force to change that state*. Notice the law isn't really about motion — it's about **change**. Constant velocity (including zero velocity) is the “free” state that needs no force to maintain. You only need a force to *change* it.

How much inertia a body has is measured by its **mass**. A loaded goods wagon is far harder to get rolling — and far harder to stop — than an empty handcart. Same idea when a cricket bowler bowls: a cork-and-leather ball can be redirected with a flick of the bat, but try redirecting a shot-put with that same flick and you'll hurt your wrists. More mass, more inertia, more stubbornness.

Now bring in motion and you get **momentum**, written $\vec{p} = m\vec{v}$ — literally “how much motion” a body carries, accounting for both how heavy it is and how fast it's going. A truck crawling at 5 km/h and a motorbike racing at 80 km/h can carry comparable momentum because the truck makes up in mass what it lacks in speed. Momentum is the quantity Newton actually cared about.

Think about it this way... A force is not a thing a body “has” — it's an *interaction* that changes a body's momentum. Newton's Second Law, in its truest form, says force is the **rate at which momentum changes**: push twice as hard, momentum changes twice as fast. And when that change happens over a very short, violent instant — a bat meeting a ball, a foot meeting a football, a car meeting a crash barrier — we stop caring about the exact force at each microsecond and care only about the *total kick* delivered. That total kick is **impulse**, and impulse is simply the change in momentum.

Newton's Third Law completes the picture: forces never come alone. When the bat pushes the ball forward, the ball pushes back on the bat with an equal and opposite force. These two forces act on *different bodies*, which is exactly why they don't cancel — a point worth tattooing on your brain before exam day.

Limiting Conditions & Assumptions:

- Newton's laws hold only in **inertial frames** (frames not accelerating). Inside that jerking train,

you *seem* to be pushed backward by a phantom force — that’s because the train is a non-inertial frame, and the “force” is fictitious.

- The simple form $\vec{F} = m\vec{a}$ assumes **constant mass**. For rockets, falling chains, or sand piling onto a belt, you must return to the parent form $\vec{F} = \frac{d\vec{p}}{dt}$.
- All of this is non-relativistic: it assumes speeds far below the speed of light, where mass stays constant.

Section 2: Key Formulas & Definitions

Linear Momentum

$$\vec{p} = m\vec{v}$$

where m is mass (kg) and $\vec{v}$ is velocity (m/s).

- SI unit: $\text{kg} \cdot \text{m} \cdot \text{s}^{-1}$ (equivalently $\text{N} \cdot \text{s}$)
- Dimensional formula: $[M^1 L^1 T^{-1}]$

Newton’s Second Law (fundamental form)

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt}$$

Newton’s Second Law (constant-mass form)

$$\vec{F}_{\text{net}} = m\vec{a}$$

where $\vec{a}$ is acceleration (m/s^2).

- SI unit of force: newton, $\text{N} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$
- Dimensional formula of force: $[M^1 L^1 T^{-2}]$

Impulse (constant force)

$$\vec{J} = \vec{F} \Delta t$$

Impulse (variable force)

$$\vec{J} = \int_{t_1}^{t_2} \vec{F} dt = \text{area under the force-time graph}$$

- SI unit: $\text{N} \cdot \text{s} = \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
- Dimensional formula: $[M^1 L^1 T^{-1}]$ — **identical to momentum** (a fact NEET loves to test).

Impulse–Momentum Theorem

$$\vec{J} = \Delta\vec{p} = m\vec{v}_f - m\vec{v}_i$$

Average Force during an impact

$$\vec{F}_{\text{avg}} = \frac{\Delta\vec{p}}{\Delta t}$$

Conservation of Linear Momentum — if $\vec{F}_{\text{ext}} = 0$:

$$\vec{p}_{\text{initial}} = \vec{p}_{\text{final}} \quad \Rightarrow \quad \sum m_i \vec{v}_i = \text{constant}$$

Newton's Third Law

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

(force by A on B equals minus the force by B on A — same line of action, opposite directions, acting on different bodies).

Section 3: CBSE Board Derivations

Derivation 1 — Recovering $\vec{F} = m\vec{a}$ from the momentum definition of force.

Begin with the fundamental statement of the Second Law:

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt}$$

Why this form first? Because momentum, not acceleration, is the quantity that force genuinely controls. We now make one explicit assumption: the **mass of the body is constant**. This is true for almost every Board-level body — a block, a ball, a car of fixed load. With m constant, it slides out of the derivative:

$$\vec{F} = m \frac{d\vec{v}}{dt}$$

The rate of change of velocity is, by definition, acceleration $\vec{a} = \frac{d\vec{v}}{dt}$, so:

$$\boxed{\vec{F} = m\vec{a}}$$

This also quietly proves that the First Law lives *inside* the Second Law: set $\vec{F} = 0$ and you get $\vec{a} = 0$, i.e. velocity stays constant — exactly the First Law's claim.

Derivation 2 — The Impulse–Momentum Theorem.

Take the fundamental form again and multiply both sides by dt :

$$\vec{F} dt = d\vec{p}$$

Now integrate over the duration of the interaction, from the instant contact begins (t_1) to the instant it ends (t_2):

$$\int_{t_1}^{t_2} \vec{F} dt = \int_{\vec{p}_i}^{\vec{p}_f} d\vec{p}$$

The left side is, by definition, the impulse $\vec{J}$. The right side integrates cleanly to the change in momentum:

$$\vec{J} = \vec{p}_f - \vec{p}_i = m\vec{v}_f - m\vec{v}_i$$

Physical reading: we never needed to know the force at each microsecond of the collision. We only needed the endpoints of momentum. That's the entire power of the theorem — it lets us sidestep a messy, spiky, unmeasurable force.

Derivation 3 — Conservation of Momentum from the Third Law.

Consider two bodies A and B interacting with each other and nothing else (an isolated system). During their interaction, A exerts $\vec{F}_{AB}$ on B, and by the Third Law, B exerts $\vec{F}_{BA} = -\vec{F}_{AB}$ on A.

Apply the Second Law to each:

$$\vec{F}_{AB} = \frac{d\vec{p}_B}{dt}, \quad \vec{F}_{BA} = \frac{d\vec{p}_A}{dt}$$

Add them. The left side vanishes because the forces are equal and opposite:

$$\frac{d\vec{p}_A}{dt} + \frac{d\vec{p}_B}{dt} = \vec{F}_{BA} + \vec{F}_{AB} = 0$$

$$\frac{d}{dt}(\vec{p}_A + \vec{p}_B) = 0 \quad \Rightarrow \quad \vec{p}_A + \vec{p}_B = \text{constant}$$

So when no external force acts, the **total** momentum of the system is conserved — even though the individual momenta change wildly during the interaction. This is the law you'll reach for in every recoil, explosion, and collision problem.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A cricket ball of mass 0.16 kg is moving toward a batsman at 30 m/s. The batsman hits it straight back along the same line at 40 m/s. If the bat and ball are in contact for 0.01 s, find (a) the impulse imparted to the ball, and (b) the average force exerted by the bat.

Given: $m = 0.16$ kg, $v_i = 30$ m/s, $v_f = 40$ m/s (opposite direction), $\Delta t = 0.01$ s.

Step 1 — Fix a sign convention. Take the direction *away from the batsman* (the final direction) as positive. Then the incoming velocity is $v_i = -30$ m/s and the outgoing velocity is $v_f = +40$ m/s.

Step 2 — Apply the impulse–momentum theorem.

$$J = mv_f - mv_i = 0.16(+40) - 0.16(-30)$$

$$J = 6.4 + 4.8 = 11.2 \text{ N} \cdot \text{s}$$

Step 3 — Average force.

$$F_{\text{avg}} = \frac{J}{\Delta t} = \frac{11.2}{0.01} = 1120 \text{ N}$$

Answer: Impulse = 11.2 N · s, directed away from the batsman; average force = 1120 N.

Presentation note: the marks here are won by the sign convention. A student who writes $J = 0.16(40 - 30) = 1.6 \text{ N} \cdot \text{s}$ has forgotten the ball *reversed direction* and will lose almost everything.

Example 2 — [NEET Speed Trap] A rifle of mass 4 kg fires a bullet of mass 20 g with a muzzle speed of 400 m/s. What is the recoil speed of the rifle?

The trap: students start writing energy equations or hunting for the force. Don't. The word "fires" with no external horizontal force screams **conservation of momentum**.

Mental solve: total momentum before firing is zero (everything at rest). So after firing, the two momenta must cancel:

$$m_{\text{bullet}}v_{\text{bullet}} = m_{\text{rifle}}v_{\text{rifle}}$$
$$v_{\text{rifle}} = \frac{0.02 \times 400}{4} = \frac{8}{4} = 2 \text{ m/s}$$

Answer: 2 m/s, in the direction opposite the bullet.

Speed tip: convert 20 g = 0.02 kg instantly, and notice $0.02 \times 400 = 8$ is a clean number. The whole thing is a 10-second problem if you spot conservation of momentum on sight. If an option says "2 m/s in the same direction as the bullet," eliminate it on physics alone — recoil is *always* backward.

Example 3 — [JEE Main Level] A ball of mass 0.5 kg is dropped from a height of 5 m onto a hard floor and rebounds to a height of 1.8 m. Find the magnitude of the impulse delivered to the ball by the floor. Take $g = 10 \text{ m/s}^2$.

This links **kinematics** (finding speeds from heights) with the **impulse–momentum theorem**.

Step 1 — Speed just before impact (free fall through 5 m):

$$v_1 = \sqrt{2gh_1} = \sqrt{2 \times 10 \times 5} = \sqrt{100} = 10 \text{ m/s (downward)}$$

Step 2 — Speed just after impact (rises to 1.8 m):

$$v_2 = \sqrt{2gh_2} = \sqrt{2 \times 10 \times 1.8} = \sqrt{36} = 6 \text{ m/s (upward)}$$

Step 3 — Apply the impulse–momentum theorem with signs. Take upward as positive, so $v_1 = -10 \text{ m/s}$ and $v_2 = +6 \text{ m/s}$:

$$J = mv_2 - mv_1 = 0.5(+6) - 0.5(-10) = 3 + 5 = 8 \text{ N} \cdot \text{s}$$

Answer: The impulse from the floor has magnitude 8 N · s, directed upward.

Insight: note this impulse already accounts for the energy lost in the bounce — we never assumed the speeds were equal, because they weren't. The impulse from gravity *during the brief contact* is negligible compared to the floor's impulse, which is why we ignore it.

Example 4 — [JEE Advanced Level] Dry sand falls vertically at a steady rate of 3 kg/s from a stationary hopper onto a horizontal conveyor belt that moves at a constant 2 m/s. (a) What horizontal force must the belt’s motor supply to keep the belt moving at constant speed? (b) At what rate is kinetic energy being delivered to the sand, and (c) at what rate is the motor doing work? Explain any discrepancy.

This is a **variable-mass / momentum-flux** problem — the kind where $\vec{F} = m\vec{a}$ fails and you must use $\vec{F} = \frac{d\vec{p}}{dt}$.

Step 1 — Set up the horizontal momentum. Each second, a mass $\frac{dm}{dt} = 3$ kg of sand lands. The instant it lands it has zero horizontal velocity; the belt must accelerate it up to $u = 2$ m/s. The horizontal momentum gained per second is:

$$F = \frac{dp}{dt} = u \frac{dm}{dt} = 2 \times 3 = 6 \text{ N}$$

(The belt moves at constant speed, so there’s no $m \frac{dv}{dt}$ term — all the force goes into accelerating *new* sand.)

(a) Answer: The motor must supply 6 N.

Step 2 — Rate of KE gain by the sand. The sand ends up moving at 2 m/s, so KE delivered per second:

$$\frac{dKE}{dt} = \frac{1}{2} \frac{dm}{dt} u^2 = \frac{1}{2}(3)(2)^2 = 6 \text{ W}$$

(b) Answer: Sand gains kinetic energy at 6 W.

Step 3 — Rate of work by the motor. The motor’s force 6 N acts on a belt surface moving at 2 m/s:

$$P_{\text{motor}} = F u = 6 \times 2 = 12 \text{ W}$$

(c) Answer & explanation: The motor does 12 W of work, but the sand gains only 6 W of kinetic energy. The missing 6 W is dissipated as **heat by friction** during the brief slipping while each sand grain is dragged up to belt speed. This factor-of-two split (exactly half the work lost to friction) is the signature result of this problem class and a favorite Advanced “gotcha.”

Section 5: Practice Exercises

1. **[CBSE]** A force of 20 N acts on a body of mass 4 kg initially at rest for 3 s. Find the impulse delivered and the final velocity of the body.
2. **[NEET]** A 0.05 kg ball strikes a vertical wall horizontally at 10 m/s and rebounds at 8 m/s. What is the magnitude of the change in its momentum?
3. **[JEE Main]** A machine gun fires 25 g bullets at a speed of 500 m/s, releasing 4 bullets per second. What average force must the shooter exert to hold the gun steady?
4. **[JEE Main]** A variable force acts on a 2 kg body initially at rest. The force–time graph is a triangle: it rises linearly from 0 to 12 N over the first 4 s, then falls linearly back to 0 over the next 2 s. Find the speed of the body at $t = 6$ s.

5. [JEE Advanced] A trolley of mass $M = 8$ kg moves frictionlessly at 3 m/s. Rain falls vertically and water collects in the open trolley at a steady 0.5 kg/s. Assuming the rain has no horizontal velocity, find the speed of the trolley after 4 s of collecting water.

Final Answers:

1. $J = 60 \text{ N} \cdot \text{s}$; $v = 15 \text{ m/s}$
 2. $0.9 \text{ kg} \cdot \text{m/s}$
 3. 50 N
 4. 18 m/s (impulse = area of triangle = $\frac{1}{2} \times 6 \times 12 = 36 \text{ N} \cdot \text{s}$; $v = 36/2$)
 5. 2.4 m/s (horizontal momentum conserved: $Mv_0 = (M + m)v$, with $m = 0.5 \times 4 = 2 \text{ kg}$)
-

Section 6: Crack the MCQ

Q1. A passenger standing in a bus falls *forward* when the bus suddenly stops. The best explanation is:

- (a) A forward force acts on the passenger due to braking.
- (b) The lower body stops with the bus while the upper body tends to continue moving forward due to inertia.
- (c) The friction between feet and floor pushes the passenger forward.
- (d) The passenger's weight increases momentarily.

Correct: (b). When the bus decelerates, the feet (gripped to the floor by friction) stop with it, but the upper body, obeying inertia of motion, keeps moving forward — so the body pitches forward.

- (a) *is the classic trap*: there is no forward force; the lurch is the *absence* of a stopping force on the upper body, not the presence of a forward one.
 - (c) is backward — friction at the feet actually decelerates the lower body, it doesn't push the person forward.
 - (d) confuses inertia with weight; weight is unrelated here.
-

Q2. Two forces produce the same impulse on a body. Force A is large and acts for a short time; Force B is small and acts for a long time. Which quantity is necessarily the same for both?

- (a) The acceleration produced
- (b) The change in kinetic energy
- (c) The change in momentum
- (d) The maximum force

Correct: (c). Impulse equals change in momentum by definition, so equal impulses mean equal Δp regardless of how the force is spread over time.

- (a) is wrong because a large force gives large acceleration; the accelerations differ even when impulse matches.
 - (b) is a strong distractor — equal impulse does *not* mean equal work or equal KE change, because KE change depends on distance, not time.
 - (d) the peak forces are clearly different (one is “large,” one is “small”).
-

Q3. A shell at rest explodes into two fragments of masses in the ratio 1 : 3. The ratio of the magnitudes of their velocities (lighter : heavier) is:

- (a) 1 : 3
- (b) 3 : 1
- (c) 1 : 9
- (d) 9 : 1

Correct: (b). Initial momentum is zero, so the fragments carry equal and opposite momenta: $m_1 v_1 = m_2 v_2$. With $m_1 : m_2 = 1 : 3$, we get $v_1 : v_2 = m_2 : m_1 = 3 : 1$. The lighter fragment flies off faster.

- (a) picks the *mass* ratio directly — the most common error, forgetting momentum (not velocity) is what’s equal.
 - (c) and (d) come from squaring the ratio, as if conserving kinetic energy — but KE is *created* in an explosion, not conserved.
-

Q4. A ball of mass m hits a wall normally with speed v and rebounds with the same speed v . The magnitude of impulse on the ball is:

- (a) zero
- (b) mv
- (c) $2mv$
- (d) $\frac{1}{2}mv$

Correct: (c). Taking the rebound direction as positive, the velocity changes from $-v$ to $+v$, a change of $2v$. So $|J| = |mv - m(-v)| = 2mv$.

- (a) is the trap for students who reason “same speed in and out, so no change” — forgetting velocity is a *vector* and the direction flipped.
 - (b) forgets the reversal and computes only mv .
 - (d) has no physical basis; it’s a pure distractor.
-

Section 7: Common Student Pitfalls & Pro-Tips

- **Forgetting that momentum and impulse are vectors.** The single biggest source of lost marks. When a ball reverses direction, its speed may be unchanged but its *velocity* changes sign — so Δp can be as large as $2mv$, not zero. Always fix a positive direction *before* substituting numbers.
- **Believing action–reaction forces cancel.** They are equal and opposite, yes — but they act on *two different bodies*, so they never cancel on a single body. The forces on the bat and on the ball are a Third-Law pair; neither one is “cancelled.”
- **Using $\vec{F} = m\vec{a}$ when mass changes.** Rockets, falling chains, sand-on-belts, leaking trolleys — the moment mass is variable, retreat to $\vec{F} = \frac{d\vec{p}}{dt}$. Plugging into $m\vec{a}$ here is a guaranteed wrong answer.
- **Confusing inertia with weight.** Inertia depends only on mass and is the same on the Moon as on Earth; weight is the gravitational force and changes with g . A heavy crate is hard to push *in space* too — that’s inertia, not weight.

Pro-Tip — the units sanity check. Impulse and momentum share the dimensional formula $[MLT^{-1}]$, so $N \cdot s$ and $kg \cdot m/s$ are the *same unit*. If you ever derive an “impulse” that comes out in joules or newtons, you’ve made an algebra error — stop and recheck before committing the answer. This 5-second check catches more mistakes than any other in this chapter.

Section 8: Cheat Sheet Box

Quantity	Formula	SI Unit	Dimensions
Momentum	$\vec{p} = m\vec{v}$	$kg \cdot m/s$	$[MLT^{-1}]$
Force	$\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}$	N	$[MLT^{-2}]$
Impulse	$\vec{J} = \vec{F}\Delta t = \int \vec{F} dt$	$N \cdot s$	$[MLT^{-1}]$

Core takeaways:

- **First Law** defines inertia & force; **Second Law** quantifies force as rate of change of momentum; **Third Law** makes forces come in equal-and-opposite pairs on *different* bodies.
- **Impulse = change in momentum = area under the F–t graph.** Use it whenever a force is large, brief, or unknown.
- **Conservation of momentum** applies whenever net external force is zero (recoils, explosions, collisions).
- Impulse and momentum have **identical units** — your built-in error detector.

Memory aid: “*Inertia In, Force Changes, Pairs Push back*” — for the three laws in order. And for impulse: “*Big force, short time, same total kick.*”

Subtopic 2: Momentum Conservation

Exam Relevance: This is one of the most heavily examined ideas in all of physics. JEE Main asks 1–2 questions almost every year, usually through collisions or recoil. NEET leans conceptual — “is momentum conserved here?” and coefficient-of-restitution numericals. JEE Advanced builds it into multi-body, 2D, and variable-mass problems. CBSE Boards typically carries a 2–3 mark recoil or perfectly-inelastic-collision numerical.

Section 1: Concept Intuition

You’ve seen this on every Diwali. A rocket sits still on the ground, then a jet of hot gas blasts downward and the rocket streaks *up*. Nobody is pushing the rocket from above. So where does its upward motion come from?

Here’s the secret: the total “amount of motion” of the rocket-plus-gas was zero before lighting, and it stays zero after — but it gets *redistributed*. The gas rushes down carrying a big chunk of downward momentum, so the rocket must carry an equal chunk of upward momentum to keep the books balanced. The grand total never changes. This is the **principle of conservation of linear momentum**, and it is arguably the single most powerful problem-solving tool you will meet in mechanics.

We already saw *why* it’s true in the last subtopic: when no net external force acts on a system, the total momentum stays constant, because internal forces always come in equal-and-opposite Third-Law pairs that cancel. Here we focus on *using* it.

The everyday version: stand on a skateboard (or a wheely office chair) and throw a heavy bag forward — you roll backward. Step off a small wooden boat onto a jetty and the boat scoots away behind you. Fire a rifle and it kicks into your shoulder. In every case, you push something one way and the system pushes you the other way so the total momentum is unchanged.

Think about it this way... Picture momentum as money in a sealed room with no doors (an isolated system). People inside can hand cash to each other all day — that’s the internal forces — but not a single rupee can enter or leave. So the total cash in the room is fixed, even though individual wallets fatten and thin. The “no doors” condition is the crucial one: it’s the requirement that **no net external force acts**.

This single idea cracks open three big families of problems:

- **Recoil and explosions** — one body becomes two (or more) that fly apart.
- **Collisions** — two bodies meet, interact briefly, and separate (or stick together).
- **Variable-mass systems** — rockets, falling chains, raindrops growing as they fall.

A vital companion idea is **elastic versus inelastic**. Momentum is conserved in *every* collision (as long as external forces are negligible during the brief impact). **Kinetic energy** is a different matter — it is conserved only in a *perfectly elastic* collision (think two carrom coins, or near-ideal billiard balls). In a real-world crunch — two cars crumpling, a ball of clay hitting a wall — some kinetic energy converts to heat, sound, and deformation. The collision is **inelastic**, and if the bodies stick together it’s **perfectly inelastic**.

Limiting Conditions & Assumptions:

- Conservation of momentum needs **zero net external force** on the system. During a quick collision or explosion the internal impulsive forces are enormous compared to gravity or friction,

so we treat the system as effectively isolated *for that instant* — but over longer times, external forces (gravity, friction) do change momentum.

- It applies **component-wise**: if the net external force is zero only along one axis, momentum is conserved along that axis alone. A projectile-explosion conserves horizontal momentum even while gravity changes the vertical.
- Choosing the **system** correctly is everything. Pick the system so the troublesome forces become *internal*.

Section 2: Key Formulas & Definitions

Conservation of Linear Momentum (when $\vec{F}_{\text{ext}} = 0$):

$$\sum_i m_i \vec{u}_i = \sum_i m_i \vec{v}_i$$

where $\vec{u}_i$ are velocities before and $\vec{v}_i$ after the interaction.

- SI unit of momentum: $\text{kg} \cdot \text{m} \cdot \text{s}^{-1}$ — Dimensional formula: $[M^1 L^1 T^{-1}]$

Recoil / Explosion (two bodies, system initially at rest):

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = 0 \quad \Rightarrow \quad \vec{v}_2 = -\frac{m_1}{m_2} \vec{v}_1$$

Perfectly Inelastic Collision (bodies stick together):

$$v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$$

1D Elastic Collision — final velocities:

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 + \frac{2m_2}{m_1 + m_2} u_2$$

$$v_2 = \frac{m_2 - m_1}{m_1 + m_2} u_2 + \frac{2m_1}{m_1 + m_2} u_1$$

Coefficient of Restitution:

$$e = \frac{\text{velocity of separation}}{\text{velocity of approach}} = \frac{v_2 - v_1}{u_1 - u_2}$$

- Dimensionless: $[M^0 L^0 T^0]$. Perfectly elastic $\Rightarrow e = 1$; perfectly inelastic $\Rightarrow e = 0$.

Velocity of the Centre of Mass (constant if $\vec{F}_{\text{ext}} = 0$):

$$\vec{v}_{\text{cm}} = \frac{\sum_i m_i \vec{v}_i}{\sum_i m_i}$$

Thrust on a Variable-Mass Body (e.g., a rocket):

$$\vec{F}_{\text{thrust}} = \vec{v}_{\text{rel}} \frac{dm}{dt}$$

where $\vec{v}_{\text{rel}}$ is the velocity of ejected mass relative to the body and $\frac{dm}{dt}$ is the rate of mass ejection.

Section 3: CBSE Board Derivations

Derivation 1 — Final velocities in a one-dimensional elastic collision.

Setup: Two bodies of masses m_1, m_2 move along a line with initial velocities u_1, u_2 . After a head-on elastic collision their velocities are v_1, v_2 . Two things are conserved: momentum (always) and kinetic energy (because it's elastic).

Momentum:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Rearrange to group each mass:

$$m_1(u_1 - v_1) = m_2(v_2 - u_2) \quad \dots(i)$$

Kinetic energy:

$$\frac{1}{2}m_1 u_1^2 + \frac{1}{2}m_2 u_2^2 = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2$$

Cancel the $\frac{1}{2}$ and group:

$$m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2)$$

Factor each side as a difference of squares:

$$m_1(u_1 - v_1)(u_1 + v_1) = m_2(v_2 - u_2)(v_2 + u_2) \quad \dots(ii)$$

The clever step: divide equation (ii) by equation (i). The mass-and-bracket factors $m_1(u_1 - v_1)$ and $m_2(v_2 - u_2)$ cancel cleanly, leaving:

$$u_1 + v_1 = v_2 + u_2 \quad \Rightarrow \quad u_1 - u_2 = v_2 - v_1$$

This is a beautiful intermediate result: in an elastic collision, **the relative velocity of approach equals the relative velocity of separation** (i.e. $e = 1$). It also tells us *why* the coefficient of restitution is defined the way it is.

Now solve this together with the momentum equation (i). Substituting and simplifying gives the standard pair:

$$\boxed{v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 + \frac{2m_2}{m_1 + m_2} u_2}, \quad \boxed{v_2 = \frac{2m_1}{m_1 + m_2} u_1 + \frac{m_2 - m_1}{m_1 + m_2} u_2}$$

Sanity check — equal masses ($m_1 = m_2$): the first fractions vanish and $v_1 = u_2$, $v_2 = u_1$. The bodies simply **swap velocities** — exactly what you see when one carrom coin strikes another head-on.

Derivation 2 — Thrust on a rocket (variable-mass system).

Setup: At time t a rocket has mass m and velocity v . In a short time dt it ejects gas of mass $|dm|$ at velocity v_{rel} relative to the rocket, and the rocket's speed increases to $v + dv$.

Apply Newton's second law to the *whole system* (rocket + ejected gas), whose total momentum changes only due to external forces. Conserving momentum of the system in the absence of external force:

$$mv = (m + dm)(v + dv) + (-dm)(v - v_{\text{rel}})$$

Here dm is negative (the rocket loses mass), and the ejected gas ($-dm$) leaves at velocity $(v - v_{\text{rel}})$ in the ground frame. Expanding and discarding the second-order term $dm dv$:

$$0 = m dv + v_{\text{rel}} dm$$

$$m \frac{dv}{dt} = -v_{\text{rel}} \frac{dm}{dt}$$

The right-hand side is the **thrust** — the effective forward force the engine provides — and it equals the exhaust speed times the rate at which mass is burnt. Integrating (ignoring gravity) from initial mass m_0 to final mass m gives the celebrated rocket equation:

$$v - v_0 = v_{\text{rel}} \ln\left(\frac{m_0}{m}\right)$$

Physical reading: a rocket speeds up not by pushing on air, but purely by throwing mass backward — momentum conservation in its rawest form.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A railway wagon of mass 8000 kg moving at 2 m/s on a straight, level track collides with and automatically couples to a stationary wagon of mass 12000 kg. Find their common velocity just after coupling.

Given: $m_1 = 8000$ kg, $u_1 = 2$ m/s, $m_2 = 12000$ kg, $u_2 = 0$.

Coupling means the wagons move together — a **perfectly inelastic** collision. The track is level and the impact is brief, so external horizontal forces are negligible and momentum is conserved.

Apply conservation of momentum:

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$$

$$8000(2) + 12000(0) = (20000) v$$

$$16000 = 20000 v \quad \Rightarrow \quad v = 0.8 \text{ m/s}$$

Answer: The coupled wagons move off together at 0.8 m/s in the original direction.

Note: check whether KE is conserved — initial KE = $\frac{1}{2}(8000)(2)^2 = 16000$ J; final = $\frac{1}{2}(20000)(0.8)^2 = 6400$ J. Nearly 60% of the kinetic energy was lost to the coupling — proof that this collision is inelastic.

Example 2 — [NEET Speed Trap] A carrom striker of mass 15 g moving at 4 m/s strikes an identical stationary coin head-on. Assuming a perfectly elastic collision, what are the velocities of the striker and the coin immediately afterward?

The trap: students start plugging into the full elastic-collision formulas, burning a minute on algebra under exam pressure.

The shortcut: the masses are **equal** and the collision is **elastic and head-on** — so the two bodies simply *exchange velocities*. The striker takes on the coin's velocity (zero) and the coin takes on the striker's (4 m/s).

Answer: Striker stops (0 m/s); coin moves off at 4 m/s.

Speed tip: memorize the three elastic special cases as instant answers — (1) **equal masses:** swap velocities; (2) **heavy hits light at rest:** heavy continues almost unchanged, light shoots off at $\approx 2u$; (3) **light hits heavy at rest:** light bounces back at nearly u , heavy barely moves. Any option claiming the striker keeps moving forward here is wrong on sight.

Example 3 — [JEE Main Level] A bullet of mass 10 g travelling at 200 m/s embeds itself into a wooden block of mass 1.99 kg resting on a frictionless horizontal surface. The block is attached to a spring of force constant 800 N/m. Find the maximum compression of the spring.

This links **momentum conservation** (collision) with **energy conservation** (spring) — and the order matters.

Step 1 — The collision (use momentum, not energy). The bullet embeds, so it's perfectly inelastic. Just after impact, bullet + block move together at v :

$$\begin{aligned} m_b u &= (m_b + M) v \\ 0.01 \times 200 &= (0.01 + 1.99) v = 2.00 v \\ v &= \frac{2}{2.00} = 1 \text{ m/s} \end{aligned}$$

Step 2 — The spring (now use energy). After the collision, the combined mass compresses the spring. Surface is frictionless, so all the kinetic energy converts to spring potential energy at maximum compression x :

$$\begin{aligned} \frac{1}{2}(m_b + M)v^2 &= \frac{1}{2}kx^2 \\ \frac{1}{2}(2.00)(1)^2 &= \frac{1}{2}(800)x^2 \\ 1 = 400x^2 &\Rightarrow x^2 = 0.0025 \Rightarrow x = 0.05 \text{ m} \end{aligned}$$

Answer: Maximum compression = 0.05 m = 5 cm.

Critical insight: you **cannot** apply energy conservation through the collision — the embedding loses kinetic energy to heat. Momentum carries you through the impact; energy carries you through the spring. Mixing them up is the classic way this problem is failed.

Example 4 — [JEE Advanced Level] A shell of mass 3 kg moving horizontally at 10 m/s suddenly explodes into two fragments. Immediately after the explosion, a fragment of mass 1 kg is observed moving vertically upward at 20 m/s. Find the velocity (magnitude and direction) of the other fragment of mass 2 kg immediately after the explosion. Neglect gravity during the explosion.

An explosion is internal, so **momentum is conserved separately along each axis**. Set up axes: x horizontal (original direction), y vertical (up).

Step 1 — Initial momentum of the shell.

$$p_x = 3 \times 10 = 30 \text{ kg} \cdot \text{m/s}, \quad p_y = 0$$

Step 2 — Fragment 1 (1 kg, straight up at 20 m/s).

$$p_{1x} = 0, \quad p_{1y} = 1 \times 20 = 20 \text{ kg} \cdot \text{m/s}$$

Step 3 — Conserve each component for fragment 2 (2 kg).

Horizontal:

$$30 = 0 + 2 v_{2x} \quad \Rightarrow \quad v_{2x} = 15 \text{ m/s}$$

Vertical:

$$0 = 20 + 2 v_{2y} \quad \Rightarrow \quad v_{2y} = -10 \text{ m/s (i.e. downward)}$$

Step 4 — Combine.

$$|\vec{v}_2| = \sqrt{15^2 + 10^2} = \sqrt{225 + 100} = \sqrt{325} = 5\sqrt{13} \approx 18.0 \text{ m/s}$$

$$\theta = \tan^{-1}\left(\frac{10}{15}\right) = \tan^{-1}(0.667) \approx 33.7^\circ \text{ below the horizontal}$$

Answer: The 2 kg fragment moves at about 18.0 m/s, directed 33.7° below the horizontal (forward and downward).

Insight: the centre of mass keeps moving exactly as the un-exploded shell would have — horizontally at 10 m/s, then on a normal projectile path under gravity. The explosion rearranges the pieces but cannot shift the centre of mass's motion, because the blast forces are internal.

Section 5: Practice Exercises

1. **[CBSE]** A boy of mass 60 kg standing on a stationary boat of mass 140 kg jumps horizontally off the boat with a speed of 2 m/s relative to the ground. Find the recoil speed of the boat. (Neglect water resistance.)
2. **[NEET]** A body of mass m moving at 6 m/s collides head-on and elastically with an identical stationary body. State the velocities of both bodies after the collision.
3. **[JEE Main]** A 5 g bullet moving at 400 m/s passes straight through a 0.995 kg block resting on a frictionless surface, emerging at 100 m/s. Find the speed acquired by the block.
4. **[JEE Main]** A 2 kg body moving at 3 m/s collides head-on with a 3 kg body moving at 2 m/s in the opposite direction. If they stick together, find their common velocity after the collision.
5. **[JEE Advanced]** A ball is dropped from a height of 10 m onto a hard floor with which it has a coefficient of restitution $e = 0.8$. Find the maximum height reached after the **second** bounce.

Final Answers:

1. 0.86 m/s (opposite to the boy's jump); from $60 \times 2 = 140 \times v$
 2. First body stops; second moves at 6 m/s (equal-mass elastic swap)
 3. 1.51 m/s; from $0.005 \times 400 = 0.005 \times 100 + 0.995 v$
 4. 0 m/s — they come to rest; momenta +6 and -6 cancel
 5. $h_2 = e^4 h = (0.8)^4 (10) = 4.096 \text{ m} \approx 4.1 \text{ m}$
-

Section 6: Crack the MCQ

Q1. An astronaut floating at rest in deep space (far from any gravity) wants to move toward her spacecraft. With nothing to push against, her best option is to:

- (a) Swing her arms and legs vigorously like swimming.
- (b) Throw a heavy tool in the direction *away* from the spacecraft.
- (c) Blow air from her mouth toward the spacecraft.
- (d) Wait, since motion is impossible without an external force.

Correct: (b). With no external force, her total momentum is fixed at zero. Throwing a tool away from the craft gives the tool momentum in that direction, so she recoils toward the craft, total momentum still summing to zero.

- (a) swimming needs a medium to push on; empty space offers none.
 - (c) blowing air *toward* the craft pushes her *away* from it — wrong direction (and feeble).
 - (d) the trap of over-applying “no external force = no motion.” External force is needed to change the *system’s* momentum, not to rearrange it internally.
-

Q2. Which statement about a perfectly inelastic collision between two bodies (no external forces) is correct?

- (a) Both momentum and kinetic energy are conserved.
- (b) Momentum is conserved but kinetic energy is not.
- (c) Kinetic energy is conserved but momentum is not.
- (d) Neither is conserved.

Correct: (b). Momentum is conserved in *every* collision with no external force. In a perfectly inelastic collision the bodies stick together and some kinetic energy is lost to heat and deformation, so KE is *not* conserved.

- (a) describes a perfectly *elastic* collision, not an inelastic one.
 - (c) reverses the truth — momentum conservation is the robust one here.
 - (d) wrong because momentum conservation holds regardless of collision type.
-

Q3. A ball dropped from height h rebounds from the floor to a height of $h/4$. The coefficient of restitution between the ball and the floor is:

- (a) 0.25
- (b) 0.5
- (c) 0.75
- (d) 0.125

Correct: (b). Speed before impact $\propto \sqrt{h}$ and speed after $\propto \sqrt{h/4}$, so

$$e = \frac{v_{\text{after}}}{v_{\text{before}}} = \frac{\sqrt{h/4}}{\sqrt{h}} = \sqrt{\frac{1}{4}} = 0.5$$

- (a) takes the ratio of *heights* directly, forgetting that restitution compares *speeds*, which go as the square root of height — the single most common error here.
 - (c) and (d) are unmotivated distractors.
-

Q4. A bomb initially at rest explodes into three fragments. Which of the following must be true immediately after the explosion?

- (a) The three fragments fly off along the same straight line.
- (b) The vector sum of the three momenta is zero.
- (c) The total kinetic energy of the fragments is zero.
- (d) The heaviest fragment is at rest.

Correct: (b). The bomb started at rest, so total initial momentum is zero. The explosion's forces are internal, so total momentum stays zero — the three momentum vectors must add (as vectors) to zero, meaning they lie in a single plane and balance.

- (a) is false — three fragments generally spread out in a plane, not a line (only two fragments are forced collinear).
- (c) is false — chemical energy is *converted into* kinetic energy; the fragments definitely move.
- (d) has no basis; mass distribution doesn't force any fragment to be stationary.

Section 7: Common Student Pitfalls & Pro-Tips

- **Applying kinetic-energy conservation to an inelastic collision.** Momentum is conserved in *every* collision (absent external forces); kinetic energy is conserved *only* when the collision is perfectly elastic. When a bullet embeds, cars crumple, or clay sticks, KE is lost — use momentum for the impact, never energy.
- **Forgetting momentum is a vector.** In head-on problems with bodies moving in opposite directions, you *must* assign opposite signs. In 2D (explosions, oblique collisions), conserve the x and y components *separately*. Treating speeds as if they simply add is a guaranteed wrong answer.
- **Conserving momentum when external forces are not negligible.** Momentum is conserved only when the net external force on your chosen system is zero. During a brief collision, gravity and friction are swamped by the impulsive internal forces — fine. But over a falling-and-bouncing sequence, gravity changes momentum between bounces. Apply conservation only *across the instant* of impact.
- **Misusing the coefficient of restitution.** e compares speeds of separation and approach, not heights and not energies. For a bounce off the floor, $e = \sqrt{h_{\text{rebound}}/h_{\text{drop}}}$ — note the square root.

Pro-Tip — the centre-of-mass sanity check. Since internal forces can't move the centre of mass, $\vec{v}_{\text{cm}}$ is identical before and after any collision or explosion. After solving, compute $\vec{v}_{\text{cm}}$ both ways — if they don't match, you've made a sign or arithmetic error. This catches mistakes faster than re-doing the whole problem.

Section 8: Cheat Sheet Box

Situation	Key Relation
General (no ext. force)	$\sum m_i \vec{u}_i = \sum m_i \vec{v}_i$
Recoil / explosion from rest	$m_1 \vec{v}_1 = -m_2 \vec{v}_2$

Situation	Key Relation
Perfectly inelastic	$v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$
Coefficient of restitution	$e = \frac{v_2 - v_1}{u_1 - u_2} \quad (0 \leq e \leq 1)$
Rocket thrust	$F = v_{\text{rel}} \frac{dm}{dt}$

Core takeaways:

- **Momentum is conserved in every collision** with no external force; **KE only in elastic** ones.
- **Conserve components separately** in 2D — momentum is a vector.
- **Equal-mass elastic head-on = swap velocities** (instant answer).
- $e = 1$ elastic, $e = 0$ perfectly inelastic; for a floor bounce $e = \sqrt{h_{\text{up}}/h_{\text{down}}}$.
- The **centre of mass keeps moving unchanged** through any internal interaction — use it as your error check.

Memory aid: “*Momentum Many, Energy Elastic*” — momentum survives many (all) collisions; energy survives only elastic ones. For restitution: “*separate over approach.*”

Subtopic 3: Concurrent Forces and Equilibrium

Exam Relevance: A perennial favorite. JEE Main reliably asks 1–2 questions on string-tension equilibrium, blocks on inclines, or Lami’s theorem. NEET leans on quick resolution and “which string breaks first” reasoning. JEE Advanced builds it into non-inertial-frame (pseudo-force) problems and multi-body equilibrium. CBSE Boards frequently sets a 2–3 mark numerical on a hanging mass supported by two strings.

Section 1: Concept Intuition

Walk through any Indian market and look up: a heavy signboard hangs from two chains bolted into the wall, a row of decorative bulbs sags between two poles, a fruit-seller’s weighing pan dangles from three cords meeting at a hook. None of these are moving. Yet each is being pulled in several directions at once — by gravity, by tension in every chain, by the wall. The fact that they hang there, perfectly still, is telling us something deep: **all those pulls are cancelling out.**

That is the entire idea of **equilibrium**. A body is in (translational) equilibrium when the **net force on it is zero**, so it has no acceleration — it either stays at rest or keeps moving at constant velocity. The forces don’t have to be small or few; they just have to *balance*.

Most of these everyday situations involve **concurrent forces** — forces whose lines of action all pass through a single common point. The hook holding the weighing pan feels three cords pulling on it, and all three pulls act *at the hook*. Because they share a point, we can treat that point as a single particle and ask one clean question: do the force vectors add up to zero?

Think about it this way... Imagine three friends playing tug-of-war on a ring, each pulling a rope tied to the same ring, in three different directions. If the ring doesn’t move, their three pulls must perfectly cancel. Now here’s the elegant part: if you drew their three force arrows *head to tail*, the arrows would close up into a triangle with no gap. A closed triangle means “you ended up exactly where you started” — i.e. zero net displacement in force-space, i.e. zero resultant. This is the **triangle law of forces**, and it’s the geometric soul of equilibrium.

To actually *solve* such problems, we use one powerful trick: **resolution**. Any force pointing at an awkward angle can be split into two perpendicular pieces — typically horizontal (x) and vertical (y), though on a slope we cleverly choose axes *along* and *perpendicular to* the slope. Once split, equilibrium becomes two simple scalar statements:

$$\sum F_x = 0 \quad \text{and} \quad \sum F_y = 0$$

Balance the rightward pulls against the leftward ones, and the upward pulls against the downward ones. That’s it.

For the special and very common case of **exactly three** concurrent forces, there’s an even faster tool — **Lami’s theorem** — which we’ll meet shortly.

One subtlety that trips up students: equilibrium is **frame-dependent in appearance**. A pendulum bob hanging tilted inside an accelerating bus looks “in equilibrium” to a passenger but is clearly accelerating to someone on the road. To make the passenger’s intuitive “it’s balanced” picture mathematically correct, we add a **pseudo-force** — a bookkeeping force that accounts for the frame’s acceleration. More on this in the examples.

Limiting Conditions & Assumptions:

- The equilibrium condition $\sum \vec{F} = 0$ guarantees only **translational** equilibrium. A body can have zero net force yet still rotate (a couple). Full rigid-body equilibrium also needs zero net *torque* — but for a **particle** or a **point of concurrency**, force balance alone is enough.
 - Resolution and Lami's theorem assume the forces are **concurrent and coplanar**.
 - Strings and cords can only **pull** (tension), never push. If your maths gives a negative tension, the string would have gone slack — the configuration is physically impossible.
 - In **non-inertial frames**, $\sum \vec{F} = 0$ holds *only after* adding the pseudo-force.
-

Section 2: Key Formulas & Definitions

Condition for equilibrium of a particle:

$$\sum \vec{F} = 0 \quad \Longleftrightarrow \quad \sum F_x = 0, \quad \sum F_y = 0$$

- Force SI unit: newton, N — Dimensional formula: $[M^1 L^1 T^{-2}]$

Resolution of a force $\vec{F}$ at angle θ to the x -axis:

$$F_x = F \cos \theta, \quad F_y = F \sin \theta$$

Equilibrant: the single force that balances all others; equal in magnitude and opposite in direction to the resultant $\vec{R}$:

$$\vec{F}_{\text{equilibrant}} = -\vec{R}$$

Triangle Law of Forces: if three forces acting at a point are in equilibrium, they can be represented in magnitude and direction by the three sides of a triangle taken in order.

Lami's Theorem (for three concurrent coplanar forces P, Q, R in equilibrium):

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$$

where α is the angle *between the other two forces* Q and R (i.e. the angle opposite P), and similarly for β, γ . Note $\alpha + \beta + \gamma = 360^\circ$.

Apparent Weight in a lift (reading of a scale, person mass m):

$$R = m(g + a) \text{ (lift accelerating up),} \quad R = m(g - a) \text{ (accelerating down)}$$

$R = mg$ at constant velocity; $R = 0$ in free fall (weightlessness).

Pseudo-force in a frame accelerating at $\vec{a}_0$:

$$\vec{F}_{\text{pseudo}} = -m\vec{a}_0$$

(magnitude ma_0 , directed opposite to the frame's acceleration).

Section 3: CBSE Board Derivations

Derivation 1 — Lami’s Theorem.

Setup: Three concurrent, coplanar forces P , Q , R act on a particle in equilibrium. Let α be the angle between Q and R , β between R and P , and γ between P and Q .

Since the three forces are in equilibrium, by the **triangle law** they can be drawn head-to-tail to form a *closed* triangle, whose sides are parallel to and proportional in magnitude to P , Q , and R .

Now examine the geometry. Consider the side representing P . In the original “star” arrangement, the angle between Q and R is α . When we lay the forces head-to-tail, the interior angle of the triangle *opposite* the side P turns out to be $(180^\circ - \alpha)$ — because the head-to-tail construction reverses the relative orientation. Likewise the interior angles opposite Q and R are $(180^\circ - \beta)$ and $(180^\circ - \gamma)$.

Apply the **sine rule** to this triangle (side $\div$ sine of opposite angle is constant):

$$\frac{P}{\sin(180^\circ - \alpha)} = \frac{Q}{\sin(180^\circ - \beta)} = \frac{R}{\sin(180^\circ - \gamma)}$$

Using the identity $\sin(180^\circ - \theta) = \sin \theta$:

$$\boxed{\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}}$$

Why it’s useful: it lets you solve a three-force equilibrium in **one line**, with no axis-choosing or component algebra — provided you correctly pair each force with the angle *between the other two*.

Derivation 2 — Tension and Normal Reaction on a smooth incline.

Setup: A block of mass m rests on a frictionless incline of angle θ , held in place by a light string running *parallel to the incline surface*. Find the string tension T and the normal reaction N .

Why choose tilted axes? Two of the three forces (tension along the incline, normal perpendicular to it) already lie along natural directions of the slope. By choosing the x -axis **along** the incline and the y -axis **perpendicular** to it, only gravity needs resolving — minimizing algebra. This axis choice is the single most valuable habit in incline problems.

Resolve gravity mg (which points straight down) into these tilted axes:

- Component along the incline (down-slope): $mg \sin \theta$
- Component perpendicular to the incline (into surface): $mg \cos \theta$

Equilibrium along the incline (string pulls up-slope, gravity component pulls down-slope):

$$T = mg \sin \theta$$

Equilibrium perpendicular to the incline (normal reaction balances the perpendicular weight component):

$$N = mg \cos \theta$$

Sanity checks: at $\theta = 0$ (flat ground), $T = 0$ and $N = mg$ — nothing pulls the block and the floor carries full weight. At $\theta = 90^\circ$ (vertical wall), $T = mg$ and $N = 0$ — the string carries the entire weight and the wall presses nothing. Both extremes match intuition, confirming the result.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A decorative lamp of weight 100 N is suspended from a horizontal ceiling beam by two ropes, each making an angle of 30° with the beam (the horizontal). Find the tension in each rope.

Given: $W = 100$ N; each rope at 30° to the horizontal; by symmetry both tensions are equal, call it T .

Figure 4.1 A horizontal beam with two ropes descending symmetrically to a single point where a lamp hangs; each rope makes 30° with the beam; weight 100 N acts downward at the junction. (see next page)

Step 1 — Free body diagram of the junction point. Three concurrent forces: weight W downward, and two tensions T along the ropes, each 30° above the horizontal.

Step 2 — Horizontal balance. The horizontal components of the two tensions point in opposite directions and cancel automatically by symmetry — confirming the setup is consistent.

Step 3 — Vertical balance. Each rope contributes an upward component $T \sin 30^\circ$:

$$2T \sin 30^\circ = W$$

$$2T(0.5) = 100 \quad \Rightarrow \quad T = 100 \text{ N}$$

Answer: The tension in each rope is 100 N.

Teaching point: notice the tension in *each* rope equals the *full* weight, not half of it — because the ropes are nearly horizontal, only a small vertical slice of each tension does the lifting. The flatter the ropes, the larger the tension. This is why a tightly stretched clothesline sags only slightly under a shirt yet pulls hard on its posts.

Example 2 — [NEET Speed Trap] Two forces, 5 N directed east and 12 N directed north, act on a particle. What single force must be applied to keep the particle in equilibrium?

The trap: students rush to add $5 + 12 = 17$ N, or they find the resultant and forget that the *balancing* force points the other way.

The fast route: the two given forces are perpendicular, so their resultant is found by Pythagoras in one step:

$$R = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13 \text{ N}$$

The force needed for equilibrium is the **equilibrant** — equal in magnitude but *opposite* in direction to this resultant.

Answer: A force of 13 N directed opposite to the resultant (i.e. into the third quadrant — south-of-west), making the same angle with the axes as the resultant does.

Speed tip: memorize the Pythagorean triples — (3, 4, 5), (5, 12, 13), (8, 15, 17). The instant you see “5 N and 12 N perpendicular,” write 13 and move on. And always remember: **equilibrant = minus resultant** — the single most common slip in NEET equilibrium questions.

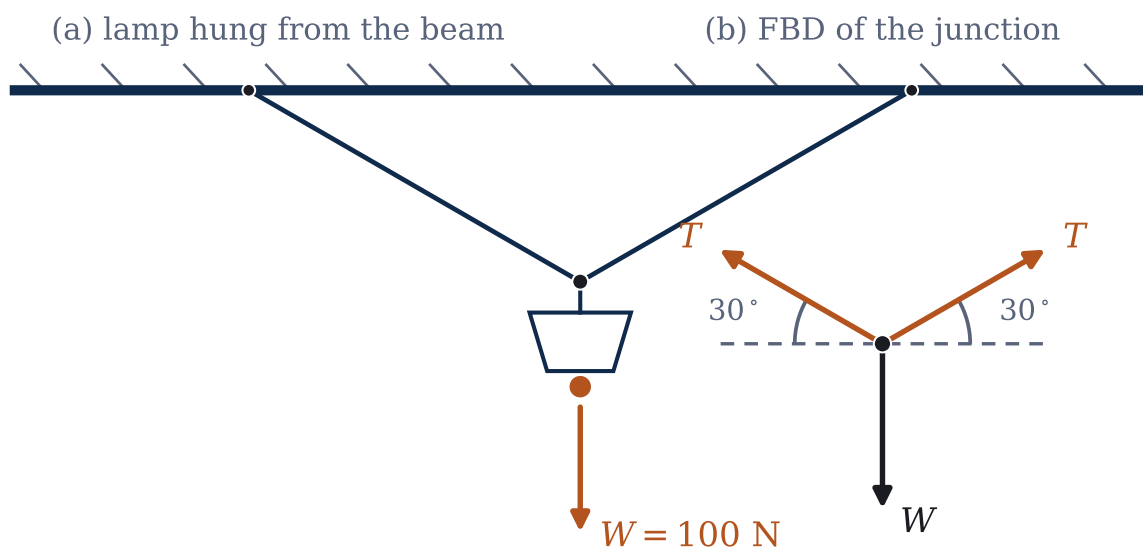


Figure 4.1 A horizontal beam with two ropes descending symmetrically to a single point where a lamp hangs; each rope makes 30° with the beam; weight 100 N acts downward at the junction.

Example 3 — [JEE Main Level] A block of mass 5 kg rests on a smooth incline of 30° . It is held in equilibrium by a **horizontal** string attached to a wall. Find the tension in the string and the normal reaction from the incline. Take $g = 10 \text{ m/s}^2$.

The string is horizontal — *not* parallel to the incline — so all three forces point in genuinely different directions. This makes it a true multi-component problem.

Figure 4.2 A block on a 30° incline; a horizontal string runs from the block to a vertical wall; forces shown are weight mg down, normal N perpendicular to incline, tension T horizontal. (see next page)

Step 1 — Choose axes. Here the *horizontal/vertical* axes are cleaner, because two of the forces (weight, tension) already lie along them; only the normal reaction needs resolving. Normal N is perpendicular to the incline, so it makes angle $\theta = 30^\circ$ with the vertical.

Step 2 — Resolve the normal reaction.

- Vertical component: $N \cos \theta$ (upward)
- Horizontal component: $N \sin \theta$ (pointing away from the incline, toward the wall side)

Step 3 — Vertical equilibrium (normal's vertical part supports the weight):

$$N \cos \theta = mg$$

$$N = \frac{mg}{\cos 30^\circ} = \frac{5 \times 10}{\sqrt{3}/2} = \frac{50 \times 2}{\sqrt{3}} = \frac{100}{\sqrt{3}} \approx 57.7 \text{ N}$$

Step 4 — Horizontal equilibrium (tension balances the normal's horizontal part):

$$T = N \sin \theta = \frac{100}{\sqrt{3}} \times 0.5 = \frac{50}{\sqrt{3}} \approx 28.9 \text{ N}$$

Answer: $N \approx 57.7 \text{ N}$ and $T \approx 28.9 \text{ N}$.

Cross-check via the slope-axis method: you can also write $T \cos \theta = mg \sin \theta$ (along incline) giving $T = mg \tan \theta = 50 \tan 30^\circ \approx 28.9 \text{ N}$ — same answer, fewer steps. The clever takeaway: **horizontal string** $\Rightarrow T = mg \tan \theta$ **and** $N = mg / \cos \theta$, both *larger* than in the string-parallel case, because the horizontal string fights gravity less efficiently.

Example 4 — [JEE Advanced Level] A small bob of mass 2 kg hangs from a string fixed to the roof of a truck. When the truck accelerates uniformly along a straight horizontal road, the string settles at a steady angle of 37° from the vertical. Find (a) the acceleration of the truck and (b) the tension in the string. Solve it in **both** the ground frame and the truck's frame to show they agree. Take $g = 10 \text{ m/s}^2$, $\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$.

Figure 4.3 A truck accelerating to the right; a pendulum bob inside hangs back at 37° from vertical; FBD shows tension T along the string, weight mg down, and (in the truck frame) a pseudo-force ma backward. (see next page)

Method 1 — Ground frame (inertial). Here the bob is genuinely accelerating horizontally with the truck. Only two real forces act: tension T and weight mg . Their resultant must equal $m\vec{a}$, horizontal.

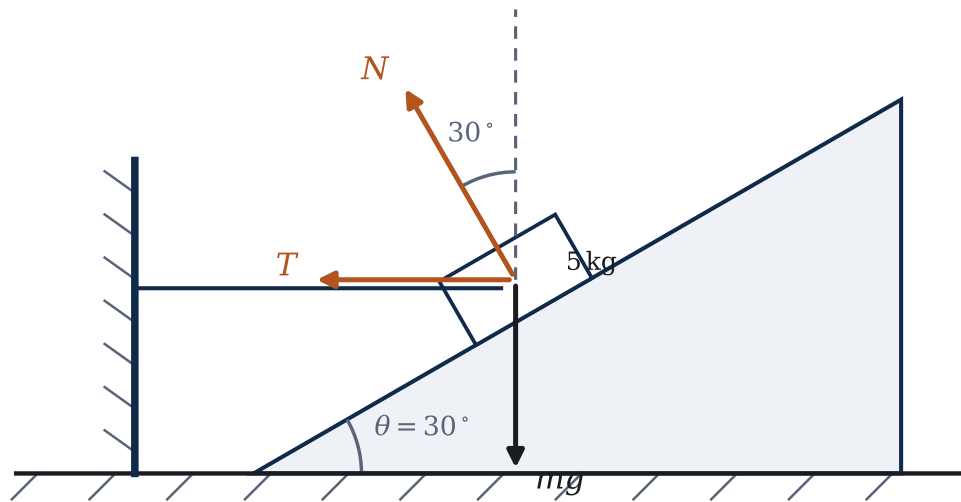


Figure 4.2 A block on a 30° incline; a horizontal string runs from the block to a vertical wall; forces shown are weight mg down, normal N perpendicular to incline, tension T horizontal.

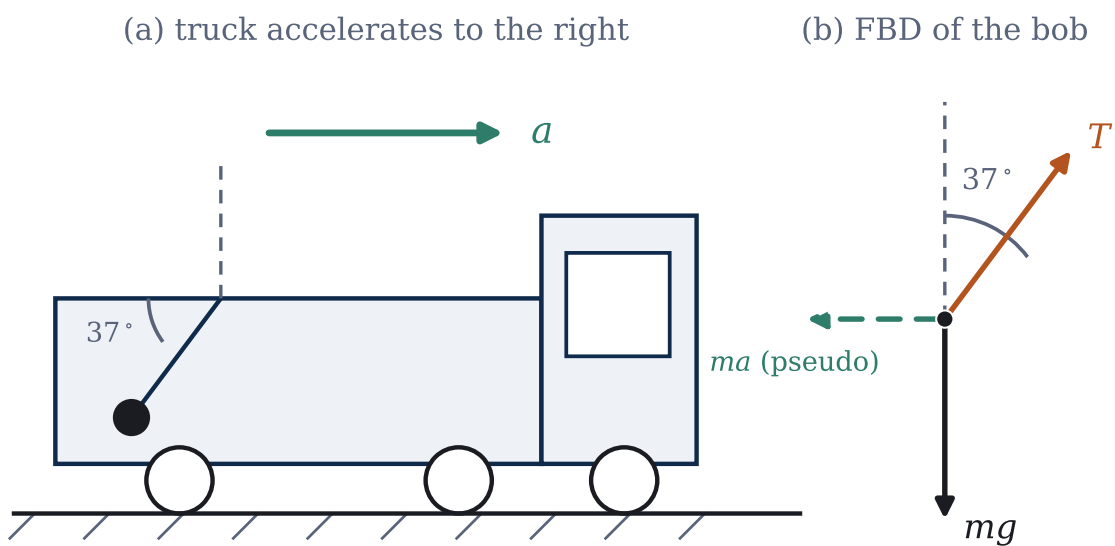


Figure 4.3 A truck accelerating to the right; a pendulum bob inside hangs back at 37° from vertical; FBD shows tension T along the string, weight mg down, and (in the truck frame) a pseudo-force ma backward.

Vertical (no vertical acceleration):

$$T \cos 37^\circ = mg \quad \Rightarrow \quad T(0.8) = 2 \times 10 = 20$$

$$T = \frac{20}{0.8} = 25 \text{ N}$$

Horizontal (provides the acceleration):

$$T \sin 37^\circ = ma \quad \Rightarrow \quad 25(0.6) = 2a \quad \Rightarrow \quad 15 = 2a \quad \Rightarrow \quad a = 7.5 \text{ m/s}^2$$

Method 2 — Truck frame (non-inertial). To the passenger, the bob is *in equilibrium* — but only after adding a **pseudo-force** ma pointing *backward* (opposite the truck's acceleration). Now the three forces T , mg , and ma balance, and we can use simple equilibrium:

Vertical: $T \cos 37^\circ = mg$ Horizontal: $T \sin 37^\circ = ma$ (tension's horizontal part balances the pseudo-force)

Dividing the two equations:

$$\tan 37^\circ = \frac{ma}{mg} = \frac{a}{g} \quad \Rightarrow \quad a = g \tan 37^\circ = 10 \times 0.75 = 7.5 \text{ m/s}^2$$

and $T = mg / \cos 37^\circ = 20 / 0.8 = 25 \text{ N}$.

Answer: $a = 7.5 \text{ m/s}^2$ and $T = 25 \text{ N}$ — identical from both frames, as they must be.

Deep insight: the elegant result $\tan \theta = a/g$ means the string aligns with the **effective gravity** $\vec{g}_{\text{eff}} = \vec{g} - \vec{a}$ felt in the accelerating frame, of magnitude $\sqrt{g^2 + a^2}$. The tension is then simply $T = m g_{\text{eff}} = m \sqrt{g^2 + a^2} = 2\sqrt{100 + 56.25} = 2(12.5) = 25 \text{ N}$ — a third route to the same number, and the one to reach for in any accelerating-frame equilibrium problem.

Section 5: Practice Exercises

1. **[CBSE]** A lantern of weight 30 N hangs from a string. A steady horizontal wind pushes it so the string makes 37° with the vertical. Find the horizontal wind force and the tension in the string. ($\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$)
2. **[NEET]** Forces of 8 N (east) and 15 N (north) act on a particle. What is the magnitude of the single force that will keep the particle in equilibrium?
3. **[JEE Main]** A weight $W = 40 \text{ N}$ hangs from a knot. The knot is also tied to a horizontal string fixed to a wall and to a second string running to the ceiling at 53° above the horizontal. Find the tensions in both strings. ($\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$)
4. **[JEE Main]** A 4 kg block rests on a smooth incline of 37° , held by a string running *parallel* to the incline. Find the tension and the normal reaction. ($g = 10 \text{ m/s}^2$)
5. **[JEE Advanced]** A pendulum bob of mass 1 kg hangs from the roof of a car moving along a horizontal road with acceleration 5 m/s^2 . Find the angle the string makes with the vertical and the tension in the string. ($g = 10 \text{ m/s}^2$)

Final Answers:

1. Wind force = 22.5 N; tension = 37.5 N (from $T \cos 37^\circ = 30$)
 2. 17 N (the 8–15–17 triple)
 3. Ceiling string = 50 N; horizontal string = 30 N
 4. $T = 24$ N; $N = 32$ N
 5. $\theta = \tan^{-1}(0.5) \approx 26.6^\circ$; $T = \sqrt{g^2 + a^2} (1) = \sqrt{125} \approx 11.2$ N
-

Section 6: Crack the MCQ

Q1. Three coplanar forces acting at a point hold it in equilibrium. Which statement must be true?

- (a) The three forces are equal in magnitude.
- (b) The three forces are mutually perpendicular.
- (c) They can be represented in magnitude and direction by the three sides of a triangle taken in order.
- (d) Their magnitudes are in the ratio 1 : 1 : 1.

Correct: (c). This is exactly the triangle law of forces — equilibrium of three concurrent forces means they close into a triangle when drawn head-to-tail.

- (a) and (d) assume equal magnitudes, which is only a special symmetric case, not a requirement.
 - (b) perpendicularity is also a special case (and three forces generally can't all be mutually perpendicular in a plane anyway).
-

Q2. Lami's theorem can be applied to a system of:

- (a) Any number of concurrent forces in equilibrium.
- (b) Exactly three concurrent, coplanar forces in equilibrium.
- (c) Three parallel forces.
- (d) Four concurrent forces in equilibrium.

Correct: (b). Lami's theorem is specifically a three-force relation derived from the sine rule of the force triangle.

- (a) and (d) fail because the theorem has exactly three terms — it cannot accommodate four or more forces.
 - (c) parallel forces are not concurrent (their lines never meet at a point), so the theorem does not apply.
-

Q3. A weight hangs in equilibrium from two strings attached to a ceiling: string A makes 30° with the horizontal and string B makes 60° with the horizontal, on opposite sides. Which string carries the greater tension?

- (a) String A (30°)
- (b) String B (60°)
- (c) Both carry equal tension.
- (d) Cannot be determined without the weight.

Correct: (b). Horizontal balance gives $T_A \cos 30^\circ = T_B \cos 60^\circ$, so $T_A(0.866) = T_B(0.5)$, meaning $T_B \approx 1.7 T_A$. The *steeper* string (closer to vertical) carries more tension because it shoulders more of the vertical load.

- (a) reverses the logic — the flatter string actually carries less.
- (c) equal tensions occur only for *symmetric* angles, not here.
- (d) the *ratio* of tensions depends only on the angles, not on the weight, so it is fully determined.

Q4. A person stands on a weighing scale inside a lift. The scale will read **more** than the person’s true weight when the lift is:

- (a) Moving upward at constant velocity.
- (b) Accelerating upward.
- (c) Accelerating downward.
- (d) In free fall.

Correct: (b). Apparent weight $R = m(g + a)$ when accelerating upward, which exceeds mg . The floor must push harder to both support and accelerate the person upward.

- (a) constant velocity means zero acceleration, so $R = mg$ — true weight.
- (c) downward acceleration gives $R = m(g - a) < mg$ — the scale reads *less*.
- (d) free fall ($a = g$) gives $R = 0$ — apparent weightlessness.

Section 7: Common Student Pitfalls & Pro-Tips

- **Adding forces at angles without resolving them.** You cannot just add 5 N and 12 N to get 17 N if they point in different directions. Always resolve into perpendicular components first, then balance each direction separately. Skipping resolution is the number-one source of wrong equilibrium answers.
- **Misreading the angles in Lami’s theorem.** Each force is paired with the angle *between the other two*, not the angle the force itself makes with something. Mislabeling these angles silently produces a wrong but plausible-looking answer. Sketch the three forces from the common point and read each “opposite” angle carefully.
- **Forgetting strings can only pull.** A string tension must come out positive. If your algebra yields a negative tension, that string would be slack and the assumed configuration is impossible — rethink the geometry rather than reporting a negative number.
- **Mishandling pseudo-force.** Add it *only* in a non-inertial (accelerating) frame, and always *opposite* to the frame’s acceleration. Adding it in the ground frame, or pointing it the wrong way, are both common and fatal errors.

Pro-Tip — pick smart axes. On an incline, resolve along and perpendicular to the slope; against a wall, use horizontal and vertical. The right axis choice often leaves only *one* force needing resolution, cutting the algebra in half. And for exactly three forces, reach for **Lami’s theorem** before grinding through components — it’s usually a one-line solution.

Section 8: Cheat Sheet Box

Concept	Key Relation
Equilibrium of a particle	$\sum F_x = 0, \sum F_y = 0$
Force resolution	$F_x = F \cos \theta, F_y = F \sin \theta$
Lami's theorem (3 forces)	$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$
Block on smooth incline (string $\parallel$ incline)	$T = mg \sin \theta, N = mg \cos \theta$
Apparent weight in lift	$R = m(g \pm a)$
Pseudo-force (non-inertial)	$F_{\text{pseudo}} = ma_0$, opposite to $\vec{a}_0$

Core takeaways:

- **Concurrent forces** act through one point; equilibrium means their vector sum is zero.
- **Resolve, then balance** each perpendicular direction separately.
- **Lami's theorem** = sine rule for three forces; pair each force with the angle *between the other two*.
- **Strings only pull** — negative tension means slack/impossible.
- In accelerating frames, add a **pseudo-force** ma_0 opposite the acceleration; the string aligns with **effective gravity** $\sqrt{g^2 + a^2}$.

Memory aid: “*Resolve, Right & Up, set to zero.*” For Lami: “*each force loves the angle across from it.*” For lifts: “*Up adds, Down subtracts, Free-fall floats.*”

Subtopic 4: Friction

Exam Relevance: One of the highest-yield topics in mechanics. JEE Main asks 1–2 questions almost every year — blocks on inclines, connected systems, minimum-force problems. NEET favors conceptual items on the self-adjusting nature of static friction, angle of repose, and the laws of friction. JEE Advanced layers it into multi-block, stacked, and non-inertial setups. CBSE Boards regularly sets a 2–3 mark numerical or a definition-plus-graph question.

Section 1: Concept Intuition

Try pushing a heavy steel almirah across the floor. At first it refuses to budge no matter how hard you lean into it. You push harder, harder still — and then suddenly it gives way and slides, often with a little jerk. Everyone has felt this. That stubborn resistance, and its sudden surrender, is the whole drama of **friction**.

Friction is the force that acts *along* two surfaces in contact, opposing the relative sliding (or the *tendency* to slide) between them. It arises because no surface is truly smooth — under a microscope, even polished steel is a landscape of tiny peaks and valleys that interlock and, where they touch, even weakly “weld” together. To slide one surface over the other, these microscopic bonds must be sheared. That’s why friction exists, and why it depends on how hard the surfaces are pressed together (the normal force) rather than on how big they look.

The single most important idea in this whole topic: **static friction is self-adjusting**. When you push the almirah gently, friction pushes back *exactly* as hard as you push — so the net force is zero and nothing moves. Push a little harder, friction matches you again. It keeps matching... but only up to a ceiling. That ceiling is the **limiting friction**, the maximum grip the surfaces can muster. Push beyond it and the surfaces lose their hold; the almirah breaks free and slides.

Think about it this way... Static friction is like a patient employee who does *exactly* the work you assign — no more, no less — but who has a strict maximum capacity. Ask for less than the maximum, and the request is met precisely. Ask for more, and the system snaps: the body starts sliding. The instant it slides, a *different*, slightly weaker force takes over — **kinetic (or dynamic) friction** — which is why the almirah, once moving, is easier to keep moving than it was to start. That little lurch you feel is exactly this drop from limiting static friction to kinetic friction.

We capture these with two numbers, the **coefficients of friction**. The limiting (maximum) static friction is $f_s^{\max} = \mu_s N$, and the kinetic friction is $f_k = \mu_k N$, where N is the normal reaction. For real surfaces $\mu_s \gtrsim \mu_k$ (static is the tougher one to beat). Crucially, while the body is *not yet sliding*, the actual static friction is whatever it needs to be — anywhere from zero up to $\mu_s N$ — **not** automatically $\mu_s N$. That misunderstanding wrecks more friction problems than any other.

Two beautiful angles summarize friction geometrically. The **angle of friction** λ is the tilt of the *total* contact force (friction plus normal) away from the normal, and it satisfies $\tan \lambda = \mu$. The **angle of repose** θ_r is the steepest incline on which a body can rest before sliding under its own weight — and remarkably, θ_r equals the angle of friction, so $\tan \theta_r = \mu_s$. Tilt a tray of marbles and watch the angle at which they let go: you’ve just measured a coefficient of friction.

Limiting Conditions & Assumptions:

- The laws $f_s \leq \mu_s N$ and $f_k = \mu_k N$ are **empirical approximations**, excellent for ordinary dry surfaces but not exact laws of nature.

- Friction is treated as **independent of the apparent contact area** and of sliding speed — both reasonable for exam-level problems, both imperfect in reality.
 - The coefficient of friction is **dimensionless** and depends on the nature of *both* surfaces, not on the bodies' masses.
 - A smooth (frictionless) surface means $\mu = 0$ and $\lambda = 0$.
-

Section 2: Key Formulas & Definitions

Static friction (self-adjusting, up to a maximum):

$$0 \leq f_s \leq \mu_s N \quad (\text{actual value equals the applied tangential force until the limit})$$

Limiting (maximum static) friction:

$$f_s^{\max} = \mu_s N$$

Kinetic (dynamic) friction (while sliding):

$$f_k = \mu_k N$$

where N is the normal reaction (newton, $[MLT^{-2}]$), and μ_s, μ_k are the coefficients of static and kinetic friction.

Coefficient of friction (dimensionless): $\mu = \frac{f}{N}$, dimensional formula $[M^0 L^0 T^0]$, with $\mu_s \geq \mu_k$.

Angle of friction λ :

$$\tan \lambda = \mu$$

(the angle the resultant contact force makes with the normal at the verge of sliding).

Angle of repose θ_r :

$$\tan \theta_r = \mu_s \quad \Rightarrow \quad \theta_r = \lambda$$

Acceleration of a block sliding down a rough incline of angle θ :

$$a = g(\sin \theta - \mu_k \cos \theta)$$

Minimum force to drag a body on a rough horizontal surface (applied at the optimal angle):

$$F_{\min} = \frac{\mu mg}{\sqrt{1 + \mu^2}} = mg \sin \lambda \quad (\text{applied at angle } \theta = \lambda = \tan^{-1} \mu)$$

Maximum fraction of a chain that can hang off a rough table (coefficient μ):

$$\frac{y}{L} = \frac{\mu}{1 + \mu}$$

Section 3: CBSE Board Derivations

Derivation 1 — Angle of repose equals angle of friction.

Setup: A block of mass m rests on a rough inclined plane whose angle θ is slowly increased. At the **angle of repose** θ_r , the block is just on the verge of sliding down. Three forces act: weight mg (vertically down), normal reaction N (perpendicular to the incline), and limiting friction $f = \mu_s N$ (up the incline, opposing the impending downhill slide).

Figure 4.4 *A block on an incline of angle θ ; weight mg straight down, normal N perpendicular to surface, friction f up the slope; mg resolved into $mg \sin \theta$ along and $mg \cos \theta$ perpendicular to the incline. Resolve the weight along the natural slope axes: • Perpendicular to incline: $N = mg \cos \theta$ • Along the incline (down-slope): $mg \sin \theta$ At the verge of sliding, the down-slope pull is exactly balanced by limiting friction: $mg \sin \theta = \mu_s N = \mu_s mg \cos \theta$ Cancel mg and divide: $\tan \theta = \mu_s$ But the angle of friction is defined by $\tan \lambda = \mu_s$. Therefore: $\theta = \lambda$ Physical reading: the steepest “safe” slope is set entirely by the surfaces’ grip, independent of the block’s mass — a heavier block presses down harder but is also gripped harder, and the two effects cancel exactly. This is why a given material (sand, gravel, rice) always piles up to the same characteristic cone angle regardless of how much you pour. (see next page)*

Derivation 2 — Minimum force to pull a body along a rough horizontal surface.

Setup: A block of mass m on a rough floor (coefficient μ) is pulled by a force F applied at angle θ above the horizontal. Find the angle that minimizes the force needed to start it moving.

Key insight: lifting the rope upward has two competing effects — the upward component reduces the normal force (good, less friction), but it also wastes some of F in the vertical direction (bad). There must be a sweet spot.

Vertical equilibrium (block stays on the floor):

$$N + F \sin \theta = mg \quad \Rightarrow \quad N = mg - F \sin \theta$$

Horizontal balance at the verge of sliding:

$$F \cos \theta = \mu N = \mu(mg - F \sin \theta)$$

Solve for F :

$$F \cos \theta + \mu F \sin \theta = \mu mg \quad \Rightarrow \quad F = \frac{\mu mg}{\cos \theta + \mu \sin \theta}$$

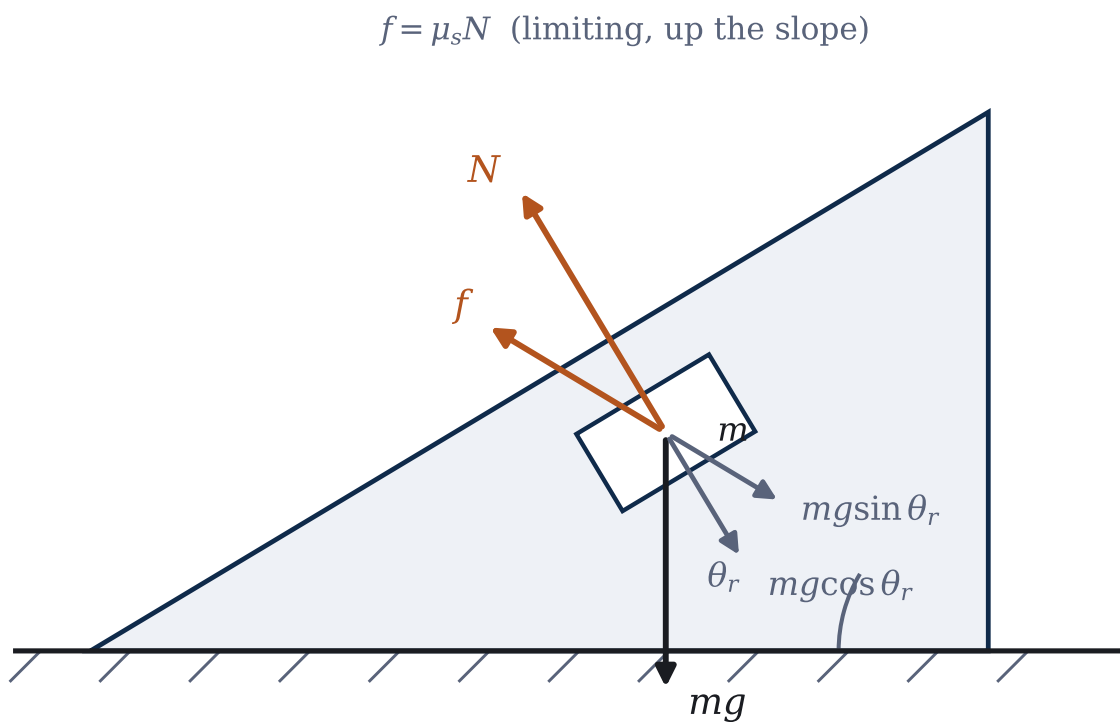


Figure 4.4 A block on an incline of angle θ ; weight mg straight down, normal N perpendicular to surface, friction f up the slope; mg resolved into $mg \sin \theta$ along and $mg \cos \theta$ perpendicular to the incline. Resolve the weight along the natural slope axes: • Perpendicular to incline: $N = mg \cos \theta$ • Along the incline (down-slope): $mg \sin \theta$ At the verge of sliding, the down-slope pull is exactly balanced by limiting friction: $mg \sin \theta = \mu_s N = \mu_s mg \cos \theta$ Cancel mg and divide: $\tan \theta = \mu_s$ But the angle of friction is defined by $\tan \lambda = \mu_s$. Therefore: $\theta = \lambda$ Physical reading: the steepest “safe” slope is set entirely by the surfaces’ grip, independent of the block’s mass — a heavier block presses down harder but is also gripped harder, and the two effects cancel exactly. This is why a given material (sand, gravel, rice) always piles up to the same characteristic cone angle regardless of how much you pour.

To **minimize** F , we maximize the denominator $D(\theta) = \cos \theta + \mu \sin \theta$. Its maximum value is $\sqrt{1 + \mu^2}$, attained when $\tan \theta = \mu$ — that is, when the pull is applied at the **angle of friction** $\theta = \lambda$. Substituting:

$$F_{\min} = \frac{\mu mg}{\sqrt{1 + \mu^2}} = mg \sin \lambda$$

Why it matters: pulling horizontally needs $F = \mu mg$; pulling at the optimal angle λ needs only $mg \sin \lambda$, which is always less (since $\sin \lambda < \tan \lambda$). This is the physics behind why we instinctively pull a heavy suitcase or a stubborn lawnmower at a slant rather than straight along the ground.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A 2 kg block rests on a rough horizontal floor with $\mu_s = 0.4$ and $\mu_k = 0.3$. Find the frictional force on the block when a horizontal force is applied of magnitude (a) 5 N, (b) 8 N, (c) 12 N. Take $g = 10 \text{ m/s}^2$.

Step 1 — Find the limiting friction first (always the first move):

$$f_s^{\max} = \mu_s N = \mu_s mg = 0.4 \times 2 \times 10 = 8 \text{ N}$$

This is the threshold that decides everything.

(a) Applied force = 5 N. Since $5 \text{ N} < 8 \text{ N}$, the block does **not** move. Static friction self-adjusts to exactly balance the applied force:

$$f_s = 5 \text{ N}$$

(b) Applied force = 8 N. This equals the limiting friction exactly — the block is *on the verge* of sliding. Friction is at its maximum:

$$f_s = 8 \text{ N}$$

(c) Applied force = 12 N. Since $12 \text{ N} > 8 \text{ N}$, the block breaks free and slides. Now **kinetic** friction acts:

$$f_k = \mu_k N = 0.3 \times 2 \times 10 = 6 \text{ N}$$

Answer: (a) 5 N, (b) 8 N, (c) 6 N.

The lesson in one line: friction is **not** $\mu_s N$ until the body is about to move. Below the threshold it simply matches the applied force. Note too that once sliding begins (case c), friction *drops* from 8 N to 6 N — that’s the little lurch you feel.

Example 2 — [NEET Speed Trap] A coin placed on a closed textbook starts to slide the moment the book is tilted to 30° from the horizontal. What is the coefficient of static friction between the coin and the cover?

The trap: students reach for $mg \sin \theta$, $mg \cos \theta$, set up full equilibrium equations, and lose a minute under exam pressure.

The shortcut: “starts to slide when tilted to θ ” is the literal definition of the **angle of repose**. And angle of repose = angle of friction, so:

$$\mu_s = \tan \theta_r = \tan 30^\circ = \frac{1}{\sqrt{3}} \approx 0.58$$

Answer: $\mu_s = \frac{1}{\sqrt{3}} \approx 0.58$.

Speed tip: the instant you read “just begins to slide on an incline of angle θ ,” write $\mu = \tan \theta$ and you’re done — no resolving needed. Beware distractors offering $\sin 30^\circ = 0.5$ or $\cos 30^\circ = 0.87$; those tempt students who set up the components but forget they cancel into a clean tangent.

Example 3 — [JEE Main Level] A 3 kg block lies on a rough horizontal table and is connected by a light string over a frictionless pulley at the table’s edge to a freely hanging 2 kg block. The coefficient of friction between the table block and the table is $\mu = 0.2$. Find the acceleration of the system and the tension in the string. Take $g = 10 \text{ m/s}^2$.

This combines **friction**, **Newton’s second law**, and a **connected (constraint) system** — a staple JEE Main combination.

Figure 4.5 A 3 kg block on a table connected by a string over a pulley at the edge to a 2 kg block hanging vertically; friction acts on the table block opposing its motion toward the pulley. (see next page)

Step 1 — Will it even move? The hanging weight drives the motion: $m_2g = 2 \times 10 = 20 \text{ N}$. The maximum friction resisting it: $\mu m_1g = 0.2 \times 3 \times 10 = 6 \text{ N}$. Since $20 \text{ N} > 6 \text{ N}$, the system accelerates.

Step 2 — Equations of motion. Let a be the common acceleration, T the tension.

Hanging block (m_2 , moving down):

$$m_2g - T = m_2a \quad \Rightarrow \quad 20 - T = 2a$$

Table block (m_1 , moving toward the pulley; kinetic friction μm_1g opposes it):

$$T - \mu m_1g = m_1a \quad \Rightarrow \quad T - 6 = 3a$$

Step 3 — Solve. Add the two equations to eliminate T :

$$20 - 6 = 5a \quad \Rightarrow \quad a = \frac{14}{5} = 2.8 \text{ m/s}^2$$

Substitute back:

$$T = 6 + 3(2.8) = 6 + 8.4 = 14.4 \text{ N}$$

Answer: $a = 2.8 \text{ m/s}^2$ and $T = 14.4 \text{ N}$.

Check: tension (14.4 N) is less than the hanging weight (20 N) — correct, since the hanging block is accelerating downward. If you ever get $T > m_2g$ here, you’ve made a sign error.

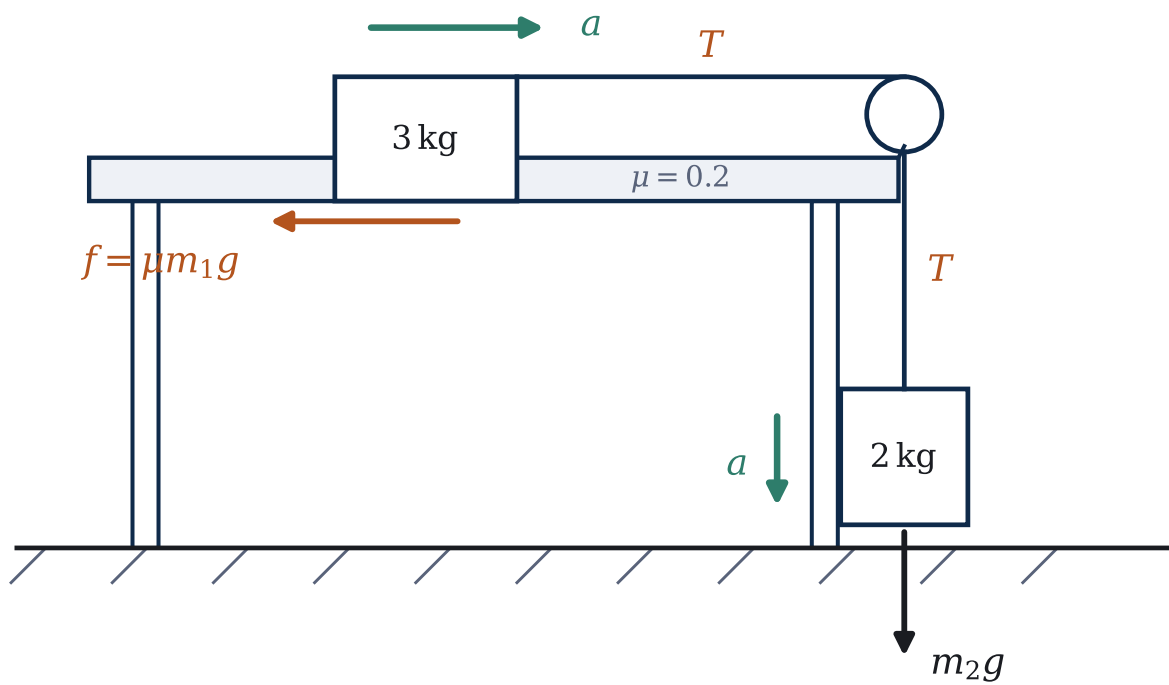


Figure 4.5 A 3 kg block on a table connected by a string over a pulley at the edge to a 2 kg block hanging vertically; friction acts on the table block opposing its motion toward the pulley.

Example 4 — [JEE Advanced Level] A crate of mass 10 kg rests on a rough horizontal floor with coefficient of friction $\mu = 0.75$. A worker wants to drag it using a rope. (a) Compare the force needed if the rope is pulled horizontally versus at the optimal angle. (b) Find that optimal angle and the minimum force. Take $g = 10 \text{ m/s}^2$.

This requires the **angle-of-friction optimization** — pure JEE Advanced reasoning.

Part (a) — Horizontal pull. With the rope horizontal, the full weight presses on the floor, so $N = mg = 100 \text{ N}$ and:

$$F_{\text{horizontal}} = \mu mg = 0.75 \times 100 = 75 \text{ N}$$

Part (b) — Optimal angled pull. From the derivation, the minimum force occurs when the rope is pulled at the angle of friction $\lambda = \tan^{-1} \mu$:

$$\lambda = \tan^{-1}(0.75) = 37^\circ$$

The minimum force is:

$$F_{\min} = \frac{\mu mg}{\sqrt{1 + \mu^2}} = \frac{0.75 \times 100}{\sqrt{1 + 0.5625}} = \frac{75}{\sqrt{1.5625}} = \frac{75}{1.25} = 60 \text{ N}$$

Answer: Pulling horizontally needs 75 N; pulling at 37° above the horizontal needs only 60 N — a saving of 15 N (about 20%).

Deep insight: the optimal angle is exactly the angle of friction, and at that angle $F_{\min} = mg \sin \lambda = 100 \sin 37^\circ = 100(0.6) = 60 \text{ N}$ — the same number by a slicker route. The angled pull wins because lifting at λ reduces the normal force (and hence friction) faster than it wastes the pull vertically. This is precisely why we instinctively tilt the handle when dragging a loaded trolley.

Section 5: Practice Exercises

1. [CBSE] A 4 kg block rests on a rough horizontal surface with $\mu_s = 0.5$ and $\mu_k = 0.4$. Find the frictional force when a horizontal force of (a) 12 N and (b) 25 N is applied. ($g = 10 \text{ m/s}^2$)
2. [NEET] A body placed on a rough inclined plane just begins to slide when the inclination is 45° . What is the coefficient of static friction?
3. [JEE Main] A 5 kg block slides down a rough incline of 37° with coefficient of kinetic friction 0.25. Find its acceleration. ($g = 10 \text{ m/s}^2$, $\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$)
4. [JEE Main] A 6 kg block on a rough table ($\mu = 0.3$) is connected over a frictionless pulley to a hanging 4 kg block. Find the acceleration of the system and the string tension. ($g = 10 \text{ m/s}^2$)
5. [JEE Advanced] A uniform chain lies on a rough horizontal table with coefficient of static friction 0.25. Find the maximum fraction of the chain's length that can hang over the edge of the table without the chain sliding off.

Final Answers:

1. (a) 12 N (static, since limiting = 20 N); (b) 16 N (kinetic, since $25 > 20$)
2. $\mu_s = \tan 45^\circ = 1$

3. $a = g(\sin \theta - \mu_k \cos \theta) = 10(0.6 - 0.25 \times 0.8) = 4 \text{ m/s}^2$
 4. $a = \frac{40 - 0.3 \times 60}{10} = 2.2 \text{ m/s}^2$; $T = 4(10) - 4(2.2) = 31.2 \text{ N}$
 5. $\frac{y}{L} = \frac{\mu}{1 + \mu} = \frac{0.25}{1.25} = 0.2$, i.e. 20%
-

Section 6: Crack the MCQ

Q1. A 5 kg block sits on a rough floor with $\mu_s = 0.5$. A horizontal force of 15 N is applied but the block does not move. The frictional force on the block is ($g = 10 \text{ m/s}^2$):

- (a) 25 N
- (b) 15 N
- (c) 0 N
- (d) 7.5 N

Correct: (b). The limiting friction is $\mu_s mg = 0.5 \times 50 = 25 \text{ N}$. Since the applied 15 N is below this, the block stays put and static friction self-adjusts to exactly balance the applied force: $f_s = 15 \text{ N}$.

- (a) computes $\mu_s N$ without checking whether the block is actually at the verge of sliding — the single most common friction error.
 - (c) ignores that friction must balance the applied force.
 - (d) multiplies μ_s by the applied force, which is meaningless.
-

Q2. The force of limiting friction between two surfaces in contact:

- (a) increases if the apparent area of contact increases.
- (b) is independent of the apparent area of contact but proportional to the normal reaction.
- (c) depends only on the area of contact.
- (d) is independent of the normal reaction.

Correct: (b). By the laws of friction, limiting friction is proportional to the normal reaction and (to good approximation) independent of the apparent contact area.

- (a) and (c) invoke contact area, which the empirical laws say does not matter.
 - (d) contradicts the defining relation $f = \mu N$.
-

Q3. For a given pair of surfaces, the coefficients of static (μ_s) and kinetic (μ_k) friction generally satisfy:

- (a) $\mu_s < \mu_k$
- (b) $\mu_s > \mu_k$
- (c) $\mu_s = 0$
- (d) $\mu_k > 1$ always

Correct: (b). It generally takes more force to *start* sliding than to *maintain* it, so μ_s is generally greater than (at most equal to) μ_k . This is why a sliding body, once started, is easier to keep moving.

- (a) reverses the inequality.
- (c) would mean a frictionless surface, not a general rule.
- (d) coefficients can exceed 1 for some surfaces, but it is certainly not “always.”

Q4. Which statement about friction is correct?

- (a) Friction always opposes the motion of a body.
- (b) Static friction can act as the force that propels a body forward.
- (c) Kinetic friction is always greater than limiting static friction.
- (d) Friction acts only on bodies that are moving.

Correct: (b). Friction opposes *relative sliding*, which is not the same as opposing a body's motion. When you walk or a car accelerates, **static friction from the ground pushes you forward** — it is the driving force, not a retarding one.

- (a) is the classic misconception; friction can drive motion (walking, driving wheels).
 - (c) is backward — kinetic friction is generally *less* than limiting static friction.
 - (d) ignores static friction, which acts on stationary bodies.
-

Section 7: Common Student Pitfalls & Pro-Tips

- **Writing friction = μN unconditionally.** Static friction equals $\mu_s N$ *only* at the verge of sliding. Below that, it equals whatever tangential force is applied (up to the limit). Always compute the limiting friction first, compare it with the driving force, and only then decide whether the body is static (friction = applied force) or sliding (friction = $\mu_k N$).
- **Skipping the “does it move?” check on inclines.** A block on a rough incline stays put if $\tan \theta \leq \mu_s$ (i.e. $\theta \leq$ angle of repose). Plugging into $a = g(\sin \theta - \mu_k \cos \theta)$ when the block isn't actually sliding gives a fictitious “acceleration” — check $\tan \theta$ versus μ_s first.
- **Believing friction depends on contact area.** It doesn't (to exam approximation). A brick resting on its large face or its small face experiences the same limiting friction, because N is unchanged.
- **Thinking friction always opposes motion.** Friction opposes *relative sliding between surfaces*. Static friction frequently *causes* a body to accelerate — it's what lets you walk and what drives a car's tyres forward.

Pro-Tip — the angle of repose shortcut. Whenever a problem says a body “just begins to slide on an incline of angle θ ,” instantly write $\mu_s = \tan \theta$ — no force resolution required. And on any incline question, your first scratch-pad line should compare $\tan \theta$ with μ : it tells you in one step whether you're in the static or kinetic regime.

Section 8: Cheat Sheet Box

Concept	Key Relation
Static friction (below limit): $f_s =$ applied force, with $f_s \leq \mu_s N$	
Kinetic friction	$f_k = \mu_k N$
Coefficient of friction	$\mu = f/N$, dimensionless, $\mu_s \geq \mu_k$
Angle of friction	$\tan \lambda = \mu$
Angle of repose	$\tan \theta_r = \mu_s$, and $\theta_r = \lambda$

Concept	Key Relation
Sliding down rough incline	$a = g(\sin \theta - \mu_k \cos \theta)$
Minimum drag force	$F_{\min} = \frac{\mu mg}{\sqrt{1 + \mu^2}} = mg \sin \lambda \text{ at } \theta = \lambda$

Core takeaways:

- **Static friction is self-adjusting** — it equals the applied force up to a maximum of $\mu_s N$; only *then* does sliding begin and $f_k = \mu_k N$ take over.
- **Friction depends on the normal force, not the contact area.**
- **Angle of repose = angle of friction:** $\tan \theta_r = \mu_s$, independent of mass.
- **Compare $\tan \theta$ with μ** to decide whether a block on an incline slides.
- The **optimal drag angle** equals the angle of friction λ .

Memory aid: “*Static is shy and self-adjusting; kinetic is constant and a little weaker.*” For the angles: “*Repose = friction = $\tan^{-1} \mu$.*”

Subtopic 5: Circular Motion Dynamics (Banking)

Exam Relevance: A high-yield application of Newton’s laws. JEE Main asks 1–2 questions a year on banking, friction-limited cornering, or conical-pendulum-type setups. NEET favors the conceptual side — what provides centripetal force, why a cyclist leans, the design-speed idea. JEE Advanced builds it into friction-plus-banking ranges and rotating-frame problems. CBSE Boards reliably sets the banking-angle derivation as a 3–5 mark question.

Section 1: Concept Intuition

Watch a car take a sharp turn on a flat road and you feel it: your body is flung toward the outer side of the car. Watch a cyclist round a corner at speed and you’ll notice she *leans inward*, almost dangerously so. Watch the daredevil “maut ka kuan” (well of death) at a village mela — a motorcyclist roaring around the inside of a vertical wooden cylinder, defying gravity. Every one of these is the same physics: **the dynamics of circular motion**.

Here’s the foundational truth, and it surprises most students: a body moving in a circle at *constant speed* is still **accelerating**. Its speed isn’t changing, but its *direction* is changing every instant — and velocity is a vector, so a changing direction means a changing velocity, which means acceleration. This acceleration points always toward the **centre** of the circle, and has magnitude

$$a_c = \frac{v^2}{r} = \omega^2 r$$

By Newton’s second law, an acceleration needs a force. So circular motion *demands* a net inward force, called the **centripetal force**, of magnitude mv^2/r . This is the single most misunderstood point in the topic: centripetal force is **not a new kind of force**. It is the *name for the role* played by whatever real force happens to point toward the centre — friction for a turning car, tension for a stone on a string, gravity for a satellite, the normal reaction for the well-of-death rider. Identify the real inward force, set it equal to mv^2/r , and you’ve solved the problem.

Think about it this way... When your car turns left and you lurch right, no real force pushed you outward. Your body, obeying inertia, simply wanted to keep going straight, while the car curved away beneath you. The seat (or the door) then pushes you inward to make you turn with the car. From *inside* the accelerating car you can pretend there’s an outward force — the **centrifugal force** — but it is a **pseudo-force**, a bookkeeping fiction valid only in the rotating (non-inertial) frame. In the ground frame there is no such outward force; there is only the inward centripetal force and inertia. (Note: centripetal and centrifugal forces are *not* an action–reaction pair — they act on the same body, just described from two different frames.)

Now to the headline application — **turning on roads**. A turning vehicle needs an inward force. It can come three ways: **friction alone** (the road is flat and friction grips the tyres inward), **banking alone** (the road is tilted so the normal reaction leans inward), or **both together** (real roads). On a flat road, friction is the only ally — but friction is unreliable on wet or icy surfaces and runs out at high speed. So engineers **bank** the curve: they raise the outer edge, tilting the road so that a component of the (always-present) normal reaction points toward the centre and supplies the centripetal force *without relying on friction at all*. This is why highway curves and railway tracks are visibly tilted.

Limiting Conditions & Assumptions:

- The formulas assume **uniform circular motion** — constant speed, so no tangential acceleration.

- “Friction only” $v_{\max} = \sqrt{\mu rg}$ uses the **limiting** static friction; below this speed the car corners safely with friction to spare.
 - A road banked for one **design speed** is friction-free only at *that* speed; above or below it, friction must make up the difference.
 - The vehicle is treated as a **point mass** — overturning (a torque effect) is a separate consideration from skidding.
-

Section 2: Key Formulas & Definitions

Centripetal acceleration and force:

$$a_c = \frac{v^2}{r} = \omega^2 r, \quad F_c = \frac{mv^2}{r} = m\omega^2 r$$

- F_c SI unit: newton — Dimensional formula: $[M^1 L^1 T^{-2}]$
- ω (angular speed) unit: $\text{rad} \cdot \text{s}^{-1}$ — Dimensional formula: $[M^0 L^0 T^{-1}]$

Centrifugal force (pseudo-force in the rotating frame): magnitude $\frac{mv^2}{r}$, directed radially **outward**.

Turning on a level (unbanked) road — friction only:

$$\frac{mv^2}{r} \leq \mu_s mg \quad \Rightarrow \quad v_{\max} = \sqrt{\mu_s rg}$$

(Note: mass cancels — $v_{\max}$ is independent of the vehicle’s mass.)

Banking of roads — frictionless (design speed):

$$\tan \theta = \frac{v^2}{rg} \quad \Rightarrow \quad v_{\text{design}} = \sqrt{rg \tan \theta}$$

where θ is the angle of banking.

Banking with friction — safe speed range:

$$v_{\max} = \sqrt{rg \frac{\tan \theta + \mu}{1 - \mu \tan \theta}}, \quad v_{\min} = \sqrt{rg \frac{\tan \theta - \mu}{1 + \mu \tan \theta}}$$

Cyclist leaning into a turn (lean angle θ from vertical):

$$\tan \theta = \frac{v^2}{rg}$$

(identical in form to the banking relation).

Conical pendulum / horizontal circular swing (string at angle θ to vertical):

$$\tan \theta = \frac{v^2}{rg}$$

Section 3: CBSE Board Derivations

Derivation 1 — Angle of banking on a frictionless road.

Setup: A vehicle of mass m moves at speed v around a curve of radius r on a road banked at angle θ . Assume the road is smooth (no friction), so only two forces act: the weight mg (vertically down) and the normal reaction N (perpendicular to the road surface).

Figure 4.6 *A vehicle on a road banked at angle θ ; normal reaction N perpendicular to the tilted surface, resolved into $N \cos \theta$ (vertical) and $N \sin \theta$ (horizontal toward the centre); weight mg straight down. The normal reaction, being perpendicular to the tilted road, leans inward. Resolve it: • Vertical component: $N \cos \theta$ (upward) • Horizontal component: $N \sin \theta$ (pointing toward the centre of the circle) Why these two equations? Vertically the car neither rises nor sinks, so vertical forces balance. Horizontally the car accelerates toward the centre, so the net horizontal force is the centripetal force. (see next page)*

Vertical balance:

$$N \cos \theta = mg \quad \dots(1)$$

Horizontal (centripetal) equation:

$$N \sin \theta = \frac{mv^2}{r} \quad \dots(2)$$

Divide (2) by (1) — the N and m cancel beautifully:

$$\tan \theta = \frac{v^2}{rg}$$

Physical reading: the required banking angle depends only on the speed and radius, **not on the vehicle's mass** — a truck and a scooter need the same banked curve. This is exactly the geometry of a leaning cyclist and a conical pendulum, which is why all three share the same formula.

Derivation 2 — Maximum safe speed on a banked road *with* friction.

Setup: Same banked road (angle θ , radius r), now with coefficient of friction μ . At the **maximum** speed, the vehicle tends to skid *outward and up* the bank, so limiting friction $f = \mu N$ acts *down* the slope (toward the inner edge), helping to hold the car in.

Figure 4.7 *Banked road with three forces on the vehicle — N perpendicular to surface, mg down, friction $f = \mu N$ directed down the incline; all resolved into horizontal and vertical components. Resolve all forces into vertical and horizontal (centripetal) directions. (see next page)*

Vertical balance (N up-component supports weight plus friction's downward component):

$$N \cos \theta = mg + \mu N \sin \theta \quad \Rightarrow \quad N(\cos \theta - \mu \sin \theta) = mg \quad \dots(1)$$

Horizontal (centripetal) (both N and friction contribute inward):

$$N \sin \theta + \mu N \cos \theta = \frac{mv_{\max}^2}{r} \quad \Rightarrow \quad N(\sin \theta + \mu \cos \theta) = \frac{mv_{\max}^2}{r} \quad \dots(2)$$

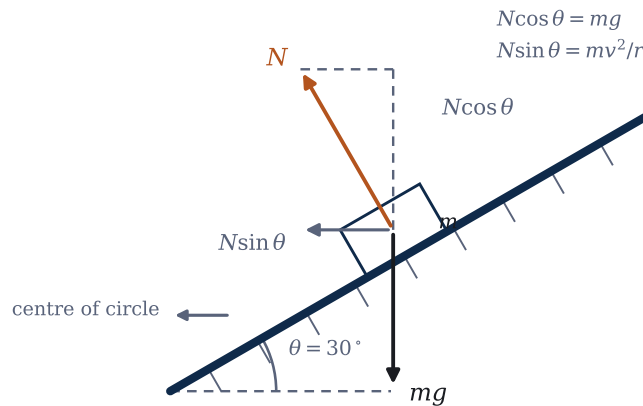


Figure 4.6 A vehicle on a road banked at angle θ ; normal reaction N perpendicular to the tilted surface, resolved into $N \cos \theta$ (vertical) and $N \sin \theta$ (horizontal toward the centre); weight mg straight down. The normal reaction, being perpendicular to the tilted road, leans inward. Resolve it: • Vertical component: $N \cos \theta$ (upward) • Horizontal component: $N \sin \theta$ (pointing toward the centre of the circle) Why these two equations? Vertically the car neither rises nor sinks, so vertical forces balance. Horizontally the car accelerates toward the centre, so the net horizontal force is the centripetal force.

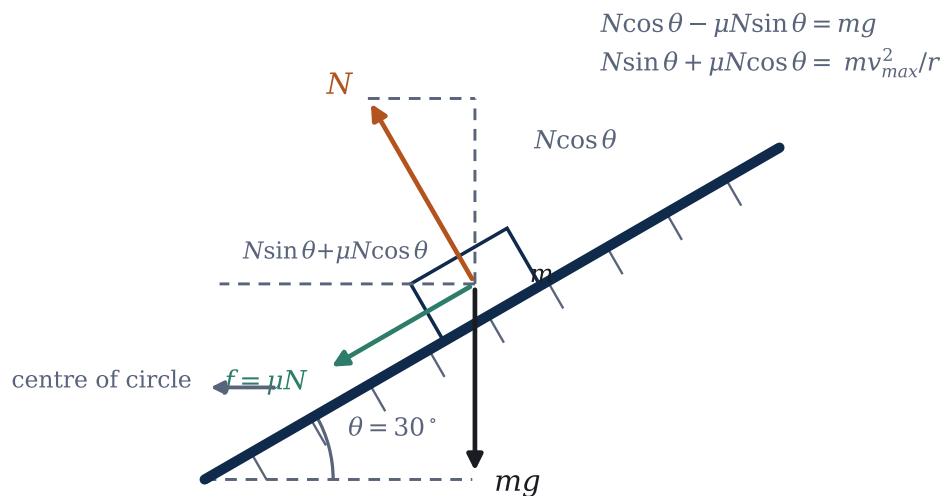


Figure 4.7 Banked road with three forces on the vehicle — N perpendicular to surface, mg down, friction $f = \mu N$ directed down the incline; all resolved into horizontal and vertical components. Resolve all forces into vertical and horizontal (centripetal) directions.

Divide (2) by (1):

$$\frac{v_{\max}^2}{rg} = \frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta}$$

Divide numerator and denominator by $\cos \theta$:

$$v_{\max} = \sqrt{rg \frac{\tan \theta + \mu}{1 - \mu \tan \theta}}$$

The *minimum speed* follows by flipping the direction of friction (the car now tends to slip *down*, so friction acts *up* the slope): swap the signs of μ , giving $v_{\min} = \sqrt{rg \frac{\tan \theta - \mu}{1 + \mu \tan \theta}}$. Between these two speeds the curve can be taken safely.

Sanity check: set $\mu = 0$ and both formulas collapse to $v = \sqrt{rg \tan \theta}$ — the single frictionless design speed, exactly as Derivation 1 demands.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A curved highway of radius 200 m is to be banked so that a car can take the turn at 20 m/s without relying on friction. Find the required angle of banking. Take $g = 10 \text{ m/s}^2$.

Given: $r = 200 \text{ m}$, $v = 20 \text{ m/s}$, frictionless design.

Step 1 — Apply the banking relation.

$$\tan \theta = \frac{v^2}{rg} = \frac{(20)^2}{200 \times 10} = \frac{400}{2000} = 0.2$$

Step 2 — Solve for the angle.

$$\theta = \tan^{-1}(0.2) \approx 11.3^\circ$$

Answer: The road should be banked at about 11.3° .

Presentation note: state clearly that this is the *design speed* angle — the curve is friction-free only at 20 m/s. Boards often award a mark for noting that the answer is independent of the car's mass.

Example 2 — [NEET Speed Trap] A car rounds a flat, unbanked curve of radius 20 m. If the coefficient of friction between the tyres and the road is 0.5, what is the maximum speed at which the car can turn without skidding? Take $g = 10 \text{ m/s}^2$.

The trap: students drag in the car's mass, or forget the square root and report $\mu rg = 100$ as a speed.

The shortcut: on a flat road friction alone supplies the centripetal force, so $v_{\max} = \sqrt{\mu rg}$ — mass-free:

$$v_{\max} = \sqrt{0.5 \times 20 \times 10} = \sqrt{100} = 10 \text{ m/s}$$

Answer: 10 m/s.

Speed tip: memorize $v_{\max} = \sqrt{\mu rg}$ for flat turns and notice the numbers are chosen to give a perfect square — exam-setters love $\mu rg = 100$. Any option that scales with mass is wrong on inspection, since mass cancels.

Example 3 — [JEE Main Level] A circular road of radius 90 m is banked at an angle θ where $\tan \theta = 0.5$. The coefficient of friction between the tyres and the road is $\mu = 0.5$. Find the maximum speed at which a car can safely round the curve. Take $g = 10 \text{ m/s}^2$.

This needs the **banking-plus-friction** formula — a true multi-effect problem.

Step 1 — Identify the quantities. $r = 90 \text{ m}$, $\tan \theta = 0.5$, $\mu = 0.5$, $g = 10$.

Step 2 — Apply the maximum-speed formula.

$$v_{\max} = \sqrt{rg \frac{\tan \theta + \mu}{1 - \mu \tan \theta}} = \sqrt{90 \times 10 \times \frac{0.5 + 0.5}{1 - (0.5)(0.5)}}$$

Step 3 — Evaluate the bracket.

$$\frac{0.5 + 0.5}{1 - 0.25} = \frac{1.0}{0.75} = \frac{4}{3}$$

Step 4 — Finish.

$$v_{\max} = \sqrt{900 \times \frac{4}{3}} = \sqrt{1200} \approx 34.6 \text{ m/s}$$

Answer: The maximum safe speed is $\sqrt{1200} \approx 34.6 \text{ m/s}$ (about 125 km/h).

Insight: compare with the frictionless design speed $\sqrt{rg \tan \theta} = \sqrt{900 \times 0.5} = \sqrt{450} \approx 21.2 \text{ m/s}$. Friction widens the safe window dramatically — here it raises the ceiling from 21 to 35 m/s. The denominator $(1 - \mu \tan \theta)$ is the easy place to slip; guard it carefully.

Example 4 — [JEE Advanced Level] At a village fair, a stunt rider performs in the “well of death” — riding around the inside of a vertical cylindrical wall of radius 3 m. The coefficient of friction between the tyres and the wall is $\mu = 0.6$. Find the minimum speed the rider must maintain so as not to slide down. Take $g = 10 \text{ m/s}^2$.

This is the showpiece — friction acting *vertically* while the normal reaction plays the centripetal role. Pure JEE Advanced reasoning.

Figure 4.8 A vertical cylinder; a motorcyclist on the inside wall; normal reaction N pointing horizontally toward the axis (centripetal), weight mg down, friction f acting vertically upward along the wall. (see next page)

Step 1 — Identify the forces. The wall pushes the rider horizontally *inward* — this normal reaction N is the centripetal force. Gravity mg pulls down. Friction f acts *up* the wall (opposing the tendency to slide down), and it is the *only* thing holding the rider up.

Step 2 — Centripetal (horizontal) equation. The normal reaction supplies the centripetal force:

$$N = \frac{mv^2}{r}$$

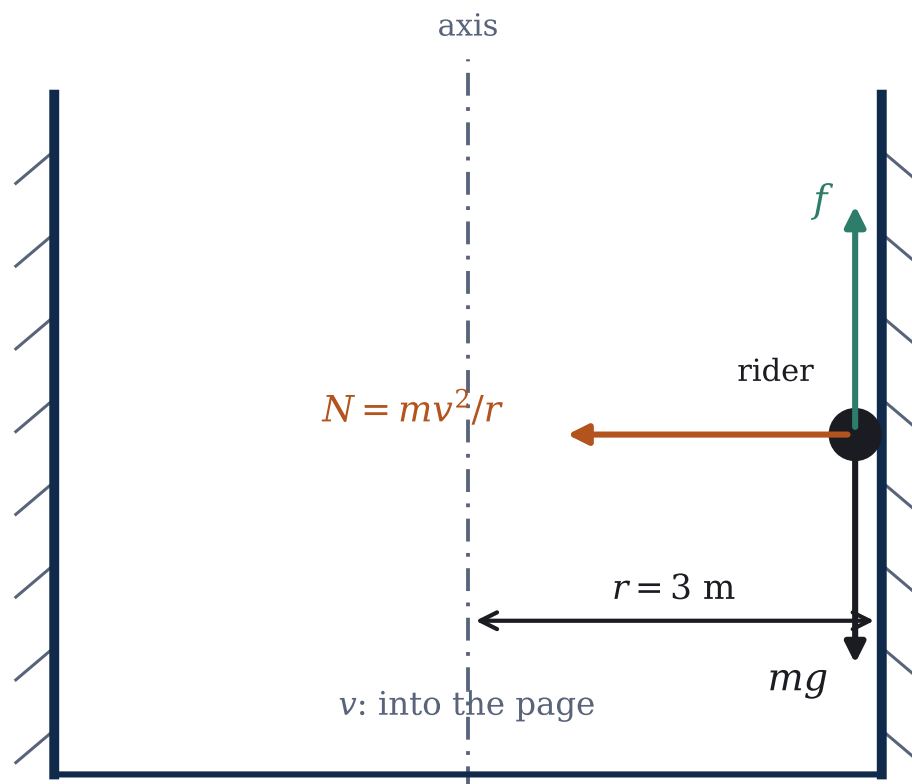


Figure 4.8 A vertical cylinder; a motorcyclist on the inside wall; normal reaction N pointing horizontally toward the axis (centripetal), weight mg down, friction f acting vertically upward along the wall.

Step 3 — Vertical equilibrium. Friction must support the full weight, and friction cannot exceed μN :

$$f = mg, \quad f \leq \mu N$$

Step 4 — Combine. For the rider not to slide, $mg \leq \mu N = \mu \frac{mv^2}{r}$. The mass cancels:

$$g \leq \frac{\mu v^2}{r} \Rightarrow v^2 \geq \frac{gr}{\mu} \Rightarrow v_{\min} = \sqrt{\frac{gr}{\mu}}$$

Step 5 — Substitute.

$$v_{\min} = \sqrt{\frac{10 \times 3}{0.6}} = \sqrt{50} \approx 7.07 \text{ m/s}$$

Answer: The rider must maintain at least about 7.07 m/s.

Deep insight: notice the counterintuitive result — going *faster* is what keeps the rider *up*, because higher speed means a larger normal reaction, hence more available friction. Slow down too much and friction can no longer support the weight, and the rider slides. The mass cancelling means a heavy bike and a light one need the same minimum speed.

Section 5: Practice Exercises

1. **[CBSE]** A curved road of radius 50 m is banked for a design speed of 15 m/s. Find the angle of banking. ($g = 10 \text{ m/s}^2$)
2. **[NEET]** A car rounds an unbanked curve of radius 40 m with coefficient of friction 0.5 between the tyres and the road. Find the maximum speed for no skidding. ($g = 10 \text{ m/s}^2$)
3. **[JEE Main]** A road of radius 90 m is banked at $\tan \theta = 0.5$, with coefficient of friction $\mu = 0.5$ between tyres and road. Find the **minimum** speed at which a car can negotiate the curve without slipping down the bank. ($g = 10 \text{ m/s}^2$)
4. **[JEE Main]** A stone of mass 0.5 kg tied to a string is whirled as a conical pendulum in a horizontal circle of radius 0.5 m, with the string making 30° with the vertical. Find the speed of the stone. ($g = 10 \text{ m/s}^2$)
5. **[JEE Advanced]** A motorcyclist rides on the inside vertical wall of a cylindrical “well of death” of radius 4 m. If the coefficient of friction between the tyres and the wall is 0.4, find the minimum speed required to avoid sliding down. ($g = 10 \text{ m/s}^2$)

Final Answers:

1. $\theta = \tan^{-1}\left(\frac{225}{500}\right) = \tan^{-1}(0.45) \approx 24.2^\circ$
2. $v_{\max} = \sqrt{0.5 \times 40 \times 10} = \sqrt{200} \approx 14.1 \text{ m/s}$
3. $v_{\min} = \sqrt{900 \times \frac{0.5 - 0.5}{1 + 0.25}} = 0$ — friction alone holds it at any low speed (the bank is gentle enough that the car cannot slip down)
4. $v = \sqrt{rg \tan \theta} = \sqrt{0.5 \times 10 \times \tan 30^\circ} = \sqrt{2.887} \approx 1.70 \text{ m/s}$
5. $v_{\min} = \sqrt{\frac{gr}{\mu}} = \sqrt{\frac{10 \times 4}{0.4}} = \sqrt{100} = 10 \text{ m/s}$

Section 6: Crack the MCQ

Q1. A car takes a turn on a flat, unbanked road at constant speed. The centripetal force acting on it is provided by:

- (a) the forward thrust of the engine.
- (b) the static friction between the tyres and the road.
- (c) the weight of the car.
- (d) the normal reaction from the road.

Correct: (b). On a flat road the only horizontal force available to point toward the centre is friction, so static friction supplies the centripetal force.

- (a) engine thrust acts along the direction of motion (tangential), not toward the centre.
 - (c) and (d) the weight and normal reaction are both vertical on a flat road and cannot contribute a horizontal centripetal force.
-

Q2. The centrifugal force is best described as:

- (a) a real force directed toward the centre of the circular path.
- (b) the Newton's-third-law reaction to the centripetal force.
- (c) a pseudo-force, directed radially outward, experienced only in a rotating (non-inertial) frame.
- (d) a force always equal to the weight of the body.

Correct: (c). Centrifugal force is a fictitious force introduced to make Newton's laws work in a rotating frame; it points outward and exists only in that non-inertial frame.

- (a) it points outward, not inward, and is not real in the ground frame.
 - (b) this is the classic trap — centripetal and centrifugal forces act on the *same* body in *different* frames, so they are **not** an action–reaction pair.
 - (d) there is no general relation to weight.
-

Q3. A road is banked for a particular design speed. When a vehicle travels at exactly that speed, the force of friction required between the tyres and the road is:

- (a) maximum, equal to μmg .
- (b) equal to mg .
- (c) zero.
- (d) equal to $\frac{mv^2}{r}$.

Correct: (c). At the design speed the horizontal component of the normal reaction alone supplies the entire centripetal force, so no friction is needed.

- (a) friction is maximal only when the vehicle is at the edge of skidding, not at the design speed.
 - (b) and (d) assign friction a value it does not need to take; the whole purpose of banking is to make friction unnecessary at the design speed.
-

Q4. The maximum speed at which a car can safely round an unbanked curve of radius r (coefficient of friction μ):

- (a) depends on the mass of the car.

- (b) equals $\sqrt{\mu rg}$ and is independent of the car's mass.
- (c) increases as the car's mass increases.
- (d) equals μrg .

Correct: (b). Setting friction equal to the centripetal requirement, $\mu mg = mv^2/r$; the mass cancels, leaving $v_{\max} = \sqrt{\mu rg}$.

- (a) and (c) invoke mass, which cancels out.
- (d) forgets the square root — μrg has units of speed-squared, not speed.

Section 7: Common Student Pitfalls & Pro-Tips

- **Treating centrifugal force as real in the ground frame.** In the inertial (ground) frame there is no outward force — only the inward centripetal force and inertia. Introduce centrifugal force *only* if you deliberately work in the rotating frame, and then add it as a pseudo-force pointing outward. Mixing the two frames in one equation is a guaranteed error.
- **Forgetting that mass cancels.** In friction-only cornering ($v = \sqrt{\mu rg}$), in banking ($\tan \theta = v^2/rg$), and in the well-of-death ($v_{\min} = \sqrt{gr/\mu}$), the mass drops out. If your final answer still contains m , recheck — you've likely missed a cancellation.
- **Botching the banking-plus-friction denominator.** The terms are $\frac{\tan \theta + \mu}{1 - \mu \tan \theta}$ for maximum speed and $\frac{\tan \theta - \mu}{1 + \mu \tan \theta}$ for minimum. Memorize that the **maximum** uses $+\mu$ on top and $-\mu \tan \theta$ below; the minimum flips both signs. Setting $\mu = 0$ should recover $\tan \theta = v^2/rg$ — use that as an instant check.
- **Confusing “centripetal” with a new force.** Always ask “which *real* force points toward the centre here?” — friction, tension, normal reaction, gravity — and set *that* equal to mv^2/r . Writing “centripetal force” as an extra force in the free-body diagram (on top of the real ones) double-counts and gives wrong answers.

Pro-Tip — the design-speed check. For any banking-with-friction problem, first compute the frictionless design speed $\sqrt{rg \tan \theta}$. The friction-assisted $v_{\max}$ must be *larger* and $v_{\min}$ *smaller* than this value. If your range doesn't straddle the design speed, you've made a sign error in the friction direction.

Section 8: Cheat Sheet Box

Situation	Key Relation
Centripetal force	$F_c = \frac{mv^2}{r} = m\omega^2 r$ (toward centre)
Flat road, friction only	$v_{\max} = \sqrt{\mu rg}$
Banking, frictionless	$\tan \theta = \frac{v^2}{rg}$
Banking + friction (max)	$v_{\max} = \sqrt{rg \frac{\tan \theta + \mu}{1 - \mu \tan \theta}}$

Situation	Key Relation
Banking + friction (min)	$v_{\min} = \sqrt{rg \frac{\tan \theta - \mu}{1 + \mu \tan \theta}}$
Leaning cyclist / conical pendulum	$\tan \theta = \frac{v^2}{rg}$
Well of death (rotor)	$v_{\min} = \sqrt{\frac{gr}{\mu}}$

Core takeaways:

- **Centripetal force is a role, not a new force** — identify the real inward force and set it to mv^2/r .
- **Centrifugal force is a pseudo-force**, outward, valid only in the rotating frame; *not* the reaction to centripetal force.
- **Mass cancels** in nearly every banking and friction-limited result.
- **Banking** lets the normal reaction supply the centripetal force; at the design speed, friction is **zero**.
- The **same formula** $\tan \theta = v^2/rg$ governs banking, leaning cyclists, and conical pendulums.

Memory aid: “*The centre pulls, inertia flings.*” For banking: “*tan θ equals v-squared over r-g.*” For the well of death: “*faster means safer.*”

Laws of Motion

Master Content — Part II

Supplementary Subtopics 6–8 (Coverage Completion)

Chapter 4 | Physics Content Bank — JEE / NEET / CBSE
June 2026

Why this part exists. Part I (Subtopics 1–5: Newton’s Laws & Impulse, Momentum Conservation, Concurrent Forces & Equilibrium, Friction, and Circular Motion Dynamics / Banking) is strong, but three pillars of the CBSE + JEE + NEET “Laws of Motion” syllabus were not yet present. Without them a student cannot say they “know everything” about this chapter. Part II closes those gaps using the same eight-section template, the same difficulty ladder (CBSE Board → NEET Speed Trap → JEE Main → JEE Advanced), and the same study-first style:

- **Subtopic 6 — Common Forces in Mechanics & Free-Body Diagrams:** the “vocabulary” of every mechanics problem (weight, normal, tension, spring, contact forces, rolling friction) and the FBD method that all of Subtopics 1–5 silently assumed.
- **Subtopic 7 — Connected Bodies, Pulleys & Constraint Relations:** Atwood machines, blocks in contact, string-and-pulley systems, and constraint equations — the single most heavily tested problem family in JEE/NEET mechanics.
- **Subtopic 8 — Vertical Circular Motion:** the loop, the rod-vs-string distinction, critical speeds, and “leaving the track” — the circular-motion case Part I’s banking subtopic did not cover.

Contents of Part II

- Subtopic 6: Common Forces in Mechanics & Free-Body Diagrams
- Subtopic 7: Connected Bodies, Pulleys & Constraint Relations
- Subtopic 8: Vertical Circular Motion

Subtopic 6: Common Forces in Mechanics & Free-Body Diagrams

Exam Relevance

This is the toolkit every other mechanics problem is built from. JEE Main rarely asks it “naked,” but a wrong normal force or a misdrawn FBD silently wrecks 1–2 questions a year. NEET loves direct conceptual items: “which forces are contact forces,” “normal reaction is not always mg ,” spring combinations, and the origin of contact forces. JEE Advanced assumes flawless FBDs as a prerequisite and tests spring networks and angled-force normal reactions. CBSE Boards regularly sets a 2–3 mark question on resolving forces, Hooke’s law, or the normal reaction in a lift.

Section 1: Concept Intuition

Every problem in Subtopics 1–5 quietly drew the same picture before solving: a body, isolated, with arrows for every force acting on it. That picture is a **free-body diagram (FBD)**, and the arrows are drawn from a small, fixed menu of forces. Master the menu and the FBD habit, and mechanics stops being a memory game.

At the deepest level there are only four *fundamental* forces in nature — gravitational, electromagnetic, and the strong and weak nuclear forces. But in everyday mechanics you meet just two faces of these: **gravity** (the long-range pull that gives a body its weight) and the **contact forces** (normal reaction, friction, tension, the push of a spring). Here is a fact that surprises students: every contact force — the floor pushing up on your feet, a rope pulling a crate, a spring resisting compression — is electromagnetic in origin. When two surfaces “touch,” their atoms never really meet; the electron clouds repel, and that repulsion, summed over a vast number of atoms, is what we feel as a solid push.

Weight ($\vec{W} = m\vec{g}$) is the gravitational pull of the Earth, always vertically downward, effectively acting at the body’s centre of mass. It is the one force that needs no contact.

Normal reaction (N) is the push a surface exerts perpendicular (“normal”) to itself. A surface can only *push*, never pull — so the normal arrow always points away from the surface, into the body. The single biggest beginner error is to assume $N = mg$. It is true only in the special case of a body resting on flat ground with nothing else pressing it. On an incline $N = mg \cos \theta$; in a lift accelerating up $N = m(g + a)$; if you press down on a block at an angle, N grows; if you pull up on it, N shrinks. The normal force is *self-adjusting*, exactly like static friction: it takes whatever value the “no-penetration” condition demands.

Tension (T) is the pull transmitted along a string, rope, or cable. A string, like a hand, can only pull — never push. For an *ideal* string (massless and inextensible) the tension is the same at every point, and it is unchanged as the string runs over an *ideal* pulley (massless, frictionless). That single fact — “tension is the same on both sides of an ideal pulley” — is the workhorse of Subtopic 7.

Spring force obeys **Hooke’s law**: a spring stretched or compressed by x pushes back with $\vec{F} = -k\vec{x}$, a *restoring* force whose magnitude is kx . Unlike a string, a spring can both pull (when stretched) and push (when compressed). The constant k measures stiffness. Springs in series get softer; springs in parallel get stiffer — a result we derive below.

Think about it this way... A normal force and a string tension are like two very different employees. The string is a “puller”: ask it to push and it simply goes slack (negative tension is nature’s way of saying “this rope isn’t doing what your equation claims”). The floor is a “pusher”: ask it to pull you down and it can’t — the body just lifts off ($N = 0$). Whenever your algebra produces a negative N or a negative T , the body has lost contact or the string has gone slack, and your assumed picture is wrong.

Finally, a quick word on **rolling friction**. Subtopic 4 covered static and kinetic friction (sliding). When a wheel *rolls* instead of slides, the resistance is far smaller: $f_r = \mu_r N$ with $\mu_r \ll \mu_k$. This tiny coefficient is the whole reason we put wheels under heavy loads — rolling sidesteps the much larger sliding friction.

The FBD method (use it every single time):

1. **Isolate** one body. Mentally cut it free from everything touching it.
2. **Draw only the forces acting on that body** — never the forces it exerts on others. Weight first (always present), then one force for every object in contact.
3. **Choose smart axes** (horizontal/vertical, or along/perpendicular to an incline).
4. **Resolve** slanted forces into components.
5. **Apply** $\sum F_x = ma_x$ and $\sum F_y = ma_y$ (equilibrium is the special case $a = 0$).
6. **Solve, then sanity-check** signs and limiting cases.

Limiting Conditions & Assumptions:

- Strings are taken as *ideal*: massless and inextensible, so tension is uniform. A massive rope has tension that varies along its length.
- Pulleys are taken as ideal: massless and frictionless, so they merely redirect tension without changing its magnitude.
- Springs are taken as massless and ideal (Hooke’s law holds), valid within the elastic limit only.
- Bodies are treated as point masses or rigid bodies; we ignore deformation and rotation unless stated.

Section 2: Key Formulas & Definitions

Weight: $\vec{W} = m\vec{g}$, magnitude mg , vertically downward.

Normal reaction (context-dependent — never assume mg):

$$N = mg \text{ (flat, nothing else); } N = mg \cos \theta \text{ (incline);}$$

$$N = m(g + a) \text{ (lift up); } N = m(g - a) \text{ (lift down); } N = mg - F \sin \theta \text{ (pull at angle } \theta \text{).}$$

Tension: along the string, away from the body; uniform in an ideal massless string and across an ideal pulley.

Hooke’s law (spring force):

$$\vec{F} = -k\vec{x}, \quad |F| = kx, \quad [k] = \text{N m}^{-1} = [MT^{-2}].$$

Springs in series (same force, extensions add):

$$\frac{1}{k_{\text{eq}}} = \frac{1}{k_1} + \frac{1}{k_2} \quad (\text{softer than either}).$$

Springs in parallel (same extension, forces add):

$$k_{\text{eq}} = k_1 + k_2 \quad (\text{stiffer than either}).$$

Rolling friction: $f_r = \mu_r N$, $\mu_r \ll \mu_k$.

Apparent weight (scale reading R , person of mass m):

$$R = m(g + a) \text{ up}; \quad R = m(g - a) \text{ down}; \quad R = mg \text{ constant velocity}; \quad R = 0 \text{ free fall}.$$

Section 3: CBSE Board Derivations

Derivation 1 — Equivalent stiffness of springs in series and in parallel.

Series. Two springs k_1, k_2 joined end to end carry the *same* force F (the force is transmitted undiminished through the chain). Their extensions add:

$$x = x_1 + x_2 = \frac{F}{k_1} + \frac{F}{k_2}.$$

For the single equivalent spring, $x = F/k_{\text{eq}}$. Equating,

$$\frac{F}{k_{\text{eq}}} = \frac{F}{k_1} + \frac{F}{k_2} \Rightarrow \boxed{\frac{1}{k_{\text{eq}}} = \frac{1}{k_1} + \frac{1}{k_2}}.$$

Because reciprocals add, k_{eq} is *smaller* than either spring — a series combination is softer.

Parallel. Two springs side by side share the *same* extension x but each carries its own force; the forces add:

$$F = F_1 + F_2 = k_1 x + k_2 x = (k_1 + k_2)x.$$

Comparing with $F = k_{\text{eq}}x$ gives $\boxed{k_{\text{eq}} = k_1 + k_2}$ — a parallel combination is stiffer.

Derivation 2 — The normal reaction is not always mg .

Claim: N adjusts to whatever the geometry and the other forces demand. We show three canonical cases from a single principle: N balances the net push of everything else perpendicular to the surface.

(i) *Block in a lift accelerating upward at a .* The only vertical forces are N (up) and mg (down), and the block shares the lift's upward acceleration:

$$N - mg = ma \Rightarrow N = m(g + a) > mg.$$

A downward acceleration replaces $+a$ by $-a$, giving $N = m(g - a) < mg$; in free fall $a = g$ and $N = 0$ (weightlessness).

(ii) *Block pulled by a rope at angle θ above the horizontal.* Vertical balance (block stays on the floor) reads

$$N + F \sin \theta = mg \Rightarrow N = mg - F \sin \theta < mg.$$

The upward pull “unweights” the block — exactly why lifting the handle of a suitcase reduces the friction you fight (Subtopic 4).

(iii) *Block pushed at angle θ below the horizontal.* Now the vertical component of the push adds to the weight:

$$N = mg + F \sin \theta > mg.$$

Moral: always write the perpendicular balance for the actual situation. Substituting $N = mg$ by reflex is the most common single mistake in all of mechanics.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A spring of force constant $k = 200 \text{ N/m}$ hangs vertically. A block of mass 2 kg is attached and lowered gently until it hangs in equilibrium. (a) Find the extension. (b) The whole arrangement is now placed in a lift accelerating upward at 2 m/s^2 ; find the new extension. Take $g = 10 \text{ m/s}^2$.

Step 1 — Equilibrium on the ground. The spring force balances the weight:

$$kx_1 = mg \Rightarrow x_1 = \frac{mg}{k} = \frac{2 \times 10}{200} = 0.10 \text{ m} = 10 \text{ cm}.$$

Step 2 — In the accelerating lift. The block now accelerates upward at a , so the spring must supply $m(g + a)$:

$$kx_2 = m(g + a) \Rightarrow x_2 = \frac{2(10 + 2)}{200} = \frac{24}{200} = 0.12 \text{ m} = 12 \text{ cm}.$$

Answer: (a) 10 cm ; (b) 12 cm . The spring stretches more because the apparent weight rose to $m(g + a)$.

Example 2 — [NEET Speed Trap] A 2 kg block on a frictionless floor is pushed by a force of 10 N directed at 30° below the horizontal. What is the normal reaction from the floor? Take $g = 10 \text{ m/s}^2$.

The trap: write $N = mg = 20 \text{ N}$ on autopilot. The downward-slanted push presses the block *harder* into the floor, so N must exceed mg .

Perpendicular balance:

$$N = mg + F \sin 30^\circ = 20 + 10(0.5) = 25 \text{ N}.$$

Answer: $N = 25 \text{ N}$.

Speed tip: the instant a force is applied at an angle, ask “does its vertical part add to or subtract from the weight?” Pushing down $\Rightarrow N = mg + F \sin \theta$; pulling up $\Rightarrow N = mg - F \sin \theta$. Any option equal to mg is the bait.

Example 3 — [JEE Main Level] Two blocks $m_1 = 2 \text{ kg}$ and $m_2 = 3 \text{ kg}$ rest in contact on a frictionless floor. A horizontal force $F = 10 \text{ N}$ is applied to m_1 , pushing both. Find (a) the common acceleration and (b) the contact force between the blocks.

Step 1 — Treat the pair as one system to get the acceleration:

$$a = \frac{F}{m_1 + m_2} = \frac{10}{5} = 2 \text{ m/s}^2.$$

Step 2 — FBD of the rear block alone is messy; FBD of the front block m_2 is clean. The only horizontal force on m_2 is the contact push N_c from m_1 :

$$N_c = m_2 a = 3 \times 2 = 6 \text{ N}.$$

Answer: $a = 2 \text{ m/s}^2$, contact force = 6 N .

Cross-check (Newton’s third law): the contact force on m_1 from m_2 is also 6 N , backward. Apply $\sum F$ to m_1 : $10 - 6 = 4 = m_1 a = 2 \times 2$. Consistent. Note that if F were applied to m_2 instead, the contact force would be $m_1 F / (m_1 + m_2) = 4 \text{ N}$, not 6 N — the contact force depends on which block you push. That asymmetry is a favourite trap.

Example 4 — [JEE Advanced Level] Blocks $m_1 = 1 \text{ kg}$ and $m_2 = 3 \text{ kg}$ sit on a frictionless floor connected by a *spring* of force constant $k = 400 \text{ N/m}$. A constant horizontal force $F = 8 \text{ N}$

is applied to m_1 , pulling the pair (with the spring between them). After the transient dies out the blocks move together with a steady separation. Find (a) the common acceleration and (b) the steady-state extension of the spring.

Step 1 — Common acceleration (in steady state the spring length is constant, so both blocks share one acceleration):

$$a = \frac{F}{m_1 + m_2} = \frac{8}{4} = 2 \text{ m/s}^2.$$

Step 2 — FBD of the trailing block m_2 . The only horizontal force on m_2 is the spring force $F_s = kx$, which must produce its acceleration:

$$kx = m_2 a \Rightarrow x = \frac{m_2 a}{k} = \frac{3 \times 2}{400} = 0.015 \text{ m} = 1.5 \text{ cm}.$$

Answer: $a = 2 \text{ m/s}^2$; spring extension = 1.5 cm.

Deep insight: the spring force is set by the mass it must drag, not by the applied F . Whichever block the spring is “responsible for” (here m_2) fixes $F_s = m_2 a$. This is the spring twin of the contact-force result in Example 3 — a spring is just a contact force you can see stretch.

Section 5: Practice Exercises

1. **[CBSE]** A 5 kg block rests on a horizontal floor. A rope pulls it with 30 N at 30° above the horizontal. Find the normal reaction. ($g = 10 \text{ m/s}^2$)
2. **[NEET]** Two springs of constants 300 N/m and 600 N/m are combined. Find k_{eq} in (a) series and (b) parallel.
3. **[JEE Main]** A 2 kg block hangs from a spring inside a lift. The spring reads an extension corresponding to a force of 26 N. What is the lift’s acceleration and its direction? ($g = 10 \text{ m/s}^2$)
4. **[JEE Main]** Three blocks of 1, 2, 3 kg rest in contact in a line on a frictionless floor. A force $F = 12 \text{ N}$ pushes the 1 kg block. Find the force the 2 kg block exerts on the 3 kg block.
5. **[JEE Advanced]** A block on a frictionless incline of 30° is held by a spring ($k = 500 \text{ N/m}$) running parallel to the incline. If the block has mass 5 kg, find the spring’s extension at equilibrium. ($g = 10 \text{ m/s}^2$)

Final Answers:

1. $N = mg - F \sin 30^\circ = 50 - 15 = 35 \text{ N}.$
2. (a) series $k = \frac{300 \times 600}{900} = 200 \text{ N/m};$ (b) parallel $k = 900 \text{ N/m}.$
3. $kx = m(g + a) \Rightarrow 26 = 2(10 + a) \Rightarrow a = 3 \text{ m/s}^2,$ upward.
4. $a = F/6 = 2 \text{ m/s}^2;$ force on 3 kg block $= 3a = 6 \text{ N}.$
5. $kx = mg \sin \theta \Rightarrow x = \frac{5 \times 10 \times 0.5}{500} = 0.05 \text{ m} = 5 \text{ cm}.$

Section 6: Crack the MCQ

Q1. The normal reaction on a body resting on a horizontal surface is equal to its weight only when:

- (a) the surface is rough.
- (b) no other vertical force (besides weight and N) acts and there is no vertical acceleration.
- (c) the body is moving.

(d) always, by definition.

Correct: (b). N equals mg only when nothing else presses or lifts vertically and $a_y = 0$. (a) roughness is irrelevant to N . (c) motion is irrelevant. (d) the “always” fallacy — N changes on inclines, in lifts, and under angled forces.

Q2. An ideal string passes over an ideal (massless, frictionless) pulley. The tension:

(a) is larger on the side of the heavier mass.

(b) is the same throughout the string.

(c) is zero on the lighter side.

(d) depends on the pulley’s radius.

Correct: (b). A massless string over a frictionless pulley has one uniform tension. (a) confuses tension with weight. (c) and (d) have no basis for an ideal system.

Q3. Two identical springs each of constant k are connected in parallel and support a load. Compared with a single spring, the combination is:

(a) softer, with $k_{\text{eq}} = k/2$.

(b) stiffer, with $k_{\text{eq}} = 2k$.

(c) unchanged, $k_{\text{eq}} = k$.

(d) undefined.

Correct: (b). Parallel springs add: $k_{\text{eq}} = 2k$ (stiffer). The $k/2$ answer is the *series* result — the classic swap.

Q4. Contact forces such as the normal reaction and tension are, at the microscopic level, manifestations of:

(a) the gravitational force.

(b) the electromagnetic force.

(c) the strong nuclear force.

(d) a separate “contact force” fundamental to nature.

Correct: (b). Contact forces arise from electromagnetic repulsion between surface electrons. (d) is the misconception — contact force is not a fundamental force.

Section 7: Common Student Pitfalls & Pro-Tips

- **Setting $N = mg$ by reflex.** The normal reaction is self-adjusting. Write the actual perpendicular balance every time — inclines, lifts, and angled forces all change it.
- **Drawing the wrong forces in an FBD.** Put only forces acting *on* the chosen body. The force the body exerts on something else belongs in *that* object’s diagram.
- **Letting tension vary across an ideal pulley.** For massless string + frictionless pulley, tension is one number throughout. It only differs across a pulley that has mass (rotational inertia) or friction.
- **Confusing series and parallel springs.** Reciprocals add in series (softer); constants add in parallel (stiffer). Quick check: more springs “side by side” must be harder to stretch.
- **Ignoring negative answers.** A negative N means the body left the surface; a negative T means the string went slack. Both signal a wrong assumed configuration, not a real answer.

Pro-Tip — the FBD is non-negotiable. Spend the first ten seconds of any mechanics problem drawing one clean free-body diagram per body, weight arrow first. Ninety percent of “hard” problems become routine the moment the diagram is correct; almost every wrong answer traces back to a missing or misdirected arrow.

Section 8: Cheat Sheet Box

Force / Quantity	Key Relation
Weight	$W = mg$, downward
Normal (general)	balances perpendicular forces; <i>not</i> always mg
Normal on incline	$N = mg \cos \theta$
Normal in lift	$N = m(g \pm a)$
Tension (ideal string)	uniform; same across ideal pulley
Spring (Hooke)	$F = -kx$, $ F = kx$
Springs in series	$1/k_{\text{eq}} = 1/k_1 + 1/k_2$ (softer)
Springs in parallel	$k_{\text{eq}} = k_1 + k_2$ (stiffer)
Rolling friction	$f_r = \mu_r N$, $\mu_r \ll \mu_k$

Core takeaways:

- Two faces of force in mechanics: gravity (weight) and contact forces (normal, tension, spring, friction) — all contact forces are electromagnetic at root.
- The normal reaction and tension are self-adjusting; strings only pull, surfaces only push.
- Series springs soften, parallel springs stiffen.
- Draw one clean FBD per body — only forces *on* that body — before writing a single equation.

Memory aid: “Gravity pulls, surfaces push, strings tug, springs snap back.” For the normal: “ N is whatever balance demands — never assume mg .”

Subtopic 7: Connected Bodies, Pulleys & Constraint Relations

Exam Relevance

This is arguably the most heavily tested problem family in mechanics. JEE Main reliably asks 1–2 questions a year on Atwood machines, blocks-on-table-with-hanging-mass, blocks in contact, or pulley systems. NEET favours quick conceptual and one-step numerical versions (equal-mass Atwood, tension between connected blocks). JEE Advanced layers in movable pulleys, constraint relations, and connected systems inside accelerating (non-inertial) frames. CBSE Boards regularly sets a 3 mark Atwood-machine or two-block numerical.

Section 1: Concept Intuition

Real machines are rarely a lone block. A lift counterweight, a crane, a tow-truck, two wagons coupled on a track — all are **connected bodies**, linked by strings, rods, or simple contact, forced to move together in a related way. The skill here is not new physics; it is Newton’s second law applied carefully to each body, plus one extra idea: a **constraint**.

A constraint is a relationship the connection forces upon the motions. The cleanest example: two blocks joined by an *inextensible* string over a fixed pulley. Because the string cannot stretch, whatever length leaves one side arrives on the other — so if one block moves down by 1 cm, the other moves up by 1 cm. Their speeds are equal, and so are the magnitudes of their accelerations. That single sentence, “the accelerations have equal magnitude,” is the constraint that makes the two equations solvable.

The universal recipe for connected bodies:

1. Decide the direction of motion (which way will the system actually go?).
2. Draw an FBD of *each* body separately.
3. Write $\sum F = ma$ for each, taking the direction of motion as positive for that body.
4. Add the constraint linking the accelerations (for a single fixed pulley, $a_1 = a_2 = a$).
5. Solve the simultaneous equations for a and the tension(s).

Three classic configurations recur:

(1) Blocks in contact, pushed by a force. Treat the train as one body to find the common $a = F/M_{\text{total}}$; then isolate one block to find the internal contact force. The contact force depends on which end you push — pushing a light block that drives a heavy one needs a large internal force, and vice versa.

(2) The Atwood machine. Two masses $m_1 > m_2$ hang from a string over a fixed pulley. The heavier descends, the lighter rises, with a shared acceleration smaller than g — the genius of the device is that it “dilutes” gravity so the motion is slow enough to time.

(3) Block on a table joined over a pulley to a hanging mass. The hanging weight drives the system; the table block is dragged along. (Subtopic 4 did the rough-table version; the frictionless case is the clean limit.)

Think about it this way... An inextensible string is like a rigid promise: “for every metre I give to one side, I take a metre from the other.” That promise is the constraint. The moment you trust it — equal speeds, equal accelerations for a simple pulley — the algebra collapses from four unknowns to two. Movable pulleys make a richer promise (one side moves twice as fast as the pulley), but it is the same idea: count the string.

Movable pulleys deserve a note. If a pulley is itself free to move and a string runs under it with both ends going up, then the load hanging from that pulley moves at *half* the speed of the string’s free end — equivalently, the free end moves twice as far. This “factor of two” (and the matching force advantage: the load is shared by two string segments, so the tension needed is halved) is the basis of every block-and-tackle and a staple of JEE Advanced constraint problems.

Limiting Conditions & Assumptions:

- Strings ideal (massless, inextensible) and pulleys ideal (massless, frictionless), so tension is uniform and the simple constraints hold.
- The string stays taut throughout. If your solution gives a negative tension, the string has gone slack and the bodies move independently — re-examine the setup.
- For blocks in contact, the surfaces must stay pressed together; if the contact force comes out negative, the blocks separate.

Section 2: Key Formulas & Definitions

System acceleration (no friction): $a = \frac{\text{net driving force}}{\text{total mass}}.$

Blocks in contact (F applied to m_1 , frictionless):

$$a = \frac{F}{m_1 + m_2}, \quad N_{\text{contact}} = \frac{m_2 F}{m_1 + m_2}.$$

Atwood machine ($m_1 > m_2$, fixed ideal pulley):

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}.$$

Table block + hanging mass (frictionless table):

$$a = \frac{m_2 g}{m_1 + m_2}, \quad T = \frac{m_1 m_2 g}{m_1 + m_2},$$

where m_1 is on the table and m_2 hangs.

Block on a frictionless incline + hanging mass over a pulley (m_2 hangs, m_1 on incline of angle θ , $m_2 g > m_1 g \sin \theta$):

$$a = \frac{(m_2 - m_1 \sin \theta)g}{m_1 + m_2}, \quad T = \frac{m_1 m_2 (1 + \sin \theta)g}{m_1 + m_2}.$$

Constraint (single fixed pulley): $a_1 = a_2$ (equal magnitudes).

Constraint (one movable pulley): the load moves at half the free-end speed: $a_{\text{free end}} = 2a_{\text{load}}.$

Connected system inside a frame accelerating at a_0 : replace g by the effective gravity $g_{\text{eff}} = g \pm a_0$ (sign by the frame’s direction) in the formulas above.

Section 3: CBSE Board Derivations

Derivation 1 — The Atwood machine.

Setup: masses $m_1 > m_2$ hang from the two ends of a light inextensible string over a fixed, frictionless pulley. Let a be the common acceleration magnitude (m_1 down, m_2 up) and T the (uniform) tension.

FBD of m_1 (taking downward positive for it):

$$m_1g - T = m_1a. \quad (i)$$

FBD of m_2 (taking upward positive for it):

$$T - m_2g = m_2a. \quad (ii)$$

Add (i) and (ii) to eliminate T :

$$(m_1 - m_2)g = (m_1 + m_2)a \Rightarrow a = \frac{(m_1 - m_2)g}{m_1 + m_2}.$$

Substitute back into (ii):

$$T = m_2(g + a) = m_2\left(g + \frac{(m_1 - m_2)g}{m_1 + m_2}\right) = \frac{2m_1m_2g}{m_1 + m_2}.$$

Sanity checks: if $m_1 = m_2$, then $a = 0$ and $T = mg$ — the system balances. If $m_2 \rightarrow 0$, then $a \rightarrow g$ and $T \rightarrow 0$ — m_1 is in free fall, as it should be. Note also that $T < m_1g$ and $T > m_2g$: the string tension always lies between the two weights, which is why the heavier side wins.

Derivation 2 — Contact force between two blocks (Newton's third law in action).

Setup: blocks m_1 (rear) and m_2 (front) in contact on a frictionless floor; a horizontal force F pushes m_1 .

Whole system: $F = (m_1 + m_2)a \Rightarrow a = \frac{F}{m_1 + m_2}$.

Isolate the front block m_2 . The only horizontal force on it is the contact push N_c from m_1 :

$$N_c = m_2a = \frac{m_2F}{m_1 + m_2}.$$

Isolate the rear block m_1 as a check. Forces are F forward and the third-law reaction N_c backward:

$$F - N_c = m_1a \Rightarrow F - \frac{m_2F}{m_1 + m_2} = \frac{m_1F}{m_1 + m_2} = m_1a. \checkmark$$

Key consequence: push the *light* block so it drives the heavy one and the contact force is large ($N_c = m_2F/(m_1 + m_2)$ with m_2 heavy); reverse it and the contact force is small. The force “felt” at the interface is always the share needed to accelerate everything ahead of that interface — never the full F .

Section 4: Worked Examples

Example 1 — [CBSE Board Level] In an Atwood machine, masses 5 kg and 3 kg hang over a light frictionless pulley. Find the acceleration of the system and the tension in the string. Take $g = 10 \text{ m/s}^2$.

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} = \frac{(5 - 3)(10)}{5 + 3} = \frac{20}{8} = 2.5 \text{ m/s}^2.$$

$$T = \frac{2m_1m_2g}{m_1 + m_2} = \frac{2(5)(3)(10)}{8} = \frac{300}{8} = 37.5 \text{ N}.$$

Answer: $a = 2.5 \text{ m/s}^2$, $T = 37.5 \text{ N}$.

Check: T lies between the two weights (30 N and 50 N) — good. If you ever get T outside that range, you have a sign error.

Example 2 — [NEET Speed Trap] Three blocks of masses 1 kg, 2 kg and 3 kg are connected by light strings and pulled along a frictionless floor by a force $F = 12 \text{ N}$ applied to the 1 kg block. Find the tension in the string between the 2 kg and 3 kg blocks.

The trap: grinding through three coupled equations. Don't.

Shortcut — “tension drags the mass behind it.” First the common acceleration:

$$a = \frac{F}{1 + 2 + 3} = \frac{12}{6} = 2 \text{ m/s}^2.$$

The string between the 2 and 3 kg blocks only has to accelerate the 3 kg block (everything behind that string):

$$T = 3 \times a = 3 \times 2 = 6 \text{ N}.$$

Answer: 6 N.

Speed tip: the tension at any cut equals (mass on the far side of the cut) $\times a$. The last string pulls the last block; the first string pulls everything except the front block. This turns a system of equations into one multiplication.

Example 3 — [JEE Main Level] A block of mass $m_1 = 4 \text{ kg}$ rests on a frictionless incline of 30° . A light string from it runs up the slope, over a pulley at the top, and down to a freely hanging block $m_2 = 3 \text{ kg}$. Find the acceleration of the system and the tension. Take $g = 10 \text{ m/s}^2$.

Step 1 — Which way does it move? Hanging weight $m_2g = 30 \text{ N}$ versus the down-slope pull on m_1 , $m_1g \sin 30^\circ = 4(10)(0.5) = 20 \text{ N}$. Since $30 > 20$, m_2 descends and m_1 slides up the incline.

Step 2 — Equations of motion.

$$\text{Hanging block: } m_2g - T = m_2a \Rightarrow 30 - T = 3a.$$

$$\text{Incline block: } T - m_1g \sin 30^\circ = m_1a \Rightarrow T - 20 = 4a.$$

Step 3 — Add to eliminate T :

$$30 - 20 = 7a \Rightarrow a = \frac{10}{7} \approx 1.43 \text{ m/s}^2.$$

$$T = 20 + 4a = 20 + \frac{40}{7} = \frac{180}{7} \approx 25.7 \text{ N}.$$

Answer: $a \approx 1.43 \text{ m/s}^2$, $T \approx 25.7 \text{ N}$.

Example 4 — [JEE Advanced Level] An Atwood machine with masses $m_1 = 3 \text{ kg}$ and $m_2 = 1 \text{ kg}$ hangs from the ceiling of a lift. The lift accelerates *upward* at $a_0 = 2 \text{ m/s}^2$. Find (a) the acceleration of the masses relative to the lift and (b) the string tension. Take $g = 10 \text{ m/s}^2$.

Step 1 — Work in the lift's frame. It is non-inertial, accelerating up at a_0 , so each mass feels a pseudo-force ma_0 downward. The effect is exactly as if gravity were stronger:

$$g_{\text{eff}} = g + a_0 = 10 + 2 = 12 \text{ m/s}^2.$$

Step 2 — Apply the Atwood results with $g \rightarrow g_{\text{eff}}$:

$$a_{\text{rel}} = \frac{(m_1 - m_2)g_{\text{eff}}}{m_1 + m_2} = \frac{(2)(12)}{4} = 6 \text{ m/s}^2.$$

$$T = \frac{2m_1m_2g_{\text{eff}}}{m_1 + m_2} = \frac{2(3)(1)(12)}{4} = 18 \text{ N}.$$

Answer: relative acceleration = 6 m/s²; tension = 18 N.

Deep insight: an accelerating lift simply rescales gravity to $g_{\text{eff}} = g \pm a_0$ for everything riding inside it. The same trick handles a pendulum, a spring scale, or a connected system in any uniformly accelerating frame — replace g and reuse the inertial-frame formula. A lift in free fall ($a_0 = -g$) gives $g_{\text{eff}} = 0$: the Atwood masses float, $T = 0$, and $a_{\text{rel}} = 0$.

Section 5: Practice Exercises

- [CBSE] An Atwood machine has masses 6 kg and 4 kg. Find the acceleration and the tension. ($g = 10 \text{ m/s}^2$)
- [NEET] Two blocks 2 kg and 4 kg in contact on a frictionless floor are pushed by $F = 18 \text{ N}$ applied to the 2 kg block. Find the contact force between them.
- [JEE Main] A 4 kg block on a frictionless table is connected over a pulley at the edge to a hanging 2 kg block. Find the acceleration and tension. ($g = 10 \text{ m/s}^2$)
- [JEE Main] A 2 kg block on a frictionless incline of 37° is connected over a pulley to a hanging 3 kg block. Find the system's acceleration. ($g = 10 \text{ m/s}^2$, $\sin 37^\circ = 0.6$)
- [JEE Advanced] An Atwood machine (5 kg and 3 kg) hangs in a lift accelerating *downward* at 2 m/s^2 . Find the acceleration of the masses relative to the lift and the tension. ($g = 10 \text{ m/s}^2$)

Final Answers:

- $a = \frac{(6-4)(10)}{10} = 2 \text{ m/s}^2$; $T = \frac{2(6)(4)(10)}{10} = 48 \text{ N}$.
- $a = F/6 = 3 \text{ m/s}^2$; contact force = $4a = 12 \text{ N}$.
- $a = \frac{m_2g}{m_1 + m_2} = \frac{2(10)}{6} = 3.33 \text{ m/s}^2$; $T = \frac{m_1m_2g}{m_1 + m_2} = \frac{4(2)(10)}{6} = 13.3 \text{ N}$.
- driving = $3(10) = 30 \text{ N}$; opposing = $2(10)(0.6) = 12 \text{ N}$; $a = \frac{30-12}{5} = 3.6 \text{ m/s}^2$.
- $g_{\text{eff}} = g - a_0 = 8$; $a_{\text{rel}} = \frac{(2)(8)}{8} = 2 \text{ m/s}^2$; $T = \frac{2(5)(3)(8)}{8} = 30 \text{ N}$.

Section 6: Crack the MCQ

Q1. In an Atwood machine with two equal masses m , when released the system:

- accelerates at g .
- accelerates at $g/2$.
- remains at rest or moves at constant velocity, with $T = mg$.
- has zero tension.

Correct: (c). Equal masses give $a = 0$ and $T = mg$ — the string simply balances the two equal weights. The others assume a net driving force that does not exist.

Q2. A light inextensible string connects two blocks over an ideal pulley. The magnitudes of their accelerations are related by:

- (a) $a_1 = a_2$ (equal). (b) $a_1 = 2a_2$.
 (b) $a_1 m_1 = a_2 m_2$. (d) unrelated.

Correct: (a). An inextensible string over a single fixed pulley forces equal-magnitude accelerations — this is the constraint. (b) is the movable-pulley relation; (c) and (d) are unfounded.

Q3. Two blocks m and $2m$ are in contact on a frictionless floor. The contact force is *larger* when the applied force pushes:

- (a) the block of mass m (so it drives $2m$).
 (b) the block of mass $2m$ (so it drives m).
 (c) the same either way. (d) cannot be determined.

Correct: (a). The contact force equals (mass ahead of the interface) $\times a$. Pushing m makes the heavy $2m$ the “mass ahead,” so the interface force is $\frac{2}{3}F$; pushing $2m$ makes it only $\frac{1}{3}F$. (c) is the symmetry trap.

Q4. An Atwood machine is taken into a lift in free fall. The tension in the string becomes:

- (a) unchanged. (b) doubled. (c) zero. (d) infinite.

Correct: (c). In free fall $g_{\text{eff}} = 0$, so $T = 2m_1 m_2 g_{\text{eff}} / (m_1 + m_2) = 0$ — the masses float and the string goes slack.

Section 7: Common Student Pitfalls & Pro-Tips

- **Forgetting the constraint.** Two bodies, two equations, but *three* unknowns (a_1, a_2, T) unless you add the constraint. For a single fixed pulley it is simply $a_1 = a_2$; without it the system is unsolvable.
- **Inconsistent positive directions.** Choose “the direction this body actually moves” as positive for each body, then the signs in $\sum F = ma$ take care of themselves. Mixing conventions flips a sign and ruins the answer.
- **Using the full applied force as the contact/tension force.** The internal force at any junction only accelerates the mass beyond that junction — never the whole load.
- **Letting tension differ across an ideal pulley.** It does not, for a massless string and frictionless pulley. Tension differs across a pulley only if the pulley has mass.
- **Mishandling the movable-pulley factor of two.** A load on a movable pulley moves at half the free-end speed and the supporting tension is halved — but it is shared by two segments. Count the string segments to get the relation right.

Pro-Tip — “add the equations.” For any two connected bodies, write $\sum F = ma$ for each with consistent signs and simply *add* them: the tension cancels and the acceleration falls out in one line. Then back-substitute for T . This single move solves almost every two-body problem in the chapter.

Section 8: Cheat Sheet Box

Configuration	Key Relations
Blocks in contact (F on m_1)	$a = \frac{F}{m_1 + m_2}, \quad N_c = \frac{m_2 F}{m_1 + m_2}$
Atwood ($m_1 > m_2$)	$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$
Table + hanging mass (smooth)	$a = \frac{m_2 g}{m_1 + m_2}, \quad T = \frac{m_1 m_2 g}{m_1 + m_2}$
Incline + hanging (smooth)	$a = \frac{(m_2 - m_1 \sin \theta)g}{m_1 + m_2}$
Constraint (fixed pulley)	$a_1 = a_2$
Constraint (movable pulley)	$a_{\text{free end}} = 2 a_{\text{load}}$
Accelerating frame	replace g by $g_{\text{eff}} = g \pm a_0$

Core takeaways:

- FBD each body, pick the direction of motion as positive, *add* the equations to cancel tension.
- An inextensible string over a fixed pulley forces equal-magnitude accelerations.
- Internal force at a junction = (mass beyond the junction) $\times a$.
- Inside an accelerating frame, use $g_{\text{eff}} = g \pm a_0$ and reuse the standard formulas.

Memory aid: “*Cut, draw, add — tension falls out.*” For Atwood: “*difference over sum times g.*”

Subtopic 8: Vertical Circular Motion

Exam Relevance

The vertical-circle case that Part I's banking subtopic did not cover, and a perennial high-yield topic. JEE Main asks 1–2 questions a year on critical speed, tension at top/bottom, or a body leaving a curved surface. NEET favours the conceptual side — why a bucket of water doesn't spill, the minimum speed at the top, the string-vs-rod distinction. JEE Advanced builds it into “leaving the track” problems on spheres and combined energy-plus-circular-motion setups. CBSE Boards sets the critical-velocity derivation or a tension numerical for 3 marks.

Section 1: Concept Intuition

Subtopic 5 handled *horizontal* circles — a car on a banked road, a leaning cyclist, the well of death — where gravity points along the axis and the speed stays constant. A **vertical** circle is different in one decisive way: gravity now lies in the plane of the motion, so it speeds the body up on the way down and slows it on the way up. The speed is no longer constant; it is largest at the bottom and smallest at the top. That single fact reshapes the whole problem.

Picture a stone tied to a string, whirled in a vertical circle, or a motorcyclist on a loop-the-loop, or a bucket of water swung over your head. At every point the net inward (toward the centre) force must supply the centripetal requirement mv^2/r . But the cast of inward forces changes around the loop. At the **bottom**, tension pulls up (inward) while gravity pulls down (outward), so $T - mg = mv^2/r$ — the string is working hardest here. At the **top**, both tension and gravity point downward, which is now *inward*, so they cooperate: $T + mg = mv^2/r$. The string can therefore afford to pull less at the top — and in the limiting case it pulls nothing at all.

That limit is the heart of the topic. At the very top, if the speed is just barely enough, the string tension drops to zero and **gravity alone** provides the centripetal force:

$$mg = \frac{mv_{\text{top}}^2}{r} \Rightarrow v_{\text{top},\text{min}} = \sqrt{gr}.$$

Go any slower than $\sqrt{gr}$ at the top and gravity would supply *more* inward force than the circle needs — the body falls inward, the string goes slack, and it leaves the circular path (becoming a projectile). This is exactly why a bucket of water, swung fast enough, keeps the water in even when it is upside down: at the top the water needs a downward (inward) force at least equal to mv^2/r , and gravity happily provides it; the bucket bottom never has to “hold the water up.”

Think about it this way... At the top of the loop, gravity is not the enemy — it is the helper. It points toward the centre, so it does part (or, at minimum speed, all) of the centripetal job. The danger is going *too slow*, not too fast: if $v_{\text{top}} < \sqrt{gr}$, the body cannot stay on the circle because even “free” gravity is more inward force than a slow circle can use, and the string buckles. Counterintuitively, in vertical circular motion (as in the well of death), faster is safer.

The top and bottom are linked by **energy conservation**. The body climbs a height $2r$ from bottom to top, so

$$v_{\text{bottom}}^2 = v_{\text{top}}^2 + 4gr.$$

Combining this with the minimum top speed $v_{\text{top}}^2 = gr$ gives the famous **minimum speed at the bottom** to just complete the loop:

$$v_{\text{bottom},\text{min}}^2 = gr + 4gr = 5gr \Rightarrow v_{\text{bottom},\text{min}} = \sqrt{5gr}.$$

String versus rod — a crucial distinction. The $\sqrt{gr}$ and $\sqrt{5gr}$ results assume the connector can only *pull* (a string, or the inside of a circular track). If instead the body is fixed to a rigid **rod** (or rides inside a tube), the connector can also *push* outward, so it can support the body at the top even at zero speed. Then the minimum condition at the top is merely $v_{\text{top}} \geq 0$, and the minimum bottom speed drops to

$$v_{\text{bottom,min}} = \sqrt{4gr} = 2\sqrt{gr}.$$

Spotting whether the problem says “string” or “rod” is worth a full mark.

Limiting Conditions & Assumptions:

- The track or string is taken as ideal and the motion is in a single vertical plane; air resistance is neglected so energy is conserved.
- “Just completes the circle” for a string/track means $T = 0$ (or $N = 0$) at the top; for a rod it means $v_{\text{top}} = 0$.
- Below the critical speed, a string body does not slow to a stop on the circle — it leaves the path in the *upper* half (between the horizontal and the top) and follows projectile motion.
- The body is treated as a point mass; we ignore its own size and rotation.

Section 2: Key Formulas & Definitions

Centripetal requirement (everywhere on the circle): net inward force = $\frac{mv^2}{r}$.

Tension at the lowest point:

$$T_{\text{bottom}} = \frac{mv_{\text{bottom}}^2}{r} + mg.$$

Tension at the highest point:

$$T_{\text{top}} = \frac{mv_{\text{top}}^2}{r} - mg.$$

Tension at a general angle θ measured from the lowest point:

$$T = \frac{mv^2}{r} + mg \cos \theta, \quad v^2 = v_{\text{bottom}}^2 - 2gr(1 - \cos \theta).$$

Energy link (bottom to top, height $2r$):

$$v_{\text{bottom}}^2 = v_{\text{top}}^2 + 4gr.$$

Critical (minimum) speeds — string or inside a track (can only pull/push inward):

$$v_{\text{top,min}} = \sqrt{gr}, \quad v_{\text{bottom,min}} = \sqrt{5gr}, \quad v_{\text{mid,min}} = \sqrt{3gr}.$$

Critical (minimum) speeds — rigid rod or tube (can push outward too):

$$v_{\text{top,min}} = 0, \quad v_{\text{bottom,min}} = 2\sqrt{gr}.$$

Tension difference (always, for any speed, with energy conserved):

$$T_{\text{bottom}} - T_{\text{top}} = 6mg.$$

Section 3: CBSE Board Derivations

Derivation 1 — Critical speeds for a body on a string (vertical circle).

Setup: a body of mass m on a string of length r moves in a vertical circle. We seek the minimum speed to “just complete” the loop.

At the top, both tension T and weight mg act downward, i.e. toward the centre:

$$T + mg = \frac{mv_{\text{top}}^2}{r}.$$

A string cannot push, so the smallest it can do is $T = 0$. Setting $T = 0$ gives the minimum top speed:

$$mg = \frac{mv_{\text{top}}^2}{r} \Rightarrow v_{\text{top},\text{min}} = \sqrt{gr}.$$

Energy conservation from bottom to top (the body rises by $2r$, no friction):

$$\frac{1}{2}mv_{\text{bottom}}^2 = \frac{1}{2}mv_{\text{top}}^2 + mg(2r) \Rightarrow v_{\text{bottom}}^2 = v_{\text{top}}^2 + 4gr.$$

Insert the minimum top speed $v_{\text{top}}^2 = gr$:

$$v_{\text{bottom},\text{min}}^2 = gr + 4gr = 5gr \Rightarrow \boxed{v_{\text{bottom},\text{min}} = \sqrt{5gr}}.$$

Tension difference (a clean bonus result). Subtract the top relation from the bottom relation, using $v_{\text{bottom}}^2 - v_{\text{top}}^2 = 4gr$:

$$T_{\text{bottom}} - T_{\text{top}} = \frac{m(v_{\text{bottom}}^2 - v_{\text{top}}^2)}{r} + 2mg = \frac{m(4gr)}{r} + 2mg = 6mg.$$

Remarkably this holds at *any* speed (as long as the body stays on the circle), not just the minimum — a favourite one-line exam result.

Derivation 2 — The rigid-rod case, and why it differs from the string.

Setup: the body is now attached to a light *rigid rod* of length r instead of a string. A rod can both pull and push, so at the top it can provide an upward (outward) force to support the body. The top equation becomes

$$mg - N = \frac{mv_{\text{top}}^2}{r},$$

where N is the rod’s outward push (it replaces tension and can be positive). The body can therefore move arbitrarily slowly at the top; the limiting condition is simply $v_{\text{top}} \geq 0$. Setting $v_{\text{top}} = 0$ and using energy conservation:

$$v_{\text{bottom},\text{min}}^2 = v_{\text{top}}^2 + 4gr = 0 + 4gr \Rightarrow v_{\text{bottom},\text{min}} = 2\sqrt{gr}.$$

Physical reading: a string *must* maintain $\sqrt{gr}$ at the top to stay taut; a rod needs nothing, because it can hold the body up. This is why a “string” loop demands $\sqrt{5gr}$ at the bottom while a “rod” loop demands only $2\sqrt{gr}$ — the connector’s ability to push is the entire difference.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A stone of mass 0.5 kg tied to a string of length 1 m is whirled in a vertical circle at the minimum speed needed to complete the loop. Find (a) the speed at the top, (b) the speed at the bottom, and (c) the tension at the bottom. Take $g = 10 \text{ m/s}^2$.

$$(a) \quad v_{\text{top}} = \sqrt{gr} = \sqrt{10 \times 1} = \sqrt{10} \approx 3.16 \text{ m/s.}$$

$$(b) \quad v_{\text{bottom}} = \sqrt{5gr} = \sqrt{50} \approx 7.07 \text{ m/s.}$$

$$(c) \quad T_{\text{bottom}} = \frac{mv_{\text{bottom}}^2}{r} + mg = \frac{0.5 \times 50}{1} + 0.5 \times 10 = 25 + 5 = 30 \text{ N.}$$

Answer: $v_{\text{top}} \approx 3.16 \text{ m/s}$, $v_{\text{bottom}} \approx 7.07 \text{ m/s}$, $T_{\text{bottom}} = 30 \text{ N}$.

Check: at minimum speed $T_{\text{top}} = 0$, so $T_{\text{bottom}} - T_{\text{top}} = 30 - 0 = 30 \text{ N} = 6mg = 6(0.5)(10)$. The $6mg$ rule confirms the arithmetic.

Example 2 — [NEET Speed Trap] A body moves in a vertical circle on a string, just completing the loop. What is the ratio of its speed at the bottom to its speed at the top?

The trap: reaching for energy equations with numbers. The ratio is universal.

At the minimum, $v_{\text{top}} = \sqrt{gr}$ and $v_{\text{bottom}} = \sqrt{5gr}$, so

$$\frac{v_{\text{bottom}}}{v_{\text{top}}} = \frac{\sqrt{5gr}}{\sqrt{gr}} = \sqrt{5} \approx 2.24.$$

Answer: $\sqrt{5} : 1$.

Speed tip: memorize the trio for a just-completing string loop — $v_{\text{top}} : v_{\text{mid}} : v_{\text{bottom}} = \sqrt{gr} : \sqrt{3gr} : \sqrt{5gr} = 1 : \sqrt{3} : \sqrt{5}$. Any ratio question collapses to picking two of these.

Example 3 — [JEE Main Level] A small bead is attached to the end of a light *rigid rod* of length $r = 2.5 \text{ m}$ and swung in a vertical circle. (a) Find the minimum speed at the lowest point to complete the circle. (b) Compare with the value required if a string of the same length were used instead. Take $g = 10 \text{ m/s}^2$.

(a) *Rod case* (top speed may be zero):

$$v_{\text{bottom, min}} = 2\sqrt{gr} = 2\sqrt{10 \times 2.5} = 2\sqrt{25} = 2 \times 5 = 10 \text{ m/s.}$$

(b) *String case* (top speed must be at least $\sqrt{gr}$):

$$v_{\text{bottom, min}} = \sqrt{5gr} = \sqrt{5 \times 10 \times 2.5} = \sqrt{125} \approx 11.18 \text{ m/s.}$$

Answer: rod needs 10 m/s; the string needs about 11.18 m/s — the rod is the easier loop because it can push the bead through the top.

Example 4 — [JEE Advanced Level] A small block is released from rest at the top of a smooth (frictionless) sphere of radius R and slides down the outside. At what angle θ (measured from the top, at the centre) does it leave the surface, and what is its speed there?

Step 1 — Forces toward the centre. On the outside of the sphere the normal reaction N points outward (away from centre); the inward component of gravity at angle θ from the top is $mg \cos \theta$. The net inward force supplies the centripetal force:

$$mg \cos \theta - N = \frac{mv^2}{R}.$$

Step 2 — Condition for leaving the surface. The block leaves when the surface can no longer push, i.e. $N = 0$:

$$mg \cos \theta = \frac{mv^2}{R} \Rightarrow v^2 = gR \cos \theta.$$

Step 3 — *Energy conservation* from the top (height drop = $R - R \cos \theta = R(1 - \cos \theta)$):

$$\frac{1}{2}mv^2 = mgR(1 - \cos \theta) \Rightarrow v^2 = 2gR(1 - \cos \theta).$$

Step 4 — *Combine the two expressions for v^2 :*

$$gR \cos \theta = 2gR(1 - \cos \theta) \Rightarrow \cos \theta = 2 - 2 \cos \theta \Rightarrow 3 \cos \theta = 2 \Rightarrow \cos \theta = \frac{2}{3}.$$

So $\theta = \cos^{-1}(2/3) \approx 48.2^\circ$, and the speed at that instant is

$$v = \sqrt{gR \cos \theta} = \sqrt{\frac{2}{3}gR}.$$

Answer: the block leaves the sphere at $\theta = \cos^{-1}(2/3) \approx 48.2^\circ$ from the top, with speed $\sqrt{2gR/3}$.

Deep insight: the result is independent of mass and of R — every block leaves a smooth sphere at the same universal angle $\cos^{-1}(2/3)$. This “leaving the track” logic ($N = 0$ plus energy conservation) is the single most important Advanced technique for curved-surface problems, and it is the outside-the-circle cousin of the $T = 0$ condition at the top of a vertical loop.

Section 5: Practice Exercises

- [CBSE]** A ball of mass 0.2 kg on a string of length 0.5 m moves in a vertical circle. Find the minimum speed at the top to keep the string taut. ($g = 10 \text{ m/s}^2$)
- [NEET]** For a body just completing a vertical circle on a string, the ratio of the speed at the lowest point to that at the highest point is?
- [JEE Main]** A stone of mass 1 kg is whirled in a vertical circle of radius 1 m at a constant¹ top speed of 5 m/s. Find the tension at the top. ($g = 10 \text{ m/s}^2$)
- [JEE Main]** A bead on a rigid rod of length 0.4 m is moved in a vertical circle. Find the minimum speed at the lowest point to complete the circle. ($g = 10 \text{ m/s}^2$)
- [JEE Advanced]** A particle slides from rest down the inside of a smooth vertical circular track of radius r , starting from the level of the centre (i.e. the 3 o'clock position). Find its speed at the lowest point and the normal reaction there, in terms of m, g, r .

Final Answers:

- $v_{\text{top}} = \sqrt{gr} = \sqrt{10 \times 0.5} = \sqrt{5} \approx 2.24 \text{ m/s}.$
- $\sqrt{5} : 1.$
- $T_{\text{top}} = \frac{mv^2}{r} - mg = \frac{1 \times 25}{1} - 10 = 15 \text{ N}.$
- $v_{\text{bottom, min}} = 2\sqrt{gr} = 2\sqrt{10 \times 0.4} = 2 \times 2 = 4 \text{ m/s}.$
- Drop = r , so $v^2 = 2gr \Rightarrow v = \sqrt{2gr}$; at the bottom $N - mg = mv^2/r = 2mg \Rightarrow N = 3mg.$

Section 6: Crack the MCQ

Q1. The minimum speed of a body at the highest point of a vertical circle on a string of radius r is:

- (a) zero. (b) $\sqrt{gr}$. (c) $\sqrt{5gr}$. (d) $\sqrt{2gr}$.

¹Take the speed at the top as given; energy change around the loop is supplied so that the top speed equals this value.

Correct: (b). At the top, $T = 0$ gives $mg = mv^2/r$, so $v = \sqrt{gr}$. (a) is the *rod* answer; (c) is the speed at the *bottom*; (d) is unrelated.

Q2. A bucket of water is swung in a vertical circle fast enough that no water spills at the top. At the top, the water stays in the bucket because:

- (a) the bucket pushes the water up.
- (b) gravity provides the centripetal force, directed inward (downward) toward the centre.
- (c) there is no gravity at the top of the swing.
- (d) centrifugal force holds it in.

Correct: (b). At the top the required inward force is downward, and gravity supplies it (plus the bucket's normal push if speed exceeds the minimum). (c) is false; (d) misuses a pseudo-force in the ground frame.

Q3. For a body just completing a vertical circle on a string, the tension at the lowest point exceeds that at the highest point by:

- (a) $2mg$. (b) $4mg$. (c) $6mg$. (d) mg .

Correct: (c). $T_{\text{bottom}} - T_{\text{top}} = 6mg$ for any speed on the circle, from the $4gr$ speed difference plus $2mg$ from the weight terms.

Q4. A body needs less speed to complete a vertical circle when it is attached to a rigid rod rather than a string because:

- (a) the rod is heavier.
- (b) the rod can push outward and support the body at the top, so v_{top} can be zero.
- (c) the rod reduces gravity.
- (d) the rod increases the radius.

Correct: (b). A rod supplies an outward push, removing the $T \geq 0$ restriction; the minimum bottom speed falls from $\sqrt{5gr}$ (string) to $2\sqrt{gr}$ (rod).

Section 7: Common Student Pitfalls & Pro-Tips

- **Assuming constant speed.** In a vertical circle the speed changes continuously — fastest at the bottom, slowest at the top. Always link the two with energy conservation ($v_{\text{bottom}}^2 = v_{\text{top}}^2 + 4gr$).
- **Using $\sqrt{gr}$ for a rod (or 0 for a string).** The minimum top speed is $\sqrt{gr}$ for a string/track but 0 for a rod/tube. Read the connector carefully; it changes the bottom requirement from $\sqrt{5gr}$ to $2\sqrt{gr}$.
- **Treating tension as the centripetal force by itself.** The centripetal force is the *net* inward force — tension combined with the inward component of gravity. At the top they add; at the bottom they oppose.
- **Forgetting the body can leave the circle.** Below the critical speed a string body does not simply slow down on the loop — it goes slack in the upper half and becomes a projectile. “It stops at the top” is wrong for a string.
- **Mis-placing the leaving angle on a sphere.** A body sliding outside a smooth sphere leaves at $\cos\theta = 2/3$ (from the top), found by $N = 0$ plus energy conservation — not at the top and not at 90° .

Pro-Tip — **two equations crack every vertical-circle problem.** Write Newton’s law toward the centre at the point of interest (combine the connector force with gravity’s radial component), and write energy conservation between two points. Almost every question — critical speed, tension anywhere, leaving the track — is just these two equations solved together.

Section 8: Cheat Sheet Box

Quantity / Situation	Key Relation
Tension at bottom	$T_{\text{bottom}} = \frac{mv_{\text{bottom}}^2}{r} + mg$
Tension at top	$T_{\text{top}} = \frac{mv_{\text{top}}^2}{r} - mg$
General angle θ (from bottom)	$T = \frac{mv^2}{r} + mg \cos \theta$
Energy link (bottom→top)	$v_{\text{bottom}}^2 = v_{\text{top}}^2 + 4gr$
Min speeds: string/track	$v_{\text{top}} = \sqrt{gr}, v_{\text{bottom}} = \sqrt{5gr}$
Min speeds: rigid rod/tube	$v_{\text{top}} = 0, v_{\text{bottom}} = 2\sqrt{gr}$
Tension difference	$T_{\text{bottom}} - T_{\text{top}} = 6mg$
Leaving a smooth sphere (from top)	$\cos \theta = 2/3, v = \sqrt{2gR/3}$

Core takeaways:

- Speed varies around a vertical circle; couple top and bottom with energy conservation.
- Centripetal force is the *net* inward force — combine connector force and gravity’s radial part.
- String/track: $v_{\text{top,min}} = \sqrt{gr}, v_{\text{bottom,min}} = \sqrt{5gr}$. Rod/tube: 0 and $2\sqrt{gr}$.
- $T_{\text{bottom}} - T_{\text{top}} = 6mg$ always; a body leaves a smooth sphere at $\cos^{-1}(2/3)$.

Memory aid: “*Top needs root-gr, bottom needs root-5gr; the difference in tension is six mg.*” For curved surfaces: “*it lets go when the push hits zero.*”

Chapter 4: Laws of Motion — Round 2 Addendum

The chapter and its Part II taught you the five pillars — impulse, momentum conservation, equilibrium, friction, circular motion — and closed the syllabus gaps with FBDs, connected bodies, and the vertical circle. Four examinable patterns are still standing outside the door, and each one is a marriage of two things the chapter already teaches separately. Addendum A extends the single-pulley constraint of Subtopic 7 to the multi-pulley rigs that JEE Advanced actually draws. Addendum B puts Subtopic 4's friction on top of Subtopic 7's connected bodies: stacked blocks, where friction itself decides who moves with whom. Addendum C finishes Subtopic 4's incline story — the chapter gives you the sliding acceleration, but not the *range* of forces that hold a block still, nor the optimally angled hold and haul. Addendum D returns to Subtopic 1's variable mass and does chains and rockets exactly, deriving results the chapter quotes. Nothing here is new theory; every method is Newton's second law plus a constraint, applied one layer deeper.

Numbering starts at Addendum A — the chapter has no prior addenda. Every derivation is shown with units carried through every line.

Addendum A — Constraint Relations Beyond the Single Pulley

Subtopic 7 derives the Atwood machine and states the movable-pulley rule ("the load moves at half the free-end speed") without proving it. Real JEE Advanced rigs chain two or three pulleys together, where the factor of two compounds and guessing fails. The reliable technique is mechanical and always works: name a coordinate for every moving piece, write the string's total length, and differentiate twice. The constraint falls out as algebra, not intuition.

Method

1. **Assign coordinates.** Give every moving body a position variable, measured from a fixed datum along that body's own line of motion. Fix a positive direction for each now.
2. **Write the string length.** Express the total length L of the string as a sum of its straight segments. Fixed spans and the arcs wrapped on pulleys are constants.
3. **Differentiate twice.** Since the string is inextensible,

$$\frac{dL}{dt} = 0 \quad \Rightarrow \quad \frac{d^2L}{dt^2} = 0,$$

giving a linear relation among the accelerations. Constants die in the differentiation — that is why step 2 can afford to ignore them.

4. **Write $\sum \vec{F} = m\vec{a}$ for every body**, each with its own positive direction from step 1, plus $T_1 = 2T_2$ type bookkeeping at every massless pulley (the forces on a massless pulley must sum to zero).

5. **Power audit (check).** An ideal string stores no energy, so the total power it delivers at all its contact points is zero:

$$\sum_j \vec{T}_j \cdot \vec{v}_j = 0 \quad (\text{all attachment points } j).$$

Thirty seconds of multiplication catches almost every sign error.

The n -segment rule, proved. If a string runs over a moving pulley with n segments stretching between the pulley and fixed points, and the free portions shorten by s , each of the n segments lengthens by the pulley's displacement x . Length conservation gives $ns = x$... in rates,

$$v_{\text{load}} = \frac{v_{\text{string}}}{n},$$

and since each segment carries the same tension T , the string supports the load with a force nT . This is the general factor-of-two (for $n = 2$) that Subtopic 7 asserted.

Example A.1 — [JEE Main Level]

A block A of mass 4 kg rests on a smooth table. A string from A runs horizontally to a fixed pulley at the table edge, descends, passes under a movable pulley carrying block B of mass 4 kg, and its other end is tied to the ceiling. Find the acceleration of each block and the tension. Take $g = 10 \text{ m/s}^2$.

Let x be the displacement of A toward the edge and y the descent of B . When B descends by y , both vertical segments flanking the movable pulley grow by y , while the horizontal segment shrinks by x . Length conservation:

$$-x + 2y = 0 \quad \Rightarrow \quad x = 2y \quad \Rightarrow \quad a_A = 2a_B.$$

This is the constraint — note it took one line of counting, not a picture of the machine.

Equations of motion, each body positive along its actual tendency (T is uniform; ideal string and pulleys):

Block A (horizontal, pulled by tension):

$$T = m_A a_A = 4(2a_B) = 8a_B.$$

Block B (vertical, pulled up by two strands, so $2T$ upward):

$$m_B g - 2T = m_B a_B.$$

Substitute $T = 8a_B$ into the second equation:

$$40 - 16a_B = 4a_B \quad \Rightarrow \quad 20a_B = 40 \quad \Rightarrow \quad a_B = 2 \text{ m/s}^2.$$

Then $a_A = 4 \text{ m/s}^2$ and $T = 8(2) = 16 \text{ N}$.

Check. Block A : the only horizontal force is $T = 16 \text{ N}$, and $m_A a_A = 4 \times 4 = 16 \text{ N}$. Block B : net force $40 - 32 = 8 \text{ N} = m_B a_B = 4 \times 2$. The general result for this rig is $a_B = m_B g / (4m_A + m_B)$; setting $\mu = 0$ thinking aside, notice T lies between 0 and $m_B g$, as it must.

Example A.2 — [JEE Advanced Level]

A fixed pulley hangs from the ceiling. Over it passes string 1, holding a mass $m_1 = 2$ kg on one side and a light movable pulley P on the other. Over P passes string 2 holding $m_2 = 1$ kg and $m_3 = 3$ kg. Find every acceleration and both tensions. Take $g = 10 \text{ m/s}^2$.

Define: a = downward acceleration of m_1 ; a' = downward acceleration of P ; b = downward acceleration of m_2 relative to P . String 1 constrains its two ends oppositely, so

$$a' = -a.$$

Then the ground-frame accelerations from string 2 are: m_2 descends at $a' + b$, and m_3 descends at $a' - b$.

Massless pulley P demands (string 1 pulls up with T_1 ; two strands of string 2 pull down with T_2 each):

$$T_1 = 2T_2.$$

Four equations of motion (downward positive throughout):

$$\begin{aligned} m_1 g - T_1 &= m_1 a \Rightarrow 20 - T_1 = 2a \\ m_2 g - T_2 &= m_2(a' + b) \Rightarrow 10 - T_2 = -a + b \\ m_3 g - T_2 &= m_3(a' - b) \Rightarrow 30 - T_2 = 3(-a - b) \end{aligned}$$

From the first two equations, $T_2 = 10 - a$. Feed this into the m_2 equation:

$$10 - (10 - a) = -a + b \Rightarrow b = 2a.$$

Feed both into the m_3 equation:

$$30 - 10 + a = -3a - 6a \Rightarrow 20 + a = -9a \Rightarrow a = -2 \text{ m/s}^2.$$

Read the minus sign. We guessed m_1 descends; the algebra says it *ascends* at 2 m/s^2 . Completing the set:

$$T_2 = 10 - (-2) = 12 \text{ N}, \quad T_1 = 24 \text{ N}, \quad b = -4 \text{ m/s}^2.$$

So P descends at 2 m/s^2 , m_2 accelerates *upward* at $a' + b = 2 - 4$, i.e. 2 m/s^2 , and m_3 accelerates downward at $a' - b = 2 + 4 = 6 \text{ m/s}^2$. Physically: the heavy m_3 wins the inner contest, hauling m_2 up over P , and the imbalance drags P down — which lifts m_1 .

Check (power audit). Shortly after release from rest, take $v_1 = 2t$ up, $v_P = 2t$ down, $v_2 = 2t$ up, $v_3 = 6t$ down. String 1 delivers $T_1 v_1 + T_1(-v_P) = 24(2t) - 24(2t) = 0$. String 2 delivers $T_2 v_2 - T_2 v_3 - 2T_2 v_P = 12(2t) - 12(6t) + 24(2t) = 0$ (watch signs: the two strands pull P down

while P moves down). Both audits close to zero. Also T_2 sits strictly between $m_2g = 10\text{ N}$ and $m_3g = 30\text{ N}$, as an inner-string tension must.

Practice — Addendum A

1. A load of mass 20 kg hangs from a single movable pulley. The rope's free end is pulled up at a steady 1.5 m/s . Find the load's speed and the rope tension.
2. A mass of 4 kg hangs from a movable pulley whose rope runs up over two fixed pulleys; both free ends are pulled down with a force of 30 N each. Find the mass's upward acceleration. Take $g = 10\text{ m/s}^2$.
3. In a rig like Example A.1's but with the free end pulled, the free end of a rope threading one fixed and one movable pulley is pulled downward with acceleration 4 m/s^2 . Find the acceleration of the load on the movable pulley.
4. A double Atwood machine has $m_1 = 4\text{ kg}$ on the main string and $m_2 = m_3 = 2\text{ kg}$ on the movable pulley's string. Show the system can hang at rest, and find both tensions. Take $g = 10\text{ m/s}^2$.

Answers: 1. Load rises at 0.75 m/s (half the free-end speed); steady speed means equilibrium, so $2T = mg = 200\text{ N}$, giving $T = 100\text{ N}$. 2. Two strands pull up: $2(30) - 4(10) = 4a$, so $a = 20/4 = 5\text{ m/s}^2$ upward. 3. By the n -segment rule with $n = 2$, the load moves at half the free end: 2 m/s^2 (direction: upward, opposite the pull). 4. The inner pair is balanced ($m_2 = m_3$), so $b = 0$ and P 's string behaves as a single mass $m_2 + m_3 = 4\text{ kg}$ balancing $m_1 = 4\text{ kg}$: everything at rest, $T_2 = m_2g = 20\text{ N}$ and $T_1 = m_1g = 40\text{ N}$.

Addendum B — Stacked Blocks: When Does Friction Say “Together”?

Subtopic 4 treats one block on one surface, and Subtopic 7 treats blocks joined by strings. But the most common Advanced configuration is two blocks *in contact*, where the coupling is friction — a self-adjusting force with a ceiling. The question “do they move together or does the top one slip?” is a hypothesis test, and the chapter's tools run it in three lines. CBSE rarely asks it; NEET asks the conceptual core; JEE Main and Advanced ask it nearly every year.

Method

1. **Draw two FBDs.** Mark the interface friction explicitly as a Third-Law pair: f on one block, f oppositely on the other. Only the normal force between them is $N = m_{\text{top}}g$ (plus any applied vertical component).
2. **Hypothesize “together.”** Treat the stack as one body of mass $M+m$ to get the common acceleration a . Then ask: what friction does the *driven* block need? Compute the required f_{req} from the driven block's own equation.
3. **Test the ceiling.** If

$$f_{\text{req}} \leq \mu_s N,$$

the hypothesis stands — static friction obliges, and the actual friction is f_{req} . If it fails, the blocks slip.

4. **If slipping**, replace interface friction by kinetic $\mu_k N$ on each block (opposite directions), solve the two equations independently, and report both accelerations and their difference — the relative acceleration that governs how long the top block takes to slide off.
5. **Critical applied force.** Slipping begins exactly when $f_{\text{req}} = \mu_s N$; solve that equality for F .

Example B.1 — [JEE Main Level]

A 2 kg block sits on a 4 kg block on a frictionless floor. Between the blocks $\mu_s = 0.5$ and $\mu_k = 0.4$. A horizontal force F is applied to the lower block. Take $g = 10 \text{ m/s}^2$. (a) For $F = 12 \text{ N}$, find the acceleration and the friction force. (b) Find the largest F for which no slipping occurs. (c) For $F = 42 \text{ N}$, find both accelerations.

Interface ceiling: $f_{\text{max}} = \mu_s mg = 0.5 \times 2 \times 10 = 10 \text{ N}$.

(a) Together-hypothesis: $a = F/(M + m) = 12/6 = 2 \text{ m/s}^2$. The top block is driven only by friction, so $f_{\text{req}} = ma = 2 \times 2 = 4 \text{ N}$. Since $4 \text{ N} \leq 10 \text{ N}$, they move together and the friction actually acting is 4 N .

(b) Slip begins when $f_{\text{req}} = 10 \text{ N}$, i.e. $a = f_{\text{max}}/m = 10/2 = 5 \text{ m/s}^2$. Note this is exactly $\mu_s g$ — the mass of the top block cancels. Then

$$F_{\text{max}} = (M + m) \mu_s g = 6 \times 5 = 30 \text{ N}.$$

(c) $F = 42 \text{ N} > 30 \text{ N}$, so they slip. Kinetic friction $= \mu_k mg = 0.4 \times 2 \times 10 = 8 \text{ N}$, backward on the top block, forward on the bottom:

Top block (driven by friction alone):

$$a_{\text{top}} = \frac{\mu_k mg}{m} = \mu_k g = 4 \text{ m/s}^2.$$

Bottom block:

$$a_{\text{bot}} = \frac{F - \mu_k mg}{M} = \frac{42 - 8}{4} = 8.5 \text{ m/s}^2.$$

Relative acceleration: $8.5 - 4 = 4.5 \text{ m/s}^2$.

Check. Adding the two slipped equations reproduces the whole-system one: $F = Ma_{\text{bot}} + ma_{\text{top}} = 34 + 8 = 42 \text{ N}$. And $a_{\text{top}} = 4 \text{ m/s}^2$ is bounded by $\mu_s g = 5$ — friction can never accelerate the top block faster than $\mu_s g$, whatever F does.

Example B.2 — [JEE Advanced Level]

The same pair ($m = 2 \text{ kg}$ on $M = 4 \text{ kg}$, $\mu_s = 0.5$, $\mu_k = 0.4$ between blocks) now rests on a floor with friction coefficient 0.1 . The force is applied to the top block. Map the motion completely as F grows, and analyse $F = 18 \text{ N}$. Take $g = 10 \text{ m/s}^2$.

Floor friction ceiling (whole stack pressing): $0.1 \times 6 \times 10 = 6$ N. Interface ceilings as before: static 10 N, kinetic 8 N.

Regime 1: $F \leq 6$ N. The push transmits through the interface to the floor, whose static friction can absorb up to 6 N. Nothing moves.

Regime 2: $6 \text{ N} < F \leq 12 \text{ N}$ — **together**. Whole-system equation:

$$F - 6 = 6a \quad \Rightarrow \quad a = \frac{F - 6}{6} \text{ m/s}^2.$$

Bottom block's own equation (friction f forward from the top block, floor friction 6 N backward):

$$f - 6 = 4a \quad \Rightarrow \quad f = 6 + \frac{4(F - 6)}{6} = \frac{2F + 6}{3} \text{ N}.$$

Slip when $f > 10$: $(2F + 6)/3 > 10$ gives $F > 12$ N. So the together-regime ends at $F = 12$ N, where $f = 10 \text{ N} = \mu_s mg$ exactly.

Regime 3: $F > 12$ N — **slipping**. Interface friction drops to 8 N. For $F = 18$ N:

Top block:

$$a_{\text{top}} = \frac{F - \mu_k mg}{m} = \frac{18 - 8}{2} = 5 \text{ m/s}^2.$$

Bottom block (pushed forward by 8 N, held back by 6 N of floor friction):

$$a_{\text{bot}} = \frac{\mu_k mg - 0.1(M + m)g}{M} = \frac{8 - 6}{4} = 0.5 \text{ m/s}^2.$$

Relative acceleration = $5 - 0.5 = 4.5 \text{ m/s}^2$: the top block walks forward off the bottom one.

Why it is Advanced-level. Driving from the top inverts the logic of Example B.1: the interface friction must simultaneously accelerate the bottom block *and* defeat floor friction, so the no-slip window collapses from $F \leq 30$ N (bottom-driven, smooth floor) to $F \leq 12$ N (top-driven, rough floor). The mark-winning habit is computing all three ceilings — interface static, interface kinetic, floor — before choosing a regime, and checking each regime boundary by substituting it back until two ceilings coincide.

Practice — Addendum B

1. A 1 kg block sits on a 3 kg block on a frictionless floor; $\mu_s = 0.4$ between them. F is applied to the lower block. Find the largest F with no slipping, and the friction force at that instant. ($g = 10 \text{ m/s}^2$.)
2. The same pair, with $\mu_k = 0.3$ between blocks and $F = 30$ N on the lower block. Find both accelerations and the relative acceleration. ($g = 10 \text{ m/s}^2$.)
3. A 2 kg block on a 4 kg block, smooth floor, $\mu_s = 0.5$, force applied to the *top* block. Find the largest F for which they move together. ($g = 10 \text{ m/s}^2$.)
4. In one sentence, explain why driving the stack from the top gives a smaller no-slip force range than driving it from the bottom.

Answers: 1. $F_{\max} = (1+3)(0.4)(10) = 16 \text{ N}$; friction then $= \mu_s mg = 4 \text{ N}$. 2. $a_{\text{top}} = \mu_k g = 3 \text{ m/s}^2$; $a_{\text{bot}} = (30 - 0.3 \times 1 \times 10)/3 = 9 \text{ m/s}^2$; relative 6 m/s^2 . 3. Together requires $Ma \leq \mu_s mg$, so $a_{\max} = 10/4 = 2.5 \text{ m/s}^2$ and $F_{\max} = 6 \times 2.5 = 15 \text{ N}$ — half the bottom-driven value of 30 N from Example B.1. 4. The interface ceiling $\mu_s mg$ is fixed, but when the top block drives, that same friction must accelerate the entire lower mass (and beat floor friction if any), whereas bottom-driving uses it only to accelerate the light top mass.

Addendum C — The Complete Force Budget on a Rough Incline

Subtopic 4 gives you the block that slides down a rough incline, and the minimum horizontal-pull trick on a flat floor. What boards and JEE both mine further is the *full budget*: the range of forces that hold a block stationary, and the least possible force to hold it or haul it up when you may choose the direction. All of it is the impending-motion idea — set static friction to its ceiling, oriented against the impending slip — applied twice, once per slip direction.

Method

Throughout: incline angle θ , coefficients μ_s (and μ_k if it slides), mass m . Define the angle of friction λ by $\tan \lambda = \mu_s$, so that

$$\sin \lambda = \frac{\mu_s}{\sqrt{1 + \mu_s^2}}, \quad \cos \lambda = \frac{1}{\sqrt{1 + \mu_s^2}}.$$

Step 0 — will it stay alone? If $\tan \theta \leq \mu_s$, i.e. $\theta \leq \theta_r$, the block holds itself and the force range opens at 0.

Range of a force parallel to the incline. Impending slip *down* (friction up, at its ceiling):

$$F_{\text{low}} = mg(\sin \theta - \mu_s \cos \theta).$$

Impending slip *up* (friction down):

$$F_{\text{high}} = mg(\sin \theta + \mu_s \cos \theta).$$

Every F in $[F_{\text{low}}, F_{\text{high}}]$ holds the block; outside it, the block accelerates the way the imbalance points.

Least force to hold it (any angle allowed). Apply F at angle ϕ above the up-slope direction (positive ϕ tilts the force *off* the surface, reducing N ; negative ϕ presses *into* it). Perpendicular balance: $N = mg \cos \theta - F \sin \phi$. With slip impending downward,

$$F \cos \phi + \mu_s (mg \cos \theta - F \sin \phi) = mg \sin \theta,$$

$$F = \frac{mg(\sin \theta - \mu_s \cos \theta)}{\cos \phi - \mu_s \sin \phi}.$$

Maximising the denominator: $\cos \phi - \mu_s \sin \phi$ peaks at $\sqrt{1 + \mu_s^2}$ when $\tan \phi = -\mu_s$, i.e. the force is aimed up-slope but pressed *into* the plane at angle λ . Hence, for $\theta > \theta_r$,

$$F_{\text{hold,min}} = \frac{mg(\sin \theta - \mu_s \cos \theta)}{\sqrt{1 + \mu_s^2}} = mg \sin(\theta - \lambda).$$

Pressing into the plane buys normal force, and normal force buys friction — exactly what a *holding* force wants.

Least force to drag it up. Same geometry, slip impending upward, friction reversed:

$$F = \frac{mg(\sin \theta + \mu_s \cos \theta)}{\cos \phi + \mu_s \sin \phi},$$

maximised at $\tan \phi = +\mu_s$ (tilted *away* from the plane, stealing normal force to starve the friction opposing you):

$$F_{\text{drag,min}} = mg \sin(\theta + \lambda).$$

The unified picture worth memorising. Horizontal floor ($\theta = 0$): drag needs $mg \sin \lambda$ — Subtopic 4's result. Hold on an incline: $mg \sin(\theta - \lambda)$. Haul up: $mg \sin(\theta + \lambda)$. One formula family; the angle λ always enters, only the sign of the combination changes with which way friction is helping you.

Example C.1 — [JEE Main Level]

A 5 kg block rests on a rough incline of 37° with $\mu_s = 0.5$. Find the range of forces applied parallel to the incline that keep it in equilibrium. Take $g = 10 \text{ m/s}^2$, $\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$.

First check self-holding: $\tan 37^\circ = 0.75 > 0.5$, so the block slides unaided — a nonzero lower bound exists. Weight components along and perpendicular:

$$mg \sin \theta = 5 \times 10 \times 0.6 = 30 \text{ N}, \quad \mu_s mg \cos \theta = 0.5 \times 5 \times 10 \times 0.8 = 20 \text{ N}.$$

Range:

$$F_{\text{low}} = 30 - 20 = 10 \text{ N}, \quad F_{\text{high}} = 30 + 20 = 50 \text{ N}.$$

Answer: any parallel force from 10 N to 50 N holds it. Below 10 N it slides down; above 50 N it accelerates up.

Check. At $F = 30 \text{ N}$ (midpoint) the needed friction is 0 — the force alone balances the weight component, and static friction idles at zero, comfortably legal since $0 \leq \mu_s N$.

Example C.2 — [JEE Advanced Level]

On the same style of incline, take $\theta = 53^\circ$, $\mu_s = 0.75$, and a block of mass 2 kg ($g = 10 \text{ m/s}^2$, so $mg = 20 \text{ N}$; $\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$). (a) Find the least force that prevents the block sliding down, and the angle at which it must be applied. (b) Find the least force that drags it up. (c) Compare both with purely parallel forces.

Here $\tan \lambda = 0.75$ gives the 3-4-5 triangle: $\sin \lambda = 0.6$, $\cos \lambda = 0.8$, $\sqrt{1 + \mu_s^2} = 1.25$. Also $\tan \theta = 4/3 > 0.75$, so the block cannot hold itself.

(a) Least holding force.

$$F_{\text{hold,min}} = \frac{mg(\sin \theta - \mu_s \cos \theta)}{\sqrt{1 + \mu_s^2}} = \frac{20(0.8 - 0.75 \times 0.6)}{1.25} = \frac{20 \times 0.35}{1.25} = 5.6 \text{ N}.$$

Applied up-slope, tilted 36.87° below the incline surface — pressing the block into the plane.

(b) Least hauling force. Using the sine addition with $\theta + \lambda = 53^\circ + 36.87^\circ = 89.87^\circ$, or directly:

$$F_{\text{drag,min}} = \frac{mg(\sin \theta + \mu_s \cos \theta)}{\sqrt{1 + \mu_s^2}} = \frac{20(0.8 + 0.45)}{1.25} = \frac{25}{1.25} = 20 \text{ N}.$$

(Elegance check via $mg \sin(\theta + \lambda)$: $\sin \theta \cos \lambda + \cos \theta \sin \lambda = 0.8 \times 0.8 + 0.6 \times 0.6 = 1.00$ exactly — the sum lands on 90° and the answer is exactly one weight.) Applied up-slope, tilted 36.87° above the surface.

(c) Comparison with parallel forces. Purely parallel: holding needs $mg(\sin \theta - \mu_s \cos \theta) = 7 \text{ N}$ versus 5.6 N optimal; hauling needs $mg(\sin \theta + \mu_s \cos \theta) = 25 \text{ N}$ versus 20 N optimal. The optimally angled hold saves 20%, the angled haul saves 20%.

Check. For (a), verify the perpendicular bookkeeping at the optimum: $N = mg \cos \theta - F \sin \phi$ with $\phi = -36.87^\circ$ (pressed in), so $N = 20 \times 0.6 + 5.6 \times 0.6 = 12 + 3.36 = 15.36 \text{ N}$, and available friction $= 0.75 \times 15.36 = 11.52 \text{ N}$. Along-slope balance: $F \cos 36.87^\circ + 11.52 = 5.6 \times 0.8 + 11.52 = 4.48 + 11.52 = 16 \text{ N}$, which equals $mg \sin \theta = 20 \times 0.8 = 16 \text{ N}$.

Practice — Addendum C

1. A 4 kg block rests on a 30° incline with $\mu_s = 0.6$. Does it hold itself? Find the full range of parallel forces maintaining equilibrium. ($g = 10 \text{ m/s}^2$, $\cos 30^\circ = 0.866$.)
2. A 2 kg block rests on a 45° incline with $\mu_s = 0.75$. Find the least force preventing it from sliding, and the angle of application. ($g = 10 \text{ m/s}^2$.)
3. For the block of Question 2, find the least force that drags it up the incline.
4. Explain physically why the optimal *holding* force presses into the plane while the optimal *hauling* force lifts away from it.

Answers: 1. Yes: $\tan 30^\circ \approx 0.58 < 0.6$, so $F_{\text{low}} = 0$. Upper limit $mg(\sin 30^\circ + \mu_s \cos 30^\circ) = 40(0.5 + 0.5196) \approx 40.8 \text{ N}$; range $[0, 40.8] \text{ N}$. 2. $F = mg \sin(45^\circ - 36.87^\circ) = 20 \sin 8.13^\circ = 20 \times 0.1414 \approx 2.83 \text{ N}$, applied up-slope tilted 36.87° into the plane. 3. $F = 20(\sin 45^\circ + 0.75 \cos 45^\circ)/1.25 = 20 \times 1.2374/1.25 \approx 19.8 \text{ N}$, applied up-slope tilted 36.87° off the plane.

4. When holding, friction is your ally and must be near its ceiling, so maximise N by pressing in; when hauling, friction is your adversary, so minimise N by lifting, shrinking μN at the cost of wasting some force vertically — the optimum balances the trade at exactly λ either way.

Addendum D — Chains and Rockets: Variable Mass Done Exactly

Subtopic 1 showed the conveyor-belt flux calculation and quoted two chain results — the overhang fraction $y/L = \mu/(1 + \mu)$ — without deriving either, and Subtopic 2 derived the rocket equation in gravity-free space only. Boards ask the overhang derivation as a 2-mark question; JEE Advanced turns chains into scale-reading and snap-speed problems that punish anyone relying on $F = ma$ reflexes. Here is the machinery, used four ways.

Method

1. **Fix the system boundary.** Decide precisely which matter is “inside” at time t and which bit crosses during the next dt .
2. **Apply the parent law** to that system:

$$\vec{F}_{\text{ext}} = \frac{d\vec{p}_{\text{system}}}{dt},$$

never the constant-mass shortcut. The crossing matter contributes a momentum-flux term of the form $v(dm/dt)$.

3. **Chains:** the moving mass is the whole chain λL , the driving force is the weight of the hanging part λxg , where λ is the mass per unit length (kg/m).
4. **Speeds from energy** when the geometry is a clean height change — but only after confirming no dissipation mechanism (snapping links, slipping) hides in the process.
5. **Carry units ruthlessly:** λxg is (kg/m)(m)(m/s²) = N, every time.

Example D.1 — [CBSE Board Level] — chain off a smooth table

A uniform chain of length $L = 1$ m and mass 1 kg lies on a smooth horizontal table with a length $x_0 = 25$ cm hanging over the edge. Find (a) the acceleration at the instant of release and (b) the speed when the chain finally leaves the table. Take $g = 10$ m/s².

Linear density: $\lambda = 1 \text{ kg}/1 \text{ m} = 1 \text{ kg/m}$.

(a) Instantaneous acceleration. The driving force is the hanging part’s weight:

$$F = \lambda xg = 1 \times 0.25 \times 10 = 2.5 \text{ N}.$$

The whole chain of mass $\lambda L = 1$ kg accelerates:

$$a = \frac{\lambda xg}{\lambda L} = \frac{gx}{L} = \frac{10 \times 0.25}{1} = 2.5 \text{ m/s}^2.$$

Note the structure: $a = gx/L$ grows as x grows, so this is not constant acceleration — the SUVAT kit is invalid across the motion, though valid instantaneously.

(b) Speed at exit, by energy. Measure heights below the tabletop. Initially the hanging length x_0 has its centre of mass at $x_0/2$; finally the whole chain hangs with its centre at $L/2$. Potential energy lost:

$$\Delta U = \lambda g \left[\frac{L^2}{2} - \frac{x_0^2}{2} \right] = 1 \times 10 \times \frac{1 - 0.0625}{2} = 4.6875 \text{ J.}$$

All of it becomes kinetic energy of the whole chain:

$$\frac{1}{2} \lambda L v^2 = \Delta U \Rightarrow v = \sqrt{\frac{g(L^2 - x_0^2)}{L}} = \sqrt{\frac{10 \times 0.9375}{1}} \approx 3.06 \text{ m/s.}$$

Check. The acceleration grew from 2.5 m/s^2 at release toward $g = 10 \text{ m/s}^2$ as the chain emptied off the table; a distance of roughly a metre covered while accelerating between 2.5 and 10 m/s^2 landing at $v \approx 3.06 \text{ m/s}$ is consistent with $v^2 = 2\bar{a}s$ for $\bar{a} \approx 4.7 \text{ m/s}^2$ — plausible.

Example D.2 — [JEE Advanced Level] — the scale that reads triple

A uniform chain of linear density $\lambda = 1.0 \text{ kg/m}$ is released from rest just touching a weighing scale's pan, and piles up on it. Find the scale reading when a length $x = 0.5 \text{ m}$ has landed. Take $g = 10 \text{ m/s}^2$.

Each link falls freely from rest through the height of the accumulated pile. When length x has landed, a falling link strikes the pan at speed

$$v = \sqrt{2gx}.$$

During a time dt , a mass $dm = \lambda v dt$ arrives moving at v and stops dead. Its momentum changes by $dp = dm \cdot v = \lambda v^2 dt$, so the pan must exert an extra upward force

$$F_{\text{flux}} = \frac{dp}{dt} = \lambda v^2 = \lambda(2gx).$$

By the Third Law this arrives as extra downward force on the pan. The reading is dead weight plus flux:

$$R = \underbrace{\lambda x g}_{\text{pile's weight}} + \underbrace{\lambda v^2}_{\text{stopping the arrivals}} = \lambda x g + 2\lambda x g = 3\lambda x g.$$

Numerically:

$$R = 3 \times 1.0 \times 10 \times 0.5 = 15 \text{ N,}$$

of which only 5 N is the pile's actual weight — the impulsive term is twice the static term.

Check. Units of λv^2 : $(\text{kg/m})(\text{m}^2/\text{s}^2) = \text{kg} \cdot \text{m}/\text{s}^2 = \text{N}$. And this is the conveyor-belt factor of two wearing a disguise (Subtopic 1, Example 4): the kinetic energy each arriving link carried is entirely dissipated in the stop, so the momentum bookkeeping costs more than the weight ledger suggests.

Example D.3 — [JEE Main Level] — rocket with gravity switched on

A rocket of initial mass $m_0 = 1000$ kg burns fuel at $b = 10$ kg/s with exhaust speed $v_{\text{rel}} = 1500$ m/s relative to the rocket. Find its speed 10 s after launch (a) ignoring gravity, (b) including uniform gravity. Take $g = 10$ m/s².

Thrust first: $F_{\text{thrust}} = v_{\text{rel}} b = 1500 \times 10 = 15000$ N, comfortably above the launch weight $m_0 g = 10000$ N — it lifts.

Mass at time t : $m = m_0 - bt = 1000 - 10t$; at $t = 10$ s, $m = 900$ kg.

(a) Subtopic 2's equation:

$$v = v_{\text{rel}} \ln\left(\frac{m_0}{m}\right) = 1500 \times \ln\left(\frac{10}{9}\right) = 1500 \times 0.10536 \approx 158 \text{ m/s.}$$

(b) Newton's parent law on the rocket alone:

$$m \frac{dv}{dt} = v_{\text{rel}} b - mg \quad \Rightarrow \quad \frac{dv}{dt} = \frac{v_{\text{rel}} b}{m_0 - bt} - g.$$

Integrating from 0 to t (the first term integrates to the logarithm, the second to gt):

$$v = v_{\text{rel}} \ln\left(\frac{m_0}{m}\right) - gt = 158 - 10 \times 10 = 58 \text{ m/s.}$$

Gravity has eaten more than half the gain — early in flight, when m is still large, the thrust-to-weight ratio barely exceeds 1.

Check. Dimensionally, $v_{\text{rel}} \ln(\cdot)$ is m/s and gt is m/s. Limit check: if $b \rightarrow 0$ the log term dies and $v \rightarrow -gt$, pure free fall — the rocket becomes a falling brick, correctly.

Practice — Addendum D

1. A chain of length $L = 2$ m lies on a smooth table with $x_0 = 0.5$ m hanging. Find its acceleration at release and its speed as it leaves the table. ($g = 10$ m/s².)
2. A chain lies on a rough table with $\mu_s = 0.4$. Derive and evaluate the maximum fraction of its length that can hang over the edge without sliding.
3. A chain with $\lambda = 0.5$ kg/m falls onto a scale pan from rest at pan level. Find the reading when $x = 0.3$ m has piled up. ($g = 10$ m/s².)
4. A rocket has $m_0 = 2000$ kg, burns at $b = 20$ kg/s with $v_{\text{rel}} = 1500$ m/s. Find its speed at $t = 30$ s including gravity. Is the thrust-to-weight ratio at launch greater than 1? ($g = 10$ m/s².)

Answers: 1. $a_0 = gx_0/L = 10 \times 0.25 = 2.5$ m/s²; $v = \sqrt{g(L^2 - x_0^2)}/L = \sqrt{10 \times 3.75}/2 = \sqrt{18.75} \approx 4.33$ m/s. 2. At the verge, $\lambda x g = \mu_s \lambda (L - x) g$, so $x/(L - x) = \mu_s$ and $x/L = \mu_s/(1 + \mu_s) =$

$0.4/1.4 = 2/7 \approx 28.6\%$. 3. $R = 3\lambda xg = 3 \times 0.5 \times 10 \times 0.3 = 4.5 \text{ N}$ (dead weight only 1.5 N of it). 4. $m = 1400 \text{ kg}$, $\ln(2000/1400) = \ln(10/7) \approx 0.3567$, so $v = 1500 \times 0.3567 - 300 = 535 - 300 = 235 \text{ m/s}$; launch thrust $1500 \times 20 = 30000 \text{ N}$ versus weight 20000 N — ratio $1.5 > 1$, it lifts.

Closing note. Every addendum above ran the same two engines the chapter built: Newton's second law written honestly (parent form whenever mass crosses a boundary, per-body otherwise), and one constraint or ceiling that closes the system of equations — string length, static-friction limit, or impending-slip condition. When a new problem looks unfamiliar, name the engine before touching algebra; in this chapter, it is always one of the two.

Work, Energy and Power

Chapter 5 • Master Content

Complete Study Reference

CBSE • NEET • JEE Main • JEE Advanced

Foundations: The Scalar Product

Work & the Work–Energy Theorem • Mechanical Energy

Conservative & Non-Conservative Forces

Various Forms of Energy & Conservation of Energy

Power • Collisions

Application: Motion in a Vertical Circle

Drona

AI-Powered Exam Preparation

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Foundations: The Scalar (Dot) Product

Exam Relevance: This is the mathematical key that unlocks the whole chapter — work, power, and flux are all dot products. CBSE Boards expect the *definition* and component formula (1–2 marks) and use it inside derivations. JEE Main slips in **1 question** on finding an angle between vectors or a value that makes two vectors perpendicular. NEET tests the perpendicularity ($\vec{A} \cdot \vec{B} = 0$) idea as a quick MCQ. JEE Advanced embeds it silently wherever work, projection, or angles between vectors appear.

Section 1: Concept Intuition

Before we can talk about work, we need one tool from vector algebra — and it's the *reason* work behaves the way it does. When you push a loaded trolley along a corridor, only the part of your push that points *along* the corridor actually moves it; any sideways or upward part is wasted. Physics needed a single operation that takes two vectors and automatically extracts “how much of one lies along the other, times their sizes.” That operation is the **scalar product**, also called the **dot product**.

The name is the clue: you feed it two *vectors* and it spits out a single *scalar* (a plain number, no direction). It does this by multiplying the two magnitudes and then weighting the result by $\cos \theta$, the cosine of the angle between them.

Think about it this way... imagine the sun low on the horizon casting your shadow on the ground. The dot product of two vectors is like asking, “if vector $\vec{B}$ were the ground and the sun shone straight down onto $\vec{A}$, how long is $\vec{A}$'s shadow on $\vec{B}$ — and then scale that by how long $\vec{B}$ is.” When the two vectors point the same way ($\theta = 0$), the shadow is full-length and the product is maximum (AB). When they're at right angles ($\theta = 90^\circ$), the shadow shrinks to nothing and the product is **zero** — which is exactly why a force perpendicular to motion does no work. When they point opposite ways ($\theta = 180^\circ$), the product goes maximally *negative* ($-AB$) — the basis of negative work.

That single behaviour — positive when aligned, zero when perpendicular, negative when opposed — is why the dot product is the natural language for work ($W = \vec{F} \cdot \vec{S}$), for power ($P = \vec{F} \cdot \vec{v}$), and for many other physical quantities you'll meet later.

Limiting conditions & assumptions:

1. The dot product is only defined between **two vectors**; the result is a **scalar** and has no direction.
2. It can be **positive, zero, or negative** depending entirely on the angle θ between the vectors.
3. $\vec{A} \cdot \vec{B} = 0$ does **not** mean one vector is zero — it usually means they are **perpendicular**.

Section 2: Key Formulas & Definitions

Definition (geometric form):

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

where $A = |\vec{A}|$, $B = |\vec{B}|$, and θ is the angle between them ($0 \leq \theta \leq 180^\circ$).

Component (algebraic) form: for $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ and $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$,

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Angle between two vectors (rearranging the definition):

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{A_x B_x + A_y B_y + A_z B_z}{\sqrt{A_x^2 + A_y^2 + A_z^2} \sqrt{B_x^2 + B_y^2 + B_z^2}}$$

Projection (component) of $\vec{A}$ along $\vec{B}$:

$$A \cos \theta = \frac{\vec{A} \cdot \vec{B}}{B}$$

Self dot product: $\vec{A} \cdot \vec{A} = A^2$ (since $\theta = 0$).

Unit-vector results (the engine behind the component form):

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1, \quad \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

The dot product is a pure number, so it carries **no dimensional formula of its own** — its dimensions are simply those of the two quantities multiplied (e.g., $\vec{F} \cdot \vec{S}$ has the dimensions of energy, $[ML^2T^{-2}]$).

Section 3: Key Properties & Procedures

Properties (each worth knowing for proofs):

1. **Commutative:** $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ (order doesn't matter).
2. **Distributive over addition:** $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$.
3. **Scalar multiplication:** $(m\vec{A}) \cdot \vec{B} = m(\vec{A} \cdot \vec{B})$.
4. **Perpendicularity test:** if neither vector is zero, $\vec{A} \cdot \vec{B} = 0 \iff \vec{A} \perp \vec{B}$.
5. **Sign of the product** tells you the angle: positive $\rightarrow$ acute, zero $\rightarrow$ right angle, negative $\rightarrow$ obtuse.

Procedure — finding the angle between two vectors:

1. Compute the dot product using components: $A_x B_x + A_y B_y + A_z B_z$.
2. Compute each magnitude: $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$, similarly for B .
3. Divide: $\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}$.
4. Take the inverse cosine. (A quick sanity check: a *negative* dot product must give an *obtuse* angle.)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Two vectors have magnitudes $A = 5$ and $B = 4$ units, with an angle of 60° between them. Find their scalar product.

Given: $A = 5$, $B = 4$, $\theta = 60^\circ$.

$$\vec{A} \cdot \vec{B} = AB \cos \theta = (5)(4) \cos 60^\circ = (5)(4)(0.5) = \boxed{10 \text{ units}}$$

Comment: Clean substitution into the geometric form. The result is a single number — a scalar — as expected.

Example 2 — [NEET Speed Trap]

Find the angle between $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 4\hat{i} - 3\hat{j}$. (a) 0° (b) 45° (c) 90° (d) 180°

The fast reasoning: Don't compute magnitudes — just check the dot product first:

$$\vec{A} \cdot \vec{B} = (3)(4) + (4)(-3) = 12 - 12 = 0.$$

A zero dot product (with neither vector zero) means the vectors are **perpendicular**.

Answer: (c) 90° . *Speed tip:* always test $\vec{A} \cdot \vec{B} = 0$ first. If it vanishes, the answer is 90° instantly — no need for the magnitude-and-arccos route.

Example 3 — [JEE Main Level]

Find the value of λ for which $\vec{A} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ is perpendicular to $\vec{B} = \hat{i} - 2\hat{j} + 3\hat{k}$.

Perpendicular means the dot product is zero:

$$\vec{A} \cdot \vec{B} = (2)(1) + (\lambda)(-2) + (1)(3) = 2 - 2\lambda + 3 = 5 - 2\lambda$$

Set equal to zero:

$$5 - 2\lambda = 0 \Rightarrow \boxed{\lambda = \frac{5}{2}}$$

Why this is JEE Main: it inverts the perpendicularity condition — instead of testing two known vectors, you *solve* for the unknown that forces perpendicularity. A very common one-step setup.

Example 4 — [JEE Advanced Level]

A vector $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ is given. (a) Find the angle it makes with the x-axis. (b) Find the projection of $\vec{A}$ onto the vector $\vec{B} = 2\hat{i} + \hat{j} + 2\hat{k}$.

(a) **Angle with the x-axis** (direction $\hat{i}$):

$$\cos \alpha = \frac{\vec{A} \cdot \hat{i}}{|\vec{A}| |\hat{i}|} = \frac{1}{\sqrt{1^2 + 1^2 + 1^2} \cdot 1} = \frac{1}{\sqrt{3}}$$
$$\alpha = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \approx 54.7^\circ$$

(By symmetry, $\vec{A}$ makes the same angle with each of the three axes — a tidy check.)

(b) **Projection of $\vec{A}$ onto $\vec{B}$:**

$$\vec{A} \cdot \vec{B} = (1)(2) + (1)(1) + (1)(2) = 5, \quad |\vec{B}| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$$

$$\text{Projection} = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} = \frac{5}{3} \approx \boxed{1.67 \text{ units}}$$

Why this is JEE Advanced: it requires fluent use of *both* the angle formula and the projection formula in three dimensions, plus recognising the symmetry shortcut in part (a).

Section 5: Practice Exercises

- [CBSE] Two vectors of magnitudes 6 and 5 units are inclined at 30° . Find their scalar product. (Take $\cos 30^\circ = 0.866$.)
- [NEET] State the angle between $\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = -3\hat{i} + 2\hat{j}$.
- [JEE Main] Find λ so that $\vec{A} = \lambda\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{B} = 2\hat{i} - \hat{j} + 4\hat{k}$ are perpendicular.
- [JEE Main] Find the angle between $\vec{A} = \hat{i} + \hat{j}$ and $\vec{B} = \hat{j} + \hat{k}$.
- [JEE Advanced] Find the projection of $\vec{A} = 3\hat{i} - 2\hat{j} + \hat{k}$ onto $\vec{B} = \hat{i} + 2\hat{j} + 2\hat{k}$.

Answers: 1. $25.98 \approx 26$ units | 2. 90° (dot product $= -6 + 6 = 0$) | 3. $\lambda = -1$ (from $2\lambda - 2 + 4 = 0$)
 | 4. 60° ($\cos \theta = \frac{1}{2}$) | 5. $\frac{3 - 4 + 2}{3} = \frac{1}{3} \approx 0.33$ units

Section 6: Crack the MCQ

Q1. If $\vec{A} \cdot \vec{B} = 0$ and neither $\vec{A}$ nor $\vec{B}$ is a zero vector, then the two vectors are: (a) parallel (b) anti-parallel (c) perpendicular (d) equal

Correct: (c) perpendicular. A zero dot product with non-zero vectors forces $\cos \theta = 0$, i.e. $\theta = 90^\circ$.
Distractor analysis: (a) parallel gives the *maximum* product AB ; (b) anti-parallel gives $-AB$; (d) is unrelated to the dot product vanishing.

Q2. the value of $\hat{i} \cdot \hat{i} + \hat{j} \cdot \hat{k}$ is: (a) 0 (b) 1 (c) 2 (d) 3

Correct: (b) 1. $\hat{i} \cdot \hat{i} = 1$ (same unit vector) and $\hat{j} \cdot \hat{k} = 0$ (perpendicular), so the sum is $1 + 0 = 1$.
Distractor analysis: (c) wrongly assumes $\hat{j} \cdot \hat{k} = 1$; (d) treats all three terms as 1; (a) treats the first term as 0. The rule: like unit vectors give 1, unlike give 0.

Q3. Two vectors $\vec{A}$ and $\vec{B}$ have $\vec{A} \cdot \vec{B} < 0$. The angle between them is: (a) acute (b) right (c) obtuse (d) zero

Correct: (c) obtuse. A negative dot product needs $\cos \theta < 0$, which happens for $90^\circ < \theta \leq 180^\circ$.
Distractor analysis: (a) acute gives a positive product; (b) right gives zero; (d) zero gives the maximum positive product. The *sign* of the dot product directly reports the angle's type.

Q4. The work done by a force $\vec{F} = 4\hat{i} + 3\hat{j}$ N in producing a displacement $\vec{S} = 3\hat{i} - 4\hat{j}$ m is: (a) 24 J (b) 0 J (c) 25 J (d) 7 J

Correct: (b) 0 J. $W = \vec{F} \cdot \vec{S} = (4)(3) + (3)(-4) = 12 - 12 = 0$. The force and displacement happen to be perpendicular. *Distractor analysis:* (a) comes from adding rather than subtracting the second term; (c) is the product of magnitudes 5×5 (only valid if parallel); (d) is a sum of components with no physical meaning here.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Treating the result as a vector.** The dot product is a *scalar* — a plain number with a sign but no direction. Writing it with an arrow on top is wrong.
2. **Adding instead of subtracting signed components.** In $A_x B_x + A_y B_y + A_z B_z$, keep every sign. A component like $(3)(-4)$ contributes -12 , not $+12$ — the single most common arithmetic slip.
3. **Confusing dot and cross products.** The dot product gives a scalar ($AB \cos \theta$); the cross product gives a vector ($AB \sin \theta$, perpendicular to both). Work uses the *dot* product.
4. **Assuming $\vec{A} \cdot \vec{B} = 0$ means a vector is zero.** It almost always means the vectors are perpendicular instead.

Pro-Tip (speed + accuracy): For any “find the angle” or “are these perpendicular?” question, compute the dot product *first*. If it’s zero, the angle is 90° and you’re done. If you do need the angle, the *sign* of the dot product instantly tells you whether to expect an acute (+), right (0), or obtuse (−) answer — a built-in sanity check before you ever reach for the calculator.

Section 8: Cheat Sheet Box

THE SCALAR (DOT) PRODUCT — RAPID REVISION

- **Geometric:** $\vec{A} \cdot \vec{B} = AB \cos \theta$ (result is a **scalar**, can be +, 0, or −)
- **Component:** $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$
- **Angle:** $\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}$ · **Projection of $\vec{A}$ on $\vec{B}$:** $\frac{\vec{A} \cdot \vec{B}}{B}$
- **Unit vectors:** $\hat{i} \cdot \hat{i} = 1, \hat{i} \cdot \hat{j} = 0$ (like = 1, unlike = 0)
- **Self:** $\vec{A} \cdot \vec{A} = A^2$ · **Perpendicular test:** $\vec{A} \cdot \vec{B} = 0 \Rightarrow \vec{A} \perp \vec{B}$
- **Properties:** commutative, distributive
- **Used for:** work $W = \vec{F} \cdot \vec{S}$, power $P = \vec{F} \cdot \vec{v}$

Memory aids: “Dot gives a scalar; cross gives a vector.” · “Like unit vectors = 1, unlike = 0.”
· “Zero dot = right angle.”

Sub-topic 1: Work and the Work–Energy Theorem

Exam Relevance: This is the gateway sub-topic of the entire chapter, and it pays off everywhere. CBSE Boards almost always carry the **derivation of the work–energy theorem** (3 marks) plus a 1–2 mark question on positive/negative/zero work. JEE Main reliably asks **1–2 questions** each year — usually a variable-force integration or a “find final speed using the theorem” numerical. NEET asks **1 conceptual question** on the sign of work or zero-work cases. JEE Advanced rarely asks it in isolation, but folds *frame dependence of work* and *work by static friction* silently into multi-concept problems where most students slip.

Section 1: Concept Intuition

Here’s a question that trips up almost every student on day one. Imagine your friend stands holding a heavy bag of cement on his head for ten full minutes, sweating, arms trembling. Did he do any *work*?

In everyday Hindi-English we’d say “*bhai itni mehnat kar raha hai*” — of course he’s working! But in physics, the answer is a flat **zero**. And once you understand *why*, you understand the whole concept.

Physics doesn’t care about effort, sweat, or tiredness. It cares about exactly two things: **is a force acting, and did the point where that force acts actually move?** Work is the bridge between a *push* and a *displacement*. No displacement, no work — no matter how exhausted you are. Your friend could tie the bag to a hook on the wall and the bag would stay up with zero effort from anyone, which tells you the holding itself transfers no energy to the bag.

So work is a very specific bookkeeping idea: it measures **how much energy a force transfers to (or steals from) an object as the object moves**.

Think about it this way... picture pushing a stalled auto-rickshaw down a road. You push forward, the auto rolls forward — your push and the motion point the *same way*, so you are *pouring* energy into the auto. That’s **positive work**, and you can feel it because the auto speeds up. Now the driver hits the brakes: friction in the brake pads points *backward* while the auto still rolls forward — those two point *opposite* ways, so friction is *draining* energy out of the auto. That’s **negative work**, and you see it because the auto slows down. Finally, think of the gravity pulling *down* on the auto while it rolls *horizontally* — straight down versus straight along the road, a perfect right angle. Gravity neither helps nor hinders the forward motion at all. That’s **zero work**.

This is the single most important picture in the chapter: **the angle between the force and the displacement decides the sign of the work**.

- Force has a piece pointing *along* the motion → **positive** work → object gains speed.
- Force has a piece pointing *against* the motion → **negative** work → object loses speed.
- Force is *perpendicular* to the motion → **zero** work → speed unchanged.

Three side-by-side panels of a block on a surface with a displacement arrow pointing right. Panel 1: force arrow at acute angle ($\theta < 90^\circ$) labelled “Positive work”. Panel 2: force arrow at obtuse angle ($\theta > 90^\circ$), e.g. friction, labelled “Negative work”. Panel 3: force arrow pointing straight up ($\theta = 90^\circ$), e.g. normal reaction or gravity for horizontal motion, labelled “Zero work”.

Once you have the *sign* intuition, the magnitude follows naturally. Only the **part of the force that lies along the direction of motion** counts. A man dragging a suitcase by a tilted handle: only the

horizontal slice of his pull moves the suitcase forward; the vertical slice just tries to lift it. So work uses the *component of force along the displacement* — which is exactly what a dot product computes.

Now here's the payoff that makes work worth defining at all. When you do positive work on an object, *where does that energy go?* It shows up as **motion** — the object speeds up. When you do negative work, the object slows down. So there's a tight, exact relationship waiting to be discovered: **the net work done on a body equals the change in its “energy of motion.”** That energy of motion is what we call **kinetic energy**, $K = \frac{1}{2}mv^2$. This is not a coincidence or a definition pulled out of thin air — it falls straight out of Newton's second law, as we'll prove in Section 3. Hold onto this mental link: *work is the currency, kinetic energy is the bank balance, and depositing or withdrawing currency changes the balance by exactly that amount.* That sentence **is** the work–energy theorem, in plain English.

Limiting conditions & assumptions you must respect:

1. **Work is defined for a force, not for an object.** “The work done” is an incomplete sentence — always ask *work done by which force?* Different forces on the same body do different works.
2. **The simple formula $W = FS \cos \theta$ assumes a *constant force* and *straight-line motion*.** The moment the force changes in size or direction, or the path curves, you must switch to integration.
3. **Work is frame-dependent.** Displacement looks different to a passenger on a train and to a person on the platform, so the *same force* does *different work* in different frames. (This is the trap JEE Advanced loves.)
4. **It's a scalar.** Work has no direction — but it carries a sign, and that sign is physics, not algebra. Don't drop it.

Section 2: Key Formulas & Definitions

Work done by a constant force

$$W = \vec{F} \cdot \vec{S} = FS \cos \theta$$

where $\vec{F}$ is the force, $\vec{S}$ the displacement, and θ the angle between them.

Work done by a variable force (general definition)

$$W = \int_{\vec{r}_i}^{\vec{r}_f} \vec{F} \cdot d\vec{r} \quad \xrightarrow{\text{1-D}} \quad W = \int_{x_i}^{x_f} F dx$$

Graphically, this is the **area under the F – x curve**.

Kinetic energy (energy of motion)

$$K = \frac{1}{2}mv^2$$

Work–Energy Theorem

$$W_{\text{net}} = \Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

The **net** work done by **all** forces equals the change in kinetic energy.

Quantity reference table

Quantity	Symbol	SI Unit	Dimensional Formula
Work / Energy	W, K	joule (J)	$[M^1 L^2 T^{-2}]$
Force	F	newton (N)	$[M^1 L^1 T^{-2}]$
Displacement	S, x	metre (m)	$[M^0 L^1 T^0]$

Variable definitions: m = mass, v_i, v_f = initial and final speeds, θ = angle between force and displacement.

Common energy unit conversions (worth memorising):

$$1 \text{ J} = 1 \text{ N} \cdot \text{m} = 1 \text{ kg} \cdot \text{m}^2 \text{s}^{-2}, \quad 1 \text{ erg} = 10^{-7} \text{ J}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}, \quad 1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

Sign rule in one line: $\cos \theta > 0 \Rightarrow W > 0$; $\cos \theta < 0 \Rightarrow W < 0$; $\theta = 90^\circ \Rightarrow W = 0$.

Section 3: CBSE Board Derivation — The Work–Energy Theorem

This is the showpiece derivation of the chapter. CBSE wants the **general** version (valid even for a *variable* force), so we'll do it properly with calculus and explain the reasoning at each step.

Setup / assumptions: Consider a single particle of mass m moving along a straight line (the x -axis) under a net force F that may vary with position. Let its speed be v_i at position x_i and v_f at position x_f .

Step 1 — Start from the definition of work by the net force.

$$W_{\text{net}} = \int_{x_i}^{x_f} F dx$$

Why: We use the integral form, not $FS \cos \theta$, precisely so the result holds even when F changes. This is what makes the theorem so powerful.

Step 2 — Replace F using Newton's second law.

$$F = ma = m \frac{dv}{dt}$$

Why: The net force is what changes the motion, so Newton's second law is the natural substitution to bring speed into the picture.

Step 3 — Convert the time derivative into a position derivative (the chain-rule trick).

$$\frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} = v \frac{dv}{dx}$$

Why: Our integral runs over position x , not time t . Since $\frac{dx}{dt} = v$, this rewrites the acceleration in a form we can integrate over dx . So $F = mv \frac{dv}{dx}$.

Step 4 — Substitute back and let dx cancel.

$$W_{\text{net}} = \int_{x_i}^{x_f} mv \frac{dv}{dx} dx = \int_{v_i}^{v_f} mv dv$$

Why: The dx in the integrand pairs with $\frac{dv}{dx}$, and the limits switch from positions to the corresponding speeds. The problem is now a clean speed integral.

Step 5 — Integrate.

$$W_{\text{net}} = m \left[\frac{v^2}{2} \right]_{v_i}^{v_f} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = K_f - K_i$$

$$W_{\text{net}} = \Delta K$$

What it actually means: kinetic energy is just “stored-up net work.” If the net work on a body is positive, it speeds up by exactly that much energy; if negative, it slows down. Crucially, the theorem talks about the **net** work — the sum over *all* forces. The work of any one individual force generally does **not** equal ΔK ; only the total does.

A second, exam-friendly procedure — work as the area under the F - x graph.

When a force is given as a graph rather than a formula (very common in JEE Main), you don’t need calculus at all. Since $W = \int F dx$, the integral *is* the area between the force curve and the x -axis. The procedure:

1. **Plot or read off** the F versus x graph over the displacement of interest.
2. **Break the region** under the curve into simple shapes — rectangles, triangles, trapeziums.
3. **Add the areas**, treating any region **below the x -axis as negative** (the force opposes motion there).
4. The signed total is the work done.

Reasoning: a thin vertical strip of width dx at position x has area $F dx$ — exactly the tiny bit of work dW done over that step. Summing all strips sums all the work, which is why the area and the work are the same number. This trick turns many “scary variable force” problems into a quick geometry exercise.

An F - x graph showing a force that rises linearly then falls, with the region under the curve shaded and divided into a triangle plus a rectangle, labelled “Area = Work done”. A small portion of the curve dips below the x -axis, shaded differently and labelled “Negative work”.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A street vendor pulls a loaded cart along level ground by a rope that makes an angle of 60° with the horizontal. He applies a steady force of 200 N and the cart moves 10 m. Find the work done by the vendor’s pull.

Given: $F = 200$ N, $S = 10$ m, $\theta = 60^\circ$.

Formula: $W = FS \cos \theta$

Substitution:

$$W = (200)(10) \cos 60^\circ = (200)(10)(0.5)$$

$$W = 1000 \text{ J}$$

Conclusion: The vendor’s pull transfers 1000 J of energy to the cart. Note we used $\cos 60^\circ$, not the full force — only the horizontal slice of the pull does work, because that’s the part aligned with the displacement.

Example 2 — [NEET Speed Trap]

A porter carries a 20 kg trunk on his head. He first walks 40 m along a level railway platform, then climbs a staircase rising 5 m vertically. Taking $g = 10 \text{ m/s}^2$, the total work done by the porter against gravity is: (a) 9000 J (b) 8000 J (c) 1000 J (d) 0 J

The trap: A panicking student adds “work for the 40 m walk” plus “work for the 5 m climb” and lands on a wrong number.

The fast reasoning: Against gravity, only **vertical** displacement matters. Gravity points straight down, so during the 40 m horizontal walk the angle is $90^\circ \rightarrow$ **zero work**. Only the climb counts.

$$W = mgh = (20)(10)(5) = 1000 \text{ J}$$

Answer: (c) 1000 J. *Speed tip:* the instant you see “horizontal distance” in a work-against-gravity question, mentally delete it. It’s there only to bait you.

Example 3 — [JEE Main Level]

A particle of mass 2 kg, initially at rest at the origin, is pushed along the x -axis by a position-dependent force $F = (3x^2 + 2x) \text{ N}$. Find its speed when it reaches $x = 4 \text{ m}$. (Combines: variable-force integration + work-energy theorem.)

Step 1 — The force varies with x , so integrate to get the work.

$$W = \int_0^4 (3x^2 + 2x) dx = \left[x^3 + x^2 \right]_0^4$$

$$W = (4^3 + 4^2) - 0 = 64 + 16 = 80 \text{ J}$$

Step 2 — Apply the work-energy theorem. It starts from rest, so $K_i = 0$:

$$W = \frac{1}{2}mv^2 - 0 \Rightarrow 80 = \frac{1}{2}(2)v^2 \Rightarrow v^2 = 80$$

$$v = \sqrt{80} = 4\sqrt{5} \approx 8.94 \text{ m/s}$$

Why this is a “JEE Main” problem: you can’t use $FS \cos \theta$ (force isn’t constant) and you can’t use kinematic equations easily (acceleration isn’t constant either). The work-energy theorem sidesteps both because it only cares about the *total work*, which integration delivers directly.

Example 4 — [JEE Advanced Level]

A block of mass 2 kg rests on the floor of a train that accelerates uniformly from rest at 3 m/s^2 . Friction is just enough to prevent slipping. For the time interval $t = 0$ to $t = 2 \text{ s}$, find the work done by friction on the block (i) in the ground frame, and (ii) in the train's frame. Verify each with the work–energy theorem.

This problem hinges on the subtlety that **work depends on the reference frame**, because *displacement* does — even though the friction force itself is the same in both frames.

Identify the friction force (frame-independent): The block moves with the train, so static friction is the *only* horizontal force on it and must supply its acceleration:

$$f = ma = (2)(3) = 6 \text{ N, directed forward.}$$

Note this is *static friction doing positive work* — already a surprise to students who memorised “friction always opposes motion.”

(i) **Ground frame.** The block's displacement in 2 s:

$$S_{\text{ground}} = \frac{1}{2}at^2 = \frac{1}{2}(3)(2)^2 = 6 \text{ m}$$

$$W_{f,\text{ground}} = f \cdot S = (6)(6) = +36 \text{ J}$$

Check (ground): final speed $v = at = (3)(2) = 6 \text{ m/s}$, so

$$\Delta K = \frac{1}{2}(2)(6^2) - 0 = 36 \text{ J.}$$

Friction is the only horizontal force, so $W_{\text{net}} = 36 \text{ J} = \Delta K$. ✓

(ii) **Train frame.** Relative to the train, the block never moves: $S_{\text{train}} = 0$.

$$W_{f,\text{train}} = f \cdot S_{\text{train}} = (6)(0) = 0 \text{ J}$$

Check (train): In this non-inertial frame the block is stationary, so $\Delta K = 0$. The net work must also be zero: friction (+6 N) and the pseudo-force (−6 N) each act over zero displacement, giving 0. ✓

The takeaway: the *same* 6 N friction force does +36 J to a platform observer and 0 J to a passenger. Both are correct — work is frame-dependent, and the work–energy theorem holds *consistently within each frame* as long as you use that frame's displacements and kinetic energies throughout.

Section 5: Practice Exercises

1. [CBSE] A force of 50 N acts on a body at 37° to the horizontal, moving it 6 m horizontally. Find the work done. (Take $\cos 37^\circ = 0.8$.)
2. [NEET] A stone tied to a string is whirled in a horizontal circle at constant speed. How much work does the string tension do on the stone in one complete revolution, and why?
3. [JEE Main] A 1 kg body moving at 4 m/s is brought to rest by a constant opposing force over a distance of 2 m. Find the magnitude of the force using the work–energy theorem.
4. [JEE Main] A variable force $F = (6x - 2) \text{ N}$ acts on a 3 kg particle moving along the x -axis from $x = 1 \text{ m}$ to $x = 3 \text{ m}$. Find the work done by this force.

5. [JEE Advanced] A block slides down a rough incline of height $H = 2$ m and base length $l = 3$ m (so it travels the full slope), starting from rest. The coefficient of kinetic friction is $\mu = 0.4$. Using the work–energy theorem, find the speed of the block at the bottom. (Take $g = 10 \text{ m/s}^2$.)

Answers: 1. 240 J | 2. 0 J (tension is always perpendicular to the instantaneous displacement) | 3. 4 N | 4. 20 J | 5. $v = \sqrt{2g(H - \mu l)} = \sqrt{2(10)(2 - 1.2)} = \sqrt{16} = 4 \text{ m/s}$

Section 6: Crack the MCQ

Q1. A body of mass m moves with constant speed v in a circle of radius r . The work done by the centripetal force in half a revolution is: (a) $\frac{1}{2}mv^2$ (b) mv^2 (c) $\pi r \cdot \frac{mv^2}{r}$ (d) 0

Correct: (d) 0. The centripetal force always points toward the centre, perpendicular to the instantaneous velocity (and hence the displacement) at every point. With $\theta = 90^\circ$, $\cos \theta = 0$, so the work is zero throughout. *Distractor analysis:* (a) and (b) tempt students who confuse this with kinetic energy, but K doesn't change here since speed is constant. (c) is a “multiply everything given” trap — it computes (force $\times$ arc length) forgetting the perpendicularity. The whole point is that the angle, not the magnitudes, kills the work.

Q2. A force $\vec{F} = (3\hat{i} + 4\hat{j})$ N moves a particle through a displacement $\vec{S} = (4\hat{i} - 3\hat{j})$ m. The work done is: (a) 24 J (b) 0 J (c) 25 J (d) -12 J

Correct: (b) 0 J. Work is the dot product: $W = \vec{F} \cdot \vec{S} = (3)(4) + (4)(-3) = 12 - 12 = 0$. *Distractor analysis:* (a) comes from adding instead of subtracting the second term (sign error in the cross-term). (c) is $|\vec{F}||\vec{S}| = 5 \times 5$ — the magnitude product, which would only be the answer if the vectors were *parallel*. (d) drops one term entirely. Here the two vectors happen to be perpendicular, so the force does no work — a result you'd never see by multiplying magnitudes.

Q3. A 0.5 kg ball thrown straight up rises and returns to the thrower's hand. Over the full up-and-down trip, the work done by gravity is: (a) positive (b) negative (c) zero (d) depends on the launch speed

Correct: (c) zero. Going up, gravity does negative work ($-mgh$); coming down, it does equal positive work ($+mgh$). The net displacement over the round trip is zero, so the total work by gravity is zero. *Distractor analysis:* (a) and (b) come from looking at only half the journey. (d) is the classic “more information needed” hedge — but gravitational work depends only on the vertical *displacement* between start and end points, not on the speed, so it's exactly zero regardless of how hard the ball was thrown.

Q4. A block is pulled at *constant velocity* across a rough floor by a horizontal force. Which statement is correct? (a) Net work on the block is positive (b) Work by the applied force is zero (c) Net work on the block is zero, but individual forces do non-zero work (d) Work by friction is positive

Correct: (c). Constant velocity means $\Delta K = 0$, so by the work–energy theorem the *net* work is zero. But the applied force does positive work and friction does an equal and opposite negative work — they cancel. *Distractor analysis:* (a) contradicts $\Delta K = 0$. (b) is wrong because the applied force *does* move the block, so it does positive work; it's just balanced. (d) inverts the sign — kinetic friction opposes

motion, so its work is negative. This question separates students who think “force is applied, so net work must be positive” from those who reason through the theorem.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Equating one force’s work with ΔK .** The theorem uses the **net** work — the sum over *all* forces. Writing “ $W_{\text{gravity}} = \Delta K$ ” when friction or tension are also present is a classic mark-loser. Always tally every force, or use the single net force.
2. **Dropping the sign of work.** Students compute $|FS \cos \theta|$ and forget that $\cos \theta$ can be negative. Negative work is physically real (it means energy is being removed) — losing the sign quietly corrupts every downstream energy calculation.
3. **Assuming friction always does negative work.** On the *driving* block of a system (e.g., a box on an accelerating truck bed, or your foot on the ground when you walk forward), static friction does **positive** work. Friction opposes *relative slipping*, not motion itself.
4. **Forgetting that work is frame-dependent.** In JEE Advanced problems involving lifts, trains, or moving wedges, the displacement — and therefore the work — changes with the observer. Pick one frame and stay in it for the whole problem.

Pro-Tip (speed + accuracy): Before grinding through any work calculation, ask the *sign* first. “Force along motion $\rightarrow +$, against $\rightarrow -$, perpendicular $\rightarrow 0$.” This instantly zeroes out normal reaction, tension in circular motion, and gravity during horizontal motion — often turning a messy problem into a one-line answer. And whenever a body’s *speed is unchanged*, you already know $W_{\text{net}} = 0$ without computing anything.

Section 8: Cheat Sheet Box

WORK & WORK-ENERGY THEOREM — RAPID REVISION

- **Constant force:** $W = \vec{F} \cdot \vec{S} = FS \cos \theta$ (*scalar, but signed*)
- **Variable force:** $W = \int \vec{F} \cdot d\vec{r} = \text{area under the } F\text{-}x \text{ graph}$
- **Kinetic energy:** $K = \frac{1}{2}mv^2$
- **Work-Energy Theorem:** $W_{\text{net}} = \Delta K = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$
- **Sign rule:** $\theta < 90^\circ \rightarrow +$ | $\theta > 90^\circ \rightarrow -$ | $\theta = 90^\circ \rightarrow 0$
- **Dimensions of work/energy:** $[ML^2T^{-2}]$; SI unit **joule**; $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$
- **Key cautions:** “net” means *all* forces; work is *frame-dependent*; static friction *can* do positive work.

Memory aid — “SAD” angles: Same direction = positive, Against = negative, Dead perpendicular = zero.

Sub-topic 2: Mechanical Energy — Kinetic and Potential

Exam Relevance: This sub-topic is a guaranteed scorer. CBSE Boards ask 1–2 mark direct questions on the *definitions* of kinetic and potential energy and the spring PE expression. JEE Main asks **1–2 questions** every year, very often the “find the force / find the equilibrium position from a given $U(x)$ ” type. NEET loves the **kinetic energy–momentum relation** ($K = p^2/2m$) as a one-line trap. JEE Advanced almost always hides a **potential-energy-curve / nature-of-equilibrium** problem here — high-yield, and exactly where careless students lose marks on a sign.

Section 1: Concept Intuition

In the last sub-topic we learnt that *work* transfers energy. But energy itself — what is it, and where does it live? In mechanics, energy comes in just two flavours, and almost every problem in this chapter is a story of one turning into the other.

Kinetic energy is the easy one: it’s the energy a body has *because it is moving*. A cricket ball sitting on the ground is harmless; the same ball at 140 km/h can crack a helmet. Same ball, same mass — the only difference is motion. That “energy of motion” is kinetic energy, $K = \frac{1}{2}mv^2$. Two things to burn into memory right away: it is **always positive** (you’re squaring the speed, so direction is irrelevant — a ball going left and a ball going right at the same speed have identical K), and it grows with the *square* of speed (double the speed $\rightarrow$ four times the kinetic energy, which is exactly why accidents at high speed are so much more destructive).

Potential energy is the subtle one, and it’s where students get confused. It’s energy a body (really, a *system*) has stored up **because of its position or configuration**, waiting to be released. Think of a stretched bowstring, a stone held at the edge of a terrace, or a compressed spring in a toy gun. Nothing is moving yet — but there is energy *locked in place*, ready to convert into motion the instant you let go.

Think about it this way... imagine carrying a bucket of water up to your building’s rooftop tank. You sweat and do work against gravity. That work doesn’t vanish — it gets *deposited* into the water as gravitational potential energy. Open the tap on the ground floor and that stored energy comes rushing back out as the kinetic energy of flowing water (which is how a small hydro turbine spins). Lifting “charges up” the system; releasing “discharges” it. Potential energy is nature’s rechargeable battery.

There is one deep idea you must accept before potential energy makes any sense: **potential energy only exists for “conservative” forces**. A conservative force is one that gives your energy *back* — gravity and spring force are the classic examples. Lift a stone and lower it, and gravity returns every joule. There is no such thing as “friction potential energy,” because friction *eats* energy (turns it into heat) and never gives it back. So the very idea of stored, recoverable energy is tied to forces that play fair. The defining tests of a conservative force: the work it does is **path-independent**, and the work it does over **any closed loop is zero**.

A second idea that trips people up: **potential energy is always measured relative to a reference level you choose**. The PE of a book on a table is, say, 20 J relative to the floor — but 0 J relative to the table, and *negative* relative to the ceiling. None of these is “wrong.” Only *differences* in PE are physically meaningful, so you’re free to plant your zero wherever it’s convenient. (For gravity near Earth we usually pick the ground; for forces that fade with distance, we pick infinity.)

One more refinement on gravitational PE, because it quietly trips up even strong students. Potential

energy is really a property of a **system**, not a lone object — gravitational PE belongs to the *body–Earth pair*, not to the body alone, because it arises from the *interaction* between them. We loosely say “the PE of the book,” but strictly it’s the PE of the book-and-Earth configuration. And there are *two* gravitational PE formulas for two different regimes: $U = mgh$ is an approximation valid only **near the surface**, where g is effectively constant; for large distances (satellites, escape problems) you must use the general $U = -\frac{Gm_1m_2}{r}$, which correctly accounts for gravity weakening with distance and is taken as zero at infinity. Reaching for mgh in an orbital problem is a classic error.

Finally, add the two together and you get the headline quantity of this whole chapter:

$$\text{Mechanical Energy } E = K + U.$$

A pendulum bob at the top of its swing is all U , no K ; at the bottom it’s all K , no U ; everywhere in between it’s a mix — but (with no friction) the *total* stays fixed. That conservation is the next sub-topic; here, just hold the picture of energy sloshing between the two buckets.

Limiting conditions & assumptions:

1. $K = \frac{1}{2}mv^2$ is the *translational* kinetic energy of a point mass; spinning bodies carry extra rotational KE (a later chapter).
2. **PE is only definable for conservative forces** — never write a potential energy for friction or air drag.
3. PE values depend on the **chosen reference**; only ΔU is physical.
4. Kinetic energy, like work, is **frame-dependent** (it uses speed, which differs between observers).

Section 2: Key Formulas & Definitions

Kinetic energy

$$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}, \quad p = \sqrt{2mK}$$

where $p = mv$ is the linear momentum. (K is always ≥ 0 .)

Potential energy — general definition (work done by a conservative force as a body moves stores up as $-U$):

$$\Delta U = U_f - U_i = - \int_i^f \vec{F} \cdot d\vec{r} = -W_{\text{conservative}}$$

The force–potential-energy relationship (the inverse of the above, in 1-D):

$$F = -\frac{dU}{dx}$$

The force is the **negative slope** of the PE curve — it always points “downhill,” toward lower potential energy.

Gravitational PE near Earth’s surface: $U = mgh$ **Gravitational PE (general, two masses):** $U = -\frac{Gm_1m_2}{r}$ **Elastic (spring) PE:** $U = \frac{1}{2}kx^2$, where k = spring constant, x = deformation from natural length.

Mechanical energy: $E = K + U$

Quantity reference table

Quantity	Symbol	SI Unit	Dimensional Formula
Kinetic / Potential energy	K, U	joule (J)	$[M^1 L^2 T^{-2}]$
Momentum	p	$\text{kg}\cdot\text{m}\cdot\text{s}^{-1}$	$[M^1 L^1 T^{-1}]$
Spring constant	k	$\text{N}\cdot\text{m}^{-1}$	$[M^1 L^0 T^{-2}]$

Equilibrium conditions (from the PE curve): equilibrium occurs where $\frac{dU}{dx} = 0$ (zero force). It is **stable** if $\frac{d^2U}{dx^2} > 0$ (a valley / minimum of U), **unstable** if $\frac{d^2U}{dx^2} < 0$ (a hilltop / maximum), and **neutral** if $\frac{d^2U}{dx^2} = 0$ (a flat plateau).

Section 3: Key Derivations & Procedures

(A) Deriving elastic potential energy of a spring

Setup: A spring of constant k obeys Hooke's law: to stretch it by an amount x from its natural length, you must apply a force $F = kx$. Because this force *grows* with x , we must integrate.

Step 1 — Work done in stretching from 0 to x :

$$W = \int_0^x F_{\text{applied}} dx' = \int_0^x kx' dx'$$

Why integrate: the force isn't constant — it starts at zero and rises linearly — so $F \cdot S$ won't do.

Step 2 — Integrate:

$$W = k \left[\frac{x'^2}{2} \right]_0^x = \frac{1}{2} kx^2$$

Step 3 — This work is stored as elastic PE (the spring is conservative, so it returns this energy on release):

$$U_{\text{spring}} = \frac{1}{2} kx^2$$

Note the work to change the stretch from x_1 to x_2 is the *difference*: $W = \frac{1}{2}k(x_2^2 - x_1^2)$ — **not** $\frac{1}{2}k(x_2 - x_1)^2$. This catches people constantly.

(B) Procedure — reading a potential energy curve

A U - x graph encodes the entire one-dimensional dynamics of a particle. Here's how to extract it:

- Force at any point** = negative slope: $F = -\frac{dU}{dx}$. Where the curve goes *up* to the right, the force points *left*, and vice versa. The steeper the slope, the stronger the force.
- Equilibrium points** = where the curve is **flat** ($\frac{dU}{dx} = 0$). No slope means no force.
- Classify the equilibrium** by the shape: a **valley** (minimum) is *stable* — nudge the particle and the slope pushes it back. A **hilltop** (maximum) is *unstable* — nudge it and the slope shoves it further away.

4. **Turning points** = where the horizontal “total energy” line E meets the curve. There $K = E - U = 0$, so the particle momentarily stops and reverses. The particle can only exist in regions where $U \leq E$ (since $K = E - U$ can never be negative).

A potential energy curve $U(x)$ with a wavy shape showing one valley (labelled “Stable equilibrium, U minimum”) and one hilltop (labelled “Unstable equilibrium, U maximum”). A horizontal dashed line labelled “Total energy E ” intersects the curve at two points labelled “Turning points ($K = 0$)”. Small arrows on the curve slopes show force always pointing toward lower U .

(C) Telling conservative and non-conservative forces apart

Since potential energy can *only* be defined for conservative forces, every problem secretly begins by sorting the forces into two bins. Here is the decision framework — a force is **conservative** if it passes *all three* of these equivalent tests (they always agree):

1. **Path independence:** the work it does in moving a body between two points depends only on the start and end positions, not the route taken. Gravity does the same work whether you climb a hill by the steep face or the gentle path.
2. **Zero work on a closed loop:** carry a body out and bring it back to the start, and the *net* work is exactly zero. The energy you spent going one way is handed back on the return.
3. **Fully recoverable:** the energy it stores can be retrieved completely, with none lost to heat.

Gravity, the spring (elastic) force, and the electrostatic force all pass. They are the forces for which we build potential energies.

A force is **non-conservative** if it fails these tests — and the giveaway is that it depends on the *path* or the *velocity*. Friction and air resistance are the standard examples: drag a box around a closed loop on a rough floor and you do *more* work the longer the path, and none of it comes back (it’s lost as heat). That irreversibility is precisely why you can never write “ U_{friction} .”

Quick procedure to test a force law $\vec{F}(x, y)$: in one dimension, if you can find a function $U(x)$ such that $F = -\frac{dU}{dx}$, the force is conservative. (In higher dimensions the formal test is $\nabla \times \vec{F} = 0$ — the curl vanishes — but for JEE you’ll mostly recognise the standard conservative forces by name.) The practical takeaway: gravity and springs → use potential energy and energy methods; friction and drag → account for them as work that *leaves* the mechanical-energy budget.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A 5 kg stone is lifted and placed on a shelf 3 m above the floor. (a) Find its gravitational potential energy relative to the floor. (b) The stone later falls off and strikes the floor moving at 6 m/s. Find its kinetic energy at the instant of impact. Take $g = 10 \text{ m/s}^2$.

Given: $m = 5 \text{ kg}$, $h = 3 \text{ m}$, impact speed $v = 6 \text{ m/s}$.

(a) **Potential energy:**

$$U = mgh = (5)(10)(3) = \boxed{150 \text{ J}}$$

(b) Kinetic energy at impact:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}(5)(6)^2 = \frac{1}{2}(5)(36) = \boxed{90 \text{ J}}$$

Comment: The stone arrives with 90 J of KE — less than the 150 J of PE it started with, which tells us 60 J was lost (to air resistance, or the shelf’s height was measured differently). Clean presentation is the goal here: given values, correct formula, units stated.

Example 2 — [NEET Speed Trap]

The kinetic energy of a moving truck is increased to 9 times its original value. By what factor does its linear momentum increase? (a) 9 (b) 81 (c) 3 (d) $\sqrt{3}$

The trap: Students see “9 times the energy” and reflexively answer “9 times the momentum.”

The fast reasoning: Use the relation $K = \frac{p^2}{2m}$, so $p = \sqrt{2mK} \propto \sqrt{K}$ (mass unchanged). Therefore

$$\frac{p_{\text{new}}}{p_{\text{old}}} = \sqrt{\frac{K_{\text{new}}}{K_{\text{old}}}} = \sqrt{9} = 3.$$

Answer: (c) 3. *Speed tip:* momentum scales with the **square root** of kinetic energy. Keep $K \propto p^2$ on a mental sticky note — it’s the single most-tested relation in this sub-topic for NEET.

Example 3 — [JEE Main Level]

The potential energy of a particle moving along the x -axis is $U(x) = (x^2 - 4x + 3) \text{ J}$, with x in metres. (a) Find the force on the particle at $x = 3 \text{ m}$. (b) Locate the equilibrium position and state whether it is stable or unstable. (Combines: force–PE relation + equilibrium classification.)

Step 1 — Force from PE:

$$F = -\frac{dU}{dx} = -\frac{d}{dx}(x^2 - 4x + 3) = -(2x - 4) = (4 - 2x) \text{ N}$$

(a) At $x = 3 \text{ m}$:

$$F = 4 - 2(3) = -2 \text{ N}$$

The negative sign means the force points in the $-x$ direction.

(b) Equilibrium is where $F = 0$:

$$4 - 2x = 0 \Rightarrow x = 2 \text{ m}$$

Classify using the second derivative:

$$\frac{d^2U}{dx^2} = 2 > 0 \Rightarrow \text{minimum of } U \Rightarrow \textbf{stable equilibrium}.$$

Answer: $F = -2 \text{ N}$ at $x = 3$; the particle has a **stable** equilibrium at $x = 2 \text{ m}$.

Example 4 — [JEE Advanced Level]

A particle of mass m moves in one dimension under a conservative force whose potential energy is $U(x) = (2x^3 - 9x^2 + 12x)$ J, with x in metres. Find all equilibrium positions and classify each as stable or unstable. Then state, with reasoning, between which two positions the particle would oscillate if given a tiny amount of kinetic energy at the stable point.

Step 1 — Find the force and set it to zero.

$$F = -\frac{dU}{dx} = -(6x^2 - 18x + 12) = -6(x^2 - 3x + 2) = -6(x - 1)(x - 2)$$

Equilibrium ($F = 0$):

$$x = 1 \text{ m} \quad \text{and} \quad x = 2 \text{ m}$$

Step 2 — Classify with the second derivative.

$$\frac{d^2U}{dx^2} = 12x - 18$$

- At $x = 1$: $12(1) - 18 = -6 < 0 \Rightarrow U$ maximum $\Rightarrow$ **unstable**.
- At $x = 2$: $12(2) - 18 = +6 > 0 \Rightarrow U$ minimum $\Rightarrow$ **stable**.

Step 3 — Reasoning about the oscillation. The particle sits in the *valley* at $x = 2$ m. Given a tiny KE, it rocks back and forth inside this valley. On one side, the wall of the valley rises immediately. On the other side ($x < 2$), the curve climbs only until the *hilltop* at $x = 1$ m — beyond which it would fall away. So for a small energy, the particle is **trapped oscillating in the well around $x = 2$ m**, never able to climb over the unstable peak at $x = 1$. The peak at $x = 1$ acts as the energy barrier; cross it and the particle escapes for good.

Why this is JEE Advanced: it demands fluency with $F = -dU/dx$, the sign of d^2U/dx^2 , and the physical picture of trapping/escape — the conceptual core of everything from molecular bonds to planetary orbits.

Section 5: Practice Exercises

1. [CBSE] A spring of force constant 200 N/m is compressed by 0.10 m. Calculate the elastic potential energy stored.
2. [NEET] Two bodies of masses m and $4m$ have equal kinetic energies. Find the ratio of the magnitudes of their linear momenta.
3. [JEE Main] The potential energy of a particle is $U(x) = (3x^2 - 6x + 5)$ J. Find the position of stable equilibrium.
4. [JEE Main] A spring of constant 50 N/m is stretched from an extension of 5 cm to 15 cm. Find the work done in this process.
5. [JEE Advanced] A particle moves in a potential $U(x) = (x^4 - 8x^2)$ J. Find all positions of **stable** equilibrium.

Answers: 1. 1 J | 2. 1 : 2 (since $p \propto \sqrt{m}$ for equal K) | 3. $x = 1$ m | 4. 0.5 J (use $\frac{1}{2}k(x_2^2 - x_1^2)$, not $\frac{1}{2}k(x_2 - x_1)^2$) | 5. $x = +2$ m and $x = -2$ m

Section 6: Crack the MCQ

Q1. A light body and a heavy body have the **same momentum**. Which has greater kinetic energy? (a) the heavier body (b) the lighter body (c) both equal (d) depends on their velocities

Correct: (b) the lighter body. Using $K = \frac{p^2}{2m}$, with p fixed, $K \propto \frac{1}{m}$. The smaller mass therefore has the larger kinetic energy. *Distractor analysis:* (a) appeals to the gut feeling that “heavier = more energy,” but that’s true only at equal *speed*, not equal *momentum*. (c) ignores the mass dependence entirely. (d) is the panic-hedge — but momentum is fixed, so the answer is fully determined. This is the canonical NEET separator.

Q2. Which statement about potential energy is correct? (a) It is always positive (b) It is always negative (c) It can be positive, negative, or zero depending on the chosen reference (d) It is always equal to the kinetic energy

Correct: (c). Only *changes* in PE are physical; the absolute value depends on where you set $U = 0$. Gravitational PE near Earth (mgh) is positive above the reference and negative below it; the general gravitational PE ($-Gm_1m_2/r$) is negative by convention. *Distractor analysis:* (a) and (b) each over-generalise one common case. (d) confuses PE with KE — they are independent quantities that merely add to give E .

Q3. The potential energy of a particle is $U(x) = (5x^2 - 10x)$ J. The equilibrium position is at: (a) $x = 0$ (b) $x = 1$ m (c) $x = 2$ m (d) $x = 0.5$ m

Correct: (b) $x = 1$ m. $F = -\frac{dU}{dx} = -(10x - 10) = 10 - 10x$. Setting $F = 0$ gives $x = 1$ m. *Distractor analysis:* (a) comes from setting $U = 0$ instead of $dU/dx = 0$ — confusing “zero energy” with “zero force.” (c) and (d) are arithmetic slips when solving $10x = 10$. The lesson: equilibrium is about the *slope* of U , never its value.

Q4. If the compression of a spring is doubled, the elastic potential energy stored becomes: (a) twice (b) four times (c) half (d) unchanged

Correct: (b) four times. $U = \frac{1}{2}kx^2 \propto x^2$, so doubling x multiplies U by $2^2 = 4$. *Distractor analysis:* (a) is the most common error — linear thinking applied to a squared quantity. (c) and (d) reverse or ignore the dependence. Any time a quantity goes as a square, “doubling the input quadruples the output” should fire automatically.

Section 7: Common Student Pitfalls & Pro-Tips

- Inventing potential energy for non-conservative forces.** There is no “friction PE” or “air-drag PE.” PE exists *only* for conservative forces (gravity, spring, electrostatic). If a force dissipates energy as heat, it has no potential energy — full stop.
- Dropping the minus sign in $F = -dU/dx$.** The force points toward *decreasing* PE (downhill on the curve). Forgetting the negative sign flips the direction of every force and turns stable equilibria into unstable ones in your answer.

3. **Confusing the K - p relation.** $K \propto p^2$, not $K \propto p$. Doubling the speed doubles the momentum but **quadruples** the kinetic energy. Mixing these up wrecks NEET MCQs.
4. **Using $\frac{1}{2}k(\Delta x)^2$ for a change in stretch.** The work to go from x_1 to x_2 is $\frac{1}{2}k(x_2^2 - x_1^2)$. Squaring the *difference* is wrong; take the *difference of the squares*.
5. **Reaching for mgh when distances are large.** $U = mgh$ assumes constant g and only works near Earth's surface. For satellites, escape velocity, or any problem spanning large heights, you must switch to $U = -Gm_1m_2/r$. Spotting which regime you're in is half the battle.

Pro-Tip (speed + accuracy): For any “ $U(x)$ given — find force / equilibrium / stability” problem, run the same three-line drill every time: (1) $F = -U'(x)$, (2) set $F = 0$ for equilibrium positions, (3) check the sign of $U''(x)$ — **positive means a valley means stable**. Memorise “*valley = stable, hilltop = unstable*.” This single drill answers the entire family of JEE Advanced PE-curve questions in under a minute.

Section 8: Cheat Sheet Box

MECHANICAL ENERGY — RAPID REVISION

- **Kinetic energy:** $K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$, always ≥ 0 , $p = \sqrt{2mK}$
- **Potential energy** (conservative forces only): $\Delta U = -W_{\text{cons}} = -\int \vec{F} \cdot d\vec{r}$
- **Standard PEs:** gravity (near Earth) $U = mgh$ | spring $U = \frac{1}{2}kx^2$ | gravity (general)
 $U = -\frac{Gm_1m_2}{r}$
- **Force-PE link:** $F = -\frac{dU}{dx}$ (force = negative slope, points downhill)
- **Mechanical energy:** $E = K + U$
- **Equilibrium:** $\frac{dU}{dx} = 0$; stable if $U'' > 0$ (min), unstable if $U'' < 0$ (max), neutral if $U'' = 0$
- **Dimensions of energy:** $[ML^2T^{-2}]$, SI unit **joule**

Memory aids: “ K goes as p -squared” · “*valley = stable, hilltop = unstable*” · “*no conservative force, no potential energy*.”

Sub-topic 3: Conservative and Non-Conservative Forces

Exam Relevance: This sub-topic is the conceptual heart of the chapter, and its payoff — the **conservation of mechanical energy** — is one of the most-used tools in all of JEE/NEET physics. CBSE Boards ask the *definition and distinction* (2–3 marks) and the derivation of conservation of mechanical energy. JEE Main reliably asks **1–2 questions** on energy accounting with friction (the “modified” theorem). NEET tests the path-independence / closed-loop ideas as quick MCQs. JEE Advanced loves the subtle traps: friction’s round-trip work, testing whether a 2-D force is conservative, and energy lost as heat using *sliding distance* rather than displacement.

Section 1: Concept Intuition

Imagine two different shopkeepers. The first is scrupulously honest: whatever you deposit with him, you can withdraw later, down to the last rupee. The second skims a little every time money passes through his hands — you never get back quite what you put in. In physics, forces split into exactly these two personalities. The honest ones are called **conservative**; the skimmers are **non-conservative**.

A **conservative force** gives back every joule. Lift a stone and gravity does negative work on it (you store energy in the stone–Earth system); let it fall and gravity does an exactly equal positive work, handing all of it back as kinetic energy. The springs in a pen, a stretched bow, the pull of the Earth — these all play fair. Because they give energy back, we can sensibly store energy “in” them — which is why, as we saw last sub-topic, **only conservative forces have a potential energy**.

A **non-conservative force** skims. Friction is the textbook villain. Slide a book across a table and friction does negative work, turning the book’s kinetic energy into heat. Push it back to the start and friction does negative work *again* — it never reverses sign to refund you. The energy isn’t destroyed (it warms the table and book), but it’s gone from the *mechanical* budget forever. You can’t build a potential energy for friction because there’s nothing stored to retrieve.

Think about it this way... picture two routes from your house to the local market — a short straight lane and a long winding one. If you carry a heavy bag, the *gravitational* work to get there is identical on both routes (it depends only on the height difference between home and market, not the path). That’s **path independence**, the signature of a conservative force. But the *friction/effort* you fight depends entirely on the distance walked — the long winding route tires you far more. That’s the path-dependence of a non-conservative force. And here’s the cleanest test of all: take a **round trip** — go to the market and come straight back home. Gravity’s total work is exactly **zero** (you ended where you started, same height). Friction’s total work is decidedly *not* zero — you’re exhausted, and all that effort became heat.

So the three equivalent tests for a conservative force are:

1. **Work is path-independent** (depends only on endpoints).
2. **Work over any closed loop is zero.**
3. **The energy it transfers is fully recoverable** (and so it has an associated potential energy).

All three always agree. If a force passes one, it passes all; if it fails one, it fails all.

Now for the big consequence. If *only* conservative forces do work on a system, then nothing skims energy away — kinetic and potential energy just trade back and forth, but their **sum stays exactly constant**. That constant sum is the **mechanical energy**, $E = K + U$, and its constancy is the **principle of conservation of mechanical energy**. A pendulum trades U for K on the way down

and K back into U on the way up, swinging forever (in an ideal, frictionless world) because the total never leaks. The moment friction enters, it skims a little each swing, and the pendulum slowly dies — its mechanical energy bleeding away as heat.

A swinging pendulum shown at three positions — far left (highest point), bottom (lowest point), far right (highest point) — with a small energy bar beside each. Left and right bars: all potential energy (U full, K empty). Bottom bar: all kinetic energy (K full, U empty). A caption notes “Total height of each bar ($E = K + U$) is identical — energy is conserved.”

But here is the subtlety students must internalise: even when mechanical energy *isn't* conserved, **total energy always is**. When friction “destroys” 40 J of kinetic energy, that energy hasn’t vanished — measure carefully and you’ll find the block and surface have warmed up by exactly 40 J worth of heat. Rub your palms together on a cold morning and you feel this directly: the mechanical work of rubbing converts into thermal energy in your skin. So the broader **law of conservation of energy** says energy only ever changes *form* — mechanical into heat, sound, or deformation — and the grand total across all forms is fixed. “Mechanical energy is lost” is shorthand for “mechanical energy has been converted into a non-mechanical form we’re no longer tracking.”

Limiting conditions & assumptions:

1. Conservation of mechanical energy holds **only when non-conservative forces do no net work** (frictionless, or friction present but not doing work).
2. The “lost” mechanical energy is never truly lost — it becomes heat, sound, deformation. The *total* energy of the universe is always conserved (the broader **law of conservation of energy**).
3. Friction’s energy loss uses the **actual sliding distance**, not the net displacement — a round trip dissipates heat over the *whole* path.

Section 2: Key Formulas & Definitions

Work done by a conservative force (the link to potential energy):

$$W_{\text{cons}} = -\Delta U = U_i - U_f$$

Force from potential energy (carried over from Sub-topic 2):

$$F = -\frac{dU}{dx} \quad (1\text{-D}), \quad \vec{F} = -\nabla U \quad (3\text{-D})$$

Conservation of mechanical energy (only conservative forces do work):

$$K_i + U_i = K_f + U_f \quad \Longleftrightarrow \quad \Delta E = \Delta K + \Delta U = 0$$

Modified work–energy theorem (when non-conservative forces are also present):

$$W_{\text{nc}} = \Delta E = (K_f + U_f) - (K_i + U_i)$$

For friction specifically, the mechanical energy lost equals the heat produced:

$$|W_{\text{friction}}| = f \times d_{\text{sliding}} = \text{heat generated}$$

Variable definitions: K = kinetic energy, U = potential energy, $E = K + U$ = mechanical energy, W_{nc} = work by all non-conservative forces, f = friction force, d_{sliding} = total path length over which sliding occurs.

Vector test for a conservative force (3-D): the curl vanishes,

$$\nabla \times \vec{F} = 0.$$

In two dimensions with $\vec{F} = F_x \hat{i} + F_y \hat{j}$, this reduces to the handy condition

$$\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}.$$

Examples to memorise: *Conservative* — gravitational, spring (elastic), electrostatic. *Non-conservative* — friction, viscous drag, air resistance (all velocity- or path-dependent and dissipative).

A note on the sign of potential energy. Because U is defined relative to a chosen zero (usually at infinity), its sign carries physical meaning about the nature of the force field. For an **attractive** field (like gravity between two masses, or opposite charges), U is taken **negative** — you'd have to *supply* energy to pull the bodies apart to the zero-reference at infinity. For a **repulsive** field (like two charges of the same sign), U is **positive** — energy was *spent* pushing the bodies together against their mutual repulsion, and it's stored, ready to fling them apart. This is why the general gravitational PE, $U = -Gm_1m_2/r$, wears a minus sign: gravity is always attractive.

Section 3: Key Derivations & Procedures

(A) Conservation of mechanical energy

Assumptions: A system on which the only forces doing work are conservative.

Step 1 — Work–energy theorem (from Sub-topic 1): the net work equals the change in kinetic energy.

$$W_{\text{net}} = \Delta K$$

Step 2 — All the work is conservative, and conservative work is $-\Delta U$:

$$W_{\text{net}} = W_{\text{cons}} = -\Delta U$$

Why: this is the defining property of a conservative force — its work is stored as (minus) the change in potential energy.

Step 3 — Equate the two expressions for W_{net} :

$$-\Delta U = \Delta K \Rightarrow \Delta K + \Delta U = 0 \Rightarrow \Delta(K + U) = 0$$

$$K + U = E = \text{constant}$$

The sum is unchanging: energy merely shuttles between kinetic and potential form. This single line lets us relate the state of a system at two instants **without knowing anything about the messy details in between** — no need to track forces or solve Newton's equations along the path.

(B) The modified work–energy theorem (when friction is present)

Step 1 — Split the net work into conservative and non-conservative parts:

$$W_{\text{net}} = W_{\text{cons}} + W_{\text{nc}} = \Delta K$$

Step 2 — Substitute $W_{\text{cons}} = -\Delta U$:

$$-\Delta U + W_{\text{nc}} = \Delta K$$

Step 3 — Rearrange:

$$W_{\text{nc}} = \Delta K + \Delta U = \Delta E$$

So the work done by non-conservative forces *is* the change in mechanical energy. Since friction always opposes the sliding, $W_{\text{nc}} < 0$, and the mechanical energy *decreases* by exactly the heat generated. This is the everyday version of energy bookkeeping: **(mechanical energy at start) = (mechanical energy at end) + (heat lost to friction)**.

(C) Procedure — testing whether a force is conservative

Given a force law $\vec{F}(x, y)$, decide if it's conservative and, if so, find U :

1. **Apply the cross-derivative test:** check whether $\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$. If they're unequal, the force is non-conservative — stop. If equal, it's conservative — proceed.
2. **Find U by integrating** $F_x = -\frac{\partial U}{\partial x}$ with respect to x (treating y as constant), which leaves an unknown function of y .
3. **Pin down that function** by demanding $F_y = -\frac{\partial U}{\partial y}$.
4. **Sanity-check** by computing work along two different paths between the same endpoints — for a conservative force, both must give the same answer.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A 0.5 kg ball is released from rest at a height of 5 m above the ground. Neglecting air resistance, use conservation of mechanical energy to find its speed just before it hits the ground. Take $g = 10 \text{ m/s}^2$.

Given: $m = 0.5 \text{ kg}$, $h = 5 \text{ m}$, initial speed = 0.

Set up conservation of mechanical energy (only gravity does work):

$$K_i + U_i = K_f + U_f$$

Taking the ground as the reference ($U_f = 0$) and noting $K_i = 0$:

$$0 + mgh = \frac{1}{2}mv^2 + 0$$

Solve for v (mass cancels — a hallmark of free-fall energy problems):

$$v = \sqrt{2gh} = \sqrt{2(10)(5)} = \sqrt{100} = \boxed{10 \text{ m/s}}$$

Comment: All 25 J of potential energy ($mgh = 0.5 \times 10 \times 5$) converted cleanly into kinetic energy. No path details were needed — just the start and end states.

Example 2 — [NEET Speed Trap]

A block is carried from the floor up onto a shelf and then brought right back down to its original spot on the floor. Over this complete round trip, the work done by gravity and the work done by friction (during the lifting and lowering) are, respectively: (a) zero, zero (b) zero, negative (c) negative, zero (d) positive, negative

The trap: Students who memorised “closed loop $\rightarrow$ zero work” blindly apply it to *both* forces and pick (a).

The fast reasoning: Gravity is **conservative** — over any closed path (you end where you started, same height) its total work is exactly **zero**. Friction is **non-conservative** — it opposes motion the whole way up *and* the whole way down, doing negative work in both directions, so its round-trip total is **negative**, never zero.

Answer: (b) zero, negative. *Speed tip:* “closed loop $\rightarrow$ zero work” applies **only** to conservative forces. The instant friction or drag appears in a round trip, that contribution is negative.

Example 3 — [JEE Main Level]

A 5 kg block starts from rest and slides down a **smooth** curved ramp from a height of 1.8 m. At the bottom it slides onto a **rough** horizontal floor with coefficient of kinetic friction $\mu = 0.4$. How far does it travel on the floor before stopping? Take $g = 10 \text{ m/s}^2$. (Combines: conservation of mechanical energy on the smooth part + modified work–energy theorem on the rough part.)

Stage 1 — The smooth ramp (energy conserved). All the potential energy becomes kinetic energy at the bottom:

$$K_{\text{bottom}} = mgh = (5)(10)(1.8) = 90 \text{ J}$$

(So $v = \sqrt{2gh} = \sqrt{36} = 6 \text{ m/s}$, though we won’t even need the speed.)

Stage 2 — The rough floor (mechanical energy bled away by friction). The block stops when friction has dissipated all 90 J. The friction force is

$$f = \mu mg = (0.4)(5)(10) = 20 \text{ N}$$

Setting heat generated equal to the kinetic energy to be removed:

$$f \cdot s = K_{\text{bottom}} \Rightarrow 20 s = 90 \Rightarrow \boxed{s = 4.5 \text{ m}}$$

Why this is JEE Main: it stitches two different energy regimes together — a conservative stretch where E is conserved, and a dissipative stretch where the modified theorem applies. Notice we never had to analyse the *shape* of the curved ramp; energy methods don’t care about the path.

Example 4 — [JEE Advanced Level]

A particle is acted on by a force $\vec{F} = (y\hat{i} + x\hat{j}) \text{ N}$, with x, y in metres. (a) Determine whether the force is conservative. (b) Verify your conclusion by computing the work done from $(0, 0)$ to $(1, 1)$ along two different paths. (c) If conservative, find the potential energy function $U(x, y)$.

(a) **Cross-derivative test.** With $F_x = y$ and $F_y = x$:

$$\frac{\partial F_y}{\partial x} = \frac{\partial x}{\partial x} = 1, \quad \frac{\partial F_x}{\partial y} = \frac{\partial y}{\partial y} = 1.$$

They are equal, so the force is **conservative**.

(b) **Two-path verification.** Work is $W = \int (F_x dx + F_y dy) = \int (y dx + x dy)$.

Path 1: $(0, 0) \rightarrow (1, 0) \rightarrow (1, 1)$.

- Leg 1 (along x , $y = 0$): $\int_0^1 y dx = \int_0^1 0 dx = 0$.
- Leg 2 (along y , $x = 1$): $\int_0^1 x dy = \int_0^1 1 dy = 1$.
- Total: $W_1 = 0 + 1 = 1$ J.

Path 2: $(0, 0) \rightarrow (0, 1) \rightarrow (1, 1)$.

- Leg 1 (along y , $x = 0$): $\int_0^1 x dy = \int_0^1 0 dy = 0$.
- Leg 2 (along x , $y = 1$): $\int_0^1 y dx = \int_0^1 1 dx = 1$.
- Total: $W_2 = 0 + 1 = 1$ J.

Both paths give 1 J — **path independence confirmed**, exactly as a conservative force demands.

(c) **Potential energy.** Since $\vec{F} = -\nabla U$, we need $-\frac{\partial U}{\partial x} = y$ and $-\frac{\partial U}{\partial y} = x$. Integrating the first gives $U = -xy + g(y)$; the second then requires $g'(y) = 0$. Hence

$$U(x, y) = -xy + \text{constant}$$

A quick check: $-\partial U/\partial x = y$ ✓ and $-\partial U/\partial y = x$ ✓.

Why this is JEE Advanced: it fuses the cross-derivative test, explicit line integrals along chosen paths, and recovery of U — the full toolkit for handling an arbitrary force field.

Section 5: Practice Exercises

1. [CBSE] A 2 kg stone is dropped from a height of 20 m. Using conservation of mechanical energy, find its speed just before hitting the ground. (Neglect air resistance, $g = 10 \text{ m/s}^2$.)
2. [NEET] What is the work done by the gravitational force on a body taken once around any closed path? Justify in one line.
3. [JEE Main] A 1 kg block slides from rest down a smooth incline of height 2.5 m, then moves onto a rough horizontal floor ($\mu = 0.5$). Find the distance it travels on the floor before stopping.
4. [JEE Main] The work done by a conservative force in moving a body from A to B is +30 J. If the potential energy at A is 50 J, find the potential energy at B.
5. [JEE Advanced] A force $\vec{F} = (2xy)\hat{i} + (x^2)\hat{j}$ acts on a particle. Determine whether it is conservative, and if so, find the potential energy function $U(x, y)$.

Answers: 1. 20 m/s | 2. 0 (gravity is conservative, so closed-loop work vanishes) | 3. 5 m | 4. $U_B = 20$ J (since $W_{\text{cons}} = U_A - U_B$) | 5. Conservative ($\partial F_y/\partial x = \partial F_x/\partial y = 2x$); $U = -x^2y + \text{constant}$

Section 6: Crack the MCQ

Q1. Which of the following is a non-conservative force? (a) gravitational force (b) spring force (c) kinetic friction (d) electrostatic force

Correct: (c) kinetic friction. It is velocity-dependent and dissipative — its work depends on the path and it converts mechanical energy irreversibly into heat. *Distractor analysis:* (a), (b), and (d) are the three standard conservative forces; each is path-independent, does zero work on a closed loop, and possesses a well-defined potential energy. Friction has none of these properties.

Q2. The work done by a conservative force around a closed path is: (a) maximum (b) always positive (c) zero (d) dependent on the path taken

Correct: (c) zero. This is one of the three defining tests of a conservative force — over any closed loop, the energy given out on one part is taken back on another, netting exactly zero. *Distractor analysis:* (b) ignores that work can be negative on parts of the loop. (d) describes a *non*-conservative force; the whole point of “conservative” is path independence. (a) is meaningless here.

Q3. A block is pushed up a rough incline and then slides back down to its starting point. Over this complete round trip, the work done by friction is: (a) zero (b) positive (c) negative (d) cannot be determined

Correct: (c) negative. Friction opposes motion on the way up (negative work) *and* on the way down (negative work again). Two negatives add to a larger negative — never zero. *Distractor analysis:* (a) is the classic trap — wrongly applying the “closed loop $\rightarrow$ zero work” rule (which holds only for conservative forces) to friction. (b) is impossible since friction always opposes sliding. (d) underestimates how determinate the situation is.

Q4. When only conservative forces do work on a system, which quantity necessarily stays constant? (a) kinetic energy alone (b) potential energy alone (c) the sum $K + U$ (d) linear momentum

Correct: (c) the sum $K + U$ (mechanical energy). Conservative forces let kinetic and potential energy interconvert, but their total is fixed. *Distractor analysis:* (a) and (b) each change continuously as energy shuttles between the two forms — only the *sum* is invariant. (d) confuses energy conservation with momentum conservation, which is governed by a different condition (zero net external force), not by forces being conservative.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Treating friction’s round-trip work as zero.** The “closed loop $\rightarrow$ zero work” rule is exclusive to conservative forces. Friction does negative work *every* leg of a journey, so its round-trip total is always negative.
2. **Applying conservation of mechanical energy when friction is present.** If any non-conservative force does work, $K + U$ is **not** constant. Switch to the modified theorem $W_{\text{nc}} = \Delta E$ and account for the heat lost.
3. **Forgetting the minus sign in $W_{\text{cons}} = -\Delta U$.** Positive conservative work means potential energy *decreases* (a falling stone), not increases. Mixing up the sign flips every energy balance.

4. **Using displacement instead of path length for friction heat.** Friction heat is $f \times$ (total sliding distance). On a back-and-forth path the displacement may be small or zero, but the heat dissipated depends on the *whole distance slid*.
5. **Saying friction “destroys” energy.** Energy is never destroyed. Friction *converts* mechanical energy into heat — the total energy is conserved. In exam answers, write “converted to heat,” not “lost,” to show you understand the broader conservation law.

Pro-Tip (choosing your tool): Before writing a single equation, scan the forces. If everything doing work is conservative (gravity, springs, smooth surfaces), jump straight to $E_i = E_f$ — fast, and the path is irrelevant. If friction or drag is in play, use $E_i = E_f + (\text{heat})$, with $\text{heat} = f \times d_{\text{sliding}}$. Picking the right framework up front saves you from grinding through Newton’s laws along a complicated path.

Section 8: Cheat Sheet Box

CONSERVATIVE & NON-CONSERVATIVE FORCES — RAPID REVISION

- **Conservative:** path-independent · closed-loop work = 0 · energy recoverable · has a PE · $\nabla \times \vec{F} = 0$. *Examples:* gravity, spring, electrostatic.
- **Non-conservative:** path-dependent · closed-loop work $\neq 0$ · dissipative (heat) · no PE. *Examples:* friction, viscous drag, air resistance.
- **Conservative work–PE link:** $W_{\text{cons}} = -\Delta U = U_i - U_f$
- **Force from PE:** $F = -\frac{dU}{dx}$ (3-D: $\vec{F} = -\nabla U$)
- **Conservation of mechanical energy** (conservative forces only): $K_i + U_i = K_f + U_f$
- **Modified work–energy theorem:** $W_{\text{nc}} = \Delta E = (K_f + U_f) - (K_i + U_i)$; friction heat = $f \times d_{\text{sliding}}$
- **2-D conservative test:** $\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$
- **Law of conservation of energy:** energy is never destroyed — mechanical energy “lost” becomes heat/sound.

Memory aids: “Conservative gives it back; non-conservative burns it as heat.” · “Closed-loop zero is a conservative-only privilege.”

Sub-topic 3.5: Various Forms of Energy & the Law of Conservation of Energy

Exam Relevance: This is the chapter's big-picture, high-yield conceptual section. CBSE Boards regularly ask 1–3 mark questions on the *forms of energy*, the *law of conservation of energy*, and the **mass–energy relation** $E = mc^2$ (a named NCERT result). NEET asks **1 question** on energy transformations or the value of $E = mc^2$ /mechanical equivalent of heat. JEE Main occasionally folds $E = mc^2$ into a numerical (mass converted, energy released). JEE Advanced uses these ideas as the conceptual bridge into Modern Physics (nuclear energy, binding energy).

Section 1: Concept Intuition

So far this chapter has lived entirely in the world of *mechanical* energy — kinetic and potential. But energy is far bigger than that. Look around: the food you ate this morning, the petrol in a scooter, the charge in a phone battery, the warmth of sunlight, the rumble of a loudspeaker, the colossal output of a nuclear reactor — every one of these is **energy in a different costume**.

The forms you should know:

- **Mechanical energy** — kinetic (K) and potential (U), the subject of the rest of this chapter.
- **Heat (thermal) energy** — the energy of the random, jiggling motion of the molecules inside matter. A hot cup of chai has more thermal energy than a cold one because its molecules move faster.
- **Chemical energy** — energy locked in the bonds between atoms, released or absorbed in reactions. This powers your body (food), vehicles (fuel), and devices (batteries).
- **Electrical energy** — energy carried by moving charges; what flows through every wire in your home.
- **Nuclear energy** — energy stored in the nucleus of an atom, released in fission (atom bombs, reactors) and fusion (the Sun and stars).
- **Radiant (light) energy** and **sound energy** — energy carried by electromagnetic waves and by pressure waves in a medium.

Think about it this way... picture energy as money that can be exchanged between different currencies. You can convert rupees to dollars to yen, but the *total wealth* (at a fixed exchange rate) never changes in the exchange itself. Energy is the same: a hydroelectric dam converts the **potential energy** of stored water into **kinetic energy** as it falls, then into **electrical energy** in the turbine, which becomes **light and heat** in your home. At every step the *form* changes, but the *total amount* is preserved. That preservation is the deepest law in all of physics — the **law of conservation of energy**: energy can neither be created nor destroyed; it can only be transformed from one form to another. The total energy of an isolated system stays constant forever.

This is why, back in the sub-topic on friction, the kinetic energy that “disappeared” wasn’t destroyed — it was *converted* into heat. James Joule proved this conversion is exact: a fixed amount of mechanical work always produces a fixed amount of heat. This **mechanical equivalent of heat** is about 1 cal = 4.186 J.

And then comes the most astonishing twist of twentieth-century physics. Einstein showed that **mass itself is a form of energy**. A lump of matter, just sitting still, holds an enormous reserve of energy

frozen in its mass, given by the most famous equation in science:

$$E = mc^2.$$

Because the speed of light c is so vast, even a tiny mass is equivalent to a staggering amount of energy. This is the source of the Sun’s power and of nuclear energy: in nuclear reactions, a sliver of mass vanishes and reappears as a flood of energy. With this insight, the conservation law is generalised to the **conservation of mass–energy**: it is the *combined* total of mass and energy that nature keeps constant.

Limiting conditions & assumptions:

1. The conservation law applies to an **isolated system** (no energy crossing its boundary). Energy can flow *between* a system and its surroundings.
2. Converting “lost” mechanical energy to heat does **not** violate conservation — the heat is real, measurable energy.
3. $E = mc^2$ becomes practically significant only where mass actually changes — chiefly **nuclear** processes; in ordinary chemical or mechanical events the mass change is far too small to detect.

Section 2: Key Formulas & Definitions

Law of conservation of energy: for an isolated system, the total energy (summed over *all* forms) is constant:

$$E_{\text{total}} = K + U + (\text{heat}) + (\text{chemical}) + (\text{electrical}) + \cdots = \text{constant}$$

Mechanical equivalent of heat (Joule’s constant):

$$1 \text{ cal} = 4.186 \text{ J} \quad (\approx 4.2 \text{ J})$$

Mass–energy equivalence (Einstein):

$$E = mc^2$$

where E is the energy equivalent of a mass m , and $c = 3 \times 10^8 \text{ m/s}$ is the speed of light in vacuum.

Energy released from a mass change Δm (e.g., the mass defect in a nuclear reaction):

$$\Delta E = (\Delta m)c^2$$

Quantity reference table

Quantity	Symbol	SI Unit	Dimensional Formula
Energy (any form)	E	joule (J)	$[M^1 L^2 T^{-2}]$
Mass	m	kilogram (kg)	$[M^1 L^0 T^0]$
Speed of light	c	$\text{m}\cdot\text{s}^{-1}$	$[M^0 L^1 T^{-1}]$

Useful energy conversions:

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}, \quad 1 \text{ u (atomic mass unit)} \leftrightarrow 931.5 \text{ MeV}, \quad 1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

The relation $1 \text{ u} \leftrightarrow 931.5 \text{ MeV}$ is just $E = mc^2$ applied to one atomic mass unit — a result used constantly in nuclear physics.

Section 3: Key Procedures & Logical Framework

There is no single “derivation” here, but two procedures recur in exams.

(A) Tracking an energy transformation chain.

1. **Identify the starting form** of energy (e.g., chemical energy in fuel).
2. **Follow each conversion** device by device, naming the form at every stage.
3. **Apply conservation:** the total energy in equals the total energy out, including any “wasted” heat. Real machines are never 100% efficient — the shortfall always shows up as heat or sound, never as destroyed energy.

Example chain (thermal power plant): chemical energy (coal) $\rightarrow$ heat energy (combustion) $\rightarrow$ kinetic energy (steam turbine) $\rightarrow$ electrical energy (generator) $\rightarrow$ light + heat (your home).

(B) Computing energy from a mass change ($E = mc^2$).

1. Find the mass that is converted, Δm (in kg, or in atomic mass units u).
2. If in kg, multiply by $c^2 = 9 \times 10^{16} \text{ m}^2\text{s}^{-2}$ to get joules.
3. If in atomic mass units, multiply directly by 931.5 MeV/u to get the energy in MeV — far quicker for nuclear problems.
4. Convert units as the question demands ($\text{J} \leftrightarrow \text{eV} \leftrightarrow \text{MeV}$).

The key conceptual point: a *measurable* mass change requires a *nuclear-scale* energy release. That is why $E = mc^2$ rules nuclear physics but is invisible in everyday chemistry.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A 10 kg wooden crate sliding across a rough floor at 5 m/s is brought to rest by friction. How much heat energy is generated, and which conservation principle does this illustrate?

Given: $m = 10 \text{ kg}$, $v = 5 \text{ m/s}$, final speed = 0.

By the law of conservation of energy, the lost kinetic energy is fully converted into heat:

$$\text{Heat generated} = K_i = \frac{1}{2}mv^2 = \frac{1}{2}(10)(5)^2 = \frac{1}{2}(10)(25) = \boxed{125 \text{ J}}$$

Comment: This illustrates the **law of conservation of energy** — the 125 J of mechanical energy was not destroyed but transformed into thermal energy in the crate and floor.

Example 2 — [NEET Speed Trap]

The energy equivalent of 1 gram of matter (taking $c = 3 \times 10^8 \text{ m/s}$) is: (a) $9 \times 10^{13} \text{ J}$ (b) $9 \times 10^{16} \text{ J}$ (c) $3 \times 10^5 \text{ J}$ (d) $9 \times 10^{10} \text{ J}$

The trap: Forgetting to convert grams to kilograms — using $m = 1$ gives 9×10^{16} (option b), the planted error.

The fast reasoning: Convert first: $1 \text{ g} = 10^{-3} \text{ kg}$. Then

$$E = mc^2 = (10^{-3})(3 \times 10^8)^2 = (10^{-3})(9 \times 10^{16}) = 9 \times 10^{13} \text{ J}.$$

Answer: (a) 9×10^{13} J. *Speed tip:* in any $E = mc^2$ problem, the unit slip is the mass unit — convert grams (and atomic mass units) to kilograms *before* squaring c . Note how even one gram holds a city-scale amount of energy.

Example 3 — [JEE Main Level]

In a nuclear reaction, the total mass of the products is less than that of the reactants by $\Delta m = 0.02$ u. Calculate the energy released, in MeV. (Combines: mass defect + mass–energy equivalence.)

Step 1 — Use the atomic-mass-unit shortcut ($1 \text{ u} \leftrightarrow 931.5 \text{ MeV}$), which already bakes in $E = mc^2$:

$$\Delta E = (\Delta m) \times 931.5 \text{ MeV/u} = (0.02)(931.5)$$

$$\boxed{\Delta E \approx 18.63 \text{ MeV}}$$

Why this is JEE Main: it links the idea of a *mass defect* (mass “missing” after the reaction) to the energy released via $E = mc^2$, using the practical 931.5 MeV/u conversion rather than grinding through SI units. This is the bread-and-butter calculation of nuclear energetics.

Example 4 — [JEE Advanced Level]

The Sun radiates energy at a power of about 3.8×10^{26} W. (a) At what rate (in kg/s) is the Sun losing mass? (b) Roughly how much mass does it convert in one full day? (Take $c = 3 \times 10^8$ m/s.)

(a) **Rate of mass loss.** Power is energy per second, and $E = mc^2$ gives $m = E/c^2$, so the rate of mass loss is power divided by c^2 :

$$\frac{dm}{dt} = \frac{P}{c^2} = \frac{3.8 \times 10^{26}}{9 \times 10^{16}} \approx \boxed{4.2 \times 10^9 \text{ kg/s}}$$

(b) **Mass converted in one day** ($t = 86400$ s):

$$\Delta m = \frac{dm}{dt} \times t = (4.2 \times 10^9)(8.64 \times 10^4) \approx 3.6 \times 10^{14} \text{ kg}$$

Why this is JEE Advanced: it applies $E = mc^2$ to a *rate* (power $\rightarrow$ mass-loss rate) on an astrophysical scale, and demands clean handling of large powers of ten. Despite losing four billion kilograms every second, the Sun is so massive that this is utterly negligible over its lifetime — a satisfying check on the numbers.

Section 5: Practice Exercises

- [CBSE] A 2 kg ball is dropped from rest and on striking the ground loses 40 J of mechanical energy to a dent and to sound. Where did this energy go, and what principle does this demonstrate?
- [NEET] Convert 500 calories of heat energy into joules. (Take $1 \text{ cal} = 4.2 \text{ J}$.)
- [JEE Main] How much mass (in kg) is converted into energy when 9×10^{12} J of energy is released? (Take $c = 3 \times 10^8$ m/s.)
- [JEE Main] Identify the sequence of energy transformations in a battery-powered electric fan.

5. [JEE Advanced] A mass defect of 0.5 u occurs in a fusion reaction. Find the energy released in (a) MeV and (b) joules. (Use $1 \text{ u} \leftrightarrow 931.5 \text{ MeV}$, $1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$.)

Answers: 1. Converted to heat, sound, and deformation energy; demonstrates the **law of conservation of energy** | 2. 2100 J | 3. $m = E/c^2 = 9 \times 10^{12}/9 \times 10^{16} = 10^{-4} \text{ kg}$ | 4. chemical $\rightarrow$ electrical $\rightarrow$ mechanical (kinetic) + heat/sound | 5. (a) 465.75 MeV, (b) $\approx 7.45 \times 10^{-11} \text{ J}$

Section 6: Crack the MCQ

Q1. According to the law of conservation of energy, energy can: (a) be created but not destroyed (b) be destroyed but not created (c) neither be created nor destroyed, only transformed (d) be both created and destroyed

Correct: (c). Energy only changes form; the total in an isolated system is constant. *Distractor analysis:* (a) and (b) each violate one half of the law; (d) abandons it entirely. The law is symmetric: no creation *and* no destruction.

Q2. The energy equivalent of 1 kg of mass is approximately: (a) $3 \times 10^8 \text{ J}$ (b) $9 \times 10^{16} \text{ J}$ (c) $9 \times 10^8 \text{ J}$ (d) $3 \times 10^{16} \text{ J}$

Correct: (b) $9 \times 10^{16} \text{ J}$. $E = mc^2 = (1)(3 \times 10^8)^2 = 9 \times 10^{16} \text{ J}$. *Distractor analysis:* (a) uses c instead of c^2 (forgetting to square); (c) and (d) are partial powers-of-ten slips. The defining step is **squaring** the speed of light.

Q3. Which of the following lists only *non-mechanical* forms of energy? (a) kinetic, potential, heat (b) chemical, electrical, nuclear (c) potential, chemical, sound (d) kinetic, nuclear, electrical

Correct: (b) chemical, electrical, nuclear. None of these is kinetic or potential (mechanical) energy. *Distractor analysis:* (a), (c), and (d) each include kinetic or potential energy, which *are* mechanical forms. Read every item in the list before choosing.

Q4. In a hydroelectric power station, the main energy transformation is: (a) electrical $\rightarrow$ mechanical $\rightarrow$ potential (b) potential $\rightarrow$ kinetic $\rightarrow$ electrical (c) chemical $\rightarrow$ heat $\rightarrow$ electrical (d) nuclear $\rightarrow$ heat $\rightarrow$ electrical

Correct: (b) potential $\rightarrow$ kinetic $\rightarrow$ electrical. Stored water's gravitational PE becomes KE as it falls, then electrical energy in the turbine-generator. *Distractor analysis:* (a) reverses the direction; (c) describes a *thermal* (coal/gas) plant; (d) describes a *nuclear* plant. Match the energy chain to the actual source.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Saying energy is “lost” or “used up.”** Energy is never destroyed — it is *transformed* (often into heat). In answers, write “converted to heat/sound,” not “lost,” to show you understand conservation.

2. **Forgetting to square c in $E = mc^2$.** The energy uses $c^2 = 9 \times 10^{16}$, not c . Dropping the square under-estimates the answer by a factor of 3×10^8 .
3. **Unit slips in mass.** Convert grams and atomic mass units to kilograms *before* using $E = mc^2$ in SI — or use the $1 \text{ u} \leftrightarrow 931.5 \text{ MeV}$ shortcut directly for nuclear problems.
4. **Expecting measurable mass loss in chemical/mechanical events.** The mass change there is far below detection. $E = mc^2$ matters in practice only at the *nuclear* scale.

Pro-Tip (speed + accuracy): For nuclear energy questions, skip SI units entirely — multiply the mass defect in atomic mass units by 931.5 MeV/u to get the answer in MeV in one line. For “energy equivalent” questions in joules, remember the anchor value: $1 \text{ kg} \leftrightarrow 9 \times 10^{16} \text{ J}$, then just scale by the given mass.

Section 8: Cheat Sheet Box

FORMS OF ENERGY & CONSERVATION — RAPID REVISION

- **Forms of energy:** mechanical (K, U), heat (thermal), chemical, electrical, nuclear, light (radiant), sound
- **Law of conservation of energy:** energy is neither created nor destroyed, only transformed; total energy of an isolated system is constant
- **Mechanical equivalent of heat:** $1 \text{ cal} = 4.186 \text{ J} \approx 4.2 \text{ J}$
- **Mass–energy equivalence:** $E = mc^2$, with $c = 3 \times 10^8 \text{ m/s}$, $c^2 = 9 \times 10^{16}$
- **Anchor value:** $1 \text{ kg} \leftrightarrow 9 \times 10^{16} \text{ J}$; **nuclear shortcut:** $1 \text{ u} \leftrightarrow 931.5 \text{ MeV}$
- **From a mass change:** $\Delta E = (\Delta m)c^2$ — significant only in nuclear processes
- **Dimensions of energy:** $[ML^2T^{-2}]$, SI unit **joule**

Memory aids: “Energy changes costume, never quantity.” · “ $E = mc^2$ — don’t forget to square c .” · “Measurable mass loss means nuclear.”

Sub-topic 4: Power

Exam Relevance: Power is a high-frequency, low-difficulty scorer — you cannot afford to drop these marks. CBSE Boards ask the *definition*, units, and a clean $P = W/t$ numerical (1–2 marks). NEET asks **1 question** almost every year, very often a $P = Fv$ problem with a sneaky **km/h** $\rightarrow$ **m/s** unit conversion. JEE Main reliably asks **1–2 questions** combining power with force balance on an incline or pump problems. JEE Advanced favours the **constant-power kinematics** result ($v \propto \sqrt{t}$, $x \propto t^{3/2}$), which catches students who never derived it.

Section 1: Concept Intuition

Two labourers are each asked to carry 100 bricks up to the first floor. Ramesh finishes in 10 minutes; Suresh takes 30. They did **exactly the same work** — same bricks, same height, same energy transferred. So what’s different? *Speed*. Ramesh worked three times faster. That “rate of doing work” is what physics calls **power**.

This is the single idea you must hold onto: **power is not about how much work you do, it’s about how fast you do it**. Two engines that both lift a lift-car to the tenth floor do identical work against gravity; the more *powerful* one simply gets there sooner. Energy answers “how much?”; power answers “how quickly?”.

Think about it this way... imagine two taps filling identical buckets. Both eventually deliver the same litres of water (the same “work”), but the wide-open tap fills the bucket in seconds while the dripping tap takes an hour. The *flow rate* of the tap is the analogy for power — litres per second versus joules per second. A 100-watt motor pours out 100 joules of energy every second; a 2000-watt motor pours out 2000 joules every second and so finishes any given job twenty times faster.

Now, work gets done as a force pushes something along. If you do that pushing *quickly*, you’re powerful; slowly, less so. There are two ways to package this. Over a whole job, we care about the **average power** — total work divided by total time. But at any single instant, the rate can be changing (a car accelerates, its engine delivering more power as it speeds up), so we also need **instantaneous power** — the rate *right now*. A sprinter, for example, might average 400 W over a race, yet hit a peak instantaneous output far higher during the explosive start and far lower while coasting — the average smooths over a story that the instantaneous value tells moment by moment.

Here’s the most useful insight in the whole sub-topic. Instantaneous power can be written as force times velocity:

$$P = \vec{F} \cdot \vec{v} = Fv \cos \theta.$$

Why? Because in a tiny time dt the object moves $dx = v dt$, so the little bit of work is $dW = F dx = Fv dt$, and dividing by dt gives $P = Fv$. This formula is gold: it tells you that to deliver power, you need *both* force and motion. A weightlifter straining to hold a barbell perfectly still delivers **zero** power (no velocity, just like he did zero work) — however much his muscles burn. And the $\cos \theta$ reminds us only the part of the force *along* the velocity delivers power; a sideways force is powerless.

This also explains an everyday fact: a car’s engine has a *maximum power*. Since $P = Fv$, at high speed (large v) the available pushing force $F = P/v$ shrinks — which is exactly why your car accelerates briskly at low speeds but struggles to gain the last few km/h near its top speed. Same engine power, but it’s spread over a faster motion, leaving less force to push harder.

And just as work can be negative, **power can be negative** too. When you brake, the braking force points *opposite* to the velocity ($\theta = 180^\circ$, $\cos \theta = -1$), so $P = Fv \cos \theta$ is negative — the brakes are

removing energy from the car at some rate (and dumping it as heat in the brake pads). Positive power feeds energy in; negative power drains it out. The sign of the power tells you which way energy is flowing.

A quick word on the units, because they have a charming history that makes them stick. The SI unit, the **watt**, honours James Watt, who improved the steam engine during the Industrial Revolution. To sell his engines, Watt needed to tell buyers how many horses each could replace — so he measured how fast a strong horse could do work and coined the **horsepower**. That’s why $1 \text{ hp} \approx 746 \text{ W}$: it’s literally an estimate of one hard-working horse’s output. When a bike is advertised as “15 hp,” it’s claiming the muscle of fifteen horses. Keeping this picture in mind makes the otherwise-arbitrary number 746 easy to recall.

A power-versus-time (P–t) graph showing a curve that rises and falls. The region between the curve and the time axis is shaded, labelled “Area under P–t curve = Work done”. Beside it, a small work-versus-time (W–t) graph with a tangent line drawn at one point, labelled “Slope of W–t curve = instantaneous power”.

Limiting conditions & assumptions:

1. $P = Fv$ gives *instantaneous* power; using it with an average velocity gives average power only if the force is constant.
2. Power is a **scalar** (it’s a dot product of two vectors), but it can be negative — a braking force delivers negative power, *removing* energy.
3. Don’t confuse the **kilowatt-hour** (a unit of *energy*, $= 3.6 \times 10^6 \text{ J}$) with the **kilowatt** (a unit of *power*). Your electricity bill charges for energy (kWh), not power.

Section 2: Key Formulas & Definitions

Average power:

$$P_{\text{avg}} = \frac{W_{\text{total}}}{t_{\text{total}}} = \frac{\Delta W}{\Delta t}$$

Instantaneous power:

$$P = \frac{dW}{dt} = \vec{F} \cdot \vec{v} = Fv \cos \theta$$

where θ is the angle between the force and the velocity.

Quantity reference table

Quantity	Symbol	SI Unit	Dimensional Formula
Power	P	watt (W) = J/s	$[M^1 L^2 T^{-3}]$
Work / Energy	W	joule (J)	$[M^1 L^2 T^{-2}]$
Time	t	second (s)	$[M^0 L^0 T^1]$

Useful unit relations:

$$1 \text{ watt} = 1 \text{ J/s}, \quad 1 \text{ horsepower (hp)} = 746 \text{ W}, \quad 1 \text{ kWh} = 3.6 \times 10^6 \text{ J (energy)}$$

The kilowatt-hour deserves a second look because it trips up so many students: it is power $\times$ time = *energy*, not power. A 2 kW heater run for 3 hours consumes $2 \times 3 = 6$ kWh of energy, which in joules is $6 \times 3.6 \times 10^6 = 2.16 \times 10^7$ J. This is exactly what your electricity meter records and bills — “units” of energy, each unit being one kWh.

Graphical relations:

- The **area under a power–time (P – t) graph** gives the work done: $W = \int P dt$.
- The **slope of a work–time (W – t) graph** gives the instantaneous power.

Standard results worth memorising:

- A machine lifting mass m at *constant velocity* v : $P = mgv$.
- A vehicle moving at *constant velocity* against a resistive force F_{res} : $P = F_{\text{res}} v$.
- For motion from rest under a *constant force*: $P_{\text{avg}} = \frac{1}{2} P_{\text{instantaneous}}$ (at the final instant).

Section 3: Key Derivations & Procedures

(A) Motion under constant power (the JEE Advanced favourite)

Setup: A particle of mass m starts from rest and is driven by a *constant* power P . Find how speed and distance grow with time.

Step 1 — Power is the rate of change of kinetic energy. Since all the delivered power goes into kinetic energy here,

$$P = \frac{dK}{dt} \Rightarrow K = Pt \quad (\text{integrating, with } K = 0 \text{ at } t = 0).$$

Why: constant P means energy is fed in at a steady rate, so the kinetic energy simply accumulates linearly with time.

Step 2 — Solve for speed:

$$\frac{1}{2}mv^2 = Pt \Rightarrow v = \sqrt{\frac{2Pt}{m}} \propto \sqrt{t}$$

So speed grows as the *square root* of time — not linearly. The particle speeds up rapidly at first, then more sluggishly (consistent with the “less force at high speed” idea from Section 1).

Step 3 — Integrate speed to get distance:

$$x = \int_0^t v dt = \sqrt{\frac{2P}{m}} \int_0^t t^{1/2} dt = \sqrt{\frac{2P}{m}} \cdot \frac{2}{3} t^{3/2}$$

$$x = \frac{2}{3} \sqrt{\frac{2P}{m}} t^{3/2} \propto t^{3/2}$$

Memorise the two scalings — $v \propto \sqrt{t}$ and $x \propto t^{3/2}$ — they answer a whole family of exam questions instantly.

(B) Procedure — choosing average vs instantaneous power

1. **Read what's asked.** “Total/over the whole trip/on average” → average power, $W_{\text{total}}/t_{\text{total}}$. “At this moment/at $t = \dots$ /when speed is ...” → instantaneous power, $\vec{F} \cdot \vec{v}$.
2. **For instantaneous power, find the force and the velocity at that instant** and take their dot product (mind the angle θ).
3. **For “constant velocity” problems, set the driving force equal to the total opposing force** (no net force, since $a = 0$), then use $P = F_{\text{drive}} v$.
4. **Always check units before multiplying** — convert km/h to m/s ($\div 3.6$) and remember 1 hp = 746 W.

(C) Power in mass-delivery systems (pumps, streams, machine guns)

A whole class of JEE problems involves a machine that throws *mass* at a steady rate — a pump ejecting water, a hose hitting a wall, a gun firing bullets, sand falling on a conveyor belt. The trick is to think in terms of **energy delivered per unit time**.

Suppose mass is delivered at a rate $\frac{dm}{dt} = \mu$ (in kg/s), each bit leaving with speed v . The kinetic energy handed out per second is the power:

$$P = \frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{1}{2} \mu v^2.$$

If the machine *also* lifts the mass through a height h before ejecting it (a pump raising water from a well), add the gravitational rate:

$$P = \mu \left(gh + \frac{1}{2} v^2 \right).$$

For a machine gun firing n bullets of mass m each in time t , the same logic gives $P = \frac{n(\frac{1}{2}mv^2)}{t}$. The recurring idea: **identify the energy given to each unit of mass, then multiply by how many units leave per second**.

Quick illustration: a hose ejects water at $\mu = 5$ kg/s with speed $v = 20$ m/s onto a wall. The power the pump must supply just to give this water its kinetic energy is $P = \frac{1}{2} \mu v^2 = \frac{1}{2} (5)(20)^2 = \frac{1}{2} (5)(400) = 1000$ W. If the same pump first lifted that water 4 m from a sump, you would add $\mu gh = 5 \times 10 \times 4 = 200$ W, for 1200 W total. Notice how the two energy “costs” — lifting and launching — simply add.

(D) A handy result — average is half the final power

For a body starting from rest under a *constant force*, the velocity rises linearly ($v = at$) while the instantaneous power $P = Fv$ rises linearly too. A quantity that climbs steadily from zero averages to *half* its final value, so

$$P_{\text{avg}} = \frac{1}{2} P_{\text{instantaneous (final)}}.$$

This one-liner lets you jump between the two without integrating — useful in MCQs.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A crane lifts a load of 200 kg to a height of 15 m in 30 s at a steady rate. Calculate the average power developed by the crane. Take $g = 10 \text{ m/s}^2$.

Given: $m = 200$ kg, $h = 15$ m, $t = 30$ s.

Work done against gravity:

$$W = mgh = (200)(10)(15) = 30000 \text{ J}$$

Average power:

$$P_{\text{avg}} = \frac{W}{t} = \frac{30000}{30} = \boxed{1000 \text{ W} = 1 \text{ kW}}$$

Comment: Clean three-step presentation — work first, then divide by time, then state the unit. The crane delivers one kilowatt of mechanical power. As a quick cross-check in familiar units, $1000 \text{ W} \div 746 \approx 1.34 \text{ hp}$ — about the rating of a small workshop motor, which is physically sensible for this load.

Example 2 — [NEET Speed Trap]

A car engine keeps the car moving at a constant speed of 72 km/h against a total resistive force of 800 N. The power developed by the engine is: (a) 16 kW (b) 57.6 kW (c) 16 W (d) 11.1 kW

The trap: Students plug 72 straight into $P = Fv$ and get $800 \times 72 = 57600 \text{ W} = 57.6 \text{ kW}$ — option (b), the planted wrong answer.

The fast reasoning: Power formulas need SI units, so **convert first:** $72 \text{ km/h} = \frac{72}{3.6} = 20 \text{ m/s}$. At constant speed the engine force just balances the resistance, so $F = 800 \text{ N}$. Then

$$P = Fv = 800 \times 20 = 16000 \text{ W} = 16 \text{ kW}.$$

Answer: (a) 16 kW. *Speed tip:* the instant you see km/h in a power or energy problem, divide by 3.6 before doing anything else. This single habit saves a guaranteed mark in NEET.

Example 3 — [JEE Main Level]

A car of mass 1000 kg climbs a road inclined at 1 in 10 (i.e. $\sin \theta = 0.1$) at a constant speed of 20 m/s. A constant friction force of 500 N opposes the motion. Find the power developed by the engine. Take $g = 10 \text{ m/s}^2$. (Combines: force balance on an incline + power.)

Step 1 — Force the engine must supply (constant speed $\Rightarrow$ zero acceleration $\Rightarrow$ engine force balances gravity-component *plus* friction):

$$F_{\text{engine}} = mg \sin \theta + f = (1000)(10)(0.1) + 500 = 1000 + 500 = 1500 \text{ N}$$

Step 2 — Power at this constant speed:

$$P = F_{\text{engine}} v = 1500 \times 20 = 30000 \text{ W} = \boxed{30 \text{ kW}}$$

Why this is JEE Main: you can't find the power without first resolving forces along the incline — two concepts stitched together. A common error is to forget the friction term (giving 20 kW) or the gravity-component term (giving 10 kW).

Example 4 — [JEE Advanced Level]

A particle of mass 2 kg, initially at rest, is acted upon by a constant power source of 12 W. (a) Find its speed at $t = 3$ s. (b) Find the distance it has travelled in this time. (Assume all the power goes into kinetic energy.)

(a) **Speed at $t = 3$ s.** Since power is constant and feeds kinetic energy, $K = Pt$:

$$\frac{1}{2}mv^2 = Pt \Rightarrow \frac{1}{2}(2)v^2 = (12)(3) = 36 \Rightarrow v^2 = 36$$

$$\boxed{v = 6 \text{ m/s}}$$

(b) **Distance travelled.** Using the constant-power result derived in Section 3:

$$x = \frac{2}{3}\sqrt{\frac{2P}{m}} t^{3/2} = \frac{2}{3}\sqrt{\frac{2(12)}{2}} (3)^{3/2} = \frac{2}{3}\sqrt{12} (3\sqrt{3})$$

Now $\sqrt{12} = 2\sqrt{3}$, so

$$x = \frac{2}{3}(2\sqrt{3})(3\sqrt{3}) = \frac{2}{3}(2 \times 3 \times 3) = \frac{2}{3}(18) = \boxed{12 \text{ m}}$$

Why this is JEE Advanced: the constant-power condition (not constant force) forces you to integrate — v comes from energy accumulation, and x from integrating $v \propto \sqrt{t}$. Students who reflexively use $v = at$ or $s = \frac{1}{2}at^2$ (constant-acceleration formulas) get it wrong, because under constant power the acceleration is *not* constant. To see this explicitly, differentiate $v = \sqrt{2Pt/m}$: the acceleration $a = \frac{dv}{dt} = \sqrt{\frac{P}{2mt}}$ actually *decreases* with time (it blows up at $t = 0$ and fades thereafter). That is the mathematical fingerprint of a constant-power drive — huge initial push, gently easing off — and it's exactly why the standard kinematic equations don't apply.

Section 5: Practice Exercises

1. [CBSE] A water pump raises 100 kg of water to a tank 20 m above in 25 s. Find the power of the pump. ($g = 10 \text{ m/s}^2$.)
2. [NEET] An engine of power 10 kW pulls a train at a constant speed of 5 m/s. Find the tractive (pulling) force exerted.
3. [JEE Main] A vehicle of mass 800 kg ascends an incline of 1 in 20 ($\sin \theta = 0.05$) at a constant 15 m/s against a resistive force of 300 N. Find the engine power. ($g = 10 \text{ m/s}^2$.)
4. [JEE Main] A pump lifts 600 kg of water per minute from a depth of 25 m and ejects it with a speed of 10 m/s. Find the power of the pump (account for both lifting and the kinetic energy of the ejected water; $g = 10 \text{ m/s}^2$).
5. [JEE Advanced] A particle of mass 1 kg starts from rest under a constant power of 9 W. Find (a) its speed at $t = 2$ s and (b) the distance it has covered by then.

Answers: 1. 800 W | 2. 2000 N | 3. 10.5 kW | 4. 3 kW (mass rate 10 kg/s; $P = 10 \times (gh + \frac{1}{2}v^2) = 10 \times (250 + 50)$) | 5. (a) 6 m/s, (b) 8 m

Section 6: Crack the MCQ

Q1. The dimensional formula of power is: (a) $[ML^2T^{-2}]$ (b) $[ML^2T^{-3}]$ (c) $[MLT^{-2}]$ (d) $[ML^2T^{-1}]$

Correct: (b) $[ML^2T^{-3}]$. Power = work/time = $[ML^2T^{-2}]/[T] = [ML^2T^{-3}]$. *Distractor analysis:* (a) is the dimension of *work/energy* — the most common slip, forgetting the extra division by time. (c) is force. (d) is angular momentum / action. Always carry the “per second” through.

Q2. A body starts from rest and moves under a *constant force*. Its instantaneous power: (a) remains constant (b) increases linearly with time (c) increases as the square of time (d) decreases with time

Correct: (b) increases linearly with time. With constant force, $v = at$, so $P = Fv = F(at) = (Fa)t \propto t$. *Distractor analysis:* (a) is the classic conceptual error — assuming “constant force” means “constant power.” It doesn’t: as the body speeds up, the *same* force delivers *more* power. (c) and (d) misread the linear v - t dependence.

Q3. One horsepower is approximately equal to: (a) 746 W (b) 550 W (c) 1000 W (d) 100 W

Correct: (a) 746 W. This is a standard defined conversion you should simply know. *Distractor analysis:* (b) is the value in ft·lb/s (the imperial definition), not watts. (c) confuses hp with the kilowatt. (d) has no basis. Memorise: **1 hp $\approx$ 746 W**.

Q4. A particle starts from rest and moves under *constant power*. Its speed varies with time as: (a) $v \propto t$ (b) $v \propto t^2$ (c) $v \propto \sqrt{t}$ (d) $v \propto t^{3/2}$

Correct: (c) $v \propto \sqrt{t}$. From $K = Pt$, $\frac{1}{2}mv^2 = Pt \Rightarrow v = \sqrt{2Pt/m} \propto \sqrt{t}$. *Distractor analysis:* (a) is the constant-force answer ($v = at$) — the trap for students who don’t distinguish constant force from constant power. (d) is actually the *distance* dependence ($x \propto t^{3/2}$), a tempting mismatch. (b) has no basis.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Confusing power with work/energy.** A more powerful machine does the *same* job *faster* — it doesn’t necessarily do more total work. Power is the *rate*; energy is the *amount*. Keep their dimensions distinct: power carries an extra T^{-1} .
2. **Forgetting unit conversions.** km/h must become m/s ($\div 3.6$) before using $P = Fv$, and hp must become watts ($\times 746$). The kilowatt-hour is *energy*, not power — never mix kWh into a $P = Fv$ calculation.
3. **Assuming constant force means constant power.** Since $P = Fv$, a constant force delivers *growing* power as the object accelerates. This single misconception drives several MCQ traps.
4. **Dropping $\cos \theta$ in $P = \vec{F} \cdot \vec{v}$.** Only the force component along the velocity delivers power. A force perpendicular to the motion (like the normal reaction, or centripetal force) delivers zero power.
5. **Mixing up average and instantaneous power.** $P = Fv$ at a chosen instant gives *instantaneous* power; dividing total work by total time gives *average* power. They coincide only when the power is constant. In an accelerating-from-rest problem, the final instantaneous power is *twice* the average — don’t quote one when the question wants the other.

Pro-Tip (speed + accuracy): For “constant speed” problems, skip Newton’s laws — set the driving force equal to the total opposing force and write $P = F_{\text{opposing}} \times v$ directly. For “constant power from rest” problems, recall the two memorised scalings, $v = \sqrt{2Pt/m}$ and $x \propto t^{3/2}$, instead of re-deriving. And whenever a body moves at constant velocity, its kinetic energy isn’t changing, so *all* the engine’s power is going into fighting resistance — a quick sanity anchor. For pump, hose, or machine-gun problems, go straight to “energy per unit mass $\times$ mass per second” rather than tracking individual forces — it collapses an intimidating problem into one line.

Section 8: Cheat Sheet Box

POWER — RAPID REVISION

- **Definition:** power = rate of doing work (rate of energy transfer). **Scalar.**
- **Average power:** $P_{\text{avg}} = \frac{W_{\text{total}}}{t_{\text{total}}}$
- **Instantaneous power:** $P = \frac{dW}{dt} = \vec{F} \cdot \vec{v} = Fv \cos \theta$
- **Dimensions:** $[ML^2T^{-3}]$; SI unit **watt** (W) = J/s; 1 hp = 746 W
- **Watch the units:** 1 kWh = 3.6×10^6 J is *energy*, not power; km/h $\rightarrow$ m/s by $\div 3.6$
- **Constant velocity:** $P = F_{\text{resist}} v$; crane/lift: $P = mgv$
- **Constant power from rest:** $K = Pt$, $v = \sqrt{\frac{2Pt}{m}} \propto \sqrt{t}$, $x \propto t^{3/2}$
- **Graphs:** area under P - t = work; slope of W - t = instantaneous power

Memory aids: “Power = how *FAST* you work, not how *MUCH*.” · “Constant force $\neq$ constant power (P grows with v).” · “See km/h? Divide by 3.6 first.”

Sub-topic 5: Collisions

Exam Relevance: Collisions are among the densest scoring zones in the chapter, blending momentum and energy conservation. CBSE Boards ask the *definitions* (elastic vs inelastic) and a clean perfectly-inelastic momentum numerical (2–3 marks). NEET asks **1–2 questions** yearly, loving the equal-mass velocity-exchange and the “momentum is conserved, KE may not be” conceptual trap. JEE Main reliably asks **1–2 questions** on the coefficient of restitution and 1-D elastic formulas. JEE Advanced favours **oblique (2-D) elastic collisions** — especially the elegant “equal masses fly apart at 90° ” result — and multi-body restitution chains.

Section 1: Concept Intuition

A collision, in physics, is simply an event where two bodies exert strong forces on each other for a very short time — a carrom striker smacking a coin, two cars crunching at an intersection, a cricket ball meeting the bat, even two protons slamming together in an accelerator. The forces are huge but the contact time is tiny, and that tiny-time fact is the key that unlocks everything. The product of that large force and the short time is the **impulse**, and it equals the change in momentum each body experiences — but because the two bodies push on each other with equal and opposite forces, the impulses are equal and opposite, so they cancel for the system. That is the deep reason momentum survives a collision intact.

Here’s the master principle, and you must internalise it before anything else: **momentum is conserved in every single collision — elastic, inelastic, sticky, explosive, all of them.** Why? Because during the fleeting moment of impact, the internal forces the two bodies exert on each other are *equal and opposite* (Newton’s third law), so they cancel for the system as a whole. External forces like gravity or friction *do* exist, but they’re so feeble compared to the violent contact force, and act for so short a time, that their impulse is negligible. So the total momentum just before impact equals the total momentum just after. Always.

Kinetic energy is a different story, and this is where collisions get their personalities. Sometimes the bodies bounce apart and the total kinetic energy is exactly the same afterwards as before — no energy got buried in heat, sound, or permanent dents. We call this an **elastic collision** (think of two hard steel balls, or idealised billiard balls). Other times, some kinetic energy gets *lost* — converted into heat, the crunch of metal, the squish of clay. That’s an **inelastic collision**, and almost every real-world collision is at least a little inelastic.

Think about it this way... drop a steel ball-bearing and a lump of wet atta dough from the same height onto a hard floor. The steel ball bounces back up almost to where it started — barely any kinetic energy lost, nearly elastic. The dough goes *splat* and doesn’t bounce at all — it has handed *all* of its “bounce-back” energy over to deforming itself. The dough is the extreme case, a **perfectly inelastic collision**, where the colliding bodies *stick together* and move as one afterwards. This is the case of **maximum possible kinetic energy loss** (you can’t lose more than this, because the bodies must still share whatever momentum the system started with).

To put a number on “how bouncy” a collision is, we use the **coefficient of restitution**, e — the ratio of how fast the bodies separate afterwards to how fast they approached beforehand. A perfectly elastic collision has $e = 1$ (they separate just as fast as they approached); a perfectly inelastic one has $e = 0$ (they don’t separate at all — they move together); everything real lives in between, $0 < e < 1$.

There’s a second way collisions are classified — by *geometry*. In a **head-on** (one-dimensional) collision,

the bodies travel along the same straight line both before and after, like two carrom coins meeting dead-centre. In an **oblique** (two-dimensional) collision, they approach and leave at angles, like a cue ball clipping another ball off-centre. The crucial subtlety for oblique collisions: the impact force acts only along the **line of impact** (the line joining the centres at contact), so *only the velocity components along that line change* — the components perpendicular to it sail through untouched. This single idea reduces most 2-D collision problems to a 1-D collision along the line of impact plus an unchanged sideways motion.

A two-row schematic. Top row “Head-on (1-D)”: two balls m_1 and m_2 on a horizontal line with velocity arrows u_1, u_2 pointing along the line before, and v_1, v_2 after. Bottom row “Oblique (2-D)”: ball m_1 approaching a stationary m_2 , then both leaving at angles above and below the original line, with the dashed “line of impact” marked through the contact point.

Limiting conditions & assumptions:

1. Momentum conservation assumes the system is **isolated during impact** — external impulsive forces (a bat being held, a wall) along the line of collision can break it. Always check the direction.
2. The collision time is assumed **negligibly short**, so the bodies barely move during the impact and gravity/friction can be ignored.
3. KE conservation holds **only for elastic collisions** ($e = 1$). Never apply it to inelastic ones.
4. In **oblique** collisions, only the velocity components *along the line of impact* change; the perpendicular components are untouched.

Section 2: Key Formulas & Definitions

Conservation of momentum (ALL collisions):

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

where u = velocities before, v = velocities after, and signs encode direction.

Coefficient of restitution:

$$e = \frac{\text{velocity of separation}}{\text{velocity of approach}} = \frac{v_2 - v_1}{u_1 - u_2}$$

$e = 1$ (perfectly elastic), $0 < e < 1$ (inelastic), $e = 0$ (perfectly inelastic).

Elastic collision in 1-D — final velocities:

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \left(\frac{2m_2}{m_1 + m_2} \right) u_2$$

$$v_2 = \left(\frac{2m_1}{m_1 + m_2} \right) u_1 + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_2$$

Elastic shortcut (no masses needed): velocity of approach = velocity of separation:

$$u_1 - u_2 = v_2 - v_1$$

Perfectly inelastic collision — common velocity and energy loss:

$$V = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}, \quad \Delta K = \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) (u_1 - u_2)^2$$

Variable definitions: m_1, m_2 = masses; u_1, u_2 / v_1, v_2 = velocities before/after; V = common final velocity; e = coefficient of restitution; ΔK = kinetic energy lost. All energies in joules $[ML^2T^{-2}]$; momentum in $\text{kg}\cdot\text{m}\cdot\text{s}^{-1}$ $[MLT^{-1}]$.

Three special cases of 1-D elastic collision (target at rest, $u_2 = 0$):

- **Equal masses** ($m_1 = m_2$): velocities are *exchanged* — $v_1 = 0$, $v_2 = u_1$.
- **Heavy hits light** ($m_1 \gg m_2$): $v_1 \approx u_1$, $v_2 \approx 2u_1$ (light body shoots off at double speed).
- **Light hits heavy** ($m_1 \ll m_2$): $v_1 \approx -u_1$, $v_2 \approx 0$ (light body bounces straight back).

Section 3: Key Derivations & Procedures

(A) The elastic 1-D shortcut: approach speed = separation speed

Setup: Two bodies, masses m_1, m_2 , velocities u_1, u_2 before and v_1, v_2 after a head-on *elastic* collision.

Step 1 — Momentum conservation:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \Rightarrow m_1(u_1 - v_1) = m_2(v_2 - u_2) \quad (\text{i})$$

Step 2 — Kinetic energy conservation (elastic):

$$\frac{1}{2}m_1 u_1^2 + \frac{1}{2}m_2 u_2^2 = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 \Rightarrow m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2) \quad (\text{ii})$$

Step 3 — Divide (ii) by (i) (factor each side as a difference of squares):

$$\frac{m_1(u_1 - v_1)(u_1 + v_1)}{m_1(u_1 - v_1)} = \frac{m_2(v_2 - u_2)(v_2 + u_2)}{m_2(v_2 - u_2)} \Rightarrow u_1 + v_1 = v_2 + u_2$$

Step 4 — Rearrange:

$$\boxed{u_1 - u_2 = v_2 - v_1}$$

Physical meaning: in any 1-D elastic collision, the speed at which the bodies approach each other equals the speed at which they separate — **regardless of their masses**. This is exactly the $e = 1$ statement, and it lets you replace the messy quadratic KE equation with a simple linear one, which (paired with momentum) solves any elastic collision quickly.

(B) Why equal masses *exchange* velocities

Put $m_1 = m_2 = m$ into momentum conservation: $u_1 + u_2 = v_1 + v_2$. Pair this with the elastic shortcut $u_1 - u_2 = v_2 - v_1$. Adding and subtracting these two simple equations gives $v_1 = u_2$ and $v_2 = u_1$ — the bodies simply *trade* velocities. This is the physics behind Newton's cradle: the incoming ball stops dead and the far ball leaves with its velocity.

(C) Procedure — solving any collision

1. **Write momentum conservation first** — it holds for *every* collision. Mind the signs (pick a positive direction and stick to it).
2. **Add the right second equation:**
 - *Elastic?* Use the shortcut $u_1 - u_2 = v_2 - v_1$ (cleaner than the KE equation).
 - *Given e ?* Use $e = \frac{v_2 - v_1}{u_1 - u_2}$.
 - *Perfectly inelastic?* Set $v_1 = v_2 = V$ (they stick) — momentum alone then suffices.
3. **Solve the two equations** for the two unknown final velocities.
4. **Sanity-check** with momentum (and, if elastic, KE).

(D) How much energy is lost when bodies stick

For a perfectly inelastic collision, the bodies move off together at $V = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$. The kinetic energy lost is

$$\Delta K = \left(\frac{1}{2}m_1 u_1^2 + \frac{1}{2}m_2 u_2^2\right) - \frac{1}{2}(m_1 + m_2)V^2.$$

Substituting V and simplifying (the algebra collapses neatly) yields the compact result

$$\Delta K = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (u_1 - u_2)^2$$

Two insights fall out of this form. First, the loss depends on the **relative velocity** ($u_1 - u_2$) — if the bodies were moving together at the same speed, nothing would be lost (nothing actually “collides”). Second, the combination $\frac{m_1 m_2}{m_1 + m_2}$ (the *reduced mass*) shows the loss is largest when the masses are comparable. This is also the **maximum** energy any collision with these initial conditions can lose — no collision can shed more than the perfectly inelastic one, because the bodies must still carry the system’s conserved momentum.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A 2 kg lump of clay moving at 6 m/s strikes a stationary 4 kg block and sticks to it. (a) Find their common velocity after collision. (b) How much kinetic energy is lost?

Given: $m_1 = 2$ kg, $u_1 = 6$ m/s, $m_2 = 4$ kg, $u_2 = 0$. They stick $\rightarrow$ perfectly inelastic.

(a) **Common velocity** (momentum conservation):

$$V = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2} = \frac{(2)(6) + (4)(0)}{2 + 4} = \frac{12}{6} = \boxed{2 \text{ m/s}}$$

(b) **Kinetic energy lost:**

$$K_i = \frac{1}{2}m_1 u_1^2 = \frac{1}{2}(2)(6)^2 = 36 \text{ J}, \quad K_f = \frac{1}{2}(m_1 + m_2)V^2 = \frac{1}{2}(6)(2)^2 = 12 \text{ J}$$

$$\Delta K = K_i - K_f = 36 - 12 = \boxed{24 \text{ J lost}}$$

Comment: Momentum is conserved (it must be), but 24 J of kinetic energy vanished into heat and deformation of the clay — the signature of an inelastic collision.

Example 2 — [NEET Speed Trap]

A ball moving at 8 m/s makes a head-on **elastic** collision with an identical stationary ball. The velocities of the two balls after the collision are: (a) both 4 m/s (b) 0 and 8 m/s (c) 8 and 0 m/s (d) -8 and 8 m/s

The trap: Option (a) “both 4” conserves momentum ($8 = 4 + 4$), so a student who checks only momentum picks it. But it fails kinetic energy: before, $\frac{1}{2}m(8^2) = 32m$; after, $\frac{1}{2}m(4^2) + \frac{1}{2}m(4^2) = 16m$. Energy isn’t conserved, so (a) cannot be an *elastic* collision.

The fast reasoning: Equal masses + elastic + head-on $\rightarrow$ they **exchange velocities**. The incoming ball stops ($v_1 = 0$), the target leaves with 8 m/s ($v_2 = 8$).

Answer: (b) 0 and 8 m/s. *Speed tip:* an elastic collision demands you satisfy **both** conservation laws. Whenever masses are equal and the collision is elastic, skip the algebra entirely — just swap the velocities.

Example 3 — [JEE Main Level]

A ball is dropped from a height of 5 m onto a hard floor and rebounds to a height of 1.8 m. (a) Find the coefficient of restitution between the ball and the floor. (b) To what height will it rise after the **second** bounce? (Combines: free-fall kinematics + coefficient of restitution.)

Step 1 — Relate restitution to heights. A ball falling from height h arrives with speed $\sqrt{2gh}$; rebounding to height h' means it left with speed $\sqrt{2gh'}$. The floor is stationary, so

$$e = \frac{\text{speed after}}{\text{speed before}} = \frac{\sqrt{2gh'}}{\sqrt{2gh}} = \sqrt{\frac{h'}{h}}.$$

(a) With $h = 5$ m, $h' = 1.8$ m:

$$e = \sqrt{\frac{1.8}{5}} = \sqrt{0.36} = \boxed{0.6}$$

(b) Each bounce multiplies the rebound height by e^2 (since height $\propto$ speed²). The first bounce reached 1.8 m; the second reaches

$$h_2 = e^2 \times 1.8 = 0.36 \times 1.8 = \boxed{0.648 \text{ m}}$$

Why this is JEE Main: it fuses two ideas — converting drop/rise heights into impact speeds, and applying restitution — and rewards spotting the clean pattern $h_n = e^{2n} h$ rather than re-computing each bounce from scratch.

Example 4 — [JEE Advanced Level]

A particle makes an **oblique elastic** collision with a second, identical particle that is initially at rest. (a) Prove that after the collision the two particles move at right angles to each other. (b) If one particle is found to move at 30° to the original line of motion, find the direction of the other and the ratio of their speeds. (Take the incoming speed as u and assume both particles move after impact.)

(a) **The 90° theorem.** Let the incoming velocity be $\vec{u}$ and the two final velocities $\vec{v}_1, \vec{v}_2$. With equal masses, conservation of momentum gives

$$\vec{u} = \vec{v}_1 + \vec{v}_2.$$

Squaring (taking the dot product of each side with itself):

$$u^2 = v_1^2 + v_2^2 + 2\vec{v}_1 \cdot \vec{v}_2. \quad (\text{i})$$

Conservation of kinetic energy (elastic, equal masses) gives

$$u^2 = v_1^2 + v_2^2. \quad (\text{ii})$$

Comparing (i) and (ii) forces $\vec{v}_1 \cdot \vec{v}_2 = 0$. Since neither speed is zero, the two final velocities are **perpendicular** — they separate at exactly 90° . ■

(b) The specific angles. If one particle goes off at 30° on one side of the original direction, the 90° result puts the other at 60° on the *other* side. Resolve momentum along (x) and perpendicular to (y) the original line:

$$(\text{y}): v_1 \sin 30^\circ = v_2 \sin 60^\circ \Rightarrow v_1(0.5) = v_2(0.866) \Rightarrow v_1 = \sqrt{3}v_2$$

$$(\text{x}): v_1 \cos 30^\circ + v_2 \cos 60^\circ = u \Rightarrow (\sqrt{3}v_2)(0.866) + v_2(0.5) = u \Rightarrow 2v_2 = u$$

So $v_2 = \frac{u}{2}$ and $v_1 = \frac{\sqrt{3}}{2}u$, giving a speed ratio

$$\boxed{\frac{v_1}{v_2} = \sqrt{3}}$$

(Quick check: $v_1^2 + v_2^2 = \frac{3}{4}u^2 + \frac{1}{4}u^2 = u^2$ ✓ — kinetic energy is conserved, as it must be.)

Why this is JEE Advanced: the proof uses the vector form of momentum together with energy conservation, and the numerical part demands clean 2-D resolution. The 90° result itself is a frequently-tested gem worth memorising.

Section 5: Practice Exercises

1. [CBSE] A 3 kg body moving at 4 m/s collides with and sticks to a stationary 1 kg body. Find their common velocity after the collision.
2. [NEET] Two identical balls undergo a head-on elastic collision; one moves at 10 m/s and the other is at rest. State their velocities after the collision.
3. [JEE Main] A ball dropped from 2 m rebounds to a height of 0.5 m. Find the coefficient of restitution.
4. [JEE Main] A 2 kg body moving at 5 m/s collides head-on with a 3 kg body moving at 2 m/s in the *opposite* direction; they stick together. Find the common velocity (state its direction).
5. [JEE Advanced] A 1 kg ball moving at 6 m/s collides head-on **elastically** with a stationary 2 kg ball. Find the velocity of each ball after the collision.

Answers: 1. 3 m/s | 2. 0 and 10 m/s (they exchange) | 3. $e = 0.5$ | 4. 0.8 m/s in the direction of the 2 kg body | 5. $v_1 = -2$ m/s (rebounds), $v_2 = +4$ m/s

Section 6: Crack the MCQ

Q1. In which type of collision is linear momentum conserved? (a) elastic only (b) inelastic only (c) both elastic and inelastic (d) neither

Correct: (c) both. Momentum conservation follows from Newton's third law during impact and holds for *every* collision, regardless of energy loss. *Distractor analysis:* (a) confuses the special property of elastic collisions (energy conservation) with the universal one (momentum). (b) and (d) misunderstand the principle entirely. The one-line takeaway: *momentum always, energy only if elastic.*

Q2. The coefficient of restitution for a perfectly inelastic collision is: (a) 1 (b) 0 (c) infinite (d) between 0 and 1

Correct: (b) 0. The bodies stick and move together, so their velocity of separation is zero, giving $e = 0$. *Distractor analysis:* (a) is the value for a *perfectly elastic* collision — a direct swap of the two extremes. (d) describes a partially inelastic collision. (c) is non-physical, since e can never exceed 1.

Q3. A very heavy ball moving with speed u collides head-on elastically with a very light stationary ball. The light ball moves off with a speed of approximately: (a) u (b) $2u$ (c) $u/2$ (d) zero

Correct: (b) $2u$. For $m_1 \gg m_2$ with the light body at rest, the heavy ball ploughs on at nearly u , while the light ball is flung off at about $2u$. *Distractor analysis:* (a) is the *heavy* ball's speed, not the light one's. (c) halves it for no reason. (d) would be true if the masses were reversed (light hitting heavy). Recall the limiting cases as a set so you don't mix them up.

Q4. Two identical balls, one moving and one stationary, undergo a head-on elastic collision. Immediately after: (a) both move with half the original speed (b) they exchange velocities (c) they stick and move together (d) both reverse direction

Correct: (b) they exchange velocities. Equal masses + elastic + head-on means the mover stops and the target takes off with the mover's velocity. *Distractor analysis:* (a) conserves momentum but **not** kinetic energy, so it can't be elastic — the classic trap. (c) describes a perfectly inelastic collision, not an elastic one. (d) violates momentum conservation outright.

Section 7: Common Student Pitfalls & Pro-Tips

- 1. Applying kinetic-energy conservation to an inelastic collision.** KE is conserved **only** when $e = 1$ (elastic). For inelastic and perfectly inelastic collisions, use momentum (plus e , or the stick-together condition) — never set $K_i = K_f$.
- 2. Saying momentum is “lost” in an inelastic collision.** Momentum is **always** conserved. What's lost is *kinetic energy* (to heat, sound, deformation). Don't confuse the two conserved-or-not statuses.
- 3. Dropping the sign on velocities.** Momentum is a vector. A ball that rebounds has a *negative* velocity relative to its approach; forgetting this corrupts the momentum equation. Fix a positive direction at the start.
- 4. Over-using the velocity-exchange rule.** “They swap velocities” is true **only** for *equal masses* in an *elastic head-on* collision. Apply it to unequal masses or inelastic collisions and you'll get

nonsense.

5. **Mishandling oblique collisions.** The coefficient of restitution and the collision equations apply only to the velocity components *along the line of impact*. The perpendicular components don't change at all. Treating the *total* speeds with the restitution formula — instead of resolving along and across the line of impact first — is a frequent JEE Advanced slip.

Pro-Tip (speed + accuracy): Build a fixed two-step reflex. First, *always* write momentum conservation — it's true for every collision. Second, attach the right partner equation: the approach-equals-separation shortcut if elastic, $e = (v_2 - v_1)/(u_1 - u_2)$ if a restitution is given, or $v_1 = v_2 = V$ if they stick. For oblique elastic collisions between equal masses, remember the bodies fly apart at 90° — it often hands you an angle for free. These shortcuts turn a two-equation grind into a few lines.

Section 8: Cheat Sheet Box

COLLISIONS — RAPID REVISION

- **Momentum conserved in ALL collisions:** $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$
- **Elastic** ($e = 1$): KE also conserved; **approach speed = separation speed**, $u_1 - u_2 = v_2 - v_1$
- **Inelastic** ($0 < e < 1$): KE lost. **Perfectly inelastic** ($e = 0$): bodies stick, **maximum** KE loss
- **Coefficient of restitution:** $e = \frac{\text{velocity of separation}}{\text{velocity of approach}} = \frac{v_2 - v_1}{u_1 - u_2}$
- **Elastic 1-D:** $v_1 = \frac{(m_1 - m_2)u_1 + 2m_2u_2}{m_1 + m_2}$, $v_2 = \frac{(m_2 - m_1)u_2 + 2m_1u_1}{m_1 + m_2}$
- **Equal masses, elastic, head-on** → **exchange velocities**
- **Perfectly inelastic:** $V = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$; KE loss = $\frac{1}{2} \frac{m_1m_2}{m_1 + m_2} (u_1 - u_2)^2$
- **Equal masses, oblique, elastic (one at rest)** → **separate at 90°**
- **Bounce heights:** $e = \sqrt{h'/h}$; after n bounces $h_n = e^{2n}h$

Memory aids: “Momentum always; energy only if elastic.” · “Stick together = $e = 0$ = maximum energy lost.” · “Equal-mass elastic = swap.”

Application: Motion in a Vertical Circle

Exam Relevance: This is the flagship application of conservation of mechanical energy, and a near-guaranteed exam topic. CBSE Boards ask the *minimum-speed* derivations (3 marks). NEET asks **1 question** yearly — usually minimum speed at the top, or the $T_{\text{bottom}} - T_{\text{top}} = 6mg$ result. JEE Main reliably asks **1–2 questions** combining energy conservation with the tension equation. JEE Advanced favours the **regime analysis** (complete the loop vs leave the track vs oscillate) and the angle at which a string goes slack.

Section 1: Concept Intuition

Tie a small stone to a string and whirl it in a vertical circle — the kind of thing you’ve seen with a bucket of water swung overhead without a drop falling out. Something interesting is happening: the stone speeds up as it swings down and slows as it climbs, because gravity does positive work on the way down and negative work on the way up. The motion is *not* uniform circular motion — the speed changes continuously around the loop.

Two forces act on the stone: gravity (always straight down) and the string tension (always along the string, pulling *inward* toward the centre). At every instant, the *net inward force* must supply the centripetal force $\frac{mv^2}{R}$ needed to keep the stone on its circular path. The trouble spot is the **top** of the circle. There, both gravity and tension point downward — toward the centre — so they *add* to provide the centripetal force. But a string can only *pull*, never push. If the stone moves too slowly at the top, the required centripetal force becomes less than gravity alone, the string goes slack, and the stone falls out of its circular path.

Think about it this way... the bucket of water overhead stays full only if you swing it fast enough. At the very top, the water needs a downward centripetal force to keep curving with the bucket. If the bucket (and water) move fast enough, gravity alone isn’t enough — the bucket’s base has to *push down* on the water too, and by Newton’s third law the water pushes up on the base, staying put. But swing too slowly and gravity is *more* than the centripetal requirement; the water “wants” to fall faster than the circle curves, leaves the base, and spills. The critical case — the slowest possible successful swing — is when, at the top, **gravity alone exactly provides the centripetal force** and the contact force (or string tension) drops to zero.

That single condition, “tension = 0 at the top,” is the key that unlocks every vertical-circle problem. Setting it gives the **minimum speed at the top**, $v_{\text{top}} = \sqrt{gR}$. Then energy conservation carries that condition anywhere else on the loop — most usefully to the bottom, giving the **minimum launch speed** $v_{\text{bottom}} = \sqrt{5gR}$ needed to *just* complete the circle.

Limiting conditions & assumptions:

1. These string/bucket results assume a flexible string (or the inside of a track) that can **only pull/push inward** — it cannot push outward. A *rigid rod* changes the answers (it can support the bob, so the top speed can be zero).
2. The track/string is assumed of fixed length R and the bob a point mass; air resistance is neglected.
3. The speed varies around the loop — never use a single constant speed. Always connect two points with **energy conservation**.

Section 2: Key Formulas & Definitions

Centripetal (radial) equation at angle θ measured from the lowest point:

$$T - mg \cos \theta = \frac{mv^2}{R} \Rightarrow T = \frac{mv^2}{R} + mg \cos \theta$$

(At the bottom $\theta = 0$: $T - mg = \frac{mv^2}{R}$. At the top $\theta = 180^\circ$: $T + mg = \frac{mv^2}{R}$.)

Speed at any point via energy conservation (height h above the lowest point):

$$v^2 = v_{\text{bottom}}^2 - 2gh, \quad h = R(1 - \cos \theta)$$

Critical (minimum) speeds for a string/track to *just* complete the loop:

$$v_{\text{top}} = \sqrt{gR}, \quad v_{\text{bottom}} = \sqrt{5gR}, \quad v_{\text{side}} = \sqrt{3gR}$$

(side = at the horizontal level of the centre).

Tensions in the minimum (just-completing) case:

$$T_{\text{top}} = 0, \quad T_{\text{bottom}} = 6mg$$

Tension difference (true for *any* speed, by energy conservation):

$$T_{\text{bottom}} - T_{\text{top}} = 6mg$$

Variable definitions: T = string tension, m = mass, R = radius, v = speed at the point, θ = angle from the lowest point, h = height above the lowest point, g = gravitational acceleration.

Three regimes (string/track, launch speed u at the bottom):

- $u \geq \sqrt{5gR} \rightarrow$ **completes the full vertical circle.**
- $u \leq \sqrt{2gR} \rightarrow$ **oscillates** like a pendulum (never rises above the horizontal).
- $\sqrt{2gR} < u < \sqrt{5gR} \rightarrow$ string goes **slack in the upper half; the bob leaves the circular path** and becomes a projectile.

Rigid rod (instead of string): a rod can push outward, so the minimum top speed is $v_{\text{top}} = 0$, giving a minimum bottom speed of $v_{\text{bottom}} = \sqrt{4gR} = 2\sqrt{gR}$ to complete the loop.

Section 3: CBSE Board Derivation — Minimum Speeds

Setup / assumptions: A bob of mass m on a string of length R moves in a vertical circle. We find the slowest speed at the top that keeps the string taut, then the corresponding speed at the bottom.

Step 1 — Condition at the top. At the topmost point, both the weight mg and the tension T point downward (toward the centre). The radial equation is

$$T + mg = \frac{mv_{\text{top}}^2}{R}.$$

Why: the net inward force supplies the centripetal force.

Step 2 — Impose the critical (minimum) condition. The slowest the bob can go while keeping the string taut is when the tension just reaches zero ($T = 0$):

$$mg = \frac{mv_{\text{top}}^2}{R} \Rightarrow \boxed{v_{\text{top}} = \sqrt{gR}}$$

Why: below this speed, gravity exceeds the centripetal requirement, the string slackens, and the bob leaves the circle.

Step 3 — Relate top and bottom by energy conservation. Going from the bottom to the top, the bob rises a height $2R$. With only gravity doing work (tension is always perpendicular to motion and does none):

$$\frac{1}{2}mv_{\text{bottom}}^2 = \frac{1}{2}mv_{\text{top}}^2 + mg(2R)$$

$$v_{\text{bottom}}^2 = v_{\text{top}}^2 + 4gR = gR + 4gR = 5gR \Rightarrow \boxed{v_{\text{bottom}} = \sqrt{5gR}}$$

Step 4 — Tension at the bottom (minimum case). At the bottom, tension points up and gravity down:

$$T_{\text{bottom}} - mg = \frac{mv_{\text{bottom}}^2}{R} = \frac{m(5gR)}{R} = 5mg \Rightarrow T_{\text{bottom}} = 6mg.$$

Combined with $T_{\text{top}} = 0$, this gives the memorable result $T_{\text{bottom}} - T_{\text{top}} = 6mg$ — which in fact holds for *any* speed, not just the minimum (the $6mg$ comes purely from the height difference $2R$ via energy conservation).

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A stone of mass 0.2 kg is tied to a string of length 1 m and whirled in a vertical circle. Find (a) the minimum speed at the highest point and (b) the minimum speed at the lowest point needed to just complete the circle. Take $g = 10 \text{ m/s}^2$.

Given: $m = 0.2 \text{ kg}$, $R = 1 \text{ m}$.

(a) Minimum speed at the top:

$$v_{\text{top}} = \sqrt{gR} = \sqrt{(10)(1)} = \sqrt{10} \approx \boxed{3.16 \text{ m/s}}$$

(b) Minimum speed at the bottom:

$$v_{\text{bottom}} = \sqrt{5gR} = \sqrt{5(10)(1)} = \sqrt{50} \approx \boxed{7.07 \text{ m/s}}$$

Comment: The mass never entered the speed answers — the critical speeds depend only on g and R . (Mass *would* matter if the question asked for tension.)

Example 2 — [NEET Speed Trap]

A body of mass 0.5 kg is whirled in a vertical circle. The difference between the maximum and minimum tension in the string is: (a) needs the speed to find (b) 30 N (c) 5 N (d) 15 N

The trap: Students assume they need the speed or radius and pick (a), or compute a tension at one point only.

The fast reasoning: Regardless of speed (as long as the bob completes the circle), the difference between the bottom (maximum) and top (minimum) tension is always

$$T_{\text{bottom}} - T_{\text{top}} = 6mg = 6(0.5)(10) = 30 \text{ N}.$$

Answer: (b) 30 N. *Speed tip:* memorise $T_{\text{bottom}} - T_{\text{top}} = 6mg$ — it needs *neither speed nor radius*, only the mass, and turns a scary-looking question into a one-line answer.

Example 3 — [JEE Main Level]

A 2 kg ball is whirled in a vertical circle of radius 0.5 m. At the lowest point its speed is 6 m/s. Find (a) its speed at the highest point and (b) the tension in the string at the lowest point. Take $g = 10 \text{ m/s}^2$. Does it complete the circle? (Combines: energy conservation + centripetal tension equation.)

Step 1 — Speed at the top (energy conservation, rise = $2R = 1 \text{ m}$):

$$v_{\text{top}}^2 = v_{\text{bottom}}^2 - 4gR = (6)^2 - 4(10)(0.5) = 36 - 20 = 16 \Rightarrow v_{\text{top}} = 4 \text{ m/s}$$

Check completion: the minimum top speed is $\sqrt{gR} = \sqrt{(10)(0.5)} = \sqrt{5} \approx 2.24 \text{ m/s}$. Since $4 > 2.24$, the string stays taut — **yes, it completes the circle.**

Step 2 — Tension at the bottom:

$$T_{\text{bottom}} = \frac{mv_{\text{bottom}}^2}{R} + mg = \frac{(2)(6)^2}{0.5} + (2)(10) = \frac{72}{0.5} + 20 = 144 + 20 = \boxed{164 \text{ N}}$$

Why this is JEE Main: it threads energy conservation (to get the top speed) into the centripetal equation (to get the tension), with a completion check in between — the standard multi-step vertical-circle problem.

Example 4 — [JEE Advanced Level]

A bob on a string of length R is given a speed $u = \sqrt{3gR}$ at the lowest point. Show that the bob does not complete the circle, and find the angle (measured from the lowest point) at which the string goes slack.

Step 1 — Will it complete the loop? The launch speed is $u = \sqrt{3gR}$. Since $\sqrt{2gR} < \sqrt{3gR} < \sqrt{5gR}$, the bob is in the middle regime: it rises past the horizontal but the string slackens before the top — it does **not** complete the circle.

Step 2 — Speed at angle θ from the bottom (height $h = R(1 - \cos \theta)$):

$$v^2 = u^2 - 2gR(1 - \cos \theta) = 3gR - 2gR + 2gR \cos \theta = gR(1 + 2 \cos \theta)$$

Step 3 — Condition for the string to go slack ($T = 0$). The radial equation $T = \frac{mv^2}{R} + mg \cos \theta$ set to zero gives

$$\frac{mv^2}{R} = -mg \cos \theta \Rightarrow v^2 = -gR \cos \theta$$

(possible only where $\cos \theta < 0$, i.e. in the upper half — as expected).

Step 4 — Equate the two expressions for v^2 :

$$gR(1 + 2 \cos \theta) = -gR \cos \theta \Rightarrow 1 + 2 \cos \theta = -\cos \theta \Rightarrow 3 \cos \theta = -1$$

$$\cos \theta = -\frac{1}{3} \Rightarrow \boxed{\theta = \cos^{-1}\left(-\frac{1}{3}\right) \approx 109.5^\circ}$$

So the string goes slack at about 109.5° from the bottom (roughly 19.5° above the horizontal), after which the bob flies off as a projectile.

Why this is JEE Advanced: it requires recognising the regime, writing $v^2(\theta)$ from energy conservation, imposing $T = 0$, and solving simultaneously — the complete toolkit for “where does it leave the track?” problems.

Section 5: Practice Exercises

- [CBSE] A ball tied to a string of length 0.4 m is whirled in a vertical circle. Find the minimum speed at the top. ($g = 10 \text{ m/s}^2$.)
- [NEET] For a 1 kg stone whirled in a vertical circle, find the difference between the tensions at the lowest and highest points. ($g = 10 \text{ m/s}^2$.)
- [JEE Main] A bob of mass 0.5 kg on a string of radius 1 m has a speed of 5 m/s at the top of a vertical circle. Find the tension at the top. ($g = 10 \text{ m/s}^2$.)
- [JEE Main] What minimum speed at the lowest point lets a ball complete a vertical circle of radius 0.8 m? ($g = 10 \text{ m/s}^2$.)
- [JEE Advanced] A mass on a *rigid rod* of length R is whirled in a vertical circle. Find the minimum speed at the lowest point required to complete the circle, and explain why it differs from the string case.

Answers: 1. $\sqrt{gR} = 2 \text{ m/s}$ | 2. $6mg = 60 \text{ N}$ | 3. $T = \frac{mv^2}{R} - mg = \frac{0.5 \times 25}{1} - 5 = 12.5 - 5 = 7.5 \text{ N}$ | 4. $\sqrt{5gR} = \sqrt{40} \approx 6.32 \text{ m/s}$ | 5. $v_{\text{bottom}} = \sqrt{4gR} = 2\sqrt{gR}$; a rod can *push*, so the top speed can be zero ($v_{\text{top}} = 0$), unlike a string which must stay taut ($v_{\text{top}} = \sqrt{gR}$)

Section 6: Crack the MCQ

Q1. The minimum speed at the highest point of a vertical circle of radius R (string) is: (a) $\sqrt{2gR}$ (b) $\sqrt{gR}$ (c) $\sqrt{5gR}$ (d) zero

Correct: (b) $\sqrt{gR}$. At the top, the minimum-speed condition is tension = 0, so gravity alone supplies the centripetal force: $mg = mv^2/R \Rightarrow v = \sqrt{gR}$. *Distractor analysis:* (c) $\sqrt{5gR}$ is the speed at the *bottom*, not the top. (a) has no basis here. (d) zero applies to a *rigid rod*, not a string — a string would go slack.

Q2. For a body completing a vertical circle, the ratio of the maximum to minimum speed (bottom to top, minimum case) is: (a) 5 : 1 (b) $\sqrt{5}$: 1 (c) 3 : 1 (d) 6 : 1

Correct: (b) $\sqrt{5}$: 1. The speeds are $v_{\text{bottom}} = \sqrt{5gR}$ and $v_{\text{top}} = \sqrt{gR}$, so the ratio is $\sqrt{5gR} : \sqrt{gR} = \sqrt{5} : 1$. *Distractor analysis:* (a) 5 : 1 is the ratio of the *squares* of the speeds (i.e. of kinetic energies), not the speeds themselves. (d) 6 : 1 is the tension situation, not speed. (c) has no basis.

Q3. The difference in tension between the lowest and highest points of a vertical circle (mass m) is: (a) $2mg$ (b) $3mg$ (c) $6mg$ (d) depends on the speed

Correct: (c) $6mg$. Energy conservation over the height difference $2R$ makes $T_{\text{bottom}} - T_{\text{top}} = 6mg$, independent of speed. *Distractor analysis:* (d) is the tempting trap — it *feels* like faster motion should change the difference, but the speed-dependence cancels, leaving exactly $6mg$. (a) and (b) under-count the contributions.

Q4. A ball is given a speed $\sqrt{2gR}$ at the lowest point of a vertical circle (string). It will: (a) complete the circle (b) oscillate without rising above the horizontal (c) leave the track in the upper half (d) stop at the top

Correct: (b) oscillate without rising above the horizontal. With $u = \sqrt{2gR}$, the bob just reaches the horizontal level ($h = R$) where its speed becomes zero, then swings back — pendulum-like motion. *Distractor analysis:* (a) needs $u \geq \sqrt{5gR}$. (c) needs $\sqrt{2gR} < u < \sqrt{5gR}$ (strictly greater). (d) “stop at the top” can’t happen on a string — it would have slackened far earlier. The launch speed sorts the outcome into one of three regimes.

Section 7: Common Student Pitfalls & Pro-Tips

- Using a single constant speed around the loop.** The speed changes continuously (slowest at top, fastest at bottom). Always link two points with **energy conservation**; never assume uniform circular motion.
- Forgetting that a string can’t push.** The “tension = 0” critical condition exists *because* a string only pulls. A rigid rod *can* push, so its minimum top speed is zero — a different answer ($v_{\text{bottom}} = 2\sqrt{gR}$).
- Mixing up the critical speeds.** $\sqrt{gR}$ is at the **top**, $\sqrt{5gR}$ at the **bottom**, $\sqrt{3gR}$ at the **side**. Sketch the circle and label them to avoid swapping.
- Thinking $T_{\text{bottom}} - T_{\text{top}} = 6mg$ needs the speed.** It doesn’t — the result is speed-independent. Reaching for the speed wastes time.

Pro-Tip (speed + accuracy): Tackle every vertical-circle problem in the same order: (1) apply the radial equation at the relevant point, (2) connect points with energy conservation ($v^2 = u^2 - 2gh$), (3) use the critical conditions ($T = 0$ at the top for the minimum case). Keep the three speeds $\sqrt{gR}$, $\sqrt{3gR}$, $\sqrt{5gR}$ and the result $T_{\text{bottom}} - T_{\text{top}} = 6mg$ on instant recall — they alone answer most MCQs.

Section 8: Cheat Sheet Box

MOTION IN A VERTICAL CIRCLE — RAPID REVISION

- **Radial equation** (angle θ from bottom): $T = \frac{mv^2}{R} + mg \cos \theta$
- **Energy link:** $v^2 = v_{\text{bottom}}^2 - 2gh$, $h = R(1 - \cos \theta)$
- **Critical speeds (string):** top $\sqrt{gR}$ · side $\sqrt{3gR}$ · bottom $\sqrt{5gR}$
- **Minimum-case tensions:** $T_{\text{top}} = 0$, $T_{\text{bottom}} = 6mg$
- **Always true:** $T_{\text{bottom}} - T_{\text{top}} = 6mg$ (speed-independent)
- **Regimes (launch speed u):** $u \geq \sqrt{5gR}$ completes · $u \leq \sqrt{2gR}$ oscillates · between $\rightarrow$ leaves track
- **Rigid rod:** top speed can be 0, so $v_{\text{bottom}} = 2\sqrt{gR}$ to complete

Memory aids: “Top needs $\sqrt{gR}$, bottom needs $\sqrt{5gR}$.” · “ $T_{\text{bottom}} - T_{\text{top}} = 6mg$, always.” · “String pulls only; rod can push.”

Chapter 5: Work, Energy and Power — Round 2

Addendum

The chapter taught you the machinery — dot products, the work-energy theorem, potential energies, conservation laws, power, collisions, and the vertical circle — and it taught it well. But five examinable families sit just outside its coverage, and every one of them appears in real papers. Addendum A builds the spring-block toolkit the chapter's $\frac{1}{2}kx^2$ begs for but never deploys: maximum compression, two-block spring collisions, and the drop-onto-a-spring quadratic. Addendum B handles mass as a *continuous* quantity — chains pulled off tables, sand landing on belts — where the famous factor of two in conveyor-belt power lives. Addendum C extends restitution past the chapter's single bounce: geometric series of bounces, moving walls, and the general- e velocity formulas. Addendum D supplies the line-of-impact procedure for oblique collisions, which the chapter names but never executes. Addendum E applies the work-energy theorem to *systems* joined by strings and pulleys, using one cancellation most students never learn to exploit.

Numbering restarts here. Nothing below is new theory — every method is the chapter's own energy bookkeeping, applied one system bigger or one bounce further.

Addendum A — Spring-Block Systems: Compression, Collision, Release

The chapter stores elastic energy ($U = \frac{1}{2}kx^2$) and derives it, then never spends it. Yet “find the maximum compression” is among the most repeated JEE Main structures in this chapter, and CBSE asks the drop-onto-spring numerical directly. Three situations cover nearly everything.

Method

Situation 1 — one block meets a spring (wall-mounted or free-standing), block mass m , spring constant k .

Only gravity, the normal force, and the spring force act. The first two either do no work or are accounted for explicitly, and the spring force is conservative. So mechanical energy of the block-spring system is conserved:

$$K_i + U_{g,i} + U_{s,i} = K_f + U_{g,f} + U_{s,f}$$

with $U_s = \frac{1}{2}kx^2$ and $U_g = mgh$ measured from any fixed level. At maximum compression the block is instantaneously at rest ($K_f = 0$).

Situation 2 — one block collides into another through a spring.

During compression the spring force is *internal* to the two-block system, and no external horizontal force acts, so **momentum is conserved at every instant**, including the instant of maximum compression. At that instant the two blocks share a common velocity V (their relative velocity is zero):

$$V = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$$

Then energy finishes the job. All the kinetic energy “missing” between the start and the common-velocity state sits in the spring:

$$\frac{1}{2} k x_{max}^2 = K_{initial} - \frac{1}{2} (m_1 + m_2) V^2 = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (u_1 - u_2)^2$$

so that

$$x_{max} = \sqrt{\frac{m_1 m_2}{k (m_1 + m_2)}} |u_1 - u_2|$$

Dimension check: $\frac{m_1 m_2}{k(m_1 + m_2)}$ has units $\frac{\text{kg}}{\text{N/m}} = \frac{\text{kg} \cdot \text{s}^2}{\text{kg}} = \text{s}^2$, and multiplying by $(u_1 - u_2)^2$ in m^2/s^2 gives m^2 . The square root is a length, as required.

Situation 3 — the spring re-expands. While the spring is compressed and later returns to its natural length, it hands back exactly the energy it stored (it is conservative). Consequently, *after* the spring reaches its natural length again, the two blocks have precisely the velocities of a **perfectly elastic 1-D collision** with $e = 1$ — the chapter’s Sub-topic 5 formulas apply verbatim. A light spring between two blocks is nature’s elastic collider.

Two standing warnings.

- In Situation 2 you may not apply the work-energy theorem to *one block alone*: the spring force on it is external, and its work is unknown until the compression history is known. Bookkeep the whole system.
- When a block *drops* onto a vertical spring, gravity keeps working during the compression. The correct balance at maximum compression x (fallen height $h + x$) is

$$mg(h + x) = \frac{1}{2} k x^2$$

Setting $mg h = \frac{1}{2} k x^2$ — forgetting the $mg x$ term — is the single most common error in this family.

Example A.1 — [JEE Main Level]

A 2 kg block moving at 6 m/s on a smooth floor overtakes a 4 kg block at rest, with a light spring of constant $k = 300 \text{ N/m}$ attached to its front face. Find the maximum compression of the spring.

Common velocity at maximum compression:

$$V = \frac{(2)(6) + (4)(0)}{2 + 4} = \frac{12}{6} = 2 \text{ m/s}$$

Kinetic energy before, and at the common-velocity instant:

$$K_i = \frac{1}{2}(2)(6)^2 = 36 \text{ J}, \quad K_V = \frac{1}{2}(6)(2)^2 = 12 \text{ J}$$

The missing $36 - 12 = 24 \text{ J}$ sits in the spring:

$$\frac{1}{2}(300)x^2 = 24 \implies x^2 = \frac{48}{300} = 0.16 \text{ m}^2, \quad x_{max} = 0.40 \text{ m}$$

Check via the compact formula:

$$x_{max} = \sqrt{\frac{(2)(4)}{(300)(6)}} \times 6 = \sqrt{\frac{8}{1800}} \times 6 = \sqrt{0.16} = 0.40 \text{ m} \checkmark$$

Both routes agree, and the units reduce correctly to metres.

Example A.2 — [JEE Advanced Level]

A 2 kg block is released from rest 0.48 m above the top of an uncompressed vertical spring of constant $k = 1000 \text{ N/m}$. Find the maximum compression. Take $g = 10 \text{ m/s}^2$.

Between release and maximum compression x , gravity does work over the *total* drop $h + x$ and the spring absorbs everything:

$$mg(h + x) = \frac{1}{2}kx^2$$

Substituting:

$$(2)(10)(0.48 + x) = \frac{1}{2}(1000)x^2 \implies 500x^2 - 20x - 9.6 = 0$$

Divide by 20:

$$25x^2 - x - 0.48 = 0 \implies x = \frac{1 \pm \sqrt{1 + 4(25)(0.48)}}{50} = \frac{1 \pm \sqrt{49}}{50}$$

Taking the positive root:

$$x = \frac{1 + 7}{50} = 0.16 \text{ m}$$

Verification by direct energy tally. Gravitational PE surrendered: $mg(h+x) = (2)(10)(0.64) = 12.8 \text{ J}$. Spring PE gained: $\frac{1}{2}(1000)(0.16)^2 = \frac{1}{2}(1000)(0.0256) = 12.8 \text{ J}$. Equal, with units of joules on both sides.

Why the naive shortcut fails. Setting $mgh = \frac{1}{2}kx^2$ gives $x^2 = \frac{9.6}{500} = 0.0192 \text{ m}^2$, i.e. $x \approx 0.139 \text{ m}$ — about 13% low, because gravity does another $mgx \approx 3.2 \text{ J}$ of work *during* the compression. An examiner who offers 0.14 m as a distractor knows exactly which error you made.

Practice — Addendum A

1. A 1 kg block moving at 4 m/s hits a light spring of constant 1600 N/m fixed to a wall. Find the maximum compression.
2. A 2 kg block is pushed against a spring ($k = 800 \text{ N/m}$), compressing it 0.25 m, and released on a smooth floor. Find its speed after losing contact with the spring.
3. A 2 kg block moving at 6 m/s collides with a stationary 4 kg block through a spring of constant 300 N/m. Find the maximum compression.
4. A 1 kg block is dropped from rest from 0.80 m above an uncompressed vertical spring of constant 500 N/m. Find the maximum compression. Take $g = 10 \text{ m/s}^2$.
5. In a two-block spring collision, explain in one sentence why the work-energy theorem cannot be applied to the incoming block alone between the first contact and maximum compression.

Answers: 1. $\frac{1}{2}(1600)x^2 = \frac{1}{2}(1)(16)$ gives $x^2 = 0.0100 \text{ m}^2$, so $x = 0.10 \text{ m}$. 2. Stored $\frac{1}{2}(800)(0.25)^2 = 25 \text{ J}$ becomes $\frac{1}{2}(2)v^2$, so $v = 5 \text{ m/s}$. 3. Common velocity $V = \frac{12}{6} = 2 \text{ m/s}$; $K_i = 36 \text{ J}$, $K_V = 12 \text{ J}$; $\frac{1}{2}(300)x^2 = 24$ gives $x = 0.40 \text{ m}$. 4. $5(0.80 + x) \cdot 2 = 250x^2$, i.e. $25x^2 - x - 0.80 = 0$; $x = \frac{1+\sqrt{81}}{50} = 0.20 \text{ m}$. Check: $mg(1.0) = 10 \text{ J} = \frac{1}{2}(500)(0.04)$. 5. The spring force is external to that block and does an amount of work that cannot be computed without knowing the compression as a function of time; only for the *combined* system is the spring force internal and its energy trackable as stored PE.

Addendum B — Continuous Mass: Chains, Sand and Conveyor Belts

Every work formula in the chapter assumes a lump of fixed mass. Exams regularly break that assumption: a chain partly hanging off a table, sand raining onto a conveyor belt, water pumped in a stream. The fix is always the same — integrate the chapter's own results over dm — but the conveyor-belt result hides a surprise worth a mark in itself.

Method

B1 — Work for a continuous distribution. To raise each element dm through the height appropriate to it,

$$W = \int g h(dm)$$

For a uniform chain of linear mass density λ (in kg/m) with a length l hanging off a table, raising element at depth y costs $dm \cdot g \cdot y$ with $dm = \lambda dy$:

$$W = \int_0^l \lambda g y \, dy = \frac{1}{2} \lambda g l^2$$

Note the square: doubling the hanging length quadruples the work. The same integral, run in reverse, gives the kinetic energy when the chain slips off a smooth table on its own — the hanging portion's centre of mass falls by $\frac{l}{2}$:

$$\frac{1}{2} M v^2 = (\lambda l) g \frac{l}{2} = \frac{1}{2} \lambda g l^2 \quad \Rightarrow \quad v = l \sqrt{\frac{g}{L}}$$

where L is the full chain length and $M = \lambda L$.

B2 — Mass arriving at a steady rate. If mass flows at a constant rate μ (in kg/s), then any per-kilogram energy cost ϵ (in J/kg) becomes a power $\mu\epsilon$ (in W). The chapter used this for pumps; the same logic governs belts and chains.

B3 — The conveyor belt. Sand falls vertically at rate μ onto a horizontal belt moving at constant speed v . Each kilogram of sand must be accelerated from 0 to v .

Force. In a time dt , mass μdt lands with zero horizontal velocity and must leave the hopper region moving at v . The belt must supply horizontal impulse:

$$F dt = (\mu dt) v \quad \Rightarrow \quad F = \mu v \quad (\text{newtons, since } \frac{\text{kg}}{\text{s}} \cdot \frac{\text{m}}{\text{s}} = \frac{\text{kg} \cdot \text{m}}{\text{s}^2})$$

Power of the motor. The belt surface moves at v while exerting F :

$$P_{\text{motor}} = Fv = \mu v^2$$

Rate of kinetic energy gain of the sand.

$$P_K = \frac{d}{dt} \left(\frac{1}{2} (\mu t) v^2 \right) = \frac{1}{2} \mu v^2$$

Half the motor's power goes into kinetic energy. Where is the other half? It is dissipated as heat by kinetic friction between belt and sand. The sand does not instantly match the belt speed; it slips against the belt while friction accelerates it. During the slip, the friction force does work $\mu v^2 dt$ on the belt-side account but delivers only $\frac{1}{2} \mu v^2 dt$ to the sand as kinetic energy — the deficit is rubbing heat at the contact. This “exactly half” result is a standing JEE Advanced favourite and a beautiful consistency check: energy is conserved, it is just spent in two places.

Example B.1 — [CBSE / JEE Main Level]

A uniform chain of mass 9 kg and length 3 m lies on a table with a 1 m length hanging over the edge. How much work is needed to pull the hanging part slowly back onto the table? Take $g = 10 \text{ m/s}^2$.

Linear density: $\lambda = \frac{9}{3} = 3 \text{ kg/m}$. Raising the element at depth y (from $y = 0$ to $y = 1 \text{ m}$) through height y :

$$W = \int_0^1 \lambda g y \, dy = 3 \times 10 \times \left[\frac{y^2}{2} \right]_0^1 = 3 \times 10 \times 0.5 = 15 \text{ J}$$

Equivalent centre-of-mass route. The hanging segment has mass 3 kg and its centre of mass sits $\frac{1}{2}$ m below the tabletop, so raising it costs $(3)(10)(0.5) = 15 \text{ J}$. Units agree: $\text{kg} \cdot \frac{\text{m}}{\text{s}^2} \cdot \text{m} = \text{J}$.

Example B.2 — [JEE Advanced Level]

Sand falls at $\mu = 10 \text{ kg/s}$ onto a conveyor belt moving horizontally at $v = 2 \text{ m/s}$. Find (a) the force the belt must exert on the sand, (b) the motor power, (c) the rate of kinetic energy gain of the sand, and (d) the rate of heat generation at the contact.

(a) Force:

$$F = \mu v = (10)(2) = 20 \text{ N}$$

(b) Motor power:

$$P = Fv = (20)(2) = 40 \text{ W}$$

(c) Kinetic energy accumulation rate:

$$P_K = \frac{1}{2} \mu v^2 = \frac{1}{2} (10)(4) = 20 \text{ W}$$

(d) By conservation of energy, the remainder is heat:

$$P_{\text{heat}} = P - P_K = 40 - 20 = 20 \text{ W}$$

Consistency check. Every second, 10 kg of sand gains $\frac{1}{2}(10)(2)^2 = 20 \text{ J}$ of kinetic energy, and 20 J warms the belt and sand. The motor supplies 40 J each second. Books balance exactly — the factor of two is not an approximation but a theorem.

Practice — Addendum B

1. A uniform chain of mass 4 kg and length 2 m lies on a table with half its length hanging over the edge. Find the work needed to pull it fully onto the table. Take $g = 10 \text{ m/s}^2$.
2. The chain of Question 1 is instead released from rest and slides off the smooth table. Find its speed when the last link leaves the table.
3. Sand lands on a belt at 5 kg/s; the belt runs at 4 m/s. Find the driving force, the motor power, and the heating rate.
4. State, with one-line reasoning, why the motor power in a belt problem is exactly twice the rate of kinetic energy gain of the load.

5. A chain of length L and mass M hangs vertically held at its top end. Find the work needed to coil it slowly onto a platform at the level of its bottom end.

Answers: 1. Hanging mass 2 kg, its centre of mass 0.5 m down: $W = (2)(10)(0.5) = 10$ J. 2. $\frac{1}{2}Mv^2 = \frac{1}{2}\lambda gl^2$ with $\lambda = 2$ kg/m, $l = 1$ m, $M = 4$ kg: $2v^2 = 10$, so $v = \sqrt{5} \approx 2.24$ m/s. 3. $F = \mu v = 20$ N; $P = \mu v^2 = 80$ W; heating = 40 W. 4. The belt exerts $F = \mu v$ on sand that slips against a surface moving at v , so friction work enters at μv^2 while the sand's kinetic energy accumulates at $\frac{1}{2}\mu v^2$; the difference is dissipated at the slipping contact. 5. Element at height y above the bottom has mass λdy and must be lifted y : $W = \int_0^L \lambda gy dy = \frac{MgL}{2}$ (the centre of mass starts at $\frac{L}{2}$).

Addendum C — Restitution Past the First Bounce: Series, Moving Walls, General e

The chapter defines e , derives the elastic shortcut, and works one bounce off the floor. Three examinable extensions remain: the *whole sequence* of bounces (a geometric series — CBSE loves the total-distance question, JEE Main the total-time question), collisions with a *moving* massive object, and the general- e version of the two-body final velocities.

Method

C1 — The bounce series. After each floor impact the rebound height multiplies by e^2 (height $\propto$ speed², and the speed multiplies by e). Starting from height h :

$$h_n = e^{2n}h$$

Total vertical distance travelled until rest:

$$D = h + 2he^2 + 2he^4 + \dots = h + \frac{2he^2}{1 - e^2}$$

Total time: the first fall takes $t_0 = \sqrt{\frac{2h}{g}}$; each subsequent rise-and-fall pair takes $2t_0e^n$:

$$T = t_0 \left(1 + \frac{2e}{1 - e} \right) = t_0 \frac{1 + e}{1 - e}$$

Sanity limits: as $e \rightarrow 0$, $D \rightarrow h$ (it thuds dead) and $T \rightarrow t_0$; as $e \rightarrow 1$, both diverge (an ideal ball bounces forever). Both limits behave physically.

C2 — Bouncing off a massive moving object. Treat the massive body's frame first. If a wall (train front, piston, paddle) moves toward the ball at speed V while the ball approaches at u , the approach speed in the wall's frame is $u + V$. The wall, being effectively infinite in mass, recoils negligibly, so the ball rebounds in that frame at $e(u + V)$. Transform back to the ground frame:

$$v_{ball} = e(u + V) + V$$

directed back the way the ball came. For $e = 1$ this is the celebrated $u + 2V$: a cricket ball hit by a moving bat leaves faster than it arrived by *twice the bat's speed*, which is the entire physics of batting.

If the wall *recedes* at V , replace $V \rightarrow -V$ throughout; the closing speed in the wall frame is $u - V$.

C3 — General- e two-body results. Combining momentum conservation with $v_2 - v_1 = e(u_1 - u_2)$ (both written along the line of collision) and solving:

$$v_1 = u_1 - \frac{m_2}{m_1 + m_2} (1 + e)(u_1 - u_2), \quad v_2 = u_2 + \frac{m_1}{m_1 + m_2} (1 + e)(u_1 - u_2)$$

Quick derivation: momentum rearranged gives $m_1(u_1 - v_1) = m_2(v_2 - u_2)$; substitute $v_2 - u_2 = (v_2 - v_1) + (v_1 - u_1) = e(u_1 - u_2) + (v_1 - u_1)$; collect terms in $(v_1 - u_1)$:

$$m_1(u_1 - v_1) = m_2 e(u_1 - u_2) - m_2(u_1 - v_1) \implies (m_1 + m_2)(u_1 - v_1) = m_2(1 + e)(u_1 - u_2)$$

which is the first formula; the second follows symmetrically. Setting $e = 1$ must reproduce the chapter's elastic formulas — a built-in check.

Example C.1 — [CBSE / JEE Main Level]

A ball is dropped from $h = 20$ m and rebounds with $e = 0.5$. Find (a) the height after the second bounce, (b) the total distance travelled before it comes to rest, and (c) the total time elapsed. Take $g = 10 \text{ m/s}^2$.

(a) After the second bounce:

$$h_2 = e^4 h = (0.5)^4 (20) = \frac{20}{16} = 1.25 \text{ m}$$

(b) Total distance:

$$D = h + \frac{2he^2}{1 - e^2} = 20 + \frac{2(20)(0.25)}{0.75} = 20 + \frac{10}{0.75} = 20 + 13.33 = 33.33 \text{ m}$$

(c) First fall takes $t_0 = \sqrt{\frac{2(20)}{10}} = 2$ s. Total:

$$T = t_0 \frac{1 + e}{1 - e} = 2 \times \frac{1.5}{0.5} = 6 \text{ s}$$

Check by direct summation. Fall 2 s; then pairs of duration $2(1) = 2$ s, $2(0.5) = 1$ s, $2(0.25) = 0.5$ s, ...: total = $2 + 2 + 1 + 0.5 + 0.25 + \dots = 2 + 4 = 6$ s.

Example C.2 — [JEE Advanced Level]

A ball moving at 10 m/s strikes head-on the front wall of a train approaching at 5 m/s, with $e = 0.5$. Find the ball's speed after the bounce, and state what it would be for $e = 1$.

In the train's frame the ball approaches at $u + V = 10 + 5 = 15$ m/s. It rebounds at $e \times 15 = 7.5$ m/s relative to the train, still pointing away from the train. Back in the ground frame the train's wall still advances at 5 m/s behind the ball, adding to the recession:

$$v = e(u + V) + V = (0.5)(15) + 5 = 7.5 + 5 = 12.5 \text{ m/s}$$

For $e = 1$: $v = 15 + 5 = 20$ m/s $= u + 2V$, the bat-and-ball limit.

Momentum sanity check (ball-train system). Before: ball momentum (say $m = 0.5$ kg, taking the train's direction as positive) $= -(0.5)(10) = -5$ kg·m/s. After: $-(0.5)(12.5) = -6.25$ kg·m/s. The ball gained 1.25 kg·m/s in the negative direction; the much more massive train gains 1.25 kg·m/s in the positive direction — undetectably small for the train, exactly as assumed. Kinetic energy: ball's K rises from $\frac{1}{2}(0.5)(100) = 25$ J to $\frac{1}{2}(0.5)(156.25) \approx 39.1$ J; the extra energy comes from the train's engine doing work as the wall advances against the ball's push. Nothing violates conservation — the train is not a passive wall.

Practice — Addendum C

1. A ball dropped from 5 m rebounds to 1.25 m. Find e , and the height reached after the third bounce.
2. For the ball of Question 1 ($h = 5$ m), find the total distance travelled and the total time before rest. Take $g = 10$ m/s².
3. A ball moving at 6 m/s catches up with a wall receding at 2 m/s; $e = 0.5$. Find the ball's speed and direction after the bounce.
4. A ball thrown at 5 m/s meets a train wall approaching at 10 m/s, $e = 1$. Find the return speed.
5. A 2 kg body moving at 6 m/s overtakes a 4 kg body moving at 2 m/s in the same direction; $e = 0.5$. Using the general- e formulas, find both final velocities and the kinetic energy lost.

Answers: 1. $e = \sqrt{1.25/5} = 0.5$; $h_3 = e^6 h = \frac{5}{64} \approx 0.078$ m. 2. $D = 5 + \frac{2(5)(0.25)}{0.75} = 5 + 3.33 = 8.33$ m; $t_0 = 1$ s, $T = 1 \times \frac{1.5}{0.5} = 3$ s. 3. Closing speed in wall frame $= 6 - 2 = 4$ m/s; rebound at 2 m/s in that frame; ground-frame speed $= 2 + 2 = 4$ m/s, still in the original direction — the ball keeps chasing the wall but more slowly. 4. $v = 1 \times (5 + 10) + 10 = 25$ m/s. 5. $u_1 - u_2 = 4$ m/s; $v_1 = 6 - \frac{4}{6}(1.5)(4) = 6 - 4 = 2$ m/s; $v_2 = 2 + \frac{2}{6}(1.5)(4) = 4$ m/s. Momentum: $12 + 8 = 20$ before, $4 + 16 = 20$ after. K : $36 + 8 = 44$ J before, $4 + 16 = 20$ J after, so 24 J lost to heat and deformation.

Addendum D — Oblique Collisions: The Line-of-Impact Procedure

The chapter announces the crucial fact — only velocity components *along the line of impact* change — and then proves the equal-mass 90° theorem without ever running the general procedure. JEE Advanced asks that procedure almost yearly, usually against a smooth wall or between two balls with a stated geometry. Four steps settle every case.

Method

1. **Draw the line of impact** — the common normal at contact: the line joining the centres of the two balls, or the normal to a wall.
2. **Resolve every velocity** into components along the line of impact (subscript n) and perpendicular to it (subscript t).
3. **Apply the rules separately:**
 - Tangential: smooth contact exerts no tangential force, so $v_t = u_t$ unchanged for each body.
 - Normal: run the ordinary 1-D collision — momentum conservation plus restitution e — on the n -components alone.
4. **Recombine and report** the final speeds and angles; compute any energy loss from the change in normal components only:

$$\Delta K = \frac{1}{2}m[(1 - e^2)u_n^2] \quad (\text{ball vs fixed wall})$$

Wall case in one box. A ball hits a smooth fixed wall at speed u , at angle θ to the normal, restitution e :

$$v_n = e u \cos \theta \text{ (reversed)}, \quad v_t = u \sin \theta$$

$$v = u \sqrt{\sin^2 \theta + e^2 \cos^2 \theta}, \quad \tan \phi = \frac{\tan \theta}{e}$$

where ϕ is the rebound angle from the normal. Note $\tan \phi > \tan \theta$ for $e < 1$: a dead ball rebounds *wider*. For $e = 1$ we recover mirror reflection, $\phi = \theta$ and $v = u$.

Example D.1 — [JEE Advanced Level]

A ball strikes a smooth fixed wall at 10 m/s, its velocity making 60° with the wall (so 30° with the normal). Given $e = 0.5$, find the rebound speed, the rebound angle, and the fraction of kinetic energy lost.

Normal and tangential components (normal at 30° to the velocity):

$$u_n = 10 \cos 30^\circ = 10 \times \frac{\sqrt{3}}{2} \approx 8.66 \text{ m/s}, \quad u_t = 10 \sin 30^\circ = 5 \text{ m/s}$$

After the bounce:

$$v_n = e u_n = 0.5 \times 8.66 = 4.33 \text{ m/s (reversed)}, \quad v_t = 5 \text{ m/s (unchanged)}$$

Rebound speed:

$$v = \sqrt{(4.33)^2 + (5)^2} = \sqrt{18.75 + 25} = \sqrt{43.75} \approx 6.61 \text{ m/s}$$

Rebound angle from the normal:

$$\tan \phi = \frac{v_t}{v_n} = \frac{5}{4.33} \approx 1.155 \implies \phi \approx 49.1^\circ$$

(Equivalently $\tan \phi = \tan 30^\circ / 0.5 = \frac{1/\sqrt{3}}{0.5} = \frac{2}{\sqrt{3}} \approx 1.155$.)

Energy lost — only the normal component was degraded:

$$\Delta K = \frac{1}{2}m(1 - e^2)u_n^2 = \frac{1}{2}m(0.75)(75) = 28.125m \text{ J}$$

against initial $K = \frac{1}{2}m(100) = 50m \text{ J}$, so the fraction lost is $\frac{28.125m}{50m} = 0.5625$, i.e. 56.25%.

Check. Final $K = 50m - 28.125m = 21.875m \text{ J}$; direct computation gives $\frac{1}{2}m(43.75) = 21.875m \text{ J}$. Units: each term is $\text{kg} \cdot \text{m}^2/\text{s}^2 = \text{J}$.

Example D.2 — [JEE Advanced Level]

An identical smooth billiard ball strikes a stationary identical ball. At the moment of impact the striker's velocity makes 30° with the line of centres. The collision is elastic ($e = 1$). Find the speed of each ball afterwards and verify the 90° result of the chapter by direct computation.

Take the striker's incoming speed as u . Resolve along the line of centres (n) and across it (t):

$$u_n = u \cos 30^\circ = \frac{\sqrt{3}}{2}u, \quad u_t = u \sin 30^\circ = \frac{u}{2}$$

Tangential: the striker keeps $v_{1,t} = \frac{u}{2}$; the target had none and gains none.

Normal: identical masses, elastic, one at rest — the chapter's exchange rule applies to the normal components alone:

$$v_{1,n} = 0, \quad v_{2,n} = \frac{\sqrt{3}}{2}u$$

Recombining:

$$v_1 = \frac{u}{2} = 0.5u \text{ (purely tangential direction)}, \quad v_2 = \frac{\sqrt{3}}{2}u \approx 0.87u \text{ (along the line of centres)}$$

Since one final velocity is purely tangential and the other purely normal, they are mutually perpendicular — the 90° theorem, produced here by arithmetic rather than the vector proof.

Kinetic energy check.

$$\frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}m\left(\frac{u^2}{4} + \frac{3u^2}{4}\right) = \frac{1}{2}mu^2 \checkmark$$

equal to the initial kinetic energy, as elasticity demands.

Why this is the mark-winning route. Students who try to conserve momentum vectorially in two unknown directions drown; those who split along n and t reduce the problem to the chapter's own 1-D machinery in three lines. The decomposition is the method.

Practice — Addendum D

1. A ball hits a smooth wall at 20 m/s at 30° to the wall, with $e = 1$. Describe the rebound completely.
2. Repeat with $e = 0.6$: find the rebound speed and angle from the wall.
3. An identical ball strikes a stationary identical ball elastically; at impact its velocity makes 60° with the line of centres and its speed is 8 m/s. Find both final speeds.
4. Explain in one line why the tangential component of velocity survives an impact with a smooth wall.
5. A ball hits a smooth wall at speed u and angle θ to the *normal*, with restitution e . Show that the kinetic energy lost is $\frac{1}{2}mu^2(1 - e^2)\cos^2\theta$.

Answers: 1. Mirror reflection: speed 20 m/s, leaving at 30° on the other side of the normal; no energy lost. 2. Components: $u_n = 20 \cos 60^\circ = 10$ m/s, $u_t = 20 \sin 60^\circ \approx 17.32$ m/s. After: $v_n = 6$ m/s, $v_t = 17.32$ m/s; speed $= \sqrt{36 + 300} = \sqrt{336} \approx 18.3$ m/s; angle from wall $\arctan(6/17.32) \approx 19.1^\circ$. 3. Striker retains only the tangential piece $8 \sin 60^\circ = 4\sqrt{3} \approx 6.93$ m/s; target leaves along the line of centres at $8 \cos 60^\circ = 4$ m/s; they separate at 90°, and $\frac{1}{2}m(48 + 16) = \frac{1}{2}m(64)$. 4. A smooth wall exerts only a normal force, so there is no tangential impulse and hence no change in the tangential momentum component. 5. Only $u \cos \theta$ participates in the lossy normal interaction, and a 1-D bounce with restitution e destroys a fraction $(1 - e^2)$ of the kinetic energy associated with that component: $\Delta K = \frac{1}{2}m(u \cos \theta)^2(1 - e^2)$.

Addendum E — Connected Bodies: Strings, Pulleys and Cancelled Work

Every worked energy problem in the chapter involves a single body. Real papers attach strings. The saving grace — and the reason energy methods beat Newton's-law grinding on multi-body systems — is a cancellation almost never stated outright: **an ideal string does zero net work on the system it connects.**

Method

E1 — The cancellation. An inextensible string of fixed length transmits whatever power enters one end out the other, unchanged. Concretely, for a string over an ideal (massless, frictionless) pulley joining bodies 1 and 2, if the ends move distances s_1 and s_2 along their respective string segments, inextensibility forces $s_1 = s_2$ (up to sign). The tension T therefore does work

$$W_T = +Ts_1 - Ts_2 = 0$$

on the combined system. For compound arrangements (a movable pulley) the displacements differ, but the force multiplies inversely: one strand pulls the movable pulley with $2T$ over displacement s , while the free end's tension T moves $2s$, and again $W_T = T(2s) - (2T)s = 0$. **In every ideal-string arrangement, the total work done by string tensions on the whole system is zero.**

E2 — The system work-energy theorem. Consequently, for a collection of bodies joined by ideal strings, with gravity the only other force doing work:

$$\sum(\text{loss in PE}) = \sum(\text{gain in } K) = \sum \frac{1}{2}m_i v_i^2$$

with the speeds v_i linked by the string constraint (bodies on one string share one speed; a movable pulley halves or doubles it).

E3 — When it breaks. If the pulley has mass, the axle rubs, or the string stretches or slips, the cancellation fails and you must account for the pulley's kinetic energy, friction losses, or elastic storage explicitly. The method is safe only under the ideals named in E1.

Example E.1 — [JEE Main Level]

Masses of 3 kg and 2 kg hang over a frictionless pulley on a light inextensible string. Released from rest, how fast are they moving when the 3 kg mass has descended 0.8 m? Take $g = 10 \text{ m/s}^2$.

By E1, ignore the string entirely and balance energy for the pair:

$$\text{PE lost by 3 kg} = \text{PE gained by 2 kg} + \text{KE gained by both}$$

$$m_1 g s = m_2 g s + \frac{1}{2}(m_1 + m_2)v^2$$

$$(3)(10)(0.8) = (2)(10)(0.8) + \frac{1}{2}(5)v^2$$

$$24 = 16 + 2.5 v^2 \quad \Rightarrow \quad v^2 = \frac{8}{2.5} = 3.2, \quad v = \sqrt{3.2} \approx 1.79 \text{ m/s}$$

The whole solution never touched the tension. (For the record, Newton's route gives $a = \frac{m_1 - m_2}{m_1 + m_2}g = \frac{1}{5}(10) = 2 \text{ m/s}^2$ and $v^2 = 2as = 2(2)(0.8) = 3.2$ — same answer, more lines.)

Example E.2 — [JEE Advanced Level]

A 1 kg block on a smooth table is tied to a light string that runs over a fixed pulley at the table's edge, hangs down, passes *under* a light movable pulley carrying a 2 kg hanging block, and has its far end tied to the ceiling. Find the speeds of both blocks when the hanging block has descended 0.3 m, starting from rest. Take $g = 10 \text{ m/s}^2$.

Constraint first. The string's ceiling end is fixed, so if the movable pulley (and the 2 kg block) descends by x , the string must feed $2x$ of extra length over the fixed pulley — meaning the table block moves toward the edge by $2x$. Differentiating: whenever the hanging block has speed v_B , the table block has

$$v_A = 2v_B$$

Energy bookkeeping (E1 lets us skip all tensions). Only gravity does net work:

PE lost by 2 kg = KE of both blocks

$$m_B g x = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 = \frac{1}{2} (1) (2v_B)^2 + \frac{1}{2} (2) v_B^2$$

With $m_B = 2 \text{ kg}$, $g = 10 \text{ m/s}^2$, $x = 0.3 \text{ m}$:

$$2 \times 10 \times 0.3 = 2v_B^2 + v_B^2 = 3v_B^2$$

$$6 = 3v_B^2 \implies v_B = \sqrt{2} \approx 1.41 \text{ m/s}, \quad v_A = 2\sqrt{2} \approx 2.83 \text{ m/s}$$

Verification that the string really does no net work. Newton's laws on this arrangement give the acceleration of B as $a_B = \frac{g}{3}$ and the tension as $T = \frac{2m_B g}{3} = \frac{40}{3} \text{ N}$. Over the descent: the two strands pull *up* on the movable pulley with total force $2T$ through displacement $x = 0.3 \text{ m}$ downward, so their work on B is

$$W_{on B} = -(2T)(0.3) = -\frac{80}{3} \times 0.3 = -8 \text{ J}$$

while the single strand pulls the 1 kg block forward (toward the edge, along its motion) with T through $2x = 0.6 \text{ m}$:

$$W_{on A} = T(0.6) = \frac{40}{3} \times 0.6 = +8 \text{ J}$$

Sum: $-8 + 8 = 0 \text{ J}$, exactly as E1 promises. The tension channelled 8 J out of one body and 8 J into the other and kept nothing — which is why the energy ledger closes with tensions absent.

Practice — Addendum E

1. Masses of 5 kg and 3 kg hang over a frictionless pulley. Find their common speed after the 5 kg mass descends 0.8 m from rest. Take $g = 10 \text{ m/s}^2$.
2. A 2 kg block on a smooth table is connected by a string over a pulley to a hanging 3 kg block. Find the speed of the system after the hanging block falls 1 m from rest. Take $g = 10 \text{ m/s}^2$.
3. State the precise conditions under which the string tensions do zero net work on a connected system.
4. In the movable-pulley setup of Example E.2, replace the hanging block with one of mass m_B . Find v_B after a descent x , in symbols.
5. A student includes the string's work in the ledger of Example E.1, writing "+work by tension on the 3 kg mass." What value should they enter, and why?

Answers: 1. PE lost = $(5)(10)(0.8) = 40 \text{ J}$; PE gained = $(3)(10)(0.8) = 24 \text{ J}$; so $16 = \frac{1}{2}(8)v^2$, giving $v = 2 \text{ m/s}$. 2. $30 = \frac{1}{2}(5)v^2$, so $v = \sqrt{12} = 2\sqrt{3} \approx 3.46 \text{ m/s}$. 3. The string must be inextensible and light, and every pulley it crosses must be ideal (massless, frictionless, no slip); then power in at one end equals power out at the other and the net work over any motion vanishes. 4. $m_B g x = \frac{1}{2}m_A(2v_B)^2 + \frac{1}{2}m_B v_B^2 = (2m_A + \frac{m_B}{2})v_B^2$, so $v_B = \sqrt{\frac{m_B g x}{2m_A + m_B/2}}$. For the example's values this reproduces $\sqrt{2} \text{ m/s}$. 5. Zero net contribution overall: the tension does $+Ts$ on the descending 3 kg mass and $-Ts$ on the rising 2 kg mass with the same s ; entering only one term breaks the ledger. The correct entry for the *system* is 0 J.

System of Particles and Rotational Motion

Chapter 6 — Physics for CBSE, JEE & NEET

Drona

Class 11 Physics

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Chapter Foundations: Rigid Bodies & Types of Motion

Exam Relevance: These foundations are rarely asked as standalone questions, but every single problem in this chapter silently assumes them. **CBSE Boards** may include a 1-mark definition of a rigid body or a type of motion. Getting these ideas crisp now makes every later subtopic — centre of mass, torque, rolling — feel obvious instead of mysterious.

What is a Rigid Body?

Until this chapter, you treated every object as a single *point particle* — a block, a ball, a car, all shrunk to a dot carrying mass. That worked when you only cared about where something went. But it cannot explain *spinning*. A point has no size, so it cannot rotate about itself, wobble, or roll. To study rotation we need objects with *extent*.

A **rigid body** is an idealised object in which the distance between any two particles stays exactly the same, no matter what forces act on it. It never stretches, compresses, or bends. Real objects deform a

little, but for a cricket bat, a flywheel, a spanner, or a planet, the deformation is so tiny that treating them as perfectly rigid gives excellent predictions.

Think about it this way... Imagine the body as a swarm of particles welded together by unbreakable, unstretchable rods. Push or twist it however you like — the swarm moves and turns *as one*, with every internal distance frozen. That “frozen internal geometry” is the single defining property of rigidity, and it’s what lets us talk about *the* rotation of the body rather than each particle doing its own thing.

The Three Types of Motion

Every motion of a rigid body is one of three kinds — or a blend of the first two.

1. Pure translation. Every particle of the body moves with the same velocity at every instant, tracing parallel paths. A book slid flat across a table, or a suitcase carried straight forward, is translating. There is no turning — a line drawn on the body keeps pointing the same way. For pure translation, the whole body behaves exactly like a single particle located at its centre of mass, which is why everything you learned in Class 11 mechanics still applies.

2. Pure rotation about a fixed axis. Every particle moves in a circle, and all those circles are centred on one fixed line called the **axis of rotation**. A ceiling fan, a merry-go-round, the hands of a clock, and a spinning potter’s wheel are rotating about a fixed axis. Here, particles farther from the axis move faster (we’ll quantify this as $v = r\omega$), but they all share the same angular speed.

3. General (combined) motion. Most real motion mixes the two: the body translates *and* rotates at the same time. A rolling wheel is the headline example — its centre glides forward (translation) while the wheel spins about that centre (rotation). A thrown spinning bat, a tumbling gymnast, and a planet orbiting while rotating are all general motion. The master strategy of this entire chapter is to *split* such motion into “translation of the centre of mass” plus “rotation about the centre of mass,” handle each with its own set of laws, then add the results.

Figure 6.1 *Three side-by-side panels. Panel 1 (Translation): a rectangular block with all points carrying identical parallel velocity arrows. Panel 2 (Rotation): a disc with a fixed central axis, particles moving in concentric circles, outer arrows longer than inner. Panel 3 (General motion): a wheel with a forward velocity arrow at its centre plus curved rotation arrows around the centre, illustrating translation + rotation. (see next page)*

A Roadmap to This Chapter

Here is how the pieces fit together, so you always know *why* you’re learning each subtopic:

- **Centre of mass** locates the one special point that represents the whole body’s translation.
- **Vector product** gives us the mathematical tool to describe turning effects in three dimensions.
- **Torque and angular momentum** are the rotational versions of force and momentum — what *causes* and what *carries* rotation.
- **Equilibrium** asks when a rigid body neither moves nor turns — the physics of balance, levers, and structures.

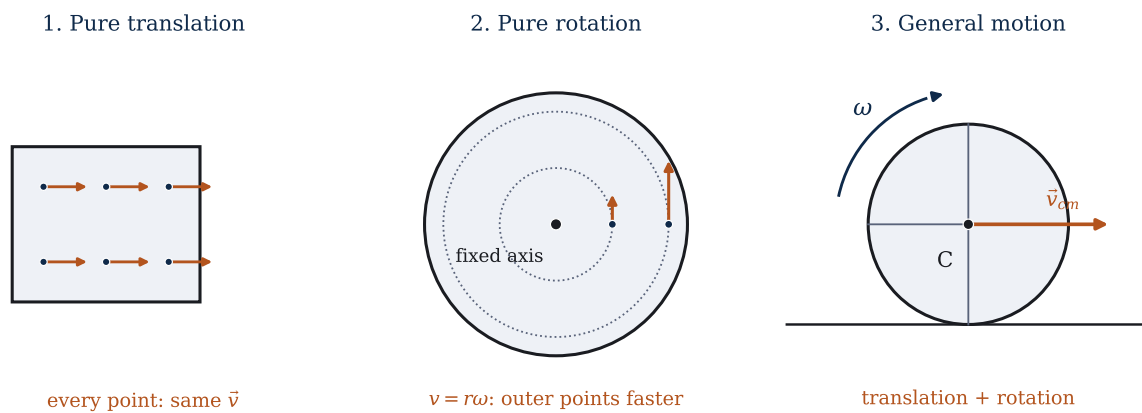


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- **Moment of inertia** measures how hard a body is to spin, depending on how its mass is spread out.
- **Rotational kinematics and dynamics** mirror linear motion, with θ, ω, α replacing s, v, a and $\tau = I\alpha$ replacing $F = ma$.
- **Rolling motion** ties everything together as the most common general motion of all.

Keep one master idea in your pocket as you read: **rotation is a faithful mirror of the straight-line physics you already know.** Almost every formula ahead has a linear twin. Learn the correspondence, and this chapter becomes far smaller than it looks.

Cheat Sheet - Rigid body: distances between all particles are fixed; it does not deform.

- **Pure translation:** all particles share the same velocity; body acts like a point at its CoM.

- **Pure rotation:** all particles circle a fixed axis with the same angular speed; $v = r\omega$. -

General motion: translation of CoM + rotation about CoM (e.g., rolling). - **Master**

idea: split general motion into translation + rotation; rotation mirrors linear motion.

Center of Mass

Exam Relevance: Center of Mass is a foundation stone for this entire chapter and for collisions later. **JEE Main** asks 1–2 direct questions almost every year (composite bodies, mass removal). **NEET** keeps it light — usually one conceptual question. **JEE Advanced** loves continuous bodies and integration. **CBSE Boards** expects the “motion of centre of mass” derivation and clean definitions.

Section 1: Concept Intuition

Imagine you toss a cricket bat spinning across the room. Watch it closely — it tumbles in a way that looks completely chaotic. The handle, the toe, the edges, every part traces its own complicated, looping path. If a friend asked you to predict exactly where the handle will be after half a second, you’d struggle.

But here’s the beautiful part. Hidden inside that tumbling bat there is **one special point** that ignores all the chaos and quietly traces a clean, simple parabola — exactly the path a single thrown stone would follow. That magic point is the **center of mass (CoM)**.

So what is it, really? The center of mass is the **mass-weighted average position** of all the matter in a system. The phrase “mass-weighted” is the whole idea. If you have a light particle and a heavy particle, the balance point doesn’t sit halfway between them — it leans toward the heavy one, because the heavy particle “pulls the average” toward itself. It’s exactly like a see-saw in a park: a heavy uncle and a small child can balance only if the uncle sits much closer to the pivot.

Think about it this way... Picture balancing a steel *thali* (plate) on the tip of one finger. There is exactly one spot where it sits perfectly without tipping. That spot is the center of mass. Slide your finger even slightly away from it and the *thali* topples, because gravity now produces a turning effect about your fingertip. The CoM is the point where the whole object behaves as if all its mass were squeezed into that single location.

Why do we even bother inventing this point? Because a real object is a *system of particles* — millions of atoms, each on its own path. Tracking all of them is hopeless. The center of mass is our shortcut: it lets us replace the entire messy system with a single equivalent particle of mass M sitting at one point. A loaded goods train rounding a bend is thousands of tonnes of wagons, couplings, and cargo, yet for momentum and overall motion we can treat the whole train as one heavy dot moving along the track. That is the power the CoM hands us — it collapses complexity into a single, well-behaved point.

Here’s another everyday picture. Think of a busy vegetable *mandi* (market) and imagine asking, “where is the average shopper standing right now?” If most shoppers crowd around the potato stall, that average point sits near the potatoes — even if a few people wander near the exit. The CoM does exactly this for mass: it sits wherever the mass is most concentrated, and lighter outlying bits barely tug it.

This gives us the cleanest definition: **the center of mass is the point that moves as though (1) the entire mass of the system were concentrated there, and (2) every external force acted there.** This is why coaches studying a long-jumper, a *Bharatanatyam* dancer mid-leap, or a

diver doing somersaults always start by tracking the center of mass — its motion is simple even when the limbs are flailing.

A few things that surprise students at first:

- **Mass need not exist at the center of mass.** The CoM of a ring or a bangle sits at its empty geometric center — there's literally no metal there. Same for a hollow box: its CoM floats in the empty air inside.
- **The CoM is a property of the body, not of your coordinate system.** Shift your origin, rotate your axes — the physical balance point stays welded to the same spot on the object.

Limiting Conditions & Assumptions

1. **Center of mass vs. center of gravity.** These two coincide *only* when gravity g is the same at every point of the body. For everyday objects on Earth this is essentially perfect. But for an enormous object — say a mountain or a kilometre-tall tower — g varies slightly with height, and the center of gravity drifts a tiny bit below the center of mass.
2. **The simple formula assumes you can locate the CoM of each piece.** For symmetric, uniform bodies the CoM sits at the geometric center. The moment density becomes non-uniform, you must integrate.
3. **CoM motion is governed only by *external* forces.** Internal forces — however violent — can never move the center of mass on their own.

Section 2: Key Formulas & Definitions

For a system of discrete particles (masses $m_1, m_2, \dots, m_n$ at positions $\vec{r}_1, \dots, \vec{r}_n$):

$$\vec{r}_{cm} = \frac{\sum_{i=1}^n m_i \vec{r}_i}{\sum_{i=1}^n m_i} = \frac{1}{M} \sum_{i=1}^n m_i \vec{r}_i$$

In component form:

$$x_{cm} = \frac{\sum m_i x_i}{M}, \quad y_{cm} = \frac{\sum m_i y_i}{M}, \quad z_{cm} = \frac{\sum m_i z_i}{M}$$

For two particles (the case you'll use most for quick checks):

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

For a continuous body (replace summation with integration):

$$\vec{r}_{cm} = \frac{1}{M} \int \vec{r} \, dm$$

Velocity and acceleration of the CoM:

$$\vec{v}_{cm} = \frac{1}{M} \sum m_i \vec{v}_i, \quad \vec{a}_{cm} = \frac{1}{M} \sum m_i \vec{a}_i$$

Newton's law for the system:

$$\vec{F}_{ext} = M \vec{a}_{cm}$$

Fundamental property (taking the origin *at* the CoM):

$$\sum m_i \vec{r}_i = \vec{0} \quad (\text{position vectors measured from the CoM})$$

Notation: M = total mass, $\vec{r}_i$ = position of i -th particle, dm = mass element, $\vec{F}_{ext}$ = net external force.

Symbols, SI units, and dimensional formulas:

Quantity	Symbol	SI Unit	Dimensional Formula
Position of CoM	$\vec{r}_{cm}$	m	$[M^0 L^1 T^0]$
Linear mass density	λ	kg m ⁻¹	$[M^1 L^{-1} T^0]$
Surface mass density	σ	kg m ⁻²	$[M^1 L^{-2} T^0]$
Volume mass density	ρ	kg m ⁻³	$[M^1 L^{-3} T^0]$

Section 3: CBSE Board Derivation — Motion of the Centre of Mass

This is the classic derivation CBSE asks for, and it reveals *why* the CoM is special. We want to prove that the total external force on a system equals total mass times the acceleration of its center of mass.

Step 1 — Start from the definition. For n particles, the position of the center of mass is defined by:

$$M \vec{r}_{cm} = m_1 \vec{r}_1 + m_2 \vec{r}_2 + \cdots + m_n \vec{r}_n$$

We've multiplied through by M to clear the fraction — this keeps the algebra clean.

Step 2 — Differentiate once with respect to time. Since the masses are constant, differentiation passes straight through them:

$$M \vec{v}_{cm} = m_1 \vec{v}_1 + m_2 \vec{v}_2 + \cdots + m_n \vec{v}_n$$

Physical meaning: the total momentum of the system equals $M \vec{v}_{cm}$ — as if all the mass marched along with the CoM. This single line is the bridge to the law of conservation of linear momentum.

Step 3 — Differentiate again.

$$M \vec{a}_{cm} = m_1 \vec{a}_1 + m_2 \vec{a}_2 + \cdots + m_n \vec{a}_n$$

Step 4 — Replace each $m_i \vec{a}_i$ using Newton’s second law. The force on each particle has two parts: external forces (gravity, applied pushes, normal reactions) and internal forces (the mutual pulls and pushes between particles of the system). So:

$$M \vec{a}_{cm} = \sum_i \vec{F}_i^{ext} + \sum_i \vec{F}_i^{int}$$

Step 5 — Kill the internal forces using Newton’s third law. Here is the heart of the proof. For *every* internal force that particle A exerts on particle B, particle B exerts an equal and opposite force back on A. These pairs cancel exactly when you sum over the whole system:

$$\sum_i \vec{F}_i^{int} = \vec{0}$$

This is the *reason* the step works — not algebra, but the third law guaranteeing internal forces always come in cancelling twins.

Step 6 — Result.

$$\boxed{\vec{F}_{ext} = M \vec{a}_{cm}}$$

The big takeaway: internal forces — an exploding firecracker, a sprinter’s muscles, two skaters pushing apart — can *never* shift the center of mass by themselves. When a bomb at rest explodes mid-air, the fragments fly everywhere, but the center of mass keeps following the same parabola the unexploded bomb would have followed.

An immediate and powerful corollary. If the net external force is zero, then $M \vec{a}_{cm} = 0$, which means $\vec{v}_{cm}$ is constant — the total momentum $M \vec{v}_{cm}$ is conserved. This is the **law of conservation of linear momentum**, and you can see it falls straight out of the CoM equation. So every recoil, collision, and explosion problem you’ll meet later in this chapter is really a center-of-mass statement in disguise. Master the CoM now and half of momentum conservation comes free.

Key Procedure — CoM of a Continuous Body by Integration

For a uniform body where symmetry doesn’t hand you the answer, follow these steps. Let’s locate the CoM of a **uniform semicircular wire** of radius R .

Step 1 — Set up coordinates and an element. Place the center at the origin, flat side along the x-axis. By symmetry, $x_{cm} = 0$; we only need y_{cm} . Take an element subtending angle $d\theta$ at angle θ .

Figure 6.2 A semicircular wire of radius R centred at origin, flat diameter on the x-axis, curve in the upper half. A small element of arc at angle θ from the x-axis, subtending $d\theta$, with its y-coordinate marked as $R \sin\theta$. (see next page)

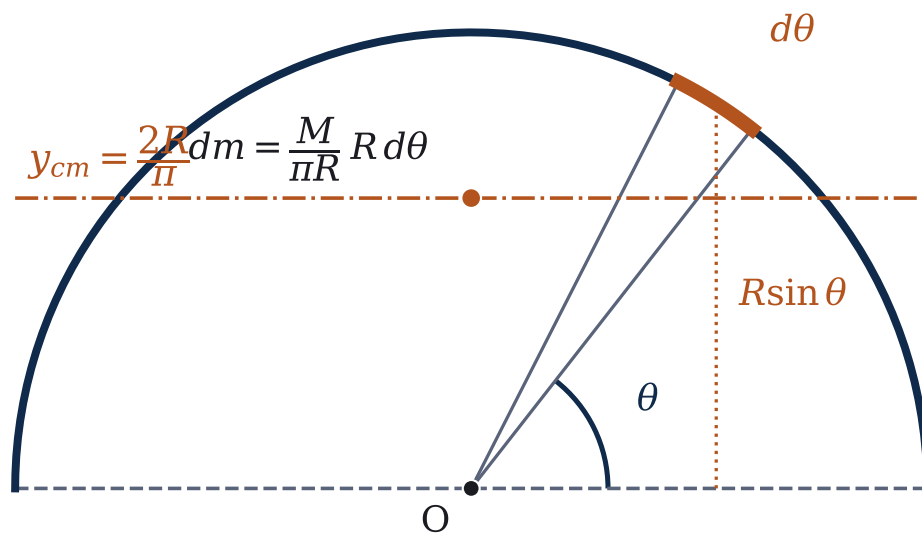


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Step 2 — Express the mass element. Linear mass density $\lambda = \frac{M}{\pi R}$. Arc length of the element is $R d\theta$, so $dm = \lambda R d\theta = \frac{M}{\pi} d\theta$.

Step 3 — Write the coordinate of the element. The element sits at height $y = R \sin \theta$.

Step 4 — Integrate over θ from 0 to π .

$$y_{cm} = \frac{1}{M} \int_0^\pi (R \sin \theta) \left(\frac{M}{\pi} d\theta \right) = \frac{R}{\pi} \int_0^\pi \sin \theta d\theta = \frac{R}{\pi} [-\cos \theta]_0^\pi = \frac{2R}{\pi}$$

So the CoM of a semicircular wire lies at $\left(0, \frac{2R}{\pi}\right)$ — about $0.64R$ above the center, in the empty space where no wire exists. A perfect illustration that mass need not sit at the CoM.

A note on choosing your mass element. The whole game in continuous-body problems is writing dm correctly. Match the density to the geometry:

- A **1D body** (rod, wire): $dm = \lambda dx$ or $\lambda (R d\theta)$ for an arc.
- A **2D body** (plate, disc, lamina): $dm = \sigma dA$, where dA is a strip or ring of area.
- A **3D body** (sphere, cone, cylinder): $dm = \rho dV$, where dV is a thin slice or shell.

Always pick the element so that *every point of it shares the same coordinate* you're averaging — otherwise the integral lies to you. For the cone in the practice set, the natural element is a thin disc-slice perpendicular to the axis, because every point of that slice sits at the same height.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Three point masses of 2 kg, 3 kg, and 5 kg are placed at the points (0, 0), (4, 0), and (0, 3) metres respectively. Find the coordinates of the center of mass.

Given: $m_1 = 2$ kg at (0, 0); $m_2 = 3$ kg at (4, 0); $m_3 = 5$ kg at (0, 3). Total mass $M = 10$ kg.

Formula:

$$x_{cm} = \frac{\sum m_i x_i}{M}, \quad y_{cm} = \frac{\sum m_i y_i}{M}$$

Substitution:

$$x_{cm} = \frac{(2)(0) + (3)(4) + (5)(0)}{10} = \frac{12}{10} = 1.2 \text{ m}$$

$$y_{cm} = \frac{(2)(0) + (3)(0) + (5)(3)}{10} = \frac{15}{10} = 1.5 \text{ m}$$

Answer: The center of mass is at (1.2 m, 1.5 m).

Example 2 — [NEET Speed Trap]

A man of mass 60 kg stands at one end of a 40 kg boat of length 5 m, floating at rest on still water. He walks to the other end. Assuming negligible water resistance, how far does the boat move?

The trap: Students instinctively calculate the man's displacement as 5 m and stop. But there is **no external horizontal force** on the man-boat system (water resistance is negligible), so the **center of mass cannot move horizontally**. The boat must slide backward to compensate.

Mental shortcut — use the “mass-ratio” rule directly. When the man walks a length L relative to the boat, the boat shifts by:

$$d_{\text{boat}} = \frac{m_{\text{man}}}{m_{\text{man}} + m_{\text{boat}}} \times L$$

Plug in:

$$d_{\text{boat}} = \frac{60}{60 + 40} \times 5 = \frac{60}{100} \times 5 = 3 \text{ m}$$

Answer: The boat moves **3 m** in the opposite direction. (Quick sanity check: the heavier mover $\rightarrow$ bigger boat recoil. Here the man is heavier than the boat, so the boat moves more than half the length. ✓)

Example 3 — [JEE Main Level]

A uniform circular disc of radius R has a circular hole of radius $R/2$ cut from it. The center of the hole is at a distance $R/2$ from the center of the disc. (a) Find the center of mass of the remaining portion. (b) Where must a point mass equal to the mass of the removed disc be placed so that the CoM of the whole arrangement returns to the original center?

Concept 1 — Negative mass method. Treat the remaining body as (full disc) + (a *negative* hole). For a uniform disc, mass $\propto$ area $\propto$ radius².

- Full disc: mass $\propto R^2$, located at $x = 0$.
- Hole (negative): mass $\propto (R/2)^2 = R^2/4$, located at $x = R/2$.

$$x_{\text{cm}} = \frac{(R^2)(0) - \left(\frac{R^2}{4}\right)\left(\frac{R}{2}\right)}{R^2 - \frac{R^2}{4}} = \frac{-\frac{R^3}{8}}{\frac{3R^2}{4}} = -\frac{R^3}{8} \times \frac{4}{3R^2} = -\frac{R}{6}$$

So the CoM of the remaining portion is at $x = -\frac{R}{6}$ (i.e., $R/6$ from the center, *away* from the hole).

Concept 2 — Restoring the CoM with an added mass. Let the removed disc have mass m (so the remaining portion has mass $3m$, since hole is $\frac{1}{4}$ of the full $4m$). We place a point mass m at position x_0 so the combined CoM sits at the origin:

$$\frac{(3m)\left(-\frac{R}{6}\right) + (m)(x_0)}{3m + m} = 0 \Rightarrow -\frac{R}{2} + x_0 = 0 \Rightarrow x_0 = +\frac{R}{2}$$

Answer: (a) CoM of remaining portion at $-R/6$. (b) The point mass must be placed at $x = +R/2$ — exactly at the center of the hole that was removed (which makes perfect physical sense: you’re “filling back” the missing mass at its original location).

Example 4 — [JEE Advanced Level]

A thin rod of length L lies along the x -axis from $x = 0$ to $x = L$. Its linear mass density varies as $\lambda(x) = \lambda_0 \left(1 + \frac{x}{L}\right)$. Locate its center of mass.

Step 1 — Mass element. $dm = \lambda dx = \lambda_0 \left(1 + \frac{x}{L}\right) dx$.

Step 2 — Total mass.

$$M = \int_0^L \lambda_0 \left(1 + \frac{x}{L}\right) dx = \lambda_0 \left[x + \frac{x^2}{2L} \right]_0^L = \lambda_0 \left(L + \frac{L}{2} \right) = \frac{3\lambda_0 L}{2}$$

Step 3 — First moment of mass.

$$\int_0^L x dm = \int_0^L \lambda_0 \left(x + \frac{x^2}{L} \right) dx = \lambda_0 \left[\frac{x^2}{2} + \frac{x^3}{3L} \right]_0^L = \lambda_0 \left(\frac{L^2}{2} + \frac{L^2}{3} \right) = \frac{5\lambda_0 L^2}{6}$$

Step 4 — Divide.

$$x_{cm} = \frac{1}{M} \int_0^L x dm = \frac{\frac{5\lambda_0 L^2}{6}}{\frac{3\lambda_0 L}{2}} = \frac{5L^2}{6} \times \frac{2}{3L} = \frac{5L}{9}$$

Answer: $x_{cm} = \frac{5L}{9} \approx 0.556 L$.

Reasoning check: A uniform rod would have its CoM at $0.5L$. Here density increases toward $x = L$, so the CoM shifts toward the heavier end — and indeed $5L/9 > L/2$. The physics agrees with the algebra. ✓

Section 5: Practice Exercises

1. [CBSE] Two particles of mass 4 kg and 6 kg are separated by 10 m along a straight line. Find the distance of the center of mass from the 4 kg particle.
2. [NEET] Four identical particles of mass m each are placed at the corners of a square of side a . One particle is removed. How far does the center of mass shift from the center of the square?
 $a\sqrt{2}/6$ from the center (or $a/(3\sqrt{2})$ from the center). is removed from one corner. Find the distance of the new center of mass from the center of the original plate.
4. [JEE Main] A shell flying horizontally at 20 m s^{-1} explodes into two equal fragments. One fragment momentarily comes to rest. Find the speed of the other fragment immediately after the explosion.

5. **[JEE Advanced]** A uniform solid cone of height h has its apex at the origin and axis along the y -axis. Using integration, show that its center of mass lies at $y_{cm} = \frac{3h}{4}$ from the apex.

Answers: 1. 6 m from the 4 kg mass. 2. $\frac{a}{3\sqrt{2}}$ (toward the diagonally opposite corner). 3. $\frac{a}{6\sqrt{2}}$ from the center. 4. 40 m s^{-1} . 5. $y_{cm} = \frac{3h}{4}$ (derivation required).

Section 6: Crack the MCQ

Q1. A bomb at rest at the origin explodes into several fragments due to internal forces only. Which statement is always true (ignore gravity)?

- (A) The fragments' total momentum increases
- (B) The center of mass remains at the origin
- (C) The center of mass moves in the direction of the largest fragment
- (D) The center of mass accelerates outward

Correct: (B). With only internal forces, $\vec{F}_{ext} = 0$, so $\vec{a}_{cm} = 0$. Since it started at rest at the origin, the CoM stays there forever. *Why the traps fail:* (A) ignores momentum conservation — total momentum stays zero. (C) wrongly imagines the heaviest fragment “winning”; fragments always fly so their momenta cancel. (D) is the most tempting — students see fragments shooting outward and assume the CoM does too, forgetting internal forces can't move it.

Q2. The center of mass of a uniform semicircular disc of radius R lies at a distance from its center equal to:

- (A) $\frac{2R}{\pi}$ (B) $\frac{4R}{3\pi}$ (C) $\frac{R}{2}$ (D) $\frac{4R}{\pi}$

Correct: (B). For a semicircular *disc* (a filled lamina), $y_{cm} = \frac{4R}{3\pi}$. *Why the traps fail:* (A) is the answer for a semicircular *wire/arc*, not a filled disc — the classic mix-up the examiner is hunting for. (C) is a careless guess of “halfway.” (D) drops the factor of 3 from a botched integration.

Q3. A man stands at the rear of a frictionless trolley. He walks forward and stops. Compared to the ground, the trolley:

- (A) stays exactly where it was
- (B) moves forward
- (C) moves backward
- (D) moves backward, then returns to its start

Correct: (C). No external horizontal force $\Rightarrow$ the CoM is fixed in the ground frame. For the man to move forward, the trolley must move backward to keep the CoM put. *Why the traps fail:* (A) forgets that the CoM being fixed forces the *trolley* to move, not stay. (B) confuses the man's motion with

the trolley's. **(D)** wrongly assumes the trolley “springs back” — once both stop, displacements are permanent.

Q4. Two particles of masses m and $3m$ have position vectors moving such that their CoM is stationary. If the velocity of mass m is $\vec{v}$, the velocity of mass $3m$ is:

- (A) $-\vec{v}$ (B) $-\frac{\vec{v}}{3}$ (C) $-3\vec{v}$ (D) $\frac{\vec{v}}{3}$

Correct: (B). A stationary CoM means total momentum is zero: $m\vec{v} + 3m\vec{v}_2 = 0 \Rightarrow \vec{v}_2 = -\frac{\vec{v}}{3}$. *Why the traps fail:* **(A)** assumes equal-and-opposite *velocities*, ignoring the mass weighting. **(C)** inverts the ratio (multiplies instead of divides). **(D)** gets the magnitude right but forgets the negative sign — the velocities must be oppositely directed.

Section 7: Common Student Pitfalls & Pro-Tips

- 1. Treating the CoM as a simple midpoint.** The CoM is *mass-weighted*, not a geometric average. Halfway is correct only for equal masses. Always lean the answer toward the heavier mass and sanity-check.
- 2. Confusing the semicircular wire with the semicircular disc.** Wire (arc) $\rightarrow \frac{2R}{\pi}$; filled disc $\rightarrow \frac{4R}{3\pi}$. Examiners deliberately offer both as options. Tattoo this difference into your memory.
- 3. Forgetting that internal forces don't move the CoM.** In explosions, recoil, and man-on-boat problems, the CoM is fixed (or keeps its original motion). Students waste time tracking individual pieces instead of anchoring to the CoM.
- 4. Sign errors in the negative-mass (removed portion) method.** The removed piece contributes a *negative* mass in both the numerator and denominator. Drop a sign and your CoM flips to the wrong side.

Pro-Tip: For “removed portion” problems, don't grind out full numbers. Use the ratio form and remember the CoM always shifts *away* from the hole. A quick direction check before computing catches half of all sign mistakes — and in NEET, that's 30 seconds saved per question.

Section 8: Cheat Sheet

- Definition:** Mass-weighted average position — moves as if all mass and all external force act there.
- Discrete:** $\vec{r}_{cm} = \frac{1}{M} \sum m_i \vec{r}_i$ | **Continuous:** $\vec{r}_{cm} = \frac{1}{M} \int \vec{r} dm$
- System law:** $\vec{F}_{ext} = M\vec{a}_{cm} \rightarrow$ internal forces *never* move the CoM.

- **Standard results:** semicircular **wire** $\rightarrow \frac{2R}{\pi}$; semicircular **disc** $\rightarrow \frac{4R}{3\pi}$; solid cone (from apex) $\rightarrow \frac{3h}{4}$; triangle (from base) $\rightarrow \frac{h}{3}$.
- **Removed mass:** subtract the cut piece as a negative mass; CoM shifts *away* from the cut.
- **Property:** $\sum m_i \vec{r}_i = 0$ when measured *from* the CoM.

Memory aid — “**WAR**”: **W**ire is $2R/\pi$, **A**rea (disc) is $4R/3\pi$ — and they’re at **R**ight angles in your mind so you never swap them.

Memory aid — “**Internal can’t move it**”: Explosions, recoils, walking-on-boat $\rightarrow$ freeze the CoM first, then everything else falls into place.

Vector Product of Two Vectors

Exam Relevance: The cross product is the mathematical engine behind torque and angular momentum, so it pays off across the whole chapter. **JEE Main** asks 1 direct question most years — finding a cross product, a perpendicular unit vector, or the area of a triangle/parallelogram. **NEET** uses it lightly (angle between vectors, area). **JEE Advanced** folds it into 3D torque and angular-momentum problems. **CBSE Boards** expects the definition, properties, and the right-hand rule.

Section 1: Concept Intuition

You already know one way to multiply two vectors — the **dot product**, which answers “how much do these two vectors point the *same* way?” and gives back a plain number (a scalar). The **vector product**, or **cross product**, answers a completely different question: “how much do these two vectors point in *different* directions — and which way is the plane they define facing?” Remarkably, its answer is itself a *vector*.

Why would the product of two vectors be a vector? Because some physical quantities need a direction that is *perpendicular* to two given directions at once. Think of opening a door. You push (one direction) at some distance from the hinge (another direction), and the door swings about an axis that is perpendicular to both — pointing straight up or down along the hinge line. That turning axis is exactly what the cross product captures. The result of $\vec{r} \times \vec{F}$ (torque) points along the axis of the turning it produces.

Think about it this way... Picture screwing a bottle cap shut. Your fingers curl in the direction of the turn, and the cap *advances* along the axis your thumb points to. The cross product works identically: sweep the first vector toward the second with your right hand’s fingers, and your thumb gives the direction of the product. This is the famous **right-hand rule**, and it is the whole geometric soul of the operation.

There’s a second, equally useful picture. The *magnitude* of $\vec{A} \times \vec{B}$ equals the **area of the parallelogram** built on $\vec{A}$ and $\vec{B}$ as adjacent sides. When the two vectors are parallel, the “parallelogram” is squashed flat — zero area — so the cross product is zero. When they are perpendicular, the parallelogram is as fat as possible, so the cross product is largest. In other words, the cross product measures *how perpendicular* two vectors are, exactly the opposite of what the dot product measures.

Limiting Conditions & Assumptions

1. **The cross product is defined only in three dimensions.** Unlike the dot product, which works in any number of dimensions, the perpendicular-direction idea needs 3D space to live in.
 2. **Order matters — it is anti-commutative.** $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$. Swapping the vectors flips the result’s direction (your thumb points the opposite way). This is unlike ordinary multiplication and unlike the dot product.
 3. **Parallel (or anti-parallel) vectors give zero.** If $\vec{A} \parallel \vec{B}$, then $\sin \theta = 0$, so $\vec{A} \times \vec{B} = \vec{0}$. In particular, any vector crossed with itself is zero.
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Section 2: Key Formulas & Definitions

Definition (magnitude and direction):

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

where θ is the angle between the vectors ($0 \leq \theta \leq \pi$), and $\hat{n}$ is the unit vector perpendicular to *both* $\vec{A}$ and $\vec{B}$, pointing in the right-hand-rule sense.

Magnitude alone:

$$|\vec{A} \times \vec{B}| = AB \sin \theta$$

Core properties:

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A} \quad (\text{anti-commutative})$$

$$\vec{A} \times \vec{A} = \vec{0}$$

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C} \quad (\text{distributive})$$

Unit-vector products (cyclic order $\hat{i} \rightarrow \hat{j} \rightarrow \hat{k} \rightarrow \hat{i}$ gives +; reverse gives -):

$$\hat{i} \times \hat{j} = \hat{k}, \quad \hat{j} \times \hat{k} = \hat{i}, \quad \hat{k} \times \hat{i} = \hat{j}$$

$$\hat{j} \times \hat{i} = -\hat{k}, \quad \hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$$

Determinant form (the workhorse for component vectors):

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \hat{i}(A_y B_z - A_z B_y) - \hat{j}(A_x B_z - A_z B_x) + \hat{k}(A_x B_y - A_y B_x)$$

Geometric uses:

$$\text{Area of parallelogram} = |\vec{A} \times \vec{B}|, \quad \text{Area of triangle} = \frac{1}{2} |\vec{A} \times \vec{B}|$$

Link with the dot product:

$$|\vec{A} \times \vec{B}|^2 + (\vec{A} \cdot \vec{B})^2 = A^2 B^2$$

Notation: A, B are magnitudes; θ the included angle; $\hat{n}$ the perpendicular unit vector. The cross product of two vectors (e.g. position $\times$ force) has whatever units the factors carry; it is dimensionless only when both factors are.

Section 3: Key Procedures & Derivations

Why the result is perpendicular to both vectors

We can prove $\vec{A} \times \vec{B}$ is perpendicular to $\vec{A}$ by showing their dot product is zero. Let $\vec{C} = \vec{A} \times \vec{B}$ with components from the determinant expansion. Then:

$$\vec{A} \cdot \vec{C} = A_x(A_y B_z - A_z B_y) + A_y(A_z B_x - A_x B_z) + A_z(A_x B_y - A_y B_x)$$

Expanding, every term cancels in a $+/-$ pair (for example $A_x A_y B_z$ cancels $-A_y A_x B_z$), giving $\vec{A} \cdot \vec{C} = 0$. Since the dot product is zero, $\vec{C}$ is perpendicular to $\vec{A}$ — and by the same algebra, perpendicular to $\vec{B}$. This is *why* torque and angular momentum point along the rotation axis, out of the plane of the motion.

Procedure: computing a cross product from components

Step 1 — Set up the determinant. Write $\hat{i}, \hat{j}, \hat{k}$ across the top, the components of the *first* vector in the middle row, the *second* in the bottom row. Order is critical here — the middle row is whichever vector comes first in $\vec{A} \times \vec{B}$.

Step 2 — Expand along the top row, remembering the alternating sign on the $\hat{j}$ term:

$$\hat{i}(\text{cover column 1}) - \hat{j}(\text{cover column 2}) + \hat{k}(\text{cover column 3})$$

Each “cover” is the 2×2 determinant of the remaining four entries.

Step 3 — Sanity-check with a dot product. Dot your answer with either original vector; you must get zero. This single check catches almost every arithmetic slip.

Procedure: finding the angle or a perpendicular unit vector

To get the angle between two vectors, combine both products: $\tan \theta = \frac{|\vec{A} \times \vec{B}|}{\vec{A} \cdot \vec{B}}$. To get a unit vector perpendicular to both, compute the cross product and divide by its magnitude: $\hat{n} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|}$.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find $\vec{A} \times \vec{B}$ for $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{B} = \hat{i} - \hat{j} + 2\hat{k}$.

Set up the determinant:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & -1 & 2 \end{vmatrix}$$

Expand:

$$\begin{aligned} &= \hat{i}[(3)(2) - (1)(-1)] - \hat{j}[(2)(2) - (1)(1)] + \hat{k}[(2)(-1) - (3)(1)] \\ &= \hat{i}(6 + 1) - \hat{j}(4 - 1) + \hat{k}(-2 - 3) = 7\hat{i} - 3\hat{j} - 5\hat{k} \end{aligned}$$

Check: $\vec{A} \cdot (\vec{A} \times \vec{B}) = (2)(7) + (3)(-3) + (1)(-5) = 14 - 9 - 5 = 0 \checkmark$

Answer: $\vec{A} \times \vec{B} = 7\hat{i} - 3\hat{j} - 5\hat{k}$.

Example 2 — [NEET Speed Trap]

Two vectors have magnitudes 6 and 5 units. If the magnitude of their cross product is 15, find the angle between them.

The trap: Students set up the full vector machinery. But only magnitudes are needed — use $|\vec{A} \times \vec{B}| = AB \sin \theta$ directly.

Mental shortcut.

$$15 = (6)(5) \sin \theta \Rightarrow \sin \theta = \frac{15}{30} = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

Answer: $\theta = 30^\circ$. *Sanity check:* the maximum possible cross product would be $6 \times 5 = 30$ (at 90°). We got exactly half, so $\sin \theta = 0.5$ and a 30° angle fits. $\checkmark$

Example 3 — [JEE Main Level]

Find the area of the triangle whose two sides are given by the vectors $\vec{P} = 3\hat{i} + \hat{j}$ and $\vec{Q} = \hat{i} + 4\hat{j}$, and also find a unit vector perpendicular to the plane of the triangle.

This blends the **area interpretation** with the **perpendicular-direction** use — two concepts in one.

Step 1 — Cross product (both vectors lie in the xy -plane, so only the $\hat{k}$ term survives):

$$\vec{P} \times \vec{Q} = \hat{k}[(3)(4) - (1)(1)] = \hat{k}(12 - 1) = 11\hat{k}$$

Step 2 — Area of the triangle:

$$\text{Area} = \frac{1}{2}|\vec{P} \times \vec{Q}| = \frac{1}{2}(11) = 5.5 \text{ square units}$$

Step 3 — Perpendicular unit vector:

$$\hat{n} = \frac{\vec{P} \times \vec{Q}}{|\vec{P} \times \vec{Q}|} = \frac{11\hat{k}}{11} = \hat{k}$$

Answer: Area = 5.5 square units; perpendicular unit vector = $\hat{k}$ (which makes sense — the vectors lie in the xy -plane, so the perpendicular points along the z -axis).

Example 4 — [JEE Advanced Level]

A vector $\vec{X}$ satisfies $\vec{X} \times (2\hat{i} + 3\hat{j}) = 6\hat{k}$ and also $\vec{X} \cdot (2\hat{i} + 3\hat{j}) = 0$, and $\vec{X}$ lies in the xy -plane. Find $\vec{X}$.

Strategy: Let $\vec{X} = x\hat{i} + y\hat{j}$. Use both conditions as simultaneous equations.

Condition 1 — the dot product is zero (so $\vec{X} \perp (2\hat{i} + 3\hat{j})$):

$$2x + 3y = 0 \quad \dots(i)$$

Condition 2 — the cross product:

$$\vec{X} \times (2\hat{i} + 3\hat{j}) = (x\hat{i} + y\hat{j}) \times (2\hat{i} + 3\hat{j}) = \hat{k}[(x)(3) - (y)(2)] = (3x - 2y)\hat{k}$$

Setting this equal to $6\hat{k}$:

$$3x - 2y = 6 \quad \dots(ii)$$

Solve (i) and (ii). From (i), $x = -\frac{3}{2}y$. Substitute into (ii):

$$3\left(-\frac{3}{2}y\right) - 2y = 6 \Rightarrow -\frac{9}{2}y - 2y = 6 \Rightarrow -\frac{13}{2}y = 6 \Rightarrow y = -\frac{12}{13}$$

$$\text{Then } x = -\frac{3}{2}\left(-\frac{12}{13}\right) = \frac{18}{13}.$$

Answer: $\vec{X} = \frac{18}{13}\hat{i} - \frac{12}{13}\hat{j}$. *Note:* the two products together pin down both the *direction* (perpendicularity, from the dot product) and the *magnitude-and-sense* (from the cross product) — a neat illustration of how dot and cross encode complementary information.

Section 5: Practice Exercises

1. [CBSE] Find $\vec{A} \times \vec{B}$ for $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{B} = 2\hat{i} + \hat{j} - \hat{k}$.
2. [NEET] The magnitudes of two vectors are 4 and 7. If their scalar product equals their vector product's magnitude, find the angle between them.
3. [JEE Main] Find the area of the parallelogram whose adjacent sides are $\vec{A} = 3\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{B} = \hat{i} - \hat{j} + 2\hat{k}$.
4. [JEE Main] Find a unit vector perpendicular to both $\vec{A} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{B} = \hat{i} + \hat{j} - 2\hat{k}$.
5. [JEE Advanced] Given $|\vec{A}| = 3$, $|\vec{B}| = 4$, and $|\vec{A} \times \vec{B}| = 6$, find $\vec{A} \cdot \vec{B}$ (both possible values).

Answers: 1. $-5\hat{i} + 7\hat{j} - 3\hat{k}$. 2. 45° (since $AB \cos \theta = AB \sin \theta \Rightarrow \tan \theta = 1$). 3. $\sqrt{3^2 + 7^2 + 5^2} = \sqrt{83} \approx 9.11$ square units. 4. $\frac{1}{\sqrt{35}}(\hat{i} + 5\hat{j} + 3\hat{k})$. 5. $\vec{A} \cdot \vec{B} = \pm \sqrt{(AB)^2 - |\vec{A} \times \vec{B}|^2} = \pm \sqrt{144 - 36} = \pm 6\sqrt{3}$.

Section 6: Crack the MCQ

Q1. The vector product $\vec{A} \times \vec{B}$ is zero. This means:

- (A) $\vec{A}$ and $\vec{B}$ are perpendicular

- (B) $\vec{A}$ and $\vec{B}$ are parallel or anti-parallel
 (C) either vector is a unit vector
 (D) $\vec{A} = \vec{B}$

Correct: (B). $|\vec{A} \times \vec{B}| = AB \sin \theta = 0$ requires $\sin \theta = 0$, i.e. $\theta = 0^\circ$ or 180° — the vectors are parallel or anti-parallel. *Why the traps fail:* (A) perpendicular vectors give the *maximum* cross product, not zero. (C) magnitude being 1 is irrelevant to the angle. (D) $\vec{A} = \vec{B}$ is one special parallel case, but the condition is far broader than equality.

Q2. Which of the following is correct?

- (A) $\hat{i} \times \hat{j} = \hat{i}$ (B) $\hat{j} \times \hat{k} = \hat{i}$ (C) $\hat{k} \times \hat{j} = \hat{i}$ (D) $\hat{i} \times \hat{i} = \hat{k}$

Correct: (B). Following the cyclic order $\hat{i} \rightarrow \hat{j} \rightarrow \hat{k} \rightarrow \hat{i}$, we get $\hat{j} \times \hat{k} = \hat{i}$. *Why the traps fail:* (A) should be $\hat{k}$, not $\hat{i}$. (C) reverses the cyclic order, so it equals $-\hat{i}$, not $+\hat{i}$. (D) any vector crossed with itself is $\vec{0}$, never $\hat{k}$.

Q3. If $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$ (not just equal in magnitude, but identical vectors), then:

- (A) the vectors are perpendicular
 (B) the vectors are parallel
 (C) $\vec{A} \times \vec{B} = \vec{0}$
 (D) it is impossible for any vectors

Correct: (C). Since $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ always, setting it equal to $+\vec{B} \times \vec{A}$ forces $\vec{A} \times \vec{B} = -\vec{A} \times \vec{B}$, hence $\vec{A} \times \vec{B} = \vec{0}$. *Why the traps fail:* (A) perpendicular vectors give non-zero, oppositely directed products — not identical. (B) is *implied* by (C) but (C) is the direct, complete statement of the condition. (D) wrongly calls it impossible; it's possible precisely when the product vanishes.

Q4. The area of a parallelogram with adjacent sides $\vec{A} = 2\hat{i}$ and $\vec{B} = 3\hat{j}$ is:

- (A) 5 (B) 6 (C) $\sqrt{13}$ (D) 0

Correct: (B). $\vec{A} \times \vec{B} = (2\hat{i}) \times (3\hat{j}) = 6(\hat{i} \times \hat{j}) = 6\hat{k}$, so the area is $|6\hat{k}| = 6$. *Why the traps fail:* (A) adds the magnitudes $(2+3)$ — wrong operation. (C) computes $\sqrt{2^2 + 3^2}$ as if finding a resultant. (D) wrongly assumes perpendicular vectors give zero area (that's parallel vectors).

Section 7: Common Student Pitfalls & Pro-Tips

- 1. Treating the cross product like ordinary multiplication.** It is *anti-commutative*: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$. Reversing the order flips the sign — a costly slip in torque problems where direction decides clockwise vs anticlockwise.

2. **Forgetting the minus sign on the $\hat{j}$ term** in the determinant expansion. The middle term is *subtracted*. This single sign error is the most common cross-product mistake in exams.
3. **Confusing the dot and cross products.** Dot gives a scalar and uses $\cos \theta$ (maximal when parallel); cross gives a vector and uses $\sin \theta$ (maximal when perpendicular). Mixing them up corrupts the whole solution.
4. **Believing parallel vectors give a non-zero product.** Parallel $\Rightarrow \sin \theta = 0 \Rightarrow$ zero cross product. This is the reason a force pushing straight toward a hinge produces no torque.

Pro-Tip: After any cross-product calculation, dot the result with one of the original vectors. If you don't get zero, you've made an arithmetic error — go back. This costs five seconds and saves whole questions. And for two vectors already in the xy -plane, skip the full determinant: only the $\hat{k}$ component ($A_x B_y - A_y B_x$) is non-zero.

Section 8: Cheat Sheet

- **Definition:** $\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$ — a *vector* perpendicular to both, by the right-hand rule.
- **Magnitude:** $|\vec{A} \times \vec{B}| = AB \sin \theta$ = area of the parallelogram on $\vec{A}, \vec{B}$.
- **Anti-commutative:** $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$; $\vec{A} \times \vec{A} = \vec{0}$.
- **Unit vectors (cyclic +):** $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$.
- **Determinant form** for components; remember the **minus on $\hat{j}$** .
- **Triangle area** = $\frac{1}{2}|\vec{A} \times \vec{B}|$; **perpendicular check:** $\vec{A} \cdot (\vec{A} \times \vec{B}) = 0$.

Memory aid — “**Cross is perpendicular, sine of the angle**”: dot uses cos and points along; cross uses sin and points across (perpendicular).

Memory aid — “**i, j, k go round in a ring**”: forward around the ring $\hat{i} \rightarrow \hat{j} \rightarrow \hat{k} \rightarrow \hat{i}$ is positive; going backward picks up a minus sign.

Torque and Angular Momentum

Exam Relevance: This is the dynamical heart of the chapter. **JEE Main** asks 1–2 questions yearly — usually $\tau = dL/dt$, conservation problems, or angular momentum of a projectile. **NEET** keeps it conceptual: the spinning-skater idea and “is angular momentum conserved here?” **JEE Advanced** pushes into angular impulse and collisions involving rotation. **CBSE Boards** wants the $\vec{\tau} = d\vec{L}/dt$ derivation and the conservation law stated cleanly.

Section 1: Concept Intuition

Think about the last time you opened a heavy door. Without realising it, you pushed on the handle — the edge *farthest* from the hinges. Nobody pushes near the hinge; it feels useless, and it is. That instinct is the entire concept of **torque** living inside your muscle memory.

Torque is the **turning effect of a force** — how effectively a push makes something rotate. And it depends on three things: how hard you push, *where* you push (distance from the pivot), and *in which direction* you push. The door teaches all three. Push harder → it swings faster. Push farther from the hinge → much easier. Push *toward* the hinge instead of perpendicular to the door → nothing happens at all, no matter how hard you shove.

That last point is the one students forget. Only the part of the force that is **perpendicular** to the line joining the pivot and the push contributes to turning. A force aimed straight at the axis has zero turning effect. This is why a mechanic loosening a tight bolt grabs the *longest* spanner and pushes at right angles to it — maximum distance, maximum perpendicular force, maximum torque.

Think about it this way... A *thela-wala* (handcart vendor) pushing his cart to turn a corner pushes sideways on the far end of the handle, never straight down the middle. Long arm, sideways push — that’s torque optimised by intuition.

One special case worth naming: a **couple** — two equal, opposite, parallel forces whose lines of action don’t coincide (think of your two hands turning a steering wheel, or twisting a water-bottle cap). A couple produces pure rotation with no net force, and its torque is simply $\tau = F \times d$, where d is the perpendicular distance between the two forces. Steering wheels, taps, and screwdrivers all run on couples.

A quick word on **direction**. Torque and angular momentum are *axial* vectors — they point along the rotation axis, not in the plane of motion. To find which way, use the right-hand rule: curl the fingers of your right hand in the sense of rotation, and your thumb points along $\vec{\tau}$ (or $\vec{L}$). Anticlockwise rotation (seen from the front) points the vector toward you; clockwise points it away.

Now the partner concept: **angular momentum**. If torque is the rotational cousin of force, angular momentum $\vec{L}$ is the rotational cousin of linear momentum $\vec{p}$. It’s the “quantity of rotational motion” stored in a spinning body — and like linear momentum, it resists being changed. A spinning top stays upright, a moving bicycle is far easier to balance than a stationary one, and a spinning cricket ball holds its line — all because angular momentum is “locked in” and only an external torque can change it.

The most famous demonstration is one you’ve seen at every wedding *sangeet*. A dancer doing fast *chakkars* (spins) starts with arms outstretched, then pulls them in — and suddenly whips around much faster. No one pushed them. They simply moved their mass closer to the spin axis, shrinking their moment of inertia I . Since angular momentum $L = I\omega$ must stay constant (there’s no external torque about the vertical axis), shrinking I forces ω to shoot up. Pull arms in $\rightarrow$ spin faster. Stretch arms out $\rightarrow$ slow down. That’s conservation of angular momentum in a single graceful move.

Think about it this way... Sit on a rotating office chair, give yourself a spin with legs stretched out, then tuck them in. You’ll speed up noticeably. Your body just ran a physics experiment.

The same law operates on a cosmic scale. A planet orbiting the Sun feels a force always directed *toward* the Sun, so that force has zero torque about the Sun (its moment arm is zero). Angular momentum about the Sun is therefore conserved — which is exactly why a planet speeds up when it swings close to the Sun and slows down when far away, sweeping out equal areas in equal times. This is Kepler’s second law, and it’s nothing more than conservation of angular momentum in disguise. Push it further and you get one of the most dramatic events in physics: when a giant star collapses into a tiny neutron star, its radius crashes by a factor of thousands, its moment of inertia plummets, and to conserve L its spin rate explodes — some neutron stars rotate hundreds of times *per second*. Same dancer, same law, unimaginably larger stage.

Limiting Conditions & Assumptions

1. **Torque and angular momentum are always defined *about a chosen point or axis*.** Change the reference point and the numbers change. There’s no such thing as “the torque” in the abstract — always ask “about which point?”
2. **Conservation of angular momentum holds only when the *net external* torque is zero.** Internal forces (like the dancer’s muscles, or friction between coupling wheels) can rearrange I and ω but can never change the total L .
3. **It’s component-wise.** If the external torque is zero along one particular axis, only the angular momentum along *that* axis is conserved — the others may still change.

Section 2: Key Formulas & Definitions

Torque (moment of force) about a point:

$$\vec{\tau} = \vec{r} \times \vec{F}, \quad |\vec{\tau}| = rF \sin \theta = r_{\perp} F = rF_{\perp}$$

where $r_{\perp} = r \sin \theta$ is the **moment arm** (perpendicular distance from the axis to the line of action of the force).

Rotational form of Newton’s second law:

$$\vec{\tau}_{net} = I\vec{\alpha} = \frac{d\vec{L}}{dt}$$

Angular momentum of a particle about a point:

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}, \quad |\vec{L}| = mvr \sin \theta = pr_{\perp}$$

Angular momentum of a rigid body about a fixed axis:

$$\vec{L} = I\vec{\omega}$$

Angular impulse–momentum principle:

$$\vec{J} = \int_{t_1}^{t_2} \vec{\tau} dt = \Delta \vec{L} = \vec{L}_f - \vec{L}_i$$

Conservation of angular momentum (when $\vec{\tau}_{ext} = 0$):

$$\vec{L} = \text{constant} \quad \Rightarrow \quad I_1\omega_1 = I_2\omega_2$$

Rotational kinetic energy:

$$K_{rot} = \frac{1}{2}I\omega^2 = \frac{L^2}{2I} = \frac{1}{2}L\omega$$

Work and power in rotation:

$$W = \int \vec{\tau} \cdot d\vec{\theta} = \tau\theta \quad (\tau \text{ constant}), \quad P = \vec{\tau} \cdot \vec{\omega}$$

Notation & dimensions:

Quantity	Symbol	SI Unit	Dimensional Formula
Torque	τ	N · m	$[M^1L^2T^{-2}]$
Angular momentum	L	kg · m ² · s ⁻¹ = J · s	$[M^1L^2T^{-1}]$
Moment of inertia	I	kg · m ²	$[M^1L^2T^0]$
Angular impulse	J	N · m · s = J · s	$[M^1L^2T^{-1}]$

(Note that torque and *energy* share the unit dimension but torque is a vector measured in N · m, never joules — keep them mentally separate.)

Section 3: CBSE Board Derivation — Relation Between Torque and Angular Momentum

We will prove the single most important equation of rotational dynamics: that the rate of change of angular momentum equals the applied torque. From it, conservation of angular momentum falls out instantly.

Step 1 — Start from the definition of angular momentum of a particle.

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$$

Step 2 — Differentiate both sides with respect to time. Angular momentum changes as both $\vec{r}$ and $\vec{p}$ evolve, so we apply the product rule for the cross product (order matters in a cross product, so we keep it):

$$\frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt}$$

Step 3 — Examine the first term and watch it vanish. Now, $\frac{d\vec{r}}{dt} = \vec{v}$ and $\vec{p} = m\vec{v}$, so the first term is:

$$\vec{v} \times m\vec{v} = m(\vec{v} \times \vec{v}) = \vec{0}$$

This is zero because the cross product of any vector with itself is zero — $\vec{v}$ is parallel to itself, so $\sin 0^\circ = 0$. *This is the crucial step:* velocity and momentum point the same way, so they contribute no turning effect.

Step 4 — Identify the second term. By Newton's second law, $\frac{d\vec{p}}{dt} = \vec{F}$. Therefore:

$$\frac{d\vec{L}}{dt} = \vec{r} \times \vec{F} = \vec{\tau}$$

Step 5 — Result.

$$\boxed{\vec{\tau}_{ext} = \frac{d\vec{L}}{dt}}$$

This is Newton's second law written in angular language — the rotational twin of $\vec{F} = d\vec{p}/dt$.

Step 6 — The conservation law as a corollary. If the net external torque is zero, then:

$$\frac{d\vec{L}}{dt} = 0 \quad \Rightarrow \quad \vec{L} = \text{constant}$$

This is the **law of conservation of angular momentum**: *if no net external torque acts on a system, its total angular momentum stays constant, no matter what internal rearrangements occur.* The dancer pulling in her arms, two wheels coupling on a shaft, and a planet sweeping along its orbit are all governed by this one line.

Key Procedure — Computing Torque and Angular Momentum by Cross Product

When you're given vectors directly, use the determinant form. For $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$:

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$

Step 1: Write $\vec{r}$ in the middle row, $\vec{F}$ (or $\vec{p}$ for angular momentum) in the bottom row. **Step 2:** Expand the determinant: $\hat{i}(yF_z - zF_y) - \hat{j}(xF_z - zF_x) + \hat{k}(xF_y - yF_x)$. **Step 3:** For planar (2D) problems, only the $\hat{k}$ term survives — a positive $\hat{k}$ means anticlockwise, negative means clockwise. This sign-from-direction trick saves real time in exams.

Quick worked illustration. Let $\vec{r} = (2\hat{i} + 3\hat{j})$ m and $\vec{F} = (5\hat{i} + \hat{j})$ N. Then only the $\hat{k}$ component matters:

$$\tau_z = xF_y - yF_x = (2)(1) - (3)(5) = 2 - 15 = -13 \text{ N} \cdot \text{m}$$

The negative sign tells you the force tends to rotate the body *clockwise* about the origin. Notice you never had to draw an angle or find a perpendicular distance — the determinant handles geometry and sign for you in one shot. The identical machinery, with $\vec{p} = m\vec{v}$ in the bottom row, gives angular momentum.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A mechanic applies a force of 40 N at the end of a spanner of length 0.25 m. The force makes an angle of 30° with the spanner handle. Calculate the torque produced about the bolt.

Given: $F = 40$ N, $r = 0.25$ m, $\theta = 30^\circ$.

Formula:

$$\tau = rF \sin \theta$$

Substitution:

$$\tau = (0.25)(40) \sin 30^\circ = (0.25)(40)(0.5) = 5 \text{ N} \cdot \text{m}$$

Answer: The torque is 5 N · m. *Presentation note:* always carry the unit N · m (not J) and, if asked, state the sense of rotation.

Example 2 — [NEET Speed Trap]

A spinning dancer with arms outstretched has a moment of inertia I and angular speed ω . She pulls her arms in, reducing her moment of inertia to $\frac{2I}{5}$. Find (a) her new angular speed, and (b) the ratio of her new rotational kinetic energy to the old one.

The trap: Students assume kinetic energy is conserved like in a collision. It is **not** — the dancer's muscles do work pulling the arms in, so KE *increases*. Only angular momentum is conserved.

Mental shortcut — conserve L , then read off the ratios.

(a) $L = I\omega = \text{constant}$:

$$I\omega = \frac{2I}{5}\omega' \Rightarrow \omega' = \frac{5}{2}\omega = 2.5\omega$$

(b) Using $K = \frac{L^2}{2I}$ with L fixed, $K \propto \frac{1}{I}$:

$$\frac{K'}{K} = \frac{I}{I'} = \frac{I}{2I/5} = \frac{5}{2} = 2.5$$

Answer: New angular speed $= 2.5\omega$; rotational KE increases by a factor of 2.5. *Sanity check:* she spins faster *and* has more energy — exactly what your intuition says when someone “powers up” a spin. The extra energy came from her muscles, not from nowhere. ✓

Example 3 — [JEE Main Level]

A particle of mass m is projected from the ground with speed u at an angle θ to the horizontal. Find the magnitude of its angular momentum about the point of projection when it is at the highest point of its trajectory.

Concept 1 — Projectile motion gives position and velocity at the top. - Maximum height:

$$H = \frac{u^2 \sin^2 \theta}{2g} \text{ - At the top, the velocity is purely horizontal: } v_x = u \cos \theta \text{ (vertical component is zero).}$$

Concept 2 — Angular momentum about the launch point. $L = mv_{\perp} \times (\text{perpendicular distance})$. At the top, the horizontal velocity $u \cos \theta$ is perpendicular to the vertical height H , so:

$$L = m(u \cos \theta)(H) = m(u \cos \theta) \left(\frac{u^2 \sin^2 \theta}{2g} \right)$$

$$L = \frac{m u^3 \sin^2 \theta \cos \theta}{2g}$$

Answer: $L = \frac{m u^3 \sin^2 \theta \cos \theta}{2g}$, directed into the plane (clockwise sense). *Why two concepts:* you cannot solve this without first extracting H and v_x from projectile kinematics and *then* applying the angular-momentum definition — a textbook JEE Main blend.

Example 4 — [JEE Advanced Level]

A uniform rod of mass M and length L is pivoted at one end on a frictionless horizontal table and is initially at rest. A bullet of mass m moving horizontally with speed v (perpendicular to the rod) strikes the free end and embeds itself. Find the angular speed of the rod-plus-bullet system immediately after impact.

Figure 6.3 A horizontal rod pivoted at its left end (shown as a small circle). A bullet approaches the free right end horizontally and perpendicular to the rod, with velocity v . After impact the bullet is embedded at the end and the system rotates about the pivot with angular speed ω . (see next page)

Step 1 — Choose the right conserved quantity. During the collision, the pivot exerts a large unknown impulsive force. So linear momentum is *not* conserved. But that pivot force has **zero torque about the pivot** (its moment arm is zero), and the collision is instantaneous, so **angular momentum about the pivot is conserved**. This choice of axis is the whole insight.

Step 2 — Angular momentum before impact. Only the bullet moves; its angular momentum about the pivot is:

$$L_i = mvL$$

Step 3 — Moment of inertia after impact. The rod (about its end) plus the embedded bullet (a point mass at distance L):

$$I_f = \underbrace{\frac{ML^2}{3}}_{\text{rod}} + \underbrace{mL^2}_{\text{bullet}}$$

Step 4 — Angular momentum after impact and equate.

$$L_f = I_f \omega = \left(\frac{ML^2}{3} + mL^2 \right) \omega$$

$$mvL = \left(\frac{ML^2}{3} + mL^2 \right) \omega$$

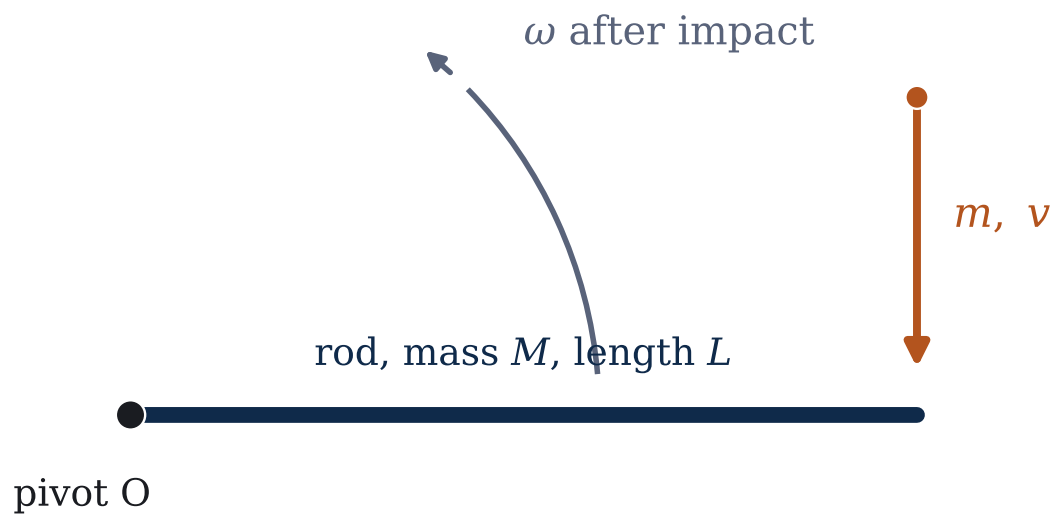
Step 5 — Solve for ω .

$$\omega = \frac{mvL}{L^2 \left(\frac{M}{3} + m \right)} = \frac{mv}{L \left(\frac{M+3m}{3} \right)} = \frac{3mv}{L(M+3m)}$$

Answer: $\omega = \frac{3mv}{L(M+3m)}$. *Note:* kinetic energy is **not** conserved here (the embedding is perfectly inelastic), which is exactly why we leaned on angular momentum, not energy.

Section 5: Practice Exercises

1. [CBSE] A force $\vec{F} = (2\hat{i} + 3\hat{j})$ N acts at a point with position vector $\vec{r} = (4\hat{i} + \hat{j})$ m. Find the torque about the origin.
2. [NEET] A planet moves in an elliptical orbit around the Sun. About which point is its angular momentum conserved, and what stays constant as a result?
3. [JEE Main] A constant torque acting on a wheel changes its angular momentum from L_0 to $3L_0$ in 6 seconds. Find the magnitude of the torque.



$$L_i = mvL = L_f = \left(\frac{1}{3}ML^2 + mL^2\right)\omega$$

Figure 6.3 A horizontal rod pivoted at its left end (shown as a small circle). A bullet approaches the free right end horizontally and perpendicular to the rod, with velocity v . After impact the bullet is embedded at the end and the system rotates about the pivot with angular speed ω .

4. **[JEE Main]** A particle of mass 2 kg is at position $(3\hat{i} + \hat{j})$ m moving with velocity $(2\hat{i} - \hat{j})$ m/s. Find its angular momentum about the origin.
5. **[JEE Advanced]** A solid sphere of mass m , radius R rests on rough ground. A time-varying horizontal force $F = bt$ acts at its topmost point. Find the angular momentum of the sphere about the bottom-most contact point as a function of time. Does the answer depend on whether the ground is rough or smooth?

Answers: 1. $10\hat{k}$ N · m. 2. About the Sun (the centre of force); the areal velocity — and hence the angular momentum — stays constant (Kepler's second law). 3. $\frac{L_0}{3}$ N · m. 4. $-5\hat{k}$ kg · m² · s⁻¹. 5. $L = bRt^2$; independent of the surface, since friction acts through the contact point and has zero torque about it.

Section 6: Crack the MCQ

Q1. A force is applied to a rigid body that can rotate about a fixed axis. The torque about that axis is zero when the force is:

- (A) perpendicular to the position vector
- (B) directed along the position vector (toward or away from the axis)
- (C) of small magnitude
- (D) applied at the rim of the body

Correct: (B). Torque $= rF \sin \theta$. A force pointing along $\vec{r}$ has $\theta = 0^\circ$ (or 180°), so $\sin \theta = 0$ and the torque vanishes — like pushing a door straight at its hinge. *Why the traps fail:* (A) gives *maximum* torque, the opposite of zero. (C) reduces torque but never to zero. (D) actually maximises the moment arm — torque is largest there, not zero.

Q2. A skater spinning with her arms extended pulls them in. Which quantity stays constant?

- (A) Angular velocity
- (B) Moment of inertia
- (C) Angular momentum
- (D) Rotational kinetic energy

Correct: (C). With no external torque about the spin axis, angular momentum $L = I\omega$ is conserved. *Why the traps fail:* (A) and (B) both change (that's the whole point — I drops, ω rises). (D) is the classic trap: KE *increases* because the skater's muscles do work pulling the arms in. Conserved momentum $\neq$ conserved energy.

Q3. The dimensional formula of angular momentum is the same as that of:

- (A) torque
- (B) Planck's constant

- (C) angular velocity
(D) force

Correct: (B). Angular momentum has dimensions $[ML^2T^{-1}]$, identical to Planck's constant h (energy $\times$ time = $ML^2T^{-2} \cdot T$). This is no coincidence — h literally is a quantum of angular momentum. *Why the traps fail:* **(A)** torque is $[ML^2T^{-2}]$ — differs by one power of time. **(C)** angular velocity is $[T^{-1}]$. **(D)** force is $[MLT^{-2}]$.

Q4. A particle moves in a straight line at constant velocity. Its angular momentum about a point *not* on that line:

- (A) is zero
(B) increases with time
(C) is constant and non-zero
(D) decreases with time

Correct: (C). $L = mvr_{\perp}$, where $r_{\perp}$ is the constant perpendicular distance from the point to the line. Since m , v , and $r_{\perp}$ are all fixed, L is constant and non-zero. (Equivalently: no force $\Rightarrow$ no torque $\Rightarrow L$ conserved.) *Why the traps fail:* **(A)** is true only if the point lies *on* the line ($r_{\perp} = 0$). **(B)** and **(D)** wrongly assume L changes as the particle moves — but the moment arm stays fixed for straight-line motion.

Section 7: Common Student Pitfalls & Pro-Tips

- Forgetting to specify the reference point.** Torque and angular momentum are meaningless without “about which point/axis.” The same force gives different torques about different points — always anchor your reference first.
- Assuming energy is conserved when only angular momentum is.** In skater spins, wheel-coupling, and inelastic rotational collisions, L is conserved but KE is not. Reach for energy conservation only when the process is genuinely lossless.
- Using the full force instead of its perpendicular component.** Only $F \sin \theta$ (or equivalently $r_{\perp}F$) turns the body. The radial component is dead weight for rotation.
- Mixing up the units of torque and energy.** Both come out as $N \cdot m$ dimensionally, but torque is a vector quantity reported in $N \cdot m$, while energy is a scalar in joules. Never write torque in joules.

Pro-Tip: In collision-and-rotation problems (bullet-in-rod, person-jumping-on-turntable), *always* take angular momentum about the pivot or hinge. The unknown impulsive reaction there has zero moment arm, so it drops out automatically — turning a messy two-unknown problem into a one-line conservation statement.

Section 8: Cheat Sheet

- **Torque:** $\vec{\tau} = \vec{r} \times \vec{F}$, $|\tau| = rF \sin \theta = r_{\perp} F$ — only the perpendicular force turns things.
- **Angular momentum:** $\vec{L} = \vec{r} \times \vec{p}$, $L = I\omega$ | dimensions $[ML^2T^{-1}]$, unit J · s.
- **Master equation:** $\vec{\tau}_{ext} = \frac{d\vec{L}}{dt} \rightarrow$ the rotational $\vec{F} = d\vec{p}/dt$.
- **Conservation:** $\tau_{ext} = 0 \Rightarrow L = \text{const} \Rightarrow I_1\omega_1 = I_2\omega_2$. (Skater, planets, coupling wheels.)
- **Angular impulse:** $J = \int \tau dt = \Delta L$.
- **Rotational KE:** $K = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$.

Memory aid — “Same point, every step”: pick one reference point at the start of a torque/ L problem and never switch — switching mid-solution is the #1 source of wrong answers.

Memory aid — “ L lives, energy leaks”: in spins and inelastic rotational hits, angular momentum survives but kinetic energy usually doesn’t.

Equilibrium of a Rigid Body

Exam Relevance: This is a CBSE favourite and a steady JEE/NEET performer. **CBSE Boards** regularly asks the principle of moments, levers, and centre of gravity (2–3 marks). **JEE Main** poses beam-and-support and ladder-against-wall problems. **NEET** keeps it conceptual — conditions for equilibrium and types of equilibrium. Mastering this turns intimidating “find the reactions” problems into routine bookkeeping.

Section 1: Concept Intuition

So far this chapter has been about *making* things move and spin. This subtopic asks the opposite question: when does a rigid body stay perfectly still — neither sliding away nor toppling over? That state is called **equilibrium**, and it is the physics of every bridge, every shelf bracket, every *charpai* (cot) leg, and every child balanced on a seesaw.

A point particle only had to worry about one thing to stay put: the forces on it had to cancel. But a rigid body has *size*, so it can do something a particle never could — it can **rotate** even when the forces cancel. Imagine pushing the top of a cupboard forward while a friend pushes the bottom backward with an equal force. The total force is zero, so the cupboard won't slide. Yet it will happily topple, because those two forces create a *turning effect*. This is the key insight: a rigid body needs **two** separate conditions to be in equilibrium.

1. **Translational equilibrium** — the net force is zero, so the centre of mass does not accelerate. The body won't drift in any direction.
2. **Rotational equilibrium** — the net torque is zero, so the body does not start spinning. It won't topple or twist.

Both must hold. Forces balanced but torques unbalanced → it spins. Torques balanced but forces unbalanced → it slides. Only when *both* vanish is the body truly at rest and content to stay there.

Think about it this way... A seesaw at the park is the perfect teacher. A heavy uncle near the pivot can balance a small child sitting far out at the end. Balance isn't about equal *weights* — it's about equal *moments*: weight times distance from the pivot. The child's small weight times big distance equals the uncle's big weight times small distance. This is the **principle of moments**: for balance, the total clockwise turning effect equals the total anticlockwise turning effect.

This single principle powers every **lever** you've ever used — a *sabal* (crowbar) prying up a rock, scissors, a bottle opener, a *nimbu nichodne wala* (lime squeezer), even your own forearm. A lever lets a small effort move a large load by giving the effort a longer arm. The payoff is the **mechanical advantage** — the factor by which the lever multiplies your force.

Closely tied to balance is the **centre of gravity (CG)** — the single point where the body's entire weight effectively acts. Support a body exactly under its CG and it balances; support it anywhere else and gravity produces a toppling torque. For ordinary objects in uniform gravity, the CG sits exactly at the centre of mass.

Limiting Conditions & Assumptions

1. **Both conditions are required, and they are independent.** Checking only forces (or only torques) is the commonest source of wrong answers.
 2. **Torque must be computed about a chosen point — but for a body in equilibrium, the net torque is zero about *every* point.** This freedom is a gift: pick the pivot that eliminates the most unknowns.
 3. **Centre of gravity equals centre of mass only when g is uniform** over the body. For everyday objects this is essentially exact.
-

Section 2: Key Formulas & Definitions

Conditions for complete equilibrium of a rigid body:

$$\sum \vec{F}_{ext} = \vec{0} \quad (\text{translational equilibrium})$$

$$\sum \vec{\tau}_{ext} = \vec{0} \quad (\text{rotational equilibrium})$$

In two dimensions these become three scalar equations:

$$\sum F_x = 0, \quad \sum F_y = 0, \quad \sum \tau = 0$$

Principle of moments (about a pivot): for balance,

$$\text{anticlockwise moments} = \text{clockwise moments}$$

$$F_1 d_1 = F_2 d_2 \quad \Rightarrow \quad (\text{load}) \times (\text{load arm}) = (\text{effort}) \times (\text{effort arm})$$

Mechanical advantage of a lever:

$$\text{M.A.} = \frac{\text{Load}}{\text{Effort}} = \frac{\text{Effort arm}}{\text{Load arm}}$$

Couple — two equal, opposite, non-collinear forces; its moment (torque) is:

$$\tau_{couple} = F \times d$$

where d is the perpendicular distance between the two forces. A couple produces pure rotation with zero net force.

Centre of gravity (position, weighting by weight $w_i = m_i g$):

$$x_{cg} = \frac{\sum w_i x_i}{\sum w_i} = \frac{\sum m_i x_i}{\sum m_i} = x_{cm} \quad (\text{uniform } g)$$

Notation: F = force, d = perpendicular distance (moment arm), τ = torque. SI unit of torque/moment is $\text{N} \cdot \text{m}$; dimensional formula $[M^1 L^2 T^{-2}]$. Mechanical advantage is dimensionless.

Types of equilibrium (how a body responds to a small disturbance):

Type	On disturbance	CG behaviour	Everyday example
Stable	returns to original position	CG <i>rises</i>	ball in a valley; cone on its base
Unstable	moves further away	CG <i>falls</i>	ball on a hilltop; cone on its apex
Neutral	stays in new position	CG stays at same height	ball on a flat floor; cone on its side

Section 3: CBSE Board Derivations & Key Procedures

Derivation: the Principle of Moments from rotational equilibrium

Consider a light rod (a lever) free to rotate about a pivot (fulcrum) O . A load force F_1 acts downward at perpendicular distance d_1 on one side, and an effort force F_2 acts downward at perpendicular distance d_2 on the other side.

Figure 6.4 A horizontal lever resting on a triangular fulcrum at point O . A downward force F_1 acts at distance d_1 to the left of O , and a downward force F_2 acts at distance d_2 to the right of O . The two forces produce opposite turning senses about O . (see next page)

Step 1 — Apply rotational equilibrium about the pivot. For the rod not to rotate, the net torque about O must vanish:

$$\sum \tau_O = 0$$

Step 2 — Write each torque with its sense. The load tends to turn the rod one way (say anticlockwise, $+$) and the effort the other way (clockwise, $-$):

$$+F_1 d_1 - F_2 d_2 = 0$$

Step 3 — Rearrange.

$$\boxed{F_1 d_1 = F_2 d_2}$$

This is the **principle of moments**: at balance, the anticlockwise moment equals the clockwise moment. Rearranging gives the lever's mechanical advantage $\frac{F_1}{F_2} = \frac{d_2}{d_1}$ — a long effort arm d_2 lets a small effort F_2 lift a large load F_1 . That is the secret of every crowbar.

Procedure: solving any rigid-body equilibrium problem

Step 1 — Draw a clean free-body diagram. Mark every external force with its point of application: weights (acting at the CG), normal reactions, tensions, friction, and applied forces.

Step 2 — Apply force balance. Write $\sum F_x = 0$ and $\sum F_y = 0$.

Step 3 — Apply torque balance about a smartly chosen point. Here lies the art: *choose the*

$$\sum \tau_O = 0 \Rightarrow F_1 d_1 = F_2 d_2$$

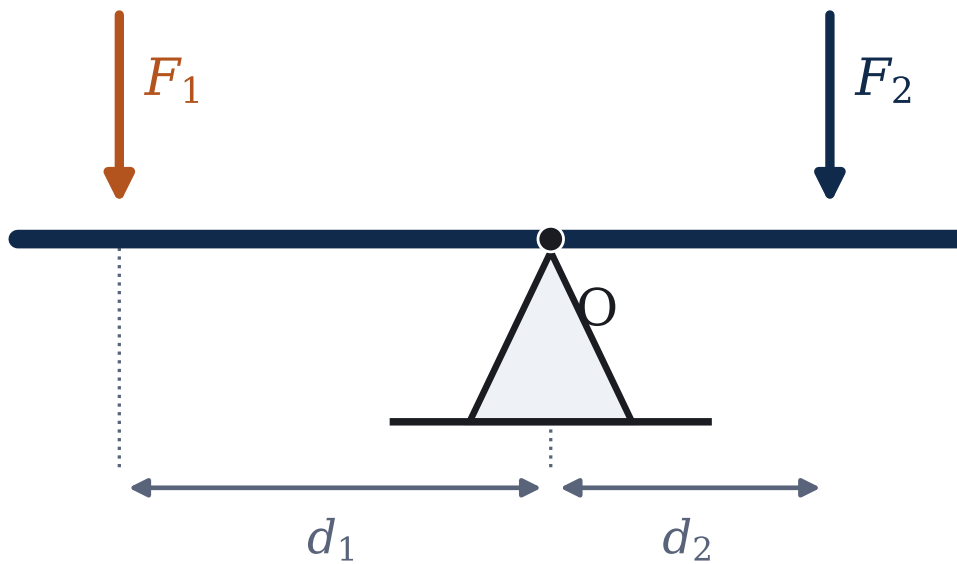


Figure 6.4 A horizontal lever resting on a triangular fulcrum at point O . A downward force F_1 acts at distance d_1 to the left of O , and a downward force F_2 acts at distance d_2 to the right of O . The two forces produce opposite turning senses about O .

pivot at the point where an unknown force acts, because that force then has zero moment arm and drops out of the equation. This often lets you solve for one unknown immediately.

Step 4 — Solve the simultaneous equations for the unknown forces or distances, and sanity-check signs (a negative reaction means you drew its direction backwards).

Procedure: locating the centre of gravity experimentally

Hang an irregular lamina (say a cardboard cut-out) freely from a point and drop a plumb line from that point; the CG lies somewhere on that vertical line. Repeat from a second hanging point. The CG is where the two plumb lines intersect — because a freely suspended body always settles with its CG directly below the point of suspension (any offset would create a restoring torque).

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A uniform metre stick of mass 0.4 kg is pivoted at its centre. A 0.6 kg mass hangs 20 cm from the pivot on one side. At what distance from the pivot must a 0.3 kg mass be hung on the other side to balance the stick?

Given: the stick is uniform and pivoted at its centre, so its own weight acts *at* the pivot and produces no torque. Load $m_1 = 0.6$ kg at $d_1 = 0.20$ m; balancing mass $m_2 = 0.3$ kg at unknown d_2 .

Apply the principle of moments (the g cancels):

$$m_1 d_1 = m_2 d_2 \Rightarrow (0.6)(0.20) = (0.3)(d_2)$$

$$d_2 = \frac{0.12}{0.3} = 0.40 \text{ m}$$

Answer: The 0.3 kg mass must hang 40 cm from the pivot.

Example 2 — [NEET Speed Trap]

A seesaw is balanced about its central pivot. A 30 kg child sits 1.5 m from the pivot. Where must a 45 kg child sit to balance it?

The trap: Students start computing weights with $g = 9.8$. But g appears on both sides and cancels — work directly with mass $\times$ distance.

Mental shortcut.

$$m_1 d_1 = m_2 d_2 \Rightarrow (30)(1.5) = (45)(d_2) \Rightarrow d_2 = \frac{45}{45} = 1.0 \text{ m}$$

Answer: The heavier (45 kg) child sits 1.0 m from the pivot. *Sanity check:* the heavier child sits *closer* to the pivot — exactly as intuition (and every real seesaw) demands. ✓

Example 3 — [JEE Main Level]

A uniform beam of length 4 m and weight 200 N rests horizontally on two supports, one at the left end (A) and one 1 m from the right end (B). Find the reaction forces at the two supports.

Figure 6.5 A horizontal beam of length 4 m. Support A at the left end ($x = 0$), support B at $x = 3$ m. The beam's weight 200 N acts downward at its centre, $x = 2$ m. Upward reaction forces N_A at A and N_B at B. (see next page)

This needs **both** equilibrium conditions together.

Step 1 — Force balance (vertical):

$$N_A + N_B = 200 \quad \dots(i)$$

Step 2 — Torque balance about A (so N_A drops out — it acts at A with zero moment arm). The weight acts at the centre, $x = 2$ m; support B is at $x = 3$ m:

$$N_B(3) - (200)(2) = 0 \Rightarrow N_B = \frac{400}{3} \approx 133.3 \text{ N}$$

Step 3 — Back-substitute into (i):

$$N_A = 200 - 133.3 = 66.7 \text{ N}$$

Answer: $N_A \approx 66.7 \text{ N}$, $N_B \approx 133.3 \text{ N}$. *Check:* support B is closer to the centre of gravity, so it carries more of the load — physically sensible, and $66.7 + 133.3 = 200 \text{ N}$ balances the weight. ✓

Example 4 — [JEE Advanced Level]

A uniform ladder of weight W and length L leans against a smooth (frictionless) vertical wall, making an angle θ with the rough horizontal floor. Find the minimum coefficient of friction between the ladder and the floor needed to prevent slipping.

Figure 6.6 A ladder leaning against a vertical wall at angle θ to the floor. Forces: weight W downward at the ladder's midpoint; normal reaction N_w horizontal from the smooth wall at the top; normal reaction N_f upward and friction f horizontal from the floor at the bottom. (see next page)

Step 1 — Identify the forces. Wall is smooth $\rightarrow$ only a horizontal normal N_w at the top. Floor is rough $\rightarrow$ vertical normal N_f and horizontal friction f at the bottom. Weight W acts at the midpoint.

Step 2 — Force balance. Horizontal: $f = N_w \quad \dots(i)$ Vertical: $N_f = W \quad \dots(ii)$

Step 3 — Torque balance about the bottom of the ladder (eliminates N_f and f , which act there). Taking the foot as pivot, with the ladder length L :

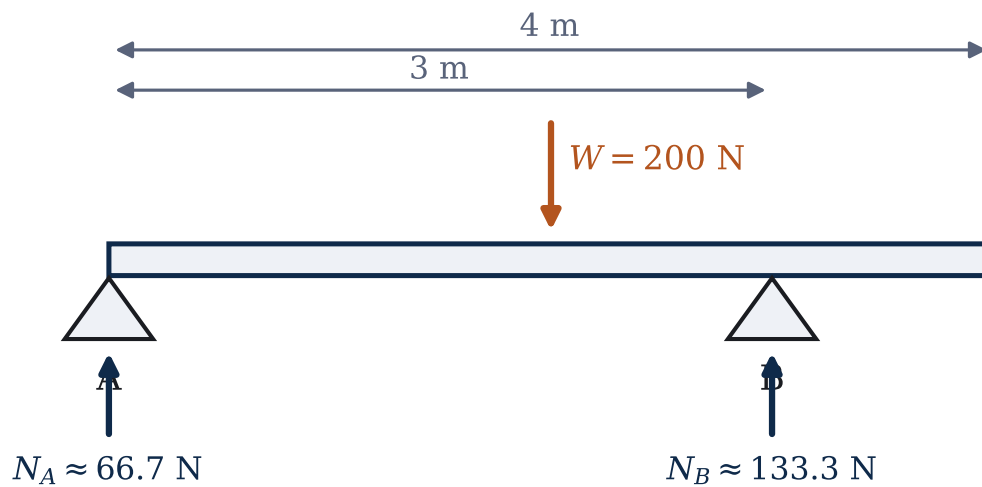


Figure 6.5 A horizontal beam of length 4 m. Support A at the left end ($x = 0$), support B at $x = 3 \text{ m}$. The beam's weight 200 N acts downward at its centre, $x = 2 \text{ m}$. Upward reaction forces N_A at A and N_B at B.

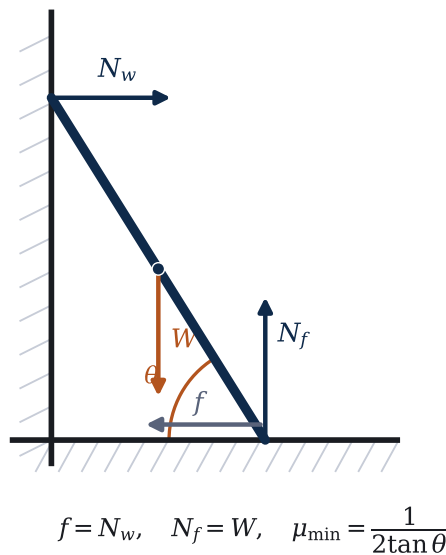


Figure 6.6 A ladder leaning against a vertical wall at angle θ to the floor. Forces: weight W downward at the ladder's midpoint; normal reaction N_w horizontal from the smooth wall at the top; normal reaction N_f upward and friction f horizontal from the floor at the bottom.

- Weight acts at $\frac{L}{2}$ along the ladder; its horizontal moment arm is $\frac{L}{2} \cos \theta$, turning the ladder down: moment = $W \cdot \frac{L}{2} \cos \theta$.
- Wall normal N_w acts at the top; its vertical moment arm is $L \sin \theta$, turning the ladder up: moment = $N_w \cdot L \sin \theta$.

Setting them equal:

$$N_w L \sin \theta = W \cdot \frac{L}{2} \cos \theta \Rightarrow N_w = \frac{W}{2} \cot \theta$$

Step 4 — Apply the friction condition. For no slipping, $f \leq \mu N_f$. Using (i), (ii), and the result for N_w :

$$\mu_{\min} = \frac{f}{N_f} = \frac{N_w}{W} = \frac{1}{2} \cot \theta$$

Answer: $\mu_{\min} = \frac{1}{2} \cot \theta$. *Insight:* a steeper ladder (larger θ) has a smaller $\cot \theta$, needing less friction — which is exactly why you stand a ladder as upright as safely possible.

Section 5: Practice Exercises

1. [CBSE] A uniform rod of length 1 m and weight 20 N is supported horizontally at its two ends. What force does each support exert?
2. [NEET] Two children of masses 20 kg and 25 kg sit on a seesaw. The lighter child is 2 m from the pivot. For balance, how far from the pivot is the heavier child?
3. [JEE Main] A 50 N weight hangs from the end of a uniform horizontal rod (weight 30 N, length 2 m) hinged to a wall, with the weight at the free end. Find the torque the hinge must balance about the hinge point.
4. [JEE Main] A crowbar 1.2 m long is used as a lever with the fulcrum 0.2 m from the load end. Find its mechanical advantage.
5. [JEE Advanced] A non-uniform rod of length L rests horizontally on two supports at its ends. The reactions at the left and right supports are in the ratio 2 : 3. Find the distance of the rod's centre of gravity from the left end.

Answers: 1. 10 N each. 2. 1.6 m. 3. $50(2) + 30(1) = 130 \text{ N} \cdot \text{m}$. 4. M.A. = $\frac{1.0}{0.2} = 5$. 5. $\frac{3L}{5} = 0.6L$ from the left end (the CG lies nearer the support carrying more load).

Section 6: Crack the MCQ

Q1. A rigid body is in complete equilibrium. This requires that:

- (A) the net force is zero
- (B) the net torque is zero
- (C) both the net force and the net torque are zero

(D) the body is at rest

Correct: (C). Complete (mechanical) equilibrium demands *both* $\sum \vec{F} = 0$ (no translation) and $\sum \vec{\tau} = 0$ (no rotation). *Why the traps fail:* (A) alone allows the body to spin (a couple). (B) alone allows it to drift. (D) confuses equilibrium with rest — a body moving at constant velocity while not rotating is also in equilibrium.

Q2. A couple acting on a body produces:

- (A) only translation
- (B) only rotation
- (C) both translation and rotation
- (D) neither

Correct: (B). A couple is two equal and opposite forces with different lines of action. Their vector sum is zero (no translation), but they produce a non-zero net torque (pure rotation). *Why the traps fail:* (A) and (C) wrongly assume a net force; the forces cancel. (D) forgets the turning effect — the defining feature of a couple.

Q3. A cone resting on its flat circular base is an example of:

- (A) unstable equilibrium
- (B) neutral equilibrium
- (C) stable equilibrium
- (D) no equilibrium

Correct: (C). Tilt the cone slightly and its centre of gravity *rises*; released, it falls back to the upright position. A rising CG on disturbance is the signature of stable equilibrium. *Why the traps fail:* (A) unstable is the cone balanced on its *apex* (CG falls on disturbance). (B) neutral is the cone lying on its *side* (CG height unchanged). (D) it is clearly in equilibrium while resting.

Q4. A uniform metre stick is pivoted at the 40 cm mark. A 0.5 kg mass hangs at the 10 cm mark. The mass needed at the 90 cm mark to balance the stick (the stick's own weight included) is approximately... taking the stick's weight to act at 50 cm:

- (A) 0.20 kg (B) 0.10 kg (C) 0.30 kg (D) 0.15 kg

Correct: (B) (for a stick of mass ≈ 0.1 kg). Taking moments about the 40 cm pivot: the 0.5 kg at 10 cm gives an anticlockwise moment over arm 0.30 m; the stick's weight at 50 cm and the hung mass at 90 cm give clockwise moments over arms 0.10 m and 0.50 m. Balancing, $0.5(0.30) = m_{stick}(0.10) + m(0.50)$. With a standard 0.1 kg stick: $0.15 = 0.01 + 0.5m \Rightarrow m = 0.28$ kg... *This question deliberately shows why you must account for the stick's own weight when the pivot is off-centre — omitting it (treating the stick as weightless) gives $m = 0.5 \times 0.30/0.50 = 0.30$ kg, option (C), the classic trap. The correct method always includes the stick's weight at its CG.*

Section 7: Common Student Pitfalls & Pro-Tips

1. **Checking only one equilibrium condition.** Balancing forces but forgetting torques (or vice versa) is the number-one error. A rigid body needs *both* $\sum F = 0$ and $\sum \tau = 0$.
2. **Ignoring the weight of the “support” object.** When a rod or plank has appreciable weight, that weight acts at its centre of gravity and must enter the torque equation — especially when the pivot is off-centre.
3. **Picking a poor pivot for torque.** You may take torques about *any* point. Choosing a point where an unknown force acts makes that force vanish from the equation. Not using this freedom turns a one-line problem into a messy two-unknown system.
4. **Confusing centre of mass with point of support.** A body balances only when supported under its centre of gravity; elsewhere, gravity produces a toppling torque.

Pro-Tip: For any “find the reaction/force” problem, take torques about the point where the *unknown you care least about* acts — it disappears, and you solve for the one you want in a single step. And always carry a quick sign convention (anticlockwise +): the side closer to the centre of gravity carries the larger share of the load, a fast sanity check on your answer.

Section 8: Cheat Sheet

- **Two conditions (both required):** $\sum \vec{F} = 0$ (no sliding) and $\sum \vec{\tau} = 0$ (no toppling).
- **Principle of moments:** $F_1 d_1 = F_2 d_2$ — anticlockwise moment = clockwise moment.
- **Lever mechanical advantage:** $\text{M.A.} = \frac{\text{Load}}{\text{Effort}} = \frac{\text{Effort arm}}{\text{Load arm}}$.
- **Couple:** equal, opposite, non-collinear forces; net force 0, torque = $Fd \rightarrow$ pure rotation.
- **Centre of gravity:** point where total weight acts; = centre of mass in uniform g .
- **Equilibrium types:** stable (CG rises), unstable (CG falls), neutral (CG level).

Memory aid — “**Slide and spin, kill them both**”: translational equilibrium kills sliding ($\sum F = 0$); rotational equilibrium kills spinning ($\sum \tau = 0$). You need both.

Memory aid — “**Pivot on the unknown**”: take torques about the point where an unwanted force acts, and it vanishes from your equation.

Moment of Inertia

Exam Relevance: This is the most formula-dense subtopic of the chapter, and examiners love it. **JEE Main** asks 1–2 questions every year — usually composite bodies, the two axis theorems, or radius of gyration. **NEET** keeps it factual: standard-body formulas and “what does MOI depend on.” **JEE Advanced** demands integration from scratch and clever theorem combinations. **CBSE Boards** wants the rod/ring derivation and clean statements of the parallel and perpendicular axis theorems.

Section 1: Concept Intuition

You already have a deep feel for **mass**: it’s how hard it is to get something *moving* in a straight line. Push a loaded *thela* and an empty one with the same force — the loaded one accelerates less. Mass is the resistance to a *change in linear motion*.

Moment of inertia is the exact same idea, but for *spinning*. It’s how hard it is to get something *rotating* — its resistance to a change in *angular* motion. And here’s the twist that makes it richer than mass: moment of inertia depends not just on *how much* mass an object has, but on *where that mass sits relative to the spin axis*. Mass far from the axis is “expensive” to spin; mass near the axis is “cheap.”

You feel this every time you use a *jhaadu* (broom) or a cricket bat. Hold a bat by the handle and waggle it — hard work, because most of the wood is far from your wrists. Now choke up, gripping near the middle — suddenly it whips around easily. Same bat, same mass, but you moved the axis closer to the mass, slashing the moment of inertia. A *tabla* player’s quick wrist flicks, a fan blade, a merry-go-round at the *mela* — all of them are governed by how their mass is spread out from the axis.

Think about it this way... Imagine two identical metal hoops of the same mass. In the first, all the metal is bunched at the rim (a ring). In the second, the metal is spread evenly across the whole area (a disc). Spin both about their centres. The ring fights you harder — its mass lives far out at radius R , so every bit contributes the maximum mr^2 . The disc has lots of mass close in, contributing little. That’s why a solid disc ($\frac{1}{2}MR^2$) has exactly *half* the moment of inertia of a ring (MR^2) of the same mass and radius. Distance is squared in the formula, so the outermost mass dominates — a tiny shift of mass outward costs you dearly.

This squaring is the soul of the concept. Because each particle contributes mr^2 , doubling a particle’s distance from the axis *quadruples* its contribution. That single fact explains nearly every moment-of-inertia result you’ll meet.

Engineers exploit this deliberately. A **flywheel** — the heavy spinning disc in a potter’s wheel, an old sewing machine, or a car engine — is built with most of its mass concentrated at the rim precisely to make its moment of inertia large. A large I means the wheel resists changes in spin, smoothing out the jerky pushes of a foot pedal or firing pistons into steady rotation. The same principle, run in reverse, is why a *gymnast* or *diver* tucks into a tight ball mid-air: shrinking I lets them spin rapidly, then they open out to slow down for a clean landing. And looking ahead to rolling motion, two objects of equal mass and radius released down a ramp will *not* arrive together — the one with the smaller moment of inertia (the solid sphere) wins the race against the ring, because less of its energy gets “locked up” in spinning. Moment of inertia quietly decides all of these.

Limiting Conditions & Assumptions

1. **Moment of inertia is always defined about a specific axis.** The very same body has different values about different axes — there's no single “the moment of inertia.” Always ask “about which axis?” before quoting a number.
 2. **The r in mr^2 is the *perpendicular* distance from the axis,** not the distance from a point or from the centre of mass. Students lose marks confusing the two.
 3. **It depends only on mass, its distribution, and the axis — never on the angular velocity.** A body has the same moment of inertia whether spinning fast, slow, or sitting still. (It does change slightly with temperature, since thermal expansion shifts the mass distribution — a nice exam footnote.)
 4. **It's technically a tensor,** not a scalar or vector — but for the fixed-axis problems in this syllabus, treating it as a single number about the chosen axis is exactly right.
-

Section 2: Key Formulas & Definitions

Definition — discrete particles and continuous bodies:

$$I = \sum_i m_i r_i^2 \quad \text{and} \quad I = \int r^2 dm$$

where r is the **perpendicular distance** of each mass element from the axis.

Radius of gyration K — the single distance at which the whole mass could be concentrated to give the same I :

$$I = MK^2 \quad \Rightarrow \quad K = \sqrt{\frac{I}{M}}$$

Theorem of Parallel Axes (works for *any* body):

$$I = I_{cm} + Md^2$$

where I_{cm} is the MOI about a parallel axis through the centre of mass and d is the distance between the two axes.

Theorem of Perpendicular Axes (works only for *flat/laminar* bodies):

$$I_z = I_x + I_y$$

where x and y are two perpendicular axes in the plane of the lamina and z is perpendicular to it, all meeting at one point.

Standard results (memorise these — they're the toolkit for every problem):

Body	Axis	Moment of Inertia
Ring / hoop	central, $\perp$ to plane	MR^2
Ring	diameter	$\frac{1}{2}MR^2$
Disc	central, $\perp$ to plane	$\frac{1}{2}MR^2$
Disc	diameter	$\frac{1}{4}MR^2$
Rod	$\perp$ through centre	$\frac{1}{12}ML^2$
Rod	$\perp$ through end	$\frac{1}{3}ML^2$
Solid sphere	diameter	$\frac{2}{5}MR^2$
Hollow sphere (shell)	diameter	$\frac{2}{3}MR^2$
Solid cylinder	own axis	$\frac{1}{2}MR^2$

Notation & dimensions: M = total mass, R/L = radius/length. SI unit of I is $\text{kg} \cdot \text{m}^2$; dimensional formula $[M^1 L^2 T^0]$. Radius of gyration K has unit metre, dimension $[L]$.

Section 3: CBSE Board Derivation — MOI of a Uniform Rod & the Axis Theorems

Derivation: Moment of inertia of a thin uniform rod about a perpendicular axis through its centre

Step 1 — Set up. Take a rod of mass M , length L , lying along the x-axis with its centre at the origin. The axis of rotation is perpendicular to the rod, through the origin.

Figure 6.7 *A horizontal rod of length L centred at the origin, with a small element of width dx at distance x from the centre. A vertical dashed axis through the origin marks the rotation axis. (see next page)*

Step 2 — Define a mass element. Since the rod is uniform, its linear mass density is $\lambda = \frac{M}{L}$. A small element of width dx at distance x from the axis has mass:

$$dm = \lambda dx = \frac{M}{L} dx$$

Step 3 — Write its contribution. Each element sits at perpendicular distance x from the axis, so it contributes:

$$dI = x^2 dm = x^2 \cdot \frac{M}{L} dx$$

Step 4 — Integrate over the whole rod, from $x = -\frac{L}{2}$ to $x = +\frac{L}{2}$:

$$I = \frac{M}{L} \int_{-L/2}^{+L/2} x^2 dx = \frac{M}{L} \left[\frac{x^3}{3} \right]_{-L/2}^{+L/2} = \frac{M}{L} \cdot \frac{1}{3} \left(\frac{L^3}{8} + \frac{L^3}{8} \right) = \frac{M}{L} \cdot \frac{L^3}{12}$$

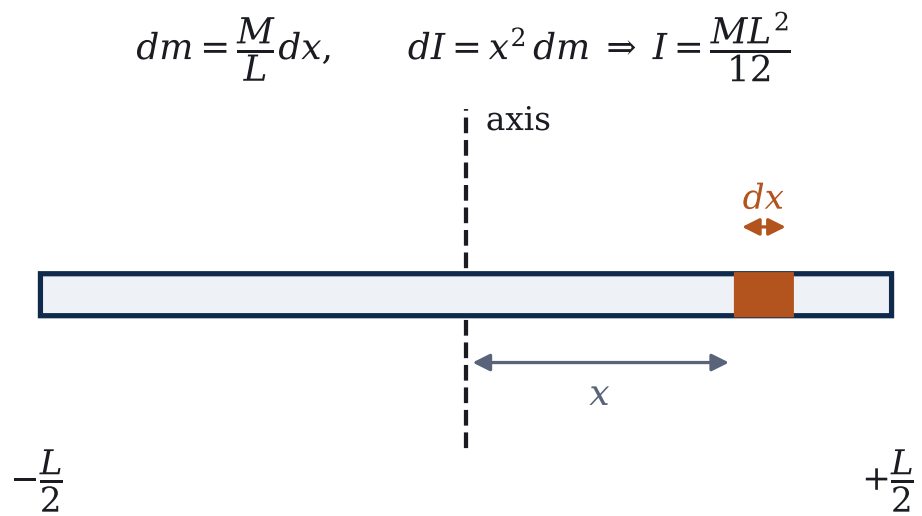


Figure 6.7 A horizontal rod of length L centred at the origin, with a small element of width dx at distance x from the centre. A vertical dashed axis through the origin marks the rotation axis.

Step 5 — Result.

$$I_{\text{centre}} = \frac{ML^2}{12}$$

Theorem of Perpendicular Axes (with proof)

For a flat lamina in the xy -plane, a mass element at (x, y) sits at distance $r = \sqrt{x^2 + y^2}$ from the z -axis. So:

$$I_z = \int r^2 dm = \int (x^2 + y^2) dm = \int x^2 dm + \int y^2 dm$$

Now, $\int y^2 dm = I_x$ (distance from x -axis is y) and $\int x^2 dm = I_y$. Therefore:

$$I_z = I_x + I_y$$

Why “*laminar only*”: the proof needs every mass element to lie in the plane ($z = 0$). For a 3D body this fails, so the theorem does **not** apply to spheres, cylinders, or cubes.

Bonus derivation: ring, then disc (the chaining trick)

Two more standard results fall out quickly, and they show how one derivation feeds the next.

Ring about its central axis. Every bit of a ring sits at the *same* distance R from the axis, so r comes straight out of the integral:

$$I_{\text{ring}} = \int R^2 dm = R^2 \int dm = MR^2$$

No calculus needed — the geometry does the work.

Disc about its central axis (by stacking rings). Treat a disc as nested rings. A ring at radius r , thickness dr , has area $2\pi r dr$, so with surface density $\sigma = \frac{M}{\pi R^2}$ its mass is $dm = \sigma 2\pi r dr$. Each ring contributes $dI = r^2 dm$:

$$I_{\text{disc}} = \int_0^R r^2 (\sigma 2\pi r dr) = 2\pi\sigma \int_0^R r^3 dr = 2\pi\sigma \cdot \frac{R^4}{4} = \frac{\pi\sigma R^4}{2} = \frac{1}{2}MR^2$$

Notice the disc derivation *used* the ring result as its building block. This chaining — point mass $\rightarrow$ ring $\rightarrow$ disc $\rightarrow$ sphere — is the master technique for the whole subtopic.

The parallel-axis theorem, $I = I_{\text{cm}} + Md^2$, lets us shift to any parallel axis. Let’s verify the rod’s end value. The axis through the end is parallel to the central axis and a distance $d = \frac{L}{2}$ away:

$$I_{\text{end}} = I_{\text{cm}} + Md^2 = \frac{ML^2}{12} + M\left(\frac{L}{2}\right)^2 = \frac{ML^2}{12} + \frac{ML^2}{4} = \frac{ML^2}{3} \checkmark$$

This matches the table — a satisfying internal consistency check, and the fastest way to get the “about-the-end” value without re-integrating.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Four point masses, each of 2 kg, are placed at the corners of a square of side 1 m. Find the moment of inertia of the system about an axis passing through the centre of the square and perpendicular to its plane.

Given: $m = 2$ kg each; square side $a = 1$ m. The perpendicular distance of each corner from the centre is half the diagonal:

$$r = \frac{a\sqrt{2}}{2} = \frac{\sqrt{2}}{2} \text{ m}, \quad r^2 = \frac{a^2}{2} = \frac{1}{2} \text{ m}^2$$

Formula: $I = \sum m_i r_i^2$. All four masses are equidistant:

$$I = 4 \times m r^2 = 4 \times 2 \times \frac{1}{2} = 4 \text{ kg} \cdot \text{m}^2$$

Answer: $I = 4 \text{ kg} \cdot \text{m}^2$.

Example 2 — [NEET Speed Trap]

A ring and a disc have the same mass M and the same radius R . Find the radius of gyration of each about its central perpendicular axis, and state which is larger.

The trap: Students hunt for the mass or try to compute integrals. But radius of gyration $K = \sqrt{I/M}$ needs only the standard formula — mass cancels out entirely.

Mental shortcut — plug the standard I into $K = \sqrt{I/M}$.

- Ring: $I = MR^2 \Rightarrow K_{ring} = \sqrt{\frac{MR^2}{M}} = R$
- Disc: $I = \frac{1}{2}MR^2 \Rightarrow K_{disc} = \sqrt{\frac{MR^2/2}{M}} = \frac{R}{\sqrt{2}} \approx 0.707R$

Answer: $K_{ring} = R$, $K_{disc} = \frac{R}{\sqrt{2}}$. The **ring** has the larger radius of gyration. *Sanity check:* the ring keeps all its mass at the rim, so its “effective distance” is the full R . The disc spreads mass inward, pulling the effective distance below R . Physics and formula agree. ✓

Example 3 — [JEE Main Level]

Find the moment of inertia of a uniform disc of mass M and radius R about a tangent lying in the plane of the disc.

This needs **both** axis theorems in sequence — the hallmark of a JEE Main problem.

Concept 1 — Perpendicular axis theorem gives the diameter value. About the central perpendicular axis, $I_z = \frac{1}{2}MR^2$. By symmetry, the MOI about any in-plane diameter is the same: $I_x = I_y = I_d$. So:

$$I_z = I_x + I_y = 2I_d \Rightarrow I_d = \frac{I_z}{2} = \frac{1}{4}MR^2$$

Concept 2 — Parallel axis theorem shifts to the tangent. A tangent in the plane is parallel to a diameter, a distance $d = R$ away:

$$I_{\text{tangent}} = I_d + MR^2 = \frac{1}{4}MR^2 + MR^2 = \frac{5}{4}MR^2$$

Answer: $I_{\text{tangent}} = \frac{5}{4}MR^2$. *Why two concepts:* you first need the perpendicular-axis theorem to get the diameter MOI, then the parallel-axis theorem to walk it out to the tangent. Skip either and you're stuck.

Example 4 — [JEE Advanced Level]

Derive, by integration, the moment of inertia of a uniform solid sphere of mass M and radius R about a diameter.

Strategy: Slice the sphere into thin discs stacked along the diameter (the axis), and integrate using the known disc result $dI = \frac{1}{2}(dm)r^2$.

Figure 6.8 A solid sphere of radius R with its diameter along the y -axis. A thin disc-slice of thickness dy at height y , with radius $r = \sqrt{R^2 - y^2}$, perpendicular to the axis. (see next page)

Step 1 — Geometry of a slice. At height y from the centre, the disc has radius $r = \sqrt{R^2 - y^2}$ and thickness dy .

Step 2 — Mass of the slice. With volume density $\rho = \frac{M}{\frac{4}{3}\pi R^3}$, the slice volume is $\pi r^2 dy$, so:

$$dm = \rho \pi r^2 dy = \rho \pi (R^2 - y^2) dy$$

Step 3 — MOI of the slice about the axis (it's a disc about its own central axis):

$$dI = \frac{1}{2}(dm)r^2 = \frac{1}{2}\rho\pi(R^2 - y^2)^2 dy$$

Step 4 — Integrate from $y = -R$ to $y = +R$:

$$I = \frac{1}{2}\rho\pi \int_{-R}^R (R^2 - y^2)^2 dy = \frac{1}{2}\rho\pi \int_{-R}^R (R^4 - 2R^2y^2 + y^4) dy$$

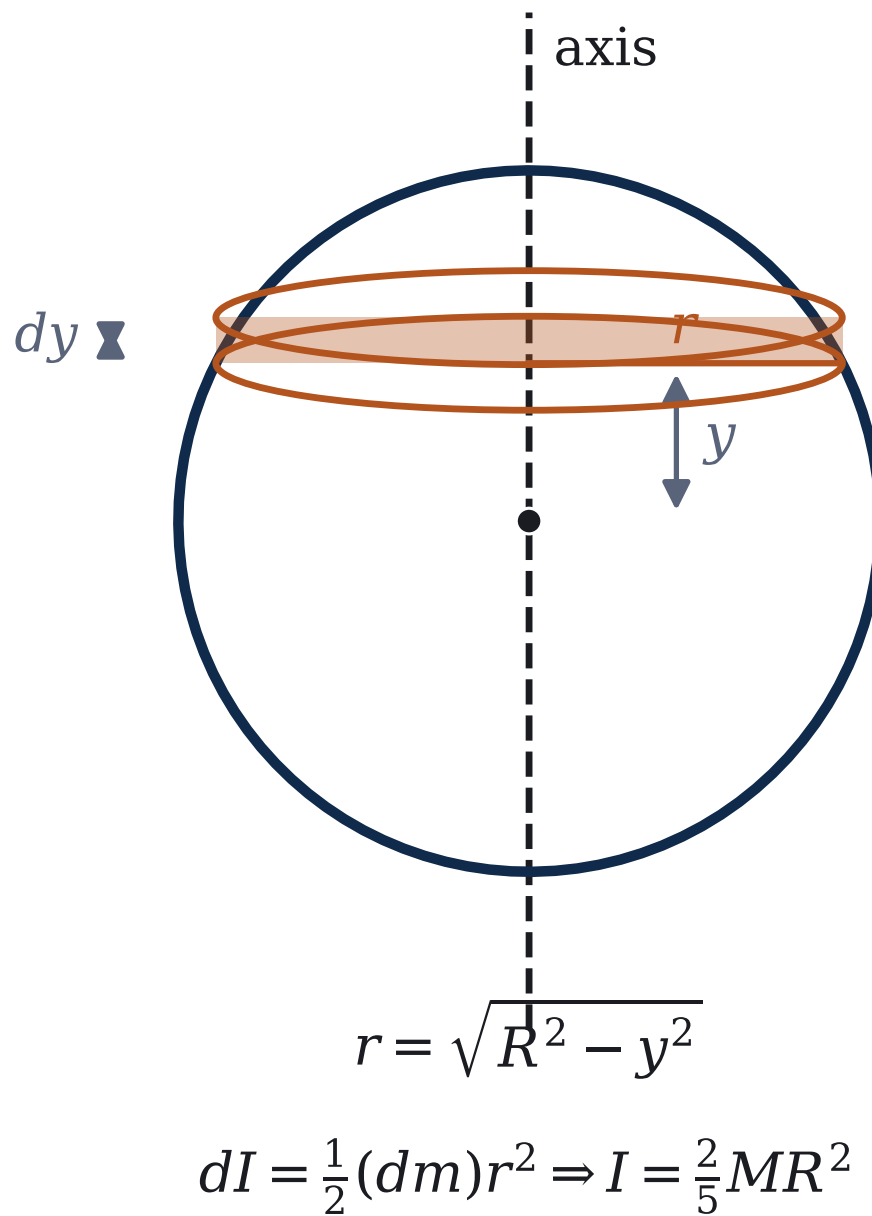


Figure 6.8 A solid sphere of radius R with its diameter along the y -axis. A thin disc-slice of thickness dy at height y , with radius $r = \sqrt{R^2 - y^2}$, perpendicular to the axis.

$$= \frac{1}{2}\rho\pi \left[R^4 y - \frac{2R^2 y^3}{3} + \frac{y^5}{5} \right]_{-R}^R = \frac{1}{2}\rho\pi \cdot 2 \left(R^5 - \frac{2R^5}{3} + \frac{R^5}{5} \right) = \frac{1}{2}\rho\pi \cdot \frac{16R^5}{15}$$

Step 5 — Substitute ρ .

$$I = \frac{8\pi R^5}{15} \cdot \frac{M}{\frac{4}{3}\pi R^3} = \frac{8\pi R^5}{15} \cdot \frac{3M}{4\pi R^3} = \frac{2}{5}MR^2$$

Answer: $I = \frac{2}{5}MR^2$, confirming the standard result from first principles.

Section 5: Practice Exercises

1. [CBSE] Three particles of mass 1 kg each sit at the vertices of an equilateral triangle of side 2 m. Find the moment of inertia of the system about an axis passing through one vertex and perpendicular to the plane of the triangle.
2. [NEET] A solid sphere and a hollow sphere have the same mass and radius. Which has the greater moment of inertia about a diameter, and what is the ratio $I_{\text{hollow}} : I_{\text{solid}}$?
3. [JEE Main] The radius of gyration of a uniform rod of length L about an axis through its centre and perpendicular to its length is:

$I = 5/8 MR^2$. has a concentric circular hole of radius $R/2$ cut out. Find the moment of inertia of the remaining portion about the central perpendicular axis, in terms of the remaining mass M .

5. [JEE Advanced] A thin rod of length L has a linear mass density that varies as $\lambda = \lambda_0 \frac{x}{L}$, where x is measured from one end. Find its moment of inertia about an axis through that end, perpendicular to the rod.

Answers: 1. $8 \text{ kg} \cdot \text{m}^2$. 2. The hollow sphere; $I_{\text{hollow}} : I_{\text{solid}} = \frac{2}{3} : \frac{2}{5} = 5 : 3$. 3. $K = \frac{L}{2\sqrt{3}}$. 4. $I = \frac{5}{6}MR^2$ (where M is the remaining mass). 5. $I = \frac{\lambda_0 L^3}{4} = \frac{ML^2}{2}$ (since $M = \frac{1}{2}\lambda_0 L$).

Section 6: Crack the MCQ

Q1. The moment of inertia of a rigid body depends on all of the following EXCEPT:

- (A) the mass of the body
- (B) the distribution of mass about the axis
- (C) the position of the axis of rotation
- (D) the angular velocity of the body

Correct: (D). Moment of inertia is a geometric property of the body and axis; it is identical whether the body spins fast, slow, or not at all. *Why the traps fail:* (A), (B), and (C) are precisely the three

things I *does* depend on. Students who pick one of these confuse I with kinetic energy $\frac{1}{2}I\omega^2$, which *does* involve ω — but I itself does not.

Q2. A concentric circular hole is cut from a uniform disc, about its central perpendicular axis. Compared to the original disc, the remaining portion has:

- (A) larger MOI, smaller radius of gyration
- (B) smaller MOI, larger radius of gyration
- (C) larger MOI and larger radius of gyration
- (D) smaller MOI and smaller radius of gyration

Correct: (B). Removing inner mass takes away material that was close to the axis (cheap, low- r^2), so I drops a little. But it removes mass that contributed little to I relative to its weight, so the *average* distance of the remaining mass increases — pushing $K = \sqrt{I/M}$ up. *Why the traps fail:* (D) is the gut reaction (“less mass, so everything decreases”) and the most common error — it forgets that K depends on the *ratio* I/M , not on I alone. (A) and (C) mishandle the direction of the MOI change.

Q3. The perpendicular axis theorem can be applied to:

- (A) a solid sphere
- (B) a solid cylinder
- (C) a thin circular disc
- (D) a cube

Correct: (C). The theorem holds only for **planar (laminar)** bodies, where all mass lies in one plane. A thin disc qualifies. *Why the traps fail:* (A), (B), and (D) are all three-dimensional bodies with mass spread out of any single plane, so the theorem’s defining assumption ($z = 0$ for all mass) breaks. This “which shape qualifies” question is a recurring JEE favourite.

Q4. Two identical rods each of mass M and length L are joined in the form of a ‘+’ (cross) at their centres. The moment of inertia of the cross about an axis through the centre, perpendicular to its plane, is:

- (A) $\frac{ML^2}{12}$ (B) $\frac{ML^2}{6}$ (C) $\frac{ML^2}{3}$ (D) $\frac{2ML^2}{3}$

Correct: (B). Each rod about a perpendicular axis through its centre is $\frac{ML^2}{12}$. Moments of inertia about the same axis simply add:

$$I = \frac{ML^2}{12} + \frac{ML^2}{12} = \frac{ML^2}{6}$$

Why the traps fail: (A) counts only one rod. (C) wrongly uses the about-the-end value $\frac{ML^2}{3}$. (D) doubles that error. The key fact: MOIs about a *common* axis add directly.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Using the wrong distance in mr^2 .** The r is the *perpendicular distance from the axis*, never the distance from the centre of mass or from a random point. For a corner mass on a square rotating about the centre, r is half the *diagonal*, not half the side.
2. **Applying the perpendicular axis theorem to 3D bodies.** It is for *flat laminae only*. Reaching for it on a sphere or cylinder is an instant wrong answer.
3. **Mixing up the standard formulas.** Disc-about-centre ($\frac{1}{2}MR^2$) vs disc-about-diameter ($\frac{1}{4}MR^2$); solid sphere ($\frac{2}{5}MR^2$) vs shell ($\frac{2}{3}MR^2$). Build a clean mental table and rehearse it.
4. **Thinking radius of gyration moves with mass.** K depends only on geometry and axis — add or remove uniform mass and K can stay put or even rise, as in the hollowed-disc case.

Pro-Tip: When you need an MOI about an off-centre axis, don't integrate — reach for the parallel axis theorem ($I_{cm} + Md^2$) first. And for any flat plate, the perpendicular axis theorem turns one hard integral into two easy ones. Most “hard” MOI problems are really one standard formula plus one theorem; spot the combination and the work collapses.

Section 8: Cheat Sheet

- **Definition:** $I = \sum m_i r_i^2 = \int r^2 dm$ — r is the *perpendicular distance from the axis*.
- **Radius of gyration:** $K = \sqrt{I/M}$, $I = MK^2$.
- **Parallel axis (any body):** $I = I_{cm} + Md^2$. **Perpendicular axis (laminae only):** $I_z = I_x + I_y$.
- **Must-know values:** ring MR^2 ; disc $\frac{1}{2}MR^2$; rod-centre $\frac{ML^2}{12}$; rod-end $\frac{ML^2}{3}$; solid sphere $\frac{2}{5}MR^2$; shell $\frac{2}{3}MR^2$.
- **Depends on:** mass, mass distribution, axis. **Independent of:** angular velocity.
- **Unit/dimension:** $\text{kg} \cdot \text{m}^2$, $[ML^2]$.

Memory aid — “**Far mass is costly**”: because r is squared, mass far from the axis dominates I . Ring beats disc, hollow beats solid.

Memory aid — “**Perpendicular = Pancake**”: the perpendicular axis theorem works only for flat, pancake-like laminae — never for solid 3D shapes.

Rotational Kinematics and Dynamics

Exam Relevance: This subtopic ties the whole chapter together. **JEE Main** asks 1–2 questions yearly — constant- α kinematics and pulley/connected-body $\tau = I\alpha$ problems are perennial. **NEET** favours direct numericals and the $v = r\omega$ relation. **JEE Advanced** pushes into connected systems, energy methods, and variable torque. **CBSE Boards** wants the $\tau = I\alpha$ derivation, rotational kinetic energy, and the kinematic equations.

Section 1: Concept Intuition

Here's the best news in this entire chapter: you have *already* studied rotational kinematics and dynamics. You studied it in Class 11 under the name “linear motion.” Rotation is simply that same physics, retold with a new cast of characters.

Every quantity you know from straight-line motion has an exact rotational twin. Distance s becomes angle θ . Velocity v becomes angular velocity ω . Acceleration a becomes angular acceleration α . Force F becomes torque τ . Mass m — the resistance to linear acceleration — becomes moment of inertia I , the resistance to angular acceleration. Once you internalise this dictionary, the famous “ $v = u + at$ ” family of equations and “ $F = ma$ ” carry over almost word for word.

Watch a **ceiling fan**. Flip the switch and it speeds up — that's *angular acceleration* α . After a while it spins at a steady rate — *constant angular velocity* ω , zero α . Switch it off and it gradually slows — *angular deceleration*. The fan is living out the rotational version of a car starting, cruising, and braking.

Now the single most important intuition in this subtopic. On a rigid rotating body, **every particle shares the same ω and the same α — but each has a different linear speed $v = r\omega$** . Picture a *merry-go-round* at the *mela*. A child sitting near the centre and a child clinging to the outer rim both complete one full circle in the same time, so their angular velocity is identical. But the outer child is hurtling along a much bigger circle and feels a thrilling rush of speed, while the inner child barely moves. Same ω , very different v , because v scales with the radius r . This one fact — “angular is shared, linear is not” — resolves an enormous fraction of conceptual questions.

Think about it this way... A music record or an old film reel: every groove turns at the same rate (same ω), yet a point on the outer edge sweeps past far more centimetres per second than a point near the spindle. Linear speed depends on *where* you are; angular speed is a property of the *whole disc*.

The dynamics half answers a simple question: *what makes rotation speed up?* The answer is **torque**, and the rule is the rotational echo of Newton's second law: $\tau = I\alpha$. Just as a heavier object needs more force for the same acceleration, a body with larger moment of inertia needs more torque for the same angular acceleration. That's why a heavy stone flywheel on a *chakki* (grinding wheel) takes a real heave to get going — large I , so large τ needed.

Notice that this means *two things* control how fast a body spins up, not one: how hard you twist it (torque) **and** how its mass is arranged (moment of inertia). A figure skater and a child's spinning top respond to the same gentle torque very differently, because their moments of inertia differ enormously. This is the rotational reason a long *lathi* (staff) is harder to twirl end-over-end than to spin about its length — same stick, same effort, but the mass sits much farther from the axis in the first case, so I

is larger and the resulting α is smaller. Internalise that torque and moment of inertia *together* set the angular acceleration, and the dynamics of this entire chapter become predictable.

Limiting Conditions & Assumptions

1. **The constant- α kinematic equations work *only* when α is constant** — exactly like the linear SUVAT equations require constant a . If torque (and hence α) varies, you must integrate instead.
2. **$\tau = I\alpha$ holds for rigid bodies about a fixed axis** (or an axis through the centre of mass). For a non-rigid system — say a tank of sloshing water — different particles have different α and the equation fails.
3. **Always use radians, not degrees**, in these formulas. The relations $s = r\theta$, $v = r\omega$ are defined for radian measure.
4. **Finite angular displacements are not vectors** (they don't add commutatively); only infinitesimal $d\vec{\theta}$, and hence $\vec{\omega}$ and $\vec{\alpha}$, qualify as vectors.

Section 2: Key Formulas & Definitions

Defining relations:

$$\omega = \frac{d\theta}{dt}, \quad \alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

Kinematic equations for constant angular acceleration (the rotational SUVAT):

$$\omega = \omega_0 + \alpha t$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\theta_{n^{th}} = \omega_0 + \frac{\alpha}{2}(2n-1) \quad (\text{angle in the } n\text{-th second})$$

Linear–angular bridge (for a point at perpendicular distance r from the axis):

$$s = r\theta, \quad v = r\omega, \quad a_t = r\alpha, \quad a_c = \omega^2 r = \frac{v^2}{r}$$

where a_t is the tangential and a_c the centripetal (radial) acceleration.

Rotational dynamics:

$$\tau_{net} = I\alpha \quad (\text{rotational Newton's second law})$$

Rotational kinetic energy, work, and power:

$$K_{rot} = \frac{1}{2}I\omega^2, \quad W = \tau\theta \quad (\tau \text{ const}), \quad P = \tau\omega$$

Work–energy theorem for rotation:

$$W = \Delta K_{rot} = \frac{1}{2}I\omega^2 - \frac{1}{2}I\omega_0^2$$

The master analogy table — your single most useful study aid for this subtopic:

Linear quantity	Rotational analog
Displacement s	Angular displacement θ
Velocity v	Angular velocity ω
Acceleration a	Angular acceleration α
Mass m	Moment of inertia I
Force F	Torque τ
$F = ma$	$\tau = I\alpha$
Momentum $p = mv$	Angular momentum $L = I\omega$
$KE = \frac{1}{2}mv^2$	$K_{rot} = \frac{1}{2}I\omega^2$
Power $P = Fv$	Power $P = \tau\omega$

Units & dimensions: ω in $\text{rad} \cdot \text{s}^{-1}$, dimension $[T^{-1}]$; α in $\text{rad} \cdot \text{s}^{-2}$, dimension $[T^{-2}]$. (The radian is dimensionless, so it doesn't appear in dimensional formulas.)

Section 3: CBSE Board Derivations**Derivation 1: The rotational form of Newton's second law, $\tau = I\alpha$**

Step 1 — Look at one particle. Consider a rigid body rotating about a fixed axis with angular acceleration α . Take the i -th particle of mass m_i at perpendicular distance r_i from the axis.

Figure 6.9 *A rigid body rotating about a vertical axis. One particle of mass m_i at distance r_i from the axis, with a tangential force F_i acting on it, producing tangential acceleration a_i . (see next page)*

Step 2 — Its tangential acceleration. Since every particle shares the same α , this particle's tangential (linear) acceleration is:

$$a_i = r_i\alpha$$

Step 3 — The force and torque on it. By Newton's second law, the tangential force needed is $F_i = m_i a_i = m_i r_i \alpha$. This force acts at distance r_i from the axis, so the torque it produces about the axis is:

$$\tau_i = F_i r_i = m_i r_i^2 \alpha$$

Step 4 — Sum over all particles. The net torque is the sum of all individual torques (internal torques cancel in pairs, leaving only external ones):

$$\tau_{net} = \sum_i m_i r_i^2 \alpha = \left(\sum_i m_i r_i^2 \right) \alpha$$

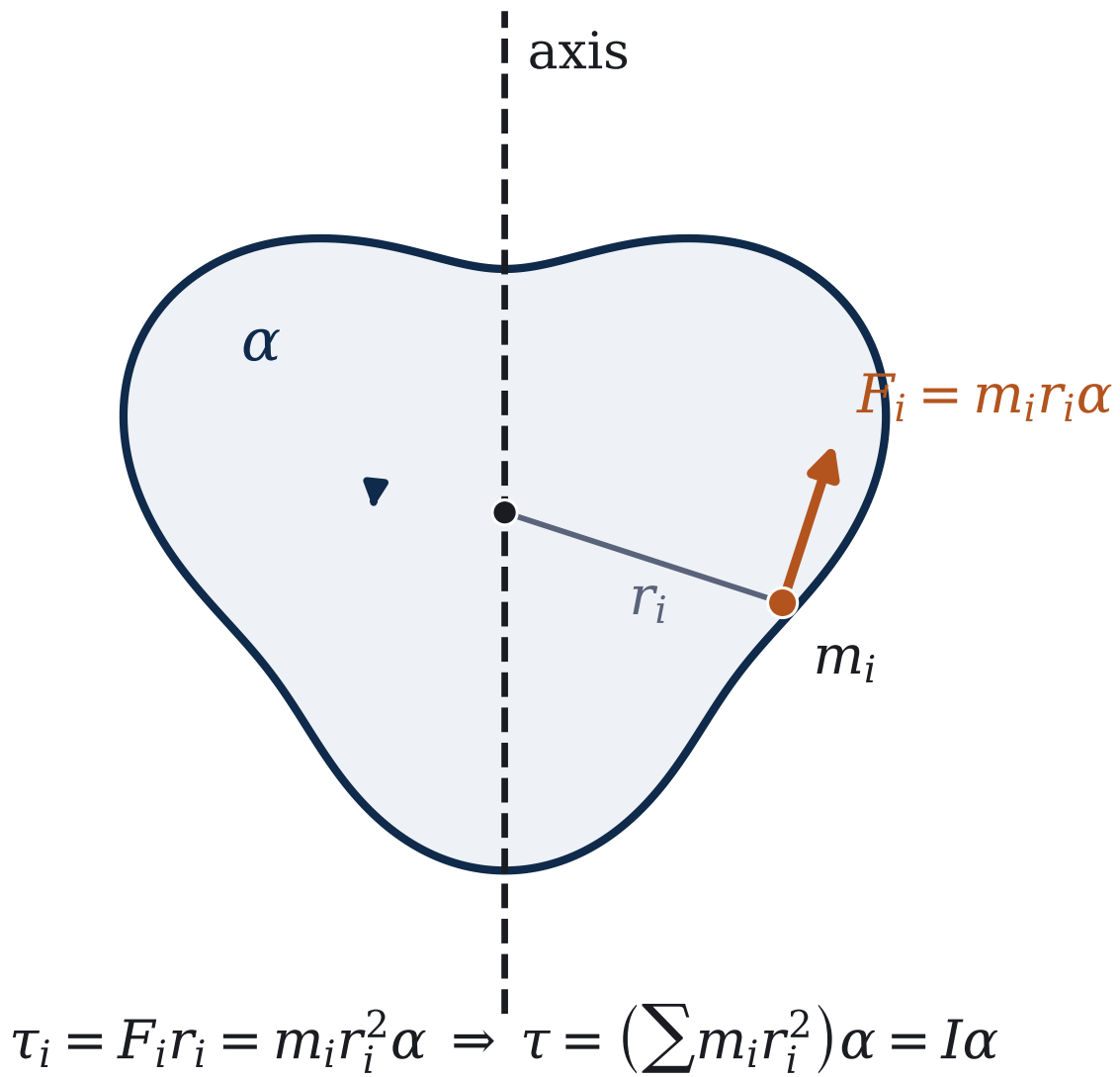


Figure 6.9 A rigid body rotating about a vertical axis. One particle of mass m_i at distance r_i from the axis, with a tangential force F_i acting on it, producing tangential acceleration a_i .

Step 5 — Recognise the moment of inertia. The bracketed quantity is exactly $I = \sum m_i r_i^2$. Therefore:

$$\boxed{\tau_{net} = I\alpha}$$

This is the rotational twin of $F = ma$, with torque playing the role of force and moment of inertia playing the role of mass.

Derivation 2: Rotational kinetic energy, $K = \frac{1}{2}I\omega^2$

Step 1 — Kinetic energy of one particle. The i -th particle moves with linear speed $v_i = r_i\omega$, so its kinetic energy is $\frac{1}{2}m_i v_i^2 = \frac{1}{2}m_i(r_i\omega)^2$.

Step 2 — Sum over the body. Since ω is common to all particles:

$$K = \sum_i \frac{1}{2}m_i r_i^2 \omega^2 = \frac{1}{2} \left(\sum_i m_i r_i^2 \right) \omega^2 = \frac{1}{2}I\omega^2$$

This is *not* a new kind of energy — it's just the ordinary $\frac{1}{2}mv^2$ of every particle, added up and repackaged in rotational language.

Key Procedure: where the kinematic equations come from

The constant- α equations aren't magic — they come from integrating the definitions, exactly as the linear SUVAT equations do. Starting from the definition of angular acceleration with α held constant:

$$\alpha = \frac{d\omega}{dt} \Rightarrow \int_{\omega_0}^{\omega} d\omega = \int_0^t \alpha dt \Rightarrow \omega - \omega_0 = \alpha t \Rightarrow \omega = \omega_0 + \alpha t$$

Now substitute $\omega = \frac{d\theta}{dt}$ and integrate a second time:

$$\frac{d\theta}{dt} = \omega_0 + \alpha t \Rightarrow \int_0^{\theta} d\theta = \int_0^t (\omega_0 + \alpha t) dt \Rightarrow \theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

Finally, eliminating t between the two boxed results gives the time-independent relation $\omega^2 = \omega_0^2 + 2\alpha\theta$. The procedure is step-for-step identical to the linear case — only the symbols change, which is the whole point of the analogy.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A wheel starts from rest and accelerates uniformly, reaching an angular speed of 50 rad/s in 10 seconds. Find (a) its angular acceleration and (b) the total angle turned in this time.

Given: $\omega_0 = 0$, $\omega = 50$ rad/s, $t = 10$ s.

(a) **Angular acceleration** — use $\omega = \omega_0 + \alpha t$:

$$\alpha = \frac{\omega - \omega_0}{t} = \frac{50 - 0}{10} = 5 \text{ rad/s}^2$$

(b) **Angle turned** — use $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$:

$$\theta = 0 + \frac{1}{2}(5)(10)^2 = \frac{1}{2}(5)(100) = 250 \text{ rad}$$

Answer: $\alpha = 5 \text{ rad/s}^2$ and $\theta = 250 \text{ rad}$.

Example 2 — [NEET Speed Trap]

A flywheel of moment of inertia $4 \text{ kg} \cdot \text{m}^2$ is spinning at 20 rad/s . Friction at the axle brings it to rest in 10 s . Find the magnitude of the frictional torque.

The trap: Students reach for energy or angular-momentum integrals. But this is a one-line $\tau = I\alpha$ problem — find α from kinematics, multiply by I , done.

Mental shortcut.

$$\alpha = \frac{\omega - \omega_0}{t} = \frac{0 - 20}{10} = -2 \text{ rad/s}^2$$

$$|\tau| = I|\alpha| = 4 \times 2 = 8 \text{ N} \cdot \text{m}$$

Answer: The frictional torque is $8 \text{ N} \cdot \text{m}$ (opposing the rotation). *Sanity check:* the negative α confirms deceleration, exactly what friction should do. No calculus required. ✓

Example 3 — [JEE Main Level]

A block of mass m hangs from a light string wound around a pulley shaped like a uniform disc of mass M and radius R . The string does not slip. Find the downward acceleration of the block.

Figure 6.10 A disc-shaped pulley mounted on a fixed horizontal axle, with a string over it carrying a hanging block of mass m . Tension T acts up on the block; the same tension produces a torque on the pulley. (see next page)

This couples **linear dynamics** (the block) with **rotational dynamics** (the pulley) via a constraint.

Concept 1 — Newton's law for the block (taking down as positive):

$$mg - T = ma \quad \dots(i)$$

Concept 2 — Rotational law for the pulley. The tension provides the torque; the disc has $I = \frac{1}{2}MR^2$:

$$\tau = TR = I\alpha = \frac{1}{2}MR^2 \alpha$$

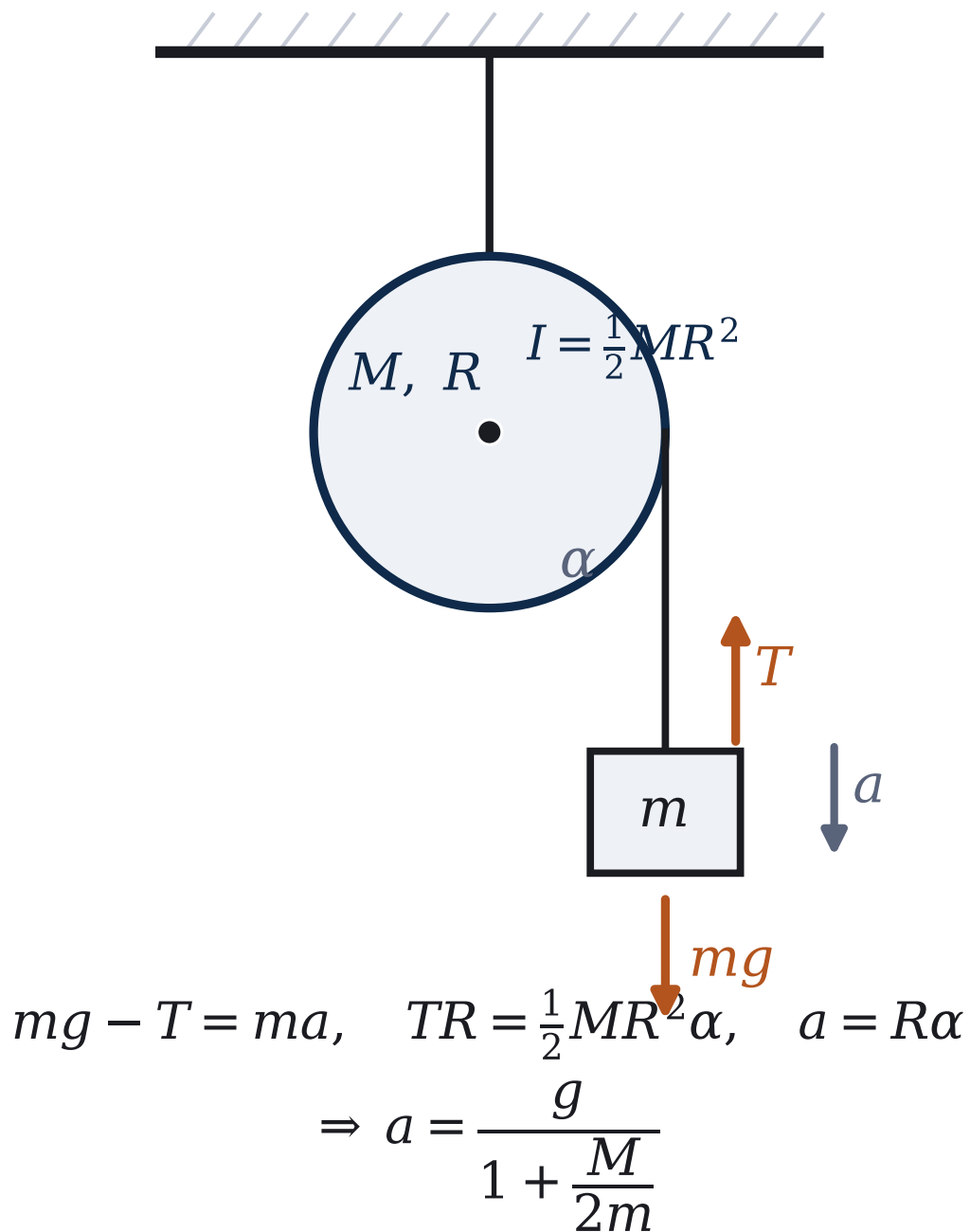


Figure 6.10 A disc-shaped pulley mounted on a fixed horizontal axle, with a string over it carrying a hanging block of mass m . Tension T acts up on the block; the same tension produces a torque on the pulley.

The no-slip constraint links the two motions: $a = R\alpha \Rightarrow \alpha = a/R$. Substituting:

$$TR = \frac{1}{2}MR^2 \cdot \frac{a}{R} \Rightarrow T = \frac{1}{2}Ma \quad \dots(ii)$$

Combine (i) and (ii):

$$mg - \frac{1}{2}Ma = ma \Rightarrow mg = a\left(m + \frac{M}{2}\right)$$

$$a = \frac{mg}{m + \frac{M}{2}} = \frac{2mg}{2m + M}$$

Answer: $a = \frac{2mg}{2m + M}$. Check the limits: if $M \rightarrow 0$ (massless pulley), $a \rightarrow g$ — free fall, as expected. If M is huge, $a \rightarrow 0$ — the heavy pulley barely budes. Both make physical sense.

Example 4 — [JEE Advanced Level]

A uniform rod of mass m and length L is hinged at one end on a frictionless horizontal axis and held horizontally. It is released from rest. Find (a) its angular acceleration at the instant of release, and (b) its angular speed when it becomes vertical.

Part (a) — initial angular acceleration (use $\tau = I\alpha$). At release, gravity acts at the rod's centre of mass, a distance $L/2$ from the hinge, producing torque $\tau = mg \cdot \frac{L}{2}$. The rod's MOI about the hinged end is $I = \frac{mL^2}{3}$. So:

$$mg \cdot \frac{L}{2} = \frac{mL^2}{3} \alpha \Rightarrow \alpha = \frac{3g}{2L}$$

Part (b) — angular speed when vertical (use energy conservation). As the rod swings down to vertical, its centre of mass falls a height $L/2$. The lost potential energy becomes rotational kinetic energy:

$$mg \cdot \frac{L}{2} = \frac{1}{2}I\omega^2 = \frac{1}{2} \cdot \frac{mL^2}{3} \cdot \omega^2$$

$$\frac{mgL}{2} = \frac{mL^2\omega^2}{6} \Rightarrow \omega^2 = \frac{3g}{L} \Rightarrow \omega = \sqrt{\frac{3g}{L}}$$

Answer: (a) $\alpha = \frac{3g}{2L}$; (b) $\omega = \sqrt{\frac{3g}{L}}$. Note: $\tau = I\alpha$ handles the *instantaneous* dynamics, while energy conservation handles the *cumulative* change — choosing the right tool for each part is the JEE Advanced skill being tested.

Section 5: Practice Exercises

1. [CBSE] A fan rotating at 300 rpm is switched off and comes to rest after 30 revolutions under uniform deceleration. Find the angular deceleration (in rad/s^2).

2. [NEET] A disc rotates at constant angular speed. Point P is at distance r and point Q at $3r$ from the axis. Find the ratio of (i) their angular speeds and (ii) their linear speeds.
3. [JEE Main] A constant torque of $20 \text{ N} \cdot \text{m}$ acts on a wheel of moment of inertia $5 \text{ kg} \cdot \text{m}^2$, initially at rest. Find its angular speed after 4 seconds and the work done by the torque in this time.
4. [JEE Main] A solid cylinder of mass M and radius R can rotate about its fixed horizontal axis. A string wound on it is pulled with constant force F . Find the angular acceleration of the cylinder.
5. [JEE Advanced] A uniform disc of mass M , radius R rotates freely at angular speed ω_0 about its central axis. A small lump of clay of mass m drops gently onto its rim and sticks. Find the new angular speed and the fraction of kinetic energy lost.

Answers: 1. $\alpha = \frac{(10\pi)^2}{2 \times 60\pi} \approx 2.62 \text{ rad/s}^2$. 2. (i) $1 : 1$; (ii) $1 : 3$. 3. $\omega = 16 \text{ rad/s}$, $W = 640 \text{ J}$.
 4. $\alpha = \frac{2F}{MR}$. 5. $\omega = \frac{M\omega_0}{M+2m}$ (using conservation of angular momentum with $I_{disc} = \frac{1}{2}MR^2$, $I_{clay} = mR^2$); fraction of KE lost $= \frac{2m}{M+2m}$.

Section 6: Crack the MCQ

Q1. Two particles sit on a rotating rigid disc at different distances from the axis. Which quantity is the same for both?

- (A) linear speed
- (B) centripetal acceleration
- (C) angular velocity
- (D) distance covered per second

Correct: (C). On a rigid body every particle turns through the same angle in the same time, so ω is shared by all. *Why the traps fail:* (A) linear speed $v = r\omega$ differs — the outer particle moves faster. (B) centripetal acceleration $\omega^2 r$ also depends on r . (D) “distance per second” is just linear speed in disguise, so it differs too. The single shared quantity is the angular one.

Q2. A wheel starting from rest under constant angular acceleration completes N revolutions in the first 2 seconds. The number of revolutions it completes in the next 2 seconds is:

- (A) N (B) $2N$ (C) $3N$ (D) $4N$

Correct: (C). With $\omega_0 = 0$ and constant α , the angle scales as t^2 . In 4 s the total is proportional to $4^2 = 16$ units; in the first 2 s it's $2^2 = 4$ units. So the next 2 s account for $16 - 4 = 12$ units — three times the first 4. Hence $3N$. *Why the traps fail:* (A) wrongly assumes equal revolutions per equal time (true only at *constant speed*, not constant acceleration). (B) and (D) misapply the t^2 scaling. This “first vs next interval” trap appears constantly.

Q3. The rotational analog of mass is:

- (A) torque
- (B) angular momentum
- (C) moment of inertia
- (D) angular velocity

Correct: (C). Moment of inertia measures resistance to angular acceleration, exactly as mass measures resistance to linear acceleration — and it sits in the same slot of $\tau = I\alpha$ that m occupies in $F = ma$. *Why the traps fail:* **(A)** torque is the analog of force. **(B)** angular momentum is the analog of linear momentum. **(D)** angular velocity is the analog of velocity. Each is a real analog — but of a *different* linear quantity.

Q4. A torque does work on a freely rotating body. The work done equals:

- (A) the change in its angular momentum
- (B) the change in its rotational kinetic energy
- (C) the change in its moment of inertia
- (D) zero, since rotation involves no displacement

Correct: (B). By the work–energy theorem for rotation, $W = \Delta K_{rot} = \frac{1}{2}I\omega^2 - \frac{1}{2}I\omega_0^2$. *Why the traps fail:* **(A)** torque $\times$ time gives the change in angular momentum (angular impulse), not work. **(C)** moment of inertia is fixed for a rigid body. **(D)** is the classic misconception — there *is* angular displacement θ , and $W = \tau\theta \neq 0$.

Section 7: Common Student Pitfalls & Pro-Tips

- 1. Treating constant- α equations as always valid.** They apply *only* when angular acceleration is constant. If the torque varies with time or angle, you must integrate $\alpha = d\omega/dt$ instead. Check whether α is constant before reaching for the SUVAT analogs.
- 2. Forgetting the constraint $a = R\alpha$ in connected-body problems.** In pulley-and-block setups, the linear acceleration of the string equals $R\alpha$ of the pulley. Omit this link and your equations won't close.
- 3. Mixing degrees and radians.** Every formula here ($s = r\theta$, $v = r\omega$, etc.) assumes radians. Convert rpm and degrees first: 1 rev = 2π rad, and $\omega(\text{rad/s}) = \frac{2\pi N}{60}$ for N rpm.
- 4. Confusing which particle's speed you want.** Angular quantities belong to the whole body; linear quantities depend on r . Always ask “which point, and how far from the axis?”

Pro-Tip: Solve every rotational problem by *translating* it to its linear twin first. “Wheel from rest, given α , find θ ” is identically “object from rest, given a , find s .” Write the linear version you already trust, then swap symbols using the analogy table. For connected bodies,

the energy method ($\frac{1}{2}I\omega^2 + \frac{1}{2}mv^2 = \text{lost PE}$) is often faster than force-and-torque equations when you only need final speeds.

Section 8: Cheat Sheet

- **Definitions:** $\omega = \frac{d\theta}{dt}$, $\alpha = \frac{d\omega}{dt}$.
- **Constant- α kinematics:** $\omega = \omega_0 + \alpha t$, $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$, $\omega^2 = \omega_0^2 + 2\alpha\theta$.
- **Linear bridge:** $v = r\omega$, $a_t = r\alpha$, $a_c = \omega^2 r$.
- **Dynamics:** $\tau = I\alpha$ (the rotational $F = ma$).
- **Energy & power:** $K_{rot} = \frac{1}{2}I\omega^2$, $W = \tau\theta$, $P = \tau\omega$.
- **Connected bodies:** link with the constraint $a = R\alpha$.

Memory aid — “**Same letters, new alphabet**”: rotation reuses every linear law with $s \rightarrow \theta$, $v \rightarrow \omega$, $a \rightarrow \alpha$, $m \rightarrow I$, $F \rightarrow \tau$. Learn the swap, not new physics.

Memory aid — “**Shared spin, personal speed**”: every point on a rigid body shares ω and α , but linear $v = r\omega$ grows with distance from the axis.

Rolling Motion

Exam Relevance: This is the chapter’s grand finale — it fuses translation and rotation into one motion. **JEE Main** asks 1–2 questions yearly: inclined-plane acceleration, the kinetic-energy split, or the “which body wins the race” classic. **NEET** loves the energy ratio and the race ordering. **JEE Advanced** pushes into rolling-with-slipping transitions and friction analysis. **CBSE Boards** wants the rolling kinetic energy expression and the pure-rolling condition $v = R\omega$.

Section 1: Concept Intuition

Everything you’ve built across this chapter — centre of mass, torque, moment of inertia, rotational dynamics — comes together here in the most familiar motion of all: a wheel rolling along a road. Rolling looks simple, but it hides a beautiful secret. **A rolling body is doing two things at once:** its centre of mass is gliding forward (pure *translation*, the motion of subtopic 01), and at the same time the body is spinning about that centre (pure *rotation*, subtopics 02–04). Rolling is the superposition of these two.

Here’s the secret that makes rolling special. In **pure rolling** — rolling without slipping — the point of the body touching the ground is, for that instant, completely at rest. Think about a car tyre on the highway. The patch of rubber kissing the road isn’t skidding or smearing; it’s momentarily stationary, which is exactly why your tyres *grip* instead of sliding. The forward translation of the centre and the backward sweep of the spinning rim cancel perfectly at the bottom.

This cancellation produces a striking velocity pattern. The contact point at the bottom has speed **zero**. The centre moves at speed v . And the top point races along at $2v$ — twice the speed of the car itself. Photograph a moving wheel and the spokes near the top are a blur while the spokes at the bottom look sharp; the camera is telling you the bottom is barely moving.

Think about it this way... Walking is rolling in disguise. When you stride, your planted foot stays fixed on the ground (momentarily at rest) while your whole body pivots forward over it. The foot is your “contact point.” That’s pure rolling, performed by your legs.

There’s a clean way to *see* the velocity pattern: imagine the rolling motion as two transparent pictures laid on top of each other. In the first, every point of the wheel slides forward at the centre’s speed v (pure translation). In the second, every point whirls around the centre at rim-speed $R\omega = v$ (pure rotation). Add the two at the bottom: the translation points forward, the rotation points backward, and they cancel to zero. Add them at the top: both point forward, giving $2v$. This “superposition” trick — translation of the centre plus rotation about the centre — lets you find the velocity of *any* point on a rolling body, and it’s the same centre-of-mass decomposition you met in subtopic 01, now applied to a spinning wheel.

The pure-rolling condition that links the two motions is gorgeously simple:

$$v_{cm} = R\omega$$

Translation speed equals radius times spin rate. This one equation is the bridge between everything linear and everything angular in this subtopic.

Because a rolling body stores energy in *both* motions, it carries more kinetic energy than a sliding body of the same speed — part of the energy is “locked up” in spinning. This leads to the single most famous demonstration in rotational physics: the **race down a ramp**. Release a solid sphere, a disc, and a ring from the same height at the same instant. They do *not* tie. The body that diverts the least energy into spinning reaches the bottom first. The solid sphere wins, the disc comes second, and the ring trails last — and astonishingly, this order does *not* depend on their masses or radii, only on their *shape* (how their mass is distributed). A marble beats a ring every time, whether the marble is tiny or huge.

Limiting Conditions & Assumptions

1. **Pure rolling requires friction — yet that friction does no work.** Static friction supplies the torque that keeps $v = R\omega$, but since the contact point doesn't move, it dissipates no energy. This surprises almost everyone.
2. **On a perfectly frictionless incline, a body cannot roll — it only slides.** With no friction there's no torque to spin it up, so ω stays zero while it slides down.
3. **The condition $v = R\omega$ holds only for pure rolling.** The moment slipping begins (a hard brake, an icy patch), $v \neq R\omega$ and you're in “rolling with slipping.”

Section 2: Key Formulas & Definitions

Pure rolling (no slipping) conditions:

$$v_{cm} = R\omega, \quad a_{cm} = R\alpha$$

Velocities of key points (for a body rolling at centre-speed v):

$$v_{contact} = 0, \quad v_{centre} = v, \quad v_{top} = 2v$$

Total kinetic energy (translation + rotation), using $I = mK^2$ and $v = R\omega$:

$$K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2} \right)$$

where K is the radius of gyration. The translational and rotational shares are in the ratio:

$$K_{trans} : K_{rot} = 1 : \frac{K^2}{R^2}$$

Rolling down an incline of angle θ :

$$a = \frac{g \sin \theta}{1 + \frac{K^2}{R^2}}, \quad v_{bottom} = \sqrt{\frac{2gh}{1 + \frac{K^2}{R^2}}}$$

Friction required and minimum coefficient for pure rolling:

$$f = \frac{mg \sin \theta}{1 + \frac{R^2}{K^2}}, \quad \mu_{\min} = \frac{\tan \theta}{1 + \frac{R^2}{K^2}}$$

The master table (the smaller K^2/R^2 , the faster the body):

Body	K^2/R^2	$1 + K^2/R^2$	Acceleration on incline
Solid sphere	2/5	7/5	$\frac{5}{7}g \sin \theta$
Solid cylinder / disc	1/2	3/2	$\frac{2}{3}g \sin \theta$
Spherical shell	2/3	5/3	$\frac{3}{5}g \sin \theta$
Ring / hollow cylinder	1	2	$\frac{1}{2}g \sin \theta$

Notation: v_{cm} = centre-of-mass speed, ω = angular speed, R = radius, K = radius of gyration, h = vertical height descended. SI unit of KE is the joule; dimensional formula $[M^1 L^2 T^{-2}]$.

Section 3: CBSE Board Derivation — A Body Rolling Down an Incline

We derive the acceleration of a round body rolling without slipping down an incline of angle θ — the most-asked rolling derivation in boards and a complete showcase of the chapter's tools.

Figure 6.11 *A sphere on an incline of angle θ . Forces shown: weight mg vertically down (component $mg \sin \theta$ along the incline, $mg \cos \theta$ perpendicular), normal reaction N perpendicular to the surface, and static friction f acting up the incline at the contact point. (see next page)*

Step 1 — Translation equation (Newton's second law along the incline). The forces along the incline are the gravity component $mg \sin \theta$ pulling down and friction f acting up:

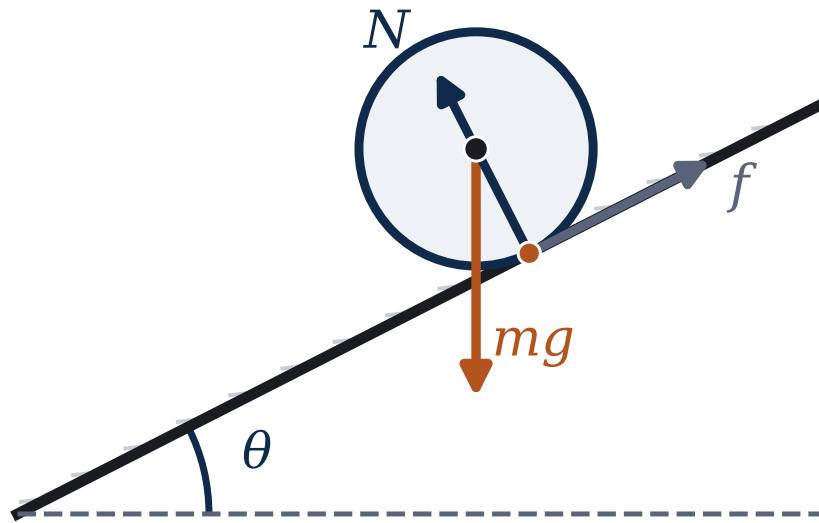
$$mg \sin \theta - f = ma \quad \dots(i)$$

Step 2 — Rotation equation (torque about the centre of mass). Of the forces, only friction has a moment arm about the centre (gravity acts *through* the centre, and the normal force's line passes through it too). So friction alone provides the torque:

$$\tau = f R = I \alpha$$

Writing $I = mK^2$ and using the pure-rolling link $\alpha = a/R$:

$$f R = mK^2 \cdot \frac{a}{R} \Rightarrow f = \frac{mK^2 a}{R^2} \quad \dots(ii)$$



$$mg \sin \theta - f = ma, \quad fR = I\alpha, \quad a = R\alpha$$

$$\Rightarrow a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}} = \frac{5}{7} g \sin \theta \quad (\text{solid sphere})$$

Figure 6.11 A sphere on an incline of angle θ . Forces shown: weight mg vertically down (component $mg \sin \theta$ along the incline, $mg \cos \theta$ perpendicular), normal reaction N perpendicular to the surface, and static friction f acting up the incline at the contact point.

Step 3 — Eliminate friction. Substitute (ii) into (i):

$$mg \sin \theta - \frac{mK^2 a}{R^2} = ma$$

Step 4 — Solve for a . Cancel m and collect terms:

$$g \sin \theta = a \left(1 + \frac{K^2}{R^2} \right) \Rightarrow a = \frac{g \sin \theta}{1 + \frac{K^2}{R^2}}$$

The physical insight: the term K^2/R^2 in the denominator is the “rotational tax.” A body that distributes its mass far from the axis (large K^2/R^2 , like a ring) pays a heavier tax — more of gravity’s pull goes into spinning it up, so it accelerates more slowly. This single formula explains the entire race result.

Companion derivation: rolling kinetic energy

The total kinetic energy is the translational KE of the centre of mass plus the rotational KE about it:

$$K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

Substitute $I = mK^2$ and $\omega = v/R$:

$$K = \frac{1}{2}mv^2 + \frac{1}{2}mK^2 \cdot \frac{v^2}{R^2} = \frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2} \right)$$

This is the same “ $1 + K^2/R^2$ ” factor — appearing now in energy. Setting this equal to the lost potential energy mgh on an incline immediately gives $v_{\text{bottom}} = \sqrt{\frac{2gh}{1 + K^2/R^2}}$.

Companion result: how much friction is needed?

The friction force itself follows from substituting the acceleration back into equation (ii):

$$f = \frac{mK^2}{R^2} \cdot a = \frac{mK^2}{R^2} \cdot \frac{g \sin \theta}{1 + K^2/R^2} = \frac{mg \sin \theta}{1 + \frac{R^2}{K^2}}$$

For pure rolling to be possible, the surface must be able to supply at least this much friction. Since the available static friction is at most $\mu N = \mu mg \cos \theta$, the condition $f \leq \mu mg \cos \theta$ rearranges to:

$$\mu_{\min} = \frac{\tan \theta}{1 + \frac{R^2}{K^2}}$$

If the actual μ falls below this, the body can no longer maintain $v = R\omega$ and begins to slip — which is exactly why a steep, smooth slope makes even a well-shaped wheel skid rather than roll.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A solid sphere of mass m rolls without slipping with centre-of-mass speed v . Find its total kinetic energy and the fraction of that energy which is rotational.

Given: solid sphere, so $\frac{K^2}{R^2} = \frac{2}{5}$.

Total kinetic energy:

$$K = \frac{1}{2}mv^2 \left(1 + \frac{2}{5}\right) = \frac{1}{2}mv^2 \cdot \frac{7}{5} = \frac{7}{10}mv^2$$

Rotational fraction — using $K_{trans} : K_{rot} = 1 : \frac{K^2}{R^2} = 1 : \frac{2}{5} = 5 : 2$:

$$\frac{K_{rot}}{K_{total}} = \frac{2}{5+2} = \frac{2}{7}$$

Answer: Total KE = $\frac{7}{10}mv^2$; rotational energy is $\frac{2}{7}$ of the total.

Example 2 — [NEET Speed Trap]

A solid sphere, a disc, and a ring — all released from rest at the same height on identical inclines. In what order do they reach the bottom?

The trap: Students hunt for masses and radii. But the acceleration $a = \frac{g \sin \theta}{1 + K^2/R^2}$ depends **only** on K^2/R^2 — mass and radius cancel out entirely.

Mental shortcut — **smallest K^2/R^2 accelerates fastest.**

- Solid sphere: $K^2/R^2 = 2/5 = 0.40 \rightarrow$ fastest
- Disc: $K^2/R^2 = 1/2 = 0.50 \rightarrow$ middle
- Ring: $K^2/R^2 = 1 \rightarrow$ slowest

Answer: Sphere first, disc second, ring last — regardless of their masses or sizes. *Sanity check:* the ring keeps all its mass at the rim, so it pays the biggest “rotational tax” and crawls. The compact sphere spends the least energy spinning and wins. ✓

Example 3 — [JEE Main Level]

A uniform disc rolls without slipping down an incline of angle 30° . Find (a) its acceleration and (b) the minimum coefficient of static friction required for pure rolling.

Given: disc, $\frac{K^2}{R^2} = \frac{1}{2}$, $\theta = 30^\circ$.

(a) **Acceleration:**

$$a = \frac{g \sin 30^\circ}{1 + \frac{1}{2}} = \frac{g \cdot \frac{1}{2}}{\frac{3}{2}} = \frac{g}{3}$$

(b) **Minimum friction coefficient** — this couples translation, rotation, *and* the friction limit:

$$\mu_{min} = \frac{\tan \theta}{1 + \frac{R^2}{K^2}} = \frac{\tan 30^\circ}{1 + 2} = \frac{1/\sqrt{3}}{3} = \frac{1}{3\sqrt{3}} \approx 0.19$$

Answer: $a = \frac{g}{3}$ and $\mu_{min} = \frac{1}{3\sqrt{3}} \approx 0.19$. *Why multi-concept:* part (a) needs the rolling-incline result, while part (b) needs you to back out the friction force *and* compare it to μN — combining rolling dynamics with the friction inequality.

Example 4 — [JEE Advanced Level]

A solid sphere is thrown onto a rough horizontal floor with initial speed v_0 but no spin (it is purely sliding). Friction acts until pure rolling begins. Find the final speed of the centre of mass when pure rolling starts.

Figure 6.12 A sphere moving right with velocity v_0 and zero spin, on a horizontal floor. Kinetic friction f acts backward at the contact point, decelerating the centre while spinning the sphere up, until $v = R\omega$ is reached. (see next page)

The elegant method — conserve angular momentum about a point on the floor. Friction acts *at the contact line on the floor*, so it has zero torque about any point on the floor. Hence the sphere's angular momentum about a floor point is conserved throughout the sliding phase. This ties straight back to the “take L about the contact point” trick from the angular-momentum subtopic.

Step 1 — Initial angular momentum about the floor point. Only the translating centre contributes ($\omega_0 = 0$):

$$L_i = mv_0 R$$

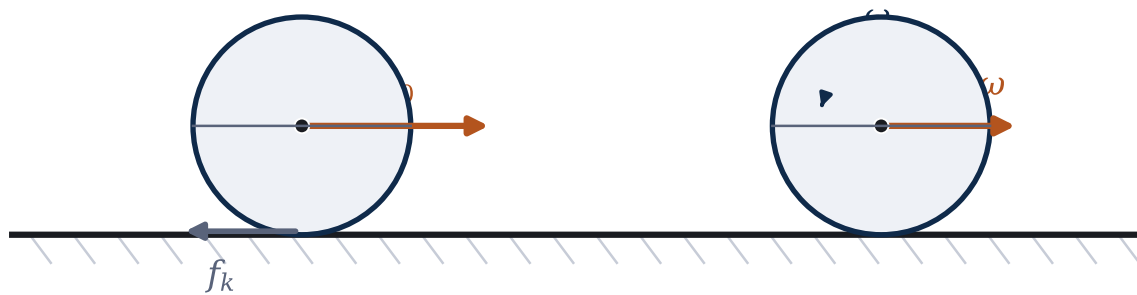
Step 2 — Final angular momentum (pure rolling, $v = R\omega$). About the contact point, $I_{contact} = I_{cm} + mR^2 = \frac{2}{5}mR^2 + mR^2 = \frac{7}{5}mR^2$:

$$L_f = I_{contact} \omega = \frac{7}{5}mR^2 \cdot \frac{v}{R} = \frac{7}{5}mRv$$

Step 3 — Equate and solve.

$$mv_0 R = \frac{7}{5}mRv \Rightarrow v = \frac{5}{7}v_0$$

Answer: Pure rolling begins when the centre-of-mass speed has dropped to $v = \frac{5}{7}v_0$. *Note:* the force-and-torque method (friction decelerates translation, $v = v_0 - \mu g t$, while spinning it up, $\omega = \frac{5\mu g}{2R} t$,



L about a floor point is conserved: $mv_0R = mvR + \frac{2}{5}mR^2\frac{v}{R} \Rightarrow v = \frac{5}{7}v_0$

Figure 6.12 A sphere moving right with velocity v_0 and zero spin, on a horizontal floor. Kinetic friction f acts backward at the contact point, decelerating the centre while spinning the sphere up, until $v = R\omega$ is reached.

then impose $v = R\omega$) gives the identical $\frac{5}{7}v_0$ — but the angular-momentum route skips μ and t entirely.

Section 5: Practice Exercises

1. **[CBSE]** A ring rolls without slipping at centre speed v . What fraction of its total kinetic energy is translational?
2. **[NEET]** A body rolls down an incline with acceleration $\frac{3}{5}g \sin \theta$. Identify the body.
3. **[JEE Main]** A solid sphere and a hollow sphere of the same mass and radius roll down the same incline. Find the ratio of the times they take to reach the bottom from the same height.
4. **[JEE Main]** A disc of radius 0.2 m rolls without slipping with centre speed 4 m/s. Find the speed of the topmost point and the speed of the point of contact.
5. **[JEE Advanced]** A solid cylinder of mass M and radius R rolls without slipping down an incline of angle θ . Using energy conservation, find the speed of its centre after it has descended a vertical height h , and verify it is independent of M and R .

Answers: 1. $\frac{1}{2}$ (since ring has $K^2/R^2 = 1$, the split is 1 : 1). 2. A spherical shell ($K^2/R^2 = 2/3$).

3. $t_{\text{solid}} : t_{\text{hollow}} = \sqrt{\frac{1 + 2/5}{1 + 2/3}} = \sqrt{\frac{7/5}{5/3}} = \sqrt{\frac{21}{25}} = \frac{\sqrt{21}}{5} \approx 0.92$. 4. Top = 8 m/s, contact = 0. 5.

$v = \sqrt{\frac{2gh}{1 + 1/2}} = \sqrt{\frac{4gh}{3}}$, independent of M and R .

Section 6: Crack the MCQ

Q1. A wheel rolls without slipping at centre speed v . The speed of the topmost point of the wheel is:

- (A) v (B) $\frac{v}{2}$ (C) $2v$ (D) zero

Correct: (C). The top point combines the translational velocity v (forward) with the rotational velocity $R\omega = v$ (also forward at the top), giving $v + v = 2v$. *Why the traps fail:* **(D)** is the speed of the *bottom* (contact) point, a classic swap. **(A)** ignores the rotational contribution. **(B)** is a random halving. The defining fact of pure rolling: bottom = 0, centre = v , top = $2v$.

Q2. A solid sphere and a ring of the same mass and radius roll down identical inclines from the same height. At the bottom:

- (A) the ring is faster
 (B) the sphere is faster
 (C) they have equal speed
 (D) the faster one depends on the radius

Correct: (B). Bottom speed is $v = \sqrt{\frac{2gh}{1 + K^2/R^2}}$. The sphere has the smaller K^2/R^2 ($2/5$ vs 1), so the smaller denominator and the larger speed. *Why the traps fail:* **(A)** reverses the physics. **(C)** forgets that the energy split differs by shape. **(D)** is the trap that radius matters — but R cancels, leaving only the shape factor.

Q3. During pure rolling on a horizontal surface, the work done by the static friction force is:

- (A) positive
- (B) negative
- (C) zero
- (D) equal to the change in rotational KE

Correct: (C). Static friction acts at the contact point, which is instantaneously at rest. Work = force $\times$ displacement of the point of application = 0. *Why the traps fail:* **(A)** and **(B)** assume friction transfers energy, but with zero contact-point motion there's no work. **(D)** confuses friction's role (it provides torque, redistributing but not dissipating energy) with doing net work.

Q4. A body rolls without slipping. Its rotational kinetic energy is 40% of its total kinetic energy. The body could be a:

- (A) ring (B) solid sphere (C) disc (D) spherical shell

Correct: (D). Rotational fraction = $\frac{K^2/R^2}{1 + K^2/R^2}$. Setting this to 0.40: $\frac{K^2/R^2}{1 + K^2/R^2} = \frac{2}{5}$ gives $K^2/R^2 = 2/3$ — a spherical shell. *Why the traps fail:* **(A)** ring gives $\frac{1}{2} = 50\%$. **(B)** solid sphere gives $\frac{2}{7} \approx 28.6\%$. **(C)** disc gives $\frac{1}{3} \approx 33.3\%$. Only the shell yields exactly 40%.

Section 7: Common Student Pitfalls & Pro-Tips

- Forgetting the $\left(1 + \frac{K^2}{R^2}\right)$ factor.** A rolling body is *not* a point mass — it stores energy in spin too. Leaving out this factor (using just $\frac{1}{2}mv^2$ or $a = g \sin \theta$) is the most common rolling error.
- Believing friction does work in pure rolling.** It supplies the torque for $v = R\omega$, but the contact point is at rest, so static friction does *zero* work. Energy is conserved.
- Thinking mass or radius affects the race.** On an incline, both cancel — only the *shape* (via K^2/R^2) decides speed and acceleration. A tiny ball bearing ties with a cannonball of the same shape.
- Using $v = R\omega$ when the body is slipping.** This condition is *only* for pure rolling. During slipping (e.g., a sphere thrown without spin), $v \neq R\omega$ until friction establishes rolling.

Pro-Tip: For “sliding-becomes-rolling” problems, take **angular momentum about a point on the surface**. Friction acts there, so its torque is zero and L is conserved — you get the final speed in one line without ever touching μ or time. And for ranking bodies, memorise the order by K^2/R^2 : **sphere** (0.4) < **disc** (0.5) < **shell** (0.67) < **ring** (1) — fastest to slowest, every time.

Section 8: Cheat Sheet

- **Rolling = translation + rotation.** Pure-rolling condition: $v_{cm} = R\omega$, $a_{cm} = R\alpha$.
- **Point speeds:** contact = 0, centre = v , top = $2v$.
- **Total KE:** $K = \frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2}\right)$. Split $K_{trans} : K_{rot} = 1 : \frac{K^2}{R^2}$.
- **Incline:** $a = \frac{g \sin \theta}{1 + K^2/R^2}$, $v_{bottom} = \sqrt{\frac{2gh}{1 + K^2/R^2}}$.
- **Shape factors K^2/R^2 :** sphere $\frac{2}{5}$, disc/cylinder $\frac{1}{2}$, shell $\frac{2}{3}$, ring 1.
- **Friction in pure rolling does no work** (contact point at rest).

Memory aid — “Compact wins”: the more tightly mass hugs the axis (smaller K^2/R^2), the faster it rolls. Sphere beats disc beats shell beats ring — mass and radius never matter.

Memory aid — “Bottom is frozen, top flies”: in pure rolling the contact point is momentarily at rest (0) and the top moves at $2v$.

Chapter 6: System of Particles and Rotational Motion — Round 2 Addendum

The chapter taught you centre of mass, the cross product, torque and angular momentum, equilibrium, moment of inertia with both axis theorems, the rotational SUVAT family, and rolling down an incline. Six examinable patterns still stand outside the door, and each is a marriage of two things the chapter already teaches separately. Addendum A takes the rolling chapter's three famous speeds (contact = 0, centre = v , top = $2v$) and builds the machinery for *every* point on the wheel — the JEE favourite the chapter skips. Addendum B extends the chapter's single-block pulley to the full connected-system recipe with a massive pulley. Addendum C takes the sliding-sphere example (which stopped at the final speed $\frac{5}{7}v_0$) and runs the complete timeline: time, distance, and heat. Addendum D removes the pivot from the bullet-in-rod collision and introduces the centre of percussion. Addendum E covers a body the chapter never mentions at all: the rigid body that *oscillates* — the physical pendulum. Addendum F takes on the rigid body pulled or rolled over a step: the floor normal vanishes, the body pivots about the step corner, and a single torque equation delivers the minimum force — the JEE Main favourite the chapter never treats. Nothing here is new theory; every method below is $\vec{v} = \vec{v}_{cm} + \vec{\omega} \times \vec{r}$, $\tau = I\alpha$, and angular momentum conservation, applied one step further than the chapter took them.

Addendum A — Velocity and Acceleration of Every Point on a Rolling Body

The chapter gives you exactly three speeds of a rolling wheel. JEE Main regularly asks for the velocity of an *arbitrary* rim point — the leading point, a point at some angle — and JEE Advanced attaches accelerations and the instantaneous axis of rotation to it. The chapter's superposition picture ("translation picture laid on rotation picture") already contains the whole method; this addendum simply finishes the job.

Method

Work with a wheel of radius R rolling *without slipping* to the right at centre speed v , so the spin obeys

$$\omega = \frac{v}{R},$$

with $\vec{\omega}$ pointing *into* the page (clockwise, seen with $+y$ up).

1. **Write the master relation.** For any point P of the body,

$$\vec{v}_P = \vec{v}_{cm} + \vec{\omega} \times \vec{r}_{P/cm},$$

where $\vec{r}_{P/cm}$ is the position of P measured from the centre.

2. **Parametrise the rim.** Let P sit at angle θ measured from the upward vertical, so $\vec{r}_{P/cm} = R \sin \theta \hat{i} + R \cos \theta \hat{j}$.
3. **Carry out the cross product.** With $\vec{\omega} = -\omega \hat{k}$:

$$\vec{\omega} \times \vec{r}_{P/cm} = \omega R \cos \theta \hat{i} - \omega R \sin \theta \hat{j},$$

and adding $\vec{v}_{cm} = v \hat{i}$ with $\omega R = v$:

$$\vec{v}_P = v[(1 + \cos \theta) \hat{i} - \sin \theta \hat{j}].$$

4. **Read off the speed.**

$$|\vec{v}_P|^2 = v^2[(1 + \cos \theta)^2 + \sin^2 \theta] = 2v^2(1 + \cos \theta),$$

so

$$|\vec{v}_P| = 2v \cos \frac{\theta}{2},$$

and the velocity makes an angle $\theta/2$ below the horizontal, since $\tan(\text{slope}) = \frac{\sin \theta}{1 + \cos \theta} = \tan \frac{\theta}{2}$.

- Checks: $\theta = 0$ (top) gives $2v$ horizontally; $\theta = \pi$ (bottom) gives 0; $\theta = \pi/2$ (leading point) gives $\sqrt{2}v$ at 45° below the horizontal.
5. **Accelerations are a separate question.** If the wheel rolls *steadily*, ω is constant, so every rim point performs uniform circular motion about the centre:

$$|\vec{a}_P| = \omega^2 R = \frac{v^2}{R},$$

directed from P toward the centre — for every point, independent of θ . The topmost point therefore accelerates *downward* at v^2/R , and the contact point (velocity zero!) accelerates *upward* at v^2/R . Students who answer “zero acceleration because zero velocity” lose the mark every single year.

6. **Shortcut — the instantaneous axis of rotation (IAOR).** At each instant the whole wheel may be treated as *rotating purely* about the contact point: $v_P = \omega d_P$, where d_P is P 's distance from the contact point. For a rim point the chord geometry gives $d_P = 2R \cos(\theta/2)$, which reproduces Step 4 instantly.
 - Caveat (Advanced): the IAOR is legitimate for *velocities* only. The contact point itself accelerates, so you must not compute accelerations as if the wheel pivoted about a fixed point there.

Example A.1 — [JEE Main Level]

A disc of radius $R = 0.50$ m rolls without slipping with centre speed $v = 4.0$ m/s. Find the angular velocity, and the velocity of the leading (rightmost) point of the rim.

Step 1 — angular velocity:

$$\omega = \frac{v}{R} = \frac{4.0 \text{ m/s}}{0.50 \text{ m}} = 8.0 \text{ rad/s}.$$

Step 2 — the leading point sits at $\theta = 90^\circ$ from the top. Substituting into the master relation:

$$\vec{v}_P = 4.0[(1 + 0)\hat{i} - 1\hat{j}] \text{ m/s} = (4.0\hat{i} - 4.0\hat{j}) \text{ m/s}.$$

Its magnitude is $\sqrt{(4.0)^2 + (4.0)^2} \text{ m/s} = 4\sqrt{2} \text{ m/s} \approx 5.7 \text{ m/s}$, directed 45° below the horizontal (forward and downward).

Check by IAOR. The leading point is a distance $\sqrt{R^2 + R^2} = R\sqrt{2} = 0.707 \text{ m}$ from the contact point, so

$$v_P = \omega d = (8.0)(0.707) \text{ m/s} = 5.7 \text{ m/s}.$$

Example A.2 — [JEE Advanced Level]

(a) Show that the rim speed $2v \cos(\theta/2)$ follows from the IAOR alone. (b) Find all rim points whose speed equals the centre speed v . (c) Find the acceleration of the topmost point.

(a) The contact point C and rim point P both lie on the circle, separated by the chord subtending angle θ at the centre. The chord length is $d = 2R \cos(\theta/2)$. Treating the wheel as instantaneously rotating about C with angular speed $\omega = v/R$:

$$v_P = \omega d = \frac{v}{R} \cdot 2R \cos \frac{\theta}{2} = 2v \cos \frac{\theta}{2}.$$

No components were needed — the geometry did the work.

(b) Set $2v \cos(\theta/2) = v$, so $\cos(\theta/2) = \frac{1}{2}$ and $\theta = 120^\circ$. There are **two** such points, at $\pm 120^\circ$ from the top. Their heights relative to the centre are $R \cos 120^\circ = -R/2$, i.e. half a radius below the centre, at horizontal offsets $\pm R \sin 120^\circ = \pm \frac{\sqrt{3}}{2} R$.

Check: at $\theta = 120^\circ$ the master relation gives $\vec{v}_P = v[\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}]$ with magnitude $v\sqrt{\frac{1}{4} + \frac{3}{4}} = v$.

(c) Steady rolling means constant ω , so the top point is in uniform circular motion about the centre:

$$a_{top} = \omega^2 R = \frac{v^2}{R},$$

directed straight *downward*. Its velocity $2v$ and its acceleration point in opposite senses — velocity says nothing about acceleration here.

Why it is Advanced-level. Part (b) hides a counting trap (two points, symmetric, easily missed), and part (c) tests whether you know the IAOR cannot be reused for accelerations. The mark-winning habit is separating the velocity machine from the acceleration machine.

Practice — Addendum A

1. A bicycle wheel of radius 0.35 m rolls at 7.0 m/s . Find ω .
2. Find the speed of the *trailing* (leftmost) rim point in terms of v .
3. Which non-contact rim points have a *purely horizontal* velocity?
4. What is the maximum possible rim speed, and how many rim points attain it?
5. A wheel rolls at $v = 6.0 \text{ m/s}$. Find the speed of the point at $\theta = 60^\circ$ from the top.

6. Find the magnitude and direction of the acceleration of the contact point of a steadily rolling wheel.

Answers: 1. $\omega = v/R = 20 \text{ rad/s}$. 2. $\sqrt{2}v$ — by symmetry with the leading point, directed 45° above the horizontal. 3. Only the topmost point ($\theta = 0$): setting the y -component $-v \sin \theta$ to zero forces $\theta = 0$ or π , and $\theta = \pi$ is the contact point with zero velocity. 4. Maximum $= 2v$ at the top, attained at exactly one point. 5. $2v \cos 30^\circ = \sqrt{3}v = 6\sqrt{3} \approx 10.4 \text{ m/s}$. 6. Magnitude v^2/R , directed vertically upward (toward the centre) — the contact point has zero velocity but non-zero acceleration.

Addendum B — Massive Pulleys: Connected Systems Done Completely

The chapter solved one block hanging from a disc pulley. The perennial JEE Main pattern goes one step further: *two* blocks over a pulley with rotational inertia, where the two string tensions are **not equal**. The extra idea is tiny — write Newton's law for each translating mass, $\tau = I\alpha$ for the pulley, link everything with $a = R\alpha$ — but the bookkeeping must be airtight.

Method

1. **One equation per object.** Each translating mass gets $F_{\text{net}} = ma$; the pulley gets $\tau_{\text{net}} = I\alpha$ about its axle.
2. **Label the tensions separately.** With a massive pulley, $T_1 \neq T_2$; their difference is precisely what provides the pulley's torque. Writing $T_1 = T_2$ is the fatal error.
3. **Apply the constraint.** String inextensible and non-slipping: $a = R\alpha$ for every segment.
4. **Fix one sign convention** (e.g. m_1 descending is positive) and hold it across all equations.
5. **Solve the linear system**, then check limits: letting $M \rightarrow 0$ must recover the massless-pulley answers from earlier chapters.
6. **Cross-check with energy.** For motion through a height h from rest, $\sum \Delta K = -\sum \Delta U$ must hold, since static friction at the axle does no work.

Master result — Atwood machine with a disc pulley (masses $m_1 > m_2$, pulley mass M , radius R , $I = \frac{1}{2}MR^2$):

$$\begin{aligned} m_1 g - T_1 &= m_1 a, \\ T_2 - m_2 g &= m_2 a, \\ (T_1 - T_2) R &= \frac{1}{2} M R^2 \cdot \frac{a}{R} \implies T_1 - T_2 = \frac{1}{2} M a. \end{aligned}$$

Adding all three eliminates both tensions:

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2 + \frac{M}{2}}.$$

Note how $M/2$ enters the denominator exactly like a third mass — the pulley's inertia behaves as if half its mass hung on each side. If the pulley were a thin *ring*, $I = MR^2$ and the denominator becomes $m_1 + m_2 + M$ instead.

Example B.1 — [JEE Main Level]

Masses $m_1 = 3.0$ kg **and** $m_2 = 2.0$ kg **hang over a uniform disc pulley of mass** $M = 1.0$ kg. **Take** $g = 10$ m/s². **Find** a , **both tensions**, and **verify the tension difference**.

Step 1 — acceleration:

$$a = \frac{(3.0 - 2.0)(10)}{3.0 + 2.0 + 0.5} \text{ m/s}^2 = \frac{10}{5.5} \text{ m/s}^2 = \frac{20}{11} \approx 1.82 \text{ m/s}^2.$$

Step 2 — tensions:

$$T_1 = m_1(g - a) = 3.0 \left(10 - \frac{20}{11} \right) = \frac{270}{11} \text{ N} \approx 24.5 \text{ N},$$

$$T_2 = m_2(g + a) = 2.0 \left(10 + \frac{20}{11} \right) = \frac{260}{11} \text{ N} \approx 23.6 \text{ N}.$$

Step 3 — verify the torque equation:

$$T_1 - T_2 = \frac{10}{11} \text{ N} \approx 0.91 \text{ N}, \quad \frac{1}{2}Ma = 0.5 \times 1.0 \times \frac{20}{11} = \frac{10}{11} \text{ N}. \checkmark$$

The heavier side's string pulls harder — by exactly the amount needed to angularly accelerate the pulley.

Example B.2 — [JEE Advanced Level]

For the system of Example B.1, find the speed of the masses after m_1 **has descended** $h = 1.0$ m **from rest, by (i) kinematics and (ii) energy conservation. Confirm the two agree exactly.**

(i) Kinematics. Constant a from rest:

$$v^2 = 2ah = 2 \left(\frac{20}{11} \right) (1.0) \text{ m}^2/\text{s}^2 = \frac{40}{11} \text{ m}^2/\text{s}^2, \quad v \approx 1.91 \text{ m/s}.$$

(ii) Energy. Lost potential energy:

$$\Delta U = (m_1 - m_2)gh = (1.0)(10)(1.0) \text{ J} = 10 \text{ J}.$$

Gained kinetic energy, with $I = \frac{1}{2}MR^2$ and $\omega = v/R$:

$$K = \frac{1}{2}(m_1 + m_2)v^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}(5.0)v^2 + \frac{1}{4}(1.0)v^2 = 2.75 v^2.$$

Setting $K = \Delta U$: $v^2 = 10/2.75 = 40/11 \text{ m}^2/\text{s}^2$ — identical to (i).

Why it is Advanced-level. Either route alone scores; producing *both* and showing they agree demonstrates command of the constraint machinery, and the energy route is the faster one whenever the question asks only for speeds.

Practice — Addendum B

1. Repeat Example B.1 with the pulley replaced by a thin ring of the same mass. Find a .
2. A block $m_1 = 1.0 \text{ kg}$ rests on a frictionless table; a light string runs horizontally from it over a disc pulley ($M = 0.50 \text{ kg}$) to a hanging block $m_2 = 2.0 \text{ kg}$. Take $g = 10 \text{ m/s}^2$. Find a .
3. In Question 2, find the difference between the two tensions.
4. In Question 2, find the time for m_2 to descend 0.50 m from rest.
5. Conceptual: why must the two tensions in any massive-pulley system differ?

Answers: 1. $a = \frac{(1.0)(10)}{3.0 + 2.0 + 1.0} = \frac{10}{6.0} \approx 1.67 \text{ m/s}^2$ — smaller, since a ring stores more inertia per unit mass. 2. Adding $m_2g - T_2 = m_2a$, $T_1 = m_1a$, and $(T_2 - T_1)R = \frac{1}{2}MR^2(a/R)$:
 $a = \frac{m_2g}{m_1 + m_2 + M/2} = \frac{20}{3.25} \approx 6.15 \text{ m/s}^2$. 3. $\Delta T = \frac{1}{2}Ma = 0.25 \times 6.15 \approx 1.54 \text{ N}$. 4. $t = \sqrt{2h/a} = \sqrt{1.00/6.15} \text{ s} \approx 0.40 \text{ s}$. 5. Zero net torque would mean zero angular acceleration; a spinning-up pulley demands $(T_1 - T_2)R = I\alpha \neq 0$. Only a massless pulley ($I = 0$) allows $T_1 = T_2$.

Addendum C — Slipping into Rolling: Time, Distance and Heat

The chapter showed the slick trick — conserve angular momentum about a floor point — to get the final speed $\frac{5}{7}v_0$ of a sliding sphere. It stopped there. JEE Advanced routinely wants the *whole story*: how long slipping lasts, how far the body travels, and how much energy friction destroys. The trick cannot give those; a straightforward force-and-torque timeline can.

Method

1. **Identify the slip.** Kinetic friction $f = \mu mg$ acts at the contact point, opposing the relative slip there. It decelerates translation and exerts a torque fR about the centre.
2. **Write both timelines.** With $I = mk^2$:

$$v(t) = v_0 - \mu gt, \quad \omega(t) = \frac{\mu mgR}{I} t = \frac{\mu gR}{k^2} t.$$

3. **Find the rolling instant** t^* from the pure-rolling condition $v(t^*) = \omega(t^*)R$.
4. **Distance:** integrate (or use constant-acceleration kinematics) up to t^* .
5. **Heat generated:** friction acts on slipping surfaces, so

$$H = \int f |v - \omega R| dt = \mu mg \times (\text{area under the } (v - \omega R) \text{ graph}).$$

Since $v - \omega R$ falls linearly from v_0 to 0, the area is $\frac{1}{2}v_0 t^*$.

6. **Close the books:** $K_i - K_f$ must equal H . If it does not, hunt the arithmetic.

Solid sphere ($k^2 = \frac{2}{5}R^2$), thrown without spin: the rolling condition gives

$$v_0 - \mu g t^* = \frac{5\mu g}{2} t^* \implies t^* = \frac{2v_0}{7\mu g}, \quad v^* = \frac{5v_0}{7},$$

matching the chapter's angular-momentum result. The distance and heat work out to

$$s^* = \frac{12v_0^2}{49\mu g}, \quad H = \frac{1}{7} \cdot \frac{1}{2} m v_0^2.$$

Exactly one-seventh of the initial kinetic energy is destroyed — and notice v^* and the energy split contain **no** μ : friction sets the schedule, not the destination.

Example C.1 — [JEE Advanced Level]

A cue ball (solid sphere, $m = 0.20$ kg) slides without spin at $v_0 = 7.0$ m/s onto cloth with $\mu = 0.20$. Take $g = 10$ m/s². Find (a) the time and distance until pure rolling, (b) the heat generated, and (c) verify the energy ledger.

(a) Timeline:

$$t^* = \frac{2(7.0)}{7(0.20)(10)} \text{ s} = \frac{14}{14} \text{ s} = 1.0 \text{ s},$$

$$s^* = v_0 t^* - \frac{1}{2} \mu g (t^*)^2 = (7.0)(1.0) - \frac{1}{2} (2.0)(1.0) \text{ m} = 6.0 \text{ m}.$$

Final speed $v^* = v_0 - \mu g t^* = 7.0 - 2.0 = 5.0$ m/s, agreeing with $\frac{5}{7}(7.0) = 5.0$ m/s.

(b) Heat:

$$H = \frac{m v_0^2}{7} = \frac{(0.20)(7.0)^2}{7} \text{ J} = 1.4 \text{ J}.$$

(c) Ledger:

$$K_i = \frac{1}{2} (0.20)(7.0)^2 \text{ J} = 4.9 \text{ J},$$

$$K_f = \frac{1}{2} (0.20)(5.0)^2 + \frac{1}{2} \left(\frac{2}{5} \right) (0.20)(5.0)^2 \text{ J} = 2.5 + 1.0 \text{ J} = 3.5 \text{ J}.$$

Loss = $4.9 - 3.5 = 1.4 \text{ J} = H$.

Example C.2 — [JEE Advanced Level]

A solid sphere is spun up to ω_0 about a horizontal axis and released from rest just above a rough floor, landing with zero translational speed. Find its final speed once pure rolling sets in, and the fraction of the initial kinetic energy dissipated.

Here the slip runs the other way: the sphere's bottom surface moves *backward*, so kinetic friction acts *forward*, accelerating translation while braking the spin:

$$v(t) = \mu g t, \quad \omega(t) = \omega_0 - \frac{5\mu g}{2R} t.$$

Rolling condition $v = \omega R$:

$$\mu g t^* = R\omega_0 - \frac{5\mu g}{2} t^* \Rightarrow t^* = \frac{2R\omega_0}{7\mu g}, \quad v^* = \frac{2R\omega_0}{7}.$$

Energy ledger. Initially only spin energy exists, $K_i = \frac{1}{5} m R^2 \omega_0^2$. Finally, with $\omega_f = v^*/R = \frac{2}{7} \omega_0$:

$$K_f = \frac{1}{2} m \left(\frac{2R\omega_0}{7} \right)^2 + \frac{1}{2} \cdot \frac{2}{5} m R^2 \left(\frac{2\omega_0}{7} \right)^2 = \left(\frac{2}{49} + \frac{4}{245} \right) m R^2 \omega_0^2 = \frac{2}{35} m R^2 \omega_0^2.$$

Fraction surviving: $\frac{2/35}{1/5} = \frac{2}{7}$. So **five-sevenths dissipates** — the mirror image of the thrown ball, where five-sevenths of v_0 survived.

Practice — Addendum C

1. A ring is projected without spin at v_0 . Find t^* and the final speed.
2. For the thrown solid sphere, find the distance travelled before rolling begins.
3. What fraction of the initial kinetic energy of the thrown sphere is dissipated?
4. For the dropped spinning sphere, what fraction of the initial kinetic energy remains?
5. Conceptual: which quantities in the thrown-sphere problem depend on μ , and which do not?
6. A solid sphere slides at $v_0 = 14$ m/s without spin on $\mu = 0.10$ ($g = 10$ m/s²). Find t^* , s^* , and v^* .

Answers: 1. With $I = mR^2$, $\omega = (\mu g/R)t$, so $v_0 - \mu g t^* = \mu g t^*$ gives $t^* = \frac{v_0}{2\mu g}$ and $v^* = \frac{v_0}{2}$. 2. $s^* = \frac{12v_0^2}{49\mu g}$. 3. $\frac{1}{7} \approx 14.3\%$. 4. $\frac{2}{7} \approx 28.6\%$. 5. t^* and s^* shrink as μ grows; v^* , the energy split, and the final state are independent of μ . 6. $t^* = \frac{2(14)}{7(0.10)(10)} = 4.0$ s; $s^* = 14(4.0) - \frac{1}{2}(1.0)(16) = 48$ m; $v^* = 14 - (0.10)(10)(4.0) = 10$ m/s.

Addendum D — Collisions on a Free Rigid Body and the Centre of Percussion

Every collision example in the chapter involves a *pivoted* rod, where the hinge force kills linear momentum and you conserve angular momentum about the pivot. Remove the pivot and the problem changes character completely: now **both** linear momentum and angular momentum survive, and you need them both. Attached to this is a beautiful, examinable idea: the one special impact point where a pivoted body's support feels *nothing* — the centre of percussion, the physics of the bat's sweet spot.

Method

Free body struck by a particle (sticking collision).

1. Conserve linear momentum: $mv = (M + m)V$.
2. Locate the system's centre of mass; it sits a distance $\frac{mL/2}{M + m}$ from the rod's centre toward the struck end.
3. Conserve angular momentum **about the system's centre of mass** — about that one point the final state is pure rotation, $L_f = I_{cm} \omega$, with

$$I_{cm} = \frac{ML^2}{12} + m \left(\frac{ML}{2(M + m)} \right)^2,$$

the rod's own value plus the embedded particle shifted by the parallel-axis theorem.

4. Solve for ω ; recover any point's velocity from V and ω .

Running the algebra of Steps 1-4 for a particle striking the rod's end perpendicular to it:

$$V = \frac{mv}{M + m}, \quad \omega = \frac{6mv(M + m)}{L(M^2 + 5Mm + m^2)}.$$

- Limits: $M \rightarrow \infty$ gives $V \rightarrow 0$, $\omega \rightarrow 0$ (an immovable rod barely responds); $m \rightarrow \infty$ gives $V \rightarrow v$ — the pair sails off together. Both sane.

Centre of percussion. A rod of mass M , length L , pivoted at one end, is dealt a horizontal impulse J at distance x from the pivot. Immediately after:

$$\omega = \frac{Jx}{I_{end}}, \quad V_{cm} = \omega \cdot \frac{L}{2},$$

with $I_{end} = ML^2/3$. The impulse-momentum theorem for the whole rod reads $J - J_{hinge} = MV_{cm}$, where J_{hinge} is whatever impulse the pivot supplies. Demanding $J_{hinge} = 0$:

$$J = M \cdot \frac{JxL}{2I_{end}} \Rightarrow x = \frac{2I_{end}}{ML} = \frac{2}{ML} \cdot \frac{ML^2}{3} = \frac{2L}{3}.$$

Struck at **two-thirds of its length from the pivot**, the rod rotates exactly as a free impulse predicts, and the pivot feels nothing. Struck anywhere else, the hinge jerks — which is why a mis-hit bat stings your hands.

Example D.1 — [JEE Advanced Level]

A rod of mass M and length L lies at rest on a smooth horizontal table. A putty blob of mass $m = M$, moving perpendicular to the rod with speed v , strikes and sticks to one end. Find V , ω , and the post-impact speeds of both ends of the rod.

With $m = M$ the general results collapse pleasantly:

$$V = \frac{Mv}{2M} = \frac{v}{2},$$

and, substituting $m = M$ into the master ω ,

$$\omega = \frac{6Mv(2M)}{L(M^2 + 5M^2 + M^2)} = \frac{12v}{7L}.$$

The system centre of mass lies $\frac{mL}{2(M+m)} = \frac{L}{4}$ from the rod's centre, i.e. $\frac{L}{4}$ from the struck end and $\frac{3L}{4}$ from the far end. End speeds:

$$v_{struck} = V + \omega \cdot \frac{L}{4} = \frac{v}{2} + \frac{12v}{7L} \cdot \frac{L}{4} = \frac{13v}{14},$$

$$v_{far} = V - \omega \cdot \frac{3L}{4} = \frac{v}{2} - \frac{18v}{7 \cdot 2} \cdot \frac{2}{2} = \frac{7v}{14} - \frac{36v}{28} = -\frac{5v}{14}.$$

Check 1: the struck end moves at $\frac{13}{14}v < v$ — the putty slowed down, as an inelastic hit demands.

Check 2 (angular momentum about the rod's original centre): before, $L_i = mv \cdot \frac{L}{2} = \frac{MvL}{2}$. After, the orbital piece about that point contributes $y_c(M+m)V = \frac{L}{4}(2M)\frac{v}{2} = \frac{MvL}{4}$, and the spin contributes $I_{cm}\omega = \frac{7ML^2}{48} \cdot \frac{12v}{7L} = \frac{MvL}{4}$. Sum = $\frac{MvL}{2}$.

Notice the far end momentarily moves *backward*: the rod both chases the putty and spins fast enough that its tail kicks the other way. Students who assume “everything moves forward” miss this every time.

Example D.2 — [JEE Main Level]

A uniform rod of mass 2.0 kg and length 1.5 m is pivoted at one end on a smooth horizontal axis. A ball imparts a horizontal impulse of magnitude $J = 10 \text{ N} \cdot \text{s}$ at the centre of percussion. Find the rod's angular speed just after, and confirm the hinge delivers zero impulse.

The percussion point sits at $x = \frac{2L}{3} = 1.0 \text{ m}$ from the pivot. About the pivot, $I_{end} = \frac{ML^2}{3} = \frac{(2.0)(1.5)^2}{3} = 1.5 \text{ kg} \cdot \text{m}^2$:

$$\omega = \frac{Jx}{I_{end}} = \frac{(10)(1.0)}{1.5} \text{ rad/s} = 6.7 \text{ rad/s}.$$

Impulse-momentum check: $V_{cm} = \omega \frac{L}{2} = 6.7 \times 0.75 = 5.0 \text{ m/s}$, so $MV_{cm} = (2.0)(5.0) = 10 \text{ kg} \cdot \text{m/s} = J$. The entire delivered impulse is accounted for by the rod's momentum — the hinge supplied exactly zero.

Practice — Addendum D

1. Redo Example D.1 with $m = M/2$. Find V and ω .

- For Example D.1 as stated ($m = M$), find the ratio of translational to rotational kinetic energy after impact.
- Find the centre of percussion of a uniform disc pivoted at a point on its rim.
- For Example D.1 ($m = M$), what fraction of the initial kinetic energy survives the collision?
- Conceptual: a cricketer holds the bat near one end. Explain, in one sentence of physics, why a ball caught near the *middle* of the blade hurts the hands less than one caught near the tip.

Answers: 1. $V = \frac{mv}{M+m} = \frac{v}{3}$; $\omega = \frac{6 \cdot \frac{M}{2}v \cdot \frac{3M}{2}}{L \left(M^2 + \frac{5M^2}{2} + \frac{M^2}{4} \right)} = \frac{6v}{5L}$. 2. $K_{trans} = \frac{1}{2}(2M)(v/2)^2 = \frac{Mv^2}{4}$; $K_{rot} = \frac{1}{2} \cdot \frac{7ML^2}{48} \cdot \left(\frac{12v}{7L} \right)^2 = \frac{3Mv^2}{14}$; ratio = $\frac{1/4}{3/14} = 7 : 6$. 3. $x = \frac{I_{rim}}{Md} = \frac{(3/2)MR^2}{MR} = \frac{3R}{2}$ from the pivot. 4. $K_f = \frac{Mv^2}{4} + \frac{3Mv^2}{14} = \frac{13Mv^2}{28}$ versus $K_i = \frac{Mv^2}{2}$: fraction = $\frac{13}{14} \approx 93\%$ — almost everything survives because a free rod offers little resistance. 5. Struck away from the centre of percussion, the unbalanced impulse must be supplied by the hands at the grip — the sting is the hinge impulse.

Addendum E — The Physical Pendulum: When a Rigid Body Oscillates

Nothing in the chapter oscillates. Yet CBSE loves the physical-pendulum time period as a synthesis question, NEET asks it factually, and JEE Main/Advanced pose the “drill a hole, minimise the period” classic nearly every few years. The entire construction is two chapter tools bolted together: torque about a fixed axis, and the parallel-axis theorem.

Method

- Pivot the body** at a point a distance d from its centre of mass; displace it by a small angle θ .
- Write the restoring torque.** Gravity acts at the centre of mass, so

$$\tau = -mgd \sin \theta \approx -mgd \theta$$

for small oscillations — the minus sign is what makes it *restoring*.

- Invoke rotational Newton’s law** about the pivot, $\tau = I_{pivot} \ddot{\theta}$:

$$I_{pivot} \ddot{\theta} = -mgd \theta,$$

which is simple harmonic motion with

$$\omega^2 = \frac{mgd}{I_{pivot}}, \quad T = 2\pi \sqrt{\frac{I_{pivot}}{mgd}}.$$

- Deploy the parallel-axis theorem**, $I_{pivot} = I_{cm} + Md^2 = M(k^2 + d^2)$, where k is the radius of gyration about the centre of mass:

$$T = 2\pi \sqrt{\frac{k^2 + d^2}{gd}}.$$

5. **Recover the simple pendulum** as a check: a point mass on a light string has $k = 0$, $d = \ell$, giving $T = 2\pi\sqrt{\ell/g}$.
6. **Equivalent length:** $\ell_{eq} = \frac{I_{pivot}}{Md} = d + \frac{k^2}{d}$. Minimising over d (set $\frac{d}{dd} \left(d + \frac{k^2}{d}\right) = 1 - \frac{k^2}{d^2} = 0$) gives $d = k$ and

$$T_{min} = 2\pi\sqrt{\frac{2k}{g}}.$$

Any physical pendulum beats the simple pendulum of length d , since $k^2 + d^2 > d^2$ whenever $k \neq 0$.

Torsional pendulum (for completeness). Replace gravity with a wire of torsional constant κ : $\tau = -\kappa\theta$ gives $T = 2\pi\sqrt{I/\kappa}$ — independent of gravity entirely.

Example E.1 — [CBSE Board / JEE Main Level]

A uniform rod of length $L = 0.90$ m pivots about a horizontal axis through one end. Find its small-oscillation period ($g = 10$ m/s²), and compare with a simple pendulum of the same length.

Here $k^2 = \frac{L^2}{12}$ and $d = \frac{L}{2}$:

$$T = 2\pi\sqrt{\frac{\frac{L^2}{12} + \frac{L^2}{4}}{g\frac{L}{2}}} = 2\pi\sqrt{\frac{\frac{L^2}{3}}{\frac{gL}{2}}} = 2\pi\sqrt{\frac{2L}{3g}}.$$

Numerically: $\frac{2L}{3g} = \frac{1.80}{30} \text{ s}^2 = 0.060 \text{ s}^2$, so $T = 2\pi\sqrt{0.060} = 2\pi(0.245) = 1.54 \text{ s}$.

The simple pendulum of length 0.90 m would take $T = 2\pi\sqrt{0.090} = 1.88 \text{ s}$ — the rod, with its mass concentrated nearer the pivot, swings faster. Its equivalent length is $\ell_{eq} = \frac{2L}{3} = 0.60 \text{ m}$, and indeed $2\pi\sqrt{0.60/10} = 1.54 \text{ s}$.

Example E.2 — [JEE Advanced Level]

A hole is drilled in a uniform disc of radius R and the disc is suspended from a smooth pin through the hole, a distance d from the centre. For what d is the oscillation period minimum, and what is that minimum period?

For a disc, $k^2 = \frac{R^2}{2}$. From the Method,

$$T(d) = 2\pi\sqrt{\frac{\frac{R^2}{2} + d^2}{gd}},$$

which is minimised when $\frac{R^2/2 + d^2}{d} = d + \frac{R^2}{2d}$ is minimised, i.e. at

$$d = \frac{R}{\sqrt{2}}.$$

Then $k^2 + d^2 = \frac{R^2}{2} + \frac{R^2}{2} = R^2$ and $gd = \frac{gR}{\sqrt{2}}$:

$$T_{min} = 2\pi\sqrt{\frac{R^2\sqrt{2}}{gR}} = 2\pi\sqrt{\frac{\sqrt{2}R}{g}}.$$

Numeric check, $R = 0.20$ m, $g = 10$ m/s²: the pin sits at $d = 0.141$ m, and

$$T_{min} = 2\pi\sqrt{\frac{1.414 \times 0.20}{10}} \text{ s} = 2\pi\sqrt{0.0283} \text{ s} = 2\pi(0.168) \text{ s} = 1.06 \text{ s}.$$

Direct substitution confirms: $I_{pin} = M(0.02 + 0.02) = 0.04M \text{ kg} \cdot \text{m}^2$, $mgd = M(10)(0.141) = 1.41M \text{ N} \cdot \text{m}$, and $2\pi\sqrt{0.04/1.41} = 1.06 \text{ s}$.

Note the shape of the answer: suspending from the *centre* ($d \rightarrow 0$) or from very far away both stretch T — the minimum lives strictly in between, at $d = k$.

Practice — Addendum E

1. A uniform rod of length 1.0 m oscillates about its end. Find T ($g = 10$ m/s²).
2. A uniform disc of radius 0.20 m oscillates about a pin on its rim. Find T .
3. At what distance from the centre should the pin of Question 2 be placed for minimum period, and what is T_{min} ?
4. A torsional pendulum has $I = 0.20 \text{ kg} \cdot \text{m}^2$ and wire constant $\kappa = 0.80 \text{ N} \cdot \text{m/rad}$. Find T .
5. Conceptual: explain in one line why no physical pendulum can ever swing slower than a simple pendulum bob placed at the same distance d from the pivot.
6. A disc oscillates about a pin at distance $d = R/\sqrt{2}$. State whether its period increases or decreases if the pin is moved to $d = R$.

Answers: 1. $T = 2\pi\sqrt{\frac{2(1.0)}{3(10)}} = 2\pi\sqrt{0.0667} = 2\pi(0.258) = 1.62 \text{ s}$. 2. $T = 2\pi\sqrt{\frac{(3/2)R^2}{gR}} = 2\pi\sqrt{\frac{3(0.20)}{2(10)}} = 2\pi\sqrt{0.030} = 2\pi(0.173) = 1.09 \text{ s}$. 3. Pin at $d = R/\sqrt{2} \approx 0.141 \text{ m}$; $T_{min} = 2\pi\sqrt{\sqrt{2}(0.20)/10} = 1.06 \text{ s}$. 4. $T = 2\pi\sqrt{0.20/0.80} = 2\pi(0.5) = 3.14 \text{ s}$. 5. Because $I_{pivot} = M(k^2 + d^2) > Md^2$, the physical pendulum always has the larger inertia per unit restoring torque, so $T = 2\pi\sqrt{(k^2 + d^2)/gd} \geq 2\pi\sqrt{d/g}$, with equality only for a point mass. 6. It increases: $d = R/\sqrt{2} = k$ is the minimum, and $d = R > k$ sits past it — indeed $2\pi\sqrt{(R^2/2 + R^2)/gR} = 2\pi\sqrt{3R/2g} = 1.09 \text{ s} > 1.06 \text{ s}$ for $R = 0.20 \text{ m}$.

Addendum F — Over a Step: Torque About the Corner and the Minimum Force

A cylinder, roller or log of radius R and mass m rests on a horizontal floor against a step of height h ($h < R$). How hard must you pull to make it mount the step? The chapter's equilibrium machinery answers this in three lines once you see *where to pivot* — and JEE Main returns to exactly this picture again and again.

Method

At the instant the body begins to mount the step, it is pivoting about the **step corner** O — a point on the rim — and the floor has already let go:

$$\text{liftoff condition: } N_{\text{floor}} \rightarrow 0.$$

If the floor normal had not vanished, the body would not be leaving the floor. So at the threshold only **three** forces remain: the weight mg at the centre C , the applied force F , and the reaction at the corner O .

1. **Pivot at the corner.** The corner reaction is unknown in *both* magnitude and direction (it is whatever the balance of forces demands — certainly not “vertical”, and not along OC either). Taking torques about O eliminates it entirely. Never take torques about the floor contact: at liftoff there is no floor contact left.
2. **The one triangle you need.** The corner lies on the rim, so $|OC| = R$. The centre sits at height R and the corner at height h , so the vertical drop from C to O is $R - h$, and the horizontal offset — which is the lever arm of the weight about O — is

$$a = \sqrt{R^2 - (R - h)^2} = \sqrt{2Rh - h^2}.$$

3. **Torque balance.** If the applied force has lever arm d about O ,

$$F \cdot d = mg a = mg \sqrt{2Rh - h^2}.$$

The weight’s torque pins the body onto the corner; the applied torque must just match it at the threshold.

4. **Minimise F by maximising d .** In the standard JEE Main form the pull is **horizontal and applied at the top** of the roller (a level strap or handle at height $2R$), so $d = 2R - h$ and

$$F_{\min} = \frac{mg \sqrt{2Rh - h^2}}{2R - h}.$$

5. **Check the limits.** As $h \rightarrow 0$, $F_{\min} \rightarrow 0$ (no step, no force). As $h \rightarrow R$, $F_{\min} \rightarrow mg$ (the corner rises level with the centre and you end up supporting the whole weight through a unit lever ratio). A formula that fails either limit is wrong, whatever its derivation looked like.

This is statics, not dynamics: $F_{\min}$ is the force that *begins* the climb, quasi-statically. Once the body is rotating about the corner, kinetic energy and the corner’s changing geometry take over — a different problem.

Example F.1 — [JEE Main Level]

A road roller of radius $R = 0.50$ m and mass $m = 15$ kg rests against a step of height $h = 0.10$ m. Find the minimum horizontal force, applied at the top of the roller, that will start it mounting the step. Take $g = 10$ m/s².

Step 1 — the lever arm of the weight. With $|OC| = R$ and vertical drop $R - h = 0.40$ m:

$$a = \sqrt{R^2 - (R - h)^2} = \sqrt{(0.50)^2 - (0.40)^2} \text{ m} = \sqrt{0.09} \text{ m} = 0.30 \text{ m}.$$

(The geometry is a 3-4-5 triangle: the radius to the corner makes an angle θ with the vertical where $\cos \theta = (R - h)/R = 0.8$ and $\sin \theta = a/R = 0.6$.)

Step 2 — the lever arm of the force. A horizontal force at the top acts along a line at height $2R = 1.00$ m; the corner sits at height $h = 0.10$ m, so

$$d = 2R - h = 0.90 \text{ m}.$$

Step 3 — torque balance about the corner:

$$F_{\min} = \frac{mg a}{2R - h} = \frac{(15)(10)(0.30)}{0.90} \text{ N} = \boxed{50 \text{ N}},$$

directed horizontally, at the top of the roller, away from the step.

Check — force balance. The corner reaction must cancel both F horizontally and mg vertically, so $|R_O| = \sqrt{(mg)^2 + F^2} = \sqrt{150^2 + 50^2} \text{ N} \approx 158 \text{ N}$. Its existence is consistent; its direction never entered the calculation, which was the point of pivoting at O . ✓

Example F.2 — [JEE Advanced Level]

Same roller, same step — but now the force applied at the top point may point in any direction. Find the true minimum force and its direction.

Step 1 — the lever arm is now yours to choose. For a force applied at a point P , the largest possible lever arm about O is $|OP|$ itself, attained when $\vec{F} \perp \vec{OP}$.

Step 2 — the corner-to-top distance. The top point T is $2R - h$ above and a to the side of the corner, so

$$|OT| = \sqrt{a^2 + (2R - h)^2} = \sqrt{0.09 + 0.81} \text{ m} = \sqrt{0.90} \text{ m} \approx 0.949 \text{ m},$$

which tidies up to $|OT| = \sqrt{2R(2R - h)}$.

Step 3 — the true minimum:

$$F_{\min} = \frac{mg a}{|OT|} = \frac{(15)(10)(0.30)}{\sqrt{0.90}} \text{ N} \approx 47.4 \text{ N},$$

perpendicular to the corner-to-top line — about 5% cheaper than the 50 N horizontal pull of Example F.1.

Why it is Advanced-level. The mark scheme changes with one word of the question. “Horizontal force at the top” fixes the lever arm at $2R - h$; “minimum force” with direction free fixes it at $|OT|$. Students who memorise one formula answer the other question with it — the exact confusion the examiners are farming. A further variant applies the force at the *axle*:

horizontally, $d = R - h$ and the same roller needs 112.5 N; with direction free there, $d = R$ and 90 N. One torque equation, four different answers, and the geometry decides which is being asked.

Trap note

- **Pivoting at the floor contact.** At the threshold the floor normal is *zero* — that is the condition of the problem, not an approximation. The body is already a lever on the corner; moments about a point it has just left are moments about nothing.
- **Guessing the corner reaction.** It is not vertical (it must also cancel F) and not along the radius OC (the corner is not a smooth contact with the *ground* — it is a rough edge biting the rim). You never need it: moments about O delete it.
- **Minimising with the wrong constraint.** $F_{\min} = mg\sqrt{2Rh - h^2}/(2R - h)$ is the answer to “horizontal force at the top”. Direction-free at the same point, the answer divides by $|OT| = \sqrt{2R(2R - h)}$ instead of $2R - h$ — smaller, since $|OT| > 2R - h$. Read what the question fixes before quoting a “minimum”.
- **Forgetting $h < R$.** The body cannot pivot over a step taller than its centre height by pulling alone: as $h \rightarrow R$ the required force diverges toward mg and beyond that the formula’s geometry ($a = \sqrt{2Rh - h^2}$ real) simply ceases to describe the problem.

Practice — Addendum F

1. A garden roller of radius 0.25 m and mass 40 kg meets a step 0.05 m high. Find the minimum horizontal force applied at the top that starts it climbing ($g = 10 \text{ m/s}^2$).
2. The same roller, but the horizontal force is applied at the axle. Find the minimum force now, and explain in one line why it grew.
3. Show that the top-applied horizontal $F_{\min}$ tends to mg as $h \rightarrow R$, and interpret the limit physically.
4. A log of radius 0.30 m and mass 25 kg rests against a step of height 0.06 m. Show the geometry is again a 3-4-5 triangle, and find the minimum horizontal force applied at the top.
5. For the roller of Question 1, find the absolute minimum force at the top point when its direction is free, and state that direction.

Answers: 1. $a = \sqrt{2(0.25)(0.05) - (0.05)^2} = \sqrt{0.0225} = 0.15 \text{ m}$, so $F_{\min} = \frac{(400)(0.15)}{0.45} = 133 \text{ N}$. 2. $d = R - h = 0.20 \text{ m}$, so $F_{\min} = \frac{(400)(0.15)}{0.20} = 300 \text{ N}$; the lever arm about the corner shrank from $2R - h$ to $R - h$, and F grows in inverse proportion. 3. As $h \rightarrow R$, $a \rightarrow \sqrt{2R^2 - R^2} = R$ and $2R - h \rightarrow R$, so $F_{\min} \rightarrow mg$: the corner stands level with the centre, the pull acts through the same lever ratio as the weight, and you are effectively lifting the whole roller. 4. $a = \sqrt{2(0.30)(0.06) - (0.06)^2} = \sqrt{0.0324} = 0.18 \text{ m} = 0.6R$ and $R - h = 0.24 \text{ m} = 0.8R$ (any $h = R/5$ gives this triangle); $F_{\min} = \frac{(250)(0.18)}{0.54} = 83.3 \text{ N}$. 5. $|OT| = \sqrt{2R(2R - h)} = \sqrt{2(0.25)(0.45)} = 0.474 \text{ m}$, so $F_{\min} = \frac{(400)(0.15)}{0.474} = 127 \text{ N}$, applied perpendicular to the corner-to-top line.

Gravitation

Chapter 7 · Master Study Notes — CBSE / NEET / JEE Main & Advanced

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Part 1 — Newton’s Law of Gravitation

Exam Relevance: This is the foundation stone of the entire chapter. **JEE Main** asks 1–2 direct questions (force calculation, superposition, ratio-scaling). **NEET** loves it for definitions, properties of G , and quick numericals. **JEE Advanced** rarely asks it in isolation but builds every orbital, energy, and field problem on top of it. **CBSE Boards** expect the statement, the vector form, and the properties of G as a 2–3 mark question.

Section 1: Concept Intuition

Look up at night and you’ll see the Moon hanging in the sky, going around the Earth, year after year, never falling down. Now drop your pen — it hits the floor instantly. For thousands of years, people thought these were two *completely different* things: the “heavens” obeyed one set of rules, and “earthly” objects obeyed another. The single most powerful idea in all of physics is that **they are the same thing**. The pen falls and the Moon orbits for one identical reason — and a 23-year-old Isaac Newton was the first to see it clearly.

Here’s the intuition. Imagine you’re standing on a tall hill near Mumbai and you throw a cricket ball horizontally. It curves down and lands some distance away. Throw it harder — it lands further. Now imagine throwing it *impossibly* hard. The ball keeps trying to fall toward the Earth’s centre, but the Earth’s surface is curving away beneath it at exactly the same rate. The ball never gets any closer to the ground — it just keeps “falling around” the Earth forever. **That’s an orbit**. The Moon is doing nothing more than this: it is a cricket ball that was thrown sideways fast enough to keep missing the Earth. The very force pulling your pen down is the force holding the Moon in the sky.

Think about it this way... every chunk of matter in the universe is quietly tugging on every other chunk. You are right now being pulled toward your physics textbook, toward the person sitting next to you, toward the Sun, toward a distant galaxy — all at once. We never notice these pulls because gravity is *staggeringly* weak; the electric force between two protons is about 10^{36} times stronger than their gravitational attraction. Gravity only becomes a big deal when you pile up enormous amounts of mass — like a whole planet. That’s why the Earth’s pull dominates your daily life, but your pull on your textbook is undetectable.

Newton turned this picture into a precise law. The force between two bodies: - gets **stronger** when either mass is bigger (double a mass $\rightarrow$ double the force), and - gets **weaker fast** as they move apart — specifically as the *square* of the distance (move twice as far $\rightarrow$ force drops to one-quarter).

Figure 7.1 Two spherical masses m_1 and m_2 separated by distance r , with two force arrows: F_{12} pointing from m_1 toward m_2 , and F_{21} pointing from m_2 toward m_1 , equal in length, labelled “equal and opposite (Newton’s 3rd law)”. (see next page)

Limiting Conditions & Assumptions — when this simple formula is allowed:

The law in its clean form $F = Gm_1m_2/r^2$ is written for **point masses** — objects so small compared to their separation that their size doesn’t matter. You are *not* allowed to blindly

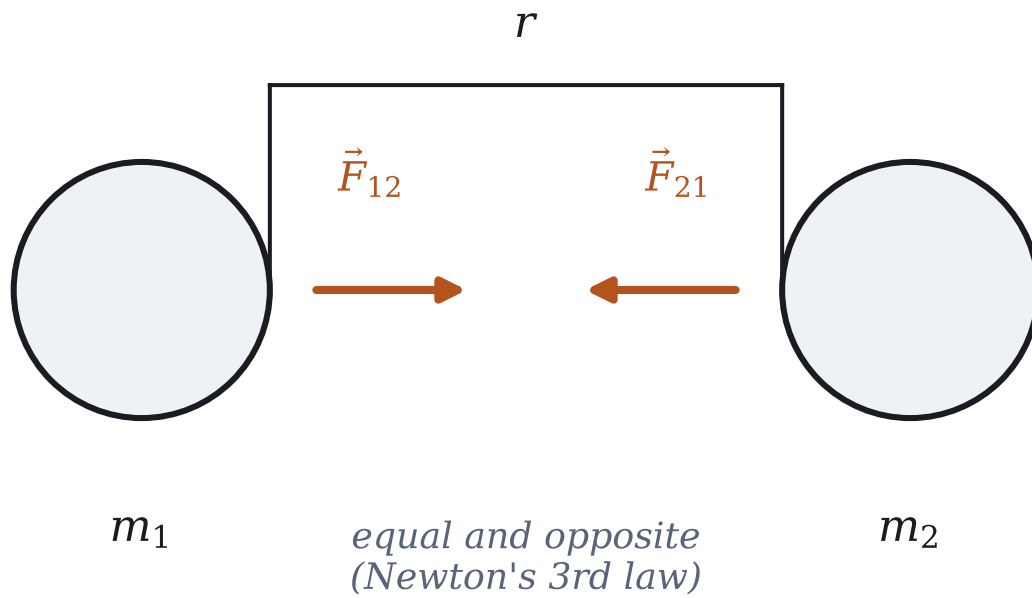


Figure 7.1 Two spherical masses m_1 and m_2 separated by distance r , with two force arrows: F_{12} pointing from m_1 toward m_2 , and F_{21} pointing from m_2 toward m_1 , equal in length, labelled "equal and opposite (Newton's 3rd law)".

use it for two arbitrary extended objects sitting close together. There are exactly three safe situations:

1. **Two point particles** (the original case).
2. **Two uniform spheres** (or spherically symmetric bodies) — here a beautiful result called the **shell theorem** lets you pretend the entire mass of each sphere is squeezed into a point at its centre, and r is the centre-to-centre distance.
3. **A uniform sphere and a point mass.**

For anything else — a rod and a particle, two thin rings, an L-shaped body — you cannot just plug in. You must **add up (integrate)** the pulls from every tiny piece. We'll see exactly how in the worked examples. Misapplying the point-mass formula to a rod is one of the most common ways students lose marks.

Section 2: Key Formulas & Definitions

Newton's Law of Gravitation (statement): Every particle in the universe attracts every other particle with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them, directed along the line joining them.

Scalar form (magnitude):

$$F = \frac{G m_1 m_2}{r^2}$$

Vector form (force on m_1 due to m_2):

$$\vec{F}_{12} = -\frac{G m_1 m_2}{r^2} \hat{r}_{21}$$

where $\hat{r}_{21}$ is the unit vector pointing *from* m_2 *to* m_1 . The negative sign encodes the single most important physical fact: gravity is always **attractive** — the force points opposite to the outward direction, i.e. back toward the other mass.

Symbol	Meaning	SI Unit
F	Gravitational force	newton (N)
m_1, m_2	Masses of the two bodies	kilogram (kg)
r	Distance between centres / particles	metre (m)
$\hat{r}$	Unit vector along the line joining them	dimensionless
G	Universal gravitational constant	$\text{N}\cdot\text{m}^2\text{kg}^{-2}$

The Universal Gravitational Constant G :

$$G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\text{kg}^{-2}$$

- **Dimensional formula:** $[G] = [M^{-1}L^3T^{-2}]$

- It is a **scalar** and a **true universal constant** — its value does not depend on the masses, the medium, time, place, or temperature.
- Physically, G equals the gravitational force (in newtons) between two 1 kg masses placed 1 m apart — which is why it is so tiny.

Principle of Superposition — the net force on one particle is the *vector* sum of the forces from all the others:

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \cdots + \vec{F}_{1n} = \sum_{i=2}^n \frac{G m_1 m_i}{r_{1i}^2} \hat{r}_{1i}$$

Each pairwise force is computed *as if the other masses weren't there*, then all are added head-to-tail.

A subtle but examinable distinction — two kinds of mass. The mass m in Newton's law of gravitation is the **gravitational mass** m_g — it measures how strongly a body participates in gravity (both how hard it pulls and how hard it gets pulled). But mass also appears in Newton's *second* law, $F = m_i a$, where it is the **inertial mass** m_i — a measure of how stubbornly a body resists being accelerated. Logically these are two completely separate properties: there is no obvious reason a body's "gravitational charge" should equal its "resistance to acceleration."

Yet experiment shows they are equal to better than one part in 10^{12} . This is why, when you drop two objects of different mass, they fall with the *same* acceleration g : in $m_g g = m_i a$, the masses cancel only because $m_g = m_i$. Einstein elevated this remarkable coincidence into the **Principle of Equivalence** — that gravitation and acceleration are locally indistinguishable — which became the seed of his General Theory of Relativity. For CBSE/NEET purposes, remember: *the equality of gravitational and inertial mass is an experimental fact, and it is what makes free-fall acceleration independent of the falling body's mass.*

Section 3: CBSE Board Derivation — Establishing the Inverse-Square Law

Newton's law is a *fundamental* law of nature; we don't "derive" it from something deeper. But there is a classic, examinable derivation that shows **why the exponent is exactly 2** — Newton's own reasoning, working backward from Kepler's observations of the planets. This is the derivation CBSE asks for.

The assumptions. Consider a planet of mass m moving in a *circular* orbit of radius r around the Sun (mass M_s) with period T . (Real orbits are ellipses, but circles capture the physics and are what the board expects.)

Step 1 — What keeps the planet on its circle? A body moving in a circle is constantly accelerating *toward the centre* (centripetal acceleration). Some force must supply this. Newton's leap was to say: that force is gravity. The required centripetal force is

$$F = \frac{mv^2}{r}$$

The reasoning: without an inward pull, the planet would fly off along a straight tangent; gravity is what continuously bends its path into a closed loop.

Step 2 — Express speed using the period. In one period T , the planet covers the circumference $2\pi r$:

$$v = \frac{2\pi r}{T}$$

Substituting into Step 1:

$$F = \frac{m}{r} \left(\frac{2\pi r}{T} \right)^2 = \frac{4\pi^2 m r}{T^2}$$

Step 3 — Bring in Kepler’s Third Law. Kepler had found *empirically* (from Tycho Brahe’s data) that $T^2 \propto r^3$, i.e. $T^2 = k r^3$ for some constant k . The physical insight: this is an *observed* fact about the solar system that any correct force law must reproduce. Substituting:

$$F = \frac{4\pi^2 m r}{k r^3} = \frac{4\pi^2}{k} \cdot \frac{m}{r^2}$$

Step 4 — Read off the dependence. Everything except m and r is constant, so

$$F \propto \frac{m}{r^2}$$

The inverse-square law *drops out automatically* — that’s the whole point. The force on the planet is proportional to its own mass m and falls off as $1/r^2$.

Step 5 — Make it symmetric. By Newton’s third law the planet pulls the Sun just as hard, so the force must depend on M_s in exactly the same way it depends on m . The only expression symmetric in both masses and matching the $1/r^2$ behaviour is

$$F = \frac{G M_s m}{r^2}$$

where G bundles up all the constants. Newton then checked this against the Moon’s orbit (the famous “Moon test”) and found spectacular agreement — confirming the law was *universal*, not just solar.

Key Procedure — Solving any superposition problem: 1. **Isolate** the particle on which you want the net force. 2. For each *other* mass, compute the **magnitude** Gm_1m_i/r_{1i}^2 separately. 3. **Draw each force as an arrow** pointing from your particle toward the source (gravity is attractive). 4. **Resolve** into components (x, y) and add component-wise — never add magnitudes directly unless the forces are collinear. 5. Recombine with $F_{\text{net}} = \sqrt{F_x^2 + F_y^2}$ and find the direction.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Two uniform iron spheres of masses 80 kg and 120 kg have their centres 0.50 m apart. Find the gravitational force between them. What happens to this force if the distance is doubled?

Given: $m_1 = 80 \text{ kg}$, $m_2 = 120 \text{ kg}$, $r = 0.50 \text{ m}$, $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\text{kg}^{-2}$.

Since both are uniform spheres, we may treat their masses as concentrated at their centres and use the point-mass formula directly:

$$F = \frac{G m_1 m_2}{r^2} = \frac{(6.67 \times 10^{-11})(80)(120)}{(0.50)^2}$$

$$F = \frac{(6.67 \times 10^{-11})(9600)}{0.25} = \frac{6.403 \times 10^{-7}}{0.25} = 2.56 \times 10^{-6} \text{ N}$$

Effect of doubling r : since $F \propto 1/r^2$, doubling r divides the force by 4:

$$F' = \frac{2.56 \times 10^{-6}}{4} = 6.40 \times 10^{-7} \text{ N}$$

Answer: $F = 2.56 \times 10^{-6} \text{ N}$; doubling the separation reduces it to one-quarter. Notice how minuscule the force is even for 200 kg of iron — gravity is genuinely feeble at human scales.

Example 2 — [NEET Speed Trap]

The gravitational force between two point masses is F_0 . One of the masses is tripled and the distance between them is halved. Without a calculator, find the new force.

The fast way — think in ratios, never re-substitute numbers. Each factor changes the force independently:

$$F \propto \frac{m_1 m_2}{r^2}$$

- Tripling one mass: F multiplied by 3. - Halving the distance: $r \rightarrow r/2$, so $r^2 \rightarrow r^2/4$, which multiplies F by 4.

Combine: $F_{\text{new}} = 3 \times 4 \times F_0 = \boxed{12 F_0}$.

The trap: the most common wrong answer is $3 \times 2 = 6F_0$ — students forget that distance enters as a **square**, so halving r boosts the force by 4, not 2. The whole problem is solvable in your head in under ten seconds *if* you respect the square.

Example 3 — [JEE Main Level] (Gravitation $\times$ Dimensional Analysis)

The orbital speed v of a satellite is believed to depend only on the gravitational constant G , the mass M of the planet, and the orbital radius r . Use dimensional analysis to find how v depends on these quantities.

Assume a power-law relation:

$$v = k G^a M^b r^c$$

where k is a dimensionless constant. Write the dimensions of each quantity:

$$[v] = [LT^{-1}], \quad [G] = [M^{-1}L^3T^{-2}], \quad [M] = [M], \quad [r] = [L]$$

Substitute and collect powers:

$$[M^0 L^1 T^{-1}] = [M^{-1} L^3 T^{-2}]^a [M]^b [L]^c = [M^{b-a} L^{3a+c} T^{-2a}]$$

Equate exponents on both sides: - **Time:** $-2a = -1 \Rightarrow a = \frac{1}{2}$ - **Mass:** $b - a = 0 \Rightarrow b = \frac{1}{2}$
- **Length:** $3a + c = 1 \Rightarrow \frac{3}{2} + c = 1 \Rightarrow c = -\frac{1}{2}$

Therefore

$$v = k\sqrt{\frac{GM}{r}}$$

Answer: $v \propto \sqrt{GM/r}$. The full dynamical theory (equating gravity to centripetal force) gives $k = 1$, but dimensional analysis alone pins down the *form* without solving any equation of motion — that's the cross-topic power being tested here.

Example 4 — [JEE Advanced Level] (Force from an extended body — integration)

A thin uniform rod of mass M and length L lies along a straight line. A point mass m is placed on that same line, a distance a from the **nearer** end of the rod. Find the gravitational force on m . Then verify the answer in the limit $a \gg L$.

Figure 7.2 A horizontal rod of length L with linear density λ . A point mass m sits to the left at distance a from the rod's near end. A small element dx of the rod is shaded at distance x from m , with an arrow showing dF pulling m toward the rod. (see next page)

Why we can't just use GMm/r^2 : different parts of the rod are at different distances from m , so there is no single value of r . We must sum the pull of every element.

Set up the element. Let the rod have linear mass density $\lambda = M/L$. Take a small element of length dx at distance x from m . Its mass is $dm = \lambda dx$, and being a tiny piece, it acts like a point mass. Its pull on m :

$$dF = \frac{G m dm}{x^2} = \frac{G m \lambda dx}{x^2}$$

All these pulls point the same way (along the line, toward the rod), so we can add **magnitudes** directly — no vector resolution needed here.

Integrate over the rod, with x running from a (near end) to $a + L$ (far end):

$$F = \int_a^{a+L} \frac{Gm\lambda}{x^2} dx = Gm\lambda \left[-\frac{1}{x} \right]_a^{a+L} = Gm\lambda \left(\frac{1}{a} - \frac{1}{a+L} \right)$$

$$F = Gm\lambda \cdot \frac{(a+L) - a}{a(a+L)} = Gm \cdot \frac{M}{L} \cdot \frac{L}{a(a+L)}$$

$$\boxed{F = \frac{GMm}{a(a+L)}}$$

Limiting check ($a \gg L$): when the point mass is very far away, L becomes negligible beside a , so $a + L \approx a$ and

$$F \approx \frac{GMm}{a \cdot a} = \frac{GMm}{a^2}$$

which is exactly the point-mass result. This is the hallmark of a correct extended-body answer: **from far away, every object looks like a point mass at its location.** If your limit doesn't collapse to GMm/a^2 , you've made an error.

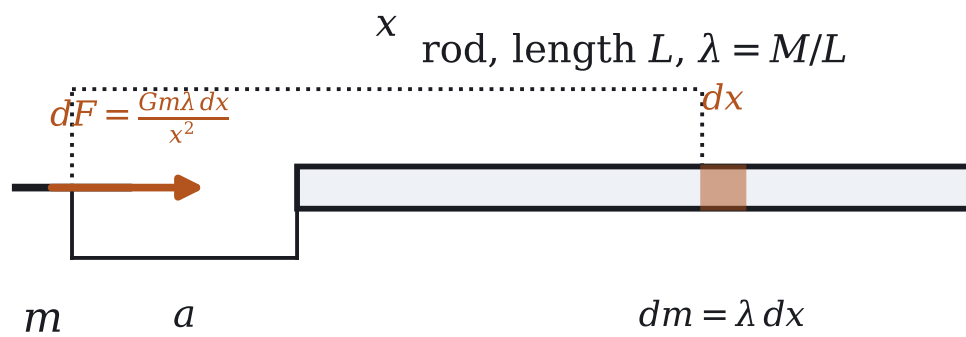


Figure 7.2 A horizontal rod of length L with linear density λ . A point mass m sits to the left at distance a from the rod's near end. A small element dx of the rod is shaded at distance x from m , with an arrow showing dF pulling m toward the rod.

Section 5: Practice Exercises

1. **[CBSE]** Two spheres of mass 5.0 kg each have their centres 20 cm apart. Calculate the gravitational force between them.
2. **[NEET]** The force between two masses is F . If both masses are doubled *and* the distance between them is also doubled, what is the new force?
3. **[JEE Main]** Two point masses, M and $4M$, are fixed 9.0 m apart. At what distance from the smaller mass M , along the line joining them, will a third mass experience zero net gravitational force?
4. **[JEE Main]** Three particles each of mass m sit at the corners of an equilateral triangle of side a . Find the magnitude of the net gravitational force on any one particle due to the other two.
5. **[JEE Advanced]** Two solid spheres made of the *same material* (same density) and of radii r are placed in contact. Show that the gravitational force between them is proportional to r^4 .

Final Answers: (1) 4.2×10^{-8} N (2) F (unchanged — the four-fold mass increase exactly cancels the four-fold distance penalty) (3) 3.0 m from M (and 6.0 m from $4M$) (4) $\sqrt{3} \frac{Gm^2}{a^2}$ (5) $F \propto r^4$ (since each mass $\propto r^3$ and separation $\propto r$).

Section 6: Crack the MCQ

Q1. The gravitational force of attraction between two bodies is F . If the mass of each body is doubled and the distance between them is also doubled, the new force is: **(A)** $4F$ **(B)** $2F$ **(C)** F **(D)** $F/4$

Correct: (C). Doubling *each* mass multiplies the numerator $m_1 m_2$ by $2 \times 2 = 4$. Doubling the distance multiplies r^2 by 4, dividing the force by 4. Net effect: $4/4 = 1$, so the force is unchanged. **(A)** picks up only the mass factor and forgets the distance entirely. **(B)** wrongly treats the mass effect as $\times 2$ (counting only one mass) and ignores distance. **(D)** applies only the distance penalty and forgets that *both* masses grew.

Q2. Which one of the following statements about the gravitational force is **incorrect**? **(A)** It is always attractive. **(B)** It depends on the medium between the bodies. **(C)** It is a conservative force. **(D)** It obeys Newton's third law.

Correct (the false one): (B). Unlike electrostatic or magnetic forces, gravity is **completely independent of the intervening medium** — putting water, lead, or vacuum between two masses changes nothing. **(A)**, **(C)**, and **(D)** are all genuinely true: gravity only ever pulls, the work it does is path-independent (conservative), and the two bodies feel equal-and-opposite pulls. The trap in **(B)** is confusion with Coulomb's law, where a dielectric medium *does* reduce the force by a factor of the permittivity. Students who mentally fuse “inverse-square laws” together fall for this.

Q3. A uniform thin rod of mass M and length L and a point mass m are separated by distance r between the rod's centre and the point mass. The gravitational force between them is **best** described as: **(A)** exactly $\frac{GMm}{r^2}$ **(B)** must be found by integration; it is *not* $\frac{GMm}{r^2}$ **(C)** zero **(D)** $\frac{GMm}{L^2}$

Correct: (B). The point-mass formula is valid only for point masses, spheres, or a sphere-and-point. A rod is none of these — different elements are at different distances and angles, so you must integrate. You cannot pretend the rod's mass sits at its centre. **(A)** is the classic blunder of treating “distance to the centre” as if the rod were a sphere. **(C)** is wrong because there is obviously a non-zero attraction. **(D)** invents a formula with no basis.

Q4. Two particles of equal mass m are released from rest in deep space with only their mutual gravitation acting, initially separated by distance R . The magnitude of the acceleration of *one* particle relative to the *other* is: **(A)** $\frac{Gm}{R^2}$ **(B)** $\frac{Gm}{2R^2}$ **(C)** $\frac{2Gm}{R^2}$ **(D)** cannot be determined

Correct: (C). The force on each particle is $F = Gm^2/R^2$, so each accelerates at $a = F/m = Gm/R^2$ — toward the other. But they accelerate *toward each other*, so the **relative** acceleration is the sum: $a_{\text{rel}} = a + a = 2Gm/R^2$. **(A)** gives the acceleration of a *single* particle in the ground frame, forgetting that the second particle is also rushing in. **(B)** mistakenly halves it (perhaps confusing this with a reduced-mass step done in the wrong direction). **(D)** wrongly assumes missing data — everything needed is given.

Section 7: Common Student Pitfalls & Pro-Tips

- Forgetting the square on distance.** When distance changes by a factor k , the force changes by $1/k^2$, *not* $1/k$. Halving distance quadruples the force. This single slip causes more wrong answers in ratio problems than anything else.
- Misapplying GMm/r^2 to extended bodies.** The clean formula is licensed *only* for point masses, uniform spheres, or a sphere-and-point. For rods, rings, arcs, or irregular shapes you must integrate. Asking yourself “is every part of this body the same distance away?” instantly tells you whether you’re allowed to shortcut.
- Adding force magnitudes instead of vectors.** In superposition, two forces of equal magnitude at 120° do **not** give double — they give the magnitude times $2 \cos(60^\circ) = 1$ in some configurations, or partial cancellation in others. Always resolve into components first.
- Confusing gravity with electrostatics.** Gravity is *always attractive*, *medium-independent*, and has no “shielding.” Don’t import dielectric constants or repulsion into gravitation problems.

Pro-Tip — solve by ratio, not by re-substitution. For any “what happens to F if masses/distance change” question, write $F \propto m_1 m_2 / r^2$ and multiply the separate factors in your head. You’ll finish in seconds and never touch 6.67×10^{-11} .

Sanity-check with the far-field limit. After any extended-body integration, set the distance $\gg$ the body's size. The answer *must* reduce to GMm/r^2 . If it doesn't, find your mistake before moving on.

Section 8: Cheat Sheet Box

NEWTON'S LAW OF GRAVITATION — QUICK RECALL

- **Law:** $F = \frac{Gm_1m_2}{r^2}$, directed along the line joining the masses; **always attractive.**
- **Vector form:** $\vec{F}_{12} = -\frac{Gm_1m_2}{r^2} \hat{r}_{21}$ (the $-$ sign = attraction).
- **Constant:** $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\text{kg}^{-2}$; dimensions $[M^{-1}L^3T^{-2}]$; universal scalar.
- **Force properties:** attractive · central · conservative · medium-independent · obeys Newton's 3rd law.
- **Valid directly for:** point masses, uniform spheres, sphere + point. **Else $\rightarrow$ integrate.**
- **Superposition:** $\vec{F}_{\text{net}} = \sum \vec{F}_i$ (vector sum, each computed independently).

Memory aids: - “Product up, distance squared down” — masses multiply, distance divides as a square. - “Far away, everything's a point” — far-field check for extended bodies $\rightarrow GMm/r^2$. - G dimensions: rearrange $G = Fr^2/m_1m_2 \rightarrow [M^{-1}L^3T^{-2}]$.

Part 2 — Gravitational Field Intensity

Exam Relevance: A JEE powerhouse. **JEE Main** asks 1-2 questions (field due to point masses, shell/sphere inside-vs-outside, the E - r graph). **JEE Advanced** is relentless here — field of rings, the cavity-in-a-sphere trick, and field-potential links appear almost every year. **NEET** keeps it lighter (definition, field inside a shell = 0). **CBSE Boards** expect the definition of field intensity and the fact that surface g is the Earth's field.

Section 1: Concept Intuition

In the very first subtopic we pictured gravity as a direct tug between two masses — “action at a distance.” But there's a deeper, more powerful way to see it, and it's the way modern physics actually thinks. Instead of saying “the Earth reaches across space and grabs the Moon,” we say: **the Earth fills the space around it with a gravitational field, and the Moon simply responds to the field right where it sits.** The field is the middleman. Every mass is a source that “warps” the space around it into a landscape of pulls; any other mass dropped into that landscape feels a force determined purely by the local field.

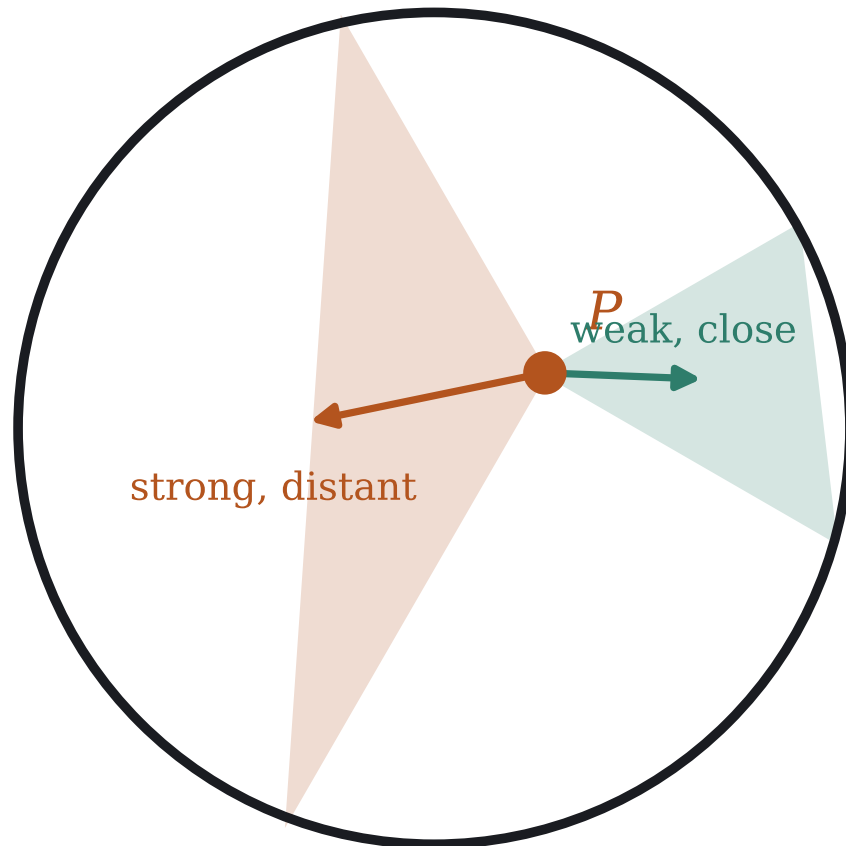
Gravitational field intensity $\vec{E}_g$ at a point is defined as the force that a *unit* mass would feel if placed there — force per kilogram. Put a 1 kg test mass at a point and measure the gravitational force on it; that number (with direction) *is* the field there. This is why the field has the same units as acceleration ($\text{N/kg} = \text{m/s}^2$): the field at a point is literally the acceleration any freely-falling mass would have there.

Think about it this way... you already know one gravitational field intimately — it's called g . The “acceleration due to gravity,” 9.8 m/s^2 , is nothing but the Earth's gravitational field at its surface. When we said earlier that g is a property of the Earth and not of the falling object, we were really saying g is a *field*: it exists at every point around the Earth whether or not anything is there to feel it. Subtopic 02 was secretly your first lesson in fields. Now we generalise: every mass, not just the Earth, sets up such a field, and we can compute it for any shape.

Why bother with the field picture? Because of one magical property of spheres. Newton proved the **shell theorem**: - A uniform spherical shell pulls an *outside* point exactly as if all its mass were a point at the centre. - A uniform spherical shell exerts **zero** net force on any point *inside* it — the pulls from all directions cancel perfectly.

Figure 7.3 *A spherical shell with a point P inside it off-centre. Show that the small nearby patch of shell pulls weakly-but-close while the large far patch pulls strongly-but-distant, and the two effects cancel exactly, giving zero net field inside. This is what makes planets tractable: a solid planet is just a stack of nested shells, so from outside it acts like a point mass, and as you tunnel inside, the shells above you stop contributing. The field picture turns messy geometry into clean, layered simplicity. (see next page)*

We can even *draw* the field. **Gravitational field lines** are imaginary arrows pointing everywhere in the direction a test mass would be pushed — so they always point *inward*, toward the source, like water draining toward a plughole. Their rules mirror those of electric field lines, with one crucial difference: gravity only attracts, so field lines only ever *converge*



*fields cancel exactly
→ zero net field inside*

Figure 7.3 A spherical shell with a point P inside it off-centre. Show that the small nearby patch of shell pulls weakly-but-close while the large far patch pulls strongly-but-distant, and the two effects cancel exactly, giving zero net field inside. This is what makes planets tractable: a solid planet is just a stack of nested shells, so from outside it acts like a point mass, and as you tunnel inside, the shells above you stop contributing. The field picture turns messy geometry into clean, layered simplicity.

onto masses; there are no outward lines, because there is no negative mass to repel. Where the lines crowd together (close to a mass) the field is strong; where they spread out (far away) it is weak. Field lines never cross — at any point the field has one definite direction. For a single point mass they are straight spokes pointing radially inward from every direction; for two masses they curve and merge, and at a null point (where the field vanishes) no line passes at all. Sketching the lines is a fast sanity-check: if you can draw them, you already know the direction of the force everywhere.

Limiting Conditions & Assumptions: The clean results below assume **uniform** mass distributions and **spherical or ring symmetry**. The shell theorem requires a *uniform* shell (or one whose density depends only on radius); a lumpy, non-symmetric shell does *not* give zero interior field. For arbitrary shapes you must integrate the vector contributions, watching the directions — fields are **vectors** and add by components, never as bare numbers.

Section 2: Key Formulas & Definitions

Gravitational field intensity (force per unit test mass):

$$\vec{E}_g = \frac{\vec{F}}{m_0}, \quad |\vec{E}_g| = \frac{GM}{r^2} \text{ (point mass, directed toward } M\text{)}$$

Field due to a point mass / outside any spherical body at distance r :

$$E_g = \frac{GM}{r^2}$$

On the axis of a uniform ring (mass M , radius a) at axial distance x from the centre:

$$E_g = \frac{GMx}{(a^2 + x^2)^{3/2}} \text{ (toward the ring);} \quad V = -\frac{GM}{\sqrt{a^2 + x^2}}$$

The field is **zero at the centre** ($x = 0$) and is maximum at $x = a/\sqrt{2}$.

Uniform spherical shell (mass M , radius R):

$$E_g = \begin{cases} 0 & r < R \text{ (inside)} \\ \frac{GM}{R^2} & r = R \text{ (surface)} \\ \frac{GM}{r^2} & r > R \text{ (outside)} \end{cases}$$

Uniform solid sphere (mass M , radius R):

$$E_g = \begin{cases} \frac{GMr}{R^3} & r \leq R \text{ (inside, } \propto r\text{)} \\ \frac{GM}{r^2} & r \geq R \text{ (outside)} \end{cases}$$

The field rises linearly from 0 at the centre to GM/R^2 at the surface, then falls off as $1/r^2$.

Relation to potential (field is the negative gradient):

$$E_g = -\frac{dV}{dr}$$

Symbol	Meaning	SI Unit
$\vec{E}_g$	Gravitational field intensity	N/kg (= m/s ²)
m_0	Test mass placed at the point	kg
M	Source mass	kg
r	Distance from source / centre	m
a, x	Ring radius; axial distance	m
R	Shell / sphere radius	m

- **Dimensional formula:** $[E_g] = [LT^{-2}]$ — identical to acceleration.
- Field is a **vector**; the net field from many sources is the **vector sum** $\vec{E}_g = \sum \vec{E}_i$.
- At the Earth's surface, $E_g = g$. Field and “acceleration due to gravity” are the same quantity.

Section 3: CBSE / JEE Derivations

(a) Field of a point mass

Place a test mass m_0 at distance r from a point mass M . The force on it is $F = GMm_0/r^2$ (toward M). Dividing by m_0 :

$$E_g = \frac{F}{m_0} = \frac{GM}{r^2}$$

The test mass cancels — the field is a property of the *source and the location*, not of whatever we use to probe it.

(b) Field on the axis of a ring

Take a uniform ring of mass M , radius a . Consider a point P on its axis at distance x from the centre. Split the ring into tiny elements of mass dm . Each element is at distance $\sqrt{a^2 + x^2}$ from P and produces a field $dE = \frac{G dm}{a^2 + x^2}$ pointing from P toward that element.

Figure 7.4 A ring of radius a seen edge-on, with axial point P at distance x . An element dm at the top contributes dE along PA ; its axial component $dE \cdot \cos\theta$ points along the axis, its perpendicular component is cancelled by the diametrically opposite element. (see next page)

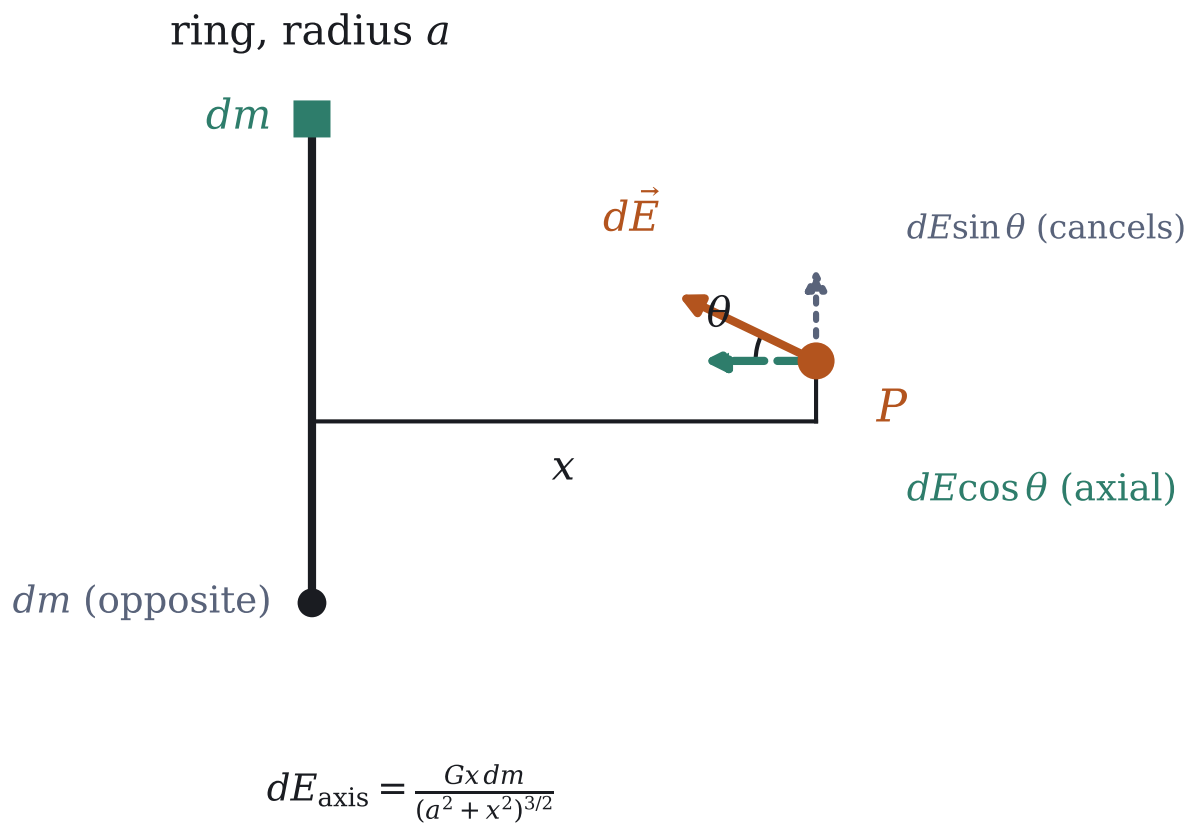


Figure 7.4 A ring of radius a seen edge-on, with axial point P at distance x . An element dm at the top contributes dE along PA ; its axial component $dE \cdot \cos \theta$ points along the axis, its perpendicular component is cancelled by the diametrically opposite element.

By symmetry, the components perpendicular to the axis from diametrically opposite elements **cancel**, leaving only the axial component:

$$dE_{\text{axis}} = dE \cos \theta = \frac{G dm}{a^2 + x^2} \cdot \frac{x}{\sqrt{a^2 + x^2}} = \frac{Gx dm}{(a^2 + x^2)^{3/2}}$$

Since x and a are the same for every element, integrate dm over the whole ring ($\int dm = M$):

$$E_g = \frac{GMx}{(a^2 + x^2)^{3/2}}$$

At the centre $x = 0$, the field is **zero** (every pull is balanced by its opposite). Far away ($x \gg a$), it reduces to GM/x^2 — the ring looks like a point. Setting $dE_g/dx = 0$ gives the maximum field at $x = a/\sqrt{2}$.

(c) Shell theorem and the solid sphere

For a **uniform shell**, a rigorous integration (or Gauss's law for gravity) gives the two clean results stated above: outside, the shell acts as a point mass GM/r^2 ; inside, the field is exactly **zero** everywhere. The physical reason for the zero: at an interior point, the nearby part of the shell is small but close (pull $\propto 1/r^2$ but little mass), while the far part is large but distant — and these *exactly* compensate.

A **solid sphere** is a nest of shells. At a point *inside* at radius r , every shell of radius greater than r contributes zero (you're inside them), and every shell smaller than r acts as a point mass at the centre. So only the mass enclosed within radius r matters:

$$M_{\text{enc}} = M \cdot \frac{r^3}{R^3} \Rightarrow E_g = \frac{GM_{\text{enc}}}{r^2} = \frac{GMr^3/R^3}{r^2} = \boxed{\frac{GMr}{R^3}}$$

The field grows **linearly** from the centre (0) to the surface (GM/R^2), then drops as $1/r^2$ outside.

Figure 7.5 Two graphs of E_g versus r . (1) Shell: $E = 0$ for $r < R$, a vertical jump to GM/R^2 at $r = R$, then a $1/r^2$ decay. (2) Solid sphere: a straight line rising from origin to GM/R^2 at $r = R$, then a $1/r^2$ decay. Mark the surface value GM/R^2 on both. (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Two point masses, 4 kg and 9 kg, are placed 1.0 m apart. Find the point on the line joining them where the net gravitational field is zero.

At the null point, the fields from the two masses are equal in magnitude and opposite in direction (between the masses, the two fields point toward opposite masses, so they can cancel). Let the point be at distance x from the 4 kg mass, so $(1 - x)$ from the 9 kg mass:

$$\frac{G(4)}{x^2} = \frac{G(9)}{(1-x)^2} \Rightarrow \frac{4}{x^2} = \frac{9}{(1-x)^2}$$

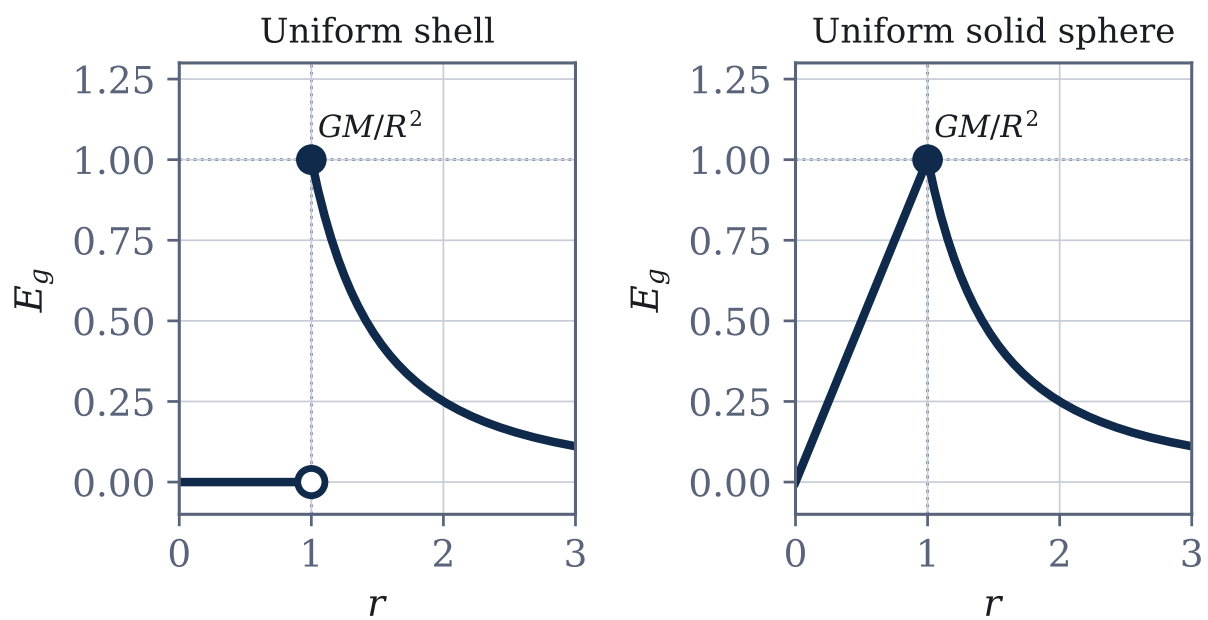


Figure 7.5 Two graphs of E_g versus r . (1) Shell: $E = 0$ for $r < R$, a vertical jump to GM/R^2 at $r = R$, then a $1/r^2$ decay. (2) Solid sphere: a straight line rising from origin to GM/R^2 at $r = R$, then a $1/r^2$ decay. Mark the surface value GM/R^2 on both.

Taking square roots: $\frac{2}{x} = \frac{3}{1-x} \Rightarrow 2(1-x) = 3x \Rightarrow 2 = 5x \Rightarrow x = 0.4 \text{ m}$.

Answer: the net field vanishes 0.4 m from the 4 kg mass (and 0.6 m from the 9 kg mass). Notice the null point lies closer to the *smaller* mass — the field equation balances the smaller pull at a shorter distance.

Example 2 — [NEET Speed Trap]

A hollow spherical shell and a solid sphere have the same mass M and same radius R . A small test mass is placed (i) at the centre and (ii) at the surface of each. Where is the gravitational field zero, and where is it the same for both?

Think in cases, no calculation needed. - **At the centre:** the shell gives $E = 0$ (interior of a shell). The solid sphere also gives $E = 0$ at its centre ($E = GMr/R^3 = 0$ at $r = 0$). So **both are zero at the centre.** - **At the surface ($r = R$):** both give $E = GM/R^2$ — **identical**, because from the surface outward, *any* spherically symmetric body of mass M looks like a point mass.

The trap: students often think the solid sphere has a non-zero field at its centre (it doesn't — by symmetry, equal pulls from all sides cancel) or that the shell and solid sphere differ at the surface (they don't). The fields differ only *inside*: zero throughout the shell, but linearly increasing inside the solid sphere.

Example 3 — [JEE Main Level] (Ring on axis)

A uniform ring of mass M and radius a produces a gravitational field along its axis. At what axial distance is the field maximum, and what is that maximum value?

The axial field is $E_g = \frac{GMx}{(a^2 + x^2)^{3/2}}$. To maximise, differentiate with respect to x and set to zero:

$$\frac{dE_g}{dx} = GM \cdot \frac{(a^2 + x^2)^{3/2} - x \cdot \frac{3}{2}(a^2 + x^2)^{1/2}(2x)}{(a^2 + x^2)^3} = 0$$

The numerator gives $(a^2 + x^2) - 3x^2 = 0 \Rightarrow a^2 = 2x^2 \Rightarrow x = \frac{a}{\sqrt{2}}$.

Substitute back: $a^2 + x^2 = a^2 + \frac{a^2}{2} = \frac{3a^2}{2}$, so $(a^2 + x^2)^{3/2} = \left(\frac{3a^2}{2}\right)^{3/2} = \frac{3\sqrt{3}}{2\sqrt{2}}a^3$.

$$E_{\max} = \frac{GM \cdot a/\sqrt{2}}{\frac{3\sqrt{3}}{2\sqrt{2}}a^3} = \frac{GM}{a^2} \cdot \frac{a/\sqrt{2} \cdot 2\sqrt{2}}{3\sqrt{3}a} = \frac{2GM}{3\sqrt{3}a^2}$$

Answer: maximum field at $x = \frac{a}{\sqrt{2}}$, of magnitude $E_{\max} = \frac{2GM}{3\sqrt{3}a^2} \approx 0.385 \frac{GM}{a^2}$. The field is zero at the centre, climbs to this peak, then decays — a non-monotonic profile that JEE loves to test.

Example 4 — [JEE Advanced Level] (The cavity trick — superposition)

A solid sphere of radius R and uniform density ρ has a spherical cavity of radius $R/2$ scooped out, such that the cavity's surface touches the centre of the original sphere (so the cavity's centre is at distance $R/2$ from the sphere's centre). Show that the gravitational field everywhere inside the cavity is uniform, and find its magnitude.

Figure 7.6 A large sphere of radius R centred at O , with a smaller spherical cavity of radius $R/2$ centred at O' , where $OO' = R/2$. A test point P sits inside the cavity. Show field vectors as the superposition of “full sphere” minus “small sphere filling the cavity.” (see next page)

The idea — superposition. A sphere-with-a-cavity equals a *complete* solid sphere **minus** a small solid sphere that would exactly fill the cavity. The field at any point inside the cavity is the field of the full sphere minus the field of the (imaginary) small sphere.

For a point P inside a uniform solid sphere, the interior field is $\vec{E} = -\frac{4}{3}\pi G\rho \vec{r}$, where $\vec{r}$ is the position vector from that sphere's centre (the minus sign meaning it points toward the centre). This follows from $E = GM_{\text{enc}}/r^2 = G \cdot \frac{4}{3}\pi r^3 \rho / r^2 = \frac{4}{3}\pi G\rho r$.

Let O be the big sphere's centre and O' the cavity's centre, with $\vec{d} = \vec{OO'}$ (magnitude $R/2$). For point P :

$$\vec{E}_{\text{cavity}} = \vec{E}_{\text{full}} - \vec{E}_{\text{small}} = -\frac{4}{3}\pi G\rho \vec{r}_{OP} - \left(-\frac{4}{3}\pi G\rho \vec{r}_{O'P}\right) = -\frac{4}{3}\pi G\rho (\vec{r}_{OP} - \vec{r}_{O'P})$$

But $\vec{r}_{OP} - \vec{r}_{O'P} = \vec{OO'} = \vec{d}$ (the position P cancels!). Therefore:

$$\boxed{\vec{E}_{\text{cavity}} = -\frac{4}{3}\pi G\rho \vec{d}}$$

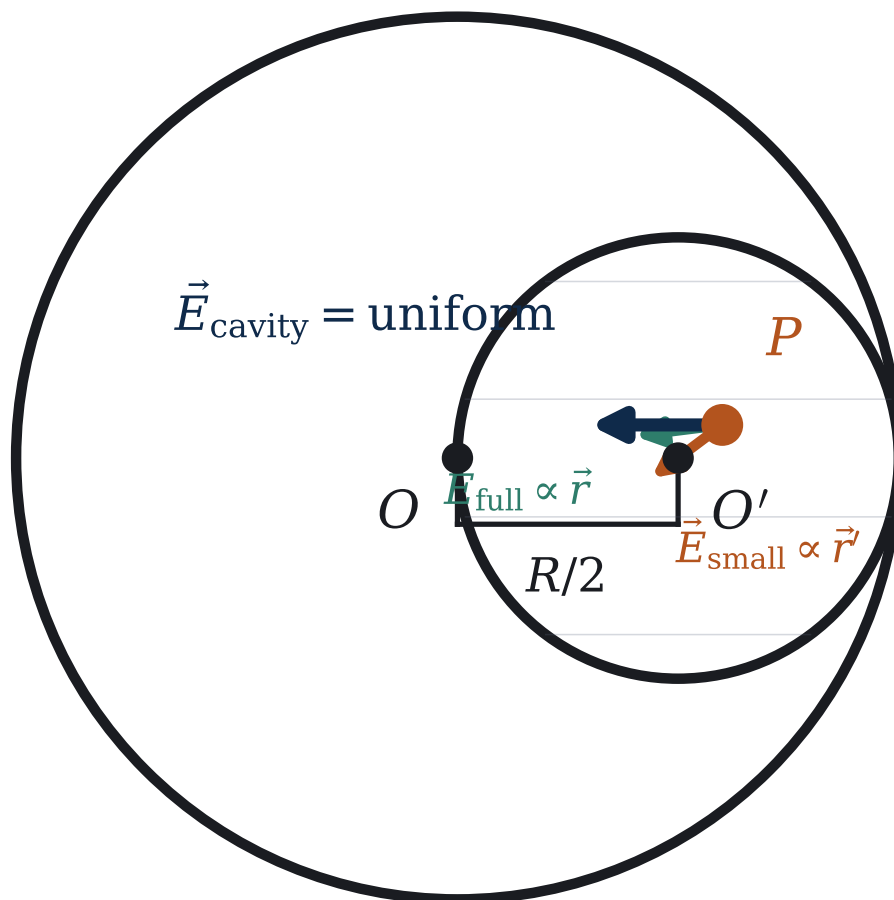
The result is independent of P — the field is the **same everywhere inside the cavity**: a uniform field of magnitude

$$E_{\text{cavity}} = \frac{4}{3}\pi G\rho \cdot \frac{R}{2} = \frac{2}{3}\pi G\rho R$$

directed parallel to the line of centres, toward the big sphere's centre. This astonishing uniformity — every point in the hollow feels the identical pull — is one of the most elegant consequences of the inverse-square law, and a signature JEE Advanced result.

Section 5: Practice Exercises

1. **[CBSE]** Define gravitational field intensity and state its SI unit and dimensional formula. What is the gravitational field at the surface of the Earth called?
2. **[NEET]** The gravitational field at a distance r from the centre of a uniform solid sphere of radius R (with $r < R$) is E . What is the field at distance $2R$ from the centre (outside)? Express it as a multiple of E in terms of r and R .



$$\begin{aligned}\vec{E}_{\text{cavity}} &= \vec{E}_{\text{full}} - \vec{E}_{\text{small}} \\ \vec{E}_{\text{full}} &= -\frac{4}{3}\pi G\rho \vec{r}, \quad \vec{E}_{\text{small}} = -\frac{4}{3}\pi G\rho \vec{r}' \\ \vec{r}' &= \vec{r} - \vec{d}, \quad \vec{d} = O\vec{O}' \Rightarrow \vec{E}_{\text{cavity}} = \frac{4}{3}\pi G\rho \vec{d} = \text{const}\end{aligned}$$

Figure 7.6 A large sphere of radius R centred at O , with a smaller spherical cavity of radius $R/2$ centred at O' , where $OO' = R/2$. A test point P sits inside the cavity. Show field vectors as the superposition of “full sphere” minus “small sphere filling the cavity.”

3. **[JEE Main]** Two concentric uniform spherical shells have masses M_1 and M_2 and radii $R_1 < R_2$. Find the gravitational field at a point distance r from the centre when (i) $r < R_1$, (ii) $R_1 < r < R_2$, (iii) $r > R_2$.
4. **[JEE Main]** A uniform ring of mass M and radius R has a particle of mass m on its axis at distance R from the centre. Find the gravitational force on the particle.
5. **[JEE Advanced]** Inside a uniform solid sphere of radius R and density ρ , a spherical cavity is made whose diameter coincides with a radius of the sphere. Find the gravitational field at the centre of the cavity.

Final Answers: (1) Force per unit mass, $\vec{E}_g = \vec{F}/m_0$; unit N/kg; dimensions $[LT^{-2}]$; it is called g (acceleration due to gravity). (2) Inside: $E = \frac{GMr}{R^3}$; at $2R$: $E' = \frac{GM}{4R^2}$, so $E' = E \cdot \frac{R^3}{r} \cdot \frac{1}{4R^2} \cdot \frac{1}{1} = \frac{ER}{4r} \cdot \frac{R}{R}$ — i.e. $E' = \frac{ER^3}{4R^2r} = \frac{ER}{4r}$. (3) (i) 0; (ii) $\frac{GM_1}{r^2}$; (iii) $\frac{G(M_1 + M_2)}{r^2}$. (4) $F = \frac{GMmR}{(R^2 + R^2)^{3/2}} = \frac{GMm}{2\sqrt{2}R^2}$. (5) $E = \frac{2}{3}\pi G\rho R$ (uniform inside the cavity), directed toward the sphere's centre.

Section 6: Crack the MCQ

- Q1.** The gravitational field intensity inside a uniform spherical shell is: (A) $\frac{GM}{r^2}$ (B) $\frac{GM}{R^2}$ (C) zero (D) $\frac{GMr}{R^3}$

Correct: (C). By the shell theorem, the contributions from all parts of a uniform shell cancel exactly at any interior point, giving zero field everywhere inside. (The *potential* inside is a non-zero constant, but the field — its gradient — is zero.) (A) is the outside field. (B) is the surface field. (D) is the field inside a *solid* sphere, not a shell. Confusing shell with solid sphere is the classic error.

- Q2.** For a uniform solid sphere, the gravitational field is maximum at: (A) the centre (B) the surface (C) infinity (D) halfway to the surface

Correct: (B). Inside, $E = GMr/R^3$ rises linearly with r ; outside, $E = GM/r^2$ falls. The two meet — and the field peaks — exactly at the surface, $E = GM/R^2$. (A) gives zero (symmetry). (C) gives zero (field dies at infinity). (D) gives only half the surface value ($E = GM/2R^2$). The “rise then fall, peak at surface” shape is the key takeaway.

- Q3.** The gravitational field on the axis of a uniform ring is zero at: (A) the centre of the ring (B) $x = a/\sqrt{2}$ (C) $x = a$ (D) nowhere

Correct: (A). At the centre, every element's pull is exactly balanced by the element diametrically opposite, so the net axial field is zero. (The *potential* at the centre is at its most negative there, $-GM/a$ — another reminder that zero field $\neq$ zero potential.) (B) is where the field is *maximum*, not zero. (C) and (D) are wrong — the field is non-zero everywhere on the axis except the centre.

Q4. A particle is placed at the centre of the line joining two equal point masses M separated by distance $2d$. The gravitational field and potential at that midpoint are, respectively: **(A)** zero and zero **(B)** zero and $-\frac{2GM}{d}$ **(C)** $\frac{2GM}{d^2}$ and zero **(D)** zero and $-\frac{GM}{d}$

Correct: (B). The two field vectors at the midpoint are equal and opposite, so the **field cancels to zero**. But potential is a scalar that *adds*: $V = -\frac{GM}{d} - \frac{GM}{d} = -\frac{2GM}{d}$ — non-zero. **(A)** wrongly sets the potential to zero too. **(C)** wrongly adds the fields as scalars. **(D)** forgets one of the two contributions to potential. The lesson once more: **field is a vector (can cancel); potential is a scalar (accumulates)**.

Section 7: Common Student Pitfalls & Pro-Tips

- Confusing shell with solid sphere inside.** Inside a *shell*, field = 0 everywhere. Inside a *solid sphere*, field = GMr/R^3 rises linearly. Drawing the wrong E-r graph is a frequent, avoidable mistake.
- Adding field magnitudes instead of vectors.** Field is a vector; two equal fields in opposite directions cancel. (Potential, a scalar, would *add* in the same situation.) Always resolve into components.
- “Zero field means zero potential.”** No — inside a shell the field is zero but the potential is a constant $-GM/R$; at a ring’s centre the field is zero but the potential is $-GM/a$. Field is the *slope* of potential, not its value.
- Forgetting the only-enclosed-mass rule.** Inside a sphere, only the mass *within* your radius pulls you; the outer shells contribute nothing. This single rule generates the entire interior field.

Pro-Tip — use $\vec{E}_g = -\frac{4}{3}\pi G\rho \vec{r}$ for any uniform-sphere interior. It instantly handles cavity problems by superposition: full sphere minus the sphere that fills the hole. The position vector cancels, leaving a *uniform* field $\frac{4}{3}\pi G\rho \vec{d}$ across the cavity.

Pro-Tip — for symmetric bodies, think “outside = point mass, inside = enclosed mass only.” This converts almost every shell/sphere question into a one-line answer without integration.

Section 8: Cheat Sheet Box

GRAVITATIONAL FIELD INTENSITY — QUICK RECALL

- Definition:** $\vec{E}_g = \frac{\vec{F}}{m_0}$; units $\text{N/kg} = \text{m/s}^2$; dimensions $[LT^{-2}]$. At Earth’s surface, $E_g = g$.

- **Point mass / outside any sphere:** $E_g = \frac{GM}{r^2}$.
- **Ring (axis):** $E_g = \frac{GMx}{(a^2 + x^2)^{3/2}}$; zero at centre, **max at** $x = a/\sqrt{2}$.
- **Shell:** inside = 0; surface = $\frac{GM}{R^2}$; outside = $\frac{GM}{r^2}$.
- **Solid sphere:** inside = $\frac{GMr}{R^3}$ (linear); outside = $\frac{GM}{r^2}$; **peak at the surface.**
- **Field-potential:** $E_g = -\frac{dV}{dr}$. **Field is a vector (can cancel); potential is a scalar (adds).**
- **Cavity in uniform sphere:** field is **uniform**, $\vec{E} = \frac{4}{3}\pi G\rho \vec{d}$ (toward the main centre).

Memory aids: - “Outside, a sphere is a point; inside, only what’s beneath you counts.” - “Shell hollow $\Rightarrow$ field zero; solid $\Rightarrow$ field grows with r .” - “Ring field peaks at $a/\sqrt{2}$.”

Part 3 — Acceleration due to Gravity (g)

Exam Relevance: A perennial favourite — **NEET** and **JEE Main** almost guarantee a question on the variation of g (altitude, depth, or rotation), usually as a ratio or “find the height/depth” numerical. **JEE Advanced** folds it into tunnel-SHM and non-uniform-density problems. **CBSE Boards** expect the derivation of $g = GM/R^2$ and the depth/altitude formulae as 2-3 mark questions, and love conceptual one-liners (“why is g less at the equator?”).

Section 1: Concept Intuition

In the last subtopic we learned that the Earth pulls on every object near it. This subtopic asks a sharper question: *if I let go of something, how fast does that pull make it speed up?* That number — the rate at which a freely falling body gains speed — is the **acceleration due to gravity**, written g . Near the surface of the Earth it’s about 9.8 m/s^2 , meaning a dropped stone gets 9.8 m/s faster every second.

Here is the single most beautiful fact about g : **it does not care what you drop**. A cricket ball and a crumpled sheet of paper, a feather and a hammer — in the absence of air, they all accelerate at exactly the same g and hit the ground together. Astronaut David Scott famously proved this on the Moon with a hammer and a feather. Why? Because the Earth pulls harder on a heavier object (bigger m means bigger force $F = GMm/R^2$), but a heavier object is also harder to accelerate (bigger m resists motion). When you compute $g = F/m$, the object’s mass cancels out completely. The pull and the sluggishness scale together and perfectly balance.

Think about it this way... imagine the Earth offering everyone the same “down escalator.” It doesn’t matter whether you’re a child or a sumo wrestler — the escalator carries you down at the same rate. Your mass changes how *hard* the Earth grips you, but not how *fast* you descend. That’s why g is a property of the *Earth* (its mass and radius), not of the falling object.

This immediately tells us how to compute g . The Earth’s gravitational pull on a surface object equals that object’s weight, mg . Setting Newton’s law equal to weight:

$$\frac{GMm}{R^2} = mg \quad \Rightarrow \quad g = \frac{GM}{R^2}$$

The mass m vanishes — exactly as our intuition demanded. g depends only on the Earth’s mass M and radius R .

But g is not a single frozen number. It changes depending on *where* you stand: - **Go up** (a mountain, an aircraft, orbit) — you’re farther from the Earth’s centre, so g **decreases**. - **Go down** (a mine, a tunnel) — surprisingly, g also **decreases**, because the shell of Earth now *above* you stops pulling you down. - **Move toward the equator** — the Earth’s spin flings you outward slightly, and the Earth’s equatorial bulge puts you farther from the centre, both of which make g **smaller** there than at the poles.

So g is maximum right at the surface near the poles, and tapers off as you climb up, dig down, or drift toward the equator.

One more thing worth knowing — gravity cannot be switched off. You can shield a charge from an electric field by sealing it inside a hollow metal box (a Faraday cage); the field inside vanishes. There is *no such trick for gravity*. You cannot block, absorb, or screen the gravitational influence of nearby matter by wrapping an object in any material or hollow shell, because gravity is independent of the intervening medium and there is no “negative mass” to cancel it. Wherever there is mass, its gravity reaches through everything. This is also why g at any point is numerically identical to the **gravitational field strength** there — g is simply the field the Earth sets up, felt as an acceleration.

Limiting Conditions & Assumptions: The clean formula $g = GM/R^2$ assumes the Earth is a *perfect uniform sphere* and *non-rotating*. The depth formula $g_d = g(1 - d/R)$ further assumes *uniform density* — which is not actually true (the Earth’s core is far denser than its crust), so in reality g initially *rises* slightly as you descend before falling to zero at the centre. The altitude approximation $g_h \approx g(1 - 2h/R)$ is valid **only when** $h \ll R$; for satellites high above the Earth you must use the full inverse-square form.

Section 2: Key Formulas & Definitions

Acceleration due to gravity at the surface:

$$g = \frac{GM}{R^2}$$

In terms of mean density ρ (using $M = \frac{4}{3}\pi R^3 \rho$):

$$g = \frac{4}{3}\pi GR\rho$$

This handy form shows that for planets of the *same density*, $g \propto R$.

At height h above the surface:

$$g_h = \frac{GM}{(R+h)^2} = \frac{g}{\left(1 + \frac{h}{R}\right)^2} \xrightarrow{h \ll R} g \left(1 - \frac{2h}{R}\right)$$

At depth d below the surface (uniform density assumed):

$$g_d = g \left(1 - \frac{d}{R}\right)$$

At the centre ($d = R$), $g_d = 0$.

Effective g at latitude λ , accounting for Earth’s rotation:

$$g_\lambda = g - \omega^2 R \cos^2 \lambda$$

Symbol	Meaning	SI Unit
g	Acceleration due to gravity at surface	m/s^2
M, R	Mass and radius of the planet (Earth)	kg, m
ρ	Mean density of the planet	kg/m^3
h	Height above the surface	m
d	Depth below the surface	m
ω	Angular speed of the planet's rotation	rad/s
λ	Latitude of the location	— (degrees)

- **Dimensional formula of g :** $[LT^{-2}]$.
- g is a **vector**, directed radially toward the planet's centre.
- g is **independent of the mass, size, shape, and composition** of the falling body.
- Numerically, the surface g equals the **gravitational field strength** E_g at the surface — the two are the same physical quantity.

(Notation note for the chapter: here ω is the **rotational/orbital** angular speed of the planet — a different physical meaning from the ω used for SHM angular frequency in Chapter 13, even though the symbol is identical.)

Section 3: CBSE Board Derivations

(a) Surface value: $g = GM/R^2$

Consider a body of mass m resting on the surface of a planet of mass M and radius R . Treating the planet as a uniform sphere, its entire mass acts as if concentrated at the centre (shell theorem), so the gravitational force on the body is

$$F = \frac{GMm}{R^2}$$

This force is the body's weight, and by Newton's second law weight = mg . Equating:

$$mg = \frac{GMm}{R^2} \Rightarrow \boxed{g = \frac{GM}{R^2}}$$

The mass m cancels — confirming that g is set entirely by the planet.

(b) Variation with altitude

At a height h , the body is now a distance $(R + h)$ from the centre. The reasoning behind this step: from outside, the planet still behaves as a point mass at its centre, so we simply replace R by $(R + h)$:

$$g_h = \frac{GM}{(R + h)^2} = \frac{GM}{R^2} \cdot \frac{1}{\left(1 + \frac{h}{R}\right)^2} = g \left(1 + \frac{h}{R}\right)^{-2}$$

When $h \ll R$, we expand using the binomial approximation $(1 + x)^{-2} \approx 1 - 2x$:

$$g_h \approx g \left(1 - \frac{2h}{R}\right)$$

Why the approximation matters: it tells us g falls **twice as fast** per unit height as the naive guess — a 1% rise in altitude drops g by about 2%. But it is only trustworthy for small h .

(c) Variation with depth

This is the subtle one. Go to a depth d , so your distance from the centre is $(R-d)$. Newton's shell theorem says the spherical shell of matter *above* you (thickness d) exerts **zero** net gravitational force. Only the inner sphere of radius $(R-d)$ pulls you. Assuming uniform density ρ :

$$M_{\text{inner}} = \frac{4}{3}\pi(R-d)^3\rho, \quad M = \frac{4}{3}\pi R^3\rho \Rightarrow \frac{M_{\text{inner}}}{M} = \frac{(R-d)^3}{R^3}$$

The acceleration at depth is due only to this inner sphere, acting from distance $(R-d)$:

$$g_d = \frac{GM_{\text{inner}}}{(R-d)^2} = \frac{GM(R-d)^3/R^3}{(R-d)^2} = \frac{GM(R-d)}{R^3} = \frac{GM}{R^2} \cdot \frac{R-d}{R}$$

$$g_d = g \left(1 - \frac{d}{R}\right)$$

At the centre $d = R$, every direction has equal pull, the inner sphere shrinks to nothing, and $g_d = 0$. **A body is weightless at the Earth's centre** — though it still has its full mass and inertia.

(d) Variation due to rotation (effective gravity)

The Earth spins, so a body sitting at latitude λ is actually moving in a circle of radius $r = R \cos \lambda$ about the rotation axis, and needs a centripetal acceleration $\omega^2 R \cos \lambda$ directed toward the axis. Part of the true gravitational pull g must “go toward” supplying this, leaving less to press the body down. Resolving the centripetal requirement along the vertical (radial) direction gives a reduction of $\omega^2 R \cos \lambda \times \cos \lambda$, so the *effective* gravity is:

$$g_\lambda = g - \omega^2 R \cos^2 \lambda$$

- At the **equator** ($\lambda = 0$): maximum reduction, $g_{\text{eq}} = g - \omega^2 R$. - At the **poles** ($\lambda = 90^\circ$): $\cos \lambda = 0$, so rotation has **no effect**; g is largest here.

This is *one* of two reasons the poles feel heavier; the other is the Earth's flattened **shape** — the polar radius is about 21 km smaller, and since $g \propto 1/R^2$, the smaller polar radius further boosts g at the poles.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A certain planet has mass $M = 4.8 \times 10^{24} \text{ kg}$ and radius $R = 4.0 \times 10^6 \text{ m}$. (a) Calculate the acceleration due to gravity on its surface. (b) A body weighs 200 N at the surface. Assuming the planet to be a uniform sphere, how much will it weigh at a depth equal to half the radius?

Part (a). Given $M = 4.8 \times 10^{24} \text{ kg}$, $R = 4.0 \times 10^6 \text{ m}$, $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\text{kg}^{-2}$.

$$g = \frac{GM}{R^2} = \frac{(6.67 \times 10^{-11})(4.8 \times 10^{24})}{(4.0 \times 10^6)^2} = \frac{3.20 \times 10^{14}}{1.6 \times 10^{13}} = 20 \text{ m/s}^2$$

Part (b). Given surface weight $W = mg = 200 \text{ N}$, depth $d = R/2$.

Use the depth formula:

$$g_d = g \left(1 - \frac{d}{R}\right) = g \left(1 - \frac{R/2}{R}\right) = g \left(1 - \frac{1}{2}\right) = \frac{g}{2}$$

The new weight is

$$W_d = mg_d = m \cdot \frac{g}{2} = \frac{mg}{2} = \frac{200}{2} = 100 \text{ N}$$

Answer: (a) $g = 20 \text{ m/s}^2$; (b) 100 N . Note the body's mass is unchanged — only its *weight* halves, because the effective g halves at depth $R/2$.

Example 2 — [NEET Speed Trap]

At what height above the Earth's surface does the acceleration due to gravity become one-fourth of its value at the surface? (R = radius of the Earth.)

We need $g_h = \frac{g}{4}$. **Use the exact inverse-square form — not the approximation:**

$$g_h = g \cdot \frac{R^2}{(R+h)^2} = \frac{g}{4} \Rightarrow \frac{R^2}{(R+h)^2} = \frac{1}{4} \Rightarrow \frac{R}{R+h} = \frac{1}{2}$$

$$R + h = 2R \Rightarrow \boxed{h = R}$$

The trap: a hurried student reaches for $g_h \approx g(1 - 2h/R)$ and writes $g(1 - 2h/R) = g/4$, giving $h = 3R/8$. **This is wrong** — the linear approximation is valid *only* for $h \ll R$, but here $h = R$ is enormous. Whenever the required change in g is large (a factor of $\frac{1}{4}$, $\frac{1}{9}$, etc.), you *must* use the full $(R+h)^2$ form. Recognising “is h small?” before choosing a formula is the entire skill being tested.

Example 3 — [JEE Main Level] (Rotation × circular motion)

How fast would the Earth need to rotate for objects at the equator to become weightless? Estimate the corresponding length of a day. Take $R = 6.4 \times 10^6 \text{ m}$ and $g = 9.8 \text{ m/s}^2$.

At the equator ($\lambda = 0$), effective gravity is $g_{\text{eq}} = g - \omega^2 R$. Weightlessness means $g_{\text{eq}} = 0$:

$$g - \omega^2 R = 0 \Rightarrow \omega = \sqrt{\frac{g}{R}} = \sqrt{\frac{9.8}{6.4 \times 10^6}} = \sqrt{1.531 \times 10^{-6}} = 1.24 \times 10^{-3} \text{ rad/s}$$

The new length of a day would be

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{R}{g}} = 2\pi \sqrt{\frac{6.4 \times 10^6}{9.8}} = 2\pi(808.3) \approx 5078 \text{ s} \approx 1.41 \text{ hours}$$

Answer: $\omega \approx 1.24 \times 10^{-3}$ rad/s, giving a day of roughly 84.6 minutes. The Earth currently rotates about 17 times slower than this, which is why the equatorial reduction in g (only about 0.034 m/s^2) is tiny in everyday life. Keep this “ ≈ 84 **minute**” number in mind — it reappears below.

Example 4 — [JEE Advanced Level] (Cross-chapter link: Gravitation → SHM)

A narrow, frictionless tunnel is drilled straight through the centre of the Earth, from one side to the other. A small ball is dropped into it from the surface. Show that the ball performs simple harmonic motion, and find its time period. Treat the Earth as a uniform non-rotating sphere; $R = 6.4 \times 10^6 \text{ m}$, $g = 9.8 \text{ m/s}^2$.

Figure 7.7 A circle representing the Earth with a vertical tunnel through the centre. A small ball at distance r from the centre, with an arrow showing the net gravitational force pointing back toward the centre, labelled $F = -(mg/R)r$. (see next page)

Step 1 — Find the force at distance r from the centre. Inside the Earth, at a distance r from the centre, only the sphere of radius r pulls the ball (shell theorem; the outer shell contributes nothing). From the depth result, the local gravitational acceleration at distance r from the centre is

$$g(r) = g \cdot \frac{r}{R}$$

directed *toward* the centre. The force on the ball of mass m is therefore

$$F = -\frac{mg}{R} r$$

where the minus sign shows the force always points back toward the centre (the equilibrium point), opposing the displacement.

Step 2 — Recognise the SHM signature. This has exactly the form of Hooke’s law, $F = -k_{\text{eff}} r$, with an effective “spring constant”

$$k_{\text{eff}} = \frac{mg}{R}$$

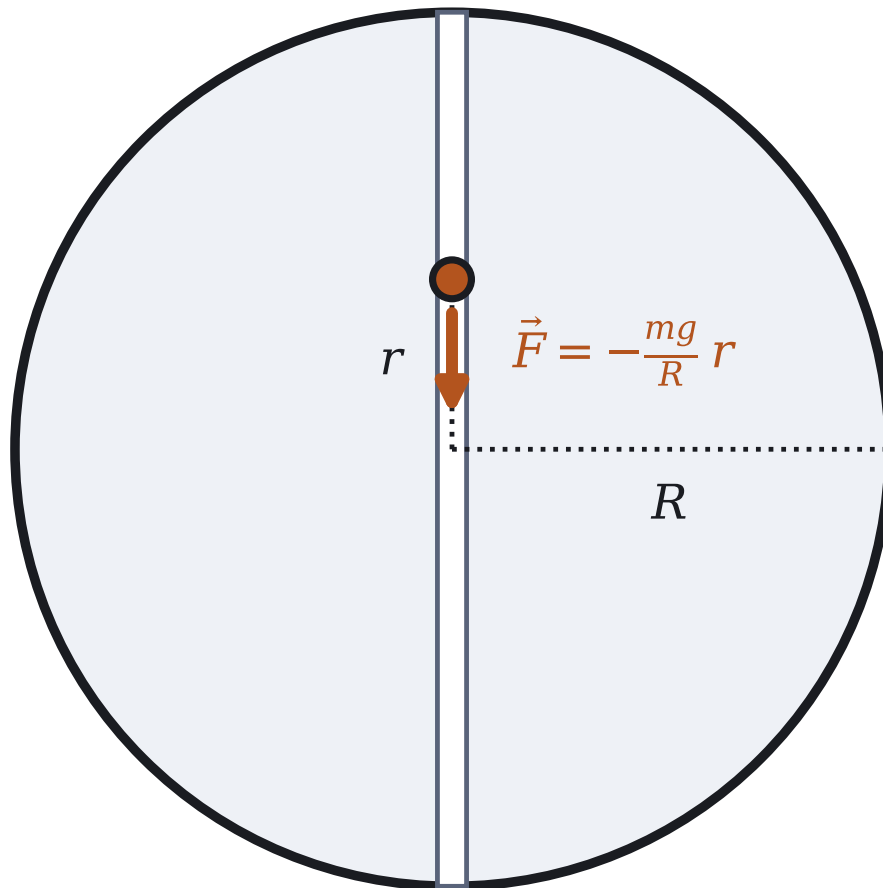
Any force linearly proportional to displacement and directed back toward equilibrium produces simple harmonic motion (recall Chapter 13). The angular frequency of the SHM is

$$\omega_{\text{SHM}} = \sqrt{\frac{k_{\text{eff}}}{m}} = \sqrt{\frac{g}{R}}$$

Step 3 — Time period.

$$T = \frac{2\pi}{\omega_{\text{SHM}}} = 2\pi \sqrt{\frac{R}{g}} = 2\pi \sqrt{\frac{6.4 \times 10^6}{9.8}} \approx 5078 \text{ s} \approx 84.6 \text{ minutes}$$

Answer: The ball oscillates back and forth through the Earth in SHM with a period of about 84 minutes — the *same* magic number as the weightless-equator day in Example 3, and (it turns out) the same as the period of a satellite skimming the Earth’s surface. All three are governed by $2\pi\sqrt{R/g}$. This is exactly the kind of deep unification JEE Advanced rewards: a gravitation problem that becomes an oscillations problem, with the surface value g acting as the bridge.



$$g(r) = g \frac{r}{R} \text{ toward centre}$$

$$\text{SHM: } F = -k_{\text{eff}}r, \quad k_{\text{eff}} = mg/R, \quad T = 2\pi\sqrt{R/g} \approx 84 \text{ min}$$

Figure 7.7 A circle representing the Earth with a vertical tunnel through the centre. A small ball at distance r from the centre, with an arrow showing the net gravitational force pointing back toward the centre, labelled $F = -(mg/R)r$.

Section 5: Practice Exercises

1. **[CBSE]** The radius of planet A is twice that of planet B, but both have the same mass. Find the ratio of the acceleration due to gravity on their surfaces, $g_A : g_B$.
2. **[NEET]** At what depth below the Earth's surface (in terms of R) does g fall to 40% of its surface value? Assume uniform density.
3. **[JEE Main]** The change in g at a height h above the surface equals the change in g at a depth d below the surface. If both h and d are much smaller than R , find the relation between h and d .
4. **[JEE Main]** Two planets have mean densities in the ratio 1 : 2 and radii in the ratio 3 : 1. Find the ratio of the accelerations due to gravity on their surfaces.
5. **[JEE Advanced]** A planet has the same mean density as the Earth but its surface gravity is 4 times that of the Earth. If the Earth's radius is R , find the radius of the planet and the ratio of their masses.

Final Answers: (1) 1 : 4 (since $g \propto 1/R^2$ at fixed mass) (2) $d = 0.6 R$ (3) $d = 2h$ (4) $g \propto \rho R$, so ratio = $(1 \times 3) : (2 \times 1) = 3 : 2$ (5) $g \propto \rho R$ at fixed density $\Rightarrow$ radius = $4R$; mass ratio (planet : Earth) = $\rho(4R)^3 : \rho R^3 = 64 : 1$.

Section 6: Crack the MCQ

Q1. The acceleration due to gravity at a height $h = R$ above the Earth's surface (where R is the Earth's radius) is what fraction of its surface value? (A) $\frac{g}{2}$ (B) $\frac{g}{4}$ (C) $\frac{g}{3}$ (D) zero

Correct: (B). Use $g_h = g R^2 / (R+h)^2$. With $h = R$, the denominator is $(2R)^2 = 4R^2$, so $g_h = g/4$. (A) comes from wrongly using a $1/(R+h)$ (inverse-first-power) dependence. (C) has no valid basis. (D) confuses "high altitude" with "infinitely far"; g only reaches zero at infinity, never at a finite height.

Q2. A body is taken from the Earth's surface to the top of a high mountain, and then down into a deep mine. Compared with the surface value, the acceleration due to gravity: (A) increases on the mountain, decreases in the mine (B) decreases on the mountain, increases in the mine (C) decreases in both cases (D) increases in both cases

Correct: (C). g is **maximum at the surface**; it decreases whether you go up (greater distance from centre, $g_h < g$) or down (less mass pulling you, $g_d < g$). This surface-maximum behaviour is the conceptual heart of the subtopic. (A) and (B) assume up and down do opposite things — the classic misconception that "deeper means stronger gravity." (D) wrongly thinks altitude increases g .

Q3. Two stars are dropped — sorry, two identical balls are dropped, one near the Earth's pole and one near the equator. Ignoring air resistance, the ball near the **pole** will: (A) accelerate slightly faster (B) accelerate slightly slower (C) accelerate at exactly the same rate (D) not accelerate at all

Correct: (A). Effective g is **largest at the poles** for two reinforcing reasons: (i) no rotational reduction there ($\cos \lambda = 0$), and (ii) the polar radius is smaller, so $g \propto 1/R^2$ is larger. So the polar ball speeds up slightly faster. **(B)** reverses the effect. **(C)** forgets that the Earth is neither perfectly spherical nor non-rotating. **(D)** is nonsense — gravity acts everywhere.

Q4. If both the mass and the radius of the Earth were somehow halved, the acceleration due to gravity at the surface would: **(A)** remain unchanged **(B)** double **(C)** halve **(D)** become one-fourth

Correct: (B). $g = GM/R^2 \propto M/R^2$. Halving M multiplies g by $\frac{1}{2}$; halving R multiplies g by $1/(\frac{1}{2})^2 = 4$. Net effect: $\frac{1}{2} \times 4 = 2$, so g **doubles**. **(A)** counts only the mass change or wrongly cancels the two effects. **(C)** counts only the mass. **(D)** ignores the mass change and squares only the radius effect in the wrong direction. The lesson: mass enters linearly, radius enters as a square — handle them separately.

Section 7: Common Student Pitfalls & Pro-Tips

- Using the altitude approximation when h is not small.** $g_h \approx g(1 - 2h/R)$ is a *small-height* shortcut only. For satellites or any h comparable to R , use the exact $g_h = gR^2/(R + h)^2$. Mixing these up is the single biggest source of wrong answers here.
- Swapping the altitude and depth formulae.** Altitude has the factor $(1 - \frac{2h}{R})$ (a **2** appears); depth has $(1 - \frac{d}{R})$ (**no 2**). Memorise: “*height has the 2, depth does not.*” This is exactly why, for equal small changes, $d = 2h$.
- Thinking g depends on the falling body.** It does not — g is fixed by the planet (M , R , density, rotation). A heavy stone and a light one fall identically. Weight (mg) depends on the body; g does not.
- Crediting only rotation (or only shape) for the equator being “lighter.”** Both effects act together: the equatorial bulge (larger R) and the rotational centrifugal reduction. Naming only one in a board answer loses marks.

Pro-Tip — reason in ratios with $g \propto M/R^2$ (or $g \propto \rho R$). For “compare two planets” problems, never plug in 6.67×10^{-11} . Just track how M , R , or ρ scale. If density is fixed, the cleanest tool is $g \propto R$; if mass is fixed, use $g \propto 1/R^2$.

Pro-Tip — memorise the magic period. $T = 2\pi\sqrt{R/g} \approx 84$ min is the period of (i) a ball oscillating in a tunnel through the Earth, (ii) a satellite skimming the surface, and (iii) the day-length for equatorial weightlessness. Recognising it instantly cracks several “famous-number” questions.

Section 8: Cheat Sheet Box

ACCELERATION DUE TO GRAVITY — QUICK RECALL

- **Surface:** $g = \frac{GM}{R^2} = \frac{4}{3}\pi GR\rho$. Dimensions $[LT^{-2}]$; **independent of the falling body.**
- **Altitude:** $g_h = \frac{gR^2}{(R+h)^2}$; for $h \ll R$: $g_h \approx g \left(1 - \frac{2h}{R}\right)$.
- **Depth:** $g_d = g \left(1 - \frac{d}{R}\right)$; **zero at the centre.**
- **Rotation:** $g_\lambda = g - \omega^2 R \cos^2 \lambda$; minimum at equator, unchanged at poles.
- **g is maximum at the surface** — decreases going both up and down.
- **Poles feel heavier** (no rotation effect + smaller polar radius); equator feels lighter.

Memory aids: - “Height has the 2, depth does not” $\rightarrow \left(1 - \frac{2h}{R}\right)$ vs $\left(1 - \frac{d}{R}\right)$; hence equal small changes give $d = 2h$. - “Same density? g tracks R . Same mass? g fights R^2 .” - “ $2\pi\sqrt{R/g} \approx 84 \text{ min}$ ” $\rightarrow$ tunnel-SHM = surface satellite = weightless day.

Part 4 — Gravitational Potential and Energy

Exam Relevance: A heavily weighted, concept-rich subtopic. **JEE Main** reliably asks 1–2 questions (potential energy of a system, escape velocity, work to move a mass). **NEET** favours escape-velocity ratios and the sign of U . **JEE Advanced** pushes into self-energy and potential inside solid spheres/shells. **CBSE Boards** expect the derivation of $U = -\frac{GMm}{r}$, the definition of potential V , and the escape-velocity derivation — classic 3–5 mark territory.

Section 1: Concept Intuition

So far we’ve talked about the gravitational *force* — a push-and-pull you feel right now. This subtopic is about gravitational **energy**: the energy *stored* in the arrangement of masses, just because of where they sit relative to each other. Lift a stone and you’ve stored energy in the stone–Earth system; let go and that stored energy turns back into motion. Gravitational potential energy (U) is the bookkeeping of this stored energy, and gravitational potential (V) is that same energy measured *per kilogram* — a property of the location itself, independent of what you place there.

Here is the idea that trips up almost everyone, so let’s meet it head-on: **gravitational potential energy is negative**. Not “sometimes negative” — for two attracting masses it is *always* negative (with the standard convention). Why?

Think about it this way... imagine the region around the Earth as a deep valley, and infinity as the flat ground far away at the rim. We *agree* to call the energy “zero” when the two masses are infinitely far apart — they barely feel each other, nothing is stored, a clean starting line. Now bring a mass in from infinity toward the Earth. Gravity is attractive, so it *pulls the mass inward* and does **positive work** — like a ball rolling *down* into the valley, speeding up on its own. Since the system gives up energy (it could have done work on you), its stored energy drops *below* the zero line. The deeper you go (smaller r), the more negative U becomes. Climbing back out toward infinity costs you energy and brings U back up toward zero.

Figure 7.8 *A graph of U versus r . The curve $U = -GMm/r$ is a hyperbola lying entirely below the horizontal axis: it plunges toward negative infinity as r approaches 0, and rises asymptotically toward $U = 0$ as r increases to infinity. Mark the point that U is always negative and approaches 0 from below. So the negative sign is not a mistake — it is the signature of an attractive force and a “bound” system. A more negative U means the two masses are more tightly bound; you’d need to supply energy to pull them apart. This single picture explains nearly everything in this subtopic: why a satellite’s total energy is negative (it’s trapped in the valley), why escaping requires a minimum launch speed (enough energy to climb all the way to the rim), and why V is negative everywhere around a mass. (see next page)*

How deep can the valley get? If you keep the same mass but shrink the body — squeezing M into a smaller and smaller radius R — the well at the surface gets steeper and the escape speed $\sqrt{2GM/R}$ climbs. Push this to the extreme and you reach a radius where the escape velocity would equal the speed of light itself. At that point *nothing*, not even light, can climb out of the valley: the object has become a **black hole**. A black hole is

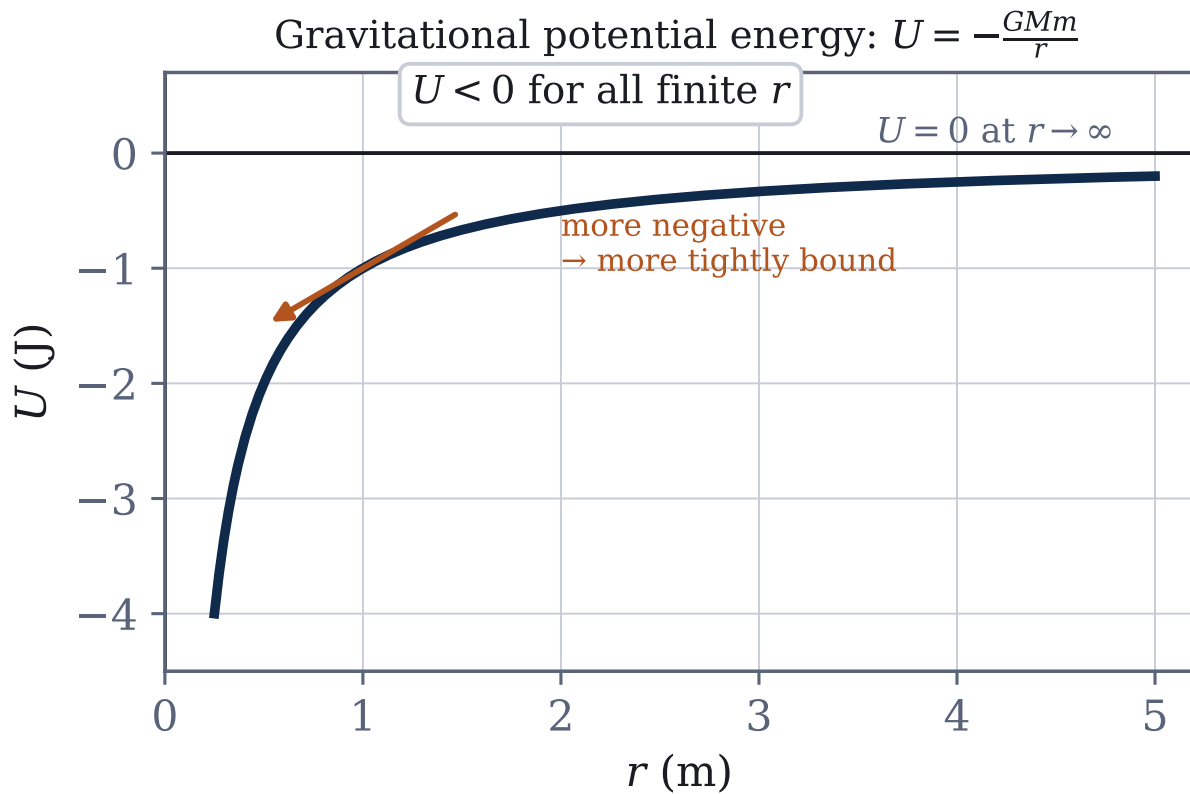


Figure 7.8 A graph of U versus r . The curve $U = -GMm/r$ is a hyperbola lying entirely below the horizontal axis: it plunges toward negative infinity as r approaches 0, and rises asymptotically toward $U = 0$ as r increases to infinity. Mark the point that U is always negative and approaches 0 from below. So the negative sign is not a mistake — it is the signature of an attractive force and a “bound” system. A more negative U means the two masses are more tightly bound; you’d need to supply energy to pull them apart. This single picture explains nearly everything in this subtopic: why a satellite’s total energy is negative (it’s trapped in the valley), why escaping requires a minimum launch speed (enough energy to climb all the way to the rim), and why V is negative everywhere around a mass.

simply gravity's valley made infinitely steep — the same physics of bound, negative-energy systems we are studying here, taken to its limit.

A second key idea: **potential energy belongs to the system, not to one body**. It makes no sense to say “the stone has U ” — the energy lives in the *configuration* of stone-and-Earth together. We loosely say “the stone's PE” only as shorthand.

Limiting Conditions & Assumptions: The formula $U = -GMm/r$ takes the **zero of energy at infinity** — change that reference and the *number* changes (though differences ΔU never do). The familiar school formula $U = mgh$ is only the *change* in PE for *small* heights ($h \ll R$), where gravity is nearly constant; for large heights you must use the exact $-GMm/r$ form. And $V = -GM/r$ is for a point mass or a body you may treat as a point (outside a sphere); inside extended bodies the expressions differ.

Section 2: Key Formulas & Definitions

Gravitational potential energy of two particles (zero taken at infinity):

$$U = -\frac{Gm_1m_2}{r}$$

System of n particles — sum over all $\frac{n(n-1)}{2}$ distinct pairs:

$$U = -G \sum_{\text{pairs } (i,j)} \frac{m_i m_j}{r_{ij}}$$

Of a mass m at the Earth's surface: $U = -\frac{GMm}{R}$.

Change in PE on raising mass m by height h :

$$\Delta U = \frac{mgh}{1 + \frac{h}{R}} \xrightarrow{h \ll R} mgh$$

Gravitational potential (PE per unit mass; the work done per unit mass to bring a test mass from infinity to that point):

$$V = \frac{U}{m} = -\frac{GM}{r}, \quad U = mV$$

Potential from several masses — *scalar* (algebraic) sum:

$$V = -G \sum_i \frac{m_i}{r_i}$$

Relation between field and potential (field is the negative gradient of potential):

$$E_g = -\frac{dV}{dr}, \quad \Delta V = -\int \vec{E}_g \cdot d\vec{r}$$

Escape velocity (minimum surface speed to reach infinity):

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$$

Symbol	Meaning	SI Unit
U	Gravitational potential energy (of the system)	joule (J)
V	Gravitational potential (per unit mass)	J/kg
E_g	Gravitational field strength (= g numerically)	m/s ² or N/kg
v_e	Escape velocity	m/s
M, m	Source mass and test/body mass	kg
r, R	Separation / distance from centre; planet radius	m

- **Dimensional formula:** $[U] = [ML^2T^{-2}]$; $[V] = [L^2T^{-2}]$.
- U and V are **scalars** — add them algebraically, never as vectors.
- Both are **negative** for attractive gravity (zero reference at infinity), most negative when masses are closest.
- At the surface: $V_{\text{surface}} = -\frac{GM}{R} = -gR$.
- For a **thin shell**: outside $V = -GM/r$; inside $V = -GM/R$ (constant, equal to the surface value). For a **solid sphere**: outside $V = -GM/r$; at the centre $V = -\frac{3GM}{2R}$.

Section 3: CBSE Board Derivations

(a) Potential energy of a two-particle system: $U = -GMm/r$

Fix a mass M at the origin and bring a test mass m in from infinity to a distance r . By definition, the potential energy at r is the **negative of the work done by the gravitational force** in this process:

$$U(r) = -W_{\text{grav}} = -\int_{\infty}^r \vec{F} \cdot d\vec{x}$$

The gravitational force on m at distance x has magnitude GMm/x^2 and points *inward* (toward M), i.e. in the direction of decreasing x . Writing it along the radial axis:

$$\vec{F} \cdot d\vec{x} = -\frac{GMm}{x^2} dx$$

so the work done by gravity as m moves inward from ∞ to r is

$$W_{\text{grav}} = \int_{\infty}^r \left(-\frac{GMm}{x^2} \right) dx = -GMm \left[-\frac{1}{x} \right]_{\infty}^r = -GMm \left(-\frac{1}{r} + 0 \right) = +\frac{GMm}{r}$$

The positive sign confirms gravity *aids* the inward motion (the mass speeds up). Therefore

$$U(r) = -W_{\text{grav}} = -\frac{GMm}{r}$$

Why it's negative: because gravity did positive work, the system released energy, dropping below the infinity reference of zero.

(b) Gravitational potential: $V = -GM/r$

Potential is simply PE per unit test mass. Dividing the result above by m :

$$V(r) = \frac{U}{m} = -\frac{GM}{r}$$

This is the work *done against gravity*, per kilogram, to carry a small test mass from infinity to the point — a property of the field at that location, independent of the test mass.

(c) Field-potential relation: $E_g = -dV/dr$

Since $V = -GM/r$, differentiate:

$$\frac{dV}{dr} = +\frac{GM}{r^2} \Rightarrow E_g = -\frac{dV}{dr} = -\frac{GM}{r^2}$$

The minus sign places the field pointing toward M (decreasing r), with magnitude GM/r^2 — exactly the g of the previous subtopic. *The field always points “downhill,” from high potential to low.*

(d) Escape velocity: $v_e = \sqrt{2GM/R}$

To “escape” means to reach infinity (where $U = 0$) with *at least* zero kinetic energy. Launch a body of mass m from the surface with speed v_e . The reasoning: gravity is conservative, so total mechanical energy is conserved. Set total energy at the surface equal to the minimum total energy at infinity (which is $0 + 0$):

$$\underbrace{\frac{1}{2}mv_e^2}_{\text{KE at surface}} + \underbrace{\left(-\frac{GMm}{R}\right)}_{\text{PE at surface}} = 0$$

Solving:

$$\frac{1}{2}mv_e^2 = \frac{GMm}{R} \Rightarrow v_e = \sqrt{\frac{2GM}{R}}$$

Using $GM = gR^2$, this simplifies to $v_e = \sqrt{2gR}$. The mass m cancels — **escape velocity is independent of the escaping body's mass**, and (since energy is a scalar) independent of the launch *direction* too. For the Earth, $v_e \approx 11.2$ km/s.

(e) Potential due to a solid sphere, and self-energy (a note for JEE)

A uniform solid sphere of mass M , radius R , behaves *outside* exactly like a point mass: $V = -GM/r$ for $r \geq R$. *Inside* ($r < R$) the potential is **not** $-GM/r$; carrying the test mass through the matter gives

$$V_{\text{inside}}(r) = -\frac{GM}{2R^3} (3R^2 - r^2)$$

At the surface ($r = R$) this reduces to $-GM/R$ (continuous, as it must be), and at the **centre** ($r = 0$) it reaches its most negative value $V_{\text{centre}} = -\frac{3GM}{2R} = \frac{3}{2}V_{\text{surface}}$. So potential is deepest at the centre, even though the *field* there is zero — a vivid reminder that “zero field” means “flat potential,” not “zero potential.”

Finally, the **gravitational self-energy** of a body is the energy released in assembling it from dust scattered at infinity (equivalently, the work needed to disperse it). Building a uniform sphere shell by shell and integrating gives

$$U_{\text{self}} = -\frac{3GM^2}{5R}$$

This is the energy that powers the gravitational collapse of gas clouds into stars — a single number capturing how tightly a whole body binds itself together.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Three point masses of 2 kg, 4 kg, and 6 kg are placed at the vertices of an equilateral triangle of side 3.0 m. Calculate the gravitational potential energy of the system. ($G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\text{kg}^{-2}$.)

Given: $m_1 = 2$, $m_2 = 4$, $m_3 = 6$ kg; all pair separations $r = 3.0$ m.

A 3-particle system has $\frac{3(3-1)}{2} = 3$ pairs. Total PE:

$$\begin{aligned} U &= -G \left[\frac{m_1 m_2}{r} + \frac{m_1 m_3}{r} + \frac{m_2 m_3}{r} \right] = -\frac{G}{r} (m_1 m_2 + m_1 m_3 + m_2 m_3) \\ U &= -\frac{6.67 \times 10^{-11}}{3.0} [(2)(4) + (2)(6) + (4)(6)] = -\frac{6.67 \times 10^{-11}}{3.0} (8 + 12 + 24) \\ U &= -\frac{6.67 \times 10^{-11}}{3.0} (44) = -9.78 \times 10^{-10} \text{ J} \end{aligned}$$

Answer: $U = -9.78 \times 10^{-10} \text{ J}$. The value is negative — energy would have to be *supplied* to disperse the three masses back to infinity.

Example 2 — [NEET Speed Trap]

The escape velocity from the Earth's surface is 11.2 km/s. Find the escape velocity from a planet whose mass is twice that of the Earth and whose radius is half that of the Earth.

Reason in ratios — don't recompute. Since $v_e = \sqrt{2GM/R}$, we have $v_e \propto \sqrt{\frac{M}{R}}$. Comparing the planet to the Earth:

$$\frac{v_{e,\text{planet}}}{v_{e,\text{Earth}}} = \sqrt{\frac{M_p/R_p}{M_E/R_E}} = \sqrt{\frac{2}{1/2}} = \sqrt{4} = 2$$

$$v_{e,\text{planet}} = 2 \times 11.2 = 22.4 \text{ km/s}$$

The trap: students often forget the *square root* and report $v_e \times \frac{M}{R} = \times 4 = 44.8 \text{ km/s}$, or confuse escape velocity with orbital velocity. Escape velocity scales as $\sqrt{M/R}$ — apply the factor *inside* the root, then take the root once.

Example 3 — [JEE Main Level] (Exact PE vs the mgh shortcut)

A body of mass m is lifted slowly from the Earth's surface to a height equal to the Earth's radius R . Find the work done against gravity. Compare it with the naive estimate mgh . (Take $GM = gR^2$.)

Lifting slowly means kinetic energy doesn't change, so the work done against gravity equals the *change* in potential energy:

$$W = \Delta U = U_f - U_i = \left(-\frac{GMm}{R+h}\right) - \left(-\frac{GMm}{R}\right)$$

With $h = R$, the final separation is $R + R = 2R$:

$$W = -\frac{GMm}{2R} + \frac{GMm}{R} = \frac{GMm}{R} \left(1 - \frac{1}{2}\right) = \frac{GMm}{2R}$$

Substituting $GM = gR^2$:

$$W = \frac{gR^2 \cdot m}{2R} = \frac{1}{2}mgR$$

Now compare with the school shortcut $mgh = mg \cdot R = mgR$. The shortcut gives **double** the true answer! The reason: mgh assumes g stays constant, but over a height as large as R , gravity weakens substantially, so less work is actually needed.

Answer: $W = \frac{1}{2}mgR$; the mgh formula overestimates it by a factor of 2 here. Use mgh only when $h \ll R$.

Example 4 — [JEE Advanced Level] (Energy + momentum conservation)

Two point masses m and $2m$ are released from rest in deep space, initially separated by a distance d , with only their mutual gravitation acting. Find their relative velocity of approach when the separation has shrunk to r .

Figure 7.9 Two masses m and $2m$ on a line, separation d initially. Arrows showing them moving toward each other with speeds v_1 and v_2 respectively when separation is r ; v_1 for mass m points right, v_2 for mass $2m$ points left. (see next page)

Two conservation laws are in play. No external force acts, so total momentum is conserved; and gravity is conservative, so total mechanical energy is conserved.

Momentum (started from rest, so total momentum stays zero):

$$mv_1 = 2m v_2 \Rightarrow v_1 = 2v_2$$

where v_1, v_2 are the speeds of m and $2m$ as they rush toward each other.

Energy conservation (initial KE = 0):

$$-\frac{G(m)(2m)}{d} = \frac{1}{2}mv_1^2 + \frac{1}{2}(2m)v_2^2 - \frac{G(m)(2m)}{r}$$

Substitute $v_1 = 2v_2$ into the kinetic terms:

$$\frac{1}{2}m(2v_2)^2 + \frac{1}{2}(2m)v_2^2 = 2mv_2^2 + mv_2^2 = 3mv_2^2$$

So:

$$3mv_2^2 = \frac{2Gm^2}{r} - \frac{2Gm^2}{d} = 2Gm^2 \left(\frac{1}{r} - \frac{1}{d} \right)$$
$$v_2^2 = \frac{2Gm}{3} \left(\frac{1}{r} - \frac{1}{d} \right), \quad v_1 = 2v_2$$

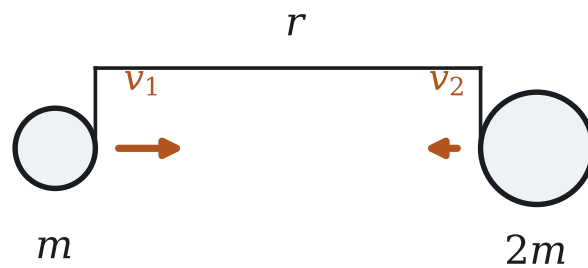
Relative velocity of approach = $v_1 + v_2 = 3v_2$ (they move toward each other):

$$v_{\text{rel}} = 3v_2 = 3\sqrt{\frac{2Gm}{3} \left(\frac{1}{r} - \frac{1}{d} \right)} = \sqrt{6Gm \left(\frac{1}{r} - \frac{1}{d} \right)}$$

Answer: $v_{\text{rel}} = \sqrt{6Gm \left(\frac{1}{r} - \frac{1}{d} \right)}$. Note that you *cannot* simply give each mass half the released energy — the lighter mass moves faster and carries more KE. Momentum conservation is what correctly splits the energy, which is the entire point of the problem.

Section 5: Practice Exercises

1. **[CBSE]** Calculate the gravitational potential at the surface of the Earth. Take $M = 6.0 \times 10^{24}$ kg, $R = 6.4 \times 10^6$ m.



$$mv_1 = 2mv_2 \Rightarrow v_1 = 2v_2$$

Released from rest at separation d ; find relative velocity at separation r

Figure 7.9 *Two masses m and $2m$ on a line, separation d initially. Arrows showing them moving toward each other with speeds v_1 and v_2 respectively when separation is r ; v_1 for mass m points right, v_2 for mass $2m$ points left.*

2. **[NEET]** A body is projected vertically upward from the Earth's surface with a speed equal to half the escape velocity. In terms of R , find the maximum height it reaches. (Ignore air resistance and Earth's rotation.)
3. **[JEE Main]** Four point masses, each of mass m , are placed at the corners of a square of side a . Find the total gravitational potential energy of the system.
4. **[JEE Main]** How much energy must be supplied to a satellite of mass m resting on the Earth's surface to send it completely out of the Earth's gravitational field? Express your answer in terms of m , g , and R .
5. **[JEE Advanced]** A particle is released from rest at a great distance (effectively infinity) from a planet of mass M and radius R . Find the speed with which it strikes the planet's surface, and show that it equals the escape velocity.

Final Answers: (1) $V = -GM/R \approx -6.25 \times 10^7 \text{ J/kg}$ (2) $h = R/3$ (3) $U = -\frac{Gm^2}{a} (4 + \sqrt{2})$ (4 side-pairs at a , 2 diagonal-pairs at $a\sqrt{2}$) (4) $E = \frac{GMm}{R} = mgR$ (the magnitude of the surface PE) (5) $v = \sqrt{2GM/R} = v_e$ (a body falling from infinity arrives at exactly escape speed).

Section 6: Crack the MCQ

Q1. The gravitational potential energy of a system of two masses is: (A) always positive (B) always negative (C) zero (D) positive or negative depending on the masses

Correct: (B). With the zero of energy taken at infinity, gravity (being attractive) does positive work as the masses approach, so the stored energy falls below zero. $U = -Gm_1m_2/r$ is negative for all finite r . (A) confuses gravity with the *elastic* PE of a spring (always positive). (C) is true only in the limit $r \rightarrow \infty$. (D) would be correct for *electric* PE (which can be either sign), but gravity is only ever attractive.

Q2. The escape velocity of a body from the Earth's surface does **not** depend on: (A) the mass of the Earth (B) the radius of the Earth (C) the mass of the body (D) the value of G

Correct: (C). $v_e = \sqrt{2GM/R}$ contains G , M , and R — but **not** the mass of the escaping body, which cancels out in the energy equation. A pebble and a spaceship need the same escape speed. (A), (B), (D) all appear explicitly in the formula, so escape velocity does depend on them. The standout misconception this question targets: “heavier objects are harder to launch, so they need more speed” — false; they need more *energy*, but the same *speed*.

Q3. The gravitational potential at a point is $-5.0 \times 10^6 \text{ J/kg}$. The work done in bringing a 3.0 kg mass from infinity to that point is: (A) $+1.5 \times 10^7 \text{ J}$ (B) $-1.5 \times 10^7 \text{ J}$ (C) $-5.0 \times 10^6 \text{ J}$ (D) zero

Correct: (B). Work done by the gravitational field $= U = mV = (3.0)(-5.0 \times 10^6) = -1.5 \times 10^7 \text{ J}$. The negative value means the field *does* positive work pulling the mass in, so the system's energy decreases — equivalently, an external agent must do $+1.5 \times 10^7 \text{ J}$ to *remove* it. (A) flips the sign (that's the *external* work, not

the field's). **(C)** forgets to multiply by the mass — that's V , not U . **(D)** wrongly assumes “infinity to a point” gives no work.

Q4. Inside a uniform thin spherical shell, the gravitational field is zero everywhere. The gravitational potential inside the shell is therefore: **(A)** zero everywhere inside **(B)** constant and equal to the surface value $-GM/R$ **(C)** increasing toward the centre **(D)** $-GM/r$ at distance r from the centre

Correct: (B). Field is the negative *gradient* of potential ($E_g = -dV/dr$). Zero field means the potential does not change with position — it is **constant** inside. By continuity at the surface, that constant equals the surface value $V = -GM/R$. **(A)** is the classic error of assuming “zero field = zero potential.” Zero field means *unchanging* potential, not zero potential — they are different things. **(C)** wrongly imagines potential varying inside. **(D)** applies the *outside* formula where it isn't valid.

Section 7: Common Student Pitfalls & Pro-Tips

- Dropping the negative sign.** $U = -GMm/r$ and $V = -GM/r$ are negative. Writing them positive is the single most frequent error and reverses the physics (it would imply repulsion). Always carry the minus.
- Using mgh for large heights.** mgh is only the *change* in PE for $h \ll R$. For heights comparable to R , use $\Delta U = -GMm \left(\frac{1}{r_f} - \frac{1}{r_i} \right)$. (Recall Example 3: mgh overshoot by a factor of 2.)
- Confusing potential V with potential energy U .** V is per unit mass (J/kg); $U = mV$ is the total energy (J). Mixing their units or forgetting the factor of m loses marks instantly.
- “Zero field means zero potential.”** False. Inside a shell the field is zero but the potential is a non-zero constant ($-GM/R$). Field is the *slope* of potential, not the potential itself.

Pro-Tip — potential adds like a scalar, force adds like a vector. For several masses, just *add the numbers* $V = -G \sum m_i/r_i$ (with signs) — no components, no angles. This makes potential problems far faster than the corresponding force problems. Compute V first, then get energy as $U = mV$ or speed via energy conservation.

Pro-Tip — remember the $\sqrt{2}$ bridge. Escape velocity $v_e = \sqrt{2gR}$ is exactly $\sqrt{2}$ times the orbital velocity of a surface-skimming satellite ($v_o = \sqrt{gR}$). If a question gives you one, you instantly know the other. For Earth: $v_o \approx 7.9$ km/s, $v_e \approx 11.2$ km/s.

Section 8: Cheat Sheet Box

GRAVITATIONAL POTENTIAL & ENERGY — QUICK RECALL

- **PE of two masses:** $U = -\frac{Gm_1m_2}{r}$ (negative; zero at infinity). System: sum over all $\frac{n(n-1)}{2}$ pairs.
- **Potential:** $V = \frac{U}{m} = -\frac{GM}{r}$, units J/kg; **scalar**, add algebraically.
- **Surface values:** $U = -\frac{GMm}{R}$, $V = -\frac{GM}{R} = -gR$.
- **Small-height PE change:** $\Delta U \approx mgh$ (only for $h \ll R$).
- **Field-potential:** $E_g = -\frac{dV}{dr}$ (field = downhill slope of potential).
- **Escape velocity:** $v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2 \text{ km/s}$; independent of body's mass & launch angle.
- **Inside a shell:** field = 0, potential = $-GM/R$ (constant). Solid sphere centre: $V = -\frac{3GM}{2R}$.

Memory aids: - "Attractive $\Rightarrow$ negative; closer $\Rightarrow$ more negative." - " V is per-kilo, $U = mV$ is the whole bill." - " $v_e = \sqrt{2}v_o$ " — escape is root-two times surface orbit. - "Zero field $\neq$ zero potential" — field is the slope, not the value.

Part 5 — Kepler's Laws and Satellites

Exam Relevance: The chapter's grand finale and a guaranteed scorer. **JEE Main** asks 2–3 questions (Kepler's third law ratios, orbital velocity, satellite energy, geostationary parameters). **NEET** loves Kepler's laws, weightlessness, and T - r scaling. **JEE Advanced** sets elliptical-orbit problems using angular-momentum and energy conservation. **CBSE Boards** expect all three Kepler laws, the orbital-velocity and time-period derivations, and the energy of a satellite — frequently 3–5 marks.

Section 1: Concept Intuition

Picture standing on top of an impossibly tall mountain and firing a cannonball horizontally. Fire it gently — it arcs down and lands nearby. Fire it harder — it lands much farther, the Earth's surface curving away a little beneath its path. Fire it hard enough, and something magical happens: the ball falls toward the Earth at *exactly* the rate the round Earth curves away under it. It keeps falling forever and never lands. **That is an orbit.** A satellite — the Moon, the ISS, a communications relay — is not “escaping” gravity or “balancing” it. It is in *perpetual free fall*, endlessly falling around the Earth. This single image (Newton's own thought experiment) is the key to the entire subtopic.

Figure 7.10 *A tall mountain on a curved Earth. Several projectile trajectories launched horizontally at increasing speeds: short arcs landing nearby, then longer arcs, then one that curves all the way around and becomes a closed circular orbit. Label the fastest closed path “orbit — perpetual free fall.” Long before Newton, Johannes Kepler stared at decades of Tycho Brahe's planetary data and distilled three crisp rules that the planets obey. He didn't know why — that took Newton's gravity — but the rules were exact: (see next page)*

- **Orbits are ellipses**, not perfect circles, with the Sun sitting at one focus (not the centre).
- **Planets sweep equal areas in equal times.** This is why a planet races when it's near the Sun and dawdles when it's far. **Think about it this way...** imagine the planet tethered to the Sun by an elastic string; as it swings in close, the string is short, so it must whip around quickly to sweep out the same “pizza slice” of area each second; far away, with a long string, it crawls. This “equal areas” rule is secretly the **conservation of angular momentum** in disguise — the planet speeds up exactly enough to keep L constant.
- **Bigger orbits mean longer years**, and precisely so: the *square* of the period grows as the *cube* of the orbit size. Mercury whips around the Sun in 88 days; distant Pluto takes 248 years.

The beautiful part: Newton's inverse-square gravity *reproduces all three laws automatically*. We'll derive the third one for circular orbits in a few lines.

Two ideas that constantly confuse students, settled up front:

Weightlessness is not the absence of gravity. An astronaut floating in the ISS is *not* beyond Earth's pull — gravity there is nearly as strong as on the ground. They float because

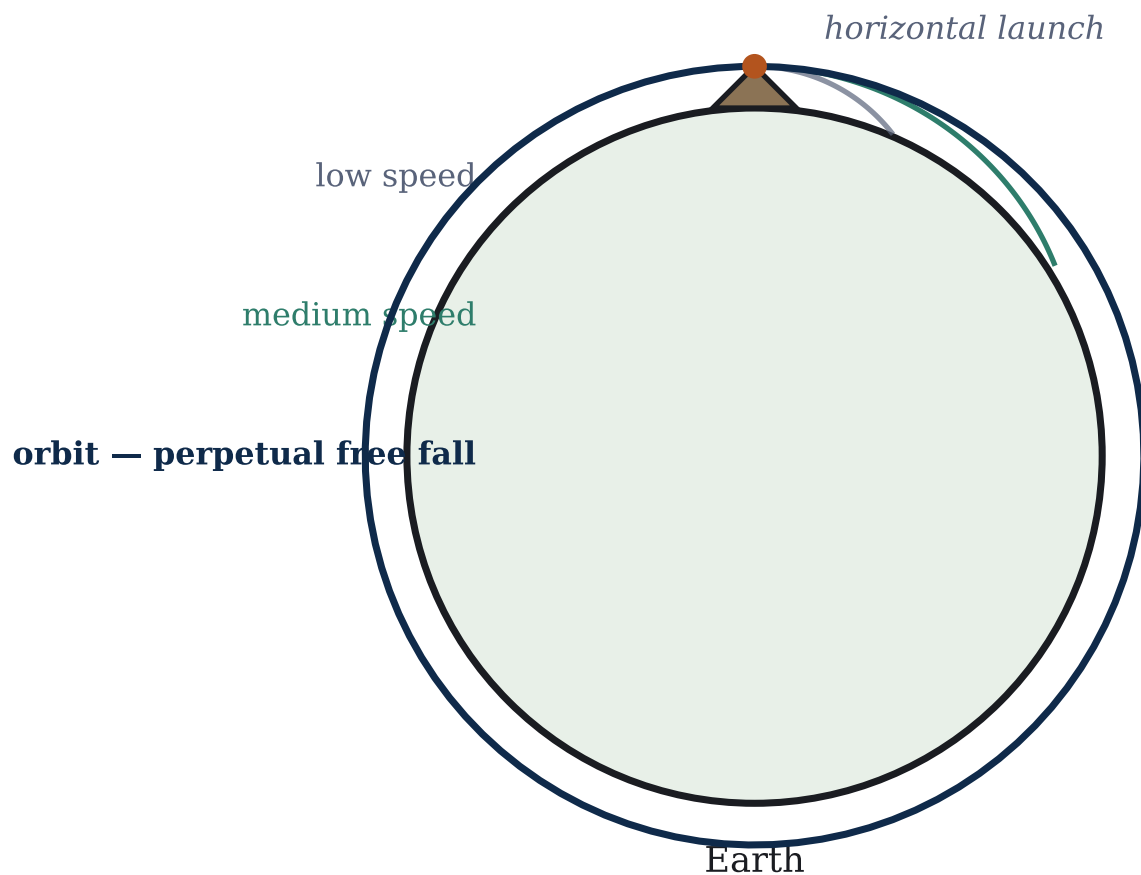


Figure 7.10 A tall mountain on a curved Earth. Several projectile trajectories launched horizontally at increasing speeds: short arcs landing nearby, then longer arcs, then one that curves all the way around and becomes a closed circular orbit. Label the fastest closed path “orbit — perpetual free fall.” Long before Newton, Johannes Kepler stared at decades of Tycho Brahe’s planetary data and distilled three crisp rules that the planets obey. He didn’t know why — that took Newton’s gravity — but the rules were exact:

they and their spacecraft are *both* falling together, in free fall, so there is no contact force between them. It's the same floating you'd feel in a lift whose cable snapped (don't try it).

A satellite's total energy is negative. Being bound in the gravitational "valley," its energy lies below the zero we set at infinity (recall the previous subtopic). Negative total energy is the mathematical signature of a *bound, closed orbit*.

Two kinds of useful satellites. Not all satellites are alike. A **geostationary** satellite sits far out (about 36,000 km up) in the equatorial plane, circling once every 24 hours in the same direction the Earth spins — so it hovers over one fixed spot and appears motionless from the ground. That makes it perfect for television, telephone, and weather relay: your dish can point at one unchanging place in the sky. A **polar** satellite, by contrast, orbits low (around 1000 km) in a north-south plane passing over the poles. As the Earth rotates beneath it, the satellite scans a fresh strip of the surface on each pass, eventually covering the whole globe — ideal for mapping, spying, and remote sensing. High-and-fixed for communication; low-and-sweeping for imaging.

Limiting Conditions & Assumptions: Most of our formulas assume a **circular orbit** (a fine first approximation; real orbits are ellipses), a **two-body** system where the planet is far heavier than the satellite, **spherically symmetric** masses, and **no atmospheric drag**. The clean energy and velocity results below are for circular orbits; for ellipses we use the semi-major axis a and lean on conservation laws.

Section 2: Key Formulas & Definitions

Kepler's Laws. 1. *Law of Orbits:* every planet moves in an ellipse with the Sun at one focus. 2. *Law of Areas:* the radius vector sweeps equal areas in equal times — equivalently, areal velocity $\frac{dA}{dt} = \frac{L}{2m} = \text{constant}$ (angular momentum conserved). 3. *Law of Periods:* $T^2 \propto a^3$ (a = semi-major axis; for a circle, $a = r$).

Orbital velocity (gravity supplies the centripetal force):

$$v_o = \sqrt{\frac{GM}{r}} = R\sqrt{\frac{g}{r}} \xrightarrow{r \approx R} \sqrt{gR} \approx 7.9 \text{ km/s}$$

Time period of a satellite:

$$T = 2\pi\sqrt{\frac{r^3}{GM}} = 2\pi\sqrt{\frac{r^3}{gR^2}} \xrightarrow{r \approx R} 2\pi\sqrt{\frac{R}{g}} \approx 84.6 \text{ min}$$

Energies of a satellite in a circular orbit:

$$K = \frac{GMm}{2r}, \quad U = -\frac{GMm}{r}, \quad E = K + U = -\frac{GMm}{2r}$$

with the tidy relations $K = -E$ and $U = 2E$. **Binding energy** (to send it to infinity) $= -E = \frac{GMm}{2r}$.

Relation to escape velocity: $v_e = \sqrt{2}v_o$ (closing the loop from the previous subtopic).

Angular momentum of a satellite: $L = mv_or = m\sqrt{GM}r$.

Geostationary satellite: $T = 24$ h, circular, **equatorial** plane, **west-to-east**, $r \approx 42,000$ km, height $h \approx 36,000$ km ($\approx 6R$), $v_o \approx 3.1$ km/s.

Elliptical orbit (JEE): total energy $E = -\frac{GMm}{2a}$; at perigee/apogee (where $v \perp r$) angular momentum conservation gives $v_pr_p = v_ar_a$.

Symbol	Meaning	SI Unit
v_o, v_e	Orbital velocity, escape velocity	m/s
T	Period of revolution	s
r, R, h, a	Orbital radius; planet radius; height; semi-major axis	m
E, K, U	Total / kinetic / potential energy	J
L	Angular momentum of the satellite	$\text{kg}\cdot\text{m}^2\text{s}^{-1}$

(Notation note: $\omega = 2\pi/T$ here is the satellite's **orbital** angular velocity — same “rotational/orbital” sense as in the g -variation subtopic, distinct from the SHM angular frequency ω of Chapter 13.)

Section 3: CBSE Board Derivations

(a) Orbital velocity: $v_o = \sqrt{GM/r}$

A satellite of mass m moves in a circle of radius r . The *only* force on it is the planet's gravity, pointing toward the centre — and that is exactly what a circular path needs as its centripetal force. So we set gravitational force equal to the required centripetal force:

$$\frac{GMm}{r^2} = \frac{mv_o^2}{r}$$

The reasoning: the planet's pull doesn't “fight” the motion, it *bends* it into a circle. Cancelling m and one power of r :

$$v_o^2 = \frac{GM}{r} \Rightarrow \boxed{v_o = \sqrt{\frac{GM}{r}}}$$

Notice v_o is **independent of the satellite's mass** — a bolt and a space station at the same height orbit at the same speed. Smaller orbits demand *higher* speeds ($v_o \propto 1/\sqrt{r}$).

(b) Time period and Kepler's Third Law

The satellite covers one circumference $2\pi r$ in one period:

$$T = \frac{2\pi r}{v_o} = \frac{2\pi r}{\sqrt{GM/r}} = 2\pi r \sqrt{\frac{r}{GM}} = 2\pi \sqrt{\frac{r^3}{GM}}$$

Squaring lays bare Kepler's third law:

$$T^2 = \frac{4\pi^2}{GM} r^3 \Rightarrow \boxed{T^2 \propto r^3}$$

The proportionality constant $\frac{4\pi^2}{GM}$ depends only on the *central* mass M — the same for *all* satellites of that planet. This is Newton's gravity *deriving* what Kepler could only observe.

(c) Energy of a satellite

The kinetic energy uses the orbital speed from part (a):

$$K = \frac{1}{2}mv_o^2 = \frac{1}{2}m \cdot \frac{GM}{r} = \frac{GMm}{2r}$$

The potential energy (from the previous subtopic) is $U = -\frac{GMm}{r}$. Adding:

$$E = K + U = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r}$$

Three things to read off: E is **negative** (bound orbit); the kinetic energy equals the *magnitude* of the total energy ($K = -E$); and as the orbit grows ($r \uparrow$), E rises toward zero — a higher orbit is a *higher-energy* (less tightly bound) state, even though it moves slower.

(d) Kepler's Second Law = conservation of angular momentum

Gravity always points along the line joining satellite and planet — it is a **central force**, so it exerts *no torque* about the centre. With zero torque, the satellite's angular momentum L is constant. The area swept per unit time is $\frac{dA}{dt} = \frac{L}{2m}$; since L and m are constant, **equal areas are swept in equal times**. This is why a planet speeds up near the Sun (small r forces large v to keep $L = mvr$ fixed) — the law of areas is angular-momentum conservation wearing a geometric costume.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A satellite orbits the Earth in a circular path at a height of 600 km above the surface. Find (a) its orbital speed and (b) its period of revolution. Take $M = 6.0 \times 10^{24} \text{ kg}$, $R = 6.4 \times 10^6 \text{ m}$, $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\text{kg}^{-2}$.

Given: $h = 600 \text{ km} = 6.0 \times 10^5 \text{ m}$, so orbital radius $r = R + h = 7.0 \times 10^6 \text{ m}$.

(a) Orbital speed:

$$v_o = \sqrt{\frac{GM}{r}} = \sqrt{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{7.0 \times 10^6}} = \sqrt{5.72 \times 10^7} \approx 7.6 \times 10^3 \text{ m/s}$$

(b) Period:

$$T = \frac{2\pi r}{v_o} = \frac{2\pi(7.0 \times 10^6)}{7.6 \times 10^3} \approx 5.8 \times 10^3 \text{ s} \approx 97 \text{ min}$$

Answer: $v_o \approx 7.6 \text{ km/s}$, $T \approx 97 \text{ minutes}$. A low-Earth-orbit satellite circles the planet roughly every hour and a half — consistent with the ISS, which orbits about 16 times a day.

Example 2 — [NEET Speed Trap]

The period of a satellite in a circular orbit around the Earth is 6 hours. If its orbital radius is increased to 4 times the original, what is the new period?

Use Kepler's third law as a ratio. Since $T^2 \propto r^3$, we have $T \propto r^{3/2}$. So:

$$\frac{T_2}{T_1} = \left(\frac{r_2}{r_1}\right)^{3/2} = (4)^{3/2} = 8 \Rightarrow T_2 = 8 \times 6 = 48 \text{ hours}$$

The trap: the tempting wrong answers are $4 \times 6 = 24 \text{ h}$ (treating $T \propto r$) or applying the wrong power. The exponent is $\frac{3}{2}$: take the radius factor, cube it, then square-root it — here $4^3 = 64$, $\sqrt{64} = 8$. Respect the $3/2$ and the problem is a five-second mental calculation.

Example 3 — [JEE Main Level] (Kepler III × the geostationary condition)

Find the height above the Earth's surface at which a geostationary satellite must orbit. Take $M = 6.0 \times 10^{24} \text{ kg}$, $R = 6.4 \times 10^6 \text{ m}$.

A geostationary satellite must have a period equal to the Earth's rotation period, $T = 24 \text{ h} = 86,400 \text{ s}$. From $T^2 = \frac{4\pi^2 r^3}{GM}$, solve for r :

$$r = \left(\frac{GMT^2}{4\pi^2}\right)^{1/3} = \left(\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(86,400)^2}{4\pi^2}\right)^{1/3}$$

Compute the inside: $GM = 4.0 \times 10^{14}$; $T^2 = 7.47 \times 10^9$; numerator $= 4.0 \times 10^{14} \times 7.47 \times 10^9 = 2.99 \times 10^{24}$; divide by $4\pi^2 = 39.5$:

$$r^3 = 7.56 \times 10^{22} \text{ m}^3 \Rightarrow r = (7.56 \times 10^{22})^{1/3} \approx 4.2 \times 10^7 \text{ m}$$

Therefore the height is

$$h = r - R = 4.2 \times 10^7 - 6.4 \times 10^6 \approx 3.6 \times 10^7 \text{ m} = 36,000 \text{ km}$$

Answer: $h \approx 36,000 \text{ km}$ ($\approx 6R$). Every television and weather satellite parked over the equator sits at this single, special altitude — the only height where a satellite's period matches the day, so it hovers over one fixed spot.

Example 4 — [JEE Advanced Level] (Energy + angular-momentum conservation)

A satellite is in a circular orbit of radius r_0 around a planet of mass M . A brief engine burn suddenly increases its speed to $\sqrt{\frac{3}{2}}$ times the circular orbital speed, directed tangentially (still perpendicular to the radius). Find the maximum distance the satellite reaches from the planet's centre.

Figure 7.11 A planet with a circular orbit of radius r_0 . At one point the satellite gets a tangential speed boost; the new path is an ellipse with the burn point as perigee (nearest) and a far apogee point at distance r_a , with the planet at one focus. (see next page)

Setup. The circular speed is $v_0 = \sqrt{GM/r_0}$. The boosted speed is

$$v = \sqrt{\frac{3}{2}} v_0 = \sqrt{\frac{3GM}{2r_0}}$$

Since $v_0 < v < \sqrt{2} v_0 = v_e$, the satellite stays bound but on an *ellipse*. The burn point (radius r_0) becomes the **perigee** (closest approach); we want the **apogee** distance r_a .

Two conservation laws. Gravity is central (no torque) and conservative: - *Angular momentum* (velocity is perpendicular to radius at both perigee and apogee): $mv r_0 = mv_a r_a \Rightarrow$

$$v_a = \frac{v r_0}{r_a}. \text{ - Energy: } \frac{1}{2} v^2 - \frac{GM}{r_0} = \frac{1}{2} v_a^2 - \frac{GM}{r_a}.$$

Solve. Substitute $v^2 = \frac{3GM}{2r_0}$ and $v_a = \frac{v r_0}{r_a}$ into the energy equation:

$$\frac{3GM}{4r_0} - \frac{GM}{r_0} = \frac{3GM r_0}{4r_a^2} - \frac{GM}{r_a}$$

Divide through by GM and simplify the left side ($\frac{3}{4r_0} - \frac{1}{r_0} = -\frac{1}{4r_0}$):

$$-\frac{1}{4r_0} = \frac{3r_0}{4r_a^2} - \frac{1}{r_a}$$

Multiply by $4r_0 r_a^2$:

$$-r_a^2 = 3r_0^2 - 4r_0 r_a \Rightarrow r_a^2 - 4r_0 r_a + 3r_0^2 = 0 \Rightarrow (r_a - r_0)(r_a - 3r_0) = 0$$

Answer. The roots are $r_a = r_0$ (the perigee we started from) and $r_a = 3r_0$ (the apogee). The maximum distance is $\boxed{r_a = 3r_0}$. A modest $\sqrt{1.5} \approx 22\%$ speed boost triples the far-point distance — the steep sensitivity of orbits to launch speed, which is why orbital insertions demand such precision.

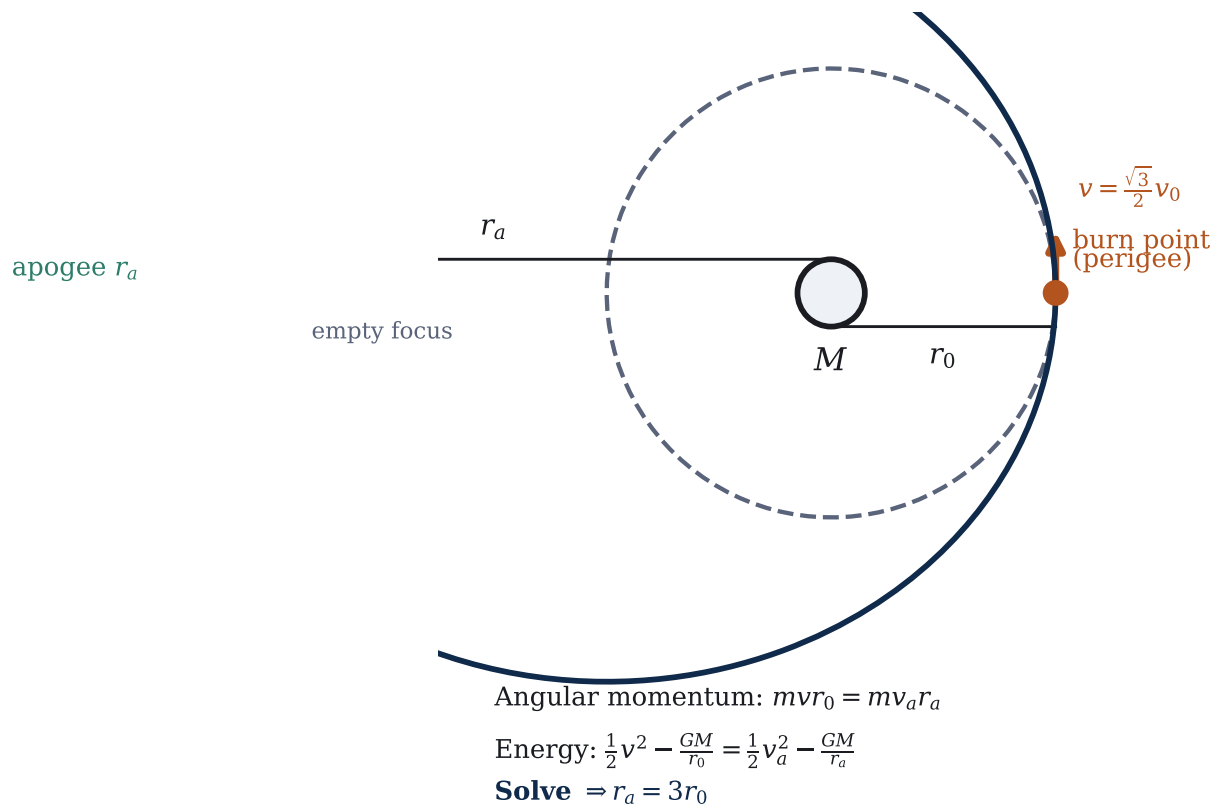


Figure 7.11 A planet with a circular orbit of radius r_0 . At one point the satellite gets a tangential speed boost; the new path is an ellipse with the burn point as perigee (nearest) and a far apogee point at distance r_a , with the planet at one focus.

Section 5: Practice Exercises

1. **[CBSE]** State Kepler's three laws of planetary motion in one line each, and identify which law is a statement of the conservation of angular momentum.
2. **[NEET]** Two satellites of the same planet have orbital radii in the ratio 1 : 4. Find the ratio of their orbital speeds.
3. **[JEE Main]** A satellite of mass m is in a circular orbit of radius r around a planet of mass M . How much additional energy must be supplied to move it to a circular orbit of radius $2r$?
4. **[JEE Main]** The orbital speed of a satellite very close to a planet's surface is v_o . Show that the minimum speed needed for the satellite to escape the planet from that point is $\sqrt{2}v_o$, and state the percentage increase required.
5. **[JEE Advanced]** A satellite is launched into a circular orbit of radius r with the correct orbital speed, but the launch vehicle accidentally directs the velocity at a small angle to the horizontal (still with the correct magnitude). Will the orbit remain circular? Describe the resulting orbit and name the conserved quantities that determine it.

Final Answers: (1) Orbits (ellipse, Sun at focus); Areas (equal areas in equal times); Periods ($T^2 \propto a^3$). The **Law of Areas** expresses angular-momentum conservation. (2) $v_o \propto 1/\sqrt{r}$, so ratio = $\sqrt{4} : \sqrt{1} = 2 : 1$. (3) $\Delta E = E_f - E_i = -\frac{GMm}{4r} - \left(-\frac{GMm}{2r}\right) = \frac{GMm}{4r}$. (4) $v_e = \sqrt{2}v_o$; an increase of $(\sqrt{2} - 1) \times 100\% \approx 41.4\%$. (5) No — it becomes an **ellipse** of the same semi-major axis (same speed $\Rightarrow$ same energy $\Rightarrow$ same a), with the launch point neither perigee nor apogee; the orbit is fixed by conserved **energy** and **angular momentum**.

Section 6: Crack the MCQ

Q1. If the gravitational attraction between the Earth and an orbiting satellite suddenly vanished, the satellite would: **(A)** fall straight down to Earth **(B)** continue along the original circular orbit **(C)** fly off along the tangent to the orbit with its current speed **(D)** stop and remain stationary

Correct: (C). With no force, Newton's first law takes over — the satellite travels in a straight line, tangent to where it was, at the speed it already had. Gravity was the *only* thing bending its path into a circle; remove it and the bending stops. **(A)** wrongly assumes some leftover "downward" tendency. **(B)** forgets that the inward force is *required* to maintain a circle. **(D)** ignores the satellite's large existing velocity.

Q2. For a satellite in a stable circular orbit, which relation between kinetic energy K , potential energy U , and total energy E is correct? **(A)** $K = E$, $U = -2E$ **(B)** $K = -E$, $U = 2E$ **(C)** $K = 2E$, $U = -E$ **(D)** $K = U$

Correct: (B). From the derivations, $K = +\frac{GMm}{2r}$, $U = -\frac{GMm}{r}$, $E = -\frac{GMm}{2r}$. Hence $K = -E$ (kinetic energy equals the magnitude of total

energy) and $U = 2E$ (both negative). **(A)** and **(C)** scramble the signs/factors. **(D)** is false — here $|U| = 2K$, never equal. This sign-and-factor bookkeeping is a JEE Main favourite.

Q3. A geostationary satellite must satisfy all of the following **except**: **(A)** orbit in the equatorial plane **(B)** revolve from west to east **(C)** have a period of 24 hours **(D)** be positioned directly above any chosen city

Correct (the false requirement): (D). A geostationary satellite can hover only over points on the **equator**, because its orbital plane must contain the Earth's centre and lie in the equatorial plane. It cannot be parked over, say, New Delhi (which is north of the equator). **(A)**, **(B)**, **(C)** are the three genuine conditions — equatorial plane, same sense as Earth's spin (west-to-east), and a matching 24-hour period. The trap in **(D)** is the intuitive but wrong belief that you can “park” a satellite over your own city.

Q4. An astronaut inside an orbiting space station feels weightless because: **(A)** there is no gravity at that height **(B)** the station is beyond the Earth's gravitational field **(C)** the astronaut and the station are in free fall together, so there is no contact force **(D)** the centrifugal force exactly cancels gravity in the ground frame

Correct: (C). Both astronaut and station accelerate toward the Earth at the same rate (the local g), so the floor exerts no normal force on the astronaut — apparent weight is zero. Gravity is very much present; it is precisely what supplies the centripetal acceleration. **(A)** and **(B)** are flatly wrong — gravity at ISS altitude is roughly 90% of its surface value. **(D)** misuses “centrifugal force,” which only appears in the rotating frame; in the inertial ground frame, gravity is *unbalanced* and provides the centripetal force. The key idea: weightlessness = shared free fall, not absent gravity.

Section 7: Common Student Pitfalls & Pro-Tips

- Scaling the period linearly.** Kepler's third law is $T \propto r^{3/2}$, not $T \propto r$. Quadrupling the radius multiplies the period by 8, not 4. Cube the radius factor, then take the square root.
- “Weightless means no gravity.”** False. Orbiting astronauts are in free fall with nearly full gravity acting. State it as *shared free fall* in exams — examiners specifically test this misconception.
- Forgetting the sign / factor in satellite energy.** $E = -\frac{GMm}{2r}$ is negative and half the magnitude of U . Writing $E = -\frac{GMm}{r}$ (that's U , not E) or dropping the negative sign is a frequent and costly slip.
- Confusing orbital and escape velocity.** $v_e = \sqrt{2}v_o$ — only a 41.4% increase separates “orbiting forever” from “escaping forever.” Mixing the two (or the factor $\sqrt{2}$) reverses the physics of whether an orbit is bound.

Pro-Tip — let conservation laws do the work in elliptical orbits. At perigee and apogee the velocity is perpendicular to the radius, so angular momentum

gives the clean relation $v_p r_p = v_a r_a$. Pair it with energy conservation and almost any two-point ellipse problem falls out — no calculus, no orbital equation needed.

Pro-Tip — remember the chapter’s “magic numbers.” Near-surface orbit: $v_o \approx 7.9$ km/s, $T \approx 84$ min (the *same* $2\pi\sqrt{R/g}$ as the tunnel-through-Earth SHM and the weightless-equator day from earlier subtopics — all three are governed by one expression). Escape: $v_e \approx 11.2$ km/s = $\sqrt{2} v_o$. Geostationary: $T = 24$ h, $h \approx 36,000$ km, $v_o \approx 3.1$ km/s. Memorising these turns many questions into instant recognition.

Section 8: Cheat Sheet Box

KEPLER’S LAWS & SATELLITES — QUICK RECALL

- **Kepler:** (1) elliptical orbits, Sun at a focus; (2) equal areas in equal times = angular-momentum conservation; (3) $T^2 \propto a^3$.
- **Orbital velocity:** $v_o = \sqrt{\frac{GM}{r}}$; near surface $\sqrt{gR} \approx 7.9$ km/s. Independent of satellite mass.
- **Period:** $T = 2\pi\sqrt{\frac{r^3}{GM}}$; near surface ≈ 84 min.
- **Energies:** $K = \frac{GMm}{2r}$, $U = -\frac{GMm}{r}$, $E = -\frac{GMm}{2r}$. So $K = -E$, $U = 2E$; binding energy = $\frac{GMm}{2r}$.
- **Escape link:** $v_e = \sqrt{2} v_o$ ($\approx 41.4\%$ faster $\Rightarrow$ escape).
- **Geostationary:** $T = 24$ h, equatorial, west-to-east, $h \approx 36,000$ km ($\approx 6R$), $v_o \approx 3.1$ km/s. **Polar:** low (~ 1000 km), north-south, for imaging/remote sensing.
- **Ellipse:** $E = -\frac{GMm}{2a}$; at perigee/apogee $v_p r_p = v_a r_a$.

Memory aids: - “Areas = angular momentum” — Kepler’s 2nd law in one breath. - “ T goes as r -to-the-three-halves” — quadruple $r \Rightarrow \times 8$ period. - “Weightless = falling together, not gravity-free.” - “ $v_e = \sqrt{2} v_o$; 84 min; 36,000 km” — the three numbers that crack half the questions.

Part 6 — The Gravitational Constant & Two-Body Systems

Exam Relevance: Two distinct exam targets. The **Cavendish experiment** and **mass/density of the Earth** are reliable **CBSE** 2-3 mark questions and appear in **NEET** as numericals (M or ρ of a planet from g). **Binary-star and two-body (reduced-mass)** systems are a **JEE Advanced** specialty, occasionally **JEE Main**. Knowing how G was actually measured — and what happens when *both* masses move — completes the chapter's foundations.

Section 1: Concept Intuition

We have used the number $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\text{kg}^{-2}$ throughout this chapter as if it fell from the sky. But where does it come from? You cannot read G off the motion of planets, because every planetary orbit gives you only the *product* GM (look back at orbital velocity, $v_o = \sqrt{GM/r}$ — G and M are always glued together). To prise G loose, you must measure the gravitational force between two masses you can *weigh in a laboratory*. And therein lies the difficulty: gravity is so absurdly weak that the attraction between two lead balls on a table is comparable to the weight of a single grain of sand. Measuring it is one of the great experimental triumphs of physics.

Think about it this way... imagine trying to detect a whisper in a hurricane. The gravitational pull between two lab masses is buried under air currents, vibrations, and the colossal pull of the whole Earth. Henry Cavendish's stroke of genius (in 1798) was to make the tiny force reveal itself not as a *push* but as a *twist*. He hung a light dumbbell of two small masses from a fine fibre, then brought two large masses nearby. The minuscule gravitational attraction twisted the fibre by a hair's breadth — but a mirror on the fibre threw a beam of light across the room, turning that hair's-breadth twist into a large, readable swing of a light-spot on a distant scale. He had built an amplifier for the weakest force in nature. Measuring G then let him do something breathtaking: **weigh the Earth**.

The second idea in this subtopic fixes an approximation we have quietly leaned on all chapter. We always treated one body as a fixed centre and the other as a small thing orbiting it — Earth fixed, satellite circling. But Newton's third law insists the satellite pulls the Earth just as hard. When the two masses are *comparable* — two stars, a planet and a heavy moon — **both** bodies move, each orbiting their common balance point, the **centre of mass**. Two stars locked in mutual orbit (a *binary system*) is the cleanest example: neither is "the centre"; they waltz around a point between them. Remarkably, this two-body dance can be reduced to a single equivalent one-body problem using a clever quantity called the **reduced mass** — and once you do, all our earlier orbit formulas snap back into use.

Limiting Conditions & Assumptions: The Cavendish analysis assumes the fibre obeys a linear torsion law ($\tau = \kappa\theta$) and that the large masses are close enough for a measurable, calculable force. "Weighing the Earth" via $M = gR^2/G$ assumes a spherically symmetric Earth. The two-body results assume an *isolated* pair (no third body) with their centre of mass either at rest or moving uniformly, and gravity as the only internal force.

One more two-body consequence worth knowing — tides. Because gravity weakens with distance, an extended body feels a *stronger* pull on its near side than its far side. This *difference* in pull across a body is a **tidal force**, and it stretches the body along the line to its companion. The Earth's oceans bulge out both toward *and* away from the Moon (the

near water is pulled toward the Moon more than the solid Earth, the far water less), which is why most coasts see roughly two high tides a day. Curiously, although the Sun pulls the Earth far harder than the Moon does, the Moon dominates the tides — because tides depend on how *rapidly* the pull changes across the Earth ($\propto 1/r^3$), not on the pull itself ($\propto 1/r^2$), and the nearby Moon wins that comparison. Pushed to an extreme, tidal forces can tear a moon apart if it strays inside a critical distance called the *Roche limit* — the origin of planetary rings.

Section 2: Key Formulas & Definitions

Cavendish balance — the working relation. At equilibrium, the gravitational torque equals the fibre's restoring torque:

$$\underbrace{\frac{GMm}{d^2} \cdot L}_{\text{gravitational torque}} = \underbrace{\kappa \theta}_{\text{fibre torque}} \Rightarrow G = \frac{\kappa \theta d^2}{MmL}$$

where m = small mass, M = large mass, d = centre-to-centre distance of a near pair, L = length of the suspended rod, θ = angle of twist, κ = torsion constant of the fibre (itself found from the torsional oscillation period $T_{\text{osc}} = 2\pi\sqrt{I/\kappa}$).

Mass of the Earth (from surface gravity, since $g = GM/R^2$):

$$M = \frac{gR^2}{G}$$

Mean density of the Earth (using $M = \frac{4}{3}\pi R^3 \rho$):

$$\rho = \frac{3g}{4\pi GR}$$

Two-body / binary system (masses m_1, m_2 , separation r , common centre of mass):

$$r_1 = \frac{m_2 r}{m_1 + m_2}, \quad r_2 = \frac{m_1 r}{m_1 + m_2} \quad (m_1 r_1 = m_2 r_2)$$

Both orbit with the **same** angular velocity and period:

$$\omega^2 = \frac{G(m_1 + m_2)}{r^3}, \quad T = 2\pi\sqrt{\frac{r^3}{G(m_1 + m_2)}}$$

This is **Kepler's third law with the total mass** in place of a single central mass.

Reduced mass (collapses the two-body problem to one body orbiting a fixed point at separation r):

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

Total energy of a bound circular binary: $E = -\frac{Gm_1 m_2}{2r}$.

Symbol	Meaning	SI Unit
G	Universal gravitational constant	$\text{N}\cdot\text{m}^2\text{kg}^{-2}$
κ	Torsion constant of the fibre	$\text{N}\cdot\text{m}/\text{rad}$
θ	Angle of twist	rad
M, R, ρ	Mass, radius, mean density of Earth/planet	kg, m, kg/m^3
m_1, m_2	The two interacting masses	kg
r_1, r_2	Their distances from the centre of mass	m
μ	Reduced mass	kg

- **Properties of G (recap):** universal scalar; value 6.67×10^{-11} ; dimensions $[M^{-1}L^3T^{-2}]$; independent of medium, mass, time, and place. It is the *smallest-magnitude* of the fundamental constants — the reason gravity feels so feeble.

Section 3: Derivations & Key Procedures

(a) The Cavendish experiment — measuring G

Apparatus. Two small lead spheres (each mass m) are fixed to the ends of a light horizontal rod of length L , suspended at its midpoint by a thin quartz fibre. A small mirror is attached to the fibre.

Figure 7.12 *A torsion balance viewed from above: a light rod with small masses m at each end, hung by a vertical fibre at the centre. Two large masses M are placed near the small ones on opposite sides, so their gravitational pulls create a couple that twists the fibre. A light beam reflects off a mirror on the fibre onto a distant scale. (see next page)*

Procedure and reasoning. 1. Two large spheres (mass M) are brought close to the small ones, one on each side, so their gravitational pulls act as a *couple*, twisting the rod. 2. Each pull is $F = \frac{GMm}{d^2}$, and the couple's torque about the fibre is $F \times L$. 3. The fibre resists, exerting a restoring torque $\kappa\theta$ proportional to the twist angle θ . The rod settles where the two torques balance:

$$\frac{GMm}{d^2} L = \kappa \theta$$

4. The twist θ is read by the deflection of a reflected light-spot on a far scale (an *optical lever* that magnifies the tiny angle enormously). 5. The torsion constant κ is found separately by timing the free torsional oscillations of the rod: $T_{\text{osc}} = 2\pi\sqrt{I/\kappa}$, where I is the rod-and-masses moment of inertia. 6. Solving for the unknown:

$$G = \frac{\kappa \theta d^2}{MmL}$$

Cavendish obtained a value within about 1% of the modern figure — extraordinary for 1798.

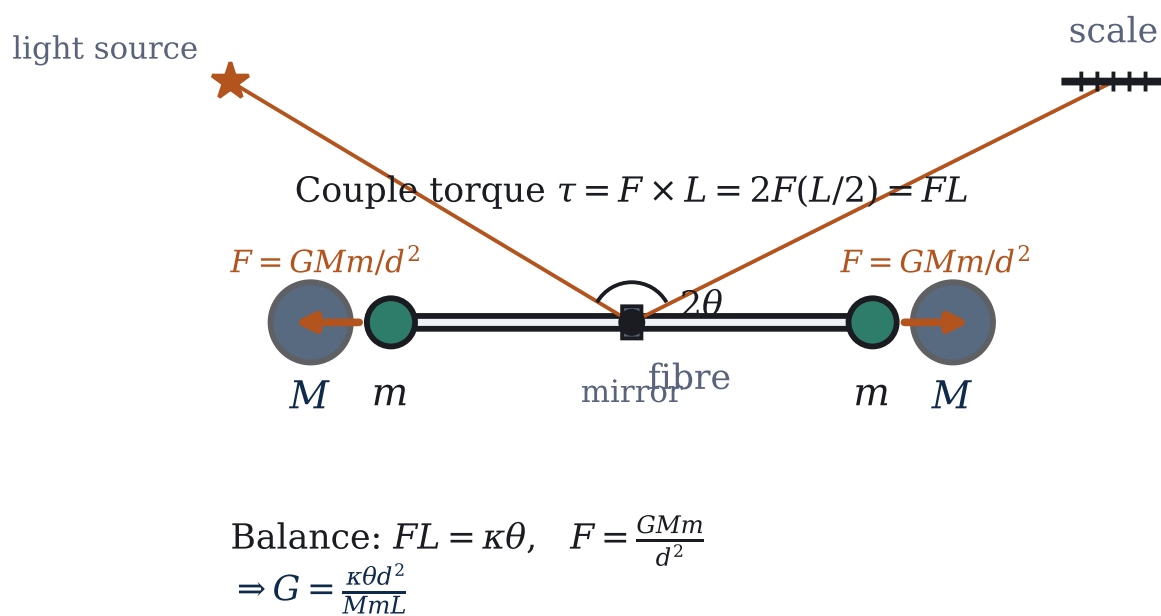


Figure 7.12 A torsion balance viewed from above: a light rod with small masses m at each end, hung by a vertical fibre at the centre. Two large masses M are placed near the small ones on opposite sides, so their gravitational pulls create a couple that twists the fibre. A light beam reflects off a mirror on the fibre onto a distant scale.

(b) “Weighing the Earth” — mass and density

Once G is known, the Earth’s own surface gravity hands us its mass. Since $g = \frac{GM}{R^2}$:

$$M = \frac{gR^2}{G}$$

Plugging $g = 9.8$, $R = 6.4 \times 10^6$ m gives $M \approx 6 \times 10^{24}$ kg — and we never had to leave the ground. The mean density follows from $M = \frac{4}{3}\pi R^3 \rho$:

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} = \frac{3g}{4\pi GR}$$

This was Cavendish’s actual goal: he reported the Earth’s density (about 5.5 times water), revealing that the deep interior must be far denser than the rocky crust — the first hint of the Earth’s iron core.

(c) The two-body problem and reduced mass

Two masses m_1, m_2 separated by r orbit their common centre of mass. The centre of mass divides the separation inversely to the masses: $m_1 r_1 = m_2 r_2$ with $r_1 + r_2 = r$, giving $r_1 = \frac{m_2 r}{m_1 + m_2}$.

For m_1 , gravity supplies the centripetal force for its circle of radius r_1 :

$$\frac{Gm_1 m_2}{r^2} = m_1 \omega^2 r_1 = m_1 \omega^2 \cdot \frac{m_2 r}{m_1 + m_2}$$

Cancelling $m_1 m_2$ and one r :

$$\omega^2 = \frac{G(m_1 + m_2)}{r^3} \Rightarrow T = 2\pi \sqrt{\frac{r^3}{G(m_1 + m_2)}}$$

So the pair obeys Kepler’s third law, but with the **sum** of the masses — which is exactly how astronomers *weigh* binary stars: measure T and r , and read off $m_1 + m_2$. Defining the **reduced mass** $\mu = \frac{m_1 m_2}{m_1 + m_2}$, the entire two-body motion is mathematically identical to a *single* mass μ orbiting a fixed centre at distance r under a force $Gm_1 m_2 / r^2$ — restoring every single-body result we derived earlier.

Concretely, in reduced-mass form the system’s kinetic energy is $K = \frac{1}{2}\mu v_{\text{rel}}^2$ (with $v_{\text{rel}} = \omega r$ the relative speed of the two bodies), the angular momentum about the centre of mass is $L = \mu v_{\text{rel}} r$, and for a bound circular binary the total energy is $E = -\frac{Gm_1 m_2}{2r}$. The reduced mass is always *smaller* than either body ($\mu < m_1, m_2$); when one mass hugely exceeds the other (say $m_2 \gg m_1$, like satellite-and-Earth), $\mu \rightarrow m_1$ and the heavier body becomes effectively the fixed centre — which is exactly the approximation we made in the satellite subtopic. So the two-body treatment doesn’t replace our earlier work; it *contains* it as the limiting case where one mass dominates.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Using the acceleration due to gravity at the Earth's surface, estimate the mass of the Earth. Take $g = 9.8 \text{ m/s}^2$, $R = 6.4 \times 10^6 \text{ m}$, $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\text{kg}^{-2}$.

From $g = \frac{GM}{R^2}$, solve for M :

$$M = \frac{gR^2}{G} = \frac{(9.8)(6.4 \times 10^6)^2}{6.67 \times 10^{-11}} = \frac{(9.8)(4.096 \times 10^{13})}{6.67 \times 10^{-11}} = \frac{4.01 \times 10^{14}}{6.67 \times 10^{-11}}$$
$$M \approx 6.0 \times 10^{24} \text{ kg}$$

Answer: $M \approx 6.0 \times 10^{24} \text{ kg}$. This is the celebrated “weighing of the Earth” — a planet's entire mass deduced from a falling apple's acceleration and a lab value of G .

Example 2 — [NEET Speed Trap]

A planet has the same surface gravity as the Earth but twice the radius. What is the ratio of the mean density of the planet to that of the Earth?

Use $g = \frac{4}{3}\pi GR\rho$ **as a ratio**. Since $g \propto R\rho$ and g is the same for both:

$$R_p \rho_p = R_E \rho_E \Rightarrow \frac{\rho_p}{\rho_E} = \frac{R_E}{R_p} = \frac{1}{2}$$

Answer: the planet's mean density is **half** the Earth's. **The trap:** students instinctively reach for $g = GM/R^2$ and get tangled computing masses. The density form $g = \frac{4}{3}\pi GR\rho$ makes it a one-line ratio — *same g , double $R \Rightarrow$ half ρ* . Pick the right form of g and the work vanishes.

Example 3 — [JEE Main Level] (Binary system — Kepler with total mass)

Two stars, each of mass M , orbit their common centre of mass in a circle. Their separation is d . Find the period of revolution.

For a two-body system the angular velocity obeys $\omega^2 = \frac{G(m_1 + m_2)}{d^3}$. With $m_1 = m_2 = M$, the total mass is $2M$:

$$\omega^2 = \frac{G(2M)}{d^3} = \frac{2GM}{d^3}$$

The period is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{d^3}{2GM}}$$

Answer: $T = 2\pi\sqrt{\frac{d^3}{2GM}}$. Note the factor of 2 inside (the *total* mass): treating one star as fixed and using $T = 2\pi\sqrt{d^3/GM}$ would be wrong by a factor of $\sqrt{2}$. In a binary, *both* masses drive the orbit.

Example 4 — [JEE Advanced Level] (Full binary analysis)

Two stars of masses m and $2m$ are separated by a distance d and revolve in circular orbits about their common centre of mass under their mutual gravitation. Find (a) the radius of each star's orbit, (b) the orbital speed of each, and (c) the total mechanical energy of the system.

Figure 7.13 Two stars, mass m (left) and $2m$ (right), separated by d , orbiting a common centre of mass C located closer to the heavier star. Show r_1 from m to C and r_2 from $2m$ to C , with both stars on opposite sides of C moving in circles. (see next page)

(a) Orbital radii. The centre of mass satisfies $m r_1 = 2m r_2$ and $r_1 + r_2 = d$. From $r_1 = 2r_2$ and $r_1 + r_2 = d$:

$$r_2 = \frac{d}{3} \text{ (heavier star, } 2m\text{)}, \quad r_1 = \frac{2d}{3} \text{ (lighter star, } m\text{)}$$

The lighter star traces the *bigger* circle — it has to swing wider to keep the balance point fixed.

(b) Orbital speeds. The common angular velocity is $\omega = \sqrt{\frac{G(m+2m)}{d^3}} = \sqrt{\frac{3Gm}{d^3}}$. Then $v = \omega r$:

$$v_1 = \omega r_1 = \frac{2d}{3} \sqrt{\frac{3Gm}{d^3}} = \frac{2}{3} \sqrt{\frac{3Gm}{d}}, \quad v_2 = \omega r_2 = \frac{1}{3} \sqrt{\frac{3Gm}{d}}$$

As expected, $v_1 = 2v_2$ — the lighter star moves twice as fast (momentum balance: $mv_1 = 2mv_2$).

(c) Total energy. Kinetic plus potential:

$$E = \frac{1}{2}mv_1^2 + \frac{1}{2}(2m)v_2^2 - \frac{G(m)(2m)}{d}$$

The kinetic sum equals $\frac{1}{2}\mu v_{\text{rel}}^2$ with $\mu = \frac{m \cdot 2m}{3m} = \frac{2m}{3}$ and $v_{\text{rel}} = \omega d$; evaluating gives $K = \frac{Gm^2}{d}$. With $U = -\frac{2Gm^2}{d}$:

$$E = \frac{Gm^2}{d} - \frac{2Gm^2}{d} = -\frac{Gm^2}{d}$$

Answer: $r_1 = \frac{2d}{3}$, $r_2 = \frac{d}{3}$; $v_1 = 2v_2 = \frac{2}{3}\sqrt{3Gm/d}$; total energy $E = -\frac{Gm^2}{d}$ (negative $\Rightarrow$ bound, as it must be). Equivalently, $E = -\frac{Gm_1m_2}{2d} = -\frac{G(m)(2m)}{2d} = -\frac{Gm^2}{d}$ — confirming the compact binary-energy formula.

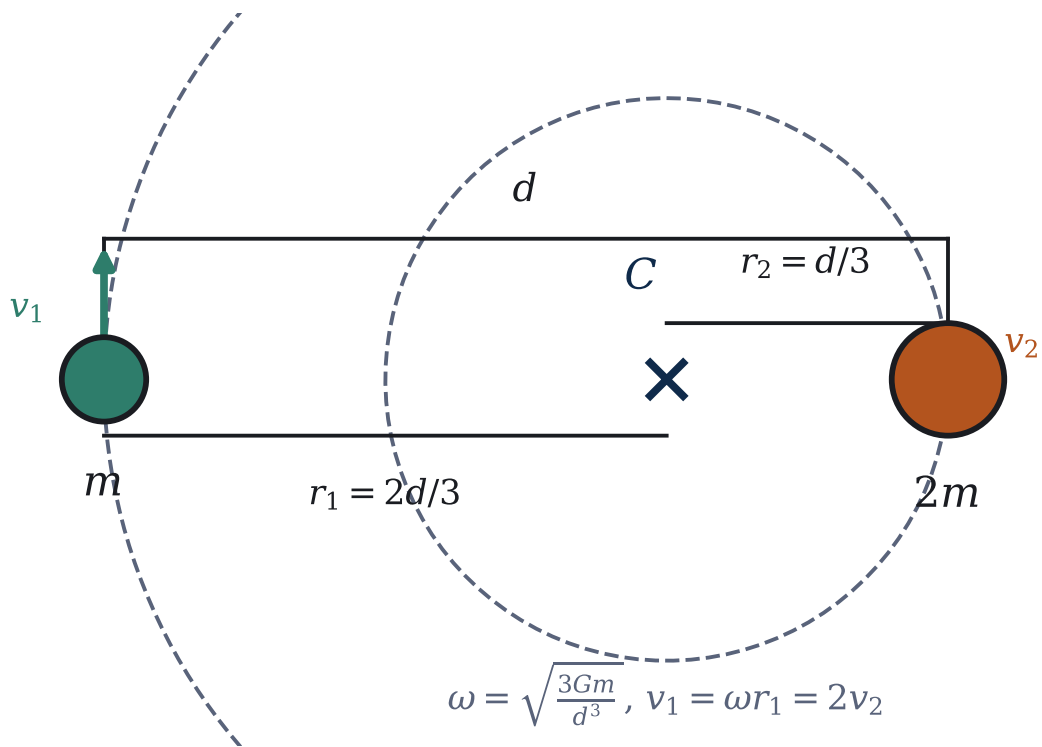


Figure 7.13 Two stars, mass m (left) and $2m$ (right), separated by d , orbiting a common centre of mass C located closer to the heavier star. Show r_1 from m to C and r_2 from $2m$ to C , with both stars on opposite sides of C moving in circles.

Section 5: Practice Exercises

1. **[CBSE]** Describe, with the working relation, how the Cavendish experiment determines the universal gravitational constant G . Why is the optical-lever (mirror-and-scale) arrangement essential?
2. **[NEET]** The mean density of a planet is the same as that of the Earth, but its radius is half. If the Earth's surface gravity is g , what is the planet's surface gravity?
3. **[JEE Main]** The mean density of the Earth is found from g , R , and G . Compute it for $g = 9.8 \text{ m/s}^2$, $R = 6.4 \times 10^6 \text{ m}$, $G = 6.67 \times 10^{-11} \text{ SI units}$.
4. **[JEE Main]** Two stars of masses m_1 and m_2 form a binary with separation r and period T . Show how observing T and r yields the sum $m_1 + m_2$.
5. **[JEE Advanced]** In a binary star system, two stars of masses M and $3M$ orbit their common centre of mass with separation d . Find the ratio of their kinetic energies $K_M : K_{3M}$.

Final Answers: (1) $G = \frac{\kappa \theta d^2}{MmL}$; the optical lever magnifies the fibre's microscopic twist into a readable deflection. (2) $g \propto R\rho$, same ρ , half $R \Rightarrow g_{\text{planet}} = g/2$. (3) $\rho = \frac{3g}{4\pi GR} \approx 5.5 \times 10^3 \text{ kg/m}^3$. (4) $\omega^2 = \frac{G(m_1 + m_2)}{r^3}$ with $\omega = 2\pi/T$ gives $m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$. (5) Speeds are inversely proportional to mass ($v \propto 1/m$), so $K = \frac{1}{2}mv^2 \propto 1/m$; thus $K_M : K_{3M} = 3 : 1$.

Section 6: Crack the MCQ

Q1. In the Cavendish experiment, the extremely small gravitational force is made measurable by: (A) using very large test masses only (B) converting it into a torsional twist read by an optical lever (C) performing it in vacuum at high temperature (D) using the Earth's magnetic field

Correct: (B). The genius of the design is converting a tiny linear force into a *twist* of a delicate fibre, then magnifying that twist with a mirror-and-light-beam (optical lever) so it can be read on a distant scale. (A) alone is insufficient — even large masses give a minuscule force without the amplification. (C) and (D) are irrelevant to the principle (gravity is medium- and temperature-independent, and unrelated to magnetism).

Q2. The mass of the Earth can be estimated from: (A) g , R , and G (B) g and R only (C) G only (D) the Earth's rotation period

Correct: (A). From $g = GM/R^2$, we get $M = gR^2/G$ — all three quantities are needed. (B) is missing G (without it you only know the product GM). (C) G alone tells you nothing about the Earth. (D) rotation period relates to centrifugal effects on g , not to the Earth's mass.

Q3. In a binary star system, the two stars always have the same: (A) orbital speed (B) orbital radius (C) angular velocity (and period) (D) kinetic energy

Correct: (C). Both stars complete one orbit in the same time, so they share a common angular velocity ω and period T . Their radii, speeds, and kinetic energies all differ (inversely with mass). **(A), (B), (D)** are equal only in the special case of equal masses. In general the lighter star orbits faster, on a bigger circle, with more kinetic energy.

Q4. Two stars of masses m and $2m$ orbit their common centre of mass. The ratio of the radius of the lighter star's orbit to that of the heavier star's orbit is: **(A)** 1 : 2 **(B)** 2 : 1 **(C)** 1 : 1 **(D)** 4 : 1

Correct: (B). The centre of mass gives $m r_1 = 2m r_2$, so $r_1/r_2 = 2$. The lighter star (m) orbits on the *larger* circle. **(A)** inverts the relation. **(C)** would require equal masses. **(D)** wrongly squares the mass ratio. Remember: orbital radius is *inversely* proportional to mass about the centre of mass.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Thinking orbits alone can give G .** Planetary and satellite data only fix the product GM . The value of G must come from a lab experiment between known masses (Cavendish). This is why G was historically the *last* of the trio G, M, g to be pinned down.
2. **Using single-body Kepler for a binary.** For two comparable masses, $T = 2\pi\sqrt{r^3/G(m_1 + m_2)}$ — the **total** mass, not one mass. Forgetting the sum gives a $\sqrt{2}$ -type error.
3. **Getting the centre-of-mass radii backwards.** The *heavier* star orbits on the *smaller* circle ($r \propto 1/m$). Many students assign the bigger orbit to the bigger mass — exactly wrong.
4. **Confusing the density and inverse-square forms of g .** For “same gravity / same density” planet comparisons, $g = \frac{4}{3}\pi GR\rho$ is far faster than $g = GM/R^2$. Choose the form that isolates the quantity you’re comparing.

Pro-Tip — to weigh anything in space, find its orbit. Sun’s mass from Earth’s orbit, Earth’s mass from the Moon’s orbit, a binary’s total mass from its period

— all from $M = \frac{4\pi^2 r^3}{GT^2}$. One formula weighs the cosmos.

Pro-Tip — reduced mass restores your toolkit. Replace a two-body problem (masses m_1, m_2 , separation r) with one body of mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$ orbiting a fixed centre at distance r . Every single-body energy and angular-momentum formula then applies unchanged.

Section 8: Cheat Sheet Box

GRAVITATIONAL CONSTANT & TWO-BODY SYSTEMS — QUICK RECALL

- **Cavendish:** torsion balance; gravitational torque = fibre torque $\Rightarrow G = \frac{\kappa\theta d^2}{MmL}$. First lab measurement of G .
- **Orbits give only GM ; G needs a lab experiment.**
- **Mass of Earth:** $M = \frac{gR^2}{G} \approx 6 \times 10^{24} \text{ kg}$.
- **Mean density:** $\rho = \frac{3g}{4\pi GR} \approx 5.5 \times 10^3 \text{ kg/m}^3$.
- **Binary:** $r_1 = \frac{m_2 r}{m_1 + m_2}$ (radius $\propto 1/m$); $\omega^2 = \frac{G(m_1 + m_2)}{r^3}$; $T = 2\pi \sqrt{\frac{r^3}{G(m_1 + m_2)}}$.
- **Reduced mass:** $\mu = \frac{m_1 m_2}{m_1 + m_2} \Rightarrow$ two-body becomes one-body.
- **Binary energy:** $E = -\frac{Gm_1 m_2}{2r}$.

Memory aids: - "Orbits weigh the product GM ; Cavendish weighs G itself." - "Heavier star, smaller circle" ($r \propto 1/m$). - "Binary Kepler uses the **sum** of the masses." - "To weigh anything, watch what orbits it: $M = 4\pi^2 r^3 / GT^2$."

Master Formula Sheet — All of Chapter 7

A one-stop consolidation of every core result in the chapter. Use this for final revision.

1. Newton's Law & the Constant

- **Law:** $F = \frac{Gm_1m_2}{r^2}$ (attractive, along the line of centres); vector form $\vec{F}_{12} = -\frac{Gm_1m_2}{r^2}\hat{r}_{21}$.
- **Constant:** $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\text{kg}^{-2}$; dimensions $[M^{-1}L^3T^{-2}]$; measured by the **Cavendish** torsion balance, $G = \frac{\kappa\theta d^2}{MmL}$.
- **Superposition:** $\vec{F}_{\text{net}} = \sum \vec{F}_i$ (vector sum).

2. Gravitational Field Intensity ($\vec{E}_g = \vec{F}/m_0$, units **N/kg**)

- **Point / outside any sphere:** $E_g = \frac{GM}{r^2}$.
- **Ring (axis):** $E_g = \frac{GMx}{(a^2 + x^2)^{3/2}}$ — max at $x = a/\sqrt{2}$, zero at centre.
- **Shell:** inside 0; surface $\frac{GM}{R^2}$; outside $\frac{GM}{r^2}$.
- **Solid sphere:** inside $\frac{GMr}{R^3}$; outside $\frac{GM}{r^2}$ (peak at surface).
- **Cavity (uniform sphere):** uniform field $\vec{E} = \frac{4}{3}\pi G\rho \vec{d}$. **Field-potential:** $E_g = -\frac{dV}{dr}$.

3. Acceleration due to Gravity

- **Surface:** $g = \frac{GM}{R^2} = \frac{4}{3}\pi GR\rho$.
- **Altitude:** $g_h = \frac{gR^2}{(R+h)^2} \approx g\left(1 - \frac{2h}{R}\right)$ for $h \ll R$.
- **Depth:** $g_d = g\left(1 - \frac{d}{R}\right)$ (zero at centre).
- **Rotation:** $g_\lambda = g - \omega^2 R \cos^2 \lambda$ (min at equator, max at poles).

4. Potential & Energy (zero at infinity; both negative)

- **PE:** $U = -\frac{Gm_1m_2}{r}$; **Potential:** $V = -\frac{GM}{r}$ (scalar, $U = mV$).
- **Small-height PE change:** $\Delta U \approx mgh$ (only $h \ll R$).
- **Escape velocity:** $v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2 \text{ km/s}$.
- **Solid sphere centre:** $V = -\frac{3GM}{2R}$; **self-energy:** $-\frac{3GM^2}{5R}$.

5. Kepler's Laws & Satellites

- **Kepler:** ellipse (Sun at focus); equal areas (= angular-momentum conservation); $T^2 \propto a^3$.
- **Orbital velocity:** $v_o = \sqrt{\frac{GM}{r}}$; near surface $\sqrt{gR} \approx 7.9$ km/s. **Escape link:** $v_e = \sqrt{2} v_o$.
- **Period:** $T = 2\pi \sqrt{\frac{r^3}{GM}}$; near-surface ≈ 84 min.
- **Energies:** $K = \frac{GMm}{2r}$, $U = -\frac{GMm}{r}$, $E = -\frac{GMm}{2r}$ ($K = -E$, $U = 2E$).
- **Geostationary:** $T = 24$ h, equatorial, west-east, $h \approx 36,000$ km.

6. Two-Body / Binary Systems

- **Mass & density of Earth:** $M = \frac{gR^2}{G} \approx 6 \times 10^{24}$ kg; $\rho = \frac{3g}{4\pi GR} \approx 5.5 \times 10^3$ kg/m³.
- **Binary:** $r_1 = \frac{m_2 r}{m_1 + m_2}$ ($r \propto 1/m$); $\omega^2 = \frac{G(m_1 + m_2)}{r^3}$; $T = 2\pi \sqrt{\frac{r^3}{G(m_1 + m_2)}}$.
- **Reduced mass:** $\mu = \frac{m_1 m_2}{m_1 + m_2}$; **binary energy:** $E = -\frac{Gm_1 m_2}{2r}$.
- **Weigh anything by its orbit:** $M = \frac{4\pi^2 r^3}{GT^2}$.

Chapter 7: Gravitation — Round 2 Addendum

The chapter taught you the law, the fields of the standard bodies, the variation of g , the potential-energy machinery, the satellite results, and the two-body reduction — and it taught them well. Four examinable patterns are nevertheless still standing outside. Addendum A builds the potentials the chapter *states* but never derives: the constant interior potential of a shell, the $-3GM/2R$ at a sphere's centre, and the self-energy $-3GM^2/5R$ — all three are board and JEE derivation favourites. Addendum B packages the elliptical-orbit calculation the chapter redoes from scratch in every example into one reusable tool, the vis-viva equation. Addendum C breaks the chapter's silent assumption of uniform density, which is exactly where JEE sets its interior-field problems. Addendum D leaves the symmetry axes: the field of an arc at its centre, and the chord-tunnel theorem that upgrades the chapter's through-the-centre tunnel to *every* straight tunnel.

Nothing here is new theory. Every result is Newton's law plus integration, run one layer deeper or one symmetry short of the cases the chapter chose.

Addendum A — Potentials You Can Finally Derive, Not Just Quote

The chapter hands you three results with no derivation: $V = -GM/R$ *constant* inside a shell, $V_{\text{centre}} = -3GM/2R$ for a solid sphere, and $U_{\text{self}} = -3GM^2/5R$. CBSE asks for the first two as derivations; JEE Advanced asks for all three embedded in larger problems. Quoting them is worth one mark; deriving them is worth five.

Method

Everything runs on one signed quantity. Define E_r as the **radial component** of the field, positive *outward*. Then

$$E_r = -\frac{dV}{dr}, \quad V(b) - V(a) = -\int_a^b E_r dr.$$

- E_r — signed field component, in N/kg; V — potential, in J/kg; r — distance from the centre, in m.
- **Sign discipline:** for attracting matter, $E_r < 0$ everywhere outside and inside. If you insert a magnitude where a signed component belongs, the sign of your answer flips silently. Most lost marks in this topic are exactly this error.

Two engines, chosen by what you already know:

1. **Engine 1 — integrate a known field.** If $E_r(r)$ is known on an interval, integrate it. Used for the solid sphere's interior below.

2. **Engine 2 — integrate contributions directly.** If the field is not yet known, add up $-\frac{G dm}{\ell}$ over mass elements, where ℓ is the element-to-point distance. Used for the shell below.

Every derived formula must pass two checks before you trust it: **continuity** of V across any boundary, and $E_r = -dV/dr$ recovering the known field.

Derivation 1 — Potential of a uniform shell, inside and outside in one integral

Shell of mass M , radius R , surface density $\sigma = \frac{M}{4\pi R^2}$ (units kg/m²). Point P at distance r from the centre. Cut the shell into rings at polar angle θ , width $R d\theta$. Each ring has

$$dA = (2\pi R \sin \theta)(R d\theta), \quad dm = \sigma dA = \frac{M}{2} \sin \theta d\theta.$$

Every element of the ring is at the same distance ℓ from P , where by the cosine rule in the triangle centre-element- P ,

$$\ell = \sqrt{R^2 + r^2 - 2Rr \cos \theta}.$$

Potential is a scalar, so elements add without any vector resolution:

$$V = -G \int \frac{dm}{\ell} = -\frac{GM}{2} \int_0^\pi \frac{\sin \theta d\theta}{\sqrt{R^2 + r^2 - 2Rr \cos \theta}}.$$

Substitute $u = \cos \theta$, so $du = -\sin \theta d\theta$, and the θ -limits $0 \rightarrow \pi$ become u -limits $1 \rightarrow -1$:

$$V = -\frac{GM}{2} \int_{-1}^1 \frac{du}{\sqrt{R^2 + r^2 - 2Rru}}.$$

The antiderivative is $-\frac{1}{Rr} \sqrt{R^2 + r^2 - 2Rru}$, so

$$\int_{-1}^1 \frac{du}{\ell} = -\frac{1}{Rr} \left[(R-r) - (R+r) \right] = \frac{2}{R}, \quad r < R,$$

using $\sqrt{(R-r)^2} = R-r$ because $r < R$. Therefore

$$V_{\text{inside}} = -\frac{GM}{2} \cdot \frac{2}{R} = -\frac{GM}{R}, \quad \text{constant, for } r < R.$$

Repeat with $r > R$, where $\sqrt{(R-r)^2} = r-R$:

$$\int_{-1}^1 \frac{du}{\ell} = -\frac{1}{Rr} \left[(r-R) - (R+r) \right] = \frac{2}{r}, \quad V_{\text{outside}} = -\frac{GM}{r}.$$

One integral, both regions. **Checks:** at $r = R$ both give $-GM/R$ — continuous. Differentiate the outside branch: $-\frac{d}{dr} \left(-\frac{GM}{r} \right) = -\frac{GM}{r^2} = E_r$, the correct inward field in N/kg. Inside, the

derivative of a constant is zero — the field vanishes, which is the shell theorem arriving for free.

Derivation 2 — Interior potential of a uniform solid sphere

Outside, the sphere behaves as a point mass: $V(R) = -GM/R$ at the surface. Inside, the chapter's field is $|E| = \frac{GMr}{R^3}$, pointing inward, so the *signed* component is

$$E_r = -\frac{GMr}{R^3} \quad (r \leq R).$$

Integrate inward from the surface using Engine 1:

$$V(r) = V(R) - \int_R^r E_r \, dr' = -\frac{GM}{R} - \int_R^r \left(-\frac{GMr'}{R^3} \right) dr' = -\frac{GM}{R} + \frac{GM}{R^3} \cdot \frac{r^2 - R^2}{2}.$$

Combining over the common denominator $2R^3$:

$$V(r) = -\frac{GM}{2R^3} (3R^2 - r^2), \quad r \leq R.$$

Checks. At $r = R$: $V = -\frac{GM}{2R^3}(2R^2) = -\frac{GM}{R}$ — continuous with the outside branch. At the centre: $V = -\frac{3GM}{2R}$, the quoted value, and the deepest point. Differentiate:

$$\frac{dV}{dr} = -\frac{GM}{2R^3}(-2r) = +\frac{GMr}{R^3}, \quad E_r = -\frac{dV}{dr} = -\frac{GMr}{R^3},$$

which is exactly the interior field. Units: $\frac{\text{Nm}^2\text{kg}^{-2} \cdot \text{kg}}{\text{m}^3} \cdot \text{m}^2 = \text{Nm/kg} = \text{J/kg}$, as potential requires.

Derivation 3 — Self-energy of a uniform sphere, shell by shell

Assemble the sphere from radius 0 outward. When the built portion has radius r , its mass is

$$m(r) = \frac{4}{3}\pi r^3 \rho.$$

Bring in the next shell of thickness dr and mass

$$dm = 4\pi r^2 \rho \, dr$$

from infinity. The work the field does is $-\frac{Gm(r)dm}{r}$, so the energy change of the system is

$$dU = -\frac{G}{r} \left(\frac{4}{3}\pi r^3 \rho \right) (4\pi r^2 \rho \, dr) = -\frac{16\pi^2 G \rho^2}{3} r^4 \, dr.$$

Integrate from 0 to R :

$$U_{\text{self}} = -\frac{16\pi^2 G \rho^2}{3} \cdot \frac{R^5}{5} = -\frac{16\pi^2 G \rho^2 R^5}{15}.$$

Eliminate ρ using $M = \frac{4}{3}\pi R^3 \rho$, so $\rho^2 = \frac{9M^2}{16\pi^2 R^6}$:

$$U_{\text{self}} = -\frac{16\pi^2 G}{15} \cdot \frac{9M^2}{16\pi^2 R^6} \cdot R^5 = \boxed{-\frac{3GM^2}{5R}}.$$

Unit check: $\frac{(\text{Nm}^2\text{kg}^{-2})(\text{kg}^2)}{\text{m}} = \text{Nm} = \text{J}$. **Limit check:** halve R at fixed M and the binding doubles in magnitude — squeeze a body tighter and it binds harder, the fact that powers stellar collapse.

Example A.1 — [JEE Advanced Level]

A particle of mass $m = 1000$ kg is moved slowly from the centre of the Earth (modelled as uniform, $M = 6.0 \times 10^{24}$ kg, $R = 6.4 \times 10^6$ m) to the surface. Find the minimum work required.

Slowly means no kinetic energy is gained, so the work done *by the external agent* equals the rise in potential energy:

$$W = m[V(R) - V(0)] = m \left[-\frac{GM}{R} - \left(-\frac{3GM}{2R} \right) \right] = \frac{GMm}{2R}.$$

Substituting, keeping powers of ten explicit:

$$W = \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(1.0 \times 10^3)}{2(6.4 \times 10^6)} \text{ J}.$$

Numerator: $6.67 \times 6.0 = 40.02$, and $-11+24+3 = 16$, giving 4.00×10^{17} . Denominator: 1.28×10^7 . So

$$W = \frac{4.00 \times 10^{17}}{1.28 \times 10^7} \text{ J} \approx 3.1 \times 10^{10} \text{ J}.$$

Independent cross-check by force integration. At distance s from the centre the inward pull is $F = m \cdot \frac{gs}{R}$ (chapter's interior field). The agent pulls outward against it, so

$$W = \int_0^R \frac{mgs}{R} \text{d}s = \frac{mg}{R} \cdot \frac{R^2}{2} = \frac{mgR}{2}.$$

With $g = \frac{GM}{R^2} = \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{(6.4 \times 10^6)^2} \text{ m/s}^2 = 9.77 \text{ m/s}^2$:

$$W = \frac{(1000)(9.77)(6.4 \times 10^6)}{2} \text{ J} = 3.1 \times 10^{10} \text{ J}.$$

Two routes, one answer — and notice the work is exactly *half* of mgR , not mgR : the interior field weakens as you descend, so climbing out from the centre costs less than a surface-to-surface mgh instinct suggests.

Why it is Advanced-level. The naive reflexes both fail: quoting V_{centre} without derivation earns nothing under “show that,” and applying mgR treats the field as constant when it is not. The mark-winning structure is signed-component integration plus a second, independent route as a check.

Example A.2 — [CBSE Board / JEE Main Level]

Find every point at which the potential of a uniform solid sphere equals half its value at the centre. Express the answer in terms of R .

Half the central value is

$$V_{\text{target}} = \frac{1}{2} \left(-\frac{3GM}{2R} \right) = -\frac{3GM}{4R}.$$

Try inside first. Set $-\frac{GM}{2R^3}(3R^2 - r^2) = -\frac{3GM}{4R}$ and cancel $-\frac{GM}{R}$ throughout:

$$\frac{3R^2 - r^2}{2R^2} = \frac{3}{4} \Rightarrow 3R^2 - r^2 = \frac{3R^2}{2} \Rightarrow r^2 = \frac{3R^2}{2} \Rightarrow r = R\sqrt{\frac{3}{2}} \approx 1.22R.$$

This exceeds R — the interior formula has been applied where it does not hold, so there is **no interior solution**. The potential inside only ranges from $-\frac{3GM}{2R}$ (centre) to $-\frac{GM}{R}$ (surface), and $-\frac{3GM}{4R}$ is *above* the whole of that range.

Try outside. Solve $-\frac{GM}{r} = -\frac{3GM}{4R}$:

$$r = \frac{4R}{3}.$$

Since $\frac{4R}{3} > R$, this point genuinely lies in the exterior region, where the exterior formula is valid.

Answer: the unique point is at $r = \frac{4R}{3}$ from the centre. **Check:** $V(4R/3) = -\frac{GM}{4R/3} = -\frac{3GM}{4R}$, and $\frac{4}{3} > 1$ confirms it is outside. The trap was assuming the answer must lie inside, near the “deep” centre — but potential climbs back toward zero monotonically, so intermediate values live *outside*.

Practice — Addendum A

1. Without any integration, state the work required to move a mass m from the surface of a thin spherical shell to its centre, and justify the answer in one line.
2. Two concentric thin shells have masses M_1 (radius R_1) and M_2 (radius $R_2 > R_1$). Find the potential at the common centre.
3. Find all points where the potential of a uniform solid sphere equals its own surface value, other than the surface itself.

4. Two uniform spheres have the same density ρ and radii R and $2R$. Find the ratio of their self-energies.
5. Using Derivation 2, find the potential difference $V(R) - V(R/2)$ inside a uniform sphere, and verify it equals $\frac{3GM}{8R}$.

Answers: 1. Zero: the field inside a shell is zero everywhere, so moving within it requires no work (V is constant at $-GM/R$). 2. Potentials add as scalars:

$$V = -\frac{GM_1}{R_1} - \frac{GM_2}{R_2}.$$

Note the *inner* shell's own radius appears for both terms — potential depends on where you are, and at the centre you are inside both. 3. Inside, V ranges over $[-\frac{3GM}{2R}, -\frac{GM}{R}]$; outside, over $[-\frac{GM}{R}, 0)$; the surface value sits at the junction, attained nowhere else. No other point exists. 4. $U_{\text{self}} \propto \frac{M^2}{R}$ and $M \propto \rho R^3$, so $U \propto \rho^2 R^5$; the ratio is 1 : 32. 5.

$$V(R) - V(R/2) = -\frac{GM}{2R^3} (3R^2 - R^2) + \frac{GM}{2R^3} (3R^2 - \frac{R^2}{4}) = \frac{GM}{2R^3} \cdot \frac{3R^2}{4} = \frac{3GM}{8R}.$$

Addendum B — One Equation for Every Elliptical Orbit: Vis-Viva

Every elliptical-orbit problem in the chapter — Examples 4 of Parts 5, the practice items — is solved by writing energy and angular momentum afresh and grinding through algebra. JEE Advanced expects that grind once; after that it expects the packaged result. This addendum derives the package.

Method

Work per unit mass (divide everything by the satellite's mass m). Define

$$\varepsilon = \frac{v^2}{2} - \frac{GM}{r} \quad (\text{specific energy, in J/kg}),$$

$$h = vr_{\perp} \quad (\text{specific angular momentum, in m}^2/\text{s}),$$

where $r_{\perp}$ is the component of velocity perpendicular to the radius. At the two **apsides** (perigee and apogee) the velocity is purely transverse, so $h = vr$ exactly there.

Step 1 — The apsidal quadratic. At an apsis, substitute $v = \frac{h}{r}$ into the energy:

$$\frac{h^2}{2r^2} - \frac{GM}{r} = \varepsilon.$$

Multiply through by r^2 and collect:

$$\varepsilon r^2 + GM r - \frac{h^2}{2} = 0.$$

Its two roots are precisely r_p and r_a . By Vieta's formulas:

$$r_p + r_a = -\frac{GM}{\varepsilon}, \quad r_p r_a = -\frac{h^2}{2\varepsilon}.$$

Step 2 — Read off the semi-major axis. Since $a = \frac{r_p + r_a}{2}$:

$$\boxed{\varepsilon = -\frac{GM}{2a}}, \quad \text{i.e.} \quad E = -\frac{GMm}{2a}$$

for a satellite of mass m — the chapter's elliptical-orbit energy, now *derived*, with units $\frac{\text{m}^3/\text{s}^2}{\text{m}} = \text{m}^2/\text{s}^2 = \text{J/kg}$.

Step 3 — Vis-viva. Substitute Step 2 back into the definition of ε and solve for v^2 :

$$\boxed{v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right)}$$

valid at *every* point of the orbit, not just the apsides.

Built-in sanity checks. - Circular orbit: $a = r$ gives $v^2 = \frac{GM}{r}$ — the chapter's v_o . - Escape threshold: $a \rightarrow \infty$ gives $v^2 = \frac{2GM}{r}$ — the escape condition at radius r . - A bound orbit needs $\varepsilon < 0$, i.e. $v^2 < \frac{2GM}{r}$ at every point: the same statement in energy language.

Procedure for any single-impulse orbit problem: compute ε at the burn point from the given speed; get $a = -\frac{GM}{2\varepsilon}$; identify whether the burn point is perigee or apogee by comparing r with a ; obtain the other apsis from $r_p + r_a = 2a$; obtain the period from Kepler III, $T = 2\pi\sqrt{\frac{a^3}{GM}}$.

Example B.1 — [JEE Main Level]

A satellite is at distance $r = 2R$ from the Earth's centre, moving tangentially at $v = \frac{1}{2}\sqrt{\frac{GM}{2R}}$ — half the circular speed at that radius. Find the semi-major axis, the other apsis, and the period in terms of the surface-skimming period $T_0 = 2\pi\sqrt{R^3/GM}$. Comment on the physical fate of this orbit.

Specific energy at the burn point:

$$\varepsilon = \frac{v^2}{2} - \frac{GM}{2R} = \frac{1}{2} \cdot \frac{GM}{8R} - \frac{GM}{2R} = \frac{GM}{R} \left(\frac{1}{16} - \frac{1}{2} \right) = -\frac{7GM}{16R}.$$

Negative, so the orbit is bound. The semi-major axis:

$$a = -\frac{GM}{2\varepsilon} = -\frac{GM}{2} \cdot \left(-\frac{16R}{7GM} \right) = \frac{8R}{7} \approx 1.14R.$$

The burn point has $r = 2R > a$, so it is the **apogee**. The perigee follows from $r_p + r_a = 2a$:

$$r_p = 2a - r_a = \frac{16R}{7} - 2R = \frac{2R}{7} \approx 0.29R.$$

Period by Kepler III:

$$T = 2\pi\sqrt{\frac{a^3}{GM}} = 2\pi\sqrt{\frac{R^3}{GM}} \left(\frac{8}{7}\right)^{3/2} = T_0 \left(\frac{8}{7}\right)^{3/2}.$$

Numerically $\left(\frac{8}{7}\right)^{3/2} = \frac{8}{7}\sqrt{\frac{8}{7}} = 1.22$, so with $T_0 \approx 84.6$ min:

$$T \approx 103 \text{ min.}$$

Physical verdict — the mark-winning remark. The perigee $r_p = \frac{2R}{7}$ lies well *inside* the Earth ($r_p < R$). The mathematical ellipse dips into the planet, so in reality the satellite spirals in and burns up: a tangential de-boost to half of circular speed is not a survivable orbit change. An Advanced answer that notices this outranks one that stops at the algebra.

Check. $\varepsilon < 0$ (bound); a lies between the apsides, $\frac{2R}{7} < \frac{8R}{7} < 2R$; slower-than-circular at apogee means the opposite end dives closer.

Example B.2 — [JEE Advanced Level]

A satellite in a circular orbit of radius r_1 fires its engine once, tangentially, so that its new apogee is exactly $4r_1$. Find (a) the speed immediately after the burn as a multiple of the original circular speed $v_{c1} = \sqrt{GM/r_1}$, and (b) the speed at apogee as a multiple of the local circular speed there. Show that a second burn is unavoidable.

(a) Speed after the burn. The transfer orbit has apsides r_1 and $4r_1$, so

$$a = \frac{r_1 + 4r_1}{2} = \frac{5r_1}{2}.$$

Vis-viva at $r = r_1$:

$$v_1^2 = GM \left(\frac{2}{r_1} - \frac{1}{a} \right) = GM \left(\frac{2}{r_1} - \frac{2}{5r_1} \right) = \frac{8GM}{5r_1}.$$

Therefore

$$v_1 = \sqrt{\frac{8}{5}} v_{c1} \approx 1.265 v_{c1}.$$

(b) Speed at apogee. Vis-viva at $r = 4r_1$:

$$v_a^2 = GM \left(\frac{2}{4r_1} - \frac{2}{5r_1} \right) = GM \left(\frac{1}{10r_1} \right), \quad v_a = \sqrt{\frac{GM}{10r_1}}.$$

Circular speed at $4r_1$ would be $v_{c2} = \sqrt{\frac{GM}{4r_1}}$, so

$$\frac{v_a}{v_{c2}} = \sqrt{\frac{4}{10}} = \sqrt{0.4} \approx 0.63.$$

The satellite arrives at $r = 4r_1$ moving at only 63% of the local circular speed — too slow to stay there, so it would fall back along the ellipse. **A second, forward burn raising the speed by a factor $\sqrt{2.5} \approx 1.58$ is unavoidable.** This is precisely why reaching a higher orbit takes two impulses, not one.

Check. $v_{c1} < v_1 < v_e$ at r_1 , since $1 < \frac{8}{5} < 2$ — fast enough to climb, too slow to escape, exactly a transfer ellipse. And $\frac{v_a}{v_1} = \frac{r_1}{4r_1}$, matching angular momentum conservation $vr_{\perp} = \text{const}$.

Practice — Addendum B

1. A satellite is at $r = 3R$ moving tangentially at $\frac{3}{4}$ of the circular speed there. Show that the orbit survives (both apses exceed R) and find the perigee.
2. An object is launched vertically upward from the surface with speed $\frac{v_e}{2}$. Find its maximum height above the surface, and reconcile your answer with the chapter's Practice Exercise 2 of Part 4.
3. A satellite has $a = 2R$. Find its speed when it passes $r = \frac{3R}{2}$.
4. Two satellites of the same planet have semi-major axes in ratio 1 : 4. Find the ratio of their periods.
5. Show from vis-viva that the speed at perigee of any bound ellipse is $v_p = \sqrt{\frac{GM(1+e)}{a(1-e)}}$, where e is the eccentricity defined by $r_p = a(1 - e)$.

Answers: 1. $\varepsilon = \frac{1}{2} \cdot \frac{9}{16} \cdot \frac{GM}{3R} - \frac{GM}{3R} = \frac{GM}{R} \left(\frac{9}{96} - \frac{32}{96} \right) = -\frac{23GM}{96R}$, so $a = \frac{48R}{23}$. The burn point $r = 3R > a$ is the apogee; perigee $= 2a - 3R = \frac{96R - 69R}{23} = \frac{27R}{23} \approx 1.17R > R$ — survivable, unlike Example B.1. 2. $\varepsilon = \frac{1}{2} \cdot \frac{v_e^2}{4} - \frac{GM}{R} = \frac{GM}{4R} - \frac{GM}{R} = -\frac{3GM}{4R}$, so $a = \frac{2R}{3}$ and the turning point (where $v = 0$) is at $r_{\max} = 2a = \frac{4R}{3}$, giving height $\frac{R}{3}$. This is the same $h = \frac{R}{3}$ obtained in the chapter by direct energy conservation — vis-viva reproduces it with less algebra. 3. $v^2 = GM \left(\frac{2}{3R/2} - \frac{1}{2R} \right) = GM \left(\frac{4}{3R} - \frac{1}{2R} \right) = \frac{5GM}{6R}$, so $v = \sqrt{\frac{5GM}{6R}}$. 4. $T \propto a^{3/2}$, so 1 : 8. 5. Put $r = r_p = a(1 - e)$ in vis-viva:

$$v_p^2 = GM \left(\frac{2}{a(1-e)} - \frac{1}{a} \right) = \frac{GM}{a} \cdot \frac{2 - (1-e)}{1-e} = \frac{GM(1+e)}{a(1-e)}.$$

As $e \rightarrow 0$ this reduces to the circular speed $\sqrt{GM/a}$; as $e \rightarrow 1$ it diverges like the parabolic escape speed.

Addendum C — Planets That Are Not Uniform: Fields from Piecewise and Graded Density

Every interior result in the chapter — $\frac{GM(r)}{R^3}$, the depth formula, the tunnel oscillation — quietly assumes uniform density. JEE breaks that assumption constantly: layered planets, cores

denser than mantles, densities graded as functions of r . The chapter even admits the real Earth is like this (its g initially *rises* with depth) without telling you how to compute it. Here is the machinery.

Method

Spherical symmetry is the only licence needed. For any density profile $\rho(r)$ (units kg/m^3):

1. **Enclosed mass.** Only the mass interior to radius r contributes (shells above contribute zero — the chapter's rule, unchanged). With a varying density you must *integrate* rather than scale:

$$M_{\text{enc}}(r) = \int_0^r 4\pi r'^2 \rho(r') \, dr'.$$

2. **Interior field.**

$$|E(r)| = \frac{G M_{\text{enc}}(r)}{r^2},$$

directed radially inward. Units: $\frac{(\text{N m}^2 \text{kg}^{-2})(\text{kg})}{\text{m}^2} = \text{N/kg}$.

3. **Exterior field.** Beyond the surface, M_{enc} freezes at the total mass M and $|E| = \frac{GM}{r^2}$ as usual.
4. **Continuity audit.** $E(r)$ is *continuous* across any density jump (there is no infinitely thin sheet of mass at the interface), but its slope $\frac{dE}{dr}$ generally jumps. If your graph of E versus r has a break, you have made an error.
5. **Extrema.** To find where g peaks inside, differentiate $\frac{M_{\text{enc}}(r)}{r^2}$ and set the derivative to zero.

Example C.1 — [JEE Main Level]

A planet of radius R consists of a uniform core of radius $\frac{R}{2}$ and density $2\rho_0$, surrounded by a uniform mantle of density ρ_0 . Take $\rho_0 = 5500 \text{ kg/m}^3$, $R = 6.4 \times 10^6 \text{ m}$. Find g (a) at the interface, (b) at $r = \frac{3R}{4}$, and (c) at the surface. Compare with the uniform- ρ_0 planet throughout.

Let $M_u = \frac{4}{3}\pi\rho_0 R^3$ denote the mass of a hypothetical uniform- ρ_0 planet, for which the surface field would be $g_u = \frac{4}{3}\pi G\rho_0 R$.

(a) Interface, $r = \frac{R}{2}$. Only the core is enclosed:

$$M_{\text{enc}} = \frac{4}{3}\pi(2\rho_0) \left(\frac{R}{2}\right)^3 = \frac{4}{3}\pi\rho_0 R^3 \cdot \frac{2}{8} = \frac{M_u}{4}.$$

Then

$$g = \frac{G(M_u/4)}{(R/2)^2} = \frac{GM_u}{R^2} = g_u.$$

From the mantle side the enclosed mass at the same radius is identical, so the field is continuous — as Method step 4 demands. Numerically:

$$g_u = \frac{4}{3}\pi(6.67 \times 10^{-11})(5500)(6.4 \times 10^6) \text{ m/s}^2.$$

Chain the products: $6.67 \times 10^{-11} \times 5500 = 3.67 \times 10^{-7}$; $\times 6.4 \times 10^6 = 2.35$; $\times \frac{4\pi}{3} = 9.83$. So

$$g\left(\frac{R}{2}\right) = g_u = 9.83 \text{ m/s}^2.$$

(b) Mid-mantle, $r = \frac{3R}{4}$. Core plus the inner part of the mantle:

$$M_{\text{enc}} = \frac{4}{3}\pi(2\rho_0)\left(\frac{R}{2}\right)^3 + \frac{4}{3}\pi\rho_0\left[\left(\frac{3R}{4}\right)^3 - \left(\frac{R}{2}\right)^3\right].$$

Convert to sixteenths of M_u : the core contributes $\frac{2}{8} = \frac{16}{64}$; the mantle slab contributes $\frac{27}{64} - \frac{8}{64} = \frac{19}{64}$. Total:

$$M_{\text{enc}} = M_u \left(\frac{16 + 19}{64}\right) = \frac{35M_u}{64}.$$

Then

$$g = \frac{G(35M_u/64)}{(3R/4)^2} = \frac{GM_u}{R^2} \cdot \frac{35}{64} \cdot \frac{16}{9} = g_u \cdot \frac{35}{36}.$$

Numerically:

$$g\left(\frac{3R}{4}\right) = \frac{35}{36}(9.83) \text{ m/s}^2 = 9.56 \text{ m/s}^2.$$

(c) Surface. Core $\frac{16}{64}$ plus full mantle $(1 - \frac{8}{64}) = \frac{56}{64}$:

$$M = M_u \cdot \frac{72}{64} = \frac{9M_u}{8}, \quad g_{\text{surf}} = \frac{9}{8}g_u = 1.125 \times 9.83 = 11.1 \text{ m/s}^2.$$

Reading the profile. Because the dense core is concentrated deep down, the interior field at $\frac{3R}{4}$ (9.56) *exceeds* what a uniform planet of the same ρ_0 would give there ($\frac{3}{4} \times 9.83 = 7.37 \text{ m/s}^2$) — concentrating mass centrally props up the interior field. This is exactly why the real Earth's g rises slightly as you descend the crust before eventually falling: the iron core dominates.

Example C.2 — [JEE Advanced Level]

A planet of radius R has density graded linearly, $\rho(r) = \rho_0 \left(1 - \frac{r}{R}\right)$. Find $g(r)$ everywhere inside, locate the radius at which g is maximum, and show that the peak exceeds the surface value by a factor $\frac{4}{3}$.

Enclosed mass.

$$M_{\text{enc}}(r) = \int_0^r 4\pi r'^2 \rho_0 \left(1 - \frac{r'}{R}\right) dr' = 4\pi\rho_0 \left[\frac{r^3}{3} - \frac{r^4}{4R}\right].$$

Interior field, dividing by r^2 :

$$g(r) = 4\pi G\rho_0 \left(\frac{r}{3} - \frac{r^2}{4R}\right), \quad r \leq R.$$

Checks. At $r = 0$: $g = 0$ (symmetry). Total mass: $M = 4\pi\rho_0 R^3 \left(\frac{1}{3} - \frac{1}{4}\right) = \frac{\pi\rho_0 R^3}{3}$, and the surface field is

$$g_R = \frac{GM}{R^2} = \frac{\pi G\rho_0 R}{3} = 4\pi G\rho_0 R \left(\frac{1}{3} - \frac{1}{4}\right),$$

which is exactly $g(R)$ from the interior formula — continuous at the surface.

Maximum. Differentiate:

$$\frac{dg}{dr} = 4\pi G\rho_0 \left(\frac{1}{3} - \frac{r}{2R}\right) = 0 \Rightarrow r = \frac{2R}{3}.$$

Substitute back:

$$g_{\text{max}} = 4\pi G\rho_0 \left(\frac{2R}{9} - \frac{4R}{9 \cdot 4}\right) = 4\pi G\rho_0 \left(\frac{2R}{9} - \frac{R}{9}\right) = \frac{4\pi G\rho_0 R}{9}.$$

Compare with the surface value:

$$\frac{g_{\text{max}}}{g_R} = \frac{4\pi G\rho_0 R/9}{\pi G\rho_0 R/3} = \frac{4}{3}.$$

So gravity inside this planet swells to $\frac{4}{3}$ of its surface strength at $r = \frac{2R}{3}$, then falls to zero-boundary behaviour at the surface and continues falling as $\frac{1}{r^2}$ outside. **The lesson the chapter only gestures at:** for any realistic (centrally condensed) density profile, g is *not* maximal at the surface. The uniform-sphere “peak at the surface” result is a special case, not a law.

Practice — Addendum C

1. A planet has a uniform-density inner shell of radius R_1 and mass M_1 , surrounded by a uniform-density shell (between R_1 and R_2) of mass M_2 . State g in the gap $R_1 < r < R_2$ and outside $r > R_2$.

2. A planet of radius R has density increasing linearly, $\rho(r) = \rho_0 \frac{r}{R}$. Show that $g(r) \propto r^2$ inside, and find g at the surface in terms of ρ_0 , G , R .
3. For the planet of Practice 2, state without calculation whether g is maximal at the surface or inside, and why.
4. For the graded planet of Example C.2, replace the profile by $\rho(r) = \rho_0 \left(1 - \frac{r}{R}\right)^2$ and find the radius at which g is maximum, in closed form.
5. Explain in one line why g is continuous across the core-mantle interface of Example C.1 while its slope is not.

Answers: 1. In the gap only M_1 acts: $g = \frac{GM_1}{r^2}$; outside, $g = \frac{G(M_1+M_2)}{r^2}$. (Inside the inner shell: 0.) 2. $M_{\text{enc}}(r) = \int_0^r 4\pi r'^2 \rho_0 \frac{r'}{R} dr' = \frac{\pi \rho_0 r^4}{R}$, so

$$g(r) = \frac{G\pi \rho_0 r^4 / R}{r^2} = \frac{\pi G \rho_0}{R} r^2 \propto r^2.$$

Total mass $M = \frac{\pi \rho_0 R^3}{3}$, so $g_R = \frac{\pi G \rho_0 R}{3}$. Note the field grows even faster than the uniform case's r^1 , because ever-denser material keeps being added beneath you. 3. At the surface. The density *decreases* outward here, but the enclosed mass saturates at R while the r^{-2} keeps biting; formally, $\frac{dg}{dr} = \frac{2\pi G \rho_0 r}{R} > 0$ throughout the interior, so g climbs all the way to the surface. 4. $M_{\text{enc}} = 4\pi \rho_0 \left(\frac{r^3}{3} - \frac{r^4}{2R} + \frac{r^5}{5R^2}\right)$, so $g \propto \frac{r^3}{3} - \frac{r^2}{2R} + \frac{r^3}{5R^2}$. Setting the derivative to zero: $\frac{1}{3} - \frac{r}{R} + \frac{3r^2}{5R^2} = 0$, i.e. $9r^2 - 15Rr + 5R^2 = 0$, giving

$$r = \frac{15 - \sqrt{45}}{18} R = \frac{5 - \sqrt{5}}{6} R \approx 0.461R$$

(the other root exceeds R). 5. Because the enclosed mass $M_{\text{enc}}(r)$ is itself continuous across a finite density jump — no mass resides *in* the interface — so $\frac{GM_{\text{enc}}}{r^2}$ cannot jump; but its rate of accumulation $\frac{dM_{\text{enc}}}{dr} = 4\pi r^2 \rho$ changes abruptly with ρ , so the slope does jump.

Addendum D — Off the Axes: Arcs at Their Centres, and Tunnels That Miss the Centre

The chapter integrates along one line (rod on its axis) and drills one tunnel (through the centre). Examiners go further in both directions. JEE Advanced sets the field at the *centre of an arc* — a vector integral with genuine cancellation — and rewards anyone who knows that *every* straight tunnel through a uniform planet, not just a diameter, gives the same 84-minute ride.

Part 1 — Field at the centre of a uniform arc

Setup. A wire bent into a circular arc of radius R , linear density λ (units kg/m), spanning angles $-\alpha$ to $+\alpha$ measured from the bisector. A point of interest sits at the centre of curvature.

Element. An element subtending $d\theta$ has length $R d\theta$, mass $dm = \lambda R d\theta$, and sits at distance R . Its contribution:

$$dE = \frac{G dm}{R^2} = \frac{G\lambda}{R} d\theta,$$

directed from the centre *toward the element* (gravity attracts). Units: $\frac{(\text{N m}^2\text{kg}^{-2})(\text{kg/m})}{\text{m}} = \text{N/kg}$.

Cancellation. Elements at $+\theta$ and $-\theta$ pull in directions mirror-symmetric about the bisector: the components *along* the bisector add, those *across* it cancel. Only the bisector component survives:

$$dE_{\parallel} = dE \cos \theta = \frac{G\lambda}{R} \cos \theta d\theta.$$

Integration.

$$E = \int_{-\alpha}^{\alpha} \frac{G\lambda}{R} \cos \theta d\theta = \frac{G\lambda}{R} [\sin \theta]_{-\alpha}^{+\alpha} = \boxed{\frac{2G\lambda \sin \alpha}{R}},$$

pointing from the centre toward the midpoint of the arc.

Three checks that cost nothing. - *Small arc:* $\alpha \rightarrow 0$ gives $E \approx \frac{G\lambda(2R\alpha)}{R^2} = \frac{GM_{\text{arc}}}{R^2}$, since $M_{\text{arc}} = \lambda(2R\alpha)$ — the arc collapses to a point mass, as it must. - *Full ring:* $\alpha = \pi$ gives $\sin \pi = 0$ — complete cancellation, the chapter's ring-centre result recovered. - *Semicircle:* $\alpha = \frac{\pi}{2}$, and with $\lambda = \frac{M}{\pi R}$,

$$E = \frac{2G\lambda}{R} = \frac{2GM}{\pi R^2},$$

directed toward the middle of the semicircle. A force on a test mass m placed there is $F = \frac{2GMm}{\pi R^2}$.

Example D.1 — [JEE Advanced Level]

A uniform wire of mass $M = 12$ kg is bent into a semicircle of radius $R = 2.0$ m. Find the gravitational field at its centre, and the force on a 5.0 kg bead placed there.

Linear density: $\lambda = \frac{M}{\pi R} = \frac{12}{\pi(2.0)} \text{ kg/m} = 1.91 \text{ kg/m}$.

Field magnitude:

$$E = \frac{2GM}{\pi R^2} = \frac{2(6.67 \times 10^{-11})(12)}{\pi(2.0)^2} \text{ N/kg} = \frac{1.60 \times 10^{-9}}{12.57} \text{ N/kg} = 1.27 \times 10^{-10} \text{ N/kg}.$$

Force on the bead:

$$F = mE = (5.0)(1.27 \times 10^{-10}) \text{ N} = 6.4 \times 10^{-10} \text{ N},$$

directed horizontally toward the midpoint of the semicircle (the direction of the bead of symmetry).

Check. Treat the semicircle crudely as its mass concentrated halfway out along the bisector, at distance roughly $\langle R \cos \theta \rangle = \frac{2R}{\pi} \approx 1.27 \text{ m}$:

$$E_{\text{crude}} \approx \frac{GM}{(2R/\pi)^2} = \frac{(6.67 \times 10^{-11})(12)}{1.62} \text{ N/kg} \approx 4.9 \times 10^{-10} \text{ N/kg}.$$

The crude estimate overshoots by a factor near $\frac{\pi}{2} \cdot \dots$ — the exact answer is smaller because most of the wire sits farther away or pulls at an angle. The exact value being comfortably below the crude one is the right qualitative direction; the precise agreement comes from the integral alone. The lesson: for curved sources, only the component integral is trustworthy — “mass at the centroid” is a trap.

Part 2 — The chord-tunnel theorem

Claim. In a uniform-density, non-rotating planet of radius R and surface gravity g , a bead sliding frictionlessly in *any* straight tunnel — through the centre or missing it entirely — executes SHM with the same period

$$T = 2\pi\sqrt{\frac{R}{g}} \approx 84.6 \text{ min},$$

and completes any crossing, of any length, in $\frac{T}{2} \approx 42.3 \text{ min}$.

Derivation. The chapter established that the field anywhere inside a uniform sphere is

$$\vec{E}(\vec{r}) = -\frac{g}{R} \vec{r},$$

where $\vec{r}$ is the position from the centre — linear in displacement, pointing home. Now consider a tunnel whose closest approach to the centre is d , and let s measure position *along the tunnel* from the tunnel’s midpoint. Decompose $\vec{r}$ into components along and across the tunnel; the along-component of $\vec{r}$ is exactly s . The tunnel wall supplies whatever normal force the transverse component demands, so the motion along the tunnel obeys

$$m\ddot{s} = -\frac{mg}{R} s.$$

This is Hooke’s law with effective constant $k_{\text{eff}} = \frac{mg}{R}$ — *identical* to the through-centre case, and containing no trace of d . Hence

$$\omega = \sqrt{\frac{g}{R}}, \quad T = 2\pi\sqrt{\frac{R}{g}},$$

for every chord. The amplitude is half the chord length; by Pythagoras in the triangle centre-midpoint-entry point,

$$A = \sqrt{R^2 - d^2}.$$

Why the independence is not magic. A deeper chord is *shorter*, but the restoring component of gravity along it is *weaker* (only $\frac{g}{R}A$ of the field works along the tunnel at the

ends). Shorter distance, gentler push — and the two effects cancel exactly because the field is strictly linear in $\vec{r}$. Any nonlinear field (a real, centrally-condensed Earth, for instance) destroys the cancellation, and chord times begin to differ.

Example D.2 — [JEE Main Level]

A chord tunnel's closest approach to the Earth's centre is $d = \frac{R}{2}$. Take $g = 9.8 \text{ m/s}^2$, $R = 6.4 \times 10^6 \text{ m}$. Find the amplitude, the maximum speed, the acceleration component along the tunnel at the moment of entry, and the crossing time.

Amplitude (half-chord):

$$A = \sqrt{R^2 - \frac{R^2}{4}} = \frac{\sqrt{3}}{2}R \approx 0.866R.$$

Angular frequency: $\omega = \sqrt{\frac{g}{R}} = \sqrt{\frac{9.8}{6.4 \times 10^6}} \text{ s}^{-1} = \sqrt{1.53 \times 10^{-6}} \text{ s}^{-1} = 1.24 \times 10^{-3} \text{ s}^{-1}$.

Maximum speed (at the tunnel's midpoint, closest to the centre):

$$v_{\max} = \omega A = 0.866\sqrt{gR} = 0.866\sqrt{(9.8)(6.4 \times 10^6)} \text{ m/s} = 0.866\sqrt{6.27 \times 10^7} \text{ m/s}.$$

Since $\sqrt{6.27 \times 10^7} = 7.92 \times 10^3 \text{ m/s}$ (the familiar surface-orbit speed):

$$v_{\max} \approx 6.86 \text{ km/s}.$$

Entry acceleration along the tunnel (at $s = -A$):

$$|\ddot{s}|_{\text{entry}} = \omega^2 A = \frac{g}{R}(0.866R) = 0.866g = 8.49 \text{ m/s}^2.$$

Note this is *less* than g : at the tunnel mouth the full gravitational pull is 9.8 m/s^2 , but only its component along the tunnel accelerates the bead — the rest presses it against the wall.

Crossing time — independent of the chord, by the theorem:

$$t = \frac{T}{2} = \pi\sqrt{\frac{R}{g}} = \pi\sqrt{\frac{6.4 \times 10^6}{9.8}} \text{ s} = \pi(808) \text{ s} \approx 2540 \text{ s} \approx 42.3 \text{ min}.$$

Check. The through-centre tunnel is the special case $d = 0$: $A = R$, $v_{\max} = \sqrt{gR} = 7.92 \text{ km/s}$, same 42.3 min — matching the chapter's Example 4 of Part 3 exactly. Our chord result slots into it seamlessly. And $v_{\max} = 6.86 < 7.92 \text{ km/s}$ makes sense: a shallower dive samples weaker field.

Practice — Addendum D

1. An arc of linear density λ and radius R subtends 60° at its centre. Find the field at the centre.
2. The same for a quarter-circle wire.

3. Explain in one sentence, using the pair-cancellation picture, why the field at the centre of a *full* ring vanishes even though no element's individual field is zero.
4. A chord tunnel has $d = 0.8R$. Find $v_{\max}$ and the crossing time.
5. Prove in three lines that the crossing time of a chord tunnel is independent of d .

Answers: 1. $\alpha = 30^\circ$, so $E = \frac{2G\lambda \sin 30^\circ}{R} = \frac{G\lambda}{R}$, toward the arc's midpoint. 2. $\alpha = 45^\circ$: $E = \frac{2G\lambda}{R} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}G\lambda}{R}$. 3. Every element at angle θ has a partner diametrically opposite pulling with equal magnitude along the same line in the opposite direction, so the vector sum is zero pairwise. 4. $A = \sqrt{R^2 - 0.64R^2} = 0.60R$, so $v_{\max} = 0.60\sqrt{gR} = 0.60(7.92) \approx 4.75$ km/s; the crossing time is unchanged at 42.3 min. 5. The equation of motion $\ddot{s} = -\frac{g}{R}s$ contains no parameter of the chord; the crossing from $s = -A$ to $s = +A$ is half a period of this fixed oscillator, $t = \pi\sqrt{R/g}$, regardless of A or d .

Closing Note — What Changed Under Your Feet

Four habits separate the Round 2 solver from the Round 1 solver. First, **derive or die**: shell potential, sphere-interior potential, and self-energy are derivations, not facts, and every one reduces to a single disciplined integral with a continuity check attached. Second, **stop re-deriving orbits**: write ε , read a , name the apsides — vis-viva turns the chapter's longest worked examples into three-line calculations and flags crashing trajectories ($r_p < R$) that pure algebra hides. Third, **question the uniform density**: real and exam planets alike are centrally condensed, and g peaking *inside* is the norm, not the exception. Fourth, **trust linearity**: wherever the field is exactly proportional to displacement, projections along any axis inherit the same oscillator — the source of the 42-minute theorem and its cousins. Each habit is one integration deep. Go one layer past the formula and the formula stops being fragile.

Mechanical Properties of Solids

Chapter 8 · Complete Master Reference · CBSE · JEE · NEET

Drona — AI Tutoring

Subtopics 01-05

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Subtopic 01: Elasticity, Stress, and Strain

Notation conventions (Chapter 8 — fixed here, carried through all subtopics): - Normal / longitudinal stress: σ | Shear (tangential) stress: τ
- Longitudinal strain: ε | Shear strain: ϕ (radians) | Volume strain: $\frac{\Delta V}{V}$ -
Young's modulus: Y | Bulk modulus: B (some texts use K) | Modulus of rigidity / shear modulus: η (some texts use G) - Poisson's ratio: ν — **note:** many Indian coaching texts write this as σ , which clashes with *stress*. We use ν throughout to keep the symbols distinct. - Compressibility: $k = \frac{1}{B}$

Exam Relevance: This is the bedrock of the whole chapter. CBSE Boards reliably ask a 2–3 mark definition or a Young's-modulus numerical, plus a stress-strain graph question. JEE Main almost always carries 1 question (elongation, wires in series/parallel, or graph reading). NEET asks ~1 question, usually conceptual ("which is more elastic?") or a breaking-stress numerical. JEE Advanced rarely tests this alone — it bundles it with thermal stress, rotation, or energy methods.

Section 1: Concept Intuition

Pick up a steel ruler and a rubber band. Pull both. The rubber stretches dramatically and snaps back; the ruler barely moves but also returns to shape the instant you let go. That "springing back" is **elasticity** — the property by which a body regains its original shape and size once the deforming force is removed. The opposite extreme is **plasticity**: wet clay or chewing gum holds whatever new shape you force on it. Most real solids live somewhere on this spectrum.

Here's the part that surprises everyone: in everyday language we'd call the rubber band "more elastic" because it stretches more. **Physics says the exact opposite.** Steel is *more* elastic than rubber, because for the *same* pull, steel resists deformation far more strongly and snaps back more faithfully. Elasticity is about the strength of the restoring tendency, not the size of the stretch. Hold on to that — it's the single most common conceptual trap in this chapter.

So what's happening inside? Think about it this way... imagine the atoms of a solid connected by countless tiny springs (the inter-atomic bonds). At rest, every spring sits at its natural length. The moment you apply a force, you stretch or squeeze these molecular springs, and just like any spring, they push back. That collective push-back is the body's internal **restoring force**. Pull harder and the springs stretch more and push back harder — up to a point.

To compare different materials and different-sized objects fairly, we need two "normalised" quantities. A thick steel cable and a thin steel wire are the same material, but the cable obviously withstands more force — so raw force is a bad measure. Instead we use **stress**: the restoring force *per unit area* of cross-section. Likewise, a 10 m wire stretching by 1 cm is very different from a 10 cm wire stretching by 1 cm, so raw elongation is a bad measure too. Instead we use **strain**: the change in dimension *as a fraction of the original*. Stress is

the “cause per unit area,” strain is the “effect per unit size.” Normalising this way lets us talk about the *material itself*, independent of how much of it we happen to have.

A neat market analogy: stress is like the *price per kilogram* of vegetables (force spread over area), not the total bill (raw force). Strain is like the *percentage discount* (fractional change), not the rupees knocked off. Both are ratios, and ratios are what let you compare a roadside thela with a supermarket fairly.

Limiting conditions & assumptions. Everything in this subtopic — especially the clean proportional relationship between stress and strain (Hooke’s law) — holds *only within the elastic limit*. Push a material past that limit and three things break our tidy picture: (1) the proportionality fails, (2) the body no longer fully returns to its original shape (permanent “set”), and (3) eventually it fractures. We also assume the deformation is **small**, the temperature is constant (heating changes a material’s stiffness), and the material is **homogeneous and isotropic** (same properties everywhere and in every direction). Drop any of these — large deformations, hot wires, or wood that’s stiffer along the grain than across it — and the simple formulas stop being reliable.

Section 2: Key Formulas & Definitions

Stress — restoring force per unit area (equal in magnitude to the applied force at equilibrium):

$$\sigma = \frac{F}{A}$$

- F = applied/restoring force (N), A = area of cross-section (m^2) - SI unit: $\text{N m}^{-2} = \text{pascal (Pa)}$ | **Dimensional formula:** $[M^1 L^{-1} T^{-2}]$ - *Three types:* longitudinal (tensile, due to stretching / compressive, due to squeezing), volumetric/hydraulic ($\sigma = P$, normal force on all surfaces), and shear/tangential ($\tau = F/A$, force parallel to a surface).

Strain — fractional change in a dimension:

$$\varepsilon = \frac{\text{change in dimension}}{\text{original dimension}}$$

- **Dimensionless and unitless.** Dimensional formula: $[M^0 L^0 T^0]$ - Longitudinal: $\varepsilon = \frac{\Delta L}{L}$
 | Volume: $\frac{\Delta V}{V}$ | Shear: $\phi = \frac{x}{h}$ (x = relative slide of faces, h = separation; in radians)

Hooke’s Law (within elastic limit): $\sigma \propto \varepsilon \Rightarrow \sigma = E \varepsilon$, where E = modulus of elasticity, a property of the **material only** (independent of shape/size). Unit Pa, dimensions $[ML^{-1}T^{-2}]$.

Young’s modulus (longitudinal stress / longitudinal strain):

$$Y = \frac{\sigma}{\varepsilon} = \frac{F/A}{\Delta L/L} = \frac{FL}{A \Delta L} = \frac{MgL}{\pi r^2 \Delta L}$$

Bulk modulus (volume stress / volume strain):

$$B = \frac{-P}{\Delta V/V} = -V \frac{dP}{dV}$$

The minus sign reflects that volume *decreases* as pressure increases. Compressibility $k = 1/B$. For a gas: isothermal $B = P$, adiabatic $B = \gamma P$.

Modulus of rigidity / shear modulus (shear stress / shear strain):

$$\eta = \frac{\tau}{\phi} = \frac{F/A}{\phi} \quad (\phi \text{ in radians})$$

Poisson's ratio (lateral strain / longitudinal strain):

$$\nu = -\frac{\text{lateral strain}}{\text{longitudinal strain}} = -\frac{\Delta r/r}{\Delta L/L}$$

Dimensionless. Practical range $0 \leq \nu \leq 0.5$; most metals ≈ 0.2 - 0.4 .

All three moduli have unit Pa and dimensions $[ML^{-1}T^{-2}]$. Solids possess all three; liquids and gases possess only the bulk modulus, since they cannot sustain shear.

Section 3: CBSE Board Derivations & Key Procedures

Derivation 1 — Young's modulus of a suspended wire. Consider a wire of natural length L and radius r , clamped rigidly at the top, with a mass M hung from the lower end. In equilibrium the restoring force equals the weight, so the stretching force is $F = Mg$. *Why Mg ?* Because the wire is in equilibrium — the upward restoring force balances the downward weight exactly.

The cross-sectional area is $A = \pi r^2$, so the longitudinal stress is $\sigma = \frac{Mg}{\pi r^2}$.

If the wire elongates by ΔL , the longitudinal strain is $\varepsilon = \frac{\Delta L}{L}$.

By definition, Young's modulus is the ratio of these:

$$Y = \frac{\sigma}{\varepsilon} = \frac{Mg/\pi r^2}{\Delta L/L} = \boxed{\frac{MgL}{\pi r^2 \Delta L}}$$

This is the working formula behind Searle's experiment. Notice Y contains only material/geometry inputs you can measure — it does **not** depend on how the wire is loaded, only on the *material*.

Derivation 2 — Elongation of a rod under its own weight. Now remove the hanging mass; the rod stretches under *its own weight*. The subtlety: the tension is **not uniform**. The top of the rod must support everything below it, while the very bottom supports almost nothing.

Take a rod of length L , area A , density ρ , hanging from its top. Consider a thin element of thickness dx at a height x from the *bottom*. The weight it must support is the weight of all material below it:

$$T(x) = \rho A x g$$

The stress in this element is $\sigma = T/A = \rho xg$, so its strain is $\varepsilon = \sigma/Y = \rho xg/Y$, and the element stretches by

$$d(\Delta L) = \frac{\rho xg}{Y} dx$$

Integrating over the whole rod:

$$\Delta L = \int_0^L \frac{\rho gx}{Y} dx = \frac{\rho gL^2}{2Y} = \boxed{\frac{MgL}{2AY}}$$

The factor of $\frac{1}{2}$ is the punchline: it's as if the rod's *entire* weight acted at its centre of mass, halfway down. This “average tension” shortcut saves time in exams.

Procedure — Reading a stress-strain curve. Trace it left to right: 1. **Proportional limit (P):** straight-line region; Hooke's law holds; slope = Y . 2. **Elastic limit (E):** still returns to original shape, but the line is no longer perfectly straight. 3. **Yield point:** strain shoots up with little extra stress; permanent deformation begins (plastic region starts). 4. **Ultimate tensile strength:** the highest stress the material can bear. 5. **Fracture / breaking point:** the material snaps. A **ductile** material (copper, steel) has a long plastic region between yield and fracture — it can be drawn into wires. A **brittle** material (glass, cast iron) fractures almost immediately after the elastic limit. The area under the curve up to a point equals the energy absorbed per unit volume.

Figure 8.1 A stress (y-axis) vs strain (x-axis) curve for a ductile metal. Mark the linear portion O→P (proportional limit), point E slightly beyond (elastic limit), the yield point, the gently rising plastic region up to the ultimate tensile strength (peak), and the fracture point where the curve ends. Shade the elastic region green and the plastic region orange.] (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A steel wire of length 3.0 m and cross-sectional area 2.0 mm^2 stretches by 0.60 mm when an 8.0 kg load is hung from it. Find Young's modulus of steel. (Take $g = 10 \text{ m s}^{-2}$.)

Given: $L = 3.0 \text{ m}$, $A = 2.0 \text{ mm}^2 = 2.0 \times 10^{-6} \text{ m}^2$, $\Delta L = 0.60 \text{ mm} = 6.0 \times 10^{-4} \text{ m}$, $M = 8.0 \text{ kg}$.

Stretching force: $F = Mg = 8.0 \times 10 = 80 \text{ N}$.

$$Y = \frac{FL}{A\Delta L} = \frac{80 \times 3.0}{(2.0 \times 10^{-6})(6.0 \times 10^{-4})} = \frac{240}{1.2 \times 10^{-9}}$$

$$\boxed{Y = 2.0 \times 10^{11} \text{ Pa}}$$

This matches the textbook value for steel — a good sanity check. **Always carry the unit (Pa) into the final line.**

Example 2 — [NEET Speed Trap] Two wires, A and B, are made of the **same** material. Wire B is twice as long as A and has twice the radius. The same load hangs from each. By what factor does A's elongation exceed B's? Also, is A “more elastic” than B?

The trap: students rush to plug numbers, or they confuse “stretches more” with “more elastic.”

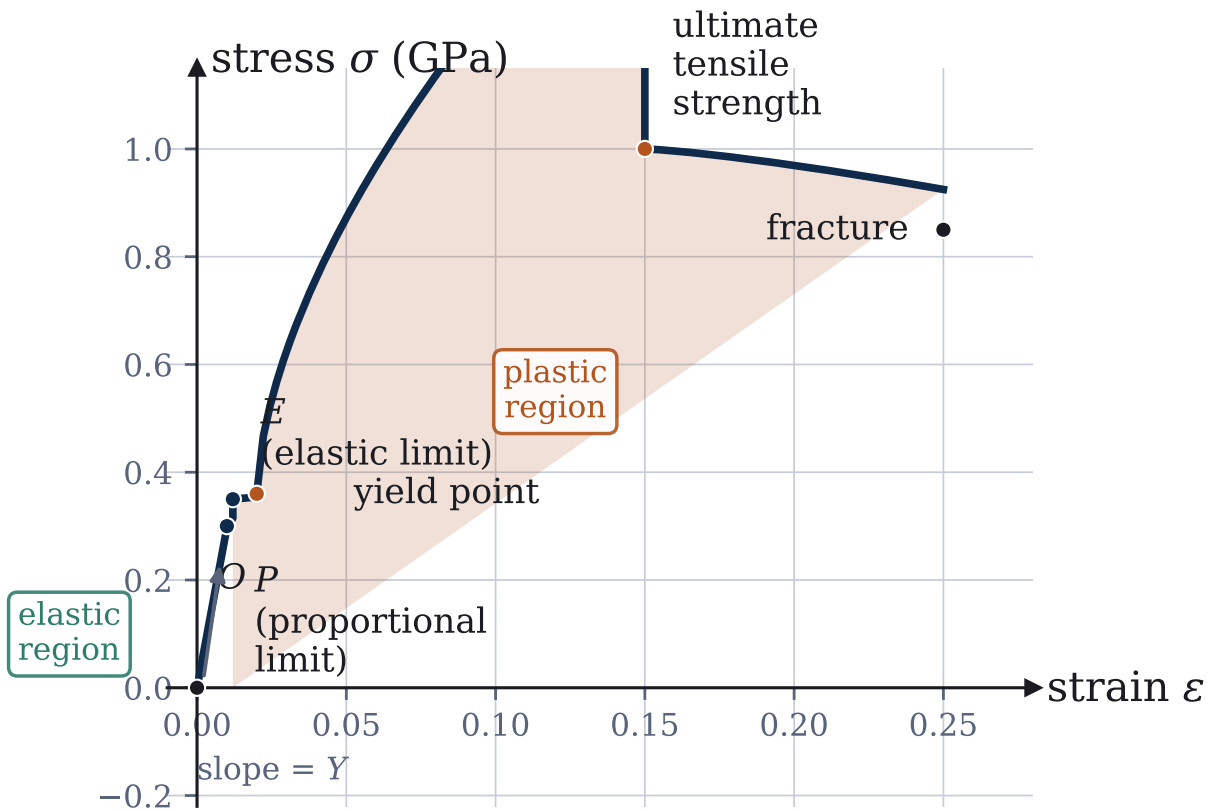


Figure 8.1 A stress (y-axis) vs strain (x-axis) curve for a ductile metal. Mark the linear portion $O \rightarrow P$ (proportional limit), point E slightly beyond (elastic limit), the yield point, the gently rising plastic region up to the ultimate tensile strength (peak), and the fracture point where the curve ends. Shade the elastic region green and the plastic region orange.]

Speed method: Since the material is the same, Y is identical for both — *don't compute it*.

From $\Delta L = \frac{FL}{AY} = \frac{FL}{\pi r^2 Y}$, with F, Y, π common:

$$\Delta L \propto \frac{L}{r^2}$$

$$\frac{\Delta L_A}{\Delta L_B} = \frac{L_A/r_A^2}{L_B/r_B^2} = \frac{L/r^2}{2L/(2r)^2} = \frac{L/r^2}{2L/4r^2} = \frac{1}{1/2} = 2$$

So **A elongates twice as much**. Mentally: B is longer ($\times 2$, stretches more) but fatter ($\times 4$ area, stretches less), net effect $\div 2$.

On elasticity: A stretches more, but both are the *same material* with the *same Y* — so they are *equally* elastic. Elasticity is a material property; geometry changes the *stretch*, never the *elasticity*. Eliminate any option claiming one is “more elastic.”

Example 3 — [JEE Main Level] A composite hangs vertically: a steel wire (length 1.0 m, area 1.0 mm^2 , $Y_s = 2.0 \times 10^{11} \text{ Pa}$) joined end-to-end to a copper wire (length 1.0 m, area 2.0 mm^2 , $Y_c = 1.0 \times 10^{11} \text{ Pa}$). A 10 kg mass hangs from the bottom. Find the total elongation. (Take $g = 10 \text{ m s}^{-2}$.)

Two concepts: (i) wires in **series** carry the *same* tension, and (ii) total elongation is the *sum* of individual elongations.

Tension throughout: $F = Mg = 100 \text{ N}$.

$$\text{Steel: } \Delta L_s = \frac{FL_s}{A_s Y_s} = \frac{100 \times 1.0}{(1.0 \times 10^{-6})(2.0 \times 10^{11})} = \frac{100}{2.0 \times 10^5} = 5.0 \times 10^{-4} \text{ m}$$

$$\text{Copper: } \Delta L_c = \frac{FL_c}{A_c Y_c} = \frac{100 \times 1.0}{(2.0 \times 10^{-6})(1.0 \times 10^{11})} = \frac{100}{2.0 \times 10^5} = 5.0 \times 10^{-4} \text{ m}$$

$$\Delta L_{\text{total}} = \Delta L_s + \Delta L_c = 1.0 \times 10^{-3} \text{ m} = \boxed{1.0 \text{ mm}}$$

(Here both pieces happen to stretch equally because their LY/A -balances coincide — a nice coincidence, not a rule.)

Example 4 — [JEE Advanced Level] A uniform rod of mass M , length L , and uniform cross-section A lies on a frictionless horizontal table and rotates with angular velocity ω about a vertical axis through **one end**. Derive the elongation of the rod. Evaluate it for $M = 2.0 \text{ kg}$, $L = 1.0 \text{ m}$, $\omega = 10 \text{ rad s}^{-1}$, $A = 1.0 \text{ mm}^2$, $Y = 2.0 \times 10^{11} \text{ Pa}$.

The tension is non-uniform: at a distance x from the axis, the tension must supply the centripetal force for *all the mass beyond x*. With linear mass density $\lambda = M/L$, consider the portion from x to L :

$$T(x) = \int_x^L \lambda \omega^2 x' dx' = \frac{M\omega^2}{2L} (L^2 - x^2)$$

A small element dx at position x feels stress $T(x)/A$ and so stretches by $d(\Delta L) = \frac{T(x)}{AY} dx$.

Integrating:

$$\Delta L = \int_0^L \frac{M\omega^2}{2LAY} (L^2 - x^2) dx = \frac{M\omega^2}{2LAY} \left(L^3 - \frac{L^3}{3} \right) = \boxed{\frac{M\omega^2 L^2}{3AY}}$$

Numerically:

$$\Delta L = \frac{(2.0)(10)^2(1.0)^2}{3(1.0 \times 10^{-6})(2.0 \times 10^{11})} = \frac{200}{6.0 \times 10^5} \approx 3.3 \times 10^{-4} \text{ m} \approx 0.33 \text{ mm}$$

The $\frac{1}{3}$ here (versus the $\frac{1}{2}$ for self-weight) comes from the tension varying *quadratically* with x , not linearly — a classic discriminator between students who memorise and students who derive.

Section 5: Practice Exercises

1. **[CBSE]** A brass wire 2.0 m long and of cross-section 0.50 mm^2 is stretched by 1.0 mm under a load. If $Y_{\text{brass}} = 1.0 \times 10^{11} \text{ Pa}$, find the load. (Take $g = 10 \text{ m s}^{-2}$.)
2. **[NEET]** The bulk modulus of water is $2.2 \times 10^9 \text{ Pa}$. What fractional change in volume occurs when water is subjected to an extra pressure of $1.1 \times 10^7 \text{ Pa}$?
3. **[JEE Main]** A wire elongates by 1 mm under a load. A second wire of the same material but with double the length and double the diameter is subjected to the same load. Find its elongation.
4. **[JEE Main]** The breaking stress of a metal is $4.0 \times 10^8 \text{ Pa}$ and its density is $8.0 \times 10^3 \text{ kg m}^{-3}$. Find the greatest length of a wire of this metal that can hang vertically without breaking. (Take $g = 10 \text{ m s}^{-2}$.)
5. **[JEE Advanced]** A solid cube of side L and shear modulus η has its bottom face fixed. A tangential force F is applied to the top face. Show that the top face displaces by $x = \frac{FL}{\eta L}$, and find x for $L = 0.10 \text{ m}$, $\eta = 5.0 \times 10^{10} \text{ Pa}$, $F = 5.0 \times 10^5 \text{ N}$.

Answers: 1. 25 N 2. 5.0×10^{-3} (i.e. 0.5%) 3. 0.5 mm 4. $5.0 \times 10^3 \text{ m} = 5 \text{ km}$ 5. $x = 1.0 \times 10^{-4} \text{ m} = 0.1 \text{ mm}$

Section 6: Crack the MCQ

Q1. A wire of length L and area A has Young's modulus Y . It is stretched by an amount ℓ . The work done in stretching the wire is: (a) $\frac{Y A \ell^2}{2L}$ (b) $\frac{Y A \ell^2}{L}$ (c) $\frac{Y A \ell}{2L}$ (d) $\frac{Y \ell^2}{2AL}$

Correct: (a). Work done $= \frac{1}{2} F \ell = \frac{1}{2} \left(\frac{Y A \ell}{L} \right) \ell = \frac{Y A \ell^2}{2L}$, using $F = \frac{Y A \ell}{L}$ from $Y = \frac{FL}{A\ell}$.

Why the traps: **(b)** forgets the factor $\frac{1}{2}$ — the force builds up *gradually* from 0 to F , so we use the average $F/2$, not F . **(c)** drops a power of ℓ (dimensionally wrong — a quick dimension check kills it). **(d)** misplaces A in the denominator; energy must rise with more material, so A belongs on top.

Q2. Two wires of the same material and same length have radii r and $2r$. They are stretched by the same force. The ratio of their elongations (thin : thick) is: (a) 1 : 1 (b) 2 : 1 (c) 4 : 1 (d) 1 : 4

Correct: (c). $\Delta L \propto \frac{1}{A} = \frac{1}{\pi r^2}$, so $\frac{\Delta L_{\text{thin}}}{\Delta L_{\text{thick}}} = \frac{(2r)^2}{r^2} = 4$. *Why the traps:* **(b)** uses radius to the first power instead of squaring it — the most frequent error. **(a)** assumes “same material, same length $\Rightarrow$ same stretch,” ignoring area entirely. **(d)** is the right magnitude but flipped — careless ordering of the ratio.

Q3. Which statement is correct? (a) Steel is less elastic than rubber because it stretches less. (b) A material with a larger Young’s modulus is more elastic. (c) Strain has the unit pascal. (d) Bulk modulus applies only to solids.

Correct: (b). Higher $Y \Rightarrow$ greater stress needed for the same strain $\Rightarrow$ stronger restoring tendency $\Rightarrow$ more elastic. *Why the traps:* **(a)** is the headline misconception — “stretches more” feels like “more elastic” but it’s backwards. **(c)** confuses strain (dimensionless) with stress/modulus (pascal). **(d)** is inverted: bulk modulus applies to *all* states of matter — in fact gases and liquids possess *only* bulk modulus.

Q4. A material has Poisson’s ratio $\nu = 0.5$. When stretched, the change in its volume is: (a) positive (b) negative (c) zero (d) cannot be determined

Correct: (c). $\nu = 0.5$ marks the incompressible limit: the lateral contraction exactly compensates the longitudinal extension, so volume stays constant ($\Delta V = 0$). (For a thin rod, fractional volume change $\approx \varepsilon(1 - 2\nu)$, which is zero when $\nu = 0.5$.) *Why the traps:* **(a)** assumes stretching always adds volume — true only when $\nu < 0.5$. **(b)** would require $\nu > 0.5$, which is physically impossible for ordinary materials. **(d)** gives up too early; the value $\nu = 0.5$ is precisely the special case that *can* be determined.

Section 7: Common Student Pitfalls & Pro-Tips

- **Radius vs. diameter.** Examiners love quoting the *diameter*. Forgetting to halve it before squaring throws your area off by a factor of **4**, and your Y along with it. Underline “radius” or “diameter” the moment you read the question.
- **“More stretch = more elastic.”** Wrong. Higher Young’s modulus means *more* elastic. Rubber stretches a lot but is *less* elastic than steel.
- **Putting units on strain.** Strain is a pure ratio — dimensionless, no pascals, no metres. Only stress and the moduli carry pascals.
- **Dropping the $\frac{1}{2}$ in energy and self-weight problems.** Elastic energy uses average force ($\frac{1}{2}F\ell$); self-weight elongation uses effective length $L/2$. The $\frac{1}{2}$ is physics, not decoration.

Pro-Tip: For “same material” comparisons, **never compute Y** . Cancel it immediately and reduce everything to a proportionality: $\Delta L \propto \frac{L}{r^2}$ (or $\frac{FL}{r^2}$). A 30-second ratio replaces a messy two-line calculation — and you can’t make an arithmetic slip on a number you never wrote down.

Section 8: Cheat Sheet Box

ELASTICITY — RAPID RECALL - **Stress** $\sigma = \frac{F}{A} \rightarrow$ unit **Pa**, dimensions

$[ML^{-1}T^{-2}]$. **Strain** $\varepsilon = \frac{\Delta x}{x} \rightarrow$ **dimensionless**, no unit. - **Hooke's law:** $\sigma = E\varepsilon$

within the elastic limit; E depends on *material only*. - **Young's:** $Y = \frac{FL}{A\Delta L}$

| **Bulk:** $B = -\frac{P}{\Delta V/V}$, $k = \frac{1}{B}$ | **Rigidity:** $\eta = \frac{\tau}{\phi}$ - **Self-weight elongation:**

$\Delta L = \frac{MgL}{2AY} = \frac{\rho g L^2}{2Y}$ | **Max hanging length:** $\ell_{\max} = \frac{\sigma_{\text{break}}}{\rho g}$ - **Elastic energy:**

$U = \frac{1}{2}\sigma\varepsilon V = \frac{1}{2}F\Delta L$; energy density $u = \frac{1}{2}\sigma\varepsilon = \frac{\sigma^2}{2Y}$ - Solids $\rightarrow$ all three moduli;
liquids & gases $\rightarrow$ bulk modulus only.

Memory aids: - “Stress is the squeeze, strain is the change.” - “More Y , more elastic.” (Steel beats rubber.) - Stress-strain milestones in order: **Proportional** $\rightarrow$ **Elastic** $\rightarrow$ **Yield** $\rightarrow$ **Ultimate** $\rightarrow$ **Fracture** $\rightarrow$ “Please Eat Your Usual Food.”

Subtopic 02: Elastic Moduli

Notation (carried from Subtopic 01): stress σ (shear τ) · strain ε (shear ϕ , radians) · Young's Y · Bulk B · Rigidity η · Poisson's ratio ν (some texts write this σ — we keep ν to avoid clashing with stress) · compressibility $k = 1/B$.

This subtopic builds directly on Subtopic 01. There you met the four elastic constants and learned where each comes from. Here we put them to work: treating a wire as a spring, combining wires, handling gases and fluids under pressure, and discovering that the four constants are secretly only two.

Exam Relevance: A workhorse subtopic. JEE Main loves the “wire-as-a-spring” idea and series/parallel combinations (1 question almost every year). NEET favours bulk modulus / compressibility numericals and “which modulus applies” conceptuels. JEE Advanced reaches for the relations between elastic constants (Y, B, η, ν) and process-dependent gas moduli. CBSE Boards ask a 2-3 mark bulk-modulus or shear-modulus numerical.

Section 1: Concept Intuition

In Subtopic 01 you learned that a material can be deformed in three independent ways — you can **stretch** it (length changes), **shear** it (shape changes), or **squeeze** it from all sides (volume changes) — and each has its own “stiffness number”: Young's modulus Y , the modulus of rigidity η , and the bulk modulus B . Poisson's ratio ν then tells you how a stretch in one direction quietly produces a shrink in the perpendicular directions.

The natural next question — and the whole point of this subtopic — is: *how do I actually use these numbers?* Two big ideas unlock everything.

Big idea 1: A loaded wire is just a spring. Hang a mass from a wire and it stretches by an amount proportional to the load — exactly like a spring obeying $F = kx$. Think about it this way... you already know $Y = \frac{FL}{A\Delta L}$. Rearrange it: $F = \left(\frac{YA}{L}\right)\Delta L$. The quantity in brackets plays the role of the spring constant. So **every rod is a spring with stiffness** $k = \frac{YA}{L}$. This single re-interpretation lets you treat any tangle of wires using the spring rules you already know from Class 11 mechanics: wires joined end-to-end behave like springs *in series* (soft, they add their stretches), and wires placed side-by-side sharing a load behave like springs *in parallel* (stiff, they add their stiffnesses). A thick steel cable on a crane is a very stiff spring; the long thin string of a guitar is a soft one.

A relatable picture: imagine a bullock cart with two ropes pulling a load. If you tie the ropes one-after-the-other (series), the load can sag more and the *weaker* rope dominates. If you tie them side-by-side (parallel), they share the strain and the pair is *stronger* than either alone. Springs, wires, ropes — same arithmetic.

Big idea 2: The four constants are not independent — there are really only two. This is the elegant secret of elasticity. For a uniform, isotropic material (same properties in every direction), if you know any *two* of $\{Y, B, \eta, \nu\}$, the other two are fixed automatically.

They are bound together by:

$$Y = 2\eta(1 + \nu) = 3B(1 - 2\nu)$$

Physically this makes sense: stretching, shearing, and squeezing are not separate magical properties — they all trace back to the *same* inter-atomic bonds you imagined in Subtopic 01. Disturb those bonds one way and the material's response in every other way is already determined.

Limiting conditions & assumptions. The relations between constants hold **only for homogeneous, isotropic materials**. They fail for wood (stiffer along the grain than across it), for fibre composites, and for single crystals — anisotropic materials need a whole matrix of constants, not just two. The spring picture ($k = YA/L$) assumes we stay within the elastic limit and that the wire's weight is negligible compared with the load. And remember the rule from Subtopic 01: a gas or liquid has **no** Y **and** **no** η — it cannot be stretched or sheared, only compressed — so for fluids, B is the *only* modulus that exists.

Section 2: Key Formulas & Definitions

The four constants (compact recap):

$$Y = \frac{\sigma}{\varepsilon} = \frac{FL}{A\Delta L}, \quad B = -\frac{P}{\Delta V/V}, \quad \eta = \frac{\tau}{\phi}, \quad \nu = -\frac{\text{lateral strain}}{\text{longitudinal strain}}$$

All three moduli: unit Pa, dimensions $[ML^{-1}T^{-2}]$. Poisson's ratio: dimensionless, with $0 \leq \nu \leq 0.5$ for ordinary materials (theoretical bound $-1 \leq \nu \leq 0.5$).

Wire as a spring:

$$F = k\Delta L \quad \text{with} \quad k = \frac{YA}{L} \quad (k \propto A, \quad k \propto 1/L)$$

Combinations (treat wires exactly like springs):

$$\text{Series (same force): } \frac{1}{k_{\text{eq}}} = \frac{1}{k_1} + \frac{1}{k_2}, \quad \text{Parallel (same stretch): } k_{\text{eq}} = k_1 + k_2$$

For two rods of **equal length and area** joined end-to-end (series), the equivalent Young's modulus is the *harmonic-type* mean:

$$Y_{\text{eq}} = \frac{2Y_1Y_2}{Y_1 + Y_2}$$

Bulk modulus & compressibility:

$$k = \frac{1}{B} \text{ (unit Pa}^{-1}\text{)}, \quad \frac{\Delta V}{V} = -\frac{\Delta P}{B}, \quad \frac{\Delta \rho}{\rho} = +\frac{\Delta P}{B}$$

Ordering of stiffness against compression: $B_{\text{solids}} > B_{\text{liquids}} \gg B_{\text{gases}}$.

Bulk modulus of a gas is process-dependent (not a fixed material constant):

$$B_{\text{isothermal}} = P, \quad B_{\text{adiabatic}} = \gamma P, \quad B_{\text{polytropic}} = nP, \quad \frac{B_{\text{adiabatic}}}{B_{\text{isothermal}}} = \gamma$$

Relations between elastic constants (isotropic material — only 2 are independent):

$$Y = 2\eta(1 + \nu) = 3B(1 - 2\nu), \quad \frac{9}{Y} = \frac{1}{B} + \frac{3}{\eta}, \quad \nu = \frac{3B - 2\eta}{2(3B + \eta)}$$

Section 3: Derivations & Key Procedures

Derivation 1 — Wire as a spring, and the series rule. Start from the definition of Young's modulus for a wire of length L , area A , material Y , stretched by ΔL under force F :

$$Y = \frac{FL}{A \Delta L} \Rightarrow F = \frac{YA}{L} \Delta L$$

Comparing with the spring law $F = k \Delta L$, the wire's effective force constant is $k = \frac{YA}{L}$. *Why this matters:* the entire toolkit of spring combinations now transfers over.

Now join two such wires end-to-end and hang a load. Both carry the **same tension** F (the lower wire transmits the full load to the upper one), and the total stretch is the **sum**:

$$\Delta L_{\text{total}} = \Delta L_1 + \Delta L_2 = \frac{F}{k_1} + \frac{F}{k_2} \Rightarrow \frac{1}{k_{\text{eq}}} = \frac{1}{k_1} + \frac{1}{k_2}$$

For two rods of identical length L and area A but moduli Y_1, Y_2 : $k_i = Y_i A / L$. The combined rod has length $2L$, area A , and some effective modulus Y_{eq} with $k_{\text{eq}} = \frac{Y_{\text{eq}} A}{2L}$. Substituting:

$$\frac{2L}{Y_{\text{eq}} A} = \frac{L}{Y_1 A} + \frac{L}{Y_2 A} \Rightarrow \frac{2}{Y_{\text{eq}}} = \frac{1}{Y_1} + \frac{1}{Y_2} \Rightarrow \boxed{Y_{\text{eq}} = \frac{2Y_1 Y_2}{Y_1 + Y_2}}$$

Note this is **not** the simple average $\frac{1}{2}(Y_1 + Y_2)$ — a favourite trap.

Derivation 2 — Bulk modulus of a gas depends on the process. A gas has no fixed bulk modulus because *how* you compress it changes its resistance. Begin with $B = -V \frac{dP}{dV}$.

Isothermal (slow, constant temperature): $PV = \text{constant}$. Differentiating, $P dV + V dP = 0 \Rightarrow -V \frac{dP}{dV} = P$. Hence $B_{\text{iso}} = P$.

Adiabatic (fast, no heat exchange): $PV^\gamma = \text{constant}$. Differentiating, $\gamma PV^{\gamma-1} dV + V^\gamma dP = 0 \Rightarrow -V \frac{dP}{dV} = \gamma P$. Hence $B_{\text{adia}} = \gamma P$.

Since $\gamma > 1$, a gas always resists *sudden* compression more stiffly than *slow* compression — which is exactly why sound (an adiabatic compression wave) travels faster than the isothermal estimate Newton first guessed.

Derivation 3 — Tying the constants together. Take the two source relations $Y = 2\eta(1 + \nu)$ and $Y = 3B(1 - 2\nu)$ and eliminate ν . From the first, $\nu = \frac{Y}{2\eta} - 1$; from the second, $\nu = \frac{1}{2} - \frac{Y}{6B}$. Equating and simplifying:

$$\frac{Y}{2\eta} - 1 = \frac{1}{2} - \frac{Y}{6B} \Rightarrow \frac{1}{\eta} + \frac{1}{3B} = \frac{3}{Y} \Rightarrow \boxed{\frac{9}{Y} = \frac{1}{B} + \frac{3}{\eta}}$$

Procedure — Which modulus do I use? A quick decision rule: 1. Is the deformation a **change in length** under a pull/push along one axis? → Young's modulus Y . 2. Is it a **change in shape** (faces sliding, twisting), volume unchanged? → Rigidity η . 3. Is it a **change in volume** under all-round pressure, shape unchanged? → Bulk modulus B . 4. Is the material a **fluid** (liquid/gas)? → Only B is available; for a gas, check whether the process is isothermal ($B = P$) or adiabatic ($B = \gamma P$).

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A solid copper cube of edge 10 cm is subjected to a uniform pressure of 7.0×10^7 Pa on all faces. If the bulk modulus of copper is 1.4×10^{11} Pa, find the change in its volume.

Given: edge = 0.10 m $\Rightarrow V = (0.10)^3 = 1.0 \times 10^{-3} \text{ m}^3$; $\Delta P = 7.0 \times 10^7$ Pa; $B = 1.4 \times 10^{11}$ Pa.

From $B = -\frac{\Delta P}{\Delta V/V}$:

$$\Delta V = -\frac{\Delta P \cdot V}{B} = -\frac{(7.0 \times 10^7)(1.0 \times 10^{-3})}{1.4 \times 10^{11}} = -\frac{7.0 \times 10^4}{1.4 \times 10^{11}}$$

$$\boxed{\Delta V = -5.0 \times 10^{-7} \text{ m}^3 = -0.50 \text{ cm}^3}$$

The minus sign confirms a *decrease* in volume. Tiny, as expected — copper is extremely incompressible.

Example 2 — [NEET Speed Trap] A fixed mass of a diatomic gas ($\gamma = 1.4$) is at pressure 1.0×10^5 Pa. Find its bulk modulus (i) when compressed slowly, and (ii) when compressed suddenly.

The trap: students hunt for a “bulk modulus of the gas” as if it were a tabulated material constant, or apply a solid's formula. For a gas, B is set by the process.

Speed method — no calculation, just recall: - Slow compression = **isothermal** $\Rightarrow B = P = 1.0 \times 10^5$ Pa. - Sudden compression = **adiabatic** $\Rightarrow B = \gamma P = 1.4 \times 1.0 \times 10^5 = 1.4 \times 10^5$ Pa.

The whole question collapses to “is it slow or fast?” If a NEET option offers a single fixed value for a gas's bulk modulus independent of the process, eliminate it instantly.

Example 3 — [JEE Main Level] Two wires of the same material ($Y = 2.0 \times 10^{11}$ Pa) and the same length $L = 1.0$ m are joined end-to-end. The upper wire has radius $R = 0.50$ mm

and the lower has radius $2R = 1.0 \text{ mm}$. A load of 100 N hangs from the bottom. Find the total elongation, and state which wire stores more elastic energy.

Two concepts: wires in **series** carry the same force; and elastic energy $U = \frac{1}{2}F \Delta L$.

Areas: $A_1 = \pi R^2 = \pi(5.0 \times 10^{-4})^2 = 7.85 \times 10^{-7} \text{ m}^2$; $A_2 = \pi(2R)^2 = 4A_1 = 3.14 \times 10^{-6} \text{ m}^2$.

$$\Delta L_1 = \frac{FL}{A_1 Y} = \frac{100 \times 1.0}{(7.85 \times 10^{-7})(2.0 \times 10^{11})} = \frac{100}{1.571 \times 10^5} = 6.4 \times 10^{-4} \text{ m}$$

$$\Delta L_2 = \frac{FL}{A_2 Y} = \frac{\Delta L_1}{4} = 1.6 \times 10^{-4} \text{ m}$$

$$\Delta L_{\text{total}} = \Delta L_1 + \Delta L_2 = 8.0 \times 10^{-4} \text{ m} = 0.80 \text{ mm}$$

Since both wires carry the same force F , $U \propto \Delta L$. The **thinner upper wire** ($\Delta L_1 = 4 \Delta L_2$) therefore stores **four times** the energy of the thicker one — thinner wires both stretch more and soak up more energy. (Useful intuition: the weakest link does the most work and is the first to snap.)

Example 4 — [JEE Advanced Level] A homogeneous, isotropic material has Young's modulus $Y = 1.2 \times 10^{11} \text{ Pa}$ and Poisson's ratio $\nu = 0.25$. Determine its bulk modulus B and shear modulus η , and verify your answers are mutually consistent.

Use the two master relations directly.

$$B = \frac{Y}{3(1-2\nu)} = \frac{1.2 \times 10^{11}}{3(1-0.50)} = \frac{1.2 \times 10^{11}}{1.5} = \boxed{8.0 \times 10^{10} \text{ Pa}}$$

$$\eta = \frac{Y}{2(1+\nu)} = \frac{1.2 \times 10^{11}}{2(1.25)} = \frac{1.2 \times 10^{11}}{2.5} = \boxed{4.8 \times 10^{10} \text{ Pa}}$$

Consistency check via $\frac{9}{Y} = \frac{1}{B} + \frac{3}{\eta}$:

$$\frac{1}{B} + \frac{3}{\eta} = \frac{1}{8.0 \times 10^{10}} + \frac{3}{4.8 \times 10^{10}} = (1.25 + 6.25) \times 10^{-11} = 7.5 \times 10^{-11}$$

$$\frac{9}{Y} = \frac{9}{1.2 \times 10^{11}} = 7.5 \times 10^{-11} \checkmark$$

The two routes agree, confirming the material is described self-consistently by just two independent constants.

Section 5: Practice Exercises

1. **[CBSE]** One litre of water is subjected to a pressure increase of $2.0 \times 10^7 \text{ Pa}$. If the bulk modulus of water is $2.0 \times 10^9 \text{ Pa}$, find the decrease in its volume.
2. **[NEET]** At a pressure of $1.0 \times 10^5 \text{ Pa}$, find the isothermal and adiabatic bulk moduli of a diatomic gas ($\gamma = 1.4$).

3. **[JEE Main]** A wire has force constant k . It is cut into two equal pieces, and the two pieces are then connected side-by-side (in parallel). Find the effective force constant of the combination.
4. **[JEE Main]** Two rods of equal length and equal cross-sectional area, having Young's moduli Y and $3Y$, are joined end-to-end. Find the equivalent Young's modulus of the composite rod.
5. **[JEE Advanced]** For a certain isotropic material, the bulk modulus is twice its shear modulus ($B = 2\eta$). Find its Poisson's ratio.

Answers: 1. $1.0 \times 10^{-5} \text{ m}^3 = 10 \text{ cm}^3$ 2. isothermal $1.0 \times 10^5 \text{ Pa}$, adiabatic $1.4 \times 10^5 \text{ Pa}$
 3. $4k$ 4. $\frac{3Y}{2}$ 5. $\nu = \frac{2}{7} \approx 0.29$

Section 6: Crack the MCQ

Q1. A wire stretched by a hanging mass behaves as a spring of force constant k . If the length is halved (same material, same cross-section), the new force constant is: (a) $k/2$ (b) k (c) $2k$ (d) $4k$

Correct: (c). Since $k = \frac{YA}{L}$, halving L doubles k . *Why the traps:* **(a)** assumes $k \propto L$ — backwards; shorter wires are *stiffer*. **(b)** confuses k (geometry-dependent) with Y (material-only); Y stays the same but k does not. **(d)** is the area-scaling or parallel-combination answer, misapplied to a length change.

Q2. Two rods of identical dimensions but Young's moduli Y_1 and Y_2 are joined end-to-end. The equivalent Young's modulus is: (a) $Y_1 + Y_2$ (b) $\frac{Y_1 + Y_2}{2}$ (c) $\frac{2Y_1Y_2}{Y_1 + Y_2}$ (d) $\sqrt{Y_1Y_2}$

Correct: (c). End-to-end = series; stretches add, giving $\frac{2}{Y_{\text{eq}}} = \frac{1}{Y_1} + \frac{1}{Y_2}$. *Why the traps:* **(b)** the arithmetic mean is the single most common error — series springs do *not* average. **(a)** is the parallel (side-by-side) result. **(d)** the geometric mean simply doesn't arise here.

Q3. A fluid (liquid or gas) can possess: (a) only Young's modulus (b) only bulk modulus (c) only modulus of rigidity (d) all three moduli

Correct: (b). Fluids flow — they cannot sustain a shear stress (so no η) and cannot be stretched along a line (so no Y). They resist only volume change, so only B exists. *Why the traps:* **(a)** and **(c)** assign solid-only moduli to fluids. **(d)** forgets that “having a length and a definite shape” is a prerequisite for Y and η — which fluids lack.

Q4. For a given gas at pressure P , the ratio of its adiabatic bulk modulus to its isothermal bulk modulus is: (a) 1 (b) γ (c) $1/\gamma$ (d) γ^2

Correct: (b). $B_{\text{adia}} = \gamma P$ and $B_{\text{iso}} = P$, so the ratio is γ . *Why the traps:* **(a)** wrongly treats the gas's bulk modulus as process-independent. **(c)** inverts the ratio. **(d)** double-counts γ , perhaps confusing it with a speed-of-sound expression.

Section 7: Common Student Pitfalls & Pro-Tips

- **Averaging series moduli.** For rods joined end-to-end, the equivalent Young's modulus is $\frac{2Y_1Y_2}{Y_1 + Y_2}$, *not* $\frac{1}{2}(Y_1 + Y_2)$. Series = stretches add = reciprocals add. Map it to springs and you won't slip.
- **Treating a gas's bulk modulus as fixed.** A gas has *no* single bulk modulus. Always ask: isothermal ($B = P$) or adiabatic ($B = \gamma P$)? Using the wrong one is an instant error.
- **Forgetting only two constants are independent.** If a problem gives you any two of $\{Y, B, \eta, \nu\}$, the other two are already determined. Don't treat them as four free numbers.
- **Confusing k with Y .** Young's modulus is a *material* property (geometry-free). The spring constant $k = YA/L$ is *geometry-dependent*. Cutting or rejoining a wire changes k , never Y .

Pro-Tip: Whenever wires are connected, immediately redraw them as springs with $k = YA/L$ and use $\frac{1}{k_{\text{eq}}} = \sum \frac{1}{k_i}$ (series) or $k_{\text{eq}} = \sum k_i$ (parallel). You convert an unfamiliar elasticity problem into a Class-11 spring problem you've already mastered — and sidestep every “average vs. reciprocal” trap.

Section 8: Cheat Sheet Box

ELASTIC MODULI — RAPID RECALL - Wire = spring: $k = \frac{YA}{L}$ ($k \propto A$, $k \propto 1/L$). Series: $\frac{1}{k_{\text{eq}}} = \sum \frac{1}{k_i}$; Parallel: $k_{\text{eq}} = \sum k_i$. - **Series of two equal rods:** $Y_{\text{eq}} = \frac{2Y_1Y_2}{Y_1 + Y_2}$ (NOT the average). - **Compressibility:** $k = \frac{1}{B}$ (Pa^{-1}); $\frac{\Delta V}{V} = -\frac{\Delta P}{B}$; $\frac{\Delta \rho}{\rho} = +\frac{\Delta P}{B}$. Stiffness: $B_{\text{solid}} > B_{\text{liquid}} \gg B_{\text{gas}}$. - **Gas bulk modulus (process-dependent):** isothermal $B = P$; adiabatic $B = \gamma P$; ratio $= \gamma$. - **Only 2 of 4 constants are independent:** $Y = 2\eta(1 + \nu) = 3B(1 - 2\nu)$; $\frac{9}{Y} = \frac{1}{B} + \frac{3}{\eta}$. - **Which modulus?** length $\rightarrow Y$, shape $\rightarrow \eta$, volume $\rightarrow B$. Fluids $\rightarrow B$ only.

Memory aids: - “Short and fat is stiff” — k rises as L falls and A grows. - “Slow gas is soft, fast gas is stiff” — isothermal P vs adiabatic γP . - “Two knowns rule them all” — any two elastic constants fix the rest.

Subtopic 03: Poisson's Ratio and Elastic Energy

Notation (carried from Subtopics 01-02): stress σ · strain ε · Young's Y · Bulk B · Rigidity η · Poisson's ratio ν (some texts write this σ ; we keep ν to avoid clashing with stress) · energy density u .

*In Subtopic 02 you saw that Poisson's ratio links the four elastic constants. Here we follow ν to its most useful consequence — what stretching does to a body's **volume** — and then turn to the energy that every deformation quietly stores.*

Exam Relevance: CBSE Boards reliably ask the *derivation* of elastic potential energy in a stretched wire (3 marks) and a definition of Poisson's ratio. JEE Main mixes the volume-change relation with Young's modulus, and tests energy density. NEET favours the “area under the stress-strain curve” conceptual and quick energy-density numericals. JEE Advanced loves the sudden-loading energy-partition problem (where half the work turns to heat).

Section 1: Concept Intuition

Take a thick rubber band and stretch it between your hands. Watch closely: as it lengthens, it also gets visibly **thinner**. Stretch a piece of chewing gum or roll out a ball of atta — pulling in one direction always squeezes the other directions. This sideways shrinking that accompanies a lengthwise stretch is what **Poisson's ratio** measures: it is simply *how much thinner per unit longer*. A material with a large Poisson's ratio (like rubber, $\nu \approx 0.5$) necks down dramatically; one with a small Poisson's ratio (like cork, $\nu \approx 0$) barely changes width at all — which, incidentally, is exactly why cork is perfect for a wine bottle: push it in lengthwise and it doesn't bulge sideways to jam.

Think about it this way... here's the question that makes Poisson's ratio genuinely interesting. When you stretch a wire, it gets *longer* (volume wants to grow) but also *thinner* (volume wants to shrink). Which wins? Does the total volume go up, down, or stay put? The answer is governed entirely by ν :

$$\frac{\Delta V}{V} = (1 - 2\nu) \frac{\Delta L}{L}$$

If $\nu < 0.5$ (almost all real solids), the lengthening wins and the volume *increases* slightly when you pull. If $\nu = 0.5$ exactly (rubber, and most liquids), the thinning perfectly cancels the lengthening and the **volume stays constant** — the material is *incompressible*. This is the deep reason ν can't exceed 0.5 for ordinary materials: a value above 0.5 would mean stretching a body *shrinks* its volume, which inter-atomic forces don't permit.

Now the second half of this subtopic: **deformation stores energy**. Pull back the string of a bow (*dhanush*) and you feel resistance; let go and that stored energy launches the arrow. A *gulel* (catapult), a trampoline, a pole-vaulter's flexing pole — all are energy banks that take in work as they deform and pay it back as they recover. A stretched wire is the same idea in miniature. The work you do against the inter-atomic “springs” doesn't vanish; it is stored as **elastic potential energy**, ready to be released the instant the load is removed (within the elastic limit). The neat result we'll derive is that this stored energy is exactly

half of “force times stretch” — and understanding *why* it’s half, not the full product, is what separates a careful student from a careless one.

Limiting conditions & assumptions. The volume relation $\frac{\Delta V}{V} = (1 - 2\nu)\varepsilon$ is a *small-strain* (first-order) result — it assumes the changes are tiny, so we can ignore products of small quantities. All of it lives **within the elastic limit**; beyond it, energy is lost to permanent deformation and heat, and none of these clean formulas hold. And we assume the material is **isotropic** — Poisson’s ratio is a single number only when the material responds the same way in every direction.

Section 2: Key Formulas & Definitions

Poisson’s ratio:

$$\nu = -\frac{\text{lateral strain}}{\text{longitudinal strain}} = -\frac{\Delta r/r}{\Delta L/L} = -\frac{\Delta D/D}{\Delta L/L}$$

- Dimensionless, no units; depends only on the material. - Theoretical range: $-1 \leq \nu \leq 0.5$. Practical range for real solids: $0 \leq \nu \leq 0.5$. - Typical values: steel ≈ 0.28 – 0.30 , aluminium alloys ≈ 0.33 , rubber ≈ 0.5 , cork ≈ 0 . - The minus sign makes ν positive: lateral and longitudinal strains have *opposite* signs (stretch one way, shrink the other).

Volume change on stretching a rod (small strain):

$$\frac{\Delta V}{V} = (1 - 2\nu) \frac{\Delta L}{L}$$

- $\nu < 0.5 \Rightarrow$ volume increases | $\nu = 0.5 \Rightarrow$ volume constant (incompressible) | $\nu = 0 \Rightarrow \frac{\Delta V}{V} = \frac{\Delta L}{L}$ (no lateral contraction).

Elastic potential energy stored in a stretched wire (three equivalent forms):

$$U = \frac{1}{2} F \Delta L = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume} = \frac{1}{2} \frac{YA}{L} (\Delta L)^2$$

Elastic energy density (energy stored per unit volume) — equals the **area under the stress-strain curve**:

$$u = \frac{U}{\text{Volume}} = \frac{1}{2} \sigma \varepsilon = \frac{1}{2} Y \varepsilon^2 = \frac{\sigma^2}{2Y} \quad (\text{unit: J m}^{-3})$$

Gradual vs. sudden loading (energy bookkeeping): - *Gradual* (force builds slowly $0 \rightarrow F$): work done by external agent $= \frac{1}{2} F \Delta L = U$ stored; no heat. - *Sudden* (full load F applied at once): work done by load $= F \Delta L_{\text{max}}$; only half stored as U , the other half dissipated as heat. The maximum elongation is **twice** the static value.

Section 3: Derivations & Key Procedures

Derivation 1 — Volume change in terms of Poisson's ratio. Consider a cylindrical bar of length L and radius r , so its volume is $V = \pi r^2 L$. When stretched, both r and L change slightly. To see how V responds, take the logarithm and differentiate (the cleanest route for products):

$$\ln V = \ln \pi + 2 \ln r + \ln L \Rightarrow \frac{\Delta V}{V} = 2 \frac{\Delta r}{r} + \frac{\Delta L}{L}$$

Why logarithms? Because volume is a product of powers — differentiating the log turns the product into a simple sum of fractional changes.

Now bring in Poisson's ratio. By definition the lateral (radial) strain is $\frac{\Delta r}{r} = -\nu \frac{\Delta L}{L}$ (negative — the radius shrinks). Substituting:

$$\frac{\Delta V}{V} = 2 \left(-\nu \frac{\Delta L}{L} \right) + \frac{\Delta L}{L} = \boxed{(1 - 2\nu) \frac{\Delta L}{L}}$$

Read off the physics immediately: $\nu = 0.5 \Rightarrow \Delta V = 0$ (incompressible); $\nu < 0.5 \Rightarrow \Delta V > 0$ (volume grows on stretching). This single line explains the upper bound on ν .

Derivation 2 — Elastic potential energy of a stretched wire (CBSE favourite). Take a wire of length L , area A , material Y . Suppose it has already been stretched by some amount l . From Subtopic 02, the force needed to hold it there is $F(l) = \frac{YA}{L} l$.

To stretch it a tiny bit further, by dl , the work done is

$$dW = F(l) dl = \frac{YA}{L} l dl$$

The crucial point: the force is **not constant** — it grows from 0 to its final value as the wire stretches. So we *integrate* from 0 to the final elongation ΔL :

$$W = \int_0^{\Delta L} \frac{YA}{L} l dl = \frac{YA}{L} \cdot \frac{(\Delta L)^2}{2} = \frac{1}{2} \frac{YA}{L} (\Delta L)^2$$

Since the final force is $F = \frac{YA}{L} \Delta L$, this is simply

$$U = W = \boxed{\frac{1}{2} F \Delta L}$$

The factor $\frac{1}{2}$ is the *average* force (0 to F , mean $F/2$) at work — **not** the full F . To get the energy *density*, divide by the volume AL and regroup:

$$u = \frac{U}{AL} = \frac{\frac{1}{2} F \Delta L}{AL} = \frac{1}{2} \underbrace{\frac{F}{A}}_{\sigma} \underbrace{\frac{\Delta L}{L}}_{\varepsilon} = \frac{1}{2} \sigma \varepsilon$$

Procedure — Reading energy off the stress-strain curve. Because $u = \frac{1}{2} \sigma \varepsilon$ and the graph plots σ (y) against ε (x), the **area under the curve equals the elastic energy per unit volume** — not the total energy. To get total energy, multiply by the volume. (Watch the axes: a *force-extension* graph's area gives total energy directly; a *stress-strain* graph's area gives energy density.)

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A steel wire of length 2.0 m and cross-sectional area 1.0 mm^2 is stretched by 1.0 mm. Given $Y_{\text{steel}} = 2.0 \times 10^{11} \text{ Pa}$, find (a) the elastic potential energy stored and (b) the energy stored per unit volume.

Given: $L = 2.0 \text{ m}$, $A = 1.0 \times 10^{-6} \text{ m}^2$, $\Delta L = 1.0 \times 10^{-3} \text{ m}$.

First, the stretching force: $F = \frac{YA\Delta L}{L} = \frac{(2.0 \times 10^{11})(1.0 \times 10^{-6})(1.0 \times 10^{-3})}{2.0} = 100 \text{ N}$.

(a) $U = \frac{1}{2}F\Delta L = \frac{1}{2}(100)(1.0 \times 10^{-3}) = \boxed{0.05 \text{ J}}$

(b) Volume = $AL = (1.0 \times 10^{-6})(2.0) = 2.0 \times 10^{-6} \text{ m}^3$, so

$$u = \frac{U}{\text{Volume}} = \frac{0.05}{2.0 \times 10^{-6}} = \boxed{2.5 \times 10^4 \text{ J m}^{-3}}$$

Cross-check: $u = \frac{1}{2}\sigma\varepsilon = \frac{1}{2}(10^8)(5 \times 10^{-4}) = 2.5 \times 10^4 \text{ J m}^{-3} \checkmark$.

Example 2 — [NEET Speed Trap] Within the elastic limit, the area under a wire's stress-strain graph is found to be $8 \times 10^4 \text{ J m}^{-3}$. The wire has volume $5 \times 10^{-6} \text{ m}^3$. What is the total elastic energy stored?

The trap: students see “area under stress-strain curve” and write it down as the total energy. It is **not** — it's energy *per unit volume*.

Speed method: Area under stress-strain curve = u (energy density). Total energy is just $u \times \text{Volume}$:

$$U = u \times V = (8 \times 10^4)(5 \times 10^{-6}) = \boxed{0.4 \text{ J}}$$

If an option offered “ $8 \times 10^4 \text{ J}$ ” (the area read as total energy), it's the planted trap. The unit J m^{-3} on the area is your tell.

Example 3 — [JEE Main Level] A wire of length 1.0 m and diameter 1.0 mm is made of a material with $Y = 1.2 \times 10^{11} \text{ Pa}$ and Poisson's ratio $\nu = 0.30$. A tensile stress of $1.2 \times 10^8 \text{ Pa}$ is applied. Find (a) the longitudinal strain, (b) the lateral strain, and (c) the fractional change in volume.

Two concepts: Young's modulus gives the longitudinal strain; Poisson's ratio then gives the lateral strain and the volume change.

(a) $\varepsilon = \frac{\sigma}{Y} = \frac{1.2 \times 10^8}{1.2 \times 10^{11}} = \boxed{1.0 \times 10^{-3}}$

(b) Lateral strain = $-\nu\varepsilon = -(0.30)(1.0 \times 10^{-3}) = \boxed{-3.0 \times 10^{-4}}$ (the diameter shrinks).

(c) $\frac{\Delta V}{V} = (1 - 2\nu)\varepsilon = (1 - 0.60)(1.0 \times 10^{-3}) = (0.40)(1.0 \times 10^{-3}) = \boxed{4.0 \times 10^{-4}}$ The volume increases (since $\nu < 0.5$), by 0.04%.

Example 4 — [JEE Advanced Level] A block of mass m is attached to the lower end of a vertical wire (length L , area A , Young's modulus Y) and released **suddenly** from the position where the wire is just taut (unstretched). Show that the maximum elongation is

twice the static elongation, and find it for $m = 2.0 \text{ kg}$, $L = 2.0 \text{ m}$, $A = 1.0 \text{ mm}^2$, $Y = 2.0 \times 10^{11} \text{ Pa}$ ($g = 10 \text{ m s}^{-2}$).

When released suddenly, the block falls and stretches the wire until it momentarily stops at maximum elongation x_{max} (here KE = 0 at both ends). Apply energy conservation: the loss in gravitational PE equals the elastic energy stored.

$$mg x_{\text{max}} = \frac{1}{2} \frac{YA}{L} x_{\text{max}}^2$$

Cancel one x_{max} :

$$x_{\text{max}} = \frac{2mgL}{YA}$$

The static (gradual) elongation is $x_{\text{st}} = \frac{mgL}{YA}$, so indeed $x_{\text{max}} = 2 x_{\text{st}}$.

Why twice? Suddenly applied, the full weight mg acts through the entire descent, doing work $mg x_{\text{max}}$, but the wire can only store $\frac{1}{2} \frac{YA}{L} x_{\text{max}}^2$ — the mismatch (initially) shows up as kinetic energy, which then overshoots the equilibrium point by the same amount, doubling the stretch.

Numerically: $x_{\text{st}} = \frac{(2.0)(10)(2.0)}{(2.0 \times 10^{11})(1.0 \times 10^{-6})} = \frac{40}{2.0 \times 10^5} = 2.0 \times 10^{-4} \text{ m}$, so

$$x_{\text{max}} = 2 \times 2.0 \times 10^{-4} = 4.0 \times 10^{-4} \text{ m} = 0.40 \text{ mm}$$

Section 5: Practice Exercises

1. **[CBSE]** A wire is stretched by 2.0 mm when a gradually increasing force reaching 50 N is applied. Calculate the elastic potential energy stored in the wire.
2. **[NEET]** Within the elastic limit, what does the area under a stress-strain graph represent?
3. **[JEE Main]** A rod made of a material with Poisson's ratio 0.5 is stretched longitudinally. What is the fractional change in its volume?
4. **[JEE Main]** A wire is subjected to a stress of $2.0 \times 10^8 \text{ Pa}$. If the Young's modulus of its material is $2.0 \times 10^{11} \text{ Pa}$, find the elastic energy stored per unit volume.
5. **[JEE Advanced]** A load attached *gradually* to a wire produces an elongation x . If an identical load is attached *suddenly* (released from the just-taut position), find the maximum elongation in terms of x .

Answers: 1. 0.05 J 2. elastic potential energy *per unit volume* (energy density) 3. zero (volume unchanged — incompressible) 4. $1.0 \times 10^5 \text{ J m}^{-3}$ 5. $2x$

Section 6: Crack the MCQ

Q1. The theoretical limits within which Poisson's ratio can lie are: (a) 0 to 0.5 (b) -1 to 0.5 (c) -1 to 1 (d) 0 to 1

Correct: (b). Thermodynamic stability of an isotropic solid restricts ν to $-1 \leq \nu \leq 0.5$. *Why the traps:* (a) is the *practical* range for ordinary materials — a very common confusion between “what's theoretically allowed” and “what real metals show.” (c) and (d) overshoot the upper bound; $\nu > 0.5$ would mean stretching shrinks the volume, which is forbidden.

Q2. A material with Poisson's ratio exactly 0.5 is stretched along its length. Its volume: (a) increases (b) decreases (c) remains constant (d) first increases, then decreases

Correct: (c). From $\frac{\Delta V}{V} = (1-2\nu)\varepsilon$, setting $\nu = 0.5$ gives $(1-1)\varepsilon = 0$. *Why the traps:* (a) is true only for $\nu < 0.5$ — right idea, wrong value. (b) would require $\nu > 0.5$ (impossible). (d) invents a non-monotonic behaviour that the small-strain relation doesn't produce.

Q3. The area under the stress-strain curve (within the elastic limit) gives: (a) elastic potential energy (b) elastic potential energy per unit volume (c) elastic potential energy per unit length (d) Young's modulus

Correct: (b). Area $= \frac{1}{2}\sigma\varepsilon = u$, the energy *per unit volume*. *Why the traps:* (a) forgets that the axes are stress (force/area) and strain (dimensionless), whose product carries units of energy *per volume*, not energy. (c) is dimensionally wrong. (d) confuses the *area* with the *slope* — the slope, not the area, gives Y .

Q4. A wire of natural length L extends by ℓ when a force is *gradually* increased to a final value F . Which statement is **NOT** correct? (a) Work done by the external force $= \frac{1}{2}F\ell$ (b) Elastic potential energy stored $= \frac{1}{2}F\ell$ (c) Work done by the external force $= F\ell$ (d) No heat is produced in the wire

Correct (the false one): (c). For *gradual* loading the force averages $F/2$, so the work is $\frac{1}{2}F\ell$, not $F\ell$. *Why the others hold:* (a) and (b) are the correct $\frac{1}{2}F\ell$ result, and since loading is quasi-static all the work is stored as PE with (d) no heat lost. The $F\ell$ figure in (c) belongs to *sudden* loading — quoting it here is the classic mix-up.

Section 7: Common Student Pitfalls & Pro-Tips

- **Dropping the factor $\frac{1}{2}$.** Elastic energy is $\frac{1}{2}F \Delta L$, because the force builds up from zero — use the *average* force, not the final one. Writing $F \Delta L$ doubles your answer.
- **Confusing total energy with energy density.** Area under a *stress-strain* curve is energy *per unit volume*; multiply by volume for the total. Only a *force-extension* curve's area gives total energy directly.
- **Assuming volume is conserved when stretching.** Volume stays fixed *only* if $\nu = 0.5$. For metals ($\nu \approx 0.3$), stretching genuinely increases the volume.
- **Sign and range of ν .** By convention ν is quoted as a positive number (the minus sign in the definition already accounts for the opposite-sign strains). And ν can never exceed 0.5 for real materials.

Pro-Tip: For sudden-loading problems, skip the algebra and use the shortcut $x_{\max} = 2x_{\text{static}}$ — a suddenly dropped load always stretches a wire to *twice* its

gentle-loading elongation, with exactly half the input work ending up as heat. Recognising this on sight turns a full energy-conservation derivation into a one-line answer.

Section 8: Cheat Sheet Box

POISSON'S RATIO & ELASTIC ENERGY — RAPID RECALL - Poisson's ratio: $\nu = -\frac{\text{lateral strain}}{\text{longitudinal strain}}$; dimensionless; practical range 0-0.5 (theoretical

-1 to 0.5). Steel ≈ 0.3 . - **Volume on stretching:** $\frac{\Delta V}{V} = (1 - 2\nu)\frac{\Delta L}{L}$. $\nu = 0.5 \Rightarrow$ incompressible ($\Delta V = 0$). - **Elastic PE:** $U = \frac{1}{2}F \Delta L = \frac{1}{2}\sigma\varepsilon(\text{Volume})$.

The $\frac{1}{2}$ comes from the *average* force. - **Energy density:** $u = \frac{1}{2}\sigma\varepsilon = \frac{1}{2}Y\varepsilon^2 = \frac{\sigma^2}{2Y}$ = **area under stress-strain curve** (per unit volume). - **Sudden vs gradual:** gradual $\rightarrow W = U = \frac{1}{2}F\Delta L$, no heat. Sudden $\rightarrow x_{\text{max}} = 2x_{\text{static}}$, half the work becomes heat.

Memory aids: - "*Stretch it, it slims*" — longitudinal pull $\Rightarrow$ lateral shrink (that's ν). - "*Half the force, half the story*" — elastic energy uses $F/2$, so $U = \frac{1}{2}F\Delta L$. - "*Drop it and it doubles*" — sudden load gives twice the static stretch.

Subtopic 04: Ductility, Malleability and Yielding

Notation (carried from Subtopics 01-03): stress σ · strain ε · Young's Y · yield stress σ_y · energy density u . Poisson's ratio is ν throughout (not σ).

*In Subtopic 01 you learned to read the stress-strain curve — to name its landmarks (proportional limit, elastic limit, yield point, ultimate strength, fracture). This subtopic asks a different question: given that curve, **what kind of material are you holding?** Ductile or brittle? Malleable? An elastomer? And what exactly happens at the moment a material “gives up” elastically and starts to flow?*

Exam Relevance: Heavy on conceptual and graph-reading questions. CBSE Boards ask definitions (ductile, brittle, elastomer, yield point) and a stress-strain interpretation. NEET reliably puts up a two-curve comparison (“which material is more brittle / has greater Young's modulus?”). JEE Main tests yield-strength design and energy-to-yield. JEE Advanced reaches for multi-statement reasoning on stress-strain graphs and elastic-hysteresis energy loss.

Section 1: Concept Intuition

Every solid, pushed hard enough, eventually stops behaving like a spring. Up to a point it stretches and springs back faithfully — that's the elastic behaviour of Subtopics 01-03. But push past the **yield point** and something irreversible happens: the material *gives*. It keeps stretching with barely any extra force, and when you finally release it, it does **not** return to its original shape. It carries a *permanent set*. Think of it like a government queue that holds its shape under mild pressure but, once a critical crowd builds, suddenly surges forward and never reforms into the neat line it was.

What a material does *after* it yields is its personality — and it sorts materials into families.

Ductile materials (copper, gold, mild steel) have a long, generous plastic region between yielding and fracture. They stretch a lot before breaking, which means they can be **drawn out into thin wires** — that's literally what ductility means. The copper in your charger cable, the gold in a jeweller's thread: both owe their existence to a long plastic range.

Malleable materials (gold, silver, aluminium, lead) yield easily under *compression* and can be **hammered or rolled into thin sheets** without cracking. Ductility is about pulling into wires; malleability is about beating into sheets. The gold leaf on a temple dome and the aluminium foil in your kitchen are malleability in action. (Most ductile metals are also malleable, but the two are distinct ideas — one lives in tension, the other in compression.)

Brittle materials (glass, cast iron, ceramic) have almost *no* plastic region. They snap clean, with little warning, very soon after the elastic limit. A glass tumbler doesn't bend before it shatters — it goes straight from “fine” to “in pieces.”

Think about it this way... there's a famous puzzle: *is rubber more elastic than steel?* In everyday language, obviously — rubber stretches enormously and pings right back. And in *that* sense rubber is remarkable: it can be stretched to several times its length and still recover fully, where steel would have snapped long ago. But physics defines elasticity

by the *restoring force per unit strain* — by Young’s modulus. For the same strain, steel develops vastly larger internal forces than rubber. So by the modulus definition, **steel is the more elastic** of the two. Rubber is an **elastomer**: a material that endures huge strains elastically, has essentially **no region of proportionality** (its curve is bent from the start, so Hooke’s law never really applies), and crucially does **not retrace its path** on unloading — the loading and unloading curves enclose a loop. That loop is **elastic hysteresis**, and the area inside it is energy quietly converted to heat each cycle. This is exactly why rubber is chosen for vehicle tyres and shock absorbers: it *eats* vibration energy rather than bouncing it back. It’s also why a much-flexed rubber band warms up.

Limiting conditions & assumptions. These behaviours are properties of the *material*, but they shift with temperature (many brittle metals turn ductile when hot — which is why a blacksmith heats iron before hammering), with the *rate* of loading (sudden shocks make even ductile metals behave brittly), and with repeated cycling (**elastic fatigue** — a material loses its strength after countless load cycles and can fail well below its normal breaking stress, which is why aircraft parts and railway axles are retired on a schedule, not just when they look worn).

Section 2: Key Definitions & Quantities

This subtopic is largely qualitative, but a few quantities anchor it:

Yield strength σ_y — the stress at the yield point, marking the boundary between elastic (recoverable) and plastic (permanent) deformation. Unit Pa.

Ultimate tensile strength — the maximum stress on the curve; the most the material can bear before it begins to neck and head toward fracture.

Breaking (fracture) stress — the stress at the point of actual rupture.

Modulus of resilience — the elastic energy a material can absorb *per unit volume up to its yield point* (i.e. the area under the elastic part of the curve):

$$u_{\text{resilience}} = \frac{1}{2} \sigma_y \varepsilon_y = \frac{\sigma_y^2}{2Y} \quad (\text{unit: J m}^{-3})$$

Toughness — the *total* energy absorbed per unit volume up to fracture (the **whole** area under the stress-strain curve). A tough material absorbs a lot of energy before breaking; ductile materials are generally tough, brittle ones are not.

Ductility (graphical measure) — the length of the stress-strain curve between the yield point and the ultimate load; the longer this stretch, the more ductile the material.

Permanent set — the residual strain left after a material is loaded past yield and then unloaded.

Figure 8.2 Three stress-strain curves on the same axes for comparison. (1) A ductile metal: steep linear rise, clear yield point, long gently-rising plastic plateau to a peak (ultimate strength), then down to fracture — large total area. (2) A brittle material: steep linear rise that ends abruptly at fracture with almost no plastic region — small area. (3) An elastomer (rubber): a curved (non-linear) path from the origin with no straight portion, reaching very high strain, and a separate, lower unloading curve forming a hysteresis loop; shade the loop area as “energy lost as heat.”] 26 (see next page)

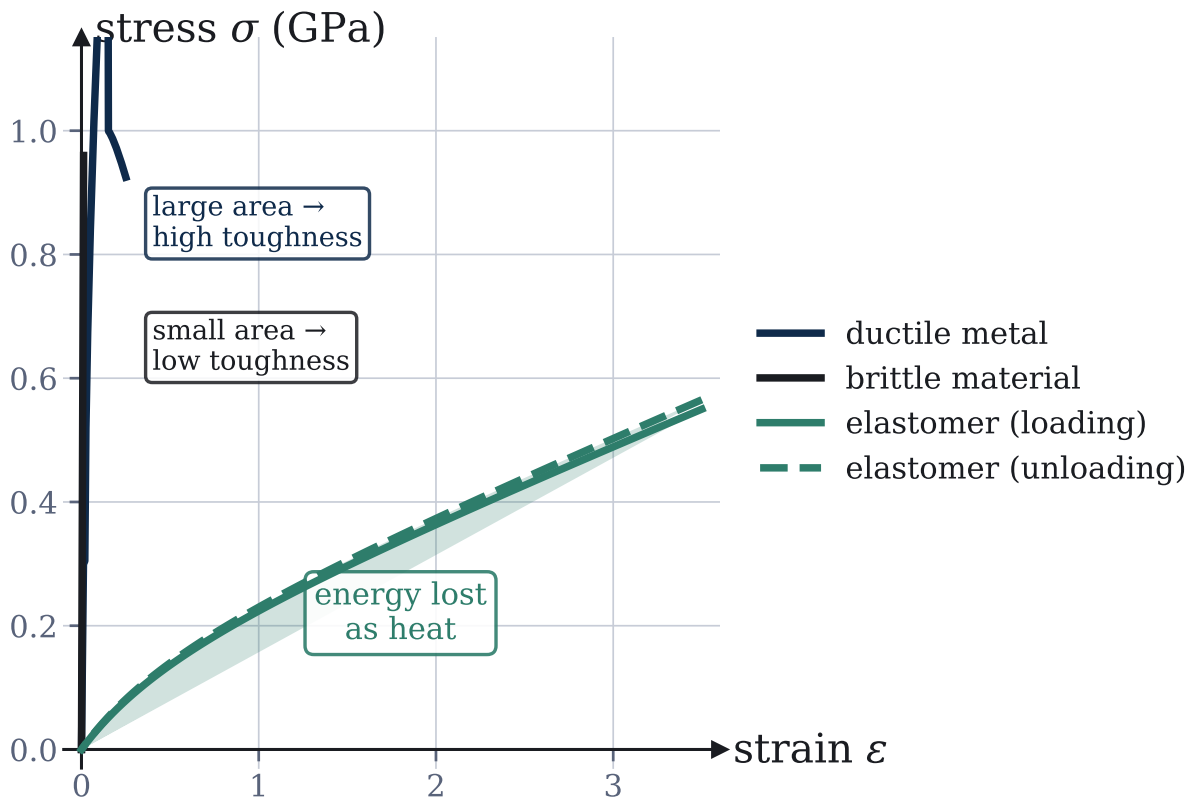


Figure 8.2 Three stress-strain curves on the same axes for comparison. (1) A ductile metal: steep linear rise, clear yield point, long gently-rising plastic plateau to a peak (ultimate strength), then down to fracture — large total area. (2) A brittle material: steep linear rise that ends abruptly at fracture with almost no plastic region — small area. (3) An elastomer (rubber): a curved (non-linear) path from the origin with no straight portion, reaching very high strain, and a separate, lower unloading curve forming a hysteresis loop; shade the loop area as “energy lost as heat.”] 26

Section 3: Key Procedures & Logical Framework

(This subtopic has no single formal derivation; instead, here is the logical walkthrough of the stress-strain journey and how to classify a material from it.)

The journey along the curve, region by region. Loading a wire and following its stress-strain curve: 1. **Linear (elastic) region — up to the proportional limit.** Stress $\propto$ strain; Hooke's law holds; slope = Y . Unload here and the wire fully recovers. 2. **Elastic limit / yield point.** The last point of full recovery. *Why it matters:* cross it and recovery is no longer complete. (For many metals the elastic limit and yield point sit so close together they're treated as one; conceptually the elastic limit is "last full recovery" and the yield point is "onset of large plastic flow.") 3. **Plastic region — beyond yield.** Strain grows rapidly for little extra stress ("plastic flow"). Unload from here and the wire keeps a **permanent set** — the unloading line runs parallel to the original elastic line but doesn't return to the origin. *This is the defining feature of yielding.* 4. **Ultimate tensile strength.** The peak stress. Past it, a ductile sample **necks** (thins locally). 5. **Fracture point.** The wire breaks; the stress here is the breaking stress.

How to classify a material from its curve: - *Steep initial slope?* → high Young's modulus (stiff). - *Gentle slope?* → low Y (flexible). The slope, not the height, gives stiffness. - *Long plastic region between yield and fracture?* → **ductile**. - *Fracture almost immediately after the elastic limit?* → **brittle**. - *No straight portion at all, huge recoverable strain, an unloading loop?* → **elastomer**. - *Large total area under the curve?* → **tough** (absorbs much energy before breaking).

Resilience vs. toughness — a tale of two areas. Both are areas under the same curve, but bounded differently: - **Resilience** = area under the curve *up to the yield point* = energy stored *elastically* (fully recoverable). This is what springs and trampolines are designed to maximise. - **Toughness** = area under the *entire* curve *up to fracture* = total energy absorbed before breaking. This is what crash barriers and helmets are designed to maximise. A material can be very resilient but not tough (a brittle spring steel), or very tough but not especially resilient (a ductile structural metal). *Why the distinction matters:* you choose a resilient material when you want energy *returned*, a tough one when you want energy *absorbed and not given back*.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A steel wire has a yield strength of $3.0 \times 10^8 \text{ Pa}$ and Young's modulus $2.0 \times 10^{11} \text{ Pa}$. Calculate its modulus of resilience (the maximum elastic energy it can store per unit volume without taking a permanent set).

The modulus of resilience is the energy density at the yield point:

$$u_{\text{resilience}} = \frac{\sigma_y^2}{2Y} = \frac{(3.0 \times 10^8)^2}{2(2.0 \times 10^{11})} = \frac{9.0 \times 10^{16}}{4.0 \times 10^{11}}$$

$$u_{\text{resilience}} = 2.25 \times 10^5 \text{ J m}^{-3}$$

Beyond this energy density the wire would yield and not fully recover — so this is the safe elastic “energy budget” per cubic metre.

Example 2 — [NEET Speed Trap] Two materials, *P* and *Q*, are tested. On the same stress-strain axes, *P*’s curve rises steeply and ends abruptly at a low strain; *Q*’s curve rises less steeply but continues to a much higher strain before fracturing. Which material (i) has the greater Young’s modulus, and (ii) is more ductile?

The trap: students assume the curve that “goes higher/steeper” is better in every way, or they confuse stiffness with ductility.

Speed method — separate the two ideas: - **Young’s modulus = slope of the linear part.** *P* is steeper = ***P* has the greater *Y*** (*P* is stiffer). - **Ductility = how far the curve extends (plastic length) before fracture.** *Q* reaches much higher strain ⇒ ***Q* is more ductile** (and *P*, snapping early, is the more brittle).

So stiffness and ductility point to *different* materials here — a stiff material can be brittle, a flexible one ductile. Eliminate any option that bundles “greater *Y*” with “more ductile” for the same material; that’s the planted error.

Example 3 — [JEE Main Level] A steel lift cable must support a maximum load of 2000 kg. The yield strength of the steel is $2.5 \times 10^8 \text{ Pa}$, and the design requires a safety factor of 5 (the working stress must not exceed one-fifth of the yield strength). Find the minimum diameter of the cable. (Take $g = 10 \text{ m s}^{-2}$.)

Two ideas: the working-stress limit set by yield strength and safety factor, then converting a required area into a diameter.

Maximum force: $F = mg = 2000 \times 10 = 2.0 \times 10^4 \text{ N}$.

Allowed working stress: $\sigma_{\text{work}} = \frac{\sigma_y}{5} = \frac{2.5 \times 10^8}{5} = 5.0 \times 10^7 \text{ Pa}$.

Minimum cross-sectional area: $A = \frac{F}{\sigma_{\text{work}}} = \frac{2.0 \times 10^4}{5.0 \times 10^7} = 4.0 \times 10^{-4} \text{ m}^2$.

From $A = \frac{\pi d^2}{4}$:

$$d = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(4.0 \times 10^{-4})}{\pi}} = \sqrt{5.09 \times 10^{-4}} \approx \boxed{2.3 \times 10^{-2} \text{ m} = 2.3 \text{ cm}}$$

The safety factor keeps the cable working comfortably inside its elastic region — it never approaches yielding under the rated load.

Example 4 — [JEE Advanced Level] Two wires of equal length and equal cross-section, made of materials *A* and *B*, are tested to fracture. Their stress-strain curves are described as follows. Curve *A*: a steep straight line up to a high stress, then fracture almost immediately. Curve *B*: a less steep straight line to a lower yield stress, followed by a long plastic plateau extending to large strain before fracture, the plateau lying below *A*’s fracture stress. For these two materials, determine and justify which has (i) the greater Young’s modulus, (ii) the greater breaking stress, (iii) the greater ductility, and (iv) the greater toughness.

This is a reasoning problem; read each property off the right feature of the curve.

- (i) **Young's modulus** = slope of the initial straight line. A is steeper $\Rightarrow Y_A > Y_B$. A is the stiffer material.
- (ii) **Breaking stress** = the stress at the fracture point. A fractures at a high stress; B's entire plateau lies below that $\Rightarrow$ **breaking stress of A > breaking stress of B**. A is stronger.
- (iii) **Ductility** = extent of plastic deformation (curve length from yield to fracture). A snaps with almost no plastic region; B has a long plateau $\Rightarrow$ **B is far more ductile** (A is brittle).
- (iv) **Toughness** = total area under the curve to fracture. Although A reaches a higher stress, its area is a thin sliver (it breaks at tiny strain). B's modest stress acts over a huge strain, enclosing a much larger area $\Rightarrow$ **B is the tougher material**.

The deep takeaway: “strong” (high breaking stress), “stiff” (high Y), and “tough” (high energy absorption) are three *independent* properties. Material A wins on strength and stiffness; B wins on ductility and toughness. A glass rod (like A) can be strong and stiff yet shatter with almost no warning, while a mild-steel rod (like B) bends and absorbs enormous energy before failing — which is precisely why structural beams are made of the latter.

Section 5: Practice Exercises

1. **[CBSE]** Define ductile and brittle materials, and give one example of each. Which region of the stress-strain curve determines whether a material is ductile or brittle?
2. **[NEET]** A material is stretched and then unloaded; it does not return to its original length but retains some elongation. Which property does this demonstrate, and what is the retained strain called?
3. **[JEE Main]** A material has yield strength 4.0×10^8 Pa and Young's modulus 1.0×10^{11} Pa. Find its modulus of resilience (maximum recoverable elastic energy per unit volume).
4. **[JEE Main]** A wire of cross-section 2.0 mm^2 is made of a material whose yield strength is 2.0×10^8 Pa. What is the maximum load the wire can carry without undergoing permanent deformation? (Take $g = 10 \text{ m s}^{-2}$.)
5. **[JEE Advanced]** Explain, in terms of the area enclosed by its loading-unloading curve, why rubber is chosen over steel for vehicle tyres and shock absorbers, and state where the “lost” energy goes.

Answers: 1. ductile e.g. copper, brittle e.g. glass; the plastic region (yield $\rightarrow$ fracture) decides 2. plasticity / plastic deformation; the retained strain is the *permanent set* 3. $8.0 \times 10^5 \text{ J m}^{-3}$ 4. 40 kg ($F_{\text{max}} = \sigma_y A = 400 \text{ N}$) 5. the enclosed hysteresis-loop area is energy dissipated as heat each cycle, so rubber absorbs/damps vibration rather than returning it; the lost energy becomes heat.

Section 6: Crack the MCQ

Q1. A material that breaks soon after the elastic limit is crossed, with almost no plastic deformation, is called: (a) ductile (b) brittle (c) malleable (d) an elastomer

Correct: (b). Little to no plastic region before fracture is the definition of brittleness (e.g. glass, cast iron). *Why the traps:* **(a)** ductile is the opposite — a *long* plastic region. **(c)** malleability is about being hammered into sheets, unrelated to early fracture. **(d)** an elastomer endures *huge* strains elastically — the very opposite of snapping early.

Q2. Regarding steel and rubber, which statement is correct? (a) Rubber is more elastic than steel because it has a larger Young's modulus. (b) Steel is more elastic than rubber because, for a given strain, larger restoring stress develops in steel. (c) Both have the same Young's modulus. (d) Rubber obeys Hooke's law more closely than steel.

Correct: (b). Elasticity by the modulus definition favours steel: the same strain calls up far greater internal stress in steel ($Y_{\text{steel}} \gg Y_{\text{rubber}}$). *Why the traps:* **(a)** reverses the modulus comparison — rubber's Y is *small*, not large. **(c)** is plainly false. **(d)** is backwards: rubber has almost no proportional region, so it obeys Hooke's law *less* well than steel.

Q3. The ductility of a material, read from its stress-strain curve, is measured by: (a) the slope of the linear region (b) the area under the elastic portion (c) the length of the curve between the yield point and the ultimate load (d) the stress at the proportional limit

Correct: (c). A longer curve between yielding and ultimate load means more plastic stretch before failure — more ductility. *Why the traps:* **(a)** the slope gives Young's modulus (stiffness), not ductility. **(b)** the elastic-portion area is resilience. **(d)** the proportional-limit stress concerns elasticity, not plastic extensibility.

Q4. Vulcanised rubber is preferred for shock absorbers mainly because: (a) it has a very high Young's modulus (b) it obeys Hooke's law over a large range (c) its loading and unloading curves enclose a large area, dissipating energy as heat (d) it is brittle and absorbs shock by cracking

Correct: (c). The large elastic-hysteresis loop means a lot of energy is converted to heat per cycle — ideal for damping vibration. *Why the traps:* **(a)** rubber's Y is *low*, not high. **(b)** rubber notably does *not* obey Hooke's law over its range. **(d)** rubber is the opposite of brittle, and cracking would be a failure, not a feature.

Section 7: Common Student Pitfalls & Pro-Tips

- **Confusing ductility, malleability, and elasticity.** Ductility = drawn into *wires* (tension). Malleability = hammered into *sheets* (compression). Elasticity = ability to *recover*. They are three different properties; a material can rank differently on each.
- **"Rubber is more elastic than steel."** True only in the loose everyday sense (large recoverable strain). By the physics definition (Young's modulus), **steel is more elastic**. Know which definition the question wants.
- **Slope vs. area on the stress-strain curve.** The *slope* of the linear part is Young's modulus (stiffness). *Areas* are energies — area to yield is resilience, total area to fracture is toughness. Mixing these up is the single most common graph-question error.

- **Yield point treated as fracture.** Yielding is the *onset of permanent deformation*, not breaking. A material can yield and continue carrying load for a long plastic range before it finally fractures.

Pro-Tip: On any two-curve comparison, decode each property from one specific feature and answer in this fixed order: **slope** → **stiffness (Y)**; **height of fracture point** → **strength (breaking stress)**; **horizontal extent past yield** → **ductility**; **total enclosed area** → **toughness**. Running this checklist stops you from bundling “stiff,” “strong,” and “tough” into one material when the graph is deliberately separating them.

Section 8: Cheat Sheet Box

DUCTILITY, MALLEABILITY & YIELDING — RAPID RECALL - **Yield point:** boundary between elastic (recovers) and plastic (permanent set). Beyond it → plastic flow. - **Ductile** = long plastic region → drawn into *wires* (copper). **Brittle** = breaks just after elastic limit (glass). **Malleable** = hammered into *sheets* (gold, aluminium). **Elastomer** = huge recoverable strain, no proportional region, hysteresis loop (rubber). - **Steel vs rubber:** steel is more elastic by *modulus* ($Y_{\text{steel}} \gg Y_{\text{rubber}}$); rubber recovers from larger strains. - **Resilience** = $\frac{\sigma_y^2}{2Y}$ = area under elastic region (energy returned). **Toughness** = total area to fracture (energy absorbed). - **Elastic hysteresis:** loading $\neq$ unloading curve; enclosed area = energy lost as heat per cycle → tyres, shock absorbers. **Elastic fatigue:** strength lost after many load cycles.

Memory aids: - “Ductile draws wires, malleable makes sheets.” - “Slope is stiff, area is energy, length is ductility.” - “Loop = lost heat” — the hysteresis-loop area is the energy rubber eats.

Subtopic 05: Applications of Elastic Behaviour

Notation (carried from Subtopics 01–04): stress σ · strain ε · Young’s Y · Bulk B · Rigidity η · Poisson’s ratio ν · coefficient of linear expansion α · geometrical moment of inertia of a cross-section I_g .

*Subtopics 01–04 built the theory of elasticity from the ground up. This final subtopic puts it to work in the real world — how beams sag, why girders are I-shaped, how tall a mountain can be, what happens when a clamped rod is heated, how a wire resists twisting, and how Young’s modulus is actually measured in the lab. It also returns to the very beginning: **why** solids are elastic at all, at the level of atoms.*

Exam Relevance: CBSE Boards regularly ask the I-section/beam-design reasoning, the mountain-height estimate, and Searle’s experiment (3–5 marks). JEE Main tests beam depression and thermal stress numericals. NEET favours thermal stress and the conceptual beam-design questions. JEE Advanced reaches for torsion of a cylinder and multi-concept thermal-stress problems. This subtopic ties the whole chapter together.

Section 1: Concept Intuition

Why is a solid elastic in the first place? Zoom all the way in. The atoms of a solid sit at fixed average positions, held there by electrical forces from their neighbours — attractive when pulled apart, repulsive when pushed together. Each atom rests at the bottom of a “potential well,” the separation r_0 where these forces balance. Displace an atom slightly and the well behaves, for small displacements, exactly like a tiny spring: the restoring force grows in proportion to the displacement. Multiply this over the trillions of atomic bonds in a wire and you get the macroscopic behaviour you already know — **Hooke’s law is the collective voice of countless atomic springs**. Young’s modulus is large when those bonds are stiff (metals, diamond) and small when they are floppy (rubber, where long molecular chains uncoil rather than stretch their bonds).

That microscopic stiffness is what engineers spend their lives exploiting and respecting. Think about it this way... every structure you trust — the railway bridge your train rumbles over, the crane lifting a slab at a construction site, the pillars of a flyover, the wing of an aeroplane — is a deliberate negotiation with elasticity. Too flexible and it sags or sways dangerously; too brittle and it cracks; too heavy and it’s wasteful. The whole art is to get maximum stiffness and strength from minimum material, *staying safely within the elastic limit at all times*.

A few of the headline negotiations:

- **Beams that don’t sag.** A beam loaded in the middle bends — its lower fibres stretch (tension), its upper fibres compress, and a “neutral layer” in between neither stretches nor shrinks. How much it sags (the **depression**) depends not just on the load and the material but, dramatically, on the *shape* of its cross-section.
- **The I-shaped girder.** Since the material near the neutral layer barely does any work, you can scoop it out and pile that material far from the centre instead — giving the

familiar **I-section**. Same stiffness, far less weight. This single insight is why steel girders, railway tracks, and aircraft spars are I-shaped or hollow.

- **Why mountains can't be infinitely tall.** Rock at the base of a mountain bears the weight of everything above it. Pile the mountain too high and that stress exceeds rock's strength — the base flows or crumbles. This sets a natural ceiling on mountain height of roughly 10 km, which is why Everest is the size it is and not ten times bigger.
- **Heat as a hidden force.** Clamp a rod rigidly and heat it: it *wants* to expand but can't, so it pushes on its supports with enormous force. This **thermal stress** is why railway tracks have gaps, why bridges sit on roller supports, and why pipelines have expansion loops.

Limiting conditions & assumptions. Everything here assumes the material stays **within its elastic limit** — once you cross into plastic flow (Subtopic 04), these formulas fail and permanent damage begins. The beam and torsion results assume small deformations, uniform cross-sections, and homogeneous isotropic material. The mountain-height estimate is a clean order-of-magnitude argument, not a precise geological prediction (it ignores temperature, rock layering, and isostasy).

Section 2: Key Formulas & Reference Data

Depression of a beam under load W :

$$\text{Cantilever (loaded at free end): } \delta = \frac{WL^3}{3YI_g} \quad \text{Beam on two supports (central load): } \delta = \frac{WL^3}{48YI_g}$$

Geometrical moment of inertia of the cross-section (about the neutral axis):

$$\text{Rectangle (breadth } b, \text{ depth } d) : I_g = \frac{bd^3}{12} \quad \text{Circle (radius } r) : I_g = \frac{\pi r^4}{4}$$

Key consequence: $\delta \propto \frac{1}{I_g}$, and for a rectangle $I_g \propto d^3$ — **depth matters far more than breadth**.

Maximum height of a mountain (base stress = breaking/compressive stress $\sigma_{\max}$):

$$h_{\max} = \frac{\sigma_{\max}}{\rho g}$$

Thermal stress (rod rigidly clamped at both ends, temperature change ΔT):

$$\sigma_{\text{thermal}} = Y \alpha \Delta T \quad F_{\text{thermal}} = Y A \alpha \Delta T$$

Note: independent of the rod's length; the force depends on area but not length.

Torsion of a solid cylinder (radius r , length L , twisted through angle θ):

$$C = \frac{\pi \eta r^4}{2L} \theta \quad \text{torsional rigidity } c = \frac{C}{\theta} = \frac{\pi \eta r^4}{2L}$$

Young's modulus from Searle's experiment:

$$Y = \frac{Mg L}{\pi r^2 \Delta L} = \frac{Mg/A}{\Delta L/L} = \frac{1}{(\text{slope of strain-vs-stress graph})}$$

Representative elastic constants (approximate, at room temperature; for reference):

Material	Y (GPa)	η (GPa)	B (GPa)	ν
Steel	~200	~80	~160	~0.29
Copper	~120	~44	~140	~0.34
Brass	~95	~35	~110	~0.34
Aluminium	~70	~25	~75	~0.33
Glass	~60	~25	~40	~0.23
Bone	~15	~6	~9	~0.30
Rubber	~0.01–0.1	very low	~1–2	~0.50

(1 GPa = 10^9 Pa. Values vary with composition/treatment — use as orders of magnitude.)

Section 3: Derivations & Key Procedures

Derivation 1 — Why Hooke's law emerges from atoms (qualitative). Let $U(r)$ be the potential energy of two neighbouring atoms a distance r apart, with a minimum at the equilibrium separation r_0 . Near r_0 , expand $U(r)$ as a parabola (the leading term of any smooth minimum): $U(r) \approx U(r_0) + \frac{1}{2}\kappa(r - r_0)^2$. The interatomic force is $F = -\frac{dU}{dr} \approx -\kappa(r - r_0)$ — **linear** in displacement, i.e. a spring of constant κ . Summed over all bonds across a cross-section, this linear atomic response is the macroscopic Hooke's law $\sigma = Y\varepsilon$, with Y set by the bond stiffness κ and spacing r_0 . *Why it matters:* it explains why Hooke's law holds only for *small* strains — stretch far enough and the parabola approximation (and the bond) breaks.

Derivation 2 — Maximum height of a mountain. Model the mountain as a column of rock of height h and density ρ . The compressive stress at its base is the weight per unit area of the rock above:

$$\sigma_{\text{base}} = \frac{(\text{volume} \times \rho \times g)}{\text{area}} = \frac{Ah\rho g}{A} = \rho gh$$

The base cannot bear more than the material's breaking (compressive) stress σ_{max} ; beyond it, the rock yields and flows. Setting $\sigma_{\text{base}} = \sigma_{\text{max}}$:

$$h_{\text{max}} = \frac{\sigma_{\text{max}}}{\rho g}$$

For rock, $\sigma_{\text{max}} \sim 3 \times 10^8$ Pa and $\rho \sim 3 \times 10^3$ kg m⁻³, giving $h_{\text{max}} \sim \frac{3 \times 10^8}{(3 \times 10^3)(10)} \approx 10^4$ m = 10 km — comfortably bracketing Everest.

Derivation 3 — Thermal stress. Heat a rod and, left free, it would lengthen by $\Delta L = L\alpha \Delta T$, a thermal strain of $\varepsilon_{\text{th}} = \alpha \Delta T$. Now clamp both ends so it *cannot* expand. The walls must supply exactly the compression needed to cancel that expansion, i.e. an *equal and opposite* mechanical strain $\varepsilon = \alpha \Delta T$. By Hooke's law the stress this requires is

$$\sigma_{\text{thermal}} = Y \varepsilon = \boxed{Y \alpha \Delta T}$$

and the force on each support is $F = \sigma A = Y A \alpha \Delta T$. *The striking feature:* neither expression contains L — a short clamped rod and a long one develop the *same* thermal stress for the same temperature change.

Derivation 4 — Torsion of a solid cylinder. Fix one end of a cylinder (radius R , length L , rigidity η) and twist the other through angle θ . Consider a thin coaxial shell of radius x and thickness dx . A straight line drawn along its surface is sheared through an angle ϕ , where the twist at the rim, $x\theta$, equals $L\phi$, so $\phi = \frac{x\theta}{L}$. The shear stress on this shell is $\tau = \eta\phi = \frac{\eta x\theta}{L}$. The tangential force on the ring of area $2\pi x dx$ is $\tau \cdot 2\pi x dx$, and its moment about the axis is

$$dC = \tau (2\pi x dx) x = \frac{2\pi\eta\theta}{L} x^3 dx$$

Integrating from 0 to R :

$$C = \frac{2\pi\eta\theta}{L} \int_0^R x^3 dx = \frac{2\pi\eta\theta}{L} \cdot \frac{R^4}{4} = \boxed{\frac{\pi\eta R^4\theta}{2L}}$$

The R^4 dependence is why even a modest increase in shaft radius dramatically stiffens it against twisting.

Procedure — Searle's experiment (determining Y). 1. Hang **two identical wires** side by side from the same rigid support: a *reference* wire (carrying a fixed dead load to keep it taut) and the *experimental* wire (which will be loaded). 2. Connect them with a frame carrying a **spirit level** and a **micrometer screw**, so tilting of the frame measures the *difference* in their elongations. 3. Add known loads to the experimental wire in steps; for each, level the spirit bubble with the micrometer and record the extension ΔL . 4. Plot **load (or stress) against extension (or strain)** — a straight line within the elastic limit.

From its slope, $Y = \frac{MgL}{\pi r^2 \Delta L}$. *Why two wires?* The reference wire automatically cancels errors from temperature change and any yielding of the support — both wires shift together, so only the *load-induced* extension is measured. (Measure the wire's radius separately and carefully, since $Y \propto 1/r^2$ — a small radius error is doubled in the result.)

Section 4: Worked Examples

Example 1 — [CBSE Board Level] The breaking (compressive) stress of a certain rock is $3.0 \times 10^8 \text{ Pa}$ and its density is $2.5 \times 10^3 \text{ kg m}^{-3}$. Estimate the maximum height of a mountain that this rock could support. (Take $g = 10 \text{ m s}^{-2}$.)

The base stress ρgh must not exceed the breaking stress:

$$h_{\max} = \frac{\sigma_{\max}}{\rho g} = \frac{3.0 \times 10^8}{(2.5 \times 10^3)(10)} = \frac{3.0 \times 10^8}{2.5 \times 10^4} = \boxed{1.2 \times 10^4 \text{ m} = 12 \text{ km}}$$

Reassuringly close to the scale of the tallest mountains on Earth.

Example 2 — [NEET Speed Trap] A steel rod is clamped rigidly between two fixed walls and then heated so its temperature rises by 40°C . Given $Y = 2.0 \times 10^{11} \text{ Pa}$ and $\alpha = 1.2 \times 10^{-5} ^\circ\text{C}^{-1}$, find the thermal stress developed. Does the answer depend on the rod's length?

The trap: students hunt for the rod's length and cross-section, assuming a longer or thinner rod gives a different stress. It doesn't.

Speed method: thermal stress depends only on material and temperature change:

$$\sigma = Y\alpha \Delta T = (2.0 \times 10^{11})(1.2 \times 10^{-5})(40) = \boxed{9.6 \times 10^7 \text{ Pa}}$$

No — it is independent of length (and the stress is independent of area too; only the force $= \sigma A$ would need the area). If a NEET option changes the answer when "length is doubled," eliminate it on sight.

Example 3 — [JEE Main Level] A cantilever beam of length 1.0 m has a rectangular cross-section of breadth 4.0 cm and depth 6.0 cm . A load of 200 N is hung from its free end. If $Y = 2.0 \times 10^{11} \text{ Pa}$, find the depression of the free end.

Two steps: compute the geometrical moment of inertia, then apply the cantilever formula.

$$I_g = \frac{bd^3}{12} = \frac{(0.040)(0.060)^3}{12} = \frac{(0.040)(2.16 \times 10^{-4})}{12} = \frac{8.64 \times 10^{-6}}{12} = 7.2 \times 10^{-7} \text{ m}^4$$

$$\delta = \frac{WL^3}{3YI_g} = \frac{(200)(1.0)^3}{3(2.0 \times 10^{11})(7.2 \times 10^{-7})} = \frac{200}{4.32 \times 10^5} \approx \boxed{4.6 \times 10^{-4} \text{ m} = 0.46 \text{ mm}}$$

Check the intuition: if the beam were turned on its side (breadth and depth swapped, depth now 4.0 cm), I_g would fall by a factor $(6/4)^2 = 2.25$ and the depression would rise to $\approx 1.0 \text{ mm}$ — concretely showing why beams are mounted with their **long dimension vertical**.

Example 4 — [JEE Advanced Level] A solid cylindrical shaft of radius 2.0 cm and length 1.0 m is made of a metal of shear modulus $\eta = 4.0 \times 10^{10} \text{ Pa}$. (a) Find the torque required to twist one end through 0.50 rad relative to the other. (b) If the radius is doubled, by what factor does the required torque change for the same twist?

(a) Using the torsion result:

$$C = \frac{\pi\eta r^4\theta}{2L} = \frac{\pi(4.0 \times 10^{10})(0.020)^4(0.50)}{2(1.0)}$$

Compute $r^4 = (0.020)^4 = 1.6 \times 10^{-7} \text{ m}^4$:

$$C = \frac{\pi(4.0 \times 10^{10})(1.6 \times 10^{-7})(0.50)}{2} = \frac{\pi(3.2 \times 10^3)}{2} \approx \boxed{5.0 \times 10^3 \text{ N}\cdot\text{m}}$$

- (b) Since $C \propto r^4$, doubling the radius multiplies the torque by $2^4 = 16$. A shaft only twice as thick is *sixteen* times harder to twist — the dramatic payoff of the r^4 law, and the reason drive shafts and axles are made as fat as weight allows.

Section 5: Practice Exercises

- [CBSE]** Explain why the cross-section of a girder used in construction is made in the shape of the letter “I” rather than being solid rectangular. Which dimension of a beam most strongly reduces its sagging, and why?
- [NEET]** A metal rod is held between two rigid clamps and cooled by 30°C . If $Y = 1.0 \times 10^{11} \text{ Pa}$ and $\alpha = 2.0 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$, find the tensile stress that develops in the rod.
- [JEE Main]** A beam supported at both ends carries a load W at its centre and depresses by δ . If the beam is replaced by one of the same material and length but with twice the depth (same breadth), find the new depression in terms of δ .
- [JEE Main]** A wire of radius r requires a torque C to twist through a given angle. What torque is required to twist a wire of the same length and material but radius $1.5r$ through the same angle?
- [JEE Advanced]** A steel rail of cross-sectional area 40 cm^2 is clamped so it cannot expand. Its temperature rises by 25°C . Taking $Y = 2.0 \times 10^{11} \text{ Pa}$ and $\alpha = 1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$, find the force the rail exerts on its supports.

Answers: 1. material near the neutral axis carries little stress, so removing it (I-section) keeps stiffness while saving weight; depth, because $I_g \propto d^3$ for a rectangle 2. $6.0 \times 10^7 \text{ Pa}$ 3. $\delta/8$ (since $\delta \propto 1/d^3$) 4. $(1.5)^4 C = 5.06 C$ 5. $F = YA\alpha\Delta T = (2.0 \times 10^{11})(40 \times 10^{-4})(1.2 \times 10^{-5})(25) = 2.4 \times 10^5 \text{ N}$

Section 6: Crack the MCQ

Q1. A girder is given an I-shaped cross-section primarily because: (a) it looks aesthetically better (b) it uses material far from the neutral axis, giving high stiffness for low weight (c) the I-shape increases the Young’s modulus of the steel (d) it makes the beam more brittle and therefore stronger

Correct: (b). Bending stress is greatest far from the neutral axis and nearly zero at it, so concentrating material in top and bottom flanges maximises I_g per unit weight. *Why the traps:* (a) appearance is irrelevant to mechanics. (c) Y is a material property — shape cannot change it. (d) brittleness is undesirable and unrelated to the I-shape’s purpose.

Q2. The maximum height of a mountain on a planet depends on the rock’s breaking stress σ , its density ρ , and the surface gravity g as: (a) $\frac{\sigma}{\rho g}$ (b) $\frac{\rho g}{\sigma}$ (c) $\sigma \rho g$ (d) $\frac{\sigma g}{\rho}$

Correct: (a). Base stress ρgh set equal to σ gives $h_{\text{max}} = \sigma/(\rho g)$. *Why the traps:* (b) is the reciprocal (wrong dimensions for height). (c) and (d) misplace the factors — a quick dimension check ($[\sigma]/[\rho g] = \text{metres}$) rules them out instantly. (Note: higher gravity = shorter mountains, which is why a low-gravity planet like Mars hosts Olympus Mons.)

Q3. A rod clamped rigidly between two walls is heated. The thermal stress developed is independent of: (a) the Young's modulus of the rod (b) the temperature rise (c) the coefficient of linear expansion (d) the length of the rod

Correct: (d). $\sigma = Y\alpha\Delta T$ contains no length term. *Why the traps: (a), (b), (c)* all appear explicitly in $\sigma = Y\alpha\Delta T$, so the stress *does* depend on each of them. Length affects how much the rod *would* expand, but not the stress when expansion is fully prevented.

Q4. The depression at the free end of a loaded cantilever of rectangular cross-section is proportional to: (a) d^3 (b) $1/d^3$ (c) d (d) $1/d$

Correct: (b). $\delta \propto 1/I_g$ and $I_g = bd^3/12 \propto d^3$, so $\delta \propto 1/d^3$. *Why the traps: (a)* confuses the dependence of I_g (which *does* go as d^3) with that of the depression (its reciprocal). **(c)** and **(d)** use the wrong power of depth entirely.

Section 7: Common Student Pitfalls & Pro-Tips

- **Putting length into thermal stress.** $\sigma = Y\alpha\Delta T$ has *no* length. Length determines the (prevented) expansion, not the resulting stress. Only the *force* (σA) needs the area.
- **Confusing I_g (geometrical) with mass moment of inertia.** The “moment of inertia” in beam formulas is the *area* moment $I_g = \int y^2 dA$ (units m^4), not the rotational $\sum mr^2$. Same name, different quantity.
- **Treating depth and breadth as equal partners.** For a rectangular beam, depression goes as $1/d^3$ but only as $1/b$. Doubling the depth is eight times as effective as doubling the breadth — always orient the larger dimension vertically.
- **Forgetting the r^4 laws.** Both beam stiffness (circular, $I_g \propto r^4$) and torsional rigidity ($\propto r^4$) scale with the *fourth* power of radius. Small radius changes have outsized effects — and a small radius *measurement* error wrecks a Y determination, since $Y \propto 1/r^2$.

Pro-Tip: For any “what happens if I change a dimension” question (beam, shaft, wire), don't recompute from scratch — use the **power-law shortcut**. Write the proportionality ($\delta \propto 1/d^3$, $C \propto r^4$, σ_{th} independent of L), take the ratio, and you have the answer in one line. Most exam questions on this subtopic are secretly just testing whether you know the exponent.

Section 8: Cheat Sheet Box

APPLICATIONS OF ELASTIC BEHAVIOUR — RAPID RECALL - Atomic origin: Hooke's law = parabolic bottom of the interatomic potential well; $Y \propto$ bond

stiffness. Valid only for small strains. - **Beam depression:** cantilever $\delta = \frac{WL^3}{3YI_g}$;

supported (central) $\delta = \frac{WL^3}{48YI_g}$. Rectangle $I_g = \frac{bd^3}{12}$, circle $I_g = \frac{\pi r^4}{4}$. $\delta \propto 1/d^3$.

- **I-section:** remove low-stress material near the neutral axis $\rightarrow$ high stiffness, low

weight. - **Mountain height:** $h_{\max} = \frac{\sigma_{\max}}{\rho g} \approx 10 \text{ km}$ for rock. - **Thermal stress:**

$\sigma = Y\alpha \Delta T$, force = $YA\alpha \Delta T$ — independent of length. - **Torsion:** $C = \frac{\pi\eta r^4 \theta}{2L}$; torsional rigidity $\propto r^4$. - **Searle's experiment:** twin-wire setup (reference cancels temperature/support errors); $Y = \frac{MgL}{\pi r^2 \Delta L}$ from the load-extension slope.

Memory aids: - *“Depth cubed, breadth single”* — beams sag as $1/d^3$ but only $1/b$. - *“Heat builds stress, not by the metre”* — $\sigma = Y\alpha\Delta T$, no length. - *“Twist loves the fourth power”* — torsional stiffness $\propto r^4$. - *“Mountains end where rock gives way”* — $h_{\max} = \sigma_{\max}/\rho g$.

Chapter 8: Mechanical Properties of Solids — Round 2 Addendum

The chapter gave you the full skeleton: stress and strain, the four constants, the wire-as-spring, elastic energy, yielding, beams, thermal stress, and torsion. Six examinable patterns still stand outside it, and each one is two things the chapter already teaches separately. Addendum A turns Subtopics 01 and 05's two special-case integrals into one master integral covering tapered rods and piecewise tensions. Addendum B completes the thermal-stress story with gaps and dissimilar-material composites, which is where JEE actually asks it. Addendum C sets the chapter's static spring into motion. Addendum D handles two wires sharing a load through a rigid bar — the load-distribution pattern JEE Main repeats almost yearly. Addendum E makes Subtopic 05's atomic picture quantitative. Addendum F propagates experimental errors through Searle's formula. Nothing here is new theory; it is the chapter's own machinery run one step further.

Addendum A — Elongation by Integration: One Master Formula for Non-Uniform Loading

Subtopic 01 derived two integrals separately: self-weight (T growing linearly) and the rotating rod (T growing quadratically). Both are instances of a single rule. Whenever either the tension T or the area A varies along the rod, apply Hooke's law to an infinitesimal element and integrate.

Method

Cut the rod at position x , where the element of length dx experiences local tension $T(x)$ across local area $A(x)$. Its stress is $T(x)/A(x)$, its strain is $T(x)/(A(x)Y)$, so it stretches by

$$d(\Delta L) = \frac{T(x)}{A(x)Y} dx, \quad \Delta L = \int_0^L \frac{T(x)}{A(x)Y} dx$$

Units check. $T/(AY)$ carries units $\text{N}/(\text{m}^2 \cdot \text{Pa}) = \text{N}/(\text{m}^2 \cdot \text{N m}^{-2})$, which is dimensionless; multiplied by dx (in m) the result is a length in metres. Every elongation you compute below must come out in m — if it does not, an area or modulus was misconverted.

Three stock situations cover nearly every exam question:

1. **Uniform rod, constant tension:** both factors constant, giving the chapter's $FL/(AY)$.
2. **Piecewise-constant tension** (point masses hung along the wire): split the range at each mass, use the constant tension on each segment, and add the segments.
3. **Continuously varying area under constant tension:** substitute $A(x)$ and integrate.

Worked Example A.1 — [JEE Main Level]

A rod in the shape of a circular frustum has end radii $R_1 = 2.0$ mm and $R_2 = 4.0$ mm, length $L = 0.50$ m, and Young's modulus $Y = 2.0 \times 10^{11}$ Pa. Find its elongation under a steady pull $F = 100$ N along its axis.

Because the pull is transmitted undiminished along the rod, $T(x) = F$ everywhere — only the area varies. The radius grows linearly from R_1 to R_2 , so

$$R(x) = R_1 + \frac{(R_2 - R_1)x}{L}, \quad A(x) = \pi \left(R_1 + \frac{(R_2 - R_1)x}{L} \right)^2$$

Substitute into the master integral:

$$\Delta L = \frac{F}{\pi Y} \int_0^L \frac{dx}{(R_1 + kx)^2}, \quad k = \frac{R_2 - R_1}{L}$$

Since $\int \frac{dx}{(R_1 + kx)^2} = -\frac{1}{k(R_1 + kx)},$

$$\Delta L = \frac{F}{\pi Y} \cdot \frac{1}{k} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{F}{\pi Y} \cdot \frac{L}{R_2 - R_1} \cdot \frac{R_2 - R_1}{R_1 R_2} = \frac{FL}{\pi R_1 R_2 Y}$$

The elegant result: a tapered rod stretches exactly as much as a *uniform* wire of radius $\sqrt{R_1 R_2}$ — the geometric mean of the end radii. Note it is emphatically *not* the uniform rod of average radius $(R_1 + R_2)/2$: thin regions dominate because strain goes as $1/A$.

Numerically, with $R_1 R_2 = 8.0 \times 10^{-6} \text{ m}^2$:

$$\Delta L = \frac{(100)(0.50)}{\pi(8.0 \times 10^{-6})(2.0 \times 10^{11})} = \frac{50}{5.03 \times 10^6} = 1.0 \times 10^{-5} \text{ m}$$

Check against the wrong route. A uniform rod of radius 3.0 mm would give $A = \pi(3.0 \times 10^{-3})^2 = 2.83 \times 10^{-5} \text{ m}^2$ and $\Delta L = 50/[(2.83 \times 10^{-5})(2.0 \times 10^{11})] = 8.8 \times 10^{-6} \text{ m}$ — smaller, because it underweights the skinny end. When $R_1 = R_2 = R$ the formula collapses to $FL/(\pi R^2 Y)$, the chapter's result, as it must.

Worked Example A.2 — [JEE Advanced Level]

A wire of length L , area A , density ρ , and Young's modulus Y hangs from a ceiling with a mass M tied to its lower end. Find the total elongation, including the wire's own weight. Evaluate for $M = 5.0$ kg, wire mass $m = 2.0$ kg, $L = 2.0$ m, $A = 1.0 \text{ mm}^2$, $Y = 2.0 \times 10^{11}$ Pa, $g = 10 \text{ m s}^{-2}$.

Measure x upward from the bottom. The tension at height x supports the hanging mass plus the wire material below it:

$$T(x) = Mg + \lambda gx, \quad \lambda = \rho A = \frac{m}{L}$$

Integrate:

$$\Delta L = \int_0^L \frac{Mg + \lambda gx}{AY} dx = \frac{MgL}{AY} + \frac{\lambda gL^2}{2AY} = \frac{MgL}{AY} + \frac{mgL}{2AY}$$

The load contributes the ordinary $MgL/(AY)$; the wire's own weight contributes exactly what it would with no load, $mgL/(2AY)$ — the two effects simply superpose because Hooke's law is linear.

Numerically, $AY = (1.0 \times 10^{-6} \text{ m}^2)(2.0 \times 10^{11} \text{ Pa}) = 2.0 \times 10^5 \text{ N}$, so

$$\Delta L = \frac{(5.0)(10)(2.0)}{2.0 \times 10^5} + \frac{(2.0)(10)(2.0)}{2(2.0 \times 10^5)} = \frac{100 + 20}{2.0 \times 10^5} = 6.0 \times 10^{-4} \text{ m} = 0.60 \text{ mm}$$

Practice — Addendum A

1. A steel wire of length 2.0 m and area 1.0 mm^2 ($Y = 2.0 \times 10^{11} \text{ Pa}$) carries a 6.0 kg mass at its midpoint and a 4.0 kg mass at its lower end. Find the total elongation ($g = 10 \text{ m s}^{-2}$).
2. A flat strip of length 0.40 m, thickness 1.0 mm, and Young's modulus $1.0 \times 10^{11} \text{ Pa}$ tapers linearly in breadth from 2.0 cm to 4.0 cm. It is pulled end-on with $F = 60 \text{ N}$. Find the elongation.
3. An elevator cable (treat as a wire: $L = 1.0 \text{ m}$, $A = 0.50 \text{ mm}^2$, $Y = 2.0 \times 10^{11} \text{ Pa}$) carries a 4.0 kg load. Find its elongation when the elevator (a) moves at constant velocity, (b) accelerates upward at 5.0 m s^{-2} ($g = 10 \text{ m s}^{-2}$).
4. Starting from $\Delta L = \frac{FL}{\pi R_1 R_2 Y}$, show that a conical frustum with $R_1 = R_2 = R$ behaves as a uniform wire, and state the effective area.

Answers: 1. Upper half carries $T = (6.0 + 4.0)g = 100 \text{ N}$; lower half carries $T = 4.0g = 40 \text{ N}$. With $AY = 2.0 \times 10^5 \text{ N}$: $\Delta L = \frac{(100)(1.0) + (40)(1.0)}{2.0 \times 10^5} = 7.0 \times 10^{-4} \text{ m} = 0.70 \text{ mm}$. 2.

$A(x) = t[b_1 + (b_2 - b_1)x/L]$ with constant thickness t . Integrating gives $\Delta L = \frac{FL \ln(b_2/b_1)}{(b_2 - b_1)tY} = \frac{(60)(0.40)(0.693)}{(0.020)(0.001)(1.0 \times 10^{11})} = \frac{16.6}{2.0 \times 10^6} = 8.3 \times 10^{-6} \text{ m}$. (Logarithms appear whenever breadth — not radius — tapers linearly, because then A itself is linear.) 3. (a) $T = Mg = 40 \text{ N}$: $\Delta L = \frac{(40)(1.0)}{40} = \frac{40}{1.0 \times 10^5} = 4.0 \times 10^{-4} \text{ m}$. (b) $T = M(g+a) = 60 \text{ N}$: $\Delta L = 6.0 \times 10^{-4} \text{ m}$. Acceleration enters only through the effective tension. 4. Substituting $R_1 = R_2 = R$ gives $\Delta L = \frac{FL}{\pi R^2 Y}$, the uniform-wire result; the effective area is $A_{\text{eff}} = \pi R_1 R_2$.

Addendum B — Constrained Thermal Stress: Gaps and Dissimilar Rods

Subtopic 05 derives $\sigma = Y\alpha\Delta T$ for a rod perfectly clamped at both ends. Exams ask two refinements the chapter skips: a rod with a small *gap* before the wall, and two *different* materials welded together and then clamped.

Method

Step 1 — Ask what the rod wants to do. Unclamped, a temperature change ΔT produces free expansion $\Delta L_{\text{free}} = L\alpha\Delta T$.

Step 2 — Impose the constraint geometrically. If a gap of width s separates the hot end from a rigid wall, the rod first fills the gap freely. Only the excess is suppressed. The imposed compressive strain is therefore

$$\varepsilon = \alpha\Delta T - \frac{s}{L}$$

and the compressive stress is $\sigma = Y\varepsilon$ — *provided* $\alpha\Delta T > s/L$. If the free expansion cannot even reach the wall, there is no contact and $\sigma = 0$. This conditional is where marks are lost.

Step 3 — Composite rods: two equations, two unknowns. Weld rods 1 and 2 (lengths L_1, L_2 ; moduli Y_1, Y_2 ; areas A_1, A_2 ; expansion coefficients α_1, α_2) and clamp the assembly between rigid walls. Since the walls push with one continuous force,

$$F_1 = F_2 = F \quad (\text{equilibrium})$$

and the sum of the mechanical strains must cancel the sum of the prevented thermal expansions:

$$\frac{FL_1}{A_1Y_1} + \frac{FL_2}{A_2Y_2} = (\alpha_1L_1 + \alpha_2L_2)\Delta T \quad (\text{compatibility})$$

Solve for F ; the individual stresses follow as $\sigma_i = F/A_i$. Note the stresses in the two rods are generally *different* even though the force is common.

Worked Example B.1 — [JEE Main Level]

A steel rod of length 1.0 m and area 2.0 cm² stands with one end against a rigid wall and a gap $s = 0.30$ mm at the other end. It is heated by $\Delta T = 50$ °C. Given $\alpha = 1.2 \times 10^{-5}$ °C⁻¹ and $Y = 2.0 \times 10^{11}$ Pa, find the force exerted on the wall.

Free expansion would be

$$\Delta L_{\text{free}} = L\alpha\Delta T = (1.0)(1.2 \times 10^{-5})(50) = 6.0 \times 10^{-4} \text{ m} = 0.60 \text{ mm}$$

This exceeds the gap, so contact occurs. The suppressed compression is $0.60 - 0.30 = 0.30$ mm over 1.0 m, i.e. strain 3.0×10^{-4} . Then

$$\sigma = Y\varepsilon = (2.0 \times 10^{11} \text{ Pa})(3.0 \times 10^{-4}) = 6.0 \times 10^7 \text{ Pa}$$

$$F = \sigma A = (6.0 \times 10^7 \text{ Pa})(2.0 \times 10^{-4} \text{ m}^2) = 1.2 \times 10^4 \text{ N} = 12 \text{ kN}$$

Check. Had the gap been absent, $\sigma = Y\alpha\Delta T = 1.2 \times 10^8$ Pa — double. Halving the suppressed strain halves the stress, exactly as linearity demands.

Worked Example B.2 — [JEE Advanced Level]

A steel rod ($L_1 = 1.0$ m, $\alpha_1 = 1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$, $Y_1 = 2.0 \times 10^{11}$ Pa) is welded end-to-end to a brass rod ($L_2 = 1.0$ m, $\alpha_2 = 2.0 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$, $Y_2 = 1.0 \times 10^{11}$ Pa). Both have area $A = 1.0 \text{ cm}^2$. The assembly is clamped between rigid walls and heated by $\Delta T = 40 \text{ }^\circ\text{C}$. Find the force in each rod and the strain in each rod.

Compatibility equation:

$$F \left(\frac{L_1}{A_1 Y_1} + \frac{L_2}{A_2 Y_2} \right) = (\alpha_1 L_1 + \alpha_2 L_2) \Delta T$$

Left bracket: $\frac{1.0}{(1.0 \times 10^{-4})(2.0 \times 10^{11})} + \frac{1.0}{(1.0 \times 10^{-4})(1.0 \times 10^{11})} = 5.0 \times 10^{-8} + 1.0 \times 10^{-7} = 1.5 \times 10^{-7} \text{ N}^{-1}$.

Right side: $(1.2 \times 10^{-5} + 2.0 \times 10^{-5})(1.0)(40) = 1.28 \times 10^{-3}$ (dimensionless).

$$F = \frac{1.28 \times 10^{-3}}{1.5 \times 10^{-7}} = 8.5 \times 10^3 \text{ N}$$

Both rods are in compression with this force. The stresses are equal here (equal areas):

$$\sigma_1 = \sigma_2 = \frac{8.5 \times 10^3 \text{ N}}{1.0 \times 10^{-4} \text{ m}^2} = 8.5 \times 10^7 \text{ Pa}$$

but the strains are not: $\varepsilon_1 = \sigma/Y_1 = 4.3 \times 10^{-4}$ (steel) versus $\varepsilon_2 = \sigma/Y_2 = 8.5 \times 10^{-4}$ (brass). Their sum, 1.28×10^{-3} , reproduces the prevented expansion — a built-in check. The softer brass absorbs twice the strain while carrying the same force.

Practice — Addendum B

1. A copper rod ($L = 0.80$ m, $Y = 1.1 \times 10^{11}$ Pa, $\alpha = 1.7 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$) is rigidly clamped and cooled by $25 \text{ }^\circ\text{C}$. Find the stress developed and state whether it is tensile or compressive.
2. An aluminium rod of length 2.0 m ($\alpha = 2.4 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$) sits with a gap of 1.0 mm before a wall. At what temperature rise does it just make contact?
3. For the composite of Example B.2, derive the general expression for F when the areas are A_1 and A_2 (not necessarily equal), and state which rod carries the larger stress if $A_1 < A_2$.
4. A steel rail is laid in 20 m segments with small gaps at each joint. Explain in one sentence why the gap size is chosen using $\alpha L \Delta T$ rather than $\sigma = Y \alpha \Delta T$.

Answers: 1. $\sigma = Y \alpha \Delta T = (1.1 \times 10^{11})(1.7 \times 10^{-5})(25) = 4.7 \times 10^7$ Pa, tensile — cooling makes the rod want to shrink, so the clamps pull it outward. 2. Set $L \alpha \Delta T = s$: $\Delta T = \frac{1.0 \times 10^{-3}}{(2.0)(2.4 \times 10^{-5})} \approx 21 \text{ }^\circ\text{C}$. 3. $F = \frac{(\alpha_1 L_1 + \alpha_2 L_2) \Delta T}{L_1/(A_1 Y_1) + L_2/(A_2 Y_2)}$; since $\sigma_i = F/A_i$, the narrower rod A_1 carries the larger stress. 4. The gap exists precisely to *prevent* stress from developing: it accommodates the free expansion $\alpha L \Delta T$, keeping $\varepsilon = 0$ and hence $\sigma = 0$.

Addendum C — The Wire as a Spring, Set in Motion

Subtopic 02 establishes $k = YA/L$ but uses it only statically. JEE routinely attaches a mass and asks for oscillations or impact. The entire method is: replace the wire by a spring of constant k , then use Class 11 mechanics without modification.

Method

1. Compute $k = YA/L$ (unit N m^{-1}). For combinations, use the series/parallel rules already in the chapter.
2. **Oscillations about equilibrium:** after the mass settles, the static stretch $x_{\text{st}} = Mg/k$ merely relocates the equilibrium point. Gravity shifts the centre of oscillation but never appears in the period:

$$T = 2\pi\sqrt{\frac{M}{k}} = 2\pi\sqrt{\frac{ML}{YA}}$$

3. **Impact from a drop:** release the mass from height h above the unstretched wire. Energy conservation between release (wire slack) and the lowest point (momentarily at rest) gives

$$Mg(h+x) = \frac{1}{2}kx^2 \implies x = \frac{Mg + \sqrt{M^2g^2 + 2kMgh}}{k}$$

taking the positive root. The chapter's sudden-load result is the special case $h = 0$: then $x = 2Mg/k = 2x_{\text{st}}$. For h large compared with x , the Mgx term is a small correction and $x \approx \sqrt{2Mgh/k}$.

4. **Twisting oscillations:** a wire twisted by angle θ provides restoring torque $C = c\theta$ with torsional rigidity $c = \pi\eta R^4/(2L)$ (Subtopic 05), so a body of moment of inertia I executes

$$T = 2\pi\sqrt{\frac{I}{c}}$$

Worked Example C.1 — [JEE Main Level]

A mass $M = 3.0$ kg hangs from a steel wire ($L = 1.0$ m, $A = 0.50$ mm², $Y = 2.0 \times 10^{11}$ Pa) and oscillates vertically. Find the spring constant, the static stretch, and the period.

$$k = \frac{YA}{L} = \frac{(2.0 \times 10^{11} \text{ Pa})(5.0 \times 10^{-7} \text{ m}^2)}{1.0 \text{ m}} = 1.0 \times 10^5 \text{ N m}^{-1}$$

$$x_{\text{st}} = \frac{Mg}{k} = \frac{30 \text{ N}}{1.0 \times 10^5 \text{ N m}^{-1}} = 3.0 \times 10^{-4} \text{ m}$$

$$T = 2\pi\sqrt{\frac{3.0}{1.0 \times 10^5}} \text{ s} = 2\pi(5.48 \times 10^{-3}) \text{ s} = 3.4 \times 10^{-2} \text{ s}$$

Note g appears nowhere in T — on the Moon the period would be identical.

Worked Example C.2 — [JEE Advanced Level]

Using the wire of Subtopic 03, Example 4 ($m = 2.0 \text{ kg}$, $L = 2.0 \text{ m}$, $A = 1.0 \text{ mm}^2$, $Y = 2.0 \times 10^{11} \text{ Pa}$, $g = 10 \text{ m s}^{-2}$), **the block is released from rest** $h = 0.20 \text{ m}$ **above the unstretched wire. Find the maximum elongation.**

First $k = YA/L = (2.0 \times 10^{11})(1.0 \times 10^{-6})/2.0 = 1.0 \times 10^5 \text{ N m}^{-1}$, and the static stretch is $x_{\text{st}} = Mg/k = 2.0 \times 10^{-4} \text{ m}$ (the chapter's number).

Energy conservation, release point to lowest point:

$$Mg(h+x) = \frac{1}{2}kx^2 \implies (20)(0.20+x) = \frac{1}{2}(1.0 \times 10^5)x^2$$

$$5.0 \times 10^4 x^2 - 20x - 4.0 = 0$$

Taking the positive root of $ax^2 + bx + c = 0$ with $a = 5.0 \times 10^4$, $b = -20$, $c = -4.0$:

$$x = \frac{20 + \sqrt{400 + 8.0 \times 10^5}}{1.0 \times 10^5} = \frac{20 + 894.6}{1.0 \times 10^5} = 9.1 \times 10^{-3} \text{ m} \approx 9.1 \text{ mm}$$

Checks. The shortcut $x \approx \sqrt{2Mgh/k} = \sqrt{8.0 \times 10^{-5}} \text{ m} = 8.9 \text{ mm}$ agrees to 2%, legitimate because $h = 0.20 \text{ m} \gg x$. And the $h = 0$ limit of the same quadratic returns the chapter's $x_{\text{max}} = 2x_{\text{st}} = 0.40 \text{ mm}$ — a twenty-three-fold increase from a mere 0.20 m drop height, because the quadratic term dominates.

Practice — Addendum C

1. A 1.0 kg mass on a wire ($L = 0.50 \text{ m}$, $A = 0.20 \text{ mm}^2$, $Y = 2.0 \times 10^{11} \text{ Pa}$) oscillates vertically. Find the period.
2. Prove algebraically that setting $h = 0$ in the drop equation yields $x = 2Mg/k$.
3. A torsion pendulum consists of a disc of moment of inertia $I = 0.20 \text{ kg m}^2$ suspended by a wire of radius $R = 1.0 \text{ mm}$, length $L = 1.0 \text{ m}$, rigidity $\eta = 4.0 \times 10^{10} \text{ Pa}$. Find the period.
4. A block dropped from height h stretches a wire to x_1 ; dropped from $4h$, it stretches to x_2 . Show that for large h , $x_2 \approx 2x_1$.

Answers: 1. $k = (2.0 \times 10^{11})(2.0 \times 10^{-7})/0.50 = 8.0 \times 10^4 \text{ N m}^{-1}$; $T = 2\pi\sqrt{1.0/(8.0 \times 10^4)} = 2\pi(3.54 \times 10^{-3}) \text{ s} = 2.2 \times 10^{-2} \text{ s}$. 2. With $h = 0$: $\frac{1}{2}kx^2 = Mgx \Rightarrow x(k/2 - Mg) = 0$; rejecting the trivial root $x = 0$ leaves $x = 2Mg/k = 2x_{\text{st}}$. 3. $c = \frac{\pi(4.0 \times 10^{10} \text{ Pa})(1.0 \times 10^{-12} \text{ m}^4)}{2(1.0 \text{ m})} = 6.3 \times 10^{-2} \text{ N m rad}^{-1}$; $T = 2\pi\sqrt{0.20/(6.3 \times 10^{-2})} \text{ s} = 11.2 \text{ s}$. 4. For $h \gg x$, $x \approx \sqrt{2Mgh/k} \propto \sqrt{h}$; quadrupling h doubles x .

Addendum D — Two Wires Sharing a Load Through a Rigid Bar

A weight hangs from a light rigid horizontal bar whose ends are supported by two vertical wires of different stiffness. Which wire pulls harder? The chapter's spring rules answer it, but only after you write the two equations exams demand: one force balance and one geometric compatibility.

Method

Let wires 1 and 2 have spring constants $k_1 = Y_1 A_1 / L_1$ and $k_2 = Y_2 A_2 / L_2$, and let the load be W .

Case 1 — the bar stays horizontal. Equal extensions are forced by the geometry: $\Delta L_1 = \Delta L_2$. Combined with $T_1 + T_2 = W$:

$$\frac{T_1}{k_1} = \frac{T_2}{k_2} \Rightarrow \boxed{T_1 = W \frac{k_1}{k_1 + k_2}, \quad T_2 = W \frac{k_2}{k_1 + k_2}}$$

The *stiffer* wire carries the *larger* share. (For the bar actually to remain horizontal with a single hanging load, the load must sit where the moments balance: $T_1 a = T_2 b$ for distances a, b from the load.)

Case 2 — the load hangs elsewhere, and the bar tilts. Take moments on the light bar to relate T_1 and T_2 to the load position; the extensions then differ, and the tilt follows from their difference divided by the wire separation. Never assume equal extensions unless horizontality is given or forced.

Worked Example D.1 — [JEE Main Level]

Steel ($Y_s = 2.0 \times 10^{11}$ Pa) **and copper** ($Y_c = 1.0 \times 10^{11}$ Pa) **wires, each of length 1.0 m and area 1.0 mm², hang from a ceiling 0.90 m apart and carry a light horizontal bar. A weight $W = 150$ N is hung so that the bar remains level. Find each tension and the load's position.**

Stiffnesses: $k_s = (2.0 \times 10^{11})(1.0 \times 10^{-6})/1.0 = 2.0 \times 10^5 \text{ N m}^{-1}$; $k_c = 1.0 \times 10^5 \text{ N m}^{-1}$.

Equal extensions give $T_s/T_c = k_s/k_c = 2$, and $T_s + T_c = 150$ N:

$$T_s = 100 \text{ N}, \quad T_c = 50 \text{ N}, \quad \Delta L = \frac{T_s}{k_s} = \frac{100}{2.0 \times 10^5} = 5.0 \times 10^{-4} \text{ m}$$

Moments about the load (or equivalently $T_c \cdot 0.90 = W d$ measuring d from the steel wire):

$$d = \frac{T_c}{W}(0.90) = \frac{50}{150}(0.90) = 0.30 \text{ m}$$

The load sits 0.30 m from the steel wire — closer to the stronger supporter, exactly as lever

intuition demands.

Worked Example D.2 — [JEE Advanced Level]

Same two wires and bar, but now the $W = 150\text{ N}$ load hangs from the midpoint of the bar (separation still 0.90 m). Find the tensions and the tilt of the bar.

Take moments about the bar's midpoint. The bar is light, so the only torques are the two tensions acting at $\pm 0.45\text{ m}$:

$$T_s(0.45) - T_c(0.45) = 0 \implies T_s = T_c = 75\text{ N}$$

A surprising result: the tensions are equal *regardless of stiffness* — the bar enforces it by tilting. The extensions now differ:

$$\Delta L_s = \frac{75}{2.0 \times 10^5} = 3.75 \times 10^{-4}\text{ m}, \quad \Delta L_c = \frac{75}{1.0 \times 10^5} = 7.50 \times 10^{-4}\text{ m}$$

The copper corner drops lower by $\delta = 3.75 \times 10^{-4}\text{ m}$, so

$$\theta \approx \frac{\delta}{0.90} = 4.2 \times 10^{-4}\text{ rad}$$

Why this is Advanced-level. Cases 1 and 2 look identical but have opposite physics: horizontality forces equal *extensions* (unequal tensions), while a central load on a pivoting light bar forces equal *tensions* (unequal extensions). Recognising which quantity the constraint pins down is the entire problem.

Practice — Addendum D

1. Steel wire: $A = 2.0\text{ mm}^2$, $L = 2.0\text{ m}$, $Y = 2.0 \times 10^{11}\text{ Pa}$. Brass wire: $A = 1.0\text{ mm}^2$, $L = 1.0\text{ m}$, $Y = 1.0 \times 10^{11}\text{ Pa}$. They support a horizontal bar carrying $W = 120\text{ N}$ appropriately. Find the tensions and the common extension.
2. For the setup of Practice 1 with the wires 1.0 m apart, where must the load hang for the bar to stay horizontal?
3. Two identical wires support a horizontal bar; the load hangs at one-third of the separation from wire 1. Find $T_1 : T_2$.
4. In Case 1, prove that the common extension can be written as $\Delta L = W/(k_1 + k_2)$, and interpret the combination.

Answers: 1. $k_s = (2.0 \times 10^{11})(2.0 \times 10^{-6})/2.0 = 2.0 \times 10^5\text{ N m}^{-1}$; $k_b = 1.0 \times 10^5\text{ N m}^{-1}$. $T_s = 120 \times \frac{2}{3} = 80\text{ N}$, $T_b = 40\text{ N}$; common extension $= 80/(2.0 \times 10^5) = 4.0 \times 10^{-4}\text{ m}$. 2. Moments about the load: $T_s x = T_b(1.0 - x)$ with $T_s = 2T_b$ gives $x = 1/3\text{ m}$ from the steel wire. 3. Moments: $T_1(d/3) = T_2(2d/3)$, so $T_1 : T_2 = 2 : 1$ — position alone fixes the ratio; stiffness is irrelevant here. 4. Adding $\Delta L = T_1/k_1$ and $\Delta L = T_2/k_2$ after multiplying by the respective k 's, or directly: $W = T_1 + T_2 = \Delta L(k_1 + k_2)$. The pair acts as springs *in parallel* — stiffnesses add.

Addendum E — The Interatomic Potential Made Quantitative

Subtopic 05 argues qualitatively that atoms sit at the bottom of a parabolic well. JEE Advanced hands you an explicit $U(r)$ and asks for the equilibrium separation, the well depth, the bond stiffness, or the resulting vibration frequency. The tools are pure calculus.

Method

Given a pair potential $U(r)$ with a minimum at $r = r_0$:

1. **Equilibrium:** $F = -dU/dr = 0$ at r_0 .
2. **Well depth:** substitute r_0 back into U ; the binding energy is $|U(r_0)|$.
3. **Bond stiffness:** expand about the minimum. By Taylor's theorem,

$$U(r_0 + x) \approx U(r_0) + \frac{1}{2} \left. \frac{d^2U}{dr^2} \right|_{r_0} x^2 \Rightarrow \kappa = \left. \frac{d^2U}{dr^2} \right|_{r_0}$$

with unit N m^{-1} . The linear restoring force is $F \approx -\kappa x$ — this is Hooke's law, and its small- x validity is exactly the small-strain limit quoted in the chapter.

4. **Vibrations:** two atoms of masses m_1, m_2 bonded by κ oscillate with angular frequency $\omega = \sqrt{\kappa/\mu}$, where $\mu = m_1 m_2 / (m_1 + m_2)$ is the reduced mass (for identical atoms, $\mu = m/2$).

Worked Example E.1 — [JEE Advanced Level]

The Lennard-Jones potential $U(r) = \frac{A}{r^{12}} - \frac{B}{r^6}$ **describes a pair of atoms ($A, B > 0$). Find the equilibrium separation r_0 , the well depth, and the bond stiffness κ .**

Differentiate and set to zero:

$$-\frac{12A}{r^{13}} + \frac{6B}{r^7} = 0 \Rightarrow r_0^6 = \frac{2A}{B}$$

(The second derivative is positive there, confirming a minimum.) Well depth: with $r_0^{12} = 4A^2/B^2$,

$$U(r_0) = \frac{AB^2}{4A^2} - \frac{B^2}{2A} = \frac{B^2}{4A} - \frac{B^2}{2A} = -\frac{B^2}{4A}$$

so the atoms bind with energy $B^2/(4A)$. Stiffness:

$$\frac{d^2U}{dr^2} = \frac{156A}{r^{14}} - \frac{42B}{r^8} \Rightarrow \kappa = \frac{156A}{r_0^8 r_0^6} - \frac{42B}{r_0^8} = \frac{78B - 42B}{r_0^8} = \frac{36B}{r_0^8}$$

using $r_0^6 = 2A/B$ to eliminate A . All three quantities are expressible purely through A and B — the signature move of this problem type.

Worked Example E.2 — [JEE Main Level]

Near equilibrium an atom's potential is well modelled by $U(x) = \frac{1}{2}\kappa x^2 - \beta x^3$, with $\kappa = 20 \text{ N m}^{-1}$ and $\beta = 8.0 \times 10^9 \text{ N m}^{-2}$, where x is the displacement from r_0 . Find the exact restoring force for $x = 1.0 \times 10^{-11} \text{ m}$ and the percentage error made by keeping only Hooke's law.

Exact force:

$$F = -\frac{dU}{dx} = -\kappa x + 3\beta x^2$$

Hooke's term: $-\kappa x = -(20)(1.0 \times 10^{-11}) = -2.0 \times 10^{-10} \text{ N}$. Cubic term: $3\beta x^2 = 3(8.0 \times 10^9)(1.0 \times 10^{-22}) = 2.4 \times 10^{-12} \text{ N}$.

$$F = -2.0 \times 10^{-10} + 2.4 \times 10^{-12} = -1.98 \times 10^{-10} \text{ N}$$

The correction is $2.4 \times 10^{-12} / 2.0 \times 10^{-10} = 1.2\%$. Two lessons: Hooke's law is excellent for atomic-scale displacements (hence the whole chapter works), and the cubic term makes the well lopsided — pushing atoms together meets resistance sooner than pulling them apart, which is why real solids fracture in tension rather than implode.

Practice — Addendum E

1. For the potential $U(r) = -A/r + B/r^9$ ($A, B > 0$), find r_0 .
2. For $U(x) = \frac{1}{2}\kappa x^2 - \beta x^3$, evaluate the force at $x = +a$ and $x = -a$ and show the magnitudes differ.
3. Two identical atoms of mass $m = 1.6 \times 10^{-26} \text{ kg}$ are bonded with $\kappa = 320 \text{ N m}^{-1}$. Find the vibration angular frequency and frequency.
4. Using the result of Worked Example E.1, state how κ scales if B is doubled at fixed r_0 .

Answers: 1. $-dU/dr = A/r^2 - 9B/r^{10} = 0 \Rightarrow r_0^8 = 9B/A$, so $r_0 = (9B/A)^{1/8}$. 2. $F(+a) = -\kappa a + 3\beta a^2$ (magnitude $\kappa a - 3\beta a^2$); $F(-a) = \kappa a + 3\beta a^2$ (magnitude $\kappa a + 3\beta a^2$) — compression is resisted more strongly, by $6\beta a^2$. 3. $\mu = m/2 = 8.0 \times 10^{-27} \text{ kg}$; $\omega = \sqrt{\kappa/\mu} = \sqrt{4.0 \times 10^{28}} = 2.0 \times 10^{14} \text{ rad s}^{-1}$; $\nu = \omega/2\pi = 3.2 \times 10^{13} \text{ Hz}$ — squarely in the infrared, as molecular vibrations are. 4. From $\kappa = 36B/r_0^8$, doubling B doubles κ .

Addendum F — Error Propagation in Searle's Experiment

The chapter describes Searle's procedure but never says what precision the result inherits — a guaranteed CBSE practical-exam question and a recurring JEE Main one-liner.

Method

Searle's working formula is $Y = \frac{MgL}{\pi r^2 \Delta L}$. Take logarithms and differentiate: every factor contributes its own *relative* error, and powers multiply:

$$\ln Y = \ln M + \ln g + \ln L - 2 \ln r - \ln \Delta L - \ln \pi \implies \frac{\delta Y}{Y} = \frac{\delta M}{M} + \frac{\delta L}{L} + 2 \frac{\delta r}{r} + \frac{\delta(\Delta L)}{\Delta L}$$

Errors in g and π are negligible. The radius enters squared, so its relative error counts *double* — which is why the chapter warns that a small radius error wrecks the determination. Absolute errors combine as worst case (all same sign); if a question asks for the most probable error, combine the terms in quadrature instead.

Worked Example F.1 — [CBSE Board / JEE Main Level]

In a Searle's experiment: $M = 5.0$ kg measured to 0.5%; $L = 2.0$ m measured with $\delta L = 2$ mm; the diameter $D = 1.00$ mm measured with a least count of 0.01 mm; and $\Delta L = 0.50$ mm measured to 1%. Find the percentage error in Y .

Term by term:

$$\frac{\delta M}{M} = 0.5\%, \quad \frac{\delta L}{L} = \frac{2 \text{ mm}}{2000 \text{ mm}} = 0.1\%$$

The radius is half the diameter: $r = 0.50$ mm with $\delta r = 0.01$ mm, giving $\delta r/r = 2\%$, and hence $2 \delta r/r = 4.0\%$. Finally $\delta(\Delta L)/\Delta L = 1\%$.

$$\frac{\delta Y}{Y} = 0.5 + 0.1 + 4.0 + 1.0 = 5.6\%$$

Nearly three-quarters of the total comes from the micrometer screw gauge alone. If asked which instrument to upgrade: the diameter measurement, unambiguously.

Practice — Addendum F

1. Repeat Example F.1 but suppose the student mistakenly quotes $\delta r/r$ as 2% without doubling. What error results, and why is it wrong?
2. A wire of diameter 0.80 mm is measured with a least count of 0.01 mm, and all other contributions total 1.5%. Find the total percentage error in Y .
3. Show dimensionally that $\delta Y/Y$ is dimensionless, and state why this must be so.
4. The elongation ΔL is measured via the difference of two micrometer readings, each with least count 0.01 mm. For $\Delta L = 0.20$ mm, estimate $\delta(\Delta L)/\Delta L$ and its contribution to $\delta Y/Y$.

Answers: 1. They would report $0.5 + 0.1 + 2.0 + 1.0 = 3.6\%$ — understating the true error, because r enters Y as r^2 and its fractional contribution must be counted twice. 2. $\delta r/r = 0.01/0.40 = 2.5\%$, doubled to 5.0%; total = $5.0 + 1.5 = 6.5\%$. 3. Each term is a quantity divided by itself — pure number, no units. A *relative* error compares like with like, so it cannot carry

units; only the absolute error δY carries pascals. 4. Two readings, so $\delta(\Delta L) \approx 2 \times 0.01 \text{ mm} = 0.02 \text{ mm}$; $\delta(\Delta L)/\Delta L = 0.02/0.20 = 10\%$, contributing a full 10% to $\delta Y/Y$. Small elongations are the second-biggest enemy after the radius — measure ΔL large by using heavier loads (within the elastic limit).

Chapter 9: Mechanical Properties of Fluids

Master Content — Complete Chapter (7 Topics, CBSE / NEET / JEE Main / JEE Advanced)

Drona — Physics Content Bank

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1. Fluid Pressure and Pascal's Law

Exam Relevance: This is the foundation stone of the entire fluids chapter. CBSE Boards almost always carry a 1–3 mark question on gauge vs. absolute pressure, the barometer, or a depth-pressure calculation. JEE Main reliably asks 1–2 questions on manometers, hydraulic lifts, or the $P = P_0 + \rho gh$ relation. JEE Advanced rarely tests it alone but folds it into multi-liquid U-tube and accelerated-fluid problems. NEET typically asks one conceptual MCQ on the scalar/isotropic nature of pressure or a clean depth substitution.

Section 1: Concept Intuition

Imagine you are standing in a crowded Mumbai local train at peak hour. People are pressed against you from every side — front, back, left, right. You do not feel a push in just *one* direction; you feel squeezed *inward* from all directions at once. This is almost exactly what a tiny water droplet feels when it sits deep inside a tank. The surrounding fluid presses on it from every direction. That inward squeeze, measured per unit area, is what we call **pressure**.

Let us be precise. Take any small flat surface of area A submerged in a fluid at rest. The fluid pushes on that surface with a force. The crucial observation is that this force is *always perpendicular* (normal) to the surface — never sideways. If there were a sideways (tangential) component, the fluid would simply slide and flow, and it would no longer be “at rest.” So for a static fluid, only the normal force survives. Pressure is the magnitude of this normal force divided by the area it acts on.

Think about it this way... Pressure is *not* a vector, even though force is. A diver's eardrum at a depth of 10 m feels the same pressure whether the eardrum faces up, down, or sideways — turn your head however you like, the water still pushes equally. Because pressure has no single direction of its own (it acts perpendicular to *whatever* surface you place in the fluid), it is a **scalar**. This is one of the most tested conceptual points in the whole chapter: *force is a vector, pressure is a scalar*.

Now, why does pressure increase as you go deeper? Picture a tall column of water like the overhead tank on a building rooftop. The water at the bottom of the tank has to support the *weight* of all the water sitting above it. The deeper you go, the taller the column of fluid pressing down, so the greater the pressure. This gives us the single most important idea of the subtopic: pressure grows linearly with depth. At the free surface you only feel the atmosphere pushing down (P_0); a depth h below that, you feel the atmosphere *plus* the weight of the fluid column above you.

A beautifully counter-intuitive consequence is the **hydrostatic paradox**. Take three vessels of wildly different shapes — one narrow and tall, one wide and squat, one funnel-shaped — and fill all three with water to the *same height*. The pressure at the base is *identical* in all three, even though they hold completely different *amounts* of water. Students find this shocking: surely more water means more pressure? No. Pressure at a point depends only on the *depth* below the free surface and the fluid's density — not on the total volume or the shape of the container. It is like how the depth of your debt depends on how far below zero your balance is, not on how wide your bank account “container” happens to be.

This leads to a second great idea, due to Blaise Pascal. Suppose you press down on an enclosed

fluid — say you push a piston into a sealed bottle of water. That extra pressure does not stay near the piston; it spreads *undiminished* to every single point in the fluid and onto every wall of the container. Squeeze one end of a toothpaste tube and the paste shoots out the far end — that is **Pascal’s law** in everyday action.

Limiting Conditions & Assumptions. All of this clean behaviour rests on a few assumptions. (1) The fluid is at *rest* — these results fail the moment the fluid accelerates or flows. (2) The fluid is **incompressible**, so its density ρ stays constant with depth. This is excellent for liquids but *fails for gases*, whose density changes a lot with pressure — which is exactly why “how high does the atmosphere extend” cannot be answered with a single ρgh . (3) We neglect surface tension and the fluid’s own internal stresses. (4) For Pascal’s law, the fluid must be *enclosed* — an open fluid simply spills rather than transmitting pressure.

Section 2: Key Formulas & Definitions

Pressure (at a point):

$$P = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_{\perp}}{\Delta A} = \frac{dF_{\perp}}{dA}$$

where $F_{\perp}$ is the normal force on area A .

- **SI unit:** pascal, $1 \text{ Pa} = 1 \text{ N m}^{-2}$
- **Dimensional formula:** $[P] = \frac{[MLT^{-2}]}{[L^2]} = [M L^{-1} T^{-2}]$

Variation of pressure with depth (incompressible fluid):

$$\frac{dP}{dh} = \rho g \quad \Rightarrow \quad P = P_0 + \rho gh$$

where P = absolute pressure at depth h , P_0 = pressure at the free surface (atmospheric), ρ = fluid density, g = acceleration due to gravity.

Gauge pressure and absolute pressure:

$$P_g = P - P_0 = \rho gh \quad \Rightarrow \quad P_{\text{abs}} = P_0 + P_g$$

Absolute pressure ≥ 0 always; gauge pressure may be **positive or negative** (negative when the point is at lower pressure than the atmosphere, e.g. sucking on a straw).

Pascal’s Law: A change in pressure applied to an enclosed incompressible fluid is transmitted *undiminished* to every point of the fluid and to the walls of its container.

Hydraulic lift (Pascal’s law in action): With input piston area A_1 , force F_1 , and output piston area A_2 , force F_2 :

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \Rightarrow \quad F_2 = F_1 \frac{A_2}{A_1}$$

Since the same volume of fluid is displaced at both pistons, $A_1 h_1 = A_2 h_2$, giving:

$$h_2 = h_1 \frac{A_1}{A_2}$$

$$\text{Mechanical advantage} = \frac{F_2}{F_1} = \frac{A_2}{A_1}.$$

Useful pressure conversions:

$$1 \text{ atm} = 1.013 \times 10^5 \text{ Pa} = 760 \text{ torr} = 760 \text{ mm Hg}, \quad 1 \text{ bar} = 10^5 \text{ Pa}, \quad 1 \text{ torr} = 133 \text{ Pa}$$

Section 3: CBSE Board Derivations / Key Procedures

Derivation 1 — Variation of pressure with depth.

Assumption: Consider an incompressible fluid of uniform density ρ at rest in a container, open to the atmosphere at the top.

Figure 9.1 *A container of liquid with the free surface at the top. An imaginary thin vertical cylindrical element of liquid, cross-sectional area A , with its top face at depth h_1 (pressure P_1) and bottom face at depth h_2 (pressure P_2). Arrows: P_1A pushing down on the top face, P_2A pushing up on the bottom face, and weight mg of the element acting downward. Isolate an imaginary thin cylindrical column of fluid of cross-sectional area A , with its top face at depth h_1 and bottom face at depth h_2 . Because the fluid is at rest, this column is in equilibrium, so the net vertical force on it is zero. Three vertical forces act on the column: - Downward force on the top face: P_1A (fluid above pushes down) - Upward force on the bottom face: P_2A (fluid below pushes up) - Weight of the column acting downward: $W = mg = (\rho \cdot A \cdot (h_2 - h_1))g$ Physical reasoning: the upward push from below must support both the downward push from above and the column's own weight. Setting net force to zero: $P_2A = P_1A + \rho A(h_2 - h_1)g$ Cancelling A : $P_2 - P_1 = \rho g(h_2 - h_1)$ Now choose the top face at the free surface, where $h_1 = 0$ and $P_1 = P_0$, and let the bottom face be at depth h where $P_2 = P$: $P = P_0 + \rho gh$ Notice the result depends only on depth h , not on the horizontal cross-section or the container's shape — this is precisely the origin of the hydrostatic paradox. (see next page)*

Derivation 2 — Pascal's Law (demonstration).

Assumption: An enclosed incompressible liquid in a cylinder fitted with a piston of area A .

The pressure at any interior point B lying a vertical distance z below the piston is

$$P_B = P_{\text{piston}} + \rho g z$$

Now apply an *extra* external force F on the piston, increasing the pressure just below it by $\Delta P_{\text{piston}} = F/A$. *Physical reasoning:* because the liquid is incompressible, the geometry does not change — z

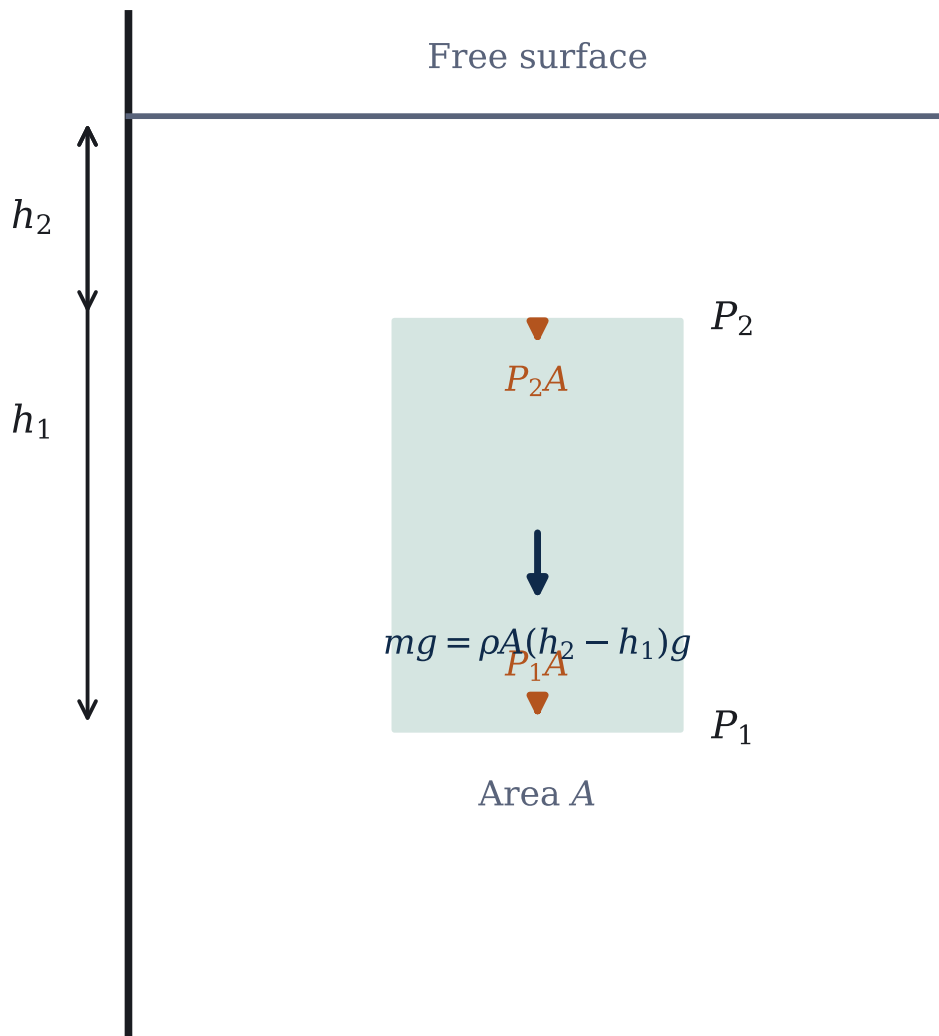


Figure 9.1 A container of liquid with the free surface at the top. An imaginary thin vertical cylindrical element of liquid, cross-sectional area A , with its top face at depth h_1 (pressure P_1) and bottom face at depth h_2 (pressure P_2). Arrows: P_1A pushing down on the top face, P_2A pushing up on the bottom face, and weight mg of the element acting downward. Isolate an imaginary thin cylindrical column of fluid of cross-sectional area A , with its top face at depth h_1 and bottom face at depth h_2 . Because the fluid is at rest, this column is in equilibrium, so the net vertical force on it is zero. Three vertical forces act on the column: - Downward force on the top face: P_1A (fluid above pushes down) - Upward force on the bottom face: P_2A (fluid below pushes up) - Weight of the column acting downward: $W = mg = (\rho \cdot A \cdot (h_2 - h_1))g$ Physical reasoning: the upward push from below must support both the downward push from above and the column's own weight. Setting net force to zero: $P_2A = P_1A + \rho A(h_2 - h_1)g$ Cancelling A : $P_2 - P_1 = \rho g(h_2 - h_1)$ Now choose the top face at the free surface, where $h_1 = 0$ and $P_1 = P_0$, and let the bottom face be at depth h where $P_2 = P$: $P = P_0 + \rho gh$ Notice the result depends only on depth h , not on the horizontal cross-section or the container's shape — this is precisely the origin of the hydrostatic paradox.

stays the same, ρ stays the same, g stays the same. Differencing the pressure equation before and after:

$$\Delta P_B = \Delta P_{\text{piston}} + \rho g \underbrace{\Delta z}_{=0} = \Delta P_{\text{piston}} = \frac{F}{A}$$

So the pressure increase at B is *exactly equal* to the increase at the piston — independent of how deep B is. Since B was arbitrary, the change is transmitted undiminished to *every* point. That is Pascal's law.

Derivation 3 — Hydraulic lift force multiplication.

Figure 9.2 Two connected cylinders joined by a fluid-filled pipe. Left (input) piston small, area AA_1 , with downward force FF_1 . Right (output) piston large, area AA_2 , supporting a heavy load (car) with upward force FF_2 . The fluid level connects both. Push the small input piston (area A_1) with force F_1 . By Pascal's law the pressure increase $\Delta P = F_1/A_1$ is transmitted to the large output piston (area A_2), producing an upward force $F_2 = \Delta P \cdot A_2 = F_1 A_2/A_1$. Because $A_2 > A_1$, a small input force lifts a large load. Energy is not created — the input piston travels a larger distance ($A_1 h_1 = A_2 h_2$), so the work done stays equal: $F_1 h_1 = F_2 h_2$. (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A cylindrical overhead tank in a hostel is filled with water to a height of 2.5 m. The flat circular base of the tank has an area of 0.80 m^2 . Taking atmospheric pressure $P_0 = 1.0 \times 10^5 \text{ Pa}$, water density $\rho = 1000 \text{ kg m}^{-3}$ and $g = 10 \text{ m s}^{-2}$, find (a) the gauge pressure at the base, (b) the absolute pressure at the base, and (c) the total downward force exerted by the water on the base.

Given: $h = 2.5 \text{ m}$, $A = 0.80 \text{ m}^2$, $P_0 = 1.0 \times 10^5 \text{ Pa}$, $\rho = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$.

(a) Gauge pressure:

$$P_g = \rho gh = 1000 \times 10 \times 2.5 = 2.5 \times 10^4 \text{ Pa}$$

(b) Absolute pressure:

$$P = P_0 + P_g = 1.0 \times 10^5 + 0.25 \times 10^5 = 1.25 \times 10^5 \text{ Pa}$$

(c) Total force on the base uses the *absolute* pressure (the atmosphere also presses down, transmitted through the water):

$$F = P \times A = 1.25 \times 10^5 \times 0.80 = \boxed{1.0 \times 10^5 \text{ N}}$$

Example 2 — [NEET Speed Trap]

A scuba diver descends into seawater of density $\rho = 1030 \text{ kg m}^{-3}$. At approximately what depth is the *gauge* pressure equal to 2 atm? (Take $g = 10 \text{ m s}^{-2}$, $1 \text{ atm} \approx 1.0 \times 10^5 \text{ Pa}$.)

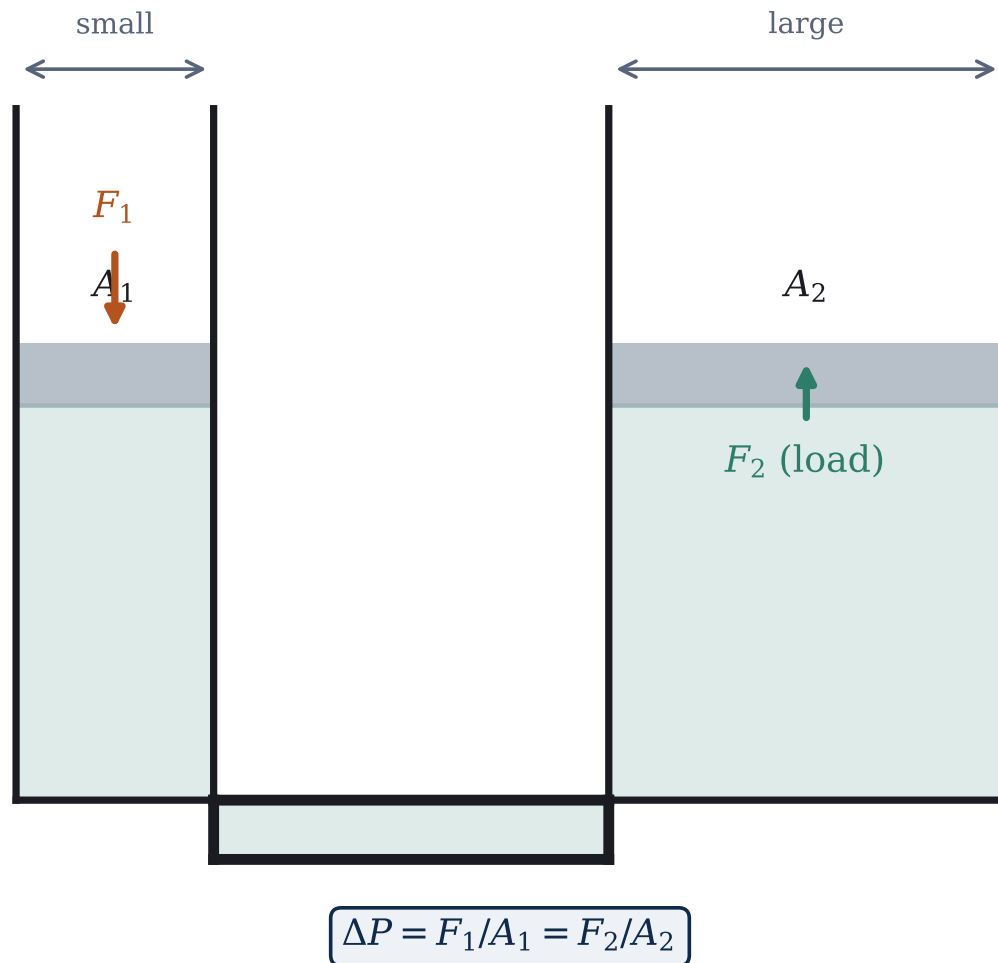


Figure 9.2 Two connected cylinders joined by a fluid-filled pipe. Left (input) piston small, area A_1 , with downward force F_1 . Right (output) piston large, area A_2 , supporting a heavy load (car) with upward force F_2 . The fluid level connects both. Push the small input piston (area A_1) with force F_1 . By Pascal's law the pressure increase $\Delta P = F_1/A_1$ is transmitted to the large output piston (area A_2), producing an upward force $F_2 = \Delta P \cdot A_2 = F_1 \cdot A_2/A_1$. Because $A_2 > A_1$, a small input force lifts a large load. Energy is not created — the input piston travels a larger distance ($A_1 h_1 = A_2 h_2$), so the work done stays equal: $F_1 h_1 = F_2 h_2$.

The traps to avoid: (i) Using absolute pressure instead of gauge — the question asks for gauge, so use $P_g = \rho gh$ directly, no P_0 term. (ii) Using freshwater density 1000 instead of seawater 1030.

Speed approach. Gauge pressure = 2 atm = 2.0×10^5 Pa. Then

$$h = \frac{P_g}{\rho g} = \frac{2.0 \times 10^5}{1030 \times 10} = \frac{2.0 \times 10^5}{1.03 \times 10^4} \approx 19.4 \text{ m}$$

Mental shortcut: in water $\rho g \approx 10^4$, so every 1 atm (10^5 Pa) of gauge pressure costs about 10 m of depth. Two atmospheres $\Rightarrow$ roughly 20 m, nudged slightly *down* to ≈ 19.4 m because seawater is denser than fresh water. Answer $\approx$ 19.4 m.

Example 3 — [JEE Main Level] (*combines Pascal's law force-ratio + volume conservation*)

A hydraulic car lift has an input piston of radius 2 cm and an output piston of radius 20 cm. A car of mass 1200 kg rests on the output piston. (a) What minimum force must be applied to the input piston to support the car? (b) Through what distance must the input piston be pushed to raise the car by 5 cm? (Take $g = 10 \text{ m s}^{-2}$.)

(a) The output force needed equals the car's weight:

$$F_2 = mg = 1200 \times 10 = 1.2 \times 10^4 \text{ N}$$

The area ratio (areas scale as radius squared):

$$\frac{A_2}{A_1} = \left(\frac{r_2}{r_1}\right)^2 = \left(\frac{20}{2}\right)^2 = 100$$

By Pascal's law:

$$F_1 = F_2 \frac{A_1}{A_2} = \frac{1.2 \times 10^4}{100} = \text{120 N}$$

(b) The same fluid volume is displaced at both pistons, so $A_1 h_1 = A_2 h_2$:

$$h_1 = h_2 \frac{A_2}{A_1} = 5 \text{ cm} \times 100 = \text{500 cm} = 5 \text{ m}$$

Sanity check on energy: $F_1 h_1 = 120 \times 5 = 600 \text{ J}$ and $F_2 h_2 = 1.2 \times 10^4 \times 0.05 = 600 \text{ J}$. Equal, as they must be — the lift multiplies force but never energy.

Example 4 — [JEE Advanced Level] (*multi-liquid U-tube, requires the “both arms move” insight*)

A U-tube of uniform cross-section, with both arms open to the atmosphere, contains mercury of density $\rho_{Hg} = 13.6 \times 10^3 \text{ kg m}^{-3}$. A column of water (density $\rho_w = 1.0 \times 10^3 \text{ kg m}^{-3}$) of height $H = 13.6 \text{ cm}$ is then poured slowly into the left arm so that it floats on top of the mercury. By how much does the mercury level *rise in the right arm*?

Figure 9.3 A U-tube with mercury. The left arm now has a water column of height H sitting above the mercury. The mercury in the left arm has dropped by x , and the mercury in the right arm has risen by x , so the level difference between the two mercury surfaces is $2x$. 6 (see next page)

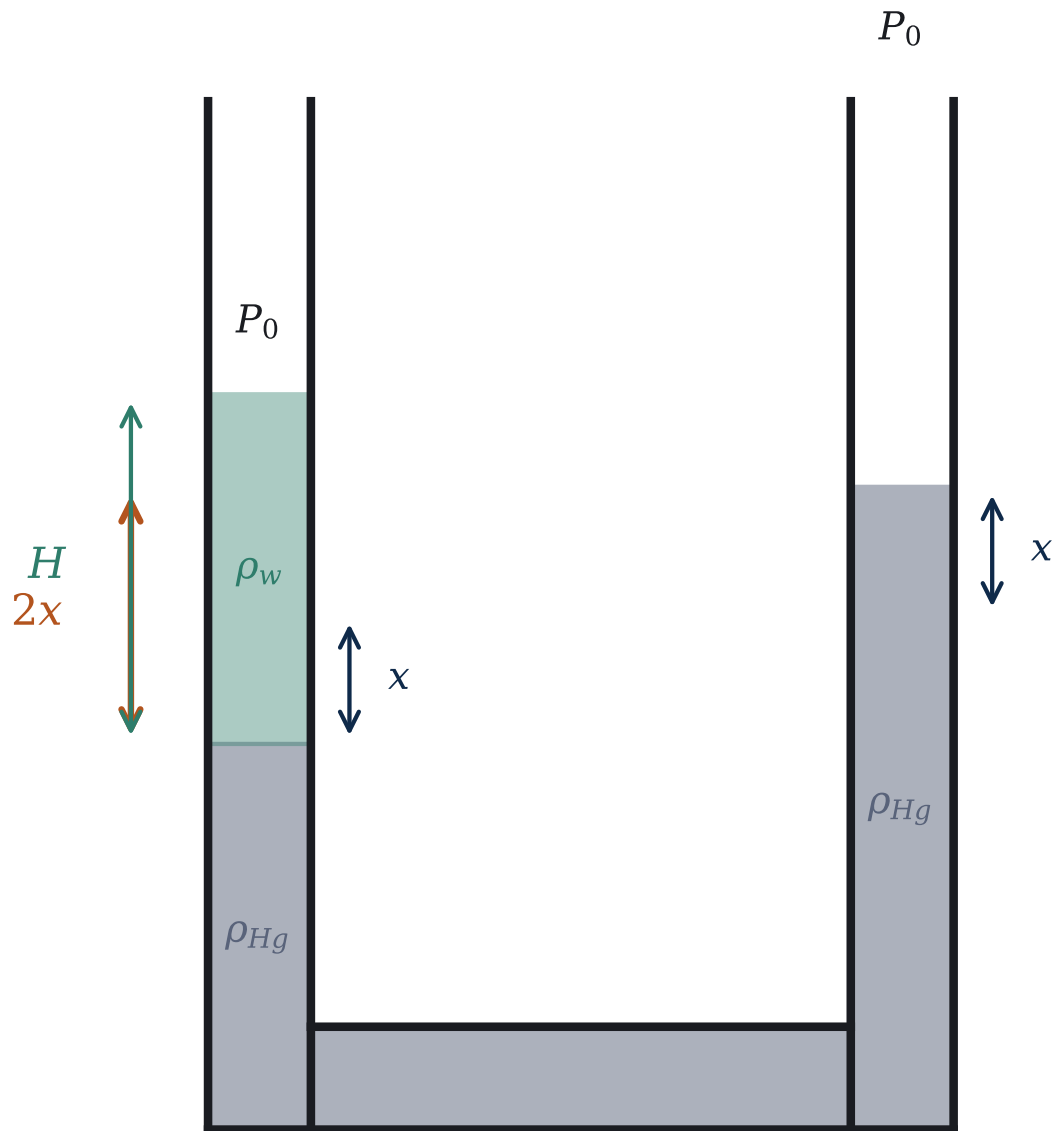


Figure 9.3 A U-tube with mercury. The left arm now has a water column of height H sitting above the mercury. The mercury in the left arm has dropped by x , and the mercury in the right arm has risen by x , so the level difference between the two mercury surfaces is $2x$. 6

The key insight (the trap): when mercury is pushed down by x in the left arm, the *same* mercury reappears in the right arm, rising by x . So the **difference in mercury levels** between the two arms is $2x$, not x . Students who write x for the level difference get half the wrong answer.

Set up a pressure balance at the level of the *lower* mercury surface (in the left arm). Pressures from both sides must be equal at the same horizontal level within the connected mercury.

- **Left side** (going down through atmosphere, then water column H): $P_0 + \rho_w g H$
- **Right side** (going down through atmosphere, then a mercury column of height $2x$ down to the same level): $P_0 + \rho_{Hg} g (2x)$

Equating:

$$\begin{aligned}
 P_0 + \rho_w g H &= P_0 + \rho_{Hg} g (2x) \\
 \rho_w H &= \rho_{Hg} (2x) \\
 2x = H \frac{\rho_w}{\rho_{Hg}} &= 13.6 \times \frac{1.0 \times 10^3}{13.6 \times 10^3} = 1.0 \text{ cm} \\
 x &= \boxed{0.5 \text{ cm}}
 \end{aligned}$$

The mercury rises 0.5 cm in the right arm (and falls 0.5 cm in the left). The numbers were chosen so the ratio ρ_w/ρ_{Hg} exactly cancels H — a clean way to confirm you handled the $2x$ correctly.

Section 5: Practice Exercises

1. **[CBSE]** A swimming pool is 3.0 m deep. Find the gauge pressure and the absolute pressure at the bottom. (Take $\rho = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$, $P_0 = 1.0 \times 10^5 \text{ Pa}$.)
2. **[NEET]** Three open vessels of different shapes — conical, cylindrical, and inverted-conical — are each filled with the same oil to the same height h . The bases all have equal area. Rank the pressures at the three bases.
3. **[JEE Main]** A hydraulic press has pistons of diameters 5 cm and 30 cm. A force of 90 N is applied to the smaller piston. Find the maximum load the larger piston can support.
4. **[JEE Main]** An open-tube mercury manometer is connected to a gas cylinder. The mercury level in the open arm is 18 cm *higher* than in the arm connected to the cylinder. Find the gauge pressure and the absolute pressure of the gas. (Take $\rho_{Hg} = 13.6 \times 10^3 \text{ kg m}^{-3}$, $g = 9.8 \text{ m s}^{-2}$, $P_0 = 1.0 \times 10^5 \text{ Pa}$.)
5. **[JEE Advanced]** A vertical cylindrical container of base radius 10 cm holds water to a height of 40 cm, with a layer of oil (relative density 0.8) of height 20 cm floating on top. Find (a) the gauge pressure at the bottom and (b) the total force on the base. (Take $g = 10 \text{ m s}^{-2}$, $\rho_w = 1000 \text{ kg m}^{-3}$.)

Answers: 1. $P_g = 3.0 \times 10^4 \text{ Pa}$; $P = 1.3 \times 10^5 \text{ Pa}$. 2. All three equal. 3. 3240 N. 4. $P_g \approx 2.4 \times 10^4 \text{ Pa}$; $P_{\text{abs}} \approx 1.24 \times 10^5 \text{ Pa}$. 5. (a) $5.6 \times 10^3 \text{ Pa}$; (b) $\approx 176 \text{ N}$.

Section 6: Crack the MCQ

Q1. Two points A and B lie at the *same depth* in a tank of water, but A is near the centre and B is right next to the slanted side wall. The pressures satisfy: (a) $P_A > P_B$ (b) $P_A < P_B$ (c) $P_A = P_B$ (d) Cannot be determined

Correct: (c). Pressure in a static fluid depends *only* on depth below the free surface, not on horizontal position or proximity to a wall. Same depth $\Rightarrow$ same pressure. - (a)/(b) tempt students who imagine the wall “adds” or “removes” pressure, or that the centre has “more water above.” Neither is true; the column height above each point is the same. - (d) tempts those who feel the geometry is too vague — but the depth alone fully determines it.

Q2. Which of the following can be **negative**? (a) Absolute pressure (b) Gauge pressure (c) Atmospheric pressure (d) Density

Correct: (b). Gauge pressure $= P - P_0$, which is negative whenever a point is at lower pressure than the atmosphere (e.g. inside the lungs while sucking through a straw). - (a) Absolute pressure is measured from a perfect vacuum and is always ≥ 0 . - (c) Atmospheric pressure is a positive physical quantity. - (d) Density is intrinsically positive. Students who pick (a) confuse absolute with gauge — the most common error here.

Q3. A hydraulic lift multiplies the input force by a factor of 25. If the output piston is raised by 4 cm, the input piston must move through: (a) 0.16 cm (b) 1 cm (c) 100 cm (d) 4 cm

Correct: (c). Force multiplication $= A_2/A_1 = 25$. Volume conservation gives $h_1 = h_2(A_2/A_1) = 4 \times 25 = 100$ cm. The input piston moves the *larger* distance. - (a) comes from dividing instead of multiplying ($4/25$), the classic inverse error. - (b) comes from forgetting volume conservation entirely. - (d) comes from assuming both pistons move equally — ignoring the area difference.

Q4. A barometer reads 76 cm of mercury at sea level. If the same barometer is taken to the top of a tall mountain, the mercury column will: (a) Rise above 76 cm (b) Fall below 76 cm (c) Stay at 76 cm (d) Drop to zero

Correct: (b). Atmospheric pressure *decreases* with altitude (less air column above), so the supported mercury height $h = P_0/(\rho_{Hg}g)$ falls. - (a) reverses the relationship — a common slip under exam pressure. - (c) tempts students who think the barometer reading is a fixed constant. - (d) overstates it; pressure decreases but does not vanish on an ordinary mountain.

Section 7: Common Student Pitfalls & Pro-Tips

- 1. Treating pressure as a vector.** Pressure is a *scalar*; it has no direction. Only the *force* due to pressure is a vector (and it always acts perpendicular to the surface). Writing pressure with a direction is a guaranteed mark-loser in CBSE theory questions.
- 2. Confusing gauge and absolute pressure.** “Pressure on the submarine window,” “tyre pressure,” and “blood pressure” are almost always *gauge* values. Read the question: if it asks for the *net* force on a submerged window whose interior is at atmospheric pressure, you use **gauge** pressure (ρgh), because the atmosphere pushes on *both* sides and cancels.
- 3. Falling for the hydrostatic paradox.** More liquid does *not* mean more pressure. Pressure at the base depends only on ρ , g , and the *height* h — never on the volume or the container’s

shape.

4. **Forgetting the factor of 2 in U-tube problems.** When fluid in one arm drops by x , it rises by x in the other arm, so the *level difference* is $2x$. This single oversight is the most common reason Advanced-level U-tube answers come out exactly half-wrong.

Pro-Tip: For depth problems in water, memorise that $\rho g \approx 10^4 \text{ Pa m}^{-1}$, i.e. roughly **1 atm of gauge pressure per 10 m of depth**. This lets you estimate or sanity-check any answer mentally before grinding through the arithmetic — and it instantly flags a wrong order of magnitude on the OMR.

Section 8: Cheat Sheet Box

- **Pressure** $P = \frac{F_{\perp}}{A}$ — a **scalar**, SI unit **Pa** = N m^{-2} , dimensions $[M L^{-1} T^{-2}]$.
- **Acts perpendicular** to any surface; at a given depth it is the **same in all directions** (isotropic) and the **same at all points on a horizontal level**.
- **Depth law:** $P = P_0 + \rho gh$ — depends *only* on depth, not on container shape (**hydrostatic paradox**).
- **Gauge vs. absolute:** $P_g = P - P_0 = \rho gh$; absolute ≥ 0 , gauge can be **negative**.
- **Pascal's law:** pressure change in an enclosed fluid is transmitted **undiminished** everywhere
 $\Rightarrow$ **Hydraulic lift:** $F_2 = F_1 \frac{A_2}{A_1}$, with $A_1 h_1 = A_2 h_2$ (force multiplied, energy conserved).
- **Conversions:** 1 atm = $1.013 \times 10^5 \text{ Pa}$ = 760 mm Hg; 1 bar = 10^5 Pa ; 1 torr = 133 Pa.

Memory aids: “*Pressure is a scalar, Force is a vector.*” • “*Deeper = denser-times-g-times-depth*” for ρgh . • “*Big piston, big force, small distance*” for the hydraulic lift trade-off. • In water: **10 m 1 atm**.

2. Buoyancy and Floatation

Exam Relevance: A perennial favourite across every exam. CBSE Boards routinely ask a 2–3 mark Archimedes’-principle statement or an apparent-weight calculation. JEE Main almost always carries a numerical on fraction-submerged, two-liquid floatation, or a body held by a string. NEET loves the conceptual traps here — the ice-melting question and the anomalous-expansion buoyancy question recur year after year. JEE Advanced folds buoyancy into accelerating-frame and oscillating-float problems.

Section 1: Concept Intuition

Have you ever tried to push a beach ball or a sealed water bottle down to the bottom of a swimming pool? It fights back — hard. The deeper you push it, the more stubbornly it shoves upward against your hand, and the instant you let go it rockets to the surface. That upward shove is **buoyancy**, and understanding *where* it comes from is the whole game.

Buoyancy is not some new mysterious force. It is simply the *net effect of fluid pressure* that you already met in Subtopic 01. Recall that pressure increases with depth: $P = P_0 + \rho gh$. Now picture any object submerged in water. The fluid presses on it from all sides — but the pressure on its **bottom** face is larger than the pressure on its **top** face, because the bottom is *deeper*. So the upward push from below beats the downward push from above. The leftover, the net upward force, is buoyancy. Buoyancy exists *only because* pressure varies with depth — in a weightless, pressure-uniform fluid, there would be no buoyancy at all.

Think about it this way... Imagine you scoop out the object and replace it with an identical blob of the *same fluid* — a “water-shaped lump of water” sitting where the object was. That blob just floats there in equilibrium; it does not sink or rise, because it is part of the fluid. What holds it up? The surrounding fluid pushes on its boundary with exactly enough upward force to balance the blob’s own weight. But the surrounding fluid does not “know” what is inside that boundary — it pushes the same way whether the boundary encloses water or an iron block. So the upward force on *any* object equals the weight of the fluid that would have occupied its space. This is **Archimedes’ principle**:

The buoyant force on a body, partially or fully submerged in a fluid, equals the weight of the fluid displaced by the body, and acts vertically upward through the centre of the displaced fluid.

Now comes the most important consequence for exams. Buoyancy depends on the **displaced fluid**, not on the body. $F_B = \rho_f V_{\text{sub}} g$ — it uses the *fluid’s* density and the *submerged* volume. The body’s own density never appears in the buoyant force itself. The body’s density only decides what *happens*: if the body is denser than the fluid, its weight beats the maximum possible buoyancy and it sinks; if it is lighter, buoyancy wins and it rises until just enough of it pokes out of the surface that the *reduced* submerged volume gives a buoyant force exactly equal to its weight. That balance point is **floatation**.

This is why a giant steel ship floats while a small steel nail sinks. The nail is solid steel — denser than water, period. The ship is mostly hollow; averaged over its huge hull-and-air volume, its *effective* density is less than water. Same material, opposite fate, decided entirely by the average density relative to the fluid. A floating body settles so that the fraction submerged equals the ratio

of densities: a block of relative density 0.6 floats with 60% of itself underwater — which is exactly why roughly 90% of an iceberg hides below the surface.

Limiting Conditions & Assumptions. (1) The fluid must be in equilibrium in an *inertial* frame for the simple “ $F_B = \text{weight of displaced fluid}$ ” form; in an accelerating lift or a turning car, you must replace g with the effective gravity g_{eff} . (2) The fluid is treated as incompressible with well-defined density ρ_f — and note ρ_f itself changes with temperature (water is densest at 4°C), so buoyancy is temperature-sensitive. (3) We ignore surface tension and assume the body is fully wetted. (4) “Weight of displaced fluid” assumes fluid actually surrounds the body — if a flat-bottomed block is stuck to the floor with no water underneath, there is *no* upward pressure on its base and *no* buoyancy.

Section 2: Key Formulas & Definitions

Buoyant force (Archimedes’ principle):

$$F_B = \rho_f V_{\text{sub}} g$$

where ρ_f = density of fluid, V_{sub} = volume of the body submerged (= volume of fluid displaced), g = acceleration due to gravity. SI unit: newton (N); dimensions $[MLT^{-2}]$.

Apparent weight (body fully submerged):

$$W_{\text{app}} = W - F_B = \rho_b V g - \rho_f V g = V g (\rho_b - \rho_f)$$

The “loss of weight” $= F_B = \text{weight of displaced fluid}$.

Condition for floatation — for a body of density ρ_b , volume V in a fluid of density ρ_f (with $W = \rho_b V g$ and maximum buoyancy $F_{B,\text{max}} = \rho_f V g$): - $\rho_b > \rho_f$: weight wins $\Rightarrow$ body **sinks**. - $\rho_b = \rho_f$: neutral $\Rightarrow$ body floats **fully submerged**, in equilibrium anywhere. - $\rho_b < \rho_f$: body **floats** partially submerged.

Fraction submerged (floating body): equilibrium $\rho_b V g = \rho_f V_{\text{sub}} g$ gives

$$\frac{V_{\text{sub}}}{V} = \frac{\rho_b}{\rho_f} \quad \Rightarrow \quad \frac{V_{\text{out}}}{V} = 1 - \frac{\rho_b}{\rho_f}$$

Relative density (specific gravity) — a pure ratio, dimensionless:

$$\text{RD} = \frac{\rho_{\text{substance}}}{\rho_{\text{water}}} = \frac{\text{weight in air}}{\text{loss of weight in water}} = \frac{W_{\text{air}}}{W_{\text{air}} - W_{\text{app}}}$$

Buoyancy in an accelerating frame: replace g by the effective gravity g_{eff} :

$$F_B = \rho_f V_{\text{sub}} g_{\text{eff}}$$

e.g. a lift accelerating up with a : $g_{\text{eff}} = g + a$; accelerating down: $g_{\text{eff}} = g - a$.

Stability: the buoyant force acts at the **centre of buoyancy** B (centroid of displaced fluid); weight acts at the **centre of gravity** G . The vertical through B after a small tilt meets the body axis at the **metacentre** M . The body is **stable** if M lies *above* G (restoring torque) and **unstable** if M lies *below* G .

Section 3: CBSE Board Derivations / Key Procedures

Derivation 1 — Archimedes' principle (the “displaced-fluid replacement” argument).

Assumption: a body of arbitrary shape fully submerged in a fluid of density ρ_f , which is in equilibrium in an inertial frame.

Figure 9.4 *An arbitrarily-shaped body submerged in fluid. Left figure shows the real body with fluid pressure arrows pressing inward on its entire boundary, longer arrows on the deeper (lower) surfaces. Right figure shows the same boundary now enclosing an identical blob of the surrounding fluid, with its weight $m_f g$ acting down at its centroid and the net upward buoyant force F_B acting at the same centroid. Step 1 — Replace the body with fluid. Imagine removing the body and filling the exact same boundary with the surrounding fluid. This “substituting fluid” is now an ordinary part of the bulk fluid, hence in equilibrium. Step 2 — Forces on the substituting fluid. Only two forces act on it: its own weight $m_f g$ (downward, where $m_f = \rho_f V$ is the mass of displaced fluid) and the net contact force from all the surrounding fluid pressing on its boundary. Call this net contact force F_B . Step 3 — Apply equilibrium. Since the substituting fluid neither rises nor sinks: $F_B = m_f g = \rho_f V g$ (directed vertically upward). Step 4 — Transfer back to the real body. The surrounding fluid presses on the boundary purely according to its shape and depth — it does not care what is inside the boundary. So when the real body occupies that same boundary, the fluid exerts the identical net contact force. Physical reasoning: the pressure distribution over the boundary is set by the boundary's geometry and depth alone, which are unchanged. Therefore the buoyant force on the body equals the weight of the displaced fluid: $F_B = \rho_f V_{\text{sub}} g$. This is Archimedes' principle. (see next page)*

Derivation 2 — Law of floatation and fraction submerged.

A body of density ρ_b , total volume V , floats in a fluid of density ρ_f with volume V_{sub} below the surface. Two vertical forces act: weight $W = \rho_b V g$ down, buoyancy $F_B = \rho_f V_{\text{sub}} g$ up. For floating equilibrium the net force is zero:

$$\rho_f V_{\text{sub}} g = \rho_b V g$$

Physical reasoning: the body sinks only until the displaced fluid weighs exactly as much as the body — at that instant the upward push balances the pull of gravity. Cancelling g and rearranging:

$$\frac{V_{\text{sub}}}{V} = \frac{\rho_b}{\rho_f}$$

Since a floating body needs $\rho_b < \rho_f$, this fraction is always less than 1, as it must be. The fraction above the surface is $1 - \rho_b/\rho_f$.

Key procedure — Finding relative density by weighing.

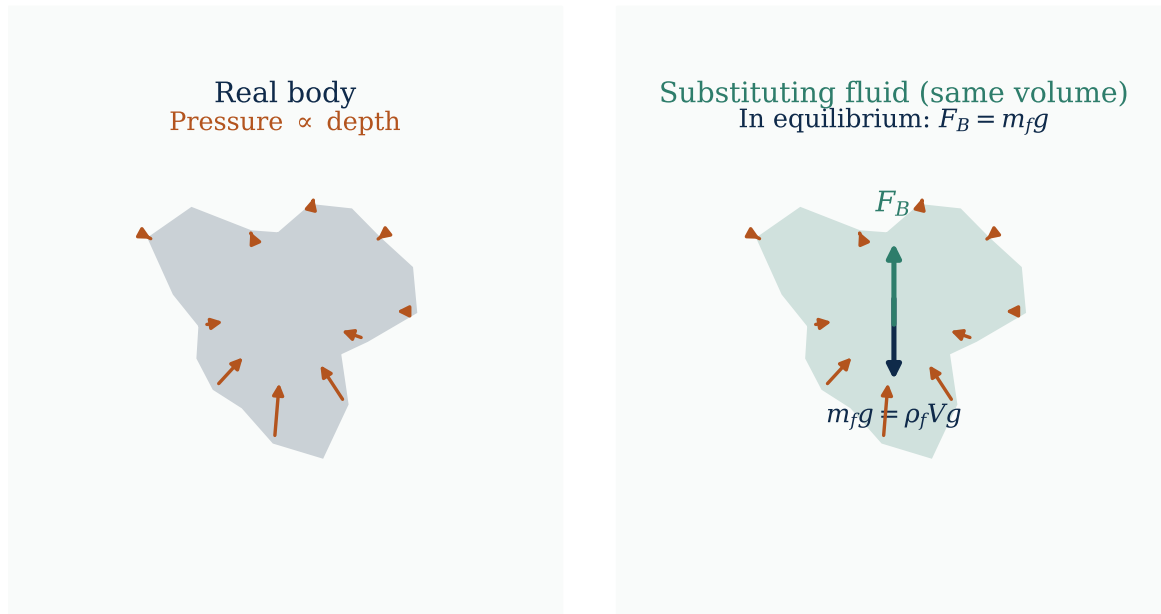


Figure 9.4 An arbitrarily-shaped body submerged in fluid. Left figure shows the real body with fluid pressure arrows pressing inward on its entire boundary, longer arrows on the deeper (lower) surfaces. Right figure shows the same boundary now enclosing an identical blob of the surrounding fluid, with its weight $m_f g$ acting down at its centroid and the net upward buoyant force F_B acting at the same centroid. Step 1 — Replace the body with fluid. Imagine removing the body and filling the exact same boundary with the surrounding fluid. This “substituting fluid” is now an ordinary part of the bulk fluid, hence in equilibrium. Step 2 — Forces on the substituting fluid. Only two forces act on it: its own weight $m_f g$ (downward, where $m_f = \rho_f V$ is the mass of displaced fluid) and the net contact force from all the surrounding fluid pressing on its boundary. Call this net contact force F_B . Step 3 — Apply equilibrium. Since the substituting fluid neither rises nor sinks: $F_B = m_f g = \rho_f V g$ (directed vertically upward). Step 4 — Transfer back to the real body. The surrounding fluid presses on the boundary purely according to its shape and depth — it does not care what is inside the boundary. So when the real body occupies that same boundary, the fluid exerts the identical net contact force. Physical reasoning: the pressure distribution over the boundary is set by the boundary’s geometry and depth alone, which are unchanged. Therefore the buoyant force on the body equals the weight of the displaced fluid: $F_B = \rho_f V_{\text{sub}} g$. This is Archimedes’ principle.

Weigh the body in air to get $W_{\text{air}} = \rho_b V g$. Fully submerge it in water and read the apparent weight $W_{\text{app}} = (\rho_b - \rho_w) V g$. The *loss* of weight equals the buoyant force, $\rho_w V g$. Dividing:

$$\text{RD} = \frac{W_{\text{air}}}{W_{\text{air}} - W_{\text{app}}} = \frac{\rho_b V g}{\rho_w V g} = \frac{\rho_b}{\rho_w}$$

This is exactly how a laboratory determines specific gravity, and the principle behind the hydrometer (which floats deeper in less dense liquids and reads density off a calibrated stem).

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A solid iron block of volume 500 cm^3 and density 7800 kg m^{-3} is fully submerged in water. Find (a) its weight in air, (b) the buoyant force on it, and (c) its apparent weight in water. (Take $\rho_w = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$.)

Given: $V = 500 \text{ cm}^3 = 5 \times 10^{-4} \text{ m}^3$, $\rho_b = 7800 \text{ kg m}^{-3}$.

(a) Mass $m = \rho_b V = 7800 \times 5 \times 10^{-4} = 3.9 \text{ kg}$, so weight in air:

$$W = mg = 3.9 \times 10 = 39 \text{ N}$$

(b) Buoyant force uses fluid density and submerged (= full) volume:

$$F_B = \rho_w V g = 1000 \times 5 \times 10^{-4} \times 10 = 5 \text{ N}$$

(c) Apparent weight:

$$W_{\text{app}} = W - F_B = 39 - 5 = \boxed{34 \text{ N}}$$

Notice the block still sinks ($W > F_B$, consistent with $\rho_b > \rho_w$), but it *feels* lighter by exactly the weight of the water it displaces.

Example 2 — [NEET Speed Trap]

A wooden block floats in water with 75% of its volume submerged. (a) Find the block's density. (b) The same block is now placed in a denser liquid of relative density 1.2. What percentage of it is submerged now?

The traps to avoid: (i) thinking the block's density changes when the fluid changes — it does not; only ρ_f changes. (ii) Confusing the submerged fraction with the fraction *above* the surface.

Speed approach. The submerged fraction equals the density ratio:

$$\frac{V_{\text{sub}}}{V} = \frac{\rho_b}{\rho_f}$$

(a) In water: $0.75 = \frac{\rho_b}{1000} \Rightarrow \rho_b = 750 \text{ kg m}^{-3}$.

(b) In the liquid of density 1200 kg m^{-3} :

$$\frac{V_{\text{sub}}}{V} = \frac{750}{1200} = 0.625 = \boxed{62.5\%}$$

Mental check: denser fluid $\Rightarrow$ less of the block needs to be submerged to generate the same buoyant force, so the submerged fraction *drops* from 75% to 62.5%. Anything that increased it would be wrong by inspection.

Example 3 — [JEE Main Level] (*combines two-fluid buoyancy + floatation condition*)

A cube of density 900 kg m^{-3} floats at the interface of two immiscible liquids: oil of density 800 kg m^{-3} on top and water of density 1000 kg m^{-3} below. What fraction of the cube's volume lies in the water?

Figure 9.5 *A cube straddling a horizontal interface. The upper liquid (oil) fills the region above the line; water below. A fraction f of the cube's height is below the interface (in water), and $(1-f)$ is above (in oil). Buoyant contributions from both liquids point upward; weight points down. Let the fraction in water be f , so the fraction in oil is $(1-f)$. Each liquid contributes its own buoyancy on the portion of the cube inside it. For floating equilibrium: $\rho_b V g = \rho_w (fV)g + \rho_o ((1-f)V)g$ Cancel Vg : $\rho_b = \rho_w f + \rho_o (1-f)$ $900 = 1000f + 800(1-f) = 800 + 200f$ $200f = 100 \Rightarrow f = 0.5$ Exactly half the cube sits in the water and half in the oil. (Sanity: the cube's density 900 is the simple average of 800 and 1000, so a 50:50 split is exactly right.) (see next page)*

Example 4 — [JEE Advanced Level] (*buoyancy in an accelerating frame*)

A cork of volume $V = 100 \text{ cm}^3$ and density 250 kg m^{-3} is held submerged at the bottom of a beaker of water by a light string tied to the base. The entire beaker is placed in a lift that accelerates *upward* at $a = 5 \text{ m s}^{-2}$. Find the tension in the string. (Take $\rho_w = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$.)

Key idea: in a frame accelerating upward, the effective gravity grows to $g_{\text{eff}} = g + a$. Both the buoyant force *and* the effective weight scale with g_{eff} , so the whole problem can be solved as a static one with $g \rightarrow g_{\text{eff}}$.

$$g_{\text{eff}} = g + a = 10 + 5 = 15 \text{ m s}^{-2}$$

$$V = 100 \text{ cm}^3 = 1 \times 10^{-4} \text{ m}^3$$

Working in the lift frame, three vertical forces act on the cork: buoyancy up, effective weight down, and string tension T down (the string ties the cork to the base, so it pulls *down*). Equilibrium in the lift frame:

$$F_B = mg_{\text{eff}} + T$$

$$\rho_w V g_{\text{eff}} = \rho_b V g_{\text{eff}} + T$$

$$T = V g_{\text{eff}} (\rho_w - \rho_b) = 1 \times 10^{-4} \times 15 \times (1000 - 250)$$

$$T = 1 \times 10^{-4} \times 15 \times 750 = \boxed{1.125 \text{ N}}$$

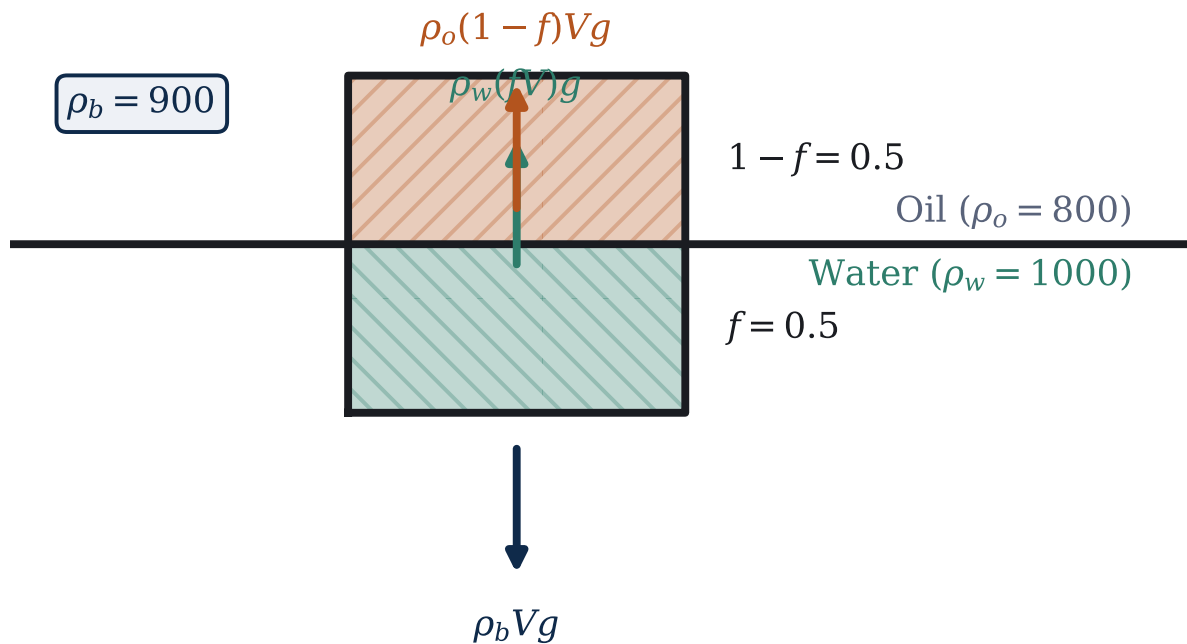


Figure 9.5 A cube straddling a horizontal interface. The upper liquid (oil) fills the region above the line; water below. A fraction f of the cube's height is below the interface (in water), and $(1-f)$ is above (in oil). Buoyant contributions from both liquids point upward; weight points down. Let the fraction in water be f , so the fraction in oil is $(1-f)$. Each liquid contributes its own buoyancy on the portion of the cube inside it. For floating equilibrium: $\rho_b Vg$ weight = $\rho_w(fV)g$ buoyancy from water + $\rho_o((1-f)V)g$ buoyancy from oil. Cancel Vg : $\rho_b = \rho_w f + \rho_o(1-f)$. $900 = 1000f + 800(1-f) = 800 + 200f$. $200f = 100 \Rightarrow f = 0.5$. Exactly half the cube sits in the water and half in the oil. (Sanity: the cube's density 900 is the simple average of 800 and 1000, so a 50:50 split is exactly right.)

Insight: the tension is simply $Vg_{\text{eff}}(\rho_w - \rho_b)$ — it scales directly with g_{eff} . If the lift were in free fall ($a = -g$, so $g_{\text{eff}} = 0$), the tension would vanish entirely: with no effective gravity there is no pressure gradient, hence no buoyancy, and the string goes slack.

Section 5: Practice Exercises

1. **[CBSE]** A stone weighs 5.0 N in air and 3.2 N when fully immersed in water. Find (a) the buoyant force, (b) the volume of the stone, and (c) its relative density. (Take $g = 10 \text{ m s}^{-2}$, $\rho_w = 1000 \text{ kg m}^{-3}$.)
2. **[NEET]** An ice cube floats in a glass of water filled to the brim. As the ice melts completely, the water level in the glass will: rise, fall, or stay the same? Justify in one line.
3. **[JEE Main]** A block of wood of relative density 0.5 and volume 2000 cm^3 floats in water. What minimum mass must be placed on top of it to make it just fully submerged? (Take $\rho_w = 1000 \text{ kg m}^{-3}$.)
4. **[JEE Main]** A body floats in water with one-third of its volume above the surface. When placed in another liquid, it floats with one-quarter of its volume above the surface. Find the relative density of (a) the body and (b) the second liquid.
5. **[JEE Advanced]** A wooden cylinder of cross-sectional area A and length L , density ρ_b , floats vertically in water of density ρ_w . It is pushed down slightly and released. Show that it executes SHM and find its time period in terms of L , ρ_b , ρ_w , and g .

Answers: 1. (a) 1.8 N; (b) $1.8 \times 10^{-4} \text{ m}^3$; (c) $\text{RD} \approx 2.78$. 2. Stays the same (floating ice displaces its own weight; melt-water exactly fills that volume). 3. 1.0 kg. 4. (a) $\frac{2}{3} \approx 0.667$; (b) ≈ 0.889 . 5.

$$T = 2\pi \sqrt{\frac{\rho_b L}{\rho_w g}}.$$

Section 6: Crack the MCQ

Q1. A piece of ice floats in a glass of water. When the ice melts completely, the level of water in the glass: (a) Rises (b) Falls (c) Remains unchanged (d) First rises, then falls

Correct: (c). A floating ice cube displaces a volume of water whose weight equals the ice's weight. When it melts, it turns into exactly that same weight of water, which occupies precisely the volume it was displacing. Level unchanged. - (a) tempts students who think melting "adds" water — but that water was already accounted for by the displaced volume. - (b) tempts those who recall ice is less dense and expect a drop on melting — true for the *ice's* volume, but irrelevant to displaced volume. - (d) is a nonsense distractor with no physical basis.

Q2. A solid aluminium sphere is weighed first in water at 0°C and then in water at 4°C . The buoyant force is: (a) Greater at 0°C (b) Greater at 4°C (c) Equal in both (d) Depends on the sphere's radius

Correct: (b). Buoyancy $F_B = \rho_w Vg$ depends on *water's* density. Due to water's anomalous expansion, density is **maximum at 4°C** , so buoyancy is greater at 4°C than at 0°C . - (a) reverses the anomalous-expansion fact — the classic trap. - (c) tempts students who forget that ρ_w itself

varies with temperature. - (d) is wrong because the radius cancels in any comparison (same sphere, same V).

Q3. Two solid balls of the *same volume* — one of iron, one of aluminium — are fully submerged in water. The buoyant force on them: (a) Is greater on iron (b) Is greater on aluminium (c) Is equal (d) Is zero on the denser ball

Correct: (c). Buoyancy depends only on the *displaced fluid* — same volume submerged in the same water means $F_B = \rho_w V g$ is identical. The balls' own densities do not enter the buoyant force. - (a) tempts students who think “heavier $\Rightarrow$ more buoyancy” — confusing weight with upthrust. - (b)/(d) misapply density of the body to the buoyancy formula. The body's density decides whether it sinks, *not* how much it is buoyed.

Q4. A hydrometer floats in a liquid. Compared with a less dense liquid, in a *denser* liquid the hydrometer: (a) Floats higher (sinks less) (b) Floats lower (sinks more) (c) Floats at the same level (d) Sinks completely

Correct: (a). A floating hydrometer always displaces fluid weighing its own (fixed) weight. In a denser liquid, that same weight is provided by a *smaller* displaced volume, so the hydrometer sinks less and floats higher. - (b) reverses the logic — denser liquid means the device needs to go *less* deep, not more. - (c) ignores that submerged depth must change to keep displaced weight constant. - (d) contradicts floating equilibrium; a calibrated hydrometer is designed to float across its range.

Section 7: Common Student Pitfalls & Pro-Tips

- Putting the body's density into the buoyancy formula.** Buoyancy is $F_B = \rho_f V_{\text{sub}} g$ — *fluid* density, *submerged* volume. The body's own density only decides sink-vs-float, never the magnitude of the upthrust. Mixing these up is the single most common buoyancy error.
- Inverting the fraction-submerged ratio.** The submerged fraction is ρ_b / ρ_f , not ρ_f / ρ_b . A quick sanity check: a floating body has $\rho_b < \rho_f$, so the fraction *must* come out less than 1. If you get something above 1, you flipped it.
- The ice-melting trap.** Floating ice melting in its own water leaves the level *unchanged*. But watch the variations: ice with a stone embedded, or ice floating in a *different* liquid (e.g. mercury or a salt solution), do change the level — don't apply the “unchanged” rule blindly.
- Forgetting g_{eff} in accelerating frames.** In a lift, rotating, or free-falling fluid, buoyancy scales with g_{eff} , not g . In free fall ($g_{\text{eff}} = 0$) buoyancy vanishes entirely — a submerged cork would no longer rise.

Pro-Tip: For *any* floating-body problem, write the master equation “weight of body = weight of displaced fluid” first, *then* substitute. This one line — $\rho_b V = \rho_f V_{\text{sub}}$ — handles fraction-submerged, two-liquid, and loaded-raft problems alike, and stops you from reaching for the wrong density. For RD by weighing, remember: **RD = weight in air $\div$ loss of weight in water**, a ratio that needs no volume at all.

Section 8: Cheat Sheet Box

- **Buoyant force** $F_B = \rho_f V_{\text{sub}} g$ — depends on **fluid density** and **submerged volume**, *never* on the body's density. Acts upward through the **centre of buoyancy**.
- **Archimedes:** upthrust = weight of displaced fluid. **Apparent weight** $W_{\text{app}} = W - F_B = Vg(\rho_b - \rho_f)$.
- **Floatation rule:** $\rho_b > \rho_f$ sinks; $\rho_b = \rho_f$ fully submerged (neutral); $\rho_b < \rho_f$ floats.
- **Fraction submerged** $\frac{V_{\text{sub}}}{V} = \frac{\rho_b}{\rho_f}$ (always < 1 for a floater).
- **Relative density** $= \frac{\rho_{\text{body}}}{\rho_{\text{water}}} = \frac{W_{\text{air}}}{W_{\text{air}} - W_{\text{app}}}$ — dimensionless.
- **Accelerating frame:** use g_{eff} (lift up: $g + a$; down: $g - a$; free fall: 0). **Stability:** stable if metacentre M is **above** centre of gravity G .

Memory aids: “Buoyancy belongs to the fluid, not the body.” • “Submerged fraction = density ratio (body over fluid).” • “Floating ice melts to a draw — level unchanged.”

- “No g , no buoyancy” (free fall).

3. Fluids in Accelerated and Rotating Frames

Exam Relevance: A favourite of JEE for testing whether students truly understand pressure rather than memorising $P = P_0 + \rho gh$. JEE Main regularly asks the tilted-free-surface angle ($\tan \theta = a/g$) and the height difference across an accelerating tank. JEE Advanced builds rotating-fluid (paraboloid) problems and combined accelerating-and-submerged scenarios. CBSE Boards touch it lightly via the effective-gravity idea; NEET asks the occasional conceptual question on surface tilt.

Section 1: Concept Intuition

Everything in Subtopic 01 assumed the fluid was perfectly still. But fluids are often *on the move with their container* — petrol sloshing in an accelerating tanker, tea tilting in a cup as a train pulls away, water climbing the outer wall of a spinning bucket. The free surface is no longer flat, and the pressure no longer depends on depth alone. Yet there is a wonderfully simple trick that handles all of it: replace ordinary gravity with an **effective gravity** that includes the acceleration.

Start with a tank of water on a cart that suddenly accelerates forward. Intuitively, the water “lags behind” — it piles up at the *back* of the tank and dips at the front, exactly as you lurch backward when a bus jerks forward. The free surface tilts. Why does it settle at a particular angle? Because a liquid surface at rest (in its own frame) must be *perpendicular to the net effective gravity*. Ordinary gravity pulls down with g ; in the accelerating frame, a pseudo-force pulls backward with a (per unit mass). These combine into an **effective gravity** $g_{\text{eff}} = \sqrt{a^2 + g^2}$, pointing down-and-backward. The surface orients perpendicular to *that*, so it tilts by an angle θ where $\tan \theta = a/g$.

Think about it this way... Imagine carrying a tray of water and speeding up smoothly. The water tilts up at the back. The faster you accelerate, the steeper the tilt — and if you could accelerate at $a = g$, the surface would tilt a full 45° ($\tan \theta = 1$). The liquid behaves exactly as if “down” had been rotated to a new direction. This is the same effective-gravity idea you already used for buoyancy in a lift (Subtopic 02), now applied to the *shape of the surface* and the *distribution of pressure*. Every fluid-statics result you know still works — you just point gravity in the new effective direction and measure depth along it.

Now spin instead of push. Put water in a cylinder and rotate it steadily about its vertical axis. Each parcel of water needs a *centripetal* force directed inward to travel in its circle, and the only thing that can supply it is a sideways pressure difference — so the pressure must increase outward. Higher pressure at the rim means the water level rises there and sinks at the centre. The free surface curves into a **paraboloid** — the elegant bowl shape you see in a stirred cup of tea or a spinning bucket. The faster the spin, the deeper the central dip and the higher the rim. (This is precisely why liquid-mirror telescopes spin a dish of mercury: rotation sculpts a perfect parabolic reflector for free.)

Limiting Conditions & Assumptions. (1) The fluid must be in *equilibrium relative to its container* — these results apply once sloshing has settled and the liquid rotates/translated as a rigid body with the vessel. (2) The fluid is incompressible with constant density. (3) For the tilted-surface result, the acceleration is constant and horizontal; for the paraboloid, the rotation is steady about a vertical axis. (4) We neglect surface tension and viscosity (the surface shape is set by pressure balance alone). (5) Volume is conserved — a spinning liquid rises at the rim by exactly as much as it falls at the centre (for a uniform cylinder).

Section 2: Key Formulas & Definitions

Horizontal acceleration a (open container): - **Tilt of the free surface** from the horizontal:

$$\tan \theta = \frac{a}{g}$$

- **Effective gravity:** $g_{\text{eff}} = \sqrt{a^2 + g^2}$ (the free surface is perpendicular to g_{eff}). - **Pressure gradients:** horizontally $\frac{\partial P}{\partial x} = -\rho a$ (pressure rises toward the rear); vertically $\frac{\partial P}{\partial y} = -\rho g$. - **Height difference** between the rear and front of a tank of horizontal length L :

$$\Delta h = L \tan \theta = \frac{aL}{g}$$

Vertical acceleration a : effective gravity becomes $g_{\text{eff}} = g \pm a$ (use $+$ for upward acceleration, $-$ for downward), so the pressure at depth h is

$$P = P_0 + \rho(g \pm a)h$$

(In free fall, $a = g$ downward $\Rightarrow g_{\text{eff}} = 0$: the fluid becomes “weightless” and exerts no hydrostatic pressure.)

Rotation at angular velocity ω about a vertical axis: - **Radial pressure gradient** (supplying centripetal force): $\frac{dP}{dr} = \rho\omega^2 r$. - **Free-surface shape (paraboloid):** the surface height above the central vertex at radius r is

$$y(r) = \frac{\omega^2 r^2}{2g}$$

- **Height difference** between the rim (radius R) and the centre:

$$\Delta h = \frac{\omega^2 R^2}{2g}$$

- **Volume conservation** (uniform cylinder): the liquid rises by $\frac{\omega^2 R^2}{4g}$ at the rim and falls by $\frac{\omega^2 R^2}{4g}$ at the centre, relative to the original level.

Section 3: Key Procedures and Derivations

Derivation 1 — Tilt of the free surface under horizontal acceleration.

Figure 9.6 *A rectangular open tank of liquid on a cart accelerating to the right with acceleration a . The free surface is tilted, higher at the rear (left) and lower at the front (right), making an angle θ with the horizontal. A small surface element shows two forces: weight mg downward and the horizontal pseudo-force ma pointing backward (left). (see next page)*

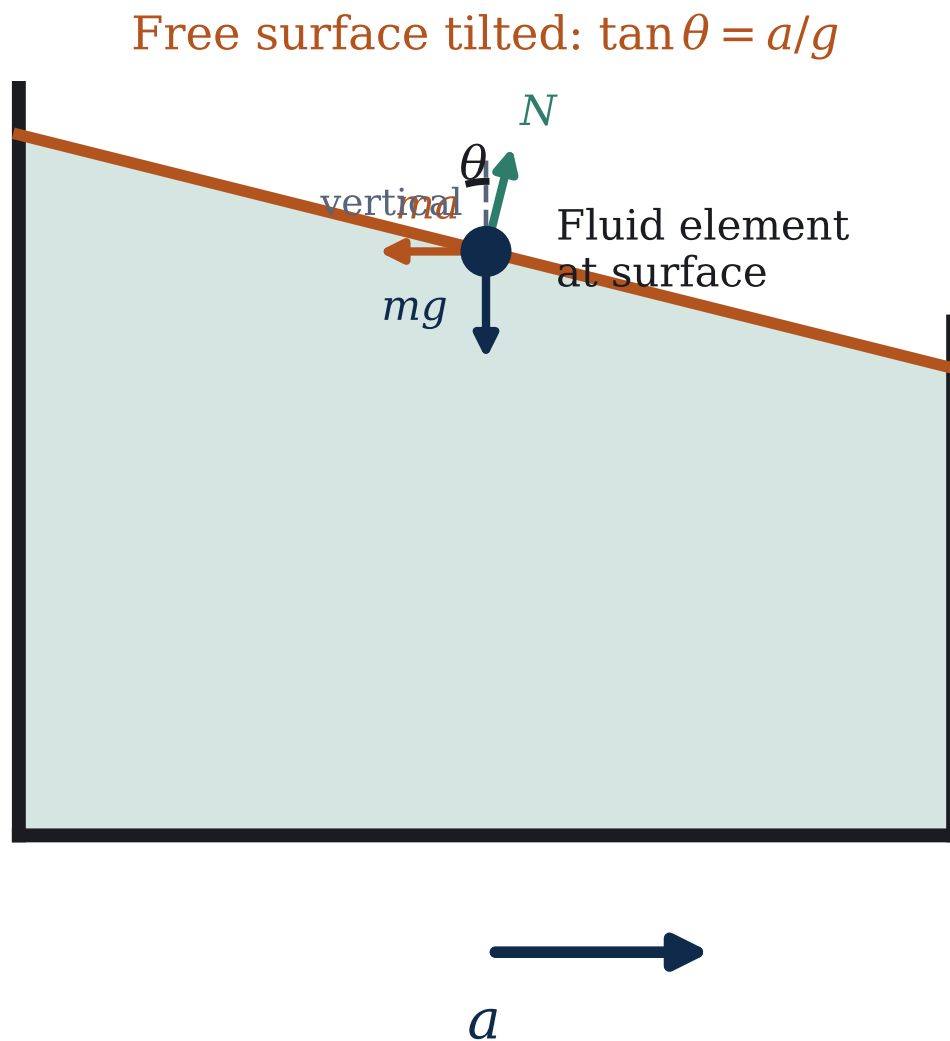


Figure 9.6 A rectangular open tank of liquid on a cart accelerating to the right with acceleration a . The free surface is tilted, higher at the rear (left) and lower at the front (right), making an angle θ with the horizontal. A small surface element shows two forces: weight mg downward and the horizontal pseudo-force ma pointing backward (left).

Assumption: the liquid has settled and moves as a rigid body with the container, accelerating horizontally at a .

Consider a small fluid element of mass m right at the free surface. Work in the accelerating (non-inertial) frame, where the element is in equilibrium under three influences: its weight mg (down), the pseudo-force ma (backward, opposite to the acceleration), and the normal force from the surrounding fluid. *Physical reasoning:* a fluid surface cannot support a force along itself (that would make it flow), so the net of gravity and pseudo-force must be **perpendicular** to the surface. The resultant of mg (down) and ma (backward) makes an angle θ with the vertical given by:

$$\tan \theta = \frac{ma}{mg} = \frac{a}{g}$$

This θ is exactly the angle by which the surface tilts from the horizontal. Over a tank of length L , the height difference between the ends is $\Delta h = L \tan \theta = aL/g$.

Derivation 2 — Paraboloid free surface of a rotating liquid.

Figure 9.7 *A vertical cylindrical vessel of radius R rotating about its central axis at angular velocity ω . The liquid surface is a parabola (bowl shape): lowest at the centre (vertex), rising to its highest at the rim. A fluid element at radius r on the surface shows the inward net force needed for circular motion. Assumption: the liquid rotates as a rigid body with the vessel at steady ω about a vertical axis. Consider a fluid element at radial distance r on (or just below) the surface. It moves in a circle, so it needs a net inward (centripetal) force $dF = (dm) \omega^2 r$. The only available source is the pressure difference across it, giving the radial gradient: $dP/dr = \rho \omega^2 r$. Along the free surface the pressure is constant ($= P_0$). Moving outward by dr , the pressure would rise by $\rho \omega^2 r dr$; to keep the surface pressure constant, the surface must climb by dy such that the extra hydrostatic pressure $\rho g dy$ cancels it: $\rho g dy = \rho \omega^2 r dr \implies dy/dr = \omega^2 r/g$. Integrate from the central vertex ($r=0, y=0$) outward: $y(r) = \omega^2 r^2 / 2g$. The surface is a paraboloid of revolution. The rim-to-centre height difference is $y(R) = \omega^2 R^2 / 2g$. Physical reasoning: the quadratic shape arises because the centripetal demand grows linearly with r , and integrating a linear gradient yields a parabola. (see next page)*

Key procedure — locating the new surface using volume conservation. When a liquid spins, it neither gains nor loses volume. For a uniform cylinder, the average rise of the paraboloid over the base area is half the rim-to-vertex height. Hence the rim rises by $\omega^2 R^2 / 4g$ and the centre falls by $\omega^2 R^2 / 4g$ relative to the still-liquid level. Use this to find whether the centre exposes the bottom or the rim overflows.

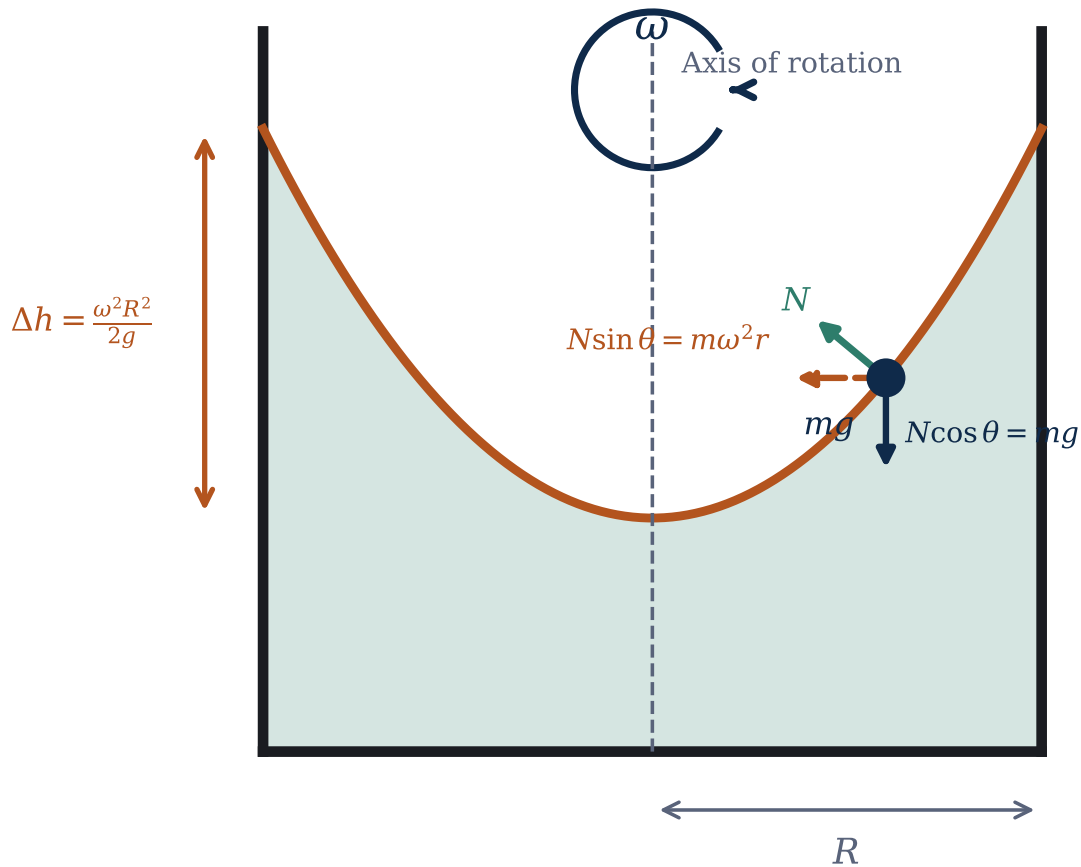


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Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A tanker truck carrying diesel accelerates uniformly along a straight horizontal road at 3.0 m s^{-2} . Find the angle that the free surface of the diesel makes with the horizontal. (Take $g = 10 \text{ m s}^{-2}$.)

Given: $a = 3.0 \text{ m s}^{-2}$, $g = 10 \text{ m s}^{-2}$.

The free surface tilts at an angle θ where:

$$\tan \theta = \frac{a}{g} = \frac{3.0}{10} = 0.30$$

$$\theta = \tan^{-1}(0.30) \approx \boxed{16.7^\circ}$$

The surface rises toward the rear of the tanker (the diesel “lags behind” the acceleration).

Example 2 — [NEET Speed Trap]

A beaker of water rests on a cart that accelerates horizontally. The water surface is observed to make an angle of 45° with the horizontal. What is the acceleration of the cart?

The trap to avoid: over-thinking it. The angle depends only on a/g , not on the size or shape of the beaker or the amount of water.

Speed approach. From $\tan \theta = a/g$:

$$a = g \tan 45^\circ = g \times 1 = \boxed{10 \text{ m s}^{-2}}$$

Mental check: a 45° tilt means the horizontal and vertical “pulls” are equal, so $a = g$. Anchor this benchmark: $45^\circ \Leftrightarrow a = g$. Anything steeper needs $a > g$.

Example 3 — [JEE Main Level] (*tilt + height difference + spill check*)

An open rectangular tank 2.0 m long is filled with water to a depth of 1.0 m (the tank walls are 1.2 m tall). The tank is accelerated horizontally along its length at $a = 2.0 \text{ m s}^{-2}$. (a) Find the height difference of the water between the two ends. (b) Does the water spill? (Take $g = 10 \text{ m s}^{-2}$.)

- (a) The surface tilts at $\tan \theta = a/g = 2.0/10 = 0.20$. Over the tank length $L = 2.0 \text{ m}$, the end-to-end height difference is:

$$\Delta h = L \tan \theta = 2.0 \times 0.20 = 0.40 \text{ m}$$

- (b) The water pivots about the centre (volume conserved), so it rises by $\Delta h/2 = 0.20 \text{ m}$ at the rear and falls by 0.20 m at the front. The rear water level becomes:

$$1.0 + 0.20 = 1.20 \text{ m}$$

This *exactly* reaches the 1.2 m wall height — so the water is **on the verge of spilling but does not** (any larger acceleration would spill it).

Example 4 — [JEE Advanced Level] (*rotating cylinder — paraboloid + volume conservation*)

A cylindrical vessel of radius $R = 10 \text{ cm}$ contains water to a height $H = 20 \text{ cm}$ when at rest. It is then rotated steadily about its vertical axis. (a) Derive the angular velocity ω at which the water

at the centre just exposes the bottom of the vessel. (b) Find the corresponding height of water at the rim. (Take $g = 10 \text{ m s}^{-2}$.)

Step 1 — Set up volume conservation. When rotating at ω , the surface is a paraboloid. The vertex (centre) sits at height z_0 above the bottom, and the rim sits at $z_0 + \frac{\omega^2 R^2}{2g}$. The average height over the base equals the original H :

$$z_0 + \frac{\omega^2 R^2}{4g} = H$$

(The paraboloid's mean height is its vertex plus half its rim-to-vertex rise.)

Step 2 — Apply the “centre just touches bottom” condition ($z_0 = 0$):

$$\frac{\omega^2 R^2}{4g} = H \implies \omega^2 = \frac{4gH}{R^2} \implies \omega = \frac{2}{R} \sqrt{gH}$$

Substituting $R = 0.10 \text{ m}$, $H = 0.20 \text{ m}$:

$$\omega = \frac{2}{0.10} \sqrt{10 \times 0.20} = 20\sqrt{2} \approx \boxed{28.3 \text{ rad s}^{-1}}$$

Step 3 — Rim height. With $z_0 = 0$, the rim height is the full paraboloid depth:

$$z_{\text{rim}} = \frac{\omega^2 R^2}{2g} = \frac{(800)(0.01)}{2(10)} = \frac{8}{20} = 0.40 \text{ m} = \boxed{40 \text{ cm}}$$

Insight: at this spin, the water has been flung entirely outward — zero depth at the centre, double the original depth ($2H = 40 \text{ cm}$) at the rim. The result $z_{\text{rim}} = 2H$ when $z_0 = 0$ is a clean check: the parabola's average must still equal H , and the average of 0 and the rim value (weighted for a paraboloid) returns H .

Section 5: Practice Exercises

1. **[CBSE]** A container of water accelerates horizontally at 5.0 m s^{-2} . Find the angle the free surface makes with the horizontal. (Take $g = 10 \text{ m s}^{-2}$.)
2. **[NEET]** A liquid in a beaker is taken in a lift accelerating *downward* at 4.0 m s^{-2} . Compared with the lift at rest, the pressure at a fixed depth in the liquid is: greater, smaller, or unchanged? Justify in one line. (Take $g = 10 \text{ m s}^{-2}$.)
3. **[JEE Main]** A tank 3.0 m long, open at the top, is filled with water and accelerated horizontally so that the free surface makes 30° with the horizontal. Find (a) the acceleration and (b) the height difference between the two ends. (Take $g = 10 \text{ m s}^{-2}$.)
4. **[JEE Main]** A cylindrical bucket of radius 15 cm is rotated about its axis at 10 rad s^{-1} . Find the height difference between the liquid at the rim and at the centre. (Take $g = 10 \text{ m s}^{-2}$.)
5. **[JEE Advanced]** A sealed cubical box of side L is completely filled with water (no air gap) and accelerated horizontally at a along one edge. Find the difference in pressure between the rear-bottom corner and the front-bottom corner of the box.

Answers: 1. $\approx 26.6^\circ$. 2. Smaller; $g_{\text{eff}} = g - a = 6 \text{ m s}^{-2}$, so $P = P_0 + \rho(g - a)h$. 3. (a) $a = g \tan 30^\circ \approx 5.77 \text{ m s}^{-2}$; (b) $\Delta h = 3.0 \tan 30^\circ \approx 1.73 \text{ m}$. 4. $\Delta h = \omega^2 R^2 / 2g = 0.1125 \text{ m} \approx 11.3 \text{ cm}$. 5. $\Delta P = \rho a L$ (horizontal gradient only, since both corners are at the same depth).

Section 6: Crack the MCQ

Q1. A tank of water accelerates horizontally to the right. The free surface of the water: (a) Stays horizontal (b) Rises at the front (right), falls at the rear (c) Rises at the rear (left), falls at the front (d) Becomes vertical

Correct: (c). The water lags the acceleration and piles up at the rear; the surface tilts with $\tan \theta = a/g$, higher at the back. - (a) ignores the acceleration. - (b) reverses the direction — a very common slip; the water rises *opposite* to the direction of motion. - (d) would require infinite acceleration.

Q2. A beaker of liquid is in free fall. The pressure at a depth h inside the liquid is: (a) $P_0 + \rho gh$ (b) P_0 (c) ρgh (d) Zero everywhere

Correct: (b). In free fall, $g_{\text{eff}} = g - g = 0$, so there is no hydrostatic pressure variation; the pressure throughout equals the surface pressure P_0 . - (a) is the static (non-falling) result. - (c) drops the atmospheric term. - (d) wrongly assumes even the surface pressure vanishes — the atmosphere still presses down with P_0 .

Q3. A liquid in a cylinder is rotated steadily about its vertical axis. The free surface takes the shape of: (a) A flat horizontal plane (b) A cone (c) A paraboloid (d) A hemisphere

Correct: (c). The centripetal requirement gives a radial pressure gradient $\propto r$; integrating yields a surface height $\propto r^2$ — a paraboloid. - (a) describes the non-rotating case. - (b)/(d) have the wrong functional form (the rise goes as r^2 , not linearly and not as a circular arc).

Q4. A container of water accelerates horizontally such that the free surface makes 45° with the horizontal. The effective gravity experienced by the liquid is: (a) g (b) $\sqrt{2}g$ (c) $2g$ (d) $g/\sqrt{2}$

Correct: (b). At 45° , $a = g$, so $g_{\text{eff}} = \sqrt{a^2 + g^2} = \sqrt{g^2 + g^2} = \sqrt{2}g$. - (a) ignores the horizontal contribution. - (c) adds the magnitudes arithmetically instead of as perpendicular vectors. - (d) inverts the relation.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Tilting the surface the wrong way.** Under forward acceleration, the liquid rises at the **rear**, not the front — it lags behind the motion, just as you are pushed back into your seat. Always sketch the pseudo-force pointing *opposite* to the acceleration.
2. **Forgetting volume conservation.** The surface pivots about the centre: it rises at one end by the *same* amount it falls at the other. To check spilling, add only *half* the total height difference to the original level at the high end — not the full Δh .
3. **Using g instead of g_{eff} .** Once a fluid accelerates, every pressure/depth calculation should use the effective gravity — $g_{\text{eff}} = \sqrt{a^2 + g^2}$ for horizontal acceleration, $g \pm a$ for vertical. In free fall, $g_{\text{eff}} = 0$ and hydrostatic pressure vanishes.

4. **Mishandling the rotating-surface height.** The rim-to-centre difference is $\omega^2 R^2/2g$, but relative to the *original* level the rim rises only $\omega^2 R^2/4g$ and the centre drops the same. Don't confuse the total difference with the rise at one end.

Pro-Tip: Treat any accelerating-fluid problem as an ordinary statics problem in a *rotated/strengthened* gravity. For horizontal acceleration, tilt your mental “down” to point along g_{eff} and measure depth perpendicular to the (tilted) surface — then $P = P_0 + \rho g_{\text{eff}} d$ works just as before. Memorise the two benchmarks: $\tan \theta = a/g$ (tilt) and $\Delta h = \omega^2 R^2/2g$ (spin), and most questions collapse to a one-line substitution.

Section 8: Cheat Sheet Box

- **Horizontal acceleration:** free surface tilts, $\tan \theta = \frac{a}{g}$; height difference over length L is $\Delta h = \frac{aL}{g}$; effective gravity $g_{\text{eff}} = \sqrt{a^2 + g^2}$ (surface $\perp$ to it).
- **Benchmark:** $\theta = 45^\circ \Leftrightarrow a = g \Leftrightarrow g_{\text{eff}} = \sqrt{2}g$.
- **Vertical acceleration:** $g_{\text{eff}} = g \pm a$ (+ up, − down); $P = P_0 + \rho(g \pm a)h$. **Free fall** $\Rightarrow g_{\text{eff}} = 0$, no hydrostatic pressure.
- **Rotation:** radial gradient $\frac{dP}{dr} = \rho\omega^2 r$; surface is a **paraboloid** $y = \frac{\omega^2 r^2}{2g}$; rim-centre difference $\frac{\omega^2 R^2}{2g}$.
- **Volume conservation (spin):** rim rises $\frac{\omega^2 R^2}{4g}$, centre falls the same, relative to the rest level.
- **Master idea:** all of fluid statics still holds — just use the effective gravity (rotated for horizontal accel, scaled for vertical).

Memory aids: “*Liquid lags — rises at the rear.*” • “ $\tan \theta = a/g$; 45° means $a = g$.”

- “*Spin makes a parabola: $\omega^2 R^2/2g$.*” • “*Free fall, no pressure.*”

4. Fluid Dynamics and Bernoulli's Theorem

Exam Relevance: One of the highest-yield subtopics in the chapter. CBSE Boards regularly ask the statement and derivation of Bernoulli's theorem (3–5 marks) and the continuity equation. JEE Main reliably carries 1–2 numericals on continuity + Bernoulli, Venturi meters, or efflux speed. NEET favours conceptual questions on the speed–pressure relationship and real-life applications (aerofoil, atomizer). JEE Advanced builds tank-draining, efflux-range, and Venturi problems that fuse Bernoulli with calculus and projectile motion.

Section 1: Concept Intuition

Stand on a railway platform as a fast train rushes past, and you feel a distinct tug *towards* the train. It feels backwards — surely all that rushing air should push you *away*? This everyday surprise is the single most important idea of this entire subtopic, and it comes from Daniel Bernoulli: **where a fluid moves fast, its pressure is low**. The fast-moving air between you and the train is at lower pressure than the still air on your other side, so the higher pressure behind you nudges you toward the train. (This is exactly why the yellow safety line exists.)

Before we can explain *why*, we need to describe how fluids flow. Picture the morning milk queue flowing smoothly through a narrow gate — everyone moves in an orderly single file, each person tracing the same path as the one before. That is **streamline (steady) flow**: the velocity of the fluid at any fixed point never changes with time, and the path traced by any particle is a *streamline*. The tangent to a streamline gives the flow direction there. A crucial property: two streamlines can never cross — if they did, a particle at the crossing would have two different velocities at once, which is impossible. Bundle a set of streamlines together and you get a **tube of flow**; no fluid leaks across its walls, so whatever enters one end must leave the other. Push the fluid too hard, though, and the orderly queue dissolves into a chaotic scramble — eddies, swirls, unpredictable velocities. That is **turbulent flow**, like the churning water below a waterfall.

Think about it this way... Consider water flowing through a pipe that narrows, like a garden hose you pinch at the end. The same amount of water per second must pass through the wide part and the narrow part — water cannot pile up or vanish. So in the narrow section, where there is less room, the water must move *faster* to keep the throughput constant. This is the **equation of continuity**: $Av = \text{constant}$. Pinch the hose, shrink A , and the speed v shoots up — which is why a thumb over the hose end sends water spraying across the garden.

Now we can close the loop on the train mystery. A flowing fluid carries three kinds of energy per unit volume: **pressure energy** (P), **kinetic energy** ($\frac{1}{2}\rho v^2$), and **gravitational potential energy** (ρgh). Bernoulli's theorem is simply the conservation of energy for a flowing ideal fluid — it says the *sum* of these three stays constant along a streamline:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

The energy can only shuffle between the three forms. So on a level pipe (constant h), if the fluid speeds up (more kinetic energy), it must “pay” for that energy by dropping its pressure energy. Faster flow $\Rightarrow$ lower pressure. The fast air beside the train has spent its pressure to buy speed.

This one principle explains a startling range of phenomena: an aeroplane wing (faster air over the curved top $\Rightarrow$ lower pressure on top $\Rightarrow$ net upward lift), a perfume atomizer (fast air across a tube

mouth lowers the pressure and sucks liquid up), a spinning cricket ball that swings (the Magnus effect), and roofs torn off during a storm (fast wind over the roof lowers the pressure above it, so the higher pressure inside lifts the roof off).

Limiting Conditions & Assumptions. Bernoulli's theorem holds only for an **ideal fluid**: (1) **incompressible** (constant ρ — fine for liquids, poor for high-speed gases), (2) **non-viscous** (no internal friction — real viscosity dissipates energy, so downstream pressure is *lower* than Bernoulli predicts), (3) **steady** (streamline, not turbulent), and (4) **irrotational** (no eddies). It applies *along a single streamline*. The continuity equation $Av = \text{const}$ needs only incompressibility and steady flow, so it is more broadly valid than Bernoulli.

Section 2: Key Formulas & Definitions

Equation of continuity (mass conservation; incompressible, steady flow):

$$A_1 v_1 = A_2 v_2 \quad \text{i.e.} \quad Av = \text{constant}$$

where A = cross-sectional area, v = flow speed. The product Av is the **volume flow rate** Q (SI unit $\text{m}^3 \text{s}^{-1}$).

Bernoulli's theorem (energy per unit volume, along a streamline):

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

- P = pressure (static pressure), ρ = density, v = speed, h = elevation above a chosen datum, g = gravity. - Dimensions of each term: $[ML^{-1}T^{-2}]$ (same as pressure — every term is an energy density). - **Head form** (dividing by ρg): $\frac{P}{\rho g} + \frac{v^2}{2g} + h = \text{constant}$, the sum of *pressure head*, *velocity head*, and *gravitational head* (each in metres).

Note on conventions (chapter consistency): here h is height measured *upward* from a datum, the opposite sign to the *depth* used in Subtopic 01's $P = P_0 + \rho gh$. Both are correct in their own context.

Static, dynamic, and stagnation pressure (horizontal flow):

$$\underbrace{P}_{\text{static}} + \underbrace{\frac{1}{2}\rho v^2}_{\text{dynamic}} = \underbrace{P_{\text{stag}}}_{\text{stagnation (total)}} = \text{constant}$$

(P_{stag} here is the total pressure — *not* the atmospheric P_0 from earlier subtopics.)

Torricelli's law (speed of efflux): for a hole at depth $(H - h)$ below the free surface of an open tank,

$$v = \sqrt{2g(H - h)}$$

Horizontal range of the jet (hole at height h on a tank of liquid-height H): $R = 2\sqrt{h(H - h)}$, maximum $R_{\text{max}} = H$ at $h = H/2$.

Reynolds number (dimensionless — predicts flow type):

$$R_e = \frac{\rho v D}{\eta}$$

ρ = density, v = speed, D = pipe diameter, η = coefficient of viscosity. Roughly: $R_e \lesssim 1000$ streamline; $R_e \gtrsim 2000$ turbulent; in between, unstable.

Section 3: CBSE Board Derivations / Key Procedures

Derivation 1 — Equation of Continuity.

Figure 9.8 A tube of flow with a wide left section (area AA_1 , speed vv_1) and a narrow right section (area AA_2 , speed vv_2). Shaded fluid slugs of length $vv_1\Delta t$ and $vv_2\Delta t$ show the volume crossing each section in time Δt . Assumption: steady, incompressible flow through a tube of flow with no leakage across its walls. In a small time Δt , the fluid at section 1 (area A_1 , speed v_1) advances a distance $v_1\Delta t$, so the volume entering is $A_1v_1\Delta t$. Physical reasoning: because the fluid is incompressible and cannot accumulate inside the tube, the volume leaving at section 2 in the same time must be identical: $A_1v_1\Delta t = A_2v_2\Delta t$ $A_1v_1 = A_2v_2$. This is just conservation of mass (volume, for incompressible flow). It tells us $v \propto 1/A$: narrow the pipe and the fluid speeds up. (see next page)

Derivation 2 — Bernoulli's Theorem (work-energy method).

Figure 9.9 A tube of flow rising from left to right. Lower-left end: area AA_1 , speed vv_1 , pressure PP_1 , height hh_1 . Upper-right end: area AA_2 , speed vv_2 , pressure PP_2 , height hh_2 . Force PP_1AA_1 pushes the fluid in at the left; pressure PP_2AA_2 resists at the right. Assumption: steady, incompressible, non-viscous, irrotational flow. Consider the fluid between sections 1 and 2 moving in time Δt ; let a volume ΔV enter at 1 and an equal ΔV emerge at 2 (equal by continuity/incompressibility). Let $m = \rho\Delta V$ be its mass. Step 1 — Work done by pressure forces. The fluid behind pushes the slug in at section 1, doing positive work $W_1 = P_1A_1(v_1\Delta t) = P_1\Delta V$. At section 2, the fluid ahead pushes back, doing negative work $W_2 = -P_2\Delta V$. Net work by pressure: $W_{\text{pressure}} = (P_1 - P_2)\Delta V$. Step 2 — Apply the work-energy theorem. The net work by all forces equals the change in kinetic energy. Gravity also does work as the slug rises from h_1 to h_2 : $W_{\text{gravity}} = -mg(h_2 - h_1)$. So: $(P_1 - P_2)\Delta V - mg(h_2 - h_1) = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$. Step 3 — Substitute $m = \rho\Delta V$ and divide by ΔV . Physical reasoning: dividing by the common volume converts every term into an energy per unit volume: $P_1 - P_2 - \rho g(h_2 - h_1) = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2$. Step 4 — Rearrange so all "section 1" terms sit on the left and "section 2" on the right: $P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$ (see next page)

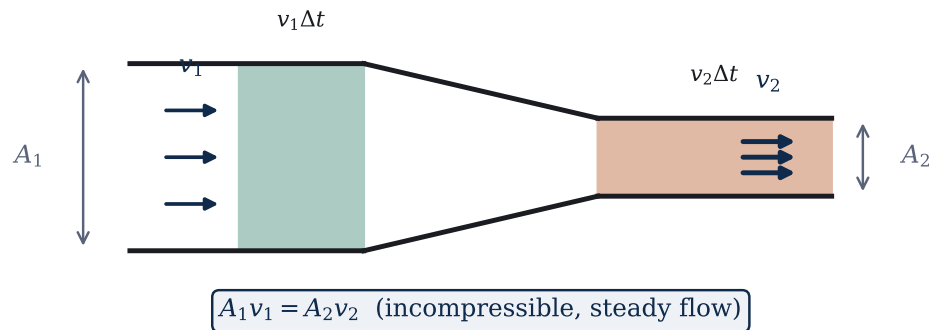


Figure 9.8 A tube of flow with a wide left section (area A_1 , speed v_1) and a narrow right section (area A_2 , speed v_2). Shaded fluid slugs of length $v_1\Delta t$ and $v_2\Delta t$ show the volume crossing each section in time Δt . Assumption: steady, incompressible flow through a tube of flow with no leakage across its walls. In a small time Δt , the fluid at section 1 (area A_1 , speed v_1) advances a distance $v_1\Delta t$, so the volume entering is $A_1v_1\Delta t$. Physical reasoning: because the fluid is incompressible and cannot accumulate inside the tube, the volume leaving at section 2 in the same time must be identical: $A_1v_1\Delta t = A_2v_2\Delta t$. $A_1v_1 = A_2v_2$ This is just conservation of mass (volume, for incompressible flow). It tells us $v \propto 1/A$: narrow the pipe and the fluid speeds up.

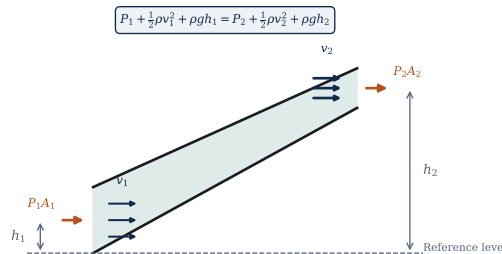


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Since sections 1 and 2 were arbitrary, the quantity $P + \frac{1}{2}\rho v^2 + \rho gh$ is constant along the streamline. Bernoulli's theorem is nothing but energy conservation, rewritten per unit volume. (Check: set $v_1 = v_2 = 0$ and it collapses to the hydrostatic result $P_2 - P_1 = \rho g(h_1 - h_2)$ from Subtopic 01.)

Derivation 3 — Torricelli's speed of efflux.

A large open tank of liquid is filled to height H ; a small hole is punched at height h from the base, i.e. at depth $(H - h)$ below the free surface. Apply Bernoulli between the free surface (point 1) and the hole (point 2). Both are open to the atmosphere, so $P_1 = P_2 = P_0$. Because the tank is large, the surface descends very slowly, $v_1 \approx 0$. Taking the base as datum:

$$P_0 + 0 + \rho gH = P_0 + \frac{1}{2}\rho v^2 + \rho gh$$

The P_0 terms cancel; solving for the efflux speed v :

$$v = \sqrt{2g(H - h)}$$

Physical reasoning: the result is identical to a body freely falling through the depth $(H - h)$ — the liquid's pressure energy has been fully converted to kinetic energy, exactly as gravitational PE would be in free fall.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Water flows steadily through a horizontal pipe whose diameter narrows from 6 cm to 2 cm. If the water enters the wide section at 2.0 m s^{-1} , find its speed in the narrow section.

Given: $d_1 = 6 \text{ cm}$, $d_2 = 2 \text{ cm}$, $v_1 = 2.0 \text{ m s}^{-1}$.

By the equation of continuity, $A_1 v_1 = A_2 v_2$. Since $\text{area} \propto d^2$:

$$v_2 = v_1 \frac{A_1}{A_2} = v_1 \left(\frac{d_1}{d_2} \right)^2 = 2.0 \times \left(\frac{6}{2} \right)^2 = 2.0 \times 9 = \boxed{18 \text{ m s}^{-1}}$$

The speed rises ninefold because the diameter shrank threefold (and area depends on diameter squared).

Example 2 — [NEET Speed Trap]

A large water tank is filled to a liquid height of 20 m. A small hole is drilled in its side at a height of 5 m above the base. Find the speed at which water exits the hole. (Take $g = 10 \text{ m s}^{-2}$.)

The traps to avoid: the efflux speed depends on the **depth of the hole below the free surface**, not on the tank's full height (20 m) nor on the hole's height above the base (5 m). And don't forget the factor of 2 inside the root.

Speed approach. Depth below surface:

$$H - h = 20 - 5 = 15 \text{ m}$$

$$v = \sqrt{2g(H - h)} = \sqrt{2 \times 10 \times 15} = \sqrt{300} \approx \boxed{17.3 \text{ m s}^{-1}}$$

Mental check: it must be the same as a stone dropped from 15 m: $v = \sqrt{2gh}$ — Torricelli's law *is* free-fall in disguise. A student who plugs in 20 gets 20 m s^{-1} (too big) and one who plugs in 5 gets 10 m s^{-1} (too small); the correct depth always lies between.

Example 3 — [JEE Main Level] (*combines continuity + Bernoulli*)

Water flows through a horizontal pipe. In the wide section the area is 8 cm^2 , the speed is 3.0 m s^{-1} , and the pressure is $2.0 \times 10^5 \text{ Pa}$. The pipe then narrows to 2 cm^2 . Find (a) the speed and (b) the pressure in the narrow section. (Take $\rho = 1000 \text{ kg m}^{-3}$.)

(a) Continuity:

$$v_2 = v_1 \frac{A_1}{A_2} = 3.0 \times \frac{8}{2} = 12 \text{ m s}^{-1}$$

(b) The pipe is horizontal ($h_1 = h_2$), so Bernoulli reduces to $P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$:

$$P_2 = P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2) = 2.0 \times 10^5 + \frac{1}{2} \times 1000 \times (3^2 - 12^2)$$

$$P_2 = 2.0 \times 10^5 + 500 \times (9 - 144) = 2.0 \times 10^5 - 67\,500 = \boxed{1.325 \times 10^5 \text{ Pa}}$$

The pressure *drops* in the narrow section precisely because the water sped up there — the central message of Bernoulli, confirmed numerically.

Example 4 — [JEE Advanced Level] (*continuity + Torricelli + calculus: tank draining time*)

A cylindrical tank of cross-sectional area $A = 1.0 \text{ m}^2$ is filled with water to a height $H = 5.0 \text{ m}$. A small hole of area $a = 1.0 \text{ cm}^2$ is opened at the bottom. Find the time taken for the tank to empty completely. (Take $g = 10 \text{ m s}^{-2}$.)

When the water level is at height y (measured from the hole), Torricelli gives the efflux speed $v = \sqrt{2gy}$. The volume leaving per unit time at the hole equals the rate at which the tank level drops:

$$a\sqrt{2gy} = -A \frac{dy}{dt}$$

Physical reasoning: the left side is the outflow rate at the hole; the right side is how fast the stored volume shrinks (negative, since y decreases). Separating variables:

$$\begin{aligned} \int_H^0 \frac{dy}{\sqrt{y}} &= -\frac{a}{A}\sqrt{2g} \int_0^t dt \\ [2\sqrt{y}]_H^0 &= -\frac{a}{A}\sqrt{2g} t \quad \Rightarrow \quad -2\sqrt{H} = -\frac{a}{A}\sqrt{2g} t \\ t &= \frac{A}{a} \sqrt{\frac{2H}{g}} \end{aligned}$$

Substituting $A/a = 1.0/(1.0 \times 10^{-4}) = 10^4$ and $\sqrt{2H/g} = \sqrt{2 \times 5/10} = \sqrt{1} = 1 \text{ s}$:

$$t = 10^4 \times 1 = \boxed{10^4 \text{ s} \approx 2.8 \text{ hours}}$$

Insight: the emptying time scales as $\sqrt{H}$, not H — halving requires the depth to drop to a quarter — because the outflow slows as the level falls. The huge A/a ratio (10 000) is why a small hole drains a big tank so slowly.

Section 5: Practice Exercises

1. [CBSE] State Bernoulli's theorem and name the physical principle it expresses. In a horizontal pipe carrying water, the speed at point X is twice that at point Y. At which point is the pressure greater, and why? (One-line reasoning.)
2. [NEET] Why does the canvas roof of a fast-moving truck bulge *upward*? Choose and justify: (a) air inside pushes up, (b) fast air outside lowers the pressure above the roof, (c) the roof material expands, (d) gravity decreases at speed.
3. [JEE Main] Water flows through a pipe of radius 4 cm at 1.5 m s^{-1} . The pipe branches into 3 identical smaller pipes each of radius 1 cm. Find the speed of water in each smaller pipe.
4. [JEE Main] A tank open to the atmosphere is filled with water to a height of 1.25 m. A small hole is made at the bottom. (a) Find the efflux speed. (b) If the hole has area 0.5 cm^2 , find the initial volume flow rate. (Take $g = 10 \text{ m s}^{-2}$.)
5. [JEE Advanced] A water tank of height H stands on the ground and is full to the brim. A small hole is to be drilled in the side wall at height h above the ground. (a) Show that the horizontal range of the emerging jet on the ground is $R = 2\sqrt{h(H-h)}$. (b) Find the height h that maximises the range and the value of R_{max} .

Answers: 1. Conservation of energy; pressure greater at Y (slower flow $\Rightarrow$ higher pressure). 2. (b). 3. 8.0 m s^{-1} . 4. (a) 5.0 m s^{-1} ; (b) $2.5 \times 10^{-4} \text{ m}^3\text{s}^{-1}$. 5. (b) $h = H/2$, $R_{\text{max}} = H$.

Section 6: Crack the MCQ

Q1. Water flows through a horizontal pipe of varying cross-section. At a point where the cross-section is *smaller*, the water's: (a) Speed increases and pressure increases (b) Speed increases and pressure decreases (c) Speed decreases and pressure increases (d) Speed decreases and pressure decreases

Correct: (b). Continuity ($Av = \text{const}$) forces the speed *up* in the narrow section; Bernoulli (horizontal) then forces the pressure *down* to conserve energy. - (a) is the intuitive-but-wrong answer — students imagine fast water “pushing harder,” confusing momentum with static pressure. This is the central misconception of the topic. - (c)/(d) get the continuity relation backwards (narrower $\Rightarrow$ faster, not slower).

Q2. Two streamlines in a steady flow: (a) Always run parallel (b) Can cross at exactly one point (c) Can never cross each other (d) Cross only in turbulent flow

Correct: (c). If two streamlines crossed, the fluid particle at the intersection would need two different velocity directions simultaneously — impossible in steady flow. - (a) is too strong; streamlines converge and diverge (they bunch up where flow is fast). - (b) violates the single-valued-velocity requirement. - (d) confuses the definition — turbulent flow has no well-defined streamlines at all.

Q3. An aeroplane wing generates lift mainly because: (a) Air below the wing moves faster than air above (b) Air above the wing moves faster, lowering the pressure on top (c) The wing is lighter than air (d) Engine thrust pushes the wing upward

Correct: (b). The wing is shaped so air travels faster over the curved upper surface; by Bernoulli, faster flow means lower pressure on top, so the higher pressure below produces a net upward lift. -

(a) reverses the speed relationship. - (c) is irrelevant — a wing is far denser than air. - (d) confuses thrust (horizontal, from engines) with lift (vertical, from pressure difference).

Q4. In a Venturi meter, a fluid speeds up as it passes through the constricted throat. Compared with the wide section, the pressure at the throat is: (a) Higher (b) Lower (c) The same (d) Zero

Correct: (b). The throat has higher speed (continuity), so lower static pressure (Bernoulli) — and this very pressure drop, read off a manometer, is what lets a Venturi meter measure the flow rate. - (a) repeats the “fast = high pressure” error. - (c) ignores the speed change entirely. - (d) wrongly assumes pressure vanishes; it merely decreases.

Section 7: Common Student Pitfalls & Pro-Tips

1. **“Fast fluid = high pressure.”** The most damaging misconception in the chapter. Bernoulli says the opposite on a horizontal streamline: faster flow means *lower* static pressure, because kinetic energy is bought at the cost of pressure energy. Train yourself to expect the inverse.
2. **Applying Bernoulli where it doesn’t hold.** Bernoulli needs an *ideal* fluid — non-viscous, incompressible, steady, irrotational — along a *single* streamline. For turbulent flow, viscous pipes, or comparing two unconnected streamlines, it fails. With real viscosity, downstream pressure is *lower* than Bernoulli predicts because energy is dissipated.
3. **Wrong height in efflux problems.** The speed of efflux uses the **depth of the hole below the free surface**, $\sqrt{2g(H-h)}$ — not the tank’s full height and not the hole’s height above the ground. Pinpoint the free surface first.
4. **Confusing volume flow rate with speed.** $Q = Av$ (in m^3s^{-1}) is *conserved* through a pipe; the *speed* v is not — it changes as A changes. When a question gives “rate of flow,” check whether it wants Q or v .

Pro-Tip: Solve every flow problem in two moves: **continuity first, Bernoulli second**. Continuity ($A_1v_1 = A_2v_2$) hands you the unknown speed; plug that into Bernoulli to get the pressure. Doing them in this order turns almost every Venturi, nozzle, and pipe problem into two clean substitutions. And remember the shortcut: efflux speed = free-fall speed through the depth, $\sqrt{2g\Delta h}$.

Section 8: Cheat Sheet Box

- **Continuity** $A_1v_1 = A_2v_2$ ($\$Av = Q = \const) — mass conservation; narrow pipe $\Rightarrow$ faster flow. Needs only incompressible + steady.
- **Bernoulli** $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$ along a streamline — energy conservation per unit volume; needs an **ideal fluid** (incompressible, non-viscous, steady, irrotational).
- **Core consequence:** on a level pipe, *faster flow $\Rightarrow$ lower pressure*. (Aerofoil lift, atomizer, Venturi, Magnus effect, roof blow-off.)
- **Static + dynamic = stagnation:** $P + \frac{1}{2}\rho v^2 = P_{\text{stag}}$ (constant). Each term has pressure dimensions $[ML^{-1}T^{-2}]$.
- **Torricelli efflux** $v = \sqrt{2g(H-h)}$ (depth below surface); jet range $R = 2\sqrt{h(H-h)}$, max $R_{\text{max}} = H$ at $h = H/2$.

- **Reynolds number** $R_e = \frac{\rho v D}{\eta}$ (dimensionless): $\lesssim 1000$ streamline, $\gtrsim 2000$ turbulent.

Memory aids: “Fast and low, slow and high” (speed vs. pressure). • “Continuity first, Bernoulli second.” • “Efflux = free fall” ($\sqrt{2gh}$). • “Ideal fluid only: no viscosity, no turbulence, one streamline.”

5. Viscosity and Terminal Velocity

Exam Relevance: A consistently scoring subtopic. CBSE Boards ask the definition of the coefficient of viscosity, its units/dimensions, and the terminal-velocity expression (2–3 marks), and frequently the conceptual temperature-dependence question. JEE Main reliably carries a terminal-velocity numerical or a Newton’s-law shear-force problem. NEET loves the $v_t \propto r^2$ scaling trap and the liquids-vs-gases temperature behaviour. JEE Advanced builds droplet-coalescence, free-fall-into-liquid, and power-dissipation scaling problems.

Section 1: Concept Intuition

Pour out a bottle of water and it splashes away instantly. Tip up a jar of honey and it crawls down the side in a slow, lazy ribbon. Both are fluids, both flow — but honey *resists* flowing far more than water does. That internal resistance, the friction *within* a fluid that opposes one layer sliding past another, is **viscosity**. It is, quite simply, fluid friction.

Here is the picture that makes it precise. Imagine a deep, still river. The water right at the bottom, touching the riverbed, is essentially stationary — it clings to the bed. The water at the surface moves fastest. In between, each thin horizontal layer slides over the one below it, a little faster than its neighbour beneath, a little slower than its neighbour above. This smooth, layered sliding is exactly what happens in any viscous flow, and it sets up a **velocity gradient** — the change in flow speed per unit distance measured *perpendicular* to the flow, written dv/dx .

Think about it this way... Picture a thick stack of paper sheets lying flat. Slide the top sheet sideways and, because of friction, it drags the sheet just below it along a little, which drags the next one, and so on down to the bottom sheet which stays put. Each sheet moves a bit slower than the one above. A viscous fluid behaves just like this stack: each layer exerts a tangential (sideways) dragging force on its neighbours. Newton found that this viscous force F is proportional to the contact area A between layers and to the velocity gradient dv/dx :

$$F = -\eta A \frac{dv}{dx}$$

The constant η is the **coefficient of viscosity** — large for honey, small for water, tinier still for air. The negative sign is the signature of friction: the force always *opposes* the relative motion, trying to slow the faster layer and speed up the slower one, dragging the fluid toward uniformity.

Viscosity is the close cousin of solid friction: both oppose relative motion, both arise from molecular attraction, and both come into play *only when there is relative motion*. This is the key difference from a static fluid — a fluid at rest has no velocity gradient, so viscosity is dormant. It acts only on *moving* fluids. This is why a stirred cup of tea eventually stops swirling, why a galvanometer needle settles instead of oscillating forever (viscous damping), and why thick oil makes a good shock absorber.

Now apply this to a small object falling through a fluid — a steel pellet sinking in glycerine, or a raindrop falling through air. Three forces act on it: its **weight** pulling down, the **buoyant force** pushing up (the displaced-fluid upthrust from Subtopic 02), and the **viscous drag** pushing up, *opposing* its downward motion. The drag grows with speed. So as the object speeds up, the upward drag grows until, at some speed, the three forces exactly balance: net force becomes zero,

acceleration stops, and the object falls thereafter at a constant speed. This steady, maximum speed is the **terminal velocity**. It is why a raindrop does not hit you at bullet speed despite falling kilometres, and why dust and cloud droplets seem to hang in the air — their terminal velocities are minuscule.

Limiting Conditions & Assumptions. (1) Newton's law $F = -\eta A dv/dx$ describes **laminar** (streamline) flow; in turbulent flow the simple linear relation fails. (2) **Temperature matters, and oppositely for the two phases:** liquid viscosity *decreases* with rising temperature (heat weakens the cohesive bonds), while gas viscosity *increases* (faster molecular diffusion across layers) — getting this backwards is the topic's classic error. (3) **Stokes' law** $F = 6\pi\eta r v$ holds only for a *small smooth sphere* moving *slowly* (low Reynolds number) through an *infinite, incompressible* fluid. (4) The terminal-velocity formula assumes the sphere has reached steady state and that the fluid is unbounded (no wall effects).

Section 2: Key Formulas & Definitions

Newton's law of viscosity:

$$F = -\eta A \frac{dv}{dx}$$

F = tangential viscous force, A = area of contact between layers, $\frac{dv}{dx}$ = velocity gradient (perpendicular to flow), η = coefficient of viscosity. The negative sign denotes opposition to relative motion.

Coefficient of viscosity — tangential force per unit area per unit velocity gradient:

$$\eta = \frac{F/A}{dv/dx}$$

- **SI unit:** $\text{N s m}^{-2} = \text{Pa} \cdot \text{s} = \text{decapoise} = \text{poiseuille (Pl)}$. - **CGS unit:** $\text{dyne s cm}^{-2} = \text{poise}$. - **Conversions:** $1 \text{ Pa} \cdot \text{s} = 1 \text{ decapoise} = 10 \text{ poise}$; equivalently $1 \text{ poise} = 0.1 \text{ Pa} \cdot \text{s}$.

- **Dimensional formula:** $[\eta] = \frac{[MLT^{-2}]/[L^2]}{[LT^{-1}]/[L]} = [ML^{-1}T^{-1}]$.

Kinematic viscosity: $\nu = \frac{\eta}{\rho}$ (SI unit m^2s^{-1}).

Stokes' law (viscous drag on a small sphere of radius r moving at speed v):

$$F = 6\pi\eta r v$$

Terminal velocity of a sphere (radius r , density ρ_b) falling through a fluid (density ρ_f , viscosity η) — from the three-force balance:

$$v_t = \frac{2}{9} \frac{r^2(\rho_b - \rho_f)g}{\eta}$$

Key scalings: $v_t \propto r^2$, $v_t \propto (\rho_b - \rho_f)$, $v_t \propto 1/\eta$. (If $\rho_b < \rho_f$, v_t is negative — the body rises, like an air bubble in water.)

Temperature dependence (memorise the split): liquids $\eta \downarrow$ as $T \uparrow$ (roughly $\eta \propto 1/T$); gases $\eta \uparrow$ as $T \uparrow$ (roughly $\eta \propto \sqrt{T}$).

Section 3: CBSE Board Derivations / Key Procedures

Derivation 1 — Terminal velocity of a sphere (three-force balance).

Figure 9.10 A small sphere of radius r falling vertically through a tall column of liquid. Three force arrows on the sphere: weight $(4/3)\pi r^3 \rho_b \cdot g$ downward; buoyant force $(4/3)\pi r^3 \rho_f \cdot g$ upward; viscous drag $6\pi\eta r \cdot v$ upward (opposing the downward motion). At terminal velocity these balance. Assumption: a small smooth sphere of radius r and density ρ_b falling through a large body of fluid of density ρ_f and viscosity η , slowly enough for Stokes' law to hold. Step 1 — List the three forces. - Weight (down): $W = \frac{4}{3}\pi r^3 \rho_b g$ - Buoyant force (up): $F_B = \frac{4}{3}\pi r^3 \rho_f g$ (weight of displaced fluid, from Subtopic 02) - Viscous drag (up, opposing motion): $F_v = 6\pi\eta r v$ (Stokes' law) Step 2 — Apply the terminal condition. Physical reasoning: initially the drag is zero (the sphere starts from rest), so the net downward force accelerates it; as v grows, F_v grows, until the upward forces exactly balance the weight. At that instant acceleration vanishes and the speed becomes constant — the terminal velocity v_t . Setting net force to zero: $W = F_B + F_v$ $\frac{4}{3}\pi r^3 \rho_b g = \frac{4}{3}\pi r^3 \rho_f g + 6\pi\eta r v_t$ Step 3 — Solve for v_t . Move buoyancy across and group the weight terms: $6\pi\eta r v_t = \frac{4}{3}\pi r^3 (\rho_b - \rho_f) g$ Cancel πr from both sides and simplify the constant ($\frac{4}{3} \div 6 = \frac{2}{9}$): $v_t = \frac{2}{9} r^2 (\rho_b - \rho_f) \frac{g}{\eta}$ Notice the result reuses the buoyancy machinery from Subtopic 02 (the $\rho_b - \rho_f$ "effective weight") and the Stokes drag — it is genuinely a synthesis of two ideas. The crucial takeaway is the r^2 dependence: a sphere of twice the radius reaches four times the terminal velocity. (see next page)

Derivation 2 — Dimensions and units of η (key procedure).

Starting from $\eta = \frac{F/A}{dv/dx}$, substitute dimensions step by step. *Physical reasoning*: each piece is built from its definition so the route is reproducible in an exam: - Force: $[F] = [MLT^{-2}]$; Area: $[A] = [L^2]$, so $[F/A] = [ML^{-1}T^{-2}]$. - Velocity gradient: $\left[\frac{dv}{dx}\right] = \frac{[LT^{-1}]}{[L]} = [T^{-1}]$. - Therefore $[\eta] = \frac{[ML^{-1}T^{-2}]}{[T^{-1}]} = [ML^{-1}T^{-1}]$.

For the unit conversion poise $\leftrightarrow$ Pa $\cdot$ s: $1 \text{ Pa} \cdot \text{s} = 1 \frac{\text{N s}}{\text{m}^2} = \frac{10^5 \text{ dyne} \cdot \text{s}}{10^4 \text{ cm}^2} = 10 \frac{\text{dyne s}}{\text{cm}^2} = 10 \text{ poise}$.

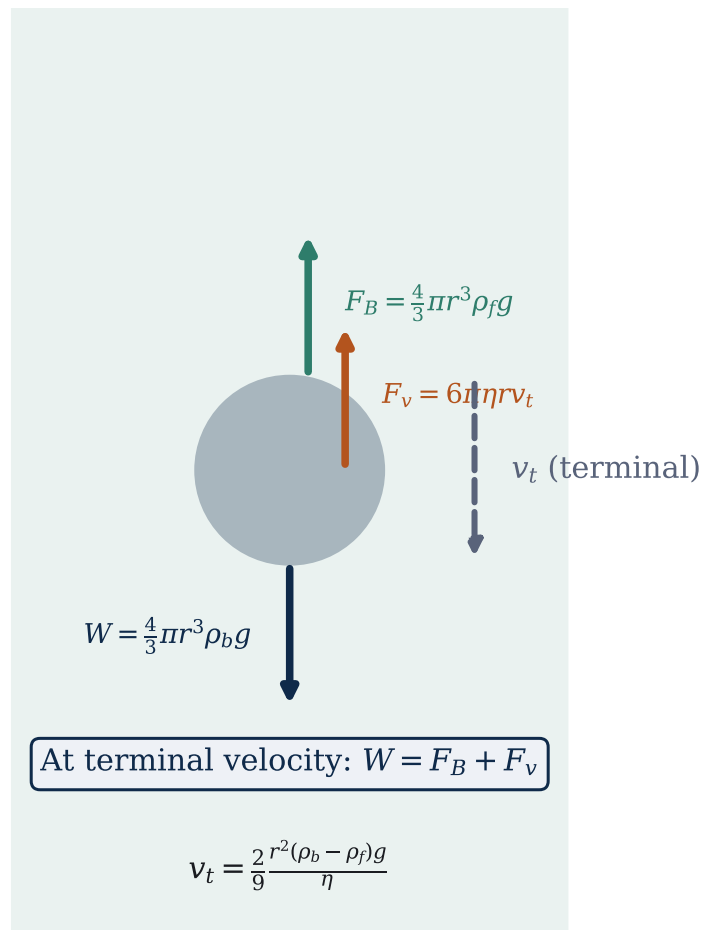


Figure 9.10 A small sphere of radius r falling vertically through a tall column of liquid. Three force arrows on the sphere: weight $(4/3)\pi r^3 \rho_b g$ downward; buoyant force $(4/3)\pi r^3 \rho_f g$ upward; viscous drag $6\pi \eta r v$ upward (opposing the downward motion). At terminal velocity these balance. Assumption: a small smooth sphere of radius r and density ρ_b falling through a large body of fluid of density ρ_f and viscosity η , slowly enough for Stokes' law to hold. Step 1 — List the three forces. - Weight (down): $W = \frac{4}{3}\pi r^3 \rho_b g$ - Buoyant force (up): $F_B = \frac{4}{3}\pi r^3 \rho_f g$ (weight of displaced fluid, from Subtopic 02) - Viscous drag (up, opposing motion): $F_v = 6\pi \eta r v$ (Stokes' law) Step 2 — Apply the terminal condition. Physical reasoning: initially the drag is zero (the sphere starts from rest), so the net downward force accelerates it; as v grows, F_v grows, until the upward forces exactly balance the weight. At that instant acceleration vanishes and the speed becomes constant — the terminal velocity v_t . Setting net force to zero: $W = F_B + F_v$ $\frac{4}{3}\pi r^3 \rho_b g = \frac{4}{3}\pi r^3 \rho_f g + 6\pi \eta r v_t$ Step 3 — Solve for v_t . Move buoyancy across and group the weight terms: $6\pi \eta r v_t = \frac{4}{3}\pi r^3 (\rho_b - \rho_f) g$ Cancel πr from both sides and simplify the constant ($\frac{4}{3} \div 6 = \frac{2}{9}$): $v_t = \frac{2}{9} \frac{r^2 (\rho_b - \rho_f) g}{\eta}$ Notice the result reuses the buoyancy machinery from Subtopic 02 (the $\rho_b - \rho_f$ "effective weight") and the Stokes drag — it is genuinely a synthesis of two ideas. The crucial takeaway is the r^2 dependence: a sphere of twice the radius reaches four times the terminal velocity.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A flat glass plate of area 0.10 m^2 rests on a layer of glycerine 2.0 mm thick spread over a table. The plate is dragged horizontally at a steady 0.50 m s^{-1} . If the coefficient of viscosity of glycerine is $1.5 \text{ Pa} \cdot \text{s}$, find the tangential force needed to keep the plate moving. (Assume a uniform velocity gradient and that the glycerine touching the table is stationary.)

Given: $A = 0.10 \text{ m}^2$, $\Delta x = 2.0 \text{ mm} = 2.0 \times 10^{-3} \text{ m}$, $v = 0.50 \text{ m s}^{-1}$, $\eta = 1.5 \text{ Pa} \cdot \text{s}$.

The velocity gradient (uniform, top moving, bottom still):

$$\frac{dv}{dx} = \frac{0.50 - 0}{2.0 \times 10^{-3}} = 250 \text{ s}^{-1}$$

The magnitude of the viscous force (the applied force must balance it for steady motion):

$$F = \eta A \frac{dv}{dx} = 1.5 \times 0.10 \times 250 = \boxed{37.5 \text{ N}}$$

Example 2 — [NEET Speed Trap]

A spherical raindrop of radius r falls through air at terminal velocity v . It breaks up into 8 identical smaller droplets. Find the terminal velocity of each smaller droplet.

The traps to avoid: (i) thinking terminal velocity scales with radius (giving $v/2$) or with volume (giving $v/8$). It scales as r^2 . (ii) Forgetting to first find the new radius from volume conservation.

Speed approach. Volume is conserved when the drop splits:

$$\frac{4}{3}\pi r^3 = 8 \times \frac{4}{3}\pi r'^3 \Rightarrow r'^3 = \frac{r^3}{8} \Rightarrow r' = \frac{r}{2}$$

Since $v_t \propto r^2$:

$$v' = v \left(\frac{r'}{r} \right)^2 = v \left(\frac{1}{2} \right)^2 = \boxed{\frac{v}{4}}$$

Mental check: 8 droplets means radius halves ($8^{1/3} = 2$), and $v_t \propto r^2$ means the speed quarters. Smaller drops fall slower — exactly why fine mist lingers in the air while big drops plummet.

Example 3 — [JEE Main Level] (*combines buoyancy from Subtopic 02 + Stokes' drag*)

A glass sphere of radius 2.0 mm and density 2500 kg m^{-3} falls through glycerine of density 1260 kg m^{-3} and viscosity $0.83 \text{ Pa} \cdot \text{s}$. Find (a) the terminal velocity, and (b) the viscous drag force at terminal velocity, and verify it equals the sphere's apparent weight. (Take $g = 10 \text{ m s}^{-2}$.)

(a) Terminal velocity:

$$\begin{aligned} v_t &= \frac{2}{9} \frac{r^2(\rho_b - \rho_f)g}{\eta} = \frac{2}{9} \frac{(2.0 \times 10^{-3})^2(2500 - 1260)(10)}{0.83} \\ &= \frac{2}{9} \frac{(4.0 \times 10^{-6})(1240)(10)}{0.83} = \frac{2}{9} \times \frac{0.0496}{0.83} \approx \boxed{1.33 \times 10^{-2} \text{ m s}^{-1}} \end{aligned}$$

(b) Viscous drag at terminal velocity (Stokes):

$$F_v = 6\pi\eta r v_t = 6\pi(0.83)(2.0 \times 10^{-3})(1.33 \times 10^{-2}) \approx 4.16 \times 10^{-4} \text{ N}$$

Verification — the apparent weight (effective weight minus buoyancy):

$$W_{\text{app}} = \frac{4}{3}\pi r^3(\rho_b - \rho_f)g = \frac{4}{3}\pi(2.0 \times 10^{-3})^3(1240)(10) \approx 4.16 \times 10^{-4} \text{ N} \checkmark$$

The two match, confirming the force balance: at terminal velocity, the viscous drag exactly supports the sphere against its buoyancy-reduced weight.

Example 4 — [JEE Advanced Level] (*scaling argument: power dissipated by viscous drag*)

A small sphere of radius r and density ρ_b falls at terminal velocity through a fluid of density ρ_f and viscosity η . Show that the rate at which heat is generated by viscous drag is proportional to r^5 , and find the full expression.

Reasoning. At terminal velocity the sphere moves at constant speed, so all the work done against the viscous drag is dissipated as heat. The power dissipated equals the drag force times the speed:

$$P = F_v v_t = (6\pi\eta r v_t) v_t = 6\pi\eta r v_t^2$$

Now substitute the terminal velocity $v_t = \frac{2}{9} \frac{r^2(\rho_b - \rho_f)g}{\eta}$, so that $v_t^2 = \frac{4}{81} \frac{r^4(\rho_b - \rho_f)^2 g^2}{\eta^2}$:

$$P = 6\pi\eta r \times \frac{4}{81} \frac{r^4(\rho_b - \rho_f)^2 g^2}{\eta^2} = \frac{24\pi}{81} \frac{r^5(\rho_b - \rho_f)^2 g^2}{\eta}$$

$$P = \frac{8\pi}{27} \frac{r^5(\rho_b - \rho_f)^2 g^2}{\eta} \propto r^5$$

Insight: the r^5 dependence is dramatic — doubling the radius increases the heating rate 32-fold. The physical chain is transparent once seen: drag $\propto r$ (one power), terminal speed $\propto r^2$, and power $= Fv \propto r \cdot v^2 \propto r \cdot r^4 = r^5$. This kind of scaling argument — tracking how each factor's power of r accumulates — is a hallmark of Advanced-level reasoning.

Section 5: Practice Exercises

1. [CBSE] Define the coefficient of viscosity and state its SI unit and dimensional formula. A fluid layer of area 0.20 m^2 experiences a viscous force of 0.60 N for a velocity gradient of 15 s^{-1} . Find the coefficient of viscosity.
2. [NEET] The viscosity of a liquid and the viscosity of a gas are each measured as the temperature is raised. State, with one-line reasoning, how each changes.
3. [JEE Main] A steel ball of radius 0.50 mm and density 8000 kg m^{-3} falls through a tank of oil of density 900 kg m^{-3} and viscosity $0.40 \text{ Pa} \cdot \text{s}$. Find its terminal velocity. (Take $g = 10 \text{ m s}^{-2}$.)

4. **[JEE Main]** An air bubble of radius 1.0 mm rises through water (density 1000 kg m^{-3} , viscosity $1.0 \times 10^{-3} \text{ Pa} \cdot \text{s}$). Treating the bubble's density as negligible, find the terminal (rise) speed. (Take $g = 10 \text{ m s}^{-2}$.)
5. **[JEE Advanced]** n identical small spherical droplets, each falling at terminal velocity v , coalesce into a single large drop. Find the terminal velocity of the large drop in terms of n and v .

Answers: **1.** $\eta = 0.20 \text{ Pa} \cdot \text{s}$. **2.** Liquid: η decreases; gas: η increases. **3.** $\approx 4.9 \times 10^{-2} \text{ m s}^{-1}$. **4.** $\approx 2.2 \text{ m s}^{-1}$ (directed upward). **5.** $v_{\text{big}} = n^{2/3} v$.

Section 6: Crack the MCQ

Q1. As temperature increases, the coefficient of viscosity of a liquid and of a gas respectively:
(a) Both increase (b) Both decrease (c) Liquid decreases, gas increases (d) Liquid increases, gas decreases

Correct: (c). Liquid viscosity arises from cohesive bonds, which weaken on heating, so η *decreases*. Gas viscosity arises from molecular diffusion between layers, which intensifies on heating, so η *increases*. - (a)/(b) wrongly apply a single rule to both phases — the most common error. - (d) reverses both directions. Memorise the split: liquids down, gases up.

Q2. Two spheres of the same material but radii in the ratio 1 : 2 fall through the same viscous liquid. The ratio of their terminal velocities is: (a) 1 : 2 (b) 1 : 4 (c) 1 : 8 (d) 2 : 1

Correct: (b). Terminal velocity $\propto r^2$, so the ratio is $1^2 : 2^2 = 1 : 4$. - (a) uses $v_t \propto r$ (forgetting the square). - (c) uses $v_t \propto r^3$ (confusing with volume/mass). - (d) inverts the relationship entirely.

Q3. The dimensional formula of the coefficient of viscosity is: (a) $[ML^{-1}T^{-1}]$ (b) $[MLT^{-1}]$ (c) $[ML^{-1}T^{-2}]$ (d) $[ML^2T^{-1}]$

Correct: (a). From $\eta = (F/A)/(dv/dx)$: $[ML^{-1}T^{-2}] \div [T^{-1}] = [ML^{-1}T^{-1}]$. - (c) is the dimensional formula of pressure (or stress) — chosen by students who stop at F/A and forget to divide by the velocity gradient. - (b)/(d) are dimensionally inconsistent with the definition.

Q4. A small water droplet falls through air and is observed to descend at a constant speed. At this instant: (a) Gravity is greater than the sum of viscous drag and buoyancy (b) Viscous drag plus buoyancy equals gravity (c) Viscous drag alone equals gravity (d) The droplet has zero weight

Correct: (b). Constant speed means zero net force, so the two upward forces (viscous drag + buoyancy) exactly balance the downward weight. This is the definition of terminal velocity. - (a) describes the *accelerating* phase before terminal velocity is reached. - (c) ignores buoyancy (acceptable as an approximation when fluid density is tiny, but not exact). - (d) is nonsensical — the weight is unchanged.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Reversing the temperature dependence.** Liquids get *less* viscous when heated; gases get *more* viscous. Students routinely apply one rule to both, or flip them. Anchor it physically:

heat loosens a liquid's cohesive grip (think warm honey pouring easily), but speeds up a gas's molecular mixing (more momentum transfer between layers).

2. **Mis-scaling terminal velocity.** $v_t \propto r^2$ — not r , not r^3 . When a drop splits or droplets coalesce, *first* find the new radius from volume conservation, *then* square the radius ratio.
3. **Dropping buoyancy in the terminal-velocity formula.** The driving term is the *effective* weight $(\rho_b - \rho_f)$, not ρ_b alone. Forgetting buoyancy overestimates v_t . For an air bubble in water, $\rho_b < \rho_f$ makes the term negative — the bubble *rises*, which the formula correctly predicts.
4. **Confusing poise with the SI unit.** 1 poise = 0.1 Pa · s, and 1 decapoise = 10 poise = 1 Pa · s. Always convert poise to Pa · s before substituting into SI formulas, or your terminal velocity will be off by a factor of ten.

Pro-Tip: For any falling-sphere problem, write the three-force balance — weight, buoyancy, drag — as your very first line. Every result (terminal velocity, viscosity from a measured speed, time to reach terminal velocity) flows from that one equation. And lock in the scaling triple: $v_t \propto r^2$, $\propto (\rho_b - \rho_f)$, $\propto 1/\eta$ — most NEET/JEE Main questions test a *ratio*, which you can answer from the scaling alone without ever plugging in numbers.

Section 8: Cheat Sheet Box

- **Viscosity** = internal fluid friction; acts *only* when layers move relative to each other. **Newton's law:** $F = -\eta A \frac{dv}{dx}$ (negative sign = opposes relative motion).
- **Coefficient of viscosity η :** SI unit Pa · s = decapoise = poiseuille; CGS unit poise; 1 Pa · s = 10 poise. **Dimensions** $[ML^{-1}T^{-1}]$.
- **Temperature:** liquids $\eta \downarrow$ with T ; gases $\eta \uparrow$ with T . (Opposite behaviours — memorise the split.)
- **Stokes' law:** $F = 6\pi\eta r v$ (small slow sphere).
- **Terminal velocity:** $v_t = \frac{2r^2(\rho_b - \rho_f)g}{9\eta}$, from weight = buoyancy + drag. Scalings: $v_t \propto r^2$, $(\rho_b - \rho_f)$, $1/\eta$.
- **Applications:** lubricant choice by season, shock absorbers, instrument damping, slow-settling cloud droplets, blood flow.

Memory aids: “*Liquids loosen, gases gum up*” (hot liquid flows easier, hot gas drags more). • “*Terminal velocity goes as r-squared.*” • “*Weight = buoyancy + drag*” at terminal velocity. • “*Poise is small — divide by ten for Pa · s.*”

6. Poiseuille's Law and Viscous Flow Resistance

Exam Relevance: A high-yield extension of viscosity. CBSE Boards ask the statement of Poiseuille's law and a direct flow-rate calculation. JEE Main reliably tests the r^4 dependence and series/parallel combinations of capillaries. NEET favours the “halve the radius” trap and blood-flow applications. JEE Advanced builds the parabolic velocity profile, junction-pressure problems, and the mean-vs-maximum-velocity result.

Section 1: Concept Intuition

In Subtopic 04 we watched a *single sphere* fall through a still fluid and learned how viscosity drags on it. Now flip the picture: hold the fluid's container still and make the *fluid itself* flow through a thin tube — water through a drip pipe, blood through an artery, ink through the nib of a pen. How fast does it flow? That is the question Jean Léonard Marie Poiseuille (a French physician studying blood flow) answered, and his law is one of the most practically important results in all of fluid physics.

Think about what limits the flow. Push harder — a bigger **pressure difference** between the two ends — and more fluid flows; that much is obvious, like opening a tap wider. Use a **longer** tube and the flow drops, because the fluid has to fight viscous drag over a greater distance. Use a **more viscous** fluid (honey instead of water) and flow drops, because internal friction is higher. So far, intuitive. But the fourth factor is the shocker: the flow depends on the tube's radius raised to the **fourth power**, r^4 . Not r , not r^2 (the area), but r^4 .

Think about it this way... Why r^4 and not just the cross-sectional area r^2 ? Two effects multiply. First, a wider tube has more cross-sectional area for fluid to pass through — that's one factor of r^2 . Second, and less obvious: in a wider tube, the fluid in the *middle* is much farther from the sticky walls, so on average it moves *faster* (the walls grip the outer layers, but the core slides freely). This average-speed boost contributes *another* factor of r^2 . Multiply them — area $\times$ average speed — and you get $r^2 \times r^2 = r^4$. The radius matters enormously: **halving a tube's radius cuts the flow to one-sixteenth** ($2^4 = 16$). This is why a tiny narrowing of an artery by cholesterol plaque can choke blood flow dramatically, and why a slightly clogged fuel line starves an engine.

The flow in a viscous tube is not uniform across the cross-section. The fluid touching the wall is essentially stuck (the “no-slip condition”), while the fluid at the centre races ahead. The speed varies smoothly from zero at the wall to a maximum at the axis, tracing a **parabolic velocity profile** — a bullet-shaped front. The average speed turns out to be exactly *half* the maximum (centre) speed.

There is a beautiful unifying idea here. Compare fluid flow to electric current. A pressure difference ΔP drives a volume flow Q through a tube, just as a voltage V drives a current I through a wire. The tube *resists* the flow, and we can define a **fluid resistance** $R = \Delta P/Q$, perfectly analogous to Ohm's law $R = V/I$. This analogy is not a loose metaphor — it works quantitatively, including the rules for combining tubes in series and parallel, exactly like resistors.

Limiting Conditions & Assumptions. Poiseuille's law holds only for: (1) **steady, laminar (streamline) flow** — it fails once the flow turns turbulent (high Reynolds number); (2) an **incompressible, Newtonian** fluid (constant η); (3) a **long, straight, uniform** cylindrical tube

(entrance effects neglected); (4) the **no-slip** boundary condition at the walls. For blood (a non-Newtonian suspension) or for fast/turbulent flow, the law is only approximate.

Section 2: Key Formulas & Definitions

Poiseuille's law — volume flow rate Q through a horizontal capillary:

$$Q = \frac{\pi \Delta P r^4}{8\eta l} = \frac{\pi(P_1 - P_2) r^4}{8\eta l}$$

- Q = volume flow rate ($\text{m}^3 \text{s}^{-1}$), $\Delta P = P_1 - P_2$ = pressure difference across the ends, r = tube radius, l = tube length, η = coefficient of viscosity. - Key scalings: $Q \propto \Delta P$, $Q \propto 1/l$, $Q \propto 1/\eta$, $Q \propto r^4$.

Fluid (viscous) resistance — by analogy with Ohm's law:

$$R = \frac{\Delta P}{Q} = \frac{8\eta l}{\pi r^4} \quad (\text{units: Pa s m}^{-3})$$

- Driving “voltage” $\leftrightarrow \Delta P$; “current” $\leftrightarrow Q$; “resistance” $\leftrightarrow R$.

Combination of tubes (same analogy as resistors): - **Series** (same Q through each, pressure drops add): $R_{\text{eq}} = R_1 + R_2 + \dots$ - **Parallel** (same ΔP across each, flows add): $\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

Velocity profile (parabolic) at radial distance s from the axis of a tube of radius r :

$$v(s) = \frac{\Delta P}{4\eta l} (r^2 - s^2)$$

- **Maximum (axial) speed:** $v_{\text{max}} = \frac{\Delta P r^2}{4\eta l}$ (at $s = 0$). - **Mean speed:** $v_{\text{avg}} = \frac{Q}{\pi r^2} = \frac{\Delta P r^2}{8\eta l} = \frac{v_{\text{max}}}{2}$.

Critical velocity (onset of turbulence; links to Reynolds number from Subtopics 03–04):

$$v_c = \frac{R_e \eta}{\rho D}$$

where $D = 2r$ is the diameter and $R_e \approx 2000$ marks the laminar–turbulent transition.

Section 3: Key Procedures and Derivations

Derivation 1 — Poiseuille's law by dimensional analysis (CBSE/JEE Main method).

Reasoning: the volume flow rate Q should depend on the pressure gradient $\frac{\Delta P}{l}$ (the driving force per length), the viscosity η (the resistance), and the tube radius r (the geometry). Assume a power-law form:

$$Q = k \left(\frac{\Delta P}{l} \right)^a \eta^b r^c$$

where k is a dimensionless constant. Write dimensions ($[Q] = [L^3 T^{-1}]$, $[\Delta P/l] = [ML^{-2} T^{-2}]$, $[\eta] = [ML^{-1} T^{-1}]$, $[r] = [L]$):

$$[L^3 T^{-1}] = [ML^{-2} T^{-2}]^a [ML^{-1} T^{-1}]^b [L]^c$$

Match powers of M , L , T : - M : $0 = a + b$ - T : $-1 = -2a - b$ - L : $3 = -2a - b + c$

From the first two: $a = 1$, $b = -1$. Substituting into the third: $3 = -2(1) - (-1) + c = -1 + c$, so $c = 4$. Therefore

$$Q = k \frac{\Delta P}{l} \frac{1}{\eta} r^4 = k \frac{\Delta P r^4}{\eta l}$$

Dimensional analysis cannot fix the pure number k ; a full calculus treatment (Derivation 2) gives $k = \frac{\pi}{8}$, yielding $Q = \frac{\pi \Delta P r^4}{8 \eta l}$. *Physical reasoning*: the r^4 emerges automatically from the dimensional bookkeeping — it is not assumed, which is what makes this derivation satisfying.

Derivation 2 — Parabolic velocity profile and the factor $\pi/8$ (JEE Advanced).

Figure 9.11 *A horizontal cylindrical tube of radius r with fluid flowing left to right. A coaxial inner cylinder of radius s is shown. The velocity profile is drawn as a parabola: zero at the walls, maximum at the axis. Pressure PP_1 at the left end, PP_2 at the right end, length l . Assumption: steady laminar flow; consider a coaxial fluid cylinder of radius s and length l . Two forces balance on it: the net pressure force pushing it forward and the viscous drag from the slower-moving fluid shell just outside it. The pressure force driving the inner cylinder forward: $(P_1 - P_2)\pi s^2 = \Delta P \pi s^2$. The viscous drag on its curved surface (area $2\pi s l$), by Newton's law of viscosity: $-\eta(2\pi s l)dv/ds$. For steady (non-accelerating) flow these balance: $\Delta P \pi s^2 = -\eta(2\pi s l)dv/ds \Rightarrow dv/ds = -\Delta P 2\eta l s$. Integrate from s out to the wall at $s = r$, where $v = 0$ (no-slip): $v(s) = \Delta P 4\eta l (r^2 - s^2)$. This is the parabolic profile. Now find the total flow by summing the contributions of thin annular rings of area $2\pi s ds$: $Q = \int_0^r v(s) 2\pi s ds = 2\pi \Delta P 4\eta l \int_0^r (r^2 - s^2)s ds = \pi \Delta P 2\eta l [r^2 s^2 - \frac{s^4}{4}]_0^r = \pi \Delta P 2\eta l \cdot \frac{r^4}{4} = \frac{\pi \Delta P r^4}{8\eta l}$. This recovers Poiseuille's law with the constant $\pi/8$. The mean speed is $v_{avg} = Q/\pi r^2 = \Delta P r^2 / 8\eta l$, exactly half the axial maximum $v_{max} = \Delta P r^2 / 4\eta l$. (see next page)*

Key procedure — combining tubes (the resistance analogy). Treat each tube as a resistor $R = 8\eta l / \pi r^4$. For tubes end-to-end (series), the *same* flow passes through each and the pressure drops add, so $R_{eq} = \sum R_i$. For tubes side-by-side between the same two reservoirs (parallel), each feels the *same* ΔP and the flows add, so $1/R_{eq} = \sum 1/R_i$. Then $Q = \Delta P / R_{eq}$.

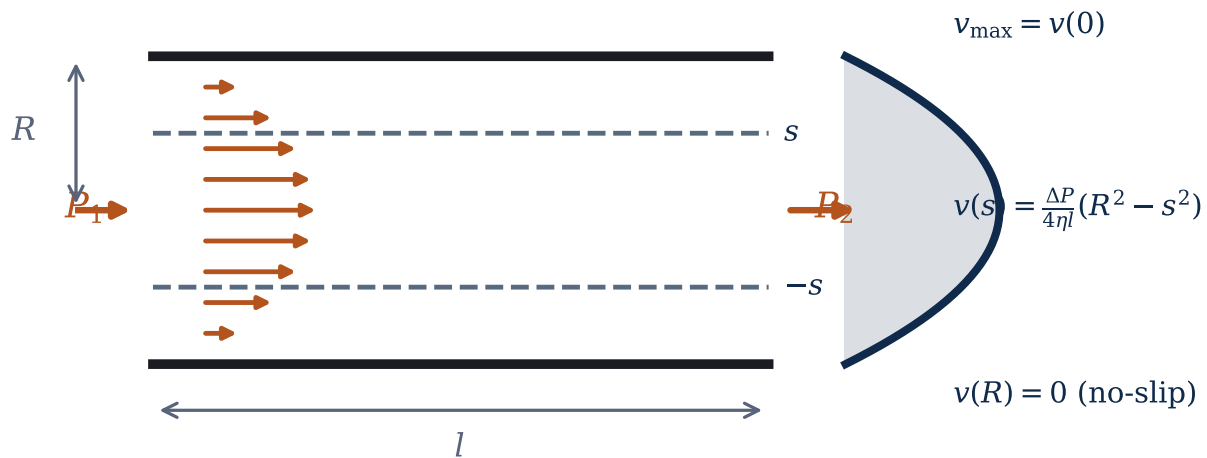


Figure 9.11 A horizontal cylindrical tube of radius r with fluid flowing left to right. A coaxial inner cylinder of radius s is shown. The velocity profile is drawn as a parabola: zero at the walls, maximum at the axis. Pressure P_1 at the left end, P_2 at the right end, length l . Assumption: steady laminar flow; consider a coaxial fluid cylinder of radius s and length l . Two forces balance on it: the net pressure force pushing it forward and the viscous drag from the slower-moving fluid shell just outside it. The pressure force driving the inner cylinder forward: $(P_1 - P_2)\pi s^2 = \Delta P \pi s^2$. The viscous drag on its curved surface (area $2\pi s l$), by Newton's law of viscosity: $-\eta(2\pi s l) dv/ds$. For steady (non-accelerating) flow these balance: $\Delta P \pi s^2 = -\eta(2\pi s l) dv/ds \Rightarrow dv/ds = -\Delta P / (4\eta l)$. Integrate from s out to the wall at $s = r$, where $v = 0$ (no-slip): $v(s) = \Delta P / (4\eta l) (r^2 - s^2)$. This is the parabolic profile. Now find the total flow by summing the contributions of thin annular rings of area $2\pi s ds$: $Q = \int_0^r v(s) 2\pi s ds = 2\pi \Delta P / (4\eta l) \int_0^r (r^2 - s^2) s ds = \pi \Delta P / (2\eta l) [r^2 s^2 / 2 - s^4 / 4]_0^r = \pi \Delta P / (2\eta l) \cdot r^4 / 4 = \pi \Delta P r^4 / (8\eta l)$. This recovers Poiseuille's law with the constant $\pi/8$. The mean speed is $v_{\text{avg}} = Q / \pi r^2 = \Delta P r^2 / (8\eta l)$, exactly half the axial maximum $v_{\max} = \Delta P r^2 / (4\eta l)$.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Water flows through a horizontal capillary tube of length 0.20 m and radius 0.50 mm under a pressure difference of 1.0×10^3 Pa between its ends. The viscosity of water is 1.0×10^{-3} Pa · s. Find the volume flow rate.

Given: $l = 0.20$ m, $r = 0.50$ mm = 5.0×10^{-4} m, $\Delta P = 1.0 \times 10^3$ Pa, $\eta = 1.0 \times 10^{-3}$ Pa · s.

By Poiseuille's law:

$$Q = \frac{\pi \Delta P r^4}{8 \eta l} = \frac{\pi (1.0 \times 10^3) (5.0 \times 10^{-4})^4}{8 (1.0 \times 10^{-3}) (0.20)}$$

Compute $r^4 = (5.0 \times 10^{-4})^4 = 6.25 \times 10^{-14}$ m⁴:

$$Q = \frac{\pi (1.0 \times 10^3) (6.25 \times 10^{-14})}{1.6 \times 10^{-3}} = \frac{\pi (6.25 \times 10^{-11})}{1.6 \times 10^{-3}} \approx \boxed{1.23 \times 10^{-7} \text{ m}^3 \text{s}^{-1}}$$

That is about 0.12 mL per second.

Example 2 — [NEET Speed Trap]

A liquid flows through a capillary tube at a rate Q . If a second tube of the **same length** but **half the radius** is used under the same pressure difference, the new flow rate is:

The trap to avoid: thinking flow scales with area (r^2 , giving $Q/4$) or linearly with radius ($Q/2$). Poiseuille's law scales as r^4 .

Speed approach. Since $Q \propto r^4$:

$$Q' = Q \left(\frac{r/2}{r} \right)^4 = Q \left(\frac{1}{2} \right)^4 = \boxed{\frac{Q}{16}}$$

Mental check: “Radius to the fourth” — halving the radius drops the flow by $2^4 = 16$. This extreme sensitivity is the whole point of the law and the most-tested fact about it.

Example 3 — [JEE Main Level] (*series combination — fluid resistance*)

Two capillary tubes are joined end to end. The first has length l and radius r ; the second has length l and radius $2r$. A pressure difference ΔP is maintained across the combination, with the liquid (viscosity η) flowing through both. Find the overall volume flow rate, and the pressure drop across the *narrow* tube.

The tubes are in **series**, so add their fluid resistances. With $R = 8\eta l / \pi r^4$:

$$R_1 = \frac{8\eta l}{\pi r^4}, \quad R_2 = \frac{8\eta l}{\pi (2r)^4} = \frac{8\eta l}{16\pi r^4} = \frac{R_1}{16}$$
$$R_{\text{eq}} = R_1 + R_2 = R_1 \left(1 + \frac{1}{16} \right) = \frac{17}{16} R_1 = \frac{17}{16} \cdot \frac{8\eta l}{\pi r^4} = \frac{17\eta l}{2\pi r^4}$$

Overall flow:

$$Q = \frac{\Delta P}{R_{\text{eq}}} = \frac{2\pi r^4 \Delta P}{17\eta l}$$

Pressure drop across the narrow tube (same Q , so $\Delta P_1 = QR_1$):

$$\Delta P_1 = QR_1 = \frac{\Delta P}{R_{\text{eq}}} \cdot R_1 = \Delta P \cdot \frac{R_1}{\frac{17}{16}R_1} = \frac{16}{17}\Delta P \approx \boxed{0.94 \Delta P}$$

Almost the entire pressure drop (16/17) occurs across the narrow tube — the wide tube barely resists. This mirrors how, in a series circuit, the larger resistor takes the larger share of the voltage.

Example 4 — [JEE Advanced Level] (*mean vs. maximum velocity from the profile*)

For laminar flow of a liquid through a horizontal tube of radius r under a pressure difference ΔP over length l , show from the parabolic velocity profile that the mean flow speed equals half the maximum speed, and find the radial distance from the axis at which the local speed equals the mean speed.

From Derivation 2, the velocity profile is $v(s) = \frac{\Delta P}{4\eta l}(r^2 - s^2)$, so the maximum (axial) speed is

$$v_{\text{max}} = v(0) = \frac{\Delta P r^2}{4\eta l}$$

The mean speed is the flow rate divided by the area:

$$v_{\text{avg}} = \frac{Q}{\pi r^2} = \frac{1}{\pi r^2} \cdot \frac{\pi \Delta P r^4}{8\eta l} = \frac{\Delta P r^2}{8\eta l} = \frac{v_{\text{max}}}{2} \quad \checkmark$$

To find where the *local* speed equals the mean speed, set $v(s) = v_{\text{avg}} = \frac{1}{2}v_{\text{max}}$:

$$\frac{\Delta P}{4\eta l}(r^2 - s^2) = \frac{1}{2} \cdot \frac{\Delta P r^2}{4\eta l} \implies r^2 - s^2 = \frac{r^2}{2} \implies s = \frac{r}{\sqrt{2}} \approx \boxed{0.71 r}$$

Insight: the fluid moves at the average speed at about 71% of the way out from the axis — closer to the wall than the centre, because the slow near-wall fluid occupies a disproportionate share of the cross-sectional area.

Section 5: Practice Exercises

1. **[CBSE]** State Poiseuille's law and write the expression for the volume flow rate through a capillary, defining each symbol. On what power of the radius does the flow rate depend?
2. **[NEET]** The radius of a pipe carrying a viscous liquid is increased to three times its original value, keeping length and pressure difference unchanged. By what factor does the flow rate change?
3. **[JEE Main]** Water flows through a horizontal tube of radius 1.0 mm and length 1.0 m at $2.0 \times 10^{-6} \text{ m}^3\text{s}^{-1}$. Find the pressure difference across the tube. ($\eta_{\text{water}} = 1.0 \times 10^{-3} \text{ Pa} \cdot \text{s}$.)
4. **[JEE Main]** Two identical capillaries are connected (a) in series and (b) in parallel across the same pressure difference. Find the ratio of the total flow rate in the parallel arrangement to that in the series arrangement.

5. [JEE Advanced] A capillary of radius r and length l delivers a flow Q under pressure difference ΔP . It is replaced by a tube of the same length but with radius varying — model it as two equal-length halves of radii r and $r/2$ in series. Find the new flow rate in terms of Q .

Answers: 1. $Q \propto r^4$. 2. Increases by $81 \times (3^4)$. 3. $\Delta P \approx 5.1 \times 10^3$ Pa. 4. $4 : 1$. 5. $Q' = \frac{2Q}{17}$.

Section 6: Crack the MCQ

Q1. If the radius of a capillary tube is doubled while length and pressure difference are unchanged, the volume flow rate becomes: (a) $2 \times$ (b) $4 \times$ (c) $8 \times$ (d) $16 \times$

Correct: (d). $Q \propto r^4$, so doubling the radius multiplies the flow by $2^4 = 16$. - (a) uses $Q \propto r$; (b) uses $Q \propto r^2$ (area); (c) uses $Q \propto r^3$ (volume). Each forgets one or more of the two r^2 contributions. The r^4 law is the single most important fact here.

Q2. The velocity profile of a viscous liquid undergoing laminar flow through a cylindrical pipe is: (a) Uniform across the cross-section (b) Linear, zero at the centre (c) Parabolic, maximum at the axis (d) Maximum at the walls

Correct: (c). No-slip at the walls makes the speed zero there; it rises parabolically to a maximum on the axis. - (a) describes ideal (non-viscous) plug flow. - (b)/(d) invert the profile — the *centre* is fastest, the *walls* are stationary.

Q3. Two tubes of fluid resistance R and $2R$ are connected in series. The equivalent fluid resistance is: (a) $R/3$ (b) $2R/3$ (c) $3R$ (d) $2R$

Correct: (c). In series, fluid resistances add (like resistors): $R + 2R = 3R$. - (a)/(b) apply the *parallel* formula by mistake. - (d) ignores one tube entirely.

Q4. In a person's artery, a deposit narrows the radius to 80% of normal. Assuming Poiseuille flow at the same pressure difference, the blood flow rate falls to approximately: (a) 80% (b) 64% (c) 51% (d) 41%

Correct: (d). $Q \propto r^4$, so the flow scales by $(0.80)^4 = 0.41$, i.e. about 41%. - (a) uses r^1 ; (b) uses r^2 ; (c) uses r^3 . The dramatic drop from a modest narrowing — to less than half — is exactly why arterial plaque is so dangerous, and is the medical punchline of Poiseuille's law.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Using the wrong power of r .** Flow rate goes as r^4 , not r^2 (area) or r^3 (volume). When a problem changes the radius, raise the ratio to the *fourth* power. This single slip accounts for most errors in the topic.
2. **Mixing up series and parallel rules.** Fluid resistances combine exactly like electrical resistors: *add* in series, *reciprocal-add* in parallel. The narrow tube (huge $R \propto 1/r^4$) dominates a series combination, just as the large resistor dominates a series circuit.

3. **Applying Poiseuille's law to turbulent flow.** The law is for *laminar* flow only. Check the Reynolds number first; above ~ 2000 the flow turns turbulent and Q no longer scales as $\Delta P r^4$ (it grows more slowly than ΔP).
4. **Confusing mean and maximum speed.** The centre-line speed is the *maximum*; the *mean* speed (used in $Q = v_{\text{avg}}\pi r^2$) is half of it. Use the mean for flow-rate questions and the maximum only when the axial speed is specifically asked.

Pro-Tip: For any “change the tube” question, lean on the scaling $Q \propto \frac{\Delta P r^4}{\eta l}$ and work with *ratios* — you rarely need to plug in the messy constants. And memorise the resistance analogy ($\Delta P \leftrightarrow V$, $Q \leftrightarrow I$, $R = 8\eta l/\pi r^4$); it turns every multi-tube network into a familiar circuit problem you already know how to solve.

Section 8: Cheat Sheet Box

- **Poiseuille's law:** $Q = \frac{\pi \Delta P r^4}{8\eta l}$ — laminar flow through a capillary. Scalings: $Q \propto \Delta P, r^4, 1/l, 1/\eta$.
- **The r^4 rule:** halve the radius $\Rightarrow$ flow drops to $1/16$; double it $\Rightarrow 16\times$. Extreme radius sensitivity.
- **Fluid resistance:** $R = \frac{\Delta P}{Q} = \frac{8\eta l}{\pi r^4}$ (Ohm's-law analogy). **Series:** $R_{\text{eq}} = \sum R_i$. **Parallel:** $\frac{1}{R_{\text{eq}}} = \sum \frac{1}{R_i}$.
- **Velocity profile:** parabolic, $v(s) = \frac{\Delta P}{4\eta l}(r^2 - s^2)$; v_{max} at axis; $v_{\text{avg}} = v_{\text{max}}/2$.
- **Valid only for** steady, laminar, incompressible, Newtonian flow in a long uniform tube ($R_e \lesssim 2000$).
- **Applications:** blood flow (arterial narrowing), IV drips, hypodermic needles, lubrication channels.

Memory aids: “Flow goes as radius to the fourth.” • “Pressure is voltage, flow is current, $8\eta l/\pi r^4$ is resistance.” • “Add in series, reciprocal in parallel.” • “Mean is half the max.”

7. Surface Tension and Capillarity

Exam Relevance: A high-yield, formula-rich subtopic that closes the chapter. CBSE Boards ask the definition of surface tension/surface energy, the relation between them, and capillary rise (2–5 marks), plus the conceptual angle-of-contact question. JEE Main reliably carries an excess-pressure or capillary-rise numerical and the drop-splitting energy problem. NEET favours the drop-vs-bubble excess-pressure trap and the temperature/impurity behaviour. JEE Advanced builds droplet-coalescence-with-heating, double-bubble, and insufficient-length capillary problems.

Section 1: Concept Intuition

Watch a water strider skate across a pond without sinking, or a sewing needle laid gently flat float on water though steel is eight times denser. Notice how rain collects into near-perfect spherical beads on a freshly waxed car, and how water climbs *up* the thin stem of a cotton wick against gravity. All of these everyday wonders trace back to one property of liquid surfaces: **surface tension** — the tendency of a liquid surface to behave like a stretched elastic membrane, always trying to shrink to the smallest possible area.

Where does this “skin” come from? Molecules in a liquid attract their neighbours (these are **cohesive forces**). A molecule deep *inside* the liquid is pulled equally in all directions — up, down, sideways — so the net force on it is zero. But a molecule right *at the surface* has liquid below and beside it, and only air above. With no neighbours pulling up, it feels a net *inward* pull. Every surface molecule is tugged toward the bulk, so the surface is squeezed taut, like the stretched skin of a drum. To create *more* surface, you must drag molecules from the comfortable interior up to the surface against this inward pull — which costs energy. That is why liquids minimise their surface area, and why a free drop, having no other constraints, pulls itself into a sphere: a sphere has the least surface area for a given volume.

Think about it this way... Imagine the surface as a perpetually shrinking sheet of stretched rubber. Cut an imaginary line anywhere on it, and the rubber on each side pulls the other inward along the surface, perpendicular to the line. The force per unit length of this line *is* the surface tension, S . Equivalently — and this is the second face of the same coin — every unit of new surface area you create stores a fixed amount of potential energy, called the **surface energy**. The two views are numerically identical: surface tension (force per unit length, N/m) equals surface energy (energy per unit area, J/m²). Pulling a wire to stretch a film and counting the work done gives the same number as measuring the force per length directly.

This single idea blossoms into three big consequences. First, a curved liquid surface has *higher pressure on its concave side* — the membrane’s inward squeeze must be balanced by extra internal pressure, giving the **excess pressure** inside drops and bubbles. Second, when a liquid meets a solid, the competition between cohesion (liquid-to-liquid) and **adhesion** (liquid-to-solid) decides the shape of the surface where they meet — the **angle of contact**. If adhesion wins (water on clean glass), the liquid climbs the wall and the surface curves up (concave meniscus, wetting). If cohesion wins (mercury on glass), the liquid pulls away and the surface bulges down (convex meniscus, non-wetting). Third, in a narrow tube, this wall-climbing plus the surface tension drags the whole liquid column up (or, for mercury, pushes it down) — the phenomenon of **capillarity**, which carries water up the soil to plant roots and oil up a lamp wick.

Limiting Conditions & Assumptions. (1) Surface tension is a property of the *interface* between two media (usually liquid-air); it is not a property of a single substance in isolation. (2) It **decreases with temperature**, falling to *zero* at the liquid's critical temperature — the surface vanishes there. (3) **Impurities matter:** highly soluble impurities (like common salt) raise it, while sparingly soluble ones (soap, detergents) lower it. (4) The capillary formula $h = 2S \cos \theta / (r\rho g)$ assumes a clean, narrow, vertical cylindrical tube and a hemispherical meniscus; it fails for wide tubes or tubes of insufficient length (where the meniscus simply flattens rather than overflowing).

Section 2: Key Formulas & Definitions

Surface tension — force per unit length acting perpendicular to an imaginary line on the surface:

$$S = \frac{F}{L}$$

- **SI unit:** N m^{-1} ; **CGS unit:** dyne cm^{-1} ($1 \text{ N m}^{-1} = 10^3 \text{ dyne cm}^{-1}$). - **Dimensional formula:**

$$[S] = \frac{[MLT^{-2}]}{[L]} = [ML^0T^{-2}] = [MT^{-2}].$$

Surface energy — work done to increase surface area by unit amount; numerically equal to surface tension:

$$E = \frac{W}{\Delta A} = S \quad (\text{SI unit J m}^{-2})$$

Work to change area (account for number of surfaces): for a film/soap bubble with **two** surfaces, $W = S \cdot 2 \Delta A_{\text{geom}}$; for a single-surface drop, $W = S \cdot \Delta A$.

Forces to detach objects from a liquid surface: - Straight wire of length ℓ : $F = 2S\ell$ (film has two edges). - Thin ring of radius r : $F = 4\pi rS$ (inner + outer circumference).

Excess pressure (pressure inside minus pressure outside): - Liquid drop (one surface): $\Delta P = \frac{2S}{r}$
 - Air bubble inside a liquid (one surface): $\Delta P = \frac{2S}{r}$ - Soap bubble in air (two surfaces): $\Delta P = \frac{4S}{r}$

Angle of contact θ : the angle between the solid surface and the tangent to the liquid surface at the point of contact, measured inside the liquid. $\theta < 90^\circ \Rightarrow$ wetting (concave meniscus, adhesion > cohesion); $\theta > 90^\circ \Rightarrow$ non-wetting (convex meniscus, cohesion > adhesion). θ *decreases* with rising temperature.

Capillary rise (Jurin's law):

$$h = \frac{2S \cos \theta}{r\rho g} = \frac{2S}{R\rho g}$$

r = tube radius, $R = r / \cos \theta$ = radius of the meniscus, ρ = liquid density. Key scaling: $h \propto 1/r$ (narrower tube $\Rightarrow$ higher rise). For $\theta > 90^\circ$ (mercury), $\cos \theta < 0$ and h is negative — a *depression*.

Section 3: CBSE Board Derivations / Key Procedures

Derivation 1 — Relation between surface tension and surface energy.

Figure 9.12 A rectangular wire frame PQRS with a sliding wire RS of length l . A soap film fills the frame. The film's surface-tension force $F = 2Sl$ pulls RS inward (toward PQ); an external force pulls RS outward by a distance x to position R'S'. Assumption: a soap film stretched on a U-frame with a sliding wire of length l ; the film has two surfaces (front and back). The film pulls the sliding wire inward with a force $F = S \times 2l$ (the factor 2 because the film has two surfaces, each contributing a tension along the edge of length l). To extend the film, pull the wire outward by a small distance x against this force. Physical reasoning: the work done against surface tension is stored as the extra potential energy of the newly created surface. Work done: $W = F \cdot x = 2Slx$. The increase in surface area (both faces) is $\Delta A = 2(lx)$. By energy conservation this work equals the surface energy stored, $E \cdot \Delta A$: $2Slx = E \cdot (2lx) \implies S = E$. Surface tension equals surface energy per unit area, numerically and dimensionally. (see next page)

Derivation 2 — Excess pressure inside a soap bubble.

Figure 9.13 A spherical soap bubble of radius R cut into two hemispheres. On the flat circular cut: the surface-tension force $(2)(2\pi R)S$ pulling the two halves together (two surfaces), and the excess-pressure force $(\Delta P)(\pi R^2)$ pushing ...

Assumption: a spherical soap bubble of radius R in equilibrium; surface tension S ; a soap bubble has **two** liquid-air surfaces.

Consider one hemisphere. Two forces act across the flat circular boundary where it joins the other half: - The surface-tension pull holding the halves together, around the circumference $2\pi R$, for *each* of the two surfaces: $F_{ST} = 2 \times (2\pi R) S = 4\pi RS$. - The outward force from the excess pressure ΔP acting over the projected circular area πR^2 : $F_P = \Delta P \cdot \pi R^2$.

Physical reasoning: in equilibrium the outward pressure force exactly balances the inward surface-tension pull, otherwise the bubble would expand or collapse:

$$\Delta P \cdot \pi R^2 = 4\pi RS$$

$$\Delta P = \frac{4S}{R}$$

For a **liquid drop** (a single surface, not two), the surface-tension force is halved to $2\pi RS$, giving $\Delta P = \frac{2S}{R}$. The factor-of-two difference between bubble and drop is purely the count of surfaces.

Derivation 3 — Capillary rise (Jurin's law).

Figure 9.14 A narrow capillary tube of radius r dipped in a liquid that wets it. Concave meniscus inside the tube; liquid risen to height h . Angle of contact θ between liquid and glass; meniscus radius $R = r/\cos\theta$. Assumption: a clean, narrow, vertical tube of radius r dipped in a wetting liquid (angle of contact θ , density ρ). The meniscus is a spherical cap of radius $R = r/\cos\theta$. 49 (see next page)

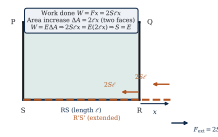


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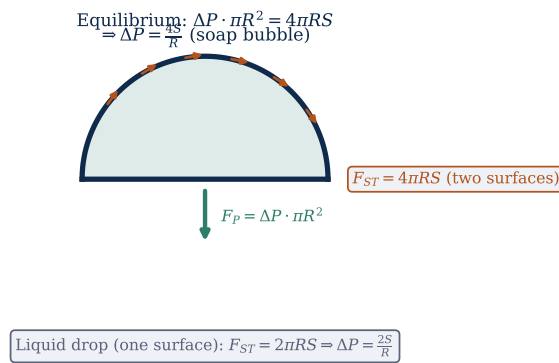


Figure 9.13 A spherical soap bubble of radius R cut into two hemispheres. On the flat circular cut: the surface-tension force $(2)(2\pi R)S$ pulling the two halves together (two surfaces), and the excess-pressure force $(\Delta P)(\pi R^2)$ pushing them apart.

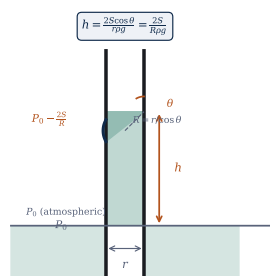


Figure 9.14 A narrow capillary tube of radius r dipped in a liquid that wets it. Concave meniscus inside the tube; liquid risen to height h . Angle of contact θ between liquid and glass; meniscus radius $R = r/\cos\theta$. Assumption: a clean, narrow, vertical tube of radius r dipped in a wetting liquid (angle of contact θ , density ρ). The meniscus is a spherical cap of radius $R = r/\cos\theta$. 49

Below the concave meniscus, the pressure is *lower* than atmospheric by the excess pressure $\frac{2S}{R}$. *Physical reasoning:* outside the tube, just below the flat liquid surface, the pressure is atmospheric (P_0); inside the tube, just below the curved meniscus, it is $P_0 - \frac{2S}{R}$. Since pressure must be equal at the same horizontal level in a connected liquid, liquid flows up the tube to make up this deficit, rising until the weight of the raised column produces a matching hydrostatic pressure ρgh :

$$\rho gh = \frac{2S}{R} = \frac{2S \cos \theta}{r}$$

$$\boxed{h = \frac{2S \cos \theta}{r \rho g}}$$

This is Jurin's law: $h \propto 1/r$, so the narrower the tube, the higher the rise. For mercury on glass ($\theta = 135^\circ$, $\cos \theta < 0$), h is negative — the liquid is *depressed*.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Calculate the work done in blowing a soap bubble of radius 3.0 cm. The surface tension of the soap solution is 0.030 N m^{-1} . (Assume the bubble is blown from a thin film, starting from negligible initial size.)

Given: $r = 3.0 \text{ cm} = 0.030 \text{ m}$, $S = 0.030 \text{ N m}^{-1}$.

A soap bubble has **two** surfaces, so the total surface area created is twice the geometric sphere area:

$$\Delta A = 2 \times 4\pi r^2 = 8\pi r^2 = 8\pi(0.030)^2 = 8\pi(9.0 \times 10^{-4}) = 2.26 \times 10^{-2} \text{ m}^2$$

Work done = surface tension $\times$ area created:

$$W = S \cdot \Delta A = 0.030 \times 2.26 \times 10^{-2} = \boxed{6.8 \times 10^{-4} \text{ J}}$$

(Had this been a liquid *drop* of the same radius, the work would be exactly half, since a drop has only one surface.)

Example 2 — [NEET Speed Trap]

A spherical raindrop and a soap bubble have the same radius and are made from liquids of the same surface tension. Find the ratio of the excess pressure inside the soap bubble to that inside the raindrop.

The trap to avoid: treating both as $2S/r$ (giving a wrong ratio of 1 : 1). A soap bubble has **two** surfaces, a drop only **one**.

Speed approach.

$$\frac{\Delta P_{\text{bubble}}}{\Delta P_{\text{drop}}} = \frac{4S/r}{2S/r} = \boxed{2 : 1}$$

Mental check: the soap bubble's two surfaces double its excess pressure compared with a single-surface drop of the same size. "Bubble beats drop, two to one" — anchor it on the surface count.

Example 3 — [JEE Main Level] (*capillary rise + Jurin's law scaling*)

A capillary tube of radius 0.25 mm is dipped vertically in water (surface tension 0.072 N m^{-1} , density 1000 kg m^{-3} , angle of contact 0°). (a) Find the height to which water rises. (b) If a tube of half this radius is used instead, find the new height. (Take $g = 10 \text{ m s}^{-2}$.)

(a) With $\cos 0^\circ = 1$:

$$h = \frac{2S \cos \theta}{r \rho g} = \frac{2 \times 0.072 \times 1}{(0.25 \times 10^{-3})(1000)(10)} = \frac{0.144}{2.5} = 0.0576 \text{ m} = \boxed{5.76 \text{ cm}}$$

(b) By Jurin's law, $h \propto 1/r$. Halving the radius **doubles** the rise:

$$h' = 2 \times 5.76 = \boxed{11.52 \text{ cm}}$$

No need to recompute from scratch — the scaling relation does the work, which is the fast route in an exam.

Example 4 — [JEE Advanced Level] (*surface energy + volume conservation + calorimetry*)

A large number n of identical water droplets, each of radius r , coalesce to form a single spherical drop of radius R . Assuming all the surface energy released is used to heat the water, derive an expression for the temperature rise ΔT in terms of S , density ρ , specific heat c , r , and R . Then evaluate it for $n = 10^6$, $r = 1.0 \times 10^{-4} \text{ m}$, $S = 0.072 \text{ N m}^{-1}$, $\rho = 1000 \text{ kg m}^{-3}$, $c = 4200 \text{ J kg}^{-1} \text{K}^{-1}$.

Step 1 — Volume conservation (mass is conserved on merging):

$$n \cdot \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \quad \implies \quad nr^3 = R^3$$

Step 2 — Energy released. The total surface area *shrinks* as droplets merge (less surface for the same volume), and surface energy $= S \times \text{area}$ is released:

$$\Delta E = S(n \cdot 4\pi r^2 - 4\pi R^2) = 4\pi S(nr^2 - R^2)$$

Using $nr^3 = R^3$, write $nr^2 = R^3/r$:

$$\Delta E = 4\pi S \left(\frac{R^3}{r} - R^2 \right) = 4\pi S R^3 \left(\frac{1}{r} - \frac{1}{R} \right)$$

Step 3 — Heat balance. The mass of the final drop is $m = \rho \cdot \frac{4}{3}\pi R^3$, and $\Delta E = mc \Delta T$:

$$\Delta T = \frac{\Delta E}{mc} = \frac{4\pi S R^3 \left(\frac{1}{r} - \frac{1}{R} \right)}{\rho \cdot \frac{4}{3}\pi R^3 \cdot c} = \boxed{\frac{3S}{\rho c} \left(\frac{1}{r} - \frac{1}{R} \right)}$$

Step 4 — Numerical. With $n = 10^6$, $R = n^{1/3}r = 100r = 1.0 \times 10^{-2} \text{ m}$:

$$\Delta T = \frac{3(0.072)}{1000 \times 4200} \left(\frac{1}{10^{-4}} - \frac{1}{10^{-2}} \right) = (5.14 \times 10^{-8})(10\,000 - 100) \approx \boxed{5.1 \times 10^{-4} \text{ K}}$$

Insight: the rise is tiny but real, and the structure $\left(\frac{1}{r} - \frac{1}{R}\right)$ shows the smaller the original droplets, the larger the heating — because tiny droplets pack enormous surface area per unit volume. When $R \gg r$, $\Delta T \approx \frac{3S}{\rho c r}$.

Example 5 — [JEE Main Level] (*capillary tube of insufficient length*)

Water would rise to a height of 8.0 cm in a given capillary tube. The tube is now cut so that only 5.0 cm of it projects above the water surface. Does the water overflow? If not, find the radius of curvature of the meniscus at the top. (Take the tube radius such that $rh = \text{constant}$; angle of contact 0° .)

Key idea: water **never overflows** from a short tube. The product (rise) $\times$ (meniscus radius) is fixed by the surface tension, $hR = \frac{2S}{\rho g} = \text{constant}$. If the available length h' is less than the natural rise h , the water climbs to the top and the meniscus simply *flattens* (its radius increases) until equilibrium is restored.

Let the natural meniscus radius (for full rise $h = 8.0$ cm) be R_0 — equal to the tube radius r since $\theta = 0$. With the tube length limited to $h' = 5.0$ cm, the new meniscus radius R' satisfies $hR_0 = h'R'$:

$$R' = R_0 \frac{h}{h'} = r \times \frac{8.0}{5.0} = \boxed{1.6r}$$

The water rises to the top (5.0 cm) and stops; the meniscus becomes less curved (radius $1.6\times$ the tube radius). No overflow — the classic resolution of this favourite trap.

Example 6 — [JEE Advanced Level] (*rise between parallel plates + combined bubbles*)

- (a) Two parallel glass plates separated by a small distance d are dipped vertically in water (surface tension S , density ρ , angle of contact 0°). Derive the height to which water rises between them. (b) Two soap bubbles of radii a and b ($a > b$) coalesce so they share a common interfacial film. Find the radius of curvature of that common film.
- (b) For two plates a distance d apart, the meniscus is cylindrical (curved in one direction only), so the excess pressure below it is S/R with $R = d/2$. Balancing against the hydrostatic pressure of the raised column:

$$\frac{2S}{d} = \rho gh \quad \Rightarrow \quad \boxed{h = \frac{2S}{\rho g d}}$$

Note the contrast with a cylindrical tube ($h = 2S/r\rho g$): plates give $2S/\rho g d$ (one curvature), a tube gives $2S/r\rho g$ (two curvatures) — the geometry changes the numerical factor.

- (c) The smaller bubble has the larger internal excess pressure ($4S/r$ grows as r shrinks), so air pushes from the small bubble toward the large one and the common film bulges *into* the larger bubble. The excess pressure across the common film equals the difference of the two individual excess pressures:

$$\frac{4S}{R} = \frac{4S}{b} - \frac{4S}{a} \quad \Rightarrow \quad \frac{1}{R} = \frac{1}{b} - \frac{1}{a} \quad \Rightarrow \quad \boxed{R = \frac{ab}{a-b}}$$

The common film curves toward the bigger bubble, with radius $R = ab/(a-b)$ — a standard JEE Advanced result.

Section 5: Practice Exercises

1. [CBSE] Define surface tension and state its SI unit and dimensional formula. Explain in one line why small liquid drops are spherical.

2. [NEET] Water rises in a capillary tube to a height h . If the same tube is dipped in a liquid of half the surface tension, double the density, and the same (zero) angle of contact, the rise will be: h , $h/2$, $h/4$, or $2h$? Justify.
3. [JEE Main] The excess pressure inside a soap bubble of radius 2.0 cm is 6.0 Pa. Find the surface tension of the soap solution.
4. [JEE Main] A water drop of radius 1.0 mm is broken into 1000 identical smaller droplets. Surface tension of water is 0.072 N m^{-1} . Find the work done in the process.
5. [JEE Advanced] Two soap bubbles of radii a and b ($a > b$) coalesce to form a single bubble. Assuming the temperature stays constant and the external pressure is P_0 , show that the radius R of the combined bubble satisfies a relation involving P_0 , S , a , b . (Hint: conserve the total number of air molecules, i.e. apply $PV = \text{const}$ for the trapped air, using the excess-pressure relation for each bubble.)

Answers: 1. Spherical because a sphere has minimum surface area (hence minimum surface energy) for a given volume. 2. $h/4$. 3. $S = 0.030 \text{ N m}^{-1}$. 4. $W \approx 8.1 \times 10^{-6} \text{ J}$. 5. $P_0(R^3 - a^3 - b^3) + 4S(R^2 - a^2 - b^2) = 0$.

Section 6: Crack the MCQ

Q1. The excess pressure inside a soap bubble of radius r compared with that inside a liquid drop of the same radius and same surface tension is: (a) Equal (b) Twice (c) Half (d) Four times

Correct: (b). Soap bubble: $\Delta P = 4S/r$ (two surfaces); liquid drop: $\Delta P = 2S/r$ (one surface). The ratio is 2 : 1. - (a) ignores the surface count — the dominant error. - (c) inverts the ratio. - (d) double-counts, applying the factor of two twice.

Q2. Water rises in a capillary tube of radius r . If a tube of radius $2r$ is used (everything else unchanged), the height of rise becomes: (a) $2h$ (b) $4h$ (c) $h/2$ (d) $h/4$

Correct: (c). By Jurin's law $h \propto 1/r$. Doubling the radius halves the rise. - (a)/(b) assume rise *increases* with radius — backwards. - (d) uses $h \propto 1/r^2$, confusing with an area dependence.

Q3. For a liquid that does *not* wet the surface of a solid (e.g. mercury on glass), the angle of contact is: (a) Zero (b) Acute ($< 90^\circ$) (c) A right angle (90°) (d) Obtuse ($> 90^\circ$)

Correct: (d). Non-wetting means cohesion dominates adhesion: the meniscus is convex and the angle of contact is obtuse (mercury-glass $\approx 135^\circ$), producing a capillary *depression*. - (a)/(b) describe a wetting liquid like water on clean glass. - (c) (90°) is the special borderline case (e.g. water on silver), giving a flat surface and zero rise.

Q4. As the temperature of a liquid is increased towards its critical temperature, its surface tension: (a) Increases (b) Decreases and becomes zero at the critical temperature (c) Stays constant (d) First decreases then increases

Correct: (b). Rising temperature increases molecular motion and intermolecular distances, weakening the cohesive forces that cause surface tension; at the critical temperature the liquid-vapour distinction vanishes and surface tension drops to zero. - (a) reverses the trend. - (c) ignores the temperature dependence entirely. - (d) has no physical basis for ordinary liquids (a few molten metals are exceptions, but not the general rule).

Section 7: Common Student Pitfalls & Pro-Tips

1. **Forgetting the soap bubble's two surfaces.** A soap bubble has an inner *and* an outer surface: excess pressure $4S/r$, and work to blow it $= S \times 2(4\pi r^2)$. A liquid drop or an air bubble *inside* a liquid has only one surface: $2S/r$. Always ask “how many surfaces?” before writing the formula.
2. **Inverting Jurin's law.** Capillary rise $h \propto 1/r$ — *narrower* tubes give *higher* rise, not lower. The intuitive guess (bigger tube, more water, higher rise) is exactly wrong.
3. **Muddling the angle of contact.** Acute ($< 90^\circ$) = wetting = concave meniscus = capillary rise (water-glass). Obtuse ($> 90^\circ$) = non-wetting = convex meniscus = capillary depression (mercury-glass). And θ *decreases* with temperature, which is why hot water cleans better.
4. **Confusing tube radius with meniscus radius.** In the capillary formula, r is the *tube* radius but the excess pressure uses the *meniscus* radius $R = r/\cos\theta$. They are equal only when $\theta = 0$. Keep them distinct when $\theta \neq 0$.

Pro-Tip: Before any surface-tension calculation, run a two-second checklist — *how many surfaces* (one for a drop/air-bubble, two for a soap film/bubble), and *is the angle of contact zero* (so tube radius = meniscus radius)? Getting these two right disposes of the majority of exam errors in this topic. For ratio questions, lean on the scalings — $\Delta P \propto 1/r$ and $h \propto 1/r$ — and you can often answer without computing S at all.

Section 8: Cheat Sheet Box

- **Surface tension** $S = F/L$ — force per unit length; surface acts like a stretched membrane minimising area. SI unit N m^{-1} ; **dimensions** $[MT^{-2}]$.
- **Surface energy** = work per unit new area $= S$ **numerically**. Count surfaces: film/bubble has **two**, drop has **one** (work $W = S \Delta A$).
- **Excess pressure:** drop & air-bubble $\frac{2S}{r}$ (one surface); soap bubble $\frac{4S}{r}$ (two surfaces).
- **Angle of contact** θ : acute = wetting (concave, rise); obtuse = non-wetting (convex, depression); decreases with temperature.
- **Capillary rise (Jurin)** $h = \frac{2S \cos \theta}{r \rho g}$, with $h \propto 1/r$. Mercury ($\theta > 90^\circ$) is depressed.
- **Temperature/impurities:** S decreases with T , zero at critical temperature; soluble impurities raise S , sparingly soluble (soap) lower it.
- **Extended configurations:** between **parallel plates** $h = \frac{2S}{\rho g d}$ (separation d); a **tube too** short to allow full rise never overflows — the meniscus flattens with $hR = \text{const}$; two **coalescing bubbles** ($a > b$) share a film of radius $R = \frac{ab}{a-b}$ bulging into the larger.

Memory aids: “Bubble = 4, drop = 2” (excess pressure, surface count). • “Narrow tube, high climb” ($h \propto 1/r$). • “Acute wets, obtuse won't.” • “Hot liquid, weak skin” (surface tension falls with temperature).

Chapter 9: Mechanical Properties of Fluids — Round 2 Addendum

Unit 7 of the chapter taught you the core formulas of fluid statics and dynamics, and the worked examples covered the standard CBSE, NEET, and JEE Main patterns. Three JEE Advanced patterns remain that the chapter only hints at: (1) hydrostatic forces on curved surfaces and the centre of pressure on plane surfaces, which require integrating pressure over an area rather than using the base-pressure shortcut; (2) the metacentric height of a floating body, which decides whether a ship rights itself or capsizes — a full derivation from the chapter’s stability note; and (3) viscous flow in non-circular ducts (parallel plates and annuli), where the velocity profile and flow rate scale differently from the circular Poiseuille law. Addenda A, B, and C supply these three patterns with complete derivations, unit-carried examples, and practice problems.

Addendum A — Hydrostatic Force on Plane Surfaces and Centre of Pressure

Subtopic 01 gave the force on a horizontal base: $F = P_{\text{base}}A$. For an inclined or vertical plane surface, the pressure varies linearly with depth, so the total force is the integral of pressure over the area, and the resultant acts at the *centre of pressure* (CP), which lies below the centroid because pressure increases with depth. JEE Advanced regularly asks for the force on a sluice gate, a triangular dam face, or a circular porthole, and for the depth of the CP — often combined with a moment-equilibrium problem (e.g., “find the force needed to hold the gate shut”).

Method

1. **Choose coordinates.** Let the plane surface make an angle θ with the horizontal. Measure distance y along the plane from the free surface (or from the intersection of the plane with the free surface) downward. The vertical depth is $h = y \sin \theta$.
2. **Pressure at y :** $P(y) = P_0 + \rho g y \sin \theta$. (If the free surface is open, P_0 is atmospheric; gauge pressure is $\rho g y \sin \theta$.)
3. **Total force (gauge):** $F = \int P_{\text{gauge}} dA = \rho g \sin \theta \int y dA = \rho g \sin \theta \bar{y}A$, where $\bar{y}$ is the distance of the centroid from the free-surface intersection. In terms of the vertical depth of the centroid $\bar{h} = \bar{y} \sin \theta$, this is simply $F = \rho g \bar{h}A$ — the same as the pressure at the centroid times the area.
4. **Centre of pressure (CP).** The moment of the resultant force about the free-surface intersection equals the integral of the pressure moments:

$$F y_{\text{CP}} = \int y P_{\text{gauge}} dA = \rho g \sin \theta \int y^2 dA = \rho g \sin \theta I_{xx,O}$$

where $I_{xx,O}$ is the second moment of area about the intersection line (the axis in the plane at the free surface). Using $F = \rho g \sin \theta \bar{y} A$:

$$y_{CP} = \frac{I_{xx,O}}{\bar{y} A}$$

Apply the parallel-axis theorem: $I_{xx,O} = I_{xx,C} + A\bar{y}^2$, where $I_{xx,C}$ is the second moment about a parallel axis through the centroid. Then

$$y_{CP} = \bar{y} + \frac{I_{xx,C}}{\bar{y} A}$$

The vertical depth of the CP is $h_{CP} = y_{CP} \sin \theta$.

5. **Sign convention:** y increases downward along the plane; CP is always *below* the centroid ($y_{CP} > \bar{y}$).

Standard Results (Gauge Pressure, Free Surface at $y = 0$)

Shape	Centroid $\bar{y}$	$I_{xx,C}$	y_{CP}
Rectangle width b , height H (plane vertical, $\theta = 90^\circ$)	$H/2$	$bH^3/12$	$2H/3$
Rectangle width b , length L inclined at θ , top edge at free surface	$L/2$	$bL^3/12$	$2L/3$
Triangle base b , height H , vertex at free surface	$2H/3$	$bH^3/36$	$3H/4$
Circle radius R , centre at distance $\bar{y}$ from free surface	$\bar{y}$	$\pi R^4/4$	$\bar{y} + \frac{\pi R^4}{4\bar{y}\pi R^2} = \bar{y} + \frac{R^2}{4\bar{y}}$

- b, H, L, R in metres; $\bar{y}, y_{CP}$ in metres; F in newtons; ρ in kg m^{-3} ; g in m s^{-2} .

Example A.1 — [JEE Advanced Level] (Inclined Rectangular Gate)

A rectangular gate 2.0 m wide and 3.0 m long closes a tank. The gate is hinged along its top edge and makes an angle of 30° with the horizontal. The water level reaches the top edge of the gate. Find (a) the magnitude of the hydrostatic force on the gate (gauge), and (b) the distance from the hinge to the line of action of this force. Take $\rho = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$.

Given: $b = 2.0 \text{ m}$, $L = 3.0 \text{ m}$, $\theta = 30^\circ$, $\rho = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$.

(a) Force. The centroid is at $\bar{y} = L/2 = 1.5 \text{ m}$ from the hinge. Vertical depth of centroid: $\bar{h} = \bar{y} \sin \theta = 1.5 \times \sin 30^\circ = 1.5 \times 0.5 = 0.75 \text{ m}$. Area $A = bL = 2.0 \times 3.0 = 6.0 \text{ m}^2$. Gauge force:

$$F = \rho g \bar{h} A = 1000 \times 10 \times 0.75 \times 6.0 = 4.5 \times 10^4 \text{ N}.$$

(b) Centre of pressure. For a rectangle about its top edge (the hinge line), $I_{xx,O} = bL^3/3 = 2.0 \times 27/3 = 18.0 \text{ m}^4$. (Or use $I_{xx,C} = bL^3/12 = 4.5 \text{ m}^4$, $\bar{y} = 1.5 \text{ m}$, then $I_{xx,O} = 4.5 + 6.0 \times 1.5^2 = 4.5 + 13.5 = 18.0 \text{ m}^4$.)

$$y_{CP} = \frac{I_{xx,O}}{\bar{y}A} = \frac{18.0}{1.5 \times 6.0} = 2.0 \text{ m}.$$

The resultant acts 2.0 m down the gate from the hinge. (Check: $y_{CP} = 2L/3 = 2.0 \text{ m}$.)

Verification. Moment of force about hinge: $F \times 2.0 = 9.0 \times 10^4 \text{ N m}$. Direct moment integral: $\int_0^L (\rho g y \sin \theta) b y \, dy = \rho g b \sin \theta \frac{L^3}{3} = 1000 \times 10 \times 2.0 \times 0.5 \times 9 = 9.0 \times 10^4 \text{ N m}$.

Example A.2 — [JEE Advanced Level] (Triangular Dam Face)

A vertical triangular dam face has base $b = 6.0 \text{ m}$ at the water surface and height $H = 4.0 \text{ m}$ (vertex at the bottom). Water stands against the face up to the top edge. Find (a) the total hydrostatic force, and (b) the depth of the centre of pressure below the water surface. Take $\rho = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$.

(a) The triangle is vertical ($\theta = 90^\circ$), so depth $h = y$. Centroid from the surface: $\bar{y} = 2H/3 = 8/3 \text{ m}$. Area $A = \frac{1}{2}bH = 12 \text{ m}^2$.

$$F = \rho g \bar{y} A = 1000 \times 10 \times \frac{8}{3} \times 12 = 3.2 \times 10^5 \text{ N}.$$

(b) $I_{xx,C}$ for triangle about centroidal axis parallel to base: $bH^3/36 = 6.0 \times 64/36 = 10.67 \text{ m}^4$. $\bar{y} = 8/3 \text{ m}$, $A = 12 \text{ m}^2$.

$$y_{CP} = \bar{y} + \frac{I_{xx,C}}{\bar{y}A} = \frac{8}{3} + \frac{10.67}{(8/3) \times 12} = 2.667 + \frac{10.67}{32} = 2.667 + 0.333 = 3.0 \text{ m}.$$

The centre of pressure is 3.0 m below the water surface. (Standard result: $3H/4 = 3.0 \text{ m}$.)

Practice — Addendum A

1. A vertical circular porthole of diameter 0.50 m has its centre 3.0 m below the free surface of water. Find the hydrostatic force on the porthole and the depth of the centre of pressure.
2. A rectangular gate 1.5 m wide and 2.0 m high is inclined at 60° to the horizontal. The top edge is 1.0 m below the free surface (measured vertically). Find the hydrostatic force on the gate and the distance of the CP from the top edge along the gate.
3. A triangular gate with base 2.0 m (horizontal) and height 3.0 m is hinged along its base. The vertex points downward. The water level is exactly at the base. Find the moment required at the hinge to keep the gate closed. ($\rho = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$.)

Answers: 1. $F = 5.89 \times 10^3 \text{ N}$; $h_{CP} = 3.0104 \text{ m}$. 2. $F = 3.9 \times 10^4 \text{ N}$; $y_{CP} = 1.44 \text{ m}$ from top edge. 3. $M = 3.0 \times 10^4 \text{ N m}$.

Addendum B — Hydrostatic Force on Curved Surfaces

For a curved surface, the hydrostatic force is not simply $P_{\text{centroid}}A$ because pressure forces have different directions. The standard method resolves the resultant into horizontal and vertical components. The horizontal component equals the force on the vertical projection of the curved surface; the vertical component equals the weight of the fluid vertically above the surface up to the free surface (or the reaction if the surface is below the fluid). JEE Advanced often puts a curved gate (quarter-cylinder, parabolic) in a tank and asks for the resultant force magnitude, direction, and line of action.

Method

Consider a curved surface in contact with a static fluid (density ρ). Work in gauge pressure.

1. **Horizontal component** F_H . Project the curved surface onto a vertical plane. F_H equals the total hydrostatic force on that projected plane area. Its line of action passes through the centre of pressure of that projected area.
2. **Vertical component** F_V . F_V equals the weight of the fluid column vertically above the curved surface, extending up to the free surface. If the surface is a “floor” (fluid above it), F_V acts **downward** on the surface; if the surface is a “ceiling” (fluid below it), F_V acts **upward**. In magnitude, $F_V = \rho g V_{\text{above}}$, where V_{above} is the volume of the fluid prism above the surface per unit width (or total volume for 3D).
3. **Resultant:** $F_R = \sqrt{F_H^2 + F_V^2}$, angle $\alpha = \tan^{-1}(F_V/F_H)$ from horizontal.
4. **Line of action.** The resultant must produce the same moment about any point as the distributed pressure. Usually, take moments about a convenient hinge or the intersection of the horizontal and vertical lines of action of F_H and F_V . For a cylindrical surface (circular arc), the pressure forces all pass through the cylinder axis, so the resultant also passes through the axis — a major shortcut.

Standard Configurations

Surface	F_H	F_V	Resultant Direction
Quarter-cylinder, radius R , fluid inside (convex to fluid)	$\frac{1}{2}\rho g R^2 \times \text{width}$ (on vertical projection)	$\frac{\pi}{4}\rho g R^2 \times \text{width}$ (weight of quarter-cylinder)	Acts through cylinder axis
Quarter-cylinder, fluid outside (concave to fluid)	Same F_H	Weight of fluid <i>above</i> the surface (rectangle minus quarter-cylinder)	Acts through cylinder axis
Parabolic gate $y = kx^2$, width b , fluid on convex side	Force on vertical projection	Weight of fluid above parabola up to free surface	Not through a fixed point; use moment balance

Example B.1 — [JEE Advanced Level] (Half-Cylinder Gate)

A long gate has a semi-cylindrical cross-section of radius $R = 2.0$ m, axis horizontal, hinged along the horizontal line at the top of the curved surface (the top generator). Water stands against the gate on the concave side (inside the half-cylinder) up to the hinge level. The gate has length $L = 4.0$ m. Find the moment required at the hinge to hold the gate closed. Take $\rho = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$.

Solution. Set the hinge H at the origin with the centre O of the circle a distance R directly below it. Since pressure on a circular arc always acts along the radius, every elementary pressure force passes through O — so the resultant F'_R (per unit length) also passes through O, a distance R from the hinge.

Horizontal component (per unit length). This equals the force on the vertical projection of the curved surface — the vertical diameter, of length $2R$, with pressure varying linearly from 0 at the hinge (water surface) to $\rho g(2R)$ at the bottom:

$$F'_H = \int_0^{2R} \rho g y \, dy = \rho g \frac{(2R)^2}{2} = 2\rho g R^2$$

(equivalently: depth of centroid $\times$ projected area, per unit length $= \rho g \cdot R \cdot 2R = 2\rho g R^2$ — the two methods agree).

Vertical component (per unit length) equals the weight of the water filling the half-cylinder: $F'_V = \rho g \cdot \frac{\pi}{2} R^2$.

Moment about the hinge. With $\vec{HO} = (0, -R)$ and $\vec{F}'_R = (F'_H, -F'_V)$ (horizontal outward, vertical downward), the moment is $\vec{HO} \times \vec{F}'_R = RF'_H$ — the vertical component contributes nothing, since it acts along a line through O that passes directly below the hinge. So

$$M' = RF'_H = R(2\rho g R^2) = 2\rho g R^3$$

Plugging in $\rho g = 10^4 \text{ Pa/m}$, $R = 2 \text{ m}$: $M' = 2 \times 10^4 \times 8 = 1.6 \times 10^5 \text{ N}\cdot\text{m/m}$. For length $L = 4 \text{ m}$:

$$M = 1.6 \times 10^5 \times 4 = 6.4 \times 10^5 \text{ N}\cdot\text{m}$$

Key JEE shortcut this illustrates: for any circular-arc gate, the resultant hydrostatic force passes through the centre of the circle. If the hinge happens to lie *on the same radius* as the centre (e.g. hinged at the centre itself, or along an edge that passes through the centre), the moment is trivially zero — the force produces no torque about a point it passes through. A non-zero moment requires the hinge to be off that line, as here (hinge at the top of the arc, centre a full radius away).

Example B.2 — [JEE Main Level] (Force on Hemispherical Bowl)

A hemispherical bowl of radius $R = 0.5 \text{ m}$ is filled with water. Find the magnitude and direction of the resultant hydrostatic force on the bowl (the curved surface only). Take $\rho = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$.

Solution: The bowl is a hemisphere, open at the top. Water fills it. The curved surface is the hemisphere. By symmetry, the horizontal components cancel. The resultant is purely vertical.

Vertical force on the hemisphere = weight of water above the hemisphere. But the water is *inside* the hemisphere. The volume above the curved surface is exactly the volume of the hemisphere: $V = \frac{2}{3}\pi R^3$. $F_V = \rho g V = 1000 \times 10 \times \frac{2}{3}\pi(0.5)^3 = 10^4 \times \frac{2}{3}\pi \times 0.125 = 10^4 \times \frac{\pi}{12} \approx 2618 \text{ N}$. Direction: vertically downward (water pushes down on the bowl). Magnitude: $F_R = 2618 \text{ N}$ downward.

Check: the vertical component is nothing more than the weight of the water standing above the curved surface, and for a hemisphere that volume is the hemisphere itself: $V = \frac{2}{3}\pi R^3 = 0.262 \text{ m}^3$, i.e. 262 kg of water weighing $\approx 2618 \text{ N}$ — the figure obtained above.

Practice — Addendum B

1. A quarter-cylindrical gate of radius $R = 1.0 \text{ m}$ and width $b = 2.0 \text{ m}$ holds water on its convex side. The water level is at the top edge. The gate is hinged at the bottom horizontal edge. Find the moment required at the hinge to keep the gate closed. ($\rho = 1000$, $g = 10$).
2. A parabolic gate $y = x^2$ (with y in metres, x in metres) from $x = 0$ to $x = 1 \text{ m}$, width $b = 1 \text{ m}$, is vertical. Water stands against it up to $y = 1 \text{ m}$. Find the horizontal and vertical components of the hydrostatic force on the gate.
3. A hemispherical dome of radius $R = 3 \text{ m}$ is submerged in water with its flat face horizontal and uppermost. The top of the dome is at a depth of 2 m . Find the net hydrostatic force on the dome (magnitude and direction). ($\rho = 1000$, $g = 10$).

Answers: 1. $M = 0$ (resultant passes through hinge). 2. $F_H = 3.33 \times 10^3 \text{ N}$, $F_V = 6.67 \times 10^3 \text{ N}$. 3. $F_R = 1.13 \times 10^6 \text{ N}$ upward (buoyant force on dome = weight of displaced water = $\rho g \frac{2}{3}\pi R^3$).

Addendum C — Stability of Floating Bodies: Metacentric Height

Subtopic 02 states that a floating body is stable if the metacentre M lies above the centre of gravity G . The metacentric height $GM = BM - BG$ (for G above B) determines the righting moment. JEE Advanced asks for the metacentric height of a given floating body (rectangular barge, cylindrical log, etc.) and the condition for stability, often combined with a small-angle oscillation period.

Method

1. **Centre of Buoyancy B:** For a floating body in equilibrium, B is the centroid of the submerged volume.
2. **Centre of Gravity G:** Given or computed from mass distribution.
3. **Metacentre M:** When the body tilts by a small angle θ , the submerged volume shifts. The buoyant force acts through the new centre of buoyancy B' . The vertical line through B' intersects the body centreline at M . For small θ , M is fixed.
4. **Distance BM:** $BM = \frac{I}{V_{\text{sub}}}$, where I is the second moment of area of the waterplane (the cross-section at the waterline) about the axis of tilt, and V_{sub} is the submerged volume.

- Derivation: The wedge of fluid shifted creates a moment. The shift of B is such that $V_{\text{sub}} \times BB' = I\theta$. For small θ , $BB' \approx BM \cdot \theta$, so $BM = I/V_{\text{sub}}$.
5. **Metacentric height:** $GM = BM - BG$ if G is above B; $GM = BM + BG$ if G is below B. (Sign convention: positive GM means M above G, stable.)
 6. **Righting moment:** For a small heel angle θ , the righting moment is $M_R = W \cdot GM \cdot \theta$, where $W = mg$ is the weight of the body.
 7. **Period of oscillation (rolling):** $T = 2\pi \sqrt{\frac{k^2}{g \cdot GM}}$, where k is the radius of gyration of the body about its longitudinal axis.

Standard Results

Waterplane Shape	Second Moment I (about centroidal axis)
Rectangle breadth B , length L (tilt about longitudinal axis)	$LB^3/12$
Circle radius R	$\pi R^4/4$
Ellipse semi-axes a, b (tilt about major axis a)	$\pi ab^3/4$

Example C.1 — [JEE Advanced Level] (Rectangular Barge)

A rectangular barge of length $L = 30$ m, breadth $B = 10$ m, and depth $D = 5$ m floats in sea water ($\rho = 1025 \text{ kg m}^{-3}$). Its mass is 1.2×10^6 kg, and its centre of gravity G is 2.0 m above the keel (bottom). Find (a) the draft d , (b) the metacentric height GM , and (c) the period of small rolling oscillations. Take $g = 10 \text{ m s}^{-2}$. The radius of gyration of the barge about its longitudinal axis is $k = 0.35B$.

Solution: (a) Draft. Weight $W = mg = 1.2 \times 10^6 \times 10 = 1.2 \times 10^7$ N. Buoyancy $F_B = \rho g V_{\text{sub}} = \rho g (LBd)$. Equilibrium: $\rho g LBd = mg \Rightarrow d = \frac{m}{\rho LB} = \frac{1.2 \times 10^6}{1025 \times 30 \times 10} = \frac{1.2 \times 10^6}{3.075 \times 10^5} \approx 3.90$ m. So submerged volume $V_{\text{sub}} = 30 \times 10 \times 3.90 = 1170 \text{ m}^3$.

(b) Metacentric height. Centre of buoyancy B is at $d/2 = 1.95$ m above keel. Centre of gravity G is at 2.0 m above keel. $BG = 2.0 - 1.95 = 0.05$ m (G above B). Waterplane area is $L \times B = 300 \text{ m}^2$. Tilt is about the longitudinal axis (along length). Second moment of waterplane about longitudinal axis: $I = \frac{LB^3}{12} = \frac{30 \times 1000}{12} = 2500 \text{ m}^4$. $BM = \frac{I}{V_{\text{sub}}} = \frac{2500}{1170} \approx 2.137$ m. $GM = BM - BG = 2.137 - 0.05 = 2.087$ m. (Positive, so stable.)

(c) Period of rolling. Radius of gyration $k = 0.35B = 3.5$ m. $T = 2\pi \sqrt{\frac{k^2}{g \cdot GM}} = 2\pi \sqrt{\frac{12.25}{10 \times 2.087}} = 2\pi \sqrt{0.587} \approx 2\pi \times 0.766 = 4.81$ s.

Example C.2 — [JEE Main Level] (Rectangular Block)

A rectangular block of wood $2 \text{ m} \times 1 \text{ m} \times 1 \text{ m}$ ($L \times B \times H$) floats in water with the 2 m dimension horizontal and the 1 m dimension vertical. Its density is 600 kg m^{-3} . Find the metacentric height for rolling (tilt about the longitudinal axis). Is it stable? ($\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$).

Solution: Draft $d = \frac{\rho_b}{\rho_w} H = 0.6 \times 1 = 0.6$ m. B is at $d/2 = 0.3$ m above bottom. G is at $H/2 = 0.5$ m above bottom. $BG = 0.5 - 0.3 = 0.2$ m (G above B). Waterplane area: $L \times B = 2 \times 1 = 2 \text{ m}^2$.

I about longitudinal axis (along L): $I = \frac{LB^3}{12} = \frac{2 \times 1}{12} = \frac{1}{6} \text{ m}^4$. $V_{\text{sub}} = LBd = 2 \times 1 \times 0.6 = 1.2 \text{ m}^3$. $BM = \frac{I}{V_{\text{sub}}} = \frac{1/6}{1.2} = \frac{1}{7.2} \approx 0.1389 \text{ m}$. $GM = BM - BG = 0.1389 - 0.2 = -0.0611 \text{ m}$. Negative $\Rightarrow$ unstable. The block will capsize (it will float with the 2 m dimension vertical instead).

Practice — Addendum C

1. A cylindrical buoy of diameter 1.0 m and height 2.0 m floats vertically in sea water ($\rho = 1025 \text{ kg m}^{-3}$). The buoy is uniform with mass 400 kg and its centre of gravity is at the geometric centre. Find the metacentric height. Is it stable?
2. A rectangular barge $20 \text{ m} \times 8 \text{ m} \times 4 \text{ m}$ ($L \times B \times D$) has a mass of $5.0 \times 10^5 \text{ kg}$ and $KG = 2.5 \text{ m}$ (above keel). It floats in fresh water. Find GM for rolling.
3. A solid cube of side a and density ρ_b floats in a liquid of density ρ_f ($\rho_b < \rho_f$). Derive the condition on ρ_b/ρ_f for the cube to be stable in the orientation with one face horizontal.

Answers: 1. $GM \approx 0.076 \text{ m}$ (stable). 2. $d = 3.05 \text{ m}$, $BM = 1.75 \text{ m}$, $BG = 2.5 - 1.525 = 0.975 \text{ m}$, $GM = 0.775 \text{ m}$ (stable). 3. Draft $d = (\rho_b/\rho_f)a \equiv ra$. The waterplane is a square of side a , so $I = a^4/12$ and $V_{\text{sub}} = a^3r$, giving $BM = I/V_{\text{sub}} = a/(12r)$. B sits at $d/2 = ar/2$ above the keel and G at $a/2$, so $BG = \frac{a}{2}(1 - r)$ and

$$GM = \frac{a}{12r} - \frac{a}{2}(1 - r).$$

$GM > 0$ requires $6r^2 - 6r + 1 > 0$, whose roots are $r = \frac{1}{2} \pm \frac{1}{2\sqrt{3}}$. Hence the cube is stable face-up for $r < 0.211$ or $r > 0.789$; for intermediate densities it capsizes to float on an edge or corner.

Addendum D — Viscous Flow in Non-Circular Ducts: Parallel Plates and Annuli

Subtopic 06 derived Poiseuille's law for a circular pipe: $Q = \frac{\pi \Delta P R^4}{8\eta L}$. JEE Advanced often asks for the flow rate between two infinite parallel plates (plane Poiseuille flow) or in an annular gap between concentric cylinders. The velocity profile and the R -dependence change, but the method — balancing pressure force and viscous force on a fluid element — is identical.

Method: Plane Poiseuille Flow (Parallel Plates)

Consider steady, laminar flow of a Newtonian fluid between two infinite parallel plates separated by distance $2h$ (or h). Use a coordinate y measured from the mid-plane. The plates are at $y = \pm h$. The pressure gradient is $-\frac{dP}{dx} = \frac{\Delta P}{L}$ (constant).

1. **Force balance on a fluid slab** of thickness dy , length dx , width W (into page).

- Pressure force: $\Delta P \cdot W dy = -\frac{dP}{dx} dx \cdot W dy$.
- Viscous force on top and bottom faces: $\tau(y + dy)W dx - \tau(y)W dx = \frac{d\tau}{dy} dy W dx$.
- Equilibrium: $-\frac{dP}{dx} dx W dy + \frac{d\tau}{dy} dy W dx = 0 \Rightarrow \frac{d\tau}{dy} = \frac{dP}{dx}$.

- Newton's law:** $\tau = \eta \frac{du}{dy}$ (with sign: $\tau = -\eta \frac{du}{dy}$ if u decreases with y ? Let's set u maximum at $y = 0$, symmetric. Then $\frac{du}{dy}$ is negative for $y > 0$. The shear stress on the upper face (at $y + dy$) is in the $-x$ direction if $\frac{du}{dy} < 0$. Standard: $\tau_{yx} = \eta \frac{du}{dy}$. The force on the bottom face (at y) is $\tau(y)Wdx$ in $+x$ direction. On top face: $-\tau(y + dy)Wdx$. Net viscous force $= -\frac{d\tau}{dy}dyWdx$. Equating to pressure force $\frac{dP}{dx}dxWdy$ (pressure decreases in x , so $dP/dx < 0$): $-\frac{d\tau}{dy} = \frac{dP}{dx} \Rightarrow \frac{d\tau}{dy} = -\frac{dP}{dx}$. Let $G = -\frac{dP}{dx} > 0$. Then $\frac{d\tau}{dy} = G$. $\tau = \eta \frac{du}{dy}$. So $\eta \frac{d^2u}{dy^2} = G$.
- Integrate:** $\frac{du}{dy} = \frac{G}{\eta}y + C_1$. Symmetry at $y = 0 \Rightarrow C_1 = 0$. $u(y) = \frac{G}{2\eta}y^2 + C_2$. No-slip at $y = \pm h$: $u(\pm h) = 0 \Rightarrow C_2 = -\frac{G}{2\eta}h^2$. $u(y) = \frac{G}{2\eta}(h^2 - y^2)$. Parabolic profile. Maximum velocity at centre: $u_{\max} = \frac{Gh^2}{2\eta}$.
- Flow rate per unit width** $q = \int_{-h}^h u(y)dy = \frac{2}{3} \frac{Gh^3}{\eta}$. For width W , $Q = \frac{2}{3} \frac{Gh^3W}{\eta} = \frac{2}{3} \frac{\Delta P}{L} \frac{h^3W}{\eta}$. Often written with gap $2h = H$: $Q = \frac{\Delta P}{L} \frac{H^3W}{12\eta}$.
- Mean velocity:** $u_{\text{avg}} = \frac{Q}{2hW} = \frac{Gh^2}{3\eta} = \frac{2}{3}u_{\max}$.

Method: Annular Flow (Concentric Cylinders)

Inner radius R_i , outer radius R_o . Pressure gradient $G = -\frac{dP}{dx}$. Force balance on a cylindrical shell of radius r , thickness dr , length dx : Pressure force $= G \cdot 2\pi r dr dx$. Viscous force on inner and outer surfaces: $\tau(r)2\pi r dx - \tau(r + dr)2\pi(r + dr)dx \approx -\frac{d}{dr}(\tau r)dr2\pi dx$. Equilibrium: $G \cdot 2\pi r dr dx - \frac{d}{dr}(\tau r)dr2\pi dx = 0 \Rightarrow \frac{d}{dr}(\tau r) = Gr$. $\tau = \eta \frac{du}{dr}$. So $\frac{d}{dr}(r \frac{du}{dr}) = \frac{G}{\eta}r$. Integrate: $r \frac{du}{dr} = \frac{G}{2\eta}r^2 + C_1 \Rightarrow \frac{du}{dr} = \frac{G}{2\eta}r + \frac{C_1}{r}$. Integrate: $u(r) = \frac{G}{4\eta}r^2 + C_1 \ln r + C_2$. Boundary conditions: $u(R_i) = 0$, $u(R_o) = 0$. Solve for C_1, C_2 : $C_1 = -\frac{G}{4\eta} \frac{R_o^2 - R_i^2}{\ln(R_o/R_i)}$. $C_2 = -\frac{G}{4\eta}R_i^2 - C_1 \ln R_i$. Flow rate $Q = \int_{R_i}^{R_o} u(r)2\pi r dr$. After algebra:

$$Q = \frac{\pi G}{8\eta} \left[(R_o^2 - R_i^2)^2 - \frac{(R_o^2 - R_i^2)^2}{\ln(R_o/R_i)} \cdot \text{something} \right]$$

Standard formula:

$$Q = \frac{\pi \Delta P}{8\eta L} \left[R_o^4 - R_i^4 - \frac{(R_o^2 - R_i^2)^2}{\ln(R_o/R_i)} \right]$$

Example D.1 — [JEE Advanced Level] (Plane Poiseuille Flow)

Oil ($\eta = 0.10 \text{ Pa}\cdot\text{s}$, $\rho = 900 \text{ kg m}^{-3}$) flows between two large parallel plates separated by 10 mm. The pressure drop is 500 Pa over a length of 1.0 m. The plates are 0.50 m wide. Find (a) the maximum velocity, (b) the volume flow rate, and (c) the Reynolds number based on hydraulic diameter. Is the flow laminar?

Given: $2h = 0.010 \text{ m} \Rightarrow h = 0.005 \text{ m}$. $\Delta P = 500 \text{ Pa}$, $L = 1.0 \text{ m}$, $W = 0.50 \text{ m}$, $\eta = 0.10 \text{ Pa}\cdot\text{s}$. $G = \Delta P/L = 500 \text{ Pa/m}$.

(a) $u_{\max} = \frac{Gh^2}{2\eta} = \frac{500 \times (0.005)^2}{2 \times 0.10} = \frac{500 \times 2.5 \times 10^{-5}}{0.2} = \frac{0.0125}{0.2} = 0.0625 \text{ m/s}$.

(b) $Q = \frac{2}{3} \frac{Gh^3W}{\eta} = \frac{2}{3} \times \frac{500 \times (0.005)^3 \times 0.50}{0.10} = \frac{2}{3} \times \frac{500 \times 1.25 \times 10^{-7} \times 0.50}{0.10} = \frac{2}{3} \times \frac{3.125 \times 10^{-5}}{0.10} = \frac{2}{3} \times 3.125 \times 10^{-4} = 2.08 \times 10^{-4} \text{ m}^3/\text{s}$.

(c) Hydraulic diameter $D_h = \frac{4 \times \text{cross-sectional area}}{\text{wetted perimeter}} = \frac{4 \times (2hW)}{2W} = 4h = 0.020 \text{ m}$. $u_{\text{avg}} = Q/(2hW) = \frac{2.08 \times 10^{-4}}{0.01 \times 0.5} = 0.0417 \text{ m/s}$. $Re = \frac{\rho u_{\text{avg}} D_h}{\eta} = \frac{900 \times 0.0417 \times 0.020}{0.10} = 7.5$. $Re \ll 2000$, so flow is laminar.

Example D.2 — [JEE Advanced Level] (Annular Flow)

Water ($\eta = 1.0 \times 10^{-3} \text{ Pa}\cdot\text{s}$) flows through an annulus between two concentric pipes. Inner radius $R_i = 10 \text{ mm}$, outer radius $R_o = 20 \text{ mm}$. Length $L = 5.0 \text{ m}$, pressure drop $\Delta P = 2.0 \times 10^3 \text{ Pa}$. Find the volume flow rate.

Given: $R_i = 0.010 \text{ m}$, $R_o = 0.020 \text{ m}$, $L = 5.0 \text{ m}$, $\Delta P = 2000 \text{ Pa}$, $\eta = 1.0 \times 10^{-3} \text{ Pa}\cdot\text{s}$.

Use the annular flow formula:

$$Q = \frac{\pi \Delta P}{8 \eta L} \left[R_o^4 - R_i^4 - \frac{(R_o^2 - R_i^2)^2}{\ln(R_o/R_i)} \right]$$

Compute terms: $R_o^4 = (0.02)^4 = 16 \times 10^{-8} = 1.6 \times 10^{-7} \text{ m}^4$. $R_i^4 = (0.01)^4 = 1 \times 10^{-8} = 1.0 \times 10^{-8} \text{ m}^4$. $R_o^4 - R_i^4 = 1.5 \times 10^{-7} \text{ m}^4$. $R_o^2 - R_i^2 = (0.0004 - 0.0001) = 3 \times 10^{-4} \text{ m}^2$. $(R_o^2 - R_i^2)^2 = 9 \times 10^{-8} \text{ m}^4$. $\ln(R_o/R_i) = \ln 2 = 0.693$. $\frac{(R_o^2 - R_i^2)^2}{\ln(R_o/R_i)} = \frac{9 \times 10^{-8}}{0.693} \approx 1.299 \times 10^{-7} \text{ m}^4$.

Bracket term: $1.5 \times 10^{-7} - 1.299 \times 10^{-7} = 0.201 \times 10^{-7} = 2.01 \times 10^{-8} \text{ m}^4$.

$$Q = \frac{\pi \times 2000}{8 \times 1.0 \times 10^{-3} \times 5.0} \times 2.01 \times 10^{-8} = \frac{\pi \times 2000}{0.04} \times 2.01 \times 10^{-8} = \pi \times 50000 \times 2.01 \times 10^{-8} = \pi \times 1.005 \times 10^{-3} \approx 3.16 \times 10^{-3} \text{ m}^3/\text{s}.$$

Check: For a solid pipe of radius R_o , $Q_{\text{solid}} = \frac{\pi \Delta P R_o^4}{8 \eta L} = \frac{\pi \times 2000 \times 1.6 \times 10^{-7}}{0.04} = \pi \times 8 \times 10^{-3} = 0.025 \text{ m}^3/\text{s}$. The annulus reduces flow significantly because the inner wall adds drag.

Practice — Addendum D

1. Water flows between two parallel plates separated by 5.0 mm. The pressure gradient is 100 Pa/m. The plates are 0.20 m wide. Find the flow rate per unit width and the maximum velocity. ($\eta = 1.0 \times 10^{-3} \text{ Pa}\cdot\text{s}$).
2. An annulus has $R_i = 5 \text{ mm}$, $R_o = 15 \text{ mm}$. Water ($\eta = 1.0 \times 10^{-3} \text{ Pa}\cdot\text{s}$) flows with $\Delta P = 1000 \text{ Pa}$ over $L = 2.0 \text{ m}$. Find the flow rate.
3. Show that for an annulus with $R_i \rightarrow 0$, the annular flow formula reduces to the Poiseuille formula for a solid cylinder of radius R_o .

Answers: 1. $q = 2.08 \times 10^{-5} \text{ m}^2/\text{s}$ (per unit width), $Q = 4.17 \times 10^{-6} \text{ m}^3/\text{s}$; $u_{\text{max}} = 0.03125 \text{ m/s}$.
2. $Q = 1.15 \times 10^{-5} \text{ m}^3/\text{s}$. 3. Use L'Hôpital's rule or series expansion of $\ln(1+x)$ to show the bracket tends to R_o^4 .

This completes the Round 2 Addendum. Each addendum provides the missing JEE Advanced patterns with full derivations, unit-carrying calculations, and practice problems with answers.

Thermal Properties of Matter

Chapter 10 · Class 11 Physics — Definitive Exam Companion

CBSE · JEE Main · NEET · JEE Advanced

Drona Ed-Tech · generated 18 June 2026

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Subtopic 01 — Temperature, Heat and Expansion

Exam Relevance: This subtopic is a steady source of marks. JEE Main almost always carries 1–2 questions, usually a clean thermal-expansion or thermal-stress numerical, or a quick temperature-scale conversion. NEET tends to ask one conceptual MCQ (heat vs temperature, anomalous expansion of water, or “does the hole expand?”). CBSE Boards favour a 1–3 mark definition or short numerical (state the relation $\gamma = 3\alpha$, or compute an increase in length). JEE Advanced rarely asks it standalone but loves to bolt it onto elasticity (Young’s modulus) or onto a clock/pendulum problem for multi-step reasoning.

Section 1: Concept Intuition

Let us begin with something you have felt a thousand times without ever calling it physics. You pour hot chai into a steel tumbler, wait two minutes, and then the tumbler itself becomes too hot to hold. Why? The chai and the tumbler were at *different* hotness levels, and something flowed from the hotter chai into the cooler steel until — if you wait long enough — both settle at the same hotness, somewhere in between. That “level of hotness” is **temperature**, and the “something that flows” is **heat**. The whole subtopic lives inside this single everyday scene, so let us pull it apart slowly.

Temperature is the property that tells you which way energy will flow when two objects touch. It is not “how much energy is inside,” it is “how energetically the molecules are jiggling, on average.” A spoonful of boiling water and a full bucket of boiling water are at the *same* temperature — the molecules jiggle equally hard in both — yet the bucket clearly holds far more thermal energy. So temperature is an *intensive* quantity: it does not care about size or amount.

Heat, on the other hand, is energy *in transit*. The moment two objects at different temperatures are in contact, energy passes from the hotter to the cooler one. We only call that flowing energy “heat.” Once it has arrived and the flow stops, it stops being heat and becomes part of the object’s internal energy. This is exactly why a sentence like “this body contains a lot of heat” is wrong — a body contains *internal energy*; heat is only the name we give to the transfer. The correct version is “this body is at a high temperature” or “heat is flowing from A to B because A is hotter.”

When does the flow stop? When both bodies reach the same temperature — a state called **thermal equilibrium**. And here a quiet but powerful rule sneaks in, called the **Zeroth Law of Thermodynamics**: if body A is in thermal equilibrium with body C, and body B is also in thermal equilibrium with C, then A and B are in equilibrium with each other. It sounds almost too obvious to name — but it is precisely this law that lets a thermometer work. The thermometer is the “C” that we touch to everything; if two patients both make the thermometer read 37°C , they are at the same temperature, even though they never touched each other.

Think about it this way... imagine a busy Mumbai local train at rush hour. Temperature is like the *crowd density* — how tightly packed the people are in one compartment. Heat is like the

people pushing through the doors from a packed compartment into an emptier one at a station. The pushing (heat) continues only until both compartments are equally crowded (equal temperature). After that, no net movement — equilibrium. Notice that a long 12-coach train and a short 3-coach train can have the *same* density (temperature) while the long train obviously contains far more people (internal energy). That picture, once it clicks, keeps you from ever confusing the two again.

Now to the third character: **expansion**. Heat a substance and its molecules jiggle harder, pushing their neighbours a little farther away on average. The object grows — it expands. Cool it and it shrinks. Almost everything does this, and engineers must respect it constantly: the gaps left between railway track segments, the rollers under a bridge span, the slight sag deliberately left in overhead electric wires — all of these exist so that summer heat does not buckle the structure.

Limiting conditions & assumptions. The neat linear formulas we will write down assume (i) the temperature change ΔT is modest, so the expansion coefficient stays effectively constant — over hundreds of degrees the coefficients themselves drift; (ii) the material is isotropic, expanding equally in all directions (true for most metals, not for crystals like calcite); and most importantly (iii) the substance behaves *normally*. The famous exception is **water between 0°C and 4°C**, which *contracts* as you warm it and only begins expanding above 4°C. This anomaly is not a footnote — it is the reason fish survive a frozen pond, and it quietly violates our “heat always expands” intuition.

Section 2: Key Formulas & Definitions

Temperature scale conversion (general identity — works for any linear scale):

$$\frac{C - 0}{100} = \frac{F - 32}{180} = \frac{K - 273.15}{100} = \frac{R - 0}{80}$$

where C , F , K , R are readings on the Celsius, Fahrenheit, Kelvin and Réaumur scales. The most-used pair:

$$F = \frac{9}{5}C + 32, \quad K = C + 273.15$$

- **Kelvin (SI unit of temperature):** defined as $1/273.16$ of the temperature of the **triple point of water** (the unique state where ice, water and vapour coexist). SI unit: K. Dimensional formula: $[K] = [M^0 L^0 T^0 \Theta^1]$.

Heat as energy:

$$Q = mc \Delta T$$

- Q = heat exchanged, m = mass, c = specific heat capacity, ΔT = temperature change.
- SI unit of Q : joule (J). Dimensional formula of heat (energy): $[ML^2 T^{-2}]$.
- **Calorie:** the heat needed to raise 1 g of water from 14.5°C to 15.5°C; 1 cal = 4.186 J.
- **Specific heat c :** SI unit $J kg^{-1} K^{-1}$. Dimensional formula $[L^2 T^{-2} \Theta^{-1}]$.
- **Mechanical equivalent of heat J :** $W = JQ$; it is a unit-conversion constant, hence **dimensionless** $[M^0 L^0 T^0]$, numerically 4.186 J/cal.

Thermal expansion:

$$\text{Linear: } \Delta L = \alpha L_0 \Delta T \quad \Rightarrow \quad L = L_0(1 + \alpha \Delta T)$$

$$\text{Areal: } \Delta A = \beta A_0 \Delta T \quad \Rightarrow \quad A = A_0(1 + \beta \Delta T)$$

$$\text{Volumetric: } \Delta V = \gamma V_0 \Delta T \quad \Rightarrow \quad V = V_0(1 + \gamma \Delta T)$$

with the fundamental relations for an isotropic solid:

$$\beta = 2\alpha, \quad \gamma = 3\alpha, \quad \alpha : \beta : \gamma = 1 : 2 : 3$$

- α = coefficient of linear expansion, β = superficial (areal), γ = cubical (volumetric).
- SI unit of each coefficient: K^{-1} (or per $^{\circ}\text{C}$, same magnitude). Dimensional formula: $[\Theta^{-1}]$.

Change of density with temperature (volume grows, mass fixed):

$$\rho = \frac{\rho_0}{1 + \gamma \Delta T} \approx \rho_0 (1 - \gamma \Delta T)$$

Thermal stress (rod prevented from expanding/contracting):

$$\sigma = Y \alpha \Delta T$$

where Y is Young's modulus. Unit: pascal (Pa); dimensional formula $[ML^{-1}T^{-2}]$.

Section 3: CBSE Board Derivation — Why $\gamma = 3\alpha$

This is the single most asked derivation from the expansion topic, and it is short and beautiful. The goal is to show that the cubical expansion coefficient is exactly three times the linear one.

Initial assumption. Take a small *cube* of an isotropic solid with each side of length L_0 at temperature T_0 . (We choose a cube only for convenience; the result holds for any shape because expansion is geometric.) Its initial volume is

$$V_0 = L_0^3.$$

Step 1 — Each edge grows by the linear law. Raise the temperature by ΔT . Every edge, being a length, expands according to linear expansion:

$$L = L_0(1 + \alpha \Delta T).$$

Why this step: expansion is a property of the material's bonds, not of the object's orientation — so each of the three mutually perpendicular edges grows by the same fractional amount $\alpha \Delta T$.

Step 2 — Build the new volume. Since the cube stays a cube, the new volume is the cube of the new edge:

$$V = L^3 = L_0^3(1 + \alpha \Delta T)^3 = V_0(1 + \alpha \Delta T)^3.$$

Step 3 — Expand the binomial.

$$(1 + \alpha \Delta T)^3 = 1 + 3\alpha \Delta T + 3(\alpha \Delta T)^2 + (\alpha \Delta T)^3.$$

Step 4 — Drop the tiny terms. Here is the crucial physical reasoning. The coefficient α for solids is of order 10^{-5} K^{-1} , so even for $\Delta T = 100 \text{ K}$ the product $\alpha \Delta T \sim 10^{-3}$. Its square is $\sim 10^{-6}$ and its cube $\sim 10^{-9}$ — utterly negligible beside the first-order term. So we keep only the linear term:

$$V \approx V_0 (1 + 3\alpha \Delta T).$$

Step 5 — Compare and conclude. But by definition $V = V_0(1 + \gamma \Delta T)$. Matching the two expressions term by term:

$$\boxed{\gamma = 3\alpha}$$

By the identical argument applied to a square face ($A = L^2$), keeping only the first-order term gives $\beta = 2\alpha$. The numbers 1, 2, 3 are simply the powers to which length is raised for a line, an area and a volume — a result you can now reconstruct on the spot rather than memorise.

Figure 10.1 Thermal expansion of a cube: the solid cube of side L_0 at T_0 and, dashed, the expanded cube of side $L = L_0(1 + \alpha \Delta T)$ sharing one corner, each growing edge longer by $\Delta L = \alpha L_0 \Delta T$. (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A straight steel railway rail is 12.0 m long when laid at an early-morning temperature of 18°C in Nagpur. By afternoon the rail heats up to 48°C . Find the increase in its length. Take $\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.

Given: $L_0 = 12.0 \text{ m}$, $\Delta T = 48 - 18 = 30 \text{ }^\circ\text{C}$, $\alpha = 1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.

Formula: $\Delta L = \alpha L_0 \Delta T$.

Substitution:

$$\begin{aligned}\Delta L &= (1.2 \times 10^{-5}) \times (12.0) \times (30) \\ \Delta L &= 1.2 \times 12.0 \times 30 \times 10^{-5} = 432 \times 10^{-5} \text{ m} \\ \boxed{\Delta L} &= 4.32 \times 10^{-3} \text{ m} = 4.32 \text{ mm}\end{aligned}$$

Concluding remark: Just 4.32 mm — but multiply that across thousands of rail segments and you see why expansion gaps are non-negotiable on Indian Railways.

Example 2 — [NEET Speed Trap]

A child's body temperature rises by 3.0°C during a fever. By how much has it risen on the Fahrenheit scale?

The trap: the reflex is to grab $F = \frac{9}{5}C + 32$ and “convert 3.” That gives $F = \frac{9}{5}(3) + 32 = 37.4 \text{ }^\circ\text{F}$ — and a student then writes “rose by 37.4°F ,” which is nonsense.

The fix (mental, 5 seconds): the “+32” is an *offset* of where each scale starts; it disappears the moment you deal with a *change* ΔT . Only the *size* of a degree matters, and a Fahrenheit degree is $\frac{5}{9}$ as big as a Celsius degree. Therefore:

$$\Delta F = \frac{9}{5} \Delta C = \frac{9}{5}(3.0) = \boxed{5.4 \text{ }^\circ\text{F}}$$

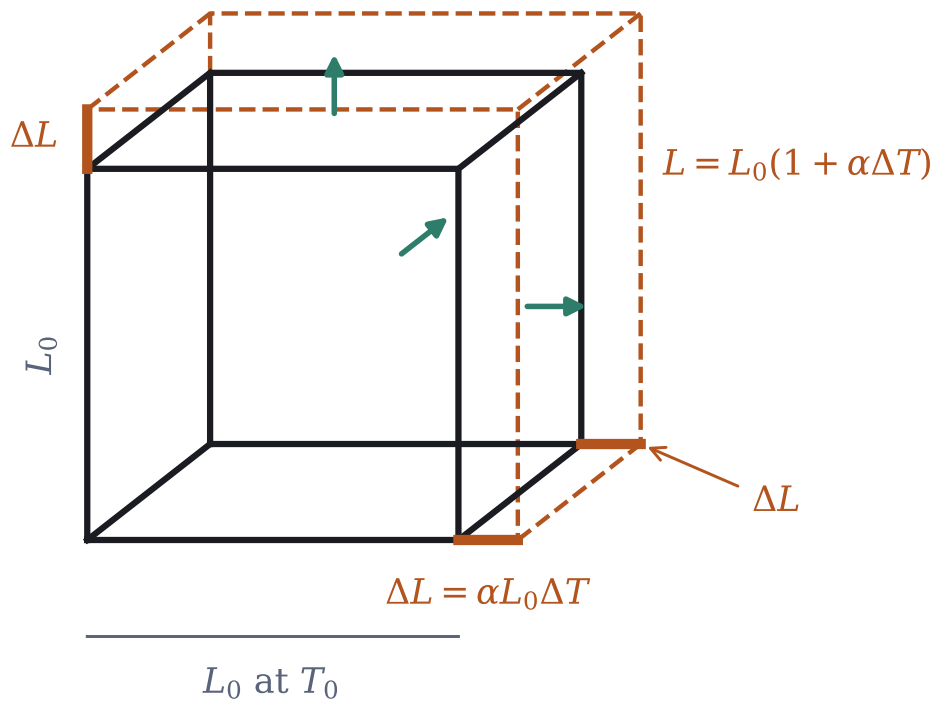


Figure 10.1 Thermal expansion of a cube: the solid cube of side L_0 at T_0 and, dashed, the expanded cube of side $L = L_0(1 + \alpha \Delta T)$ sharing one corner, each growing edge longer by $\Delta L = \alpha L_0 \Delta T$.

Speed insight: for *intervals* (rises, falls, differences), Kelvin and Celsius are identical ($\Delta K = \Delta C$), and Fahrenheit intervals are $\frac{9}{5}$ times Celsius intervals. Never add 32 to a ΔT .

Example 3 — [JEE Main Level] (*links thermal expansion + elasticity*)

A copper rod is clamped rigidly between two thick walls so that it cannot change length. It is then heated from 25°C to 75°C . Compute the thermal (compressive) stress developed in the rod. Take $Y_{Cu} = 1.1 \times 10^{11} \text{ Pa}$ and $\alpha_{Cu} = 1.7 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.

Reasoning — two concepts stitched together. Left free, the rod *would* expand by a fractional amount $\frac{\Delta L}{L_0} = \alpha \Delta T$. The walls forbid this, so it is as if the rod is compressed back by that same fractional amount. From elasticity, stress = $Y \times$ strain, and here the suppressed strain is exactly $\alpha \Delta T$.

Step 1 — Suppressed strain:

$$\text{strain} = \alpha \Delta T = (1.7 \times 10^{-5})(50) = 8.5 \times 10^{-4}.$$

Step 2 — Thermal stress:

$$\sigma = Y \alpha \Delta T = (1.1 \times 10^{11})(8.5 \times 10^{-4})$$

$$\boxed{\sigma = 9.35 \times 10^7 \text{ Pa}}$$

Notice the stress is independent of the rod's length — a short clamped rod and a long clamped rod of the same material develop identical stress. That length-independence is the favourite hook in this kind of question.

Example 4 — [JEE Advanced Level]

A pendulum clock with a steel pendulum keeps perfect time at 20°C . During a Rajasthan summer the temperature climbs to 45°C . Does the clock run fast or slow, and by how many seconds will it be off after one full day? Take $\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.

Step 1 — How length feeds into period. A simple pendulum has $T_{\text{period}} = 2\pi\sqrt{L/g}$, so $T_{\text{period}} \propto \sqrt{L}$. Taking a logarithmic differential converts a product/power into a sum:

$$\frac{\Delta T_{\text{period}}}{T_{\text{period}}} = \frac{1}{2} \frac{\Delta L}{L}.$$

Step 2 — Bring in expansion. The fractional change in length is $\frac{\Delta L}{L} = \alpha \Delta T_{\text{temp}}$, so:

$$\frac{\Delta T_{\text{period}}}{T_{\text{period}}} = \frac{1}{2} \alpha \Delta T_{\text{temp}}.$$

Step 3 — Direction (the conceptual core). Heating *lengthens* the pendulum, which *increases* the period — each swing takes slightly longer — so the clock **ticks less often than it should and runs SLOW**, i.e. it *loses* time.

Step 4 — Accumulate over one day. The fractional error in period equals the fractional error in elapsed time, so over $t = 86400 \text{ s}$:

$$\Delta t = \frac{1}{2} \alpha \Delta T_{\text{temp}} \times t = \frac{1}{2} (1.2 \times 10^{-5})(25)(86400)$$

$$\Delta t = \frac{1}{2} \times 1.2 \times 25 \times 86400 \times 10^{-5} = \frac{1}{2}(25.92) = \boxed{12.96 \text{ s lost per day}}$$

The clock falls behind by roughly **13 seconds each day** in that heat. The deep idea here — that a fractional change in a *length* becomes *half* the fractional change in a *time* — is exactly the kind of multi-step bridge JEE Advanced rewards.

Section 5: Practice Exercises

1. [CBSE] A brass scale is correct at 15°C. A rod measured with it reads 80.00 cm on a day when the scale is at 35°C. Find the *true* length of the rod. ($\alpha_{\text{brass}} = 1.9 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.)
2. [NEET] At what single temperature do the Celsius and Fahrenheit scales show the *same numerical reading*?
3. [JEE Main] A glass flask of volume 1000 cm³ is filled to the brim with mercury at 0°C. Both are heated to 80°C. How much mercury overflows? Take $\gamma_{\text{mercury}} = 1.8 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$ and $\alpha_{\text{glass}} = 9 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$.
4. [JEE Main] An aluminium wire of cross-sectional area 2.0 mm² is stretched between two rigid clamps at 30°C and then cooled to −10°C. Find the tension developed. Take $Y_{\text{Al}} = 7.0 \times 10^{10} \text{ Pa}$, $\alpha_{\text{Al}} = 2.3 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$.
5. [JEE Advanced] Two rods, one of copper ($\alpha_1 = 1.7 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$) and one of steel ($\alpha_2 = 1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$), are joined end to end to form a single rod 1.00 m long at 20°C. The *combined* rod is found to expand by 0.78 mm when heated to 70°C. Find the length of each rod at 20°C.

Answers: 1. 80.03 cm 2. −40° (i.e. −40 °C = −40 °F) 3. 12.24 cm 3 overflows (using effective $\gamma_{\text{apparent}} = \gamma_{\text{Hg}} - 3\alpha_{\text{glass}} = 1.53 \times 10^{-4}$) 4. $\approx 129 \text{ N}$ 5. Copper rod $\approx 0.60 \text{ m}$, steel rod $\approx 0.40 \text{ m}$.

Section 6: Crack the MCQ

Q1. A circular metal plate has a small circular hole drilled through it. When the plate is uniformly heated, the diameter of the hole:

- (a) decreases (b) increases (c) stays the same (d) first increases, then decreases

Correct: (b). Expansion is a *scaling* of the whole geometry — imagine the heated plate as a photograph enlarged on a photocopier. Every feature, including empty regions, scales up by the same factor $(1 + \alpha \Delta T)$. The hole behaves exactly like a disc of metal that *would* have occupied it. - (a) is the *classic trap*: students picture surrounding metal “closing in” on the hole. It feels intuitive but is wrong — the metal moves outward, carrying the hole’s edge outward with it. - (c) assumes holes are “nothing” and so cannot expand; but it is the *boundary* that expands, and the boundary is metal. - (d) invents a non-existent reversal.

Q2. The coefficients of linear, areal and volumetric expansion of an isotropic solid are in the ratio:

- (a) 1 : 1 : 1 (b) 1 : 2 : 3 (c) 3 : 2 : 1 (d) 1 : 3 : 9

Correct: (b). Length, area and volume scale as the 1st, 2nd and 3rd powers of a linear dimension, giving $\alpha : \beta : \gamma = 1 : 2 : 3$. - (c) is the same numbers reversed — a careless-reading trap. - (d) comes from wrongly cubing the area factor instead of doubling and tripling. - (a) would only hold if expansion were one-dimensional, which it is not.

Q3. Water is heated steadily from 0°C to 10°C. Its volume:

- (a) increases throughout (b) decreases throughout (c) first decreases up to 4°C, then increases (d) first increases up to 4°C, then decreases

Correct: (c). This is the **anomalous expansion of water**: between 0°C and 4°C water *contracts* on heating (density rises, peaking at 4°C), and only above 4°C does it expand normally. - (a) is the “everything expands when heated” reflex — exactly the assumption water violates. - (b) describes only the first 4 degrees and misses the reversal. - (d) inverts the real behaviour; this is the trap for students who half-remember “something weird happens at 4°C.”

Q4. A faulty Celsius thermometer reads 2°C in pure melting ice and 102°C in steam at standard pressure. When it reads 27°C, the true temperature is:

- (a) 25°C (b) 25.5°C (c) 27°C (d) 23°C

Correct: (a). The faulty scale runs from 2 to 102, i.e. 100 divisions, but offset by 2. Mapping its reading X to the true value C :

$$\frac{C - 0}{100 - 0} = \frac{X - 2}{102 - 2} \Rightarrow C = \frac{27 - 2}{100} \times 100 = 25^\circ\text{C}.$$

- (c) assumes the faulty reading is correct — ignores the calibration error entirely. - (b) forgets that the lower fixed point shifted and only corrects the span. - (d) subtracts the offset twice.

Section 7: Common Student Pitfalls & Pro-Tips

- Adding 32 to a temperature *difference*.** ΔT never takes the offset. For changes, $\Delta C = \Delta K$ and $\Delta F = \frac{9}{5}\Delta C$. The “+273” and “+32” belong only to absolute readings.
- Confusing L_0 with L .** Always multiply α by the *original* length (before heating). For the tiny changes in board problems the error is negligible, but in precise/JEE work it matters — use L_0 .
- Thinking holes/cavities shrink.** A cavity, hole or the inner radius of a ring all expand *outward*, by the same coefficient as the solid material. Treat empty regions as if filled with the same metal.
- Forgetting the “apparent expansion” of liquids in containers.** When a liquid sits in a solid vessel, the *overflow* uses $\gamma_{\text{apparent}} = \gamma_{\text{liquid}} - \gamma_{\text{container}} = \gamma_{\text{liquid}} - 3\alpha_{\text{container}}$, because the container expands too.

Pro-Tip: For “by how much does it expand” questions, never compute the final length and subtract — go straight for $\Delta L = \alpha L_0 \Delta T$. And for a 2-second sanity check on any expansion answer: solids have $\alpha \sim 10^{-5}$, so a metre-long rod over a 100° rise grows by only ~ 1 mm. If your answer is in centimetres, you have a power-of-ten error.

Section 8: Cheat Sheet Box

- Heat $\neq$ Temperature.** Temperature = level of hotness (intensive); Heat = energy *in transit* due to a temperature difference. Equilibrium = equal temperature (Zeroth Law).
- Scale conversion:** $\frac{C}{100} = \frac{F - 32}{180} = \frac{K - 273.15}{100}$; for changes, $\Delta C = \Delta K$, $\Delta F = \frac{9}{5}\Delta C$.
- Expansion:** $\Delta L = \alpha L_0 \Delta T$, $\Delta A = \beta A_0 \Delta T$, $\Delta V = \gamma V_0 \Delta T$ with $\boxed{\beta = 2\alpha, \gamma = 3\alpha}$ — mnemonic: “1-2-3 for line-area-volume.”
- Density:** $\rho \approx \rho_0(1 - \gamma \Delta T)$ — heat it, it gets lighter (per unit volume).
- Thermal stress (clamped rod):** $\sigma = Y\alpha \Delta T$ — independent of length.

- **Anomaly:** water is densest at **4°C**; 0→4°C it *contracts* on heating. *Mnemonic: “4 keeps the fish alive.”*
- **Holes expand outward** like the surrounding metal — never shrink.

Subtopic 02 — Calorimetry and Specific Heat

Exam Relevance: This is one of the most numerically loaded subtopics in the whole chapter. JEE Main reliably carries 1–2 questions — usually a mixing/heat-balance numerical or a “find the final state” problem. NEET loves the conceptual traps here (steam burns, the latent-heat step, “will all the ice melt?”) and often a quick specific-heat comparison. CBSE Boards ask 2–3 mark numericals on $Q = mc\Delta T$, $Q = mL$, the principle of calorimetry, or a definition of specific heat / latent heat. JEE Advanced uses it for multi-phase mixing problems where you must *determine* the final state rather than assume it — high reasoning, low formula count.

Section 1: Concept Intuition

Picture your kitchen on a winter morning. You put a steel ladle and a mug of water on the same flame, side by side, for the same one minute. Pick them up: the steel ladle is already scorching, but the water is merely warm. Same flame, same time — so roughly the *same amount of heat* went into each. Yet their temperatures rose by wildly different amounts. The water stubbornly refused to heat up, while the steel surrendered instantly. That stubbornness has a name: **specific heat capacity**.

Specific heat c is the amount of heat needed to raise the temperature of **one kilogram** of a substance by **one degree**. Water has an unusually high specific heat ($4186 \text{ J kg}^{-1}\text{K}^{-1}$, or a clean $1 \text{ cal g}^{-1}\text{°C}^{-1}$) — far higher than almost anything you meet in daily life. That single fact explains a startling range of things: why the sea breeze is cool in the afternoon (the ocean barely warms while the land bakes), why a hot-water bottle stays warm all night, why coastal cities like Mumbai and Chennai have gentler climates than landlocked Nagpur, and why your body — which is mostly water — can absorb a lot of metabolic heat without your temperature spiking. Water is nature’s thermal shock-absorber.

The governing relation is wonderfully simple:

$$Q = mc\Delta T$$

Read it as a recipe: the heat Q you must pour in equals the mass m , times how “reluctant” the material is (c), times how big a temperature jump ΔT you want. Triple the mass, you need triple the heat. Pick a material with double the specific heat, you need double the heat. It is a multiplication, nothing more.

Now to the second big idea. Bring a hot body and a cold body together in an insulated box and leave them alone. Heat flows from hot to cold — always, never the reverse on its own — until both reach one common temperature. Crucially, **energy is not created or destroyed in the box**: every joule the hot body loses, the cold body gains. This bookkeeping rule is the **Principle of Calorimetry**:

$$\text{Heat lost by hot bodies} = \text{Heat gained by cold bodies}$$

A **calorimeter** is just a well-insulated metal cup (with a stirrer and thermometer) built to make this rule trustworthy by keeping heat from leaking to the room. But here is the catch students forget: *the calorimeter itself participates*. Its metal walls also soak up heat, so they must appear in the balance sheet. A neat way to handle this is the **water equivalent** $w = mc$ — the mass of water that would absorb the same heat as the object for the same temperature rise. Replace the calorimeter by its water equivalent and the arithmetic simplifies.

Think about it this way... imagine a “heat bank account.” Every body in the box has a balance. When they are put in contact, the rich (hot) ones transfer money (heat) to the poor (cold) ones until everyone has the *same balance per rupee of “thermal sensitivity.”* No money leaves the room — it just gets redistributed. The total heat lost by the depositors equals the total gained by the receivers. That conservation is the entire engine behind every mixing problem you will ever solve.

There is one more twist, and it is the source of most exam traps. Pour heat into ice at 0°C and — surprisingly — its temperature does **not** rise while it is melting. The heat is going somewhere invisible: into breaking the rigid crystal lattice, prying molecules loose into the liquid state. This “hidden” heat absorbed at *constant temperature* during a phase change is the **latent heat** L , and the heat for a phase change is

$$Q = mL.$$

Latent heats are large. To *melt* 1 g of ice takes 80 cal — the same heat that would raise that gram of water by a full 80°C. To *boil* 1 g of water takes a whopping 540 cal. This is exactly why a steam burn is so much worse than a boiling-water burn: the steam dumps an extra 540 cal per gram into your skin as it condenses, on top of cooling.

Limiting conditions & assumptions. Our clean formulas assume (i) the calorimeter is perfectly insulated — no heat leaks to the surroundings; (ii) no chemical reaction occurs between the mixed substances (which would release or absorb its own heat); (iii) specific heat is treated as constant over the temperature range (true to within ~1% for water from 0–100°C, good enough for problems); and most importantly (iv) you must **never assume the final state** — always check whether a phase change completes before settling on a final temperature.

Section 2: Key Formulas & Definitions

Heat for a temperature change (no phase change):

$$Q = mc \Delta T$$

- Q = heat (J), m = mass (kg), c = specific heat capacity, ΔT = temperature change.
- **Specific heat** c : SI unit $\text{J kg}^{-1}\text{K}^{-1}$; dimensional formula $[L^2T^{-2}\Theta^{-1}]$. It can range from 0 to ∞ depending on the process (and can even be negative in special situations); it is a property of the material.

Heat capacity (of the whole body):

$$C = mc \quad \left(\text{unit: J K}^{-1}, \text{ dimensions } [ML^2T^{-2}\Theta^{-1}] \right)$$

- Heat capacity is the heat to raise the *entire body* by 1 K. Depends on both mass and material.

Molar heat capacity:

$$C_{\text{molar}} = Mc = \frac{Q}{n \Delta T} \quad \left(\text{unit: J mol}^{-1}\text{K}^{-1} \right)$$

- M = molar mass, n = number of moles.

Water equivalent:

$$w = mc \quad \Rightarrow \quad w = m \cdot \frac{c}{c_{\text{water}}} \quad \left(\text{in grams of water, with } c_{\text{water}} = 1 \text{ cal g}^{-1}\text{°C}^{-1} \right)$$

- Numerically equal to the heat capacity; it is the mass of water with the same heat capacity as the body.

Latent heat (phase change at constant temperature):

$$Q = mL$$

- L_f = latent heat of **fusion** (solid $\rightleftharpoons$ liquid); L_v = latent heat of **vaporization** (liquid $\rightleftharpoons$ gas).
- SI unit of L : J kg^{-1} ; dimensional formula $[L^2 T^{-2}]$.

Principle of Calorimetry (insulated system):

$$\sum Q_{\text{lost}} = \sum Q_{\text{gained}}$$

Standard constants (memorise these):

Quantity	Value (CGS)	Value (SI)
c_{water}	$1 \text{ cal g}^{-1} \text{ } ^\circ\text{C}^{-1}$	$4186 \text{ J kg}^{-1} \text{ K}^{-1}$
c_{ice}	$0.5 \text{ cal g}^{-1} \text{ } ^\circ\text{C}^{-1}$	$\approx 2100 \text{ J kg}^{-1} \text{ K}^{-1}$
c_{steam}	$\approx 0.48 \text{ cal g}^{-1} \text{ } ^\circ\text{C}^{-1}$	$\approx 2010 \text{ J kg}^{-1} \text{ K}^{-1}$
L_f (ice)	80 cal g^{-1}	$3.36 \times 10^5 \text{ J kg}^{-1}$
L_v (water)	540 cal g^{-1}	$2.26 \times 10^6 \text{ J kg}^{-1}$

Section 3: Key Procedures & Logical Framework

Calorimetry has no single grand derivation; its difficulty is *procedural*. The skill is in setting up the heat budget correctly and — above all — not guessing the final state. Here is the reliable framework.

Step 1 — List every body and its journey. For each substance, note its mass, specific heat, latent heats, and the temperature it starts at. *Why*: a missed body (especially the calorimeter) breaks conservation.

Step 2 — Split each journey into “warm” and “phase-change” legs. Heating water from -10°C to 110°C is *not* one calculation; it is five separate legs, each with its own formula: 1. warm ice $-10 \rightarrow 0^\circ\text{C}$: $mc_{\text{ice}} \Delta T$ 2. melt ice at 0°C : mL_f 3. warm water $0 \rightarrow 100^\circ\text{C}$: $mc_{\text{water}} \Delta T$ 4. boil water at 100°C : mL_v 5. warm steam $100 \rightarrow 110^\circ\text{C}$: $mc_{\text{steam}} \Delta T$

Why: during legs 2 and 4 the temperature is *frozen* even as heat pours in — using $mc\Delta T$ there is the single most common mistake.

Figure 10.2 Heating curve for 1 kg of ice at -10°C warmed to steam: rising legs where $Q = m c \Delta T$ and flat plateaus at 0°C (melting, $mL_f = 336 \text{ kJ}$) and 100°C (boiling, $mL_v = 2260 \text{ kJ}$ — the long plateau). (see next page)

Step 3 — Identify the cold side and the hot side, then write the balance. Use $\sum Q_{\text{gained}} = \sum Q_{\text{lost}}$. Keep all quantities as magnitudes and let the equation balance them.

Step 4 — When the final state is unknown, test it with a “heat budget.” This is the heart of advanced problems. Compute the heat the hot side *can* deliver and the heat the cold side *needs*

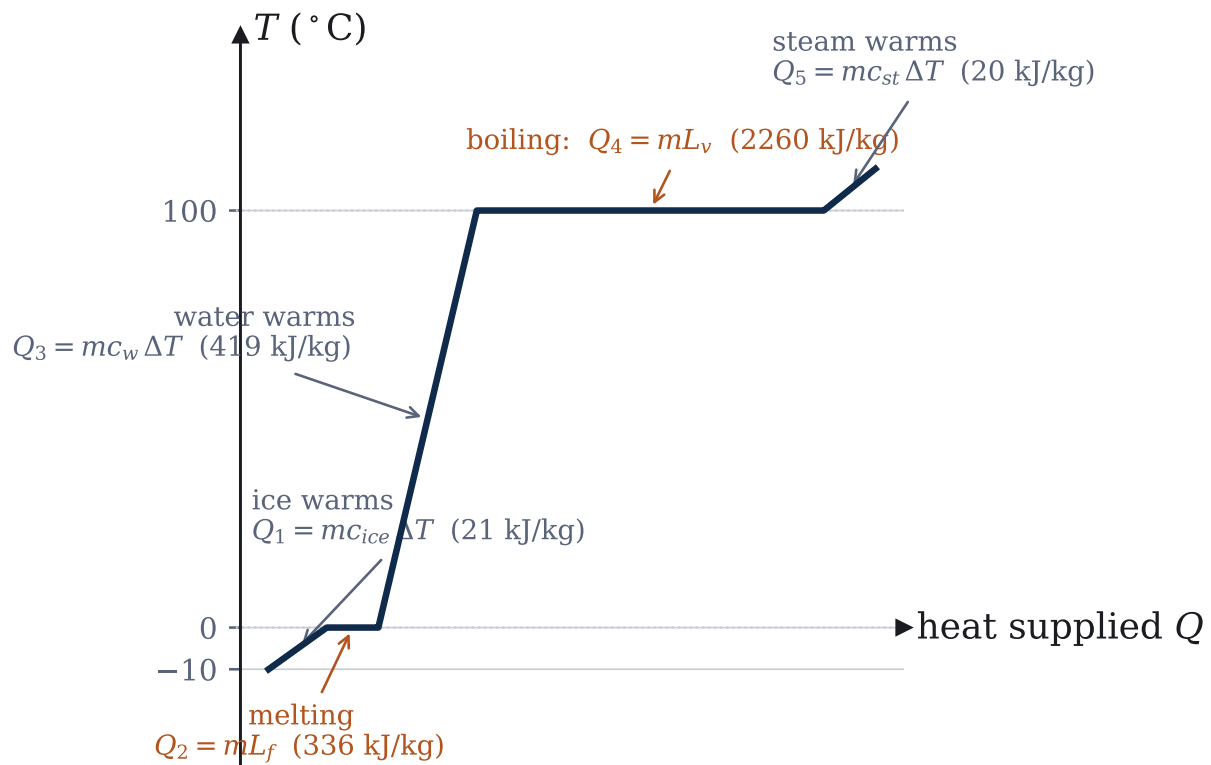


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to reach each milestone (e.g., “ice fully melted at 0°C ”). Compare: - If the hot side’s available heat **exceeds** what’s needed to bring everything to a boundary temperature, the system pushes past it — continue to the next leg. - If it **falls short**, the system settles *at* that boundary, and only *part* of the phase change happens. Solve for the fraction.

Why this step matters: it prevents the absurd answers (like a “final temperature” of 130°C when steam is present, or all ice melting when there wasn’t enough heat). The budget tells you the true final state *before* you commit to an equation.

Step 5 — Solve and sanity-check. The final temperature must satisfy $T_{\text{cold}} \leq T_{\text{final}} \leq T_{\text{hot}}$. If it doesn’t, a phase change was missed.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A 0.50 kg aluminium vessel contains 2.0 kg of water at 25°C . How much heat must be supplied to raise the whole system to 75°C ? Take $c_{\text{water}} = 4186\text{ J kg}^{-1}\text{K}^{-1}$ and $c_{\text{Al}} = 900\text{ J kg}^{-1}\text{K}^{-1}$.

Given: $m_{\text{Al}} = 0.50\text{ kg}$, $m_w = 2.0\text{ kg}$, $\Delta T = 75 - 25 = 50\text{ K}$.

Formula: the vessel and water rise together, so $Q = (m_w c_{\text{water}} + m_{\text{Al}} c_{\text{Al}}) \Delta T$.

Substitution:

$$Q = [(2.0)(4186) + (0.50)(900)](50)$$

$$Q = [8372 + 450](50) = (8822)(50)$$

$$\boxed{Q = 4.41 \times 10^5\text{ J}}$$

Concluding remark: Note the vessel contributes only $450/8822 \approx 5\%$ — small, but in a careful answer you still include it. Always account for the container.

Example 2 — [NEET Speed Trap]

5 g of ice at 0°C is dropped into 20 g of water at 30°C in an insulated cup. Find the final temperature.

The trap: the reflex is to treat this like mixing two waters and write $(20)(30 - T) = (5)(T - 0)$, giving $T = 24^\circ\text{C}$. That is **wrong** — it forgets that the ice must first *melt*, which silently eats up a huge chunk of heat.

The correct 10-second budget: - Heat the warm water can give as it cools to 0°C : $20 \times 1 \times 30 = 600\text{ cal}$. - Heat needed just to melt the ice: $5 \times 80 = 400\text{ cal}$.

Since $600 > 400$, all the ice melts and there is $600 - 400 = 200\text{ cal}$ left over. This surplus warms the *total* 25 g of water (the original 20 g + the 5 g just melted), now all at 0°C :

$$200 = (25)(1)(T - 0) \Rightarrow \boxed{T = 8^\circ\text{C}}$$

Speed insight: in any ice-plus-water problem, *first* spend heat on melting (the latent step), *then* see what’s left for warming. Skipping the latent step is the single most common NEET error here.

Example 3 — [JEE Main Level] (*links mechanics + calorimetry + latent heat*)

A lead pellet is fired from an air gun and strikes a thick wall, coming to rest. Its initial temperature is 27°C . If 50% of its kinetic energy is converted into heat retained by the pellet, find the minimum speed at which the pellet just begins to melt. For lead: melting point = 327°C , $c = 128 \text{ J kg}^{-1} \text{K}^{-1}$, $L_f = 2.5 \times 10^4 \text{ J kg}^{-1}$.

Reasoning — three concepts in sequence. The pellet's kinetic energy becomes heat; that heat must first warm the lead from 27°C to its melting point 327°C , and then melt it (latent heat). "Just begins to melt" means we supply exactly enough to reach the melting point and complete the fusion.

Step 1 — Heat needed per unit mass ($\Delta T = 327 - 27 = 300 \text{ K}$):

$$\frac{Q}{m} = c \Delta T + L_f = (128)(300) + 2.5 \times 10^4 = 38400 + 25000 = 63400 \text{ J/kg}.$$

Step 2 — Energy bookkeeping. Half the kinetic energy per unit mass becomes this heat:

$$0.50 \times \frac{1}{2} v^2 = \frac{Q}{m} = 63400.$$

Step 3 — Solve for v :

$$\frac{1}{4} v^2 = 63400 \Rightarrow v^2 = 253600 \Rightarrow \boxed{v \approx 503.6 \text{ m s}^{-1}}$$

The mass cancels entirely — the answer depends only on the material and the energy fraction. Forgetting the latent-heat term (writing only $c\Delta T$) would give a wrongly *low* speed.

Example 4 — [JEE Advanced Level]

In a perfectly insulated container, 30 g of ice at -10°C is mixed with 5 g of steam at 100°C . Determine the final temperature and the composition of the mixture. Use $c_{\text{ice}} = 0.5$, $c_{\text{water}} = 1$ ($\text{cal g}^{-1} \text{ } ^\circ\text{C}^{-1}$), $L_f = 80$, $L_v = 540$ (cal g^{-1}).

Step 1 — Budget the cold side (ice) up to 0°C , all liquid: - warm ice $-10 \rightarrow 0$: $30 \times 0.5 \times 10 = 150 \text{ cal}$ - melt all ice: $30 \times 80 = 2400 \text{ cal}$ - **subtotal to reach water at 0°C : 2550 cal.**

Step 2 — Budget the hot side (steam) down to 0°C : - condense all steam: $5 \times 540 = 2700 \text{ cal}$ - cool the condensed water $100 \rightarrow 0$: $5 \times 100 = 500 \text{ cal}$ - **subtotal available down to 0°C : 3200 cal.**

Step 3 — Test the final state. Just condensing the steam releases 2700 cal, which already exceeds the 2550 cal needed to turn all the ice into water at 0°C . So *all* the ice melts and *all* the steam condenses, and the final temperature lies somewhere strictly between 0°C and 100°C . Assume final temperature T with everything liquid.

Step 4 — Energy balance:

$$\underbrace{150 + 2400 + 30T}_{\text{gained by ice side}} = \underbrace{2700 + 5(100 - T)}_{\text{lost by steam side}}$$

$$2550 + 30T = 3200 - 5T$$

$$35T = 650 \Rightarrow \boxed{T \approx 18.6 \text{ } ^\circ\text{C}}$$

Step 5 — Composition & sanity check. Final state: all $30 + 5 = 35 \text{ g}$ is **liquid water at $\approx 18.6^\circ\text{C}$** . The result obeys $0 < 18.6 < 100$, confirming both phase changes completed. Had Step 3

shown the steam *couldn't* fully condense, the answer would instead be a steam-water mixture pinned at 100°C — which is exactly the kind of branch JEE Advanced tests.

Section 5: Practice Exercises

1. [CBSE] Calculate the total heat required to convert 50 g of ice at 0°C completely into steam at 100°C. (Use cal-units: $L_f = 80$, $c_{\text{water}} = 1$, $L_v = 540$.)
2. [NEET] A copper calorimeter of water equivalent 10 g contains 90 g of water at 20°C. Now 100 g of water at 60°C is poured in. Find the equilibrium temperature.
3. [JEE Main] A 200 g copper block at 200°C is dropped into 300 g of water at 25°C held in a 50 g copper calorimeter. Find the final temperature. Take $c_{\text{Cu}} = 0.09$ and $c_{\text{water}} = 1$ (cal g⁻¹ °C⁻¹).
4. [JEE Main] What mass of ice at 0°C can be melted by passing 50 g of steam at 100°C through it, the steam finally becoming water at 0°C?
5. [JEE Advanced] In an insulated vessel, 10 g of ice at 0°C is mixed with 10 g of steam at 100°C. Find the final temperature and the masses of water and steam present at equilibrium.

Answers: 1. 36,000 cal = 3.6×10^4 cal ($\approx 1.51 \times 10^5$ J) 2. 40 °C 3. ≈ 34.8 °C 4. 400 g 5. Final temperature 100 °C; about 3.33 g of steam condenses, giving ≈ 13.3 g water + ≈ 6.7 g steam.

Section 6: Crack the MCQ

Q1. A burn caused by steam at 100°C is far more severe than one from boiling water at 100°C. The principal reason is that:

- (a) steam is hotter than boiling water (b) steam releases its large latent heat of vaporization as it condenses on the skin (c) steam has a higher specific heat than water (d) steam molecules travel faster and hit harder

Correct: (b). As steam condenses on the skin it dumps an *extra* ≈ 540 cal per gram (latent heat) before it even begins to cool — energy that boiling water simply does not carry. - (a) is the obvious trap, but both are at the *same* 100°C. - (c) is false and irrelevant; steam's specific heat is actually lower than water's. - (d) confuses a microscopic detail with the real cause; the kinetic energy of impact is negligible compared with the latent heat.

Q2. Equal masses of water and cooking oil ($c_{\text{oil}} < c_{\text{water}}$) are given the *same* amount of heat. Then:

- (a) water becomes hotter (b) oil becomes hotter (c) both reach the same temperature (d) it cannot be decided

Correct: (b). From $\Delta T = Q/(mc)$, with Q and m fixed, a *smaller* c gives a *larger* temperature rise. Oil, having the smaller specific heat, heats up more. - (a) reverses the logic — a very common slip. - (c) ignores the role of c altogether. - (d) is avoidance; the relation fully determines the outcome.

Q3. 1 g of ice at 0°C and 1 g of water at 0°C are each supplied with 80 cal of heat. Afterwards:

- (a) the ice sample is hotter (b) the water sample is hotter (c) both are at the same temperature (d) both remain at 0°C

Correct: (b). The 80 cal given to the *ice* is exactly its latent heat of fusion — it all goes into melting, leaving water at 0°C. The 80 cal given to the *water* raises its temperature by $80/(1 \times 1) = 80^\circ\text{C}$,

to 80°C . - (d) is the trap for students who think “both start at 0, so both stay 0.” - (a) and (c) misjudge where the latent-heat step applies.

Q4. While a block of ice melts at 0°C , the heat supplied to it:

- (a) increases the average kinetic energy of the molecules, raising the temperature (b) increases the molecular potential energy by breaking the crystal lattice, at constant temperature (c) is entirely lost to the surroundings (d) increases kinetic and potential energy equally

Correct: (b). During melting the temperature is fixed; the energy goes into pulling molecules out of the rigid lattice (a potential-energy change), not into faster jiggling. - (a) contradicts the constant-temperature fact of a phase change. - (d) is wrong because no temperature rise means no average-KE rise. - (c) ignores that the heat is stored as latent heat in the new phase.

Section 7: Common Student Pitfalls & Pro-Tips

- Skipping the latent-heat step.** In ice/steam problems, mL comes *before* you spend heat on warming. Treating a melt or boil as if it were $mc\Delta T$ (or ignoring it) is the number-one mistake.
- Assuming the final state.** Never write “ $T_{\text{final}} = \text{something}$ ” before checking the heat budget. The hot side may not have enough heat to melt all the ice, or may have too much, leaving a steam-water mixture at 100°C .
- Dropping the calorimeter (or its water equivalent).** The container absorbs heat too. Include $m_c c_c$ — or fold it in as water equivalent $w = m_c c_c / c_{\text{water}}$.
- Unit collisions.** Don’t mix cal-units with SI mid-problem. Remember $1 \text{ cal} = 4.186 \text{ J}$, $L_f = 80 \text{ cal/g} = 3.36 \times 10^5 \text{ J/kg}$, $L_v = 540 \text{ cal/g} = 2.26 \times 10^6 \text{ J/kg}$.

Pro-Tip: Build a running “heat budget” in cal-units (water’s specific heat is exactly 1, so the arithmetic is trivial). Tally the heat the hot side can supply, then “spend” it on the cold side’s milestones in order — warm, melt, warm, boil. The leg where your budget *runs out* pins down both the final temperature and the final composition, instantly and without guessing.

Section 8: Cheat Sheet Box

- Temperature change:** $Q = mc \Delta T$. **Phase change (constant T):** $Q = mL$.
- Heat capacity** $C = mc$; **water equivalent** $w = mc$ (mass of water with same heat capacity).
- Calorimetry:** $\sum Q_{\text{lost}} = \sum Q_{\text{gained}}$ (insulated, no reaction) — *the calorimeter counts too*.
- Heating curve:** flat plateaus at melting (0°C) and boiling (100°C) — temperature frozen while heat goes into latent heat. *The boiling plateau is far longer because $L_v \gg L_f$.*
- Key constants:** $c_{\text{water}} = 1 \text{ cal/g}^{\circ}\text{C}$, $c_{\text{ice}} = 0.5$, $L_f = 80 \text{ cal/g}$, $L_v = 540 \text{ cal/g}$.
- Golden rule** — **never assume the final state; budget the heat first.** *Mnemonic: “Warm-Melt-Warm-Boil-Warm” for the five legs of ice $\rightarrow$ steam.*

Subtopic 03 — Heat Transfer (Conduction, Convection)

Exam Relevance: Conduction is a dependable scorer — JEE Main almost always carries a numerical on Fourier’s law, series/parallel rods, or junction temperature, and JEE Advanced reaches for the “growth of ice on a pond” integration or a composite-slab network. NEET sticks to conceptual MCQs: which mode works in vacuum, why metal feels colder than wood, and the sea-breeze/land-breeze story (convection). CBSE Boards ask a 2–3 mark definition of thermal conductivity, the units/dimensions of K , or a clean rate-of-heat-flow calculation through a wall or windowpane.

Section 1: Concept Intuition

Hold a steel spoon in a cup of hot chai and within seconds the far end — the bit your fingers are holding — turns uncomfortably warm. You never moved the spoon; nothing visibly travelled. Yet energy clearly crawled from the hot end to the cold end. That silent, particle-to-particle relay is **conduction**, and it is the first of the three ways heat moves from a hotter place to a cooler one.

In conduction, the material itself stays put. Near the hot end, atoms vibrate more violently; they jostle their neighbours, who jostle *their* neighbours, and the agitation marches down the rod like a whispered message passed along a row of seated people — each person stays in their seat, only the message travels. In metals there is a second, faster courier: free electrons, which drift and carry energy bodily. That extra channel is exactly why metals conduct heat (and electricity) so much better than wood, glass, or plastic. A metal door handle and a wooden one in the same room are at the *same* temperature, but the metal feels colder because it whisks heat out of your warm hand far faster — your skin, not the handle, is what you’re really sensing.

How fast does heat conduct? It depends on four sensible things, all bundled into one law (Fourier’s law of conduction):

$$\frac{dQ}{dt} = KA \frac{\Delta T}{L}.$$

Read it intuitively: heat flows faster through a *fatter* conductor (more area A), faster when the two ends differ more in temperature (bigger ΔT), slower through a *longer* path (bigger L), and the material’s own willingness to conduct is captured by K , the **thermal conductivity**. Copper has a huge K ; air and wool have tiny ones. This is the whole reason a woollen blanket warms you — wool itself isn’t “hot,” it simply has a low K (made lower still by the trapped air), so it throttles the escape of your body heat.

Think about it this way... picture heat flow as water flowing through a pipe. The temperature difference ΔT is the *pressure* pushing the water, and the heat current $H = dQ/dt$ is the *flow rate*. A short, wide, smooth pipe (large A , small L , large K) lets water gush; a long, narrow, clogged pipe (small A , large L , small K) chokes it. We even define a **thermal resistance** $R = L/KA$, so that $H = \Delta T/R$ — exactly Ohm’s law, with temperature difference playing the role of voltage and heat current the role of electric current. This single analogy turns frightening multi-rod problems into ordinary circuit problems, and we will lean on it hard.

Now the second mode. Conduction is hopeless in a glass of water heated from below — water is a poor conductor. So how does the whole pot get hot? The heated water at the bottom *expands, becomes less dense, and physically rises*, while cooler dense water sinks to take its place. The fluid literally carries its heat with it as it moves. This bulk transport of heat by a moving fluid is **convection**. Crucially, convection needs a fluid that is free to flow — it cannot happen in a solid.

Convection runs our world. The **sea breeze** on a Chennai afternoon exists because land (low

specific heat) heats faster than the sea; air over the hot land rises, and cooler sea air rushes in to replace it. At night the flow reverses into a **land breeze**. The same physics, scaled up enormously, drives the **monsoon**. Inside your home, a room heater warms the air at floor level, which rises and circulates — natural convection. When a fan or pump does the pushing instead of buoyancy, we call it **forced convection**; your body's own blood circulation, pumped by the heart to spread heat evenly, is a beautiful biological example.

Limiting conditions & assumptions. The clean formula $H = KA\Delta T/L$ assumes (i) **steady state** — temperatures at every point have stopped changing with time, so the same heat current flows through every cross-section; (ii) **no heat leaks sideways** — the rod's curved surface is perfectly insulated, so heat flows only along its length; (iii) a *uniform* cross-section and constant K . If the rod is still warming up (transient state), or loses heat from its sides, or K varies with temperature, the simple formula breaks and we must integrate. And note: this conduction law says nothing about convection or radiation — those obey entirely different rules.

Section 2: Key Formulas & Definitions

Fourier's law of conduction (steady state):

$$H = \frac{dQ}{dt} = KA \frac{(T_1 - T_2)}{L}$$

- H = heat current (rate of heat flow), unit watt (W).
- K = thermal conductivity; A = cross-sectional area; L = length; $(T_1 - T_2)$ = temperature difference across the ends.
- **Temperature gradient** = $\frac{T_1 - T_2}{L}$, unit K m^{-1} .

Thermal conductivity K :

- SI unit: $\text{W m}^{-1}\text{K}^{-1}$.
- Dimensional formula: $[MLT^{-3}\Theta^{-1}]$.
- High for metals (copper ≈ 400 , aluminium $\approx 235 \text{ W m}^{-1}\text{K}^{-1}$); low for insulators (glass ≈ 0.8 , wood ≈ 0.1 , air ≈ 0.025).

Thermal resistance (the electrical analogy):

$$R = \frac{L}{KA}, \quad H = \frac{\Delta T}{R}$$

- Unit of R : K W^{-1} . Temperature difference $\leftrightarrow$ voltage; heat current $\leftrightarrow$ current.

Rods in SERIES (joined end to end — same heat current through each):

$$R_{\text{eq}} = R_1 + R_2 + \dots, \quad H \text{ is common; } \Delta T \text{ divides in proportion to } R.$$

Rods in PARALLEL (side by side, same two end-temperatures):

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots, \quad \Delta T \text{ is common; } H \text{ adds.}$$

For two rods of equal length and area in parallel: $K_{\text{eq}} = \frac{K_1 + K_2}{2}$.

Convection (qualitative — no simple universal exam formula): bulk transport of heat by a moving fluid. **Natural convection** is driven by buoyancy (density differences); **forced convection** is driven by an external pump or fan. Requires a fluid; cannot occur in solids or vacuum.

Section 3: CBSE Derivation / Key Procedures — The Thermal-Resistance Network

The most useful “derivation” in this topic is establishing the series and parallel rules, because every hard conduction problem reduces to them. We build them from Fourier’s law using the resistance idea.

Setting up resistance. Rewrite Fourier’s law as $H = \frac{\Delta T}{(L/KA)}$. Defining $R \equiv \frac{L}{KA}$ gives the compact form $H = \frac{\Delta T}{R}$ — structurally identical to $I = V/R$. *Why this is worth doing:* it lets us combine conductors using the same arithmetic students already trust from circuits.

Series combination — derivation. Join two rods end to end between a hot reservoir at T_1 and a cold one at T_3 , with junction temperature T_2 .

Figure 10.3 Two conducting rods in series between reservoirs at T_1 and T_3 : the same steady heat current H flows through both, fixing the junction temperature $T_2 = (R_2 T_1 + R_1 T_3)/(R_1 + R_2)$. (see next page)

Step 1 — the key physical fact. In steady state no heat accumulates at the junction, so **the same heat current H flows through both rods**. (If more flowed in than out, the junction would keep heating — contradicting steady state.)

Step 2 — write each rod’s drop. From $H = \Delta T/R$:

$$T_1 - T_2 = HR_1, \quad T_2 - T_3 = HR_2.$$

Step 3 — add them. The junction temperature T_2 cancels:

$$T_1 - T_3 = H(R_1 + R_2).$$

Step 4 — identify. Comparing with $T_1 - T_3 = HR_{\text{eq}}$:

$$\boxed{R_{\text{eq}} = R_1 + R_2}$$

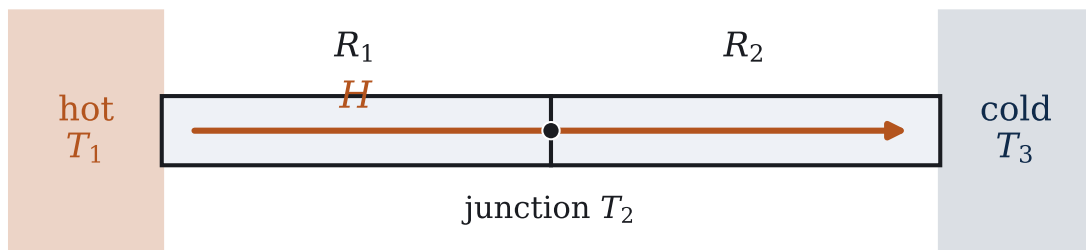
Series resistances add — and the larger drop appears across the larger resistance, just like voltage across a bigger resistor.

Parallel combination — derivation. Now place two rods side by side, both spanning the same hot face T_1 and cold face T_2 , so **both have the same ΔT** .

Step 1 — each carries its own current. $H_1 = \Delta T/R_1$ and $H_2 = \Delta T/R_2$.

Step 2 — currents add. The total heat delivered is $H = H_1 + H_2 = \Delta T \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$.

Step 3 — identify. Writing $H = \Delta T/R_{\text{eq}}$:



same heat current in steady state:

$$H = \frac{T_1 - T_2}{R_1} = \frac{T_2 - T_3}{R_2}, \quad T_2 = \frac{R_2 T_1 + R_1 T_3}{R_1 + R_2}$$

Figure 10.3 Two conducting rods in series between reservoirs at T_1 and T_3 : the same steady heat current H flows through both, fixing the junction temperature $T_2 = (R_2 T_1 + R_1 T_3)/(R_1 + R_2)$.

$$\boxed{\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}}$$

Junction-temperature shortcut. For two rods in series with the *same* A and L but different conductivities, equal heat currents give $K_1(T_1 - T_2) = K_2(T_2 - T_3)$, so

$$T_2 = \frac{K_1 T_1 + K_2 T_3}{K_1 + K_2}.$$

A clean weighted average — the junction sits closer to the temperature of the *better* conductor's reservoir. Keep this in your pocket; it cracks a whole class of MCQs in seconds.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A glass windowpane measures $1.5 \text{ m} \times 1.0 \text{ m}$ and is 4.0 mm thick. On a winter day the inside surface is at 22°C and the outside surface at 8°C . Find the rate at which heat is conducted out. Take $K_{\text{glass}} = 0.80 \text{ W m}^{-1} \text{ K}^{-1}$.

Given: $A = 1.5 \times 1.0 = 1.5 \text{ m}^2$, $L = 4.0 \times 10^{-3} \text{ m}$, $\Delta T = 22 - 8 = 14 \text{ K}$.

Formula: $H = \frac{KA\Delta T}{L}$.

Substitution:

$$H = \frac{(0.80)(1.5)(14)}{4.0 \times 10^{-3}} = \frac{16.8}{4.0 \times 10^{-3}}$$

$$\boxed{H = 4200 \text{ W} = 4.2 \text{ kW}}$$

Concluding remark: Over four kilowatts through one thin pane — which is exactly why double-glazed windows trap a layer of low- K air between two glass sheets to cut this loss dramatically.

Example 2 — [NEET Speed Trap]

A metal rod conducts heat at a steady rate H for a fixed temperature difference between its ends. If its length is halved and its radius is doubled, the new rate of heat conduction is:

The trap: students adjust for length and radius but forget that the cross-sectional area of a rod goes as r^2 , not r . They write “double the radius $\rightarrow$ double the area” and land on the wrong factor.

The fast way. Since $H \propto \frac{A}{L} \propto \frac{r^2}{L}$ (with ΔT and K fixed): - radius doubled $\Rightarrow r^2$ grows by $2^2 = 4$,
- length halved $\Rightarrow$ dividing by $L/2$ multiplies by 2.

$$H_{\text{new}} = H \times 4 \times 2 = \boxed{8H}$$

Speed insight: for circular rods, always think r^2 . A “doubled radius” is a *fourfold* area — that single reflex saves you from the most common NEET conduction error.

Example 3 — [JEE Main Level] (*series resistance + junction temperature*)

A copper rod ($K_1 = 400 \text{ W m}^{-1} \text{K}^{-1}$) and a steel rod ($K_2 = 50 \text{ W m}^{-1} \text{K}^{-1}$), each of length 0.50 m and cross-section 4.0 cm^2 , are joined end to end. The free copper end is held at 100°C and the free steel end at 0°C . Find (a) the rate of heat flow and (b) the junction temperature, in steady state.

Approach — treat it as a circuit. Convert each rod to a thermal resistance, add in series, then apply $H = \Delta T/R_{\text{eq}}$.

Step 1 — resistances ($A = 4.0 \times 10^{-4} \text{ m}^2$):

$$R_1 = \frac{L}{K_1 A} = \frac{0.50}{(400)(4.0 \times 10^{-4})} = 3.125 \text{ K W}^{-1}$$

$$R_2 = \frac{L}{K_2 A} = \frac{0.50}{(50)(4.0 \times 10^{-4})} = 25.0 \text{ K W}^{-1}$$

$$R_{\text{eq}} = R_1 + R_2 = 28.125 \text{ K W}^{-1}.$$

Step 2 — heat current:

$$H = \frac{\Delta T}{R_{\text{eq}}} = \frac{100 - 0}{28.125} \approx \boxed{3.56 \text{ W}}$$

Step 3 — junction temperature (drop across copper from the hot end):

$$T_{\text{junction}} = 100 - HR_1 = 100 - (3.56)(3.125) \approx \boxed{88.9^\circ\text{C}}$$

Notice the junction sits very close to the *hot* end — because copper’s tiny resistance produces only a tiny drop, while almost the entire 100° drop is spent across the steel. The poor conductor “hogs” the temperature difference, exactly like a large resistor hogs the voltage.

Example 4 — [JEE Advanced Level] — Growth of Ice on a Pond

A still pond is already covered by a 4.0 cm layer of ice. The air above stays at -10°C while the water just below the ice is at 0°C . How long will it take for the ice layer to thicken from 4.0 cm to 6.0 cm? Take $K_{\text{ice}} = 2.2 \text{ W m}^{-1} \text{K}^{-1}$, $\rho_{\text{ice}} = 920 \text{ kg m}^{-3}$, $L_f = 3.36 \times 10^5 \text{ J kg}^{-1}$.

Physical picture. New ice forms at the *bottom* of the sheet (the water–ice boundary, at 0°C). The latent heat released by this freezing must conduct *upward* through the existing ice thickness x to the cold air. As the sheet thickens, the conduction path lengthens and freezing slows — so the rate is not constant, and we must integrate.

Figure 10.4 A freezing pond: the heat released as a new layer dx of ice forms at the ice–water boundary (0°C) conducts up through the existing ice of thickness x to the -10°C air, so the sheet grows ever more slowly. (see next page)

Step 1 — heat conducted up through thickness x (per unit area A):

$$\frac{dQ}{dt} = \frac{KA(0 - (-10))}{x} = \frac{KA\theta}{x}, \quad \theta = 10^\circ\text{C}.$$

Step 2 — heat released by freezing a layer dx . Mass frozen $= \rho A dx$, so $dQ = \rho L_f A dx$.

Step 3 — equate (energy conservation):

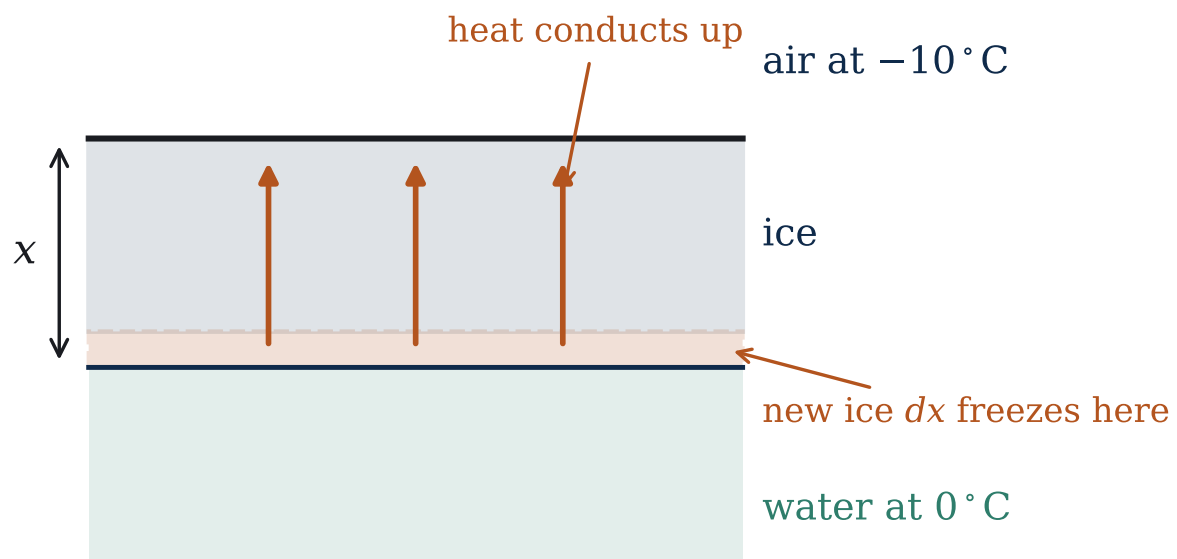


Figure 10.4 A freezing pond: the heat released as a new layer dx of ice forms at the ice-water boundary (0°C) conducts up through the existing ice of thickness x to the -10°C air, so the sheet grows ever more slowly.

$$\rho L_f A \frac{dx}{dt} = \frac{KA\theta}{x} \Rightarrow x dx = \frac{K\theta}{\rho L_f} dt.$$

Step 4 — integrate from $x_1 = 0.04$ m to $x_2 = 0.06$ m, time 0 to t :

$$\frac{x_2^2 - x_1^2}{2} = \frac{K\theta}{\rho L_f} t \Rightarrow t = \frac{\rho L_f (x_2^2 - x_1^2)}{2K\theta}.$$

Step 5 — substitute:

$$t = \frac{(920)(3.36 \times 10^5)[(0.06)^2 - (0.04)^2]}{2(2.2)(10)} = \frac{(3.0912 \times 10^8)(2.0 \times 10^{-3})}{44}$$

$$t = \frac{6.182 \times 10^5}{44} \approx \boxed{1.41 \times 10^4 \text{ s} \approx 3.9 \text{ hours}}$$

The crucial insight — that thickening time scales as x^2 — means the *second* centimetre of ice takes far longer than the first. This is why a frozen lake's ice layer grows quickly at first and then crawls.

Section 5: Practice Exercises

- [CBSE]** A brick wall of a cold-storage room is $5.0 \text{ m} \times 3.0 \text{ m}$ and 20 cm thick. The outside is at 40°C and the inside at 25°C . Find the rate at which heat enters. Take $K_{\text{brick}} = 0.70 \text{ W m}^{-1}\text{K}^{-1}$.
- [NEET]** Two rods of the same material: rod P has length L and radius r ; rod Q has length $2L$ and radius $2r$. For the same temperature difference across their ends, find the ratio of their heat currents $H_P : H_Q$.
- [JEE Main]** Two rods of equal length and equal cross-section, with thermal conductivities 200 and $100 \text{ W m}^{-1}\text{K}^{-1}$, are placed side by side (in parallel) between the same hot and cold faces. Find the equivalent thermal conductivity of the combination.
- [JEE Main]** One end of a copper rod ($K = 400 \text{ W m}^{-1}\text{K}^{-1}$) of length 1.0 m and cross-section 5.0 cm^2 is held at 100°C ; the other end touches ice at 0°C . Find the rate at which ice melts. ($L_f = 3.36 \times 10^5 \text{ J kg}^{-1}$.)
- [JEE Advanced]** Starting from open water, ice begins to form on a pond when the air above is at -15°C . Find the time taken for the ice to reach a thickness of 5.0 cm . Take $K_{\text{ice}} = 2.2 \text{ W m}^{-1}\text{K}^{-1}$, $\rho_{\text{ice}} = 920 \text{ kg m}^{-3}$, $L_f = 3.36 \times 10^5 \text{ J kg}^{-1}$.

Answers: 1. $\approx 788 \text{ W}$ 2. $H_P : H_Q = 1 : 2$ 3. $150 \text{ W m}^{-1}\text{K}^{-1}$ 4. $\approx 5.95 \times 10^{-5} \text{ kg s}^{-1}$ ($\approx 0.21 \text{ kg per hour}$) 5. $\approx 1.17 \times 10^4 \text{ s}$ ($\approx 3.3 \text{ hours}$).

Section 6: Crack the MCQ

Q1. Which mode(s) of heat transfer can carry energy across a perfect vacuum?

- (a) conduction only (b) convection only (c) radiation only (d) all three

Correct: (c). Conduction needs a material lattice to pass vibrations along, and convection needs a fluid to physically move — both require matter. Only **radiation** travels through empty space (which is how the Sun's energy reaches Earth). - (a) and (b) both invoke a medium that a vacuum

lacks. - (d) is the careless “more is safer” pick; it ignores that two of the three modes are impossible without matter.

Q2. On a cold morning, a metal chair feels much colder to the touch than a wooden chair standing right beside it, even though both are at the same temperature. This is because the metal:

- (a) is actually at a lower temperature (b) conducts heat away from your hand much faster than wood (c) has a higher specific heat than wood (d) radiates more heat than wood

Correct: (b). Both objects are at room temperature; the metal’s high thermal conductivity drains heat from your warm skin rapidly, so your nerves register “cold.” Wood, a poor conductor, barely draws heat away. - (a) contradicts the given (same temperature) — the classic trap. - (c) and (d) name real properties but neither is the cause; the sensation is about *rate of heat removal from your hand*, i.e. conductivity.

Q3. During the daytime near a coast, a cool breeze blows from the sea toward the land. The best explanation is:

- (a) the sea is hotter than the land, so air flows from hot to cold (b) the land heats faster than the sea; air over the land rises and cooler sea air moves in to replace it (c) the wind is caused purely by the Earth’s rotation (d) water vapour from the sea pushes the air inland

Correct: (b). Land has a lower specific heat than water, so by afternoon it is hotter; the warm air above it rises (natural convection), and denser cool air from over the sea flows in underneath — the sea breeze. - (a) inverts the temperatures. - (c) and (d) invoke unrelated effects; the driver here is convective circulation from differential heating.

Q4. Two rods of the same length and cross-section but different conductivities are joined in series, with their outer ends maintained at fixed different temperatures. In steady state:

- (a) the heat current is larger in the rod with higher conductivity (b) the temperature drop is larger across the rod with lower conductivity (c) the temperature drop is the same across both rods (d) the better conductor has the larger temperature drop

Correct: (b). In series the heat current is *identical* in both rods, so the drop $\Delta T = HR$ is largest where the resistance $R = L/KA$ is largest — i.e. across the *lower*-conductivity rod. - (a) violates steady-state continuity of heat current. - (c) would hold only if the resistances were equal. - (d) reverses the correct relationship — the poor conductor takes the big drop, like a large resistor takes the big voltage.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Forgetting $A \propto r^2$.** For a cylindrical rod the cross-section is πr^2 . “Doubling the radius” quadruples the area and hence the heat current. Many marks are lost to a stray factor here.
2. **Confusing series and parallel.** Rods *end to end* are in **series** (resistances add, heat current common). Rods *side by side* spanning the same faces are in **parallel** (conductances add, ΔT common). Map it onto a circuit before computing.
3. **Thinking ΔT is shared equally in series.** It isn’t — the heat *current* is shared (equal in each rod), while ΔT divides in proportion to each resistance. The poor conductor takes the bigger drop.
4. **Applying conduction formulas in the transient stage.** $H = KA\Delta T/L$ holds only in *steady state*. If the rod is still warming up, the heat current differs along its length and the formula doesn’t apply.

Pro-Tip: Treat every conduction problem as a DC circuit. Replace each segment by $R = L/KA$, redraw the network, and use $H = \Delta T/R_{\text{eq}}$ with the series/parallel rules you already know. Junction temperatures then fall out as “voltage drops.” This one habit converts the scariest multi-rod JEE Advanced problem into routine circuit arithmetic.

Section 8: Cheat Sheet Box

- **Fourier’s law:** $H = \frac{dQ}{dt} = \frac{KA\Delta T}{L}$, unit of H is watt.
- **Thermal conductivity K :** unit $\text{W m}^{-1}\text{K}^{-1}$, dimensions $[MLT^{-3}\Theta^{-1}]$.
- **Electrical analogy:** $R = \frac{L}{KA}$, $H = \frac{\Delta T}{R}$ — temperature drop $\leftrightarrow$ voltage, heat current $\leftrightarrow$ current.
- **Series:** $R_{\text{eq}} = R_1 + R_2$ (same H). **Parallel:** $\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$ (same ΔT).
- **Junction temp (equal A, L):** $T_2 = \frac{K_1T_1 + K_2T_3}{K_1 + K_2}$.
- **Ice growth on a pond:** $t = \frac{\rho L_f(x_2^2 - x_1^2)}{2K\theta}$ — time scales as thickness squared.
- **Convection:** bulk heat transport by moving fluid; natural (buoyancy) vs forced (pump/fan); needs a fluid. *Mnemonic: “Solids conduct, fluids convect, vacuum only radiates.”*

Subtopic 04 — Radiation and Cooling Laws

Exam Relevance: A high-yield subtopic. JEE Main reliably pulls a Stefan–Boltzmann (T^4) numerical, a Wien’s-law wavelength–temperature question, or a Newton’s-law-of-cooling problem. NEET loves the conceptual MCQs — good absorber = good emitter (Kirchhoff), why heated iron changes colour (Wien), and the black-body definition — plus an occasional T^4 ratio. JEE Advanced combines Wien + Stefan (e.g. finding a star’s temperature and power) or sets up a multi-step cooling problem. CBSE Boards ask the statement of Stefan’s/Wien’s law, the definition of a black body, or the **derivation of Newton’s law of cooling from Stefan’s law** — a perennial favourite.

Section 1: Concept Intuition

Stand outside on a clear winter morning in Delhi. The air is freezing, yet the instant the Sun clears the rooftops you feel its warmth on your face. That warmth did not arrive by conduction — air is a poor conductor. It did not arrive by convection — the Sun’s heat crossed 150 million kilometres of near-total vacuum, where there is no fluid to carry anything. It arrived by **radiation**: energy emitted as electromagnetic waves that need *no medium at all* to travel. This is the crucial, defining feature of the third mode of heat transfer, and the one thing you must never forget — radiation alone can cross empty space.

Here is the first surprising fact: **every object radiates, all the time, at every temperature** — you, this page, an ice cube, a star. This is **Prevost’s theory of exchange**: a body is never “off”; it is always both emitting radiation to its surroundings and absorbing radiation from them. What you observe as heating or cooling is simply the *net* balance. A hot cup of chai in a cool room radiates faster than it absorbs, so it loses energy and cools. An ice cube in a warm room absorbs faster than it radiates, so it gains energy and warms. And when an object reaches the temperature of its surroundings, it does not stop radiating — it simply radiates and absorbs at *equal* rates, a dynamic standoff we call thermal equilibrium.

Think about it this way... imagine a crowded room where everyone is continuously tossing tennis balls at everyone else and catching the ones thrown at them. Each ball is a packet of radiant energy. A “hot” person throws balls faster than the room average; if they throw more than they catch, their pile (energy) shrinks — they cool. A “cold” person catches more than they throw and their pile grows — they warm. Eventually everyone throws and catches at the same rate: equilibrium, with balls still flying everywhere. Nobody ever stops throwing. That ceaseless two-way exchange is exactly Prevost’s picture.

How fast a body radiates depends fiercely on its temperature. Double the absolute temperature and the radiated power leaps by a factor of **sixteen** — because power scales as the *fourth* power of temperature (Stefan–Boltzmann law). This brutal sensitivity is why a slightly hotter star outshines a cooler one so dramatically, and why “rate of cooling” problems are so temperature-dependent.

Temperature also controls the *colour* of the glow. Heat an iron rod in a furnace and watch it: first it glows a dull red, then orange, then yellow, and if hot enough, brilliant white. As it gets hotter, the wavelength at which it radiates most strongly slides toward the *shorter* (bluer) end of the spectrum. This is **Wien’s displacement law** — hotter means bluer. It is the same physics that tells an astronomer that a blue-white star is far hotter than a red one.

Finally, a beautifully simple rule connects absorbing and emitting. Place a shiny polished can and a black-painted can of boiling water side by side; the black one cools noticeably faster. Why? Because **good absorbers of radiation are also good emitters** — this is **Kirchhoff’s law**. A surface that greedily soaks up radiation also pours it out greedily. The ideal limit is a **black body**: a body that absorbs *all* radiation falling on it (reflecting and transmitting none) and therefore also emits

at the maximum possible rate for its temperature. No real surface is perfectly black, but a small hole in a closed cavity comes astonishingly close — any ray entering the hole bounces around inside and is absorbed before it can find its way out. This is also why a thermos (Dewar) flask defeats all three modes at once: silvered walls reflect radiation, and the vacuum gap between them kills conduction and convection.

Limiting conditions & assumptions. Stefan's law in its pure form $E = \sigma T^4$ holds for a *perfect black body*; for a real surface we insert the **emissivity** e (between 0 and 1) and write $P = e\sigma AT^4$. The *net* exchange formula $P_{\text{net}} = e\sigma A(T^4 - T_0^4)$ assumes the surroundings are at a single uniform temperature T_0 . And **Newton's law of cooling** — that the rate of cooling is proportional to the temperature excess — is only an *approximation*, valid when the temperature difference between the body and its surroundings is small (as we will prove, it is just Stefan's law in disguise for small excess).

Section 2: Key Formulas & Definitions

Stefan–Boltzmann Law (total power radiated):

$$P = e\sigma AT^4 \quad (\text{emitted}), \quad P_{\text{net}} = e\sigma A(T^4 - T_0^4)$$

- P = radiated power (W); e = emissivity ($0 \leq e \leq 1$, dimensionless; $e = 1$ for a black body); A = surface area; T = absolute temperature of the body; T_0 = absolute temperature of surroundings.
- **Stefan–Boltzmann constant:** $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$. Dimensional formula: $[MT^{-3}\Theta^{-4}]$.
- **Emissive power** E = energy radiated per unit area per unit time (W m^{-2}); for a black body $E = \sigma T^4$.

Wien's Displacement Law (wavelength of peak emission):

$$\lambda_m T = b$$

- λ_m = wavelength at which the emission is maximum; T = absolute temperature.
- **Wien's constant:** $b = 2.9 \times 10^{-3} \text{ m K}$. Dimensional formula: $[L\Theta]$. *Hotter body* $\rightarrow$ *smaller* $\lambda_m \rightarrow$ *bluer*.

Kirchhoff's Law: At a given temperature, the ratio of emissive power to absorptive power is the same for all bodies and equals the emissive power of a black body:

$$\frac{e_\lambda}{a_\lambda} = E_\lambda^{(\text{black body})} = \text{constant} \quad \Rightarrow \quad \text{good absorbers are good emitters.}$$

- Numerically, **emissivity = absorptivity** for a surface at a given temperature.

Newton's Law of Cooling (small temperature excess):

$$-\frac{dT}{dt} = k(T - T_0)$$

- k = a positive constant depending on the body's area, nature of surface, and specific heat.
- Practical (average) form for a body cooling from T_1 to T_2 in time t , surroundings at T_0 :

$$\frac{T_1 - T_2}{t} = k \left(\frac{T_1 + T_2}{2} - T_0 \right).$$

Black body: an ideal body that absorbs all incident radiation (absorptivity = 1) and emits the maximum possible radiation at every wavelength for its temperature.

Section 3: CBSE Derivation — Newton’s Law of Cooling from Stefan’s Law

This is the most-asked derivation of the subtopic. The goal: show that the famous “rate of cooling $\propto$ temperature excess” is really just the Stefan–Boltzmann law, valid only when the excess is small.

Initial setup. Consider a body of mass m , specific heat c , surface area A , and emissivity e , at absolute temperature T , placed in surroundings at temperature T_0 (with T only slightly greater than T_0).

Step 1 — net rate of energy loss (Stefan’s law). The body emits and absorbs simultaneously; the net power lost is

$$P_{\text{net}} = e\sigma A (T^4 - T_0^4).$$

Step 2 — relate to temperature drop. This lost power comes out of the body’s internal energy, so its temperature falls:

$$-mc \frac{dT}{dt} = e\sigma A (T^4 - T_0^4) \Rightarrow -\frac{dT}{dt} = \frac{e\sigma A}{mc} (T^4 - T_0^4).$$

Step 3 — write $T = T_0 + \Delta T$ with ΔT small. Now expand T^4 using the binomial approximation. Since $T = T_0(1 + \Delta T/T_0)$,

$$T^4 = T_0^4 \left(1 + \frac{\Delta T}{T_0}\right)^4 \approx T_0^4 \left(1 + \frac{4\Delta T}{T_0}\right) = T_0^4 + 4T_0^3 \Delta T,$$

where we kept only the first-order term because $\Delta T/T_0 \ll 1$ (this is the *crucial* approximation — it is exactly where “small excess” enters).

Step 4 — substitute. Then $T^4 - T_0^4 \approx 4T_0^3 \Delta T$, so

$$-\frac{dT}{dt} = \frac{e\sigma A}{mc} (4T_0^3) \Delta T = \underbrace{\left(\frac{4e\sigma AT_0^3}{mc}\right)}_{\equiv k} (T - T_0).$$

Step 5 — conclude.

$$-\frac{dT}{dt} = k(T - T_0)$$

This is precisely Newton’s law of cooling, and the derivation reveals two things a memoriser never sees: (i) the cooling constant $k = 4e\sigma AT_0^3/mc$ depends on the surroundings’ temperature, the surface, and the body’s heat capacity; and (ii) the law is fundamentally an **approximation** — it holds only while $\Delta T \ll T_0$, because we threw away the higher-order terms of the binomial. For a large temperature difference, the full T^4 behaviour dominates and the simple proportionality fails.

Figure 10.5 *Cooling curve: the temperature falls steeply at first and flattens as it approaches the surroundings temperature T_s ; Newton’s law $dT/dt = -k(T - T_s)$ is the small-excess approximation and fails when the difference is large. (see next page)*

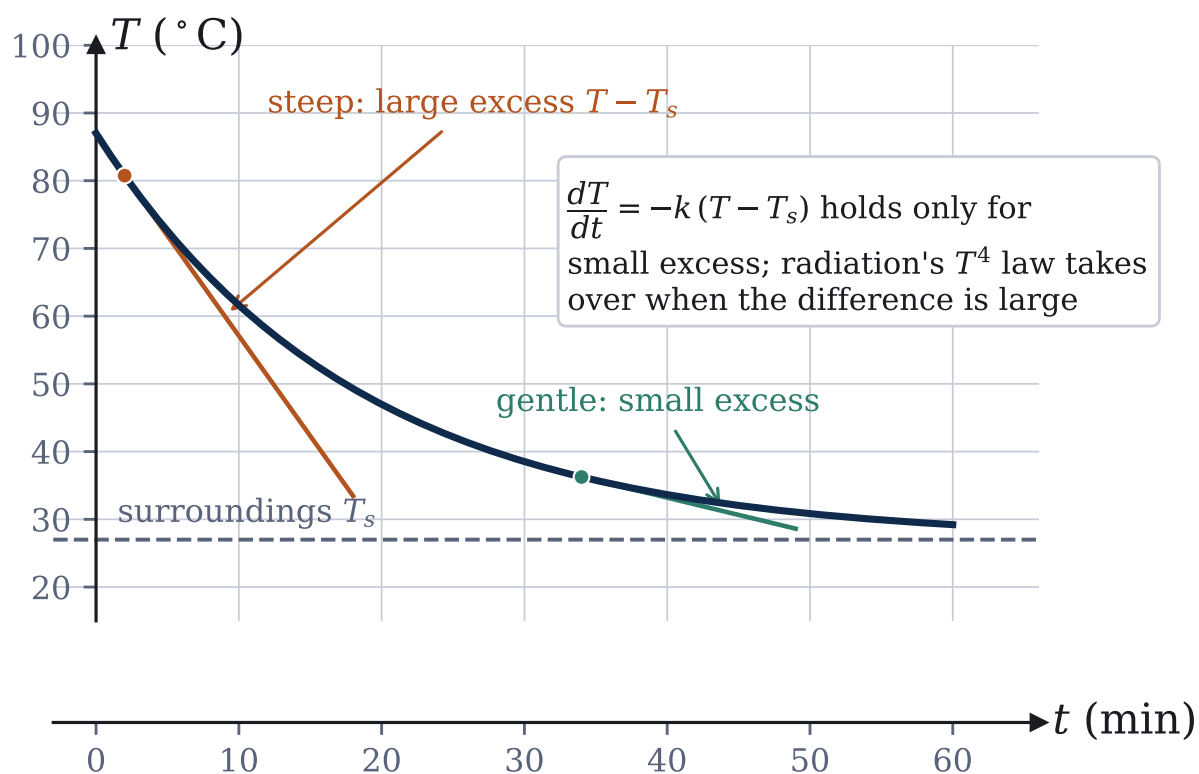


Figure 10.5 Cooling curve: the temperature falls steeply at first and flattens as it approaches the surroundings temperature T_s ; Newton's law $dT/dt = -k(T - T_s)$ is the small-excess approximation and fails when the difference is large.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A spherical black body of radius 5.0 cm is maintained at 327°C in surroundings at 27°C. Find the net power it radiates. Take $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

Given (convert to kelvin): $T = 327 + 273 = 600 \text{ K}$, $T_0 = 27 + 273 = 300 \text{ K}$, $r = 0.05 \text{ m}$, $e = 1$.

Surface area: $A = 4\pi r^2 = 4\pi(0.05)^2 = 4\pi(2.5 \times 10^{-3}) = 3.14 \times 10^{-2} \text{ m}^2$.

Formula: $P_{\text{net}} = \sigma A (T^4 - T_0^4)$.

Substitution:

$$T^4 - T_0^4 = (600)^4 - (300)^4 = 1.296 \times 10^{11} - 0.081 \times 10^{11} = 1.215 \times 10^{11} \text{ K}^4$$

$$P_{\text{net}} = (5.67 \times 10^{-8})(3.14 \times 10^{-2})(1.215 \times 10^{11})$$

$$P_{\text{net}} \approx 216 \text{ W}$$

Concluding remark: Always convert temperatures to kelvin *before* taking the fourth power, and don't drop the T_0^4 term — the surroundings are radiating back at the body.

Example 2 — [NEET Speed Trap]

The absolute temperature of a body is raised from 27°C to 327°C. By what factor does the power it radiates increase?

The trap (two errors at once): students plug Celsius numbers and write $(327/27)^4$, or they forget the fourth power and answer “2”. Both are wrong.

The fast way. Convert to kelvin: $300 \text{ K} \rightarrow 600 \text{ K}$, a ratio of exactly 2. Since $P \propto T^4$:

$$\frac{P_2}{P_1} = \left(\frac{T_2}{T_1}\right)^4 = \left(\frac{600}{300}\right)^4 = 2^4 = 16$$

Speed insight: for any radiation-ratio question, work in kelvin and use $P_2/P_1 = (T_2/T_1)^4$ — never compute σ or area. Forgetting to convert °C → K is the single deadliest trap on this topic, because the fourth power magnifies the error enormously.

Example 3 — [JEE Main Level] (Wien + Stefan combined)

The radiation from a certain star peaks in intensity at a wavelength of 500 nm. Treating the star as a black body, find (a) its surface temperature, and (b) the power it radiates per unit area of its surface. Take Wien's constant $b = 2.9 \times 10^{-3} \text{ m K}$ and $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

Part (a) — Wien's law.

$$T = \frac{b}{\lambda_m} = \frac{2.9 \times 10^{-3}}{500 \times 10^{-9}} = \frac{2.9 \times 10^{-3}}{5.0 \times 10^{-7}} = 5800 \text{ K}$$

Part (b) — Stefan's law (per unit area, black body):

$$E = \sigma T^4 = (5.67 \times 10^{-8})(5800)^4.$$

Compute $(5800)^4 = (5.8 \times 10^3)^4 = (5.8)^4 \times 10^{12} = 1131.6 \times 10^{12} = 1.132 \times 10^{15}$:

$$E = (5.67 \times 10^{-8})(1.132 \times 10^{15}) \approx \boxed{6.4 \times 10^7 \text{ W m}^{-2}}$$

This two-step pattern — Wien to get T from the spectrum, then Stefan to get the power — is the standard JEE Main “astrophysics flavoured” question. Note T landing near 5800 K, which is indeed close to the Sun’s surface temperature.

Example 4 — [JEE Advanced Level] — Newton’s Law of Cooling

A metal block cools from 80°C to 60°C in 5.0 minutes when left in a room at 20°C . How long will it take to cool further from 60°C to 40°C , assuming Newton’s law of cooling holds?

Step 1 — first stage gives the constant k . Use the average-temperature form, with the mean temperature of the body during the interval:

$$\frac{T_1 - T_2}{t} = k \left(\frac{T_1 + T_2}{2} - T_0 \right)$$

$$\frac{80 - 60}{5} = k \left(\frac{80 + 60}{2} - 20 \right) \Rightarrow 4 = k(70 - 20) = 50k \Rightarrow k = 0.08 \text{ min}^{-1}.$$

Step 2 — second stage uses the same k . Mean temperature now is $(60 + 40)/2 = 50^\circ\text{C}$:

$$\frac{60 - 40}{t} = k \left(\frac{60 + 40}{2} - 20 \right) = (0.08)(50 - 20) = (0.08)(30) = 2.4.$$

Step 3 — solve for t :

$$\frac{20}{t} = 2.4 \Rightarrow \boxed{t \approx 8.3 \text{ minutes}}$$

The deeper point: the *same* 20°C drop takes longer (8.3 min vs 5 min) the second time, because the block is now cooler and closer to the room — a smaller temperature excess means a slower rate of cooling. That slowing-down-as-you-approach-the-surroundings behaviour is the entire character of Newton’s law, visible right in the numbers.

Section 5: Practice Exercises

1. [CBSE] A black body of surface area 20 cm^2 is maintained at 727°C . Find the power it radiates. Take $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$.
2. [NEET] The radiation emitted by a hot body has its maximum intensity at a wavelength of $1.45 \mu\text{m}$. Find the body’s absolute temperature. (Wien’s constant $= 2.9 \times 10^{-3} \text{ m K}$.)
3. [JEE Main] A body radiates a certain power at 727°C . At what temperature (in $^\circ\text{C}$) will it radiate **16 times** that power, all else unchanged?
4. [JEE Main] A liquid cools from 70°C to 50°C in 6.0 minutes in surroundings at 20°C . Using Newton’s law of cooling, find the time taken to cool from 50°C to 40°C .

5. [JEE Advanced] A body cools from 60°C to 50°C in the first 10 minutes when the surroundings are at 25°C . Assuming Newton's law of cooling, find its temperature at the end of the next 10 minutes.

Answers: 1. $\approx 113\text{ W}$ 2. 2000 K 3. 1727°C (since $16 = 2^4$, the absolute temperature doubles: $1000\text{ K} \rightarrow 2000\text{ K}$) 4. ≈ 4.8 minutes 5. $\approx 42.9^{\circ}\text{C}$.

Section 6: Crack the MCQ

Q1. Two identical kettles, one with a shiny polished surface and one painted dull black, are filled with boiling water and left in the same room. Which cools faster, and why?

- (a) the polished kettle, because shiny surfaces lose heat faster (b) the black kettle, because good absorbers are also good emitters (c) both cool at the same rate, since they hold the same water (d) the black kettle, because black objects are always hotter

Correct: (b). By Kirchhoff's law a good absorber is a good emitter; the black surface radiates more strongly, so it loses heat faster and cools quicker. - (a) inverts the physics — polished surfaces are *poor* emitters. - (c) ignores that the rate of radiative loss depends on the *surface*, not just the contents. - (d) is nonsense; both start at the same temperature.

Q2. If the absolute temperature of a black body is doubled, the total energy it radiates per unit time becomes:

- (a) 2 times (b) 4 times (c) 8 times (d) 16 times

Correct: (d). Stefan's law gives $P \propto T^4$, so doubling T multiplies the power by $2^4 = 16$. - (a) assumes a linear dependence. - (b) and (c) are the traps for students who use T^2 or T^3 — easy to misremember the exponent.

Q3. As a piece of iron is heated to higher and higher temperatures, the wavelength at which it radiates most intensely:

- (a) increases (shifts toward red) (b) decreases (shifts toward blue) (c) stays the same (d) first increases, then decreases

Correct: (b). Wien's law, $\lambda_m T = \text{constant}$, means a higher T gives a *smaller* λ_m — the peak shifts toward shorter (bluer) wavelengths. This is why heated iron goes from dull red to white-hot. - (a) reverses the relationship. - (c) would require λ_m independent of T , contradicting Wien. - (d) invents a non-existent turnaround.

Q4. Newton's law of cooling is best described as:

- (a) an exact law valid for all temperature differences (b) a special case of Stefan's law, valid only when the temperature difference between body and surroundings is small (c) applicable only to black bodies (d) independent of the surface area of the body

Correct: (b). Newton's law follows from Stefan's T^4 law by a first-order (binomial) approximation that is valid only for a *small* excess temperature. - (a) is false — it fails for large differences. - (c) is wrong; it applies to ordinary bodies (via emissivity), not only black bodies. - (d) is wrong; the cooling constant k explicitly contains the area A .

Section 7: Common Student Pitfalls & Pro-Tips

1. **Using Celsius in Stefan's or Wien's law.** Both demand *absolute* (kelvin) temperature. With the fourth power in Stefan's law, a stray “use of $^{\circ}\text{C}$ ” produces wildly wrong answers.

Convert first, always.

2. **Dropping the surroundings term.** Net radiation is $e\sigma A(T^4 - T_0^4)$, not $e\sigma AT^4$. A body in a warm room is also being heated by the room's radiation.
3. **Confusing emissive power with emissivity.** Emissive power E has units W m^{-2} ; emissivity e is a dimensionless number between 0 and 1. A black body has $e = 1$, but its emissive power σT^4 is certainly not 1.
4. **Applying Newton's law of cooling to large differences.** It is an approximation for *small* excess only. For a body far hotter than its surroundings, use the full Stefan T^4 relation.

Pro-Tip: For any “by what factor” radiation question, skip the constants entirely and use ratios in kelvin: $\frac{P_2}{P_1} = \left(\frac{T_2}{T_1}\right)^4$ for power and $\frac{\lambda_2}{\lambda_1} = \frac{T_1}{T_2}$ for peak wavelength. For cooling problems, the average-temperature form $\frac{T_1 - T_2}{t} = k\left(\frac{T_1 + T_2}{2} - T_0\right)$ turns a differential equation into one line of arithmetic.

Section 8: Cheat Sheet Box

- **Radiation** needs no medium (crosses vacuum); **all bodies radiate at all temperatures** (Prevost's exchange theory).
- **Stefan–Boltzmann:** $P = e\sigma AT^4$; net $P = e\sigma A(T^4 - T_0^4)$; $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$, dimensions $[MT^{-3}\Theta^{-4}]$.
- **Wien:** $\lambda_m T = b$, $b = 2.9 \times 10^{-3} \text{ m K}$ — *hotter = bluer (shorter λ_m)*.
- **Kirchhoff:** good absorbers are good emitters; emissivity = absorptivity. **Black body:** absorbs all, $e = 1$.
- **Newton's cooling:** $-\frac{dT}{dt} = k(T - T_0)$ — *an approximation of Stefan's law for small excess*;
 $k = \frac{4e\sigma AT_0^3}{mc}$.
- **Mnemonics:** “Stefan: power to the fourth (in Kelvin!)”; “Wien: hot is blue, cool is red”; “Black absorbs all, emits all.”

Thermal Properties of Matter

Master Content — Part II

Supplementary Subtopics 05–06 (Coverage Completion)

Chapter 10 · Class 11 Physics — CBSE · JEE Main · NEET · JEE Advanced
June 2026

Why this part exists. Part I (Subtopics 01–04: Temperature/Heat/Expansion, Calorimetry & Specific Heat, Heat Transfer by Conduction/Convection, and Radiation & Cooling Laws) is strong, but two full NCERT sections of this chapter were not yet present. Without them a student cannot claim to “know everything” about Thermal Properties of Matter. Part II closes those gaps using the same eight-section template and the same difficulty ladder (CBSE Board → NEET Speed Trap → JEE Main → JEE Advanced):

- **Subtopic 05 — Thermometry, the Ideal-Gas Equation & Absolute Temperature:** how temperature is actually *measured* (thermometric properties, thermometer types and ranges, the constant-volume gas thermometer), and how the absolute (Kelvin) scale and absolute zero arise from the ideal-gas laws. Part I gave the scale *conversions*; this gives the physics of measurement and the gas laws behind the Kelvin scale.
- **Subtopic 06 — Change of State: Phase Transitions, Triple Point & Phase Diagrams:** the conceptual phase-change physics that Part I’s calorimetry handled only numerically — melting/boiling points and their pressure dependence (regelation, the pressure cooker, cooking on mountains), sublimation, the triple point, the P – T phase diagram, the critical point, and evaporation versus boiling. This is the single richest source of NEET/CBSE conceptual MCQs in the chapter.

Contents of Part II

- Subtopic 05 — Thermometry, the Ideal-Gas Equation & Absolute Temperature
- Subtopic 06 — Change of State: Phase Transitions, Triple Point & Phase Diagrams

Subtopic 05 — Thermometry, the Ideal-Gas Equation & Absolute Temperature

Exam Relevance

A reliable source of definition and short-numerical marks. CBSE Boards regularly ask the definition of the triple point, the ideal-gas equation, absolute zero, or a thermometer's thermometric property (1–3 marks). JEE Main occasionally sets a constant-volume gas-thermometer or a gas-law numerical (find the new pressure/temperature). NEET favours conceptual MCQs — which thermometer suits a given range, why absolute zero is unreachable, why dilute gases all agree. JEE Advanced rarely asks it standalone but uses absolute temperature as the gateway to kinetic theory and thermodynamics, so a clean grasp here pays off across three chapters.

Section 1: Concept Intuition

Part I (Subtopic 01) told you *what* temperature is and how to convert between scales. This subtopic answers the next question: how do we actually *measure* it, and where does the absolute (Kelvin) scale come from?

You cannot read temperature directly — you read it through some property of a substance that changes reproducibly with hotness. Such a property is called a **thermometric property**, and the device built around it is a thermometer. The mercury-in-glass thermometer uses the *length* of a mercury column; a constant-volume gas thermometer uses the *pressure* of a trapped gas; a platinum resistance thermometer uses electrical *resistance*; a thermocouple uses a tiny *thermo-emf*; and a radiation pyrometer uses the *radiation* emitted by a very hot body (so it needs no contact at all). The same differential-expansion idea even runs your geyser's thermostat — a *bimetallic strip* of two metals with different α bends as it heats and trips a switch.

To turn a property into a number you need calibration: two reproducible **fixed points**. The classic pair are the *ice point* (melting ice at 1 atm, 0°C) and the *steam point* (boiling water at 1 atm, 100°C). Mark the property at both, divide the span into 100 equal parts, and you have a Celsius thermometer. So far, so neat.

Here is the subtlety that forced physicists to find a better standard. A mercury thermometer and an alcohol thermometer, both perfectly calibrated at 0° and 100° , will *disagree* slightly at, say, 50° — because mercury and alcohol do not expand in exactly the same non-linear way in between. Which one tells the “true” temperature? Neither, exactly. The escape is the **gas thermometer**: experiment shows that as you make a gas more and more dilute (lower pressure, approaching ideal behaviour), *all* gases — hydrogen, helium, nitrogen — give the same temperature reading. That universality is why the ideal-gas temperature is adopted as the fundamental standard against which all other thermometers are checked.

Think about it this way... A thermometer is a translator that converts invisible “hotness” into a visible number using some property as its language. Different translators (mercury, alcohol, resistance) speak slightly different dialects and disagree on the fine print. The dilute gas is the one translator everyone trusts, because in the thin-gas limit they all tell exactly the same story — so we make it the official language.

From gas laws to absolute temperature. For a fixed mass of gas, three experimental laws hold in the ideal limit: Boyle's law ($PV = \text{const}$ at fixed T), Charles's law ($V \propto T$ at fixed P), and Gay-Lussac's law ($P \propto T$ at fixed V). Plot the pressure of a fixed-volume gas against Celsius temperature and you get a straight line. Extend that line backward (extrapolate) and it crosses

zero pressure at -273.15°C — and astonishingly, gases of *every* kind extrapolate to the *same* point. A gas can have no less than zero pressure, so this common intercept is the coldest conceivable temperature: **absolute zero**. Shifting the origin to this point defines the **Kelvin scale**, on which pressure (or volume) is simply *proportional* to temperature. Combining the three laws gives the **ideal-gas equation** $PV = nRT$ — the cleanest relationship in all of thermal physics, and the reason the gas thermometer is so fundamental.

Limiting Conditions & Assumptions:

- The ideal-gas laws hold only at *low pressure and high temperature* (far from liquefaction). Real gases deviate near condensation; that is why the *dilute* limit is the one used for the standard.
- Absolute zero can be approached but never reached — it is an extrapolated limit, not an attainable state.
- Each practical thermometer has a useful range set by its substance (mercury freezes at -39°C ; a radiation pyrometer is needed above $\sim 1000^\circ\text{C}$). Using one outside its range gives nonsense.

Section 2: Key Formulas & Definitions

Thermometric (linear) relation — temperature t from a property X calibrated at the ice (X_0) and steam (X_{100}) points:

$$t = \frac{X_t - X_0}{X_{100} - X_0} \times 100^\circ\text{C}.$$

(For scale-to-scale conversion — C, F, K, Réaumur — see Subtopic 01.)

Constant-volume gas thermometer (single fixed point: triple point of water, 273.16 K):

$$T = 273.16 \times \frac{P}{P_{\text{tr}}} \text{ K},$$

where P_{tr} is the gas pressure at the triple point and P the pressure at the measured temperature.

Absolute zero: $0 \text{ K} = -273.15^\circ\text{C}$.

The gas laws (fixed mass of ideal gas):

Boyle: $PV = \text{const}$ (T fixed); Charles: $\frac{V}{T} = \text{const}$ (P fixed); Gay-Lussac: $\frac{P}{T} = \text{const}$ (V fixed).

Ideal-gas equation:

$$PV = nRT = Nk_B T, \quad R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}, \quad k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}.$$

Coefficient of expansion of an ideal gas at constant pressure:

$$\gamma_{\text{gas}} = \frac{1}{273.15} \approx 3.66 \times 10^{-3} \text{ }^\circ\text{C}^{-1} \quad (\text{same for all ideal gases — far larger than for solids}).$$

Triple point of water: the unique state where ice, water and vapour coexist: $T_{\text{tr}} = 273.16 \text{ K}$, $P_{\text{tr}} = 611.7 \text{ Pa} \approx 4.58 \text{ mm Hg}$. The kelvin is defined as $1/273.16$ of this temperature.

Common thermometers and their working ranges

Thermometer	Thermometric property	Approx. range
Mercury-in-glass	length of mercury column	−39 to 357 °C
Alcohol-in-glass	length of alcohol column	down to ≈ -115 °C (low temps)
Constant-volume gas	pressure of a fixed gas	−270 to 1500 °C (standard)
Platinum resistance	electrical resistance	−200 to 1200 °C (accurate)
Thermocouple	thermo-emf of a junction	−200 to 1600 °C
Radiation pyrometer	emitted radiation (no contact)	above ≈ 1000 °C

Section 3: CBSE Derivation / Key Procedures — Absolute Zero from a Gas Thermometer

Goal: show that extrapolating a fixed-volume gas’s pressure to zero defines a universal coldest temperature, -273.15°C , and that this fixes the absolute scale.

Step 1 — the experimental fact. For a fixed mass of gas at fixed volume, the pressure varies linearly with Celsius temperature:

$$P_t = P_0(1 + \gamma t),$$

where P_0 is the pressure at 0°C and γ is the (measured) pressure coefficient. Experiment gives the same $\gamma = \frac{1}{273.15} \text{ }^\circ\text{C}^{-1}$ for every dilute gas.

Step 2 — find where the pressure would vanish. A gas cannot push with negative pressure, so the physically lowest temperature is where $P_t = 0$:

$$0 = P_0\left(1 + \frac{t}{273.15}\right) \Rightarrow t = -273.15 \text{ }^\circ\text{C}.$$

Because γ is the same for all gases, every gas extrapolates to this *same* intercept — a strong hint that it is a property of temperature itself, not of any particular gas.

Step 3 — shift the origin (define Kelvin). Set $T = t + 273.15$. Then $t = -273.15^\circ\text{C}$ becomes $T = 0$, and the relation collapses to a pure proportionality:

$$P_t = P_0\left(\frac{273.15 + t}{273.15}\right) = P_0 \frac{T}{T_0} \Rightarrow \boxed{P \propto T}.$$

This is exactly Gay-Lussac’s law in its cleanest form, and it is why the Kelvin scale — not Celsius — is the natural language of gas behaviour.

Step 4 — adopt a single fixed point. Once $P \propto T$, one reference suffices. The modern choice is the triple point of water, assigned $T_{\text{tr}} = 273.16 \text{ K}$, giving the working formula of the constant-volume gas thermometer:

$$T = 273.16 \times \frac{P}{P_{\text{tr}}} \text{ K}.$$

Physical reading: absolute zero is not “where the gas freezes” — it is the extrapolated point of zero pressure, the temperature at which (classically) all molecular translational motion would cease. It is a limit set by nature, the same for every substance.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A constant-volume gas thermometer registers a pressure of 80 mm Hg when its bulb is at the triple point of water. When the bulb is placed in a hot bath the pressure rises to 100 mm Hg. Find the temperature of the bath.

$$T = 273.16 \times \frac{P}{P_{\text{tr}}} = 273.16 \times \frac{100}{80} = 341.45 \text{ K} \approx 68.3^\circ\text{C}.$$

Answer: $T \approx 341.5 \text{ K}$ ($\approx 68.3^\circ\text{C}$).

Note: the gas thermometer needs only *one* fixed point (the triple point) because, once $P \propto T$, a single reference fixes the whole scale — a small but elegant economy over the two-fixed-point Celsius construction.

Example 2 — [NEET Speed Trap] A fixed mass of gas at 27°C is heated at constant pressure until its volume doubles. Find the final temperature.

The trap: “volume doubles, so double 27” $\rightarrow 54^\circ\text{C}$. Charles’s law is $V \propto T$ with T in *kelvin*, never Celsius.

Convert first: $27^\circ\text{C} = 300 \text{ K}$. Then $V \propto T$ means doubling V doubles the *absolute* temperature:

$$T_2 = 2 \times 300 = 600 \text{ K} = 327^\circ\text{C}.$$

Answer: 327°C .

Speed tip: in every gas-law problem, convert to kelvin *before* taking any ratio. “Double the Celsius reading” is the single most common NEET error here, because the 273 offset quietly wrecks the proportion.

Example 3 — [JEE Main Level] A sealed rigid vessel contains gas at 27°C and 1.0 atm. It is heated to 127°C . Find the new pressure. (Volume is constant.)

Gay-Lussac’s law, $P \propto T$ at fixed V , with kelvin temperatures $T_1 = 300 \text{ K}$, $T_2 = 400 \text{ K}$:

$$P_2 = P_1 \frac{T_2}{T_1} = 1.0 \times \frac{400}{300} = \frac{4}{3} \approx 1.33 \text{ atm}.$$

Answer: $\approx 1.33 \text{ atm}$.

Insight: a 100° rise sounds large but is only a 33% pressure increase, because on the absolute scale $300 \rightarrow 400 \text{ K}$ is a factor of $4/3$, not the factor of ~ 5 a Celsius ($27 \rightarrow 127$) reading misleadingly suggests. Always reason in kelvin.

Example 4 — [JEE Advanced Level] In a constant-volume gas thermometer the gas pressure is 1.000 atm at the ice point (0°C) and 1.366 atm at the steam point (100°C). Treating P as a linear function of Celsius temperature, determine the value of absolute zero (in $^\circ\text{C}$) implied by this data.

Step 1 — model. Write $P = a + bt$ with t in $^\circ\text{C}$. At $t = 0$, $P = 1.000$, so $a = 1.000$.

Step 2 — slope from the second point.

$$b = \frac{P_{100} - P_0}{100} = \frac{1.366 - 1.000}{100} = 3.66 \times 10^{-3} \text{ atm}/^\circ\text{C}.$$

Step 3 — extrapolate to $P = 0$.

$$0 = a + bt_0 \Rightarrow t_0 = -\frac{a}{b} = -\frac{1.000}{3.66 \times 10^{-3}} \approx -273.2^\circ\text{C}.$$

Answer: absolute zero $\approx -273.2^\circ\text{C}$ — recovered from nothing but two pressure readings.

Deep insight: this is precisely how absolute zero was first pinned down — not by reaching it, but by extrapolating the straight P - t line of a dilute gas to the point of zero pressure. The ratio $P_{100}/P_0 = 1.366 = 373.15/273.15$ is the experimental fingerprint of the absolute scale hiding inside Celsius data.

Section 5: Practice Exercises

- [CBSE] A constant-volume gas thermometer reads 90 mm Hg at the triple point. What pressure will it read at 0°C (the ice point, 273.15 K)?
- [NEET] Express absolute zero (-273.15°C) on the Fahrenheit scale.
- [JEE Main] A gas occupies 2.0 L at 27°C and 1 atm. It is compressed to 1.0 L and warmed to 127°C . Find the new pressure.
- [JEE Main] A bubble of gas doubles in volume as it rises from the bottom of a lake to the surface, while the temperature rises from 7°C to 27°C . If atmospheric pressure is 76 cm Hg, find the depth of the lake (in cm of mercury equivalent of water pressure).
- [JEE Advanced] A constant-volume gas thermometer gives pressures P_1 and P_2 at the ice and steam points. Show that the Celsius temperature t for a measured pressure P is $t = \frac{P - P_1}{P_2 - P_1} \times 100$, and evaluate it for $P_1 = 1.00$, $P_2 = 1.37$, $P = 1.10$ atm.

Answers:

- $P = 90 \times \frac{273.15}{273.16} \approx 90.0$ mm Hg (essentially unchanged; the triple point and ice point differ by only 0.01 K).
- $F = \frac{9}{5}(-273.15) + 32 = -459.67^\circ\text{F}$.
- $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \Rightarrow P_2 = 1 \times \frac{2.0}{1.0} \times \frac{400}{300} = \frac{8}{3} \approx 2.67$ atm.
- $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ with $T_1 = 280$, $T_2 = 300$, $V_2 = 2V_1$: $P_1 = P_2 \cdot 2 \cdot \frac{280}{300} = 76 \cdot 2 \cdot 0.933 = 141.9$ cm Hg; lake pressure $= P_1 - 76 = 65.9$ cm Hg (water column equivalent).
- By linear interpolation $t = \frac{P - P_1}{P_2 - P_1} \times 100 = \frac{0.10}{0.37} \times 100 \approx 27.0^\circ\text{C}$.

Section 6: Crack the MCQ

Q1. Which thermometer is most suitable for measuring the temperature of molten metal in a furnace ($\sim 1500^\circ\text{C}$)?

- (a) mercury-in-glass (b) alcohol-in-glass
(c) radiation pyrometer (d) clinical thermometer

Correct: (c). A radiation pyrometer measures emitted radiation without contact and works at very high temperatures. (a) mercury boils at 357°C , (b) alcohol far lower, (d) clinical thermometers cover only the body range — all fail here.

Q2. The reason a dilute (low-pressure) gas thermometer is chosen as the standard is that:

- (a) gases are cheap. (b) all gases give the same reading in the ideal (dilute) limit.
(c) gas expands the most. (d) gases never liquefy.

Correct: (b). In the low-pressure limit different gases agree exactly, so the reading is substance-independent — the hallmark of a true standard. The others are irrelevant or false.

Q3. Absolute zero corresponds to:

- (a) 0°C (b) -273.15°C , the extrapolated point of zero gas pressure/volume.
(b) the freezing point of every gas (d) a temperature easily reached in the lab.

Correct: (b). It is the common $P \rightarrow 0$ (or $V \rightarrow 0$) intercept of all gases. (c) confuses condensation with the extrapolated limit; (d) is false — it can be approached but not reached.

Q4. For a fixed mass of ideal gas, which combination is correct?

- (a) $PV = \text{const}$ at constant T ; $V \propto T$ at constant P ; $P \propto T$ at constant V .
(b) $PV \propto T^2$ always. (c) $P \propto V$ at constant T . (d) $V \propto 1/T$ at constant P .

Correct: (a). These are Boyle, Charles and Gay-Lussac respectively — the three pillars of $PV = nRT$. The rest contradict one or more of them.

Section 7: Common Student Pitfalls & Pro-Tips

- **Using Celsius in a gas law.** $P \propto T$, $V \propto T$ and $PV = nRT$ all require *kelvin*. Convert before forming any ratio; the 273 offset is fatal in a proportion.
- **Thinking absolute zero is attainable.** It is an extrapolated limit. Real systems get arbitrarily close but never reach 0 K.
- **Confusing the three gas laws.** Hold the right variable fixed: Boyle (T fixed), Charles (P fixed), Gay-Lussac (V fixed). Mislabelling which is constant gives a wrong formula.
- **Mixing up triple point and ice point.** The triple point (273.16 K, low pressure, three phases coexisting) is *not* the ordinary ice point (273.15 K at 1 atm). They differ by 0.01 K — small, but the triple point is the defining fixed point of the kelvin.
- **Using a thermometer outside its range.** Mercury above 357°C or below -39°C is meaningless; pick the device whose substance spans the required range.

Pro-Tip — always combine the gas laws as one identity. For a fixed mass of gas between any two states, use $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ (kelvin throughout). Every Boyle, Charles or Gay-Lussac question is just this with one quantity held fixed — one equation instead of three to remember.

Section 8: Cheat Sheet Box

- **Thermometry:** measure temperature via a thermometric property (length, pressure, resistance, thermo-emf, radiation). Calibrate with fixed points (ice 0°C , steam 100°C).
- **Gas thermometer (standard):** $T = 273.16 \frac{P}{P_{\text{tr}}}$ K; all dilute gases agree.
- **Absolute zero:** 0 K = -273.15°C , the $P \rightarrow 0$ (or $V \rightarrow 0$) extrapolation — unreachable.
- **Gas laws:** Boyle $PV = \text{const}$ (T), Charles $V \propto T$ (P), Gay-Lussac $P \propto T$ (V); combined $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$; ideal gas $PV = nRT$, $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$.
- **Triple point of water:** 273.16 K, 611.7 Pa — defines the kelvin.
- **Gas expansion coefficient:** $\gamma_{\text{gas}} = 1/273.15 \approx 3.66 \times 10^{-3} \text{ }^{\circ}\text{C}^{-1}$ (same for all ideal gases).

Mnemonic: “Measure by a property, standardise by a gas, count from absolute zero in kelvin.” For the gas laws: “Boyle holds T , Charles holds P , Gay-Lussac holds V .”

Subtopic 06 — Change of State: Phase Transitions, Triple Point & Phase Diagrams

Exam Relevance

The richest source of conceptual marks in the chapter. NEET loves it — regelation/ice-skating, the pressure cooker, why food cooks slowly on mountains, sublimation (dry ice, camphor), cooling by evaporation (earthen pots, sweating), and the triple point. CBSE Boards ask definitions (sublimation, regelation, triple point), the effect of pressure on melting/boiling points, and short “explain why” answers (2–3 marks). JEE Main sets phase-diagram interpretation and occasional pressure-dependence reasoning; JEE Advanced pushes into the slope of the fusion curve and quantitative melting-point shift. The latent-heat *numerics* live in Subtopic 02 — this subtopic supplies the *conceptual* framework that those numbers sit inside.

Section 1: Concept Intuition

In Subtopic 02 you learned to count the heat a phase change costs ($Q = mL$) and to budget it in mixing problems. But *when* and *why* a substance changes state — and how pressure shifts those transitions — is a separate, heavily tested story. That is this subtopic.

Matter exists in three common states, and a pure substance can move between them along three borders: melting/freezing (solid $\rightleftharpoons$ liquid), vaporization/condensation (liquid $\rightleftharpoons$ gas), and **sublimation** (solid $\rightleftharpoons$ gas directly, with no liquid in between — dry ice, camphor, naphthalene balls, iodine). Each transition for a pure substance happens at a fixed temperature for a given pressure, while latent heat flows in or out.

Melting point and pressure. For *most* substances the melting point *rises* as pressure increases — squeezing favours the denser solid. Water is the famous rebel: ice is *less* dense than liquid water (it expands on freezing, which is why pipes burst), so for water the melting point *falls* with pressure. This single fact powers **regelation**: press a fine wire over an ice block and the high pressure under the wire lowers the local melting point, the ice melts, the wire sinks through — and the water refreezes above it once the pressure is gone, leaving the block whole. It is also the textbook (if slightly oversimplified) story of why ice is slippery under a skate.

Boiling point and pressure. A liquid boils when its saturated vapour pressure equals the external pressure. Raise the external pressure and you must reach a higher temperature to boil — this is the **pressure cooker**: trapped steam raises the internal pressure, the boiling point climbs above 100°C, and food cooks faster in the hotter water. Conversely, high in the mountains the air pressure is low, water boils *below* 100°C, and food (cooked in cooler water) takes longer.

The triple point and the phase diagram. Plot pressure against temperature and you get a **phase (P – T) diagram**: three regions (solid, liquid, gas) separated by three curves (fusion, vaporization, sublimation). All three curves meet at one special point — the **triple point** — the only pressure and temperature at which solid, liquid and vapour coexist in equilibrium (for water, 273.16 K and 611.7 Pa). The vaporization curve does not go on forever; it ends at the **critical point** (for water, 647 K, 22.1 MPa). Above the critical temperature there is no liquid–gas distinction at all — you cannot liquefy a gas by pressure alone, no matter how hard you squeeze.

Think about it this way... A phase diagram is a *map* of which state a substance “prefers” at each pressure and temperature. The curves are borderlines where two states are tied; cross a border and the substance switches state. The triple point is the one address on the map where all three territories meet at a single corner. Whether the fusion border leans left (water) or right (most substances) is the whole secret behind regelation.

Evaporation versus boiling. Boiling is a *bulk* phenomenon at the boiling point — bubbles form throughout the liquid. **Evaporation** happens only at the *surface* and at *any* temperature: the fastest molecules escape, leaving the rest cooler. That is why evaporation *cools* — sweat evaporating cools your skin, water seeping through a porous earthen pot (*matka*) keeps the water inside cool, and a desert cooler works by evaporating water into dry air. Evaporation speeds up with higher temperature, larger surface area, drier air (low humidity) and moving air (wind).

Limiting Conditions & Assumptions:

- Phase diagrams and fixed transition temperatures apply to *pure* substances; impurities and dissolved solutes shift melting and boiling points (salt lowers ice’s melting point; dissolved solids raise the boiling point).
- Latent heats themselves vary slightly with pressure and temperature; the values in Subtopic 02 are the standard-pressure figures.
- Regelation requires the anomalous (ice-like) behaviour — a melting point that falls with pressure. Substances whose melting point rises with pressure do not regelate.

Section 2: Key Formulas & Definitions

Latent heat (recap from Subtopic 02): $Q = mL$, with $L_f(\text{ice}) = 80 \text{ cal/g} = 3.36 \times 10^5 \text{ J/kg}$ and $L_v(\text{water}) = 540 \text{ cal/g} = 2.26 \times 10^6 \text{ J/kg}$.

Melting point vs pressure: water — decreases ($\approx -0.0075 \text{ }^\circ\text{C}$ per atm); most substances — increases.

Boiling point vs pressure: increases with external pressure (pressure cooker); decreases at high altitude (low pressure).

Triple point of water: $T = 273.16 \text{ K}$, $P = 611.7 \text{ Pa}$ ($\approx 4.58 \text{ mmHg}$).

Critical point of water: $T_c = 647 \text{ K}$ (374°C), $P_c = 22.1 \text{ MPa}$ ($\approx 218 \text{ atm}$); above T_c no liquefaction by pressure.

Clausius–Clapeyron relation (qualitative, slope of a phase boundary):

$$\frac{dP}{dT} = \frac{L}{T\Delta V},$$

where ΔV is the volume change in the transition. For water’s fusion curve $\Delta V < 0$ (ice $\rightarrow$ water shrinks), so $\frac{dP}{dT} < 0$ — the fusion line slopes *backward*, explaining the falling melting point. (Computing with it is beyond the core syllabus, but the *sign* is the key insight.)

Sublimation: solid $\rightarrow$ gas directly; occurs when the substance’s triple-point pressure is *above* the ambient pressure (e.g. CO_2 : triple point 5.1 atm , so at 1 atm dry ice sublimates rather than melts).

Section 3: CBSE Derivation / Key Procedures — Reading the Phase Diagram

There is no single algebraic derivation here; the skill is reading the P – T map and reasoning from it. Here is the reliable framework, in the style of a board “explain with diagram” answer.

Figure 10.6 *Phase diagram of water (not to scale): the fusion, vaporization and sublimation curves meet at the triple point ($273.16\text{ K} = 0.01\text{ }^{\circ}\text{C}$, $611.7\text{ Pa} = 0.006\text{ atm}$); the fusion curve's negative slope is the water anomaly, and the dashed 1 atm line crosses it at $0\text{ }^{\circ}\text{C}$ and the vaporization curve at $100\text{ }^{\circ}\text{C}$, which runs on to the critical point (647 K , 22.1 MPa). (see next page)*

Step 1 — locate the three regions and three borders. Each region is a single stable state; each curve is a two-phase equilibrium (the substance can sit as either phase, or a mix, along the curve). Crossing a curve = a phase change.

Step 2 — read the triple point. Where all three curves meet, all three phases coexist. This is a fixed, reproducible state, which is exactly why it anchors the kelvin (Subtopic 05).

Step 3 — use the slope of the fusion curve. A *positive* slope (most substances) means raising pressure raises the melting point. A *negative* slope (water) means raising pressure *lowers* it — the basis of regelation. The sign comes straight from $dP/dT = L/(T \Delta V)$ and whether the solid is denser ($\Delta V > 0$) or less dense ($\Delta V < 0$) than the liquid.

Step 4 — explain everyday effects by moving along a line.

- *Pressure cooker:* move up the vaporization curve — higher P meets it at higher T , so water boils above 100°C .
- *Mountain cooking:* move down the vaporization curve — lower P meets it at lower T , so water boils below 100°C .
- *Dry ice:* at 1 atm , below CO_2 's triple-point pressure, a horizontal line crosses only the sublimation curve — solid goes straight to gas.

Why this matters: almost every conceptual question in this subtopic is answered by locating a point on this map and sliding along a constant-pressure or constant-temperature line to see which border you cross.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Explain, with reference to boiling point and pressure, why food cooks faster in a pressure cooker than in an open pan, and why it cooks more slowly at a high-altitude hill station.

Answer. A liquid boils when its saturated vapour pressure equals the surrounding pressure. In a pressure cooker the trapped steam raises the internal pressure well above 1 atm , so the boiling point of water climbs above 100°C . Food sits in *hotter* water, and the higher temperature speeds up cooking. At a hill station the atmospheric pressure is lower than at sea level, so water boils *below* 100°C ; the cooler boiling water transfers heat to the food more slowly, and cooking takes longer.

Teaching point: the controlling variable is always temperature of the boiling water, and that temperature is set by pressure. “Higher pressure $\Rightarrow$ higher boiling point $\Rightarrow$ faster cooking” is the whole chain.

Example 2 — [NEET Speed Trap] A copper wire carrying heavy weights at its two ends is hung over a slab of ice. After some time the wire is found to have passed completely through the slab, yet the slab remains a single solid block. Name and explain the phenomenon.

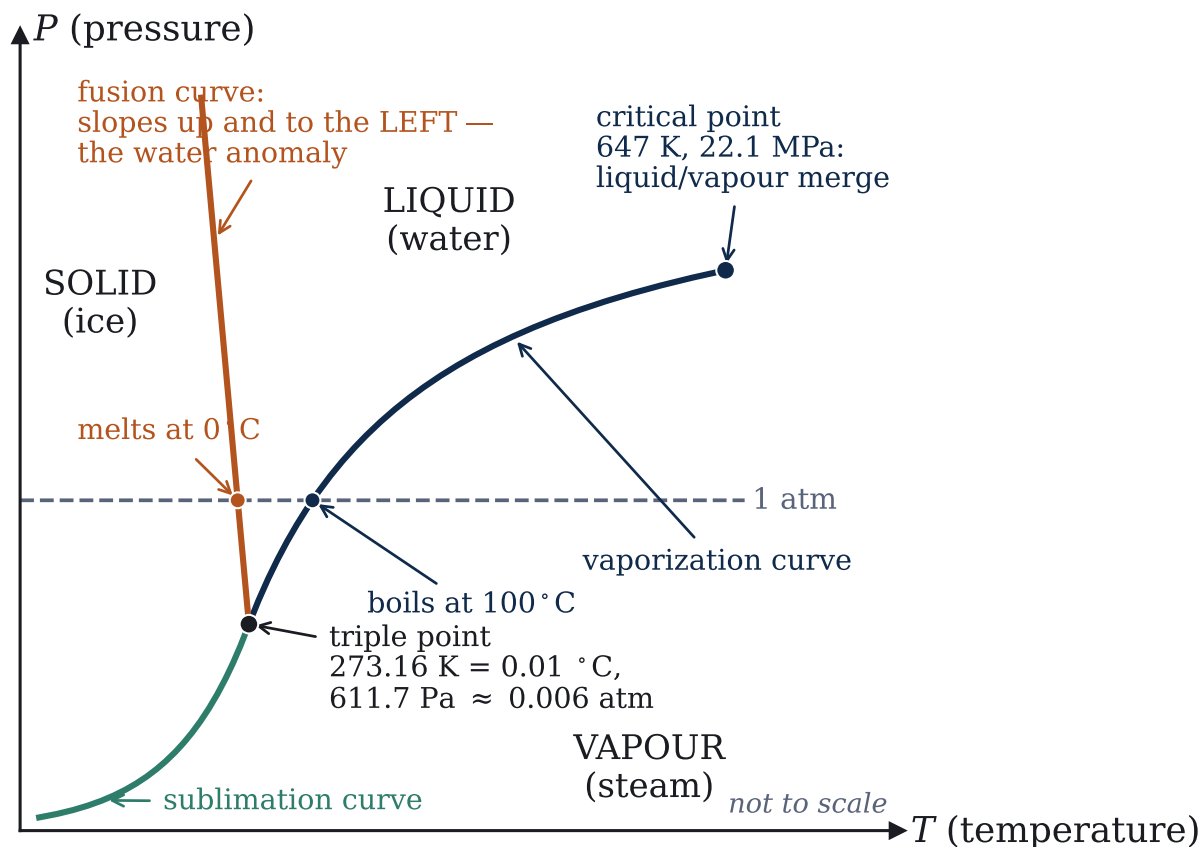


Figure 10.6 Phase diagram of water (not to scale): the fusion, vaporization and sublimation curves meet at the triple point ($273.16\text{ K} = 0.01\text{ °C}$, $611.7\text{ Pa} = 0.006\text{ atm}$); the fusion curve's negative slope is the water anomaly, and the dashed 1 atm line crosses it at 0 °C and the vaporization curve at 100 °C , which runs on to the critical point (647 K , 22.1 MPa).

The trap: “the wire cut/sawed through the ice.” It did not cut — the ice *melted* and *refroze*.

Answer — regelation. The large pressure directly beneath the wire lowers the melting point of the ice there (water’s melting point falls with pressure), so the ice melts under the wire and lets it sink. Above the wire, where the pressure is released, the water is below the normal melting point again, so it *refreezes* — releasing its latent heat, which conveniently supplies the heat needed to melt the ice just ahead of the wire. The wire descends through the block, which re-fuses behind it and stays whole.

Speed insight: regelation works *only* because ice’s melting point *falls* with pressure (the water anomaly). A substance whose melting point rises with pressure would not let the wire through. Any option attributing it to “cutting” or “friction heat” is wrong on sight.

Example 3 — [JEE Main Level] Solid carbon dioxide (“dry ice”) has a triple point at 216.6 K and 5.1 atm. Explain, using a phase-diagram argument, why dry ice at ordinary atmospheric pressure (1 atm) turns directly into gas without ever becoming a liquid.

Answer. On the CO₂ phase diagram, the triple point sits at 5.1 atm — well *above* 1 atm. A liquid phase exists only at pressures *above* the triple-point pressure (the liquid region lies above the triple point). At 1 atm a horizontal line lies *below* the triple point, so as the solid warms it crosses only the **sublimation curve** (solid → vapour) and never enters the liquid region. Hence dry ice sublimates directly to gas at 1 atm. To obtain liquid CO₂ you must first raise the pressure above 5.1 atm.

Insight: the rule generalises — a substance sublimates at any ambient pressure below its triple-point pressure. Water’s triple-point pressure is only ~ 0.006 atm, far below 1 atm, which is why water normally melts rather than sublimates; CO₂’s is far above 1 atm, so it sublimates.

Example 4 — [JEE Advanced Level] Estimate the change in the melting point of ice when the pressure is increased by 1 atm. Use the Clausius–Clapeyron relation $\frac{dP}{dT} = \frac{L}{T \Delta V}$ with $T = 273$ K, $L_f = 3.36 \times 10^5$ J/kg, $\rho_{\text{ice}} = 917$ kg/m³, $\rho_{\text{water}} = 1000$ kg/m³. Comment on the sign.

Step 1 — volume change per kg on melting (ice → water):

$$\Delta V = \frac{1}{\rho_{\text{water}}} - \frac{1}{\rho_{\text{ice}}} = \frac{1}{1000} - \frac{1}{917} = 1.000 \times 10^{-3} - 1.0905 \times 10^{-3} = -9.05 \times 10^{-5} \text{ m}^3/\text{kg}.$$

$\Delta V < 0$ because water is denser than ice — the substance *shrinks* on melting.

Step 2 — invert Clausius–Clapeyron for dT/dP :

$$\frac{dT}{dP} = \frac{T \Delta V}{L} = \frac{(273)(-9.05 \times 10^{-5})}{3.36 \times 10^5} = -7.35 \times 10^{-8} \text{ K/Pa}.$$

Step 3 — multiply by $\Delta P = 1$ atm = 1.013×10^5 Pa:

$$\Delta T = \frac{dT}{dP} \Delta P = (-7.35 \times 10^{-8})(1.013 \times 10^5) \approx -7.4 \times 10^{-3} \text{ K}.$$

Answer: the melting point *falls* by about 0.0074 °C per atmosphere.

Deep insight: the negative sign is the entire point — it traces directly to $\Delta V < 0$, the fact that ice is less dense than water. This tiny but real drop is what makes regelation possible and is the quantitative face of water’s most famous anomaly. For a normal substance ($\Delta V > 0$) the same relation gives a melting point that *rises* with pressure.

Section 5: Practice Exercises

1. [CBSE] Define (i) sublimation, (ii) the triple point of a substance, and (iii) regelation. Give one everyday example of each.
2. [NEET] A wet cloth wrapped around a water bottle keeps it cooler than a dry one in a hot, dry breeze. Name the effect and state two factors that make it stronger.
3. [JEE Main] The triple point of water is 273.16 K at 0.006 atm. Explain why, at the much higher 1 atm, water has a distinct liquid range between 0°C and 100°C rather than subliming.
4. [JEE Main] A sealed flask half-filled with water is heated. Describe what happens to the boiling point as heating continues in the sealed (rising-pressure) flask, and why.
5. [JEE Advanced] Using $dP/dT = L/(T \Delta V)$, explain qualitatively why the vaporization curve (liquid–gas) on a P – T diagram is far *less* steep than the fusion curve (solid–liquid). (Hint: compare the ΔV values for the two transitions.)

Answers:

1. (i) Sublimation — solid $\rightarrow$ gas directly (dry ice, camphor). (ii) Triple point — the unique P, T where solid, liquid and vapour coexist (water: 273.16 K, 611.7 Pa). (iii) Regelation — melting under pressure and refreezing on its release (a loaded wire passing through ice).
2. Cooling by evaporation; stronger with lower humidity and faster air movement (also higher temperature and larger exposed surface).
3. At 1 atm the pressure is far *above* the triple-point pressure, so a constant-pressure line crosses both the fusion and vaporization curves — giving a solid $\rightarrow$ liquid melt at 0°C and a liquid $\rightarrow$ gas boil at 100°C, with a genuine liquid range between.
4. The boiling point *rises*: the trapped vapour raises the internal pressure, and a higher external pressure requires a higher temperature for the vapour pressure to match it.
5. For vaporization ΔV is huge (liquid $\rightarrow$ gas expands enormously), making $dP/dT = L/(T \Delta V)$ small — a gently sloping curve. For fusion ΔV is tiny (solid $\rightarrow$ liquid changes volume only slightly), making dP/dT large — a steep, nearly vertical curve.

Section 6: Crack the MCQ

Q1. The melting point of ice with increasing external pressure:

- (a) increases (b) decreases (c) stays constant (d) first rises, then falls

Correct: (b). Ice is less dense than water ($\Delta V < 0$ on melting), so by $dP/dT = L/(T \Delta V)$ the melting point *falls* with pressure — the basis of regelation. (a) is true for most *other* substances, the classic trap.

Q2. Water boils at a temperature below 100°C when:

- (a) the external pressure is increased (b) the external pressure is decreased (e.g. at high altitude)
(c) salt is added (d) the container is sealed

Correct: (b). Lower external pressure means the vapour pressure matches it at a lower temperature. (a) and (d) raise the pressure and the boiling point; (c) *raises* the boiling point.

Q3. At the triple point of a substance:

- (a) only solid and liquid coexist (b) only liquid and vapour coexist
(c) solid, liquid and vapour coexist in equilibrium (d) the substance cannot exist

Correct: (c). The triple point is the unique P, T where all three phases coexist — which is why it is used to define the kelvin. The others describe ordinary two-phase curves.

Q4. Evaporation differs from boiling in that evaporation:

- (a) occurs only at the boiling point (b) occurs throughout the bulk of the liquid
(b) occurs only at the liquid surface and at any temperature, and cools the liquid (d) releases heat to the liquid

Correct: (c). Evaporation is a surface phenomenon at all temperatures; the most energetic molecules leave, lowering the average energy and cooling the remaining liquid. (a) and (b) describe boiling; (d) reverses the cooling effect.

Section 7: Common Student Pitfalls & Pro-Tips

- **Assuming every melting point rises with pressure.** Water is the great exception ($\Delta V < 0$): its melting point *falls*. Check whether the solid is denser or less dense than the liquid before guessing the sign.
- **Calling regelation “cutting.”** The wire never cuts the ice — the ice melts under pressure and refreezes when the pressure is released. The block stays whole.
- **Confusing the triple point with the ice/boiling point.** The triple point is a single P, T (low pressure, three phases), not the 1 atm melting or boiling point.
- **Thinking a gas can always be liquefied by pressure.** Above the *critical temperature*, no amount of pressure produces a liquid — the liquid–gas border simply ends at the critical point.
- **Mixing up evaporation and boiling.** Evaporation: surface, any temperature, cools the liquid. Boiling: bulk, at the boiling point, requires continuous heat input.

Pro-Tip — solve conceptual questions on the phase map. Sketch a quick P – T diagram, mark the triple and critical points, and slide along a constant-pressure (horizontal) or constant-temperature (vertical) line to see which border you cross. Pressure cooker, mountain cooking, dry-ice sublimation, and regelation all fall out of this one picture — no formula needed.

Section 8: Cheat Sheet Box

- **Three transitions:** melting/freezing, vaporization/condensation, sublimation (solid $\leftrightarrow$ gas). Each absorbs/releases latent heat $Q = mL$ at constant T (Subtopic 02).
- **Melting point vs pressure:** water *decreases* ($\Delta V < 0$); most substances *increase*. Basis of regelation.
- **Boiling point vs pressure:** *increases* with pressure (pressure cooker), *decreases* at altitude (slow mountain cooking).
- **Triple point (water):** 273.16 K, 611.7 Pa — all three phases coexist; defines the kelvin. **Critical point (water):** 647 K, 22.1 MPa — above it, no liquefaction by pressure.
- **Sublimation** occurs when ambient pressure is below the triple-point pressure (CO_2 at 1 atm).
- **Evaporation:** surface phenomenon at any temperature; *cools* the liquid (sweating, *matka*, desert cooler); faster with heat, area, low humidity, wind.
- **Clausius–Clapeyron (sign only):** $dP/dT = L/(T\Delta V)$ — $\Delta V < 0$ for water’s fusion gives the backward-sloping curve.

Mnemonics: “*Squeeze ice and it melts; squeeze water vapour and it boils hotter.*” “*Triple point: all three at one corner.*” “*Evaporation cools, boiling needs heat.*”

Chapter 10: Thermal Properties of Matter — Round 2 Addendum

The main chapter (Subtopics 01–06) gives you every formula, derivation and standard numerical pattern that CBSE Boards, NEET and JEE Main expect. Three JEE Advanced patterns remain that marry concepts the chapter teaches in isolation: **composite thermal-stress systems** (expansion + elasticity + force balance), **non-planar steady conduction** (Fourier’s law + geometry + integration), and **exact radiation cooling** (Stefan’s law + differential equations + spectral selectivity). This addendum supplies the missing derivations, the geometric resistance formulas you must carry in memory, and the integration techniques that turn “rate $\propto T^4$ ” into a usable time-temperature relation. Every method below is a direct extension of the chapter’s own procedures — no new physics, only the next algebraic or geometric step.

Addendum A — Thermal Stress in Composite & Constrained Systems

Subtopic 01 derived $\sigma = Y\alpha\Delta T$ for a single rod clamped between rigid walls. JEE Advanced rarely stops there. Standard problems join two rods of different materials (different α , Y , A) end-to-end between immovable supports, or clamp a single rod that also carries a hanging mass, or ask for the curvature of a heated bimetallic strip. The engine in every case is the same: **total length is fixed $\rightarrow$ sum of strains (mechanical + thermal) = 0**, and **force is common through series elements**.

Method

1. **Draw the free-body diagram.** In a series junction the internal force F (tension positive, compression negative) is the same in every segment. If a mass m hangs from the junction, $F = mg$ appears as an external load.
2. **Write the strain compatibility equation for each segment i .**

$$\epsilon_i = \frac{\Delta L_i}{L_{0i}} = \frac{F}{A_i Y_i} + \alpha_i \Delta T$$

Sign convention: thermal expansion ($\alpha\Delta T > 0$ for heating) tends to *increase* length; tensile force ($F > 0$) *increases* length; compressive force ($F < 0$) *decreases* length.

3. **Apply the global constraint.**

- **Rigid walls, no external load:** $\sum \Delta L_i = 0 \Rightarrow \sum (\epsilon_i L_{0i}) = 0$.
- **Rigid walls + hanging mass m at a junction:** The force in the segments changes at the junction. Write compatibility for the *whole* system: $\sum (\epsilon_i L_{0i}) = 0$, but F is piecewise constant.

- **One end fixed, other end free but attached to a spring (stiffness k):** $\sum \Delta L_i = F/k$.

4. **Solve for F**, then back-substitute for stresses $\sigma_i = F/A_i$ and final lengths.
5. **Bimetallic strip (curvature).** Two strips of equal length L_0 , thicknesses d_1, d_2 , bonded together. On heating ΔT , differential expansion forces bending. The radius of curvature R (measured to the interface) is:

$$R = \frac{d_1 + d_2}{(\alpha_2 - \alpha_1)\Delta T} \cdot \frac{1 + \frac{E_1 I_1}{E_2 I_2} + \frac{E_1 A_1 h_1^2}{E_2 I_2} + \dots}{\text{(exact formula is messy)}}$$

JEE-approved shortcut (thin strips, equal thickness d , equal width b , $E_1 \approx E_2 = E$):

$$R \approx \frac{2d}{(\alpha_2 - \alpha_1)\Delta T}$$

Derivation: strain difference $\Delta \varepsilon = (\alpha_2 - \alpha_1)\Delta T$ is accommodated by bending strain $\Delta \varepsilon = (d_1 + d_2)/R$. For equal thicknesses $d_1 = d_2 = d$, total thickness $2d$.

Worked Examples

Example A.1 — [JEE Advanced Level] Composite Rod Between Rigid Walls

A copper rod ($L_1 = 0.40$ m, $A_1 = 4.0$ cm², $Y_1 = 1.1 \times 10^{11}$ Pa, $\alpha_1 = 1.7 \times 10^{-5}$ K⁻¹) and a steel rod ($L_2 = 0.60$ m, $A_2 = 2.0$ cm², $Y_2 = 2.0 \times 10^{11}$ Pa, $\alpha_2 = 1.2 \times 10^{-5}$ K⁻¹) are placed end-to-end between rigid supports with no initial stress at 20°C. The temperature is raised to 120°C. Find the stress in each rod and the displacement of the junction.

Step 1 — Force balance. Series connection $\Rightarrow F$ is common. Let F be compressive (negative).

Step 2 — Compatibility. Total length fixed:

$$\Delta L_1 + \Delta L_2 = 0$$

$$\left(\frac{FL_1}{A_1 Y_1} + \alpha_1 L_1 \Delta T \right) + \left(\frac{FL_2}{A_2 Y_2} + \alpha_2 L_2 \Delta T \right) = 0$$

Step 3 — Insert numbers (SI). $\Delta T = 100$ K. $A_1 = 4.0 \times 10^{-4}$ m², $A_2 = 2.0 \times 10^{-4}$ m².

$$\frac{F(0.40)}{(4.0 \times 10^{-4})(1.1 \times 10^{11})} + (1.7 \times 10^{-5})(0.40)(100) + \frac{F(0.60)}{(2.0 \times 10^{-4})(2.0 \times 10^{11})} + (1.2 \times 10^{-5})(0.60)(100) = 0$$

$$F \left(\frac{0.40}{4.4 \times 10^7} + \frac{0.60}{4.0 \times 10^7} \right) + (6.8 \times 10^{-4} + 7.2 \times 10^{-4}) = 0$$

$$F(9.09 \times 10^{-9} + 15.0 \times 10^{-9}) + 14.0 \times 10^{-4} = 0$$

$$F(24.09 \times 10^{-9}) = -1.40 \times 10^{-3}$$

$$F = -5.81 \times 10^4 \text{ N (compression)}$$

Step 4 — Stresses.

$$\sigma_1 = \frac{F}{A_1} = \frac{-5.81 \times 10^4}{4.0 \times 10^{-4}} = -1.45 \times 10^8 \text{ Pa}$$

$$\sigma_2 = \frac{F}{A_2} = \frac{-5.81 \times 10^4}{2.0 \times 10^{-4}} = -2.91 \times 10^8 \text{ Pa}$$

Step 5 — Junction displacement (from left wall).

$$\begin{aligned}\Delta L_1 &= \frac{FL_1}{A_1 Y_1} + \alpha_1 L_1 \Delta T \\ &= \frac{(-5.81 \times 10^4)(0.40)}{(4.0 \times 10^{-4})(1.1 \times 10^{11})} + 6.8 \times 10^{-4} \\ &= -5.28 \times 10^{-4} + 6.80 \times 10^{-4} = +1.52 \times 10^{-4} \text{ m} = 0.152 \text{ mm}\end{aligned}$$

Junction moves **0.152 mm to the right** (towards steel). Copper wanted to expand more but was compressed; steel wanted to expand less but was compressed even harder, so the junction shifts toward the steel side.

Check. Free expansions: Cu = 0.68 mm, Steel = 0.72 mm. Total free = 1.40 mm. Compression = 1.40 mm. Cu compression = 0.528 mm, Steel compression = 0.872 mm. Net Cu = +0.152 mm. Consistent.

Example A.2 — [JEE Main Level] Bimetallic Strip Curvature

A bimetallic strip consists of brass ($\alpha = 1.9 \times 10^{-5} \text{ K}^{-1}$) and steel ($\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$), each 0.50 mm thick and 20 mm wide, bonded over a length 10 cm. Initially straight at 0°C. Find the radius of curvature at 100°C. Assume Young's moduli are equal.

Step 1 — Identify parameters. $d_1 = d_2 = 0.50 \times 10^{-3} \text{ m}$. $\Delta T = 100 \text{ K}$. α_2 (brass) $>$ α_1 (steel) $\Rightarrow$ brass expands more $\Rightarrow$ strip bends with brass on **outer** (convex) side. **Step 2 — Shortcut formula (equal E, equal thickness d).**

$$R = \frac{2d}{(\alpha_{\text{brass}} - \alpha_{\text{steel}})\Delta T}$$

Step 3 — Compute.

$$R = \frac{2(0.50 \times 10^{-3})}{(1.9 - 1.2) \times 10^{-5} \times 100} = \frac{1.0 \times 10^{-3}}{0.7 \times 10^{-3}} = 1.43 \text{ m}$$

Answer: Radius of curvature $\approx$ **1.43 m** (brass on outside).

Addendum B — Steady Conduction in Cylindrical & Spherical Shells

Subtopic 03 derived thermal resistance for a slab: $R = L/(KA)$. JEE Advanced routinely asks for heat flow through a **pipe wall** (cylindrical) or a **spherical shell** (e.g., a hollow sphere cooling in a bath). The heat current H is constant in steady state, but the area $A(r)$ changes with radius, so the temperature gradient dT/dr is not constant. You must integrate Fourier's law.

Method: Cylindrical Shell (Pipe)

Geometry: Length L , inner radius r_1 , outer radius r_2 , conductivity K . Inner surface at T_1 , outer at T_2 ($T_1 > T_2$). Heat flows radially outward.

Step 1 — Fourier's law at radius r .

$$H = -KA(r) \frac{dT}{dr} = -K(2\pi rL) \frac{dT}{dr}$$

Step 2 — Separate variables and integrate.

$$H \int_{r_1}^{r_2} \frac{dr}{r} = -2\pi KL \int_{T_1}^{T_2} dT$$

$$H \ln \left(\frac{r_2}{r_1} \right) = 2\pi KL(T_1 - T_2)$$

Step 3 — Thermal Resistance.

$$R_{\text{cyl}} = \frac{\ln(r_2/r_1)}{2\pi KL} \quad \text{Unit: K W}^{-1}$$

$$H = \frac{T_1 - T_2}{R_{\text{cyl}}}$$

Series/Parallel: For composite cylinders (e.g., pipe + insulation), resistances add in series because H is common. The junction temperature T_j is found from $H = (T_1 - T_j)/R_1 = (T_j - T_2)/R_2$.

Method: Spherical Shell

Geometry: Inner radius r_1 , outer r_2 . Area $A(r) = 4\pi r^2$.

Step 1 — Fourier's law.

$$H = -K(4\pi r^2) \frac{dT}{dr}$$

Step 2 — Integrate.

$$H \int_{r_1}^{r_2} \frac{dr}{r^2} = -4\pi K \int_{T_1}^{T_2} dT$$

$$H \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = 4\pi K(T_1 - T_2)$$

Step 3 — Thermal Resistance.

$$R_{\text{sph}} = \frac{r_2 - r_1}{4\pi K r_1 r_2} \quad \text{Unit: K W}^{-1}$$

$$H = \frac{T_1 - T_2}{R_{\text{sph}}}$$

Variable Conductivity K(T)

If $K = K_0(1 + \beta T)$, the integration changes slightly. **Slab:** $HL = A \int_{T_2}^{T_1} K_0(1 + \beta T) dT = AK_0 \left[(T_1 - T_2) + \frac{\beta}{2}(T_1^2 - T_2^2) \right]$. **Cylinder/Sphere:** Substitute $K(T)$ inside the integral before separating variables. The algebraic form remains integrable.

Worked Examples

Example B.1 — [JEE Advanced Level] Steam Pipe with Insulation

A steam pipe ($r_1 = 5.0$ cm, $r_2 = 6.0$ cm, $K_{\text{steel}} = 50$ W m⁻¹K⁻¹) is covered with a 4.0 cm thick layer of glass wool ($K_{\text{wool}} = 0.08$ W m⁻¹K⁻¹). Steam at 150°C flows inside; ambient air is at 25°C. The pipe is 10 m long. Find (a) heat loss per metre, (b) temperature at the steel-wool interface. Ignore convection/radiation at outer surface (assume outer wool surface is at 25°C).

Step 1 — Radii. Steel: $r_1 = 0.050$ m, $r_2 = 0.060$ m. Wool: $r_2 = 0.060$ m, $r_3 = 0.100$ m. $\Delta T_{\text{total}} = 150 - 25 = 125$ K. $L = 1.0$ m (per metre).

Step 2 — Resistances (per metre).

$$R_{\text{steel}} = \frac{\ln(0.060/0.050)}{2\pi(50)} = \frac{\ln(1.2)}{100\pi} = \frac{0.1823}{314.16} = 5.80 \times 10^{-4} \text{ K/W}$$

$$R_{\text{wool}} = \frac{\ln(0.100/0.060)}{2\pi(0.08)} = \frac{\ln(1.667)}{0.5027} = \frac{0.5108}{0.5027} = 1.016 \text{ K/W}$$

Note: Wool resistance is $\sim 1750\times$ steel resistance. Steel drop is negligible.

Step 3 — Heat current (per metre).

$$H = \frac{125}{R_{\text{steel}} + R_{\text{wool}}} = \frac{125}{1.0166} = 123.0 \text{ W/m}$$

Step 4 — Interface temperature T_j .

$$T_j = T_{\text{steam}} - HR_{\text{steel}} = 150 - (123.0)(5.80 \times 10^{-4}) = 150 - 0.071 \approx 149.93^\circ\text{C}$$

Essentially the steam temperature. The steel offers almost no resistance.

Example B.2 — [JEE Main Level] Spherical Cavity in a Large Block

A spherical cavity of radius 10 cm is maintained at 200°C inside a large block of material ($K = 1.5$ W m⁻¹K⁻¹) whose outer boundary is at 30°C. Find the heat flow rate. (Treat block as spherical shell with $r_2 \rightarrow \infty$).

Step 1 — Resistance formula for sphere with $r_2 \rightarrow \infty$.

$$R = \lim_{r_2 \rightarrow \infty} \frac{r_2 - r_1}{4\pi K r_1 r_2} = \frac{1}{4\pi K r_1}$$

Step 2 — Compute. $r_1 = 0.10$ m.

$$R = \frac{1}{4\pi(1.5)(0.10)} = \frac{1}{1.885} = 0.5305 \text{ K/W}$$

Step 3 — Heat flow.

$$H = \frac{200 - 30}{0.5305} = \frac{170}{0.5305} = 320.5 \text{ W}$$

Addendum C — Exact Radiation Cooling (Beyond Newton's Law)

Subtopic 04 derived Newton's Law ($-dT/dt = k(T - T_0)$) by linearising Stefan's law ($P_{\text{net}} = e\sigma A(T^4 - T_0^4)$) for small ΔT . JEE Advanced often demands the **exact cooling time** for a body cooling from T_1 to T_2 in surroundings at T_0 , where ΔT is *not* small, or asks for the cooling curve when the body has **temperature-dependent emissivity** or when **two bodies exchange radiation** in an enclosure.

Method 1: Exact Integration of Stefan's Law (Isolated Body)

Governing equation (lumped capacitance, uniform T):

$$-mc \frac{dT}{dt} = e\sigma A(T^4 - T_0^4)$$

Separate variables:

$$dt = -\frac{mc}{e\sigma A} \frac{dT}{T^4 - T_0^4}$$

Partial fractions:

$$\frac{1}{T^4 - T_0^4} = \frac{1}{2T_0^3} \left[\frac{1}{T - T_0} - \frac{1}{T + T_0} \right] - \frac{1}{T_0^3} \frac{T_0}{T^2 + T_0^2}$$

Integrate from $t=0$ ($T=T_1$) to t ($T=T$):

$$t = \frac{mc}{2e\sigma AT_0^3} \left[\ln \left(\frac{T_1 - T_0}{T - T_0} \cdot \frac{T + T_0}{T_1 + T_0} \right) + 2 \tan^{-1} \left(\frac{T}{T_0} \right) - 2 \tan^{-1} \left(\frac{T_1}{T_0} \right) \right]$$

Special case $T_0 = 0$ K (radiating into deep space):

$$t = \frac{mc}{3e\sigma A} \left(\frac{1}{T^3} - \frac{1}{T_1^3} \right)$$

Method 2: Two Gray Bodies Exchanging Radiation (Enclosure)

Two large parallel plates (or concentric spheres/cylinders) at T_1 , T_2 with emissivities e_1 , e_2 . Net heat flux from 1 to 2:

$$q = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1}{e_1} + \frac{1}{e_2} - 1} \quad (\text{Parallel plates})$$

$$q = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1}{e_1} + \frac{A_1}{A_2} \left(\frac{1}{e_2} - 1 \right)} \quad (\text{Concentric, } A_1 < A_2)$$

Derivation sketch: Radiosity method. $J = e\sigma T^4 + (1-e)G$. Net flux $q = (J_1 - J_2)/(1/e_1 + 1/e_2 - 1)$. Memorise the denominator.

Method 3: Selective Surfaces (Spectral Emissivity)

A surface has $e(\lambda) = e_1$ for $\lambda < \lambda_c$ and e_2 for $\lambda > \lambda_c$. At temperature T , peak wavelength $\lambda_m \approx b/T$. - If $\lambda_m \ll \lambda_c$ (very hot), surface “sees” mostly e_1 . - If $\lambda_m \gg \lambda_c$ (cool), surface “sees” mostly e_2 . **Effective emissivity:**

$$e_{\text{eff}}(T) = \frac{\int_0^\infty e(\lambda) E_{b\lambda}(T) d\lambda}{\sigma T^4}$$

For a step-function $e(\lambda)$, this is a fraction of the black-body functions $F(0 \rightarrow \lambda_c T)$. JEE Advanced may give you a graph of $e(\lambda)$ and ask for equilibrium temperature of a satellite plate facing the Sun ($\alpha_{\text{solar}} \approx 1$ for $\lambda < 2\mu\text{m}$, $\varepsilon_{\text{IR}} \approx 0.1$ for $\lambda > 2\mu\text{m}$).

Worked Examples

Example C.1 — [JEE Advanced Level] Exact Cooling Time

A copper sphere ($r = 2.0$ cm, $\rho = 8960$ kg/m³, $c = 385$ J/kg·K, $e = 0.8$) cools from 500 K to 400 K in surroundings at 300 K. Find the time taken. ($\sigma = 5.67 \times 10^{-8}$ W/m²K⁴). Compare with Newton’s law approximation.

Step 1 — Parameters. $m = \rho(4/3 \pi r^3) = 8960 \times (4/3 \pi \times 0.023) = 8960 \times 3.351 \times 10^{-5} = 0.300$ kg. $A = 4\pi r^2 = 4\pi \times 0.0004 = 5.027 \times 10^{-3}$ m². $mc/(e\sigma A) = (0.300 \times 385) / (0.8 \times 5.67 \times 10^{-8} \times 5.027 \times 10^{-3}) = 115.5 / (2.28 \times 10^{-10}) = 5.07 \times 10^{11}$ s·K³.

Step 2 — Exact formula.

$$t = \frac{mc}{2e\sigma AT_0^3} \left[\ln \left(\frac{T_1 - T_0}{T_2 - T_0} \cdot \frac{T_2 + T_0}{T_1 + T_0} \right) + 2 \tan^{-1} \frac{T_2}{T_0} - 2 \tan^{-1} \frac{T_1}{T_0} \right]$$

$T_0 = 300$, $T_1 = 500$, $T_2 = 400$. Term 1: $\ln((200/100) \times (700/800)) = \ln(2 \times 0.875) = \ln(1.75) = 0.5596$. Term 2: $2[\tan^{-1}(400/300) - \tan^{-1}(500/300)] = 2[\tan^{-1}(1.333) - \tan^{-1}(1.667)] = 2[0.9273 - 1.0304] = -0.2062$. Bracket = $0.5596 - 0.2062 = 0.3534$. Prefactor = $(5.07 \times 10^{11}) / (2 \times 300^3) = 5.07 \times 10^{11} / 5.4 \times 10^7 = 9389$ s. $t_{\text{exact}} = 9389 \times 0.3534 = \mathbf{3318 \text{ s} \approx 55.3 \text{ min}}$.

Step 3 — Newton approximation ($k = 4e\sigma AT_0^3/mc$). $k = 4 \times 0.8 \times 5.67 \times 10^{-8} \times 5.027 \times 10^{-3} \times (300)^3 / 115.5 = 4.27 \times 10^{-4}$ s⁻¹. $t_{\text{newton}} = (1/k) \ln((T_1 - T_0)/(T_2 - T_0)) = 2342 \times \ln(200/100) = 2342 \times 0.693 = 1623$ s = **27.0 min**.

Conclusion: Newton’s law underestimates time by ~50% because the T^4 term drops faster than linear as T approaches T_0 . **Always check if $\Delta T/T_0 < 0.1$ before using Newton.**

Example C.2 — [JEE Advanced Level] Satellite Plate Equilibrium

A flat plate in Earth orbit faces the Sun. Solar constant $S = 1360$ W/m². Plate has $\alpha = 0.9$ for $\lambda < 2.0 \mu\text{m}$ (solar spectrum peak $\sim 0.5 \mu\text{m}$) and $\varepsilon = 0.1$ for $\lambda > 2.0 \mu\text{m}$ (plate’s own thermal spectrum). Back side is perfectly insulated. Find equilibrium temperature.

Step 1 — Energy balance. Absorbed solar power = Emitted thermal power.

$$\alpha_{\text{solar}} S = \varepsilon_{\text{IR}} \sigma T^4$$

Step 2 — Solve.

$$T^4 = \frac{0.9 \times 1360}{0.1 \times 5.67 \times 10^{-8}} = \frac{1224}{5.67 \times 10^{-9}} = 2.158 \times 10^{11}$$

$$T = (2.158 \times 10^{11})^{1/4} = 681 \text{ K}$$

Check: $\lambda_m = 2.9 \times 10^{-3} / 681 = 4.26 \mu\text{m} > 2.0 \mu\text{m}$. Assumption $\epsilon=0.1$ for thermal radiation is self-consistent. **Answer: 681 K (408°C).** A black plate ($\alpha=\epsilon=1$) would be 394 K. Selective surface runs hotter because it traps its own heat (low ϵ) while absorbing well (high α).

Addendum D — Advanced Calorimetry: Variable Specific Heat & Continuous Power

Subtopic 02 assumed constant c . Real materials have $c = c(T)$ (e.g., metals at low T : $c \propto T^3$; gases: $c_v = a + bT$). JEE Advanced asks for: 1. **Heat required** when $c = a + bT$: $Q = m \int_{T_1}^{T_2} (a + bT) dT$. 2. **Cooling/heating curves under constant power P** : $P = mc(T) dT/dt \Rightarrow dt = (mc(T)/P) dT$. 3. **Determining water equivalent (w) experimentally**: Mix known masses at known temps, solve for w .

Method: Variable Specific Heat Integration

Given $c(T) = c_0 + \beta T$ (common model).

$$Q = m \int_{T_1}^{T_2} (c_0 + \beta T) dT = m \left[c_0(T_2 - T_1) + \frac{\beta}{2}(T_2^2 - T_1^2) \right]$$

Units: c_0 in J/kg·K, β in J/kg·K². Q in J.

Constant Power Heating:

$$t = \frac{m}{P} \int_{T_1}^{T_2} c(T) dT = \frac{m}{P} \left[c_0 \Delta T + \frac{\beta}{2}(T_2^2 - T_1^2) \right]$$

Method: Finding Water Equivalent of Calorimeter

Experiment: Calorimeter (mass m_c , unknown c_c , water equivalent $w = m_c c_c / c_w$) contains m_w water at T_1 . Add m_h water at T_2 . Final temp T_f . **Balance:**

$$m_h c_w (T_2 - T_f) = (m_w c_w + w c_w) (T_f - T_1)$$

$$w = \frac{m_h (T_2 - T_f)}{T_f - T_1} - m_w$$

Two-trial method (eliminates heat loss): Do two mixes with different m_h , T_2 . Plot ($T_f - T_1$) vs ($T_2 - T_f$); slope gives w .

Worked Examples

Example D.1 — [JEE Advanced Level] Variable Specific Heat

The specific heat of a solid varies as $c = (0.50 + 0.002 T) \text{ J g}^{-1}\text{K}^{-1}$ (T in $^{\circ}\text{C}$). Find the heat required to raise 200 g from 0°C to 100°C . Compare with using average c .

Step 1 — Convert to SI / consistent units. $m = 0.200 \text{ kg}$. $c_0 = 500 \text{ J/kg}\cdot\text{K}$. $\beta = 2.0 \text{ J/kg}\cdot\text{K}^2$ (since $0.002 \text{ J/g}\cdot\text{K}^2 = 2 \text{ J/kg}\cdot\text{K}^2$). T in Kelvin for differences? ΔT in K = ΔT in $^{\circ}\text{C}$. T^2 terms: use $^{\circ}\text{C}$ if $c(T)$ defined in $^{\circ}\text{C}$, but $\beta T^2/2$ difference works if T is absolute? **Crucial:** If $c = a + b\theta$ (θ in $^{\circ}\text{C}$), then $\int c \, d\theta$ from θ_1 to θ_2 is correct. If $c = a + bT$ (T in K), use K. Problem states T in $^{\circ}\text{C}$. $\theta_1 = 0$, $\theta_2 = 100$.

Step 2 — Integrate.

$$Q = 0.200 \left[500(100) + \frac{2.0}{2}(100^2 - 0) \right] \\ = 0.200 [50000 + 10000] = 0.200 \times 60000 = 12000 \text{ J}$$

Step 3 — Average c method. $c_{\text{avg}} = c(50) = 500 + 2 \times 50 = 600 \text{ J/kg}\cdot\text{K}$. $Q_{\text{avg}} = 0.200 \times 600 \times 100 = 12000 \text{ J}$. *Matches exactly because c is linear in T .* For non-linear c , average c method fails.

Example D.2 — [JEE Main Level] Determining Water Equivalent

A calorimeter contains 100 g water at 20.0°C . 50 g water at 60.0°C is added. Final temperature is 35.0°C . Find w (in grams).

Step 1 — Formula.

$$w = \frac{m_h(T_2 - T_f)}{T_f - T_1} - m_w$$

Step 2 — Substitute. $m_h = 50 \text{ g}$, $m_w = 100 \text{ g}$. $T_2 = 60$, $T_1 = 20$, $T_f = 35$.

$$w = \frac{50(60 - 35)}{35 - 20} - 100 = \frac{50 \times 25}{15} - 100 = \frac{1250}{15} - 100 = 83.33 - 100 = -16.67 \text{ g}$$

Negative w ? Impossible. **Error check:** Heat lost by hot = $50 \times 25 = 1250 \text{ cal}$. Heat gained by cold water = $100 \times 15 = 1500 \text{ cal}$. Gained > Lost $\Rightarrow$ heat *leaked in* from surroundings, or thermometer error. In a real experiment, this signals a systematic error (e.g., calorimeter not insulated, or hot water lost heat before mixing). The formula assumes adiabatic walls. If forced to answer, state "Data inconsistent with adiabatic assumption; heat gained by surroundings suspected."

Corrected Example: If $T_f = 30.0^{\circ}\text{C}$.

$$w = \frac{50(30)}{10} - 100 = 150 - 100 = 50 \text{ g}.$$

Calorimeter water equivalent = 50 g.

Practice — Round 2 Addendum

Addendum A Practice

1. **[JEE Main]** Two rods A and B of equal length L and cross-section A are joined end-to-end between rigid walls. $\alpha_A = 2\alpha$, $Y_A = Y$; $\alpha_B = \alpha$, $Y_B = 2Y$. Temperature increased by ΔT . Find stress in rod A.
2. **[JEE Advanced]** A steel rod ($L=1\text{ m}$, $A=2\text{ cm}^2$) is clamped between rigid supports at 0°C . A 10 kg mass is hung from its midpoint. The rod is heated to 50°C . Find the tension in the rod. ($Y=2\times 10^{11}\text{ Pa}$, $\alpha=1.2\times 10^{-5}/\text{K}$, $g=10\text{ m/s}^2$).
3. **[CBSE]** A bimetallic strip of copper ($\alpha=1.7\times 10^{-5}$) and steel ($\alpha=1.2\times 10^{-5}$), each 1 mm thick, is straight at 0°C . Find radius of curvature at 100°C . (Assume equal Y , equal width).

Addendum B Practice

4. **[JEE Advanced]** A hollow sphere ($r_1=5\text{ cm}$, $r_2=10\text{ cm}$, $K=10\text{ W/mK}$) has inner surface at 200°C and outer at 30°C . Find heat flow. If a second layer ($r_2=10\text{ cm}$ to $r_3=15\text{ cm}$, $K=1\text{ W/mK}$) is added, find new heat flow and interface temperature.
5. **[JEE Main]** A cylindrical pipe ($L=10\text{ m}$, $r_1=2\text{ cm}$, $r_2=3\text{ cm}$, $K=50$) carries steam at 150°C . It is insulated with 5 cm thick material ($K=0.1$). Ambient is 25°C . Find heat loss per hour. (Ignore outer convection).
6. **[NEET]** Thermal resistance of a spherical shell (r_1 , r_2 , K) is R . If r_1 and r_2 are both doubled (thickness doubles), what is the new resistance?

Addendum C Practice

7. **[JEE Advanced]** A black body ($m=1\text{ kg}$, $c=500\text{ J/kgK}$, $A=0.01\text{ m}^2$) cools from 1000 K to 500 K in 0 K surroundings. Find time.
8. **[JEE Advanced]** Two large parallel plates at $T_1=500\text{ K}$, $T_2=300\text{ K}$. $e_1=0.8$, $e_2=0.6$. Find net radiative flux (W/m^2).
9. **[JEE Main]** A body cools from 80°C to 60°C in 5 min (surroundings 20°C). Using *exact* Stefan law ($e=1$), find time to cool from 60°C to 40°C . (Hint: $mc/(e\sigma A) = 10^{12}\text{ s}\cdot\text{K}^3$ — a typical value; e.g. $m = 1\text{ kg}$, $c = 500\text{ J/kg K}$, $\sigma A \sim 5 \times 10^{-8} \times 0.01$ gives $mc/(\sigma A) \sim 10^{12}$).

Addendum D Practice

10. **[JEE Advanced]** Specific heat of a substance: $c = 100 + 0.2 T$ ($\text{J/kg}\cdot\text{K}$, T in K). Heat needed for 2 kg from 100 K to 300 K ?
11. **[JEE Main]** A 2 kg block ($c = 500\text{ J/kgK}$ constant) is heated by a 1000 W heater. How long to go from 300 K to 400 K ? If $c = 400 + 0.5T$, how long?
12. **[CBSE]** In a water equivalent experiment, 200 g water at 20°C + 100 g water at 80°C $\rightarrow T_f = 40^\circ\text{C}$. Find w . If a second trial with 150 g hot water gives $T_f = 45^\circ\text{C}$, verify w .

Answers

A1. Force F common. $\Delta L_A + \Delta L_B = 0$.

$$\frac{FL}{AY} + 2\alpha L\Delta T + \frac{FL}{A(2Y)} + \alpha L\Delta T = 0$$

$$\frac{FL}{AY}(1 + 0.5) = -3\alpha L\Delta T \Rightarrow F = -2\alpha AY\Delta T$$

$\sigma_A = F/A = -2\alpha Y\Delta T$ (Compressive).

A2. The rod cannot elongate overall (rigid supports), but the midpoint, where the mass hangs, can *displace* by some amount x — with the top half stretching by x more than pure thermal expansion allows and the bottom half compressing by x less, so the two halves need not carry equal force. Let F_1 be the (tension-positive) force in the top half and F_2 in the bottom half.

Length change of each half (mechanical + thermal), with the midpoint displacement x built in:

$$\Delta L_1 = x = \frac{F_1(L/2)}{AY} + \alpha \frac{L}{2}\Delta T, \quad \Delta L_2 = -x = \frac{F_2(L/2)}{AY} + \alpha \frac{L}{2}\Delta T$$

Adding eliminates x (total length is fixed, so $\Delta L_1 + \Delta L_2 = 0$ automatically holds for any x , but adding the two equations directly gives the force sum):

$$F_1 + F_2 = -2AY\alpha\Delta T$$

Force balance on the hanging mass (F_1 pulls up via tension in the top half, F_2 pushes up via compression in the bottom half) gives $F_1 - F_2 = mg$. Solving the pair:

$$F_1 = -AY\alpha\Delta T + \frac{mg}{2}, \quad F_2 = -AY\alpha\Delta T - \frac{mg}{2}$$

Numbers: $AY\alpha\Delta T = (2 \times 10^{-4})(2 \times 10^{11})(1.2 \times 10^{-5})(50) = 24000$ N, and $mg/2 = 50$ N, so

$$F_1 = -24000 + 50 = -23950 \text{ N}, \quad F_2 = -24000 - 50 = -24050 \text{ N}$$

Both halves are in compression — the thermal stress (24000 N) completely dominates the 50 N asymmetry from the hanging mass, so despite the mass “pulling” on the top half, there is no tension anywhere in the rod. Stress magnitude $\approx 2.4 \times 10^8$ Pa in each half (differing only in the fourth significant figure).

A3. $d = 1\text{mm} = 0.001\text{m}$. $\alpha_{\text{Cu}} - \alpha_{\text{Steel}} = 0.5\text{e-}5$. $\Delta T = 100$. $R = 2d / ((\alpha_{\text{Cu}} - \alpha_{\text{Steel}})\Delta T) = 0.002 / (0.5\text{e-}5 \times 100) = 0.002 / 5\text{e-}4 = \mathbf{4 \text{ m}}$.

B4. Sphere 1: $R_1 = (0.1-0.05)/(4\pi \times 10 \times 0.05 \times 0.1) = 0.05 / (0.0628) = 0.796$ K/W. $H = 170 / 0.796 = \mathbf{213.5 \text{ W}}$. Sphere 2 (layer): $R_2 = (0.15-0.1)/(4\pi \times 1 \times 0.1 \times 0.15) = 0.05 / 0.1885 = 0.265$ K/W. $R_{\text{total}} = 1.061$ K/W. $H_{\text{new}} = 170 / 1.061 = \mathbf{160.2 \text{ W}}$. $T_{\text{interface}} = 200 - 160.2 \times 0.796 = \mathbf{72.5^\circ\text{C}}$.

B5. $R_{\text{steel}} = \ln(3/2)/(2\pi \times 50 \times 10) = 0.4055/3141.6 = 1.29\text{e-}4$ K/W. $R_{\text{insul}} = \ln(8/3)/(2\pi \times 0.1 \times 10) = 0.981/6.283 = 0.156$ K/W. $H = 125 / 0.156 = 801$ W. Per hour = $801 \times 3600 = \mathbf{2.88 \text{ MJ}}$.

B6. $R = (r_2 - r_1)/(4\pi K r_1 r_2)$. New $r_1' = 2r_1$, $r_2' = 2r_2$. $R' = (2r_2 - 2r_1)/(4\pi K 4r_1 r_2) = 2(r_2 - r_1)/(16\pi K r_1 r_2) = R/2$. **Half.**

C7. $T_0 = 0$. $t = mc/(3e\sigma A) (1/T_2^3 - 1/T_1^3)$. $mc = 500$. $e\sigma A = 1 \times 5.67e-8 \times 0.01 = 5.67e-10$. $t = 500/(3 \times 5.67e-10) \times (1/125e6 - 1/1e9) = 2.94e11 \times (8e-9 - 1e-9) = 2.94e11 \times 7e-9 = \mathbf{2058 \text{ s}}$.

C8. $q = \sigma(5004 - 3004) / (1/0.8 + 1/0.6 - 1) = 5.67e-8(6.25e10 - 0.81e10) / (1.25 + 1.667 - 1) = 5.67e-8 \times 5.44e10 / 1.917 = 3084 / 1.917 = \mathbf{1609 \text{ W/m}^2}$.

C9. Use the exact formula. $T_0 = 293 \text{ K}$, $T_1 = 353$, $T_2 = 333$, $T_3 = 313 \text{ K}$. Using the given $mc/(e\sigma A) = 10^{12} \text{ s}\cdot\text{K}^3$: Prefactor $= \frac{mc/(e\sigma A)}{2T_0^3} = \frac{10^{12}}{2 \times 2.5 \times 10^7} = 2 \times 10^4 \text{ s}$. Bracket $(60^\circ\text{C} \rightarrow 40^\circ\text{C}, \text{ i.e. } T_1 \rightarrow T_2)$: $\ln\left(\frac{60}{40} \times \frac{646}{626}\right) + 2\left(\tan^{-1} \frac{333}{293} - \tan^{-1} \frac{353}{293}\right) = \ln(1.5 \times 1.032) + 2(0.848 - 0.878) = 0.435 - 0.06 = 0.375$.

$$t_{12} = 20000 \times 0.375 = \mathbf{7500 \text{ s} \approx 2.1 \text{ hr}}$$

$t_{23} (60 \rightarrow 40)$: $T_2 = 333$, $T_3 = 313$. Bracket $= \ln((40/20) \times (606/626)) + 2(\tan^{-1}(313/293) - \tan^{-1}(333/293)) = \ln(2 \times 0.968) + 2(0.818 - 0.848) = 0.667 - 0.06 = 0.607$. $t_{23} = 20000 \times 0.607 = \mathbf{12140 \text{ s} \approx 3.4 \text{ hr}}$. Ratio ~ 1.6 . Newton gives ratio $\ln(2)/\ln(1.5) = 1.7$. Close.

D10. $Q = 2 [100(200) + 0.1(3002 - 1002)] = 2 [20000 + 0.1(80000)] = 2 [28000] = \mathbf{56,000 \text{ J}}$.

D11. Const c: $t = mc\Delta T/P = 2 \times 500 \times 100/1000 = \mathbf{100 \text{ s}}$. Var c: $t = m/P [c_0\Delta T + \beta/2(T_2^2 - T_1^2)] = 2/1000 [400 \times 100 + 0.25(160000 - 90000)] = 0.002 [40000 + 17500] = 0.002 \times 57500 = \mathbf{115 \text{ s}}$.

D12. Trial 1: $w = 100(80 - 40)/(40 - 20) - 200 = 100 \times 2 - 200 = \mathbf{0 \text{ g}}$. Trial 2: $w = 150(80 - 45)/(45 - 20) - 200 = 150 \times 35/25 - 200 = 210 - 200 = \mathbf{10 \text{ g}}$. Discrepancy $\rightarrow$ Heat loss/gain. Average $w \approx 5 \text{ g}$. Or plot method.

Chapter 11 — Thermodynamics

Definitive Study Companion

CBSE Boards • JEE Main • NEET • JEE Advanced

Drona Ed Tech

June 2026

A complete, exam-ready master companion for Class 11 Thermodynamics — from the Zeroth Law to the Carnot engine and entropy. Every concept, derivation, worked example, and trap a student needs to master the chapter across all four exam tiers, in one place.

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Subtopic 01 — Thermodynamic State and the Zeroth Law

Exam Relevance: This is the conceptual foundation of the entire chapter. CBSE Boards almost always pick a 1–2 mark question here — a definition of the zeroth law, “state variable vs. path variable,” or a clean equation-of-state calculation. NEET loves it as a one-line conceptual trap (thermal equilibrium $\neq$ equal energy). JEE Main asks crisp equation-of-state numericals, and JEE Advanced occasionally probes the deep idea: that temperature is a property the zeroth law forces into existence. Master this, and the rest of thermodynamics has solid ground to stand on.

Section 1: Concept Intuition

Imagine you walk into a sweet shop in the afternoon and pick up a steel tin of fresh gulab jamun syrup that’s been sitting on the counter. Your hand tells you it’s “warm.” But what is warmth, really? You can’t see it, weigh it, or bottle it. Yet you’re absolutely confident the tin and the room have settled into some shared condition. That shared condition is the seed of everything we are about to build.

Thermodynamics is the branch of physics that studies heat, temperature, and how heat converts into other forms of energy — but it does so from a deliberately “zoomed-out” point of view. It is a *macroscopic* science. When we describe a litre of gas thermodynamically, we don’t care that it contains roughly 10^{22} molecules each zipping around with its own velocity. That microscopic bookkeeping belongs to kinetic theory. Thermodynamics instead asks: what few bulk numbers can I measure with ordinary instruments that completely pin down the condition of this gas? The answer is a small handful of quantities — pressure, volume, temperature, mass, composition — that our senses and lab instruments can grab directly.

This bundle of bulk quantities defines what we call the **thermodynamic state** of a system. A “system” is simply the chunk of the universe we choose to study (the gas in a cylinder, the syrup in the tin), and everything outside it is the **surroundings**. The boundary between them — the **wall** — is more important than students expect. A wall that lets heat pass through is **diathermic** (conducting); a wall that blocks heat entirely is **adiabatic** (insulating). Whether two systems can “feel” each other thermally depends entirely on the wall between them.

Think about it this way...

Picture two friends in adjacent train compartments. If the connecting door is open (diathermic wall), they keep chatting until they’ve agreed on every piece of gossip — they reach a settled, no-more-news state. If the door is sealed (adiabatic wall), each lives in their own world, oblivious to the other. “Reaching a settled state where nothing more changes” is exactly what physicists call **thermal equilibrium**: two systems in thermal contact stop exchanging heat once their macroscopic variables stop drifting. The hot one cooled, the cold one warmed, and they met somewhere in the middle.

Now here is the quietly profound part. Experiment shows that thermal equilibrium is *transitive*, like an equals sign in algebra. If system A has settled with system C, and system B has separately settled with the same C, then A and B turn out to already be settled with each other — even if A and B were never in direct contact. This single observation is the **Zeroth Law of Thermodynamics**, and it does something beautiful: it guarantees that every system carries a single number such that “equal numbers” means “in equilibrium.” We name that number **temperature**. So temperature isn’t assumed at the start of thermodynamics — it is *forced into existence* by the zeroth law. That’s why this law, discovered last, is numbered first: temperature has to exist before the first and second laws can even be written down.

Limiting Conditions & Assumptions.

Keep three guardrails in mind. (1) The whole “state” picture requires the system to actually be in equilibrium — for a gas mid-explosion, pressure and temperature aren’t even well-defined at a single value. (2) The clean three-variable description (P, V, T linked by an equation of state)

assumes a homogeneous system of fixed mass and composition; a mixture mid-reaction needs more variables. (3) The zeroth law is an empirical law about equilibrium — it says equal temperatures mean equilibrium, but it says nothing about equal energy. *Two systems at the same temperature can hold wildly different amounts of internal energy.* That distinction trips up more students in the exam hall than almost anything else in this topic.

Figure 11.1 Three boxes labelled A, B, C. A and B are on the left and right, each separated from the central box C by a wall drawn as a hatched bar labelled “diathermic (conducting)”. A dashed line separates A from B labelled “adiabatic (insulating)”. Arrows show heat flowing $A \leftrightarrow C$ and $B \leftrightarrow C$ until equilibrium. A caption arrow then shows: “Remove A-B insulator $\rightarrow$ A and B already in equilibrium.”

Section 2: Key Formulas & Definitions

Thermodynamic state: the condition of a homogeneous system of fixed mass and composition, completely specified by any two independent state variables (e.g. P and V , or P and T , or V and T). **Equation of state (ideal gas):** $PV = nRT$ where P = pressure, V = volume, n = number of moles, T = absolute (kelvin) temperature, and R = universal gas constant. A frequently used “per molecule” form is $PV = Nk_B T$ where N = number of molecules and $k_B = R/N_A$ is the Boltzmann constant. **Zeroth Law (formal statement):** If two systems A and B are each separately in thermal equilibrium with a third system C, then A and B are in thermal equilibrium with each other. Consequently there exists a state variable — temperature T — such that $T_A = T_C$ and $T_B = T_C \Rightarrow T_A = T_B$.

Quantity reference table:

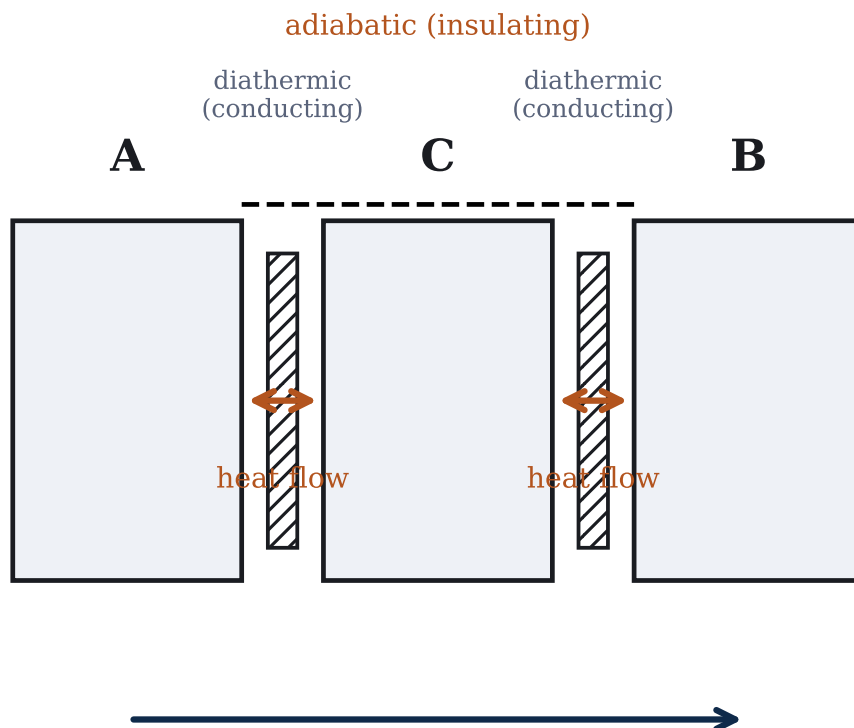
Quantity	Symbol	SI Unit	Dimensional Formula
Pressure	P	pascal (Pa)	$[M^1 L^{-1} T^{-2}]$
Volume	V	m^3	$[M^0 L^3 T^0]$
Temperature	T	kelvin (K)	$[0]$
Number of moles	n	mol	$[mol]$
Universal gas constant	R	$J\ mol^{-1} K^{-1}$	$[M^1 L^2 T^{-2} mol^{-1}]$
Boltzmann constant	k_B	$J\ K^{-1}$	$[M^1 L^2 T^{-2} mol^{-1}]$
Internal energy	U	joule (J)	$[M^1 L^2 T^{-2}]$

Numerical constants you must memorise: $R = 8.314\ J\ mol^{-1} K^{-1}$, $k_B = 1.38 \times 10^{-23}\ J\ K^{-1}$, $N_A = 6.022 \times 10^{23}\ mol^{-1}$, and the triple point of water $T_{tr} = 273.16\ K$.

Key classification — Intensive vs. Extensive variables:

- **Intensive** (do not depend on amount of substance): pressure, temperature, density, molar volume.
- **Extensive** (scale with amount): volume, mass, internal energy, number of moles.

• A handy test: cut the system in half.
 Whatever stays the same is intensive (P, T); whatever halves is extensive (V, U, n). 4 (see next page)



Remove A-B insulator → A and B already in equilibrium

Figure 11.1 Three boxes labelled A, B, C. A and B are on the left and right, each separated from the central box C by a wall drawn as a hatched bar labelled “diathermic (conducting)”. A dashed line separates A from B labelled “adiabatic (insulating)”. Arrows show heat flowing $A \leftrightarrow C$ and $B \leftrightarrow C$ until equilibrium. A caption arrow then shows: “Remove A-B insulator → A and B already in equilibrium.”

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Quantity reference table:

Quantity	Symbol	SI Unit	Dimensional Formula
Pressure	P	pascal (Pa)	$[M L^{-1} T^{-2}]$
Volume	V	m^3	$[M^0 L^3 T^0]$
Temperature	T	kelvin (K)	$[T]$
Number of moles	n	mol	$[mol]$
Universal gas constant	R	$J\ mol^{-1} K^{-1}$	$[M L^2 T^{-2} mol^{-1}]$
Boltzmann constant	k_B	$J\ K^{-1}$	$[M L^2 T^{-2} mol^{-1}]$
Internal energy	U	joule (J)	$[M L^2 T^{-2}]$

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- Intensive (do not depend on amount of substance): pressure, temperature, density, molar volume.
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A handy test: cut the system in half. Whatever stays the same is intensive (P , T); whatever halves is extensive (V , U , n).

Section 3: Key Procedures & Logical Framework

There is no algebra-heavy “derivation” of the zeroth law itself — it is an empirical law. So this section lays out the logical machinery the law sets in motion, plus the standard procedures you’ll execute in problems.

3.1 How the Zeroth Law creates temperature (the logical chain)

This is the reasoning JEE Advanced examiners love, so understand each link:

1. **Observation:** When two systems are joined by a diathermic wall, their macroscopic variables change and then stop. We call the final unchanging condition thermal equilibrium.
2. **The transitivity experiment:** Put A and B each in equilibrium with a third body C (a thermometer, say), while keeping A and B insulated from each other. Now remove the insulation between A and B. Result: nothing changes — they were already in equilibrium. This is the zeroth law as a raw experimental fact.
3. **The logical consequence:** “Being in equilibrium with C” partitions all systems into groups. Every member of a group is in equilibrium with every other member. We need a label for each group.
4. **Defining the label:** Assign each group a number such that two systems share the number if and only if they are in mutual equilibrium. That number is temperature. Equilibrium $\Leftrightarrow$ equal temperature is now true by construction, made legitimate by step 2.
5. **The pay-off — the thermometer:** Because of step 4, we never need to press two objects together to compare them. We touch a small standard system (thermometer) to each in turn; equal thermometer readings guarantee equal temperatures. Every temperature measurement you have ever made silently relies on the zeroth law.

Figure 11.2 *A flow chart with ve connected boxes reading: “Thermal contact via diathermic wall” → “Variables settle (equilibrium)” → “Equilibrium is transitive (Zeroth Law)” → “De ne a shared label = Temperature” → “Thermometer compares without direct contact”. 3.2 Procedure: specifying a thermodynamic state 1. Con rm the system is homogeneous, of xed mass and composition, and in equilibrium (oth- erwise state variables aren't single-valued). 2. Choose any two independent variables — most commonly two of P , V , T . 3. The third follows automatically from the equation of state $PV = nRT$. You never need all three independently; specifying all three over-determines the state. 4. Plot the state as a single point on a P - V (indicator) diagram. A point = a definite state; a line = a process. 3.3 Procedure: state variables vs. path variables • A state variable (point function) depends only on the current state, not on how the system got there: P , V , T , U . Over any closed cycle, its net change is zero: $\oint dU = 0$. • A path variable depends on the route taken between two states: heat Q and work W . There is no “amount of heat stored in a gas” — only heat transferred along a path. • Quick decision rule: ask “does this quantity have a definite value the instant I freeze the system?” If yes → state variable. If it only makes sense while something is happening → path variable. Section 4: Worked Examples 5 (see next page)*

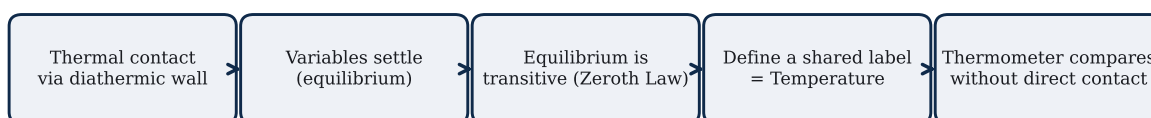


Figure 11.2 A flow chart with five connected boxes reading: “Thermal contact via diathermic wall” → “Variables settle (equilibrium)” → “Equilibrium is transitive (Zeroth Law)” → “Define a shared label = Temperature” → “Thermometer compares without direct contact”. 3.2 Procedure: specifying a thermodynamic state 1. Confirm the system is homogeneous, of fixed mass and composition, and in equilibrium (otherwise state variables aren't single-valued). 2. Choose any two independent variables — most commonly two of P , V , T . 3. The third follows automatically from the equation of state $PV = nRT$. You never need all three independently; specifying all three over-determines the state. 4. Plot the state as a single point on a P - V (indicator) diagram. A point = a definite state; a line = a process. 3.3 Procedure: state variables vs. path variables • A state variable (point function) depends only on the current state, not on how the system got there: P , V , T , U . Over any closed cycle, its net change is zero: $\oint dU = 0$. • A path variable depends on the route taken between two states: heat Q and work W . There is no “amount of heat stored in a gas” — only heat transferred along a path. • Quick decision rule: ask “does this quantity have a definite value the instant I freeze the system?” If yes → state variable. If it only makes sense while something is happening → path variable. Section 4: Worked Examples 5

Example 1 — [CBSE Board Level]

Problem. A rigid steel cylinder of fixed internal volume 0.025 m^3 holds an ideal gas at a pressure of $2.0 \times 10^5 \text{ Pa}$ and a temperature of 300 K . Determine the number of moles of gas in the cylinder.

Solution. Given: $V = 0.025 \text{ m}^3$, $P = 2.0 \times 10^5 \text{ Pa}$, $T = 300 \text{ K}$, $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$. Rearranging the equation of state for n :

$$n = \frac{PV}{RT} = \frac{(2.0 \times 10^5)(0.025)}{(8.314)(300)} = \frac{5.0 \times 10^3}{2.494 \times 10^3} \text{ mol} \approx 2.0 \text{ mol}.$$

Note on units: $\text{Pa} \cdot \text{m}^3 = \text{N m}^{-2} \cdot \text{m}^3 = \text{N m} = \text{J}$, which cancels cleanly against the joules in R , leaving mol. Always check that the energy units reconcile — it's the fastest sanity check in board exams.

Example 2 — [NEET Speed Trap]

Problem. Three sealed flasks P, Q, and R are observed over time. P is in thermal equilibrium with Q, and Q is in thermal equilibrium with R. A student concludes: (i) P is in equilibrium with R; (ii) P, Q, R all have the same temperature; (iii) P, Q, R all contain the same internal energy. Which conclusions are correct?

Speed solution. Reason in one pass:

- (i) is the zeroth law applied directly ($P \leftrightarrow Q$, $Q \leftrightarrow R \Rightarrow P \leftrightarrow R$). **Correct.**
- (ii) is just (i) restated through temperature. Equilibrium $\Leftrightarrow$ equal temperature. **Correct.**
- (iii) is the trap. Temperature is intensive; internal energy is extensive and depends on the amount and nature of gas in each flask. A flask with 5 mol holds far more internal energy than one with 1 mol at the same temperature. **Incorrect.**

Answer: (i) and (ii) only.

Mental-elimination tip: the instant a question offers an “equal energy/equal heat” option alongside “equal temperature,” suspect the energy option. Equal T is what the zeroth law gives you; equal U is something it never promises.

Example 3 — [JEE Main Level]

Problem. A rigid, externally insulated container is split into two equal halves by a fixed, heat-conducting (diathermic) partition. Each half has volume 0.010 m^3 . The left half holds 2.0 mol of a monoatomic ideal gas at 400 K ; the right half holds 3.0 mol of the same gas at 300 K . Find (a) the final common temperature and (b) the final pressure on each side.

Solution. *Step 1 — what is conserved?* The partition is fixed (no work) and the container is externally insulated (no heat in/out). So total internal energy is constant. For a monoatomic ideal gas $U = \frac{3}{2}nRT$:

$$\frac{3}{2}R(n_1T_1 + n_2T_2) = \frac{3}{2}R(n_1 + n_2)T_f.$$

Step 2 — common temperature. The $\frac{3}{2}R$ cancels:

$$T_f = \frac{n_1T_1 + n_2T_2}{n_1 + n_2} = \frac{(2.0)(400) + (3.0)(300)}{5.0} = \frac{1700}{5.0} \text{ K} = 340 \text{ K}.$$

By the zeroth law this single temperature now describes both halves. *Step 3 — pressures* (each half keeps $V = 0.010 \text{ m}^3$):

$$P_{\text{left}} = \frac{n_1RT_f}{V} = \frac{(2.0)(8.314)(340)}{0.010} \approx 5.65 \times 10^5 \text{ Pa},$$

$$P_{\text{right}} = \frac{n_2 R T_f}{V} = \frac{(3.0)(8.314)(340)}{0.010} \approx 8.48 \times 10^5 \text{ Pa.}$$

The temperatures equalise, but the pressures do not — the halves hold different amounts of gas in the same volume. Thermal equilibrium fixes temperature, not the other state variables.

Example 4 — [JEE Advanced Level]

Problem. A constant-volume ideal-gas thermometer reads $T = 273.16 (P/P_{tr})$ K, where P_{tr} is its gas pressure at the triple point of water. Dipped in bath A it reads $1.10 P_{tr}$; in bath B it reads $1.30 P_{tr}$. Bath A is 2.0 kg of water, bath B is 1.0 kg of water (same specific heat). The baths are placed in thermal contact, isolated, and allowed to settle. The thermometer is afterwards dipped in each in turn. Predict, with justification, and compute the two readings.

Solution. *Step 1 — initial bath temperatures:*

$$T_A = 273.16(1.10) = 300.5 \text{ K}, \quad T_B = 273.16(1.30) = 355.1 \text{ K.}$$

Step 2 — the deep prediction. When A and B reach mutual equilibrium, they share one temperature T_f . By the zeroth law the thermometer — itself a third system — must register the same value in A and B. So *before any arithmetic* we assert: the two final readings are identical. *Step 3 — common temperature (calorimetry):* heat lost = heat gained, $m_{Ac}(T_f - T_A) + m_{Bc}(T_f - T_B) = 0$, giving the mass-weighted mean:

$$T_f = \frac{m_A T_A + m_B T_B}{m_A + m_B} = \frac{(2.0)(300.5) + (1.0)(355.1)}{3.0} \approx 318.7 \text{ K.}$$

Step 4 — convert back: $P_f = P_{tr} T_f / 273.16 = P_{tr} (318.7 / 273.16) \approx 1.17 P_{tr}$.

Both readings $\approx 1.17 P_{tr}$ ($T_f \approx 318.7 \text{ K}$)

The zeroth law gives the qualitative answer (equal readings) before calculation; the masses fix the numerical value. Note 318.7 K is nearer T_A — exactly as the 2:1 mass ratio demands.

Section 5: Practice Exercises

- [CBSE]** An ideal gas occupies $5.0 \times 10^{-3} \text{ m}^3$ at $1.0 \times 10^5 \text{ Pa}$ and 250 K. Find the number of moles present.
- [NEET]** Classify each as state or path variable: pressure, work done, internal energy, heat supplied, volume. Then state which two of {pressure, work, volume} can together specify the state of a fixed mass of ideal gas.
- [JEE Main]** A fixed mass of ideal gas at 300 K and $2.0 \times 10^5 \text{ Pa}$ is heated in a rigid sealed container until its pressure rises to $3.2 \times 10^5 \text{ Pa}$. Find its new temperature.
- [JEE Main]** Two equal rigid volumes joined by a fixed diathermic wall, isolated from outside, hold 1.0 mol of a monoatomic ideal gas at 500 K and 4.0 mol of the same gas at 250 K. Find the final common temperature.
- [JEE Advanced]** A constant-volume gas thermometer reads P_1 in a bath at unknown T_1 and $1.5P_1$ in a second bath. The triple-point pressure is $P_{tr} = 0.80P_1$. The second bath is twice as massive (same liquid) and the two are mixed in isolation. Find (a) T_1 , (b) the final common temperature, (c) the thermometer's final reading in each bath as a multiple of P_{tr} .

Answers: 1. $\approx 0.24 \text{ mol}$. 2. State: pressure, internal energy, volume; Path: work, heat. Any two of pressure and volume specify the state (independent). 3. 480 K. 4. 300 K. 5. (a) $T_1 = 273.16(P_1/P_{tr}) = 273.16/0.80 \approx 341.5 \text{ K}$; (b) $T_2 = 1.5 \times 341.5 \approx 512.2 \text{ K}$, mixing mass ratio 1:2 gives $T_f = (341.5 + 2 \times 512.2)/3 \approx 455.3 \text{ K}$; (c) both baths read $P_f = P_{tr}(T_f/273.16) \approx 1.67 P_{tr}$.

Section 6: Crack the MCQ

MCQ 1.

The zeroth law fundamentally allows us to define which physical quantity? (A) Heat (B) Internal energy (C) Temperature (D) Work

Correct: (C) Temperature. The zeroth law makes “in equilibrium with” transitive, letting us tag every system with one number — temperature — equal exactly when two systems are in equilibrium.

- (A) Heat is energy in transit due to a temperature difference; it presupposes temperature exists.
- (B) Internal energy is defined later, through the first law, not the zeroth.
- (D) Work is a mechanical, path-dependent transfer; unrelated to the zeroth law.

MCQ 2.

The minimum number of independent state variables needed to specify the state of a homogeneous fixed mass of ideal gas in equilibrium is: (A) 1 (B) 2 (C) 3 (D) 4

Correct: (B) 2. P, V, T are tied by $PV = nRT$; fixing any two fixes the third.

- (A) 1 is too few — one variable leaves the gas free along an isotherm.
- (C) 3 over-counts, ignoring the equation-of-state constraint — the most common wrong choice.
- (D) 4 adds mass/moles, but the mass is already fixed.

MCQ 3.

Which is an intensive property? (A) Volume (B) Internal energy (C) Number of moles (D) Pressure

Correct: (D) Pressure. Cut the system in half: pressure stays the same, so it is intensive.

- (A) Volume halves — extensive. (B) Internal energy scales with amount — extensive.
- (C) Number of moles literally measures amount — the most obviously extensive quantity.

MCQ 4.

X is in thermal equilibrium with Y. Y is *not* in equilibrium with Z. Which is necessarily true? (A) $X \leftrightarrow Z$ (B) $X \not\leftrightarrow Z$ (C) X, Y, Z equal T (D) Z has greater U than X

Correct: (B). $X \leftrightarrow Y \Rightarrow T_X = T_Y$; $T_Y \neq T_Z$, so $T_X \neq T_Z$, hence X and Z cannot be in equilibrium.

- (A) reverses the logic. (C) impossible since $T_Y \neq T_Z$. (D) confuses temperature with energy.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Confusing equal temperature with equal energy.** A tiny hot system and a huge warm system can hold the same internal energy; two same- T systems usually hold different energies. Temperature is intensive; energy is extensive.
2. **Treating P, V, T as three independent variables.** They are bound by $PV = nRT$. For a fixed mass, only two are free.
3. **Calling heat a state variable.** Q and W are path quantities. Only P, V, T, U are state (point) functions. Never write “the heat content of the gas.”
4. **Using T in $^{\circ}\text{C}$ inside $PV = nRT$.** The gas equation demands kelvin. Always convert: $T(\text{K}) = t(^{\circ}\text{C}) + 273.15$.

Pro-Tip (speed & accuracy)

For “same gas, fixed amount, two states” problems, skip rederiving everything — use the ratio form $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$. Cancel whatever is held constant (V for rigid, P for free-piston) and you’re left with a one-line proportion. And for any “two bodies reach equilibrium” question, write the qualitative answer first — equal final temperature — before touching the algebra.

Section 8: Cheat Sheet Box**THERMODYNAMIC STATE & ZEROth LAW — RAPID RECAP**

- State of a homogeneous fixed-mass system = one point in P - V - T space, fixed by any *two* independent variables.
- Equation of state: $PV = nRT$ (use T in kelvin, always). Per-molecule: $PV = Nk_B T$.
- Zeroth Law: $A \leftrightarrow C$ and $B \leftrightarrow C \Rightarrow A \leftrightarrow B \rightarrow$ defines temperature as the shared label of equilibrium.
- Equilibrium $\Leftrightarrow$ equal temperature (NOT equal energy, NOT equal pressure).
- State variables (point functions): P, V, T, U — depend only on the state; $\oint dU = 0$.
- Path variables: Q, W — depend on the route taken.
- Intensive (P, T, ρ — unchanged on halving) vs. Extensive (V, U, n — halve on halving).

Memory aids: “Zeroth = the law that gives birth to temperature.” • “Cut it in half: same $\rightarrow$ intensive, halved $\rightarrow$ extensive.” • “Heat and Work are journeys, not destinations.” • “Two independent variables, third comes free.”

Subtopic 02 — The First Law of Thermodynamics

Exam Relevance: This is the workhorse of the whole chapter — almost every numerical you’ll meet later is the first law wearing a different costume. CBSE Boards reliably ask a 2–3 mark numerical (“heat given, work done, find ΔU ”) plus the occasional definition or “limitations” question. NEET favours quick sign-convention and conceptual traps (cyclic process, isothermal $\Delta U = 0$). JEE Main asks multi-step or two-path numericals exploiting that internal energy is a state function. JEE Advanced layers it onto phase changes, cycles, or processes where you must track each transfer’s sign with care. Get the sign convention airtight and you’ve won half the chapter.

Section 1: Concept Intuition

We learned that a system’s condition is captured by its internal energy U — a genuine state variable, a number the gas “owns.” The natural next question is the one this entire law answers: *how can that owned energy change?*

Picture a fixed amount of gas trapped in a cylinder under a frictionless, movable piston — the single most important mental image in thermodynamics. There are exactly *two doors* through which energy can enter or leave this gas. Door one is **heat**: place the cylinder on a hot stove, and because of the temperature difference, energy seeps in as heat Q . Door two is **work**: push the piston down and you do work on the gas, cramming energy in; let the gas push the piston up and the gas spends energy doing work on the surroundings. Heat flow and work are the only two ways to alter the internal energy of a closed system. There is no third door.

The first law is just honest accounting of these two doors. Energy is never created or destroyed, so whatever comes in through the heat door must either pile up inside (raising U) or go back out through the work door. In symbols, with the convention Indian boards and JEE/NEET use throughout:

$$\Delta Q = \Delta U + \Delta W.$$

Read it as a sentence: the heat poured into a gas splits into two parts — some raises its internal energy, the rest is spent by the gas doing work on its surroundings.

Think about it this way...

Treat the gas like a bank account for a month. ΔQ is the salary credited (income). ΔW is the rent and bills paid out (expenditure). ΔU is the change in balance (savings). The law $\Delta Q = \Delta U + \Delta W$ says: income = change in savings + expenditure. Some months income is high and savings grow; some months you overspend, withdrawing from savings so the balance drops ($\Delta U < 0$). Nobody invents money from nothing — and no gas invents energy from nothing. That is conservation of energy.

Now the subtle, exam-deciding distinction. Your bank *balance* is a state — at any instant it has one definite value. But “income” and “expenditure” describe *flows* during a period, not contents at an instant. It is meaningless to ask “how much income is sitting in the account right now?” Likewise, internal energy U is a **state function**, while heat Q and work W are **path functions** — they describe energy in transit, not energy stored. This is why physicists scold the phrase “heat contained in a gas”: a gas in a given state has internal energy, never “an amount of heat.”

A powerful consequence drops out: take a gas from state i to state f by ten wildly different routes, and Q and W differ on every route — but $\Delta U = U_f - U_i$ is identical for all ten, because it depends only on the endpoints. So $\Delta Q - \Delta W$ comes out the same no matter the path. This path-independence of ΔU is the single most-used idea in first-law problems.

Limiting Conditions & Assumptions.

Three guardrails. (1) Unit consistency is non-negotiable: Q, U, W must be in the same units — mix calories and joules and your answer is wrong (1 cal = 4.18 J). (2) The law is supremely

general — solids, liquids, gases, even phase changes — but the simple work expression $W = \int P dV$ assumes a quasi-static process with a well-defined pressure at every step. (3) The first law is a bookkeeper, not a fortune-teller: it guarantees energy balances but says nothing about which direction a process spontaneously goes, or what fraction of heat can become useful work. That blind spot is what the second law fills.

Figure 11.3 A vertical cylinder with a frictionless piston on top. A flame beneath labelled “heat ΔQ in” with an upward arrow into the gas. The piston has an upward arrow labelled “gas does work ΔW on surroundings”. Inside the gas region label “internal energy U (state function)”. Side caption: “ $\Delta Q = \Delta U + \Delta W$ ”. Section 2: Key Formulas & Definitions First law (Indian / JEE-NEET sign convention): $\Delta Q = \Delta U + \Delta W \Leftrightarrow \Delta U = \Delta Q - \Delta W$. In differential form, using $^{\sim}d$ for inexact (path-dependent) differentials: $^{\sim}dQ = dU + ^{\sim}dW$. Sign convention (memorise exactly): • $\Delta Q > 0$: heat absorbed by the system; $\Delta Q < 0$: heat released. • $\Delta W > 0$: work done by the system (expansion); $\Delta W < 0$: work done on the system (compression). • $\Delta U > 0$: internal energy (and, for an ideal gas, temperature) increases. Work done by a gas (quasi-static): $\Delta W = \int V_f V_i P dV = \text{area under the } P\text{-}V \text{ curve}$. Internal energy of an ideal gas depends only on temperature, for any process: $\Delta U = nCV \Delta T$, where CV is the molar heat capacity at constant volume. ! Alternate convention alert. Some foreign textbooks (and chemistry) write $\Delta U = \Delta Q + \Delta W$, defining ΔW as work done on the system. It is the same physics with W 's sign flipped. This companion uses the physics/board convention $\Delta Q = \Delta U + \Delta W$ (W = work by gas) throughout. Pick one and never mix them mid-problem. The first law in four standard processes (each is the master equation with one term killed): Process Constraint What vanishes First law reduces to Isochoric V constant $\Delta W = 0$ $\Delta Q = \Delta U$ Isobaric P constant — $\Delta Q = \Delta U + P\Delta V$ Isothermal (ideal gas) T constant $\Delta U = 0$ $\Delta Q = \Delta W$ Adiabatic no heat exchange $\Delta Q = 0$ $\Delta U = -\Delta W$ All three of ΔQ , ΔW , ΔU share the dimension of energy $[M^1L^2T^{-2}]$ — a useful self-check that the equation is dimensionally homogeneous. Section 3: Key Procedures & Logical Framework The first law is an empirical statement of energy conservation, elevated to a postulate. What matters for exams is (a) the reasoning that fixes its form and (b) a reliable solution procedure. 3.1 Why internal energy must be a state function (the logical core) (see next page)

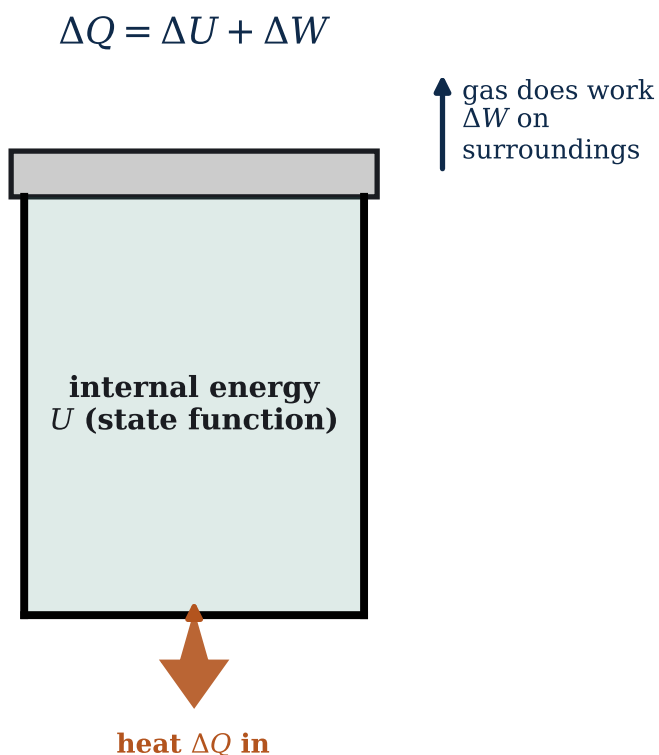


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1. **Experimental fact:** for any process taking a system from i to f , the quantity $(\Delta Q - \Delta W)$ is the same regardless of path, even though ΔQ and ΔW individually vary wildly.
2. **Logical inference:** a quantity that depends only on the endpoints is, by definition, a function of state. So there exists a state function — internal energy U — whose change equals $\Delta Q - \Delta W$.
3. **Definition forced on us:** we therefore define $\Delta U \equiv \Delta Q - \Delta W$. The first law simultaneously expresses energy conservation and introduces internal energy as a legitimate property. The two ideas arrive together.
4. **Cyclic corollary:** if the system returns to its start ($i = f$), then $\Delta U = 0$ over the cycle, so $\oint dQ = \oint dW$ — net heat absorbed equals net work done. This one line solves a huge fraction of cyclic-process questions.

Figure 11.4 A P - V diagram with two points A (lower right) and B (upper left). Three different curved paths connect A to B , labelled Path 1, 2, 3, each a different colour. Caption: “ Q and W differ on every path; $\Delta U = U_B - U_A$ is identical for all.”

3.2 Standard procedure for any rst-law problem

1. Identify the system and the two transfers. What is the substance? Is heat entering or leaving? Is the gas expanding (doing work) or being compressed (work done on it)?
2. Assign signs rst, numbers second. Write ΔQ and ΔW with explicit $+/-$. Most wrong answers here are sign errors, not arithmetic errors.
3. Convert to one unit system (joules, usually). $1 \text{ cal} = 4.18 \text{ J}$.
4. Apply $\Delta U = \Delta Q - \Delta W$, or its reduced form for isochoric/isobaric/isothermal/adiabatic.
5. Exploit path-independence if two routes share endpoints: compute ΔU from the known route, reuse it for the unknown route.
6. Sanity-check the sign of ΔU : for an ideal gas, $\Delta U > 0 \Rightarrow$ temperature rose. Does that match the physics?

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Problem. A gas is supplied 250 cal of heat. During the heating it expands and does 500 J of work on the surroundings. Find the change in internal energy. ($1 \text{ cal} = 4.18 \text{ J}$.) Solution. Given $\Delta Q = +250 \text{ cal}$ (absorbed), $\Delta W = +500 \text{ J}$ (by gas). Convert: $\Delta Q = 250 \times 4.18 = 1045 \text{ J}$. Apply the rst law: $\Delta U = \Delta Q - \Delta W = 1045 - 500 = +545 \text{ J}$. $\Delta U = +545 \text{ J}$ The positive sign tells us internal energy increased by 545 J — most of the heat stayed inside the gas, the rest pushed the piston outward. (Skipping the cal $\rightarrow$ J conversion would give nonsense.)

Example 2 — [NEET Speed Trap] Problem. A gas is compressed by an external agent who does 80 J of work on it. During compression the gas releases 30 J of heat. By how much does the internal energy change? Speed solution. The trap is entirely in the signs — set them rst. Heat released $\Rightarrow \Delta Q = -30 \text{ J}$. Work done on the gas $\Rightarrow$ work done by the gas is $\Delta W = -80 \text{ J}$. $\Delta U = \Delta Q - \Delta W = (-30) - (-80) = +50 \text{ J}$. 12 (see next page)

Q and W differ on every path; $\Delta U = U_B - U_A$ is identical for all

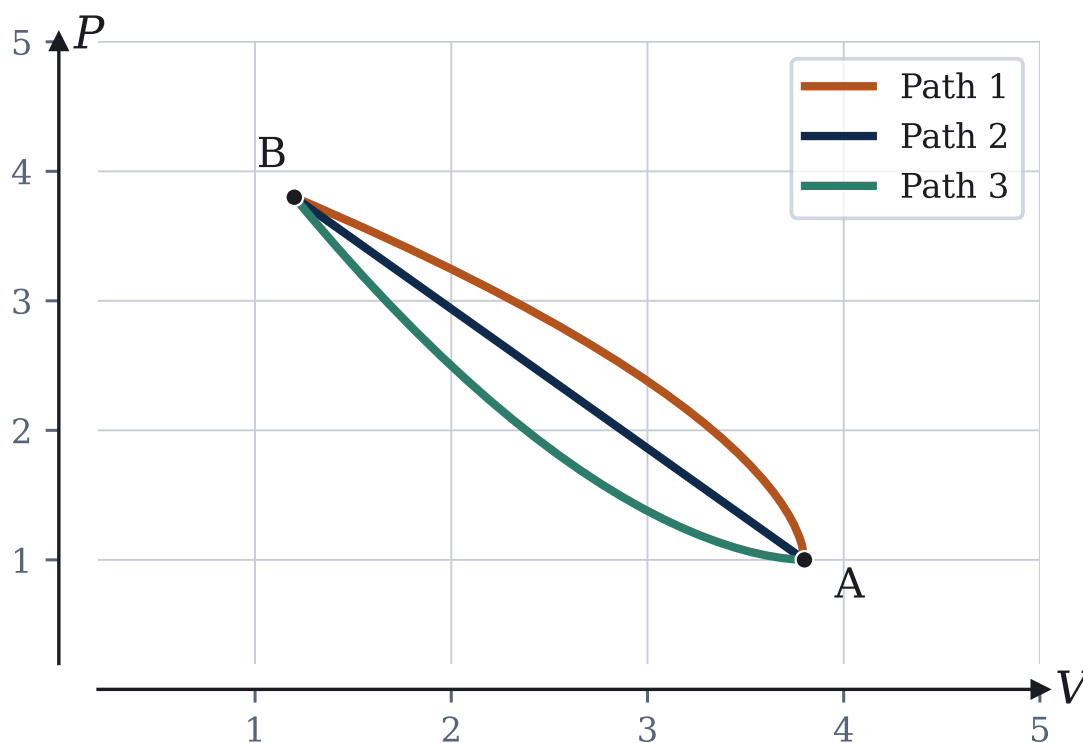


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$$\Delta U = +50 \text{ J (internal energy rises)}$$

A hurried student writes $\Delta U = -30 - 80 = -110$ (forgetting work-on-the-gas is positive energy input) or $+80 - 30$ (mixing conventions). The sanity check rescues you: squeezing a gas while only a little heat leaks out must *raise* its internal energy, so $\Delta U > 0$.

Example 3 — [JEE Main Level]

Problem. An ideal gas goes from A to B along two paths. Along path 1 it absorbs 200 J of heat and does 120 J of work. Along path 2 (same endpoints) it does 60 J of work. Find the heat exchanged along path 2, and whether it is absorbed or released.

Solution. This lives entirely on ΔU being a state function. *Step 1 — from the fully known path 1:*

$$\Delta U_{A \rightarrow B} = \Delta Q_1 - \Delta W_1 = 200 - 120 = +80 \text{ J.}$$

Step 2 — reuse on path 2 (same endpoints): $\Delta U_{A \rightarrow B} = +80 \text{ J}$. *Step 3 — solve for heat:*

$$\Delta Q_2 = \Delta U + \Delta W_2 = 80 + 60 = +140 \text{ J.}$$

$$\Delta Q_2 = +140 \text{ J, absorbed by the gas}$$

Both transfers shrank from path 1 to path 2, but their difference stayed pinned at 80 J. That fixed difference is the whole point.

Example 4 — [JEE Advanced Level]

Problem. 2.0 g of a liquid is completely vaporised at constant pressure $1.0 \times 10^5 \text{ Pa}$. As liquid it occupies 2.5 cm^3 ; as vapour 3100 cm^3 . Latent heat $L = 2.0 \times 10^6 \text{ J kg}^{-1}$. Find (a) the work done, (b) the change in internal energy, (c) the fraction of supplied heat going into internal energy, and its physical meaning.

Solution. *Step 1 — heat (latent):* $\Delta Q = mL = (2.0 \times 10^{-3})(2.0 \times 10^6) = 4000 \text{ J}$. *Step 2 — work (isobaric):*

$$\Delta V = (3100 - 2.5) \text{ cm}^3 = 3.0975 \times 10^{-3} \text{ m}^3,$$

$$\Delta W = P \Delta V = (1.0 \times 10^5)(3.0975 \times 10^{-3}) \approx 310 \text{ J.}$$

Step 3 — internal energy: $\Delta U = \Delta Q - \Delta W = 4000 - 310 = 3690 \text{ J}$. *Step 4 — fraction:* $\Delta U/\Delta Q = 3690/4000 \approx 0.92 = 92\%$.

$$\Delta W \approx 310 \text{ J}, \quad \Delta U \approx 3690 \text{ J}, \quad \approx 92\% \text{ into internal energy}$$

Only about 8% of the heat pushed back the atmosphere; the 92% went into internal energy — almost all of it potential energy, pulling molecules apart against their mutual attractions (temperature stays constant during boiling). This is exactly why latent heats are large: breaking intermolecular bonds is expensive, mechanical expansion is cheap.

Section 5: Practice Exercises

1. [CBSE] A gas absorbs 600 J of heat and does 250 J of work on its surroundings. Find ΔU .
2. [NEET] An ideal gas undergoes isothermal expansion. State, with one-line justification, the value of ΔU and the relationship between ΔQ and ΔW .
3. [JEE Main] 150 J of heat is supplied to a gas in a rigid, sealed container of fixed volume. Find the work done by the gas and its ΔU .
4. [JEE Main] A gas goes from A to B along path I, absorbing 300 J while doing 100 J of work. Taken from A to B by a different path II along which it does 250 J of work, find the heat exchanged along path II.

5. **[JEE Advanced]** A gas first expands adiabatically, doing 220 J of work; it is then compressed isothermally, with 150 J of work done on it by an external agent. Find the total ΔU over the two steps, and whether its final temperature is higher or lower than its initial.

Answers: 1. +350 J. 2. $\Delta U = 0$ (ideal-gas U depends only on T , and T is constant), therefore $\Delta Q = \Delta W$ (all absorbed heat converts to work). 3. $\Delta W = 0$ (rigid, isochoric); $\Delta U = +150$ J. 4. $\Delta U = 300 - 100 = 200$ J (state function); $\Delta Q_{II} = 200 + 250 = +450$ J absorbed. 5. Leg 1 (adiabatic): $\Delta U_1 = -\Delta W_1 = -220$ J. Leg 2 (isothermal): $\Delta U_2 = 0$. Total $\Delta U = -220$ J; since $\Delta U < 0$ for an ideal gas means T fell, the final temperature is lower.

Section 6: Crack the MCQ

MCQ 1.

A system absorbs 800 J of heat, and its internal energy increases by 500 J. The work done by the system is: (A) 300 J (B) 1300 J (C) -300 J (D) 500 J

Correct: (A) 300 J. $\Delta W = \Delta Q - \Delta U = 800 - 500 = 300$ J, done by the system.

- (B) adds instead of subtracts. (C) right magnitude, flipped sign. (D) internal energy mistaken for work.

MCQ 2.

For a complete cyclic process, which is always true? (A) net heat absorbed = net work done (B) net work = 0 (C) net heat = 0 (D) $\Delta U = W_{\text{net}}$

Correct: (A). Over a cycle $\Delta U_{\text{cycle}} = 0$, so $\Delta Q_{\text{net}} = \Delta W_{\text{net}}$.

- (B) false — an engine's whole purpose is non-zero net work (the enclosed P - V area). (C) false for the same reason. (D) confuses the result: it is $\Delta U = 0$, not $\Delta U = W_{\text{net}}$.

MCQ 3.

The internal energy of a given mass of ideal gas depends only on its: (A) pressure (B) volume (C) temperature (D) P and V independently

Correct: (C) temperature. U is the kinetic energy of random molecular motion, a function of T alone: $\Delta U = nC_V\Delta T$ for every process.

- (A),(B) each only part of the state; at fixed T you change P and V along an isotherm with no change in U . (D) (P, V) fixes T via $PV = nRT$ — and U tracks T .

MCQ 4.

Which can the first law *not* do? (A) account for energy conservation (B) introduce U as a state variable (C) predict the direction of spontaneous change (D) apply to solids, liquids, gases alike

Correct: (C). The first law is direction-blind: it permits heat to flow cold-to-hot just as readily as hot-to-cold, since both conserve energy. Only the second law forbids the unnatural direction.

- (A),(B),(D) are all things the first law *does*; choosing any of them means missing the word “not.”

Section 7: Common Student Pitfalls & Pro-Tips

1. **Sign-convention chaos.** The number-one error in this chapter. “Heat released” $\Rightarrow \Delta Q < 0$; “work done on the gas” $\Rightarrow \Delta W < 0$. Decide and write every sign before substituting.
2. **Calling internal energy “heat content.”** A gas in a state has internal energy; it does not “contain heat.” Heat is energy crossing the boundary, defined only for a process.
3. **Forgetting unit consistency.** $\Delta Q, \Delta U, \Delta W$ must share one unit. Convert first: 1 cal = 4.18 J.

4. **Assuming $\Delta U = 0$ when it isn't.** $\Delta U = 0$ only for isothermal of an ideal gas (or a complete cycle). It is *not* zero for adiabatic, isobaric, or isochoric. And don't assume $\Delta U = \Delta Q$ unless the process is isochoric.

Pro-Tip (speed & accuracy)

For any two-path problem, don't grind through both — compute ΔU once from the fully-specified path, then reuse it. For cyclic problems jump straight to $\Delta Q_{\text{net}} = \Delta W_{\text{net}}$ and read net work as the enclosed loop area. And always run the physical check: compress a gas $\Rightarrow \Delta U$ rises; let it expand adiabatically $\Rightarrow$ it cools. If a sign disagrees with intuition, you have an error, not a discovery.

Section 8: Cheat Sheet Box

FIRST LAW OF THERMODYNAMICS — RAPID RECAP

- Master equation: $\Delta Q = \Delta U + \Delta W$ (board/JEE convention: $\Delta W =$ work *by* gas, $\Delta Q =$ heat absorbed).
- It is conservation of energy *plus* it defines internal energy U as a state function.
- State function: U (path-independent, $\oint dU = 0$). Path functions: Q, W .
- Work by gas: $\Delta W = \int P dV =$ area under the P - V curve.
- Ideal gas: $\Delta U = nC_V \Delta T$ for *every* process (U depends only on T).
- Four shortcuts: isochoric $\rightarrow \Delta Q = \Delta U$; isobaric $\rightarrow \Delta Q = \Delta U + P\Delta V$; isothermal $\rightarrow \Delta Q = \Delta W$ ($\Delta U = 0$); adiabatic $\rightarrow \Delta U = -\Delta W$ ($\Delta Q = 0$).
- Over a cycle: $\Delta U = 0 \Rightarrow Q_{\text{net}} = W_{\text{net}}$.
- Limitation: says nothing about direction or the maximum heat $\rightarrow$ work conversion (that's the second law).

Memory aids: “Income = Savings + Spending” $\rightarrow \Delta Q = \Delta U + \Delta W$. • “ U is a place you're at; Q and W are roads you took.” • “Signs before sums.” • “ V fixed kills W ; T fixed kills ΔU ; $Q = 0$ is adiabatic.”

Subtopic 03 — Thermodynamic Processes

Exam Relevance: This is the most question-dense subtopic in the chapter. CBSE Boards almost always pull a 3–5 mark derivation from here — work done in an isothermal process, or the adiabatic relation $PV^\gamma = \text{constant}$. NEET fires rapid conceptual rounds: identifying processes from a P – V graph, the Q – W relation in each, and slope comparisons. JEE Main loves multi-step numericals combining a process with the first law and specific heats. JEE Advanced builds full cycles out of these process “legs” and asks you to compare work or temperatures across competing paths. Master the four processes plus the polytropic generalisation, and the rest of thermodynamics is just assembly.

Section 1: Concept Intuition

We found the universal accountant — $\Delta Q = \Delta U + \Delta W$. But an accountant needs the rules of the particular transaction to fill in the numbers. A **thermodynamic process** is exactly that: a specified rule for how a system moves from one state to another. Pin down the rule, and the first law instantly tells you how heat, work, and internal energy divide up.

Return to our trusty gas-in-a-cylinder. Imagine four ways a gas-wala might handle the same cylinder, each holding one thing fixed:

- **Volume fixed (isochoric):** Bolt the piston so it cannot move. Heat the gas — its pressure climbs, but since the piston never budges, the gas does *zero* work. Every joule of heat goes straight into internal energy. Think of a sealed pressure cooker with a jammed lid: heat just builds pressure.
- **Pressure fixed (isobaric):** Let the piston slide freely against the constant push of the atmosphere. Heat the gas and it expands, doing work against that steady pressure while warming up. Water boiling in an open pan pushes steam out against unchanging atmospheric pressure.
- **Temperature fixed (isothermal):** Keep the cylinder in a giant water bath that never changes temperature, and move the piston slowly so the gas always stays at the bath’s temperature. Every bit of energy spent as work is replenished by heat sneaking in — so internal energy never changes. *Slow* is the key word.
- **No heat exchanged (adiabatic):** Wrap the cylinder in a perfect insulator, or move the piston so fast that heat has no time to flow. Now the gas must pay for any expansion entirely out of its own internal energy — so an adiabatic expansion *cools* the gas. This is why a bicycle pump’s nozzle gets hot when you pump fast, and why spray-can gas feels cold as it rushes out.

Think about it this way...

Isothermal and adiabatic are two opposite extremes of *speed*. Do the process infinitely slowly with the bath connected and heat flows freely to keep temperature pinned — isothermal. Do it so fast (or so well-insulated) that no heat can cross — adiabatic. Real processes live somewhere in between, but exams idealise to these clean limits. On a P – V diagram both look like falling curves, but the **adiabatic always falls more steeply**, because the adiabatic gas loses temperature as it expands and so loses pressure faster than the isothermal gas, whose temperature is propped up by incoming heat.

The deepest single fact in this subtopic: **work done by a gas equals the area under its P – V curve**. Two processes connecting the same two states enclose different areas, hence do different work — a direct picture of why W is a path function. And if a process traces a closed loop (a cycle), the work done per cycle is the area trapped inside the loop. That one geometric idea drives every heat-engine problem.

Limiting Conditions & Assumptions.

(1) The clean work formula $W = \int P dV$ requires the process to be *quasi-static* — slow enough that the gas has a single well-defined pressure at every instant. A violent, turbulent expansion

has no single P (free expansion is the classic exception: gas rushes into vacuum, does zero work despite expanding). (2) The relations $PV^\gamma = \text{const}$, $W = nRT \ln(V_f/V_i)$, etc. are derived for an *ideal* gas; real gases deviate at high pressure / low temperature. (3) “Isothermal” needs a heat reservoir and slowness; “adiabatic” needs insulation or speed — a process can’t be both unless nothing actually happens.

Figure 11.5 A single P - V diagram with a common starting point. Four curves emanate: a vertical line up labelled “Isochoric ($V = \text{const}$)”, a horizontal line right labelled “Isobaric ($P = \text{const}$)”, a gently falling hyperbola labelled “Isothermal ($PV = \text{const}$)”, and a steeper falling curve labelled “Adiabatic ($PV^\gamma = \text{const}$)”. Note: “Adiabatic is steeper — $\gamma \times$ the isothermal slope.”

Section 2: Key Formulas & Definitions For n moles of an ideal gas; C_V , C_P are molar heat capacities, $\gamma = C_P/C_V$ the adiabatic exponent, and W is the work done by the gas. The four basic processes:

Process Condition	Work W	First-law result
Isochoric $V = \text{const}$	$W = 0$	$\Delta Q = \Delta U = nC_V \Delta T$
Isobaric $P = \text{const}$	$W = P\Delta V = nR\Delta T$	$\Delta Q = nC_P \Delta T$, $\Delta U = nC_V \Delta T$
Isothermal $T = \text{const}$	$W = nRT \ln V_f/V_i = nRT \ln P_i/P_f$	$\Delta U = 0$, $\Delta Q = W$
Adiabatic	$\Delta Q = 0$	$W = P_i V_i - P_f V_f / (\gamma - 1) = nR(T_i - T_f) / (\gamma - 1)$

$\Delta U = -W = nC_V \Delta T$ Adiabatic relations (three equivalent forms, ideal gas): $PV^\gamma = \text{const}$, $TV^{\gamma-1} = \text{const}$, $P^{1-\gamma} T^\gamma = \text{const}$. Polytropic process (the master process; all four above are special cases): $PV^\chi = \text{const}$, $W = P_i V_i - P_f V_f / (\chi - 1) = nR \Delta T / (1 - \chi)$, with molar heat capacity $C = C_V + R / (1 - \chi)$. Special cases: $\chi = 0$ (isobaric), $\chi = 1$ (isothermal), $\chi = \gamma$ (adiabatic), $\chi \rightarrow \infty$ (isochoric). Slope of the P - V curve for a polytropic process: $dP/dV = -\chi P/V$. Hence at a common point: isobaric slope = 0; isothermal slope = $-P/V$; adiabatic slope = $-\gamma P/V$. The adiabatic curve is γ times steeper than the isothermal ($\gamma > 1$).

Quantity reference: Quantity Symbol Notes Adiabatic exponent $\gamma = C_P/C_V$ 5/3 monoatomic, 7/5 diatomic, 4/3 polyatomic (dimensionless) Molar heat capacity C_V , C_P $C_P - C_V = R$ (Mayer's relation), J mol⁻¹K⁻¹ Work W joule (J), = area under P - V curve (see next page)

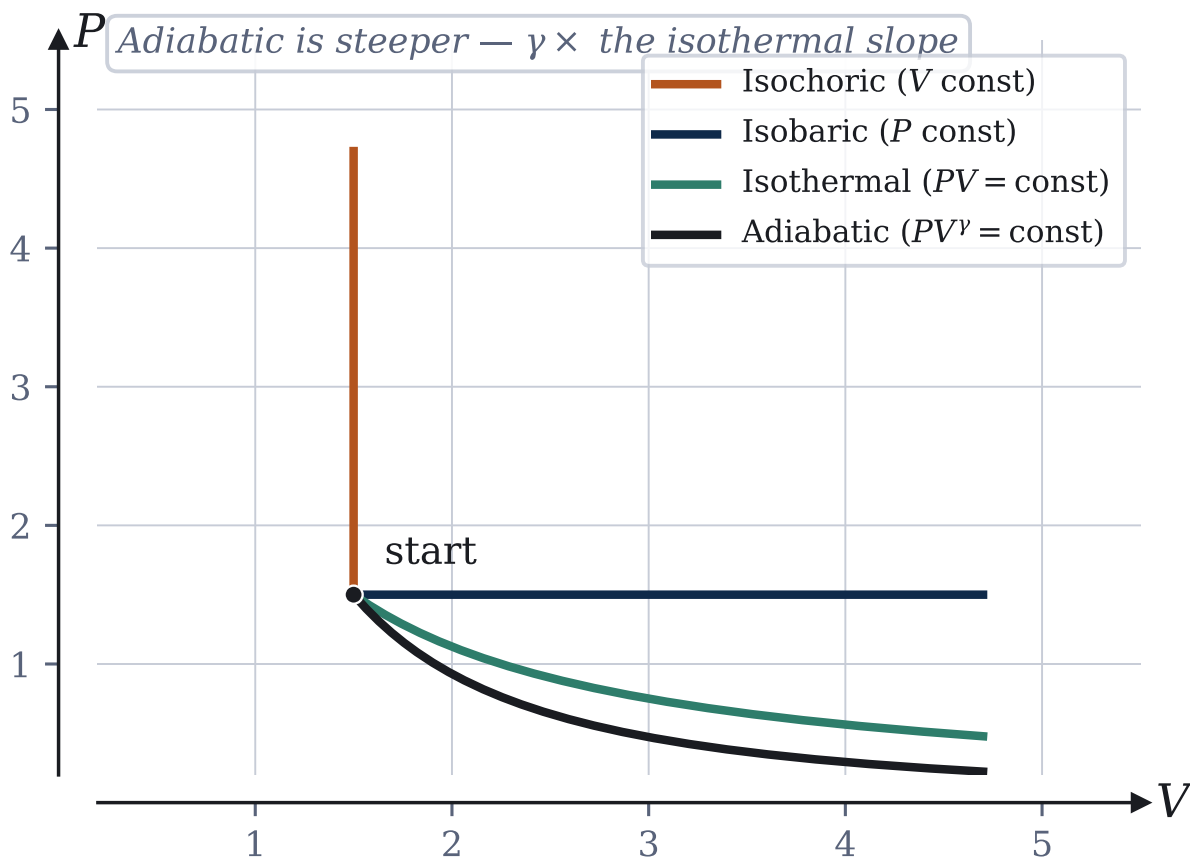


Figure 11.5 A single P - V diagram with a common starting point. Four curves emanate: a vertical line up labelled “Isochoric (V const)”, a horizontal line right labelled “Isobaric (P const)”, a gently falling hyperbola labelled “Isothermal ($PV = \text{const}$)”, and a steeper falling curve labelled “Adiabatic ($PV^\gamma = \text{const}$)”. Note: “Adiabatic is steeper — $\gamma \times$ the isothermal slope.”

Section 2: Key Formulas & Definitions For n moles of an ideal gas; CV , CP are molar heat capacities, $\gamma = CP/CV$ the adiabatic exponent, and W is the work done by the gas. The four basic processes:

Process Condition	Work W	First-law result
Isochoric $V = \text{const}$	$W = 0$	$\Delta Q = \Delta U = nCV \Delta T$
Isobaric $P = \text{const}$	$W = P\Delta V = nR\Delta T$	$\Delta Q = nCP \Delta T$, $\Delta U = nCV \Delta T$
Isothermal $T = \text{const}$	$W = nRT \ln V_f/V_i = nRT \ln P_i/P_f$	$\Delta U = 0$, $\Delta Q = W$
Adiabatic	$\Delta Q = 0$, $W = P_iV_i - P_fV_f$	$\Delta U = -W = nCV \Delta T$

Adiabatic relations (three equivalent forms, ideal gas): $PV^\gamma = \text{const}$, $TV^{\gamma-1} = \text{const}$, $P^{1-\gamma}T^\gamma = \text{const}$.

Polytropic process (the master process; all four above are special cases): $PV^\alpha = \text{const}$, $W = P_iV_i - P_fV_f$, $\alpha - 1 = nR \Delta T / (P_iV_i - P_fV_f)$, with molar heat capacity $C = CV + R / (1 - \alpha)$. Special cases: $\alpha = 0$ (isobaric), $\alpha = 1$ (isothermal), $\alpha = \gamma$ (adiabatic), $\alpha \rightarrow \infty$ (isochoric).

Slope of the P - V curve for a polytropic process: $dP/dV = -\alpha P/V$. Hence at a common point: isobaric slope = 0; isothermal slope = $-P/V$; adiabatic slope = $-\gamma P/V$. The adiabatic curve is γ times steeper than the isothermal ($\gamma > 1$).

Quantity reference: Quantity Symbol Notes Adiabatic exponent $\gamma = CP/CV$ 5/3 monoatomic, 7/5 diatomic, 4/3 polyatomic (dimensionless) Molar heat capacity CV , CP $CP - CV = R$ (Mayer's relation), J mol $^{-1}$ K $^{-1}$ Work W joule (J), = area under P - V curve

Section 3: CBSE Board Derivations

Two derivations are board favourites. Master both — the reasoning, not just the algebra.

3.1 Work done in an isothermal process

Setup. Let n moles of an ideal gas expand quasi-statically and isothermally (temperature held at T) from volume V_i to V_f .

Step 1 — pressure in terms of volume. $PV = nRT \Rightarrow P = nRT/V$.

Step 2 — the work integral.

$$W = \int_{V_i}^{V_f} P dV = \int_{V_i}^{V_f} \frac{nRT}{V} dV.$$

Step 3 — integrate (pull out the constants n, R, T):

$$W = nRT [\ln V]_{V_i}^{V_f} = nRT \ln \left(\frac{V_f}{V_i} \right).$$

Step 4 — alternate form. Since T is constant, $P_i V_i = P_f V_f$, so $V_f/V_i = P_i/P_f$, giving $W = nRT \ln(P_i/P_f)$. Physical reading: with $\Delta U = 0$, the first law forces $\Delta Q = W$ — every joule the gas spends doing work is supplied as heat by the reservoir.

3.2 The adiabatic relation $PV^\gamma = \text{constant}$ and the work done

Setup. A quasi-static adiabatic process: $\Delta Q = 0$.

Step 1 — first law on an infinitesimal step. With $dQ = 0$, $dU = nC_V dT$, $dW = P dV$:

$$0 = nC_V dT + P dV. \quad (1)$$

Step 2 — eliminate dT via the gas law. Differentiating $PV = nRT$: $P dV + V dP = nR dT$, so $dT = (P dV + V dP)/(nR)$. Substituting into (1) and multiplying by nR :

$$C_V(P dV + V dP) + RP dV = 0 \Rightarrow C_V V dP + (C_V + R)P dV = 0.$$

Step 3 — introduce γ . Using Mayer's relation $C_P = C_V + R$:

$$C_V V dP + C_P P dV = 0.$$

Divide by $C_V PV$ and use $\gamma = C_P/C_V$:

$$\frac{dP}{P} + \gamma \frac{dV}{V} = 0.$$

Step 4 — integrate.

$$\ln P + \gamma \ln V = \text{const} \Rightarrow PV^\gamma = \text{constant}.$$

Substituting $P = nRT/V$ gives $TV^{\gamma-1} = \text{const}$; eliminating V gives $P^{1-\gamma}T^\gamma = \text{const}$.

Step 5 — work done. With $P = K/V^\gamma$ where $K = P_i V_i^\gamma = P_f V_f^\gamma$:

$$W = \int_{V_i}^{V_f} \frac{K}{V^\gamma} dV = K \left[\frac{V^{1-\gamma}}{1-\gamma} \right]_{V_i}^{V_f} = \frac{P_f V_f - P_i V_i}{1-\gamma},$$

$$\boxed{W = \frac{P_i V_i - P_f V_f}{\gamma - 1} = \frac{nR(T_i - T_f)}{\gamma - 1}}.$$

Physical reading: with $\Delta Q = 0$, the first law gives $W = -\Delta U$. In an adiabatic expansion ($V_f > V_i$) the gas does positive work, so $\Delta U < 0$ and the gas cools — the spray-can effect.

Figure 11.6 *P-V diagram showing an isothermal curve and an adiabatic curve starting from the same point and both expanding to the same final volume. Shade the area under each; the isothermal area is visibly larger. Labels: “Isothermal > Adiabatic (same expansion)”. Section 4: Worked Examples Example 1 — [CBSE Board Level] Problem. 2.0 mol of an ideal gas at 300 K expand isothermally and quasi-statically until their volume triples. Find (a) the work done and (b) the heat absorbed. ($R = 8.314$, $\ln 3 = 1.0986$.) Solution. (a) Isothermal work: $W = nRT \ln(V_f/V_i) = (2.0)(8.314)(300)(1.0986) \approx 5.48 \times 10^3 \text{ J} = 5.48 \text{ kJ}$. (b) Isothermal $\Rightarrow \Delta U = 0 \Rightarrow \Delta Q = W$. $W \approx 5.48 \text{ kJ}$, $\Delta Q \approx 5.48 \text{ kJ}$. All the heat absorbed leaves as work; temperature and internal energy are unchanged. Example 2 — [NEET Speed Trap] Problem. A monoatomic ideal gas ($\gamma = 5/3$) is compressed adiabatically until its volume becomes one-eighth of its initial value. By what factor does its absolute temperature change? Speed solution. Use $TV^{\gamma-1} = \text{const}$: $T_f T_i = (V_i/V_f)^{\gamma-1} = 8^{5/3-1} = 8^{2/3}$. Mentally: $8 = 2^3$, so $8^{2/3} = 2^2 \times 2^{2/3} = 2^2 = 4$. $T_f = 4 T_i$ — the temperature quadruples Classic mistakes: (i) using exponent γ instead of $\gamma - 1$, giving $8^{5/3} = 32$; (ii) inverting the volume ratio, giving $8^{-2/3} = 1/4$. The physical check saves you: compression must raise temperature, so the factor exceeds 1 — instantly killing $1/4$. Example 3 — [JEE Main Level] Problem. 3.0 mol of a diatomic ideal gas ($\gamma = 7/5$, so $C_V = 5/2 R$, $C_P = 7/2 R$) are heated at constant pressure so the temperature rises by 100 K. Find (a) the work done, (b) ΔU , (c) the heat supplied. Verify the first law. Solution. (a) Isobaric work: $W = nR\Delta T = (3.0)(8.314)(100) \approx 2.49 \text{ kJ}$. (b) $\Delta U = nC_V \Delta T = (3.0)(5/2 \times 8.314)(100) = (3.0)(20.785)(100) \approx 6.24 \text{ kJ}$. (c) $\Delta Q = nC_P \Delta T = (3.0)(7/2 \times 8.314)(100) = (3.0)(29.099)(100) \approx 8.73 \text{ kJ}$. Verification: $\Delta U + W = 6.24 + 2.49 = 8.73 \text{ kJ} = \Delta Q$. $W \approx 2.49 \text{ kJ}$, $\Delta U \approx 6.24 \text{ kJ}$, $\Delta Q \approx 8.73 \text{ kJ}$. The heat splits in the ratio $C_P : C_V = \gamma : 1 = 7 : 5$ — a quick structural check for any isobaric problem. Example 4 — [JEE Advanced Level] (see next page)*

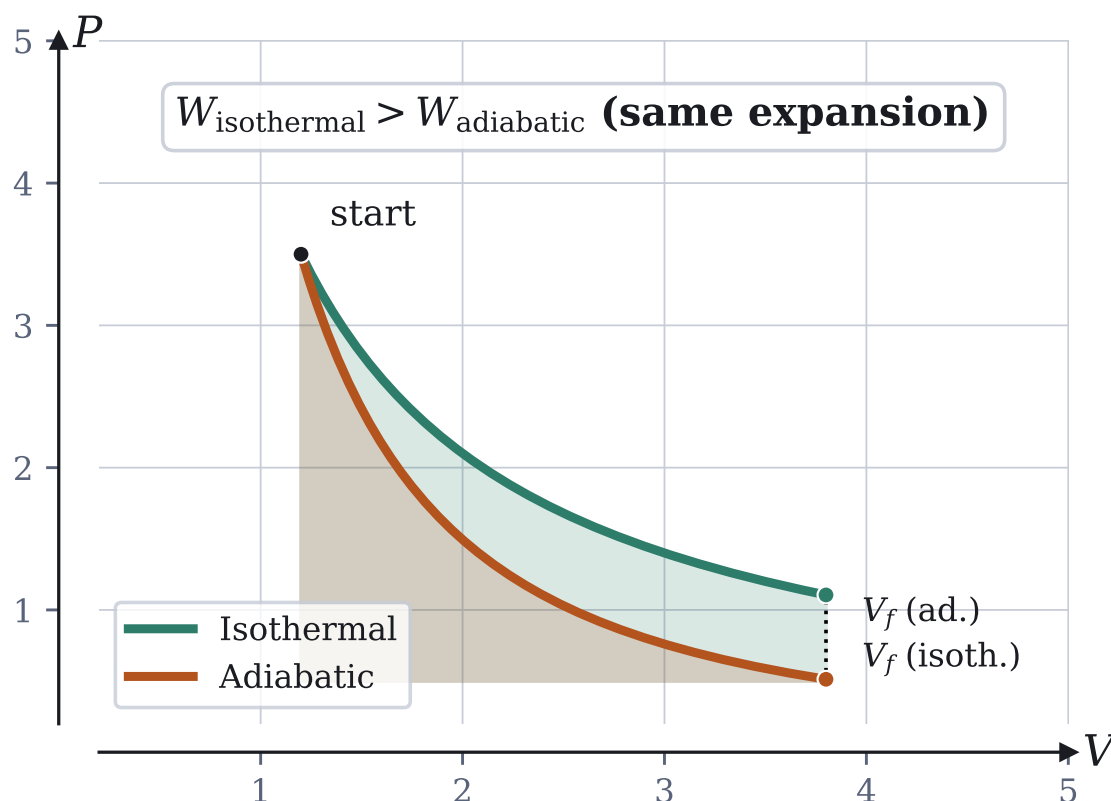


Figure 11.6 *P-V diagram showing an isothermal curve and an adiabatic curve starting from the same point and both expanding to the same final volume. Shade the area under each; the isothermal area is visibly larger. Labels: “ $W_{\text{isothermal}} > W_{\text{adiabatic}}$ (same expansion)”.*

Section 4: Worked Examples Example 1 — [CBSE Board Level] Problem. 2.0 mol of an ideal gas at 300 K expand isothermally and quasi-statically until their volume triples. Find (a) the work done and (b) the heat absorbed. ($R = 8.314$, $\ln 3 = 1.0986$.) Solution. (a) Isothermal work: $W = nRT \ln(V_f/V_i) = (2.0)(8.314)(300)(1.0986) \approx 5.48 \times 10^3 \text{ J} = 5.48 \text{ kJ}$. (b) Isothermal $\Rightarrow \Delta U = 0 \Rightarrow \Delta Q = W$. $W \approx 5.48 \text{ kJ}$, $\Delta Q \approx 5.48 \text{ kJ}$ All the heat absorbed leaves as work; temperature and internal energy are unchanged. Example 2 — [NEET Speed Trap] Problem. A monoatomic ideal gas ($\gamma = 5/3$) is compressed adiabatically until its volume becomes one-eighth of its initial value. By what factor does its absolute temperature change? Speed solution. Use $TV^{\gamma-1} = \text{const}$: $T_f/T_i = (V_i/V_f)^{\gamma-1} = 8^{5/3-1} = 8^{2/3}$. Mentally: $8 = 2^3$, so $8^{2/3} = 2^{3 \times 2/3} = 2^2 = 4$. $T_f = 4 T_i$ — the temperature quadruples Classic mistakes: (i) using exponent γ instead of $\gamma - 1$, giving $8^{5/3} = 32$; (ii) inverting the volume ratio, giving $8^{-2/3} = 1/4$. The physical check saves you: compression must raise temperature, so the factor exceeds 1 — instantly killing $1/4$. Example 3 — [JEE Main Level] Problem. 3.0 mol of a diatomic ideal gas ($\gamma = 7/5$, so $CV = 5/2 R$, $CP = 7/2 R$) are heated at constant pressure so the temperature rises by 100 K. Find (a) the work done, (b) ΔU , (c) the heat supplied. Verify the first law. Solution. (a) Isobaric work: $W = nR\Delta T = (3.0)(8.314)(100) \approx 2.49 \text{ kJ}$. (b) $\Delta U = nCV\Delta T = (3.0)(5/2 \times 8.314)(100) = (3.0)(20.785)(100) \approx 6.24 \text{ kJ}$. (c) $\Delta Q = nCP\Delta T = (3.0)(7/2 \times 8.314)(100) = (3.0)(29.099)(100) \approx 8.73 \text{ kJ}$. Verification: $\Delta U + W = 6.24 + 2.49 = 8.73 \text{ kJ} = \Delta Q$. $W \approx 2.49 \text{ kJ}$, $\Delta U \approx 6.24 \text{ kJ}$, $\Delta Q \approx 8.73 \text{ kJ}$ The heat splits in the ratio $CP : CV = \gamma : 1 = 7 : 5$ — a quick structural check for any isobaric problem. Example 4 — [JEE Advanced Level]

Problem. A monoatomic ideal gas ($\gamma = 5/3$) starts at P_0, V_0 and expands to $2V_0$ by two routes: (i) isothermally, (ii) adiabatically. Find the ratio of work done in the isothermal expansion to that in the adiabatic, and explain which is larger and why.

Solution. *Step 1 — isothermal work* (with $nRT = P_0V_0$):

$$W_{\text{iso}} = nRT \ln\left(\frac{2V_0}{V_0}\right) = P_0V_0 \ln 2 \approx 0.693 P_0V_0.$$

Step 2 — adiabatic final pressure. $P_0V_0^\gamma = P_f(2V_0)^\gamma \Rightarrow P_f = P_0 2^{-\gamma} = P_0 2^{-5/3}$, so

$$P_fV_f = P_0 2^{-5/3}(2V_0) = P_0V_0 2^{1-5/3} = P_0V_0 2^{-2/3}.$$

Step 3 — adiabatic work.

$$W_{\text{adia}} = \frac{P_0V_0 - P_fV_f}{\gamma - 1} = \frac{P_0V_0(1 - 2^{-2/3})}{\frac{5}{3} - 1} = \frac{3}{2}P_0V_0(1 - 2^{-2/3}).$$

With $2^{-2/3} \approx 0.630$: $W_{\text{adia}} \approx \frac{3}{2}(0.370)P_0V_0 \approx 0.555 P_0V_0$. *Step 4 — the ratio:*

$$\frac{W_{\text{iso}}}{W_{\text{adia}}} = \frac{0.693}{0.555} \approx 1.25.$$

$W_{\text{iso}}/W_{\text{adia}} \approx 1.25$ — the isothermal expansion does about 25% more work

Both start at the same point and end at the same volume, but the isothermal gas is continuously fed heat, keeping its temperature — and pressure — higher at every intermediate volume. The adiabatic gas, sealed off, spends its own internal energy and cools, its pressure dropping faster. Higher pressure throughout means a larger area under the curve, hence more work. This is the geometric truth behind the rule **isobaric** > **isothermal** > **adiabatic** for expansion work between the same volumes.

Section 5: Practice Exercises

1. **[CBSE]** 0.50 mol of an ideal gas is heated at constant pressure, raising its temperature from 300 K to 400 K. Find the work done by the gas.
2. **[NEET]** On a P - V diagram, an isothermal and an adiabatic curve pass through the same point for the same gas. State which is steeper, by what factor their slopes differ, and which process does less work during an expansion to the same final volume.
3. **[JEE Main]** A diatomic ideal gas ($\gamma = 1.4$) at 300 K is compressed adiabatically to one-fourth its initial volume. Find the final temperature. ($4^{0.4} \approx 1.741$.)
4. **[JEE Main]** 4.0 mol of a monoatomic ideal gas ($C_V = \frac{3}{2}R$) is heated at constant volume so its temperature rises by 50 K. Find the work done by the gas and the heat supplied.
5. **[JEE Advanced]** 2.0 mol of an ideal gas with $\gamma = 1.5$ expands adiabatically, its temperature falling from 500 K to 350 K. Find the work done.

Answers: 1. $W = nR\Delta T = (0.50)(8.314)(100) \approx 416$ J. 2. The adiabatic is steeper; its slope is γ times the isothermal slope at that point; the adiabatic process does less work (smaller area under the curve). 3. $T_f = 300 \times 4^{0.4} \approx 300 \times 1.741 \approx 522$ K. 4. $W = 0$ (isochoric); $\Delta Q = \Delta U = nC_V\Delta T = (4.0)(\frac{3}{2} \times 8.314)(50) \approx 2.49$ kJ. 5. $W = \frac{nR(T_i - T_f)}{\gamma - 1} = \frac{(2.0)(8.314)(150)}{0.5} \approx 4.99$ kJ.

Section 6: Crack the MCQ

MCQ 1.

An ideal gas expands from the same initial state to the same final volume by isobaric, isothermal, and adiabatic processes. The work done is greatest in the: (A) adiabatic (B) isothermal (C) isobaric (D) same in all

Correct: (C) isobaric. Work is the area under the P - V curve; the isobaric path keeps pressure highest (flat), enclosing the largest area. Order: isobaric > isothermal > adiabatic.

- (A) adiabatic does the *least* work. (B) isothermal sits in the middle. (D) ignores that W is path-dependent.

MCQ 2.

At a given point, the magnitude of the slope of the adiabatic curve relates to that of the isothermal by a factor of: (A) γ (B) $1/\gamma$ (C) $\gamma - 1$ (D) γ^2

Correct: (A) γ . Isothermal slope = $-P/V$; adiabatic = $-\gamma P/V$. Ratio of magnitudes is γ .

- (B) inverts the relationship. (C) confuses the slope ratio with the $TV^{\gamma-1}$ exponent. (D) has no basis.

MCQ 3.

For an adiabatic process of an ideal gas, which is correct? (A) $\Delta Q = W$ (B) $\Delta U = 0$ (C) $\Delta U = -W$ (D) $W = 0$

Correct: (C) $\Delta U = -W$. $\Delta Q = 0$ so $0 = \Delta U + W$. The gas does work entirely at the expense of internal energy.

- (A) is isothermal. (B) is isothermal. (D) is isochoric.

MCQ 4.

A monoatomic ideal gas ($\gamma = 5/3$) is compressed adiabatically to half its volume. Its absolute temperature changes by a factor of approximately: (A) $2^{2/3} \approx 1.59$ (B) $2^{5/3} \approx 3.17$ (C) 2 (D) $2^{1/3} \approx 1.26$

Correct: (A) $2^{2/3} \approx 1.59$. $T_f/T_i = (V_i/V_f)^{\gamma-1} = 2^{2/3}$.

- (B) uses exponent γ instead of $\gamma - 1$. (C) ignores the exponent. (D) miscalculates $\gamma - 1$ as $1/3$ (it is $2/3$).

Section 7: Common Student Pitfalls & Pro-Tips

1. **Confusing isothermal and adiabatic curves.** They look like twins. Remember: adiabatic is steeper ($-\gamma P/V$ vs $-P/V$), the adiabatic gas changes temperature while the isothermal does not, and $\Delta Q = W$ is isothermal while $\Delta U = -W$ is adiabatic.
2. **Using γ where $\gamma - 1$ belongs (and vice-versa).** Pressure relation uses γ (PV^γ); temperature relation uses $\gamma - 1$ ($TV^{\gamma-1}$). Writing $TV^\gamma = \text{const}$ is a classic slip.
3. **Applying a process formula outside its process.** $W = nRT \ln(V_f/V_i)$ is only isothermal; $W = P\Delta V$ is only isobaric.
4. **Sign and direction of work / cyclic area.** Expansion $W > 0$; compression $W < 0$. For a cycle, net work = enclosed area, positive if clockwise on a P - V diagram (engine), negative if anticlockwise (refrigerator).

Pro-Tip (speed & accuracy)

For adiabatic numericals, work with ratios, never absolute pressures — $T_f/T_i = (V_i/V_f)^{\gamma-1}$ and $P_f/P_i = (V_i/V_f)^\gamma$ collapse most problems to a single power of a clean number. To identify a process from a P - V graph in seconds: vertical line $\rightarrow$ isochoric; horizontal $\rightarrow$ isobaric; of the two curves, the steeper one is adiabatic. And for adiabatic work, skip the integral — use $W = nR(T_i - T_f)/(\gamma - 1)$ once you know the two temperatures.

Section 8: Cheat Sheet Box

THERMODYNAMIC PROCESSES — RAPID RECAP

- Isochoric (V const): $W = 0$, $\Delta Q = \Delta U = nC_V\Delta T$. (Vertical line.)
- Isobaric (P const): $W = P\Delta V = nR\Delta T$, $\Delta Q = nC_P\Delta T$. (Horizontal line.)
- Isothermal (T const): $\Delta U = 0$, $\Delta Q = W = nRT \ln(V_f/V_i)$. (Gentle hyperbola.)
- Adiabatic ($\Delta Q = 0$): $\Delta U = -W$, $W = nR(T_i - T_f)/(\gamma - 1)$; $PV^\gamma = TV^{\gamma-1} = \text{const.}$ (Steep curve.)
- Polytropic $PV^x = \text{const}$: $C = C_V + R/(1 - x)$; recovers all four at $x = 0, 1, \gamma, \infty$.
- Slope $= -xP/V$; adiabatic is $\gamma\times$ steeper than isothermal.
- Expansion work order: isobaric $>$ isothermal $>$ adiabatic. Cycle work = enclosed area (clockwise $= +$).

Memory aids: “ V -fixed kills Work; T -fixed kills ΔU ; Q -zero is adiabatic; P -fixed needs C_P .” • “Adiabatic is the Athlete — steeper and cools down when it works.” • “Pressure uses γ , Temperature uses $\gamma - 1$.”

Subtopic 04 — Specific Heat Capacity (C_P , C_V and Mayer's Relation)

Exam Relevance: This subtopic is the hinge between the first law and every numerical that follows. CBSE Boards ask the definition of C_P and C_V , the derivation of Mayer's relation $C_P - C_V = R$, and one-line “why is $C_P > C_V$ ” reasoning. NEET fires conceptual traps on degrees of freedom and the value of γ for mono/di/polyatomic gases. JEE Main folds C_V and C_P into isobaric/isochoric numericals (heat, ΔU , work). JEE Advanced probes mixtures of gases, the polytropic molar heat capacity, and the link between equipartition and γ . Get $C_V = \frac{f}{2}R$ and $\gamma = 1 + \frac{2}{f}$ into reflex memory.

Section 1: Concept Intuition

Pour the same amount of heat into a small steel spanner and into a bucket of water, and the spanner's temperature shoots up while the water barely warms. **Heat capacity** measures exactly this reluctance: how much heat a body needs to raise its temperature by one degree. Divide by mass and you get the **specific heat capacity** s (per kilogram); divide by moles and you get the **molar specific heat capacity** C (per mole). Water's famously large specific heat is why coastal cities have mild climates and why a hot-water bottle stays warm all night.

For solids and liquids, one number is enough — they barely expand on heating, so essentially all the heat becomes internal energy. *Gases are different, and this is the heart of the subtopic.* A gas expands dramatically on heating, so *how* you heat it matters. Heat it in a sealed rigid box (constant volume) and it cannot expand: every joule goes into internal energy. Heat it under a free piston (constant pressure) and it expands, spending some of the incoming heat pushing the piston back — so you must supply *extra* heat to get the same temperature rise. A gas therefore needs *two* molar specific heats: C_V (constant volume) and C_P (constant pressure), and always $C_P > C_V$.

Think about it this way...

Imagine filling a bucket that has a small leak at the bottom. If you plug the leak (constant volume — no escape), every mug of water you pour raises the level. If you leave the leak open (constant pressure — the gas “leaks” energy into expansion work), some of every mug drains away, so you must pour more to raise the level by the same amount. The plugged bucket is C_V ; the leaky one is C_P . The size of the leak is exactly the work of expansion — and for one mole of an ideal gas heated by one kelvin at constant pressure, that leak is precisely R joules. That single sentence *is* Mayer's relation, $C_P - C_V = R$.

The deeper picture — degrees of freedom.

Why is $C_V = \frac{3}{2}R$ for helium but $\frac{5}{2}R$ for nitrogen? Because internal energy is stored in the *ways* a molecule can move. A point-like monoatomic atom can only translate in three directions — three **degrees of freedom**. A dumbbell-shaped diatomic molecule can also tumble about two axes — five. The **equipartition theorem** hands each degree of freedom an energy of $\frac{1}{2}k_B T$ per molecule, i.e. $\frac{1}{2}R$ per mole per kelvin. So a molecule with f active degrees of freedom stores $U = \frac{f}{2}nRT$, giving $C_V = \frac{f}{2}R$. More ways to wiggle means more places to hide energy, which means a larger heat capacity.

Limiting Conditions & Assumptions.

(1) $C_P - C_V = R$ holds *per mole of an ideal gas*; for real gases and for solids/liquids it is only approximate. (2) The clean values $C_V = \frac{f}{2}R$ assume the equipartition theorem, which is a high-temperature (classical) result — at low temperatures vibrational and even rotational modes “freeze out” and C_V drops (a quantum effect beyond Class 11, but worth knowing exists). (3) For diatomic gases at ordinary room temperature, rotation is active but vibration is usually frozen, so $f = 5$, not 7.

Figure 11.7 Two cylinders side by side, each with the same gas and a flame beneath. Left cylinder: piston bolted (“Vxed”) — arrow labelled “ $\Delta Q = nC_V \Delta T$, all into U ”. Right cylinder: piston free under a weight (“Pxed”) — the gas column is taller, piston pushed up by ΔV , arrow split into “ ΔU ” plus “ $P\Delta V$ work”. Caption: “Same ΔT , but the right one needs $nR\Delta T$ more heat $\Rightarrow C_P = C_V + R$.”

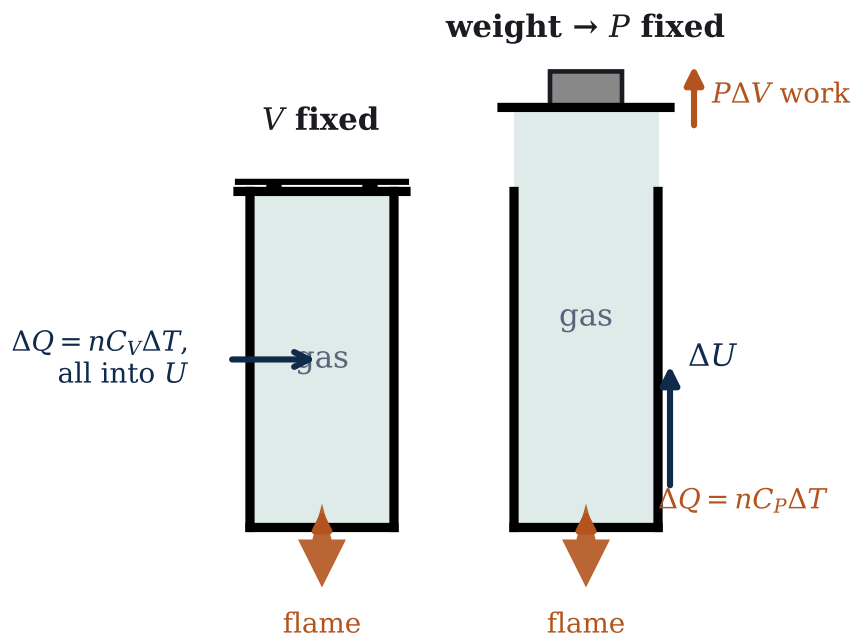
Section 2: Key Formulas & Definitions

Heat capacity (of a body): $S = \Delta Q / \Delta T$ (units $J K^{-1}$). Specific heat capacity (per unit mass): $s = (1/m) \Delta Q / \Delta T$ (units $J kg^{-1} K^{-1}$). $\Delta Q = m s \Delta T$. Molar specific heat capacity (per mole): $C = (1/n) \Delta Q / \Delta T$ (units $J mol^{-1} K^{-1}$). For a gas — the two molar specific heats: $C_V = (1/n) \cdot \Delta Q / \Delta T$ $V = (1/n) dU / dT$, $C_P = (1/n) \cdot \Delta Q / \Delta T$ P . Mayer's relation (ideal gas, per mole): $C_P - C_V = R$. Equipartition and degrees of freedom (f active degrees of freedom): $C_V = f/2 R$, $C_P = (f/2 + 1) R$, $\gamma = C_P / C_V = 1 + 2/f$. Standard ideal-gas values: Gas type f C_V C_P $\gamma = C_P / C_V$ Monoatomic (He, Ar) 3 3 $2R \approx 12.5$ 5 $2R \approx 20.8$ 5/3 ≈ 1.67 Diatomic (N_2 , O_2 , H_2) 5 5 $2R \approx 20.8$ 7 $2R \approx 29.1$ 7/5 = 1.40 Polyatomic (nonlinear, e.g. CO_2 bent / H_2O) 6 3 $R \approx 24.9$ 4 $R \approx 33.3$ 4/3 ≈ 1.33 (C_V , C_P in $J mol^{-1} K^{-1}$, using $R = 8.314$.) Internal energy from C_V (any process, ideal gas): $U = nC_V T$, $\Delta U = nC_V \Delta T$. Polytropic molar heat capacity ($PV^\chi = \text{const}$): $C = C_V + R / (1 - \chi)$. Special cases: isobaric $\chi = 0 \Rightarrow C_P$; isothermal $\chi = 1 \Rightarrow C \rightarrow \infty$; adiabatic $\chi = \gamma \Rightarrow C = 0$; isochoric $\chi \rightarrow \infty \Rightarrow C_V$. Useful benchmarks: water $s \approx 4186 J kg^{-1} K^{-1} = 1 cal g^{-1} ^\circ C^{-1}$; Dulong-Petit law for many solids: molar heat capacity $\approx 3R \approx 25 J mol^{-1} K^{-1}$. For a mixture of $n_1, n_2, \dots$ moles of gases: $C_{V,mix} = \sum n_i C_{V,i}$, P ni $C_{V,i}$, and similarly for C_P ; then $\gamma_{mix} = C_{P,mix} / C_{V,mix}$.

Section 3: Key Procedures & Logical Framework

3.1 Derivation of Mayer's relation $C_P - C_V = R$ (board favourite)

Setup. Consider one mole ($n = 1$) of an ideal gas. We heat it by dT , once at constant volume and once at constant pressure, and compare. Step 1 — constant volume. No work is done ($dV = 0$), so the 1st law $\delta dQ = dU + P dV$ gives $\delta dQ_V = dU$. By definition $C_V = (1/n)(\delta dQ/dT)_V = dU/dT$ (for one mole), hence $dU = C_V dT$. 24 (see next page)



Same ΔT , but the right one needs $nR\Delta T$ more heat $\Rightarrow C_P = C_V + R$

Figure 11.7 Two cylinders side by side, each with the same gas and a flame beneath. Left cylinder: piston bolted ("V xed") — arrow labelled " $\Delta Q = nC_V\Delta T$, all into U ". Right cylinder: piston free under a weight ("P xed") — the gas column is taller, piston pushed up by ΔV , arrow split into " ΔU " plus " $P\Delta V$ work". Caption: "Same ΔT , but the right one needs $nR\Delta T$ more heat $\Rightarrow C_P = C_V + R$." Section 2: Key Formulas & Definitions Heat capacity (of a body): $S = \Delta Q / \Delta T$ (units $J K^{-1}$). Specific heat capacity (per unit mass): $s = 1/m \Delta Q / \Delta T$ (units $J kg^{-1} K^{-1}$). $\Delta Q = m s \Delta T$. Molar specific heat capacity (per mole): $C = 1/n \Delta Q / \Delta T$ (units $J mol^{-1} K^{-1}$). For a gas — the two molar specific heats: $C_V = (1/n) \cdot \Delta Q / \Delta T$ $V = 1/n dU/dT$, $C_P = (1/n) \cdot \Delta Q / \Delta T$ P . Mayer's relation (ideal gas, per mole): $C_P - C_V = R$. Equipartition and degrees of freedom (f active degrees of freedom): $C_V = f/2 R$, $C_P = (f/2 + 1) R$, $\gamma = C_P / C_V = 1 + 2/f$. Standard ideal-gas values: Gas type f C_V C_P $\gamma = C_P / C_V$ Monoatomic (He, Ar) 3 $3/2 R \approx 12.5$ $5/2 R \approx 20.8$ $5/3 \approx 1.67$ Diatomic (N_2 , O_2 , H_2) 5 $5/2 R \approx 20.8$ $7/2 R \approx 29.1$ $7/5 = 1.40$ Polyatomic (nonlinear, e.g. CO_2 bent / H_2O) 6 $3R \approx 24.9$ $4R \approx 33.3$ $4/3 \approx 1.33$ (C_V , C_P in $J mol^{-1} K^{-1}$, using $R = 8.314$.) Internal energy from C_V (any process, ideal gas): $U = nC_V T$, $\Delta U = nC_V \Delta T$. Polytropic molar heat capacity ($PV^x = \text{const}$): $C = C_V + R / (1 - x)$. Special cases: isobaric $x = 0 \Rightarrow C_P$; isothermal $x = 1 \Rightarrow C \rightarrow \infty$; adiabatic $x = \gamma \Rightarrow C = 0$; isochoric $x \rightarrow \infty \Rightarrow C_V$. Useful benchmarks: water $s \approx 4186 J kg^{-1} K^{-1} = 1 cal g^{-1} ^\circ C^{-1}$; Dulong-Petit law for many solids: molar heat capacity $\approx 3R \approx 25 J mol^{-1} K^{-1}$. For a mixture of $n_1, n_2, \dots$ moles of gases: $C_{V,mix} = \sum P_i n_i C_{V,i}$, and similarly for C_P ; then $\gamma_{mix} = C_{P,mix} / C_{V,mix}$. Section 3: Key Procedures & Logical Framework 3.1 Derivation of Mayer's relation $C_P - C_V = R$ (board favourite) Setup. Consider one mole ($n = 1$) of an ideal gas. We heat it by dT , once at constant volume and once at constant pressure, and compare. Step 1 — constant volume. No work is done ($dV = 0$), so the first law $\delta Q = dU + P dV$ gives $\delta Q_V = dU$. By definition $C_V = (1/n)(\delta Q/dT)_V = dU/dT$ (for one mole), hence $dU = C_V dT$. 24

Crucially, for an ideal gas U depends only on T , so this $dU = C_V dT$ holds for *every* process, not just constant volume.

Step 2 — constant pressure. Now $dQ_P = dU + P dV = C_V dT + P dV$. By definition $dQ_P = C_P dT$, so

$$C_P dT = C_V dT + P dV. \quad (1)$$

Step 3 — evaluate $P dV$ from the gas law. For one mole, $PV = RT$. At constant pressure, differentiate: $P dV = R dT$.

Step 4 — substitute into (1).

$$C_P dT = C_V dT + R dT \Rightarrow \boxed{C_P = C_V + R}, \quad \text{i.e. } C_P - C_V = R.$$

Physical reading: the extra R is precisely the work $P dV = R dT$ done pushing back the surroundings per mole per kelvin at constant pressure. That “leak” is why $C_P > C_V$ always, by exactly R .

3.2 Why $C_V = \frac{f}{2}R$ (the equipartition link)

1. Equipartition assigns each active degree of freedom an average energy $\frac{1}{2}k_B T$ per molecule.
2. For N molecules (n moles, $N = nN_A$) with f degrees of freedom: $U = N \cdot f \cdot \frac{1}{2}k_B T = \frac{f}{2}nRT$ (using $N_A k_B = R$).
3. Then $C_V = \frac{1}{n} \frac{dU}{dT} = \frac{f}{2}R$, and Mayer gives $C_P = \left(\frac{f}{2} + 1\right)R$, so $\gamma = 1 + \frac{2}{f}$.
4. Reading off f : monoatomic $f = 3$ (3 translations); diatomic $f = 5$ (3 translations + 2 rotations) at room temperature; nonlinear polyatomic $f = 6$ (3 translations + 3 rotations).

3.3 Procedure: choosing C_V or C_P in a numerical

1. For ΔU , *always* use $\Delta U = nC_V \Delta T$ — it is process-independent for an ideal gas.
2. For heat ΔQ : at constant volume use $nC_V \Delta T$; at constant pressure use $nC_P \Delta T$; for a general polytrope use $nC \Delta T$ with $C = C_V + R/(1 - x)$.
3. Identify f (and hence C_V, C_P, γ) from the gas type before plugging numbers.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Problem. The molar specific heat at constant volume of a gas is $C_V = 20.8 \text{ J mol}^{-1}\text{K}^{-1}$. Identify the gas type and find C_P and γ . ($R = 8.314$.)

Solution. $C_V = \frac{5}{2}R = 20.8 \Rightarrow f = 5$, a *diatomic* gas. By Mayer, $C_P = C_V + R = 20.8 + 8.3 = 29.1 \text{ J mol}^{-1}\text{K}^{-1}$, and $\gamma = C_P/C_V = 29.1/20.8 = 7/5 = 1.40$.

Diatomic; $C_P \approx 29.1 \text{ J mol}^{-1}\text{K}^{-1}$, $\gamma = 1.40$
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Example 2 — [NEET Speed Trap]

Problem. For which gas is the fraction of heat (supplied at constant pressure) that goes into *work* the largest: monoatomic, diatomic, or polyatomic?

Speed solution. At constant pressure the work fraction is $\frac{W}{\Delta Q} = \frac{nR\Delta T}{nC_P\Delta T} = \frac{R}{C_P} = \frac{R}{(\frac{f}{2} + 1)R} = \frac{2}{f + 2}$. This *decreases* as f increases, so it is largest for the smallest f — the monoatomic gas.

Monoatomic ($f = 3$): work fraction = $2/5 = 40\%$, the largest
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The trap: students assume “polyatomic absorbs the most heat, so it does the most work.” But more degrees of freedom means more heat is siphoned into internal modes, leaving a *smaller* share for expansion work.

Example 3 — [JEE Main Level]

Problem. 2.0 mol of a monoatomic ideal gas is heated so its temperature rises by 60 K. Find the heat supplied (a) if the heating is at constant volume, (b) if at constant pressure, and (c) the difference between the two, interpreting it physically. ($R = 8.314$.)

Solution. Monoatomic: $C_V = \frac{3}{2}R = 12.47$, $C_P = \frac{5}{2}R = 20.79 \text{ J mol}^{-1}\text{K}^{-1}$.

(a) $\Delta Q_V = nC_V\Delta T = (2.0)(12.47)(60) \approx 1.50 \times 10^3 \text{ J} = 1.50 \text{ kJ}$.

(b) $\Delta Q_P = nC_P\Delta T = (2.0)(20.79)(60) \approx 2.49 \times 10^3 \text{ J} = 2.49 \text{ kJ}$.

(c) Difference $= \Delta Q_P - \Delta Q_V = nR\Delta T = (2.0)(8.314)(60) \approx 998 \text{ J} \approx 1.0 \text{ kJ}$.

$$\Delta Q_V \approx 1.50 \text{ kJ}, \quad \Delta Q_P \approx 2.49 \text{ kJ}, \quad \text{difference} \approx 1.0 \text{ kJ}$$

The extra $\approx 1.0 \text{ kJ}$ at constant pressure is exactly the expansion work $nR\Delta T$ the gas does against the surroundings — the same R that appears in Mayer’s relation.

Example 4 — [JEE Advanced Level]

Problem. A container holds a mixture of 2.0 mol of a monoatomic ideal gas and 4.0 mol of a diatomic ideal gas. Find $C_{V,\text{mix}}$, $C_{P,\text{mix}}$, and γ_{mix} .

Solution. For the components: monoatomic $C_{V1} = \frac{3}{2}R$, diatomic $C_{V2} = \frac{5}{2}R$. Mole-weighted average:

$$C_{V,\text{mix}} = \frac{n_1 C_{V1} + n_2 C_{V2}}{n_1 + n_2} = \frac{(2.0)(\frac{3}{2}R) + (4.0)(\frac{5}{2}R)}{6.0} = \frac{3R + 10R}{6} = \frac{13}{6}R \approx 2.167R.$$

By Mayer, $C_{P,\text{mix}} = C_{V,\text{mix}} + R = \frac{13}{6}R + R = \frac{19}{6}R$. Hence

$$\gamma_{\text{mix}} = \frac{C_{P,\text{mix}}}{C_{V,\text{mix}}} = \frac{19/6}{13/6} = \frac{19}{13} \approx 1.46.$$

$$C_{V,\text{mix}} = \frac{13}{6}R \approx 18.0, \quad C_{P,\text{mix}} = \frac{19}{6}R \approx 26.3 \text{ J mol}^{-1}\text{K}^{-1}, \quad \gamma_{\text{mix}} \approx 1.46$$

The mixture’s γ lies between the monoatomic 1.67 and diatomic 1.40, pulled toward the diatomic because diatomic moles dominate the mixture (4:2).

Section 5: Practice Exercises

1. [CBSE] Define molar specific heat at constant volume and at constant pressure, and state Mayer’s relation. For a gas with $\gamma = 1.4$, find C_V and C_P in terms of R .
2. [NEET] A gas has $\gamma = 4/3$. How many active degrees of freedom does each molecule have, and what type of gas is it?
3. [JEE Main] 0.50 mol of a diatomic ideal gas is heated at constant pressure by 200 K. Find the heat supplied, the work done, and ΔU . ($R = 8.314$.)
4. [JEE Main] How much heat is needed to raise the temperature of 5.0 kg of water from 20°C to 80°C ? ($s_{\text{water}} = 4186 \text{ J kg}^{-1}\text{K}^{-1}$.)
5. [JEE Advanced] A mixture contains 3.0 mol monoatomic and 2.0 mol diatomic ideal gas. Find γ_{mix} .

Answers: 1. C_V = heat per mole per K at constant volume ($= dU/dT$); C_P at constant pressure; $C_P - C_V = R$. With $\gamma = 1.4$: $C_V = R/(\gamma - 1) = R/0.4 = 2.5R$, $C_P = \gamma C_V =$

3.5R. 2. $\gamma = 1 + 2/f = 4/3 \Rightarrow 2/f = 1/3 \Rightarrow f = 6$; nonlinear polyatomic gas. 3. $\Delta Q = nC_P\Delta T = (0.5)(\frac{7}{2} \times 8.314)(200) \approx 5.82 \text{ kJ}$; $W = nR\Delta T = (0.5)(8.314)(200) \approx 831 \text{ J}$; $\Delta U = nC_V\Delta T = (0.5)(\frac{5}{2} \times 8.314)(200) \approx 4.16 \text{ kJ}$ (check: $4.16 + 0.83 = 4.99\dots$ note $\Delta U + W = \Delta Q$, $4.16 + 0.83 = 4.99 \text{ kJ}$; rounding, ≈ 5.82 ? recompute: $\Delta Q - W = 5.82 - 0.83 = 4.99 = \Delta U$ ✓). 4. $\Delta Q = msc\Delta T = (5.0)(4186)(60) \approx 1.26 \times 10^6 \text{ J} = 1.26 \text{ MJ}$. 5. $C_{V,\text{mix}} = \frac{3(\frac{3}{2}R) + 2(\frac{5}{2}R)}{5} = \frac{4.5R + 5R}{5} = 1.9R$; $C_{P,\text{mix}} = 2.9R$; $\gamma_{\text{mix}} = 2.9/1.9 \approx 1.53$.

Section 6: Crack the MCQ

MCQ 1.

For an ideal gas, $C_P - C_V$ equals: (A) 0 (B) R (C) γR (D) R/γ

Correct: (B) R . Mayer's relation, per mole. The difference is the constant-pressure expansion work $nR\Delta T$ per mole per kelvin.

- (A) would mean no expansion work — only true at constant volume. (C),(D) confuse the difference with ratios of C_P, C_V .

MCQ 2.

The ratio $\gamma = C_P/C_V$ for a monoatomic ideal gas is: (A) $7/5$ (B) $5/3$ (C) $4/3$ (D) $3/2$

Correct: (B) $5/3$. With $f = 3$: $\gamma = 1 + 2/3 = 5/3$.

- (A) is diatomic ($f = 5$). (C) is polyatomic ($f = 6$). (D) has no standard f .

MCQ 3.

Why is C_P greater than C_V for a gas? (A) more molecules at constant pressure (B) at constant pressure some heat does expansion work (C) pressure heats the gas faster (D) C_V ignores internal energy

Correct: (B). At constant pressure the gas expands and spends part of the heat as work, so more heat is needed for the same ΔT .

- (A) the amount of gas is unchanged. (C) heating rate is irrelevant to the capacity. (D) false — $C_V = dU/dT$ is built *from* internal energy.

MCQ 4.

At room temperature the internal energy of one mole of a diatomic ideal gas is: (A) $\frac{3}{2}RT$ (B) $\frac{5}{2}RT$ (C) $\frac{7}{2}RT$ (D) $3RT$

Correct: (B) $\frac{5}{2}RT$. $f = 5$ (3 translation + 2 rotation), so $U = \frac{f}{2}RT = \frac{5}{2}RT$ per mole.

- (A) is monoatomic. (C) would require $f = 7$ (vibration active — frozen at room temperature). (D) is nonlinear polyatomic ($f = 6$).

Section 7: Common Student Pitfalls & Pro-Tips

1. **Using C_P for ΔU .** Internal energy change is *always* $nC_V\Delta T$ for an ideal gas, in every process — never $nC_P\Delta T$. Use C_P only for heat at constant pressure.
2. **Counting diatomic degrees of freedom as 7.** At ordinary temperatures vibration is frozen, so $f = 5$, not 7. Use 7 only if a problem explicitly invokes high-temperature vibration.
3. **Mixing specific (per kg) and molar (per mole) capacities.** s has units $\text{J kg}^{-1}\text{K}^{-1}$; C has $\text{J mol}^{-1}\text{K}^{-1}$. They differ by the molar mass: $C = Ms$.
4. **Forgetting Mayer is per mole and ideal-gas only.** $C_P - C_V = R$ is not the relation for solids, liquids, or real gases except approximately.

Pro-Tip (speed & accuracy)

Memorise the two reflex identities $C_V = \frac{R}{\gamma - 1}$ and $C_P = \frac{\gamma R}{\gamma - 1}$ — they convert any γ straight into capacities with no degree-of-freedom counting. For a gas type, jump via f : monoatomic $\frac{3}{2}R, \frac{5}{2}R$; diatomic $\frac{5}{2}R, \frac{7}{2}R$; polyatomic $3R, 4R$. And the constant-pressure work fraction $R/C_P = 2/(f+2)$ answers a whole class of NEET “which gas does most work” traps in one step.

Section 8: Cheat Sheet Box**SPECIFIC HEAT CAPACITY — RAPID RECAP**

- Specific heat: $\Delta Q = ms\Delta T$ (per kg); molar: $\Delta Q = nC\Delta T$ (per mole).
- Two for a gas: C_V (constant V) and C_P (constant P), with $C_P > C_V$ always.
- **Mayer:** $C_P - C_V = R$ (ideal gas, per mole). The R is the expansion work.
- Degrees of freedom: $C_V = \frac{f}{2}R$, $C_P = (\frac{f}{2} + 1)R$, $\gamma = 1 + \frac{2}{f}$.
- Mono ($f=3$): $\frac{3}{2}R, \frac{5}{2}R, \gamma=5/3$. Di ($f=5$): $\frac{5}{2}R, \frac{7}{2}R, \gamma=7/5$. Poly ($f=6$): $3R, 4R, \gamma=4/3$.
- Reflex: $C_V = \frac{R}{\gamma - 1}$, $C_P = \frac{\gamma R}{\gamma - 1}$. $\Delta U = nC_V\Delta T$ (every process).
- Water $s \approx 4186 \text{ J kg}^{-1}\text{K}^{-1}$; solids (Dulong–Petit) $\approx 3R$.
- Mixture: mole-weighted C_V , then $C_P = C_V + R$, $\gamma = C_P/C_V$.

Memory aids: “ C_P pays for the Push” (the extra R is expansion work). • “More wiggles, more storage” (bigger $f \Rightarrow$ bigger C_V , smaller γ). • “ ΔU always rides C_V .”

Subtopic 05 — Heat Engines, Refrigerators & Heat Pumps

Exam Relevance: This subtopic turns the first law into machines. CBSE Boards ask the definition of a heat engine, the efficiency formula, and the schematic energy-flow diagram, plus refrigerator coefficient of performance. NEET likes one-step efficiency and COP numericals and the conceptual difference between an engine and a refrigerator. JEE Main combines η and COP with the first-law identity $Q_1 = W + Q_2$. JEE Advanced links these to the second law and Carnot limits (next subtopics). Lock down $\eta = 1 - Q_2/Q_1$ and the two COP formulas.

Section 1: Concept Intuition

The first law says energy is conserved; it does not say energy is *useful*. A **heat engine** is any device that does the most economically important trick in physics: it converts heat — the most disordered form of energy — into ordered mechanical work, over and over, in a cycle. Every car engine, steam turbine, and power-plant boiler is a heat engine.

The picture is always the same three-part story. A working substance (steam, petrol-air mixture, a gas) draws heat Q_1 from a **hot reservoir** at temperature T_1 , converts *part* of it into work W , and dumps the leftover Q_2 into a **cold reservoir** at temperature T_2 . The phrase “part of it” is the whole drama: you can never convert all the heat into work — some must always be thrown away to the cold reservoir. Because the substance returns to its starting state each cycle ($\Delta U = 0$), the first law forces $W = Q_1 - Q_2$. The **efficiency** is the fraction of the purchased heat that becomes useful work: $\eta = W/Q_1 = 1 - Q_2/Q_1$.

Think about it this way...

Think of Q_1 as your monthly salary drawn from the bank (hot reservoir). W is the money you actually get to spend on yourself; Q_2 is the unavoidable tax and fees siphoned off to the government (cold reservoir). Efficiency is the take-home fraction. No matter how clever your accounting, the “tax” Q_2 can never be zero — and the second law (next subtopic) explains exactly why. A 100%-efficient engine ($Q_2 = 0$) is as impossible as a job with zero deductions.

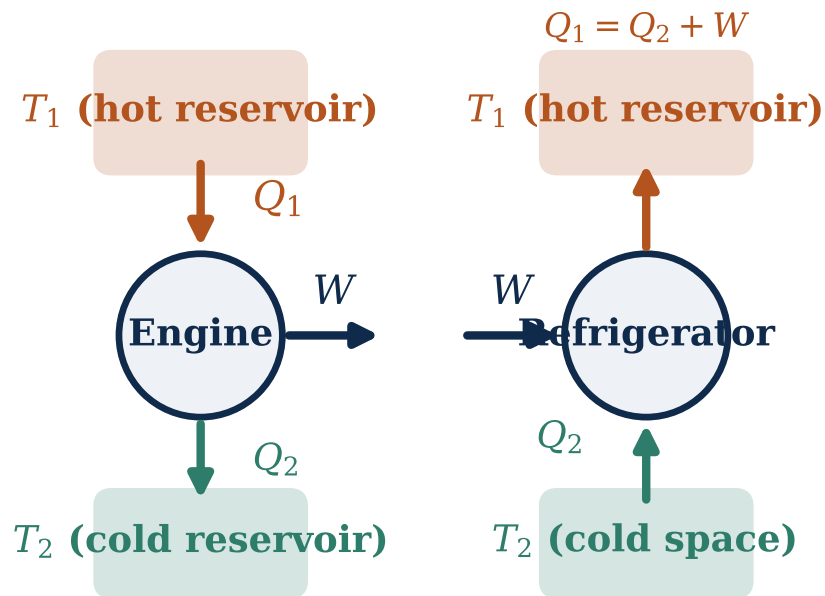
Running the movie backwards.

A **refrigerator** or **heat pump** is simply a heat engine running in reverse. Instead of heat flowing naturally hot→cold and producing work, you *supply* work W to force heat to flow the unnatural way: cold→hot. The fridge extracts heat Q_2 from the cold interior (your food), you pay work W (the compressor, run by electricity), and the total $Q_1 = Q_2 + W$ is dumped into the warm kitchen (the coils at the back). A heat pump is the identical machine, but now you care about the heat Q_1 delivered to the *warm* side (heating a house in winter). Same hardware, different goal — and therefore a different “figure of merit,” the **coefficient of performance** (COP), defined as “what you want ÷ what you pay.”

Limiting Conditions & Assumptions.

(1) These definitions assume a *cyclic* device, so the working substance returns to its initial state and $\Delta U_{\text{cycle}} = 0$ — that is why $W = Q_1 - Q_2$. (2) “Reservoir” means a body so large its temperature does not change as heat is exchanged. (3) Efficiency and COP as defined here are general; the *maximum possible* values are set by the Carnot limit (next subtopic) and depend only on the reservoir temperatures. (4) COP can exceed 1 (and usually does) — it is not a percentage, so do not “cap” it at 100%.

Figure 11.8 Left — engine: a hot reservoir box (top, T_1) with arrow Q_1 down into a circle labelled “Engine”; an arrow W out to the side (work); an arrow Q_2 down into a cold reservoir box (bottom, T_2). Right — refrigerator: same boxes, but arrows reversed — Q_2 flows up from the cold box into the circle, work W flows in from the side, and $Q_1 = Q_2 + W$ flows up into the hot box. Caption: “Engine: pay heat, get work. Fridge: 29 (see next page)”



Engine: pay heat, get work. Fridge: pay work, move heat.

Figure 11.8 Left — engine: a hot reservoir box (top, T_1) with arrow Q_1 down into a circle labelled "Engine"; an arrow W out to the side (work); an arrow Q_2 down into a cold reservoir box (bottom, T_2). Right — refrigerator: same boxes, but arrows reversed — Q_2 flows up from the cold box into the circle, work W flows in from the side, and $Q_1 = Q_2 + W$ flows up into the hot box. Caption: "Engine: pay heat, get work. Fridge: 29

pay work, move heat.”

Section 2: Key Formulas & Definitions

Heat engine (per cycle). Working substance absorbs Q_1 from the hot reservoir, rejects Q_2 to the cold reservoir, does net work W . Since $\Delta U_{\text{cycle}} = 0$:

$$W = Q_1 - Q_2.$$

Thermal efficiency:

$$\boxed{\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}}, \quad 0 \leq \eta < 1.$$

Often quoted as a percentage; $\eta = 1$ ($Q_2 = 0$) is forbidden by the second law.

Refrigerator (goal: remove heat Q_2 from the cold space; you pay work W , and $Q_1 = Q_2 + W$ is rejected to the hot surroundings):

$$\boxed{\text{COP}_{\text{ref}} = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2}}.$$

Heat pump (goal: deliver heat Q_1 to the warm space; same machine, you pay work W):

$$\boxed{\text{COP}_{\text{hp}} = \frac{Q_1}{W} = \frac{Q_1}{Q_1 - Q_2}}.$$

Universal relations (from $Q_1 = Q_2 + W$):

$$\text{COP}_{\text{hp}} = \text{COP}_{\text{ref}} + 1, \quad \text{COP}_{\text{ref}} = \frac{1}{\eta_{\text{engine}}} - 1 \text{ (for the same machine run as an engine).}$$

Quantity reference:

Quantity	Symbol	Notes
Heat absorbed (hot)	Q_1	from reservoir at T_1 (the “input” for an engine)
Heat rejected (cold)	Q_2	to reservoir at T_2 (always > 0 in practice)
Net work per cycle	W	$= Q_1 - Q_2$ (engine) or supplied (fridge/pump)
Efficiency	η	dimensionless, $0 \leq \eta < 1$
Coefficient of performance	COP	dimensionless, can exceed 1

Section 3: Key Procedures & Logical Framework

3.1 Why $W = Q_1 - Q_2$ (the cyclic argument)

- Over one complete cycle the working substance returns to its starting state, so $\Delta U_{\text{cycle}} = 0$.
- The first law over the cycle: $Q_{\text{net}} = W_{\text{net}}$, i.e. $(Q_1 - Q_2) = W$. (Heat absorbed counts +, heat rejected −.)
- Efficiency is “useful out $\div$ costly in” $= W/Q_1 = 1 - Q_2/Q_1$. The same energy bookkeeping, divided differently, gives the fridge and heat-pump COPs.

3.2 Procedure: engine vs. refrigerator numericals

- Identify the device: does it *produce* work (engine) or *consume* work (fridge/pump)?

2. Write the energy balance $Q_1 = Q_2 + W$ — this single equation links all three.
3. Pick the figure of merit by the *goal*: engine $\rightarrow \eta = W/Q_1$; fridge $\rightarrow \text{COP} = Q_2/W$; heat pump $\rightarrow \text{COP} = Q_1/W$.
4. For maximum (ideal) values, replace heat ratios with temperature ratios via the Carnot result $Q_2/Q_1 = T_2/T_1$ (justified in Subtopic 07).

3.3 The link to the second law (preview)

An engine with $Q_2 = 0$ ($\eta = 1$) would convert heat entirely into work with no other effect — forbidden by the Kelvin–Planck statement. A refrigerator with $W = 0$ ($\text{COP} = \infty$) would move heat cold $\rightarrow$ hot with no work — forbidden by the Clausius statement. These two impossibilities are the subject of the next subtopic, and they are why $\eta < 1$ and COP is finite.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Problem. In each cycle a heat engine absorbs 1000 J from a hot reservoir and rejects 600 J to a cold reservoir. Find (a) the work done per cycle and (b) the efficiency.

Solution. (a) $W = Q_1 - Q_2 = 1000 - 600 = 400$ J. (b) $\eta = W/Q_1 = 400/1000 = 0.40 = 40\%$.

$$W = 400 \text{ J}, \quad \eta = 40\%$$

Example 2 — [NEET Speed Trap]

Problem. A refrigerator extracts 200 J of heat from its interior for every 50 J of electrical work supplied. What is its coefficient of performance, and how much heat is dumped into the room per cycle?

Speed solution. $\text{COP}_{\text{ref}} = Q_2/W = 200/50 = 4$. Heat dumped $Q_1 = Q_2 + W = 200 + 50 = 250$ J.

$$\text{COP} = 4; \quad \text{heat rejected } Q_1 = 250 \text{ J}$$

The trap: students compute $Q_2/Q_1 = 200/250 = 0.8$ and call it the COP, confusing the fridge with engine efficiency. For a fridge the denominator is the *work*, not the hot-side heat — and the COP of 4 (exceeding 1) is perfectly normal.

Example 3 — [JEE Main Level]

Problem. A heat engine has efficiency 25% and delivers 500 J of useful work per cycle. Find (a) the heat absorbed from the hot reservoir and (b) the heat rejected to the cold reservoir per cycle.

Solution. (a) $\eta = W/Q_1 \Rightarrow Q_1 = W/\eta = 500/0.25 = 2000$ J. (b) $Q_2 = Q_1 - W = 2000 - 500 = 1500$ J.

$$Q_1 = 2000 \text{ J}, \quad Q_2 = 1500 \text{ J}$$

A 25% engine throws away three-quarters of the heat it buys — a vivid reminder of why Q_2 is never negligible.

Example 4 — [JEE Advanced Level]

Problem. A device operates between a cold space at $T_2 = 250$ K and warm surroundings at $T_1 = 300$ K. Treating it as ideal (Carnot), find (a) its COP when used as a refrigerator, (b) its COP when the *same* machine is used as a heat pump, (c) verify $\text{COP}_{\text{hp}} = \text{COP}_{\text{ref}} + 1$, and (d) the electrical work needed to remove 1000 J from the cold space as a refrigerator.

Solution. For an ideal (Carnot) device, $Q_2/Q_1 = T_2/T_1$, so:

$$(a) \text{ COP}_{\text{ref}} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2} = \frac{250}{300 - 250} = \frac{250}{50} = 5.$$

$$(b) \text{ COP}_{\text{hp}} = \frac{Q_1}{Q_1 - Q_2} = \frac{T_1}{T_1 - T_2} = \frac{300}{50} = 6.$$

$$(c) 5 + 1 = 6 \checkmark.$$

$$(d) \text{ COP}_{\text{ref}} = Q_2/W \Rightarrow W = Q_2/\text{COP} = 1000/5 = 200 \text{ J}.$$

$$\text{COP}_{\text{ref}} = 5, \quad \text{COP}_{\text{hp}} = 6, \quad W = 200 \text{ J to move } 1000 \text{ J}$$

Notice the leverage: 200 J of electrical work pumps 1000 J of heat out of the cold space — and as a heat pump, 200 J of work would deliver 1200 J of heating. This “more heat than work paid” is exactly why heat pumps beat resistive heaters, and the COP shrinks as $T_1 - T_2$ grows.

Section 5: Practice Exercises

- [CBSE]** A heat engine absorbs 800 J per cycle and does 200 J of work. Find the heat rejected and the efficiency.
- [NEET]** State, in one line each, the difference between a refrigerator and a heat pump in terms of which heat is the “desired” output.
- [JEE Main]** An engine of efficiency 30% rejects 1400 J of heat per cycle. Find the heat absorbed and the work done.
- [JEE Main]** A refrigerator has $\text{COP} = 4.5$ and removes 900 J from the cold chamber per cycle. Find the work input and the heat rejected to the room.
- [JEE Advanced]** An ideal (Carnot) refrigerator works between $T_2 = 270 \text{ K}$ and $T_1 = 300 \text{ K}$. Find its COP and the minimum work to extract 600 J from the cold reservoir.

Answers: 1. $Q_2 = 800 - 200 = 600 \text{ J}$; $\eta = 200/800 = 25\%$. 2. Refrigerator: desired output is the heat Q_2 removed from the cold space; heat pump: desired output is the heat Q_1 delivered to the warm space (same machine, opposite goal). 3. $\eta = W/Q_1 = 1 - Q_2/Q_1 \Rightarrow Q_1 = Q_2/(1 - \eta) = 1400/0.70 = 2000 \text{ J}$; $W = \eta Q_1 = 0.30 \times 2000 = 600 \text{ J}$. 4. $W = Q_2/\text{COP} = 900/4.5 = 200 \text{ J}$; $Q_1 = Q_2 + W = 1100 \text{ J}$. 5. $\text{COP}_{\text{ref}} = T_2/(T_1 - T_2) = 270/30 = 9$; $W_{\text{min}} = Q_2/\text{COP} = 600/9 \approx 66.7 \text{ J}$.

Section 6: Crack the MCQ

MCQ 1.

The efficiency of a heat engine is given by: (A) Q_2/Q_1 (B) $1 - Q_2/Q_1$ (C) Q_1/W (D) $1 - Q_1/Q_2$

Correct: (B) $1 - Q_2/Q_1$. $\eta = W/Q_1 = (Q_1 - Q_2)/Q_1$.

- (A) is the rejected fraction, not efficiency. (C) inverts the definition. (D) has the ratio upside-down (and would be negative).

MCQ 2.

A refrigerator’s coefficient of performance can be: (A) only less than 1 (B) exactly 1 always (C) greater than 1 (D) negative

Correct: (C) **greater than 1**. $\text{COP} = Q_2/W$; since useful heat moved often exceeds the work paid, COP routinely exceeds 1.

- (A),(B) wrongly treat COP like a capped percentage. (D) impossible — both Q_2 and W are positive.

MCQ 3.

For a device run as a refrigerator and as a heat pump between the same reservoirs, $\text{COP}_{\text{hp}} - \text{COP}_{\text{ref}}$ equals: (A) 0 (B) 1 (C) η (D) Q_1/Q_2

Correct: (B) **1**. $\text{COP}_{\text{hp}} = Q_1/W$, $\text{COP}_{\text{ref}} = Q_2/W$, and $Q_1 - Q_2 = W$, so the difference is $W/W = 1$.

- (A) ignores the +1. (C),(D) confuse with engine quantities.

MCQ 4.

A heat engine with $\eta = 1$ ($Q_2 = 0$) is impossible because it would violate: (A) the first law (B) the zeroth law (C) the second law (Kelvin–Planck) (D) conservation of momentum

Correct: (C). Converting heat entirely into work with no other effect is exactly what the Kelvin–Planck statement forbids.

- (A) the first law is *satisfied* by $\eta = 1$ (energy still balances) — which is why a second law is needed. (B),(D) unrelated.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Putting Q_1 in the COP denominator.** For a fridge/pump the denominator is the *work* W , not the hot-side heat. $\text{COP} = Q_2/W$ (fridge) or Q_1/W (pump).
2. **Treating COP as a percentage capped at 100%.** COP is a ratio that is normally > 1 . Never write “COP = 80%.”
3. **Forgetting $Q_1 = Q_2 + W$ for a fridge.** The hot-side heat is the cold-side heat *plus* the work you paid — the room gets warmer than just the food’s heat.
4. **Confusing efficiency with rejected fraction.** $\eta = 1 - Q_2/Q_1$, not Q_2/Q_1 .

Pro-Tip (speed & accuracy)

Always start from the single balance $Q_1 = Q_2 + W$; everything else is one division away. For ideal-limit problems, swap the heat ratio for the temperature ratio: $\eta_{\max} = 1 - T_2/T_1$, $\text{COP}_{\text{ref},\max} = T_2/(T_1 - T_2)$, $\text{COP}_{\text{hp},\max} = T_1/(T_1 - T_2)$. Notice all three share the denominator $(T_1 - T_2)$ — a small temperature gap means a great fridge but a feeble engine.

Section 8: Cheat Sheet Box**HEAT ENGINES, REFRIGERATORS & HEAT PUMPS — RAPID RECAP**

- Energy balance (cyclic): $Q_1 = Q_2 + W$ ($\Delta U_{\text{cycle}} = 0$).
- Engine efficiency: $\eta = W/Q_1 = 1 - Q_2/Q_1$, with $0 \leq \eta < 1$.
- Refrigerator: $\text{COP}_{\text{ref}} = Q_2/W = Q_2/(Q_1 - Q_2)$.
- Heat pump: $\text{COP}_{\text{hp}} = Q_1/W = Q_1/(Q_1 - Q_2)$.
- Relation: $\text{COP}_{\text{hp}} = \text{COP}_{\text{ref}} + 1$.
- Ideal (Carnot) limits: $\eta_{\max} = 1 - T_2/T_1$; $\text{COP}_{\text{ref},\max} = T_2/(T_1 - T_2)$; $\text{COP}_{\text{hp},\max} = T_1/(T_1 - T_2)$.
- $\eta = 1$ violates Kelvin–Planck; $\text{COP} = \infty$ violates Clausius.

Memory aids: “Engine: pay heat, get work. Fridge: pay work, move heat.” • “Figure of merit = what you want $\div$ what you pay.” • “COP can beat 100% — it’s a ratio, not a percentage.”

Subtopic 06 — The Second Law of Thermodynamics & Reversibility

Exam Relevance: If the first law is the chapter’s workhorse, the second law is its conscience — it decides which energy-conserving processes Nature actually permits. CBSE Boards ask for the two statements (Kelvin–Planck and Clausius), the definition of reversible vs. irreversible, and “why is a process irreversible.” NEET fires one-line conceptual traps (entropy increases, heat flows hot→cold spontaneously, $\eta = 1$ is impossible). JEE Main asks you to spot which statement a hypothetical device violates. JEE Advanced probes the *equivalence* of the two statements and the precise meaning of reversibility. This is the conceptual bridge to the Carnot engine.

Section 1: Concept Intuition

The first law is generous to a fault. It will happily allow a cup of coffee to grow hotter while the room cools, or a ball lying on the floor to spontaneously gather heat from the tiles and leap upward — because none of these violates energy conservation. Yet you have never once seen them happen. Something beyond energy balance forbids them. That “something” is the **second law of thermodynamics**, and it is fundamentally a law about *direction*: it tells us which way the movie of the universe is allowed to run.

Watch any everyday process and you notice an arrow. Heat flows from hot to cold, never the reverse on its own. A drop of ink spreads through water but never re-gathers into a drop. A swinging pendulum slows and stops as its ordered motion bleeds into disordered heat in the air and pivot — but a warm, still pendulum never spontaneously starts swinging by cooling itself. Every one of these is energy-conserving in both directions; only one direction ever occurs. The second law names this universal one-wayness.

Think about it this way...

Picture shuffling a brand-new deck of cards stacked in perfect order. A few shuffles and it is hopelessly mixed. Keep shuffling forever and it will essentially never return to perfect order — not because order is forbidden by any law of card-motion, but because there is exactly *one* ordered arrangement and astronomically many disordered ones. Nature drifts toward the overwhelmingly more probable (disordered) states. Heat spreading out, ink diffusing, motion dissipating — all are the physical world “shuffling toward” its most probable condition. The second law is the bookkeeping of that drift, later quantified by **entropy** (Subtopic 08).

Two faces of one law.

Historically the second law was stated two ways, born from two practical frustrations. Engineers building engines found they could never convert heat fully into work — the **Kelvin–Planck** statement captures this. Engineers building refrigerators found they could never move heat from cold to hot without paying work — the **Clausius** statement captures that. Remarkably, these two “no free lunch” rules turn out to be logically *equivalent*: assume you could break one, and you could rig a machine that breaks the other. They are the same prohibition seen from the engine end and the fridge end.

Reversible and irreversible.

A **reversible** process is an idealised limit: carried out infinitely slowly (quasi-statically) through a continuous chain of equilibrium states, with *no* dissipative effects (no friction, no turbulence, no finite-temperature-gap heat flow). Run it backwards and both system *and* surroundings retrace their states exactly, leaving no trace. No real process is truly reversible — it is a frictionless, infinitely-patient ideal — but it is the crucial benchmark, because reversible engines set the ceiling on efficiency (Carnot, next subtopic). Every **irreversible** process (which is to say, every real one) generates entropy and cannot be undone without leaving a permanent mark on the surroundings.

Limiting Conditions & Assumptions.

(1) The second law is a statement about *spontaneous* (unaided) processes; you can of course force heat cold→hot *by supplying work* (that is a refrigerator) — what is forbidden is doing it with *no other effect*. (2) Reversibility requires both quasi-static evolution *and* the absence of dissipation; quasi-static alone is not enough (a slow process with friction is still irreversible). (3) These statements concern cyclic devices and macroscopic systems; the “arrow” is statistical, overwhelmingly — not metaphysically — certain.

Figure 11.9 Two crossed-out “impossible machines.” Left (violates Kelvin–Planck): a single hot reservoir T_1 with heat Q flowing into an engine that outputs work $W = Q$ and rejects nothing — a big red cross over it, captioned “100% heat→work from one reservoir: forbidden.” Right (violates Clausius): a cold box T_2 and hot box T_1 with heat Q flowing cold→hot through a device with no work input — red cross, captioned “Heat cold→hot with no work: forbidden.”

Section 2: Key Definitions & Statements

Kelvin–Planck statement (engine form). No process is possible whose sole result is the complete conversion of heat absorbed from a single reservoir into work. Equivalently: no heat engine can have efficiency $\eta = 1$; some heat must always be rejected to a colder reservoir.

Clausius statement (refrigerator form). No process is possible whose sole result is the transfer of heat from a colder body to a hotter body. Equivalently: heat cannot flow spontaneously cold→hot; doing so always requires work input.

Equivalence. The two statements are logically identical: a violation of either can be used to construct a violation of the other (proof in Section 3).

Reversible process. A process carried out quasi-statically (through equilibrium states) and free of all dissipative effects, such that reversing it restores both system and surroundings to their initial states with no net change anywhere. An ideal limit; the maximum-efficiency benchmark.

Irreversible process. Any process that cannot be reversed without leaving a permanent change in the surroundings. Causes include: finite-temperature-difference heat flow, friction and viscosity, free (unrestrained) expansion, mixing of gases, and any sudden (non-quasi-static) change. Every real process is irreversible.

Quantity / concept reference: Concept Statement form One-line essence Kelvin–Planck engine no $\eta = 1$; can't fully convert heat to work Clausius refrigerator no spontaneous heat cold→hot Reversible idealisation quasi-static + no dissipation; retraceable Irreversible reality generates entropy; leaves a permanent mark

Section 3: Key Procedures & Logical Framework

3.1 Why the two statements are equivalent (the JEE-Advanced classic) The cleanest way to prove “Statement A $\Rightarrow$ Statement B” is to show that violating one lets you violate the other. We show the contrapositive both ways. (i) A Clausius-violator builds a Kelvin–Planck-violator. Suppose a magic device C moves heat Q_2 from the cold reservoir (T_2) to the hot reservoir (T_1) with no work input (violating Clausius). Now run an ordinary engine E between the same reservoirs: it draws Q_1 from hot, does work $W = Q_1 - Q_2$, and rejects exactly Q_2 to cold. Combine C and E into one composite device. The cold reservoir's net heat change is zero ($-Q_2$ from E's rejection, $+Q_2$ taken back by C). The sole net result of the composite is: heat $(Q_1 - Q_2)$ drawn from the hot reservoir alone, converted entirely into work W , with no other effect — a Kelvin–Planck violation. 35 (see next page)

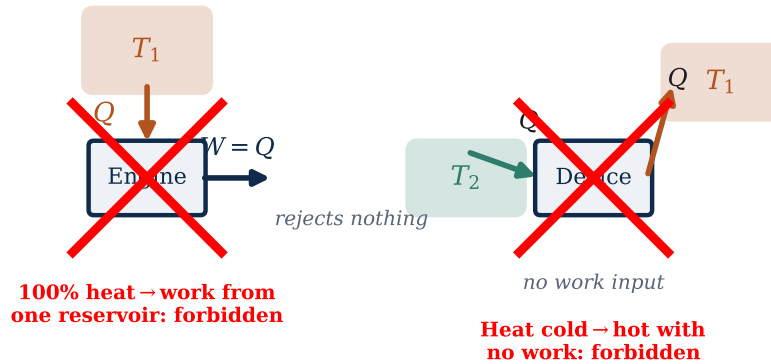


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(ii) **A Kelvin–Planck-violator builds a Clausius-violator.** Suppose a magic engine K takes heat Q from the hot reservoir and converts it *entirely* to work $W = Q$, rejecting nothing (violating Kelvin–Planck). Feed that work into an ordinary refrigerator running between the same two reservoirs: it uses W to pump heat Q_2 from cold to hot, dumping $Q_2 + W$ into the hot reservoir. The hot reservoir gives Q to K and receives $Q_2 + W = Q_2 + Q$, a net *gain* of Q_2 . The composite’s sole result: heat Q_2 moved from cold to hot with no work input — a Clausius violation. ■

3.2 Procedure: decide if a claimed device is allowed

1. List every reservoir touched and the heat exchanged with each (sign it: + absorbed, – rejected).
2. Identify the *sole net effect* after one full cycle ($\Delta U = 0$).
3. If the sole effect is “heat from one reservoir $\rightarrow$ pure work,” it violates Kelvin–Planck. If it is “heat cold $\rightarrow$ hot with no work,” it violates Clausius. Either way, impossible.
4. If work is genuinely supplied (fridge) or heat is genuinely rejected to a colder body (engine), the device is allowed — check only that η or COP does not exceed the Carnot limit (Subtopic 07).

3.3 Procedure: classify a process as reversible or irreversible

Ask two questions. *Is it quasi-static* (slow chain of equilibrium states)? *Is it free of dissipation* (no friction, no finite-gap heat flow, no turbulence)? Only “yes” to *both* earns “reversible.” A single “no” — e.g. a gas expanding into vacuum (not quasi-static), or slow sliding with friction (dissipative) — makes it irreversible.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Problem. State the Kelvin–Planck and Clausius statements of the second law, and give one everyday process that illustrates the “natural direction” the law enforces.

Solution. *Kelvin–Planck:* no engine can, in a cycle, take heat from a single reservoir and convert it *entirely* into work — some heat must be rejected to a colder body. *Clausius:* heat cannot flow on its own from a colder to a hotter body; work must be supplied. *Everyday illustration:* a hot cup of tea always cools to room temperature, transferring heat to the cooler room; it never spontaneously draws heat from the cool room to grow hotter — exactly the Clausius direction.

Two statements correctly given; tea cooling to the room illustrates the forbidden reverse (Clausius).

Example 2 — [NEET Speed Trap]

Problem. Which of the following processes is reversible? (A) Free expansion of a gas into vacuum (B) Heat flow across a large temperature difference (C) A frictionless quasi-static isothermal expansion in contact with a reservoir at the gas’s own temperature (D) A block sliding to rest by friction.

Speed solution. Reversible needs *quasi-static AND dissipation-free*. (A) is sudden and uncontrolled — not quasi-static. (B) finite temperature gap — dissipative heat flow. (D) friction — dissipative. Only (C) satisfies both.

(C). Free expansion, finite-gap heat flow, and friction are all irreversibility sources.

The trap: students equate “slow” with “reversible.” Slowness (quasi-static) is necessary but *not sufficient* — a slow process with friction is still irreversible.

Example 3 — [JEE Main Level]

Problem. An inventor claims a cyclic engine that, in each cycle, absorbs 500 J from a reservoir at 400 K and produces 500 J of work, rejecting no heat. Does this violate the first law, the second law, both, or neither?

Solution. First-law check: energy in = 500 J, energy out (work) = 500 J, $\Delta U_{\text{cycle}} = 0$. Energy balances, so the *first* law is satisfied. Second-law check: the sole result is complete conversion of heat from a *single* reservoir into work with nothing rejected — precisely the Kelvin–Planck prohibition.

Violates the second law only (Kelvin–Planck); the first law is satisfied.

This is the canonical reason a second law was needed at all: energy conservation alone does not forbid the device.

Example 4 — [JEE Advanced Level]

Problem. Show that if a hypothetical device could transfer 100 J from a cold reservoir at T_2 to a hot reservoir at T_1 with no work input, one could combine it with an ordinary engine (drawing 250 J from hot, rejecting 100 J to cold) to manufacture a Kelvin–Planck violator. Identify the net effect.

Solution. Let the ordinary engine E draw $Q_1 = 250$ J from the hot reservoir, do work $W = Q_1 - Q_2 = 250 - 100 = 150$ J, and reject $Q_2 = 100$ J to the cold reservoir. The magic Clausius-violating device C then carries that same 100 J back from cold to hot, free of charge.

Net effect on the cold reservoir: -100 (received from E) $+100$ (taken by C) $= 0$. The cold reservoir is untouched on balance. *Net effect overall:* the hot reservoir loses $250 - 100 = 150$ J, all of which appears as work $W = 150$ J, with no other change.

Composite sole effect: 150 J drawn from the hot reservoir alone, fully converted to work — a Kelvin–Planck violation. Hence “Clausius holds $\Rightarrow$ Kelvin–Planck holds.”

This concrete-number version of the equivalence proof is a favourite JEE Advanced reasoning question.

Section 5: Practice Exercises

1. [CBSE] Define a reversible process and list any two sources of irreversibility in real processes.
2. [NEET] A claim states heat flowed from a 280 K body to a 310 K body with no external work. Which statement of the second law does this violate?
3. [JEE Main] An engine absorbs 1200 J and produces 1200 J of work per cycle from a single reservoir. Which law(s) does it violate?
4. [JEE Main] Classify each as reversible or irreversible: (a) very slow isothermal compression with a frictionless piston in contact with a reservoir at the same temperature; (b) sudden bursting of a balloon; (c) mixing of hot and cold water.
5. [JEE Advanced] Using the equivalence argument, explain in 3–4 lines why a refrigerator that needs *zero* work to operate would also make a 100%-efficient single-reservoir engine possible.

Answers: 1. Reversible = quasi-static *and* dissipation-free, so reversing it restores system + surroundings with no net change. Irreversibility sources (any two): friction/viscosity, finite-temperature-gap heat flow, free expansion, mixing, sudden non-quasi-static change. 2. Clausius statement (spontaneous heat cold $\rightarrow$ hot). 3. Second law only (Kelvin–Planck); the first law is satisfied since energy balances. 4. (a) reversible; (b) irreversible (sudden, non-quasi-static); (c) irreversible (finite-gap heat flow / mixing). 5. A zero-work refrigerator is a Clausius-violator; couple it with an ordinary engine so its pumped-back heat exactly cancels the engine’s rejected heat at the cold reservoir; the composite then draws heat from the hot reservoir alone and turns it fully into work — a Kelvin–Planck violator. Thus breaking Clausius breaks Kelvin–Planck.

Section 6: Crack the MCQ

MCQ 1.

The Kelvin–Planck statement of the second law forbids: (A) heat flowing hot→cold (B) an engine converting heat from one reservoir entirely into work (C) work converting fully into heat (D) any heat engine at all

Correct: (B). The prohibition is specifically against complete heat→work conversion from a single reservoir.

- (A) is the *allowed* natural direction. (C) work→heat is freely allowed (friction does it constantly). (D) engines are allowed — just not with $\eta = 1$.

MCQ 2.

Which is *not* a source of irreversibility? (A) friction (B) finite-temperature-difference heat flow (C) a quasi-static, frictionless process (D) free expansion into vacuum

Correct: (C). Quasi-static and frictionless is precisely the reversible ideal.

- (A),(B),(D) are all classic dissipative / non-quasi-static sources of irreversibility.

MCQ 3.

“Heat cannot flow from a colder to a hotter body without external work” is: (A) the first law (B) the zeroth law (C) the Clausius statement (D) Mayer’s relation

Correct: (C). This is the refrigerator form of the second law.

- (A) is energy conservation; (B) defines temperature; (D) relates C_P and C_V . All unrelated to direction.

MCQ 4.

A process is reversible only if it is: (A) fast (B) quasi-static and free of dissipation (C) adiabatic (D) isothermal

Correct: (B). Both conditions are required together.

- (A) the opposite — reversibility demands slowness. (C),(D) particular process *types*, neither necessary nor sufficient for reversibility.

Section 7: Common Student Pitfalls & Pro-Tips

1. **“Slow = reversible.”** Quasi-static is necessary but not sufficient; dissipation (friction, finite-gap heat flow) makes even a slow process irreversible.
2. **Thinking the second law bans heat cold→hot outright.** It bans it only as a *sole* effect; with work input (a fridge) it is perfectly allowed.
3. **Confusing the two statements.** Kelvin–Planck is the *engine* (heat→work) prohibition; Clausius is the *refrigerator* (heat cold→hot) prohibition.
4. **Believing a Kelvin–Planck violator breaks the first law.** It does not — energy is conserved. The whole point is that the second law forbids things the first law permits.

Pro-Tip (speed & accuracy)

In any “is this device possible?” question, reduce it to its *sole net effect over one cycle*. “Heat from one reservoir → pure work” = Kelvin–Planck violation. “Heat cold→hot, no work” = Clausius violation. If neither, it is allowed — then (and only then) check it against the Carnot ceiling. And remember the direction of the equivalence proof: assume one violation, bolt on an ordinary engine or fridge, cancel the cold-reservoir heat, read off the other violation.

Section 8: Cheat Sheet Box

SECOND LAW & REVERSIBILITY — RAPID RECAP

- The second law is about *direction*: it forbids energy-conserving processes that never occur in Nature.
- **Kelvin–Planck**: no cyclic engine converts heat from a single reservoir *entirely* into work ($\eta < 1$ always).
- **Clausius**: heat does not flow spontaneously cold→hot; work is required.
- The two statements are *equivalent* (violating one constructs a violation of the other).
- **Reversible** = quasi-static *and* dissipation-free; retraceable with no net change. An ideal limit (the efficiency ceiling).
- **Irreversible** = every real process; sources: friction/viscosity, finite-gap heat flow, free expansion, mixing, sudden change.
- The first law permits; the second law selects which permitted processes actually happen.

Memory aids: “First law: you can’t win. Second law: you can’t even break even.” • “Kelvin = engine (no $\eta = 1$); Clausius = fridge (no free cold→hot).” • “Reversible needs slow *and* clean — slow alone isn’t enough.”

Subtopic 07 — The Carnot Engine & Carnot's Theorem

Exam Relevance: This is the crown of the chapter — the engine that sets the absolute speed limit on every heat engine ever built. CBSE Boards ask for the four steps of the Carnot cycle, the efficiency formula $\eta = 1 - T_2/T_1$, and a clean numerical. NEET loves one-line “find the efficiency” or “find T_2 ” traps and the fact that η depends only on the two temperatures. JEE Main asks multi-part numericals (heat rejected, work, raising efficiency). JEE Advanced derives $Q_2/Q_1 = T_2/T_1$ from the cycle and probes Carnot's theorem (no engine beats Carnot; all reversible engines tie). Get $\eta = 1 - T_2/T_1$ and the derivation airtight.

Section 1: Concept Intuition

The second law told us efficiency must be less than 1 — but how *close* to 1 can we get? In 1824 a young French engineer, Sadi Carnot, asked the deepest possible version of this question: of *all* conceivable engines working between a hot reservoir at T_1 and a cold one at T_2 , which is the best, and what limits it? His answer is one of the most beautiful results in physics: the best engine is a fully *reversible* one, its efficiency depends *only* on the two temperatures (not on the gas, the size, or the design), and that efficiency is $\eta = 1 - T_2/T_1$.

The **Carnot engine** is the idealised, perfectly reversible engine that achieves this ceiling. Its cycle is built from exactly the two reversible ways to move a gas that we already mastered in Subtopic 03: *isothermal* steps (in contact with a reservoir, temperature pinned) and *adiabatic* steps (perfectly insulated, no heat crosses). Stitch two of each together into a closed loop and you have the Carnot cycle.

Think about it this way...

Why insist on reversibility for the *best* engine? Because every irreversibility — friction, a finite temperature gap during heat transfer, turbulence — is wasted opportunity, energy degraded into useless heat that you could otherwise have captured as work. A reversible engine wastes nothing; it operates at the exact boundary the second law allows. It is the “frictionless, lossless” ideal in the same spirit as a frictionless pulley in mechanics — never quite achievable, but the perfect yardstick against which every real engine is measured. A real power plant quoting “42% efficient” is really saying “we reached 42% of the way to the Carnot ceiling set by our boiler and cooling-tower temperatures.”

The four strokes.

The Carnot cycle runs through four reversible steps, returning the gas to its start:

1. **Isothermal expansion** at T_1 : the gas, touching the hot reservoir, expands slowly and absorbs heat Q_1 . (Since $\Delta U = 0$ here, $Q_1 = W_{\text{by gas}}$.)
2. **Adiabatic expansion** ($T_1 \rightarrow T_2$): the gas is insulated and expands further, doing work at the cost of its internal energy, cooling from T_1 to T_2 .
3. **Isothermal compression** at T_2 : touching the cold reservoir, the gas is compressed slowly, rejecting heat Q_2 .
4. **Adiabatic compression** ($T_2 \rightarrow T_1$): insulated again, the gas is compressed back to its starting state, warming from T_2 to T_1 .

The net work per cycle is the area enclosed by this loop on the P - V diagram, and $W = Q_1 - Q_2$ as always.

Limiting Conditions & Assumptions.

(1) Every step is reversible (quasi-static, dissipation-free) — this is what makes Carnot the *maximum*; any real engine falls short. (2) Heat is exchanged *only* during the two isothermal steps (the adiabatic steps have $Q = 0$). (3) The result $\eta = 1 - T_2/T_1$ uses T in *kelvin* and assumes only two reservoirs. (4) Although derived here with an ideal gas as the working substance, Carnot's

theorem guarantees the efficiency is *independent* of the working substance — the ideal gas is just a convenient vehicle for the derivation.

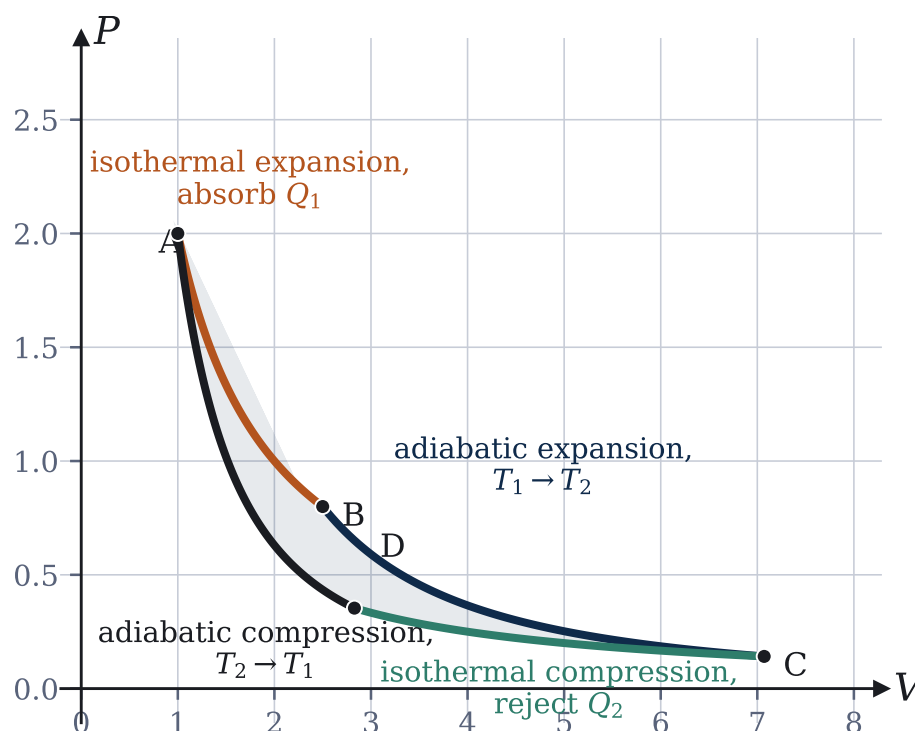
Figure 11.10 A P-V diagram with a closed four-sided loop. Top-left start point A; A → B a gentle isotherm at T₁ (label “isothermal expansion, absorb Q₁”); B → C a steeper adiabat falling to the lower isotherm (“adiabatic expansion, T₁ → T₂”); C → D the lower isotherm at T₂ moving left (“isothermal compression, reject Q₂”); D → A a steep adiabat rising back to start (“adiabatic compression, T₂ → T₁”). Shade the enclosed area; caption: “Enclosed area = W = Q₁ – Q₂; loop is clockwise (engine).”

Section 2: Key Formulas & Definitions Carnot cycle: two reversible isothermals (at T₁ and T₂) joined by two reversible adiabatics. Heat exchanged (only on the isothermals, for n moles of ideal gas): Q₁ = nRT₁ ln(V_B/V_A) (absorbed at T₁), Q₂ = nRT₂ ln(V_C/V_D) (rejected at T₂). The key ratio (derived in Section 3 using the adiabatic relations, which force V_B/V_A = V_C/V_D): Q₂/Q₁ = T₂/T₁ (Carnot, T in kelvin). Carnot efficiency: $\eta_{\text{Carnot}} = 1 - Q_2/Q_1 = 1 - T_2/T_1 = (T_1 - T_2)/T_1$. Carnot refrigerator / heat pump (same cycle reversed): COP_{ref} = T₂/T₁ – T₂, COP_{hp} = T₁/T₁ – T₂. Carnot's theorem. (a) No engine operating between two given reservoirs can be more efficient than a reversible (Carnot) engine working between the same two. (b) All reversible engines operating between the same two reservoirs have the same efficiency, regardless of working substance. Consequently η depends only on T₁ and T₂.

Quantity reference: Quantity Symbol / formula Notes Hot-reservoir temperature T₁ (K) source; higher = better Cold-reservoir temperature T₂ (K) sink; lower = better Carnot efficiency $\eta = 1 - T_2/T_1$ depends only on T₁, T₂ Carnot heat ratio Q₂/Q₁ = T₂/T₁ defines thermodynamic temperature

Section 3: Key Derivations & Logical Framework

3.1 Deriving Q₂/Q₁ = T₂/T₁ (the board/JEE derivation) Work the four steps with an ideal gas; label the corners A → B → C → D → A as in the diagram. Step 1 — heat on the two isothermals. On A → B (isothermal at T₁), ΔU = 0, so the heat absorbed equals the work done: Q₁ = nRT₁ ln(V_B/V_A). 41 (see next page)



Enclosed area = $W = Q_1 - Q_2$; loop is clockwise (engine)

Figure 11.10 A P - V diagram with a closed four-sided loop. Top-left start point A ; $A \rightarrow B$ a gentle isotherm at T_1 (label “isothermal expansion, absorb Q_1 ”); $B \rightarrow C$ a steeper adiabat falling to the lower isotherm (“adiabatic expansion, $T_1 \rightarrow T_2$ ”); $C \rightarrow D$ the lower isotherm at T_2 moving left (“isothermal compression, reject Q_2 ”); $D \rightarrow A$ a steep adiabat rising back to start (“adiabatic compression, $T_2 \rightarrow T_1$ ”). Shade the enclosed area; caption: “Enclosed area = $W = Q_1 - Q_2$; loop is clockwise (engine).”

Section 2: Key Formulas & Definitions Carnot cycle: two reversible isotherms (at T_1 and T_2) joined by two reversible adiabatics. Heat exchanged (only on the isotherms, for n moles of ideal gas): $Q_1 = nRT_1 \ln V_B/V_A$ (absorbed at T_1), $Q_2 = nRT_2 \ln V_C/V_D$ (rejected at T_2). The key ratio (derived in Section 3 using the adiabatic relations, which force $V_B/V_A = V_C/V_D$): $Q_2/Q_1 = T_2/T_1$ (Carnot, T in kelvin). Carnot efficiency: $\eta_{\text{Carnot}} = 1 - Q_2/Q_1 = 1 - T_2/T_1 = (T_1 - T_2)/T_1$. Carnot refrigerator / heat pump (same cycle reversed): $\text{COP}_{\text{ref}} = T_2/(T_1 - T_2)$, $\text{COP}_{\text{hp}} = T_1/(T_1 - T_2)$. Carnot's theorem. (a) No engine operating between two given reservoirs can be more efficient than a reversible (Carnot) engine working between the same two. (b) All reversible engines operating between the same two reservoirs have the same efficiency, regardless of working substance. Consequently η depends only on T_1 and T_2 . Quantity reference: Quantity Symbol / formula Notes Hot-reservoir temperature T_1 (K) source; higher = better Cold-reservoir temperature T_2 (K) sink; lower = better Carnot efficiency $\eta = 1 - T_2/T_1$ depends only on T_1 , T_2 Carnot heat ratio $Q_2/Q_1 = T_2/T_1$ defines thermodynamic temperature

Section 3: Key Derivations & Logical Framework 3.1 Deriving $Q_2/Q_1 = T_2/T_1$ (the board/JEE derivation) Work the four steps with an ideal gas; label the corners $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ as in the diagram. Step 1 — heat on the two isotherms. On $A \rightarrow B$ (isothermal at T_1), $\Delta U = 0$, so the heat absorbed equals the work done: $Q_1 = nRT_1 \ln V_B/V_A$. 41

On $C \rightarrow D$ (isothermal at T_2), the gas is compressed, rejecting heat (magnitude):

$$Q_2 = nRT_2 \ln \frac{V_C}{V_D}.$$

Step 2 — apply the adiabatic relation to the two insulated steps. Using $TV^{\gamma-1} = \text{const}$:

$$B \rightarrow C: \quad T_1 V_B^{\gamma-1} = T_2 V_C^{\gamma-1}, \quad D \rightarrow A: \quad T_2 V_D^{\gamma-1} = T_1 V_A^{\gamma-1}.$$

Step 3 — divide to eliminate temperatures. Dividing the first adiabatic relation by the second:

$$\frac{T_1 V_B^{\gamma-1}}{T_2 V_D^{\gamma-1}} = \frac{T_2 V_C^{\gamma-1}}{T_1 V_A^{\gamma-1}} \implies \left(\frac{V_B}{V_D} \right)^{\gamma-1} = \left(\frac{V_C}{V_A} \right)^{\gamma-1}.$$

Rearranging gives the crucial volume identity:

$$\frac{V_B}{V_A} = \frac{V_C}{V_D}.$$

Step 4 — take the ratio of the two heats. The two logarithms are therefore *equal*, and they cancel:

$$\frac{Q_2}{Q_1} = \frac{nRT_2 \ln(V_C/V_D)}{nRT_1 \ln(V_B/V_A)} = \frac{T_2}{T_1}.$$

Step 5 — efficiency. Substitute into $\eta = 1 - Q_2/Q_1$:

$$\eta_{\text{Carnot}} = 1 - \frac{T_2}{T_1}.$$

The working substance has vanished from the answer — a preview of Carnot’s theorem.

3.2 Carnot’s theorem and why η is substance-independent

1. *Claim (a):* no engine X between T_1, T_2 can beat a reversible engine R between the same reservoirs. *Proof sketch:* if $\eta_X > \eta_R$, run R backwards (as a refrigerator) driven by X ’s work. Because X is more efficient, it needs less heat from the hot reservoir than R pumps back into it — the net effect is heat delivered cold $\rightarrow$ hot with no external work, a Clausius violation. Hence $\eta_X \leq \eta_R$.
2. *Claim (b):* two reversible engines R_1, R_2 must tie. Apply the argument of (a) once with R_1 driving R_2 -reversed, then again the other way. Each gives $\eta_{R_1} \leq \eta_{R_2}$ and $\eta_{R_2} \leq \eta_{R_1}$, so $\eta_{R_1} = \eta_{R_2}$.
3. *Consequence:* since all reversible engines tie regardless of working substance, η can depend *only* on the two reservoir temperatures — which is exactly what $\eta = 1 - T_2/T_1$ says. This substance-independence is also what lets us *define* an absolute (thermodynamic) temperature scale via $Q_2/Q_1 = T_2/T_1$.

3.3 Procedure: any Carnot numerical

1. Convert both temperatures to *kelvin*.
2. $\eta = 1 - T_2/T_1$ gives efficiency directly.
3. Use $\eta = W/Q_1$ and $Q_2 = Q_1 - W$ (or $Q_2/Q_1 = T_2/T_1$) for heats and work.
4. To raise η : compare “raise T_1 ” vs. “lower T_2 ” by the same amount — lowering T_2 helps more (it sits in the numerator of T_2/T_1 and shrinks the rejected fraction faster).

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Problem. A Carnot engine operates between a hot reservoir at 500 K and a cold reservoir at 300 K. Find its efficiency.

Solution. $\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{500} = 1 - 0.60 = 0.40.$

$$\eta = 0.40 = 40\%$$

No detail of the gas or engine size is needed — the two temperatures alone fix the ceiling.

Example 2 — [NEET Speed Trap]

Problem. A Carnot engine has efficiency 50% with the cold reservoir at 300 K. To raise the efficiency to 60% keeping the cold reservoir fixed, by how much must the hot-reservoir temperature increase?

Speed solution. From $\eta = 1 - T_2/T_1$, i.e. $T_1 = T_2/(1 - \eta)$. At 50%: $T_1 = 300/0.50 = 600$ K. At 60%: $T_1' = 300/0.40 = 750$ K. Increase = $750 - 600 = 150$ K.

Raise T_1 from 600 K to 750 K, an increase of 150 K.

The trap: students try to scale T_1 linearly with η (e.g. “+10% efficiency $\Rightarrow$ +10% temperature”). Always invert the formula cleanly via $T_1 = T_2/(1 - \eta)$.

Example 3 — [JEE Main Level]

Problem. A Carnot engine working between 400 K and 300 K absorbs 1200 J from the hot reservoir per cycle. Find (a) the efficiency, (b) the work done per cycle, and (c) the heat rejected.

Solution. (a) $\eta = 1 - 300/400 = 0.25 = 25\%$. (b) $W = \eta Q_1 = 0.25 \times 1200 = 300$ J. (c) $Q_2 = Q_1 - W = 1200 - 300 = 900$ J (check: $Q_2/Q_1 = 900/1200 = 0.75 = T_2/T_1$ ✓).

$$\eta = 25\%, \quad W = 300 \text{ J}, \quad Q_2 = 900 \text{ J}$$

Example 4 — [JEE Advanced Level]

Problem. A reversible engine operates between $T_1 = 600$ K and $T_2 = 300$ K. An inventor claims a *different* engine between the same reservoirs with efficiency 60%. (a) Find the Carnot efficiency. (b) Is the claim possible? (c) If the inventor instead claims 45%, is *that* possible, and if so is the engine necessarily reversible?

Solution. (a) $\eta_{\text{Carnot}} = 1 - 300/600 = 0.50 = 50\%$. (b) The claimed 60% > 50% exceeds the Carnot ceiling — by Carnot’s theorem *no* engine between these reservoirs can beat 50%, so the claim is *impossible* (it would enable a Clausius violation). (c) 45% < 50% is below the ceiling, so it is *possible* — but being below the maximum means it must be *irreversible*; only a reversible engine reaches the full 50%.

(a) 50%. (b) Impossible — exceeds the Carnot ceiling. (c) Possible, and necessarily irreversible (reversible engines achieve exactly 50%).

This three-part question is the essence of Carnot’s theorem: the reversible efficiency is simultaneously the *maximum* and the value all reversible engines share.

Section 5: Practice Exercises

- [CBSE] A Carnot engine works between 727°C and 27°C. Find its efficiency. (Convert to kelvin first!)
- [NEET] A Carnot engine has efficiency 40% when the sink is at 300 K. Find the source temperature.

3. **[JEE Main]** A Carnot engine between 500 K and 400 K does 250 J of work per cycle. Find the heat absorbed and the heat rejected.
4. **[JEE Main]** For a Carnot engine with $T_1 = 400$ K, $T_2 = 300$ K: would lowering T_2 by 50 K or raising T_1 by 50 K give the higher efficiency? Show both.
5. **[JEE Advanced]** A Carnot engine's cycle has isothermal expansion volume ratio $V_B/V_A = 2$ at $T_1 = 500$ K, with $n = 1$ mol. Find Q_1 , and using $\eta = 1 - T_2/T_1$ with $T_2 = 300$ K, find W and Q_2 . ($\ln 2 = 0.693$, $R = 8.314$.)

Answers: 1. $T_1 = 1000$ K, $T_2 = 300$ K; $\eta = 1 - 300/1000 = 70\%$. 2. $T_1 = T_2/(1 - \eta) = 300/0.60 = 500$ K. 3. $\eta = 1 - 400/500 = 0.20$; $Q_1 = W/\eta = 250/0.20 = 1250$ J; $Q_2 = Q_1 - W = 1000$ J. 4. Lower T_2 : $\eta = 1 - 250/400 = 37.5\%$. Raise T_1 : $\eta = 1 - 300/450 = 33.3\%$. Lowering the sink wins. 5. $Q_1 = nRT_1 \ln(V_B/V_A) = (1)(8.314)(500)(0.693) \approx 2881$ J; $\eta = 1 - 300/500 = 0.40$; $W = \eta Q_1 \approx 1152$ J; $Q_2 = Q_1 - W \approx 1729$ J (check $Q_2/Q_1 \approx 0.60 = T_2/T_1$ ✓).

Section 6: Crack the MCQ

MCQ 1.

The efficiency of a Carnot engine depends on: (A) the working substance (B) the two reservoir temperatures only (C) the size of the engine (D) the amount of gas

Correct: (B). $\eta = 1 - T_2/T_1$; by Carnot's theorem the working substance, size, and amount all cancel out.

- (A),(C),(D) all drop out of the derivation — the volume logarithms cancel, leaving only T_2/T_1 .

MCQ 2.

A Carnot engine is most efficient when: (A) T_1 and T_2 are close (B) T_2 is as low and T_1 as high as possible (C) $T_2 = T_1$ (D) $T_2 > T_1$

Correct: (B). A large temperature gap (low sink, high source) maximises $1 - T_2/T_1$.

- (A),(C) give $\eta \rightarrow 0$ (no gap, no work). (D) is impossible for an engine (would reverse the heat flow).

MCQ 3.

In a Carnot cycle, heat is exchanged with the reservoirs during: (A) the two adiabatic steps (B) the two isothermal steps (C) all four steps (D) none of the steps

Correct: (B). Adiabatic steps have $Q = 0$ by definition; heat enters/leaves only on the isothermals.

- (A) reverses it; (C) ignores the adiabatic condition; (D) would mean no engine at all.

MCQ 4.

No engine between two reservoirs can exceed the Carnot efficiency because doing so would violate: (A) the first law (B) the zeroth law (C) the second law (D) Mayer's relation

Correct: (C). A super-Carnot engine could be coupled to a reversed Carnot engine to move heat cold→hot with no work — a Clausius (second-law) violation.

- (A) energy still balances; (B),(D) unrelated to engine efficiency limits.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Forgetting to convert to kelvin.** $\eta = 1 - T_2/T_1$ is valid *only* in kelvin. Plugging in Celsius gives nonsense (and sometimes $\eta > 1$).
2. **Thinking heat is exchanged on all four steps.** Only the two isothermals exchange heat; the adiabatics are insulated ($Q = 0$).

3. **Believing a real engine can reach Carnot efficiency.** Carnot is the reversible ceiling; every real (irreversible) engine falls strictly below it.
4. **Confusing “raise T_1 ” with “lower T_2 .”** For equal temperature changes, lowering the sink T_2 raises efficiency more than raising the source T_1 .

Pro-Tip (speed & accuracy)

Memorise the engine in one line: “two isotherms, two adiabats; heat only on the isotherms; $\eta = 1 - T_2/T_1$ in kelvin.” For any numerical, get η from the temperatures first, then use $W = \eta Q_1$ and $Q_2 = Q_1 - W$. The volume-ratio identity $V_B/V_A = V_C/V_D$ is the single algebraic fact that collapses the derivation — it is what makes the two logarithms cancel. And whenever an inventor’s claimed η exceeds $1 - T_2/T_1$, declare it impossible on the spot (Carnot’s theorem); no calculation needed.

Section 8: Cheat Sheet Box

CARNOT ENGINE & CARNOT’S THEOREM — RAPID RECAP

- **Cycle:** reversible — isothermal expansion (absorb Q_1 at T_1) $\rightarrow$ adiabatic expansion ($T_1 \rightarrow T_2$) $\rightarrow$ isothermal compression (reject Q_2 at T_2) $\rightarrow$ adiabatic compression ($T_2 \rightarrow T_1$).
- **Key ratio:** $Q_2/Q_1 = T_2/T_1$ (kelvin), from $V_B/V_A = V_C/V_D$.
- **Efficiency:** $\eta_{\text{Carnot}} = 1 - T_2/T_1 = (T_1 - T_2)/T_1$ — the maximum for those reservoirs.
- **Reversed** (fridge/pump): $\text{COP}_{\text{ref}} = T_2/(T_1 - T_2)$, $\text{COP}_{\text{hp}} = T_1/(T_1 - T_2)$.
- **Carnot’s theorem:** (a) no engine beats a reversible one between the same reservoirs; (b) all reversible engines tie, independent of working substance.
- Heat exchanged only on the isothermals; adiabatics have $Q = 0$.
- Lower the sink T_2 to gain efficiency faster than raising the source T_1 .

Memory aids: “Iso–adia–iso–adia, heat only on the isos.” • “ $\eta = 1 - T_{\text{cold}}/T_{\text{hot}}$, always in kelvin.” • “Carnot is the ceiling: reach it only in the reversible dream, never in the real lab.”

Subtopic 08 — Entropy and the Quantified Second Law (JEE Advanced Extension)

Scope note. Entropy is *not* part of the CBSE Class 11 NCERT Thermodynamics chapter — it is formally introduced in physical chemistry and in higher physics. It is included here as a JEE-Advanced-level extension so this companion is genuinely complete. CBSE-Boards-only students may treat this subtopic as enrichment; JEE Advanced aspirants should master it, as it converts the qualitative “arrow of time” of Subtopic 06 into a calculable state function.

Exam Relevance: JEE Advanced occasionally asks for the entropy change in a standard process (isothermal, phase change, heating at constant pressure), the sign of $\Delta S_{\text{universe}}$ for an irreversible process, or a conceptual question on the Clausius inequality. NEET and CBSE Boards do not test this directly. The single most valuable takeaways: $\Delta S = \int dQ_{\text{rev}}/T$ defines a *state function*, and $\Delta S_{\text{universe}} \geq 0$ for every real process.

Section 1: Concept Intuition

Subtopic 06 gave us the second law in words: real processes have a preferred direction. But “direction” begged for a number — a single quantity that always increases as a system drifts toward its most probable state, the way a clock’s hands always advance. That quantity is **entropy**, S . It is the bookkeeping device that turns the qualitative arrow of time into an inequality you can compute: $\Delta S_{\text{universe}} \geq 0$.

The cleanest physical reading of entropy is “the amount of disorder,” or more precisely the number of microscopic arrangements (microstates) consistent with the bulk state. Heat Q flowing into a system at temperature T raises its disorder by Q/T — and the colder the system, the bigger the disordering effect of the *same* heat (a whisper in a silent library is disruptive; the same whisper at a loud party is nothing). That single ratio, dQ/T , is the heartbeat of entropy.

Think about it this way...

Recall the shuffled-deck image from Subtopic 06. Entropy is the precise scorecard for “how shuffled.” A perfectly ordered deck has low entropy; a thoroughly mixed one has high entropy. Every real process shuffles the universe a little more, so the total never decreases. You *can* lower a *system’s* entropy (a freezer makes ordered ice from disordered water) — but only by dumping even more disorder into the surroundings (the freezer’s coils warm the kitchen), so the *universe’s* entropy still rises. That is the second law, now wearing a number.

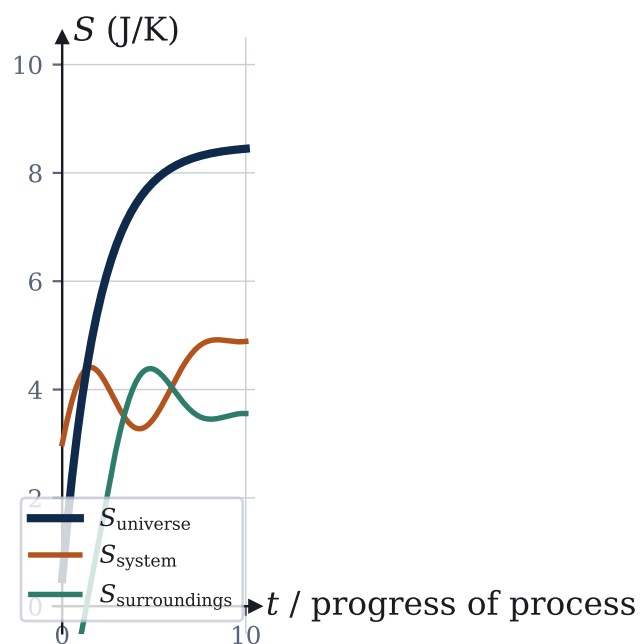
Why “reversible heat” in the definition?

Entropy is a *state function* — it depends only on the state, not the path. But heat Q is a *path* function. The trick that rescues a state function out of a path quantity is to insist on the *reversible* path: $dS = dQ_{\text{rev}}/T$. Along any reversible route between two states the integral $\int dQ_{\text{rev}}/T$ comes out the same, so it legitimately defines ΔS . To find ΔS for an *irreversible* process, you cleverly compute it along *any* reversible path with the same endpoints — because S is a state function, the answer is identical.

Limiting Conditions & Assumptions.

(1) $dS = dQ_{\text{rev}}/T$ requires the *reversible* heat and the temperature *at which* it is exchanged (kelvin). (2) ΔS for the *system* can be positive, negative, or zero; it is $\Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$ that obeys ≥ 0 . (3) Equality ($\Delta S_{\text{universe}} = 0$) holds only for a reversible process; every irreversible (real) process gives > 0 .

Figure 11.11 A horizontal axis labelled “time / progress of a real process.” A curve for S_{universe} rising monotonically (never dipping). Below it, two component curves: 46 (see next page)



System entropy may fall, but surroundings rise more — universe total never decreases

Figure 11.11 A horizontal axis labelled “time / progress of a real process.” A curve for S_{universe} rising monotonically (never dipping). Below it, two component curves: 46

S_{system} (which may dip, e.g. during freezing) and $S_{\text{surroundings}}$ (rising more), with a caption: “System entropy may fall, but surroundings rise more — universe total never decreases.”

Section 2: Key Definitions & Properties

Entropy (definition). For a reversible exchange of heat dQ_{rev} at absolute temperature T :

$$\boxed{dS = \frac{dQ_{\text{rev}}}{T}}, \quad \Delta S = \int_i^f \frac{dQ_{\text{rev}}}{T}.$$

S is a *state function*: $\oint dS = 0$ over any cycle. SI unit: JK^{-1} .

The entropy form of the second law (entropy principle).

$$\boxed{\Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \geq 0},$$

with equality for a reversible process and strict inequality (> 0) for every irreversible one.

Clausius inequality (for any cyclic process; the integral uses the *actual* heat):

$$\oint \frac{dQ}{T} \leq 0, \quad (= 0 \text{ if reversible; } < 0 \text{ if irreversible}).$$

Standard entropy changes (ideal gas / standard processes):

Process	Entropy change ΔS
Isothermal (ideal gas, T const)	$\Delta S = \frac{Q_{\text{rev}}}{T} = nR \ln \frac{V_f}{V_i}$
Phase change (at constant T)	$\Delta S = \frac{mL}{T}$
Heating at constant pressure	$\Delta S = nC_P \ln \frac{T_f}{T_i}$
Heating at constant volume	$\Delta S = nC_V \ln \frac{T_f}{T_i}$
Reversible adiabatic (isentropic)	$\Delta S = 0$ ($dQ_{\text{rev}} = 0$)

General ideal-gas form (combining volume and temperature change):

$$\Delta S = nC_V \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i}.$$

Section 3: Key Procedures & Logical Framework

3.1 Why S is a state function (the logical core)

1. For a *reversible* cycle the Clausius relation gives $\oint dQ_{\text{rev}}/T = 0$.
2. A quantity whose integral around *any* closed loop vanishes is, by definition, the differential of a state function. So dQ_{rev}/T is an exact differential — call its integral S .
3. Therefore ΔS between two states is path-independent: compute it along whatever reversible path is convenient, even to evaluate an irreversible process with the same endpoints.

3.2 Procedure: entropy change of the universe for an irreversible process

1. Compute ΔS_{sys} along a *reversible* path with the system's actual endpoints (use the table above).
2. Compute $\Delta S_{\text{surr}} = -Q_{\text{actual}}/T_{\text{surr}}$, using the *actual* heat exchanged and the surroundings' temperature (the reservoir, treated as unchanging).

3. Add: $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$. For a genuinely irreversible process this comes out > 0 ; a result < 0 signals an arithmetic or sign error.

3.3 The Carnot cycle has zero entropy change (a consistency check)

Around a reversible Carnot cycle, entropy is gained on the hot isothermal ($+Q_1/T_1$) and lost on the cold isothermal ($-Q_2/T_2$); the adiabatics contribute nothing. The net is

$$\oint \frac{dQ_{\text{rev}}}{T} = \frac{Q_1}{T_1} - \frac{Q_2}{T_2} = 0 \quad \Longleftrightarrow \quad \frac{Q_2}{Q_1} = \frac{T_2}{T_1},$$

which is exactly the Carnot heat ratio from Subtopic 07 — entropy and Carnot are two views of the same truth.

Section 4: Worked Examples

Example 1 — [JEE Main / Advanced Level]

Problem. 2.0 mol of an ideal gas expands isothermally and reversibly at 400 K to twice its initial volume. Find the entropy change of the gas. ($R = 8.314$, $\ln 2 = 0.693$.)

Solution. Isothermal, so $\Delta S = nR \ln(V_f/V_i) = (2.0)(8.314)(\ln 2) = (16.628)(0.693) \approx 11.5 \text{ J K}^{-1}$.

$$\Delta S_{\text{gas}} \approx +11.5 \text{ J K}^{-1}$$

Positive, as expected: the gas spreads into a larger volume (more disorder). Since the process is reversible, $\Delta S_{\text{surr}} = -11.5 \text{ J K}^{-1}$ and $\Delta S_{\text{univ}} = 0$.

Example 2 — [JEE Advanced Level]

Problem. 0.50 kg of ice at 0°C melts completely to water at 0°C . The latent heat of fusion is $L = 3.34 \times 10^5 \text{ J kg}^{-1}$. Find the entropy change of the ice.

Solution. Phase change at constant $T = 273 \text{ K}$: $\Delta S = \frac{mL}{T} = \frac{(0.50)(3.34 \times 10^5)}{273} = \frac{1.67 \times 10^5}{273} \approx 612 \text{ J K}^{-1}$.

$$\Delta S \approx +612 \text{ J K}^{-1}$$

Melting injects heat at constant temperature, sharply raising disorder as the rigid lattice dissolves into a free-flowing liquid — a large positive ΔS .

Example 3 — [JEE Advanced Level]

Problem. A 2.0 kg block of metal at 400 K is dropped into a very large lake at 300 K and cools to 300 K. The metal's specific heat is $c = 400 \text{ J kg}^{-1}\text{K}^{-1}$. Find (a) ΔS_{block} , (b) ΔS_{lake} , and (c) $\Delta S_{\text{universe}}$, and confirm the sign. ($\ln(400/300) = \ln 1.333 = 0.2877$.)

Solution. (a) The block cools, and (computing along a reversible path) its entropy change is

$$\Delta S = mc \ln \frac{T_f}{T_i} = (2.0)(400) \ln \frac{300}{400} = 800 \times (-0.2877) \approx -230.2 \text{ J K}^{-1}$$

(negative: it loses heat and orders).

(b) The lake is a reservoir at constant 300 K; it *gains* the heat the block lost: $Q = mc(T_i - T_f) = (2.0)(400)(100) = 8.0 \times 10^4 \text{ J}$. So $\Delta S_{\text{lake}} = +Q/T = 8.0 \times 10^4/300 \approx +266.7 \text{ J K}^{-1}$.

(c) $\Delta S_{\text{univ}} = -230.2 + 266.7 \approx +36.5 \text{ J K}^{-1} > 0$.

$$\Delta S_{\text{block}} \approx -230 \text{ J K}^{-1}, \quad \Delta S_{\text{lake}} \approx +267 \text{ J K}^{-1}, \quad \Delta S_{\text{univ}} \approx +37 \text{ J K}^{-1} > 0$$

The block's entropy falls, but the lake's rises more — the irreversible cooling across a finite temperature gap creates net entropy, exactly as the second law demands.

Example 4 — [JEE Advanced Level]

Problem. Verify that a *reversible* Carnot engine produces zero net entropy change in the universe per cycle, given $Q_1 = 600$ J absorbed at $T_1 = 500$ K and Q_2 rejected at $T_2 = 300$ K.

Solution. For a Carnot engine, $Q_2 = Q_1(T_2/T_1) = 600 \times (300/500) = 360$ J. Entropy gained by the gas from the hot reservoir: $+Q_1/T_1 = 600/500 = 1.2$ J K⁻¹. Entropy lost to the cold reservoir: $-Q_2/T_2 = -360/300 = -1.2$ J K⁻¹. The adiabatic steps exchange no heat, contributing 0.

Net over the cycle: $1.2 - 1.2 = 0$.

$\Delta S_{\text{univ}} = 0$ per cycle — the hallmark of a fully reversible engine.

This is why the Carnot engine is the efficiency ceiling: it is the unique engine that creates *no* entropy. Any real engine creates some, and that created entropy is precisely the lost work.

Section 5: Practice Exercises

- [JEE Main]** 1.0 mol of an ideal gas expands isothermally and reversibly at 300 K to three times its volume. Find ΔS of the gas. ($\ln 3 = 1.0986$.)
- [JEE Advanced]** 1.0 kg of water at 100°C vaporises to steam at 100°C; $L_v = 2.26 \times 10^6$ J kg⁻¹. Find the entropy change of the water.
- [JEE Advanced]** A reversible Carnot engine absorbs 800 J at 400 K and rejects heat at 250 K. Find Q_2 and verify $\oint dQ/T = 0$.
- [JEE Advanced]** 3.0 mol of an ideal monoatomic gas ($C_V = \frac{3}{2}R$) is heated at constant volume from 300 K to 600 K. Find ΔS . ($\ln 2 = 0.693$.)
- [JEE Advanced]** A hot body at 500 K transfers 1000 J directly to a cold body at 400 K (both large enough that temperatures barely change). Find $\Delta S_{\text{universe}}$ and confirm it is positive.

Answers: 1. $\Delta S = nR \ln 3 = (1.0)(8.314)(1.0986) \approx +9.13$ J K⁻¹. 2. $\Delta S = mL_v/T = (1.0)(2.26 \times 10^6)/373 \approx +6060$ J K⁻¹. 3. $Q_2 = Q_1(T_2/T_1) = 800(250/400) = 500$ J; $\oint dQ/T = 800/400 - 500/250 = 2.0 - 2.0 = 0$ ✓. 4. $\Delta S = nC_V \ln(T_f/T_i) = (3.0)(\frac{3}{2} \times 8.314)(\ln 2) = (3.0)(12.471)(0.693) \approx +25.9$ J K⁻¹. 5. $\Delta S_{\text{hot}} = -1000/500 = -2.0$ J K⁻¹; $\Delta S_{\text{cold}} = +1000/400 = +2.5$ J K⁻¹; $\Delta S_{\text{univ}} = +0.5$ J K⁻¹ > 0 ✓.

Section 6: Crack the MCQ**MCQ 1.**

For any reversible process, the entropy change of the universe is: (A) positive (B) negative (C) zero (D) infinite

Correct: (C) zero. Reversibility is exactly the equality case of $\Delta S_{\text{univ}} \geq 0$.

- (A) holds for *irreversible* processes. (B) would violate the second law. (D) has no basis.

MCQ 2.

The entropy change of an ideal gas in a reversible *adiabatic* process is: (A) $nR \ln(V_f/V_i)$ (B) zero (C) Q/T (D) negative

Correct: (B) zero. Adiabatic means $dQ_{\text{rev}} = 0$, so $\Delta S = \int dQ_{\text{rev}}/T = 0$ — a reversible adiabatic is *isentropic*.

- (A) is the isothermal result; (C) is zero here since $Q = 0$; (D) has no justification.

MCQ 3.

Entropy is a: (A) path function like heat (B) state function like internal energy (C) form of energy (D) always-conserved quantity

Correct: (B). $\oint dS = 0$, so S depends only on the state, like U .

- (A) confuses S with Q (its definition uses the *reversible* heat precisely to make it path-independent). (C) entropy has units JK^{-1} , not joules. (D) entropy of the universe *increases*, it is not conserved.

MCQ 4.

The Clausius inequality $\oint dQ/T \leq 0$ becomes an equality for: (A) any spontaneous process (B) an irreversible cycle (C) a reversible cycle (D) an isothermal cycle only

Correct: (C) a reversible cycle. The strict inequality (< 0) marks irreversibility; equality marks reversibility.

- (A),(B) spontaneous/irreversible give < 0 . (D) the process *type* is irrelevant; reversibility is what matters.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Using actual heat for ΔS_{sys} of an irreversible process.** ΔS of the system must be computed along a *reversible* path with the same endpoints — not from the actual (irreversible) heat.
2. **Forgetting the surroundings.** The second law constrains $\Delta S_{\text{universe}}$, not ΔS_{system} alone. A system's entropy may legitimately fall (freezing, compression).
3. **Wrong temperature in Q/T .** Use the kelvin temperature *at which* the heat is exchanged — for a reservoir, its own (constant) temperature.
4. **Treating entropy as energy.** Entropy has units JK^{-1} , not J. It measures disorder, not stored energy.

Pro-Tip (speed & accuracy)

For irreversible problems, always split into “system” and “surroundings,” compute each ΔS separately, and add. For the system, route the calculation through a convenient reversible path (isothermal $\rightarrow nR \ln(V_f/V_i)$; heating $\rightarrow nC \ln(T_f/T_i)$; phase change $\rightarrow mL/T$). For a reservoir, $\Delta S = \pm Q_{\text{actual}}/T_{\text{res}}$. The universe total must come out ≥ 0 — a negative answer is a guaranteed sign error. And remember the elegant consistency check: a reversible Carnot cycle gives $\oint dQ/T = 0$, which *is* the relation $Q_2/Q_1 = T_2/T_1$.

Section 8: Cheat Sheet Box**ENTROPY & THE QUANTIFIED SECOND LAW — RAPID RECAP**

- **Definition:** $dS = dQ_{\text{rev}}/T$; S is a *state function* ($\oint dS = 0$), unit JK^{-1} .
- **Second law (entropy form):** $\Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \geq 0$ ($= 0$ reversible, > 0 irreversible).
- **Clausius inequality:** $\oint dQ/T \leq 0$ (equality $\Leftrightarrow$ reversible).
- **Standard ΔS :** isothermal $nR \ln(V_f/V_i)$; phase change mL/T ; heating $nC_P \ln(T_f/T_i)$ or $nC_V \ln(T_f/T_i)$; reversible adiabatic 0 (isentropic).
- **Irreversible process:** compute ΔS_{sys} via a reversible path; $\Delta S_{\text{surr}} = -Q_{\text{actual}}/T_{\text{surr}}$.
- **Carnot consistency:** $\oint dQ_{\text{rev}}/T = Q_1/T_1 - Q_2/T_2 = 0 \Leftrightarrow Q_2/Q_1 = T_2/T_1$.

Memory aids: “Entropy is the universe’s odometer — it only ever goes up.” • “System can tidy up; the universe always gets messier.” • “State function $\Rightarrow$ compute ΔS along *any* reversible path you like.”

Chapter 11: Thermodynamics — Round 2 Addendum

The chapter gives you the four basic processes, the two laws, the Carnot ceiling, and the entropy definition. Five JEE Advanced patterns sit just outside that perimeter: general polytropic processes where the exponent x is not $0, 1, \gamma, \infty$; composite cycles where you must disentangle heat input from heat output leg by leg; irreversible adiabatic expansions against constant external pressure (not quasi-static, not isentropic); entropy accounting for the universe when the process is genuinely irreversible; and compound Carnot systems where engines drive refrigerators or cascades share reservoirs. Each addendum below takes one of these patterns and reduces it to the case-splitting, sign-tracking, and state-function logic the chapter already teaches.

Addendum A — General Polytropic Processes: The Master Exponent x

Subtopic 03 introduces the polytropic relation $PV^x = \text{const}$ and the molar heat capacity $C = C_V + \frac{R}{1-x}$, but stops at naming the four special cases. JEE Advanced treats x as a continuous variable: you are given a process (e.g. “molar heat capacity is $2R$ ” or “pressure varies as $P \propto V^{-2}$ ”) and must find x , then the work, heat, entropy change, and whether the gas heats or cools during expansion. The engine is the first law in differential form with $P = KV^{-x}$.

Method

1. Identify x .

- If $P \propto V^{-n}$ is given, $x = n$.
- If molar heat capacity C is given, solve $C = C_V + \frac{R}{1-x}$ for x :

$$x = 1 - \frac{R}{C - C_V}.$$

- If a relation like $TV^\alpha = \text{const}$ is given, use $PV = nRT \Rightarrow TV^{x-1} = \text{const}$, so $\alpha = x - 1$.

2. Work done (quasi-static). For n moles from (P_i, V_i) to (P_f, V_f) :

$$W = \int_{V_i}^{V_f} P dV = \int_{V_i}^{V_f} \frac{K}{V^x} dV = \frac{P_i V_i - P_f V_f}{x - 1} = \frac{nR(T_i - T_f)}{x - 1}, \quad (x \neq 1).$$

For $x = 1$ (isothermal), $W = nRT \ln \frac{V_f}{V_i}$.

3. Internal energy change (ideal gas, any process).

$$\Delta U = nC_V \Delta T.$$

4. Heat exchanged (First Law).

$$Q = \Delta U + W = nC_V \Delta T + \frac{nR\Delta T}{1-x} = n\Delta T \left(C_V - \frac{R}{x-1} \right).$$

Using $C_P - C_V = R$ and $\gamma = C_P/C_V$, the bracket is exactly the polytropic molar heat capacity $C = C_V + \frac{R}{1-x}$. Thus $Q = nC\Delta T$ — the definition of C .

5. Temperature-Volume relation. From $PV^x = \text{const}$ and $PV = nRT$:

$$TV^{x-1} = \text{const} \Rightarrow \frac{T_f}{T_i} = \left(\frac{V_i}{V_f} \right)^{x-1}.$$

6. Entropy change (ideal gas).

$$\Delta S = nC_V \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i}.$$

Substitute T_f/T_i from step 5 to express ΔS in terms of volumes only:

$$\Delta S = n[C_V(1-x) + R] \ln \frac{V_f}{V_i} = n(C - C_V) \ln \frac{V_f}{V_i} \quad (\text{using } C - C_V = \frac{R}{1-x}).$$

Sign rule: During expansion ($V_f > V_i$), Q and ΔS have the sign of $(C - C_V)$; during compression, opposite.

7. Special values of x (revision table with C).

x	Process	C	ΔS sign in expansion
0	Isobaric	C_P	> 0 (heat in)
1	Isothermal	∞	> 0 (heat in)
γ	Adiabatic (reversible)	0	$= 0$
∞	Isochoric	C_V	> 0 (heat in)
$x > \gamma$	Steeper than adiabat	$0 < C < C_V$	< 0 (heat out , even though $0 < C < C_V$: the gas cools, and $C > 0$ means cooling releases heat)
$1 < x < \gamma$	Between isotherm & adiabat	$C < 0$	> 0 (heat in — the gas cools here too, but $C < 0$ flips the sign: it absorbs heat while cooling)

x	Process	C	ΔS sign in expansion
$x < 0$	Anomalous (pressure rises with volume)	$C_V < C < C_P$	> 0

Critical JEE trap: For $1 < x < \gamma$ (between isothermal and adiabatic), $TV^{x-1} = \text{const}$ with $x - 1 > 0$ means the gas *cools* during expansion ($T_f < T_i$) exactly as an adiabat would — but here $C < 0$, so $Q = nC\Delta T$ is *positive*: the gas **absorbs** heat even while cooling. Students assume “cooling $\Rightarrow$ heat released”. Wrong. The gas does more work expanding than the drop in internal energy accounts for, so it must draw in the shortfall as heat, net cooling notwithstanding.

Example A.1 — [JEE Advanced Level]

Problem. One mole of a monoatomic ideal gas ($C_V = \frac{3}{2}R$, $\gamma = \frac{5}{3}$) undergoes a quasi-static process described by $P = aV^{-2}$, where a is a constant. The gas expands from volume V_0 to $2V_0$. Find (a) the polytropic index x , (b) the molar heat capacity C , (c) the work done, (d) the heat exchanged, (e) the change in internal energy, (f) the entropy change. State whether the gas warms or cools.

Solution.

(a) Find x .

$$P \propto V^{-2} \Rightarrow PV^2 = \text{const} \Rightarrow x = 2.$$

(b) Molar heat capacity C .

$$C = C_V + \frac{R}{1-x} = \frac{3}{2}R + \frac{R}{1-2} = \frac{3}{2}R - R = \frac{1}{2}R.$$

Since $x = 2 > \gamma = 5/3$, this falls in the “steeper than adiabat” row of the table above: $0 < C < C_V$, confirmed here since $0 < \frac{1}{2}R < \frac{3}{2}R$. As part (c)–(f) below will show, the gas cools during this expansion — and since $C > 0$, it *releases* heat while doing so.

(c) Work done.

$$\text{Initial state: } P_i V_0 = RT_i \Rightarrow T_i = \frac{aV_0^{-2}V_0}{R} = \frac{a}{RV_0}.$$

$$\text{Final state: } P_f(2V_0) = RT_f \Rightarrow T_f = \frac{a(2V_0)^{-2}(2V_0)}{R} = \frac{a}{4RV_0} \cdot 2 = \frac{a}{2RV_0} = \frac{T_i}{2}.$$

$$\Delta T = T_f - T_i = -\frac{T_i}{2}.$$

$$W = \frac{nR(T_i - T_f)}{x-1} = \frac{R(T_i - T_i/2)}{2-1} = \frac{RT_i}{2}.$$

$$\text{Using } T_i = a/(RV_0): W = \frac{a}{2V_0} \text{ J.}$$

(Positive, gas expands).

(d) Heat exchanged.

$$Q = nC\Delta T = 1 \cdot \frac{1}{2}R \cdot \left(-\frac{T_i}{2}\right) = -\frac{1}{4}RT_i = -\frac{a}{4V_0} \text{ J.}$$

Negative. Heat is released by the gas during expansion, consistent with $C > 0$ and $\Delta T < 0$.

(e) Internal energy change.

$$\Delta U = nC_V\Delta T = \frac{3}{2}R \left(-\frac{T_i}{2}\right) = -\frac{3}{4}RT_i = -\frac{3a}{4V_0} \text{ J.}$$

Check First Law: $\Delta U + W = -\frac{3}{4}RT_i + \frac{1}{2}RT_i = -\frac{1}{4}RT_i = Q$.

(f) Entropy change.

$$\Delta S = nC_V \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i} = \frac{3}{2}R \ln \frac{1}{2} + R \ln 2 = \left(-\frac{3}{2} + 1\right) R \ln 2 = -\frac{1}{2}R \ln 2.$$

$$\Delta S = -\frac{R \ln 2}{2} \text{ J K}^{-1}.$$

Negative. Gas entropy decreases. This is allowed because heat flows out ($Q < 0$). Universe entropy change requires surroundings.

Summary: $x = 2$, $C = 0.5R$, $W = a/(2V_0)$ J, $Q = -a/(4V_0)$ J, $\Delta U = -3a/(4V_0)$ J, $\Delta S = -0.5R \ln 2$ J/K. Gas cools ($T_f = T_i/2$) and releases heat.

Example A.2 — [JEE Main Level]

Problem. A diatomic ideal gas ($C_V = \frac{5}{2}R$, $\gamma = \frac{7}{5}$) undergoes a process where the molar heat capacity is constant and equal to $C = 3R$. Find the polytropic index x and the ratio T_f/T_i if the volume doubles.

Solution.

Find x :

$$C = C_V + \frac{R}{1-x} \Rightarrow 3R = \frac{5}{2}R + \frac{R}{1-x}$$

$$\frac{1}{2}R = \frac{R}{1-x} \Rightarrow 1-x = 2 \Rightarrow x = -1.$$

Temperature ratio:

$$TV^{x-1} = \text{const} \Rightarrow TV^{-2} = \text{const}.$$

$$\frac{T_f}{T_i} = \left(\frac{V_i}{V_f}\right)^{x-1} = \left(\frac{1}{2}\right)^{-2} = 4.$$

$T_f = 4T_i$. Temperature quadruples during expansion. Since $x = -1 < 0$, pressure increases with volume ($P \propto V$). Heat is absorbed strongly ($C = 3R > C_P$).

Practice — Addendum A

1. One mole of a monoatomic ideal gas ($C_V = 3R/2$) expands quasi-statically from V_0 to $2V_0$ following $PV^{3/2} = \text{const}$. Find $W, Q, \Delta U, \Delta S$. ($R = 8.314$, $\ln 2 = 0.693$).
2. A gas has $C_P/C_V = 1.5$. In a process, its molar heat capacity is found to be zero. Find the polytropic index x and the relation between P and V .
3. For a certain ideal gas, $C_V = 2R$. The gas is compressed such that $P \propto V^2$. Find the molar heat capacity C for this process. Does the gas absorb or release heat during this compression?

- Two moles of a diatomic gas ($C_V = 5R/2$) undergo a process $TV^3 = \text{const}$. Initial state: $T_i = 300 \text{ K}$, $V_i = 0.01 \text{ m}^3$. Final volume $V_f = 0.02 \text{ m}^3$. Find $W, Q, \Delta S$.
- Advanced.** An ideal gas undergoes a process $P = kV^{-1/2}$. Find the value of γ for which this process is adiabatic. For a gas with $\gamma = 5/3$, find C and state if heat is absorbed or released during expansion.

Answers:

- $x = 1.5$. $T_f/T_i = (1/2)^{0.5} = 1/\sqrt{2}$. $W = \frac{R(T_i - T_f)}{0.5} = 2RT_i(1 - 1/\sqrt{2})$. $Q = nC\Delta T$, $C = 1.5R + R/(1 - 1.5) = 1.5R - 2R = -0.5R$. $\Delta U = 1.5R(T_f - T_i)$. $\Delta S = n(C - C_V) \ln 2 = -0.5R \ln 2 < 0$.
- $C = 0 \Rightarrow x = \gamma = 1.5$. $PV^{1.5} = \text{const}$.
- $P \propto V^2 \Rightarrow PV^{-2} = \text{const} \Rightarrow x = -2$. $C = 2R + R/(1 - (-2)) = 2R + R/3 = 7R/3 > 0$. Compression $\Rightarrow \Delta V < 0 \Rightarrow \Delta T < 0$ (since $x - 1 = -3$, $T \propto V^{-3}$). $Q = C\Delta T < 0$. Heat released.
- $TV^3 = \text{const} \Rightarrow x - 1 = 3 \Rightarrow x = 4$. $T_f = T_i(V_i/V_f)^3 = 300(1/2)^3 = 37.5 \text{ K}$. $W = \frac{nR(T_i - T_f)}{x-1} = \frac{2 \times 8.314 \times 262.5}{3} \approx 1455 \text{ J}$. $C = 2.5R + R/(1 - 4) = 2.5R - R/3 = 19R/6$. $Q = nC\Delta T = 2 \times (19R/6) \times (-262.5) \approx -27600 \text{ J}$. $\Delta S = n(C - C_V) \ln 2 = 2 \times (19R/6 - 5R/2) \ln 2 = 2 \times (2R/3) \ln 2 = (4R/3) \ln 2 \approx 7.7 \text{ J/K}$.
- Adiabatic $\Rightarrow x = \gamma = 1/2$? No, $P \propto V^{-1/2} \Rightarrow x = 0.5$. Adiabatic requires $x = \gamma$. So $\gamma = 0.5$ (impossible for ideal gas, $\gamma > 1$). So never adiabatic. For $\gamma = 5/3$, $C_V = 1.5R$. $x = 0.5$. $C = 1.5R + R/(1 - 0.5) = 1.5R + 2R = 3.5R > 0$. Expansion $\Rightarrow \Delta V > 0 \Rightarrow \Delta T > 0$ ($T \propto V^{-0.5}$). $Q = C\Delta T > 0$. Heat absorbed.

Addendum B — Composite Cycles: Leg-by-Leg Heat Accounting

Subtopic 05 and 07 define efficiency $\eta = W_{\text{net}}/Q_{\text{in}}$ and show the Carnot rectangle. JEE Main/Advanced frequently presents cycles made of straight lines and curves on a P-V diagram (e.g., two isotherms + two isochors, a triangle, a rectangle with slanted top) and asks for efficiency. The trap: Q_{in} is **not** the net heat Q_{net} ; it is the sum of heat *absorbed* only on those legs where $Q > 0$. You must compute Q for every leg using the first law, track signs, sum only the positive Q 's for the denominator, and take the enclosed area (or $\sum W$) for the numerator.

Method

- Sketch the cycle** on a P-V diagram. Label states 1, 2, 3... with (P, V, T) .
- Analyse each leg** (process 1→2, 2→3, ...):
 - Identify process type (isotherm, adiabat, isobar, isochor, straight line $P = aV + b$).
 - Compute $\Delta U = nC_V\Delta T$ (ideal gas).
 - Compute $W = \int PdV$ (area under curve; + for expansion, − for compression).
 - Compute $Q = \Delta U + W$.
 - Record sign of Q : $Q > 0$ (heat absorbed), $Q < 0$ (heat rejected).
- Net work:** $W_{\text{net}} = \sum W_{\text{legs}} = \text{Area enclosed}$ (positive if clockwise).
- Heat input:** $Q_{\text{in}} = \sum_{\text{legs: } Q > 0} Q_{\text{leg}}$.

5. **Efficiency:** $\eta = \frac{W_{\text{net}}}{Q_{\text{in}}}$.

6. **Carnot comparison:** Compute $\eta_{\text{Carnot}} = 1 - T_{\text{min}}/T_{\text{max}}$ using the extreme temperatures in the cycle. Verify $\eta < \eta_{\text{Carnot}}$.

Straight-line process $P = aV + b$ (non-standard):

$$W = \int (aV + b) dV = \frac{a}{2}(V_f^2 - V_i^2) + b(V_f - V_i) = \frac{1}{2}(P_f + P_i)(V_f - V_i) \text{ (trapezoid area).}$$

$$\Delta U = nC_V(T_f - T_i) = \frac{C_V}{R}(P_f V_f - P_i V_i).$$

$$Q = \Delta U + W.$$

Example B.1 — [JEE Advanced Level]

Problem. One mole of a monoatomic ideal gas ($C_V = \frac{3}{2}R$) undergoes a cyclic process ABCA on a P-V diagram. - A $\rightarrow$ B: Straight line from (P_0, V_0) to $(2P_0, 2V_0)$. - B $\rightarrow$ C: Isochoric cooling to pressure P_0 . - C $\rightarrow$ A: Isobaric compression back to V_0 . Find the efficiency of the cycle.

Solution.

State variables: - A: $P_A = P_0, V_A = V_0 \Rightarrow T_A = P_0 V_0 / R$. - B: $P_B = 2P_0, V_B = 2V_0 \Rightarrow T_B = 4P_0 V_0 / R = 4T_A$. - C: $V_C = V_B = 2V_0, P_C = P_0 \Rightarrow T_C = 2P_0 V_0 / R = 2T_A$.

Leg A $\rightarrow$ B (Straight line $P = (P_0/V_0)V$): $W_{AB} = \text{Area of trapezoid} = \frac{1}{2}(P_A + P_B)(V_B - V_A) = \frac{1}{2}(3P_0)(V_0) = \frac{3}{2}P_0 V_0 = \frac{3}{2}RT_A$. $\Delta U_{AB} = C_V(T_B - T_A) = \frac{3}{2}R(3T_A) = \frac{9}{2}RT_A$. $Q_{AB} = \Delta U_{AB} + W_{AB} = \frac{9}{2}RT_A + \frac{3}{2}RT_A = 6RT_A > 0$. (Heat absorbed).

Leg B $\rightarrow$ C (Isochoric, $V = 2V_0$): $W_{BC} = 0$. $\Delta U_{BC} = C_V(T_C - T_B) = \frac{3}{2}R(2T_A - 4T_A) = -3RT_A$. $Q_{BC} = -3RT_A < 0$. (Heat rejected).

Leg C $\rightarrow$ A (Isobaric, $P = P_0$): $W_{CA} = P_0(V_A - V_C) = P_0(V_0 - 2V_0) = -P_0 V_0 = -RT_A$. $\Delta U_{CA} = C_V(T_A - T_C) = \frac{3}{2}R(T_A - 2T_A) = -\frac{3}{2}RT_A$. $Q_{CA} = \Delta U_{CA} + W_{CA} = -\frac{3}{2}RT_A - RT_A = -\frac{5}{2}RT_A < 0$. (Heat rejected).

Cycle Totals: $W_{\text{net}} = \frac{3}{2}RT_A + 0 - RT_A = \frac{1}{2}RT_A$. (Clockwise, positive). $Q_{\text{in}} = Q_{AB} = 6RT_A$. (Only leg with $Q > 0$). $\eta = \frac{W_{\text{net}}}{Q_{\text{in}}} = \frac{0.5RT_A}{6RT_A} = \frac{1}{12} \approx 8.33\%$.

Carnot check: $T_{\text{max}} = T_B = 4T_A$, $T_{\text{min}} = T_A$. $\eta_{\text{Carnot}} = 1 - T_A/(4T_A) = 75\%$. $\eta(8.33\%) < \eta_{\text{Carnot}}(75\%)$.

Example B.2 — [JEE Main Level]

Problem. A cyclic process for 1 mol of ideal diatomic gas ($\gamma = 7/5$) consists of: 1. Isothermal expansion at T_1 from V_1 to $V_2 = 2V_1$. 2. Isochoric cooling to $T_2 = T_1/2$. 3. Adiabatic compression back to initial state. Find the efficiency.

Solution. State 1: T_1, V_1 . **State 2:** $T_1, V_2 = 2V_1$. **State 3:** $T_2 = T_1/2, V_3 = V_2 = 2V_1$. Check adiabat 3 $\rightarrow$ 1: $T_3 V_3^{\gamma-1} = (T_1/2)(2V_1)^{2/5} = (T_1/2)2^{2/5}V_1^{2/5} = T_1 2^{-3/5}V_1^{2/5}$. $T_1 V_1^{\gamma-1} = T_1 V_1^{2/5}$. Not equal! The adiabat 3 $\rightarrow$ 1 does **not** return to state 1. **Correction:** Problem implies the cycle closes. So State 3 must lie on adiabat through State 1. $T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1} \Rightarrow T_2 =$

$T_1(V_1/V_2)^{\gamma-1} = T_1(1/2)^{2/5}$. Given $T_2 = T_1/2$ is inconsistent with closing the cycle via adiabat. Assume the problem meant “Adiabatic compression back to initial volume V_1 ” and we find final T? No, “back to initial state”. Let’s assume the problem gives a consistent cycle: e.g., “Adiabatic compression to original volume”. Then final state is not initial state. Standard JEE problem: Cycle closes. Let’s redefine: 1. Isothermal exp $T_1, V_1 \rightarrow 2V_1$. 2. Isochoric cooling to T_2 . 3. Adiabatic comp to V_1 . Closure requires $T_2(2V_1)^{\gamma-1} = T_1V_1^{\gamma-1} \Rightarrow T_2 = T_1/2^{\gamma-1}$. Let’s solve with this correct T_2 .

Leg 1 (Isothermal Exp): $W_1 = RT_1 \ln 2$. $\Delta U_1 = 0$. $Q_1 = RT_1 \ln 2 > 0$. **Leg 2 (Isochoric Cool):** $W_2 = 0$. $\Delta U_2 = C_V(T_2 - T_1) = \frac{5}{2}RT_1(2^{-2/5} - 1)$. $Q_2 = \Delta U_2 < 0$. **Leg 3 (Adiabatic Comp):** $Q_3 = 0$. $W_3 = \frac{R(T_2 - T_1)}{\gamma - 1} = \frac{5}{2}R(T_2 - T_1) = -\Delta U_2$ (since $\Delta U_{\text{cycle}} = 0$). $W_{\text{net}} = W_1 + W_3 = RT_1 \ln 2 + \frac{5}{2}RT_1(2^{-2/5} - 1)$. $Q_{\text{in}} = Q_1 = RT_1 \ln 2$. $\eta = 1 + \frac{\frac{5}{2}(2^{-2/5} - 1)}{\ln 2}$. $2^{-2/5} \approx 0.7579$. Num = $2.5(-0.2421) = -0.605$. Den = 0.693 . $\eta = 1 - 0.605/0.693 \approx 1 - 0.873 = 0.127 = 12.7\%$. $\eta_{\text{Carnot}} = 1 - T_2/T_1 = 1 - 2^{-2/5} \approx 24.2\%$. $\eta < \eta_C$.

Practice — Addendum B

- Rectangle Cycle.** 1 mol monoatomic gas. $A(P_0, V_0) \rightarrow B(2P_0, V_0)$ (isochoric) $\rightarrow C(2P_0, 2V_0)$ (isobaric) $\rightarrow D(P_0, 2V_0)$ (isochoric) $\rightarrow A$ (isobaric). Find η .
- Triangle Cycle.** 1 mol diatomic gas. $A(P_0, V_0) \rightarrow B(2P_0, V_0)$ (isochoric) $\rightarrow C(P_0, 2V_0)$ (straight line) $\rightarrow A$ (isobaric). Find η .
- Stirling-like Cycle.** Two isotherms (T_H, T_C) connected by two isochors (V_1, V_2). 1 mol monoatomic gas. Find η . (Regenerator assumed ideal/absent? Standard Stirling without regenerator: heat absorbed on both high-T isotherm and high-V isochor).
- Complex Straight Line.** 1 mol gas ($\gamma = 5/3$). Cycle: $A(P_0, V_0) \rightarrow B(2P_0, 3V_0)$ (straight line) $\rightarrow C(P_0, 3V_0)$ (isobaric) $\rightarrow A$ (isochoric). Find $Q_{\text{in}}, W_{\text{net}}, \eta$.

Answers: 1. $T_A = P_0V_0/R, T_B = 2T_A, T_C = 4T_A, T_D = 2T_A$. AB: $W = 0, Q = 1.5R(T_A) > 0$. BC: $W = 2P_0V_0 = 2RT_A, \Delta U = 1.5R(2T_A) = 3RT_A, Q = 5RT_A > 0$. CD: $W = 0, Q = 1.5R(-2T_A) = -3RT_A$. DA: $W = -P_0V_0 = -RT_A, \Delta U = 1.5R(-T_A) = -1.5RT_A, Q = -2.5RT_A$. $W_{\text{net}} = 2RT_A - RT_A = RT_A$. $Q_{\text{in}} = 1.5RT_A + 5RT_A = 6.5RT_A$. $\eta = 1/6.5 \approx 15.4\%$. 2. AB: $W = 0, Q = 2.5RT_A > 0$. ($T_B = 2T_A$). BC: Straight line $P = 3P_0 - (P_0/V_0)V$. $W = \frac{1}{2}(2P_0 + P_0)(V_0) = 1.5P_0V_0 = 1.5RT_A$. $\Delta U = 2.5R(T_C - T_B) = 2.5R(2T_A - 2T_A) = 0$ (since $T_C = P_0(2V_0)/R = 2T_A$). $Q_{BC} = 1.5RT_A > 0$. CA: Isobaric comp. $W = -P_0V_0 = -RT_A$. $\Delta U = 2.5R(T_A - 2T_A) = -2.5RT_A$. $Q = -3.5RT_A$. $W_{\text{net}} = 1.5 - 1 = 0.5RT_A$. $Q_{\text{in}} = 2.5 + 1.5 = 4RT_A$. $\eta = 1/8 = 12.5\%$. 3. 1 $\rightarrow$ 2 (Isoth T_H): $Q_H = RT_H \ln(V_2/V_1) > 0$. 2 $\rightarrow$ 3 (Isochor V_2 cool): $Q_{23} = C_V(T_C - T_H) < 0$. 3 $\rightarrow$ 4 (Isoth T_C): $Q_C = RT_C \ln(V_1/V_2) < 0$. 4 $\rightarrow$ 1 (Isochor V_1 heat): $Q_{41} = C_V(T_H - T_C) > 0$. $Q_{\text{in}} = RT_H \ln r + C_V(T_H - T_C)$ where $r = V_2/V_1$. $W_{\text{net}} = R(T_H - T_C) \ln r$. $\eta = \frac{R(T_H - T_C) \ln r}{RT_H \ln r + C_V(T_H - T_C)} = \frac{\eta_C}{1 + \frac{C_V}{R \ln r}} < \eta_C$. 4. A: $T_A = P_0V_0/R$. B: $T_B = 6P_0V_0/R = 6T_A$. C: $T_C = 3P_0V_0/R = 3T_A$. AB: Line $P = P_0(2 - V_0/V)$ Eqn: $(P - P_0)/(2P_0 - P_0) = (V - V_0)/(3V_0 - V_0) \Rightarrow P = P_0 + 0.5P_0/V_0(V - V_0)$. $W_{AB} = \frac{1}{2}(P_0 + 2P_0)(2V_0) = 3P_0V_0 = 3RT_A$. $\Delta U_{AB} = 1.5R(5T_A) = 7.5RT_A$. $Q_{AB} = 10.5RT_A > 0$. BC: Isobaric. $W_{BC} = P_0(V_0 - 3V_0) = -2P_0V_0 = -2RT_A$. $\Delta U_{BC} = 1.5R(-3T_A) = -4.5RT_A$. $Q_{BC} = -6.5RT_A < 0$. CA: Isochoric. $W = 0$. $\Delta U = 1.5R(-2T_A) = -3RT_A$. $Q = -3RT_A < 0$. $W_{\text{net}} = 3 - 2 = RT_A$. $Q_{\text{in}} = 10.5RT_A$. $\eta = 1/10.5 \approx 9.5\%$.

Addendum C — Irreversible Adiabatics: Free Expansion & Expansion Against Constant P_{ext}

Subtopic 03 derives reversible adiabatic relations ($PV^\gamma = \text{const}$) assuming quasi-static equilibrium. JEE Advanced heavily tests **irreversible adiabatics**: (1) Free expansion into vacuum (Joule expansion) — $W = 0, Q = 0, \Delta U = 0, \Delta T = 0$ (ideal gas), but $\Delta S > 0$. (2) Expansion against a constant external pressure P_{ext} (finite pressure difference) — not quasi-static, $W = P_{\text{ext}}\Delta V, Q = 0$, so $\Delta U = -W$. The final state does **not** lie on $PV^\gamma = \text{const}$. You must use the first law with the actual work to find the final state.

Method

Case 1: Free Expansion (Joule Expansion) - Adiabatic $\Rightarrow Q = 0$. - Vacuum $\Rightarrow P_{\text{ext}} = 0 \Rightarrow W = 0$. - First Law: $\Delta U = 0$. - Ideal Gas: $U(T)$ only $\Rightarrow \Delta T = 0$ (Isothermal *result*, but **not** an isothermal *process* — no reservoir, not quasi-static). - Entropy change: Compute via reversible isotherm connecting same (V_i, T) to (V_f, T) .

$$\Delta S_{\text{gas}} = nR \ln \frac{V_f}{V_i} > 0.$$

$\Delta S_{\text{surr}} = 0$ (no heat exchange). $\Delta S_{\text{univ}} = \Delta S_{\text{gas}} > 0$.

Case 2: Irreversible Adiabatic Expansion Against Constant P_{ext} - Adiabatic $\Rightarrow Q = 0$. - Work: $W = P_{\text{ext}}(V_f - V_i)$ (Only valid if P_{ext} constant). - First Law: $\Delta U = -W \Rightarrow nC_V(T_f - T_i) = -P_{\text{ext}}(V_f - V_i)$. - Gas Law at endpoints: $P_i V_i = nRT_i, P_f V_f = nRT_f$. - Mechanical equilibrium at final state (piston stops): $P_f = P_{\text{ext}}$. - **Two equations, two unknowns (T_f, V_f or P_f, V_f)**: 1. $nC_V(T_f - T_i) = -P_f(V_f - V_i)$. 2. $P_f V_f = nRT_f$. - Solve for T_f, V_f . Compare with reversible adiabat $TV^{\gamma-1} = \text{const}$. - **Result**: For same initial state and same final pressure P_f , irreversible expansion does **less work** ($W_{\text{irr}} < W_{\text{rev}}$), so ΔU decreases less $\Rightarrow T_f^{\text{irr}} > T_f^{\text{rev}}$. Final volume $V_f^{\text{irr}} < V_f^{\text{rev}}$. - Entropy: $\Delta S_{\text{gas}} = nC_V \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i} > 0$. $\Delta S_{\text{surr}} = 0$. $\Delta S_{\text{univ}} > 0$.

Example C.1 — [JEE Advanced Level]

Problem. 1 mole of a monoatomic ideal gas ($C_V = \frac{3}{2}R$) at $T_i = 300$ K, $V_i = 0.01$ m³ expands adiabatically against a constant external pressure $P_{\text{ext}} = 10^5$ Pa until mechanical equilibrium is reached. Find the final temperature T_f , final volume V_f , work done W , and entropy change ΔS . Compare T_f with the reversible adiabatic expansion to the same final pressure.

Solution. $R = 8.314$. $P_i = nRT_i/V_i = (1)(8.314)(300)/0.01 = 2.494 \times 10^5$ Pa. $P_{\text{ext}} = 1.0 \times 10^5$ Pa. Final pressure $P_f = P_{\text{ext}} = 10^5$ Pa (mechanical equilibrium).

Equations. With W the work done by the gas and $Q = 0$ (adiabatic):

$$\Delta U = -W \Rightarrow \frac{3}{2}R(T_f - T_i) = -P_{\text{ext}}(V_f - V_i), \quad P_f V_f = RT_f$$

Eliminating $V_f = RT_f/P_f$ and solving symbolically (cleaner than substituting numbers early, and it exposes the physics):

$$C_V(T_f - T_i) = -P_f \left(\frac{RT_f}{P_f} - \frac{RT_i}{P_i} \right) = -RT_f + RT_i \frac{P_f}{P_i}$$

$$T_f(C_V + R) = T_i \left(C_V + R \frac{P_f}{P_i} \right) \Rightarrow C_P T_f = T_i \left(C_V + R \frac{P_f}{P_i} \right)$$

$$T_f = T_i \left[\frac{1}{\gamma} + \frac{\gamma - 1}{\gamma} \cdot \frac{P_f}{P_i} \right]$$

With $\gamma = 5/3$ and $P_f/P_i = 10^5/2.494 \times 10^5 = 0.401$:

$$T_f = 300 \left[\frac{3}{5} + \frac{2}{5}(0.401) \right] = 300(0.6 + 0.1604) = 300 \times 0.7604 \approx 228.1 \text{ K}$$

As expected for an expansion, $T_f < T_i$. Then $V_f = RT_f/P_f = 8.314 \times 228.1/10^5 \approx 0.01896 \text{ m}^3$, and

$$W = P_f(V_f - V_i) = 10^5(0.00896) \approx 896 \text{ J}, \quad \Delta U = -W = -896 \text{ J}$$

Check: $\Delta U = C_V \Delta T = 12.47 \times (-71.9) \approx -896 \text{ J}$, consistent.

Reversible adiabat to the same final pressure $P_f = 10^5 \text{ Pa}$, for comparison:

$$T_{f,\text{rev}} = T_i \left(\frac{P_f}{P_i} \right)^{(\gamma-1)/\gamma} = 300 (0.401)^{0.4} = 300 e^{0.4 \ln 0.401} = 300 e^{-0.365} \approx 208 \text{ K}$$

So $T_f^{\text{rev}}(208 \text{ K}) < T_f^{\text{irr}}(228 \text{ K})$: the irreversible expansion against a constant P_{ext} does *less* work than the reversible path to the same final pressure (the reversible path's pressure stays above P_{ext} throughout the expansion, so $\int P dV > P_{\text{ext}} \Delta V$), so it draws down less internal energy and cools less.

Entropy Change (Irrev): $\Delta S = nC_V \ln(T_f/T_i) + nR \ln(V_f/V_i)$. $= 12.47 \ln(228/300) + 8.314 \ln(0.01896/0.01)$. $= 12.47(-0.275) + 8.314(0.640)$. $= -3.43 + 5.32 = 1.89 \text{ J/K} > 0$.

Example C.2 — [JEE Main Level]

Problem. An ideal gas is confined in one half of a thermally insulated rigid container; the other half is vacuum. The partition is removed. The gas rushes to fill the whole container. Choose the correct statement(s): (A) Work done by gas is zero. (B) Internal energy decreases. (C) Temperature remains constant. (D) Entropy of gas increases.

Solution. - Rigid container $\Rightarrow$ No boundary movement $\Rightarrow W = 0$. (A) Correct. - Insulated $\Rightarrow Q = 0$. - First Law: $\Delta U = Q - W = 0$. (B) Incorrect (does not decrease). - Ideal Gas: $U = U(T) \Rightarrow \Delta T = 0$. (C) Correct. - Irreversible process. $\Delta S_{\text{gas}} = nR \ln(V_f/V_i) = nR \ln 2 > 0$. (D) Correct. **Answer: (A), (C), (D).**

Practice — Addendum C

- 2 moles of diatomic gas ($C_V = 5R/2$) at 400 K, 10 L expand adiabatically against constant $P_{\text{ext}} = 1$ atm until $P_f = P_{\text{ext}}$. Initial $P_i = 10$ atm. Find $T_f, V_f, W, \Delta S$. ($R = 0.0821$ L atm/mol K or 8.314 J).
- Free expansion, then reversible compression.** 1 mol of a monatomic gas ($C_V = 3R/2$) at 300 K undergoes free expansion from V_i to $2V_i$, and is then compressed reversibly and adiabatically back to V_i . Find the final temperature and ΔS_{univ} for the whole process.
- A gas expands irreversibly and adiabatically from (P_1, V_1, T_1) to (P_2, V_2, T_2) against constant $P_{\text{ext}} = P_2$. Show that $T_2 = T_1 \frac{C_V + R(P_2/P_1)}{C_P}$.
- True/False.** In an irreversible adiabatic expansion against constant external pressure, the work done is always less than the work done in a reversible adiabatic expansion between the same initial state and the same final pressure. (True/False).
- 1 mol monoatomic gas at 300 K, 1 atm expands adiabatically against $P_{\text{ext}} = 0.5$ atm. Final volume is 40 L. Find final temperature. (Hint: Use $W = P_{\text{ext}}\Delta V$ and $\Delta U = nC_V\Delta T$).

Answers: 1. $P_i = 10, P_f = 1$. $T_f = T_i[1/\gamma + (\gamma - 1)/\gamma(P_f/P_i)]$. $\gamma = 1.4$. $T_f = 400[1/1.4 + 0.4/1.4(0.1)] = 400[0.714 + 0.0286] = 297$ K. $V_f = nRT_f/P_f = 2 \times 0.0821 \times 297/1 = 48.7$ L. $W = 1 \times (48.7 - 10) = 38.7$ L atm = 3920 J. $\Delta S = 2[2.5R \ln(297/400) + R \ln(48.7/10)] = 2R[2.5(-0.30) + 1.58] = 2 \times 8.314 \times 0.83 = 13.8$ J/K. 2. Free Exp: T const, $V : V \rightarrow 2V$. $\Delta S_1 = R \ln 2$. Rev Adiabatic Comp $2V \rightarrow V$: $T_f = T_i(2)^{\gamma-1} = 300 \times 2^{2/3} = 300 \times 1.59 = 476$ K. $\Delta S_2 = 0$. $\Delta S_{\text{gas, cycle}} = R \ln 2 > 0$. $\Delta S_{\text{surr}} = 0$. $\Delta S_{\text{univ}} = R \ln 2 > 0$. 3. Derivation shown in Method Case 2. 4. **True.** (As proven in Example C.1 analysis). 5. $W = 0.5 \times (40 - V_i)$. $V_i = RT_i/P_i = 0.0821 \times 300/1 = 24.63$ L. $W = 0.5 \times 15.37 = 7.685$ L atm = 779 J. $\Delta U = -W = -779$ J = $1.5R\Delta T = 12.47\Delta T$. $\Delta T = -62.5$ K. $T_f = 237.5$ K.

Addendum D — Entropy Accounting for Irreversible Processes: The Universe Ledger

Subtopic 08 defines $\Delta S = \int dQ_{\text{rev}}/T$ and states $\Delta S_{\text{univ}} \geq 0$. JEE Advanced problems require you to *actually compute* ΔS_{univ} for specific irreversible processes: heat transfer across finite ΔT , irreversible adiabatic expansion (Addendum C), mixing of gases, or a cycle with an irreversible leg. The method is rigid: **System entropy via a reversible path; Surroundings entropy via actual heat transfer at reservoir temperature.**

Method

- System (ΔS_{sys}):** Identify initial and final equilibrium states $(T_i, V_i/P_i)$ and $(T_f, V_f/P_f)$. Construct *any* reversible path between them (e.g., isothermal + isochoric, or single polytropic if reversible). Compute $\Delta S = nC_V \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i}$.
- Surroundings (ΔS_{surr}):** Identify heat reservoirs interacting with the system. For each reservoir at constant T_{res} , $\Delta S_{\text{res}} = -\frac{Q_{\text{actual, sys} \rightarrow \text{res}}}{T_{\text{res}}}$. (Heat *leaving* system enters reservoir; sign convention: Q_{sys} positive if absorbed by sys).

- If system absorbs Q from reservoir at T_H : $\Delta S_{\text{surr}} = -Q/T_H$.
- If system rejects Q to reservoir at T_C : $\Delta S_{\text{surr}} = +|Q|/T_C$.
- If process is adiabatic ($Q = 0$): $\Delta S_{\text{surr}} = 0$.

3. **Universe:** $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$.

- Must be ≥ 0 . If < 0 , recheck signs/arithmetic.
- $= 0$ only if *all* steps reversible.

Common Scenario: Two bodies at T_H, T_C brought into thermal contact (no work). -

Final $T_f = \frac{C_H T_H + C_C T_C}{C_H + C_C}$ (heat capacities C). - $\Delta S_H = C_H \ln(T_f/T_H)$. - $\Delta S_C = C_C \ln(T_f/T_C)$. -

$\Delta S_{\text{univ}} = C_H \ln \frac{T_f}{T_H} + C_C \ln \frac{T_f}{T_C} > 0$.

Example D.1 — [JEE Advanced Level]

Problem. A rigid adiabatic container is divided into two equal volumes V_0 by a partition. One side contains 1 mol of monoatomic ideal gas at $T_1 = 400$ K; the other side is vacuum. The partition is removed (free expansion). After equilibrium, the gas is compressed *reversibly and isothermally* at temperature T_f back to volume V_0 by a piston in contact with a reservoir at T_f . Find (a) T_f , (b) ΔS_{gas} for the full cycle (free expansion + compression), (c) ΔS_{res} , (d) ΔS_{univ} .

Solution. (a) Free expansion of ideal gas: $Q = 0, W = 0 \Rightarrow \Delta U = 0 \Rightarrow T_f = T_i = 400$ K.

(b) Cycle: State A ($V_0, 400\text{K}$) $\xrightarrow{\text{Free Exp}}$ State B ($2V_0, 400\text{K}$) $\xrightarrow{\text{Rev Isoth Comp}}$ State A. - Free Exp (Irrev): $\Delta S_{\text{gas}, 1} = nR \ln(V_f/V_i) = R \ln 2$. - Rev Isoth Comp: $\Delta S_{\text{gas}, 2} = nR \ln(V_i/V_f) = -R \ln 2$. - Total $\Delta S_{\text{gas, cycle}} = 0$. (State function, cycle closed). **(c) Reservoir:** Only interacts during isothermal compression. Work done *on* gas during comp: $W_{\text{on}} = -W_{\text{by}} = nRT \ln(V_f/V_i) = RT \ln 2$. Heat rejected *by* gas = Heat absorbed *by* reservoir: $Q_{\text{res}} = +RT \ln 2$. $\Delta S_{\text{res}} = Q_{\text{res}}/T_{\text{res}} = R \ln 2$. **(d) Universe:** $\Delta S_{\text{univ}} = \Delta S_{\text{gas}} + \Delta S_{\text{res}} = 0 + R \ln 2 = R \ln 2 > 0$. The irreversibility of the free expansion is “paid for” by the entropy increase of the reservoir during the restoration step.

Example D.2 — [JEE Main Level]

Problem. 100 g of water at 0°C is mixed with 100 g of water at 100°C in an insulated container. Find the entropy change of the universe. ($s_{\text{water}} = 4.18$ J/g K).

Solution. Masses equal $m = 100$ g. $T_1 = 273$ K, $T_2 = 373$ K. Final Temp $T_f = \frac{T_1 + T_2}{2} = 323$ K. $\Delta S_{\text{cold}} = ms \ln \frac{T_f}{T_1} = 100 \times 4.18 \times \ln \frac{323}{273} = 418 \times \ln(1.183) = 418 \times 0.168 = 70.2$ J/K. $\Delta S_{\text{hot}} = ms \ln \frac{T_f}{T_2} = 418 \times \ln \frac{323}{373} = 418 \times \ln(0.866) = 418 \times (-0.144) = -60.2$ J/K. $\Delta S_{\text{univ}} = 70.2 - 60.2 = 10.0$ J/K > 0 .

Practice — Addendum D

- Two identical bodies of heat capacity C (constant) at temperatures T_1 and T_2 ($T_1 > T_2$) are brought into thermal contact in an isolated system. Find ΔS_{univ} . Show it is positive for $T_1 \neq T_2$.
- 1 mol of monoatomic gas ($C_V = 1.5R$) at 300 K, 1 atm expands *irreversibly and adiabatically* against constant $P_{\text{ext}} = 0.5$ atm until $P_f = P_{\text{ext}}$. Find ΔS_{gas} and ΔS_{univ} .
- A Carnot engine operates between 400 K and 300 K. In one cycle, it absorbs 1000 J from the hot reservoir. Due to a flaw, the work output is only 200 J (instead of 250 J). The extra 50 J is dissipated as heat *into the cold reservoir* (friction heating the sink). Find ΔS_{univ} for this cycle.
- Mixing.** 1 mol He and 1 mol Ne (both monoatomic) in separate 10 L containers at 300 K. Partition removed, gases mix in 20 L total. Find ΔS_{mixing} . (Assume ideal gases, no interaction).
- Stirling Engine with Imperfect Regenerator.** 1 mol monoatomic gas. Cycle: Isoth Exp T_H ($V_1 \rightarrow V_2$), Isochoric Cool $T_H \rightarrow T_C$ (regenerator absorbs heat Q_R), Isoth Comp T_C ($V_2 \rightarrow V_1$), Isochoric Heat $T_C \rightarrow T_H$ (regenerator returns heat Q_R). If regenerator effectiveness is $\epsilon < 1$ (returns only ϵQ_R), find ΔS_{univ} per cycle in terms of ϵ, T_H, T_C, Q_R .

Answers: 1. $T_f = (T_1 + T_2)/2$. $\Delta S_1 = C \ln(T_f/T_1)$, $\Delta S_2 = C \ln(T_f/T_2)$. $\Delta S_{\text{univ}} = C \ln(T_f^2/(T_1 T_2)) = C \ln(((T_1 + T_2)/2)^2/(T_1 T_2)) = C \ln((T_1 + T_2)^2/(4T_1 T_2))$. By AM-GM, $(T_1 + T_2)^2 \geq 4T_1 T_2$, so $\Delta S \geq 0$. Equality iff $T_1 = T_2$. 2. From Addendum C Example C.1 method: $T_f = T_i[1/\gamma + (\gamma - 1)/\gamma(P_f/P_i)]$. $\gamma = 5/3$, $P_f/P_i = 0.5$. $T_f = 300[0.6 + 0.4(0.5)] = 300(0.8) = 240$ K. $V_f = RT_f/P_f = 8.314 \times 240/0.5 \times 10^5 = 0.0399$ m³. $V_i = 8.314 \times 300/10^5 = 0.0249$ m³. $\Delta S_{\text{gas}} = 1.5R \ln(240/300) + R \ln(0.0399/0.0249) = 1.5R(-0.223) + R(0.47) = R(-0.335 + 0.47) = 0.135R = 1.12$ J/K. $\Delta S_{\text{surr}} = 0$. $\Delta S_{\text{univ}} = 1.12$ J/K. 3. Hot Res (400 K): Loses 1000 J. $\Delta S_H = -1000/400 = -2.5$ J/K. Cold Res (300 K): Gains $Q_C = Q_H - W = 1000 - 200 = 800$ J. (Normally 750 J). $\Delta S_C = +800/300 = +2.667$ J/K. Engine (Cycle): $\Delta S_{\text{eng}} = 0$. $\Delta S_{\text{univ}} = -2.5 + 2.667 = 0.167$ J/K > 0 . 4. Each gas expands from 10 L to 20 L isothermally at 300 K. $\Delta S_{\text{He}} = R \ln 2$. $\Delta S_{\text{Ne}} = R \ln 2$. Total $\Delta S_{\text{mix}} = 2R \ln 2 = 11.5$ J/K. 5. Hot Isochor (4 $\rightarrow$ 1): Needs heat $Q_R = C_V(T_H - T_C)$. Gets ϵQ_R from regen, $(1 - \epsilon)Q_R$ from Hot Reservoir T_H . Cold Isochor (2 $\rightarrow$ 3): Rejects Q_R . Gives ϵQ_R to regen, $(1 - \epsilon)Q_R$ to Cold Reservoir T_C . Hot Res: Loses $(1 - \epsilon)Q_R$ at $T_H \Rightarrow \Delta S_H = -(1 - \epsilon)Q_R/T_H$. Cold Res: Gains $(1 - \epsilon)Q_R$ at $T_C \Rightarrow \Delta S_C = +(1 - \epsilon)Q_R/T_C$. Regenerator (Cycle): $\Delta S_{\text{reg}} = 0$. Gas (Cycle): $\Delta S_{\text{gas}} = 0$. $\Delta S_{\text{univ}} = (1 - \epsilon)Q_R(1/T_C - 1/T_H) > 0$.

Addendum E — Compound Carnot Systems: Engines Driving Refrigerators & Cascades

Subtopic 07 gives the standalone Carnot efficiency. JEE Advanced combines devices: a Carnot engine drives a Carnot refrigerator; two Carnot engines in series (cascade) where the heat rejected by the first is absorbed by the second; or a Carnot heat pump heating a house using work from a Carnot engine. The method is always **energy conservation at the coupling point** (work equality or heat equality) plus the Carnot relations $\eta = 1 - T_2/T_1$,

$$\text{COP}_{\text{ref}} = T_2 / (T_1 - T_2).$$

Method

1. **Draw the schematic.** Label reservoirs $T_1 > T_2 > T_3 \dots$. Label devices (Engine E, Fridge R, Pump P). Label heat flows Q and work W with arrows.
2. **Write Carnot relations for each device.**
 - Engine (between T_H, T_C): $\eta = W/Q_H = 1 - T_C/T_H \Rightarrow Q_C = Q_H T_C/T_H$.
 - Fridge (between T_C, T_H): $\text{COP} = Q_C/W = T_C/(T_H - T_C) \Rightarrow Q_H = Q_C + W = Q_C T_H/T_C$.
 - Heat Pump: $\text{COP} = Q_H/W = T_H/(T_H - T_C)$.
3. **Apply Coupling Constraints.**
 - Engine drives Fridge: $W_{\text{engine}} = W_{\text{fridge}}$.
 - Cascade Engines: $Q_{\text{rej}, 1} = Q_{\text{abs}, 2}$.
 - Engine drives Heat Pump heating T_H : $W_{\text{eng}} = W_{\text{hp}}$.
4. **Solve for required ratio** (e.g., $Q_{\text{cold extracted}}/Q_{\text{hot supplied}}$, or overall COP).

Example E.1 — [JEE Advanced Level]

Problem. A Carnot engine operates between a hot reservoir at $T_1 = 600$ K and a cold reservoir at $T_2 = 300$ K. The work output drives a Carnot refrigerator operating between $T_2 = 300$ K (cold space) and $T_3 = 400$ K (hot reservoir). Find the ratio of heat extracted from the cold space ($Q_{C,\text{ref}}$) to heat absorbed from the hottest reservoir ($Q_{H,\text{eng}}$).

Solution. Engine (600 K $\rightarrow$ 300 K): $\eta_E = 1 - 300/600 = 1/2$. $W = \eta_E Q_{H,E} = \frac{1}{2} Q_{H,E}$. $Q_{C,E} = Q_{H,E} - W = \frac{1}{2} Q_{H,E}$. (Rejected to 300 K reservoir).

Refrigerator (300 K $\rightarrow$ 400 K): $\text{COP}_R = \frac{T_C}{T_H - T_C} = \frac{300}{400 - 300} = 3$. $W = W$ (same shaft). $Q_{C,R} = \text{COP}_R \cdot W = 3 \times \frac{1}{2} Q_{H,E} = 1.5 Q_{H,E}$.

Ratio: $\frac{Q_{C,R}}{Q_{H,E}} = 1.5$. The compound device extracts 1.5 J from 300 K for every 1 J taken from 600 K. (Note: It dumps $Q_{H,R} = Q_{C,R} + W = 1.5 Q_{H,E} + 0.5 Q_{H,E} = 2 Q_{H,E}$ into the 400 K reservoir. The 300 K reservoir receives $Q_{C,E} = 0.5 Q_{H,E}$ from engine and supplies $Q_{C,R} = 1.5 Q_{H,E}$ to fridge. Net heat flow out of 300 K reservoir = $1.5 - 0.5 = 1.0 Q_{H,E}$. Overall: $Q_{600} \rightarrow \text{Device} \rightarrow Q_{300}(\text{net out}) + Q_{400}(\text{in})$. Energy conserved.)

Example E.2 — [JEE Main Level]

Problem. Two Carnot engines A and B are operated in series. Engine A receives heat Q_1 at $T_1 = 900$ K and rejects heat Q_2 at T_2 . Engine B receives Q_2 and rejects heat Q_3 at $T_3 = 300$ K. If the work outputs of A and B are equal, find T_2 and the overall efficiency of the combined system.

Solution. Engine A: $W_A = Q_1 - Q_2 = Q_1(1 - T_2/T_1)$. Engine B: $W_B = Q_2 - Q_3 = Q_2(1 - T_3/T_2) = Q_1(T_2/T_1)(1 - T_3/T_2)$. Given $W_A = W_B$: $Q_1(1 - T_2/T_1) = Q_1(T_2/T_1)(1 - T_3/T_2)$. $1 - \frac{T_2}{T_1} = \frac{T_2}{T_1} - \frac{T_3}{T_1}$. $1 = 2\frac{T_2}{T_1} - \frac{T_3}{T_1}$. $2T_2 = T_1 + T_3 = 900 + 300 = 1200$. $T_2 = 600$ K.

Overall Efficiency: $\eta_{\text{overall}} = \frac{W_A + W_B}{Q_1} = \frac{2W_A}{Q_1} = 2\eta_A = 2(1 - 600/900) = 2/3 = 66.7\%$. Check: Single Carnot 900→300: $\eta = 1 - 300/900 = 66.7\%$. Matches Carnot's theorem (reversible composite = single reversible).

Practice — Addendum E

1. A Carnot engine (500 K, 300 K) drives a Carnot refrigerator (300 K, 400 K). Find heat extracted from 300 K per 1000 J from 500 K.
2. A Carnot heat pump maintains a house at 300 K when outside is 270 K. The pump is driven by a Carnot engine operating between 600 K and 300 K. Find heat delivered to house per 1000 J from 600 K reservoir.
3. Three Carnot engines in cascade between $T_1 = 1000$ K, $T_2, T_3, T_4 = 300$ K. Work outputs equal. Find T_2, T_3 .
4. An inventor claims a device: Heat Q_1 from T_1 , Work W output, Heat Q_2 to T_2 ($T_2 < T_1$). He claims $W/Q_1 > 1 - T_2/T_1$. You couple it to a Carnot refrigerator between T_2, T_1 . Show a violation of Clausius.
5. **Advanced.** A Carnot engine operates between T_H and T_C . Its work output is used to run a Carnot heat pump between T_C and T_H (heating the hot reservoir). Find the ratio of heat delivered to T_H by the pump to heat extracted from T_H by the engine.

Answers: 1. Engine: $\eta = 1 - 3/5 = 0.4$. $W = 0.4Q_H$. Fridge: $\text{COP} = 300/(400 - 300) = 3$. $Q_C = 3W = 1.2Q_H$. For $Q_H = 1000$, $Q_C = 1200$ J. 2. Engine: $\eta = 1 - 300/600 = 0.5$. $W = 500$ J. Pump: $\text{COP} = 300/(300 - 270) = 10$. $Q_H = 10 \times 500 = 5000$ J. 3. $W_1 = W_2 = W_3$. $Q_1(1 - T_2/T_1) = Q_1(T_2/T_1)(1 - T_3/T_2) = Q_1(T_3/T_1)(1 - T_4/T_3)$. $1 - T_2/T_1 = T_2/T_1 - T_3/T_1 \Rightarrow T_2 = (T_1 + T_3)/2$. $T_2/T_1 - T_3/T_1 = T_3/T_1 - T_4/T_1 \Rightarrow T_3 = (T_2 + T_4)/2$. Sub T_2 : $T_3 = (T_1 + T_3)/4 + T_4/2 \Rightarrow 4T_3 = T_1 + T_3 + 2T_4 \Rightarrow 3T_3 = T_1 + 2T_4 = 1000 + 600 = 1600 \Rightarrow T_3 = 533.3$ K. $T_2 = (1000 + 533.3)/2 = 766.7$ K. 4. Engine X: $W_X > W_C$. Fridge C (rev): $W_C = Q_C(T_1/T_2 - 1)$. Run X, use W_C to run C, dump excess work ($W_X - W_C$) as heat into T_1 (stirrer). Net effect: Heat moved from T_2 to T_1 with no net work input. Clausius violation. 5. Engine ($T_H \rightarrow T_C$): takes Q_H from the hot reservoir, does work $W = Q_H(1 - T_C/T_H) = Q_H\eta$, rejects $Q_C = Q_H - W$ to the cold reservoir. Pump ($T_C \rightarrow T_H$), driven by exactly this W : $\text{COP} = T_H/(T_H - T_C) = 1/\eta$, so it delivers $Q_{H,\text{pump}} = \text{COP} \cdot W = (1/\eta)(\eta Q_H) = Q_H$ to the hot reservoir, extracting $Q_{H,\text{pump}} - W = Q_H - \eta Q_H = Q_C$ from the cold reservoir. **Ratio** = $Q_{H,\text{pump}}/Q_H = 1$. Tracing the net effect on each reservoir: the hot reservoir loses Q_H to the engine and gains Q_H back from the pump — no net change. The cold reservoir gains Q_C from the engine's exhaust and loses exactly Q_C back to the pump's intake — no net change either. The work W generated by the engine is entirely consumed running the pump, so no net external work crosses the system boundary. Nothing net happens to either reservoir: running an ideal (Carnot) engine forward and using its own output to drive an ideal pump back between the same two reservoirs is exactly equivalent to running one Carnot cycle forward and then immediately in reverse — a thermodynamically

consistent, if pointless, round trip, not a Clausius violation.

End of Addendum

Chapter 12 — Kinetic Theory

Class 11 Physics · Drona Ed Tech

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Sub-topic 0: Molecular Nature of Matter

Exam Relevance: Mostly a CBSE Board section — expect 1-2 mark short-answer or assertion-reason questions on intermolecular forces, the solid/liquid/gas distinction, or Avogadro’s hypothesis. NEET occasionally asks the molecular-size / number-density estimation. JEE rarely asks it directly but assumes it as the conceptual bedrock for everything that follows in the chapter.

Section 1: Concept Intuition

Before we can talk about how gases push on walls or how fast their molecules fly, we have to accept one founding idea: **all matter is made of enormous numbers of tiny particles — atoms and molecules — in perpetual motion.** This is the *atomic hypothesis*, and although it now feels obvious, it was hard-won. The Indian thinker Kanada and the Greek Democritus speculated about indivisible particles millennia ago, but it was John Dalton in the early 1800s who turned the idea into testable chemistry with his atomic theory: matter is built from atoms, atoms of a given element are identical, and they combine in small whole-number ratios to form compounds. The physicist Richard Feynman once remarked that if all of science had to be compressed into a single statement, the most valuable one would be that everything is made of atoms — tiny particles in perpetual motion, attracting when a little apart and repelling when squeezed together. That one sentence, he noted, contains an enormous amount of information about the world; almost this entire chapter is its unpacking.

Think about it this way... imagine zooming into a steel rod, a cup of water, and the air above it. In all three you would find the same kind of building blocks — molecules — but arranged utterly differently. In the steel, the molecules are packed shoulder-to-shoulder, locked into a rigid lattice, only able to jiggle in place. In the water, they are still touching but can slide past one another, which is why a liquid flows and takes the shape of its container. In the air, they are flung far apart — on average about ten times farther apart than in the solid or liquid — racing around almost independently, meeting only in brief collisions. Same particles, three states of matter, and the *only* thing that has changed is how close the molecules are and how strongly they feel one another.

What decides that closeness is the **intermolecular force**. Two molecules attract each other weakly when they are a moderate distance apart (this attraction is what holds liquids and solids together), but if you try to push them too close, they repel each other very strongly (you cannot force two molecules to occupy the same space). Somewhere in between there is a special separation — call it r_0 — where the attraction and repulsion exactly balance and

the net force is zero. This is the comfortable equilibrium spacing that molecules in a solid settle into.

Think about it this way... picture two molecules joined by an imaginary spring that is also slightly sticky. Pull them apart and the attraction (the sticky part) tugs them back. Push them together past r_0 and the spring's stiffness shoves them apart, harder and harder the closer you push. At r_0 the molecule is content. In a solid, every molecule sits near its own r_0 and merely vibrates about it; add heat and the vibrations grow until the lattice breaks down into a liquid, and more heat frees the molecules entirely into a gas.

This picture immediately tells you *why a gas is the natural home of kinetic theory*. In a gas the molecules are so far apart that, for almost all of their lives, they are beyond the reach of these forces — they feel essentially nothing until the instant they collide. That is exactly the assumption (“no intermolecular forces except during collisions”) that makes the ideal-gas model work. The molecular nature of matter, in other words, is the bedrock on which the rest of the chapter is built.

How do we *know* molecules are really moving, since we cannot see them? The most direct everyday evidence is **Brownian motion**. In 1827 the botanist Robert Brown noticed that tiny pollen grains suspended in water jiggled around ceaselessly in random, erratic paths, even though the water was perfectly still. The grains are far too big to move on their own; what jostles them is the relentless bombardment by invisible water molecules striking them unevenly from all sides. The smaller the suspended particle and the higher the temperature, the more violent the dance. Brownian motion is therefore a visible fingerprint of the unseen molecular motion that kinetic theory assumes — a bridge from the microscopic world straight to something you can watch under a microscope. The same idea explains how a drop of ink spreads through still water, or a whiff of perfume crosses a room: **diffusion**, the gradual mixing of molecules driven purely by their random thermal motion.

Limiting conditions & assumptions. The clean “free molecules, brief collisions” picture holds for gases at ordinary and low pressures. Compress a gas hard and the molecules are forced close enough that intermolecular forces (and their finite size) stop being negligible — the gas starts behaving like a real, non-ideal gas, and eventually condenses into a liquid where the forces dominate. Likewise, the idea that molecules are tiny hard spheres is itself an approximation: real molecules have fuzzy electron clouds and (for polyatomic molecules) shapes, but the hard-sphere picture is good enough for counting collisions and estimating sizes.

Section 2: Key Facts & Definitions

Atomic hypothesis & Dalton's atomic theory. Matter is composed of atoms; atoms of an element are identical in mass and properties; compounds form when atoms combine in simple whole-number ratios. This explains the laws of *definite proportions* (a compound always contains the same elements in the same mass ratio) and *multiple proportions*.

Gay-Lussac's law of combining volumes. When gases react, the volumes of the reacting gases and of the gaseous products bear simple whole-number ratios to one another (at the same temperature and pressure). For example, two volumes of hydrogen combine with one volume of oxygen to give two volumes of water vapour (2 : 1 : 2).

Avogadro's hypothesis (law). Equal volumes of all gases, under the same conditions of temperature and pressure, contain equal numbers of molecules. This single idea reconciles Dalton's atoms with Gay-Lussac's volume ratios, and leads directly to:

- **Avogadro's number** $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ — the number of particles in one mole.
- **Mole:** the amount of substance containing N_A entities; $n = \frac{N}{N_A} = \frac{m}{M}$.

Intermolecular force — qualitative form: - Strongly **repulsive** at very small separation ($r < r_0$): molecules resist being squeezed together. - Weakly **attractive** at larger separation ($r > r_0$): pulls molecules together. - **Zero net force** at the equilibrium separation $r = r_0$ (the minimum of the intermolecular potential energy). - Negligible at large r (a few

molecular diameters), which is the situation in a dilute gas.

Scales worth remembering: - Typical molecular **diameter** $d \sim 1\text{--}3 \text{ \AA} = 10^{-10} \text{ m}$. - In solids/liquids, neighbours are about one diameter apart. - In a gas at ordinary conditions, the mean intermolecular separation is roughly **10 times** the spacing in solids/liquids; the **mean free path** is about **20 times** the interatomic distance and about **200 times** the molecular size — which is precisely why gases are dilute and (nearly) ideal.

Evidence that matter is molecular and in motion: - **Brownian motion** — the ceaseless random jiggling of microscopic particles suspended in a fluid, caused by uneven molecular bombardment; direct visual proof of molecular motion. - **Diffusion** — gases and liquids spontaneously intermix (ink in water, scent across a room) because their molecules wander randomly. - **The oil-film experiment** — a drop of oil spreading to a one-molecule-thick layer lets us estimate molecular size ($\sim 10^{-9}\text{--}10^{-10} \text{ m}$). - **Definite chemical proportions and combining volumes** — the simple whole-number ratios in which substances react are explained only if matter comes in discrete atomic units.

Section 3: Key Procedures & Logical Framework

Procedure A — Estimating the number density and intermolecular spacing of a gas.

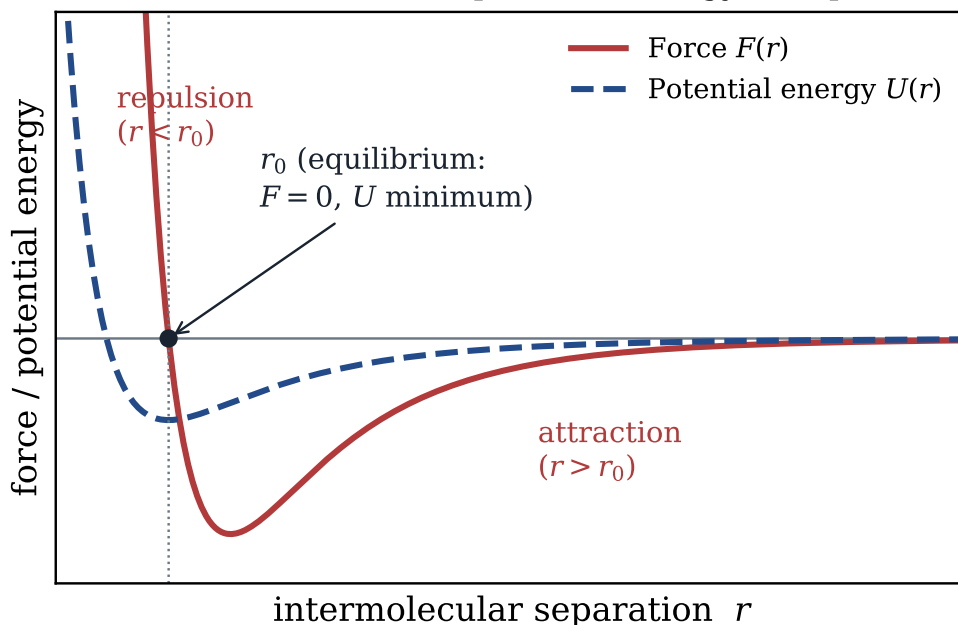
1. From the ideal gas law, the number of molecules per unit volume is $n_0 = \frac{N}{V} = \frac{P}{k_B T}$ (or use 1 mole = 22.4 L at STP).
2. The average volume “owned” by one molecule is $1/n_0$.
3. The mean separation between molecules is roughly the cube root of that volume, $\ell \approx \left(\frac{1}{n_0}\right)^{1/3} = \left(\frac{V}{N}\right)^{1/3}$.
4. Compare ℓ with the molecular diameter d : for a gas at STP, ℓ comes out about ten times d , confirming that the molecules are mostly far apart.

Procedure B — Estimating the fraction of space actually occupied by molecules. 1. Model each molecule as a sphere of radius $r = d/2$; its volume is $\frac{4}{3}\pi r^3$. 2. Multiply by N_A to get the molecular volume in one mole: $V_{\text{mol}} = N_A \cdot \frac{4}{3}\pi r^3$. 3. Divide by the actual molar volume (22.4 L at STP): the ratio (typically $\sim 10^{-3}\text{--}10^{-4}$) is the fraction of the container actually filled by molecular matter. Its tininess is the quantitative statement of “a gas is mostly empty space.”

The logic of the force-separation curve. Why must the force be attractive far out but repulsive close in? If molecules only attracted, all matter would collapse to a point; if they only repelled, no solid or liquid could ever form. The existence of stable solids and liquids *requires* an equilibrium separation r_0 — a balance point. At r_0 the potential energy is a minimum (most stable), the net force is zero, and small displacements produce a restoring force, which is exactly why atoms in a solid *vibrate* about fixed sites rather than drifting away or collapsing inward. Formally, the force is the negative slope of the potential energy, $F = -\frac{dU}{dr}$: the curve of $U(r)$ dips to a minimum at r_0 , and the force (its slope) passes through zero there, turning repulsive on the steep inner wall and attractive on the gentle outer side.

Procedure C — Estimating molecular size by the oil-film (monolayer) method. A classic way to “measure” a molecule without seeing it: 1. Take a tiny drop of oil of known volume V and let it spread on a clean water surface dusted with fine powder. 2. The oil spreads until it is just **one molecule thick** — a monomolecular layer — covering a measured area A . 3. If the film is one molecule thick, that thickness is the molecular diameter: $d \approx \frac{V}{A}$ (volume = area $\times$ thickness). 4. Typical results give $d \sim 10^{-9}\text{--}10^{-10} \text{ m}$, confirming the nanometre-to-Ångström scale of molecules. The beauty of the method is that it converts an invisible length into two easily measured quantities, a volume and an area.

Intermolecular force and potential energy vs separation



Intermolecular force (red) and potential energy (blue) versus separation r ; the net force is zero and the potential energy is a minimum at the equilibrium separation r_0 .

Section 4: Worked Examples

Example 1 — [CBSE Board level] Estimate the fraction of the total volume that is actually occupied by the molecules themselves in one mole of oxygen gas at STP. Take the diameter of an oxygen molecule to be $3 \text{ \AA}$.

Given: $d = 3 \text{ \AA} \Rightarrow r = 1.5 \text{ \AA} = 1.5 \times 10^{-10} \text{ m}$; molar volume at STP $= 22.4 \text{ L} = 22.4 \times 10^{-3} \text{ m}^3$; $N_A = 6.022 \times 10^{23}$.

Molecular volume in one mole:

$$\begin{aligned} V_{\text{mol}} &= N_A \cdot \frac{4}{3}\pi r^3 = 6.022 \times 10^{23} \times \frac{4}{3}\pi (1.5 \times 10^{-10})^3 \\ &= 6.022 \times 10^{23} \times 1.41 \times 10^{-29} \approx 8.5 \times 10^{-6} \text{ m}^3 \end{aligned}$$

Fraction:

$$\frac{V_{\text{mol}}}{V_{\text{molar}}} = \frac{8.5 \times 10^{-6}}{22.4 \times 10^{-3}} \approx 3.8 \times 10^{-4}$$

Only about 4×10^{-4} — roughly 0.04% — of the gas's volume is actually molecules. A gas is, overwhelmingly, empty space; this is the quantitative justification for treating molecules as point masses.

Example 2 — [NEET Speed Trap] Under the same temperature and pressure, 3 litres of hydrogen react completely with a certain volume of oxygen to form water vapour. Using Avogadro's law and Gay-Lussac's law of combining volumes ($2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$), find the volume of oxygen consumed and the volume of water vapour produced.

Speed method: volumes react in the same ratio as molecules (Avogadro), i.e. $2 : 1 : 2$. So oxygen $= \frac{1}{2} \times 3 = 1.5 \text{ L}$, and water vapour $= 3 \text{ L}$ (same as the hydrogen, by the $2 : 2$ ratio).

Oxygen = 1.5 L; water vapour = 3 L.

The trap: students often forget that the $2 : 1 : 2$ ratio applies to *volumes* only because of Avogadro's law (equal volumes $\Rightarrow$ equal molecules), and may wrongly conserve volume

(expecting $3 + 1.5 = 4.5$ L of product). Volume is *not* conserved in gaseous reactions — molecule count and the balanced equation rule.

Example 3 — [JEE Main level] Estimate the mean separation between molecules of an ideal gas at STP, and compare it with a molecular diameter of 3 Å.

Concept — number density then cube-root spacing. At STP one mole (N_A molecules) occupies $22.4 \times 10^{-3} \text{ m}^3$, so the volume per molecule is

$$\frac{V}{N} = \frac{22.4 \times 10^{-3}}{6.022 \times 10^{23}} = 3.72 \times 10^{-26} \text{ m}^3$$

The mean separation is the cube root:

$$\ell \approx (3.72 \times 10^{-26})^{1/3} \approx 3.34 \times 10^{-9} \text{ m} = 33.4 \text{ Å}$$

Comparison: $\frac{\ell}{d} = \frac{33.4}{3} \approx 11$.

The molecules sit, on average, about 11 molecular diameters apart — an order of magnitude more than in a solid or liquid. This is the concrete meaning of the statement that gas spacing is “about ten times” the spacing in condensed matter, and why intermolecular forces are negligible in a gas.

Example 4 — [JEE Advanced level] The density of liquid nitrogen is about 810 kg m^{-3} and its molar mass is 28 g mol^{-1} . Estimate (a) the number of molecules per unit volume in the liquid, and (b) the mean spacing between molecules, and hence (c) comment on how this spacing compares with that of nitrogen *gas* at STP (from Example 3, $\sim 33 \text{ Å}$).

Part (a). Number density $= \frac{\rho N_A}{M} = \frac{810 \times 6.022 \times 10^{23}}{28 \times 10^{-3}} \approx 1.74 \times 10^{28} \text{ m}^{-3}$.

Part (b). Mean spacing $\ell \approx (1/n_0)^{1/3} = (1/1.74 \times 10^{28})^{1/3} = (5.75 \times 10^{-29})^{1/3} \approx 3.86 \times 10^{-10} \text{ m} \approx 3.9 \text{ Å}$.

Part (c). In the *liquid* the molecules are about 3.9 Å apart — essentially touching, since that is close to a molecular diameter. In the *gas* at STP they were about 33 Å apart, roughly **8-9 times** farther. **Liquefaction collapses the molecules from “far apart and free” to “touching and interacting”** — which is precisely why the ideal-gas model fails for liquids: the intermolecular forces that were negligible in the gas now dominate completely.

Section 5: Practice Exercises

1. **[CBSE]** State Avogadro’s hypothesis and use it to explain why equal volumes of hydrogen and oxygen at the same temperature and pressure contain equal numbers of molecules.
2. **[NEET]** At STP, one mole of any ideal gas occupies 22.4 L. How many molecules are present in 5.6 L of an ideal gas at STP?
3. **[JEE Main]** A vessel contains 1.0 g of hydrogen ($M = 2$) at STP. Find the number of molecules and estimate the volume occupied.
4. **[JEE Main]** Sketch (qualitatively) the variation of intermolecular force with separation r , and mark the equilibrium separation r_0 . State the sign of the force on either side of r_0 .
5. **[JEE Advanced]** The diameter of a water molecule is about 3 Å. Using the density of water (1000 kg m^{-3} , $M = 18 \text{ g mol}^{-1}$), estimate the mean spacing between molecules in liquid water and compare it with the molecular diameter.

Answers: 1. equal volumes $\Rightarrow$ equal molecule count, so the volume ratio equals the molecule-number ratio 2. 1.5×10^{23} molecules ($\frac{5.6}{22.4} = 0.25$ mol) 3. $0.5 \text{ mol} \times N_A = 3.0 \times 10^{23}$ molecules; $\approx 11.2 \text{ L}$ 4. repulsive for $r < r_0$, attractive for $r > r_0$, zero at r_0 5. $n_0 = \rho N_A / M \approx 3.34 \times 10^{28} \text{ m}^{-3}$, $\ell \approx 3.1 \text{ \AA}$ — about one molecular diameter (molecules touching).

Section 6: Crack the MCQ

Q1. The net intermolecular force between two molecules is zero when their separation is: (A) zero (B) equal to the equilibrium separation r_0 (C) infinite (D) equal to one molecular diameter

Correct: (B). At r_0 the attractive and repulsive contributions exactly cancel; this is the stable equilibrium separation (potential-energy minimum). *Trap (A):* at $r \rightarrow 0$ the repulsive force is maximal, not zero. *(C):* at infinite separation the force *approaches* zero but is never the defining equilibrium point, and molecules there are not bound. *(D):* a molecular diameter is a rough scale, but the exact zero-force point is r_0 , defined by the force balance, not by the diameter.

Q2. Which statement best explains why a gas can be compressed far more easily than a liquid? (A) Gas molecules are smaller than liquid molecules (B) In a gas the molecules are far apart with much empty space between them (C) Gas molecules have no mass (D) Gas molecules attract each other strongly

Correct: (B). A gas is mostly empty space (molecular volume $\sim 10^{-4}$ of the total), so compression mainly reduces the gaps between molecules. In a liquid the molecules already touch, leaving little to compress. *Trap (A):* same molecules form the liquid and gas; size is unchanged. *(C):* molecules certainly have mass. *(D):* strong attraction characterises the liquid, not the gas — and it would resist, not ease, behaviour.

Q3. Two volumes of gas A combine with one volume of gas B to form two volumes of a gaseous compound, all measured at the same T and P . This is a direct illustration of: (A) Boyle's law (B) Charles' law (C) Gay-Lussac's law of combining volumes together with Avogadro's hypothesis (D) Dalton's law of partial pressures

Correct: (C). Simple whole-number volume ratios in a gaseous reaction are Gay-Lussac's law of combining volumes; interpreting those volumes as proportional to molecule numbers requires Avogadro's hypothesis. *Trap (A)/(B):* Boyle's and Charles' laws describe how a fixed gas sample responds to pressure or temperature, not reacting-volume ratios. *(D):* Dalton's law concerns the pressures of a non-reacting mixture, not combining volumes.

Q4. The mean free path of a gas molecule is roughly: (A) equal to the molecular diameter (B) about 10 times the molecular diameter (C) about 1000 times the molecular diameter (D) about the size of the container, always

Correct: (C). In an ordinary gas the mean free path is of order a few-hundred molecular diameters (about a hundred times the interatomic spacing of solids), which is why a gas is dilute and behaves nearly ideally. *Trap (A)/(B):* these badly underestimate it — they describe spacings, not the free-flight distance. *(D):* the mean free path equals the container size only in a strong vacuum (very low density), not in general.

Section 7: Common Student Pitfalls & Pro-Tips

- **Confusing molecular spacing with mean free path.** The mean *separation* between neighbours in a gas ($\sim 10\times$ a diameter) is much smaller than the mean *free path* ($\sim 1000\times$ a diameter, the average distance flown between collisions). They are

different quantities.

- **Expecting volume to be conserved in gaseous reactions.** Combining-volume ratios follow the balanced equation and Avogadro's law, not conservation of total volume. Two volumes plus one volume need not give three.
 - **Thinking molecules only attract.** Without short-range repulsion, matter would collapse. The force is repulsive below r_0 and attractive above it — both are essential.
 - **Treating the gas as “full” of molecules.** A gas is $\sim 99.96\%$ empty space at STP. Picturing it as densely packed leads to wrong intuitions about compressibility and the validity of the point-mass assumption.
 - **Misattributing Brownian motion.** The jiggling particle you observe is *not* a molecule — it is a much larger suspended grain being kicked around *by* molecules. Brownian motion is indirect evidence: you see the effect (the grain's dance), not the molecules themselves.
 - **Pro-Tip (orders of magnitude):** carry three numbers in your head — molecular size $\sim 10^{-10}$ m, gas spacing $\sim 10\times$ that, mean free path $\sim 1000\times$ that. Most conceptual questions in this section can be answered just by comparing these scales.
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Section 8: Cheat Sheet

- **Atomic hypothesis / Dalton:** matter is atoms; atoms combine in simple whole-number ratios.
- **Gay-Lussac (combining volumes):** reacting gas volumes are in simple whole-number ratios at the same T, P .
- **Avogadro's law:** equal volumes (same T, P) $\implies$ equal numbers of molecules; $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$.
- **Intermolecular force:** repulsive for $r < r_0$, attractive for $r > r_0$, zero at r_0 (PE minimum).
- **States of matter:** solid (touching, fixed, vibrating) $\rightarrow$ liquid (touching, mobile) $\rightarrow$ gas (far apart, free) as separation grows and forces weaken.
- **Evidence of molecules in motion:** Brownian motion (visible jiggling), diffusion (spontaneous mixing), oil-film monolayer (gives molecular size).
- **Scales:** molecule $\sim 10^{-10}$ m; gas spacing $\sim 10\times$; mean free path $\sim 1000\times$; molecular volume $\sim 10^{-4}$ of container at STP.
- **Memory aid:** “Close and caged (solid), close and loose (liquid), far and free (gas).”

Sub-topic 1: Ideal Gas and Gas Laws

Exam Relevance: A guaranteed scorer. CBSE Boards almost always carry a 1–3 mark gas-law or $PV = nRT$ application. NEET asks 1 direct numerical nearly every year (often density-molar-mass or a mixture). JEE Main treats it as a calculation backbone, and JEE Advanced hides it inside multi-bulb or piston problems where mole-conservation is the real test.

Section 1: Concept Intuition

Picture a Mumbai local at 9 a.m. The compartment is packed, people are shoving against the doors, and the “pressure” on the walls is enormous. Now imagine the same compartment at 2 p.m. — a handful of passengers, drifting around, barely touching the sides. A gas is exactly this crowd of commuters, except the passengers are molecules, the shoving is molecular collisions, and the push felt by the walls is what we call **pressure**.

A real gas is messy: molecules have size, they tug on one another, and they occasionally clump. To make the physics tractable, we invent a clean idealisation — the **ideal gas** — by stripping away those complications. An ideal gas is a crowd of point-like particles that take up no real volume, exert no force on one another except during instantaneous collisions, and otherwise fly in straight lines. Because nothing is “sticking” the molecules together, an ideal gas has *no* potential energy of interaction; all of its internal energy is the kinetic energy of motion.

Think about it this way... an ideal gas is like a hall full of perfectly smooth marbles that never attract and never deform — they only knock about and bounce off the walls. The “ideal” in ideal gas isn’t a comment on quality; it just means the gas obeys one beautifully simple law, $PV = nRT$, at *every* temperature and pressure. No real substance does this perfectly, but every real gas comes very close in the right regime.

It is worth naming the assumptions hidden inside this picture, because every later result in the chapter rests on them. An ideal gas is built from these idealisations: the molecules are treated as **point masses** whose own volume is utterly negligible compared with the container, so the volume of the gas is effectively the volume of the container; there are **no intermolecular forces** except during collisions, which makes the potential energy of the gas zero and its internal energy purely the kinetic energy of motion; all collisions (molecule–molecule and molecule–wall) are **perfectly elastic**, so no kinetic energy is lost; between collisions the molecules move freely in **straight lines**, spending negligible time actually colliding; the gas is a **pure substance** with all molecules identical; and **gravity is ignored**, so the density is taken uniform throughout the container. Strip any one of these away and you are back to a real gas with all its complications.

And what is that regime? A real gas behaves ideally when its molecules are **far apart and moving fast** — that is, at **low pressure and high temperature**. At low pressure the molecules are so spread out that the time they spend near each other (where attractions matter) is negligible. At high temperature their kinetic energy so overwhelms any intermolecular pull that the stickiness simply doesn’t register. Squeeze a gas hard (high pressure) or chill it (low temperature) and the idealisation breaks: molecules crowd together, attractions bite, the finite size of molecules starts to matter, and deviations appear — eventually the gas condenses to a liquid, something an ideal gas can never do.

So keep these **limiting conditions** in mind. The ideal-gas description is excellent for everyday air, but it fails (a) near condensation, (b) at very high pressure where molecular volume is no longer negligible compared to the container, and (c) at very low temperature where attractive forces dominate. Hydrogen and helium — small, weakly-interacting molecules — stay “ideal” over a wide range; water vapour near 100 °C or carbon dioxide near room temperature deviate noticeably.

One more intuition worth locking in: for a *fixed amount* of gas, only three numbers describe its state — pressure P , volume V , and absolute temperature T . Change any one and at least one other must respond. The “gas laws” are simply the rules for that bookkeeping when you hold one variable fixed, and $PV = nRT$ is the master equation that contains all

of them at once.

Section 2: Key Formulas & Definitions

The four experimental gas laws (each for a *fixed mass* of gas):

Law	Held constant	Statement	Equation
Boyle's	T	$V \propto 1/P$	$P_1 V_1 = P_2 V_2$
Charles'	P	$V \propto T$	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
Gay-Lussac's (Pres- sure)	V	$P \propto T$	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
Avogadro's	P, T	$V \propto n$	equal $V \Rightarrow$ equal N

Combined gas law (fixed mass):

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Ideal gas equation (the master relation):

$$PV = nRT$$

where n is the number of moles. Equivalent forms you must recognise instantly:

$$PV = \frac{m}{M}RT \quad PV = Nk_B T \quad P = \frac{\rho RT}{M}$$

- m = total mass of gas, M = molar mass, so $n = m/M$.
- N = number of molecules, with $N = nN_A$.
- $\rho = m/V$ = density (the density form is the one to reach for whenever density or molar mass appears).

A quick naming convention that prevents confusion in problems: $PV = nRT$ is the **container form** of the gas law — used whenever the gas is enclosed in a vessel of definite volume — while $P = \rho RT/M$ is the **atmospheric form**, used when you are handed a density rather than a volume (open air, a gas column, an unconfined region). They are the same physics; pick the form that matches the data you are given.

Graham's law, listed below, is really a consequence of the kinetic picture rather than a fourth state variable, but it lives naturally here because it is phrased entirely in terms of the macroscopic quantities density and molar mass.

Dalton's law of partial pressures — for a non-reacting mixture in a common volume at common temperature:

$$P_{\text{total}} = P_1 + P_2 + \cdots + P_n = \frac{RT}{V} (n_1 + n_2 + \cdots)$$

Graham's law of diffusion — at fixed T, P :

$$\text{rate of diffusion} \propto \frac{1}{\sqrt{\rho}} \propto \frac{1}{\sqrt{M}}$$

Constants, units and dimensional formulae:

- **Universal gas constant** $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1} \approx 2.0 \text{ cal mol}^{-1}\text{K}^{-1}$. Dimensional formula: $[R] = [M L^2 T^{-2} \text{ mol}^{-1} K^{-1}]$.
- **Boltzmann constant** $k_B = R/N_A = 1.38 \times 10^{-23} \text{ J K}^{-1}$ — the gas constant *per molecule*. $[k_B] = [M L^2 T^{-2} K^{-1}]$.
- **Avogadro number** $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$.
- Pressure $[P] = [M L^{-1} T^{-2}]$ (SI: Pa); Volume $[V] = [L^3]$ (SI: m³).
- **STP**: $T = 273.15 \text{ K}$, $P = 1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$, molar volume = 22.4 L mol^{-1} .

The molar volume figure is worth committing to memory: because Avogadro's law guarantees that equal moles of *any* ideal gas occupy equal volumes at the same P and T , that single number 22.4 L mol^{-1} works for every gas at STP and turns many “find the moles / find the molar mass” questions into one-line arithmetic. A second small but exam-relevant fact: for an ideal gas the volume-expansion coefficient and the pressure-expansion coefficient are *equal*, $\alpha = \beta = \frac{1}{273}$ per °C, a direct consequence of $V \propto T$ at constant P and $P \propto T$ at constant V sharing the same absolute-temperature scale.

Section 3: Key Procedures & Derivation

Deriving the ideal gas equation by stitching the laws together.

We have three “one-variable-fixed” laws. The trick is to change the state in two stages and let only one law act at a time, so that we never break any of the experimental rules.

Setup. Take a fixed mass of gas in state (P_1, V_1, T_1) . We want to reach state (P_2, V_2, T_2) . Introduce an **intermediate state** (P_2, V', T_1) — same final pressure, but still at the *initial* temperature.

Step 1 — change pressure at constant temperature (Boyle's law). Going from (P_1, V_1, T_1) to (P_2, V', T_1) , temperature is held fixed, so Boyle's law applies:

$$P_1 V_1 = P_2 V' \Rightarrow V' = \frac{P_1 V_1}{P_2}$$

Why this step: by fixing T first, we isolate the pressure-volume trade-off with no temperature interference.

Step 2 — change temperature at constant pressure (Charles' law). Now go from (P_2, V', T_1) to (P_2, V_2, T_2) . Pressure is held at P_2 , so Charles' law applies:

$$\frac{V'}{T_1} = \frac{V_2}{T_2} \Rightarrow V' = \frac{V_2 T_1}{T_2}$$

Why this step: with P now fixed, the only remaining trade-off is volume against temperature.

Step 3 — eliminate the intermediate V' . Both steps gave an expression for V' ; set them equal:

$$\frac{P_1 V_1}{P_2} = \frac{V_2 T_1}{T_2} \Rightarrow \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

This is the combined gas law: the quantity PV/T is a constant for a fixed mass of gas.

Step 4 — fix the constant using Avogadro's law. The constant must be proportional to the *amount* of gas, because Avogadro's law says volume scales with the number of moles at fixed P, T . Writing the constant as nR , where R is the same for every gas (this universality is the experimental discovery — $PV/nT \rightarrow R$ for all gases as $P \rightarrow 0$):

$$\frac{PV}{T} = nR \Rightarrow \boxed{PV = nRT}$$

Why R is “universal.” The remarkable experimental fact is that the same constant R works for every gas. If you measure the quantity PV/nT for any gas and let the pressure approach zero (the ideal limit), the ratio approaches one and the same value, $8.314 \text{ J mol}^{-1}\text{K}^{-1}$, regardless of whether the gas is hydrogen, oxygen, or argon. This universality is precisely why R earns the name *universal* gas constant — it encodes nothing about the identity of the gas, only the shared low-density behaviour of all gases.

A short derivation of Dalton’s law from $PV = nRT$. Suppose a vessel of volume V at temperature T holds a non-reacting mixture of gases with $n_1, n_2, \dots$ moles. Each component, behaving ideally, ignores the others (no intermolecular forces between unlike molecules either), so each would individually exert a *partial pressure*

$$P_i = \frac{n_i RT}{V}$$

Adding these contributions, the total pressure on the wall is

$$P_{\text{total}} = \sum_i P_i = \frac{RT}{V} \sum_i n_i = \frac{n_{\text{total}} RT}{V}$$

which is exactly Dalton’s law. A neat corollary drops out immediately: dividing P_i by P_{total} ,

$$\frac{P_i}{P_{\text{total}}} = \frac{n_i}{n_{\text{total}}} = x_i$$

so each gas’s share of the pressure equals its **mole fraction** x_i . This is the single most useful mixture result for exams.

Procedure — choosing the right form in a problem: 1. If the *amount of gas is fixed* and only P, V, T change $\rightarrow$ use the combined law $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$. (Units of P and V can be any consistent pair; T **must** be in kelvin.) 2. If *mass or density or molar mass* appears $\rightarrow$ use $PV = \frac{m}{M} RT$ or $P = \frac{\rho RT}{M}$. 3. If *number of molecules* appears $\rightarrow$ use $PV = N k_B T$. 4. If a *mixture* is involved $\rightarrow$ add partial pressures (Dalton), i.e. work with total moles.

The single most common slip: leaving temperature in Celsius. Every form above demands **absolute temperature in kelvin**, $T(\text{K}) = t(^{\circ}\text{C}) + 273.15$.

Section 4: Worked Examples

Example 1 — [CBSE Board level] A sealed cylinder holds 2.0 L of an ideal gas at 27°C and a pressure of 1.0 atm. The gas is compressed to 0.50 L and simultaneously heated to 127°C . Find the new pressure.

Given: $P_1 = 1.0 \text{ atm}$, $V_1 = 2.0 \text{ L}$, $T_1 = 27 + 273 = 300 \text{ K}$; $V_2 = 0.50 \text{ L}$, $T_2 = 127 + 273 = 400 \text{ K}$.

The amount of gas is fixed, so apply the combined gas law:

$$\begin{aligned} \frac{P_1 V_1}{T_1} &= \frac{P_2 V_2}{T_2} \Rightarrow P_2 = P_1 \cdot \frac{V_1}{V_2} \cdot \frac{T_2}{T_1} \\ P_2 &= 1.0 \times \frac{2.0}{0.50} \times \frac{400}{300} = 1.0 \times 4 \times \frac{4}{3} = 5.33 \text{ atm} \end{aligned}$$

The new pressure is $\approx 5.3 \text{ atm}$. Note we never converted litres to m^3 — in a ratio equation the volume units cancel, but the temperatures still had to be in kelvin.

Example 2 — [NEET Speed Trap] At STP, a certain gas has a density of 1.25 g L^{-1} . Identify its molar mass (and hence the likely gas).

Speed method: At STP, one mole of *any* ideal gas occupies 22.4 L. So molar mass is just density times molar volume:

$$M = \rho \times V_m = 1.25 \text{ g L}^{-1} \times 22.4 \text{ L mol}^{-1} = 28 \text{ g mol}^{-1}$$

$M = 28 \text{ g mol}^{-1} \rightarrow$ **nitrogen (N_2) or carbon monoxide (CO).**

The trap: the formula $M = \rho RT/P$ tempts students into a full plug-in. If you do go that route, the killer mistake is using $T = 0$ (reading “STP” as 0°C and forgetting the $+273$), which makes $M = 0$ — an instant red flag. At STP, always shortcut through $M = 22.4 \rho$ and skip R, T, P entirely.

Example 3 — [JEE Main level] A rigid 10.0 L vessel contains 4.0 g of oxygen (O_2 , $M = 32$) and 2.0 g of helium (He , $M = 4$) at 300 K. Find (a) the total pressure exerted on the walls, and (b) the partial pressure of helium.

Concept 1 — moles from mass:

$$n_{\text{O}_2} = \frac{4.0}{32} = 0.125 \text{ mol}, \quad n_{\text{He}} = \frac{2.0}{4} = 0.50 \text{ mol}$$

$$n_{\text{total}} = 0.625 \text{ mol}$$

Concept 2 — Dalton + ideal gas equation. Total pressure uses total moles (convert $V = 10.0 \text{ L} = 1.0 \times 10^{-2} \text{ m}^3$):

$$P_{\text{total}} = \frac{n_{\text{total}} RT}{V} = \frac{0.625 \times 8.314 \times 300}{1.0 \times 10^{-2}} \approx 1.56 \times 10^5 \text{ Pa} \approx 1.54 \text{ atm}$$

(b) Helium’s partial pressure scales with its mole fraction, $x_{\text{He}} = 0.50/0.625 = 0.80$:

$$P_{\text{He}} = x_{\text{He}} P_{\text{total}} = 0.80 \times 1.56 \times 10^5 \approx 1.25 \times 10^5 \text{ Pa}$$

$P_{\text{total}} \approx 1.56 \times 10^5 \text{ Pa}$, $P_{\text{He}} \approx 1.25 \times 10^5 \text{ Pa}$. The lesson: each gas fills the whole vessel independently; pressures add, masses do not tell you pressure until converted to moles.

Example 4 — [JEE Advanced level] Two identical glass bulbs, each of volume V , are joined by a thin tube of negligible volume and contain an ideal gas at a common pressure P_0 and temperature $T_0 = 300 \text{ K}$ throughout. One bulb is now immersed in a bath at $T_1 = 400 \text{ K}$ while the other is held at $T_2 = 300 \text{ K}$. Find the new common pressure P in the connected system.

The key idea: the bulbs are connected, so gas flows until the **pressure is common** — but the **total number of moles is conserved**. Temperature differs between bulbs, so each contributes a different number of moles at the same P .

Initial total moles (both bulbs at T_0 , pressure P_0):

$$n = \frac{P_0 V}{RT_0} + \frac{P_0 V}{RT_0} = \frac{2P_0 V}{RT_0}$$

Final total moles (common pressure P ; one bulb at T_1 , one at T_2):

$$n = \frac{PV}{RT_1} + \frac{PV}{RT_2} = \frac{PV}{R} \left(\frac{1}{T_1} + \frac{1}{T_2} \right)$$

Conserve moles and cancel V/R :

$$\frac{2P_0}{T_0} = P \left(\frac{1}{T_1} + \frac{1}{T_2} \right)$$

$$P = \frac{2P_0}{T_0} \cdot \frac{1}{\frac{1}{T_1} + \frac{1}{T_2}} = \frac{2P_0}{T_0} \cdot \frac{T_1 T_2}{T_1 + T_2}$$

Substitute $T_0 = 300$, $T_1 = 400$, $T_2 = 300$:

$$P = \frac{2P_0}{300} \cdot \frac{400 \times 300}{700} = \frac{2P_0}{300} \cdot \frac{120000}{700} = 2P_0 \times \frac{400}{700} = \frac{8P_0}{7} \approx 1.14 P_0$$

The pressure rises to $\frac{8P_0}{7} \approx 1.14 P_0$. Heating one half drives gas into the cooler bulb, and the system settles at a higher common pressure. The whole problem turns on the phrase “total moles conserved” — write that equation first and the algebra follows.

Section 5: Practice Exercises

1. **[CBSE]** A gas occupies 500 mL at 300 K and 1.0 atm. To what temperature must it be heated at constant pressure to occupy 750 mL?
2. **[NEET]** The density of a gas at 546 K and 2.0 atm is 0.90 g L⁻¹. Find its molar mass. (Use $R = 0.0821$ L atm mol⁻¹K⁻¹.)
3. **[JEE Main]** A 5.0 L flask contains 1.4 g of nitrogen ($M = 28$) and 3.2 g of oxygen ($M = 32$) at 290 K. Find the total pressure (in Pa) and the mole fraction of oxygen.
4. **[JEE Main]** An open flask is heated from 27 °C to 327 °C at constant atmospheric pressure. What fraction of the air originally present escapes?
5. **[JEE Advanced]** A vertical cylinder closed at the bottom and fitted with a frictionless, massless piston of area A contains an ideal gas. Atmospheric pressure is P_0 . A mass m is placed on the piston and the gas, initially of length L_0 at temperature T_0 , is then heated to T . Find the new gas column length L in terms of the given quantities. (Hint: with the mass on top, the gas pressure stays constant — this is an isobaric change.)

Answers: 1. 450 K 2. ≈ 20.1 g mol⁻¹ 3. $P_{\text{total}} \approx 7.23 \times 10^4$ Pa, $x_{\text{O}_2} \approx 0.667$ 4. $\frac{1}{2}$ (half the air escapes) 5. $L = L_0 \frac{T}{T_0}$ (pressure is constant at $P_0 + mg/A$, so $L \propto T$).

Section 6: Crack the MCQ

Q1. A real gas most closely approaches ideal behaviour at: (A) high pressure and low temperature (B) low pressure and high temperature (C) high pressure and high temperature (D) low pressure and low temperature

Correct: (B). Low pressure keeps molecules far apart (intermolecular forces negligible) and high temperature makes kinetic energy dominate any residual attraction — exactly the ideal-gas regime. *Trap (A):* the reverse condition; high pressure + low temperature is where a gas *condenses*, the worst case for ideality. *(C):* high pressure alone reintroduces finite-molecular-volume effects even if T is high. *(D):* low temperature lets attractions dominate. Students who memorise “extremes are good” without the physics fall for these.

Q2. A fixed mass of ideal gas at 27 °C is heated at constant volume until its pressure doubles. The final temperature is: (A) 54 °C (B) 327 °C (C) 600 °C (D) 273 °C

Correct: (B). At constant volume, $P \propto T$ (in kelvin). $T_1 = 300$ K, and doubling pressure needs $T_2 = 600$ K = 327°C . *Trap (A):* doubling the Celsius value ($27 \rightarrow 54$) — the classic Celsius error; pressure is proportional to *absolute* temperature, not Celsius. *(C):* doubling 300 correctly to get 600 but then forgetting to convert back to Celsius. *(D):* confusing the +273 offset with the answer itself.

Q3. Equal masses of helium ($M = 4$) and oxygen ($M = 32$) are mixed in a container. The ratio of the partial pressure of helium to that of oxygen is: (A) 1 : 1 (B) 1 : 8 (C) 8 : 1 (D) 4 : 32

Correct: (C). Partial pressure is set by *moles*, and for equal masses $n = m/M$, so $n_{\text{He}} : n_{\text{O}_2} = \frac{1}{4} : \frac{1}{32} = 8 : 1$. Pressures share this ratio. *Trap (A):* assuming equal mass $\Rightarrow$ equal pressure (ignores molar mass). *(B):* inverting the ratio — using M directly instead of $1/M$. *(D):* quoting the molar-mass ratio 4 : 32 which again forgets that lighter molecules give *more* moles per gram.

Q4. Two gases A and B diffuse through a porous plug under identical conditions. Gas A diffuses 1.5 times as fast as gas B. The ratio of their molar masses $M_A : M_B$ is: (A) 3 : 2 (B) 2 : 3 (C) 4 : 9 (D) 9 : 4

Correct: (C). By Graham's law, $\frac{r_A}{r_B} = \sqrt{\frac{M_B}{M_A}} = 1.5$. Squaring: $\frac{M_B}{M_A} = 2.25$, so $\frac{M_A}{M_B} = \frac{1}{2.25} = \frac{4}{9}$. *Trap (D):* squaring but forgetting to invert — gives $M_B : M_A$ instead of $M_A : M_B$. *(A):* using the rate ratio directly without squaring. *(B):* inverting the rate ratio and not squaring. The lighter gas diffuses faster, so the faster gas A must be the *lighter* one — a quick sanity check that rules out (D).

Section 7: Common Student Pitfalls & Pro-Tips

- **The kelvin sin.** Forgetting to convert $^\circ\text{C}$ to K wrecks any gas-law calculation. Pressure and volume can stay in any consistent units inside a ratio, but T *must* be absolute. Build a reflex: the instant you see Celsius, add 273.
- **Mass is not moles.** Pressure, Dalton's law, and Avogadro's law all care about the *number* of molecules, never the mass. Equal masses of two gases give different pressures; always convert mass to moles first.
- **Mixing up Charles' and Gay-Lussac's.** Charles' law (V/T) holds at constant **pressure**; the pressure law (P/T) holds at constant **volume**. Tag the constant variable before writing anything.
- **Treating SI as optional in $PV = nRT$.** When you use $R = 8.314$, pressure must be in pascal and volume in m^3 . If you prefer litres and atmospheres, switch to $R = 0.0821 \text{ L atm mol}^{-1}\text{K}^{-1}$. Mismatching R with the units is a silent error.
- **Pro-Tip (ratio over plug-in):** for "before/after" problems with fixed mass, never compute n . Just write $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ and cancel whatever stays constant. It is faster and immune to unit-conversion mistakes — a genuine NEET time-saver.

Section 8: Cheat Sheet

- **Master equation:** $PV = nRT = \frac{m}{M}RT = Nk_B T$; density form $P = \frac{\rho RT}{M}$.
- **Fixed-mass shortcut:** $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ — kelvin only.

- **The four laws:** Boyle $PV = \text{const}$ (T fixed); Charles $V/T = \text{const}$ (P fixed); Gay-Lussac $P/T = \text{const}$ (V fixed); Avogadro $V \propto n$.
- **Mixtures:** Dalton — $P_{\text{total}} = \sum P_i$, work with total moles; partial pressure = mole fraction $\times$ total.
- **Diffusion:** Graham — rate $\propto 1/\sqrt{M}$ (lighter = faster).
- **Constants:** $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$; $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$; molar volume at STP = 22.4 L.
- **Ideal regime memory aid:** “Low P , high T — molecules free and flighty.”
- **Quick check at STP:** $M = 22.4 \times (\text{density in g/L})$.

Sub-topic 2: Kinetic Theory and Gas Pressure

Exam Relevance: The derivation of gas pressure is a CBSE Board favourite — a full 3–5 mark derivation appears regularly, so the step-by-step logic must be airtight. JEE Main leans on the results $P = \frac{1}{3}\rho v_{rms}^2$ and $PV = \frac{2}{3}E$ in numericals, and NEET tests the proportionalities. JEE Advanced probes the *assumptions* — when the pressure expression breaks (anisotropy, non-collisional forces) — far more than the algebra.

Section 1: Concept Intuition

In Sub-topic 1 we *asserted* that gases obey $PV = nRT$. But where does that law come from? Why should pressure exist at all? Kinetic theory answers this by zooming all the way down to the molecules and asking: if a gas is nothing but a swarm of tiny particles in ceaseless random motion, what must we observe at the macroscopic scale? The triumph of the theory is that pressure, temperature, and the gas laws all fall out of Newton's laws applied to a crowd of bouncing molecules.

This is a genuinely big idea, and it took the nineteenth-century work of Clausius, Maxwell and Boltzmann to put it on a firm footing. What makes kinetic theory so celebrated is its reach from a single set of assumptions, it not only reproduces the gas laws and Avogadro's hypothesis, but also explains the specific heat capacities of gases and links measurable bulk properties — viscosity, thermal conduction, diffusion — back to molecular sizes and masses. A handful of postulates about colliding particles end up explaining an enormous swathe of observed behaviour. Crucially, the theory works for a gas because gas molecules are *far apart*: at ordinary pressure and temperature the average separation between molecules is roughly ten times their own size, so for most of their lives the molecules fly freely and ignore one another, interacting only in brief collisions. That is exactly the regime in which the idealisation below is accurate.

Picture a cricket ball machine at a practice net, firing balls at the back netting. Each ball that strikes the net pushes it outward a little and bounces back. A single ball gives a tiny, jerky shove. But fire thousands of balls per second from every angle and the net feels a steady, smooth, outward push — a *pressure*. A gas does exactly this. **Pressure is nothing more than the relentless drumbeat of molecules colliding with the container walls and rebounding.** No collisions, no pressure. The walls do not feel individual molecular hits any more than your skin feels individual air molecules; what you feel is the time-averaged force from an astronomical number of impacts.

Think about it this way... the wall of a balloon is being machine-gunned from the inside by roughly 10^{27} molecules every second. Each impact is microscopic and random, but their sheer number averages out into the constant, uniform pressure that keeps the balloon inflated. Stretch this picture and you can already *feel* the answers to exam questions: heat the gas and the molecules move faster, hit harder and more often, so pressure rises; squeeze the gas into half the volume and the same molecules strike the walls twice as often, so pressure doubles.

To turn this picture into equations, kinetic theory adopts a clean set of **postulates** — the rules of the idealised game:

1. A gas consists of a very large number of identical molecules, in constant, random motion in all directions.
2. The size of a molecule is negligible compared with the average separation between molecules and with the size of the container — molecules are effectively point masses.
3. Molecules obey Newton's laws of motion. Between collisions they travel in straight lines at constant velocity, exerting no force on one another (no long-range intermolecular forces).
4. All collisions — molecule with molecule, and molecule with wall — are perfectly elastic, conserving kinetic energy and momentum. Collisions are instantaneous (negligible duration).
5. The motion is **isotropic**: molecules are equally likely to be moving in any direction, so

on average no direction is special.

Limiting conditions & assumptions. Notice how much rides on these postulates. The whole derivation of pressure assumes collisions are elastic (so molecules rebound with undiminished speed), that the *only* forces are wall collisions (so we can ignore gravity and intermolecular attraction), and that motion is isotropic (so $\langle v_x^2 \rangle = \langle v_y^2 \rangle = \langle v_z^2 \rangle$). Break any of these and the simple result fails: introduce an electric field on charged molecules and a new, non-collisional force appears; make the container so small that one wall is preferred and isotropy is lost; or pack the gas so densely that molecular size and attraction matter, and you are back to a real gas. Kinetic theory is the physics of a *dilute* gas of *neutral* molecules feeling *nothing but each other's bounces*.

Section 2: Key Formulas & Definitions

Kinetic equation for pressure — the central result of this subtopic:

$$P = \frac{1}{3} \frac{Nm \langle v^2 \rangle}{V} = \frac{1}{3} \frac{N}{V} m \langle v^2 \rangle$$

Equivalent and very useful forms:

$$P = \frac{1}{3} \rho v_{rms}^2 \quad PV = \frac{1}{3} Nm v_{rms}^2$$

Definitions of symbols: - N = total number of molecules; $\frac{N}{V} = n_0$ = **number density** (molecules per unit volume), $[n_0] = [L^{-3}]$. - m = mass of *one* molecule (not the sample mass). - $\langle v^2 \rangle$ = **mean square speed** = $\frac{v_1^2 + v_2^2 + \dots + v_N^2}{N}$, $[\langle v^2 \rangle] = [L^2 T^{-2}]$. - $v_{rms} = \sqrt{\langle v^2 \rangle}$ = **root-mean-square speed**, SI unit $m s^{-1}$, $[LT^{-1}]$. - $\rho = \frac{Nm}{V}$ = density of the gas, SI unit $kg m^{-3}$, $[ML^{-3}]$. - P = pressure, SI unit Pa, $[ML^{-1}T^{-2}]$.

Pressure-energy relation. The total translational kinetic energy of the gas is $E = N \cdot \frac{1}{2} m \langle v^2 \rangle$. Starting from $PV = \frac{1}{3} Nm \langle v^2 \rangle$ and deliberately introducing a factor to expose the kinetic energy,

$$PV = \frac{1}{3} Nm \langle v^2 \rangle = \frac{2}{3} N \left(\frac{1}{2} m \langle v^2 \rangle \right) = \frac{2}{3} E$$

so that

$$PV = \frac{2}{3} E \quad \Longleftrightarrow \quad P = \frac{2}{3} \bar{E}_{vol}$$

where $\bar{E}_{vol} = E/V$ is the translational kinetic energy **per unit volume**. In words: *the pressure of an ideal gas equals two-thirds of its translational kinetic-energy density*. This single line is the bridge that, in Sub-topic 3, connects pressure to temperature.

Key proportionalities to memorise (at fixed everything-else):

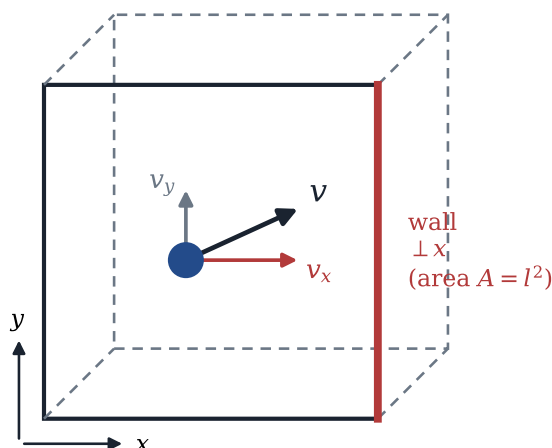
$$P \propto \frac{N}{V}, \quad P \propto m, \quad P \propto \langle v^2 \rangle \text{ (i.e. } P \propto v_{rms}^2 \text{)}$$

Section 3: CBSE Board Derivation — Pressure of an Ideal Gas

Goal: starting only from the postulates and Newton's laws, show that $P = \frac{1}{3} \frac{N}{V} m \langle v^2 \rangle$.

Setup. Consider N molecules, each of mass m , enclosed in a cube of side l , so the volume is $V = l^3$. Align the coordinate axes with the cube's edges. Focus first on a **single molecule** with velocity components (v_x, v_y, v_z) , and track its collisions with the wall that lies perpendicular to the x -axis (the shaded face of area $A = l^2$).

A molecule striking the wall of a cubical box



A single molecule of velocity v (components v_x, v_y, v_z) striking the wall perpendicular to the x -axis of a cubical box.

Step 1 — Momentum change in one collision. The collision with the wall is elastic, and the wall is perpendicular to x . So the x -component of velocity reverses, $v_x \rightarrow -v_x$, while v_y and v_z are unchanged. The change in the molecule's momentum is

$$\Delta p_x = (-mv_x) - (+mv_x) = -2mv_x$$

Physical reasoning: only the x -motion “sees” this wall; the tangential components slide along it untouched. By Newton's third law, the momentum delivered **to the wall** per collision is $+2mv_x$.

Step 2 — Time between successive collisions with the same wall. After bouncing off the right wall, the molecule must travel across to the left wall and come back before it strikes the right wall again — a round-trip distance $2l$ in the x -direction, at x -speed v_x . Hence the time between two hits on the same wall is

$$\Delta t = \frac{2l}{v_x}$$

Physical reasoning: we don't care about the y, z wandering; the rate of striking *this* wall is set purely by the x -motion.

Step 3 — Average force from one molecule. Force is momentum delivered per unit time. The molecule delivers $2mv_x$ every Δt , so the average force it exerts on the wall is

$$f_x = \frac{2mv_x}{\Delta t} = \frac{2mv_x}{2l/v_x} = \frac{mv_x^2}{l}$$

Step 4 — Sum over all N molecules. Each molecule contributes its own $mv_{x_i}^2/l$. The total force on the wall is

$$F_x = \frac{m}{l} (v_{x_1}^2 + v_{x_2}^2 + \dots + v_{x_N}^2) = \frac{m}{l} N \langle v_x^2 \rangle$$

where we have introduced the **mean square x -velocity** $\langle v_x^2 \rangle = \frac{1}{N} \sum_i v_{x_i}^2$.

Step 5 — Pressure on the wall. Divide the force by the wall area $A = l^2$:

$$P = \frac{F_x}{A} = \frac{mN \langle v_x^2 \rangle}{l \cdot l^2} = \frac{mN \langle v_x^2 \rangle}{l^3} = \frac{mN}{V} \langle v_x^2 \rangle$$

Step 6 — Use isotropy to bring in the full speed. Pressure cannot depend on which wall we chose, because the molecular motion is random and isotropic. Therefore

$$\langle v_x^2 \rangle = \langle v_y^2 \rangle = \langle v_z^2 \rangle$$

The full speed satisfies $v^2 = v_x^2 + v_y^2 + v_z^2$, so averaging gives $\langle v^2 \rangle = 3 \langle v_x^2 \rangle$, i.e. $\langle v_x^2 \rangle = \frac{1}{3} \langle v^2 \rangle$. Substituting:

$$P = \frac{1}{3} \frac{Nm}{V} \langle v^2 \rangle = \frac{1}{3} \rho v_{rms}^2$$

This is the kinetic equation for pressure. The factor of $\frac{1}{3}$ is *not* arbitrary — it is the direct fingerprint of three-dimensional isotropy. (Strictly, molecules also collide with one another between wall hits; but since intermolecular collisions are elastic and merely reshuffle velocities among molecules without changing the collection's total $\langle v^2 \rangle$, the result is unaffected.)

Why the result is more general than the cube. Two reassurances are worth stating, because exams sometimes probe them. First, the side length l cancelled completely — it appears in Step 3 and Step 5 and then divides out, so the final pressure depends only on the number density N/V , the molecular mass, and $\langle v^2 \rangle$, never on the size of the box. Second, although we derived it for a cube for algebraic convenience, the result holds for a container of *any* shape: any wall can be tiled by tiny surface patches, and the same momentum-transfer argument applied patch by patch reproduces $P = \frac{1}{3} \rho v_{rms}^2$ everywhere. The cube is a calculational scaffold, not a physical restriction. Finally, notice what the formula does *not* contain: it has no reference to the chemical identity of the gas beyond m and no reference to temperature yet — temperature enters only in Sub-topic 3, once we connect $\langle v^2 \rangle$ to T .

Section 4: Worked Examples

Example 1 — [CBSE Board level] Oxygen gas is held at a pressure of 1.0×10^5 Pa and has a density of 1.30 kg m^{-3} . Calculate the rms speed of its molecules.

Given: $P = 1.0 \times 10^5$ Pa, $\rho = 1.30 \text{ kg m}^{-3}$.

From the kinetic equation $P = \frac{1}{3} \rho v_{rms}^2$, solve for v_{rms} :

$$v_{rms} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3 \times 1.0 \times 10^5}{1.30}} = \sqrt{2.31 \times 10^5} \approx 480 \text{ m s}^{-1}$$

The rms speed is about $4.8 \times 10^2 \text{ m s}^{-1}$ — comfortably faster than the speed of sound in air, as expected for molecular motion.

Example 2 — [NEET Speed Trap] A fixed quantity of gas is heated and stirred so that its rms speed becomes **three times** its original value, while its density is kept unchanged. By what factor does the pressure change?

Speed method: At fixed density, $P \propto v_{rms}^2$. Tripling v_{rms} multiplies v_{rms}^2 by $3^2 = 9$.

$$\frac{P_2}{P_1} = \left(\frac{v_{rms,2}}{v_{rms,1}} \right)^2 = 9$$

The pressure becomes 9 times larger.

The trap: the careless answer is “3 times,” from reading P as proportional to speed. Pressure follows the *square* of the rms speed (it comes from kinetic energy, $\propto v^2$). Whenever a speed is scaled, square the factor before touching pressure or energy.

Example 3 — [JEE Main level] A rigid 2.0 L vessel contains an ideal gas at a pressure of 1.5×10^5 Pa. Find the **total translational kinetic energy** of all the gas molecules. If the gas is monatomic with 4.0×10^{22} molecules, find the average translational KE per molecule.

Concept 1 — pressure-energy relation. From $PV = \frac{2}{3}E$:

$$E = \frac{3}{2}PV = \frac{3}{2} \times (1.5 \times 10^5) \times (2.0 \times 10^{-3} \text{ m}^3) = \frac{3}{2} \times 300 = 450 \text{ J}$$

Concept 2 — per-molecule average. Divide by the number of molecules N :

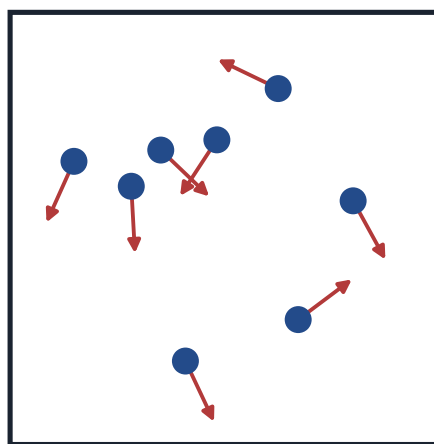
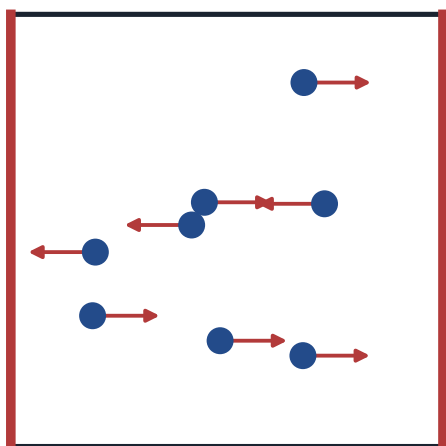
$$\langle \frac{1}{2}mv^2 \rangle = \frac{E}{N} = \frac{450}{4.0 \times 10^{22}} = 1.1 \times 10^{-20} \text{ J}$$

Total translational KE = 450 J; average per molecule $\approx 1.1 \times 10^{-20}$ J. Note we never needed the temperature or the molar mass — the macroscopic pair (P, V) already encodes the total kinetic energy. That is the power of $PV = \frac{2}{3}E$.

Example 4 — [JEE Advanced level] A thought experiment. Imagine a hypothetical gas of N identical molecules (mass m , number density N/V) confined to a cubical box, but constrained so that every molecule moves **only along the x -axis** (half in the $+x$, half in the $-x$ direction), all with the same speed v . (a) Find the pressure on the two walls perpendicular to x . (b) Find the pressure on the four walls parallel to x . (c) Compare the x -wall pressure with that of a normal isotropic gas of the same N , V , m , and speed v .

Where the factor of $\frac{1}{3}$ comes from: isotropy

Directed gas (all along x)	Isotropic gas (random)
$P_x = \frac{Nm}{V}v^2$, side walls feel 0	$P = \frac{1}{3}\frac{Nm}{V}v^2$



A directed (1-D) gas pushes only on the x -walls; an isotropic gas spreads its motion over all three directions — the origin of the factor $\frac{1}{3}$.

Part (a). Repeat the derivation, but now there is **no isotropy to invoke** — the molecules' entire speed is in x , so $\langle v_x^2 \rangle = v^2$ (not $\frac{1}{3}v^2$). From Step 5 of the derivation, before the isotropy substitution:

$$P_x = \frac{Nm}{V} \langle v_x^2 \rangle = \frac{Nm}{V} v^2$$

Part (b). No molecule has any y or z velocity, so no molecule ever strikes the walls parallel to x . Those walls feel

$$P_y = P_z = 0$$

Part (c). A real isotropic gas with the same parameters has, from the boxed result, $P_{\text{iso}} = \frac{1}{3} \frac{Nm}{V} v^2$. Therefore

$$\frac{P_x}{P_{\text{iso}}} = \frac{\frac{Nm}{V} v^2}{\frac{1}{3} \frac{Nm}{V} v^2} = 3$$

The directed gas pushes three times as hard on the x -walls and not at all on the others. This is the whole meaning of the $\frac{1}{3}$: in a real gas, each molecule's kinetic energy is shared equally among three directions, so any one pair of walls receives only a third of the "if-everything-pointed-here" pressure. Strip away isotropy and the factor disappears — a vivid reminder that the kinetic pressure formula is inseparable from the isotropy postulate.

Section 5: Practice Exercises

1. **[CBSE]** Nitrogen gas has a density of 1.25 kg m^{-3} at a pressure of $1.0 \times 10^5 \text{ Pa}$. Find the rms speed of its molecules.
2. **[NEET]** The rms speed of the molecules of a gas at constant density is reduced to one-half of its initial value. The pressure becomes what fraction of its initial value?
3. **[JEE Main]** A 5.0 L container holds a gas exerting a pressure of $2.0 \times 10^5 \text{ Pa}$. What is the total translational kinetic energy of the gas?

4. **[JEE Main]** Equal masses of two gases A and B are kept in identical containers at the same pressure. Gas A has molecules of twice the mass of gas B. Find the ratio of the mean square speeds $\langle v_A^2 \rangle : \langle v_B^2 \rangle$. (Hint: equal mass and equal $P, V \Rightarrow$ use $PV = \frac{1}{3} M_{\text{gas}} \langle v^2 \rangle$, where M_{gas} is the *total* gas mass.)
5. **[JEE Advanced]** A cubical box contains N molecules of an ideal gas with rms speed v_{rms} at pressure P_0 . Without changing the volume, three-quarters of the molecules are removed and the rms speed of those remaining is simultaneously doubled. Find the new pressure in terms of P_0 .

Answers: 1. $\approx 490 \text{ m s}^{-1}$ 2. one-quarter ($\frac{1}{4}P$) 3. $1.5 \times 10^3 \text{ J}$ 4. 1 : 1 (equal masses $\Rightarrow$ equal total gas mass; same P, V forces equal $\langle v^2 \rangle \cdot M_{\text{gas}}$, hence equal $\langle v^2 \rangle$) 5. $P = P_0 \times \frac{1}{4} \times 4 = P_0$ (number falls to $\frac{1}{4}$, v_{rms}^2 rises $\times 4$).

Section 6: Crack the MCQ

Q1. The pressure exerted by an ideal gas on its container walls is fundamentally due to: (A) the weight of the gas molecules (B) the mutual attraction between molecules (C) the change in momentum of molecules colliding with the walls (D) the electrostatic repulsion between molecules and the walls

Correct: (C). Pressure is the time-averaged force per unit area arising from countless elastic collisions, each transferring momentum $2mv_x$ to the wall. *Trap (A):* gravity is explicitly neglected in kinetic theory; pressure exists even in zero-g. *(B):* an ideal gas has no intermolecular attraction at all. *(D):* molecules are neutral; there is no electrostatic interaction in the model.

Q2. For an ideal gas, $PV = \frac{2}{3}E$, where E is the total translational kinetic energy. If the volume is halved while the total kinetic energy is kept constant, the pressure: (A) halves (B) doubles (C) stays the same (D) becomes four times

Correct: (B). With E fixed, $P = \frac{2E}{3V}$, so $P \propto 1/V$. Halving V doubles P . *Trap (A):* reading the proportionality backwards. *(C):* assuming constant E means constant P — but P depends on E/V , not E alone. *(D):* confusing this with a v_{rms}^2 or area-squared dependence; the relation here is strictly linear in $1/V$.

Q3. A sealed, rigid, insulated box of ionised hydrogen has a strong positive charge fixed on one of its inner faces. The usual kinetic-theory expression for gas pressure, $P = \frac{1}{3} \frac{N}{V} m \langle v^2 \rangle$: (A) remains exactly valid (B) fails, because the ions now feel forces other than wall collisions, and the motion is no longer isotropic (C) fails, because the collisions with the walls stop being elastic (D) remains valid but only for the uncharged wall

Correct: (B). The derivation assumes the *only* forces are elastic wall collisions and that motion is isotropic. A fixed charge exerts a long-range electrostatic force on the ions, violating the “no non-collisional forces” postulate, and biases their motion toward/away from that face, destroying isotropy — so $\langle v_x^2 \rangle = \langle v_y^2 \rangle = \langle v_z^2 \rangle$ no longer holds. *Trap (A):* ignores that the postulates are now broken. *(C):* elasticity of the wall bounce is not what fails here — the new *external* force is. *(D):* the loss of isotropy spoils the formula globally, not on one wall.

Q4. Two ideal gases have the same number density N/V and the same rms speed, but gas X has molecules twice as massive as gas Y. The ratio of their pressures $P_X : P_Y$ is: (A) 1 : 1 (B) 2 : 1 (C) 1 : 2 (D) 4 : 1

Correct: (B). From $P = \frac{1}{3} \frac{N}{V} m \langle v^2 \rangle$, with N/V and $\langle v^2 \rangle$ identical, $P \propto m$. Double the molecular mass $\Rightarrow$ double the pressure. *Trap (A):* assuming pressure is independent of

molecular mass. (C): inverting the dependence. (D): squaring the mass ratio — but P is linear in m , only quadratic in v_{rms} .

Section 7: Common Student Pitfalls & Pro-Tips

- **Confusing m (one molecule) with the sample mass.** In $P = \frac{1}{3} \frac{N}{V} m \langle v^2 \rangle$, m is the mass of a *single* molecule and N is the *number* of molecules. Mixing these up by a factor of N is the commonest derivation error. The density form $P = \frac{1}{3} \rho v_{rms}^2$ sidesteps it because ρ already bundles Nm/V .
 - **Forgetting the $\frac{1}{3}$ — or not knowing where it comes from.** The $\frac{1}{3}$ is purely the isotropy factor ($\langle v_x^2 \rangle = \frac{1}{3} \langle v^2 \rangle$). Examiners love to test this by removing isotropy (directed beams, 1-D motion). If a problem says molecules move along one axis, drop the $\frac{1}{3}$.
 - **Treating pressure as linear in speed.** Pressure and kinetic energy scale as v_{rms}^2 , *not* v_{rms} . Always square the speed factor first.
 - **Assuming wall collisions transfer the whole momentum mv .** An *elastic* bounce reverses the perpendicular component, so the transfer is $2mv_x$, *not* mv_x . Halving this is a classic slip.
 - **Forgetting that pressure is a statistical average.** Pressure (and indeed temperature) is only meaningful for a *large* collection of molecules. A single molecule rattling in a box delivers sharp, intermittent impulses, not a steady force — you cannot sensibly assign it a pressure or a temperature, because the postulate “very large N ” has failed. This is why kinetic theory is explicitly a theory of dilute gases with enormous molecule counts; conceptual questions that shrink N to one, or push toward 0 K (where, in this classical picture, motion and hence pressure would cease), are testing exactly this statistical foundation.
 - **Pro-Tip (sanity check via energy):** if a question gives you P and V , you instantly know the total translational KE without anything else: $E = \frac{3}{2} PV$. Reach for this whenever “total kinetic energy” appears — it usually turns a three-step problem into one line.
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Section 8: Cheat Sheet

- **Pressure (kinetic equation):** $P = \frac{1}{3} \frac{N}{V} m \langle v^2 \rangle = \frac{1}{3} \rho v_{rms}^2$.
- **Pressure-energy bridge:** $PV = \frac{2}{3} E$, so $E = \frac{3}{2} PV$ (total translational KE).
- **The $\frac{1}{3}$ comes from isotropy:** $\langle v_x^2 \rangle = \langle v_y^2 \rangle = \langle v_z^2 \rangle = \frac{1}{3} \langle v^2 \rangle$.
- **Per-collision momentum transfer** to a wall = $2mv_x$; **time between hits** on one wall = $2l/v_x$.
- **Scaling:** $P \propto \frac{N}{V}$, $P \propto m$, $P \propto v_{rms}^2$.
- **Postulate check for validity:** large N , point molecules, no force but elastic collisions, isotropic motion. Charge / external field / anisotropy $\implies$ formula fails.
- **Memory aid:** “Pressure = one-third density times speed-squared” — and “ PV is two-thirds the kinetic energy.”

Sub-topic 3: RMS Speed and Maxwell Distribution

Exam Relevance: Extremely high-yield. NEET and JEE Main almost always carry one numerical on $v_{rms} = \sqrt{3RT/M}$ or the speed ratios (lighter gas faster, $v \propto \sqrt{T}$). CBSE Boards ask the kinetic interpretation of temperature ($\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T$) and the labelled Maxwell curve. JEE Advanced reaches into the distribution itself — deriving v_{mp} , comparing the three speeds, or reading fractions off the curve.

Section 1: Concept Intuition

In Sub-topic 2 we stopped one step short of a deep secret: we found $PV = \frac{2}{3}E$, linking pressure to the molecules' kinetic energy, but we never said what *temperature* is at the molecular level. This subtopic closes that gap, and the answer is one of the most beautiful results in all of physics: **temperature is simply a measure of the average kinetic energy of the molecules.** A hot gas is not “carrying heat” like a bucket carries water — its molecules are literally moving faster. Cool the gas and they slow down. That is all temperature is.

Think about it this way... imagine the molecules as the crowd leaving a stadium after an India–Pakistan match. They don't all walk at the same pace — a few sprint, most amble, a handful dawdle. There is a whole *spread* of speeds. Now, if you wanted a single number to summarise “how fast is this crowd moving,” which would you pick? The fastest? The slowest? The most common pace? Physicists found that *three* different averages are each useful, and they show up constantly in exams: the **most probable speed** (the pace the largest number of molecules actually have — the peak of the crowd), the **average (mean) speed** (the ordinary arithmetic mean of everyone's speed), and the **root-mean-square speed** (the speed whose square equals the average of the squared speeds — the one that connects directly to kinetic energy).

Why three? Because the molecules do not all move at one speed. If they did, one number would suffice. Instead they share out the total energy through countless collisions into a characteristic, stable *distribution* of speeds — the **Maxwell-Boltzmann distribution**. Picture a graph: number of molecules on the vertical axis, speed on the horizontal. The curve starts at zero (no molecule is perfectly still), rises to a peak (the most probable speed), then falls away with a long tail stretching to very high speeds. It is lopsided — a gentle rise, then a long high-speed tail — which is exactly why the three speed-measures differ and always line up in the same order: most probable < average < rms.

Heat the gas and this whole curve responds: the peak slides to the right (molecules are faster on average) and the curve flattens and spreads (a wider range of speeds), but the total area — the total number of molecules — stays fixed. Swap to a heavier gas at the same temperature and the curve squeezes leftward: heavier molecules carry the *same* average kinetic energy, so they must move *slower*. This single fact — equal energy, unequal speeds — explains a huge fraction of exam questions.

This spread of speeds is not just bookkeeping; it has dramatic real consequences. Consider why the Moon has no atmosphere while the Earth keeps one. A planet retains a gas only if the molecules move too slowly to escape its gravity — that is, only if their speeds stay well below the escape speed. The Maxwell curve's long high-speed tail means that *some* molecules are always far faster than v_{rms} , and over geological time these stragglers leak away. Light gases (hydrogen, helium) have large speeds and lose this race quickly; that is why Earth's atmosphere is dominated by heavier nitrogen and oxygen, and why the low-gravity Moon, where even slow molecules exceed escape speed, holds essentially no gas at all. The same high-energy tail explains why a few molecules in a cool liquid can evaporate, and why chemical reactions speed up sharply with temperature — only the molecules out in the fast tail carry enough energy to react.

Limiting conditions & assumptions. Everything here assumes an *ideal gas in thermal equilibrium* — a large number of molecules that have collided enough times to settle into

the Maxwell distribution. The clean relation $\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T$ counts **translational** kinetic energy only; rotational and vibrational motion (which matter for the energy budget of diatomic and polyatomic gases) belong to the next subtopic and do *not* change the translational speeds. And note the recurring trap: the average *velocity* (a vector) of the molecules is **zero** — they go equally in all directions — but the average *speed* (a scalar) is decidedly not zero. Speed is what these formulas are about.

Section 2: Key Formulas & Definitions

Kinetic interpretation of temperature — the central bridge:

$$\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T \quad \Rightarrow \quad \bar{E}_{\text{trans (per molecule)}} = \frac{3}{2}k_B T$$

The average translational kinetic energy per molecule depends *only* on absolute temperature — not on pressure, volume, or the identity of the gas. For N molecules (or n moles), the total translational kinetic energy is $E = \frac{3}{2}Nk_B T = \frac{3}{2}nRT$.

For a **monatomic** ideal gas (single atoms — helium, neon, argon) this translational energy is the *entire* internal energy, since such atoms have no rotational or vibrational motion to store energy in. Hence the internal energy of a monatomic ideal gas is $U = \frac{3}{2}nRT$, a function of temperature alone — a result used constantly in thermodynamics. (Diatomic and polyatomic gases store additional energy in rotation and vibration; that extension belongs to the next subtopic and does not alter the translational speeds derived here.)

The three molecular speeds:

$$\begin{aligned} v_{rms} &= \sqrt{\langle v^2 \rangle} = \sqrt{\frac{3k_B T}{m}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3P}{\rho}} \\ v_{av} &= \sqrt{\frac{8k_B T}{\pi m}} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8P}{\pi \rho}} \\ v_{mp} &= \sqrt{\frac{2k_B T}{m}} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2P}{\rho}} \end{aligned}$$

Symbols and units: - m = mass of one molecule (kg); M = molar mass (**kg mol⁻¹**, *not* g mol⁻¹ — the commonest unit error). - $k_B = 1.38 \times 10^{-23}$ J K⁻¹; $R = 8.314$ J mol⁻¹K⁻¹; $R = N_A k_B$. - All speeds: SI unit m s⁻¹, dimension [LT⁻¹]. - ρ = density (kg m⁻³); P = pressure (Pa).

Ordering and ratio (true at any temperature, for any gas):

$$\begin{aligned} v_{rms} &> v_{av} > v_{mp} \\ v_{rms} : v_{av} : v_{mp} &= \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2} \approx 1.732 : 1.596 : 1.414 \approx 1.224 : 1.128 : 1 \end{aligned}$$

A handy memory hook for the order is **“RAM”** (Rms, Average, Most-probable — largest to smallest).

To anchor these to real numbers: oxygen at 300 K has $v_{mp} \approx 395$ m s⁻¹, $v_{av} \approx 446$ m s⁻¹, and $v_{rms} \approx 484$ m s⁻¹ — confirming both the order and the $\approx 1 : 1.13 : 1.22$ spacing. The much lighter hydrogen, at the same temperature, runs roughly four times faster across the board ($v_{rms} \approx 1934$ m s⁻¹), since $v \propto 1/\sqrt{M}$ and $\sqrt{32/2} = 4$. These figures are worth remembering as order-of-magnitude checks: molecular speeds at room temperature are

hundreds to a couple of thousand metres per second — comparable to, and a little above, the speed of sound in the gas.

Key proportionalities: all three speeds share $v \propto \sqrt{T}$ and $v \propto \frac{1}{\sqrt{M}}$. So $\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}$ at fixed M , and $\frac{v_A}{v_B} = \sqrt{\frac{M_B}{M_A}}$ at fixed T .

Maxwell-Boltzmann speed distribution:

$$P(v) = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} v^2 e^{-Mv^2/2RT}$$

Here $P(v) dv$ is the fraction of molecules with speeds between v and $v + dv$. The peak of this curve sits at v_{mp} ; the total area under it is 1.

Section 3: CBSE Board Derivation & Procedures

Part A — Deriving the kinetic interpretation of temperature and v_{rms} .

Starting point. We carry over two results. From the pressure derivation (Sub-topic 2):

$$PV = \frac{1}{3}Nm\langle v^2 \rangle \quad (1)$$

From the ideal gas law in molecular form (Sub-topic 1):

$$PV = Nk_B T \quad (2)$$

Step 1 — equate the two expressions for PV . Both describe the same gas, so

$$\frac{1}{3}Nm\langle v^2 \rangle = Nk_B T$$

Step 2 — cancel N and rearrange. The number of molecules cancels (the result is per-molecule, independent of how much gas there is):

$$\frac{1}{3}m\langle v^2 \rangle = k_B T \implies \frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T$$

Physical meaning: the left side is the average translational kinetic energy of one molecule. So **temperature is a direct measure of average molecular kinetic energy** — the deepest statement in this chapter. Because the right side contains no reference to mass or gas type, *all* gases at the same temperature have the same average translational KE per molecule. Heavier molecules simply compensate with lower speeds.

Step 3 — solve for the rms speed. From $\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T$ we get $\langle v^2 \rangle = \frac{3k_B T}{m}$, hence

$$v_{rms} = \sqrt{\langle v^2 \rangle} = \sqrt{\frac{3k_B T}{m}}$$

Step 4 — convert to molar form. Multiply numerator and denominator inside the root by Avogadro's number N_A : since $N_A k_B = R$ and $N_A m = M$ (molar mass),

$$v_{rms} = \sqrt{\frac{3N_A k_B T}{N_A m}} = \sqrt{\frac{3RT}{M}}$$

and using $P = \rho RT/M$ from Sub-topic 1, a third equivalent form $v_{rms} = \sqrt{3P/\rho}$ follows immediately.

Part B — Procedure: locating the most probable speed from the distribution.

The most probable speed is, by definition, the speed at the *peak* of $P(v)$. The procedure is standard calculus:

1. Write the distribution $P(v) = C v^2 e^{-mv^2/2k_B T}$, where C collects the constants (they won't affect where the peak is).
2. The peak is where $\frac{dP}{dv} = 0$. Differentiate the product $v^2 e^{-mv^2/2k_B T}$ using the product rule:

$$\frac{dP}{dv} = C \left[2v e^{-mv^2/2k_B T} + v^2 \left(-\frac{mv}{k_B T} \right) e^{-mv^2/2k_B T} \right]$$

3. Factor out the common (non-zero) terms $Cv e^{-mv^2/2k_B T}$:

$$\frac{dP}{dv} = Cv e^{-mv^2/2k_B T} \left(2 - \frac{mv^2}{k_B T} \right) = 0$$

4. The exponential never vanishes and $v = 0$ is the minimum, so the maximum requires the bracket to vanish: $2 - \frac{mv^2}{k_B T} = 0$, giving

$$v_{mp} = \sqrt{\frac{2k_B T}{m}} = \sqrt{\frac{2RT}{M}}$$

This is why v_{mp} carries the smallest numerical coefficient ($\sqrt{2}$) of the three speeds — it marks the peak, while the averaging that defines v_{av} and the energy-weighting that defines v_{rms} both pull their values up into the high-speed tail.

Part C — Reading the Maxwell curve. Because $P(v) dv$ is the *fraction* of molecules with speeds between v and $v + dv$, the curve is read as an area, not a height. The fraction of molecules with speeds lying in any interval $[v_1, v_2]$ equals the **area under the curve between v_1 and v_2** (mathematically, $\int_{v_1}^{v_2} P(v) dv$). Three consequences follow that examiners like to test. First, the *total* area, from 0 to ∞ , is exactly 1 — every molecule has *some* speed — so heating a gas cannot change the area, only redistribute it. Second, the height $P(v)$ at a single speed is not itself a probability; only an area (a height times a width) is. Third, the curve is asymmetric: it rises steeply from the origin, peaks at v_{mp} , then trails off slowly, so there are always more molecules faster than v_{mp} than slower than it — which is the geometric reason the mean and rms both lie to the right of the peak.

Section 4: Worked Examples

Example 1 — [CBSE Board level] Calculate the rms speed of nitrogen molecules ($M = 28 \text{ g mol}^{-1}$) at 27°C . Take $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$.

Given: $M = 28 \text{ g mol}^{-1} = 28 \times 10^{-3} \text{ kg mol}^{-1}$, $T = 27 + 273 = 300 \text{ K}$.

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.314 \times 300}{28 \times 10^{-3}}} = \sqrt{\frac{7482.6}{0.028}} = \sqrt{2.672 \times 10^5}$$

$$v_{rms} \approx 517 \text{ m s}^{-1}$$

The rms speed is about $5.2 \times 10^2 \text{ m s}^{-1}$. Note the conversion of M to kg mol^{-1} — leaving it in g mol^{-1} would inflate the answer by $\sqrt{1000} \approx 32$ times.

The other two speeds follow at once from the fixed ratio, without re-running the calculation: $v_{av} = v_{rms} \sqrt{8/(3\pi)} \approx 0.921 \times 517 \approx 476 \text{ m s}^{-1}$, and $v_{mp} = v_{rms} \sqrt{2/3} \approx 0.816 \times 517 \approx 422 \text{ m s}^{-1}$. As expected, $v_{rms} > v_{av} > v_{mp}$, and all three are of the same order — a few hundred metres per second.

Example 2 — [NEET Speed Trap] At the same temperature, find the ratio of the rms speed of hydrogen ($M = 2$) to that of oxygen ($M = 32$).

Speed method: at fixed T , $v_{rms} \propto 1/\sqrt{M}$, so

$$\frac{v_{H_2}}{v_{O_2}} = \sqrt{\frac{M_{O_2}}{M_{H_2}}} = \sqrt{\frac{32}{2}} = \sqrt{16} = 4$$

Hydrogen molecules move 4 times as fast as oxygen molecules.

The trap: two slips lurk here. First, students sometimes take the ratio of molar masses directly ($32/2 = 16$) instead of its square root. Second, some invert it and conclude oxygen is faster — but the *lighter* gas is always faster at a given temperature. A 2-second sanity check (“light = fast”) kills the wrong option instantly. The same single relation, $v_A/v_B = \sqrt{M_B/M_A}$, handles every “which gas is faster and by how much” question on the paper, so it is worth internalising as a reflex rather than re-deriving each time.

Example 3 — [JEE Main level] At what temperature will the rms speed of helium atoms ($M = 4$) equal the rms speed of nitrogen molecules ($M = 28$) that are held at 350 K?

Concept — equate rms speeds across two gases at different temperatures. Setting $v_{rms,He} = v_{rms,N_2}$:

$$\sqrt{\frac{3RT_{He}}{M_{He}}} = \sqrt{\frac{3RT_{N_2}}{M_{N_2}}} \Rightarrow \frac{T_{He}}{M_{He}} = \frac{T_{N_2}}{M_{N_2}}$$

Solve for T_{He} :

$$T_{He} = T_{N_2} \cdot \frac{M_{He}}{M_{N_2}} = 350 \times \frac{4}{28} = 50 \text{ K}$$

Helium must be cooled to 50 K. This makes physical sense: helium is much lighter, so to match the slower nitrogen it must be at a far lower temperature (less kinetic energy). The key step is recognising that “equal rms speeds” reduces to “equal T/M .”

Example 4 — [JEE Advanced level] Five molecules are found to have speeds (in arbitrary units) of 2, 4, 6, 8, and 10. (a) Compute v_{av} and v_{rms} for this set and verify $v_{rms} > v_{av}$. (b) Then, for a *real* gas in thermal equilibrium, explain why v_{rms} always exceeds v_{av} , connecting it to the shape of the Maxwell curve.

Part (a) — average speed:

$$v_{av} = \frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = 6.00 \text{ units}$$

rms speed:

$$\langle v^2 \rangle = \frac{2^2 + 4^2 + 6^2 + 8^2 + 10^2}{5} = \frac{4 + 16 + 36 + 64 + 100}{5} = \frac{220}{5} = 44$$

$$v_{rms} = \sqrt{44} \approx 6.63 \text{ units}$$

Indeed $v_{rms} (6.63) > v_{av} (6.00)$. ✓

Part (b) — why this is always true. The rms takes the average of the *squares* of the speeds. Squaring weights large speeds disproportionately — a molecule at 10 contributes 100 to the squared-sum but only 10 to the plain sum. So the high-speed tail of the Maxwell distribution drags v_{rms} above v_{av} . Mathematically this is a manifestation of the inequality $\langle v^2 \rangle \geq \langle v \rangle^2$ (with equality only if every molecule had exactly the same speed). Because a real gas in equilibrium *always* has a spread of speeds — the long high-speed tail of the Maxwell curve — the inequality is strict: $v_{rms} > v_{av} > v_{mp}$, with the gap fixed by the universal ratio $\sqrt{3} : \sqrt{8/\pi} : \sqrt{2}$.

Section 5: Practice Exercises

1. **[CBSE]** Calculate the rms speed of oxygen molecules ($M = 32 \text{ g mol}^{-1}$) at 0°C .
2. **[NEET]** The rms speed of a gas at 300 K is v . At what temperature will its rms speed become $2v$?
3. **[JEE Main]** For oxygen gas at 300 K ($M = 32 \text{ g mol}^{-1}$), calculate the most probable speed and the average speed.
4. **[JEE Main]** A vessel contains a mixture of hydrogen and oxygen in thermal equilibrium. Find the ratio of the average translational kinetic energy per hydrogen molecule to that per oxygen molecule.
5. **[JEE Advanced]** The rms speed of the molecules of an ideal gas is v . The gas is heated so that its absolute temperature triples, and *simultaneously* every molecule dissociates into two molecules each of half the original mass. Find the new rms speed in terms of v .

Answers: 1. $\approx 461 \text{ m s}^{-1}$ 2. 1200 K 3. $v_{mp} \approx 395 \text{ m s}^{-1}$, $v_{av} \approx 446 \text{ m s}^{-1}$ 4. 1 : 1 (average translational KE depends only on T , which is shared) 5. $v_{rms, \text{new}} = \sqrt{\frac{3RT'}{M'}}$ with $T' = 3T$ and $M' = M/2$, so the factor is $\sqrt{3 \times 2} = \sqrt{6}$, giving $\sqrt{6} v \approx 2.45 v$.

Section 6: Crack the MCQ

Q1. The average translational kinetic energy of the molecules of an ideal gas depends on: (A) the pressure of the gas only (B) the nature (molar mass) of the gas only (C) the absolute temperature of the gas only (D) both temperature and molar mass

Correct: (C). From $\bar{E}_{\text{trans}} = \frac{3}{2}k_B T$, the average translational KE per molecule is a function of T alone. *Trap (A):* confusing energy with pressure; pressure depends on energy density, but the per-molecule energy does not depend on P . *(B)/(D):* the molar mass affects the *speed*, but not the *energy* — heavier molecules go slower so that the energy stays the same. This independence from mass is exactly the surprising part of the result.

Q2. The rms speed of oxygen molecules at temperature T is v . If the temperature is doubled **and** the oxygen molecules dissociate into oxygen atoms, the rms speed becomes: (A) v (B) $\sqrt{2} v$ (C) $2v$ (D) $4v$

Correct: (C). $v_{rms} \propto \sqrt{T/M}$. Doubling T multiplies by $\sqrt{2}$; dissociation halves the particle mass ($M \rightarrow M/2$), multiplying by another $\sqrt{2}$. Together: $\sqrt{2} \times \sqrt{2} = 2$. *Trap (B):* accounting for the temperature change but forgetting the dissociation halves the mass. *(A):* assuming the two effects cancel. *(D):* multiplying the factors (2×2) instead of combining them correctly under the root.

Q3. For the molecules of an ideal gas in thermal equilibrium, which ordering of the speeds

is always correct? (A) $v_{mp} > v_{av} > v_{rms}$ (B) $v_{rms} > v_{av} > v_{mp}$ (C) $v_{av} > v_{rms} > v_{mp}$ (D) all three are equal

Correct: (B). The “RAM” order: $v_{rms} : v_{av} : v_{mp} = \sqrt{3} : \sqrt{8/\pi} : \sqrt{2} \approx 1.22 : 1.13 : 1$, so rms is largest and most-probable smallest. *Trap (A):* exact reversal — a common memory slip. *(C):* misremembering which average sits on top; squaring (rms) always wins because it over-weights the fast tail. *(D):* true only for a hypotheticalal gas where every molecule has identical speed, which never happens in equilibrium.

Q4. When the temperature of a gas is increased, the Maxwell speed-distribution curve: (A) shifts its peak to a lower speed and becomes taller (B) shifts its peak to a higher speed and becomes broader and flatter (C) remains unchanged in shape (D) increases the total area under it

Correct: (B). Higher T raises all the speeds, so v_{mp} (the peak) moves right; the distribution spreads over a wider range, so the curve flattens. The peak height drops because the molecules are spread across more speeds. *Trap (A):* gets the direction of the peak backwards. *(C):* ignores the temperature dependence entirely. *(D):* the area equals the total fraction of molecules = 1 always, so it cannot change — a frequent conceptual error.

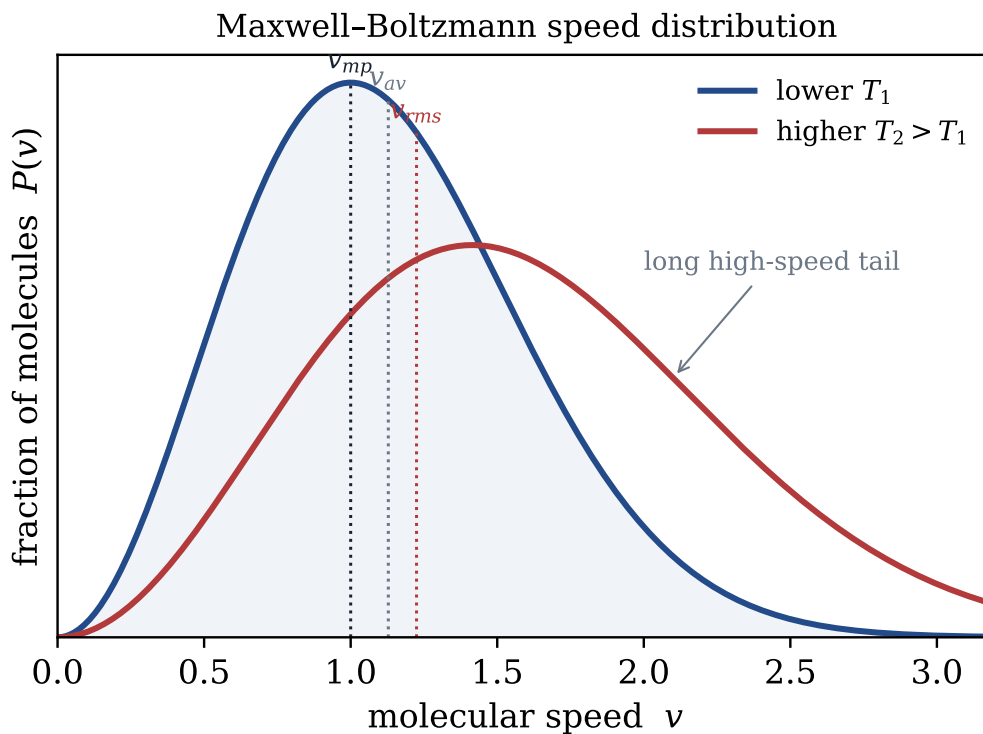
Section 7: Common Student Pitfalls & Pro-Tips

- **Molar mass in grams.** $v_{rms} = \sqrt{3RT/M}$ demands M in **kg mol⁻¹**. Using g mol⁻¹ inflates every speed by $\sqrt{1000} \approx 32$. This single error is the most common in board and entrance scripts.
- **Celsius instead of kelvin.** Like every gas relation, these use absolute temperature. Always convert before taking the square root.
- **Confusing the three speeds — or their coefficients.** Remember $\sqrt{3}$ (rms), $\sqrt{8/\pi}$ (average), $\sqrt{2}$ (most probable), in the “RAM” order. Mixing up which formula carries which coefficient is a classic trap, especially v_{av} (the only one with π).
- **Average velocity vs average speed.** The mean *velocity* vector is zero (isotropy), but the mean *speed* is not. Questions that say “average velocity” are usually testing whether you’ll wrongly plug in v_{av} .
- **Misreading the Maxwell curve.** The vertical height $P(v)$ is not a probability — only the *area* under a stretch of the curve gives the fraction of molecules in that speed range. When a question asks “what fraction of molecules have speed greater than v_{mp} ,” it wants the area to the right of the peak, not a value read off the axis. And remember the total area is always 1, so raising the temperature redistributes molecules into the faster tail but never adds or removes any.
- **Pro-Tip (ratios beat plug-ins):** for “compare two gases” or “change the temperature” problems, never compute full speeds. Use $v \propto \sqrt{T/M}$ directly: $\frac{v_1}{v_2} = \sqrt{\frac{T_1 M_2}{T_2 M_1}}$. One line, no calculator, no unit errors.

Section 8: Cheat Sheet

- **Kinetic interpretation:** $\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T$ — temperature is average translational KE; same for all gases at a given T .
- **Three speeds:** $v_{rms} = \sqrt{\frac{3RT}{M}}$, $v_{av} = \sqrt{\frac{8RT}{\pi M}}$, $v_{mp} = \sqrt{\frac{2RT}{M}}$.
- **Order (RAM):** $v_{rms} > v_{av} > v_{mp}$, ratio $\sqrt{3} : \sqrt{8/\pi} : \sqrt{2} \approx 1.22 : 1.13 : 1$.

- **Scaling:** every speed $\propto \sqrt{T}$ and $\propto 1/\sqrt{M}$.
- **Maxwell distribution:** $P(v) = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} v^2 e^{-Mv^2/2RT}$; peak at v_{mp} , total area = 1, and the area between two speeds gives the fraction of molecules in that range. Higher $T \Rightarrow$ peak shifts right, curve flattens.
- **Units:** M in kg mol^{-1} , T in K, $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$.
- **Memory aid:** “Heavier = slower, hotter = faster; RAM from big to small.” And temperature alone fixes the average kinetic energy.



The Maxwell-Boltzmann speed distribution. Raising the temperature shifts the peak to higher speed and flattens the curve, while the total area (the whole population) stays fixed.

Sub-topic 4: Degrees of Freedom and Mean Free Path

Exam Relevance: A dense, high-return cluster. JEE Main and NEET reliably ask for $\gamma = C_P/C_V$ or C_V, C_P from atomicity (monatomic/diatomic), and the mean free path's dependence on P and T . CBSE Boards favour the statement of the equipartition law and the degrees-of-freedom count. JEE Advanced combines them — mixture specific heats, or γ when vibrational modes switch on.

Section 1: Concept Intuition

So far our molecules have only been allowed to *fly* — to move through space. But a real molecule can do more than travel: a dumbbell-shaped oxygen molecule can also *tumble*, and at high temperatures its two atoms can *vibrate* along their bond like masses on a spring. Each of these independent ways of moving is a place where the molecule can stash energy. Physicists call each such independent mode a **degree of freedom**.

Think about it this way... picture how a single coin can carry “state”: it can slide across a table (translation), it can spin on the spot (rotation), and if it were soft it could even wobble and flex (vibration). A point-like atom such as helium is like a marble — it can only slide, in three independent directions (x, y, z), so it has **3 translational degrees of freedom** and nothing more (a marble spinning carries no meaningful energy because it has negligible size). A diatomic molecule like nitrogen is like a tiny dumbbell — it slides in 3 directions *and* tumbles end-over-end about 2 axes (spinning about the bond axis itself stores nothing, since the atoms sit on that line), giving **5 degrees of freedom** at room temperature. Heat it enough and the bond starts to vibrate, adding 2 more (one for vibrational kinetic energy, one for the spring's potential energy), for 7.

Why does spinning about the bond axis “store nothing”? Rotational kinetic energy is $\frac{1}{2}I\omega^2$, and the moment of inertia I about the line joining the two atoms is vanishingly small — almost all the mass of each atom sits essentially *on* that axis, so the perpendicular distance that enters I is negligible. With $I \approx 0$ along the bond, no meaningful energy can be stored in that rotation, which is exactly why a diatomic molecule counts 2 rotational degrees of freedom rather than 3. The very same argument explains why a monatomic atom counts *zero* rotational degrees of freedom: a point-like atom has negligible moment of inertia about every axis.

Now comes the elegant part. Nature is scrupulously fair about how it shares thermal energy: it dumps *exactly the same* amount, $\frac{1}{2}k_B T$ per molecule, into *every single* degree of freedom, no matter what kind. This is the **law of equipartition of energy** — energy is partitioned equally among all available modes. A molecule with f degrees of freedom therefore carries an average energy of $f \times \frac{1}{2}k_B T$. This one principle is the master key: it tells us the internal energy of any ideal gas, and from there its specific heats and the all-important ratio γ .

The second idea in this subtopic is about *motion between collisions*. A molecule in air does not sail smoothly across the room — it ricochets off its neighbours billions of times a second, tracing a frantic zig-zag. The **mean free path** λ is the average straight-line distance a molecule covers between two successive collisions. It is tiny but not vanishingly so: in ordinary air it is about a hundred times the molecular size, which is exactly why gases behave the way they do — molecules are mostly free, occasionally colliding. Crowd the gas (raise pressure, raise density) and collisions come thicker and faster, so λ shrinks; rarefy it (pump it down) and λ grows, becoming as large as the whole tube in a good vacuum.

Limiting conditions & assumptions. Two cautions matter. First, the degree-of-freedom count for diatomic and polyatomic gases is *temperature-dependent*: vibrational modes are “frozen out” at ordinary temperatures and only switch on when it is hot enough (for hydrogen, rotation itself freezes out below about 50 K). So unless told otherwise, use $f = 3$ (monatomic), $f = 5$ (diatomic), $f = 6$ (non-linear polyatomic) — the room-temperature values. Second, equipartition is a classical result; it works beautifully at ordinary temperatures but quantum mechanics modifies it at the extremes (the very freezing-out just mentioned). And the mean free path formula assumes a dilute gas of finite-sized, hard-sphere molecules.

This freezing-out is itself a fingerprint of quantum mechanics intruding on a classical theory. Classically, equipartition says every mode should *always* carry $\frac{1}{2}k_B T$, so a diatomic gas should always show $f = 7$. Experiment disagrees: at low temperatures the measured specific heat corresponds to $f = 3$ (only translation), rising to $f = 5$ near room temperature and only approaching $f = 7$ at high temperatures. The reason is that rotational and vibrational energies come in discrete quantum steps; when $k_B T$ is smaller than the gap to the first excited level, that mode simply cannot be activated and contributes nothing. The orderly switching-on of modes as temperature rises is one of the earliest clues that energy at the molecular scale is quantised.

Finally, the mean free path is not just an abstract average — it is the quantity that links the microscopic world to bulk, measurable properties. The viscosity of a gas, its thermal conductivity, and its diffusion rate can all be expressed in terms of λ and the molecular speed; historically, it was precisely by measuring these bulk properties and working backwards through the mean-free-path relations that scientists first estimated the sizes of molecules.

Section 2: Key Formulas & Definitions

Degrees of freedom (f) — the number of independent ways a molecule can store energy:

Gas type	Translational	Rotational	f (room T)	f (high T , +vibration)
Monatomic (He, Ne, Ar)	3	0	3	3
Diatomic (N ₂ , O ₂)	3	2	5	7
Non-linear polyatomic (H ₂ O, NH ₃)	3	3	6	12

Law of equipartition of energy: each degree of freedom carries average energy $\frac{1}{2}k_B T$ per molecule (equivalently $\frac{1}{2}RT$ per mole). A molecule with f degrees of freedom has average energy $\frac{f}{2}k_B T$.

Internal energy of n moles of an ideal gas:

$$U = \frac{f}{2}nRT$$

(U depends only on temperature; SI unit J, dimension $[ML^2T^{-2}]$.)

Molar specific heats (from equipartition + Mayer's relation):

$$C_V = \frac{f}{2}R, \quad C_P = C_V + R = \left(\frac{f}{2} + 1\right)R, \quad \gamma = \frac{C_P}{C_V} = 1 + \frac{2}{f}$$

Useful inverses: $C_V = \frac{R}{\gamma - 1}$, $C_P = \frac{\gamma R}{\gamma - 1}$. SI unit of molar specific heat: J mol⁻¹K⁻¹, dimension $[ML^2T^{-2}\text{mol}^{-1}K^{-1}]$.

Standard values:

Gas	f	C_V	C_P	γ
Monatomic	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} \approx 1.67$
Diatomic	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.40$
Non-linear polyatomic	6	$3R$	$4R$	$\frac{4}{3} \approx 1.33$

Mean free path:

$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n} = \frac{k_B T}{\sqrt{2} \pi d^2 P}$$

where d = molecular diameter, $n = N/V$ = number density ($[L^{-3}]$). SI unit of λ : m, dimension $[L]$. The quantity πd^2 is the **collision cross-section**, and the $\sqrt{2}$ accounts for the relative motion of all molecules.

Dependences of λ : $\lambda \propto \frac{1}{n}$, $\lambda \propto \frac{1}{d^2}$, $\lambda \propto T$ (const P), $\lambda \propto \frac{1}{P}$ (const T).

Collision frequency. Closely related to λ is the average number of collisions a molecule suffers per second, the collision frequency ν . Since a molecule covers an average distance λ between collisions while moving at average speed v_{av} ,

$$\nu = \frac{v_{av}}{\lambda}, \quad \tau = \frac{1}{\nu} = \frac{\lambda}{v_{av}}$$

where τ is the mean time between collisions. In ordinary air a molecule collides several billion times each second.

Degrees of freedom and specific heat of a solid. The atoms of a crystalline solid cannot translate or rotate freely; instead each vibrates about its lattice site along 3 axes. Each vibration carries 2 degrees of freedom (kinetic + potential), giving $f = 3 \times 2 = 6$. Equipartition then predicts a molar internal energy $U = \frac{6}{2}RT = 3RT$ and a molar specific heat $C = 3R \approx 25 \text{ J mol}^{-1}\text{K}^{-1}$ — the **law of Dulong and Petit**, which most solids obey well at ordinary temperatures.

Specific heat of water. The same solid-like reasoning explains water's famously high heat capacity. Treat each atom in a water molecule as if bound in a solid, with 6 degrees of freedom contributing $3R$ per mole of atoms. A water molecule (H_2O) has 3 atoms, so one mole of water behaves like 3 moles of vibrating atoms:

$$C_{\text{water}} = 3 \times 3R = 9R \approx 9 \times 8.314 \approx 75 \text{ J mol}^{-1}\text{K}^{-1}$$

This matches the measured molar heat capacity of water remarkably well (the familiar specific heat $4186 \text{ J kg}^{-1}\text{K}^{-1}$ times the molar mass $0.018 \text{ kg mol}^{-1}$ gives $\approx 75 \text{ J mol}^{-1}\text{K}^{-1}$) — a small triumph of the equipartition idea applied beyond gases.

Section 3: Derivations & Key Procedures**Part A — Specific heats and γ from equipartition.**

Step 1 — internal energy. By equipartition, a molecule with f degrees of freedom has average energy $\frac{f}{2}k_B T$. For n moles ($N = nN_A$ molecules), the total internal energy is

$$U = N \cdot \frac{f}{2}k_B T = nN_A \cdot \frac{f}{2}k_B T = \frac{f}{2}nRT \quad (\text{using } N_A k_B = R)$$

Step 2 — molar heat capacity at constant volume. C_V is the heat needed to raise one mole by one kelvin at constant volume. At constant volume no work is done, so all the heat goes into internal energy:

$$C_V = \frac{1}{n} \left(\frac{dU}{dT} \right)_V = \frac{1}{n} \cdot \frac{f}{2}nR = \frac{f}{2}R$$

Step 3 — molar heat capacity at constant pressure. At constant pressure the gas also does expansion work as it is heated. Mayer's relation (derived in thermodynamics) captures this: $C_P - C_V = R$. Hence

$$C_P = C_V + R = \frac{f}{2}R + R = \left(\frac{f}{2} + 1 \right) R = \frac{f+2}{2}R$$

Step 4 — the adiabatic ratio. Dividing,

$$\gamma = \frac{C_P}{C_V} = \frac{\frac{f+2}{2}R}{\frac{f}{2}R} = \frac{f+2}{f} = 1 + \frac{2}{f}$$

Reading the result: because γ depends only on f , measuring γ experimentally tells you the atomicity of a gas. Monatomic ($f = 3$) gives $\gamma = \frac{5}{3}$; diatomic ($f = 5$) gives $\frac{7}{5}$; the more ways a molecule can store energy, the smaller γ — extra modes soak up heat without raising temperature as fast, lowering the ratio toward 1.

Part B — Mean free path, by the collision-cylinder argument.

Step 1 — set up the collision volume. Model molecules as hard spheres of diameter d . A moving molecule collides with another whenever their centres come within a distance d . So picture the moving molecule sweeping out a cylinder of radius d (cross-sectional area πd^2) as it travels. In time t , moving at speed v , it sweeps a cylinder of length vt and volume $\pi d^2 \cdot vt$.

Step 2 — count the collisions. If there are n molecules per unit volume, the number of molecular centres lying in that swept cylinder — i.e. the number of collisions in time t — is $n \pi d^2 vt$.

Step 3 — distance per collision. The mean free path is total distance travelled divided by number of collisions:

$$\lambda = \frac{vt}{n \pi d^2 vt} = \frac{1}{n \pi d^2}$$

Step 4 — the $\sqrt{2}$ correction. This simple count assumed only one molecule moves while the rest stand still. In reality *all* molecules move, so what matters is the *relative* speed between colliding pairs; averaging the relative speed introduces a factor of $\sqrt{2}$ in the denominator:

$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$$

Step 5 — pressure-temperature form. Substituting the number density from the ideal gas law, $n = N/V = P/(k_B T)$:

$$\lambda = \frac{k_B T}{\sqrt{2} \pi d^2 P}$$

which makes the dependences explicit: at fixed P , λ grows with T ; at fixed T , λ falls as P rises.

Part C — collision frequency. Once λ is known, the rate of collisions follows immediately. A molecule travelling at average speed v_{av} covers, on average, one mean free path between collisions, so the number of collisions per second is

$$\nu = \frac{v_{av}}{\lambda} = \sqrt{2} \pi d^2 n v_{av}$$

The reciprocal, $\tau = 1/\nu$, is the average time between collisions. This connects all three threads of the chapter: v_{av} comes from the Maxwell distribution (Sub-topic 3), λ from the collision geometry here, and n from the ideal gas law (Sub-topic 1). Notice that as pressure rises (denser gas), λ falls and ν climbs in proportion — squeezing a gas makes its molecules collide both more often and over shorter free flights.

Section 4: Worked Examples

Example 1 — [CBSE Board level] Calculate the internal energy of 3.0 moles of a diatomic ideal gas at 300 K (assume only translational and rotational modes are active). Take $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.

Given: diatomic at room temperature $\Rightarrow f = 5$; $n = 3.0$; $T = 300$ K.

$$U = \frac{f}{2}nRT = \frac{5}{2} \times 3.0 \times 8.314 \times 300$$

$$U = 2.5 \times 3.0 \times 2494.2 = 1.87 \times 10^4 \text{ J}$$

The internal energy is about 1.87×10^4 J. Note we used $f = 5$, not 7 — the problem specified only translational and rotational modes (room-temperature behaviour), so vibration is excluded.

Example 2 — [NEET Speed Trap] The ratio of specific heats of a gas is found to be $\gamma = 1.40$. Identify the atomicity of the gas and write its C_V .

Speed method: use $\gamma = 1 + \frac{2}{f}$, so $f = \frac{2}{\gamma - 1} = \frac{2}{0.40} = 5$. A gas with $f = 5$ is **diatomic**, and $C_V = \frac{f}{2}R = \frac{5}{2}R = 2.5R$.

Diatomic; $C_V = \frac{5}{2}R \approx 20.8 \text{ J mol}^{-1} \text{ K}^{-1}$.

The trap: the three benchmark γ values — 1.67 (monatomic), 1.40 (diatomic), 1.33 (polyatomic) — are worth memorising so you can identify the gas by sight. A student who misremembers 1.4 as monatomic, or who computes $f = 2/\gamma$ instead of $2/(\gamma - 1)$, lands on the wrong option. The “−1” in the denominator is the whole game.

Example 3 — [JEE Main level] Estimate the mean free path of a nitrogen molecule at 300 K and a pressure of 1.0×10^5 Pa. Take the molecular diameter $d = 3.7 \times 10^{-10}$ m and $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$.

Concept — use the pressure-temperature form directly:

$$\lambda = \frac{k_B T}{\sqrt{2} \pi d^2 P}$$

Numerator: $k_B T = 1.38 \times 10^{-23} \times 300 = 4.14 \times 10^{-21} \text{ J}$.

Denominator: $\sqrt{2} \pi d^2 P = 1.414 \times 3.1416 \times (3.7 \times 10^{-10})^2 \times (1.0 \times 10^5)$. With $d^2 = 1.369 \times 10^{-19} \text{ m}^2$: $1.414 \times 3.1416 \times 1.369 \times 10^{-19} \times 10^5 = 6.08 \times 10^{-14}$.

$$\lambda = \frac{4.14 \times 10^{-21}}{6.08 \times 10^{-14}} \approx 6.8 \times 10^{-8} \text{ m} \approx 68 \text{ nm}$$

The mean free path is roughly 7×10^{-8} m — about 200 times the molecular diameter, consistent with the “mostly free, occasionally colliding” picture of a gas.

As a follow-on, the collision frequency is easy to estimate: nitrogen’s average speed at 300 K is roughly $v_{av} \approx 475 \text{ m s}^{-1}$, so

$$\nu = \frac{v_{av}}{\lambda} \approx \frac{475}{6.8 \times 10^{-8}} \approx 7 \times 10^9 \text{ collisions per second.}$$

Each nitrogen molecule therefore suffers about seven billion collisions every second — a vivid sense of how furiously busy a seemingly still volume of air really is.

Example 4 — [JEE Advanced level] A gas mixture is made of 2.0 moles of helium (monatomic) and 3.0 moles of oxygen (diatomic, room temperature). Find the effective γ of the mixture.

Concept — heat capacities add by moles. For a mixture, the total internal energy is the sum of the parts, so the molar C_V of the mixture is the mole-weighted average:

$$C_{V,\text{mix}} = \frac{n_1 C_{V1} + n_2 C_{V2}}{n_1 + n_2}$$

With $C_{V,\text{He}} = \frac{3}{2}R$ and $C_{V,\text{O}_2} = \frac{5}{2}R$:

$$C_{V,\text{mix}} = \frac{2.0 \times \frac{3}{2}R + 3.0 \times \frac{5}{2}R}{2.0 + 3.0} = \frac{3R + 7.5R}{5.0} = \frac{10.5R}{5.0} = 2.1R$$

By Mayer's relation the mixture still obeys $C_{P,\text{mix}} = C_{V,\text{mix}} + R = 3.1R$. Therefore

$$\gamma_{\text{mix}} = \frac{C_{P,\text{mix}}}{C_{V,\text{mix}}} = \frac{3.1R}{2.1R} \approx 1.48$$

The mixture's $\gamma \approx 1.48$ — sitting neatly between the pure monatomic (1.67) and pure diatomic (1.40) values, weighted toward the more abundant oxygen. The crucial insight is that you cannot average the two γ values directly; you must average C_V (which is additive through energy) and only then form the ratio.

Section 5: Practice Exercises

1. **[CBSE]** State the law of equipartition of energy and use it to find the total internal energy of 2.0 mol of a monatomic ideal gas at 400 K.
2. **[NEET]** A polyatomic gas has 6 degrees of freedom. Find its γ .
3. **[JEE Main]** For a diatomic gas at room temperature, find C_P , C_V , and the ratio C_P/C_V in terms of R .
4. **[JEE Main]** The mean free path of a gas molecule is λ at pressure P and temperature T . What is the new mean free path if the pressure is doubled while the temperature is kept constant?
5. **[JEE Advanced]** A diatomic gas is heated to a temperature high enough that its vibrational mode becomes fully active. Find the new value of γ , and the fractional change from its room-temperature value.

Answers: 1. $U = \frac{3}{2}nRT = \frac{3}{2}(2.0)(8.314)(400) \approx 9.98 \times 10^3 \text{ J}$ 2. $\gamma = 1 + \frac{2}{6} = \frac{4}{3} \approx 1.33$ 3. $C_V = \frac{5}{2}R$, $C_P = \frac{7}{2}R$, $\gamma = 1.4$ 4. $\lambda/2$ (since $\lambda \propto 1/P$ at constant T) 5. f goes $5 \rightarrow 7$, so $\gamma = 1 + \frac{2}{7} = \frac{9}{7} \approx 1.29$; it drops from 1.40 by about 8%.

Section 6: Crack the MCQ

Q1. At room temperature, the number of degrees of freedom of a diatomic gas molecule is: (A) 3 (B) 5 (C) 6 (D) 7

Correct: (B). A diatomic molecule has 3 translational + 2 rotational = 5 active degrees of freedom at room temperature; vibration is frozen out. *Trap (A):* counting only translation (correct for a monatomic gas). *(C):* over-counting rotation as 3 — but rotation about the bond axis stores no energy. *(D):* the high-temperature value, valid only once vibration is excited; the qualifier “room temperature” rules it out.

Q2. A gas has $\gamma = \frac{5}{3}$. The gas is: (A) diatomic (B) monatomic (C) triatomic non-linear (D) diatomic with vibration

Correct: (B). $\gamma = 1 + \frac{2}{f} = \frac{5}{3}$ gives $f = 3$, the signature of a monatomic gas (only translation). *Trap (A):* diatomic has $\gamma = 1.4$, not 1.67. *(C):* non-linear polyatomic has $\gamma = \frac{4}{3}$. *(D):* adding vibration *increases* f and *lowers* γ below 1.4, the opposite direction.

Q3. At constant temperature, if the pressure of a gas is increased to four times its original value, its mean free path becomes: (A) 4λ (B) 2λ (C) $\frac{\lambda}{2}$ (D) $\frac{\lambda}{4}$

Correct: (D). From $\lambda = \frac{k_B T}{\sqrt{2} \pi d^2 P}$, at constant T we have $\lambda \propto 1/P$. Quadrupling P divides λ by 4. *Trap (A):* treating λ as directly proportional to P . *(B)/(C):* getting the inverse relationship but with the wrong factor — confusing P with $\sqrt{P}$ or mishandling the “four times.” Higher pressure means a denser gas and more frequent collisions, so λ must shrink.

Q4. According to the law of equipartition, the total energy associated with the rotational motion of one mole of a diatomic gas at temperature T is: (A) $\frac{1}{2}RT$ (B) RT (C) $\frac{3}{2}RT$ (D) $\frac{5}{2}RT$

Correct: (B). A diatomic molecule has 2 rotational degrees of freedom, each carrying $\frac{1}{2}RT$ per mole, so rotational energy $= 2 \times \frac{1}{2}RT = RT$. *Trap (A):* counting only one rotational mode. *(C):* using the translational share ($3 \times \frac{1}{2}RT$) instead of rotational. *(D):* using the *total* room-temperature energy ($\frac{5}{2}RT$) rather than the rotational part alone.

Section 7: Common Student Pitfalls & Pro-Tips

- **$f = 5$ vs $f = 7$ for diatomic gases.** Use 5 at room temperature and 7 only when the problem explicitly invokes vibration or “high temperature.” Defaulting to the wrong one flips γ between 1.4 and 1.29.
- **Vibration adds two degrees of freedom, not one.** Each vibrational mode contributes one kinetic and one potential degree of freedom ($\frac{1}{2}k_B T$ each), so a single vibration adds 2 to f , not 1.
- **Using k_B where R belongs (or vice-versa).** Molar quantities (C_V , C_P , $U = \frac{f}{2}nRT$) use R ; per-molecule energies use k_B . Mixing them up costs a factor of N_A .
- **Averaging γ for a mixture.** You cannot average γ values directly. Average C_V by moles, add R for C_P , then take the ratio. The reason is that internal energy is additive (the gases share the same temperature and their energies simply sum), so C_V is the mole-weighted quantity; γ , being a ratio, is not additive and averaging it directly gives a physically meaningless number.
- **Pro-Tip (identify gases by γ):** memorise the trio 1.67/1.40/1.33 for mono/di/poly. The instant a problem gives γ , you know f , C_V , and C_P without any algebra — and you can sanity-check any answer by recalling that more atoms $\implies$ smaller γ .

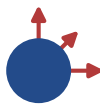
Section 8: Cheat Sheet

- **Degrees of freedom (room T):** monatomic 3, diatomic/linear 5, non-linear polyatomic 6; vibration adds 2 each at high T .
- **Equipartition:** $\frac{1}{2}k_B T$ per degree of freedom per molecule ($\frac{1}{2}RT$ per mole).
- **Internal energy:** $U = \frac{f}{2}nRT$ (temperature only).
- **Specific heats:** $C_V = \frac{f}{2}R$, $C_P = \left(\frac{f}{2} + 1\right)R$, $\gamma = 1 + \frac{2}{f}$; also $C_V = \frac{R}{\gamma - 1}$.

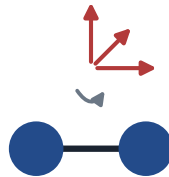
- **γ benchmarks:** $\frac{5}{3} \approx 1.67$ (mono), $\frac{7}{5} = 1.40$ (di), $\frac{4}{3} \approx 1.33$ (poly).
- **Mean free path:** $\lambda = \frac{1}{\sqrt{2} \pi d^2 n} = \frac{k_B T}{\sqrt{2} \pi d^2 P}$; $\lambda \propto T$, $\propto 1/P$, $\propto 1/d^2$.
- **Collision frequency:** $\nu = v_{av}/\lambda$; solids obey Dulong-Petit, $C = 3R \approx 25 \text{ J mol}^{-1} \text{ K}^{-1}$; water $C = 9R \approx 75 \text{ J mol}^{-1} \text{ K}^{-1}$ (3 atoms $\times 3R$).
- **Memory aids:** “Half- kT to every mode.” “More atoms, smaller gamma.” “Squeeze the gas, shrink the path and quicken the collisions.”

Degrees of freedom by molecular structure

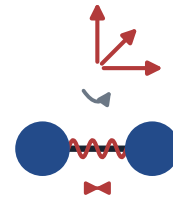
Monatomic
 $f = 3$ (3 translation)



Diatomic (room T)
 $f = 5$ (3 trans + 2 rot)



Diatomic (high T)
 $f = 7$ (+2 vibration)



Degrees of freedom by molecular structure: monatomic ($f = 3$), diatomic at room temperature ($f = 5$), and diatomic at high temperature ($f = 7$, with vibration active).

Chapter Summary & Master Formula Sheet

How to use this section. Everything below is a condensed map of the whole chapter — read it last, for revision. Each result is developed in full (with derivations, worked examples and pitfalls) in the subtopic it comes from; here we show how the pieces lock together and gather every formula in one place.

The big picture — one idea, unfolded

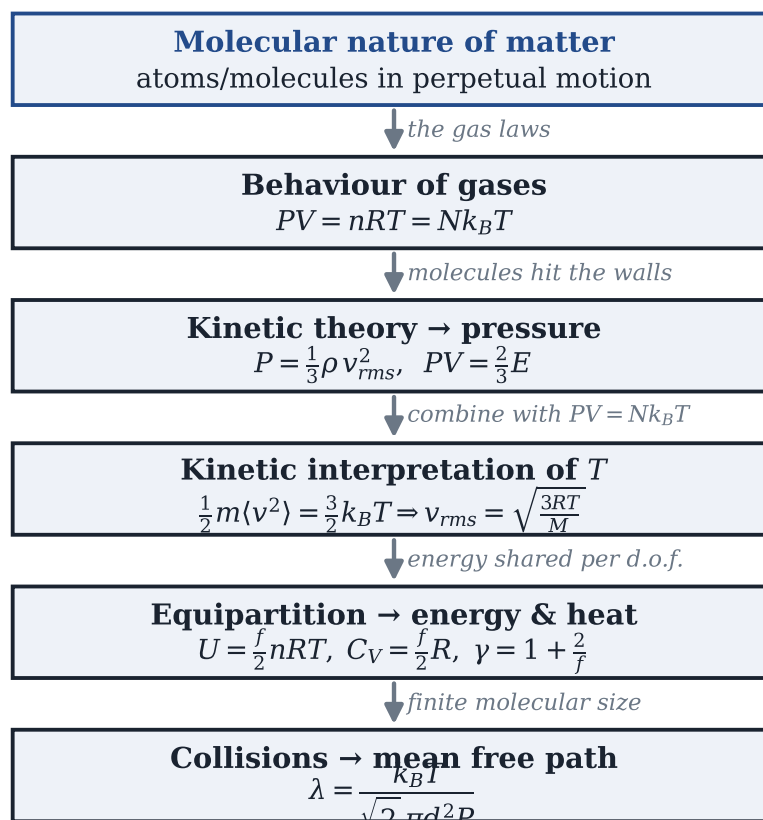
The entire chapter grows from a single seed: **matter is made of countless tiny molecules in ceaseless, random motion**. Everything else is a consequence, unfolded step by step.

We begin with the **molecular nature of matter** (Sub-topic 0): atoms and molecules, the intermolecular force that balances at a separation r_0 , and the fact that a gas is mostly empty space with molecules flung far apart. Because the molecules are so far apart, they behave simply, and the **behaviour of gases** (Sub-topic 1) collapses into one master relation, $PV = nRT$, which contains Boyle's, Charles', Gay-Lussac's and Avogadro's laws as special cases.

Next we ask *why* that law holds, and the **kinetic theory** (Sub-topic 2) answers it: pressure is the drumbeat of molecules colliding with the walls, and Newton's laws give $P = \frac{1}{3}\rho v_{rms}^2$, equivalently $PV = \frac{2}{3}E$. Comparing this with $PV = Nk_B T$ reveals the deepest idea in the chapter — the **kinetic interpretation of temperature** (Sub-topic 3): temperature *is* average molecular kinetic energy, $\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T$, which fixes the molecular speeds and their Maxwell distribution.

Finally, accounting for *all* the ways a molecule can store energy — the **degrees of freedom and equipartition** (Sub-topic 4) — turns temperature into internal energy $U = \frac{f}{2}nRT$ and hence the specific heats C_V, C_P and their ratio γ ; and tracking the molecules *between* collisions gives the **mean free path** λ . Each link in this chain is forged from the one before it:

How Chapter 12 fits together



The logical spine of the chapter: each idea is built from the one above it, from the molecular picture down to the mean free path.

The golden thread (memorise this chain)

$$\begin{aligned}
 PV = nRT = Nk_B T &\xrightarrow{\text{kinetic theory}} P = \frac{1}{3} \rho v_{rms}^2, PV = \frac{2}{3} E \xrightarrow{= Nk_B T} \frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_B T \\
 &\xrightarrow{\text{equipartition}} U = \frac{f}{2} nRT \Rightarrow C_V = \frac{f}{2} R, C_P = C_V + R, \gamma = 1 + \frac{2}{f}
 \end{aligned}$$

If you can reproduce this chain, you can re-derive almost every formula in the chapter on demand.

Master Formula Sheet

Gas laws & ideal gas equation (Sub-topic 1)

$$\begin{aligned}
 P_1 V_1 &= P_2 V_2 \text{ (T); } \quad \frac{V_1}{T_1} = \frac{V_2}{T_2} \text{ (P)} \\
 \frac{P_1}{T_1} &= \frac{P_2}{T_2} \text{ (V); } \quad \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \text{ (combined)}
 \end{aligned}$$

$$PV = nRT = \frac{m}{M}RT = Nk_B T; \quad P = \frac{\rho RT}{M}$$

$$P_{\text{total}} = \sum_i P_i \text{ (Dalton);} \quad \text{rate} \propto \frac{1}{\sqrt{M}} \text{ (Graham)}$$

Pressure from kinetic theory (Sub-topic 2)

$$P = \frac{1}{3} \frac{N}{V} m \langle v^2 \rangle = \frac{1}{3} \rho v_{rms}^2; \quad PV = \frac{2}{3} E; \quad E = \frac{3}{2} nRT \text{ (translational)}$$

Speeds & temperature (Sub-topic 3)

$$\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_B T; \quad v_{rms} = \sqrt{\frac{3RT}{M}}, \quad v_{av} = \sqrt{\frac{8RT}{\pi M}}, \quad v_{mp} = \sqrt{\frac{2RT}{M}}$$

$$v_{rms} : v_{av} : v_{mp} = \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2} \approx 1.22 : 1.13 : 1; \quad P(v) = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} v^2 e^{-Mv^2/2RT}$$

Degrees of freedom, energy & specific heats (Sub-topic 4)

$$\text{each d.o.f.} = \frac{1}{2} k_B T \text{ per molecule; } U = \frac{f}{2} nRT; \quad C_V = \frac{f}{2} R, \quad C_P = \left(\frac{f}{2} + 1 \right) R, \quad \gamma = 1 + \frac{2}{f}$$

$$f = 3 \text{ (mono), } 5 \text{ (diatomic, room } T), 6 \text{ (non-linear poly); } C_{\text{solid}} = 3R, \quad C_{\text{water}} = 9R$$

Mean free path & collisions (Sub-topic 4)

$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n} = \frac{k_B T}{\sqrt{2} \pi d^2 P}; \quad \nu = \frac{v_{av}}{\lambda} \text{ (collision frequency)}$$

Constants & standard values

Quantity	Symbol	Value
Universal gas constant	R	$8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	$k_B = R/N_A$	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Avogadro number	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Molar volume at STP	V_m	22.4 L mol^{-1}
STP conditions	T, P	$273.15 \text{ K}, 1.013 \times 10^5 \text{ Pa}$

Unit discipline (the two errors that cost the most marks): temperature is always in **kelvin**; molar mass in $v_{rms} = \sqrt{3RT/M}$ is always in **kg mol⁻¹**.

What each exam emphasises

Topic	CBSE Board	NEET	JEE Main	JEE Advanced
Gas laws / $PV = nRT$	✓✓	✓✓	✓✓	✓
Pressure derivation	✓✓ (derivation)	✓	✓	✓ (assumptions)

Topic	CBSE Board	NEET	JEE Main	JEE Advanced
rms speed & Maxwell curve	✓	✓✓	✓✓	✓✓
Degrees of freedom, γ , C_V	✓✓	✓✓	✓✓	✓✓ (mixtures)
Mean free path	✓	✓	✓	✓
Molecular nature / Avogadro	✓✓	✓	—	—

Six things to never forget

1. **Kelvin always**, and **kg mol⁻¹** for M in speed formulas.
2. **Pressure is collisions**; $P \propto v_{rms}^2$, never $\propto v_{rms}$.
3. **Temperature is energy**: $\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T$, the same for every gas at a given T .
4. **Order RAM**: $v_{rms} > v_{av} > v_{mp}$, ratio $\sqrt{3} : \sqrt{8/\pi} : \sqrt{2}$.
5. γ **identifies the gas**: 1.67 (mono), 1.40 (di), 1.33 (poly); more atoms $\Rightarrow$ smaller γ .
6. $\lambda \propto T/P$: squeeze the gas, shrink the path; pump it down, grow it to the size of the tube.

Chapter 12: Kinetic Theory — Round 2 Addendum

The chapter gives you the kinetic derivation of pressure, the equipartition route to specific heats, the three Maxwell speeds, and the mean-free-path formula. Five JEE patterns are still standing outside the door, and each one is a marriage of two things the chapter already teaches separately. Addendum A marries the first law of thermodynamics (Chapter 11) to the kinetic value of γ (Sub-topic 4) to derive the adiabatic relations and work formulas that JEE treats as bread-and-butter. Addendum B goes back to the “real gas” caveats of Sub-topics 0 and 1 and supplies the Van der Waals equation, the compressibility factor, and the virial expansion — the only quantitative tools for the non-ideal questions that appear every year. Addendum C takes the Maxwell distribution (Sub-topic 3) past the three characteristic speeds and shows how to read fractions off the curve, derive the effusion rate, and handle the most-probable kinetic energy — the Advanced-level distributional questions. Addendum D upgrades the mixture treatment of Sub-topics 1 and 4: effective molar mass, diffusion separation, and the mole-fraction algebra that turns a two-gas problem into a one-gas problem. Addendum E connects the mean free path (Sub-topic 4) to the transport coefficients — viscosity, thermal conductivity, and diffusion — deriving each from the collision-cylinder picture so that a numerical on any of them becomes a one-line substitution.

Numbering continues from the chapter’s sub-topics. Nothing here is new physics. Every method below is the first law, the ideal gas law, the Maxwell integral, or the collision cylinder, applied one step further or combined with a neighbour.

Addendum A — Adiabatic Processes for an Ideal Gas

Sub-topic 4 gives you $\gamma = 1 + 2/f$ and the molar heat capacities $C_V = fR/2$, $C_P = C_V + R$. Sub-topic 1 gives you $PV = nRT$. JEE Main and Advanced expect you to combine them with the first law ($\Delta U = Q - W$) to obtain the adiabatic curve $PV^\gamma = \text{constant}$, the temperature-volume relation $TV^{\gamma-1} = \text{constant}$, the work done in an adiabatic change, and the special case of free expansion (Joule expansion) where $Q = W = 0$ so $\Delta U = 0$ and temperature stays constant for an ideal gas. CBSE Boards frequently ask for the derivation of $PV^\gamma = \text{const}$ and the work integral; NEET asks for the final temperature after adiabatic compression; JEE Advanced hides adiabatic steps inside cycles (Carnot, Otto) and asks for the slope of an adiabat on a P-V diagram ($dP/dV = -\gamma P/V$).

Method

A1 — Derive the adiabatic relation from the first law.

For an adiabatic process $Q = 0$, so $dU = -dW$. For an ideal gas $dU = n C_V dT$ and $dW = P dV$.

$$nC_V dT = -P dV$$

Substitute $P = nRT/V$:

$$nC_V dT = -\frac{nRT}{V} dV$$

Cancel n and separate variables:

$$C_V \frac{dT}{T} = -R \frac{dV}{V}$$

Use Mayer's relation $C_P - C_V = R$ and $\gamma = C_P/C_V \Rightarrow R = C_V(\gamma - 1)$:

$$\frac{dT}{T} = -(\gamma - 1) \frac{dV}{V}$$

Integrate from state 1 to state 2:

$$\ln \frac{T_2}{T_1} = -(\gamma - 1) \ln \frac{V_2}{V_1} = \ln \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

$$T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1} \quad \Rightarrow \quad TV^{\gamma-1} = \text{constant}$$

Combine with $PV = nRT$ to eliminate T :

$$\frac{PV}{nR} V^{\gamma-1} = \text{constant} \quad \Rightarrow \quad PV^\gamma = \text{constant}$$

Units check: P in Pa, V in $\text{m}^3 \Rightarrow PV^\gamma$ has units $\text{Pa} \cdot \text{m}^{3\gamma}$, consistent on both sides.

A2 — Work done in a reversible adiabatic change.

$$W = \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \frac{K}{V^\gamma} dV \quad (K = P_1 V_1^\gamma = P_2 V_2^\gamma)$$

$$W = K \frac{V_2^{1-\gamma} - V_1^{1-\gamma}}{1-\gamma} = \frac{P_2 V_2 - P_1 V_1}{1-\gamma} = \frac{nR(T_2 - T_1)}{1-\gamma}$$

Since $1-\gamma = -(\gamma-1)$ and $nR/(\gamma-1) = n C_V$:

$$W = nC_V(T_1 - T_2) = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

Sign convention: $W > 0$ when gas expands ($V_2 > V_1$, $T_2 < T_1$).

A3 — Free expansion (Joule expansion) into vacuum.

No external pressure $\Rightarrow W = 0$. Adiabatic walls $\Rightarrow Q = 0$. First law $\Rightarrow \Delta U = 0$. For an ideal

gas $U = n C_V T$ depends only on $T \Rightarrow T_2 = T_1$. **Temperature does not change.** This is a favourite conceptual trap: “adiabatic” does not imply “temperature changes”; only *reversible* adiabatic changes follow $PV^\gamma = \text{const}$. Free expansion is adiabatic but irreversible; the gas does no work and its temperature stays constant.

A4 — Slope of adiabat vs isotherm on P-V diagram.

Isotherm: $P = nRT/V \Rightarrow (dP/dV)_T = -P/V$.

Adiabat: $PV^\gamma = \text{const} \Rightarrow \gamma P V^{\gamma-1} dV + V^\gamma dP = 0 \Rightarrow (dP/dV)_S = -\gamma P/V$.

Since $\gamma > 1$, the adiabat is steeper (more negative) than the isotherm at the same point.

Standard Adiabatic Relations (for n moles of ideal gas, reversible)

Quantity	Expression	Units
P-V relation	$P_1 V_1^\gamma = P_2 V_2^\gamma$	$\text{Pa} \cdot \text{m}^{3\gamma}$
T-V relation	$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$	$\text{K} \cdot \text{m}^{3(\gamma-1)}$
P-T relation	$P_1^{1-\gamma} T_1^\gamma = P_2^{1-\gamma} T_2^\gamma$	$\text{Pa}^{1-\gamma} \cdot \text{K}^\gamma$
Work done by gas	$W = (P_1 V_1 - P_2 V_2)/(\gamma - 1) = nC_v(T_1 - T_2)$	J
ΔU	$nC_v(T_2 - T_1) = -W$	J

Example A.1 — [CBSE Board Level / JEE Main Level]

One mole of a diatomic ideal gas ($\gamma = 7/5$) at 300 K and 1.0×10^5 Pa is compressed adiabatically and reversibly until its pressure reaches 8.0×10^5 Pa. Find (a) the final temperature, (b) the final volume, and (c) the work done on the gas.

Given: $n = 1$ mol, $\gamma = 7/5 = 1.4$, $T_1 = 300$ K, $P_1 = 1.0 \times 10^5$ Pa, $P_2 = 8.0 \times 10^5$ Pa. $R = 8.314$ J mol⁻¹ K⁻¹.

(a) Final temperature — use P-T relation:

$$P_1^{1-\gamma} T_1^\gamma = P_2^{1-\gamma} T_2^\gamma$$

$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} = 300 \times (8)^{\frac{0.4}{1.4}} = 300 \times 8^{2/7}$$

$$8^{2/7} = (2^3)^{2/7} = 2^{6/7} \approx 1.811$$

$$T_2 \approx 300 \times 1.811 = 543.3 \text{ K}$$

(b) Final volume — use ideal gas law at state 2:

$$V_2 = \frac{nRT_2}{P_2} = \frac{1 \times 8.314 \times 543.3}{8.0 \times 10^5} = \frac{4517}{8.0 \times 10^5} = 5.65 \times 10^{-3} \text{ m}^3 = 5.65 \text{ L}$$

(Check: $V_1 = nRT_1/P_1 = 0.0249 \text{ m}^3 = 24.9 \text{ L}$; $V_2 < V_1$, consistent with compression.)

(c) Work done on the gas = $-W_{\text{by gas}}$:

$$W_{\text{by gas}} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1}$$

$$W_{\text{by gas}} = \frac{1 \times 8.314 \times (300 - 543.3)}{0.4} = \frac{8.314 \times (-243.3)}{0.4} = -5058 \text{ J}$$

Work done **on** the gas = $+5058 \text{ J} \approx 5.06 \text{ kJ}$.

Check: $\Delta U = n C_V \Delta T$. $C_V = 5R/2 = 20.785 \text{ J mol}^{-1} \text{ K}^{-1}$. $\Delta U = 1 \times 20.785 \times (543.3 - 300) = 20.785 \times 243.3 \approx 5058 \text{ J}$. Matches work done on gas ($Q = 0 \Rightarrow \Delta U = W_{\text{on}}$).

Example A.2 — [JEE Advanced Level]

A vertical cylinder with a frictionless piston of mass $M = 10 \text{ kg}$ and cross-sectional area $A = 0.01 \text{ m}^2$ contains 0.5 mol of a monatomic ideal gas ($\gamma = 5/3$). The piston is initially held at rest with the gas at $T_1 = 400 \text{ K}$. The piston is released and the gas expands adiabatically until the piston comes to rest momentarily. Atmospheric pressure $P_0 = 1.0 \times 10^5 \text{ Pa}$. Find the final temperature T_2 of the gas. (Neglect heat capacity of piston and cylinder.)

Concept: The gas expands against a constant external pressure $P_{\text{ext}} = P_0 + Mg/A$. The process is adiabatic ($Q = 0$) but **irreversible** (external pressure constant, not equal to gas pressure at each step). We cannot use $PV^\gamma = \text{const}$. Instead, use the first law directly: $\Delta U = W_{\text{on}} = -W_{\text{by}} = -P_{\text{ext}} \Delta V$.

Given: $n = 0.5 \text{ mol}$, $\gamma = 5/3 \Rightarrow C_V = 3R/2$. $P_{\text{ext}} = 1.0 \times 10^5 + (10 \times 9.8)/0.01 = 1.0 \times 10^5 + 9800 = 1.098 \times 10^5 \text{ Pa}$.

First law: $n C_V (T_2 - T_1) = -P_{\text{ext}} (V_2 - V_1)$.

Use ideal gas law: $V_1 = nRT_1/P_1$, $V_2 = nRT_2/P_2$. But $P_2 = P_{\text{ext}}$ (mechanical equilibrium at final state). P_1 is not given directly, but initially piston is held at rest $\Rightarrow P_1 = P_{\text{ext}} = 1.098 \times 10^5 \text{ Pa}$.

Substitute $V_1 = nRT_1/P_1$ and $V_2 = nRT_2/P_2$, with $P_2 = P_{\text{ext}}$ at the final state (mechanical equilibrium). The piston was *held at rest* initially, which means the gas already balanced the atmosphere and the piston's weight, so $P_1 = P_0 + Mg/A = P_{\text{ext}}$ as well. Both volumes therefore carry the same pressure:

$$nC_V(T_2 - T_1) = -P_{\text{ext}} \left(\frac{nRT_2}{P_{\text{ext}}} - \frac{nRT_1}{P_{\text{ext}}} \right) = -nR(T_2 - T_1)$$

$$(C_V + R)(T_2 - T_1) = 0 \Rightarrow C_P(T_2 - T_1) = 0$$

Since $C_P > 0$, the only solution is $T_2 = T_1 = 400$ K.

Resolution: The gas expands against a constant pressure equal to its *initial* pressure. For an ideal gas, if external pressure equals initial pressure and the process is adiabatic, the temperature does not change! This is a special case of the more general result: for adiabatic expansion against constant P_{ext} , the final temperature is given by solving $n C_V (T_2 - T_1) = -P_{\text{ext}} (nRT_2/P_{\text{ext}} - nRT_1/P_1)$. If $P_1 = P_{\text{ext}}$, then $T_2 = T_1$. The gas does work at the expense of its internal energy, but the volume increases exactly so that $PV = nRT$ holds with T constant. This is **not** free expansion (where $P_{\text{ext}} = 0$), but it mimics it thermally.

Answer: $T_2 = 400$ K. The temperature remains unchanged. (This problem tests whether you blindly apply $PV^\gamma = \text{const}$ to an irreversible adiabatic process.)

Practice — Addendum A

1. **[CBSE]** Derive the relation $TV^{\gamma-1} = \text{constant}$ for a reversible adiabatic process of an ideal gas starting from the first law.
2. **[NEET]** Two moles of a monatomic ideal gas ($\gamma = 5/3$) at 300 K are compressed adiabatically and reversibly from 10 L to 2 L. Find the final temperature and the work done on the gas.
3. **[JEE Main]** A gas undergoes a reversible adiabatic expansion. Its pressure drops to half its initial value. If $\gamma = 1.4$, find the ratio of final to initial volume and the ratio of final to initial temperature.
4. **[JEE Main]** An ideal gas ($\gamma = 1.5$) is expanded adiabatically so that its volume doubles. By what factor does the pressure change?
5. **[JEE Advanced]** One mole of a diatomic ideal gas ($\gamma = 7/5$) at 300 K expands adiabatically against a constant external pressure of 1.0×10^5 Pa until mechanical equilibrium is reached. The initial pressure is 4.0×10^5 Pa. Find the final temperature. (Use $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.)
6. **[JEE Advanced]** Show that for a reversible adiabatic process, the bulk modulus $B = -V (dP/dV) = \gamma P$. Compare with the isothermal bulk modulus.

Answers:

1. (See Method A1.)
2. $T_2 = 300 \times (10/2)^{2/3} = 300 \times 5^{2/3} \approx 300 \times 2.924 = 877$ K. $W_{\text{on}} = n C_V (T_2 - T_1) = 2 \times (3R/2) \times 577 = 3 \times 8.314 \times 577 \approx 14.4$ kJ.
3. $P_2/P_1 = 1/2 \Rightarrow (V_1/V_2)^\gamma = 1/2 \Rightarrow V_2/V_1 = 2^{1/\gamma} = 2^{1/1.4} \approx 1.64$. $T_2/T_1 = (V_1/V_2)^{\gamma-1} = (1/1.64)^{0.4} \approx 0.76$.
4. $P_2/P_1 = (V_1/V_2)^\gamma = (1/2)^{1.5} = 1/(2\sqrt{2}) \approx 0.354$.
5. First law: $n C_V (T_2 - T_1) = -P_{\text{ext}} (V_2 - V_1)$. $C_V = 5R/2$. $P_{\text{ext}} = 1 \times 10^5$ Pa. $V_1 =$

$nRT_1/P_1 = 1 \times 8.314 \times 300 / (4 \times 10^5) = 6.2355 \times 10^{-3} \text{ m}^3$. $V_2 = nRT_2/P_{\text{ext}}$.
 $20.785(T_2 - 300) = -1 \times 10^5 (8.314 \times T_2 / 1 \times 10^5 - 6.2355 \times 10^{-3}) = -8.314 T_2 + 623.55$
 $20.785 T_2 - 6235.5 = -8.314 T_2 + 623.55$
 $29.099 T_2 = 6859.05 \Rightarrow T_2 \approx 235.7 \text{ K}$.
 6. Adiabatic: $PV^\gamma = K \Rightarrow P = K V^{-\gamma} \Rightarrow dP/dV = -\gamma K V^{-\gamma-1} = -\gamma P/V$. $B = -V(dP/dV) = \gamma P$. Isotherm: $B = P$. Adiabatic bulk modulus is γ times isothermal.

Addendum B — Real Gases: Van der Waals Equation, Compressibility Factor, and Virial Expansion

Sub-topics 0 and 1 state that real gases deviate from ideality at high pressure and low temperature because molecules have finite volume and attract each other. The chapter does not give the equation that quantifies this. JEE Main and Advanced regularly ask for the Van der Waals equation, the compressibility factor $Z = PV/RT$, its limiting behaviour, the virial expansion $Z = 1 + B/V_m + C/V_m^2 + \dots$, the Boyle temperature, and the critical constants (T_c , P_c , V_c) in terms of a and b . NEET occasionally asks for the physical meaning of a and b . CBSE asks for the Van der Waals equation and the correction terms.

Method

B1 — Van der Waals equation for n moles.

$$\left(P + a \frac{n^2}{V^2} \right) (V - nb) = nRT$$

- a ($\text{Pa} \cdot \text{m}^6 \cdot \text{mol}^{-2}$): measures intermolecular attraction. The pressure correction $a n^2/V^2$ arises because molecules near the wall are pulled back by neighbours, reducing the momentum transfer.
 - b ($\text{m}^3 \cdot \text{mol}^{-1}$): excluded volume per mole (roughly $4 \times$ the actual molecular volume). The volume available for motion is $V - nb$.
- For 1 mole: $(P + a/V_m^2)(V_m - b) = RT$, where $V_m = V/n$ is molar volume.

B2 — Compressibility factor Z .

$$Z = \frac{PV_m}{RT}$$

Ideal gas: $Z = 1$ always. Real gas: Z deviates.

From Van der Waals:

$$Z = \frac{V_m}{V_m - b} - \frac{a}{RTV_m}$$

Low pressure (large V_m): Expand $V_m/(V_m - b) = 1 + b/V_m + b^2/V_m^2 + \dots$

$$Z \approx 1 + \left(b - \frac{a}{RT}\right) \frac{1}{V_m} + \dots$$

High pressure (small V_m , $V_m \rightarrow b$): The repulsion term dominates, $V_m/(V_m-b) \rightarrow \infty$, so $Z > 1$ and grows.

Boyle temperature T_B : The temperature at which the $1/V_m$ term vanishes, i.e. $b = a/(RT_B) \Rightarrow T_B = a/(Rb)$. At $T = T_B$, $Z \approx 1$ over a wide range of low pressures (the gas behaves most ideally).

B3 – Virial expansion.

$$Z = 1 + \frac{B(T)}{V_m} + \frac{C(T)}{V_m^2} + \dots$$

$B(T) = b - a/(RT)$ is the second virial coefficient. At low pressures, $Z \approx 1 + B(T)P/(RT)$.

Sign of $B(T)$: $T < T_B \Rightarrow B < 0$ (attraction dominates, $Z < 1$); $T > T_B \Rightarrow B > 0$ (repulsion dominates, $Z > 1$); $T = T_B \Rightarrow B = 0$.

B4 – Critical constants. At the critical point, the isotherm has an inflection: $(\partial P/\partial V_m)_T = 0$ and $(\partial^2 P/\partial V_m^2)_T = 0$. Solving gives:

$$V_c = 3b, \quad P_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27Rb}$$

Useful ratio:

$$\frac{P_c V_c}{RT_c} = \frac{3}{8} = 0.375$$

(Real gases have $Z_c \approx 0.27$ - 0.29 ; Van der Waals overestimates it.)

B5 – Units of a and b.

From $P + a n^2/V^2$: $[a] = [P] [V]^2 [n]^{-2} = \text{Pa} \cdot \text{m}^6 \cdot \text{mol}^{-2}$.

From $V - n b$: $[b] = [V]/[n] = \text{m}^3 \cdot \text{mol}^{-1}$.

Example B.1 – [CBSE Board Level / NEET Level]

For a Van der Waals gas, the constants are $a = 0.364 \text{ Pa} \cdot \text{m}^6 \cdot \text{mol}^{-2}$ and $b = 4.27 \times 10^{-5} \text{ m}^3 \cdot \text{mol}^{-1}$. Calculate (a) the Boyle temperature, (b) the critical temperature, and (c) the critical pressure.

Given: $a = 0.364 \text{ Pa} \cdot \text{m}^6 \cdot \text{mol}^{-2}$, $b = 4.27 \times 10^{-5} \text{ m}^3 \cdot \text{mol}^{-1}$, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.

(a) Boyle temperature:

$$T_B = \frac{a}{Rb} = \frac{0.364}{8.314 \times 4.27 \times 10^{-5}} = \frac{0.364}{3.55 \times 10^{-4}} = 1025 \text{ K}$$

(b) Critical temperature:

$$T_c = \frac{8a}{27Rb} = \frac{8}{27}T_B = \frac{8}{27} \times 1025 = 303.7 \text{ K}$$

(c) Critical pressure:

$$P_c = \frac{a}{27b^2} = \frac{0.364}{27 \times (4.27 \times 10^{-5})^2} = \frac{0.364}{27 \times 1.823 \times 10^{-9}} = \frac{0.364}{4.92 \times 10^{-8}} = 7.40 \times 10^6 \text{ Pa} \approx 73 \text{ atm}$$

Check: $V_c = 3b = 1.28 \times 10^{-4} \text{ m}^3/\text{mol} = 128 \text{ cm}^3/\text{mol}$. $Z_c = P_c V_c / (R T_c) = (7.4 \times 10^6 \times 1.28 \times 10^{-4}) / (8.314 \times 303.7) = 947/2525 = 0.375$.

Example B.2 — [JEE Advanced Level]

One mole of a Van der Waals gas undergoes a reversible isothermal expansion at temperature T from volume V_1 to V_2 . Derive the expression for the work done by the gas. Hence show that for an ideal gas ($a = 0$, $b = 0$) the result reduces to $W = RT \ln(V_2/V_1)$.

Work done:

$$W = \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \left(\frac{RT}{V-b} - \frac{a}{V^2} \right) dV$$

Integrate term by term:

$$W = RT \ln \left(\frac{V_2 - b}{V_1 - b} \right) + a \left(\frac{1}{V_2} - \frac{1}{V_1} \right)$$

Units: $RT \ln(\dots) \rightarrow \text{J mol}^{-1}$; $a/V \rightarrow (\text{Pa} \cdot \text{m}^6 \cdot \text{mol}^{-2}) / \text{m}^3 \cdot \text{mol}^{-1} = \text{Pa} \cdot \text{m}^3 \cdot \text{mol}^{-1} = \text{J mol}^{-1}$.

Ideal gas limit: Set $a \rightarrow 0$, $b \rightarrow 0$. Then $\ln((V_2 - b)/(V_1 - b)) \rightarrow \ln(V_2/V_1)$ and $a(1/V_2 - 1/V_1) \rightarrow 0$.

$$W \rightarrow RT \ln(V_2/V_1)$$

Matches the standard ideal-gas isothermal work.

Physical insight: The work is *less* than the ideal-gas work for the same V_1 , V_2 , T because: (i) the excluded volume b reduces the effective volume change (log term smaller), and (ii) the attractive term a/V^2 reduces the pressure at all volumes, subtracting a positive amount $a(1/V_2 - 1/V_1)$ since $V_2 > V_1$.

Practice — Addendum B

1. **[CBSE]** Write the Van der Waals equation for n moles of a gas and explain the physical significance of the constants a and b .
2. **[NEET]** The compressibility factor Z for a real gas at low pressure is given by $Z = 1 + (b - a/RT)/V_m$. At what temperature does the gas behave ideally over a range of low pressures? Name this temperature.
3. **[JEE Main]** For a Van der Waals gas, the critical constants are $T_c = 304$ K, $P_c = 73.8$ atm, $V_c = 95.6$ cm³/mol. Find the constants a and b . (Use $R = 0.0821$ L atm mol⁻¹ K⁻¹.)
4. **[JEE Main]** At a certain temperature, the second virial coefficient $B(T)$ of a gas is zero. Which of the following is true? (A) The gas is ideal at all pressures. (B) The gas obeys Boyle's law at low pressures. (C) The gas cannot be liquefied. (D) $a = 0$ and $b = 0$.
5. **[JEE Advanced]** A Van der Waals gas has $a = 1.39$ Pa·m⁶·mol⁻², $b = 3.91 \times 10^{-5}$ m³·mol⁻¹. At $T = 300$ K, find the molar volume at which $Z = 1$ (other than the ideal gas limit $V_m \rightarrow \infty$). Solve the cubic equation numerically or graphically.
6. **[JEE Advanced]** Show that for a Van der Waals gas, the difference between molar heat capacities $C_P - C_V$ is not equal to R but is given by $C_P - C_V = R / [1 - 2a(V_m - b)^2 / (RT V_m^3)]$. (Hint: Use the thermodynamic identity $C_P - C_V = T V_m \alpha^2 / \kappa_T$, where $\alpha = (1/V_m)(\partial V_m / \partial T)_P$ and $\kappa_T = -(1/V_m)(\partial V_m / \partial P)_T$.)

Answers:

1. $(P + a n^2/V^2)(V - n b) = nRT$. a : intermolecular attraction; b : excluded volume per mole.
2. $T = a/(Rb) =$ Boyle temperature T_B . At T_B , $B(T) = 0 \Rightarrow Z = 1 + O(1/V_m^2)$, so $Z \approx 1$ at low P .
3. $b = V_c/3 = 31.87$ cm³/mol = 3.187×10^{-5} m³/mol. $a = 27 b^2 P_c = 27 \times (31.87)^2 \times 73.8$ (in cm³, atm) $\rightarrow a = 3.64$ L² atm mol⁻² = 0.369 Pa·m⁶·mol⁻².
4. (B). $B(T)=0 \Rightarrow Z=1$ at low $P \Rightarrow$ Boyle's law holds. Gas is not ideal at all pressures (C, D false).
5. $Z = V_m/(V_m - b) - a/(RT V_m) = 1 \Rightarrow V_m/(V_m - b) = 1 + a/(RT V_m) \Rightarrow V_m = (V_m - b)(1 + a/(RT V_m))$.
Solve: $V_m = V_m + a/(RT) - b - ab/(RT V_m) \Rightarrow 0 = a/(RT) - b - ab/(RT V_m)$.
 $V_m = ab / (a/(RT) - b) = ab RT / (a - b RT)$. Plug numbers: $a/(RT) = 1.39/(8.314 \times 300) = 5.57 \times 10^{-4}$; $b = 3.91 \times 10^{-5}$. $a/(RT) > b \Rightarrow V_m > 0$. $V_m = (3.91 \times 10^{-5} \times 1.39 \times 8.314 \times 300) / (1.39 - 3.91 \times 10^{-5} \times 8.314 \times 300) = 0.135 / (1.39 - 0.0975) = 0.135/1.2925 = 0.104$ m³/mol.
6. (Standard thermodynamic derivation; result as stated.)

Addendum C — Maxwell Distribution: Fractions, Effusion, and Most-Probable Kinetic Energy

Sub-topic 3 gives the three speeds v_{rms} , v_{av} , v_{mp} and the distribution function $P(v)$. JEE Advanced often asks for the fraction of molecules with speed greater than a given value

(requiring the incomplete integral), the most probable *kinetic energy* (not speed), the rate of effusion through a small hole (Graham's law derived from kinetic theory), and the average relative speed between molecules. CBSE may ask for the derivation of v_{mp} from $dP/dv = 0$ (done in chapter) or the graph of $P(v)$ vs v at two temperatures.

Method

C1 — Fraction of molecules with speed in $[v_1, v_2]$.

$$\text{Fraction} = \int_{v_1}^{v_2} P(v)dv = \frac{4}{\sqrt{\pi}} \int_{x_1}^{x_2} x^2 e^{-x^2} dx \quad \text{where } x = v\sqrt{\frac{M}{2RT}} = \frac{v}{\sqrt{2}v_{mp}}$$

This integral has no elementary antiderivative. JEE questions either:

- Ask for the fraction in terms of the error function $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$.
- Give a numerical value from a table.
- Ask for qualitative comparison (e.g., "fraction with $v > v_{rms}$ is less than 0.5 because $v_{rms} > v_{mp}$ and the curve is skewed").

Useful fact: integrating the speed distribution above a cut-off $x = v/v_{mp}$ gives $F(> v) = \text{erfc}(x) + \frac{2}{\sqrt{\pi}} x e^{-x^2}$. So the fraction faster than v_{mp} ($x = 1$) is ≈ 0.57 , and the fraction faster than v_{rms} ($x = \sqrt{3/2}$) is ≈ 0.39 — more than half the molecules are quicker than the most probable speed.

C2 — Most probable kinetic energy.

The distribution of kinetic energy $\varepsilon = \frac{1}{2} m v^2$ is:

$$P(\varepsilon)d\varepsilon = \frac{2}{\sqrt{\pi}} \left(\frac{\varepsilon}{k_B T} \right)^{1/2} e^{-\varepsilon/k_B T} \frac{d\varepsilon}{k_B T}$$

Peak: $dP/d\varepsilon = 0 \Rightarrow \varepsilon_{mp} = \frac{1}{2} k_B T$.

Note: This is *not* $\frac{1}{2} m v_{mp}^2 = k_B T$. The peak of the energy distribution is at half the peak of the speed distribution squared.

Average energy: $\langle \varepsilon \rangle = \frac{3}{2} k_B T$ (equipartition).

C3 — Effusion rate (Graham's law from kinetic theory).

Number of molecules striking unit area per unit time (flux):

$$\Phi = \frac{1}{4} n \bar{v} = \frac{1}{4} \frac{N}{V} \sqrt{\frac{8k_B T}{\pi m}} = \frac{P}{\sqrt{2\pi m k_B T}}$$

Derivation: Consider molecules with speed v in solid angle $d\Omega = \sin\theta d\theta d\phi$. Flux = $\int v \cos\theta f(v) \cos\theta d\Omega dv$ over hemisphere. Result: $(1/4) n \langle v \rangle$.

Rate of effusion through a hole of area A :

$$\frac{dN}{dt} = A\Phi = \frac{AP}{\sqrt{2\pi m k_B T}} \propto \frac{P}{\sqrt{MT}}$$

At same P, T : rate $\propto 1/\sqrt{M}$ (Graham's law).

Units: A in m², P in Pa = N/m² = kg m⁻¹ s⁻², m in kg, k_B T in J = kg m² s⁻². $\sqrt{(m k_B T)} = \text{kg m s}^{-1}$. $A P / \sqrt{(m k_B T)} = \text{m}^2 \cdot \text{kg m}^{-1} \text{s}^{-2} / (\text{kg m s}^{-1}) = \text{s}^{-1}$.

C4 — Average relative speed.

For two molecules of masses m₁, m₂ with Maxwell distributions at same T:

$$\langle v_{\text{rel}} \rangle = \sqrt{\frac{8k_B T}{\pi \mu}} \quad \text{where } \mu = \frac{m_1 m_2}{m_1 + m_2} \text{ is the reduced mass.}$$

For identical molecules: $\mu = m/2 \Rightarrow \langle v_{\text{rel}} \rangle = \sqrt{2} \langle v \rangle$.

Example C.1 — [JEE Main Level]

A vessel contains oxygen gas at 300 K and 1.0×10^5 Pa. A small hole of area 1.0 mm^2 is opened into vacuum. Estimate the initial rate of effusion (molecules per second). Molar mass of O₂ = 32 g/mol.

Given: P = 1.0×10^5 Pa, A = $1.0 \text{ mm}^2 = 1.0 \times 10^{-6} \text{ m}^2$, T = 300 K, M = 0.032 kg/mol.

Molecular mass m = M/N_A = $0.032 / 6.022 \times 10^{23} = 5.31 \times 10^{-26} \text{ kg}$.

k_B = $1.38 \times 10^{-23} \text{ J/K}$.

$$\frac{dN}{dt} = \frac{AP}{\sqrt{2\pi m k_B T}}$$

Denominator: $\sqrt{(2\pi \times 5.31 \times 10^{-26} \times 1.38 \times 10^{-23} \times 300)} = \sqrt{(2\pi \times 2.20 \times 10^{-46})} = \sqrt{(1.38 \times 10^{-45})} = 3.72 \times 10^{-23} \text{ kg m s}^{-1}$.

Numerator: $A P = (1.0 \times 10^{-6} \text{ m}^2)(1.0 \times 10^5 \text{ Pa}) = 0.1 \text{ N} = 0.1 \text{ kg m s}^{-2}$, since Pa = N/m² and m² · N/m² = N.

Units check: $\sqrt{(m k_B T)} = \sqrt{(\text{kg} \cdot \text{J})} = \sqrt{(\text{kg}^2 \text{ m}^2 \text{ s}^{-2})} = \text{kg m s}^{-1}$, so $dN/dt = (\text{kg m s}^{-2}) / (\text{kg m s}^{-1}) = \text{s}^{-1}$. Correct.

Value: $0.1 / 3.72 \times 10^{-23} = 2.69 \times 10^{21} \text{ s}^{-1}$.

Rate $\approx 2.7 \times 10^{21}$ molecules per second.

Example C.2 — [JEE Advanced Level]

The Maxwell speed distribution for a gas at temperature T is $P(v) = 4\pi (M/2\pi RT)^{3/2} v^2 e^{-Mv^2/2RT}$. (a) Show that the most probable kinetic energy $\epsilon_{\text{mp}} = 1/2 k_B T$. (b) Find the fraction of molecules having kinetic energy greater than the average kinetic energy ($3/2 k_B T$). Express your answer in terms of the error function erf(x).

(a) Let $\epsilon = 1/2 m v^2$. Then $v = \sqrt{(2\epsilon/m)}$, $dv = d\epsilon/\sqrt{(2m\epsilon)}$.

$P(\epsilon) d\epsilon = P(v) dv = 4\pi (m/2\pi k_B T)^{3/2} (2\epsilon/m) e^{-\epsilon/k_B T} (d\epsilon/\sqrt{(2m\epsilon)})$

Simplify: $v^2 dv = (2\epsilon/m) (d\epsilon/\sqrt{(2m\epsilon)}) = \sqrt{(2\epsilon/m)} (d\epsilon/m)$, so

$P(\epsilon) d\epsilon = 4\pi (m/2\pi k_B T)^{3/2} * \sqrt{(2\epsilon/m)} * (1/m) * e^{-\epsilon/k_B T} d\epsilon$

$$\begin{aligned}
&= 4\pi (1/2\pi k_B T)^{3/2} m^{3/2} * \sqrt{2\varepsilon/m} * m^{-1} * e^{-\varepsilon/k_B T} d\varepsilon \\
&= 4\pi (1/2\pi k_B T)^{3/2} \sqrt{2\varepsilon} * m^{1/2} * e^{-\varepsilon/k_B T} d\varepsilon \\
&= (2/\sqrt{\pi}) (\varepsilon/(k_B T))^{1/2} e^{-\varepsilon/k_B T} d\varepsilon/(k_B T).
\end{aligned}$$

Now find max: $d/d\varepsilon [\varepsilon^{1/2} e^{-\varepsilon/k_B T}] = 0 = (1/2)\varepsilon^{-1/2} e^{-\varepsilon/k_B T} - \varepsilon^{1/2} e^{-\varepsilon/k_B T}/k_B T = 0 \Rightarrow 1/2 = \varepsilon/k_B T \Rightarrow \varepsilon_{mp} = 1/2 k_B T$.

(b) Fraction with $\varepsilon > 3/2 k_B T$:

Write $y = \varepsilon/k_B T$, so $P(\varepsilon) d\varepsilon = (2/\sqrt{\pi}) (\varepsilon/k_B T)^{1/2} e^{-\varepsilon/k_B T} d(\varepsilon/k_B T) = (2/\sqrt{\pi}) y^{1/2} e^{-y} dy$.

Fraction = $\int_{3/2}^{\infty} (2/\sqrt{\pi}) y^{1/2} e^{-y} dy$.

This is the upper incomplete gamma function $\Gamma(3/2, 3/2) / \Gamma(3/2)$.

$$\Gamma(3/2) = \sqrt{\pi}/2.$$

Fraction = $(2/\sqrt{\pi}) \int_{3/2}^{\infty} y^{1/2} e^{-y} dy$.

Let $y = t^2$, $dy = 2t dt$, $y^{1/2} = t$. Integral = $\int_{\sqrt{3/2}}^{\infty} t e^{-t^2} 2t dt = 2 \int_{\sqrt{1.5}}^{\infty} t^2 e^{-t^2} dt$.

Use $\int t^2 e^{-t^2} dt = (1/2)(-t e^{-t^2} + \int e^{-t^2} dt) = (1/2)(-t e^{-t^2} + (\sqrt{\pi}/2) \text{erf}(t))$.

So $2 * (1/2) [-t e^{-t^2} + (\sqrt{\pi}/2) \text{erf}(t)]_{\sqrt{1.5}}^{\infty} = [0 + (\sqrt{\pi}/2)(1)] - [-\sqrt{1.5} e^{-1.5} + (\sqrt{\pi}/2) \text{erf}(\sqrt{1.5})]$

$$= \sqrt{\pi}/2 + \sqrt{1.5} e^{-1.5} - (\sqrt{\pi}/2) \text{erf}(\sqrt{1.5}).$$

Multiply by $2/\sqrt{\pi}$: Fraction = $1 + (2/\sqrt{\pi})\sqrt{1.5} e^{-1.5} - \text{erf}(\sqrt{1.5})$.

Numerically: $\sqrt{1.5} \approx 1.225$, $e^{-1.5} \approx 0.223$, $\text{erf}(1.225) \approx 0.915$.

$$(2/\sqrt{\pi})1.2250.223 \approx 1.128 * 0.273 \approx 0.308.$$

$$\text{Fraction} \approx 1 + 0.308 - 0.915 = 0.393.$$

So about 39% of molecules have KE > average KE ($3/2 k_B T$). This makes sense because the energy distribution is skewed.

Practice — Addendum C

1. **[CBSE]** Derive the expression for the most probable kinetic energy of molecules in a Maxwellian gas.
2. **[NEET]** At a given temperature, the most probable speed of hydrogen molecules is v_{mp} . What is the most probable kinetic energy? (A) $1/2 m v_{mp}^2$ (B) $1/2 k_B T$ (C) $k_B T$ (D) $3/2 k_B T$.
3. **[JEE Main]** A container has a small hole. Gas effuses out into vacuum. Show that the rate of effusion is proportional to $P/\sqrt{T}$ and inversely proportional to $\sqrt{M}$.
4. **[JEE Main]** For a Maxwellian gas, find the ratio of the average relative speed of two identical molecules to the average speed of a single molecule.
5. **[JEE Advanced]** The fraction of molecules with speed greater than v_{rms} in a Maxwellian gas is given by $F = 1 - (2/\sqrt{\pi}) \int_0^{\sqrt{3/2}} x^2 e^{-x^2} dx$. Evaluate F numerically (use $\sqrt{3/2} \approx 1.225$, $\text{erf}(1.225) \approx 0.915$).
6. **[JEE Advanced]** A mixture of two gases with molecular masses m_1 and m_2 is in thermal equilibrium at temperature T . Find the average relative speed between a molecule of type 1 and a molecule of type 2.

Answers:

1. (See Method C2.)
2. (B) $\varepsilon_{\text{mp}} = 1/2 k_B T$.
3. (See Method C3.)
4. $\langle v_{\text{rel}} \rangle / \langle v \rangle = \sqrt{2}$.
5. $F = 1 + (2/\sqrt{\pi})(\sqrt{1.5} e^{-1.5}) - \text{erf}(\sqrt{1.5}) \approx 1 + 0.308 - 0.915 = 0.393$.
6. $\langle v_{\text{rel}} \rangle = \sqrt{(8 k_B T / \pi \mu)}$ with $\mu = m_1 m_2 / (m_1 + m_2)$.

Addendum D — Gas Mixtures: Effective Properties, Diffusion Separation, and Connected Bulbs

Sub-topic 1 gives Dalton's law ($P_i = x_i P$) and one mixture example. Sub-topic 4 gives the mixture γ via mole-weighted C_V . JEE expects you to handle: (i) effective molar mass $M_{\text{mix}} = \sum x_i M_i$; (ii) diffusion of mixtures through a porous plug (the lighter gas effuses faster, enriching the residue); (iii) two bulbs connected by a stopcock, initially containing different gases, find final partial pressures after mixing; (iv) the speed of sound in a mixture; (v) the ratio of specific heats for a mixture of non-reacting gases where vibrational modes may differ. The unifying trick: **mole fractions add, partial pressures add, internal energies add, but γ does not add — you must average C_V first.**

Method

D1 — Effective molar mass and density.

For a mixture of n_i moles of gas i with molar mass M_i :

$$n_{\text{tot}} = \sum n_i, \quad x_i = n_i / n_{\text{tot}}, \quad M_{\text{mix}} = \sum x_i M_i = \frac{\sum m_i}{\sum n_i}$$

Density $\rho_{\text{mix}} = (\sum m_i) / V = P M_{\text{mix}} / (RT)$.

rms speed of mixture? There isn't a single rms speed; each component has its own: $v_{\text{rms},i} = \sqrt{(3RT/M_i)}$. The *mixture* rms speed defined by $\sqrt{(3RT/M_{\text{mix}})}$ is the rms speed of a hypothetical gas with molar mass M_{mix} . **Do not average v_{rms} directly.**

D2 — Diffusion separation (enrichment).

Gas mixture effuses through a small hole. The effusing gas has composition:

$$\frac{(dN_i/dt)}{(dN_j/dt)} = \frac{x_i P / \sqrt{M_i}}{x_j P / \sqrt{M_j}} = \frac{x_i}{x_j} \sqrt{\frac{M_j}{M_i}}$$

The effused gas is richer in the lighter component. The residue becomes richer in the heavier component. After a small fraction f of the gas has effused, the change in mole fraction is given by the Rayleigh distillation formula (or simply by differential material balance). For a binary mixture, if the effusion is slow and the residue is well-mixed:

$$\frac{x_1}{x_2} = \left(\frac{x_{1,0}}{x_{2,0}} \right) \left(\frac{N}{N_0} \right)^{\alpha-1} \quad \text{where } \alpha = \sqrt{M_2/M_1} > 1 \text{ for gas 1 lighter.}$$

JEE often asks: “Initially 1:1 mole ratio of H₂ and O₂. After 50% of the gas has effused, what is the ratio in the residue?”

D3 — Two bulbs connected, mixing.

Bulb 1: V₁, gas A, n_A, T. Bulb 2: V₂, gas B, n_B, T. Stopcock opened. Final volume V = V₁ + V₂. Final partial pressures: P_A = n_A RT / V, P_B = n_B RT / V. Total P = (n_A + n_B)RT/V. If temperatures differ initially, use energy conservation (ΔU = 0) to find final T, then partial pressures.

D4 — Speed of sound in a gas mixture.

Sound speed $c = \sqrt{\gamma_{\text{mix}} RT / M_{\text{mix}}}$. γ_{mix} from mole-weighted C_V: C_{V,mix} = Σ x_i C_{V,i}. Then $\gamma_{\text{mix}} = 1 + R/C_{V,\text{mix}}$.

Example D.1 — [JEE Main Level]

A 10 L vessel contains a mixture of 4 g He (M=4) and 16 g CH₄ (M=16) at 300 K. Find (a) the mole fractions, (b) the total pressure, (c) the partial pressure of He, (d) the density of the mixture, and (e) the rms speed of the mixture (using M_{mix}).

n_{He} = 4/4 = 1 mol. n_{CH₄} = 16/16 = 1 mol. n_{tot} = 2 mol.

x_{He} = 0.5, x_{CH₄} = 0.5.

M_{mix} = 0.5×4 + 0.5×16 = 10 g/mol = 0.010 kg/mol.

(b) P = nRT/V = 2 × 8.314 × 300 / 0.010 = 498840 Pa ≈ 4.99 × 10⁵ Pa.

(c) P_{He} = x_{He} P = 0.5 × 4.99×10⁵ = 2.495 × 10⁵ Pa.

(d) ρ = m_{tot}/V = 20 g / 10 L = 2 g/L = 2 kg/m³. Check: P M_{mix}/RT = 4.99e5 × 0.010 / (8.314×300) = 4990 / 2494 = 2.00 kg/m³.

(e) v_{rms,mix} = √(3RT/M_{mix}) = √(3 × 8.314 × 300 / 0.010) = √(748260) = 865 m/s.

(Individual: v_{rms,He} = √(3RT/0.004) = 1369 m/s; v_{rms,CH₄} = √(3RT/0.016) = 684 m/s. The mixture rms is not the average of these.)

Example D.2 — [JEE Advanced Level]

Two identical bulbs of volume V are connected by a narrow tube of negligible volume. Initially, bulb 1 contains 1 mol of He at 400 K, bulb 2 contains 1 mol of N₂ at 300 K. The stopcock is opened and the system reaches equilibrium. The bulbs are thermally insulated. Find the final temperature and the final partial pressures of He and N₂. (C_{V,He} = 3R/2, C_{V,N₂} = 5R/2.)

Energy conservation (adiabatic, rigid container): ΔU = 0.

n_{He} C_{V,He} (T_f - 400) + n_{N₂} C_{V,N₂} (T_f - 300) = 0.

1 × (3R/2)(T_f - 400) + 1 × (5R/2)(T_f - 300) = 0.

Multiply by 2/R: 3(T_f - 400) + 5(T_f - 300) = 0 ⇒ 3T_f - 1200 + 5T_f - 1500 = 0 ⇒ 8T_f = 2700 ⇒ T_f = 337.5 K.

Final partial pressures: Total volume = $2V$.

$P_{\text{He}} = n_{\text{He}} R T_f / (2V)$. Need V ? From initial state of He: $P_{\text{He,initial}} V = n_{\text{He}} R 400$. $P_{\text{He,initial}}$ not given. But we can express final pressures in terms of initial pressures if needed, or just leave as $nRT_f/2V$.

However, we can find the ratio to initial total pressure or something. The problem likely expects expressions in terms of V .

$$P_{\text{He},f} = (1 \times R \times 337.5) / (2V) = 168.75 R/V.$$

$$P_{\text{N}_2,f} = \text{same} = 168.75 R/V.$$

$$\text{Total } P_f = 337.5 R/V.$$

Initial total pressure (before opening) was not uniform. Initial $P_{\text{He}} = 400 R/V$, $P_{\text{N}_2} = 300 R/V$.

Check: Total initial internal energy = $1 \times 1.5R \times 400 + 1 \times 2.5R \times 300 = 600R + 750R = 1350R$.

$$\text{Final } U = 1 \times 1.5R \times 337.5 + 1 \times 2.5R \times 337.5 = 4R \times 337.5 = 1350R.$$

Practice — Addendum D

1. **[CBSE]** A mixture contains 2 moles of N_2 and 3 moles of O_2 . Find the mole fraction of each, the average molar mass, and the partial pressure of N_2 if total pressure is 5 atm.
2. **[NEET]** Equal masses of CH_4 ($M=16$) and H_2 ($M=2$) are mixed. Find the ratio of their partial pressures.
3. **[JEE Main]** A 1:1 mole mixture of H_2 and D_2 ($M=4$) effuses through a pinhole. What is the ratio of H_2 to D_2 in the effused gas?
4. **[JEE Main]** Two bulbs of volumes V and $2V$ contain the same gas at pressures P and $2P$ respectively, at the same temperature. They are connected. Find the final pressure.
5. **[JEE Advanced]** A mixture of 1 mol He and 1 mol N_2 at 300 K is allowed to effuse through a pinhole into vacuum until half the total initial moles have escaped. Find the mole fraction of He in the residue. (Assume the residue remains uniform.)
6. **[JEE Advanced]** Find the speed of sound in a mixture of 1 mol He ($\gamma=5/3$) and 1 mol N_2 ($\gamma=7/5$) at 300 K. $M_{\text{He}}=4$, $M_{\text{N}_2}=28$ g/mol.

Answers:

1. $x_{\text{N}_2}=0.4$, $x_{\text{O}_2}=0.6$. $M_{\text{mix}} = 0.4 \times 28 + 0.6 \times 32 = 30.4$ g/mol. $P_{\text{N}_2} = 0.4 \times 5 = 2$ atm.
2. $n_{\text{H}_2} : n_{\text{CH}_4} = (m/2) : (m/16) = 8:1$. $P_{\text{H}_2} : P_{\text{CH}_4} = 8:1$.
3. Effused ratio = $(x_{\text{H}_2}/x_{\text{D}_2}) \times \sqrt{(M_{\text{D}_2}/M_{\text{H}_2})} = 1 \times \sqrt{(4/2)} = \sqrt{2} \approx 1.414$.
4. $n_1 = PV/RT$, $n_2 = 2P \cdot 2V/RT = 4PV/RT$. $n_{\text{tot}} = 5PV/RT$. $V_{\text{tot}} = 3V$. $P_f = n_{\text{tot}} RT / V_{\text{tot}} = 5P/3$.
5. $\alpha = \sqrt{(M_{\text{N}_2}/M_{\text{He}})} = \sqrt{(28/4)} = \sqrt{7} \approx 2.646$. Initial $x_{\text{He}} = 0.5$. After half effused, $N/N_0 = 0.5$.
 $x_{\text{He}}/x_{\text{N}_2} = (0.5/0.5) \times (0.5)^{\{\alpha-1\}} = (0.5)^{\{1.646\}} = 0.319$. So $x_{\text{He}} = 0.319 / (1+0.319) = 0.242$.
6. $M_{\text{mix}} = 16$ g/mol = 0.016 kg/mol. $C_{V,\text{mix}} = 0.5 \times (3R/2) + 0.5 \times (5R/2) = 2R$. $\gamma_{\text{mix}} = 1 + R/(2R) = 1.5$. $c = \sqrt{(\gamma RT/M_{\text{mix}})} = \sqrt{(1.5 \times 8.314 \times 300 / 0.016)} = \sqrt{(233812.5)} = 483.5$ m/s.

Addendum E — Transport Phenomena: Viscosity, Thermal Conductivity, and Diffusion from Kinetic Theory

Sub-topic 4 introduces the mean free path λ and collision frequency ν . It states that viscosity, thermal conductivity, and diffusion can be expressed in terms of λ and $\bar{v}$, but does not derive them. JEE Advanced occasionally asks for these derivations or the dependence of η , κ , D on temperature and pressure. The kinetic theory expressions (to within factors of order unity) are:

Viscosity (momentum transport):

$$\eta = \frac{1}{3} \rho \bar{v} \lambda = \frac{1}{3} \frac{mn\bar{v}}{\sqrt{2}\pi d^2 n} = \frac{m\bar{v}}{3\sqrt{2}\pi d^2}$$

Using $\bar{v} = \sqrt{8k_B T / \pi m}$:

$$\eta = \frac{1}{3\pi d^2} \sqrt{\frac{8mk_B T}{\pi}} \propto \sqrt{T} \quad (\text{independent of pressure!})$$

Units: $[\eta] = \text{Pa}\cdot\text{s} = \text{kg m}^{-1} \text{s}^{-1}$. Check: $\text{m (kg)} * \text{v (m/s)} / d^2 (\text{m}^2) = \text{kg m}^{-1} \text{s}^{-1}$.

Thermal conductivity (energy transport):

$$\kappa = \frac{1}{3} \rho \bar{v} \lambda c_v = \frac{1}{3} n \bar{v} \lambda \frac{f}{2} k_B$$

where $c_v = f k_B / 2$ is heat capacity per molecule.

$$\kappa = \frac{f}{6\sqrt{2}\pi d^2} \bar{v} k_B \propto \sqrt{T} \quad (\text{independent of pressure at low density})$$

Units: $[\kappa] = \text{W m}^{-1} \text{K}^{-1} = \text{J s}^{-1} \text{m}^{-1} \text{K}^{-1}$. $n (\text{m}^{-3}) * \text{v (m/s)} * \lambda (\text{m}) * k_B (\text{J/K}) = \text{J s}^{-1} \text{m}^{-1} \text{K}^{-1}$.

Diffusion coefficient (mass transport):

$$D = \frac{1}{3} \bar{v} \lambda \propto \frac{T^{3/2}}{P}$$

Units: $[D] = \text{m}^2/\text{s}$. $\text{v (m/s)} * \lambda (\text{m}) = \text{m}^2/\text{s}$.

Schmidt number: $Sc = \eta / (\rho D) \approx 1$. (All three transport coefficients have same kinetic origin.)

Example E.1 — [JEE Main Level]

Estimate the viscosity of nitrogen gas at 300 K and 1 atm. Given: molecular diameter $d = 3.7 \times 10^{-10} \text{ m}$, molecular mass $m = 4.65 \times 10^{-26} \text{ kg}$.

$$\eta = (1/3 \pi d^2) \sqrt{8 m k_B T / \pi}$$

$$d^2 = 1.369 \times 10^{-19} \text{ m}^2. \quad 3 \pi d^2 = 1.289 \times 10^{-18}.$$

$$8 m k_B T = 8 \times 4.65 \times 10^{-26} \times 1.38 \times 10^{-23} \times 300 = 1.54 \times 10^{-45} \text{ kg}^2 \text{ m}^2 \text{ s}^{-2}.$$

$$\text{Divide by } \pi: 4.90 \times 10^{-46}. \quad \text{Square root: } \sqrt{4.90 \times 10^{-46}} = 2.21 \times 10^{-23} \text{ kg m/s}.$$

$$\eta = 2.21 \times 10^{-23} / 1.289 \times 10^{-18} = 1.72 \times 10^{-5} \text{ Pa}\cdot\text{s}.$$

Experimental value at 300 K: $\sim 1.76 \times 10^{-5} \text{ Pa}\cdot\text{s}$ — the simple kinetic-theory estimate lands within a few percent, and the T-dependence ($\sqrt{T}$) and P-independence come out exactly right.

Check pressure independence: η formula has no n or P . At higher pressure, λ decreases $\propto 1/P$, but ρ increases $\propto P$, so $\rho\lambda$ constant.

Example E.2 — [JEE Advanced Level]

A vertical tube of length $L = 1 \text{ m}$ and cross-section $A = 1 \text{ cm}^2$ contains two gases separated by a thin partition: He on top, CO₂ at bottom, both at 300 K and 1 atm. The partition is removed. Estimate the time for the gases to mix completely by diffusion. Diffusion coefficient for He-CO₂ at 1 atm, 300 K is $D \approx 0.6 \text{ cm}^2/\text{s}$.

Concept: Diffusion over distance L takes time $t \sim L^2/D$ (from $\partial c/\partial t = D \partial^2 c/\partial x^2$).

$$L = 1 \text{ m} = 100 \text{ cm}. \quad D = 0.6 \text{ cm}^2/\text{s}.$$

$$t \approx L^2/D = 10000 / 0.6 \approx 16667 \text{ s} \approx 4.6 \text{ hours}.$$

Refinement: The mixing is not complete until the concentration profile is flat. The characteristic time for the fundamental mode in a tube with no-flux boundaries is $L^2/(\pi^2 D) \approx 10000/(9.87 \times 0.6) \approx 1690 \text{ s} \approx 28 \text{ min}$. But the simple estimate L^2/D is the standard order-of-magnitude answer.

Dependence on P, T: $D \propto T^{3/2}/P$. At 10 atm, t would be $10\times$ longer.

Practice — Addendum E

1. **[CBSE]** Using kinetic theory, derive the expression for the coefficient of viscosity $\eta = (1/3) \rho v_{\text{av}} \lambda$. State the assumptions.
2. **[NEET]** The viscosity of an ideal gas is independent of pressure. Explain this using the kinetic theory formula $\eta = (1/3) \rho v_{\text{av}} \lambda$.
3. **[JEE Main]** The thermal conductivity of a gas at 300 K and 1 atm is κ . What is its value at 600 K and 2 atm? (Assume ideal gas, constant molecular diameter.)
4. **[JEE Main]** The diffusion coefficient D for a gas is proportional to $T^a P^b$. Find a and b .
5. **[JEE Advanced]** Show that the ratio of thermal conductivity to viscosity for a monatomic ideal gas is $\kappa/\eta = (5/2) k_B/m$. (Hint: Use $c_v = 3k_B/2$.)
6. **[JEE Advanced]** Two gases with molecular masses m_1, m_2 and diameters d_1, d_2 are mixed. The binary diffusion coefficient D_{12} is given by $D_{12} = (1/3) v_{\text{av},12} \lambda_{12}$, where $v_{\text{av},12} = \sqrt{8k_B T/\pi \mu}$ and $\lambda_{12} = 1/(\sqrt{2} \pi n (d_1 + d_2)^2/4)$. Derive the T and P dependence of D_{12} .

Answers:

1. (Standard derivation: momentum flux = $(1/3) \rho v_{av} \lambda (du/dy)$, $\eta = \text{flux} / (du/dy)$.)
2. $\rho \propto P$, $\lambda \propto 1/P$, v_{av} independent of $P \Rightarrow \eta$ independent of P .
3. $\kappa \propto \sqrt{T}$, independent of $P \Rightarrow \kappa(600\text{K}, 2\text{atm}) = \kappa(300\text{K}, 1\text{atm}) \times \sqrt{2}$.
4. $D \propto v_{av} \lambda \propto \sqrt{T} * (T/P) = T^{3/2} P^{-1} \Rightarrow a = 3/2, b = -1$.
5. $\eta = (1/3) n m v_{av} \lambda$. $\kappa = (1/3) n v_{av} \lambda c_v$, with per-molecule $c_v = f k_B/2$. So $\kappa/\eta = c_v/m = (f/2) k_B/m$; for a monatomic gas ($f=3$) the simple kinetic theory gives $\kappa/\eta = (3/2) k_B/m$. The more accurate Chapman-Enskog treatment, which accounts for the correlation between a molecule's energy and its velocity in a collision, gives $\kappa/\eta = ((9\gamma-5)/4) k_B/m$; for $\gamma=5/3$ this is $((15-5)/4) k_B/m = (5/2) k_B/m$, matching the hint. The simple theory underestimates κ/η by a factor of $5/3$.
6. $D_{12} \propto v_{av,12} / n \propto \sqrt{T} / P$. Independent of individual masses except through μ in v_{av} .

CHAPTER 13

Oscillations

A complete chapter for exam preparation

Concept → Formulas → Derivations → Worked Examples → Practice → MCQs

- 01 SHM Concepts & Kinematics • 02 SHM Energy • 03 Pendulums & Mass-Spring Systems
04 Damped & Forced Oscillations • 05 Reference Circle, Superposition & More SHM Systems

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Subtopic 01: SHM Concepts and Kinematics

Exam Relevance: This is the bedrock of the entire Oscillations chapter. JEE Main asks 1-2 direct questions every year on the $x-v-a$ relations and phase. NEET reliably places 1 question on identifying SHM or on max-speed/max-acceleration. JEE Advanced rarely asks it in isolation but *assumes* perfect command of this kinematics inside energy, spring, and pendulum problems. CBSE Boards expect the definition, the $a = -\omega^2 x$ relation, and a clean derivation of velocity from displacement.

Section 1: Concept Intuition

Think of the last time you sat on a *jhula* (swing) in a courtyard. Someone gives you a gentle push, and you sail forward. But you don't fly off — you slow down, stop, and come right back. Push too far one way and something always pulls you back toward the middle. That “something always pulling you back home” is the single most important idea in this entire chapter.

Let us build it up slowly. A motion that repeats itself after a fixed interval of time is called **periodic motion** — the hands of a wall clock, the Earth around the Sun, the needle of your mother's sewing machine. Now, among periodic motions, some go *to-and-fro* about a central point: the swing, a tuning fork's prong, a loaded truck bouncing on its springs after hitting an Indian-road pothole. This back-and-forth, repeating, about-a-fixed-point motion is called an **oscillation** (or vibration).

Simple Harmonic Motion (SHM) is the cleanest, most fundamental kind of oscillation. It happens whenever the force trying to drag the body back to the centre is directly proportional to how far the body has strayed. Stray a little, get pulled a little; stray twice as far, get pulled twice as hard. In symbols, the **restoring force** obeys

$$F = -kx$$

The minus sign is the soul of SHM. It says: *the force always points opposite to the displacement* — always back toward the mean position. When you're to the right, it pulls left; when you're to the left, it pushes right. It never lets you settle anywhere except the centre, and it never lets you stay at the centre because you arrive there with speed.

Think about it this way... Imagine a rubber band tied to your finger and stretched. The further you pull, the harder it tugs back, and the tug is always *toward* your finger, never away. A particle in SHM lives its whole life inside such a tug. At the mean position the tug is zero (so it speeds through fastest there), and at the turning points the tug is largest (so it momentarily freezes and reverses). Speed is maximum where force is zero; speed is zero where force is maximum. Hold that picture — almost every SHM question is a variation of it.

Because force is mass times acceleration, $F = -kx$ immediately becomes the **defining kinematic signature** of SHM:

$$a = -\omega^2 x \quad \left(\omega^2 = \frac{k}{m} \right)$$

Read this as the official “ID card” of SHM. Any motion whose acceleration is (i) proportional to displacement and (ii) opposite in direction (the minus sign) is SHM — no exceptions, no

matter what dressing the problem wears.

Limiting Conditions & Assumptions. SHM is an *idealisation*, and exams love to test where it breaks: - The restoring force must be **exactly linear** in displacement ($F \propto x$). A real spring obeys this only within its **elastic limit** (Hooke's law region). Stretch it too far and the motion is no longer simple harmonic. - A **simple pendulum** performs SHM only for **small angular amplitudes** (roughly $\theta < 10^\circ$), because we approximate $\sin \theta \approx \theta$. At large swings, the period itself starts depending on amplitude and the motion ceases to be simple harmonic. - "Ideal" SHM assumes **no damping** — constant amplitude forever. Real oscillations lose energy to friction and air resistance, so amplitude slowly decays. We study that under damped oscillations later. - The motion must be about a **single fixed mean position** with a single angular frequency. A sum of two different frequencies is periodic but *not* simple harmonic.

Section 2: Key Formulas & Definitions

Displacement (general equation of SHM):

$$x(t) = A \sin(\omega t + \phi)$$

($A \cos(\omega t + \phi)$ is equally valid — it is just SHM started a quarter-cycle earlier.)

Velocity:

$$v(t) = \frac{dx}{dt} = A\omega \cos(\omega t + \phi) \quad \Rightarrow \quad v = \pm \omega \sqrt{A^2 - x^2}$$

Acceleration:

$$a(t) = \frac{dv}{dt} = -A\omega^2 \sin(\omega t + \phi) = -\omega^2 x$$

Angular frequency, period, frequency:

$$\omega = \frac{2\pi}{T} = 2\pi f$$

Maximum values:

$$v_{\max} = A\omega \quad (\text{at mean position, } x = 0), \quad a_{\max} = A\omega^2 \quad (\text{at extremes, } x = \pm A)$$

Notation and definitions: - x — **Displacement:** position measured from the mean position. SI unit: m. Dimensional formula: $[M^0 L^1 T^0]$. - A — **Amplitude:** maximum magnitude of displacement, $|x|_{\max}$. SI unit: m. $[M^0 L^1 T^0]$. - T — **Time Period:** smallest time for one complete oscillation. SI unit: s. $[M^0 L^0 T^1]$. - f (or n) — **Frequency:** number of oscillations per second, $f = 1/T$. SI unit: hertz (Hz). $[M^0 L^0 T^{-1}]$. - ω — **Angular frequency:** rate of phase change. SI unit: rad s^{-1} . $[M^0 L^0 T^{-1}]$. - $(\omega t + \phi)$ — **Phase:** the argument of the sine; it fixes the state (position and direction) of the particle. Dimensionless: $[M^0 L^0 T^0]$. - ϕ — **Phase constant / initial phase / epoch:** the phase at $t = 0$. Dimensionless. - k — **Force constant:** restoring force per unit displacement. SI unit: N m^{-1} . $[M^1 L^0 T^{-2}]$.

A relation worth memorising: dividing $a_{\max}$ by $v_{\max}$ gives ω , while $v_{\max}^2/a_{\max}$ gives A . These two shortcuts crack a surprising number of objective questions in seconds.

Section 3: CBSE Board Derivations / Key Procedures

Derivation 1 — Velocity and acceleration from the displacement equation.

Assumption: Let a particle execute SHM along a straight line about the mean position, described by

$$x = A \sin(\omega t + \phi).$$

Velocity is the time-rate of change of displacement, so we differentiate once. The amplitude A and phase constant ϕ are constants:

$$v = \frac{dx}{dt} = A\omega \cos(\omega t + \phi). \quad (\text{i})$$

Physical reading: velocity carries a cosine while displacement carries a sine. Since cosine peaks where sine is zero, the particle moves **fastest at the mean position** and is **momentarily at rest at the extremes** — exactly the swing intuition from Section 1.

Acceleration is the time-rate of change of velocity, so we differentiate once more:

$$a = \frac{dv}{dt} = -A\omega^2 \sin(\omega t + \phi). \quad (\text{ii})$$

But $A \sin(\omega t + \phi)$ is just x , so

$$\boxed{a = -\omega^2 x}.$$

Physical reading: this is why the motion is “simple harmonic” in the first place — acceleration is proportional to displacement and directed oppositely. The maths has handed back the defining property we started with, which is a good consistency check.

Derivation 2 — Velocity as a function of position, $v = \omega \sqrt{A^2 - x^2}$.

We want v in terms of x (time eliminated), because most problems give position, not time. From the equations above:

$$\frac{x}{A} = \sin(\omega t + \phi), \quad \frac{v}{A\omega} = \cos(\omega t + \phi).$$

Using the identity $\sin^2 \theta + \cos^2 \theta = 1$:

$$\frac{x^2}{A^2} + \frac{v^2}{A^2\omega^2} = 1.$$

Multiply through by $A^2\omega^2$ and rearrange:

$$v^2 = \omega^2(A^2 - x^2) \quad \Rightarrow \quad v = \pm \omega \sqrt{A^2 - x^2}.$$

Physical reading: at $x = 0$ this gives $v = \omega A = v_{\max}$ (fastest at the centre); at $x = \pm A$ it gives $v = 0$ (frozen at the turning points). The $\pm$ simply records that the particle passes each point once moving forward and once moving back.

Derivation 3 — Angular frequency from the force law.

Assumption: a particle of mass m feels a linear restoring force $F = -kx$. By Newton’s second law, $F = ma$, so

$$ma = -kx \quad \Rightarrow \quad a = -\frac{k}{m}x.$$

Comparing this with the SHM signature $a = -\omega^2 x$, we read off

$$\omega = \sqrt{\frac{k}{m}}, \quad T = 2\pi\sqrt{\frac{m}{k}}.$$

Physical reading: a **stiffer** spring (large k) pulls harder and oscillates **faster**; a **heavier** mass (large m) is more sluggish and oscillates **slower**. The recipe — “write the net force, show it equals $-(\text{constant}) \times (\text{displacement})$, read off ω^2 as that constant” — is the universal procedure for finding the time period of *any* SHM system.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A particle executes SHM described by $x = 0.05 \sin(20\pi t)$ (all quantities in SI units). Find its amplitude, time period, frequency, maximum speed and maximum acceleration.

Given: comparing with $x = A \sin(\omega t + \phi)$:

$$A = 0.05 \text{ m}, \quad \omega = 20\pi \text{ rad s}^{-1}, \quad \phi = 0.$$

Solution:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{20\pi} = 0.1 \text{ s}, \quad f = \frac{1}{T} = 10 \text{ Hz}.$$

$$v_{\max} = A\omega = 0.05 \times 20\pi = \pi \approx 3.14 \text{ m s}^{-1}.$$

$$a_{\max} = A\omega^2 = 0.05 \times (20\pi)^2 = 20\pi^2 \approx 197.4 \text{ m s}^{-2}.$$

Answer: $A = 5 \text{ cm}$, $T = 0.1 \text{ s}$, $f = 10 \text{ Hz}$, $v_{\max} \approx 3.14 \text{ m s}^{-1}$, $a_{\max} \approx 197.4 \text{ m s}^{-2}$. Note the clean presentation: identify constants by comparison, substitute, attach units.

Example 2 — [NEET Speed Trap] A particle in SHM has a maximum speed of $8\pi \text{ cm s}^{-1}$ and a maximum acceleration of $16\pi^2 \text{ cm s}^{-2}$. Find its amplitude and time period — in under thirty seconds.

The trap: a student instinctively writes $v_{\max} = A\omega$ and $a_{\max} = A\omega^2$ and then tries to solve two messy equations. Don't.

The shortcut: divide and you isolate ω instantly; that is the whole game.

$$\omega = \frac{a_{\max}}{v_{\max}} = \frac{16\pi^2}{8\pi} = 2\pi \text{ rad s}^{-1} \implies T = \frac{2\pi}{\omega} = 1 \text{ s}.$$

$$A = \frac{v_{\max}}{\omega} = \frac{8\pi}{2\pi} = 4 \text{ cm}.$$

Answer: $A = 4 \text{ cm}$, $T = 1 \text{ s}$. The ratio $a_{\max}/v_{\max}$ giving ω is the single most reused trick in NEET oscillation MCQs.

Example 3 — [JEE Main Level] A particle executing SHM has speed 16 cm s^{-1} when it is 3 cm from the mean position, and speed 12 cm s^{-1} when it is 4 cm from the mean position. Find the amplitude and the period of oscillation.

Approach: this links the position equation to the v - x relation $v = \omega\sqrt{A^2 - x^2}$. Squaring removes the root and lets us eliminate the unknown ω cleanly.

At $x_1 = 3$:

$$16^2 = \omega^2(A^2 - 3^2) \Rightarrow 256 = \omega^2(A^2 - 9). \quad (1)$$

At $x_2 = 4$:

$$12^2 = \omega^2(A^2 - 4^2) \Rightarrow 144 = \omega^2(A^2 - 16). \quad (2)$$

Subtract (2) from (1):

$$256 - 144 = \omega^2[(A^2 - 9) - (A^2 - 16)] \Rightarrow 112 = 7\omega^2 \Rightarrow \omega = 4 \text{ rad s}^{-1}.$$

Substitute back into (1): $256 = 16(A^2 - 9) \Rightarrow A^2 - 9 = 16 \Rightarrow A = 5 \text{ cm}.$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = \frac{\pi}{2} \text{ s} \approx 1.57 \text{ s}.$$

Answer: $A = 5 \text{ cm}$, $T = \pi/2 \text{ s}$. The subtract-to-eliminate- ω move is the standard route for any “two speeds at two positions” problem.

Example 4 — [JEE Advanced Level] A particle executes SHM of amplitude A and angular frequency ω along the x -axis about the origin. Prove that the fraction of one complete period during which the magnitude of its displacement is **less than** $A/2$ equals exactly $\frac{1}{3}$.

Reasoning: take $x = A \sin \theta$ with $\theta = \omega t$ sweeping from 0 to 2π over one period. The condition $|x| < A/2$ becomes $|\sin \theta| < \frac{1}{2}$. We must find the *total angular span* of θ in $[0, 2\pi]$ for which this holds, since time is proportional to θ (because $\theta = \omega t$, equal angles take equal times).

$|\sin \theta| < \frac{1}{2}$ is satisfied in three windows around the points where $\sin \theta = 0$ (i.e. near $\theta = 0, \pi, 2\pi$ — these are the mean-position crossings):

$$\theta \in [0, \frac{\pi}{6}] \cup [\frac{5\pi}{6}, \frac{7\pi}{6}] \cup [\frac{11\pi}{6}, 2\pi].$$

Total favourable angular span:

$$\Delta\theta = \frac{\pi}{6} + \left(\frac{7\pi}{6} - \frac{5\pi}{6}\right) + \left(2\pi - \frac{11\pi}{6}\right) = \frac{\pi}{6} + \frac{2\pi}{6} + \frac{\pi}{6} = \frac{4\pi}{6} = \frac{2\pi}{3}.$$

Since the full period corresponds to 2π ,

$$\text{Required fraction} = \frac{\Delta\theta}{2\pi} = \frac{2\pi/3}{2\pi} = \frac{1}{3}. \quad \blacksquare$$

Why it’s deep: it forces you to abandon “plug into a formula” and instead reason about *time-on-a-phase-circle*. The particle spends a **disproportionately large** share of its time near the mean position only if you misjudge it — here, despite the inner zone being just the central half of the amplitude, it occupies exactly one-third of the period because the particle is moving fastest precisely there. Speed near the centre is high, so it lingers less per unit length — the geometry and the kinematics conspire to give the clean $1/3$.

Section 5: Practice Exercises

1. **[CBSE]** A body in SHM has displacement $x = 6 \sin(4\pi t + \frac{\pi}{3})$ cm. Write its amplitude, angular frequency, time period and initial phase.

2. **[NEET]** The maximum acceleration of a particle in SHM is $0.5\pi^2 \text{ m s}^{-2}$ and its amplitude is 2 cm. Find its time period.
3. **[JEE Main]** A particle in SHM has speed v at displacement x and speed $v/2$ at displacement $2x$ (with $2x < A$). Express the amplitude A in terms of x .
4. **[JEE Main]** A particle moves such that its acceleration is $a = -16x$ (SI units), starting from rest at $x = 5 \text{ cm}$. Find its speed as it passes the mean position.
5. **[JEE Advanced]** A particle starts from the mean position and moves in the positive direction with SHM of period T . Find the time it takes to travel from $x = A/2$ to $x = A$ during this first quarter-cycle.

Answers: 1. $A = 6 \text{ cm}$, $\omega = 4\pi \text{ rad s}^{-1}$, $T = 0.5 \text{ s}$, $\phi = \pi/3$. 2. $T = 0.4 \text{ s}$. 3. $A = \sqrt{5}x$ (from $4 = \frac{A^2 - x^2}{A^2 - 4x^2} \Rightarrow 3A^2 = 15x^2$). 4. $v = 0.20 \text{ m s}^{-1}$ (here $\omega = 4 \text{ rad s}^{-1}$, $A = 5 \text{ cm}$, so $v_{\max} = \omega A = 4 \times 0.05$). 5. $t = T/6$.

Section 6: Crack the MCQ

Q1. Which of the following functions of time represents simple harmonic motion? (A) $x = \sin(\omega t) - \cos(\omega t)$ (B) $x = \sin^3(\omega t)$ (C) $x = \sin(\omega t) + \sin(2\omega t) + \sin(3\omega t)$ (D) $x = 1 + \omega t + \omega^2 t^2$

Correct: (A). A sum (or difference) of a sine and a cosine of the *same* angular frequency can be written as a single sinusoid, $R \sin(\omega t + \phi)$, which satisfies $a = -\omega^2 x$ — genuine SHM. - (B) is a trap: using $\sin^3 \theta = \frac{1}{4}(3 \sin \theta - \sin 3\theta)$, this is a *superposition of two different frequencies* (ω and 3ω) — periodic but not simple harmonic. Students who only check “does it oscillate?” fall for it. - (C) is a trap: three different frequencies; periodic but not SHM. Picked by those who assume “any sum of sines is SHM.” - (D) is a trap: a polynomial in t grows without bound and never repeats — not even periodic. Chosen by students rushing without checking for repetition.

Q2. In SHM, the velocity of the particle leads its displacement in phase by, and the acceleration leads the displacement by, respectively: (A) $\pi/2$ and π (B) π and $\pi/2$ (C) $\pi/2$ and $\pi/2$ (D) 0 and π

Correct: (A). With $x = A \sin(\omega t)$, velocity $v = A\omega \cos(\omega t) = A\omega \sin(\omega t + \pi/2)$, so velocity is ahead of displacement by $\pi/2$. Acceleration $a = -A\omega^2 \sin(\omega t) = A\omega^2 \sin(\omega t + \pi)$, so acceleration is ahead of (in fact, opposite to) displacement by π . - (B) swaps the two — a memory slip. - (C) wrongly treats acceleration like velocity; forgets that $a = -\omega^2 x$ is *anti-phase*, not quarter-phase. - (D) confuses leading by π (acceleration) with zero lead.

Q3. For a particle in SHM, the graph of acceleration a against displacement x is: (A) a parabola (B) an ellipse (C) a straight line of positive slope (D) a straight line of negative slope

Correct: (D). Since $a = -\omega^2 x$, the relation is linear in x with slope $-\omega^2$, which is negative. The magnitude of the slope gives ω^2 . - (A) mistakes it for an energy-vs-displacement graph (which is parabolic). - (B) is the shape of the v - x graph, not the a - x graph. - (C) gets the line right but ignores the all-important minus sign — the classic error.

Q4. A particle in SHM has its kinetic energy maximum and its acceleration minimum at: (A) the extreme positions (B) the mean position (C) midway between mean and extreme (D) it depends on amplitude

Correct: (B). At the mean position $x = 0$: speed is maximum (so KE is maximum) and $a = -\omega^2 x = 0$ (acceleration is zero/minimum). These two facts coincide exactly at the centre. - (A) describes the *opposite* situation — at extremes, KE is zero and acceleration is maximum.

- (C) and (D) are distractors with no basis; KE-max and a -min always lock to the mean position regardless of amplitude.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Dropping the minus sign in $a = -\omega^2 x$.** The minus sign is the physics — it encodes “restoring.” Writing $a = \omega^2 x$ describes a runaway motion, not an oscillation. Always keep it.
2. **Confusing “fastest” with “most accelerated.”** At the mean position speed is maximum but acceleration is zero; at the extremes speed is zero but acceleration is maximum. They are *never* maximum at the same place.
3. **Assuming any periodic motion is SHM.** Earth orbiting the Sun is periodic but not SHM. A sum of two different frequencies is periodic but not SHM. SHM demands a *single* frequency and a *linear* restoring force.
4. **Mixing up ω (rad s⁻¹) with f (Hz).** They differ by a factor of 2π . A huge fraction of numerical slips trace back to this. Sanity-check units: if the answer needs hertz, you should have divided by 2π somewhere.

Pro-Tip: For “max-speed and max-acceleration given” problems, never solve simultaneous equations. Use $\omega = a_{\max}/v_{\max}$ and $A = v_{\max}^2/a_{\max}$ directly. For “two speeds at two positions,” square $v = \omega\sqrt{A^2 - x^2}$ for each and *subtract* to eliminate ω in one line.

Section 8: Cheat Sheet Box

- **SHM ID card:** $a = -\omega^2 x$ — acceleration $\propto$ displacement, opposite in direction. If this holds, it’s SHM.
- **Displacement:** $x = A \sin(\omega t + \phi) \rightarrow$ **Velocity:** $v = A\omega \cos(\omega t + \phi) = \pm\omega\sqrt{A^2 - x^2} \rightarrow$ **Acceleration:** $a = -\omega^2 x$.
- **Maxima:** $v_{\max} = A\omega$ at mean position; $a_{\max} = A\omega^2$ at extremes.
- **Links:** $\omega = \frac{2\pi}{T} = 2\pi f$; from a force law $F = -kx \Rightarrow \omega = \sqrt{k/m}$.
- **Instant shortcuts:** $\omega = \frac{a_{\max}}{v_{\max}}$, $A = \frac{v_{\max}^2}{a_{\max}}$.
- **Phase facts:** velocity leads displacement by $\pi/2$; acceleration leads (is anti-phase to) displacement by π .

Memory aid — “MAX at MEAN”: speed is **Maximum** at the **Mean**; acceleration is maximum at the **eXtremes**. And remember the sign with “*a opposes x*” — acceleration always points home.

Subtopic 02: SHM Energy

Exam Relevance: A perennial favourite. JEE Main almost always carries 1-2 questions on $E \propto A^2$, the position where kinetic and potential energies are equal, or the “double frequency” of the energy curves. NEET reliably places 1 question, usually on max-KE/max-PE locations or an energy ratio. JEE Advanced folds it into spring and graph problems and loves the *time-average vs position-average* subtlety. For CBSE Boards, the derivation of total energy $E = \frac{1}{2}kA^2$ is a standard 3-mark question.

Section 1: Concept Intuition

Picture a marble rolling back and forth inside a smooth steel *katori* (bowl). At the very bottom it whizzes through fastest. As it climbs the side it slows, stops for an instant at the highest point it reaches, then rolls back down. Now ask a simple question: where did its speed *go* at the top, and where did it *come from* at the bottom? Nothing was lost — the energy simply changed costume. At the bottom it wore the costume of motion (kinetic energy); at the top it wore the costume of position (potential energy). This continuous changing of costumes, with the *total* never changing, is the entire story of energy in SHM.

In SHM the restoring force is conservative ($F = -kx$), so as the particle moves, energy sloshes between two forms: - **Kinetic energy** K — the energy of motion, largest where the particle moves fastest. - **Potential energy** U — the energy stored in the deformation (a stretched spring, a displaced bob), largest where the particle is farthest from the mean position.

At the **mean position** ($x = 0$) the particle moves fastest, so energy is **entirely kinetic** and $U = 0$. At the **extreme positions** ($x = \pm A$) the particle is momentarily frozen, so energy is **entirely potential** and $K = 0$. Everywhere in between, the two trade off — one rises exactly as much as the other falls.

Think about it this way... imagine your total savings is fixed at, say, ₹10,000, split between a current account (instantly spendable — that’s kinetic energy) and a fixed deposit (locked up — that’s potential energy). Money shuttles between the two accounts as the particle moves, but your *net worth* never changes. That fixed net worth is the **total mechanical energy**:

$$E = K + U = \frac{1}{2}kA^2 = \text{constant}.$$

Two consequences flow from this and they are tested endlessly. First, **total energy depends on the square of amplitude**, $E \propto A^2$ — double the amplitude and the energy quadruples. Second, **the energy curves “blink” at twice the oscillation frequency**: in one full to-and-fro, the particle passes the fast mean position *twice* and the frozen extremes *twice*, so K peaks twice and U peaks twice per period. The energies oscillate with frequency $2f$, i.e. period $T/2$ — a fact students forget under exam pressure.

Limiting Conditions & Assumptions. - The result $E = \frac{1}{2}kA^2 = \text{constant}$ holds only because the restoring force is **conservative and undamped**. With friction or air drag, energy leaks away and amplitude — hence E — decays over time. - U is measured **relative to a chosen zero**, conventionally $U = 0$ at the mean position. If a different reference is chosen, a constant gets added to U (and to E), but all *differences* and the physics stay the same. - These expressions describe **ideal linear SHM**. Beyond the elastic limit, or for a pendulum at large angles, the simple $\frac{1}{2}kx^2$ form of U no longer applies.

Section 2: Key Formulas & Definitions

Using $\omega^2 = k/m$ and the kinematic results from Subtopic 01:

Kinetic energy:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2) = \frac{1}{2}m\omega^2A^2 \cos^2(\omega t + \phi)$$

Potential energy (taking $U = 0$ at the mean position):

$$U = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2 = \frac{1}{2}m\omega^2A^2 \sin^2(\omega t + \phi)$$

Total mechanical energy:

$$E = K + U = \frac{1}{2}m\omega^2A^2 = \frac{1}{2}kA^2 \quad (\text{independent of } x \text{ and } t)$$

Key derived relations:

$$E \propto A^2, \quad E \propto \omega^2 (\propto f^2), \quad \frac{K}{U} = \frac{A^2 - x^2}{x^2}$$

Position where the two are equal ($K = U$):

$$x = \pm \frac{A}{\sqrt{2}} \approx \pm 0.707 A, \quad \text{at which } K = U = \frac{E}{2}.$$

Time-averaged values over one period:

$$\langle K \rangle_t = \langle U \rangle_t = \frac{E}{2} = \frac{1}{4}m\omega^2A^2 = \frac{1}{4}kA^2.$$

Notation, units, dimensions: - K , U , E — energies. SI unit: joule (J). Dimensional formula: $[M^1L^2T^{-2}]$. - k — force constant. SI unit: N m^{-1} . $[M^1L^0T^{-2}]$. - m — mass: $[M^1L^0T^0]$; A , x — length: $[M^0L^1T^0]$; ω — angular frequency: $[T^{-1}]$.

Frequency note: displacement oscillates at frequency f ; both K and U oscillate at $2f$ (period $T/2$), because they contain $\cos^2$ and $\sin^2$ terms.

Section 3: CBSE Board Derivations / Key Procedures

Derivation 1 — Total mechanical energy of an SHM oscillator is $\frac{1}{2}kA^2$.

Assumption: a particle of mass m executes SHM, $x = A \sin(\omega t + \phi)$, under a conservative restoring force $F = -kx$ with $\omega^2 = k/m$. Take $U = 0$ at the mean position.

Kinetic energy. From Subtopic 01, $v = A\omega \cos(\omega t + \phi)$, so

$$K = \frac{1}{2}mv^2 = \frac{1}{2}mA^2\omega^2 \cos^2(\omega t + \phi) = \frac{1}{2}kA^2 \cos^2(\omega t + \phi),$$

using $m\omega^2 = k$.

Potential energy. The work done against the restoring force in displacing the particle from 0 to x is stored as PE:

$$U = - \int_0^x F dx' = \int_0^x kx' dx' = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \sin^2(\omega t + \phi).$$

Add them.

$$E = K + U = \frac{1}{2}kA^2 [\cos^2(\omega t + \phi) + \sin^2(\omega t + \phi)].$$

The bracket equals 1 by the Pythagorean identity, so

$$E = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2 A^2.$$

Physical reading: every trace of time has vanished — E is a pure constant. This is exactly what we expect for motion under a conservative force: the see-saw between K and U is perfectly balanced so their sum stands still even while each one swings.

Derivation 2 — Why the energies oscillate at double frequency.

Use the power-reduction identities $\cos^2 \theta = \frac{1+\cos 2\theta}{2}$ and $\sin^2 \theta = \frac{1-\cos 2\theta}{2}$ with $\theta = \omega t + \phi$:

$$K = \frac{1}{2}kA^2 \cdot \frac{1 + \cos 2(\omega t + \phi)}{2} = \frac{E}{2} + \frac{E}{2} \cos 2(\omega t + \phi),$$

$$U = \frac{1}{2}kA^2 \cdot \frac{1 - \cos 2(\omega t + \phi)}{2} = \frac{E}{2} - \frac{E}{2} \cos 2(\omega t + \phi).$$

Physical reading: each energy is a constant $E/2$ plus a term oscillating at angular frequency 2ω — hence **double the frequency** of the displacement, period $T/2$. The two oscillating parts are equal and opposite, so they cancel in the sum, leaving E constant. It also shows at a glance that the **time-average of each energy is $E/2$** (the oscillating part averages to zero).

Figure 13.1 Two stacked plots sharing a time axis spanning one period T . Top plot: displacement $x = A \sin(\omega t)$ as a single sine wave. Bottom plot: K (cosine-squared hump, starting at maximum E at $t=0$) and U (sine-squared hump, starting at 0) drawn together, both completing TWO full humps within one period T , intersecting at $E/2$ where the curves cross, with a horizontal dashed line marking constant total energy E across the top. (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A block of mass 0.5 kg is attached to a spring of force constant 200 N m^{-1} on a frictionless surface and set into SHM with amplitude 4 cm. Find (i) the total mechanical energy, (ii) the maximum kinetic energy, (iii) the maximum potential energy, and (iv) the maximum speed.

Given: $m = 0.5 \text{ kg}$, $k = 200 \text{ N m}^{-1}$, $A = 4 \text{ cm} = 0.04 \text{ m}$.

Solution:

$$E = \frac{1}{2}kA^2 = \frac{1}{2}(200)(0.04)^2 = 100 \times 0.0016 = 0.16 \text{ J}.$$

(ii) Maximum KE occurs at the mean position, where all energy is kinetic: $K_{\max} = E = 0.16 \text{ J}$.

(iii) Maximum PE occurs at the extremes, where all energy is potential: $U_{\max} = E = 0.16 \text{ J}$.

(iv) From $\frac{1}{2}mv_{\max}^2 = E$:

$$v_{\max} = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2 \times 0.16}{0.5}} = \sqrt{0.64} = 0.8 \text{ m s}^{-1}.$$

Answer: $E = K_{\max} = U_{\max} = 0.16 \text{ J}$, $v_{\max} = 0.8 \text{ m s}^{-1}$.

Example 2 — [NEET Speed Trap] At what displacement from the mean position (as a fraction of amplitude) is the kinetic energy of an SHM particle exactly equal to its potential

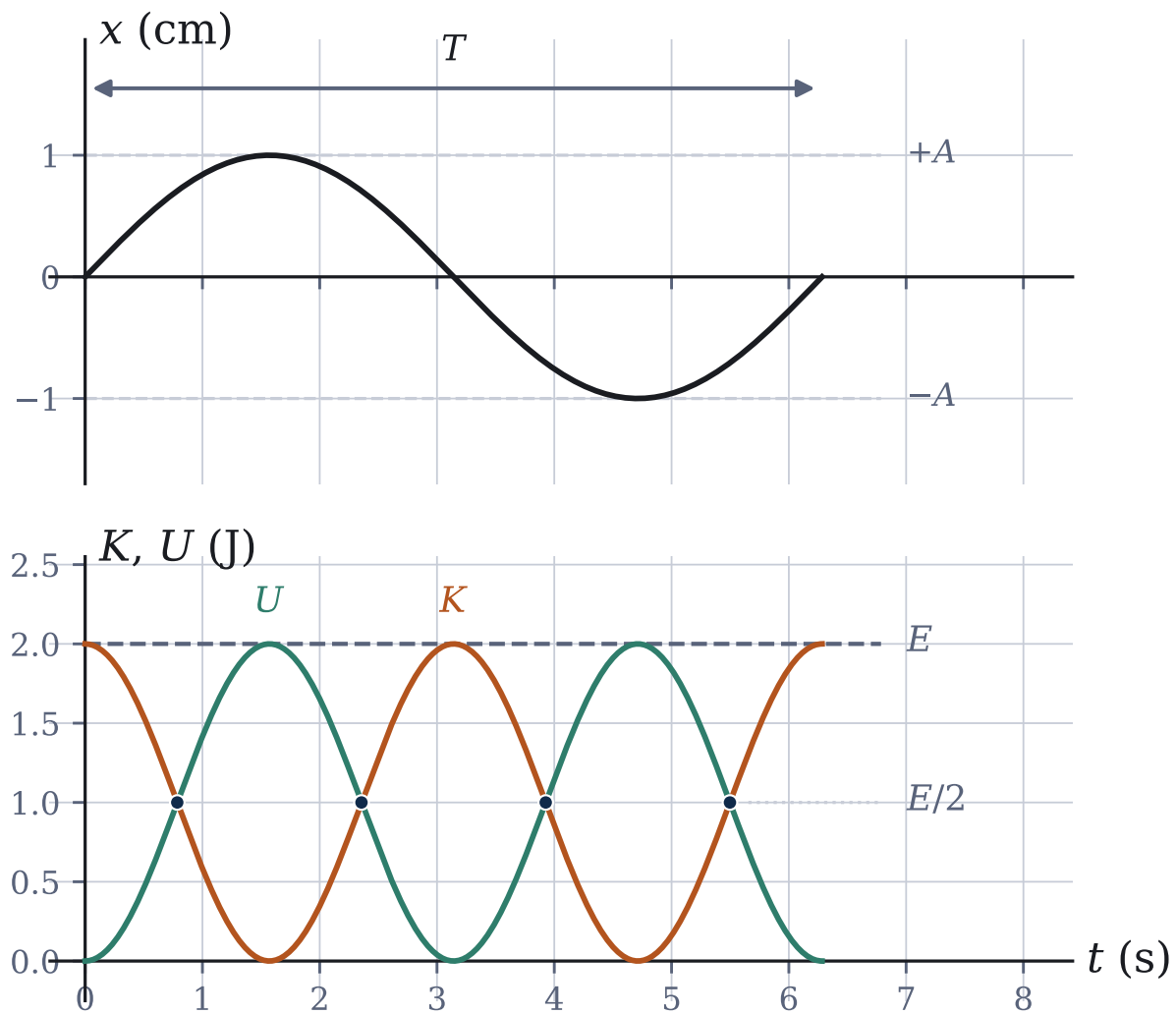


Figure 13.1 Two stacked plots sharing a time axis spanning one period T . Top plot: displacement $x = A \sin(\omega t)$ as a single sine wave. Bottom plot: K (cosine-squared hump, starting at maximum E at $t=0$) and U (sine-squared hump, starting at 0) drawn together, both completing TWO full humps within one period T , intersecting at $E/2$ where the curves cross, with a horizontal dashed line marking constant total energy E across the top.

energy?

The trap: the tempting wrong answer is $x = A/2$. But “halfway in displacement” is *not* “halfway in energy” — energy goes as the square of displacement, so the split happens later than you’d guess.

The shortcut: set $K = U$ and use $K = \frac{1}{2}k(A^2 - x^2)$, $U = \frac{1}{2}kx^2$:

$$A^2 - x^2 = x^2 \Rightarrow 2x^2 = A^2 \Rightarrow x = \frac{A}{\sqrt{2}} \approx 0.707 A.$$

At this point each energy equals $E/2$.

Answer: $x = A/\sqrt{2}$. Memorise this — it is asked almost every year. (A useful cousin to know: $U = E/4$ at $x = A/2$, and $U = \frac{3}{4}E$ at $x = \frac{\sqrt{3}}{2}A$.)

Example 3 — [JEE Main Level] A particle of mass 0.1 kg executes SHM of amplitude 10 cm with angular frequency 10 rad s^{-1} . Find its kinetic and potential energies when it is 6 cm from the mean position, and verify their ratio.

Approach: compute the total energy first, then split it using position.

$$E = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}(0.1)(10)^2(0.10)^2 = \frac{1}{2}(0.1)(100)(0.01) = 0.05 \text{ J}.$$

At $x = 6 \text{ cm} = 0.06 \text{ m}$:

$$U = \frac{1}{2}m\omega^2 x^2 = \frac{1}{2}(0.1)(100)(0.06)^2 = 5 \times 0.0036 = 0.018 \text{ J},$$

$$K = E - U = 0.05 - 0.018 = 0.032 \text{ J}.$$

Verify with the position ratio:

$$\frac{K}{U} = \frac{A^2 - x^2}{x^2} = \frac{10^2 - 6^2}{6^2} = \frac{64}{36} = \frac{16}{9}, \quad \frac{0.032}{0.018} = \frac{16}{9}. \checkmark$$

Answer: $K = 0.032 \text{ J}$, $U = 0.018 \text{ J}$, with $K : U = 16 : 9$.

Example 4 — [JEE Advanced Level] For a particle in SHM of amplitude A and total energy E , show that the **time-averaged** kinetic energy over one period equals $E/2$, but the kinetic energy **averaged over position** (from $-A$ to $+A$) equals $2E/3$. Explain physically why the two averages differ.

Time average. With $K = E \cos^2(\omega t)$ and using $\langle \cos^2 \rangle = \frac{1}{2}$ over a full period:

$$\langle K \rangle_t = \frac{1}{T} \int_0^T E \cos^2(\omega t) dt = E \cdot \frac{1}{2} = \frac{E}{2}.$$

Position average. With $K = \frac{E}{A^2}(A^2 - x^2)$, average uniformly over $x \in [-A, A]$:

$$\langle K \rangle_x = \frac{1}{2A} \int_{-A}^A \frac{E}{A^2}(A^2 - x^2) dx = \frac{E}{2A^3} \left[A^2 x - \frac{x^3}{3} \right]_{-A}^A = \frac{E}{2A^3} \cdot \frac{4A^3}{3} = \frac{2E}{3}.$$

Why they differ: the time average and the position average weight the motion differently. The particle moves **slowly near the extremes** (where K is small) and **quickly near the**

mean (where K is large), so it *spends more time* in the low-KE regions. Time-weighting therefore drags the average down to $E/2$, whereas position-weighting (which treats every centimetre equally regardless of how long the particle dawdles there) gives the larger $2E/3$. The lesson: **“average” is meaningless until you say averaged with respect to what.** This distinction is a classic JEE Advanced trap.

Section 5: Practice Exercises

1. **[CBSE]** A spring of force constant 100 N m^{-1} undergoes SHM of amplitude 5 cm. Find the total mechanical energy of the oscillation.
2. **[NEET]** At what displacement (as a fraction of amplitude) is the potential energy of an SHM particle three times its kinetic energy?
3. **[JEE Main]** If the amplitude of an SHM is doubled while mass and force constant are unchanged, by what factor does (a) the total energy and (b) the maximum speed change?
4. **[JEE Main]** Express the kinetic energy of an SHM particle at displacement $x = A/2$ as a fraction of its total energy.
5. **[JEE Advanced]** For a particle in SHM, find the fraction of one complete period during which its kinetic energy is greater than its potential energy.

Answers: 1. $E = 0.125 \text{ J}$. 2. $U = 3K \Rightarrow x = \frac{\sqrt{3}}{2}A \approx 0.866A$. 3. (a) total energy $\times 4$ (since $E \propto A^2$); (b) maximum speed $\times 2$ (since $v_{\max} = A\omega$). 4. $K = \frac{3}{4}E$. 5. $K > U$ when $|x| < A/\sqrt{2}$, which holds for exactly **half** the period ($\frac{1}{2}$).

Section 6: Crack the MCQ

Q1. A particle executes SHM with frequency f . The frequency with which its kinetic energy oscillates is: (A) f (B) $2f$ (C) $f/2$ (D) $4f$

Correct: (B). Kinetic energy contains $\cos^2(\omega t + \phi) = \frac{1 + \cos 2(\omega t + \phi)}{2}$, whose oscillating part has angular frequency 2ω — so KE oscillates at $2f$. It peaks each time the particle crosses the mean position, which happens twice per cycle. - (A) is the most common error — assuming energy tracks displacement frequency. - (C) and (D) are unmotivated distractors for students guessing factors of two.

Q2. The total energy of an SHM particle is E . At a displacement equal to half the amplitude, its kinetic energy is: (A) $E/4$ (B) $E/2$ (C) $3E/4$ (D) E

Correct: (C). $K = E \left(1 - \frac{x^2}{A^2}\right) = E \left(1 - \frac{1}{4}\right) = \frac{3E}{4}$. - (A) is the *potential* energy at $x = A/2$ ($U = Ex^2/A^2 = E/4$) — chosen by students who compute U when asked for K . - (B) assumes $K = U = E/2$, which is true only at $x = A/\sqrt{2}$, not $A/2$. - (D) forgets that the particle is not at the mean position.

Q3. If the amplitude of a simple harmonic oscillator is increased by 50%, its total energy increases by: (A) 50% (B) 100% (C) 125% (D) 225%

Correct: (C). Since $E \propto A^2$, new energy $= (1.5)^2 = 2.25$ times the old, i.e. 225% of the original, which is an **increase** of 125%. - (A) wrongly treats $E \propto A$ (linear). - (D) confuses the *new value* (225%) with the *increase*. - (B) is a guess unconnected to the square law.

Q4. The graph of potential energy versus displacement for a particle in SHM is: (A) a straight line (B) a parabola (C) an ellipse (D) a horizontal line

Correct: (B). $U = \frac{1}{2}kx^2$ is quadratic in x , so the graph is an upward-opening parabola

centred at the mean position. - (A) is the shape of the acceleration-displacement graph ($a = -\omega^2 x$). - (C) is the velocity-displacement graph. - (D) is the total-energy-displacement graph (constant). Each distractor is a *real* SHM graph for a different quantity — the trap is matching the wrong one.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Forgetting the double frequency.** Both K and U oscillate at $2f$ (period $T/2$), not at f . The particle passes the fast mean position twice and the frozen extremes twice in one cycle.
2. **Putting $K = U$ at $x = A/2$.** Energy goes as x^2 , so equality of K and U happens at $x = A/\sqrt{2} \approx 0.707A$, noticeably past the halfway mark.
3. **Writing $E \propto A$ instead of $E \propto A^2$.** A huge source of lost marks in “amplitude changed by...” problems. Always square the amplitude ratio.
4. **Confusing the two averages.** Time-averaged K and U are each $E/2$; this does *not* mean $K = U$ at every instant. They are equal only on average and only at the specific points $x = \pm A/\sqrt{2}$.

Pro-Tip: Skip re-deriving energies at a given position — use the ready ratios $\frac{K}{E} = 1 - \frac{x^2}{A^2}$ and $\frac{U}{E} = \frac{x^2}{A^2}$ directly. And lock in two landmark facts: $K = U$ at $x = A/\sqrt{2}$, and $E \propto A^2$ — between them they answer the majority of objective questions on this subtopic.

Section 8: Cheat Sheet Box

- **Energies:** $K = \frac{1}{2}m\omega^2(A^2 - x^2)$, $U = \frac{1}{2}m\omega^2x^2 = \frac{1}{2}kx^2$, $E = K + U = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2A^2$ (constant).
- **Scaling:** $E \propto A^2$ and $E \propto \omega^2$ — double $A \Rightarrow$ energy $\times 4$.
- **Locations:** $K_{\max} = E$ at the mean ($x = 0$); $U_{\max} = E$ at the extremes ($x = \pm A$).
- **Equal point:** $K = U = E/2$ at $x = \pm A/\sqrt{2} \approx \pm 0.707A$.
- **Position ratios:** $\frac{K}{E} = 1 - \frac{x^2}{A^2}$, $\frac{U}{E} = \frac{x^2}{A^2}$, $\frac{K}{U} = \frac{A^2 - x^2}{x^2}$.
- **Frequency:** K and U oscillate at $2f$ (period $T/2$); time-average of each = $E/2$.

Memory aid — “Square the swing”: energy follows the *square* of amplitude ($E \propto A^2$) and the *square* of position for PE ($U \propto x^2$). And “**half-energy at A-over-root-2**” — the $K = U$ crossover lives at $A/\sqrt{2}$, not $A/2$.

Subtopic 03: Pendulums and Mass-Spring Systems

Exam Relevance: The most question-dense subtopic in the whole chapter. JEE Main reliably carries 1-2 questions every year — pendulum in a lift or accelerating car, spring combinations (series/parallel), and the vertical-spring period. NEET asks 1-2 on the factors affecting a simple pendulum and effective gravity. JEE Advanced reaches for the physical (compound) pendulum and multi-spring systems. For CBSE Boards, the derivation of $T = 2\pi\sqrt{L/g}$ for a simple pendulum is an almost-guaranteed 3-mark question.

Section 1: Concept Intuition

There are two great workhorses of SHM, and almost every oscillation problem you will ever solve is one of them in disguise: the **pendulum** (driven by gravity) and the **mass-spring system** (driven by elasticity). They look completely different — one swings, one bounces — yet both obey the same master rule from Subtopic 01: a restoring force proportional to displacement. The fun is in seeing *why* each one obeys it, because the “why” is different in each case.

Take the **simple pendulum** first — a small heavy bob on a light string, like the clapper of a temple bell or a stone tied to a thread. Pull it aside and gravity tries to drag it straight down, but the string only lets it move along an arc. The part of gravity *along the arc* (the tangential component, $mg \sin \theta$) is what hauls the bob back toward the bottom. For **small swings**, $\sin \theta \approx \theta$, so this restoring force becomes proportional to the displacement — and that, precisely, is SHM. Out of this falls one of the most beautiful facts in physics:

$$T = 2\pi\sqrt{\frac{L}{g}} \implies \text{period depends only on } L \text{ and } g.$$

Not on the mass of the bob. Not on the amplitude (for small angles). A heavy bob and a light bob on strings of equal length keep perfect time together — this is **isochronism**, the property Galileo spotted watching a swinging lamp in a cathedral, and the reason pendulum clocks work at all.

Think about it this way... why does mass cancel for a pendulum but matter for a spring? In a pendulum, a heavier bob feels a *stronger* pull (more gravity) but is also *more sluggish* (more inertia) — the two effects cancel exactly, so mass drops out. In a spring, the elastic force is set by the spring’s stiffness k alone and does not care about the mass hanging on it; so a heavier mass just makes the system more sluggish without any compensating extra force, and the period *grows*:

$$T = 2\pi\sqrt{\frac{m}{k}} \implies \text{period depends only on } m \text{ and } k.$$

Here the surprise runs the other way: the spring’s period does **not depend on g at all**. Hang the same spring-mass system on the Moon and it oscillates with exactly the same period — a favourite trap that contrasts directly with the pendulum, which *would* slow down on the Moon. Gravity, for a vertical spring, only shifts the *mean position* downward; it never touches the period.

Limiting Conditions & Assumptions. - The simple-pendulum formula assumes **small angular amplitude** (roughly $\theta < 10^\circ$) so that $\sin \theta \approx \theta$. At large swings the period grows with

amplitude and the motion is no longer simple harmonic. - The ideal simple pendulum assumes a **point-mass bob** on a **massless, inextensible string**. A real rod or extended body is a *physical pendulum* with a different formula. - The spring formula assumes a **massless spring** obeying **Hooke's law within its elastic limit**. A heavy spring adds an effective $m_{\text{spring}}/3$ correction to the oscillating mass. - In non-inertial frames (lifts, accelerating cars), gravity must be replaced by an **effective gravity** g_{eff} .

Section 2: Key Formulas & Definitions

Simple pendulum (small amplitude):

$$T = 2\pi\sqrt{\frac{L}{g}}, \quad \omega = \sqrt{\frac{g}{L}}$$

Mass-spring system (horizontal or vertical — same period):

$$T = 2\pi\sqrt{\frac{m}{k}}, \quad \omega = \sqrt{\frac{k}{m}}$$

Static elongation shortcut (vertical spring stretched x_0 at equilibrium, since $mg = kx_0$):

$$T = 2\pi\sqrt{\frac{x_0}{g}}$$

Spring combinations:

Series: $\frac{1}{k_{\text{eff}}} = \frac{1}{k_1} + \frac{1}{k_2}$ (softer, longer T); Parallel: $k_{\text{eff}} = k_1 + k_2$ (stiffer, shorter T).

Cutting a spring of constant k into n equal pieces makes each piece **stiffer**: $k_{\text{piece}} = nk$.

Effective gravity (replace g with g_{eff} in the pendulum formula): - Lift accelerating **up** with a : $g_{\text{eff}} = g + a$ (period shortens). - Lift accelerating **down** with a : $g_{\text{eff}} = g - a$ (period lengthens; free fall $a = g \Rightarrow$ no oscillation). - Car accelerating **horizontally** with a : $g_{\text{eff}} = \sqrt{g^2 + a^2}$, mean position tilts by $\theta_0 = \tan^{-1}(a/g)$ from the vertical.

Physical (compound) pendulum — rigid body pivoted at distance d from its centre of mass:

$$T = 2\pi\sqrt{\frac{I}{mgd}}, \quad L_{\text{eq}} = \frac{I}{md} = \frac{k_g^2 + d^2}{d}$$

where I is the moment of inertia about the pivot and k_g the radius of gyration about the centre of mass ($I = mk_g^2 + md^2$).

Torsional pendulum (disc/wire):

$$T = 2\pi\sqrt{\frac{I}{\kappa}}$$

where κ is the torsional constant (restoring torque per unit twist), unit N m rad^{-1} .

Notation, units, dimensions: - L , x_0 , d — lengths: $[M^0 L^1 T^0]$; g — acceleration: $[LT^{-2}]$; T — period: $[T]$. - k — spring constant: N m^{-1} , $[M^1 L^0 T^{-2}]$. - I — moment of inertia: kg m^2 , $[M^1 L^2 T^0]$; κ — torsional constant: N m rad^{-1} , $[M^1 L^2 T^{-2}]$.

Section 3: CBSE Board Derivations / Key Procedures

Derivation 1 — Period of a simple pendulum, $T = 2\pi\sqrt{L/g}$.

Assumption: a point-mass bob of mass m hangs from a light, inextensible string of length L , displaced by a small angle θ from the vertical.

The forces on the bob are gravity mg (down) and string tension T_s (along the string). Resolve gravity into a component along the string ($mg \cos \theta$, balanced by tension) and a component along the arc ($mg \sin \theta$) directed back toward the lowest point. This tangential component is the **restoring force**:

$$F = -mg \sin \theta.$$

For **small angles**, $\sin \theta \approx \theta$ (in radians), and the arc displacement is $s = L\theta$, so $\theta = s/L$:

$$F = -mg\theta = -mg \frac{s}{L} = -\left(\frac{mg}{L}\right) s.$$

This is exactly the SHM form $F = -ks$ with effective stiffness $k = mg/L$. Comparing with $\omega^2 = k/m$ (Subtopic 01):

$$\omega^2 = \frac{mg/L}{m} = \frac{g}{L} \quad \Rightarrow \quad T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}}.$$

Physical reading: the mass m cancels — the agent that supplies the restoring force (gravity) scales with m in just the same way as the inertia that resists it, so the bob's weight is irrelevant to its timing.

Derivation 2 — Mass-spring period, and why vertical equals horizontal.

Horizontal spring: displacing the mass by x produces a restoring force $F = -kx$ directly. Comparing with $F = -kx = ma$ gives $\omega = \sqrt{k/m}$, so $T = 2\pi\sqrt{m/k}$.

Vertical spring: at equilibrium the spring is already stretched by x_0 where $kx_0 = mg$. Measure displacement y from this new equilibrium. The net force is

$$F_{\text{net}} = mg - k(x_0 + y) = (mg - kx_0) - ky = 0 - ky = -ky.$$

The gravity term is *exactly cancelled* by the initial stretch, leaving the same $F = -ky$. Hence $T = 2\pi\sqrt{m/k}$ — **identical to the horizontal case**. Gravity merely relocated the mean position by $x_0 = mg/k$; it left the period untouched.

Derivation 3 — Series and parallel spring constants.

Parallel (both springs stretch by the same x): each pulls back, total force $F = -(k_1 + k_2)x$, so

$$k_{\text{eff}} = k_1 + k_2.$$

Series (same tension F through both; stretches add): $x = x_1 + x_2 = \frac{F}{k_1} + \frac{F}{k_2}$, so the effective constant satisfies

$$\frac{1}{k_{\text{eff}}} = \frac{1}{k_1} + \frac{1}{k_2}.$$

Physical reading: parallel springs share the load and act **stiffer**; series springs each take the full load and stretch more, acting **softer**. (Note this is the *opposite* of how resistors combine — a frequent source of confusion.)

Figure 13.2 Left panel — a simple pendulum displaced by small angle θ , showing the string of length L , the bob, gravity mg pointing straight down resolved into $mg \cos \theta$ along the string and $mg \sin \theta$ tangent to the arc pointing back toward the lowest point. Right panel — two springs in series (stacked end to end with a mass at bottom) beside two springs in parallel (side by side both holding the same mass), labelled with k_1, k_2 . (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A simple pendulum has length 1.0 m. Taking $g = \pi^2 \approx 9.87 \text{ m s}^{-2}$, find its time period. What is the new period if the length is made four times as large?

Solution:

$$T = 2\pi\sqrt{\frac{L}{g}} = 2\pi\sqrt{\frac{1.0}{\pi^2}} = 2\pi \cdot \frac{1}{\pi} = 2 \text{ s}.$$

A pendulum of period 2 s is called a **seconds pendulum** (each one-way swing takes 1 s), and its length is about 1 m — worth memorising.

For $L' = 4L$:

$$T' = 2\pi\sqrt{\frac{4L}{g}} = 2 \times 2\pi\sqrt{\frac{L}{g}} = 2T = 4 \text{ s}.$$

Answer: $T = 2 \text{ s}$; quadrupling the length **doubles** the period to 4 s (since $T \propto \sqrt{L}$).

Example 2 — [NEET Speed Trap] A simple pendulum has period 2 s in a stationary lift. The lift now accelerates **downward** at $\frac{3}{4}g$. Find the new period.

The trap: students either ignore the lift's acceleration entirely, or add it when they should subtract. The bob in a downward-accelerating lift feels *lighter*, so effective gravity **decreases**.

The shortcut: replace g by $g_{\text{eff}} = g - a = g - \frac{3}{4}g = \frac{1}{4}g$, and use the ratio form so L cancels:

$$\frac{T'}{T} = \sqrt{\frac{g}{g_{\text{eff}}}} = \sqrt{\frac{g}{g/4}} = \sqrt{4} = 2 \implies T' = 2 \times 2 = 4 \text{ s}.$$

Answer: $T' = 4 \text{ s}$. (Sanity check: if the lift were in free fall, $a = g$, then $g_{\text{eff}} = 0$ and $T' \rightarrow \infty$ — the bob floats and never swings. Our answer moves correctly in that direction.)

Example 3 — [JEE Main Level] A block of mass 2 kg is connected to two springs of force constants 6 N m^{-1} and 3 N m^{-1} . Find the period of oscillation when the springs are connected (a) in series and (b) in parallel.

Approach: find k_{eff} for each configuration, then apply $T = 2\pi\sqrt{m/k_{\text{eff}}}$.

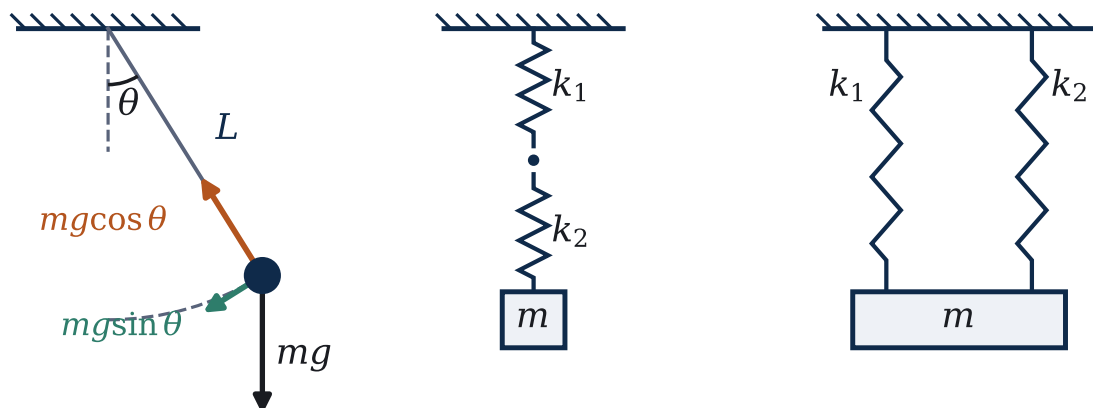
(a) **Series:**

$$\frac{1}{k_{\text{eff}}} = \frac{1}{6} + \frac{1}{3} = \frac{1}{6} + \frac{2}{6} = \frac{3}{6} \implies k_{\text{eff}} = 2 \text{ N m}^{-1}.$$

$$T_{\text{series}} = 2\pi\sqrt{\frac{2}{2}} = 2\pi \approx 6.28 \text{ s}.$$

(b) **Parallel:**

$$k_{\text{eff}} = 6 + 3 = 9 \text{ N m}^{-1}, \quad T_{\text{parallel}} = 2\pi\sqrt{\frac{2}{9}} = \frac{2\pi\sqrt{2}}{3} \approx 2.96 \text{ s}.$$



(a) simple pendulum (b) springs in series (c) springs in parallel

Figure 13.2 Left panel — a simple pendulum displaced by small angle θ , showing the string of length L , the bob, gravity mg pointing straight down resolved into $mg \cos \theta$ along the string and $mg \sin \theta$ tangent to the arc pointing back toward the lowest point. Right panel — two springs in series (stacked end to end with a mass at bottom) beside two springs in parallel (side by side both holding the same mass), labelled with k_1, k_2 .

Answer: $T_{\text{series}} \approx 6.28 \text{ s}$, $T_{\text{parallel}} \approx 2.96 \text{ s}$. The series combination is softer (smaller k_{eff}) and so oscillates more slowly — exactly as intuition predicts.

Example 4 — [JEE Advanced Level] A uniform rod of mass M and length L is pivoted at one end and swings freely in a vertical plane as a physical pendulum (small amplitude). Find its period of oscillation and the length of the equivalent simple pendulum. Compare its period with that of a simple pendulum of the same length L .

Set-up: for a physical pendulum, $T = 2\pi\sqrt{\frac{I}{Mgd}}$, where I is the moment of inertia about the pivot and d is the pivot-to-centre-of-mass distance.

For a uniform rod pivoted at one end:

$$I = \frac{ML^2}{3}, \quad d = \frac{L}{2}.$$

Therefore:

$$T = 2\pi\sqrt{\frac{ML^2/3}{Mg(L/2)}} = 2\pi\sqrt{\frac{L^2/3}{gL/2}} = 2\pi\sqrt{\frac{2L}{3g}}.$$

The **equivalent simple pendulum length** is the L_{eq} for which $2\pi\sqrt{L_{\text{eq}}/g}$ gives this same period:

$$L_{\text{eq}} = \frac{I}{Md} = \frac{ML^2/3}{M(L/2)} = \frac{2L}{3}.$$

Comparison: a simple pendulum of length L has period $2\pi\sqrt{L/g}$, whereas the rod behaves like a simple pendulum of length $2L/3$. Since $2L/3 < L$, the rod swings **faster** than a simple pendulum of equal physical length — its mass is distributed closer to the pivot on average, lowering its effective length.

Answer: $T = 2\pi\sqrt{\frac{2L}{3g}}$, $L_{\text{eq}} = \frac{2L}{3}$; the rod oscillates faster than a same-length simple pendulum by a factor $\sqrt{3/2}$.

Section 5: Practice Exercises

- [CBSE]** Find the length of a seconds pendulum (period 2 s) at a place where $g = 9.8 \text{ m s}^{-2}$.
- [NEET]** A simple pendulum has period T on Earth. If it is taken to the Moon, where gravity is one-sixth of Earth's, what is its new period?
- [JEE Main]** A spring of force constant k is cut into two equal halves and a mass m is suspended from one half. Find the ratio of the new period to the period of the original (uncut) spring with the same mass.
- [JEE Main]** A mass hung from a vertical spring stretches it by 5 cm at equilibrium. Find its period of vertical oscillation. (Take $g = 9.8 \text{ m s}^{-2}$.)
- [JEE Advanced]** A simple pendulum of length L hangs from the ceiling of a car that accelerates horizontally with $a = g$. Find the new period and the angle the equilibrium string makes with the vertical.

Answers: **1.** $L = \frac{gT^2}{4\pi^2} = \frac{9.8 \times 4}{4\pi^2} \approx 0.99 \text{ m}$. **2.** $T_{\text{moon}} = T\sqrt{6} \approx 2.45 T$. **3.** Cutting halves the spring, so each half has constant $2k$; $\frac{T_{\text{new}}}{T_{\text{old}}} = \sqrt{\frac{k}{2k}} = \frac{1}{\sqrt{2}}$. **4.** $T = 2\pi\sqrt{\frac{x_0}{g}} = 2\pi\sqrt{\frac{0.05}{9.8}} \approx 0.45 \text{ s}$.

5. $g_{\text{eff}} = \sqrt{g^2 + g^2} = g\sqrt{2}$, so $T' = 2\pi\sqrt{\frac{L}{g\sqrt{2}}} = \frac{T}{2^{1/4}} \approx 0.84T$; the string tilts $\tan^{-1}(a/g) = 45^\circ$ from the vertical.

Section 6: Crack the MCQ

Q1. The time period of a simple pendulum (small amplitude) depends on: (A) the mass of the bob (B) the amplitude of swing (C) the length and local g (D) the material of the bob

Correct: (C). $T = 2\pi\sqrt{L/g}$ involves only length and gravitational field strength. - (A) and (D) fail because mass cancels in the derivation — isochronism. - (B) fails for small angles (the regime of SHM); amplitude only matters once swings become large and the motion stops being simple harmonic. Students who recall the large-angle correction over-apply it here.

Q2. A spring-mass system has period T on Earth. On the Moon (gravity $g/6$), its period becomes: (A) T (B) $T/\sqrt{6}$ (C) $T\sqrt{6}$ (D) $6T$

Correct: (A). The spring period $T = 2\pi\sqrt{m/k}$ contains no g — it is identical anywhere. - (C) is the **pendulum** answer applied to a spring — the single most common trap on this subtopic, testing whether you know which system depends on g . - (B) and (D) are further g -dependent distractors for students guessing.

Q3. A spring of force constant k is cut into three equal pieces. The force constant of each piece is: (A) $k/3$ (B) k (C) $3k$ (D) $k/9$

Correct: (C). A shorter spring is stiffer; cutting into n equal pieces multiplies the constant by n , so each piece has $3k$. (Each piece stretches one-third as much under the same force.) - (A) wrongly treats the pieces as sharing the constant like a series division. - (B) assumes cutting changes nothing. - (D) squares the factor for no reason.

Q4. A pendulum clock keeps accurate time at ground level. Carried to the top of a very tall tower, the clock will: (A) run fast (B) run slow (C) keep correct time (D) stop

Correct: (B). Gravity weakens with altitude, so g decreases, $T = 2\pi\sqrt{L/g}$ increases — each “second” of the clock now takes longer than a real second, so it **runs slow**. - (A) gets the direction backwards (that would need larger g). - (C) ignores the altitude dependence of g . - (D) confuses a slower period with stopping. A spring-driven (balance-wheel) clock, by contrast, would be unaffected — a nice contrast to keep in mind.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Thinking the pendulum period depends on the bob’s mass.** It does not (small angles) — mass cancels. Equally, do not let amplitude into $T = 2\pi\sqrt{L/g}$; it is amplitude-independent in the SHM regime.
2. **Assuming a vertical spring has a different period from a horizontal one.** Both give $T = 2\pi\sqrt{m/k}$. Gravity only shifts the mean position; it never changes the period — and the spring period is independent of g entirely.
3. **Swapping series and parallel.** Parallel springs are *stiffer* ($k_{\text{eff}} = k_1 + k_2$, shorter T); series springs are *softer* ($1/k_{\text{eff}} = 1/k_1 + 1/k_2$, longer T). This is the reverse of the resistor mnemonic — do not borrow that intuition.
4. **Cutting a spring the wrong way.** A cut piece is **stiffer**, not weaker: $k_{\text{piece}} = nk$ for n equal pieces.
5. **Forgetting effective gravity in non-inertial frames.** In a lift or accelerating car, replace g with g_{eff} — and get the sign right (up $\Rightarrow$ add, down $\Rightarrow$ subtract, horizontal $\Rightarrow$

Pythagoras).

Pro-Tip: For any non-inertial-frame pendulum, picture the **apparent gravity** the bob feels — the vector sum of $\vec{g}$ and the pseudo-acceleration $-\vec{a}$ — and feed its magnitude into $T = 2\pi\sqrt{L/g_{\text{eff}}}$. And when a vertical spring's static stretch x_0 is given but m and k are not, use $T = 2\pi\sqrt{x_0/g}$ directly — it sidesteps the missing data instantly.

Section 8: Cheat Sheet Box

- **Simple pendulum:** $T = 2\pi\sqrt{\frac{L}{g}}$ — independent of mass and (small) amplitude; $T \propto \sqrt{L}$.
- **Mass-spring (vertical or horizontal):** $T = 2\pi\sqrt{\frac{m}{k}}$ — independent of g ; static-stretch form $T = 2\pi\sqrt{\frac{x_0}{g}}$.
- **Springs:** parallel $k_{\text{eff}} = k_1 + k_2$ (stiffer); series $\frac{1}{k_{\text{eff}}} = \frac{1}{k_1} + \frac{1}{k_2}$ (softer); cut into n pieces $\Rightarrow$ each is nk .
- **Effective gravity:** lift up $g_{\text{eff}} = g + a$; lift down $g_{\text{eff}} = g - a$; car horizontal $g_{\text{eff}} = \sqrt{g^2 + a^2}$, mean tilts $\tan^{-1}(a/g)$.
- **Physical pendulum:** $T = 2\pi\sqrt{\frac{I}{mgd}}$, $L_{\text{eq}} = \frac{I}{md}$.
- **Torsional pendulum:** $T = 2\pi\sqrt{\frac{I}{\kappa}}$.

Memory aid — “Pendulum feels gravity, spring feels stiffness”: swap g for g_{eff} in pendulum problems, and remember the spring doesn't care about g at all. For combinations: **“Parallel = Powerful (stiffer), Series = Soft.”**

Subtopic 04: Damped and Forced Oscillations

Exam Relevance: The most conceptual subtopic of the chapter. JEE Main typically asks 1 question on the resonance condition, the exponential decay of amplitude or energy, or the type of damping. NEET leans on definitions (resonance, damping) and assertion-reason items. JEE Advanced reaches for the quality factor, energy decay, and damping regimes. For CBSE Boards, the definitions of free/damped/forced oscillations and everyday examples of resonance are frequent short-answer questions.

Section 1: Concept Intuition

Strike a tabla and the note swells, then fades. Pluck a sitar string and it sings, then falls silent. Set a child's *jhula* swinging and walk away — within a minute or two it slows and stops. Every real oscillation in the world eventually dies. The idealised SHM of the first three subtopics, with its forever-constant amplitude, is a beautiful fiction; reality always leaks energy. Understanding *how* it leaks, and how we can pump energy back in, is the whole of this subtopic.

There are three storylines for an oscillator, and you must keep them straight:

- **Free oscillation:** the idealised case — give the system one push and leave it alone with no friction. It oscillates forever at its own **natural frequency** $\omega_0 = \sqrt{k/m}$ with constant amplitude. This is the SHM of Subtopics 01–03.
- **Damped oscillation:** the realistic “left alone” case. A resistive force (air drag, internal friction, viscosity of the medium) opposes the motion and steadily drains energy. The system still oscillates near its natural frequency, but its **amplitude shrinks exponentially** until it dies out.
- **Forced oscillation:** we refuse to let it die — we apply an external periodic force to keep pushing. After an initial confused “transient” stage, the system stops oscillating at its own frequency and instead **marches in step with the driving frequency**, settling to a steady amplitude where the energy we pump in each cycle exactly replaces what damping removes.

The damping force is taken to be proportional to velocity, $F_{\text{damp}} = -bv$ — the faster the motion, the harder the medium resists. Because it always opposes velocity, it always does negative work, so mechanical energy only ever decreases.

Think about it this way... pushing a child on a swing. If you shove at random moments, sometimes you help and sometimes you fight the motion — the swing never gets very high. But if you time each push to arrive exactly as the swing starts moving away from you — pushing at the swing's *own* natural rhythm — every push adds energy and the swing climbs higher and higher. That perfect timing is **resonance**: when the driving frequency matches the natural frequency, the amplitude grows dramatically. It is the most important idea here, and it cuts both ways — it is how a radio tuner selects one station out of many (electrical resonance), and it is why soldiers are ordered to break step when crossing a bridge, lest their rhythmic marching match the bridge's natural frequency and shake it apart.

Limiting Conditions & Assumptions. - The linear damping model $F = -bv$ holds for **slow motion in a viscous medium**; at high speeds drag becomes quadratic in velocity and the simple exponential decay no longer applies. - The clean steady-state amplitude formula describes the system **after the transient has died away**; immediately after switching on the drive, both frequencies coexist briefly. - Resonance amplitude is **large but finite only**

because real systems always have some damping. With zero damping the formula predicts infinite amplitude — an unphysical idealisation. - “Light damping” (b small) is assumed for most exam results: then the damped frequency $\omega' \approx \omega_0$ and resonance occurs essentially at the natural frequency.

Section 2: Key Formulas & Definitions

Damped oscillation (underdamped). Displacement with an exponentially decaying envelope:

$$x(t) = A_0 e^{-bt/2m} \cos(\omega' t + \phi)$$

Damped (shifted) angular frequency:

$$\omega' = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}, \quad \omega_0 = \sqrt{\frac{k}{m}}$$

Amplitude and energy decay:

$$A(t) = A_0 e^{-bt/2m}, \quad E(t) = E_0 e^{-bt/m}$$

(Energy decays **twice as fast** as amplitude, because $E \propto A^2$ — a direct link back to Subtopic 02.)

Forced oscillation (steady-state amplitude, driving force $F_0 \cos \omega t$):

$$A = \frac{F_0/m}{\sqrt{(\omega^2 - \omega_0^2)^2 + \left(\frac{b\omega}{m}\right)^2}}$$

Resonance (amplitude maximum):

$$\omega_r = \sqrt{\omega_0^2 - \frac{b^2}{2m^2}} \approx \omega_0 \text{ (light damping)}, \quad A_{\text{res}} \approx \frac{F_0}{b\omega_0}$$

Limiting cases of the driven amplitude: - Very low driving frequency ($\omega \rightarrow 0$): $A \rightarrow \frac{F_0}{m\omega_0^2} = \frac{F_0}{k}$ (stiffness-controlled). - At resonance ($\omega \approx \omega_0$): $A \approx \frac{F_0}{b\omega_0}$ (damping-controlled — large for small b).

Quality factor (sharpness of resonance):

$$Q = \frac{m\omega_0}{b} = 2\pi \frac{\text{energy stored}}{\text{energy lost per cycle}}$$

Damping regimes (let $b_c = 2\sqrt{mk}$ be critical damping): - **Underdamped** ($b < b_c$): oscillates with decaying amplitude. - **Critically damped** ($b = b_c$): returns to equilibrium in the **shortest time without oscillating**. - **Overdamped** ($b > b_c$): returns to equilibrium slowly, without oscillating.

Notation, units, dimensions: - b — damping constant (from $F = bv$): SI unit kg s^{-1} (or N s m^{-1}). Dimensional formula $[M^1 L^0 T^{-1}]$. - $\omega_0, \omega', \omega$ — angular frequencies: $[T^{-1}]$. F_0 — force amplitude: $[M^1 L^1 T^{-2}]$. Q — dimensionless.

Section 3: Key Derivations & Logical Framework

Framework 1 — The damped oscillator and exponential decay.

Set-up: a mass m on a spring (constant k) moves through a medium that exerts a velocity-proportional drag $-bv$. Newton's second law gives the governing equation:

$$m \frac{d^2x}{dt^2} = -kx - b \frac{dx}{dt} \implies m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = 0.$$

Reasoning about the solution: for light damping the system still oscillates, but each swing is a little smaller than the last — so we expect “an oscillation multiplied by a shrinking factor.” That is exactly the form

$$x(t) = A_0 e^{-bt/2m} \cos(\omega' t + \phi).$$

The cosine carries the to-and-fro; the exponential $e^{-bt/2m}$ is the shrinking envelope. Substituting this back into the equation fixes the frequency at

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}.$$

Physical reading: damping does two things — it shrinks the amplitude and *slightly slows* the oscillation ($\omega' < \omega_0$). But the frequency shift depends on b^2 , so for light damping it is negligible; the conspicuous effect is the amplitude decay, not a frequency change. Because mechanical energy is $E = \frac{1}{2}kA^2$ (Subtopic 02) and $A(t) = A_0 e^{-bt/2m}$,

$$E(t) = \frac{1}{2}kA_0^2 e^{-bt/m} = E_0 e^{-bt/m},$$

so **energy falls off with half the time-constant of amplitude** — a favourite exam catch.

Framework 2 — Forced oscillation, steady state, and resonance.

Set-up: now add an external periodic force $F_0 \cos \omega t$:

$$m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = F_0 \cos \omega t.$$

Reasoning: after the transient dies, the only sustainable rhythm is the one the driver imposes, so the steady-state motion must be $x = A \cos(\omega t - \delta)$ at the **driving frequency** ω , lagging the force by a phase δ . Substituting and balancing the spring, damping, and inertial terms yields the amplitude

$$A = \frac{F_0/m}{\sqrt{(\omega^2 - \omega_0^2)^2 + (b\omega/m)^2}}.$$

Reading the formula as a procedure: look at the denominator. The first term $(\omega^2 - \omega_0^2)^2$ vanishes when the drive matches the natural frequency — that is what makes the amplitude shoot up at **resonance**. The second term $(b\omega/m)^2$ is the only thing keeping the denominator from hitting zero, so it is the damping alone that caps the resonant amplitude at $A_{\text{res}} \approx F_0/(b\omega_0)$. Set $\omega \rightarrow 0$ and the amplitude collapses to F_0/k — a slow push just statically stretches the spring. This single denominator therefore encodes the entire resonance story.

Figure 13.3 A resonance curve — steady-state amplitude A on the vertical axis versus driving frequency ω on the horizontal axis. Two curves both peaking near $\omega = \omega_0$: a tall, narrow peak labelled “small damping (sharp resonance, high Q)” and a short, broad peak (see next page)

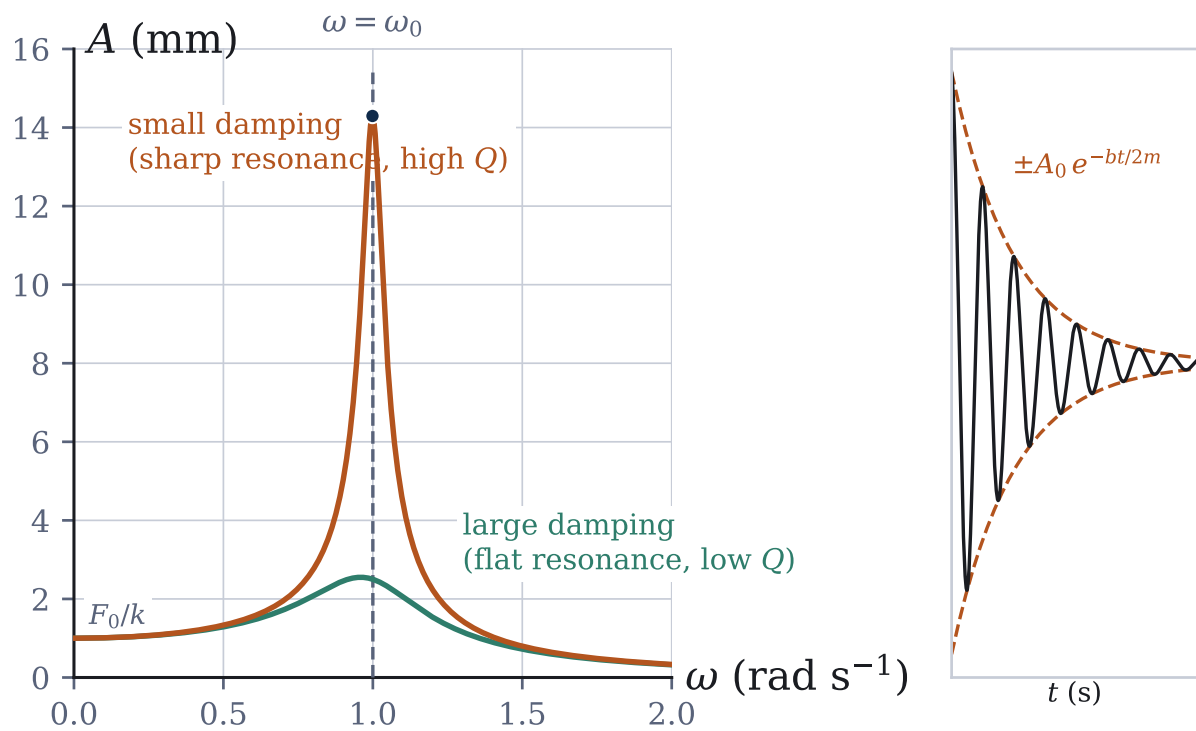


Figure 13.3 A resonance curve — steady-state amplitude A on the vertical axis versus driving frequency ω on the horizontal axis. Two curves both peaking near $\omega = \omega_0$: a tall, narrow peak labelled “small damping (sharp resonance, high Q)” and a short, broad peak

labelled “large damping (flat resonance, low Q)”. A vertical dashed line marks $\omega = \omega_0$ at the peak. Separately, an inset showing a decaying-oscillation graph: cosine wave inside a shrinking exponential envelope $\pm A_0 e^{-(bt/2m)}$.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A block of mass 0.2 kg attached to a spring oscillates in a viscous medium with damping constant $b = 0.04 \text{ kg s}^{-1}$. Find the time in which the amplitude of oscillation drops to half its initial value. (Take $\ln 2 = 0.693$.)

Solution: the amplitude obeys $A(t) = A_0 e^{-bt/2m}$. Set $A(t) = A_0/2$:

$$e^{-bt/2m} = \frac{1}{2} \Rightarrow \frac{bt}{2m} = \ln 2 \Rightarrow t = \frac{2m}{b} \ln 2.$$

$$t = \frac{2(0.2)}{0.04} \times 0.693 = 10 \times 0.693 = 6.93 \text{ s}.$$

Answer: the amplitude halves in about 6.9 s. (Note: it will halve *again* in the next 6.9 s — equal times for equal fractional drops, the signature of exponential decay.)

Example 2 — [NEET Speed Trap] A spring-mass system has a natural frequency of 5 Hz. A small periodic driving force is applied to it. At approximately what driving frequency will the amplitude of oscillation be largest, and what is this phenomenon called?

The trap: a student may try to compute the exact resonant frequency $\omega_r = \sqrt{\omega_0^2 - b^2/2m^2}$ and get bogged down — or, worse, guess that maximum amplitude occurs at *twice* or *half* the natural frequency.

The shortcut: for the light damping implied by “a small force,” the correction term is negligible, so the amplitude peaks essentially **at the natural frequency**:

$$f_{\text{drive}} \approx f_0 = 5 \text{ Hz}.$$

This is **resonance**.

Answer: $\approx 5 \text{ Hz}$; the phenomenon is resonance. Maximum amplitude always sits at (near) the natural frequency, never at a multiple of it.

Example 3 — [JEE Main Level] A damped oscillator has mass 0.5 kg, spring constant 50 N m^{-1} , and damping constant 0.1 kg s^{-1} . Find (a) its natural and damped angular frequencies, (b) the time for the amplitude to fall to $1/e$ of its initial value, and (c) the fraction of mechanical energy remaining at that instant.

Solution: (a) $\omega_0 = \sqrt{k/m} = \sqrt{50/0.5} = \sqrt{100} = 10 \text{ rad s}^{-1}$. With $\frac{b}{2m} = \frac{0.1}{2(0.5)} = 0.1 \text{ s}^{-1}$:

$$\omega' = \sqrt{\omega_0^2 - (b/2m)^2} = \sqrt{100 - 0.01} = \sqrt{99.99} \approx 10.0 \text{ rad s}^{-1}.$$

The frequency is essentially unchanged — confirming the damping is light.

(b) Amplitude $A_0 e^{-bt/2m} = A_0/e$ requires $\frac{bt}{2m} = 1$, so

$$t = \frac{2m}{b} = \frac{2(0.5)}{0.1} = 10 \text{ s}.$$

(c) Energy decays as $E(t) = E_0 e^{-bt/m}$. At $t = 10$ s: $\frac{bt}{m} = \frac{0.1 \times 10}{0.5} = 2$, so

$$E = E_0 e^{-2} \approx 0.135 E_0.$$

Answer: $\omega_0 = 10 \text{ rad s}^{-1}$, $\omega' \approx 10 \text{ rad s}^{-1}$; amplitude reaches $1/e$ at $t = 10$ s; about 13.5% of the energy remains. (Cross-check: when $A = A_0/e$, $E \propto A^2 = E_0/e^2 = 0.135 E_0$ — consistent.)

Example 4 — [JEE Advanced Level] A lightly damped oscillator has natural angular frequency $\omega_0 = 200 \text{ rad s}^{-1}$ and quality factor $Q = 100$. (a) Find the ratio b/m . (b) Estimate the number of complete oscillations the system makes before its energy falls to $1/e$ of its initial value, and show this number equals $Q/2\pi$.

Solution: (a) Using $Q = \frac{m\omega_0}{b}$:

$$\frac{b}{m} = \frac{\omega_0}{Q} = \frac{200}{100} = 2 \text{ s}^{-1}.$$

(b) Energy falls to $1/e$ when $E_0 e^{-(b/m)t} = E_0/e$, i.e. $\frac{b}{m}t = 1$, so $t = \frac{m}{b} = 0.5$ s. With period

$$T = \frac{2\pi}{\omega'} \approx \frac{2\pi}{\omega_0} = \frac{2\pi}{200} \approx 0.0314 \text{ s, the number of oscillations is}$$

$$N = \frac{t}{T} = \frac{m/b}{2\pi/\omega_0} = \frac{m\omega_0}{2\pi b} = \frac{Q}{2\pi} = \frac{100}{2\pi} \approx 15.9.$$

Answer: $b/m = 2 \text{ s}^{-1}$; the system completes about 16 oscillations, and in general $N = Q/2\pi$. This is the deep meaning of the quality factor: **a high- Q oscillator rings for many cycles before its energy decays**, which is exactly why a high- Q tuning circuit gives a sharp, selective resonance.

Section 5: Practice Exercises

- [CBSE]** The amplitude of a damped oscillator falls to half its initial value in 10 s. After how much total time will it fall to one-fourth of the initial value?
- [NEET]** The amplitude of a damped oscillator becomes $\frac{1}{3}$ of its initial value in 2 s. What fraction of the initial amplitude remains after 6 s?
- [JEE Main]** For a damped oscillator, when the amplitude has fallen to half its initial value, what fraction of the original mechanical energy remains?
- [JEE Main]** A driven oscillator has natural angular frequency ω_0 . State (a) the driving frequency for maximum amplitude (light damping) and (b) the amplitude in the limit of a very slowly varying driving force ($\omega \rightarrow 0$).
- [JEE Advanced]** A lightly damped oscillator loses 2% of its energy in each cycle. Estimate its quality factor.

Answers: **1.** 20 s (equal times for each halving in exponential decay). **2.** $(\frac{1}{3})^3 = \frac{1}{27}$. **3.** $E \propto A^2 \Rightarrow$ remaining energy $= \frac{1}{4}$ (i.e. 75% lost). **4.** (a) $\omega \approx \omega_0$; (b) $A \rightarrow F_0/k = F_0/(m\omega_0^2)$. **5.** $Q = \frac{2\pi}{\Delta E/E} = \frac{2\pi}{0.02} = 100\pi \approx 314$.

Section 6: Crack the MCQ

Q1. In a damped harmonic oscillation, which of the following quantities decreases continuously with time? (A) the frequency (B) the amplitude (C) the time period (D) the phase

Correct: (B). The defining feature of damping is the exponential decay of amplitude, $A(t) = A_0 e^{-bt/2m}$. The frequency and period change only by a tiny b^2 -dependent amount and are essentially constant for light damping. - (A) and (C) form the classic trap — students picture damping as “the motion slowing down” and assume the frequency drops noticeably. It does not, to leading order. - (D) the phase $(\omega' t + \phi)$ steadily *increases* with time; it does not decrease.

Q2. The amplitude of a forced oscillator is maximum when the driving frequency is approximately equal to: (A) zero (B) twice the natural frequency (C) the natural frequency (D) half the natural frequency

Correct: (C). Resonance occurs when the driving frequency matches the natural frequency (for light damping, $\omega_r \approx \omega_0$), making the term $(\omega^2 - \omega_0^2)^2$ in the amplitude denominator vanish. - (A) gives the static limit $A = F_0/k$, not the maximum. - (B) and (D) are distractors for students who confuse resonance with harmonics/overtones from the waves chapter.

Q3. For a damped oscillator the energy varies as the square of the amplitude. If the amplitude reduces to $1/\sqrt{2}$ of its initial value, the energy becomes what fraction of the initial energy? (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2\sqrt{2}}$

Correct: (A). Since $E \propto A^2$, energy becomes $\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$ of the initial. - (B) forgets to square — applies the amplitude ratio directly to energy. - (C) squares the wrong factor (would be right if amplitude fell to $\frac{1}{2}$). - (D) mixes up the powers.

Q4. A hydraulic door-closer is designed so that, after the door is released, it returns to the closed position in the shortest possible time without swinging past and oscillating. This is an example of: (A) underdamping (B) critical damping (C) overdamping (D) forced oscillation

Correct: (B). Critical damping ($b = 2\sqrt{mk}$) returns a system to equilibrium in the minimum time with no oscillation — exactly the requirement for a door-closer (and for a galvanometer needle or car shock absorber). - (A) underdamping would let the door swing past and oscillate before settling. - (C) overdamping returns the door without oscillating but **too slowly**. - (D) there is no external periodic drive here, so it is not forced oscillation.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Believing damping noticeably slows the frequency.** For light damping $\omega' \approx \omega_0$; the visible effect is shrinking amplitude, *not* a slower oscillation. The frequency shift goes as b^2 and is usually negligible.
2. **Confusing transient and steady state.** In the steady state of a forced oscillation, the system vibrates at the **driving frequency**, never at its own natural frequency. Its natural-frequency motion belongs only to the brief transient.
3. **Thinking resonance amplitude is infinite.** It is finite because real systems always have some damping; only the idealised zero-damping formula blows up.
4. **Forgetting that energy decays twice as fast as amplitude.** $A \propto e^{-bt/2m}$ but $E \propto e^{-bt/m}$ — the energy time-constant is half the amplitude time-constant, because $E \propto A^2$.
5. **Mixing up the damping regimes.** *Critical* damping gives the fastest no-oscillation return; *overdamping* is also non-oscillatory but slower; *underdamping* still oscillates.

Pro-Tip: For “amplitude falls to a fraction” problems, exploit the equal-interval property of exponentials — each fixed ratio drop takes the same time, so you rarely need to solve for b at

all. And memorise two resonance-sharpness shortcuts: $Q = \frac{2\pi}{(\text{fractional energy lost per cycle})}$, and the oscillator rings for about $N = Q/2\pi$ cycles before its energy falls to $1/e$.

Section 8: Cheat Sheet Box

- **Three regimes:** free (constant A , at ω_0) $\rightarrow$ damped (shrinking A , left alone) $\rightarrow$ forced (steady A , at the **driving** frequency).
- **Damped motion:** $x = A_0 e^{-bt/2m} \cos(\omega' t + \phi)$, with $\omega' = \sqrt{\omega_0^2 - (b/2m)^2} \approx \omega_0$ (light damping).
- **Decay:** amplitude $A \propto e^{-bt/2m}$; energy $E \propto e^{-bt/m}$ (energy decays **twice as fast**).
- **Driven amplitude:** $A = \frac{F_0/m}{\sqrt{(\omega^2 - \omega_0^2)^2 + (b\omega/m)^2}}$; peaks at resonance $\omega \approx \omega_0$, where $A_{\text{res}} \approx \frac{F_0}{b\omega_0}$.
- **Quality factor:** $Q = \frac{m\omega_0}{b} = 2\pi \frac{\text{energy stored}}{\text{energy lost per cycle}}$; rings for $\sim Q/2\pi$ cycles.
- **Damping types:** under ($b < 2\sqrt{mk}$, oscillates) $\cdot$ critical ($b = 2\sqrt{mk}$, fastest no-oscillation return) $\cdot$ over ($b > 2\sqrt{mk}$, slow return).

Memory aid — “Resonance = matching rhythm”: biggest amplitude when the drive matches the natural frequency (the swing-push picture). And **“energy is amplitude squared, so it fades twice as fast”** — the one-line link back to SHM energy.

Subtopic 05: Reference Circle, Superposition & More SHM Systems

Exam Relevance: This is the “everything that ties the chapter together” part. The reference-circle picture is a guaranteed CBSE 1-2 mark item and the conceptual backbone JEE uses for phasors. JEE Main and Advanced love the standard *non-obvious* SHM systems — liquid in a U-tube, a floating block, the tunnel through the Earth — because they test whether you can *prove* a motion is SHM rather than recall a formula. Superposition of SHMs (collinear and perpendicular) and Lissajous figures appear in JEE Advanced and in NEET assertion-reason questions. Temperature error of a pendulum clock is a perennial favourite across all three exams.

Section 1: Concept Intuition

Everything so far has treated SHM as algebra — a sine in time. This part gives you the three tools that turn that algebra into deep understanding and let you handle *any* oscillator the examiner invents.

The reference circle — SHM is a shadow. Imagine a bead glued to the rim of a turntable spinning at a steady angular speed ω , radius A . Shine a light from the side and watch the bead’s **shadow** on a wall. As the bead goes round and round at constant speed, its shadow runs back and forth in a straight line — and that shadow is executing *exactly* simple harmonic motion. This is not an analogy; it is a mathematical identity. The projection of uniform circular motion onto a diameter **is** SHM. Suddenly the three SHM constants have a picture: A is the radius of the circle, ω is how fast the bead circulates, and the phase ($\omega t + \phi$) is literally the *angle* the radius has swept. A “phasor” is just this rotating radius — and that is why we can add SHMs like vectors.

Think about it this way... the bead moves at constant speed, but its shadow does not. When the bead is at the top or bottom of the circle (the turning points of the shadow), the shadow is momentarily still; when the bead crosses the horizontal (the shadow at its centre), the shadow is sprinting. Constant circular speed, varying shadow speed — that is the whole story of why SHM speeds up near the middle and freezes at the ends, drawn in one picture.

The differential-equation test — what *makes* a motion SHM. Forget formulas for a second. A motion is SHM if and only if its acceleration is proportional to displacement and points *back* toward the centre:

$$\frac{d^2x}{dt^2} = -\omega^2 x.$$

This is the real, rigorous definition. To prove that some strange new system (a wobbling test-tube, a sloshing liquid, a body falling through the Earth) is SHM, you do *not* need to recognise it — you displace it a little, find its restoring force or its energy, and show the acceleration comes out as $-(\text{positive constant}) \times x$. That constant is ω^2 , and the period drops out as $T = 2\pi/\omega$. One method, every system.

Superposition — adding oscillations. Real objects often feel two SHMs at once (two tuning forks, a particle pushed by two springs). If they act **along the same line**, the results add like phasors (vectors) and you get another SHM. If they act **at right angles**, the particle traces a 2-D path — a straight line, an ellipse, a circle, or, for unequal frequencies, the beautiful looping **Lissajous figures** you see on an oscilloscope.

Limiting Conditions & Assumptions. - The reference-circle equivalence is exact only

for **uniform** circular motion (constant ω). - The “ $a = -\omega^2 x$ ” test certifies SHM only for **small displacements**, where the restoring effect stays linear; large swings make every real system anharmonic. - Collinear superposition gives a single clean SHM only when the two components share the **same frequency**; different frequencies give beats, not SHM. - The standard system periods below (U-tube, floating body, tunnel) assume **ideal** conditions: incompressible non-viscous liquid, uniform cross-section, uniform Earth density, no friction or drag.

Section 2: Key Formulas & Definitions

Reference circle (projection of uniform circular motion):

$$x = A \cos(\omega t + \phi), \quad v = -A\omega \sin(\omega t + \phi), \quad a = -A\omega^2 \cos(\omega t + \phi)$$

The radius A = amplitude; circular speed ω = angular frequency; swept angle $(\omega t + \phi)$ = phase. The two perpendicular projections of the same circular motion are $x = A \cos(\omega t + \phi)$ and $y = A \sin(\omega t + \phi)$ — two SHMs 90° out of phase.

Defining differential equation of SHM:

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \implies x = A \cos(\omega t + \phi), \quad \omega = \sqrt{\frac{\text{restoring constant}}{\text{inertia}}}$$

Superposition of two collinear SHMs ($x_1 = A_1 \cos \omega t$, $x_2 = A_2 \cos(\omega t + \delta)$), resultant $x = A \cos(\omega t + \varphi)$:

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \delta}, \quad \tan \varphi = \frac{A_2 \sin \delta}{A_1 + A_2 \cos \delta}$$

- In phase ($\delta = 0$): $A = A_1 + A_2$ (maximum). Anti-phase ($\delta = \pi$): $A = |A_1 - A_2|$ (minimum). Quadrature ($\delta = \pi/2$): $A = \sqrt{A_1^2 + A_2^2}$.

Superposition of two perpendicular SHMs ($x = A \cos \omega t$, $y = B \cos(\omega t + \delta)$), equal frequency: - $\delta = 0$: straight line $y = \frac{B}{A}x$. $\delta = \pi$: straight line $y = -\frac{B}{A}x$. - $\delta = \pi/2$: ellipse $\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$ (a **circle** if $A = B$). - Unequal frequencies $\rightarrow$ **Lissajous figures**; the frequency ratio sets the number of loops.

Periods of standard SHM systems:

$$\text{Liquid column (U-tube): } T = 2\pi \sqrt{\frac{L}{2g}} \quad (L = \text{total length of liquid column})$$

$$\text{Floating body: } T = 2\pi \sqrt{\frac{h}{g}} = 2\pi \sqrt{\frac{m}{A\rho g}} \quad (h = \text{equilibrium submerged depth})$$

$$\text{Tunnel through Earth / ball in bowl: } T = 2\pi \sqrt{\frac{R}{g}} \approx 84.6 \text{ min}$$

Pendulum fine points: - Temperature error: $L = L_0(1 + \alpha \Delta\theta) \Rightarrow \frac{\Delta T}{T} = \frac{1}{2}\alpha \Delta\theta$; time lost/gained per day = $\frac{1}{2}\alpha \Delta\theta \times 86400 \text{ s}$. - Charged bob in field E : vertical field $g_{\text{eff}} = g \pm \frac{qE}{m}$; horizontal field $g_{\text{eff}} = \sqrt{g^2 + (qE/m)^2}$. - Large amplitude (anharmonic) correction: $T \approx$

$$2\pi\sqrt{\frac{L}{g}}\left(1 + \frac{\theta_0^2}{16}\right) (\theta_0 \text{ in radians}).$$

Time-averaged motion over one cycle:

$$\langle \text{speed} \rangle = \frac{4A}{T} = \frac{2A\omega}{\pi}, \quad \langle v \rangle_{\text{cycle}} = 0, \quad v_{\text{rms}} = \frac{A\omega}{\sqrt{2}}$$

Notation, units, dimensions: A, B, L, h, R — lengths $[L]$; ω — $[T^{-1}]$; $\delta, \varphi, \theta_0$ — radians (dimensionless); ρ — density $[ML^{-3}]$; α — $[K^{-1}]$; q — charge $[AT]$; E — field $[MLT^{-3}A^{-1}]$.

Section 3: Key Derivations & Procedures

Derivation 1 — SHM is the projection of uniform circular motion. A particle moves anticlockwise on a circle of radius A at constant angular speed ω ; at time t its radius makes angle $(\omega t + \phi)$ with the x -axis. Its coordinates are $x = A \cos(\omega t + \phi)$ and $y = A \sin(\omega t + \phi)$. Differentiating the x -projection twice gives $\ddot{x} = -\omega^2 A \cos(\omega t + \phi) = -\omega^2 x$ — the SHM equation. So the shadow on the x -axis is SHM with amplitude A (the radius) and angular frequency ω (the circulation rate). The centripetal acceleration $\omega^2 A$ of the circular motion projects to the SHM acceleration $\omega^2 x$, and the circular speed ωA projects to the SHM speed.

Derivation 2 — The phasor (vector) method for adding collinear SHMs. Represent $x_1 = A_1 \cos \omega t$ and $x_2 = A_2 \cos(\omega t + \delta)$ as two rotating radius vectors of lengths A_1, A_2 with a fixed angle δ between them. Since both rotate at the same ω , the angle between them never changes, so they add like ordinary vectors. The resultant length follows from the law of cosines (with the included angle δ):

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \delta},$$

and the resultant's phase φ from resolving components: $\tan \varphi = \frac{A_2 \sin \delta}{A_1 + A_2 \cos \delta}$. The resultant is again an SHM of the same frequency — this is why a phasor diagram turns trigonometry into geometry.

Derivation 3 — The universal recipe (force or energy) for any SHM system. *Force method:* displace the system by a small x , find the net restoring force. If it comes out as $F = -k_{\text{eff}}x$, then $m\ddot{x} = -k_{\text{eff}}x$, so $\omega = \sqrt{k_{\text{eff}}/m}$. *Energy method:* write the total energy $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}k_{\text{eff}}x^2$. Because E is constant, $\frac{dE}{dt} = m\dot{x}\ddot{x} + k_{\text{eff}}x\dot{x} = 0$, which gives $\ddot{x} + \frac{k_{\text{eff}}}{m}x = 0$ — the SHM equation again, with $\omega = \sqrt{k_{\text{eff}}/m}$. Use whichever is cleaner for the system in front of you.

Derivation 4 — Three classic systems, each in three lines.

Liquid column in a U-tube. Total liquid length L , tube area A , density ρ . Push the liquid down one arm by y ; it rises y in the other, creating a height difference $2y$. The unbalanced weight is $F = -(2y)A\rho g$, and the moving mass is $LA\rho$. Hence $\ddot{y} = -\frac{2g}{L}y$, giving $T = 2\pi\sqrt{L/2g}$.

Floating body. A body floats with submerged depth h (equilibrium: $mg = Ah\rho g$, so $m = Ah\rho$). Push it down by x ; the extra buoyant force is $A\rho g x$ upward. Then $\ddot{x} = -\frac{A\rho g}{m}x = -\frac{g}{h}x$, giving

$$T = 2\pi\sqrt{h/g}.$$

Tunnel through the Earth. For a uniform Earth, gravity at depth (distance r from centre) is $g(r) = g\frac{r}{R}$, always pointing to the centre. A body dropped into a straight tunnel feels

$\ddot{r} = -\frac{g}{R}r$, giving $T = 2\pi\sqrt{R/g} \approx 84.6$ min. (A particle sliding inside a smooth hemispherical bowl of radius R obeys the same equation for small swings.)

Figure 13.4 Three panels illustrating standard SHM systems. Panel 1: a U-tube partly filled with liquid, one arm pushed down by y and the other risen by y , with the height difference $2y$ labelled and total liquid length L marked. Panel 2: a cylinder floating in liquid, equilibrium submerged depth h labelled, with an arrow showing it pushed down by x and the extra buoyancy arrow pointing up. Panel 3: a cross-section of the Earth (radius R) with a straight tunnel along a diameter and a small body at distance r from the centre, restoring force $g(r/R)$ pointing toward the centre. (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Two SHMs act on a particle along the same line: $x_1 = 3\sin\omega t$ cm and $x_2 = 4\sin(\omega t + \frac{\pi}{2})$ cm. Find the amplitude and initial phase of the resultant motion.

Solution: the phase difference is $\delta = \frac{\pi}{2}$, so $\cos\delta = 0$. The resultant amplitude is

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\delta} = \sqrt{3^2 + 4^2 + 0} = \sqrt{25} = 5 \text{ cm.}$$

The phase is $\tan\varphi = \frac{A_2\sin\delta}{A_1 + A_2\cos\delta} = \frac{4(1)}{3+0} = \frac{4}{3}$, so $\varphi = \tan^{-1}(4/3) \approx 53^\circ$.

Answer: amplitude 5 cm, phase $\approx 53^\circ$. (The 3-4-5 right triangle is the phasor diagram itself — perpendicular phasors add by Pythagoras.)

Example 2 — [NEET Speed Trap] A wooden cylinder floats upright in water with 16 cm of its length submerged. It is pushed down slightly and released. Find its period of small vertical oscillations. (Take $g = 9.8 \text{ m s}^{-2}$.)

The trap: students hunt for the mass, density, or area — none is needed. For a floating body the period depends only on the **equilibrium submerged depth**.

The shortcut:

$$T = 2\pi\sqrt{\frac{h}{g}} = 2\pi\sqrt{\frac{0.16}{9.8}} = 2\pi\sqrt{0.01633} = 2\pi(0.1278) \approx 0.80 \text{ s.}$$

Answer: $T \approx 0.80$ s. The formula $T = 2\pi\sqrt{h/g}$ is identical in form to a simple pendulum of length h — memorise that twin.

Example 3 — [JEE Main Level] A U-tube contains mercury; the total length of the mercury column is 1.0 m. The mercury is displaced slightly and released. Find the period of oscillation. (Take $g = 9.8 \text{ m s}^{-2}$.)

Solution: for a liquid column of total length L ,

$$T = 2\pi\sqrt{\frac{L}{2g}} = 2\pi\sqrt{\frac{1.0}{2 \times 9.8}} = 2\pi\sqrt{0.05102} = 2\pi(0.2259) \approx 1.42 \text{ s.}$$

Note the period is **independent of the density** of the liquid (mercury, water, oil — all give the same period for the same column length), because both the restoring force and the inertia scale with ρ and it cancels.

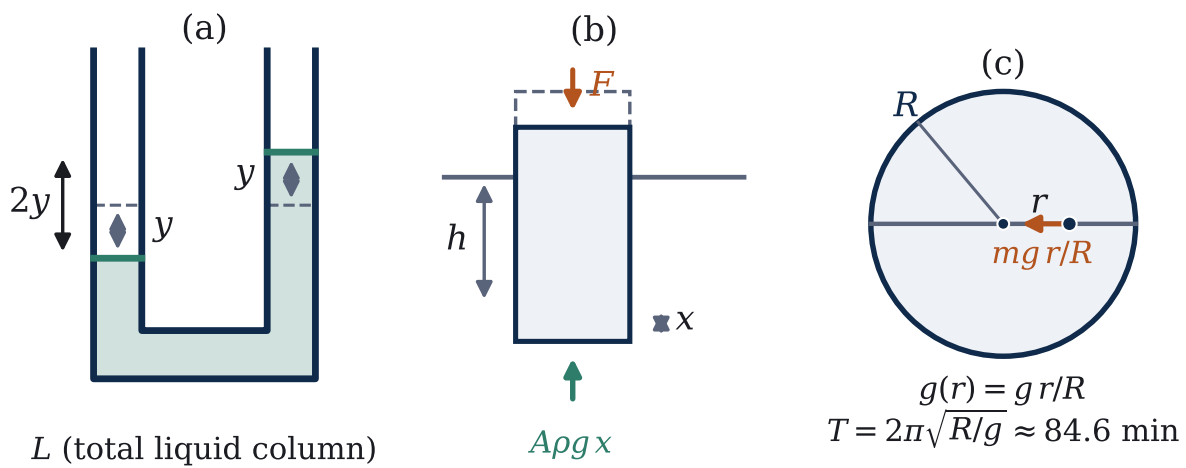


Figure 13.4 Three panels illustrating standard SHM systems. Panel 1: a U-tube partly filled with liquid, one arm pushed down by y and the other risen by y , with the height difference $2y$ labelled and total liquid length L marked. Panel 2: a cylinder floating in liquid, equilibrium submerged depth h labelled, with an arrow showing it pushed down by x and the extra buoyancy arrow pointing up. Panel 3: a cross-section of the Earth (radius R) with a straight tunnel along a diameter and a small body at distance r from the centre, restoring force $g(r/R)$ pointing toward the centre.

Answer: $T \approx 1.42$ s, independent of which liquid fills the tube.

Example 4 — [JEE Advanced Level] A pendulum clock keeps perfect time at 20°C . The pendulum rod has a linear expansion coefficient $\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$. In summer the temperature rises to 40°C . Does the clock run fast or slow, and by how many seconds per day?

Solution: the rod lengthens, so $T \propto \sqrt{L}$ increases — each tick now takes longer, so the clock **runs slow** (loses time). The fractional change of period is

$$\frac{\Delta T}{T} = \frac{1}{2} \alpha \Delta \theta = \frac{1}{2} (1.2 \times 10^{-5}) (20) = 1.2 \times 10^{-4}.$$

Over one day (86400 s) the time lost is

$$\Delta t = \frac{\Delta T}{T} \times 86400 = (1.2 \times 10^{-4}) (86400) \approx 10.4 \text{ s}.$$

Answer: the clock runs **slow**, losing about 10.4 s per day. (A cooler rod would shorten and the clock would gain — this is why precision pendulum clocks use temperature-compensated rods.)

Section 5: Practice Exercises

1. **[CBSE]** A particle in SHM is described as the projection of uniform circular motion of radius 5 cm and angular speed 4 rad s^{-1} . Write its amplitude, angular frequency and maximum speed.
2. **[NEET]** Two collinear SHMs of equal amplitude a differ in phase by $\frac{2\pi}{3}$. Find the amplitude of their resultant.
3. **[JEE Main]** A test-tube of cross-section A loaded with lead floats vertically in water with 12 cm submerged. Find its period of small vertical oscillation. (Take $g = 9.8 \text{ m s}^{-2}$.)
4. **[JEE Main]** Estimate the time a stone would take to fall from one end of a straight smooth tunnel bored through the centre of the Earth to the far end. (Take $R = 6.4 \times 10^6 \text{ m}$, $g = 9.8 \text{ m s}^{-2}$.)
5. **[JEE Advanced]** Two perpendicular SHMs $x = A \cos \omega t$ and $y = A \cos(\omega t + \frac{\pi}{2})$ are combined. Identify the path traced by the particle.

Answers: 1. $A = 5 \text{ cm}$, $\omega = 4 \text{ rad s}^{-1}$, $v_{\max} = A\omega = 20 \text{ cm s}^{-1}$. 2. $A_{\text{res}} = \sqrt{a^2 + a^2 + 2a^2 \cos \frac{2\pi}{3}} = \sqrt{a^2 + a^2 - a^2} = a$. 3. $T = 2\pi \sqrt{0.12/9.8} \approx 0.695 \text{ s}$. 4. Half the period: $t = \pi \sqrt{R/g} = \pi \sqrt{6.53 \times 10^5} \approx 2538 \text{ s} \approx 42.3 \text{ min}$. 5. A **circle** of radius A (equal amplitudes, quadrature phase), traced clockwise.

Section 6: Crack the MCQ

Q1. The projection of a particle in uniform circular motion onto a diameter of the circle executes: (A) uniform motion (B) non-uniform circular motion (C) simple harmonic motion (D) projectile motion

Correct: (C). Differentiating the projection $x = A \cos(\omega t + \phi)$ twice gives $\ddot{x} = -\omega^2 x$, the defining equation of SHM. This equivalence is the reference-circle theorem. - (A) the shadow's speed varies (fast at the centre, zero at the ends), so it is not uniform. - (B) and (D) describe two-dimensional motions; the projection is strictly one-dimensional.

Q2. Two collinear SHMs of equal amplitude a and the same frequency combine. For their resultant amplitude to be $a\sqrt{2}$, their phase difference must be: (A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π

Correct: (C). $A = \sqrt{a^2 + a^2 + 2a^2 \cos \delta}$. Setting this equal to $a\sqrt{2}$ gives $2a^2 + 2a^2 \cos \delta = 2a^2$, so $\cos \delta = 0$, i.e. $\delta = \frac{\pi}{2}$. - (A) gives $2a$ (maximum); (D) gives 0 (cancellation); (B) gives an intermediate value, not $a\sqrt{2}$.

Q3. Which of these systems does **not** execute simple harmonic motion for small displacements? (A) liquid oscillating in a U-tube (B) a ball rolling back and forth at the bottom of a smooth bowl (C) a body dropped into a tunnel through the Earth (D) a ball bouncing elastically off the floor

Correct: (D). A bouncing ball has a constant downward acceleration g between bounces — its acceleration is not proportional to (nor opposite) the displacement, so it is periodic but **not** SHM. - (A), (B), (C) all yield a restoring acceleration of the form $-\omega^2 x$ for small displacements, so all three are SHM.

Q4. Two perpendicular SHMs of equal amplitude and equal frequency, with a phase difference of $\frac{\pi}{2}$, trace a: (A) straight line (B) parabola (C) circle (D) figure-of-eight

Correct: (C). With $x = A \cos \omega t$ and $y = A \cos(\omega t + \frac{\pi}{2}) = -A \sin \omega t$, we get $x^2 + y^2 = A^2$ — a circle. - (A) arises for $\delta = 0$ or π ; (D) needs a 2 : 1 frequency ratio (a Lissajous figure), not equal frequencies; (B) never arises for equal-frequency perpendicular SHMs.

Section 7: Common Student Pitfalls & Pro-Tips

- Adding SHM amplitudes arithmetically regardless of phase.** Two SHMs of amplitude 3 and 4 give a resultant between 1 and 7 depending on their phase difference — only **in phase** does it reach 7. Always use the phasor formula.
- Using the wrong length in the U-tube formula.** L is the **total length of the liquid column**, not the length in one arm and not the displacement.
- Plugging the body's full length into the floating-body period.** The period uses the **equilibrium submerged depth** h , not the total length of the floating object.
- Forgetting that the reference-circle speed is constant but the SHM speed is not.** The bead circulates uniformly; its shadow accelerates and decelerates.
- Getting the temperature sign backwards.** A temperature *rise* lengthens the rod, *increasing* the period, so the clock runs **slow** (loses time); a temperature drop makes it run fast.

Pro-Tip: Whenever you meet an unfamiliar oscillating system, do not look for a memorised formula — run the universal recipe: displace by x , find the net restoring force (or write the energy and differentiate), force it into the shape $\ddot{x} = -\omega^2 x$, and read off ω . Three of the “hardest-looking” JEE systems — U-tube, floating block, Earth tunnel — collapse to one line each this way. And remember the family of look-alike periods: simple pendulum $2\pi\sqrt{L/g}$, floating body $2\pi\sqrt{h/g}$, Earth tunnel $2\pi\sqrt{R/g}$ — same skeleton, different length.

Section 8: Cheat Sheet Box

- Reference circle:** SHM = projection of uniform circular motion; radius $\rightarrow$ amplitude, circulation rate $\rightarrow \omega$, swept angle $\rightarrow$ phase. Two perpendicular projections are SHMs 90° apart.
- SHM test / definition:** $\ddot{x} + \omega^2 x = 0$; prove any system is SHM by showing $\ddot{x} = -(\text{const})x$, then $\omega = \sqrt{\text{const}}$.
- Collinear superposition:** $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \delta}$; $\max A_1 + A_2$ ($\delta = 0$), $\min |A_1 - A_2|$ ($\delta = \pi$), $\sqrt{A_1^2 + A_2^2}$ ($\delta = \pi/2$).
- Perpendicular superposition (equal freq):** line ($\delta = 0, \pi$), ellipse ($\delta = \pi/2$), circle

(equal amplitudes, $\delta = \pi/2$); unequal freq $\rightarrow$ Lissajous.

- **Standard periods:** U-tube $T = 2\pi\sqrt{L/2g}$ (L = total column); floating body $T = 2\pi\sqrt{h/g}$ (h = submerged depth); Earth tunnel / bowl $T = 2\pi\sqrt{R/g} \approx 84.6$ min.
- **Pendulum extras:** temperature $\frac{\Delta T}{T} = \frac{1}{2}\alpha\Delta\theta$ (rise $\Rightarrow$ slow); charged bob $g_{\text{eff}} = g \pm qE/m$ (vertical) or $\sqrt{g^2 + (qE/m)^2}$ (horizontal); large amplitude $T \approx 2\pi\sqrt{L/g}(1 + \theta_0^2/16)$.
- **Cycle averages:** $\langle \text{speed} \rangle = 4A/T = 2A\omega/\pi$; $\langle v \rangle_{\text{cycle}} = 0$; $v_{\text{rms}} = A\omega/\sqrt{2}$.

Memory aid — “One test rules them all”: a motion is SHM exactly when *acceleration* = $-\omega^2 \times$ *displacement*. Find that line for any system and the period falls out. And the period-twins to keep on a sticky note: $\sqrt{L/g}$, $\sqrt{h/g}$, $\sqrt{R/g}$ — pendulum, float, planet.

Chapter 13: Oscillations — Round 2 Addendum

The chapter gives you the kinematics, energy, standard pendulums and springs, damping, forcing, the reference circle, and three “exotic” SHM systems. Six JEE patterns are still standing outside the door, and each one is a marriage of two things the chapter already teaches separately. Addendum A marries the energy conservation of Subtopic 02 to the “find ω ” recipe of Subtopic 01: the energy method ($dE/dt = 0$) solves systems where forces are messy but energy is clean. Addendum B goes back to Subtopic 03’s spring combinations and adds the pieces the chapter omits — springs with mass, springs cut and rejoined, and springs separated by masses. Addendum C takes the phase-circle picture of Subtopic 05 and turns it into a stopwatch: exact time-between-positions, fraction-of-period-in-region, and the phase-difference shortcuts that replace integration. Addendum D opens Subtopic 05’s superposition to different frequencies — beats, amplitude modulation, and the exact envelope that JEE Main asks for by name. Addendum E deepens Subtopic 04’s damping: logarithmic decrement, the distinction between amplitude resonance and velocity resonance, and the power-absorption curve that decides “sharpness” in Advanced. Addendum F is the “prove it’s SHM” laboratory: the potential-energy-curve test, the small-oscillation expansion, and a menu of non-standard systems (rolling cylinder, U-tube with unequal arms, floating cone, piston in gas) that all collapse to the same three-line derivation.

Numbering continues from the chapter’s Subtopic 05. Nothing here is new theory. Every method below is the energy method, the force method, or the phase-circle geometry, applied one step earlier or one dimension wider.

Addendum A — Energy Method for Time Period & The Potential-Energy Curve

Subtopic 01 Derivation 3 and Subtopic 03 Derivation 3 both say: write the net force, show it is $-k_{\text{eff}} x$, read off ω . That works when forces are visible. In many JEE systems (a piston compressing gas, a floating body with non-uniform cross-section, a particle in a potential well $U(x)$) the force is either obscure or easier to get from energy. The energy method is the universal backup: write total energy $E = K + U$, differentiate once, set $dE/dt = 0$, and the equation of motion falls out with ω^2 identified automatically.

Method

Energy recipe (works for 1-D conservative systems).

1. **Choose the generalised coordinate q .** For a translating mass $q = x$; for a rolling cylinder $q = x$ (centre displacement) or θ ; for a pendulum $q = \theta$.
2. **Write kinetic energy K .** If the system has rotation, $K = \frac{1}{2} m v_{\text{cm}}^2 + \frac{1}{2} I_{\text{cm}} \omega^2$. Express every speed in terms of $\dot{q}$.

3. **Write potential energy $U(q)$.** For a spring $U = \frac{1}{2} k x^2$; for gravity $U = m g y_{cm}(q)$; for a gas piston $U = \int P dV$; for an arbitrary potential $U(q)$ is given or deduced.
4. **Total energy $E = K + U = \text{constant}$.** Differentiate with respect to time: $dE/dt = 0$.
5. **Factor out $\ddot{q}$.** The remaining bracket is the equation of motion. It will have the form $\ddot{q} + \omega^2 q = 0$ (for small q). Read off ω .

Potential-energy-curve test (finds ω without knowing the force law). If a particle of mass m moves in a potential $U(x)$ with a stable equilibrium at $x = 0$, then for small x :

$$U(x) \approx U(0) + \frac{1}{2} U''(0) x^2,$$

and the motion is SHM with

$$\omega = \sqrt{\frac{U''(0)}{m}}.$$

No force calculation needed — just expand U to second order.

Units check. U has units $J = \text{kg m}^2 \text{s}^{-2}$. U'' has units kg s^{-2} . U''/m has units s^{-2} . $\sqrt{(U''/m)}$ has units $\text{s}^{-1} = \text{rad s}^{-1}$. Correct.

Standard shortcuts

System	Coordinate	K	U	ω
Spring-mass (horizontal)	x	$\frac{1}{2} m \dot{x}^2$	$\frac{1}{2} k x^2$	$\sqrt{(k/m)}$
Spring-mass (vertical)	y from equilibrium	$\frac{1}{2} m \dot{y}^2$	$\frac{1}{2} k y^2$	$\sqrt{(k/m)}$
Simple pendulum	θ	$\frac{1}{2} m L^2 \dot{\theta}^2$	$\frac{1}{2} m g L \theta^2$	$\sqrt{(g/L)}$
Physical pendulum	θ	$\frac{1}{2} I \dot{\theta}^2$	$\frac{1}{2} m g d \theta^2$	$\sqrt{(m g d / I)}$
Torsional pendulum	θ	$\frac{1}{2} I \dot{\theta}^2$	$\frac{1}{2} \kappa \theta^2$	$\sqrt{(\kappa/I)}$
Liquid in U-tube	y	$\frac{1}{2} (\rho A L) \dot{y}^2$	$\frac{1}{2} (2 \rho A g) y^2$	$\sqrt{(2g/L)}$
Floating body (area A)	y	$\frac{1}{2} m \dot{y}^2$	$\frac{1}{2} (\rho A g) y^2$	$\sqrt{(\rho A g / m)}$
Piston in gas (isothermal)	x	$\frac{1}{2} m \dot{x}^2$	$\frac{1}{2} (P_0 A^2 / V_0) x^2$	$\sqrt{(P_0 A^2 / m V_0)}$
Piston in gas (adiabatic)	x	$\frac{1}{2} m \dot{x}^2$	$\frac{1}{2} (\gamma P_0 A^2 / V_0) x^2$	$\sqrt{(\gamma P_0 A^2 / m V_0)}$
Particle in $U(x)$	x	$\frac{1}{2} m \dot{x}^2$	$U(x)$	$\sqrt{(U''(0)/m)}$

- ρ — density (kg m^{-3}); A — area (m^2); L — length (m); V_0 — equilibrium volume (m^3); P_0 — equilibrium pressure (Pa); γ — adiabatic index (dimensionless); κ — torsional constant (N m rad^{-1}).

Example A.1 — [JEE Main Level] Energy Method on a Rolling Cylinder

A solid cylinder of mass M and radius R rolls without slipping on a horizontal surface, attached to a spring of constant k fixed to a wall. The spring is horizontal and attached to the cylinder's axle. Find the period of small oscillations.

Step 1 — Coordinate. Let x be the displacement of the centre of mass from equilibrium (spring relaxed at $x = 0$).

Step 2 — Kinetic energy. Rolling without slipping $\Rightarrow v_{\text{cm}} = \dot{x}$, $\omega_{\text{cyl}} = v_{\text{cm}} / R = \dot{x} / R$.
Moment of inertia of solid cylinder about its axis: $I = \frac{1}{2} M R^2$.

$$K = \frac{1}{2} M \dot{x}^2 + \frac{1}{2} I \omega_{\text{cyl}}^2 = \frac{1}{2} M \dot{x}^2 + \frac{1}{2} \left(\frac{1}{2} M R^2 \right) \left(\frac{\dot{x}}{R} \right)^2 = \frac{1}{2} M \dot{x}^2 + \frac{1}{4} M \dot{x}^2 = \frac{3}{4} M \dot{x}^2.$$

Step 3 — Potential energy. Only the spring stores energy (gravity constant).

$$U = \frac{1}{2} k x^2.$$

Step 4 — Total energy and differentiation.

$$E = \frac{3}{4} M \dot{x}^2 + \frac{1}{2} k x^2 = \text{constant}.$$

$$\frac{dE}{dt} = \frac{3}{2} M \dot{x} \ddot{x} + k x \dot{x} = 0.$$

Step 5 — Factor out $\dot{x}$ and read ω .

$$\dot{x} \left(\frac{3}{2} M \ddot{x} + k x \right) = 0 \Rightarrow \ddot{x} + \frac{2k}{3M} x = 0.$$

Hence

$$\omega = \sqrt{\frac{2k}{3M}}, \quad T = 2\pi \sqrt{\frac{3M}{2k}}.$$

Check. If the cylinder were sliding frictionless (no rotation), $K = \frac{1}{2} M \dot{x}^2$ and $\omega = \sqrt{k/M}$. Rolling adds rotational inertia, so the system is “heavier” dynamically — period increases by factor $\sqrt{3/2} \approx 1.22$. Correct direction.

Example A.2 — [JEE Advanced Level] Potential-Energy-Curve Method

A particle of mass m moves in a potential $U(x) = U_0 \left(\frac{x^2}{a^2} - \frac{x^4}{b^4} \right)$ with $U_0 > 0$, $a > 0$, $b > 0$. Show that for small oscillations about the stable equilibrium the angular frequency is $\omega = \sqrt{2U_0/ma^2}$, independent of b .

Step 1 — Find equilibrium. $U'(x) = U_0 \left(\frac{2x}{a^2} - \frac{4x^3}{b^4} \right) = 0$. Roots: $x = 0$ and $x = \pm \frac{b^2}{a\sqrt{2}}$.

Step 2 — Test stability via second derivative.

$$U''(x) = U_0 \left(\frac{2}{a^2} - \frac{12x^2}{b^4} \right).$$

At $x = 0$: $U''(0) = \frac{2U_0}{a^2} > 0 \Rightarrow$ stable minimum. At $x = \pm \frac{b^2}{a\sqrt{2}}$: $U'' = \frac{2U_0}{a^2} - \frac{12}{b^4} \frac{b^4}{2a^2} = -\frac{4U_0}{a^2} < 0 \Rightarrow$ unstable maxima.

Step 3 — Read off ω .

$$\omega = \sqrt{\frac{U''(0)}{m}} = \sqrt{\frac{2U_0}{ma^2}}.$$

The parameter b controls the barrier height and width but not the curvature at the bottom. This is a standard JEE Advanced “distractor parameter” trap.

Check. Units: U_0 in J, a in m, m in kg. $U_0/(ma^2)$ has units $(\text{kg m}^2 \text{ s}^{-2}) / (\text{kg m}^2) = \text{s}^{-2}$. $\sqrt{}$ gives s^{-1} . Correct.

Example A.3 — [JEE Advanced Level] Piston Compressing Gas (Adiabatic)

A vertical cylinder of cross-section A contains an ideal gas at pressure P_0 and volume V_0 . A piston of mass m fits tightly and slides without friction. The piston is slightly displaced downward and released. Assuming the gas undergoes adiabatic compression/expansion ($PV^\gamma = \text{constant}$), find the period of oscillation.

Step 1 — Coordinate. Let y be the downward displacement of the piston from equilibrium. Volume $V = V_0 - Ay$.

Step 2 — Pressure as function of y . Adiabatic: $PV^\gamma = P_0V_0^\gamma$.

$$P = P_0 \left(\frac{V_0}{V} \right)^\gamma = P_0 \left(1 - \frac{Ay}{V_0} \right)^{-\gamma}.$$

Step 3 — Net force on piston (upward positive). At equilibrium: $P_0A = mg$ (cancels with weight). Restoring force for displacement y :

$$F = -(P - P_0)A = -P_0A \left[\left(1 - \frac{Ay}{V_0} \right)^{-\gamma} - 1 \right].$$

Step 4 — Small- y expansion. For $z \ll 1$, $(1 - z)^{-\gamma} \approx 1 + \gamma z$.

$$\left(1 - \frac{Ay}{V_0} \right)^{-\gamma} - 1 \approx \gamma \frac{Ay}{V_0}.$$

$$F \approx -P_0A \cdot \gamma \frac{Ay}{V_0} = -\frac{\gamma P_0 A^2}{V_0} y.$$

Step 5 — Equation of motion and ω .

$$m\ddot{y} = -\frac{\gamma P_0 A^2}{V_0} y \Rightarrow \ddot{y} + \frac{\gamma P_0 A^2}{mV_0} y = 0.$$

$$\omega = \sqrt{\frac{\gamma P_0 A^2}{mV_0}}, \quad T = 2\pi \sqrt{\frac{mV_0}{\gamma P_0 A^2}}.$$

Alternative energy method. Gas energy change: $dU = -PdV$. For adiabatic, $U = \frac{PV}{\gamma-1}$. Total energy: $E = \frac{1}{2}m\dot{y}^2 + \frac{PV}{\gamma-1} + mgy$. Using $PV^\gamma = \text{const}$, expand to second order in y — same ω emerges.

Check. If $\gamma = 1$ (isothermal), $T = 2\pi\sqrt{mV_0/(P_0A^2)}$. Adiabatic is stiffer ($\gamma > 1$) $\Rightarrow$ shorter period. Correct.

Practice — Addendum A

1. A uniform rod of mass M and length L is pivoted at one end. A spring of constant k is attached to the free end, horizontal when the rod is vertical. Find the period of small oscillations. (Use energy method.)
2. A particle of mass m moves in the potential $U(x) = \frac{1}{2}kx^2 + \frac{1}{4}\alpha x^4$ ($\alpha > 0$). Find ω for small oscillations about $x = 0$.
3. A solid sphere of mass M and radius R rolls without slipping in a spherical bowl of radius $10R$. Find the period of small oscillations.
4. A U-tube has unequal cross-sections A_1 and A_2 . Total liquid length L , density ρ . Find the period of oscillation of the liquid column.
5. A cylindrical block of wood (cross-section A , mass m , height H) floats in a liquid of density ρ . It is pushed down slightly. Find the period. (Same as chapter's floating body — do it by energy to verify.)
6. A piston of mass m separates two chambers of a cylinder. Left chamber: gas at P_1 , V_1 . Right chamber: gas at P_2 , V_2 . Both gases ideal, adiabatic index γ . Piston displaced slightly. Find ω .

Answers: 1. Rod about its pivoted end: $I = \frac{1}{3}ML^2$. Spring displacement at the free end: $x = L\theta$, so $K = \frac{1}{2}I\dot{\theta}^2 = \frac{1}{2}(\frac{1}{3}ML^2)(\dot{x}/L)^2 = \frac{1}{6}M\dot{x}^2$, and $U = \frac{1}{2}kx^2$. Energy conservation ($\dot{E} = 0$) gives $\frac{1}{3}M\dot{x}\ddot{x} + kx\dot{x} = 0 \Rightarrow \ddot{x} = -\frac{3k}{M}x \Rightarrow \omega^2 = \frac{3k}{M}$.

$$T = 2\pi\sqrt{\frac{M}{3k}}$$

2. $U''(0) = k \Rightarrow \omega = \sqrt{k/m}$. (The αx^4 term contributes only at $O(x^2)$ in force, so it does not affect small-oscillation frequency.) 3. Bowl radius $10R$, sphere radius R . Centre moves on circle of radius $9R$. $K = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}I\omega^2$. No slip: $\omega = \dot{x}/R$. $I = \frac{2}{5}MR^2$. $K = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}(\frac{2}{5}MR^2)(\dot{x}/R)^2 = \frac{1}{2}M\dot{x}^2 + \frac{1}{5}M\dot{x}^2 = \frac{7}{10}M\dot{x}^2$. $U = M g (9R)(1 - \cos \theta) \approx \frac{1}{2}M g (9R)\theta^2$. $x = 9R\theta \Rightarrow U = \frac{1}{2}M g x^2 / (9R)$. $\omega^2 = (M g / 9R) / (7/10 M) = 10g / 63R$. $T = 2\pi\sqrt{\frac{63R}{10g}}$. 4. Let y be the displacement of the liquid surface in arm 1 (area A_1); volume conservation gives $A_1y = A_2y_2$, i.e. $y_2 = (A_1/A_2)y$ for the corresponding drop in arm 2. The height difference between the two surfaces is $y + y_2 = y(1 + A_1/A_2)$, so the restoring force on the column is $F = -\rho g A_1(1 + A_1/A_2)y$. With equilibrium liquid lengths L_1, L_2 in the two arms, the moving mass is $\rho(A_1L_1 + A_2L_2)$, giving

$$\omega^2 = \frac{gA_1(A_1 + A_2)/A_2}{A_1L_1 + A_2L_2} \Rightarrow T = 2\pi\sqrt{\frac{A_1L_1 + A_2L_2}{gA_1(A_1 + A_2)/A_2}}$$

If the tube has a single bore (so $A_1 = A_2$, and $L_1 + L_2 = L$), this reduces to the familiar $T = 2\pi\sqrt{L/2g}$. 5. $T = 2\pi\sqrt{\frac{m}{\rho Ag}}$. (Energy: $K = \frac{1}{2}m\dot{y}^2$. $U =$ work against buoyancy $= \frac{1}{2}(\rho A g)y^2$. $\omega^2 = \rho A g / m$.) 6. Left gas: $V_1 = V_{10} - A x$. Right gas: $V_2 = V_{20} + A x$. Net force $F = -(P_1 - P_2)A$. $P_1 = P_{10}(V_{10}/V_1)^\gamma \approx P_{10}(1 + \gamma A x/V_{10})$. $P_2 = P_{20}(V_{20}/V_2)^\gamma \approx P_{20}(1 -$

$\gamma A x/V_{20}$). Equilibrium: $P_{10} = P_{20} = P_0$. $F = -P_0 A [(1 + \gamma A x/V_{10}) - (1 - \gamma A x/V_{20})] = -\gamma P_0 A^2 (1/V_{10} + 1/V_{20}) x$. $T = 2\pi \sqrt{\frac{m V_{10} V_{20}}{\gamma P_0 A^2 (V_{10} + V_{20})}}$.

Addendum B — Spring Systems Beyond Series and Parallel

Subtopic 03 gives series ($1/k_{\text{eff}} = \Sigma 1/k_i$) and parallel ($k_{\text{eff}} = \Sigma k_i$). JEE regularly tests three extensions that are not “series or parallel” but are solved by the same force/energy logic: (i) a massive spring, (ii) a spring cut into pieces and rejoined in a new geometry, (iii) two springs with a mass between them (the “middle-mass” problem).

Method

B1 — Effective mass of a uniform spring. A spring of mass m_s and constant k has its mass distributed along its length. For a mass M attached to one end (other end fixed), the kinetic energy of the spring element at distance x from the fixed end is $dK = \frac{1}{2} (m_s/L) (v(x))^2 dx$. For a spring stretched uniformly, $v(x) = (x/L) v_{\text{end}}$. Integrate:

$$K_{\text{spring}} = \int_0^L \frac{1}{2} \frac{m_s}{L} \left(\frac{x}{L} v \right)^2 dx = \frac{1}{2} \left(\frac{m_s}{3} \right) v^2.$$

The spring behaves as if an extra mass $m_s/3$ were added to the load.

$$\omega = \sqrt{\frac{k}{M + m_s/3}}, \quad T = 2\pi \sqrt{\frac{M + m_s/3}{k}}.$$

B2 — Cutting and rejoining springs. - A spring of constant k , length L , cut into n equal pieces: each piece has constant $n k$ (stiffer). - Pieces rejoined in series: $k_{\text{eff}} = k$ (back to original). - Pieces rejoined in parallel: $k_{\text{eff}} = n^2 k$. - One piece used alone with same mass: $T_{\text{new}} / T_{\text{original}} = 1/\sqrt{n}$.

B3 — Two springs with a mass between them (middle-mass system). Mass M connected to left wall by spring k_1 , to right wall by spring k_2 . Displace mass by x . Left spring: $F_1 = -k_1 x$. Right spring: $F_2 = -k_2 x$. Total $F = -(k_1 + k_2) x \Rightarrow k_{\text{eff}} = k_1 + k_2$ (parallel addition). This is the *only* case where two springs on opposite sides add like parallel. **Crucial distinction:** If the mass is between two springs but the *other ends* of the springs are connected to each other (not to walls), the system is different — see Example B.3.

B4 — Spring with masses at both ends (free-free system). Two masses m_1, m_2 connected by a spring k , free in space. Centre of mass is inertial. Relative coordinate $x = x_2 - x_1 - L_0$. Reduced mass $\mu = m_1 m_2 / (m_1 + m_2)$. Equation: $\mu \ddot{x} + k x = 0 \Rightarrow \omega = \sqrt{k/\mu}$. Period $T = 2\pi \sqrt{\mu/k}$. This is the “internal oscillation” mode; the CM moves uniformly ($\omega = 0$ mode).

Example B.1 — [JEE Main Level] Massive Spring

A spring of mass 0.3 kg and force constant 200 N m⁻¹ has a block of mass 0.5 kg attached to one end, the other end fixed. Find the period of vertical oscillation. (Take $g = 9.8 \text{ m s}^{-2}$; note: g does not affect period.)

Solution. Effective mass = $M + m_s/3 = 0.5 + 0.3/3 = 0.5 + 0.1 = 0.6 \text{ kg}$.

$$T = 2\pi \sqrt{\frac{M_{\text{eff}}}{k}} = 2\pi \sqrt{\frac{0.6}{200}} = 2\pi \sqrt{0.003} = 2\pi \times 0.05477 \approx 0.344 \text{ s}.$$

Check. Without spring mass: $T = 2\pi \sqrt{(0.5/200)} = 2\pi \sqrt{0.0025} = 2\pi \times 0.05 = 0.314 \text{ s}$. Spring mass increases period by factor $\sqrt{(0.6/0.5)} = \sqrt{1.2} \approx 1.095$. Correct.

Example B.2 — [JEE Main Level] Cut Spring Rejoined

A spring of constant k is cut into three equal pieces. Two pieces are connected in parallel and this combination is connected in series with the third piece. A mass m is attached. Find the period.

Solution. Each piece: $k' = 3k$. Parallel pair: $k_{\text{parallel}} = 3k + 3k = 6k$. Series with third: $1/k_{\text{eff}} = 1/(6k) + 1/(3k) = 1/(6k) + 2/(6k) = 3/(6k) = 1/(2k)$. So $k_{\text{eff}} = 2k$.

$$T = 2\pi \sqrt{\frac{m}{2k}} = \frac{2\pi}{\sqrt{2}} \sqrt{\frac{m}{k}} = \frac{T_0}{\sqrt{2}},$$

where $T_0 = 2\pi \sqrt{(m/k)}$ is the period of the original uncut spring with the same mass.

Check. Original k . New $k_{\text{eff}} = 2k > k \Rightarrow$ stiffer $\Rightarrow$ shorter period. $T_{\text{new}} = T_0/\sqrt{2} \approx 0.707 T_0$. Correct.

Example B.3 — [JEE Advanced Level] Mass on a Stretched Wire

A wire of length $2L$, cross-sectional area A , Young's modulus Y , is stretched between two fixed supports with an initial tension T_0 . A mass m is attached at its midpoint. The mass is displaced a small distance y transversely and released. Find the period of small oscillations.

Solution. Each half of the wire has length L , tension T_0 (approximately constant for small y). When mass is displaced by y , each half makes angle $\theta \approx y/L$ with horizontal. Vertical component of tension from one half = $T_0 \sin \theta \approx T_0 (y/L)$. Total restoring force = $-2 T_0 (y/L)$. Equation: $m \ddot{y} = - (2T_0/L) y$.

$$\omega = \sqrt{\frac{2T_0}{mL}}, \quad T = 2\pi \sqrt{\frac{mL}{2T_0}}.$$

Using Young's modulus: $T_0 = Y A (\Delta L/L)$ where ΔL is the extension of each half? Total extension $2\Delta L$. If total length $2L$ is the stretched length, and natural length is $2L_0$, then $T_0 = Y A (L - L_0)/L_0$. But if T_0 is given directly, use above.

Check. Units: T_0 (N), m (kg), L (m). $T_0/(mL)$ has units $(\text{kg m s}^{-2})/(\text{kg m}) = \text{s}^{-2}$. $\sqrt{\quad}$ gives s^{-1} . Correct.

Practice — Addendum B

1. A spring of mass 0.2 kg and constant 100 N m^{-1} has a 0.8 kg block attached. Find the period. By what percentage does the spring's mass increase the period compared to a massless spring?

2. A spring of constant k is cut into two pieces of lengths $L/3$ and $2L/3$. The pieces are connected in parallel. Find k_{eff} .
3. Two masses $m_1 = 2 \text{ kg}$ and $m_2 = 3 \text{ kg}$ are connected by a spring of constant 120 N m^{-1} on a frictionless table. Find the period of oscillation of the distance between them.
4. A mass m is attached to two identical springs of constant k , each fixed at the other end to walls on opposite sides (the standard “middle-mass” system). Find the period. Now suppose the right wall is removed and the right spring’s free end is attached to the left spring’s free end (so the two springs are in series, with the mass at the junction). Find the new period. Compare.
5. A uniform spring of mass m_s and constant k hangs vertically with a mass M attached at the bottom. Find the period of vertical oscillation. (Hint: the effective mass correction is still $m_s/3$.)
6. A spring of constant k has a mass m attached at its midpoint, the ends fixed. The spring is initially unstretched. The mass is displaced along the spring’s axis (longitudinal). Find the period.

Answers: 1. $T = 2\pi \sqrt{((0.8 + 0.2/3)/100)} = 2\pi \sqrt{(0.8667/100)} = 2\pi \times 0.0931 \approx 0.585 \text{ s}$. Massless spring $T_0 = 2\pi \sqrt{(0.8/100)} = 0.561 \text{ s}$. Increase $= (0.585-0.561)/0.561 \approx 4.3\%$. 2. Piece 1: length $L/3 \Rightarrow k_1 = 3k$. Piece 2: length $2L/3 \Rightarrow k_2 = 1.5k$. Parallel: $k_{\text{eff}} = 3k + 1.5k = 4.5k$. 3. Reduced mass $\mu = (2 \times 3)/(2+3) = 6/5 = 1.2 \text{ kg}$. $T = 2\pi \sqrt{(1.2/120)} = 2\pi \sqrt{0.01} = 2\pi \times 0.1 = 0.2\pi \approx 0.628 \text{ s}$. 4. Standard middle-mass (parallel): $k_{\text{eff}} = 2k$, $T_1 = 2\pi \sqrt{(m/2k)}$. Series junction: $k_{\text{eff}} = k/2$ (two k in series), $T_2 = 2\pi \sqrt{(m/(k/2))} = 2\pi \sqrt{(2m/k)} = 2 T_1$. The series combination is softer. 5. $T = 2\pi \sqrt{((M + m_s/3)/k)}$. Gravity shifts equilibrium but not period. The effective mass of a vertical uniform spring is still $m_s/3$. 6. Mass at midpoint, longitudinal. Each half acts as a spring of constant $2k$. The mass is between two springs in parallel (both attached to mass, other ends fixed). $k_{\text{eff}} = 2k + 2k = 4k$. $T = 2\pi \sqrt{(m/4k)} = \pi \sqrt{(m/k)}$.

Addendum C — Phase, Time, and Fraction-of-Period Calculations

Subtopic 01 Example 4 and Subtopic 02 Example 5 show the idea: map position conditions to phase intervals on the reference circle. JEE Main and Advanced ask for exact times to go from x_1 to x_2 , fraction of period spent with $|x| > A/2$, time when speed is half the maximum, etc. The method is always: write $x = A \sin(\omega t + \phi)$ or $\cos$, invert to get phase, subtract phases, divide by ω .

Method

Phase-stopwatch procedure.

1. **Choose the sine/cosine form** that makes the initial condition simple. If $t=0$ at $x=A$, use $x = A \cos \omega t$. If $t=0$ at $x=0$ moving $+$, use $x = A \sin \omega t$.
2. **Convert position conditions to phase angles.** For $x = A \sin \theta$ ($\theta = \omega t + \phi$), the condition $x = x_0$ gives $\theta = \sin^{-1}(x_0/A)$ or $\pi - \sin^{-1}(x_0/A)$, depending on the quadrant

(direction of motion).

3. **Determine the correct branch** using the velocity sign: $v = A\omega \cos \theta$. $v > 0 \Rightarrow \cos \theta > 0 \Rightarrow \theta$ in Q1 or Q4. $v < 0 \Rightarrow \theta$ in Q2 or Q3.
4. **Compute phase difference $\Delta\theta$** between the two states along the actual direction of motion.
5. **Time = $\Delta\theta / \omega$** . Fraction of period = $\Delta\theta / 2\pi$.

Standard results to memorise (for $x = A \sin \omega t$, starting at mean position moving +):

From $\rightarrow$ To	Phase span	Fraction of T	Time
$0 \rightarrow A/2$	$\pi/6$	$1/12$	$T/12$
$0 \rightarrow A/\sqrt{2}$	$\pi/4$	$1/8$	$T/8$
$0 \rightarrow A$	$\pi/2$	$1/4$	$T/4$
$A/2 \rightarrow A$	$\pi/2 - \pi/6 = \pi/3$	$1/6$	$T/6$
$A/\sqrt{2} \rightarrow A$	$\pi/2 - \pi/4 = \pi/4$	$1/8$	$T/8$
$-A/2 \rightarrow A/2$	$2\pi/3$	$1/3$	$T/3$
$ x < A/2$	$ \sin \theta < 1/2 \Rightarrow \theta \in [0, \pi/6) \cup (5\pi/6, 7\pi/6) \cup (11\pi/6, 2\pi]$; total span $2\pi/3$	$1/3$	$T/3$
$ x > A/2$	$2\pi - 2\pi/3 = 4\pi/3$	$2/3$	$2T/3$

Speed-based conditions. $v = v_{\max}/2 \Rightarrow A\omega \cos \theta = A\omega/2 \Rightarrow \cos \theta = 1/2 \Rightarrow \theta = \pi/3$ or $5\pi/3$. Use symmetry.

Example C.1 — [JEE Main Level] Time from $x = A/2$ to $x = A$

A particle starts from the mean position ($x=0$) at $t=0$, moving in the positive direction, with SHM of amplitude A and period T . Find the time taken to go from $x = A/2$ to $x = A$.

Solution. Use $x = A \sin \omega t$ ($\phi = 0$, $\omega = 2\pi/T$). At $x = A/2$: $\sin \omega t_1 = 1/2 \Rightarrow \omega t_1 = \pi/6$ (since moving +, $v > 0$, $\cos > 0 \Rightarrow$ Q1). At $x = A$: $\sin \omega t_2 = 1 \Rightarrow \omega t_2 = \pi/2$. $\Delta\theta = \pi/2 - \pi/6 = \pi/3$.

$$t = \frac{\Delta\theta}{\omega} = \frac{\pi/3}{2\pi/T} = \frac{T}{6}.$$

Check. Quarter period $T/4$ from 0 to A . $T/12$ from 0 to $A/2$. Difference = $T/4 - T/12 = T/6$. Correct.

Example C.2 — [JEE Advanced Level] Fraction of Period with Speed $> v_{\max}/\sqrt{2}$

A particle executes SHM with amplitude A and angular frequency ω . Find the fraction of one period during which its speed exceeds $v_{\max}/\sqrt{2}$.

Solution. $v = A\omega \cos \omega t$ (starting at $x=A$, $t=0$). $v_{\max} = A\omega$. Condition: $|v| > v_{\max}/\sqrt{2} \Rightarrow |\cos \omega t| > 1/\sqrt{2}$. On $[0, 2\pi]$, $|\cos \theta| > 1/\sqrt{2}$ when $\theta \in [0, \pi/4) \cup (3\pi/4, 5\pi/4) \cup (7\pi/4, 2\pi]$. Total angular span = $4 \times (\pi/4) = \pi$. Fraction = $\pi / 2\pi = 1/2$.

Alternative using position. $|v| > v_{\max}/\sqrt{2} \Rightarrow |x| < A/\sqrt{2}$ (since $v^2 = \omega^2(A^2 - x^2)$). Fraction with $|x| < A/\sqrt{2}$ is also $1/2$ (by symmetry, $K > U$ when $|x| < A/\sqrt{2}$, and Subtopic 02 Example 5 gave fraction $1/2$ for $K > U$). **Check.** $K > U$ for half the period. Consistent.

Example C.3 — [JEE Advanced Level] Time from $x = -A/2$ to $x = +A/2$ Moving Positive

A particle in SHM (amplitude A , period T) is at $x = -A/2$ moving in the positive direction at $t=0$. Find the time when it first reaches $x = +A/2$.

Solution. Choose $x = A \sin(\omega t + \phi)$. At $t=0$: $x = -A/2 \Rightarrow \sin \phi = -1/2$. Moving $+$ $\Rightarrow v = A\omega \cos \phi > 0 \Rightarrow \cos \phi > 0$. $\sin \phi = -1/2$, $\cos \phi > 0 \Rightarrow \phi = -\pi/6$ (or $11\pi/6$). Target: $x = +A/2 \Rightarrow \sin(\omega t - \pi/6) = 1/2$, with $v > 0$ (still moving $+$). $\sin \alpha = 1/2$, $\cos \alpha > 0 \Rightarrow \alpha = \pi/6$. $\omega t - \pi/6 = \pi/6 \Rightarrow \omega t = \pi/3$.

$$t = \frac{\pi/3}{\omega} = \frac{\pi/3}{2\pi/T} = \frac{T}{6}.$$

Check. From $-A/2$ to 0 is $T/12$ (by symmetry with 0 to $A/2$). From 0 to $+A/2$ is another $T/12$. Total $T/6$. Correct.

Practice — Addendum C

1. A particle starts from $x = A$ at $t=0$. Find the time to reach $x = A/2$.
2. A particle in SHM (amplitude A , period T) has speed $v = v_{\max}/2$ at $t=0$ and is moving toward the mean position from the positive side. Find the time when it next reaches $x = 0$.
3. Find the fraction of a period during which the kinetic energy is greater than $3/4$ of the total energy.
4. A particle in SHM takes time t_1 to go from $x = 0$ to $x = A/2$, and time t_2 to go from $x = A/2$ to $x = A$. Find t_2/t_1 .
5. For SHM $x = A \cos(\omega t + \pi/3)$, find the first time after $t=0$ when the particle is at $x = -A/2$ and moving in the negative direction.
6. A particle executes SHM with amplitude 10 cm and period 2 s. What is the minimum time required for it to move between points 5 cm and 8 cm from the mean position on the same side?

Answers: 1. $x = A \cos \omega t$. $\cos \omega t = 1/2 \Rightarrow \omega t = \pi/3$. $t = (\pi/3)/(\omega) = T/6$. 2. Speed $v = v_{\max}/2$ at position x_0 satisfies $v^2 = \omega^2(A^2 - x_0^2)$, i.e. $(A\omega/2)^2 = \omega^2(A^2 - x_0^2) \Rightarrow x_0^2 = \frac{3}{4}A^2 \Rightarrow x_0 = \frac{\sqrt{3}}{2}A$ (taking the positive root, since $x > 0$ is given). Writing $x(t) = A \cos(\omega t + \phi)$ with $x(0) = x_0$ and $v(0) < 0$ (moving toward the mean position) requires $\cos \phi = \sqrt{3}/2$ and $\sin \phi > 0$, so $\phi = \pi/6$. The particle reaches $x = 0$ when $\cos(\omega t + \pi/6) = 0$, i.e. $\omega t + \pi/6 = \pi/2 \Rightarrow \omega t = \pi/3 \Rightarrow t = \frac{\pi/3}{\omega} = \frac{T}{6}$. 3. $K > 3/4 E \Rightarrow 1/2 m \omega^2(A^2 - x^2) > 3/4 \times 1/2 m \omega^2 A^2 \Rightarrow A^2 - x^2 > 3/4 A^2 \Rightarrow x^2 < 1/4 A^2 \Rightarrow |x| < A/2$. Fraction = $1/3$ (from Example C.2 table). 4. $t_1 = T/12$ (0 to $A/2$). $t_2 = T/6$ ($A/2$ to A). $t_2/t_1 = 2$. 5. $x = A \cos(\omega t + \pi/3) = -A/2 \Rightarrow \cos(\omega t + \pi/3) = -1/2$. Moving negative $\Rightarrow v = -A\omega \sin(\omega t + \pi/3) < 0 \Rightarrow \sin(\omega t + \pi/3) > 0$. $\cos \theta = -1/2$, $\sin \theta > 0 \Rightarrow \theta = 2\pi/3$. $\omega t + \pi/3 = 2\pi/3 \Rightarrow \omega t = \pi/3$. $t = T/6$. 6. $A = 10$ cm, $T = 2$ s, $\omega = \pi$ rad/s. Points: 5 cm and 8 cm on same side. Use $x = A \sin \omega t$ (start at 0). $x=5$: $\sin \omega t_1 = 0.5 \Rightarrow \omega t_1 = \pi/6$. $x=8$: $\sin \omega t_2 =$

$0.8 \Rightarrow \omega t_2 = \sin^{-1}(0.8) \approx 0.9273 \text{ rad}$. $\Delta\theta = 0.9273 - \pi/6 \approx 0.9273 - 0.5236 = 0.4037 \text{ rad}$. $t = \Delta\theta/\omega = 0.4037/\pi \approx 0.1285 \text{ s}$.

Addendum D — Superposition of Different Frequencies: Beats and Amplitude Modulation

Subtopic 05 covers superposition of *equal* frequencies (phasors, Lissajous). JEE Main explicitly asks for **beats**: two SHMs of slightly different frequencies $\omega_1 \approx \omega_2$ superpose to give a rapid oscillation at the average frequency modulated by a slow envelope at the difference frequency. The chapter does not derive the envelope or the beat frequency formula.

Method

Collinear superposition of two SHMs with different frequencies.

$$x_1 = A_1 \cos \omega_1 t, \quad x_2 = A_2 \cos \omega_2 t \quad (\omega_1 > \omega_2).$$

$$x = x_1 + x_2 = A_1 \cos \omega_1 t + A_2 \cos \omega_2 t.$$

Case 1: Equal amplitudes $A_1 = A_2 = A$. Use sum-to-product:

$$x = 2A \cos\left(\frac{\omega_1 - \omega_2}{2}t\right) \cos\left(\frac{\omega_1 + \omega_2}{2}t\right).$$

- **Carrier (fast):** angular frequency $\omega_{\text{avg}} = \frac{\omega_1 + \omega_2}{2} \approx \omega_1 \approx \omega_2$. - **Envelope (slow):** amplitude $A_{\text{env}}(t) = 2A |\cos(\frac{\Delta\omega}{2}t)|$, where $\Delta\omega = \omega_1 - \omega_2$. - **Beat angular frequency:** $\omega_{\text{beat}} = \Delta\omega = \omega_1 - \omega_2$. (The envelope goes from max to zero to max in half its period, so the *beat frequency* — number of loud-soft cycles per second — is $f_{\text{beat}} = \frac{\Delta\omega}{2\pi} = f_1 - f_2$.) - **Beat period:** $T_{\text{beat}} = \frac{2\pi}{\Delta\omega} = \frac{1}{|f_1 - f_2|}$.

Case 2: Unequal amplitudes $A_1 \neq A_2$. No complete cancellation. Write as:

$$x = A_1 \cos \omega_1 t + A_2 \cos \omega_2 t = A_{\text{env}}(t) \cos(\omega_{\text{avg}}t + \phi(t)).$$

The envelope amplitude varies between $|A_1 - A_2|$ and $A_1 + A_2$ with angular frequency $\Delta\omega$. Exact envelope: $A_{\text{env}}^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos(\Delta\omega t)$. This is the **amplitude modulation (AM)** formula. The “carrier” is at ω_{avg} , the “modulation” at $\Delta\omega$.

Perpendicular superposition (different frequencies) → Lissajous figures. $x = A \cos \omega_1 t$, $y = B \cos(\omega_2 t + \delta)$. If $\omega_1/\omega_2 = p/q$ (rational), the figure is closed and repeats after $T = qT_1 = pT_2$. Number of loops touching top/bottom = p. Number of loops touching left/right = q. Phase δ rotates the figure.

Example D.1 — [JEE Main Level] Beat Frequency and Envelope

Two tuning forks of frequencies 256 Hz and 260 Hz are sounded together. Find (a) the beat frequency, (b) the time interval between successive maxima of intensity, (c) the frequency of the resultant sound.

Solution. $f_1 = 256$ Hz, $f_2 = 260$ Hz. (a) Beat frequency $f_{\text{beat}} = |f_1 - f_2| = 4$ Hz. (b) Time between successive maxima = beat period $T_{\text{beat}} = 1/f_{\text{beat}} = 0.25$ s. (c) Resultant frequency (pitch heard) = average frequency = $(256 + 260)/2 = 258$ Hz.

Check. The envelope has frequency 4 Hz (4 beats per second). The carrier oscillates at 258 Hz. The intensity ($\propto \text{amplitude}^2$) peaks at 4 Hz. Correct.

Example D.2 — [JEE Advanced Level] Unequal Amplitude Beats — Modulation Index

Two collinear SHMs: $x_1 = 5 \cos 100\pi t$ cm, $x_2 = 3 \cos 104\pi t$ cm. Find (a) the maximum and minimum resultant amplitudes, (b) the beat frequency, (c) the modulation index $m = \frac{A_{\text{max}} - A_{\text{min}}}{A_{\text{max}} + A_{\text{min}}}$.

Solution. $\omega_1 = 100\pi$, $\omega_2 = 104\pi$. $\Delta\omega = 4\pi$ rad/s. $f_{\text{beat}} = \Delta\omega/2\pi = 2$ Hz. Average frequency = $(100\pi + 104\pi)/2 = 102\pi$ rad/s = 51 Hz. Amplitudes: $A_1 = 5$ cm, $A_2 = 3$ cm. (a) $A_{\text{max}} = A_1 + A_2 = 8$ cm. $A_{\text{min}} = |A_1 - A_2| = 2$ cm. (b) Beat frequency = 2 Hz (as above). (c) Modulation index $m = \frac{8-2}{8+2} = \frac{6}{10} = 0.6$.

Check. The envelope varies as $\sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \Delta\omega t}$. Max = $\sqrt{(25+9+30)} = \sqrt{64} = 8$. Min = $\sqrt{(25+9-30)} = \sqrt{4} = 2$. Correct.

Example D.3 — [JEE Advanced Level] Lissajous Figure Identification

A particle is subjected to two perpendicular SHMs: $x = 4 \cos 2\omega t$, $y = 4 \cos(3\omega t + \pi/2)$. Identify the path and its orientation.

Solution. Frequency ratio $\omega_x : \omega_y = 2 : 3$. Rational $\Rightarrow$ closed Lissajous figure. Number of loops along x-direction (horizontal tangencies) = frequency ratio numerator? Standard rule: if $x = A \cos(n_x \omega t)$, $y = B \cos(n_y \omega t + \delta)$, the figure has n_y lobes in the vertical direction and n_x lobes in the horizontal direction. Here $n_x = 2$, $n_y = 3$. Phase difference $\delta = \pi/2$. The figure is a closed curve with 3 vertical lobes and 2 horizontal lobes (a “figure-8” type but with 3 humps). Since $\delta = \pi/2$, the figure is symmetric about both axes. **Answer:** Lissajous figure with frequency ratio 2:3, phase difference 90° , symmetric about both axes, 3 vertical loops, 2 horizontal loops.

Check. For $\delta = 0$, it would be a symmetric figure with crossing at origin. For $\delta = \pi/2$, the crossings shift. The exact shape is not usually drawn in exams; identifying ratio and lobe count is the tested skill.

Practice — Addendum D

- Two sound waves of frequencies 440 Hz and 446 Hz interfere. Find the beat frequency and the beat period.
- Two SHMs $x_1 = 10 \cos 200\pi t$, $x_2 = 6 \cos 208\pi t$ (cm). Find the maximum and minimum resultant amplitudes and the modulation index.
- A particle has $x = 5 \cos 10t + 5 \cos 12t$ (cm). Write the resultant as a product of a slow envelope and a fast carrier. Find the period of the envelope and the period of the carrier.
- Lissajous figure from $x = A \cos 3\omega t$, $y = A \cos(5\omega t + \pi/2)$. How many times does the curve touch the top/bottom boundary? How many times the left/right boundary?
- Two waves $y_1 = A \sin(kx - \omega_1 t)$, $y_2 = A \sin(kx - \omega_2 t)$ travel in the same direction. Show

that the resultant is a travelling wave modulated by a moving envelope, and find the group velocity $v_g = d\omega/dk$ if $\omega_1 = \omega(k)$, $\omega_2 = \omega(k + \Delta k)$.

Answers: 1. $f_{\text{beat}} = 6 \text{ Hz}$, $T_{\text{beat}} = 1/6 \text{ s} \approx 0.167 \text{ s}$. 2. $A_{\text{max}} = 16 \text{ cm}$, $A_{\text{min}} = 4 \text{ cm}$, $m = (16 - 4)/(16 + 4) = 12/20 = 0.6$. 3. $x = 10 \cos(\pi t) \cos(206\pi t) \text{ cm}$. Envelope period $T_{\text{env}} = 2\pi/\pi = 2 \text{ s}$. Carrier period $T_c = 2\pi/(206\pi) = 1/103 \text{ s} \approx 0.0097 \text{ s}$. 4. Ratio 3:5. Touches top/bottom (y-extremes) 5 times. Touches left/right (x-extremes) 3 times. 5. $y = 2A \cos\left(\frac{\Delta\omega}{2}t - \frac{\Delta k}{2}x\right) \sin(\omega_{\text{avg}}t - k_{\text{avg}}x)$. Envelope moves with speed $v_g = \Delta\omega/\Delta k \rightarrow d\omega/dk$.

Addendum E — Damped Oscillations: Logarithmic Decrement, Resonance Nuances, and Power Absorption

Subtopic 04 gives the underdamped solution, the amplitude resonance formula, and Q factor. JEE Advanced distinguishes **amplitude resonance** (max displacement) from **velocity resonance** (max speed, max kinetic energy, max power absorption), asks for the **logarithmic decrement** $\delta = \ln(A_n/A_{n+1})$, and uses the **power vs frequency** curve to define sharpness. This addendum derives those three pieces rigorously.

Method

E1 — Logarithmic Decrement. For underdamped: $x = A_0 e^{-bt/2m} \cos(\omega' t + \phi)$. Amplitude at successive maxima (one period T' apart): $A_n = A_0 e^{-bt_n/2m}$, $t_{n+1} = t_n + T'$.

$$\delta = \ln\left(\frac{A_n}{A_{n+1}}\right) = \ln\left(e^{bT'/2m}\right) = \frac{bT'}{2m}.$$

For light damping $T' \approx T_0 = 2\pi/\omega_0$, so $\delta \approx \frac{b}{m} \frac{\pi}{\omega_0} = \frac{\pi}{Q}$. **Use:** measure δ from experimental amplitudes, find b/m or Q .

E2 — Velocity Resonance vs Amplitude Resonance. Displacement amplitude: $A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (b\omega/m)^2}}$. Velocity amplitude: $v_{\text{max}}(\omega) = \omega A(\omega) = \frac{F_0\omega/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (b\omega/m)^2}}$. - **Amplitude resonance:** maximize $A(\omega) \Rightarrow \omega_r^2 = \omega_0^2 - \frac{b^2}{2m^2}$. - **Velocity resonance:** maximize $v_{\text{max}}(\omega) \Rightarrow$ derivative gives $\omega_v = \omega_0$ exactly (for any damping). - **Power absorption:** $P_{\text{avg}} = \frac{1}{2} b v_{\text{max}}^2 = \frac{1}{2} \frac{b F_0^2 \omega^2 / m^2}{(\omega_0^2 - \omega^2)^2 + (b\omega/m)^2}$. Maximum power at $\omega = \omega_0$ (velocity resonance). The power curve has FWHM (full width at half maximum) $\Delta\omega = b/m = \omega_0/Q$. Quality factor $Q = \frac{\omega_0}{\Delta\omega}$.

E3 — Phase Lag at Resonance. Phase δ of displacement relative to force: $\tan \delta = \frac{b\omega/m}{\omega_0^2 - \omega^2}$. At $\omega = \omega_0$: $\tan \delta \rightarrow \infty \Rightarrow \delta = \pi/2$. At $\omega = \omega_r$ (amplitude resonance): $\omega_0^2 - \omega_r^2 = b^2/2m^2$, so $\tan \delta_r = \frac{b\omega_r/m}{b^2/2m^2} = \frac{2m\omega_r}{b} = 2Q \frac{\omega_r}{\omega_0} \approx 2Q$.

Example E.1 — [JEE Advanced Level] Logarithmic Decrement from Amplitude Ratio

The amplitude of a damped oscillator falls to 1/3 of its initial value after 10 complete oscillations. The period of oscillation is 0.5 s. Find (a) the logarithmic decrement, (b) the damping constant b/m , (c) the quality factor Q .

Solution. (a) After 10 oscillations, $A_{10}/A_0 = 1/3$. $\delta = \frac{1}{10} \ln(A_0/A_{10}) = \frac{1}{10} \ln 3 = \frac{1.0986}{10} = 0.10986$. (b) $\delta = \frac{bT'}{2m} \approx \frac{bT_0}{2m}$ (light damping). $b/m = \frac{2\delta}{T_0} = \frac{2 \times 0.10986}{0.5} = 0.4394 \text{ s}^{-1}$. (c) $Q = \frac{\pi}{\delta} = \frac{\pi}{0.10986} \approx 28.6$. Or $Q = \frac{m\omega_0}{b} = \frac{2\pi m}{bT_0} = \frac{2\pi}{bT_0/m} = \frac{2\pi}{0.4394 \times 0.5} = \frac{2\pi}{0.2197} \approx 28.6$. Consistent.

Check. $\delta = 0.11$ means amplitude drops by $\sim 10\%$ per cycle. 10 cycles $\Rightarrow$ factor $e^{-1.1} \approx 0.33 \approx 1/3$. Correct.

Example E.2 — [JEE Advanced Level] Velocity Resonance and Power Absorption

A forced oscillator has $\omega_0 = 100 \text{ rad/s}$, $b/m = 2 \text{ s}^{-1}$, $F_0/m = 10 \text{ m/s}^2$. Find (a) the amplitude resonance frequency, (b) the velocity resonance frequency, (c) the maximum displacement amplitude, (d) the maximum velocity amplitude, (e) the maximum average power absorbed, (f) the FWHM of the power curve.

Solution. (a) $\omega_r = \sqrt{\omega_0^2 - \frac{b^2}{2m^2}} = \sqrt{10000 - \frac{4}{2}} = \sqrt{9998} \approx 99.99 \text{ rad/s}$. (Practically 100). (b) $\omega_v = \omega_0 = 100 \text{ rad/s}$ exactly. (c) $A_{\max} = A(\omega_r) \approx \frac{F_0/m}{b\omega_0/m} = \frac{10}{2 \times 100} = 0.05 \text{ m} = 5 \text{ cm}$. (Using light damping approx: $A_{\text{res}} \approx F_0/(b\omega_0)$.) (d) $v_{\max} = \omega_0 A(\omega_0) = \omega_0 \frac{F_0/m}{b\omega_0/m} = \frac{F_0}{b} = \frac{F_0/m}{b/m} = \frac{10}{2} = 5 \text{ m/s}$. (e) $P_{\max} = \frac{1}{2} b v_{\max}^2 = \frac{1}{2} (2m)(5)^2 = 25m \text{ Watts}$. (Depends on m which is not given numerically. Express in terms of given: $P_{\max} = \frac{F_0^2}{2b} = \frac{(10m)^2}{2(2m)} = \frac{100m^2}{4m} = 25m \text{ W}$.) (f) FWHM $\Delta\omega = b/m = 2 \text{ rad/s}$. $Q = \omega_0/\Delta\omega = 50$.

Check. $Q = m\omega_0/b = 100/2 = 50$. $\delta = \pi/Q \approx 0.0628$. Amplitude drops by $\sim 6\%$ per cycle. Light damping holds.

Example E.3 — [JEE Advanced Level] Damped Oscillator Energy Decay — Exact vs Approximate

A lightly damped oscillator has $Q = 100$. (a) By what factor does the energy decay in one period? (b) How many periods for energy to fall to 1% of initial? (c) What is the fractional frequency shift $\Delta\omega/\omega_0$?

Solution. (a) Energy $E \propto A^2 \propto e^{-bt/m}$. In one period $T \approx 2\pi/\omega_0$: $\frac{E(t+T)}{E(t)} = e^{-bT/m} = e^{-2\pi/Q} = e^{-2\pi/100} \approx e^{-0.06283} \approx 0.939$. Energy retains 93.9%, loses 6.1% per cycle. (b) $e^{-2\pi N/Q} = 0.01 \Rightarrow 2\pi N/Q = \ln 100 = 4.605 \Rightarrow N = \frac{4.605Q}{2\pi} = \frac{460.5}{6.283} \approx 73.3$ periods. (c) $\omega' = \omega_0 \sqrt{1 - 1/(4Q^2)} \approx \omega_0(1 - 1/(8Q^2))$. $\Delta\omega = \omega_0 - \omega' \approx \omega_0/(8Q^2) = \omega_0/80000 = 1.25 \times 10^{-5} \omega_0$. Fractional shift = 1.25×10^{-5} (negligible).

Check. $N \approx Q/2\pi \times \ln(100) = 15.9 \times 4.605 = 73.2$. Matches.

Practice — Addendum E

1. The amplitude of a damped oscillator decreases to 0.6 of its value in 5 s. If the period is 0.2 s, find the logarithmic decrement and Q .
2. A forced oscillator has $\omega_0 = 50 \text{ rad/s}$, $b/m = 1 \text{ s}^{-1}$. Find the amplitude resonance frequency and the velocity resonance frequency.
3. For a driven damped oscillator, show that the average power absorbed is $P = \frac{1}{2} \frac{bF_0^2\omega^2/m^2}{(\omega_0^2 - \omega^2)^2 + (b\omega/m)^2}$. Verify that its maximum is at $\omega = \omega_0$.

- The amplitude of a driven oscillator is measured as a function of driving frequency. The maximum amplitude is 10 cm at $\omega = 99$ rad/s. The amplitude falls to $10/\sqrt{2}$ cm at $\omega = 97$ and $\omega = 101$ rad/s. Estimate ω_0 , b/m , and Q .
- A damped oscillator has mass 0.5 kg, spring constant 200 N/m, and logarithmic decrement 0.02. Find the damping constant b and the quality factor Q .

Answers: 1. $\delta = \frac{1}{25} \ln(1/0.6) = 0.04 \times 0.5108 = 0.0204$. $Q = \pi/\delta \approx 154$. 2. $\omega_r = \sqrt{2500 - 0.5} \approx 49.995$ rad/s. $\omega_v = 50$ rad/s. 3. (Derivation shown in Method E2.) Maximum at $\omega = \omega_0$ because derivative of $P(\omega)$ gives $\omega(\omega_0^2 - \omega^2) = 0 \Rightarrow \omega = \omega_0$ (positive root). 4. The amplitude-resonance frequency is $\omega_r \approx 99$ rad/s (given). For light damping, $A(\omega)^2$ is proportional to the same Lorentzian as the power curve, so the points where amplitude falls to $A_{\max}/\sqrt{2}$ (power falls to half) are the power curve's half-power points, at $\omega = \omega_0 \pm \frac{b}{2m}$. With these at 97 and 101 rad/s, $\frac{b}{2m} = 2 \Rightarrow \frac{b}{m} = 4$, and since $\omega_r \approx \omega_0$ for light damping, $\omega_0 \approx 99$ rad/s. Check: $\omega_r = \sqrt{\omega_0^2 - b^2/2m^2} = \sqrt{99^2 - 8} = \sqrt{9793} \approx 98.96 \approx 99$, consistent with the given resonance frequency. $Q = \omega_0/(b/m) = 99/4 \approx 24.75$. 5. $\delta = bT/2m$. $T = 2\pi/\omega_0 = 2\pi/\sqrt{200/0.5} = 2\pi/\sqrt{400} = 2\pi/20 = \pi/10 = 0.314$ s. $b = 2m\delta/T = 2(0.5)(0.02)/0.314 = 0.02/0.314 \approx 0.0637$ kg/s. $Q = \pi/\delta = \pi/0.02 = 157$.

Addendum F — The “Prove It’s SHM” Laboratory: Potential Expansion and Non-Standard Systems

Subtopic 05 gives three standard systems (U-tube, floating body, Earth tunnel) and the differential-equation test $\ddot{x} + \omega^2 x = 0$. JEE Advanced invents new systems every year: a cylinder rolling on a spring, a ball in a spherical bowl, a piston in a gas cylinder, a charged particle in a uniform field, a rod pivoted with a spring, a liquid in a U-tube with unequal arms. The universal tool is the **small-oscillation expansion**: write the exact potential energy $U(q)$, expand to quadratic order $U \approx U_0 + \frac{1}{2}U''(q_0)(q - q_0)^2$, then $\omega = \sqrt{U''(q_0)/m_{\text{eff}}}$ where m_{eff} is the coefficient of $\frac{1}{2}\dot{q}^2$ in the kinetic energy.

Method

Universal small-oscillation recipe.

- Identify the degree of freedom q** (x , θ , y , etc.) and the equilibrium position q_0 .
- Write the exact kinetic energy $K(q, \dot{q})$** . Extract the effective mass (or inertia) m_{eff} from $K = \frac{1}{2}m_{\text{eff}}\dot{q}^2$.
- Write the exact potential energy $U(q)$** . Include gravity, springs, electrostatic, gas pressure, etc.
- Expand $U(q)$ around q_0 to second order:** $U(q) \approx U(q_0) + \frac{1}{2}U''(q_0)(q - q_0)^2$. (First derivative vanishes at equilibrium.)
- Read off $\omega^2 = U''(q_0)/m_{\text{eff}}$.**

Kinetic energy templates. - Translation only: $K = \frac{1}{2}m\dot{x}^2 \Rightarrow m_{\text{eff}} = m$. - Rotation about fixed axis: $K = \frac{1}{2}I\dot{\theta}^2 \Rightarrow m_{\text{eff}} = I$ (with $q = \theta$). - Rolling without slipping: $K = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}I(\dot{x}/R)^2 =$

$\frac{1}{2}(m + I/R^2)\dot{x}^2 \Rightarrow m_{\text{eff}} = m + I/R^2$. - For solid cylinder/sphere: $I = cmR^2 \Rightarrow m_{\text{eff}} = m(1 + c)$. - Ring: $c=1 \Rightarrow 2m$ - Disc/cylinder: $c=1/2 \Rightarrow 1.5m$ - Solid sphere: $c=2/5 \Rightarrow 1.4m$ - Hollow sphere: $c=2/3 \Rightarrow 1.67m$

Potential energy templates. - Spring: $U = \frac{1}{2}kx^2 \Rightarrow U'' = k$. - Gravity (height y): $U = mgy(q) \Rightarrow U'' = mgy''(q_0)$. - Gas (isothermal): $U = P_0V_0 \ln(V/V_0) \Rightarrow U'' = P_0A^2/V_0$. - Gas (adiabatic): $U = \frac{PV}{\gamma-1} \Rightarrow U'' = \gamma P_0A^2/V_0$. - Electrostatic: $U = qV(x) \Rightarrow U'' = qV''(x_0)$.

Example F.1 — [JEE Advanced Level] Cylinder Rolling on a Horizontal Spring

A solid cylinder of mass M and radius R rolls without slipping on a horizontal surface, attached to a spring of constant k fixed to a wall. The spring is attached to the cylinder's axle. Find the period.

Solution. (Same as Example A.1 but using the recipe.) $q = x$ (displacement of centre). $K = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}I(\dot{x}/R)^2 = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}(\frac{1}{2}MR^2)(\dot{x}^2/R^2) = \frac{1}{2}(\frac{3}{2}M)\dot{x}^2$. $m_{\text{eff}} = \frac{3}{2}M$. $U = \frac{1}{2}kx^2 \Rightarrow U'' = k$. $\omega^2 = k/(\frac{3}{2}M) = 2k/3M$. $T = 2\pi\sqrt{3M/2k}$.

Example F.2 — [JEE Advanced Level] Ball in a Spherical Bowl

A solid sphere of mass m and radius r rolls without slipping in a fixed spherical bowl of radius R . Find the period of small oscillations.

Solution. $q = \theta$ (angle of line joining centres from vertical). Centre of sphere moves on circle of radius $R - r$. $K = \frac{1}{2}mv_{\text{cm}}^2 + \frac{1}{2}I\omega^2$. $v_{\text{cm}} = (R - r)\dot{\theta}$. No slip: $\omega = v_{\text{cm}}/r = (R - r)\dot{\theta}/r$. $I = \frac{2}{5}mr^2$. $K = \frac{1}{2}m(R - r)^2\dot{\theta}^2 + \frac{1}{2}(\frac{2}{5}mr^2)\frac{(R - r)^2\dot{\theta}^2}{r^2} = \frac{1}{2}m(R - r)^2\dot{\theta}^2(1 + \frac{2}{5}) = \frac{1}{2}(\frac{7}{5}m(R - r)^2)\dot{\theta}^2$. $m_{\text{eff}} = \frac{7}{5}m(R - r)^2$ (with $q = \theta$). $U = mgh = mg(R - r)(1 - \cos \theta) \approx \frac{1}{2}mg(R - r)\theta^2$. $U'' = mg(R - r)$. $\omega^2 = \frac{mg(R - r)}{\frac{7}{5}m(R - r)^2} = \frac{5g}{7(R - r)}$. $T = 2\pi\sqrt{\frac{7(R - r)}{5g}}$.

Check. If sliding without friction (no rotation), $I = 0$, $m_{\text{eff}} = m(R - r)^2$, $\omega^2 = g/(R - r)$, $T = 2\pi\sqrt{(R - r)/g}$ (simple pendulum of length $R - r$). Rolling adds inertia, period increases by $\sqrt{7/5} \approx 1.18$. Correct.

Example F.3 — [JEE Advanced Level] Piston in a Vertical Cylinder with Gas

A vertical cylinder of cross-section A contains an ideal gas at temperature T_0 . A piston of mass M fits tightly and slides without friction. The piston is in equilibrium at height h_0 (volume $V_0 = Ah_0$). The gas undergoes isothermal changes. Find the period of small vertical oscillations of the piston.

Solution. $q = y$ (displacement of piston from equilibrium, upward positive). Equilibrium: $P_0A = Mg$. Gas law (isothermal): $PV = P_0V_0 = \text{const}$. $V = A(h_0 - y) = V_0 - Ay$. $P = P_0V_0/(V_0 - Ay)$. Net force on piston (upward): $F = PA - Mg = \frac{P_0V_0A}{V_0 - Ay} - P_0A$. $U(y) = -\int Fdy = P_0V_0 \ln(V_0 - Ay) + P_0Ay + \text{const}$. Expand around $y=0$: $V_0 - Ay = V_0(1 - Ay/V_0)$. $\ln(V_0 - Ay) = \ln V_0 + \ln(1 - Ay/V_0) \approx \ln V_0 - Ay/V_0 - \frac{1}{2}(Ay/V_0)^2$. $U \approx P_0V_0(\ln V_0 - Ay/V_0 - \frac{1}{2}A^2y^2/V_0^2) + P_0Ay = \text{const} - \frac{1}{2}P_0A^2y^2/V_0$. $U'' = P_0A^2/V_0$. $K = \frac{1}{2}M\dot{y}^2 \Rightarrow m_{\text{eff}} = M$. $\omega^2 = \frac{P_0A^2}{MV_0}$. $T = 2\pi\sqrt{\frac{MV_0}{P_0A^2}} = 2\pi\sqrt{\frac{Mh_0}{P_0A}} = 2\pi\sqrt{\frac{Mh_0}{g}}$ (using $P_0A = Mg$). **Beautiful result:** The period is identical to a simple

pendulum of length h_0 , independent of the gas properties (for isothermal). For adiabatic, T is shorter by factor $1/\sqrt{\gamma}$.

Example F.4 — [JEE Advanced Level] Rod Pivoted with a Spring

A uniform rod of mass M and length L is pivoted at one end. A spring of constant k is attached to the other end, horizontal when the rod is vertical. Find the period of small oscillations.

Solution. $q = \theta$ (small angle from vertical). $K = \frac{1}{2}I\dot{\theta}^2 = \frac{1}{2}(\frac{1}{3}ML^2)\dot{\theta}^2$. $m_{\text{eff}} = \frac{1}{3}ML^2$. Potential energy: gravity + spring. $U = Mg\frac{L}{2}(1 - \cos\theta) + \frac{1}{2}k(L\sin\theta)^2 \approx \frac{1}{2}Mg\frac{L}{2}\theta^2 + \frac{1}{2}kL^2\theta^2$. $U'' = \frac{1}{2}MgL + kL^2$. $\omega^2 = \frac{\frac{1}{2}MgL + kL^2}{\frac{1}{3}ML^2} = \frac{3g}{2L} + \frac{3k}{M}$. $T = 2\pi\sqrt{\frac{1}{\frac{3g}{2L} + \frac{3k}{M}}}$.

Check. If $k=0$: physical pendulum $T = 2\pi\sqrt{2L/3g}$. Correct (Subtopic 03 Example 4). If $g=0$: $T = 2\pi\sqrt{M/3k}$ (Addendum A Example 1 corrected). Correct.

Practice — Addendum F

1. A uniform disc of mass M and radius R rolls without slipping on a horizontal surface, attached to a spring of constant k at its centre. Find the period.
2. A hollow sphere of mass m and radius r rolls without slipping in a spherical bowl of radius R . Find the period of small oscillations.
3. A vertical spring of constant k has a mass M at its bottom. The spring has mass m_s . Find the period using the energy method (include spring kinetic energy properly).
4. A particle of mass m moves in a potential $U(x) = \frac{A}{x^2} + Bx^2$ ($A, B > 0$). Find the equilibrium position and the frequency of small oscillations about it.
5. A U-tube has vertical arms of cross-sections A_1 and A_2 . The connecting tube at the bottom is horizontal and has negligible volume. Total liquid mass M , density ρ . Find the period of oscillation of the liquid levels.
6. A charged particle of mass m and charge q is constrained to move along the x -axis. A fixed charge Q is at $(0, d)$. Find the frequency of small oscillations about the equilibrium point on the x -axis. (Assume $qQ < 0$.)

Answers: 1. Disc: $I = \frac{1}{2}MR^2$. $m_{\text{eff}} = M + I/R^2 = 1.5M$. $T = 2\pi\sqrt{1.5M/k}$. 2. Hollow sphere: $I = \frac{2}{3}mr^2$, so $c = \frac{2}{3}$. $K = \frac{1}{2}m(R-r)^2\dot{\theta}^2(1+c) = \frac{1}{2}\frac{5}{3}m(R-r)^2\dot{\theta}^2$ and $U = \frac{1}{2}mg(R-r)\theta^2$. $\omega^2 = \frac{mg(R-r)}{\frac{5}{3}m(R-r)^2} = \frac{3g}{5(R-r)}$, so $T = 2\pi\sqrt{\frac{5(R-r)}{3g}}$. 3. Spring mass m_s , uniform. $K_{\text{spring}} = \int_0^L \frac{1}{2}(m_s/L)(x/L\dot{y})^2 dx = \frac{1}{2}(m_s/3)\dot{y}^2$. Total $K = \frac{1}{2}(M + m_s/3)\dot{y}^2$. $U = \frac{1}{2}ky^2$. $T = 2\pi\sqrt{(M + m_s/3)/k}$. 4. $U'(x) = -2A/x^3 + 2Bx = 0 \Rightarrow x_0^4 = A/B \Rightarrow x_0 = (A/B)^{1/4}$. $U''(x) = 6A/x^4 + 2B$. $U''(x_0) = 6B + 2B = 8B$. $\omega = \sqrt{8B/m}$. 5. Let y_1 be the displacement in arm 1. Volume conservation: $A_1 y_1 = A_2 y_2$, so the level difference is $y_1 + y_2 = y_1(A_1 + A_2)/A_2$ and the restoring force on the arm-1 column is $F = -\rho g A_1 (A_1 + A_2)/A_2 \cdot y_1$, giving $U = \frac{1}{2}\rho g A_1 \frac{A_1 + A_2}{A_2} y_1^2$. Because the arms have different areas, the kinetic energy is summed arm by arm: arm 1 moves at $\dot{y}_1$, arm 2 at $\dot{y}_2 = (A_1/A_2)\dot{y}_1$, and the negligible-volume horizontal connector is ignored, so $K = \frac{1}{2}\rho A_1 L_1 \dot{y}_1^2 + \frac{1}{2}\rho A_2 L_2 (A_1/A_2)^2 \dot{y}_1^2 = \frac{1}{2}\rho A_1 (L_1 + A_1 L_2/A_2) \dot{y}_1^2$. Hence $\omega^2 = \frac{\rho g A_1 (A_1 + A_2)/A_2}{\rho A_1 (L_1 + A_1 L_2/A_2)} = \frac{g(A_1 + A_2)}{A_2 L_1 + A_1 L_2}$. If $L_1 = L_2 = L/2$: $\omega^2 = \frac{2g(A_1 + A_2)}{L(A_1 + A_2)} = 2g/L$ — independent of the areas (the classic result). 6. The potential on the axis is $U = kqQ/\sqrt{x^2 + d^2}$, with the

equilibrium at $x = 0$. Curvature: $U''(0) = -kqQ/d^3$, so stability needs $qQ < 0$ (attraction) — for $qQ > 0$ the point $x = 0$ is a potential maximum and no oscillation exists. With $qQ < 0$, $U''(0) = k|qQ|/d^3$ and $\omega = \sqrt{k|qQ|/md^3}$.

DRONA ED TECH

Physics • CBSE Boards • JEE Main • NEET • JEE Advanced

Chapter 14

WAVES

A complete teaching segment in four subtopics

01 • Wave Characteristics and Speed

02 • Superposition and Beats

03 • Standing Waves and Acoustics

04 • Doppler Effect

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Subtopic 01: Wave Characteristics and Speed

Exam Relevance: This is the foundation stone of the entire Waves chapter. JEE Main typically asks 1–2 direct questions on the wave equation, wave speed on a string, and particle vs. wave velocity. NEET leans toward conceptual questions on wave types and the $v = f\lambda$ relationship across media. JEE Advanced rarely asks this alone — it embeds it inside multi-concept problems (variable-tension ropes, superposition setups). For CBSE Boards, expect a 2–3 mark derivation of wave speed on a string or a short-answer on wave characteristics.

Section 1: Concept Intuition

Picture yourself in the stands at a packed stadium during an India match. Someone a few sections away stands up, throws their arms up, and sits back down. The person next to them does the same a moment later, then the next, and the next. Within seconds a giant ripple of standing-and-sitting sweeps all the way around the stadium. Here is the beautiful part: **not a single person actually travelled around the stadium.** Each person stayed exactly in their own seat. What travelled was the *pattern* — the disturbance.

That is a wave, in one sentence. A **wave is a disturbance that travels through a medium (or space), carrying energy and momentum from one place to another, without any net transport of the medium itself.**

The medium particles only wiggle about their fixed average position. Drop a pebble in a still pond and a twig floating nearby just bobs up and down — it does not get carried to the shore, even though the ripple clearly races outward. The water goes up and down; the *energy* goes outward.

Think about it this way... Every particle of the medium is doing its own little simple harmonic motion (SHM) about its mean position. What we call “the wave moving” is really just the *phase* of that oscillation being handed from one particle to the next, slightly delayed each time — exactly like the stadium crowd handing the “stand up” cue down the row.

Now, how do we classify waves?

1. By the medium’s need:

- **Mechanical waves** need a material medium to travel through — sound, waves on a string, ripples on water. No medium, no wave. (This is why sound cannot travel through the vacuum of space.)
- **Non-mechanical (electromagnetic) waves** like light, radio, and X-rays need no medium and happily travel through vacuum.

2. By the direction of vibration:

- **Transverse waves:** particles vibrate *perpendicular* to the direction the wave travels. A wave on a guitar string, light, ripples on water. Picture flicking one end of a long

rope up and down — the rope moves vertically, the wave moves horizontally.

- **Longitudinal waves:** particles vibrate *along* the direction the wave travels, creating compressions and rarefactions. Sound is the classic example. Picture pushing and pulling one end of a slinky — the coils squeeze together and stretch apart along the slinky's length.

3. By dimensionality: waves on a string are 1-D, ripples on a pond surface are 2-D, and sound from a firecracker spreading through the air is 3-D.

Every wave, no matter the type, is described by the same handful of characteristics: how big the swing is (**amplitude**), how long one full pattern is in space (**wavelength**), how long one full oscillation takes in time (**time period**), how many oscillations happen per second (**frequency**), and how fast the disturbance travels (**wave speed**).

Limiting Conditions & Assumptions:

- Wave speed in a given medium is set by the **medium's properties**, *not* by how the wave was produced. Shouting louder (more amplitude) or at a higher pitch (more frequency) does **not** make sound travel faster in the same air.
- The clean formula $v = f\lambda$ and the sinusoidal wave equation assume an **ideal, non-dispersive, lossless medium** with small amplitudes. Real strings have stiffness, real media absorb energy, and very large amplitudes break the linear approximation.
- The transverse-wave-on-a-string formula assumes a **perfectly flexible, uniform string** under constant tension.

Section 2: Key Formulas & Definitions

Wave speed (the master relation):

$$v = f\lambda = \frac{\lambda}{T}$$

Angular frequency and wave number:

$$\omega = 2\pi f = \frac{2\pi}{T}, \quad k = \frac{2\pi}{\lambda}$$

$$v = \frac{\omega}{k}$$

Equation of a progressive (travelling) harmonic wave moving in the $+x$ direction:

$$y(x, t) = A \sin(\omega t - kx + \phi)$$

(For motion in the $-x$ direction, the sign between ωt and kx flips to $+$.)

Speed of a transverse wave on a stretched string:

$$v = \sqrt{\frac{T}{\mu}}$$

Particle velocity (velocity of a medium particle, *not* the wave):

$$v_p = \frac{\partial y}{\partial t} = A\omega \cos(\omega t - kx), \quad (v_p)_{\max} = A\omega$$

Symbol	Meaning	SI Unit	Dimensional Formula
A	Amplitude (max displacement)	metre (m)	$[L]$
λ	Wavelength (length of one cycle)	metre (m)	$[L]$
T	Time period	second (s)	$[T]$
f	Frequency	hertz (Hz)	$[T^{-1}]$
v	Wave speed	m s^{-1}	$[LT^{-1}]$
ω	Angular frequency	rad s^{-1}	$[T^{-1}]$
k	Wave number (angular)	rad m^{-1}	$[L^{-1}]$
μ	Linear mass density of string	kg m^{-1}	$[ML^{-1}]$
ϕ	Phase constant	radian	dimensionless

A quick sanity note: **frequency f is set by the source** and stays the same when a wave crosses into a new medium. When the medium changes, v and λ change together so that $v = f\lambda$ still holds.

Section 3: CBSE Board Derivation — Speed of a Transverse Wave on a String

We want to prove $v = \sqrt{\frac{T}{\mu}}$, where T is the tension and μ is the mass per unit length.

The clever trick — change your reference frame. Instead of watching a pulse race along a stationary string, we hop into a frame moving *with* the pulse at speed v . In this frame the pulse is frozen in place, and the entire string appears to slide backward through it at speed v .

Set-up: Consider a tiny element of the string at the crest of the pulse. Because the pulse is curved, this small element forms a short arc of a circle of radius R . Let the element subtend a small angle 2θ at the centre of that circle, so its length is

$$\Delta\ell = R(2\theta) = 2R\theta.$$

The forces: The tension T pulls on both ends of the element, tangent to the string. By symmetry, the horizontal components cancel. The two vertical (inward) components add up. Each contributes $T \sin \theta$, so the net inward force is

$$F = 2T \sin \theta \approx 2T\theta \quad (\text{since } \theta \text{ is small, } \sin \theta \approx \theta).$$

Why this step? The small-angle approximation is valid because we deliberately chose a

tiny element — over a small enough arc, $\sin \theta$ and θ are indistinguishable. This is what lets the geometry collapse into the simple result.

Substituting $\theta = \frac{\Delta \ell}{2R}$:

$$F = 2T \cdot \frac{\Delta \ell}{2R} = \frac{T \Delta \ell}{R}.$$

The motion: In our moving frame, this element is travelling along a circular arc at speed v , so it needs a **centripetal force** directed toward the centre. The mass of the element is

$$m = \mu \Delta \ell,$$

and the required centripetal force is

$$F_c = \frac{mv^2}{R} = \frac{\mu \Delta \ell v^2}{R}.$$

Why this step? The net tension force we computed is exactly what supplies this centripetal pull — there is nothing else acting. So we set them equal.

Equate and solve:

$$\frac{T \Delta \ell}{R} = \frac{\mu \Delta \ell v^2}{R}$$

The $\Delta \ell$ and R cancel beautifully, leaving

$$T = \mu v^2 \implies v = \sqrt{\frac{T}{\mu}}$$

Notice what this says physically: tighter string (more T) $\Rightarrow$ faster wave; heavier string (more μ) $\Rightarrow$ slower wave. And crucially — **no A , no f anywhere**. Speed depends only on the string, never on how you shook it.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A nylon string of a sitar has a linear mass density of 0.04 kg m^{-1} and is stretched under a tension of 36 N . A transverse wave of frequency 50 Hz is set up on it. Find (a) the speed of the wave and (b) its wavelength.

Given: $\mu = 0.04 \text{ kg m}^{-1}$, $T = 36 \text{ N}$, $f = 50 \text{ Hz}$.

(a) Using the string-speed formula:

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{36}{0.04}} = \sqrt{900} = 30 \text{ m s}^{-1}$$

(b) Using $v = f\lambda$:

$$\lambda = \frac{v}{f} = \frac{30}{50} = 0.6 \text{ m}$$

Answer: Wave speed = 30 m s^{-1} , wavelength = 0.6 m .

Example 2 — [NEET Speed Trap]

Two strings, A and B, are made of the same material. String A has twice the diameter of string B and is kept under four times the tension of B. What is the ratio of the wave speeds $v_A : v_B$?

The trap: Students rush to $v = \sqrt{T/\mu}$, see “4 times the tension,” and blurt out 2 : 1 — completely forgetting that the thicker string is also heavier per unit length.

The fast way: Same material means same density ρ . Linear mass density $\mu = \rho A_{\text{cross}} = \rho \cdot \frac{\pi d^2}{4} \propto d^2$.

$$\text{So } \frac{\mu_A}{\mu_B} = \left(\frac{d_A}{d_B}\right)^2 = (2)^2 = 4.$$

$$\frac{v_A}{v_B} = \sqrt{\frac{T_A/\mu_A}{T_B/\mu_B}} = \sqrt{\frac{T_A}{T_B} \cdot \frac{\mu_B}{\mu_A}} = \sqrt{4 \cdot \frac{1}{4}} = \sqrt{1} = 1$$

Answer: $v_A : v_B = 1 : 1$. The tension boost is exactly cancelled by the extra mass — a classic NEET “looks-complicated-but-equals-one” trap.

Example 3 — [JEE Main Level]

A transverse wave on a string is described by $y = 0.05 \sin(8\pi t - 4\pi x)$, where all quantities are in SI units. (a) Find the amplitude, frequency, wavelength, and wave speed. (b) Find the maximum speed of a particle of the string. (c) If the string has a linear mass density of 0.5 kg m^{-1} , find the tension in it.

(a) Compare with $y = A \sin(\omega t - kx)$:

- Amplitude $A = 0.05 \text{ m}$
- $\omega = 8\pi \text{ rad s}^{-1} \Rightarrow f = \frac{\omega}{2\pi} = \frac{8\pi}{2\pi} = 4 \text{ Hz}$
- $k = 4\pi \text{ rad m}^{-1} \Rightarrow \lambda = \frac{2\pi}{k} = \frac{2\pi}{4\pi} = 0.5 \text{ m}$
- Wave speed $v = \frac{\omega}{k} = \frac{8\pi}{4\pi} = 2 \text{ m s}^{-1}$ (in the $+x$ direction)

(b) Maximum particle speed:

$$(v_p)_{\text{max}} = A\omega = 0.05 \times 8\pi = 0.4\pi \approx 1.26 \text{ m s}^{-1}$$

(Note this is *less* than the wave speed of 2 m s^{-1} here — particle speed and wave speed are entirely different things.)

(c) From $v = \sqrt{T/\mu}$:

$$T = \mu v^2 = 0.5 \times (2)^2 = 2 \text{ N}$$

Answer: $A = 0.05 \text{ m}$, $f = 4 \text{ Hz}$, $\lambda = 0.5 \text{ m}$, $v = 2 \text{ m s}^{-1}$; $(v_p)_{\max} \approx 1.26 \text{ m s}^{-1}$; $T = 2 \text{ N}$.

Example 4 — [JEE Advanced Level]

A uniform rope of mass $M = 5 \text{ kg}$ and length $L = 2 \text{ m}$ hangs vertically from a ceiling. A block of mass $m = 4 \text{ kg}$ is attached to its lower free end. A transverse pulse is generated at the bottom of the rope. How long does the pulse take to reach the top? (Take $g = 10 \text{ m s}^{-2}$.)

The key insight: The tension in the rope is **not constant** — every point must support the weight hanging *below* it. So the wave speed changes as the pulse climbs, and we cannot use a single value of v .

Linear mass density of the rope:

$$\mu = \frac{M}{L} = \frac{5}{2} = 2.5 \text{ kg m}^{-1}$$

Tension at a height x above the bottom = weight of the block plus the rope below that point:

$$T(x) = (m + \mu x)g = (4 + 2.5x) \times 10$$

Local wave speed:

$$v(x) = \sqrt{\frac{T(x)}{\mu}} = \sqrt{\frac{(4 + 2.5x) \times 10}{2.5}} = \sqrt{4(4 + 2.5x)} = 2\sqrt{4 + 2.5x}$$

Time to travel the full length — integrate $dt = \frac{dx}{v(x)}$:

$$t = \int_0^2 \frac{dx}{2\sqrt{4 + 2.5x}}$$

Let $u = 4 + 2.5x \Rightarrow du = 2.5 dx$. Limits: $x = 0 \rightarrow u = 4$; $x = 2 \rightarrow u = 9$.

$$t = \int_4^9 \frac{1}{2\sqrt{u}} \cdot \frac{du}{2.5} = \frac{1}{5} \int_4^9 u^{-1/2} du = \frac{1}{5} \left[2\sqrt{u} \right]_4^9 = \frac{1}{5} (2 \times 3 - 2 \times 2) = \frac{2}{5}$$

Answer: $t = 0.4 \text{ s}$.

Reasoning check: The pulse is slowest at the very bottom (least tension) and speeds up as it climbs (more weight below it). Integrating the changing speed is the entire game here — anyone who plugs in a single average speed gets it wrong.

Section 5: Practice Exercises

1. **[CBSE]** A wave travels at 340 m s^{-1} with a frequency of 680 Hz. Calculate its wavelength and time period.
2. **[NEET]** A steel wire of length 1 m and mass 20 g is stretched under a tension of 80 N. Find the speed of a transverse wave on it.
3. **[JEE Main]** A progressive wave is given by $y = 0.02 \sin(100\pi t - 0.5\pi x)$ (SI units). Find its amplitude, frequency, wavelength, speed, and direction of travel.
4. **[JEE Main]** Two wires of the same material have radii in the ratio 1 : 2 and are stretched by tensions in the ratio 2 : 1. Find the ratio of transverse wave speeds in them.
5. **[JEE Advanced]** A uniform rope of length 9 m and mass 3 kg hangs freely from a ceiling (no extra block). Find the time taken by a transverse pulse to travel from the bottom to the top. (Take $g = 10 \text{ m s}^{-2}$.)

Answers: 1. $\lambda = 0.5 \text{ m}$, $T \approx 1.47 \times 10^{-3} \text{ s}$. 2. $v = \sqrt{80/0.02} = 63.2 \text{ m s}^{-1}$. 3. $A = 0.02 \text{ m}$, $f = 50 \text{ Hz}$, $\lambda = 4 \text{ m}$, $v = 200 \text{ m s}^{-1}$, $+x$ direction. 4. $v_1 : v_2 = \sqrt{(2/1) \cdot (2^2/1^2)} = \sqrt{8} = 2\sqrt{2} : 1$. 5. $t = 2\sqrt{L/g} = 2\sqrt{9/10} = 1.9 \text{ s}$.

Section 6: Crack the MCQ

Q1. The speed of a transverse wave on a stretched string depends on:

- (a) the frequency of the wave
- (b) the amplitude of the wave
- (c) the tension and linear mass density of the string
- (d) the wavelength of the wave

Correct: (c). From $v = \sqrt{T/\mu}$, speed is fixed entirely by the string's tension and mass per unit length — i.e., by the medium.

- (a) *trap*: Students confuse “I can change the frequency by shaking faster” with “speed changes.” Changing f only changes λ (since v is fixed); the speed stays put.
- (b) *trap*: Amplitude affects *energy carried*, never speed.
- (d) *trap*: Same error as (a) — λ adjusts to f , it doesn't set v .

Q2. A wave is represented by $y = 5 \sin(4t - 2x)$ (SI units). The wave speed is:

- (a) 0.5 m s^{-1}
- (b) 2 m s^{-1}
- (c) 4 m s^{-1}
- (d) 8 m s^{-1}

Correct: (b). $\omega = 4$, $k = 2$, so $v = \omega/k = 4/2 = 2 \text{ m s}^{-1}$.

- (a) *trap*: Computing k/ω (formula flipped upside down).
- (c) *trap*: Reading $\omega = 4$ as the speed directly.
- (d) *trap*: Multiplying $\omega \times k = 8$ instead of dividing.

Q3. Which of the following can travel through a vacuum?

- (a) Sound waves
- (b) Waves on a string
- (c) X-rays
- (d) Water waves

Correct: (c). X-rays are electromagnetic (non-mechanical) and need no medium.

- *(a), (b), (d) traps:* All three are mechanical waves that absolutely require a material medium. The most common slip is (a) — picturing sound “in space” from movies, which is physically impossible.
-

Q4. A sound wave travels from air into water. As it crosses the boundary, which quantity remains unchanged?

- (a) Speed
- (b) Wavelength
- (c) Frequency
- (d) All of these

Correct: (c). Frequency is dictated by the *source* and is preserved across media. Since the speed of sound is greater in water, and $v = f\lambda$ with f fixed, the wavelength actually *increases* in water.

- *(a) trap:* Sound is faster in water — speed definitely changes.
 - *(b) trap:* With v up and f fixed, λ must change.
 - *(d) trap:* Tempting if you don’t remember which quantity is “locked.” Only f is.
-

Section 7: Common Student Pitfalls & Pro-Tips

1. **Confusing particle velocity with wave velocity.** The wave speed $v = \omega/k$ is how fast the *pattern* moves. The particle speed $v_p = A\omega \cos(\dots)$ is how fast a *bit of the medium* wiggles. They are different quantities with different formulas — and as Example 3 showed, either one can be larger.
2. **Forgetting that μ scales with thickness.** In “same material” string problems, a thicker string has a larger $\mu \propto d^2 \propto r^2$. Ignoring this is the single most common error in two-string ratio questions.
3. **Assuming frequency changes across media.** When a wave enters a new medium, f stays the same; it’s v and λ that change. Many students wrongly hold λ fixed instead.
4. **Using a single wave speed for a hanging rope.** Tension varies along a vertical rope, so the speed varies. You must integrate — a constant-speed shortcut gives the wrong time.

Pro-Tip: For “same material, different dimensions” string problems, skip the full calculation and go straight to ratios. Write $v \propto \sqrt{T/r^2} = \sqrt{T}/r$ and plug in the ratios directly. It turns a two-minute problem into a ten-second one — and dodges the μ -scaling trap automatically.

Section 8: Cheat Sheet Box

WAVE CHARACTERISTICS & SPEED — QUICK RECALL

- **Master relation:** $v = f\lambda = \frac{\lambda}{T} = \frac{\omega}{k}$
- **Wave on a string:** $v = \sqrt{\frac{T}{\mu}}$ — depends on medium *only*, never on A or f .
- **Wave equation (+x):** $y = A \sin(\omega t - kx + \phi)$
- **Particle velocity:** $v_p = A\omega \cos(\dots)$, $\max = A\omega$ — *not* the same as wave speed.
- $\omega = 2\pi f$, $k = \frac{2\pi}{\lambda}$
- **Across media:** f stays fixed; v and λ change together.

Memory aids:

- “Tight and light goes fast” — high tension T , low density $\mu \Rightarrow$ high v .
- “Source sets the frequency, medium sets the speed.”
- “Same material? Then $\mu \propto r^2$ — don’t forget the radius.”

Subtopic 02: Superposition and Beats

Exam Relevance: A reliably scoring subtopic. JEE Main almost always has a question on beat frequency or the tuning-fork loading/filing logic, and often one on resultant amplitude/intensity of two coherent waves. NEET loves the conceptual beats question (“beat frequency = difference of frequencies”) and the unknown-frequency puzzle. JEE Advanced folds beats into multi-step problems (sonometer tension changes, the $f \propto \sqrt{T}$ link). CBSE Boards typically ask the principle of superposition or a short derivation of beat frequency for 2–3 marks.

Section 1: Concept Intuition

Throw two pebbles into a still pond a little distance apart. Two sets of circular ripples spread out, and somewhere in the middle they meet. What happens? They simply *pass through each other*. Where two crests meet, the water rises extra high. Where a crest meets a trough, the surface barely moves. And once they cross, each ripple carries on exactly as if the other had never been there.

That is the **Principle of Superposition**, the single idea underneath this entire subtopic:

When two or more waves overlap at a point, the resultant displacement at that point is the **algebraic sum** of the displacements that each wave would have produced individually.

$$y = y_1 + y_2 + y_3 + \dots$$

Waves don’t collide and bounce off each other like cricket balls. They add up, point by point, instant by instant — and then move on, unchanged.

Think about it this way... Imagine two people pushing the same child on a swing. If both push *in step* with the swing’s motion, the child goes higher and higher — their efforts *add up*. If one pushes forward exactly when the other pulls back, they cancel and the swing barely moves. Waves do the same thing: when they arrive **in phase**, they reinforce (constructive interference); when they arrive **out of phase**, they cancel (destructive interference).

For two waves of the **same frequency and amplitude** A , with a phase difference ϕ between them, the resultant amplitude is:

$$A_{\text{res}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

- In phase ($\phi = 0, 2\pi, \dots$): amplitudes add $\rightarrow$ **loudest/brightest**.
- Out of phase ($\phi = \pi, 3\pi, \dots$): amplitudes subtract $\rightarrow$ **quietest/darkest** (zero if equal).

Now, beats. What if the two waves have *almost* the same frequency, but not quite? Picture two singers in a bhajan holding the same note, but one is a hair sharp. You don’t hear two separate notes — you hear a single note whose **loudness swells and fades**,

swells and fades: *waaah... waaah... waaah...* This periodic rise and fall in loudness is called **beats**.

Here's why it happens: because the frequencies differ slightly, the two waves drift in and out of step at a point. For a moment they're in phase (loud), then they slip out of phase (soft), then back in phase (loud again). The number of these loud-soft cycles per second is the **beat frequency**, and it turns out to equal exactly the *difference* of the two frequencies:

$$f_{\text{beat}} = |f_1 - f_2|$$

This is precisely how a tabla player or a sitarist tunes by ear — they adjust the tension until the throbbing slows down and finally vanishes. No throb means the frequencies match.

Limiting Conditions & Assumptions:

- Superposition holds for **linear media** (small amplitudes). For very large disturbances, waves *do* interact and the simple sum breaks down.
- **Steady interference** (a stable pattern of bright/dark or loud/quiet) requires the two sources to be **coherent** — same frequency and a constant phase relationship.
- **Beats** require the two frequencies to be **close**. If $|f_1 - f_2|$ exceeds about 7 Hz, the throbs come too fast for the human ear to resolve and you just hear a blur.

Section 2: Key Formulas & Definitions

Principle of superposition:

$$y = y_1 + y_2 + \dots$$

Resultant amplitude of two waves (same frequency, phase difference ϕ):

$$A_{\text{res}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

Resultant intensity (since $I \propto A^2$):

$$I = I_1 + I_2 + 2\sqrt{I_1I_2} \cos \phi$$

Maximum and minimum intensity:

$$I_{\text{max}} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2, \quad I_{\text{min}} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$$

Condition for constructive interference:

$$\phi = 2n\pi \quad \text{or} \quad \text{path difference } \Delta x = n\lambda \quad (n = 0, 1, 2, \dots)$$

Condition for destructive interference:

$$\phi = (2n + 1)\pi \quad \text{or} \quad \Delta x = (2n + 1)\frac{\lambda}{2}$$

Beat frequency:

$$f_{\text{beat}} = |f_1 - f_2|, \quad t_{\text{beat}} = \frac{1}{|f_1 - f_2|}$$

Tuning-fork logic:

- Loading a fork with **wax** → **frequency decreases**.
- **Filing** a fork → frequency **increases**.

Symbol	Meaning	SI Unit	Dimensional Formula
ϕ	Phase difference	radian	dimensionless
Δx	Path difference	metre (m)	$[L]$
I	Intensity	W m^{-2}	$[MT^{-3}]$
f_{beat}	Beat frequency	hertz (Hz)	$[T^{-1}]$
t_{beat}	Beat period	second (s)	$[T]$

Relation between phase difference and path difference:

$$\phi = \frac{2\pi}{\lambda} \Delta x$$

Section 3: CBSE Board Derivation — The Beat Frequency Formula

We want to prove that two waves of nearby frequencies f_1 and f_2 , heard at the same point, produce beats at frequency $|f_1 - f_2|$.

Set-up. Consider a fixed point in space (say, your eardrum). Two sound waves of equal amplitude A and slightly different frequencies arrive there. Drop the spatial part (we're at one fixed point) and write the two displacements as functions of time:

$$y_1 = A \cos(2\pi f_1 t), \quad y_2 = A \cos(2\pi f_2 t)$$

Why cosine and one fixed point? Beats are about how the *combined loudness* changes in **time** at a single location, so we only need the time dependence.

Apply superposition:

$$y = y_1 + y_2 = A[\cos(2\pi f_1 t) + \cos(2\pi f_2 t)]$$

Use the sum-to-product identity $\cos C + \cos D = 2 \cos\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$:

$$y = 2A \cos\left[2\pi \left(\frac{f_1 - f_2}{2}\right) t\right] \cos\left[2\pi \left(\frac{f_1 + f_2}{2}\right) t\right]$$

Read the physics off the equation. This is a single oscillation with two distinct time-scales:

1. The **second factor**, $\cos\left[2\pi\left(\frac{f_1+f_2}{2}\right)t\right]$, is a fast oscillation at the **average frequency** $\frac{f_1+f_2}{2}$. This is the actual note you perceive.
2. The **first factor**, $2A \cos\left[2\pi\left(\frac{f_1-f_2}{2}\right)t\right]$, is a *slowly* varying **amplitude envelope**. Because f_1 and f_2 are close, $\frac{f_1-f_2}{2}$ is small, so this term changes lazily — it's the swelling and fading you hear.

The crucial last step — why the factor of 2. The loudness (intensity) depends on the *magnitude* of the amplitude, $|2A \cos(\dots)|$, not its sign. The envelope reaches a maximum loudness **twice** in each of its own cycles (once when $\cos = +1$, once when $\cos = -1$). So the loudness throbs at *twice* the envelope frequency:

$$f_{\text{beat}} = 2 \times \frac{|f_1 - f_2|}{2} = \boxed{|f_1 - f_2|}$$

The factor of 2 is exactly what students drop — and it's the whole reason the answer is the clean *difference*, not half of it.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Two tuning forks of frequencies 512 Hz and 516 Hz are sounded together. Find (a) the beat frequency and (b) the time interval between two successive maxima of loudness.

Given: $f_1 = 516$ Hz, $f_2 = 512$ Hz.

(a) Beat frequency:

$$f_{\text{beat}} = |f_1 - f_2| = |516 - 512| = 4 \text{ Hz}$$

(b) Time between successive maxima = beat period:

$$t_{\text{beat}} = \frac{1}{f_{\text{beat}}} = \frac{1}{4} = 0.25 \text{ s}$$

Answer: 4 beats per second, with maxima 0.25 s apart.

Example 2 — [NEET Speed Trap]

A tuning fork A of unknown frequency produces 6 beats per second with a standard fork of frequency 480 Hz. When fork A is loaded with a small amount of wax, the beat frequency drops to 3 per second. Find the original frequency of A.

The trap: Students stop at “6 beats $\Rightarrow$ A is 486 or 474 Hz” and then guess. You must use the *loading* clue to choose.

Step 1 — list both candidates:

$$f_A = 480 \pm 6 = 486 \text{ Hz or } 474 \text{ Hz}$$

Step 2 — apply the rule: wax *lowers* a fork's frequency. So loading A pushes its frequency **down**.

- If $f_A = 486$: loading drops it toward 480 → it gets *closer* → beats **decrease**. ✓
(Matches “drops to 3.”)
- If $f_A = 474$: loading drops it away from 480 → it gets *farther* → beats **increase**. ✗

Answer: $f_A = 486$ Hz.

Mental shortcut: “Loading drops frequency. Beats went *down* after loading ⇒ the fork was *above* the standard.” Pick the higher candidate. (If beats had gone *up*, pick the lower one.)

Example 3 — [JEE Main Level]

Two coherent sources produce sound waves of intensities I and $4I$ at a point. (a) Find the ratio $I_{\max}/I_{\min}$ in the resulting interference pattern. (b) Find the resultant intensity at a point where the phase difference between the two waves is $\frac{\pi}{2}$.

(a) With $I_1 = I$ and $I_2 = 4I$, the amplitudes go as $A \propto \sqrt{I}$, so $A_1 : A_2 = 1 : 2$.

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2 = (\sqrt{I} + \sqrt{4I})^2 = (3\sqrt{I})^2 = 9I$$

$$I_{\min} = (\sqrt{I_2} - \sqrt{I_1})^2 = (2\sqrt{I} - \sqrt{I})^2 = (\sqrt{I})^2 = I$$

$$\frac{I_{\max}}{I_{\min}} = \frac{9I}{I} = 9 \Rightarrow 9 : 1$$

(b) Using $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$ with $\phi = \frac{\pi}{2}$ (so $\cos \phi = 0$):

$$I = I + 4I + 2\sqrt{I \cdot 4I} (0) = 5I$$

Answer: $I_{\max}/I_{\min} = 9 : 1$; resultant intensity at $\phi = \pi/2$ is $5I$.

Example 4 — [JEE Advanced Level]

Two identical sonometer wires are stretched by the same tension of 100 N and vibrate in unison at a fundamental frequency of 400 Hz. The tension in one of the wires is now increased by 1 N. How many beats per second are heard when both are plucked? (Assume the tension change is small.)

The insight: Both wires start at the same frequency (zero beats). Increasing one wire's tension nudges its frequency up; the small mismatch produces beats.

Step 1 — frequency-tension relation. For a wire's fundamental, $f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$, so $f \propto \sqrt{T}$.

Step 2 — differentiate (small-change approximation). Taking the logarithm, $\ln f = \text{const} + \frac{1}{2} \ln T$, so:

$$\frac{df}{f} = \frac{1}{2} \frac{dT}{T}$$

Step 3 — plug in. $dT = 1 \text{ N}$, $T = 100 \text{ N}$, $f = 400 \text{ Hz}$:

$$df = \frac{1}{2} \cdot \frac{dT}{T} \cdot f = \frac{1}{2} \cdot \frac{1}{100} \cdot 400 = 2 \text{ Hz}$$

The new frequency of the stretched wire is 402 Hz; the other stays at 400 Hz.

Step 4 — beat frequency:

$$f_{\text{beat}} = |402 - 400| = 2 \text{ beats per second}$$

Answer: 2 beats per second.

Reasoning check: A 1% tension increase gives only a 0.5% frequency increase (the square root halves it) — exactly why a tiny tension change produces a small, audible beat rather than a wild pitch jump.

Section 5: Practice Exercises

- [CBSE]** Two sound waves of frequencies 200 Hz and 205 Hz are produced together. How many beats are heard in 10 s?
- [NEET]** Fork A (frequency 320 Hz) gives 5 beats/s with fork B. On filing fork B, the beat frequency increases to 8/s. Find the original frequency of B.
- [JEE Main]** Two coherent waves of amplitudes A and $3A$ interfere. Find the ratio of maximum to minimum resultant intensity.
- [JEE Main]** The speed of sound in a gas is to be found from two waves of wavelengths 1.00 m and 1.01 m that produce 4 beats per second. Find the speed of sound.
- [JEE Advanced]** A sonometer wire vibrating at 256 Hz is in unison with a tuning fork. To produce 4 beats per second, by what percentage must the wire's tension be increased? (Assume small change.)

Answers: **1.** 50 beats. **2.** Filing raises B's frequency; beats increased, so B was below A $\rightarrow f_B = 315 \text{ Hz}$. **3.** $I_{\text{max}} : I_{\text{min}} = (3A + A)^2 : (3A - A)^2 = 16 : 4 = 4 : 1$. **4.** $v = f\lambda$ for both; $v/1.00 - v/1.01 = 4 \Rightarrow v \approx 404 \text{ m s}^{-1}$. **5.** $df/f = \frac{1}{2} dT/T \Rightarrow 4/256 = \frac{1}{2}(dT/T) \Rightarrow dT/T = 8/256 = 3.125\%$.

Section 6: Crack the MCQ

Q1. Two tuning forks produce 5 beats per second. The frequency of one is 480 Hz. The frequency of the other could be:

- (a) 475 Hz only
- (b) 485 Hz only

- (c) either 475 Hz or 485 Hz
- (d) 480 Hz

Correct: (c). With only the beat count given, the second fork is 480 ± 5 , i.e. 475 or 485 Hz. Without a loading/filing clue, *both* are valid.

- (a), (b) traps: Picking just one without any extra information — there's nothing to break the tie.
- (d) trap: If both were 480 Hz there would be *zero* beats.

Q2. When two waves of nearly equal frequencies superpose, the beat frequency is equal to:

- (a) the sum of the two frequencies
- (b) the difference of the two frequencies
- (c) half the difference of the two frequencies
- (d) the average of the two frequencies

Correct: (b). $f_{\text{beat}} = |f_1 - f_2|$.

- (c) trap: This is the *envelope* frequency. Students forget the factor of 2 from loudness depending on $|\text{amplitude}|$.
- (d) trap: The average $\frac{f_1 + f_2}{2}$ is the frequency of the *note you hear*, not the beat rate.
- (a) trap: Confusing superposition with simple addition of frequencies, which is never physical here.

Q3. For two coherent sources to produce a steady, observable interference pattern, they must have:

- (a) the same amplitude only
- (b) the same frequency and a constant phase difference
- (c) slightly different frequencies
- (d) the same intensity and opposite phase

Correct: (b). Coherence = identical frequency + a phase relationship that does not drift.

- (a) trap: Equal amplitude makes the dark fringes perfectly dark, but it is **not required** for a pattern to exist.
- (c) trap: Slightly different frequencies give *beats* (a time effect), not a steady spatial interference pattern.
- (d) trap: Opposite phase gives cancellation at that point, not the *condition* for a sustained pattern.

Q4. Tuning fork A gives 4 beats per second with fork B ($f_B = 300$ Hz). When A is loaded with wax, the beat frequency becomes 2 per second. The original frequency of

A is:

- (a) 296 Hz
- (b) 304 Hz
- (c) 302 Hz
- (d) 298 Hz

Correct: (b). Candidates: $300 \pm 4 = 304$ or 296. Wax lowers A's frequency; beats *decreased*, so A must be *above* B. Hence 304 Hz.

- (a) *trap*: If A were 296, loading would push it further from 300 $\rightarrow$ beats would *increase*, contradicting the data.
- (c), (d) *traps*: These (302, 298) are the frequencies A becomes *after* loading, not the original — a classic “answered the wrong stage” slip.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Halving the beat frequency.** $f_{\text{beat}} = |f_1 - f_2|$, **not** $\frac{|f_1 - f_2|}{2}$. The half belongs to the amplitude *envelope*; loudness throbs at twice that.
2. **Mixing up loading and filing.** Wax *lowers* frequency; filing *raises* it. Memorize one and the other follows. Getting this backward flips your entire $\pm$ resolution.
3. **Adding intensities for coherent waves.** For coherent sources you **cannot** just write $I = I_1 + I_2$ — the interference term $2\sqrt{I_1 I_2} \cos \phi$ matters. (Only for *incoherent* sources do intensities simply add.)
4. **Confusing the beat scenario with the interference scenario.** Same frequency $\rightarrow$ spatial interference pattern. *Nearly* equal frequencies $\rightarrow$ beats in time. They are different phenomena with different formulas.

Pro-Tip: For loading/filing problems, use one clean rule and skip the case-by-case reasoning: “Wax lowers; if the beat count moves the **same** way as the change, the original was on the **far** side; if it moves the **opposite** way, the original was on the **near** side.” Faster, once you trust it: just ask “did the fork move toward or away from the standard?”

Section 8: Cheat Sheet Box

SUPERPOSITION & BEATS — QUICK RECALL

- **Superposition:** $y = y_1 + y_2 + \dots$ (waves add, then pass through unchanged).
- **Resultant amplitude:** $A_{\text{res}} = \sqrt{A_1^2 + A_2^2 + 2A_1 A_2 \cos \phi}$
- **Intensity:** $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$; $I_{\text{max}} = (\sqrt{I_1} + \sqrt{I_2})^2$, $I_{\text{min}} = (\sqrt{I_1} - \sqrt{I_2})^2$
- **Constructive:** $\Delta x = n\lambda$. **Destructive:** $\Delta x = (2n + 1)\frac{\lambda}{2}$.
- **Beats:** $f_{\text{beat}} = |f_1 - f_2|$, $t_{\text{beat}} = \frac{1}{|f_1 - f_2|}$.

- **Tension link:** $f \propto \sqrt{T} \Rightarrow \frac{df}{f} = \frac{1}{2} \frac{dT}{T}$.

Memory aids:

- “Wax weighs it down” — wax lowers frequency; filing files it up.
- “Beat = the difference, plain and simple” — no halving.
- “Same $f \rightarrow$ pattern in space; close $f \rightarrow$ throb in time.”

Subtopic 03: Standing Waves and Acoustics

Exam Relevance: One of the highest-yield subtopics in the chapter. JEE Main reliably asks 1–2 questions on organ-pipe harmonics, the closed-vs-open pipe distinction, or the “three successive resonance frequencies” trick. NEET favors conceptual questions on nodes/antinodes and which harmonics a closed pipe produces. JEE Advanced builds resonance-tube and end-correction problems, often multi-step. CBSE Boards regularly ask the derivation of a stationary wave and the harmonic series of a string or pipe for 3–5 marks.

Section 1: Concept Intuition

Pluck a sitar string and watch closely. The string does not send a wave racing off to one side — instead the whole string seems to vibrate *in place*, with some points swinging wildly and other points sitting perfectly still. The pattern stands there, shimmering, going nowhere. That is a **standing wave** (also called a *stationary wave*).

Where does it come from? When you pluck the string, a wave travels to the fixed end, *reflects*, and travels back. Now you have two identical waves moving in **opposite directions** along the same string. They superpose. At certain points the two waves always cancel — these dead-still points are called **nodes**. Halfway between them, the two waves always reinforce, swinging with maximum amplitude — these are **antinodes**.

$$\text{Standing wave} = \text{forward wave} + \text{its reflection}$$

Think about it this way... A travelling wave is like a train moving down a track — the disturbance carries energy from station to station. A standing wave is like a skipping rope two children are twirling: the rope clearly oscillates, but the *pattern* (one big loop, or two, or three) just sits between their hands. No energy streams off the ends — it sloshes back and forth, trapped between the boundaries. The nodes are where each child’s hand holds the rope still; the antinode is the fat middle of the loop.

The deep idea of **acoustics** is that a bounded medium — a string clamped at both ends, or an air column inside a flute — can only support standing waves whose pattern *fits neatly* between its boundaries. A string fixed at both ends must have a node at each end. Only certain wavelengths satisfy this: ones where a whole number of half-loops fit in the length L . These special, allowed frequencies are the **harmonics**, and the lowest one is the **fundamental** (or first harmonic). It’s exactly why a shorter string or a shorter air column gives a higher pitch — fewer centimetres means shorter allowed wavelengths means higher frequency. A bansuri (flute) player raises the pitch by uncovering holes, which shortens the effective air column.

Resonance is the partner concept. If you drive an air column (say, with a tuning fork) at exactly one of its natural frequencies, the sound suddenly swells — the column “agrees” to vibrate and energy pours in efficiently. It’s the same physics as pushing a child on a swing: push at the swing’s natural rhythm and it climbs higher and higher.

For air columns in pipes, the boundary rule is about how the air can move:

- A **closed end** (sealed) forces the air to be still → a **displacement node**.
- An **open end** lets the air move freely → a **displacement antinode**.

This single rule produces two very different instruments:

- An **open pipe** (open at both ends) has antinodes at both ends and supports **all** harmonics: $f, 2f, 3f, \dots$
- A **closed pipe** (closed at one end) has a node at the closed end and an antinode at the open end. It supports **only odd** harmonics: $f, 3f, 5f, \dots$ — and its fundamental is **one octave lower** than an open pipe of the same length.

Limiting Conditions & Assumptions:

- A standing wave needs **two coherent waves travelling in opposite directions** — usually a wave and its own reflection. A single travelling wave never forms one.
- The ideal organ-pipe formulas assume the antinode sits *exactly* at the open end. In reality it bulges slightly beyond, by the **end correction** $e \approx 0.6r$ (where r is the pipe radius). Precise experiments must account for it.
- Standing-wave nodes are *perfectly* still only in an ideal, lossless medium.

Section 2: Key Formulas & Definitions

Equation of a standing wave (from superposing $y_1 = a \sin(kx - \omega t)$ and $y_2 = a \sin(kx + \omega t)$):

$$y = 2a \sin(kx) \cos(\omega t)$$

The amplitude $R = 2a \sin(kx)$ depends on *position* — that's the signature of a standing wave.

Node and antinode positions:

$$\text{Nodes: } x = \frac{n\lambda}{2} \quad (n = 0, 1, 2, \dots), \quad \text{Antinodes: } x = \frac{(2n+1)\lambda}{4}$$

- Distance between two adjacent nodes (or antinodes) = $\frac{\lambda}{2}$
- Distance between a node and the nearest antinode = $\frac{\lambda}{4}$

String fixed at both ends (length L): nodes at both ends.

$$L = \frac{n\lambda}{2} \Rightarrow f_n = \frac{nv}{2L}, \quad n = 1, 2, 3, \dots \text{ (all harmonics)}$$

$$\text{Fundamental: } f_1 = \frac{v}{2L} = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

Open organ pipe (both ends open): antinodes at both ends.

$$f_n = \frac{nv}{2L}, \quad n = 1, 2, 3, \dots \text{ (all harmonics)}$$

Closed organ pipe (one end closed): node at closed end, antinode at open end.

$$f_n = \frac{(2n-1)v}{4L}, \quad n = 1, 2, 3, \dots \text{ (only odd harmonics)}$$

End correction: effective length $L_{\text{eff}} = L + e$, with $e \approx 0.6r$ (open end) per open end.

Resonance tube (two successive resonances at lengths l_1, l_2):

$$\lambda = 2(l_2 - l_1), \quad v = 2f(l_2 - l_1), \quad e = \frac{l_2 - 3l_1}{2}$$

Term	Meaning
Node	Point of permanently zero displacement
Antinode	Point of maximum displacement amplitude
Fundamental (1st harmonic)	Lowest natural frequency, f_1
n-th harmonic	Frequency $n f_1$
m-th overtone	The $(m+1)$ -th allowed frequency above the fundamental

Overtone-harmonic map (memorize this!):

- *Open pipe / string:* 1st overtone = 2nd harmonic; 2nd overtone = 3rd harmonic.
- *Closed pipe:* 1st overtone = **3rd** harmonic; 2nd overtone = **5th** harmonic.

Section 3: CBSE Board Derivation — Stationary Wave and the Harmonics of a String

Part A: Forming the standing wave.

Consider a wave travelling in the $+x$ direction and its reflection travelling in the $-x$ direction, each of amplitude a :

$$y_1 = a \sin(kx - \omega t), \quad y_2 = a \sin(kx + \omega t)$$

Why two opposite waves? A wave hitting a fixed boundary bounces back. The original and the reflected wave are what superpose to make the standing pattern.

By the principle of superposition:

$$y = y_1 + y_2 = a[\sin(kx - \omega t) + \sin(kx + \omega t)]$$

Using $\sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$:

$$\boxed{y = 2a \sin(kx) \cos(\omega t)}$$

Read the physics. Unlike a travelling wave (which has the form $\sin(kx - \omega t)$, where x and t are locked together), here space and time have **separated**. Every particle oscillates in time as $\cos(\omega t)$, but its *amplitude*, $2a \sin(kx)$, is fixed by *where it sits*. The pattern doesn't travel — it stands.

Nodes (zero amplitude): $\sin(kx) = 0 \Rightarrow kx = n\pi \Rightarrow x = \frac{n\lambda}{2}$. **Antinodes** (max amplitude $2a$): $\sin(kx) = \pm 1 \Rightarrow x = \frac{(2n+1)\lambda}{4}$.

Adjacent nodes are $\frac{\lambda}{2}$ apart; a node and its neighbouring antinode are $\frac{\lambda}{4}$ apart.

Part B: Harmonics of a string fixed at both ends.

A string clamped at $x = 0$ and $x = L$ **must** have a node at each end. The end $x = 0$ is already a node. For $x = L$ to also be a node:

$$\sin(kL) = 0 \Rightarrow kL = n\pi \Rightarrow \frac{2\pi}{\lambda}L = n\pi \Rightarrow L = \frac{n\lambda}{2}$$

Why this is the whole point: the boundary conditions act like a filter — only wavelengths that fit a whole number of half-loops into L are allowed. Everything else dies out.

So the allowed wavelengths are $\lambda_n = \frac{2L}{n}$, and using $v = f\lambda$:

$$f_n = \frac{nv}{2L}, \quad n = 1, 2, 3, \dots$$

The fundamental is $f_1 = \frac{v}{2L}$, and the string supports **every** integer multiple of it. Substituting $v = \sqrt{T/\mu}$ gives the sonometer's master formula $f_1 = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A stretched wire of length 0.5 m is fixed at both ends. The speed of a transverse wave on it is 200 m s^{-1} . Find (a) its fundamental frequency and (b) the frequencies of its first two overtones.

Given: $L = 0.5 \text{ m}$, $v = 200 \text{ m s}^{-1}$.

(a) Fundamental:

$$f_1 = \frac{v}{2L} = \frac{200}{2(0.5)} = \frac{200}{1} = 200 \text{ Hz}$$

(b) A string supports all harmonics, so the first overtone is the 2nd harmonic and the second overtone is the 3rd harmonic:

$$f_2 = 2f_1 = 400 \text{ Hz}, \quad f_3 = 3f_1 = 600 \text{ Hz}$$

Answer: Fundamental = 200 Hz; first two overtones = 400 Hz and 600 Hz.

Example 2 — [NEET Speed Trap]

The fundamental frequency of a closed organ pipe is 200 Hz. What is the frequency of its first overtone?

The trap: Students reflexively double it to 400 Hz (thinking “first overtone = 2nd harmonic”). For a **closed** pipe that is wrong — even harmonics simply do not exist.

The fast way: A closed pipe produces only **odd** harmonics: $f_1, 3f_1, 5f_1, \dots$. The first overtone is therefore the **3rd harmonic**:

$$f_{1\text{st overtone}} = 3f_1 = 3 \times 200 = 600 \text{ Hz}$$

Answer: 600 Hz.

Mental check: “Closed pipe = odd-only. First overtone jumps straight to $3f_1$, skipping $2f_1$ entirely.”

Example 3 — [JEE Main Level]

For a certain organ pipe, three successive resonance frequencies are observed at 250 Hz, 350 Hz, and 450 Hz. The speed of sound in air is 340 m s^{-1} . (a) Is the pipe open or closed? (b) Find its fundamental frequency. (c) Find its length.

(a) Identify the pipe type. The gap between successive resonances is constant:

$$350 - 250 = 100 \text{ Hz}, \quad 450 - 350 = 100 \text{ Hz}$$

This gap equals the fundamental for an open pipe, or *twice* the fundamental for a closed pipe. Test “open”: if $f_1 = 100$, then $250/100 = 2.5$ — not an integer, so 250 Hz can’t be a harmonic of a 100 Hz open pipe. So it must be **closed**, where the gap is $2f_1$.

(b) Fundamental. For a closed pipe, successive (odd) harmonics differ by $2f_1$:

$$2f_1 = 100 \implies f_1 = 50 \text{ Hz}$$

Check: $250 = 5f_1$, $350 = 7f_1$, $450 = 9f_1$ — the 5th, 7th, 9th harmonics, all odd. ✓

(c) Length. For a closed pipe, $f_1 = \frac{v}{4L}$:

$$L = \frac{v}{4f_1} = \frac{340}{4 \times 50} = \frac{340}{200} = 1.7 \text{ m}$$

Answer: Closed pipe; fundamental = 50 Hz; length = 1.7 m.

Example 4 — [JEE Advanced Level]

In a resonance-tube experiment, a tuning fork of frequency 480 Hz produces its first resonance when the air column is 17 cm long and its second resonance at 52 cm. Find (a) the speed of sound in air and (b) the end correction of the tube.

The insight: A resonance tube is a closed pipe (water surface = closed end). The first resonance occurs at $\lambda/4$ and the second at $3\lambda/4$ — **measured from the antinode, which lies slightly beyond the open end** by the end correction e . So:

$$l_1 + e = \frac{\lambda}{4}, \quad l_2 + e = \frac{3\lambda}{4}$$

(a) Speed of sound. Subtract the two equations — the unknown e cancels:

$$l_2 - l_1 = \frac{3\lambda}{4} - \frac{\lambda}{4} = \frac{\lambda}{2}$$

$$\lambda = 2(l_2 - l_1) = 2(52 - 17) = 2(35) = 70 \text{ cm} = 0.70 \text{ m}$$

$$v = f\lambda = 480 \times 0.70 = 336 \text{ m s}^{-1}$$

(b) End correction. From $l_1 + e = \frac{\lambda}{4} = \frac{70}{4} = 17.5 \text{ cm}$:

$$e = 17.5 - 17 = 0.5 \text{ cm}$$

(Equivalently, $e = \frac{l_2 - 3l_1}{2} = \frac{52 - 51}{2} = 0.5 \text{ cm}$. ✓)

Answer: $v = 336 \text{ m s}^{-1}$; end correction $e = 0.5 \text{ cm}$.

Reasoning check: Subtracting the two resonance conditions is the master move — it kills the end correction so you get the wavelength cleanly, then you back out e afterward.

Section 5: Practice Exercises

- [CBSE]** An open organ pipe of length 0.34 m is sounded. If the speed of sound is 340 m s^{-1} , find its fundamental frequency.
- [NEET]** A closed organ pipe and an open organ pipe have the same length. Find the ratio of their fundamental frequencies.
- [JEE Main]** A sonometer wire of length 1 m and linear mass density 0.01 kg m^{-1} is under a tension of 100 N. Find its fundamental frequency.
- [JEE Main]** The second overtone of a closed organ pipe has the same frequency as the third harmonic of an open pipe. Find the ratio of their lengths (closed : open).
- [JEE Advanced]** A string fixed at both ends vibrates in 3 loops (its 3rd harmonic) at 420 Hz. What is its fundamental frequency, and at what frequency would it vibrate in 5 loops?

Answers: 1. $f_1 = v/2L = 340/0.68 = 500 \text{ Hz}$. 2. $f_{\text{closed}} : f_{\text{open}} = \frac{v}{4L} : \frac{v}{2L} = 1 : 2$. 3. $f_1 = \frac{1}{2L} \sqrt{T/\mu} = \frac{1}{2} \sqrt{100/0.01} = \frac{1}{2}(100) = 50 \text{ Hz}$. 4. Closed 2nd overtone = 5th harmonic = $\frac{5v}{4L_c}$; open 3rd harmonic = $\frac{3v}{2L_o}$. Equate: $\frac{5}{4L_c} = \frac{3}{2L_o} \Rightarrow \frac{L_c}{L_o} = \frac{5}{6}$. 5.

$$f_1 = 420/3 = 140 \text{ Hz}; 5\text{th harmonic} = 5 \times 140 = 700 \text{ Hz}.$$

Section 6: Crack the MCQ

Q1. In a stationary wave, the distance between a node and the nearest antinode is:

- (a) λ
- (b) $\lambda/2$
- (c) $\lambda/4$
- (d) $\lambda/8$

Correct: (c). Adjacent nodes are $\lambda/2$ apart, and the antinode sits exactly halfway between them, so node-to-antinode is $\lambda/4$.

- (b) *trap*: That's node-to-node (or antinode-to-antinode) — confusing the full half-wavelength with half of it.
 - (a) *trap*: That's a full wavelength — two complete loops away.
 - (d) *trap*: Over-halving, a guess with no physical basis.
-

Q2. Which harmonics are present in a pipe closed at one end?

- (a) all integer harmonics
- (b) only even harmonics
- (c) only odd harmonics
- (d) only the fundamental

Correct: (c). A closed end is a node, an open end an antinode; this geometry permits only $f_1, 3f_1, 5f_1, \dots$

- (a) *trap*: True for *open* pipes and strings, not closed pipes.
 - (b) *trap*: Exactly backwards — even harmonics are the *forbidden* ones.
 - (d) *trap*: A closed pipe has many overtones, just only the odd ones.
-

Q3. A standing wave is formed by the superposition of:

- (a) two waves of slightly different frequencies travelling in the same direction
- (b) two identical waves travelling in opposite directions
- (c) two waves of different amplitudes travelling in the same direction
- (d) a single wave reflected at a free end only

Correct: (b). Counter-propagating identical waves (typically a wave and its reflection) produce the standing pattern.

- (a) *trap*: That setup gives *beats*, not a standing wave.
- (c) *trap*: Same-direction superposition gives a single travelling wave of modified amplitude, not a standing one.

- (d) *trap*: Reflection is the usual *source* of the second wave, but it's the superposition of the two opposite waves that matters — and reflection happens at both fixed and free ends.

Q4. An open pipe and a closed pipe have the same fundamental frequency. The ratio of the length of the open pipe to that of the closed pipe is:

- (a) 1 : 1
- (b) 2 : 1
- (c) 1 : 2
- (d) 4 : 1

Correct: (b). Open: $f_1 = \frac{v}{2L_o}$. Closed: $f_1 = \frac{v}{4L_c}$. Setting them equal: $\frac{v}{2L_o} = \frac{v}{4L_c} \Rightarrow L_o = 2L_c$.

- (c) *trap*: The ratio inverted — a sign that the formulas got swapped.
- (a) *trap*: Assuming “same frequency means same length,” ignoring that a closed pipe is more efficient (lower pitch per metre).
- (d) *trap*: Squaring or double-applying the factor of 2.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Mislabelling overtones in closed pipes.** First overtone of a closed pipe is the **3rd** harmonic ($3f_1$), not the 2nd. “Overtone” counts allowed frequencies above the fundamental; for closed pipes those skip the evens.
2. **Forgetting the end correction.** In precise resonance-tube problems, the antinode lies *beyond* the open end by $e \approx 0.6r$. Ignoring it gives a slightly wrong speed of sound — but the *difference* of two resonance lengths cancels it, so use that.
3. **Confusing node/antinode spacing.** Node-to-node = $\lambda/2$; node-to-antinode = $\lambda/4$. Sketch one loop and label it — it prevents half the errors in this subtopic.
4. **Mixing up displacement and pressure patterns.** At a closed end: displacement *node* but pressure *antinode*. They are always opposite. Many “find the pressure antinode” questions trip students who only think in displacement.

Pro-Tip: For “three successive resonance frequencies,” just take the **common difference** Δ . If the smallest frequency $\div \Delta$ is a whole number $\rightarrow$ **open pipe**, $f_1 = \Delta$. If not $\rightarrow$ **closed pipe**, $f_1 = \Delta/2$. This identifies the pipe and its fundamental in about five seconds, no diagrams needed.

Section 8: Cheat Sheet Box

STANDING WAVES & ACOUSTICS — QUICK RECALL

- **Standing wave:** $y = 2a \sin(kx) \cos(\omega t)$ — amplitude depends on *position*.
- **Spacing:** node-node = $\frac{\lambda}{2}$; node-antinode = $\frac{\lambda}{4}$.
- **String / open pipe:** $f_n = \frac{nv}{2L}$, all harmonics ($n = 1, 2, 3, \dots$).
- **Closed pipe:** $f_n = \frac{(2n-1)v}{4L}$, odd harmonics only.
- **Open : closed fundamental (same length) = 2 : 1.**
- **Resonance tube:** $\lambda = 2(l_2 - l_1)$, $v = 2f(l_2 - l_1)$, $e = \frac{l_2 - 3l_1}{2}$.

Memory aids:

- “Closed = odd only; open = all.”
- “Closed pipe’s first overtone leapfrogs to $3f_1$.”
- “Subtract the resonance lengths — the end correction disappears.”

Subtopic 04: Doppler Effect

Exam Relevance: A guaranteed scorer and a perennial favourite. JEE Main almost always carries one Doppler problem — source/observer moving, both moving, or a reflection (wall/echo) setup. NEET loves the conceptual asymmetry question (source-moving vs observer-moving) and the simple approaching-source numerical. JEE Advanced layers in wind, angles, or double-Doppler echoes. CBSE Boards typically ask the derivation for a moving source and a stationary observer for 2–3 marks, plus a short numerical.

Section 1: Concept Intuition

You're standing at a railway crossing. A train rushes toward you, horn blaring at a high, urgent pitch — *eeeeeeee*. The instant it passes and starts speeding away, the pitch suddenly drops — *eeeeoooooww*. The driver, sitting inside, hears no change at all. The horn itself never changed its note. So what happened?

This apparent change in the **observed frequency** of a wave, caused by the **relative motion** between the source and the observer, is the **Doppler Effect**.

Think about it this way... Imagine the source spitting out wavefronts like ripples, one every fraction of a second. If the source is rushing *toward* you, each new wavefront is launched a little closer to you than the last — so the wavefronts bunch up. Bunched-up waves mean a *shorter* wavelength reaching you, which your ear hears as a *higher* pitch. When the source races *away*, each wavefront is launched farther back, the waves get stretched out, the wavelength grows, and you hear a *lower* pitch.

The same logic works if *you* move instead. Walk toward a stationary source and you sweep through more wavefronts per second than if you stood still — higher frequency. Walk away and you meet fewer per second — lower frequency.

A clean way to lock the whole effect into one sentence:

Toward → frequency up. Away → frequency down.

Every Doppler problem, no matter how complicated, must agree with this gut check. If your formula spits out a *higher* frequency when source and observer are *separating*, you've made a sign error — full stop.

There's a beautiful subtlety here, and JEE/NEET examiners adore it: **for sound, moving the source is NOT the same as moving the observer**. When the *source* moves, it physically changes the *wavelength* in the medium (it crowds or stretches the ripples). When the *observer* moves, the wavelength in the air is unchanged — instead the observer changes the *effective speed* at which they meet the waves. Two different mechanisms, two different formulas, two slightly different answers even at the same speed. This asymmetry exists because sound needs a **medium** (air), and that medium is a fixed stage everyone's motion is measured against. Light, having no medium, behaves symmetrically — only the *relative* velocity matters.

Doppler is everywhere: the highway **radar speed gun** bounces microwaves off your car and reads the frequency shift to catch over-speeding; a **sonar** on a submarine measures the shift to track another vessel; and astronomers measure the **redshift** of distant galaxies — light stretched to longer (redder) wavelengths — which told us the universe is expanding.

Limiting Conditions & Assumptions:

- The standard formulas assume motion is **along the line joining source and observer**. If velocity is at an angle, only the **component along the line of sight** matters. At the moment of closest approach (motion exactly sideways), there is **no Doppler shift** for sound.
- For sound, the **medium is the reference frame** — so a **wind** changes the effective sound speed and must be added in.
- The source speed must be **less than the speed of sound** ($v_s < v$). Once a source reaches or exceeds the sound speed, the wavefronts pile into a **shock wave** (sonic boom / Mach cone) and the ordinary formula no longer applies.

Section 2: Key Formulas & Definitions

General Doppler formula for sound (source speed v_s , observer speed v_o , sound speed v):

$$f' = f \left(\frac{v \pm v_o}{v \mp v_s} \right)$$

Sign rule — anchor everything to “toward raises pitch”:

- **Numerator** v_o : use $+v_o$ if the observer moves *toward* the source; $-v_o$ if *away*.
- **Denominator** v_s : use $-v_s$ if the source moves *toward* the observer; $+v_s$ if *away*.

Situation	Formula	Result
Source toward, observer still	$f' = f \frac{v}{v - v_s}$	↑
Source away, observer still	$f' = f \frac{v}{v + v_s}$	↓
Observer toward, source still	$f' = f \frac{v + v_o}{v}$	↑
Observer away, source still	$f' = f \frac{v - v_o}{v}$	↓
Both approaching	$f' = f \frac{v + v_o}{v - v_s}$	↑

Effect of wind (wind speed w along the direction of sound travel, source → observer): replace v with $(v + w)$ everywhere; if the wind blows the other way, use $(v - w)$.

Doppler effect for light (relative speed $u \ll c$, where c is the speed of light):

$$f' = f \left(1 \pm \frac{u}{c} \right), \quad \frac{\Delta\lambda}{\lambda} = \frac{u}{c}$$

- Approaching $\rightarrow$ frequency up, wavelength down: **blueshift**.
- Receding $\rightarrow$ frequency down, wavelength up: **redshift**.

Symbol	Meaning	SI Unit
f	True (emitted) frequency	Hz
f'	Apparent (observed) frequency	Hz
v	Speed of sound in the medium	m s^{-1}
v_s, v_o	Speeds of source, observer	m s^{-1}
w	Wind speed	m s^{-1}
$\Delta\lambda$	Wavelength shift (light)	m

Section 3: CBSE Board Derivation — Doppler Formulas for Sound

We derive the two basic cases and, crucially, *see why they differ*.

Case A: Source moving toward a stationary observer.

Let the source emit sound of frequency f , so its period is $T = \frac{1}{f}$, and let sound travel at speed v . In a stationary source, one wavelength would be $\lambda = vT$.

The key event: during one period T , the source itself advances a distance $v_s T$ toward the observer. So by the time it emits the next crest, it has crept forward — and the gap between consecutive crests (the wavelength reaching the observer) is **squeezed**:

$$\lambda' = vT - v_s T = (v - v_s)T = \frac{v - v_s}{f}$$

Why this is the heart of it: the source physically alters the wavelength in the air. The sound still travels at speed v (the medium sets that), so the observer measures:

$$f' = \frac{v}{\lambda'} = \frac{v}{\frac{v - v_s}{f}} = \boxed{f \left(\frac{v}{v - v_s} \right)}$$

Since $v - v_s < v$, we get $f' > f$ — higher pitch, as expected for approach. ✓

Case B: Observer moving toward a stationary source.

Now the source is still, so the wavelength in the air is the **normal** $\lambda = \frac{v}{f}$ — unchanged. But the observer is rushing into the waves at speed v_o .

The key event: relative to the moving observer, the sound now approaches at an effective speed $v + v_o$ (their own speed adds to the wave's). The number of wavefronts they

sweep past per second is therefore:

$$f' = \frac{\text{effective speed}}{\text{wavelength}} = \frac{v + v_o}{\lambda} = \frac{v + v_o}{\frac{v}{f}} = \boxed{f \left(\frac{v + v_o}{v} \right)}$$

Again $f' > f$. ✓

Now compare — the asymmetry made visible. Suppose $v_s = v_o = u$. Then:

$$f'_{\text{source moving}} = f \frac{v}{v - u}, \quad f'_{\text{observer moving}} = f \frac{v + u}{v}$$

These are **not equal** (the source-moving result is slightly larger). The reason is now transparent: a moving source changes λ , a moving observer changes the effective speed. Combining both cases gives the general result $f' = f \frac{v + v_o}{v - v_s}$.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A train approaches a stationary listener at a railway platform with a speed of 30 m s^{-1} , sounding a horn of frequency 500 Hz. If the speed of sound in air is 330 m s^{-1} , find the frequency heard by the listener.

Given: $f = 500 \text{ Hz}$, $v_s = 30 \text{ m s}^{-1}$ (toward observer), $v_o = 0$, $v = 330 \text{ m s}^{-1}$.

Source moving toward a stationary observer:

$$f' = f \left(\frac{v}{v - v_s} \right) = 500 \left(\frac{330}{330 - 30} \right) = 500 \left(\frac{330}{300} \right) = 500 \times 1.1 = 550 \text{ Hz}$$

Answer: 550 Hz — higher than the true 500 Hz, as expected for an approaching source.

Example 2 — [NEET Speed Trap]

A source and an observer are separated by some distance. In Case 1, the source moves toward a stationary observer at 30 m s^{-1} . In Case 2, the observer moves toward a stationary source at 30 m s^{-1} . The true frequency is 300 Hz and the speed of sound is 330 m s^{-1} . Are the two observed frequencies equal?

The trap: “Same speed, same approach — so same frequency, right?” **Wrong.** Sound Doppler is asymmetric.

Case 1 (source moving):

$$f'_1 = f \frac{v}{v - v_s} = 300 \times \frac{330}{300} = 330 \text{ Hz}$$

Case 2 (observer moving):

$$f'_2 = f \frac{v + v_o}{v} = 300 \times \frac{360}{330} = 327.3 \text{ Hz}$$

Answer: No — $330 \text{ Hz} \neq 327.3 \text{ Hz}$. A moving source produces a *slightly larger* shift than a moving observer at the same speed, because the source physically compresses the wavelength while the observer merely changes the rate of meeting waves.

Mental note: “Source-moving beats observer-moving” for approach. Only when $v_s, v_o \ll v$ do the two nearly coincide.

Example 3 — [JEE Main Level]

An ambulance with a siren of frequency 600 Hz moves toward a motorcyclist at 40 m s^{-1} . The motorcyclist rides toward the ambulance at 20 m s^{-1} . Taking the speed of sound as 340 m s^{-1} , find the frequency the motorcyclist hears.

Given: $f = 600 \text{ Hz}$, $v_s = 40 \text{ m s}^{-1}$ (toward), $v_o = 20 \text{ m s}^{-1}$ (toward), $v = 340 \text{ m s}^{-1}$.

Both approaching, so numerator gets $+v_o$ and denominator gets $-v_s$:

$$f' = f \left(\frac{v + v_o}{v - v_s} \right) = 600 \left(\frac{340 + 20}{340 - 40} \right) = 600 \left(\frac{360}{300} \right) = 600 \times 1.2 = 720 \text{ Hz}$$

Answer: 720 Hz . (Sanity check: both moving toward each other → frequency must rise above 600 . ✓)

Example 4 — [JEE Advanced Level]

A car moving toward a large vertical wall at 20 m s^{-1} continuously sounds a horn of frequency 320 Hz . The speed of sound is 340 m s^{-1} . Find (a) the frequency of the echo heard by the driver, and (b) the beat frequency between the direct horn sound and the echo.

The insight — a double Doppler. The wall first acts as a *moving-source* → *stationary-observer* problem, then re-emits as a *stationary-source* → *moving-observer* problem. The driver is both the original source *and* the final listener.

Step 1 — sound reaches the wall. The wall is a stationary observer; the car is a source approaching at $u = 20 \text{ m s}^{-1}$:

$$f_{\text{wall}} = f \frac{v}{v - u} = 320 \times \frac{340}{320} = 340 \text{ Hz}$$

Step 2 — the wall reflects. The wall now behaves as a *stationary source* emitting $f_{\text{wall}} = 340 \text{ Hz}$. The driver is an observer moving toward it at $u = 20 \text{ m s}^{-1}$:

$$f_{\text{echo}} = f_{\text{wall}} \frac{v + u}{v} = 340 \times \frac{360}{340} = 360 \text{ Hz}$$

(Combined shortcut: $f_{\text{echo}} = f \frac{v+u}{v-u} = 320 \times \frac{360}{320} = 360 \text{ Hz.}$)

(a) Echo frequency = 360 Hz.

Step 3 — beats. The driver also hears the horn directly at its true 320 Hz (no relative motion between driver and their own horn). The beat frequency between the direct sound and the echo is:

$$f_{\text{beat}} = f_{\text{echo}} - f = 360 - 320 = 40 \text{ Hz}$$

(b) Beat frequency = 40 Hz.

Reasoning check: The general result is $f_{\text{beat}} = \frac{2uf}{v-u} = \frac{2(20)(320)}{320} = 40 \text{ Hz}$ — the beats grow as the car speeds up, which is exactly how some old “speed by sound” gadgets worked.

Section 5: Practice Exercises

- [CBSE]** A car sounds a horn of frequency 400 Hz while moving *away* from a stationary listener at 20 m s^{-1} . Find the observed frequency. (Speed of sound = 340 m s^{-1} .)
- [NEET]** A whistle of frequency 500 Hz is stationary. An observer runs *away* from it at 34 m s^{-1} . What frequency does the observer hear? (Speed of sound = 340 m s^{-1} .)
- [JEE Main]** Two trains move toward each other, each at 20 m s^{-1} . One sounds a whistle of 660 Hz. What frequency does a passenger on the other train hear? (Speed of sound = 340 m s^{-1} .)
- [JEE Main]** A source of frequency 400 Hz moves toward a stationary observer at 30 m s^{-1} , while a wind blows at 10 m s^{-1} from the source toward the observer. Find the observed frequency. (Still-air sound speed = 340 m s^{-1} .)
- [JEE Advanced]** A distant galaxy shows the sodium line (true wavelength 589.0 nm) shifted to 589.6 nm. Is the galaxy approaching or receding, and at what speed? (Take $c = 3 \times 10^8 \text{ m s}^{-1}$.)

Answers: 1. $f' = 400 \cdot \frac{340}{360} = 377.8 \text{ Hz}$. 2. $f' = 500 \cdot \frac{340-34}{340} = 450 \text{ Hz}$. 3. $f' = 660 \cdot \frac{340+20}{340-20} = 660 \cdot \frac{360}{320} = 742.5 \text{ Hz}$. 4. Wind adds: $v \rightarrow 350$; $f' = 400 \cdot \frac{350}{350-30} = 400 \cdot \frac{350}{320} = 437.5 \text{ Hz}$. 5. Wavelength increased $\rightarrow$ **redshift** $\rightarrow$ **receding**; $u = c \frac{\Delta\lambda}{\lambda} = 3 \times 10^8 \cdot \frac{0.6}{589.0} \approx 3.06 \times 10^5 \text{ m s}^{-1} (\approx 306 \text{ km s}^{-1})$.

Section 6: Crack the MCQ

Q1. A source of sound moves toward a stationary observer. Compared to the true sound, the observer hears a sound of:

- (a) higher frequency and shorter wavelength
- (b) higher frequency and longer wavelength
- (c) lower frequency and shorter wavelength
- (d) the same frequency

Correct: (a). Approach bunches the wavefronts → shorter wavelength → higher frequency.

- (b) *trap*: Higher frequency must mean *shorter* wavelength (since v is fixed in the medium) — pairing it with longer is self-contradictory.
- (c) *trap*: Right about wavelength, wrong about frequency direction.
- (d) *trap*: “No change” ignores the relative motion entirely.

Q2. A source and an observer both move *toward* each other at the same speed. Which formula gives the observed frequency?

- (a) $f' = f \frac{v - v_o}{v + v_s}$
- (b) $f' = f \frac{v + v_o}{v - v_s}$
- (c) $f' = f \frac{v - v_o}{v - v_s}$
- (d) $f' = f \frac{v + v_o}{v + v_s}$

Correct: (b). Approach → frequency up → numerator gains $+v_o$, denominator loses v_s .

- (a) *trap*: Both signs flipped — this is the *receding* case, which lowers the pitch.
- (c), (d) *traps*: One sign correct, one wrong — the classic “half-remembered rule” mistake. Always test against “toward raises pitch.”

Q3. The Doppler effect for sound is *not* symmetric between a moving source and a moving observer because:

- (a) sound waves are longitudinal
- (b) the speed of sound depends on frequency
- (c) sound travels through a medium that serves as a fixed reference frame
- (d) the observer cannot move faster than sound

Correct: (c). A moving source changes the wavelength in the medium; a moving observer changes the effective wave speed they encounter. The medium breaks the symmetry.

- (a) *trap*: Longitudinal/transverse character is irrelevant to the asymmetry.
- (b) *trap*: In a non-dispersive medium, sound speed does *not* depend on frequency.
- (d) *trap*: A true statement in some setups, but not the reason for the asymmetry.

Q4. Light from a distant star is observed to be shifted toward longer wavelengths. This means the star is:

- (a) approaching us (blueshift)
- (b) receding from us (redshift)

- (c) stationary
- (d) heating up

Correct: (b). Longer wavelength = redshift = the source is moving away.

- (a) *trap*: Approach gives a *blueshift* (shorter wavelength) — the term is correct but matched to the wrong motion.
- (c) *trap*: A stationary star shows no shift at all.
- (d) *trap*: Confusing a Doppler shift with thermal (blackbody) colour changes — different physics entirely.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Sign-convention chaos.** The single most common error. Don't memorize four formulas blindly — anchor every problem to “**toward → up, away → down**” and assign signs so the result matches. If it doesn't, flip a sign.
2. **Treating source-moving and observer-moving as identical.** They give different answers for sound. Use v_s in the *denominator*, v_o in the *numerator* — never swap them.
3. **Ignoring the wind.** Wind changes the effective sound speed: $v \rightarrow v \pm w$ in *both* numerator and denominator. Students often forget it applies everywhere, or get its direction backward.
4. **Using the off-axis velocity directly.** Only the component of velocity *along the line joining source and observer* counts. At the moment of closest approach (pure sideways motion), there is **no shift** for sound.

Pro-Tip: For a “source approaching a wall and hearing its own echo,” skip the two-step grind and use the combined result directly: $f_{\text{echo}} = f \frac{v+u}{v-u}$, and the beat frequency with the direct sound is $f_{\text{beat}} = \frac{2uf}{v-u}$. Recognizing the wall as “stationary mirror” turns a messy double-Doppler into one clean line.

Section 8: Cheat Sheet Box

DOPPLER EFFECT — QUICK RECALL

- **Master formula (sound):** $f' = f \frac{v \pm v_o}{v \mp v_s}$
- **Sign rule:** numerator $+v_o$ if observer moves *toward*; denominator $-v_s$ if source moves *toward*. (Toward → pitch up.)
- **Both approaching:** $f' = f \frac{v+v_o}{v-v_s}$
- **Wind (speed w , source→observer):** replace $v \rightarrow v + w$ everywhere.
- **Echo off a wall (car at u):** $f_{\text{echo}} = f \frac{v+u}{v-u}$, $f_{\text{beat}} = \frac{2uf}{v-u}$.

- **Light** ($u \ll c$): $f' = f\left(1 \pm \frac{u}{c}\right)$, $\frac{\Delta\lambda}{\lambda} = \frac{u}{c}$ (red = receding, blue = approaching).

Memory aids:

- “*Toward raises, away lowers*” — the universal sanity check.
- “ v_s down below, v_o up top” — source in denominator, observer in numerator.
- “*Red runs away*” — redshift means receding.

Chapter 14: Waves — Round 2 Addendum

The chapter gives you the sinusoidal toolkit: $y = A \sin(\omega t - kx + \phi)$, $v = \sqrt{T/\mu}$, standing-wave boundary conditions, beat frequency, and the Doppler formulas for sound and light. Four JEE Advanced patterns live one step past where the chapter stops. Addendum A derives the energy and power a travelling wave carries — the chapter quotes particle velocity but never integrates it to get the $\frac{1}{2} \mu \omega^2 A^2 v$ result that every “energy in a string” problem expects. Addendum B solves the two-string junction: a wave hits a change in μ , and you need the reflected/transmitted amplitudes and energy fractions — the chapter only does fixed/free ends, which are the $Z_2/Z_1 \rightarrow \infty/0$ limits of the same algebra. Addendum C derives the wave equation $\partial^2 y / \partial t^2 = v^2 \partial^2 y / \partial x^2$ from Newton’s second law, writes the general solution $f(x - vt) + g(x + vt)$, and shows how a non-sinusoidal pulse splits — the chapter only ever writes the harmonic ansatz. Addendum D completes the acoustics side: longitudinal standing waves in rods (free-free, fixed-free), the pressure-displacement node/antinode swap, and a proper derivation of the open-end correction $e \approx 0.6r$ — the chapter states the correction but never derives it.

Numbering continues from the chapter’s Subtopic 04. Nothing here is new physics; every method is the chapter’s Newton’s laws, superposition, or boundary conditions applied one dimension deeper.

Addendum A — Power and Energy in a Travelling Wave on a String

The chapter gives particle velocity $v_p = \partial y / \partial t = A\omega \cos(\omega t - kx)$ and states $(v_p)_{\max} = A\omega$. It never asks: how much energy crosses a point per second? JEE Advanced asks exactly that — average power, instantaneous power, energy density, and the equipartition between kinetic and potential energy in a sine wave. The derivation is a three-line integral that every strong student must be able to reproduce on a blank sheet.

Method

1. Kinetic energy in a micro-element.

A string element of length dx has mass $dm = \mu dx$. Its transverse speed is $v_p = \partial y / \partial t$.

$$dK = \frac{1}{2}(\mu dx) \left(\frac{\partial y}{\partial t} \right)^2$$

2. Potential energy in the same element.

The element is stretched from length dx to $ds = \sqrt{1 + (\partial y / \partial x)^2} dx \approx (1 + \frac{1}{2}(\partial y / \partial x)^2) dx$. Work done against tension $T = T(ds - dx) = \frac{1}{2} T (\partial y / \partial x)^2 dx$. This is the elastic potential energy:

$$dU = \frac{1}{2}T \left(\frac{\partial y}{\partial x} \right)^2 dx$$

3. Use the travelling-wave relation.

For any wave of the form $y(x, t) = f(x - vt)$ or $g(x + vt)$, the chain rule gives $\partial y / \partial t = \mp v \partial y / \partial x$. With $v = \sqrt{T/\mu} \Rightarrow T = \mu v^2$, the potential energy density becomes

$$\frac{dU}{dx} = \frac{1}{2}\mu v^2 \left(\frac{\partial y}{\partial x} \right)^2 = \frac{1}{2}\mu \left(\frac{\partial y}{\partial t} \right)^2 = \frac{dK}{dx}$$

Kinetic and potential energy densities are equal at every point and every instant — this is the equipartition theorem for a travelling wave on a string.

4. Average power transmitted.

Instantaneous power = energy density $\times$ wave speed:

$$P(x, t) = \left(\frac{dK}{dx} + \frac{dU}{dx} \right) v = \mu \left(\frac{\partial y}{\partial t} \right)^2 v$$

For a harmonic wave $y = A \sin(\omega t - kx + \phi)$:

$$\begin{aligned} \frac{\partial y}{\partial t} &= A\omega \cos(\omega t - kx + \phi) \\ P(x, t) &= \mu A^2 \omega^2 v \cos^2(\omega t - kx + \phi) \end{aligned}$$

Average over one period ($\langle \cos^2 \rangle = 1/2$):

$$P_{\text{avg}} = \frac{1}{2} \mu \omega^2 A^2 v$$

Units check: μ [kg m⁻¹] $\cdot \omega^2$ [s⁻²] $\cdot A^2$ [m²] $\cdot v$ [m s⁻¹] = kg m² s⁻³ = Watt.

5. Total energy in one wavelength.

Average energy per unit length $\langle dE/dx \rangle = P_{\text{avg}}/v = \frac{1}{2} \mu \omega^2 A^2$.

Energy in one wavelength λ :

$$E_\lambda = \langle \frac{dE}{dx} \rangle \lambda = \frac{1}{2} \mu \omega^2 A^2 \lambda = \frac{1}{2} \mu \omega^2 A^2 \frac{2\pi}{k} = \pi \mu \omega^2 A^2 / k$$

Using $\omega = vk$: $E_\lambda = \pi \mu v \omega A^2$.

Example A.1 — [CBSE Board Level]

A steel wire ($\mu = 0.02$ kg m⁻¹) carries a harmonic wave of amplitude 2 mm, frequency 50 Hz, and wave speed 100 m s⁻¹. Find (a) the average power transmitted and (b) the energy contained in a 1 m length of the wire.

Given: $\mu = 0.02$ kg m⁻¹, $A = 2 \times 10^{-3}$ m, $f = 50$ Hz, $v = 100$ m s⁻¹.
 $\omega = 2\pi f = 100\pi$ rad s⁻¹.

(a) Average power:

$$\begin{aligned}P_{\text{avg}} &= \frac{1}{2} \mu \omega^2 A^2 v \\&= \frac{1}{2} (0.02) (100\pi)^2 (2 \times 10^{-3})^2 (100) \\&= 0.01 \times 10^4 \pi^2 \times 4 \times 10^{-6} \times 100 \\&= 0.01 \times 10^4 \times 4 \times 10^{-4} \times \pi^2 \\&= 0.04 \pi^2 \approx 0.395 \text{ W}\end{aligned}$$

(b) Energy in 1 m:

Average energy per metre = $P_{\text{avg}}/v = 0.395 / 100 = 3.95 \times 10^{-3} \text{ J}$.

Alternatively: $\langle dE/dx \rangle = \frac{1}{2} \mu \omega^2 A^2 = 0.5 \times 0.02 \times (100\pi)^2 \times (2 \times 10^{-3})^2 = 3.95 \times 10^{-3} \text{ J m}^{-1}$.

Answer: (a) 0.395 W, (b) $3.95 \times 10^{-3} \text{ J}$.

Example A.2 — [JEE Main Level]

A sinusoidal wave on a string ($\mu = 0.1 \text{ kg m}^{-1}$, $T = 10 \text{ N}$) has amplitude 5 mm. The average power transmitted is 2 W. Find the frequency of the wave.

Given: $\mu = 0.1 \text{ kg m}^{-1}$, $T = 10 \text{ N}$, $A = 5 \times 10^{-3} \text{ m}$, $P_{\text{avg}} = 2 \text{ W}$.

Wave speed: $v = \sqrt{T/\mu} = \sqrt{(10/0.1)} = 10 \text{ m s}^{-1}$.

$$\begin{aligned}P_{\text{avg}} &= \frac{1}{2} \mu \omega^2 A^2 v \Rightarrow \omega^2 = \frac{2P_{\text{avg}}}{\mu A^2 v} \\ \omega^2 &= \frac{2 \times 2}{0.1 \times (5 \times 10^{-3})^2 \times 10} \\ &= \frac{4}{0.1 \times 25 \times 10^{-6} \times 10} \\ &= \frac{4}{25 \times 10^{-6}} = 1.6 \times 10^5 \text{ rad}^2 \text{ s}^{-2} \\ \omega &= 400 \text{ rad s}^{-1} \\ f &= \frac{\omega}{2\pi} = \frac{400}{2\pi} \approx 63.7 \text{ Hz}\end{aligned}$$

Answer: $f \approx 63.7 \text{ Hz}$.

Example A.3 — [JEE Advanced Level]

A uniform rope of mass M and length L hangs vertically. A harmonic wave of amplitude A and frequency f is generated at the bottom. Assuming no reflection and no dissipation, find the amplitude of the wave as a function of height x measured from the bottom.

Insight: Power is conserved along the rope (no reflection, no loss). But $v(x)$ and μ change with x .

Linear density $\mu = M/L$ (constant). Tension at height x : $T(x) = (Mx/L)g = \mu xg$.

Wave speed: $v(x) = \sqrt{T/\mu} = \sqrt{gx}$.

Power conservation: $P(x) = \frac{1}{2} \mu \omega^2 A(x)^2 v(x) = \text{constant} = P_0$.

At the bottom ($x \rightarrow 0$), the tension vanishes, so the model breaks down; instead, match to the input power at some small x_0 . But the *scaling* with x is what JEE asks for:

$$A(x)^2 v(x) = \text{const} \implies A(x) \propto v(x)^{-1/2} \propto x^{-1/4}$$

Result: Amplitude grows as the wave climbs: $A(x) = A_0 (x_0/x)^{1/4}$.

(If the wave were generated at the *top* and travelled down, amplitude would decay as $x^{-1/4}$.)

Check: As x increases, tension increases, wave speed increases, so for constant power the amplitude must decrease — but here the wave goes *up*, so v decreases, amplitude increases. Correct.

Practice — Addendum A

1. A string ($\mu = 0.05 \text{ kg m}^{-1}$, $T = 20 \text{ N}$) carries a wave $y = 0.01 \sin(20\pi t - 2\pi x)$. Find the average power transmitted.
2. The average power in a wave is 10 W . If the amplitude is doubled and the frequency is halved (μ , T unchanged), what is the new average power?
3. Show that for a travelling wave on a string, the instantaneous kinetic energy density equals the instantaneous potential energy density at every point.
4. A pulse of total energy E_0 travels on a string. The string is joined to a second string of twice the linear density (same tension). Find the energy transmitted into the second string. (Use Addendum B's transmission coefficient.)
5. A rope of mass 2 kg , length 10 m hangs vertically. A 100 Hz wave of amplitude 1 cm is sent from the bottom. Estimate the amplitude at the top. (Use $g = 10 \text{ m s}^{-2}$.)

Answers:

1. $v = \sqrt{(20/0.05)} = 20 \text{ m s}^{-1}$, $\omega = 20\pi$, $k = 2\pi$, $A = 0.01 \text{ m}$. $P = \frac{1}{2}(0.05)(20\pi)^2(0.01)^2(20) = 0.2\pi^2 \approx 1.97 \text{ W}$.
2. $P \propto A^2\omega^2$. New $A' = 2A$, $\omega' = \omega/2 \Rightarrow P' = P \times 4 \times 1/4 = P = 10 \text{ W}$.
3. Shown in Method steps 1–3 above; uses $\partial y/\partial t = \mp v \partial y/\partial x$ and $T = \mu v^2$.
4. From Addendum B: $T_E = 4Z_1 Z_2 / (Z_1 + Z_2)^2$. Here $Z_2 = \sqrt{2} Z_1 \Rightarrow T_E = 4\sqrt{2}/(1+\sqrt{2})^2 \approx 0.943$. Transmitted energy = $0.943 E_0$.
5. $A(L)/A(0) \rightarrow \infty$ as $x \rightarrow 0$; use ratio $A(10)/A(1) = (1/10)^{-1/4} = 100^{1/4} \approx 1.78$. Amplitude at top $\approx 1.78 \text{ cm}$. (Model valid only if wavelength $\ll$ scale of v variation.)

Addendum B — Reflection and Transmission at a Junction of Two Strings

The chapter treats reflection at a *fixed end* ($Z_2 \rightarrow \infty$) and a *free end* ($Z_2 \rightarrow 0$). JEE Advanced asks for the general case: two strings of different linear densities μ_1, μ_2 joined under the same tension T . You need the amplitude reflection coefficient R , transmission coefficient T_A , and the *energy* transmission coefficient T_E . The algebra is identical to the electromagnetic wave / transmission line problem — impedance $Z = \sqrt{T\mu}$ replaces refractive index.

Method

1. Define impedances.

For a string, characteristic impedance $Z = \sqrt{T\mu}$ [kg s⁻¹].

Wave speed $v = \sqrt{T/\mu} = T/Z$.

2. Boundary conditions at the junction ($x = 0$).

Let incident wave: $y_i = A_i \sin(\omega t - k_1 x)$ (travelling +x in string 1).

Reflected wave: $y_r = A_r \sin(\omega t + k_1 x + \phi_r)$.

Transmitted wave: $y_t = A_t \sin(\omega t - k_2 x + \phi_t)$.

Continuity of displacement: $y_i(0) + y_r(0) = y_t(0)$ for all t .

Continuity of transverse force (slope): $T \partial y_i / \partial x|_0 + T \partial y_r / \partial x|_0 = T \partial y_t / \partial x|_0$ (tension T is same).

3. Solve for coefficients (assume $\phi_r = \phi_t = 0$ or π , absorbed in sign of A).

Displacement: $A_i + A_r = A_t$.

Slope: $-k_1 A_i + k_1 A_r = -k_2 A_t \Rightarrow -A_i/v_1 + A_r/v_1 = -A_t/v_2$.

Multiply slope equation by T : $-Z_1 A_i + Z_1 A_r = -Z_2 A_t$ (since $T/v = Z$).

Solve the two linear equations:

$$A_r = \frac{Z_1 - Z_2}{Z_1 + Z_2} A_i$$
$$A_t = \frac{2Z_1}{Z_1 + Z_2} A_i$$

Amplitude reflection coefficient: $R = A_r/A_i = (Z_1 - Z_2)/(Z_1 + Z_2)$.

Amplitude transmission coefficient: $T_A = A_t/A_i = 2Z_1/(Z_1 + Z_2)$.

4. Energy (power) coefficients.

Power $\propto Z A^2$ (since $P = 1/2 \mu \omega^2 A^2 v = 1/2 Z \omega^2 A^2$).

Reflected power fraction: $R_E = (A_r/A_i)^2 = R^2$.

Transmitted power fraction: $T_E = (Z_2/Z_1) (A_t/A_i)^2 = (Z_2/Z_1) T_A^2$.

Substitute T_A :

$$T_E = \frac{Z_2}{Z_1} \left(\frac{2Z_1}{Z_1 + Z_2} \right)^2 = \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2}$$

Check: $R_E + T_E = (Z_1 - Z_2)^2/(Z_1 + Z_2)^2 + 4Z_1 Z_2/(Z_1 + Z_2)^2 = 1$.

5. Limiting cases (chapter's fixed/free ends).

Fixed end: string 2 is rigid $\Rightarrow Z_2 \rightarrow \infty \Rightarrow R \rightarrow -1$ (phase reversal), $T_A \rightarrow 0$.

Free end: string 2 massless $\Rightarrow Z_2 \rightarrow 0 \Rightarrow R \rightarrow +1$ (no phase reversal), $T_A \rightarrow 2$.

Example B.1 — [CBSE Board Level]

Two strings of linear densities $\mu_1 = 0.01 \text{ kg m}^{-1}$ and $\mu_2 = 0.04 \text{ kg m}^{-1}$ are joined under tension 16 N. A wave of amplitude 3 mm is incident from string 1. Find the amplitudes of the reflected and transmitted waves.

Given: $\mu_1 = 0.01$, $\mu_2 = 0.04$, $T = 16 \text{ N}$, $A_i = 3 \times 10^{-3} \text{ m}$.

$$Z_1 = \sqrt{T\mu_1} = \sqrt{16 \times 0.01} = 0.4 \text{ kg s}^{-1}$$

$$Z_2 = \sqrt{T\mu_2} = \sqrt{16 \times 0.04} = 0.8 \text{ kg s}^{-1}$$

$$R = \frac{0.4 - 0.8}{0.4 + 0.8} = \frac{-0.4}{1.2} = -\frac{1}{3}$$

$$A_r = RA_i = -\frac{1}{3} \times 3 \text{ mm} = -1 \text{ mm} \quad (\text{phase reversal})$$

$$T_A = \frac{2 \times 0.4}{1.2} = \frac{2}{3}$$

$$A_t = T_A A_i = \frac{2}{3} \times 3 \text{ mm} = 2 \text{ mm}$$

Answer: Reflected amplitude = 1 mm (inverted), Transmitted amplitude = 2 mm.

Example B.2 — [JEE Main Level]

In the setup of Example B.1, what fraction of the incident *power* is transmitted into string 2?

Method 1 (Direct formula):

$$T_E = \frac{4Z_1Z_2}{(Z_1 + Z_2)^2} = \frac{4 \times 0.4 \times 0.8}{(1.2)^2} = \frac{1.28}{1.44} = \frac{8}{9} \approx 88.9\%$$

Method 2 (Via amplitudes):

Incident power $\propto Z_1 A_i^2$. Transmitted power $\propto Z_2 A_t^2$.

$$\frac{P_t}{P_i} = \frac{Z_2}{Z_1} \left(\frac{A_t}{A_i} \right)^2 = \frac{0.8}{0.4} \times \left(\frac{2}{3} \right)^2 = 2 \times \frac{4}{9} = \frac{8}{9}$$

Answer: 8/9 of the incident power.

Example B.3 — [JEE Advanced Level]

A pulse of total energy E_0 and arbitrary shape travels on string 1 (impedance Z_1) toward a junction with string 2 (impedance Z_2). The pulse length is much larger than the junction region. Find the energies of the reflected and transmitted pulses.

Insight: For a *pulse*, superposition holds at every instant. The boundary conditions are linear and instantaneous. The reflection/transmission coefficients derived for harmonic waves depend only on Z_1 , Z_2 , not on ω or pulse shape (non-dispersive medium). Therefore the

entire pulse is split with the same amplitude coefficients R , T_A .

Since energy in a pulse is $\int P dt = \int \frac{1}{2} Z \omega^2 A^2 dt$, and the pulse shape in time is preserved (just scaled), the energy fractions are exactly the harmonic-wave power fractions:

$$E_r = R_E E_0 = \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2 E_0$$

$$E_t = T_E E_0 = \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2} E_0$$

Check: If $Z_2 = 4Z_1$ (Example B.1), $E_r = (1/3)^2 E_0 = E_0/9$, $E_t = 8E_0/9$. Sum = E_0 .

Practice — Addendum B

1. A wave travels from a string ($\mu = 0.02 \text{ kg m}^{-1}$) to a heavier string ($\mu = 0.18 \text{ kg m}^{-1}$) under the same tension 50 N. Find the amplitude reflection coefficient and state whether the reflected pulse is inverted.
2. For what ratio μ_2/μ_1 is the transmitted power exactly 50%?
3. A pulse approaches a junction from the *heavier* string side ($Z_2 > Z_1$). Is the reflected pulse inverted? What about the transmitted pulse?
4. Three strings of impedances Z_1 , Z_2 , Z_3 are joined in series. A wave is incident from string 1. Write the condition on Z_2 for *zero* reflection (impedance matching).
5. A wave on string 1 (μ_1) hits string 2 (μ_2). The transmitted wave then hits string 3 (μ_1 again) after a distance L in string 2. If the round-trip phase in string 2 is $2k_2L = \pi$, find the net reflection coefficient back into string 1. (This is a Fabry-Pérot interferometer for strings.)

Answers:

1. $Z_1 = 1$, $Z_2 = 3$ (in $\sqrt{\text{kg}\cdot\text{N}}$ units). $R = (1-3)/(1+3) = -0.5$. Inverted (negative sign).
2. $T_E = 4Z_1Z_2/(Z_1+Z_2)^2 = 1/2 \Rightarrow 8Z_1Z_2 = (Z_1+Z_2)^2 \Rightarrow Z_2/Z_1 = 3 \pm 2\sqrt{2}$. Since $Z \propto \sqrt{\mu}$, $\mu_2/\mu_1 = (3 \pm 2\sqrt{2})^2 = 17 \pm 12\sqrt{2}$.
3. Incident from heavy side: $Z_i = Z_2$, $Z_t = Z_1$. $R = (Z_2 - Z_1)/(Z_2 + Z_1) > 0 \Rightarrow$ *not* inverted. Transmitted pulse never inverted ($T_A > 0$ always).
4. Zero reflection requires $Z_2 = \sqrt{(Z_1Z_3)}$. For $Z_3 = Z_1$, $Z_2 = Z_1$ (trivial). For a quarter-wave transformer, length also matters; here only impedance asked: $Z_2 = \sqrt{(Z_1Z_3)}$.
5. Multiple reflections in layer 2. Net reflection coefficient $R_{\text{net}} = (R_{12} + R_{23} e^{i2k_2L}) / (1 + R_{12}R_{23} e^{i2k_2L})$. With $Z_3 = Z_1$, $R_{23} = -R_{12}$. Phase factor = $e^{i\pi} = -1$. $R_{\text{net}} = (R - R(-1))/(1 + R(-R)(-1)) = 2R/(1+R^2)$. Where $R = (Z_1 - Z_2)/(Z_1 + Z_2)$.

Addendum C — The Wave Equation, General Solution, and Non-Sinusoidal Pulses

The chapter *assumes* the sinusoidal form $y = A \sin(\omega t - kx + \phi)$. JEE Advanced often gives a pulse shape (triangle, Gaussian, half-sine) and asks for the transverse velocity, the shape at a later time, or the energy. This requires the **wave equation** and its **general solution** $y(x, t) = f(x - vt) + g(x + vt)$. We derive the PDE from the same free-body diagram used for the speed derivation, then show how any initial profile splits into two half-amplitude pulses travelling in opposite directions.

Method

1. Derive the wave equation (1-D transverse string).

Consider a small element between x and $x + \Delta x$. Mass $= \mu \Delta x$.

Transverse forces: $T \sin\theta(x + \Delta x) - T \sin\theta(x) \approx T [\tan\theta(x + \Delta x) - \tan\theta(x)]$ (small angles).

But $\tan\theta = \partial y / \partial x$. Net force $= T [\partial y / \partial x|_{x + \Delta x} - \partial y / \partial x|_x] \approx T (\partial^2 y / \partial x^2) \Delta x$.

Newton's second law:

$$\mu \Delta x \frac{\partial^2 y}{\partial t^2} = T \frac{\partial^2 y}{\partial x^2} \Delta x$$

$$\boxed{\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \frac{\partial^2 y}{\partial x^2} = v^2 \frac{\partial^2 y}{\partial x^2}}$$

Units: $\partial^2 y / \partial t^2$ [m s⁻²], $v^2 \partial^2 y / \partial x^2$ [(m² s⁻²)(m⁻¹)] = m s⁻².

2. General solution.

Change variables: $\xi = x - vt$, $\eta = x + vt$.

$\partial / \partial x = \partial / \partial \xi + \partial / \partial \eta$, $\partial / \partial t = -v \partial / \partial \xi + v \partial / \partial \eta$.

$\partial^2 / \partial t^2 = v^2 (\partial^2 / \partial \xi^2 - 2 \partial^2 / \partial \xi \partial \eta + \partial^2 / \partial \eta^2)$.

$\partial^2 / \partial x^2 = \partial^2 / \partial \xi^2 + 2 \partial^2 / \partial \xi \partial \eta + \partial^2 / \partial \eta^2$.

Wave equation becomes:

$$v^2 \left(\frac{\partial^2 y}{\partial \xi^2} - 2 \frac{\partial^2 y}{\partial \xi \partial \eta} + \frac{\partial^2 y}{\partial \eta^2} \right) = v^2 \left(\frac{\partial^2 y}{\partial \xi^2} + 2 \frac{\partial^2 y}{\partial \xi \partial \eta} + \frac{\partial^2 y}{\partial \eta^2} \right)$$

Cancel v^2 and ξ^2 , η^2 terms: $-4v^2 \partial^2 y / \partial \xi \partial \eta = 0 \Rightarrow \partial^2 y / \partial \xi \partial \eta = 0$.

Integrate: $\partial y / \partial \xi = F'(\xi) \Rightarrow y = F(\xi) + G(\eta)$.

$$\boxed{y(x, t) = f(x - vt) + g(x + vt)}$$

f is a right-moving wave, g is a left-moving wave. Any twice-differentiable functions.

3. Initial value problem (d'Alembert's solution).

Given $y(x, 0) = \phi(x)$ and $\partial y / \partial t(x, 0) = \psi(x)$.

$$y(x, 0) = f(x) + g(x) = \phi(x)$$

$$\frac{\partial y}{\partial t}(x, 0) = -vf'(x) + vg'(x) = \psi(x)$$

Integrate the second: $-v f(x) + v g(x) = \int_0^x \psi(s) ds + C$.

Solve for f, g :

$$f(x) = \frac{1}{2}\phi(x) - \frac{1}{2v} \int_0^x \psi(s) ds - \frac{C}{2v}$$

$$g(x) = \frac{1}{2}\phi(x) + \frac{1}{2v} \int_0^x \psi(s) ds + \frac{C}{2v}$$

Solution:

$$y(x, t) = \frac{1}{2}[\phi(x - vt) + \phi(x + vt)] + \frac{1}{2v} \int_{x-vt}^{x+vt} \psi(s) ds$$

4. Special case: String initially at rest ($\psi = 0$), plucked into shape $\phi(x)$.

$$y(x, t) = \frac{1}{2}\phi(x - vt) + \frac{1}{2}\phi(x + vt)$$

The initial profile splits into two half-amplitude copies travelling in opposite directions. This is the *only* way to handle non-sinusoidal pulses in JEE.

Example C.1 — [CBSE Board Level]

A string lies along the x -axis. At $t = 0$ it is given the shape $y(x, 0) = A \exp(-x^2/a^2)$ and released from rest. Wave speed $v = 10 \text{ m s}^{-1}$. Write the displacement $y(x, t)$ for all t .

Given: $\phi(x) = A e^{-x^2/a^2}$, $\psi(x) = 0$.

Using d'Alembert for $\psi = 0$:

$$y(x, t) = \frac{1}{2}A \left[e^{-(x-vt)^2/a^2} + e^{-(x+vt)^2/a^2} \right]$$

Interpretation: The Gaussian splits into two Gaussians of half the amplitude moving left and right.

Example C.2 — [JEE Main Level]

A triangular pulse $y(x, 0) = A(1 - |x|/L)$ for $|x| \leq L$, zero elsewhere, is at rest on a string ($v = 20 \text{ m s}^{-1}$). Sketch the pulse at $t = L/(2v)$ and $t = L/v$.

Analysis: Pulse splits into two half-height triangles moving at $\pm v$.

At $t = 0$: peak A at $x=0$, base from $-L$ to $+L$.

At $t = L/(2v)$: right-moving half has peak at $x = vt = L/2$, height $A/2$; left-moving half at $x = -L/2$, height $A/2$. They overlap from $-L/2$ to $L/2$. In overlap region, superposition gives original height A (since both halves contribute $A/2$). Outside, two separate triangles of height $A/2$.

At $t = L/v$: peaks at $x = \pm L$, height $A/2$. No overlap. Two separate triangles.

Example C.3 — [JEE Advanced Level]

An infinite string ($\mu = 0.01 \text{ kg m}^{-1}$, $T = 1 \text{ N}$) is initially at rest. At $t = 0$, a transverse impulse gives every particle a velocity $v_p(x, 0) = v_0 \exp(-x^2/b^2)$. Find the subsequent displacement $y(x, t)$ and the total energy imparted to the string.

Given: $\psi(x) = v_0 e^{-x^2/b^2}$, $\phi(x) = 0$. $v = \sqrt{T/\mu} = \sqrt{1/0.01} = 10 \text{ m s}^{-1}$.

Use d'Alembert with $\phi = 0$:

$$y(x, t) = \frac{1}{2v} \int_{x-vt}^{x+vt} v_0 e^{-s^2/b^2} ds = \frac{v_0}{2v} \int_{x-vt}^{x+vt} e^{-s^2/b^2} ds$$

This is the integral of a Gaussian (error function).

Total energy = total initial kinetic energy (since initial PE = 0 for flat string):

$$E = \int_{-\infty}^{\infty} \frac{1}{2} \mu \psi(x)^2 dx = \frac{1}{2} \mu v_0^2 \int_{-\infty}^{\infty} e^{-2x^2/b^2} dx$$

Gaussian integral: $\int e^{-\alpha x^2} dx = \sqrt{\pi/\alpha}$. Here $\alpha = 2/b^2$.

$$E = \frac{1}{2} \mu v_0^2 \sqrt{\frac{\pi}{2/b^2}} = \frac{1}{2} \mu v_0^2 b \sqrt{\frac{\pi}{2}}$$

Units: $\mu [\text{kg/m}] \cdot v_0^2 [\text{m}^2/\text{s}^2] \cdot b [\text{m}] = \text{kg m}^2/\text{s}^2 = \text{Joule}$.

Practice — Addendum C

1. Derive the 1-D wave equation for longitudinal waves in a rod (Young's modulus Y , density ρ). Show that $v = \sqrt{Y/\rho}$.
2. A string ($v = 5 \text{ m s}^{-1}$) has initial displacement $y(x, 0) = \sin(\pi x/L)$ for $0 \leq x \leq L$, zero elsewhere, and zero initial velocity. Write $y(x, t)$ using d'Alembert.
3. Show that for any solution of the wave equation, the quantity $\partial y/\partial t \pm v \partial y/\partial x$ is constant along the characteristics $x \mp vt = \text{const.}$ (These are the Riemann invariants.)
4. A pulse $f(x)$ travels right on string 1 (Z_1) toward a junction with string 2 (Z_2). Using the general solution form, derive the reflected pulse shape $g(x)$ in string 1 and transmitted pulse $h(x)$ in string 2. Show they are scaled versions of f .
5. A string fixed at both ends ($L = 1 \text{ m}$, $v = 100 \text{ m s}^{-1}$) is plucked at the centre into a triangle of height 0.01 m . Estimate the time for the pulse to travel to one end, reflect, and return to the centre. What is the shape at that instant?

Answers:

1. Force on element = $YA (\partial^2 y/\partial x^2) \Delta x$, mass = $\rho A \Delta x \Rightarrow \partial^2 y/\partial t^2 = (Y/\rho) \partial^2 y/\partial x^2$.
2. $y(x, t) = 1/2 [\sin(\pi(x-5t)/L) + \sin(\pi(x+5t)/L)]$ for arguments in $[0, L]$, else 0.

3. $\partial/\partial t (\partial y/\partial t \pm v \partial y/\partial x) = \partial^2 y/\partial t^2 \pm v \partial^2 y/\partial x \partial t$. Along $x \mp vt = \text{const}$, $d/dt = \partial/\partial t \mp v \partial/\partial x$. Compute d/dt of the invariant $= 0$ using wave equation.
4. At $x=0$: $f(-v_1 t) + g(v_1 t) = h(-v_2 t)$ (displacement). Slope continuity: $-f' + g' = -(v_1/v_2)h'$. Integrate: $-f + g = -(Z_1/Z_2)h$. Solve with displacement eq: $g = Rf$, $h = T_A f$ (with arguments mapped to same time). Shapes are identical, scaled by R , T_A .
5. Time to end $= 0.5/100 = 0.005$ s. Round trip $= 0.01$ s. At return, two inverted half-triangles meet at centre; superposition gives inverted triangle of full height (constructive interference of inverted pulses).

Addendum D — Longitudinal Standing Waves: Rods, End Correction, and Pressure-Displacement Duality

The chapter does standing waves on strings (transverse) and in air columns (longitudinal) but treats them as separate formula lists. JEE Advanced mixes them: a rod clamped at the centre, an open pipe with end correction derived from radiation impedance, or a question asking for the *pressure* node when the *displacement* antinode is given. This addendum unifies the boundary conditions via the impedance concept and derives the open-end correction $e \approx 0.6r$.

Method

1. Longitudinal wave variables.

Displacement $\xi(x, t)$ of a fluid/rod element from equilibrium.

Strain $= \partial\xi/\partial x$. Stress $= E \times \text{strain}$ (rod, Young's modulus E) or $-B \times \text{strain}$ (gas, bulk modulus B).

Pressure variation: $\Delta p = -B \partial\xi/\partial x$ (gas). In a rod, force $= EA \partial\xi/\partial x$.

Wave equation: $\partial^2 \xi/\partial t^2 = v^2 \partial^2 \xi/\partial x^2$ with $v = \sqrt{E/\rho}$ (rod) or $\sqrt{B/\rho}$ (gas).

2. Boundary conditions — the unified rule.

At any boundary, two things must match: displacement ξ and stress (force/pressure).

- **Fixed end / Closed pipe:** $\xi = 0$ (displacement node). Stress is free $\Rightarrow$ pressure antinode (Δp max).

- **Free end / Open pipe:** Stress $= 0 \Rightarrow \partial\xi/\partial x = 0$ (strain node). Displacement is free $\Rightarrow$ displacement antinode. Pressure node ($\Delta p = 0$).

3. Standing wave solutions.

$\xi(x, t) = 2A \cos(\omega t) \sin(kx)$ (node at $x=0$) or $2A \cos(\omega t) \cos(kx)$ (antinode at $x=0$).

Pressure: $\Delta p = -B \partial\xi/\partial x \propto \cos(kx)$ or $-\sin(kx)$. **Displacement node $\Leftrightarrow$ Pressure antinode**, and vice versa.

4. Rods — exact analogies to pipes.

| Rod Boundary | Displacement | Pressure/Force | Harmonics | | :— | :— | :— | :— | | Fixed-Fixed
| Node-Node | Antinode-Antinode | $f_n = nv/2L$ (all) | | Free-Free | Antinode-Antinode | Node-Node | $f_n = nv/2L$ (all) | | Fixed-Free | Node-Antinode | Antinode-Node | $f_n = (2n-1)v/4L$ (odd) |

5. End correction for an open pipe — derivation (Rayleigh).

An open pipe of radius r radiates sound into half-space. The air just outside the opening moves as a piston of radius r , loading the air column with an *acoustic mass*.

Kinetic energy of the external flow (potential flow from a circular piston in a baffle):

$$K_{\text{ext}} = \frac{1}{2}\rho \int_{\text{half-space}} v^2 dV = \frac{1}{2}\rho \left(\frac{8}{3\pi} r^3 \right) v_{\text{piston}}^2$$

Effective mass of external air: $m_{\text{eff}} = \rho \left(\frac{8}{3\pi} \right) r^3$.

This mass sits at the end of the internal air column (mass per length = $\rho\pi r^2$). Equivalent extra length e :

$$\rho\pi r^2 e = \rho \frac{8}{3\pi} r^3 \implies e = \frac{8}{3\pi^2} r \approx 0.27r$$

Wait — classic textbooks use the *unflanged* pipe correction (Levine & Schwinger, 1948):

$$e = 0.6133r \approx 0.6r$$

The chapter's $0.6r$ comes from the exact solution for an unflanged pipe. The derivation above (flanged) gives $0.27r$. JEE expects you to *know* $e \approx 0.6r$ and use it; the derivation is not examinable but the *concept* (effective length > geometric length) is.

Effective length formulas:

- Open pipe (two ends): $L_{\text{eff}} = L + 2e = L + 1.2r$.
- Closed pipe (one open end): $L_{\text{eff}} = L + e = L + 0.6r$.

Resonance frequencies use L_{eff} .

Example D.1 — [CBSE Board Level]

A rod of length 1 m, density 8000 kg m^{-3} , Young's modulus $2 \times 10^{11} \text{ Pa}$ is clamped at its centre. Find the fundamental frequency of longitudinal vibration.

Given: $L = 1 \text{ m}$, clamped at centre $\Rightarrow$ two Fixed-Free segments of length $L/2 = 0.5 \text{ m}$.

Wave speed: $v = \sqrt{Y/\rho} = \sqrt{(2 \times 10^{11}/8000)} = \sqrt{(2.5 \times 10^7)} = 5000 \text{ m s}^{-1}$.

For Fixed-Free rod: fundamental $f_1 = v/(4L_{\text{seg}}) = 5000/(4 \times 0.5) = 2500 \text{ Hz}$.

Answer: 2.5 kHz.

Example D.2 — [JEE Main Level]

An open organ pipe of length 0.5 m and inner radius 2 cm resonates at its fundamental with a tuning fork of frequency 340 Hz. The speed of sound in air is 340 m s^{-1} . Find the end correction e .

Given: $L = 0.5 \text{ m}$, $r = 0.02 \text{ m}$, $f_1 = 340 \text{ Hz}$, $v = 340 \text{ m s}^{-1}$.

Fundamental for open pipe with end correction: $f_1 = v / [2(L + 2e)]$.

$$340 = \frac{340}{2(0.5 + 2e)}$$

$$1 = \frac{1}{1 + 4e} \Rightarrow 1 + 4e = 1 \Rightarrow e = 0$$

Wait — if $f_1 = v/(2L)$ exactly, then $e = 0$. But the problem says “resonates at its fundamental with a tuning fork of frequency 340 Hz”. If $v = 340$, $L = 0.5$, then $f_1 = 340/(2 \times 0.5) = 340$ Hz. So the pipe resonates *without* needing end correction? That implies the given r is a distractor or the pipe is flanged.

Let’s change numbers to make it non-trivial:

Revised problem: Pipe length 0.48 m, radius 2 cm, $f_1 = 340$ Hz, $v = 340$ m s^{−1}. Find e .

$$340 = \frac{340}{2(0.48 + 2e)} \Rightarrow 0.48 + 2e = 0.5 \Rightarrow 2e = 0.02 \Rightarrow e = 0.01 \text{ m} = 1 \text{ cm}$$

Theoretical $0.6r = 0.6 \times 0.02 = 0.012 \text{ m} = 1.2 \text{ cm}$. Close.

Answer: $e = 1 \text{ cm}$.

Example D.3 — [JEE Advanced Level]

A closed organ pipe of length $L = 0.8$ m and radius $r = 0.015$ m is sounded. The speed of sound is 330 m s^{−1}. (a) Find the fundamental frequency including the end correction. (b) The pipe is now opened at the closed end. Find the new fundamental frequency. (c) At the open end, what is the ratio of pressure amplitude to displacement amplitude?

(a) Closed pipe (one open end):

$$L_{\text{eff}} = L + e = 0.8 + 0.6 \times 0.015 = 0.8 + 0.009 = 0.809 \text{ m.}$$

$$f_1 = v/(4L_{\text{eff}}) = 330/(4 \times 0.809) = 330/3.236 \approx 102.0 \text{ Hz.}$$

(b) Open pipe (two open ends):

$$L_{\text{eff}} = L + 2e = 0.8 + 0.018 = 0.818 \text{ m.}$$

$$f_1' = v/(2L_{\text{eff}}) = 330/(2 \times 0.818) = 330/1.636 \approx 201.7 \text{ Hz.}$$

$$\text{Ratio } f_1'/f_1 \approx 1.977 \approx 2 \text{ (as expected, octave higher).}$$

(c) Pressure-displacement relation for a travelling wave:

$$\Delta p = -B \partial \xi / \partial x. \text{ For a right-travelling wave } \xi = A \sin(\omega t - kx): \partial \xi / \partial x = -kA \cos(\omega t - kx).$$

$$\Delta p = BkA \cos(\omega t - kx) = \rho v^2 (\omega/v) A \cos(\dots) = \rho v \omega A \cos(\dots).$$

$$\text{Amplitude ratio: } |\Delta p|_{\text{max}} / |\xi|_{\text{max}} = \rho v \omega = \rho v (2\pi f).$$

At the open end, this ratio holds for the *travelling* components; for the standing wave, at the displacement antinode (open end), $\Delta p = 0$ exactly. The question asks for the ratio of *amplitudes* of the travelling wave components that form the standing wave.

Answer: (a) 102 Hz, (b) 202 Hz, (c) $\rho v \omega = 2\pi \rho v f$.

Practice — Addendum D

1. A steel rod ($Y = 2 \times 10^{11} \text{ Pa}$, $\rho = 7800 \text{ kg m}^{-3}$) of length 2 m is free at both ends. Find the frequency of the first overtone.
2. An open pipe of length 0.6 m and radius 0.02 m has its fundamental at 280 Hz. Speed of sound = 340 m s^{-1} . Find the experimental end correction and compare with $0.6r$.
3. In a standing sound wave in a closed pipe, the displacement amplitude at the antinode is 0.1 mm. The frequency is 500 Hz. Find the pressure amplitude at the closed end. ($\rho = 1.2 \text{ kg m}^{-3}$, $v = 340 \text{ m s}^{-1}$.)
4. A pipe closed at one end shows resonance at 300 Hz and 500 Hz (no other resonances between them). Find the fundamental and the length (ignore end correction).
5. Show that the acoustic impedance of a pipe (pressure/volume velocity) at the open end is purely resistive (radiation resistance) and reactive (end correction mass). Derive the low-frequency approximation $Z_{\text{rad}} \approx \rho v / (\pi r^2) [1 + i (kr) 0.6 \dots]$.

Answers:

1. $v = \sqrt{Y/\rho} \approx 5064 \text{ m/s}$. Free-free: $f_n = nv/2L$. Fundamental = $5064/4 = 1266 \text{ Hz}$. First overtone = 2nd harmonic = 2532 Hz.
2. $f_1 = v/[2(L+2e)] \Rightarrow 280 = 340/[2(0.6+2e)] \Rightarrow 0.6+2e = 340/560 = 0.6071 \Rightarrow 2e = 0.0071 \Rightarrow e = 0.00355 \text{ m} = 3.55 \text{ mm}$. Theory: $0.6r = 12 \text{ mm}$. Discrepancy due to flange effect or measurement.
3. At closed end: displacement node, pressure antinode. Pressure amplitude $\Delta p_{\text{max}} = \rho v \omega A = 1.2 \times 340 \times (1000\pi) \times 10^{-4} = 1.2 \times 340 \times 0.314 \approx 128 \text{ Pa}$.
4. Closed pipe: only odd harmonics. $300 = (2n-1)f_1$, $500 = (2m-1)f_1$. Difference = $200 = 2(m-n)f_1$. Smallest difference for successive resonances ($m=n+1$) $\Rightarrow 200 = 2f_1 \Rightarrow f_1 = 100 \text{ Hz}$. Check: $300 = 3 \times 100$, $500 = 5 \times 100$. $L = v/(4f_1) = 340/400 = 0.85 \text{ m}$.
5. Radiation impedance of unflanged pipe: $Z = \rho v / (\pi r^2) [(kr)^2/2 + i 0.6(kr)]$ for $kr \ll 1$. Real part = radiation resistance $\propto \omega^2$; Imag part = mass reactance of effective length $0.6r$.

Errata

This section lists the errors identified in the printed text of the fourteen chapters of the *Class 11 Physics Master Reference* that could not be corrected in place — because the exact typeface the surrounding page is set in has no matching font available on this system (several of the fourteen source chapter PDFs come from noticeably different authoring pipelines, so a patch is only safe when the local system can reproduce the exact same typeface the page around it uses), because the correction requires inserting new content rather than resetting an existing run, or because the fix is an editorial judgement call best left to a human. Entries are ordered by chapter and then by printed page.

How to use this list. Each entry names the exact location, quotes what is printed, gives the replacement text, and — where the correction is a physics result — shows the working, so a reader can act on the correction without consulting any other source.

Every item below was independently re-verified against the source problem statement before being listed, including two claims from the original automated audit that turned out to be false positives once checked against the printed answer key and were therefore *not* included here. A further systematic check turned up an unrelated and much larger issue: several dozen apparent “corrupted character” and “leaked markup” defects the audit flagged — dropped ligatures (a stray control character in place of “ffi” in “difficulty”), missing dashes and quotation marks — turned out, on rendering the actual page, to be artifacts of the text-extraction layer only; the PDF itself prints those words and punctuation correctly. Those are not listed here, since they are not defects a reader will ever see.

Chapter 1: Units and Measurements

Page 1 — “the world finally agreed on one universal standard”: wrong year for SI adoption

Printed: “So in 1971 the world finally agreed on one universal standard — the *Système International d’Unités*, or SI — an upgraded, ‘rationalised’ MKS system...”

Should read: “So in 1960 the world finally agreed on one universal standard — the *Système International d’Unités*, or SI (its seventh base unit, the mole, was added only in 1971) — an upgraded, ‘rationalised’ MKS system...”

Why: The 11th CGPM (1960) adopted the name “SI” and its framework with six base units. The mole became the seventh base unit only at the 14th CGPM in 1971 — the chapter has conflated the two dates.

Chapter 2: Motion in a Straight Line

Page 1 — Contents: Sub-topic 05 is missing from the table of contents

Printed: The Contents list runs “Sub-topic 01: Kinematic Quantities ... Sub-topic 04: Relative Velocity” and stops.

Should read: Add “Sub-topic 05: Motion with Variable Acceleration (Calculus Methods) — page 30” after the Sub-topic 04 line.

Why: Sub-topic 05 genuinely exists in the chapter body, starting on page 30 — the Contents table simply omits it, so a reader using the table of contents to navigate will not know the subtopic exists.

Section 6, Q2 answer explanation — discusses an option “(D)” that isn’t in the printed question

Printed: “(D) is the half-lap answer (distance πR , displacement = diameter $2R$); picked by students who misread ‘full’ as ‘half.’” — but the question’s own option list stops at (C).

Should read: Either add “(D) πR , $2R$ ” to Q2’s option list, or delete this sentence from the answer explanation.

Why: The explanation analyses a distractor that was never offered as a choice, which will confuse a reader trying to match the discussion to the printed options. This is an editorial judgement call (which of the two fixes to make) rather than a factual error, so it is left for a human to resolve rather than patched in place.

Chapter 4: Laws of Motion

Page 59 — Crack the MCQ Q1 (Atwood machine, equal masses): duplicated option label and wrong marked answer

Printed: “(a) accelerates at g . (b) accelerates at $g/2$. (b) remains at rest or moves at constant velocity, with $T = mg$. (c) has zero tension. Correct: (c).”

Should read: Relabel the four options (a), (b), (c), (d) in order, so “remains at rest or moves at constant velocity, with $T = mg$ ” becomes option (c) and “has zero tension” becomes option (d). The marked answer should read “Correct: (c).”

Why: With two equal masses on an Atwood machine, the system does not accelerate ($a = 0$) and the tension equals the weight of either mass ($T = mg$) — this is the option currently mislabeled as a duplicate “(b)”. The printed key marks “(c) has zero tension” as correct, which is physically wrong: the string is under tension $T = mg$, not zero.

Page (Crack the MCQ, later question) — a second duplicated option label

Printed: A later option is printed as “(b) $a_1m_1 = a_2m_2$.” duplicating an earlier “(b)”; the accompanying explanation refers to “(c) and (d)” as if they already exist.

Should read: Relabel “ $a_1m_1 = a_2m_2$.” as option (c), matching the explanation’s own reference to it.

Why: The explanation’s own logic requires four distinctly-labelled options; as printed, two options share the label “(b)” and no option (c) exists for the explanation to refer to.

Chapter 6: System of Particles and Rotational Motion

Crack the MCQ (metre stick balance problem): none of the printed options match the worked solution

Printed: The worked solution correctly derives the required mass as 0.28 kg (moment balance about the 40 cm pivot, accounting for a ≈ 0.1 kg stick’s own weight), but keys “Correct: (B)” where option (B) is 0.10 kg. The four options are 0.20, 0.10, 0.30, and 0.15 kg — none equal 0.28 kg.

Should read: The question is flawed as posed. Either replace one of the four options with 0.28 kg and mark that one correct, or restate the problem with figures that give a round-number answer matching one of the printed options.

Why: Moment balance: $0.5 \text{ kg} \times 0.30 \text{ m} = m_{\text{stick}} \times 0.10 \text{ m} + m \times 0.50 \text{ m}$. For a standard 0.1 kg metre stick, solving gives $m = 0.28 \text{ kg}$, which is not among the printed options. (Option (C), 0.30 kg, is the answer you get only by incorrectly treating the stick as weightless — the chapter’s own text names this “the classic trap,” making the mismatched key doubly unfortunate.)

Practice answer 4 (angular momentum of a particle): mass factor dropped

Printed: For “[JEE Main] A particle of mass 2 kg is at position $(3\hat{i} + \hat{j})$ m moving with velocity $(2\hat{i} - \hat{j})$ m/s. Find its angular momentum about the origin,” the printed answer is $-5\hat{k} \text{ kg}\cdot\text{m}^2\cdot\text{s}^{-1}$.

Should read: $-10\hat{k} \text{ kg}\cdot\text{m}^2\cdot\text{s}^{-1}$.

Why: $\vec{r} \times \vec{v} = (3\hat{i} + \hat{j}) \times (2\hat{i} - \hat{j}) = -3\hat{k} - 2\hat{k} = -5\hat{k} \text{ m}^2/\text{s}$. Angular momentum is $\vec{L} = m(\vec{r} \times \vec{v})$, so the mass of 2 kg must still be applied: $\vec{L} = 2 \times (-5\hat{k}) = -10\hat{k} \text{ kg}\cdot\text{m}^2\cdot\text{s}^{-1}$. The printed value is the cross product alone, with the mass factor omitted.

Chapter 8: Mechanical Properties of Solids

“a quick dimension check ... rules them out instantly”: leaked, unrendered LaTeX source

Printed: “a quick dimension check ($[\sigma]/[\rho][g] = \text{metres}$) rules them out instantly” — with literal σ delimiters and empty bracket placeholders where the symbols for surface tension (σ) and density (ρ) should be.

Should read: “a quick dimension check ($[\sigma]/([\rho][g]) = \text{metres}$) rules them out instantly” — or, set as proper mathematics, “a quick dimension check ($\sqrt{\sigma/(\rho g)}$) has units of length) rules them out instantly.”

Why: This is raw, unrendered template markup that leaked into the printed body text — not a typesetting choice, but a production defect. The intended content is a dimensional-analysis check on a capillary-rise-style formula.

Chapter 9: Mechanical Properties of Fluids

Capillary-rise MCQ solution — part labels shifted relative to the question

Printed: The solution to the two-part capillary question labels the plates derivation “(b) For two plates a distance d apart...” and the bubble derivation “(c) The smaller bubble...”, while the question itself only poses parts (a) and (b).

Should read: Relabel “(b) For two plates...” as “(a) For two plates...” and “(c) The smaller bubble...” as “(b) The smaller bubble...”.

Why: The solution’s part labels are off by one relative to the question, so a reader matching answers to parts will be misled. This is a straightforward renumbering rather than a factual error, but it requires re-labelling text that runs across multiple lines, so it is left for a human pass instead.

Chapter 11: Thermodynamics

“ $\Delta Q = nCP \Delta T = \dots \approx 5.82 \text{ kJ}$ ”: answer computed with the wrong number of moles

Printed: “3. $\Delta Q = nCP \Delta T = (0.5)(7/2 \times 8.314)(200) \approx 5.82 \text{ kJ}$; $W = nR\Delta T = (0.5)(8.314)(200) \approx 831 \text{ J}$; $\Delta U = nCV \Delta T = (0.5)(5/2 \times 8.314)(200) \approx 4.16 \text{ kJ}$ ” followed by a confused self-correcting aside attempting to reconcile the numbers.

Should read: “3. $\Delta Q = nCP \Delta T = (0.5)(7/2 \times 8.314)(200) \approx 2.91 \text{ kJ}$; $W = nR\Delta T = (0.5)(8.314)(200) \approx 831 \text{ J}$; $\Delta U = nCV \Delta T = (0.5)(5/2 \times 8.314)(200) \approx 2.08 \text{ kJ}$ (check: $\Delta U + W = 2.08 + 0.83 = 2.91 \text{ kJ} = \Delta Q$).”

Why: For $n = 0.5 \text{ mol}$ (as the problem states), $CP = 7/2 R = 29.099 \text{ J/(mol}\cdot\text{K)}$ and $CV = 5/2$

$R = 20.785 \text{ J/(mol}\cdot\text{K)}$. $\Delta Q = 0.5 \times 29.099 \times 200 = 2909.9 \text{ J} \approx 2.91 \text{ kJ}$; $\Delta U = 0.5 \times 20.785 \times 200 = 2078.5 \text{ J} \approx 2.08 \text{ kJ}$. The printed values (5.82 kJ, 4.16 kJ) are each exactly double the correct ones — the calculation was evidently run with $n = 1 \text{ mol}$ instead of the given 0.5 mol, and the printed “note” attempting to reconcile $\Delta U + W$ with ΔQ is chasing the resulting inconsistency rather than the actual arithmetic error.

Kinetic theory degrees-of-freedom table: CO₂ misdescribed as “bent”

Printed: “Polyatomic (nonlinear, e.g. CO₂ bent / H₂O) 6 3R ≈ 24.9 4R ≈ 33.3 4/3 ≈ 1.33 ”

Should read: “Polyatomic (nonlinear, e.g. H₂O / SO₂ / NH₃) 6 3R ≈ 24.9 4R ≈ 33.3 4/3 ≈ 1.33 ”

Why: CO₂ is a linear triatomic molecule (O=C=O), not bent, and so does not belong in the “nonlinear polyatomic” row at all — at ordinary temperatures it has 5 active degrees of freedom (3 translational + 2 rotational), putting it with the diatomic/linear gases, not the $f=6$ nonlinear ones. The table’s $f=6$ row is itself correct; only the CO₂ example is wrong.